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MEETING ABSTRACTS

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January 23–25, 2025

Moscone West | San Francisco, CA | Online

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Editor: Ryan Gentzler, MD

Managing Editor: Krystal Ingram

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The abstracts contained within are published as an online supplement to *Journal of Clinical Oncology* (*JCO*) and carry *JCO* citations. The format for citation of abstracts is as follows: J Clin Oncol 43, 2025 (suppl 4; abstr 1).

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1 Poster Session

Trends in incidence and survival in anal adenocarcinoma: A SEER study of racial and age disparities. First Author: Supriya Peshin, Norton Community Hospital, Norton, VA

Background: Anal adenocarcinoma, a rare but increasingly diagnosed malignancy, has demonstrated rising incidence rates, warranting a deeper analysis of its epidemiology and outcomes across demographic factors like race and age. **Methods:** A retrospective analysis of data from SEER database (2000-2020) was conducted. Incidence rates per 100,000 individuals and 5-year survival outcomes were compared across racial groups (White, Black, Asian, Hispanic) and age categories (≤50 years, 51-70 years, >70 years). Additionally, trends were analyzed across two decades: 2000-2010 and 2010-2020. This study investigates epidemiological trends of anal adenocarcinoma, focusing on variations in incidence and survival outcomes by race and age, with specific comparisons between the periods 2000-2010 and 2010-2020. **Results:** Incidence of anal adenocarcinoma increased from 0.7 per 100,000 in 2000 to 1.6 per 100,000 in 2020. From 2000 to 2010, incidence increased rose 0.7 to 1.0 per 100,000, and from 2010 to 2020, it rose from 1.0 to 1.6 per 100,000. Rates were highest among Black individuals, increasing from 1.5 per 100,000 (2000-2010) to 2.3 (2010-2020), followed by White (1.2 to 1.5), Hispanic (0.8 to 1.0), and Asian populations (0.3 to 0.5). Age-related outcomes showed that in 2000-2010, patients aged ≤50 years had a 5-year survival rate of 75% versus 55% in age 51-70 versus 35% for >70 years. In 2010-2020, these rates dropped to 70%, 50%, and 30% in respective age groups. Racial disparities in survival were evident: Black patients had a 5-year survival rate of 40% (2000-2010) and 35% (2010-2020), significantly lower than White patients, who had rates of 65% (2000-2010) and 60% (2010-2020). **Conclusions:** The incidence of anal adenocarcinoma has steadily increased from 2000 to 2020, particularly among the Black population, while survival rates have declined. Additionally, survival rates are lower in individuals older than 70 years. Highlighting disparities in incidence and outcomes, further research is essential to uncover the underlying causes and develop targeted interventions to mitigate risks and improve patient outcomes. Keywords: anal adenocarcinoma, epidemiology, race, age, survival rates, disparities. Research Sponsor: None.

This table summarizes the key trends in anal adenocarcinoma incidence and survival by race and age over the two decades from 2000 to 2020. The increases in incidence, particularly among Black individuals, and the worsening survival rates, especially in older and Black populations, are notable findings from the study.

Category	2000-2010	2010-2020	Change (2000-2020)
Incidence per 100,000 (Overall)	0.7 → 1.0	1.0 → 1.6	+0.9
Incidence by Race			
- White	1.2 → 1.5	1.5 → 1.5	+0.3
- Black	1.5 → 2.3	2.3 → 2.3	+0.8
- Hispanic	0.8 → 1.0	1.0 → 1.0	+0.2
- Asian	0.3 → 0.5	0.5 → 0.5	+0.2
5-Year Survival by Age			
- ≤50 years	75%	70%	-5%
- 51-70 years	55%	50%	-5%
- >70 years	35%	30%	-5%
5-Year Survival by Race			
- White	65%	60%	-5%
- Black	40%	35%	-5%

2 Poster Session

Trends and disparities in palliative care utilization among patients with anal cancer: A ten-year retrospective study. First Author: Ayobami Gbenga Olafimihan, John H. Stroger, Jr. Hospital of Cook County, Chicago, IL

Background: Anal cancer patients often experience high symptom burden and psychosocial distress. Early integration of palliative care (PC) in their management can improve health-related quality of life. We examined the trends and predictors of PC utilization among hospitalized anal cancer patients in the US. **Methods:** A retrospective longitudinal study using the NIS database (2010-2019) was conducted. Using joinpoint regression and multivariable logistic regression, trends and factors associated with PC receipt were assessed. **Results:** The overall prevalence of PC utilization in the cohort of about 70,000 admissions with anal cancer was 6.9%. PC consultations increased from 3,372 to 11,081 per 100,000 anal cancer hospitalization (p-trend <0.001) with an average annual percentage change of 12.8%. Individuals ≥ 60 years with anal cancer had lower odds (Adjusted odds ratio (AOR): 0.81; 95% CI: 0.68-0.97) of receiving PC compared to their counterparts < 60 years. Patients in the second (AOR: 1.31; 95% CI: 1.05-1.62), and fourth (AOR: 1.29; 95% CI: 1.01–1.64) median household national income quartiles had about 30% greater likelihood of utilizing PC relative to those in the first income quartile. Patients in teaching hospitals had 40% higher likelihood (AOR: 1.40; 95% CI: 1.14-1.71) of PC in comparison to patients in non-teaching hospitals. Relative to patients who had a routine discharge home or with self-care, those discharged to facilities or with home health care were five-fold more likely (AOR: 5.04; 95% CI: 4.09-6.20) to receive PC. Those who died during hospitalization were also more likely to utilize PC (AOR: 33.9; 95% CI: 24.9-46.1). Other factors associated with PC receipt were non-elective admissions and higher Comorbidity burden. **Conclusions:** Though the trends in PC utilization have improved over the years, it remains suboptimal. Older patients were less likely to receive PC. Targeted interventions are needed to narrow the identified disparities for optimal utilization of PC in this patient population. Research Sponsor: None.

Factors predicting the use of palliative care among hospitalized anal cancer patients.[a]

		Unadjusted Odds Ratio (95% CI)	Adjusted Odds Ratio (95% CI)
Age	*"60 years and above "vs "Less than 60"*	0.98 (0.86-1.12)	**0.81 (0.68-0.97)**
Hospital region	*South vs Northeast*	1.04 (0.84-1.29)	**1.33 (1.04-1.69)**
	West vs Northeast	**1.28 (1.02-1.61)**	**1.39 (1.06-1.82)**
Hospital Teaching Status	*Teaching vs Nonteaching*	**1.36 (1.16-1.60)**	**1.40 (1.14-1.71)**
Patient disposition	*Transfer to facility/home health care vs Routine discharge*	**4.98 (4.14-5.99)**	**5.04 (4.09-6.20)**
	Died vs Routine discharge	**39.4 (30.2-51.4)**	**33.9 (24.9-46.1)**
Admission type	*Non-elective vs Elective*	**2.97 (2.39-3.68)**	**2.91 (2.28-3.70)**
Median Household income national quartiles	*Quartile 2 vs Quartile 1*	1.07 (0.90-1.29)	**1.31 (1.05-1.62)**
	Quartile 4 vs Quartile 1	1.17 (0.96-1.42)	**1.29 (1.01-1.64)**

[a]CI = confidence intervals.

3 Poster Session

Trends in surgery and overall survival in non-metastatic anal cancer: A population-based analysis. First Author: Cathy Eng, Department of Medicine, Division of Hematology and Oncology, Vanderbilt Ingram Cancer Center, Nashville, TN

Background: The incidence of anal cancer has recently exceeded > 10,000 patients/year in the United States. For non-metastatic squamous cell carcinoma of the anal canal (SCCA), concurrent chemoradiotherapy (CRT) followed by abdominoperineal resection (APR) as salvage surgery for non-responders is a key treatment approach. Over the past two decades, findings from phase III trials (RTOG 98-11 and ACT II) have increased our knowledge about appropriate treatment approaches. ACT II indicated that delayed assessment of tumor response and consideration of APR up until 26 weeks is appropriate, as many patients demonstrate delayed response to treatment. Our analysis examines whether current evidence has influenced real-world clinical practice by analyzing trends in rates of primary APR and overall survival (OS) outcomes. **Methods:** We conducted a retrospective cohort study using data from the Surveillance, Epidemiology, and End Results (SEER) registry, including patients diagnosed with non-metastatic SCCA between 2004 and 2020. Patients with T0 or T1N0 stage disease were excluded. Descriptive statistics were generated, and chi-square tests were used to compare characteristics between patients who underwent surgery and those who did not. Temporal trends in surgery and radiation therapy were analyzed using the Cochrane-Armitage trend test. Multivariable logistic regression was employed to identify factors associated with surgery uptake, while OS trends were analyzed using Kaplan-Meier and Cox proportional hazard models. **Results:** A total of 16,718 patients were included, with 33.1% requiring salvage APR following CRT. The proportion of patients requiring surgery significantly declined from 46.6% in 2004 to 31.1% in 2020 ($p < 0.001$). Factors associated with surgery included younger age, male gender, and non-Hispanic Black race ($p < 0.001$). Kaplan-Meier analysis revealed significant improvements in OS over the study period (log-rank p = 0.0002), with more recent years (2016-2020) associated with significantly better OS compared to earlier periods (2004-2007, HR = 0.77, $p < 0.001$). **Conclusions:** The significant decrease in rates of salvage APR for non-metastatic SCCA likely reflects the global impact of recent phase III trials. Meanwhile, OS rates have been steadily improving. These findings suggest improved overall knowledge regarding treatment strategies and highlight the real-world impact of clinical investigation in non-metastatic squamous cell carcinoma of the anal canal. Research Sponsor: None.

4 Poster Session

Preliminary phase 1/1b dose expansion results of the bifunctional EGFR/TGFβ inhibitor ficerafusp alfa (BCA101) with pembrolizumab in patients with squamous cell carcinoma of the anal canal. First Author: Van K. Morris II, The University of Texas MD Anderson Cancer Center, Houston, TX

Background: Anti-PD-1 monotherapy has limited efficacy in patients (pts) with advanced squamous cell cancer of the anal canal (SCAC). In KN-158, pembrolizumab (P) was reported to have an ORR of 10%, DCR of 25%, and a 12-month PFS rate of 15% in patients with incurable SCAC. TGFβ promotes immunosuppression within the tumor microenvironment, and inhibition of this pathway may improve tumor susceptibility to immune checkpoint blockade.Ficerafusp alfa (F)is a bifunctional EGFR/TGFβ inhibitor with manageable safety and favorable preliminary anti-tumor efficacy in advanced solid tumors (ESMO 2022 731MO). We evaluated the combination of F+P in pts with chemotherapy-refractory locally advanced/unresectable or metastatic SCAC. **Methods:** This single-arm, multicenter dose expansion from an ongoing phase I/Ib trial enrolled pts with ICI-naïve SCAC that was locally advanced/unresectable or metastatic, who had received 1-2 prior lines of chemotherapy. Pts received ficerafusp alfa (1500 mg IV on days 1, 8, 15) and pembrolizumab (200 mg IV on day 1) every 21-day cycle. The primary endpoint was safety (CTCAE v5), and secondary endpoints were radiographic response (RECIST 1.1), duration of response, progression-free survival, and overall survival. **Results:** Among 26 pts treated, 21 (81%) were female, and median age was 61 (range: 43-82). As of 7/2024, 22 pts are evaluable for efficacy. ORR was 32% (95% CI 13.9-54.9%). Six of the seven responses were confirmed, including one pt with a radiographic PR with pathologic CR to F+P. Seven pts (32%) achieved SD as best response, resulting in DCR of 64%. Median PFS was 2.8 months (95% CI 1.4-14.8) and the PFS rate at 12 months was 36%. G3 treatment-related adverse events were observed in 10/26 pts (38%, most common: Anemia (19%) and acneiform rash (12%)). TRAEs led to permanent discontinuation of study treatment in 2 pts (8%). The most common TRAEs of any grade were acneiform rash (13/26, 50%), epistaxis (46%) and headache (38%). **Conclusions:** The addition of ficerafusp alfa demonstrated encouraging efficacy, with increased ORR, DCR, and 12-month PFS relative to historical precedent with pembrolizumab alone. Safety and tolerability were acceptable. Further investigation of this combination in pts with advanced SCAC is warranted. Updated data will be presented. Study funded by Bicara Therapeutics Inc. with access to pembrolizumab in collaboration with Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA (NCT04429542). Clinical trial information: NCT04429542. Research Sponsor: Bicara Therapeutics.

5 Poster Session

Definitive local therapy in solitary liver metastatic anal squamous cell carcinoma. First Author: Teerada Siripoon, Mayo Clinic Rochester, Rochester, MN

Background: Metastatic anal squamous cell carcinoma is rare and associated with a poor prognosis. There is limited evidence on the impact of local interventions on survival outcomes in patients with metastatic disease. We aimed to evaluate the survival of patients with liver-only metastatic anal cancer following definitive local management of liver lesions. **Methods:** This is a single-institution retrospective cohort study of patients with liver-limited metastatic anal cancer who underwent curative-intent local therapy. Clinical characteristics were collected and analyzed. The Kaplan-Meier method was used to calculate disease-free survival (DFS) and overall survival (OS). Prognostic factors associated with DFS and OS were assessed using Cox-proportional hazards models. **Results:** Twenty-five patients were included, median age was 59 years, and 92% were female. None of the patients had known HIV infection, 92% had HPV-positive disease and 76% had metachronous liver metastases following locoregional disease. All patients received chemoradiation for the primary lesion Unilobar hepatic involvement was present in 76% of the patients, with 56% having a single metastatic lesion. Pre-intervention systemic therapy was provided to 52% of the patients, including 8 who received carboplatin and paclitaxel, with a median duration of 3 months and ORR of 69%. Definitive treatment of metastatic disease included surgical resection (60%), non-surgical interventions (20%), and multimodal therapy (20%). All patients had viable tumor in the pathological specimen. With a median follow-up of 22 months, median DFS was 7.3 months, and median OS was 51.3 months. An ECOG performance status of 2, poorly differentiated histology, and extrahepatic recurrence were identified as significant prognostic factors in the univariate analysis, but not in the multivariate analysis. **Conclusions:** Our results demonstrate potential for long-term survival in liver-limited metastatic anal squamous cell carcinoma amenable to local therapeutic intervention combined with systemic therapy. These findings warrant further prospective study of local therapy in oligometastatic anal cancer. Research Sponsor: None.

Characteristics	N=25 (%)
Median number of liver metastases (range), lesions	1 (1-20)
- 1	14 (56)
- 2-3	8 (32)
- >3	3 (12)
Median maximal diameter of liver metastases (range), cm	3 (0.9-11.9)
Best response of liver metastases to chemotherapy (N=13)	
- Complete response	2 (15)
- Partial response	7 (54)
- Stable disease	3 (23)
- Progressive disease	1 (8)
Surgical procedure (N=20)	
- Right/left hemi-hepatectomy	4 (20)
- Segmentectomy	14 (70)
- Wedge resection	2 (10)
Non-surgical procedure (N=10)	
- Local ablation	5 (50)
- Radiotherapy	3 (30)
- Other	2 (20)
Hepatic surgical margins (N=20)	
- Positive margins	0
- Close margins (<1mm)	2 (10)
- Negative margins	18 (90)
Recurrent disease	
- Yes	20 (80)
o Intrahepatic	14 (70)
o Extrahepatic	6 (30)
- No	5 (20)

6 Poster Session

GOBLET platform study: Preliminary safety and tumor response results for the relapsed anal carcinoma cohort in patients treated with pelareorep and atezolizumab. First Author: Dominik Paul Modest, Department of Hematology, Oncology and Cancer Immunology, Charité-Universitätsmedizin Berlin, Berlin, Germany

Background: Patients with relapsed unresectable squamous cell carcinoma of the anal canal (SCCA) have limited treatment options after failure of standard first-line therapy. While checkpoint inhibitor monotherapy is possible, response rates are low (10-24%). Therapies that induce an inflammatory tumor microenvironment may improve the susceptibility of SCCA to immunotherapy. Pelareorep (pela) is a non-genetically modified, intravenously administered reovirus that stimulates the expansion of tumor-infiltrating lymphocyte (TIL) clones and simultaneously makes tumors visible to the immune system by upregulating interferon-induced gene expression. Pela-based combination therapies have demonstrated activity in metastatic breast and pancreatic cancer. In the GOBLET platform study, the efficacy of pela plus atezolizumab (atezo) in SCCA is being evaluated. **Methods:** GOBLET is a phase 1/2, open-label, non-randomized, Simon two-stage platform study in patients with advanced or metastatic GI cancers. SCCA patients in the trial are treated with pela plus atezo. Ten evaluable patients will be enrolled in Stage 1 and an additional 18 evaluable patients will be enrolled in Stage 2 if the prespecified Stage 1 success criterion is met ($\geq$2 responses). The coprimary endpoints are tolerability of the treatment regimen and objective response rate (based on RECIST v1.1). Safety data are reviewed in an ongoing manner by an independent Data Safety Monitoring Board (DSMB). In addition, tumor and blood samples are being collected to assess TIL clonal expansion and other biomarkers. **Results:** Twelve patients are currently evaluable for tumor response in the SCCA cohort. Four out of 12 patients have achieved an objective response to therapy including 1 prolonged complete response ($>$15 months) and 3 partial responses- resulting in an ORR of 33.3%. Final ORR, progression-free survival, and overall survival results are pending. The DSMB has identified no safety signals and has recommended that enrollment continues without modification. TIL clonal expansion in the blood has been observed at treatment cycle 4 in all 3 responding patients for whom data are available. **Conclusions:** Twelve patients are currently evaluable for tumor response in the SCCA cohort. Four out of 12 patients have achieved an objective response to therapy including 1 prolonged complete response ($>$15 months) and 3 partial responses resulting in an ORR of 33.3%. Final ORR, progression-free survival, and overall survival results are pending. The DSMB has identified no safety signals and has recommended that enrollment continues without modification. TIL clonal expansion in the blood has been observed at treatment cycle 4 in all 3 responding patients for whom data are available. Clinical trial information: 2020-003996-16. Research Sponsor: None.

7 Poster Session

A single institution retrospective study evaluating mitomycin dosing during chemoradiation in anal squamous cell carcinoma. First Author: Jenny Gong, University Hospitals, Case Western Reserve University, Cleveland, OH

Background: Mitomycin (MMC) is used concurrently during chemoradiation (CRT) treatment for non-metastatic anal squamous cell carcinoma (SCCA). The most effective dosing of MMC is not established, and due to concerns for tolerability, many have adopted a dose-reduction strategy. This study evaluates differences in treatment response and recurrence between MMC dosing in this setting. **Methods:** Demographics, age, smoking history, staging, and treatment history were collected for patients with non-metastatic SCCA treated with CRT from January 2011-September 2024 at a tertiary cancer center. Chi-square analyses compared categorical variables, and ANOVA tests were used to compare treatment response and recurrence between 3 MMC doses (12mg/m^2 x 1 dose with cap of 20mg, 10mg/m^2 x 2 doses, 10mg/m^2 x 1 dose). **Results:** There were 124 patients, 6 of whom were excluded due to lack of follow up. Of the 118 included, 70% were female, 16% Black, 3% Hispanic/Latino, and 58% current/former smokers. Median age was 61 years (31-98). Thirty-three (28%) pts were treated with MMC 12mg/m^2, 71 (60%) with 10mg/m^2 x2 doses, and 14 (12%) with 10mg/m^2 x1 dose. Stage I included 18% of 12mg/m2, 7% of 10mg/m^2 x2 doses, and 7% of 10mg/m^2 x1 dose; stage II included 18% of 12mg/m2, 37% of 10mg/m^2 x2 doses, and 50% of 10mg/m^2 x1 dose; stage III included 64% of 12mg/m2, 56% of 10mg/m^2 x2 doses, and 43% of 10mg/m^2 x1 dose. A total of 39.7% were treated with MMC/capecitabine and 60.3% with MMC/fluorouracil. Complete response was achieved in 91% receiving the 12mg/m^2, 75% receiving 10mg/m^2 x2 dose, and 71% receiving 10mg/m^2 x1 dose (p=0.13, Table 1). Using logistic regression with MMC dose, stage, age, and smoking status as independent predictors, the odds of achieving complete response were lower in 10mg/m^2 x2 doses (p=0.03) and 10mg/m^2 x1 dose (p=0.03) compared to 12mg/m^2. Local, regional, or metastatic recurrence occurred in 15% of pts receiving 12mg/m^2, 17% of 10mg/m^2 x2 doses, and 7% of 10mg/m^2 x1 dose. No significant difference was seen in local (p=0.52), regional (p=0.93), metastatic (p=0.87), or overall recurrence (p=0.65) between the MMC doses. **Conclusions:** Odds of achieving complete response was greater with 12mg/m^2 dosing. Local, regional, and metastatic recurrence were similar among the different MMC dosing cohorts. Differences in tolerance and side effects should be delineated to guide individualized dosing of MMC. Research Sponsor: None.

8 Poster Session

Retrospective study evaluating the genomic landscape of anal squamous cell carcinoma using liquid biopsy. First Author: Madison Conces, Department of Oncology, University Hospitals Seidman Cancer Center, Case Western Reserve University, Cleveland, OH

Background: Advanced-stage squamous cell carcinoma of the anus (aSCCA) is a rare malignancy. Diagnostic biopsies are often inadequate for tissue-based genomic profiling, thus genomic profiles are largely unexplored in this setting. We assessed liquid biopsy NGS results of aSCCA patients (pts) and described genomic profiles by age and sex. To our knowledge, we are the first to describe genetic alterations by liquid biopsy in this patient population. **Methods:** Pts with aSCCA who underwent circulating tumor DNA (ctDNA) NGS with Guardant360 assays from 2017 to 2024 were included. Frequency of alterations were assessed based on sex and age (18-49 years (yrs), $\geq$50 yrs). Co-occurrence patterns were assessed using cBioPortal. Demographics were extracted from test requisition forms. Statistical analyses via two sided t-tests were performed with significance defined as $p<0.05$. **Results:** 646 pts with aSCCA had non-synonymous ctDNA alterations detected; 69.2% (N=447) were female and 30.8% (N=199) male. Median age was 66 yrs (32-94) with 9.9% (N=64) $<$50 yrs and 90.1% (N=582) $\geq$50 yrs. Most frequently altered genes were *PIK3CA* (17.3%), *TP53* (9.8%), *ATM* (7.8%), *EGFR* (4.3%), and *BRCA2* (4.2%). More unique genes were altered in $\geq$50 yrs cohort [53 mutations, 38 genes with copy number amplification (CNA)] vs $<$50 yrs cohort (30 mutations, 66 CNA). Most frequent genes altered in $<$50 yrs cohort were *TP53* (17.6%), *PIK3CA* (13.2%), *EGFR* (11.0%), *APC* (4.4%), and *MET* (4.4%). *PIK3CA* (16.7%), *TP53* (12.2%), *ATM* (8.8%), *BRCA2* (5.5%), and *EGFR* (3.4%) were most frequent in $\geq$50 yrs cohort. More genes were altered in females (64 mutations, 32 CNA) vs. males (56 mutations, 41 CNA). Most frequent alterations in females were *PIK3CA* E545K (3.4%), *PIK3CA* CNA (2.8%), *PIK3CA* E542K (1.9%), and *ATM* CNA (1.0%) vs in male pts being *PIK3CA* E545K (4.8%), *PIK3CA* CNA (3.3%), *PIK3CA* E542K (3.3%), *EGFR* CNA (1.6%), *TERT* promoter (1.0%). Overall, *PIK3CA* and *TP53* alterations were mutually exclusive (odds ratio 0.43, $p<0.001$). **Conclusions:** Liquid biopsy is feasible in patients with aSCCA. Most frequently detected alterations and CNAs were in *PIK3CA* and *TP53* in aSCCA which is consistent with prior studies evaluating aSCCA tumor NGS in tissue-based assays. Alterations varied across age and sex. *PIK3CA* E545K and E542K alterations were detected frequently in this population and have on-label therapies for non-aSCCA indications which may be a source for clinical consideration in this cancer type in the future. This data illustrates the importance of liquid biopsy NGS to also screen for clinical trial enrollment in a disease which is poorly responsive to standard therapy. Additional research is warranted to further elucidate genomic profiles for clinical applications of this rare tumor. Research Sponsor: None.

9 **Poster Session**

Biomarkers associated with recurrence after salvage surgery for localized anal cancer (AC). First Author: Mir Lim, The University of Texas MD Anderson Cancer Center, Houston, TX

Background: While most patients with localized AC are cured by chemoradiation (cRT), some recur and require salvage surgery (SS). No adjuvant therapies are effective after SS. We profiled cRT-refractory AC tumors from SS to identify biomarkers linked to recurrence. **Methods:** Using an IRB-approved protocol, patients with localized, cRT-refractory AC were classified as recurrent (R; N=19) or non-recurrent (NR; N=10) according to outcome after SS. Digital spatial profiling whole transcriptome analysis (GeoMx) on FFPE AC evaluated expression of 18000+ genes on 40 regions of interest (ROI) for R vs 25 ROIs for NR AC cores, separated into "tumor" (panCK+) and "tumor microenvironment" (TME; panCK-). log2 transformed RNA counts were compared for expression between ROIs of R and NR AC (limma). For GSEA hallmark pathway analysis, normalized enrichment scores (NES) were calculated using genes with differential gene expression. Expression of immune biomarkers using multiplex immunofluorescence (mIF) were quantified (N/mm2) and compared with a student's t-test. **Results:** With a median follow up of 26.6 months, median OS (29.1 months vs NR; $p < .001$) was shorter for R AC. For NR AC, higher expression of genes associated with immune activation (p-adj$<$ .0001 for all) - B2M (fold change (FC) 33.0), HLA-DPA1 (FC 30.1), and HLA-DPB1 (FC 15.0) – and higher NES for "interferon alpha response" (NES 3.2) and "interferon gamma response" (NES 3.1) (FDR $<$ .05 for both) were observed. R AC had higher expression of genes (p-adj $<$ .0001 for all) regulating TGF-beta signaling – LY6K (FC 10.3) and BCAR3 (FC 3.6), and higher "EMT" (NES 2.1) and "TGF-beta" (NES 1.6) GSEA signatures (FDR$<$ .05 for both). On mIF, greater CD3+CD8+ T cells were observed in tumor (148 vs 42 N/mm2; p=.02) and TME (8.2 vs 2.8 N/mm2; p=.008) for NR AC. Increased Thy1+ cancer-associated fibroblasts were noted in tumor (4.2 vs 1.2 N/mm2; p=.04) and in TME (149 vs 69 N/mm2; p=.04) for R AC. **Conclusions:** Signatures of immune activation were observed with gene expression and mIF in AC patients cured by SS. R AC had higher immune suppressing profiles. These data support developing novel therapeutics that promote immune activation in trials of patients with cRT-refractory AC at high risk for recurrence after SS. Research Sponsor: None.

10 **Poster Session**

Tumor tissue modified viral (TTMV)-HPV DNA as a biomarker of treatment efficacy for definitive chemoradiation in anal squamous cell carcinoma: A step towards truly personalized therapy. First Author: Shubhi Agarwal, Mayo Clinic Florida, Jacksonville, FL

Background: Cell-free circulating tumor tissue modified viral DNA (ctDNA) has been used as a biomarker to detect recurrence for anal squamous cell carcinoma (ASCC). However, its value in informing treatment modification during definitive chemoradiation (CRT) has yet to be explored. We evaluated ctDNA response during and at completion of definitive CRT to establish patterns of disease response and inform future studies to personalize treatment. **Methods:** From 11/2022 to 6/2024, 13 consecutive patients with biopsy proven, non-metastatic ASCC received definitive CRT and had ctDNA testing using a commercially available droplet digital quantitative PCR assay. ctDNA testing was done at baseline (pre-CRT), week 4 of CRT, and/or end of CRT, 1-3 months post-CRT, and every 6 months thereafter. CRT included concurrent 5FU/MMC (or capecitabine/MMC) and radiotherapy according to RTOG 0529. Treatment response assessment included: physical exam, anoscopy, and imaging with PET/CT, CT C/A/P +/- MRI abdomen and pelvis. **Results:** 13 patients had ctDNA testing and 11 patients had HPV-positive ASCC and tested positive for ctDNA. The 11 patients with detectable baseline ctDNA were included in this analysis. The median age at diagnosis was 63 years old. Staging included: 1- Stage IIA, 7- Stage IIB, 1- Stage IIIA, 2- Stage IIIC. 1 patient was cN0, while 11 were cN1. With a median follow-up from completion of CRT of 13 months, 4 patients have persistent or recurrent disease: 2 with distant metastases, 1 with persistent local disease (1 month post-CRT), and 1 with local and distant recurrence. Baseline value of ctDNA did not correlate to stage at diagnosis, rate of clearance during CRT, or likelihood of recurrence. 5 of 11 patients cleared ctDNA by week 4 of CRT. Of 5 patients who cleared ctDNA by week 4 of CRT, 4 remain NED and 1 developed distant metastases. Of 6 patients with persistent ctDNA at week 4 of CRT, 3 have persistent or recurrent disease- 1 local persistence 1 month post-CRT, 1 local and distant recurrence 6 months post-CRT, and 1 with distant recurrence 12 months post- CRT. Among 6 patients who cleared ctDNA by the final week of CRT, 1 patient developed recurrent disease. Among the 4 patients in the series with residual/recurrent disease, 3 of them had not cleared ctDNA by the end of CRT. Of 10 patients with ctDNA testing at 1 month post-CRT, 8 had no detectable ctDNA. 2 patients with persistent (1) or recurrent (1) ctDNA at 1 month post-CRT developed disease recurrence- 1 with distant recurrence and 1 with local and distant recurrence. **Conclusions:** ctDNA assessment during CRT may provide insight into disease response that can predict patterns of failure and facilitate personalized therapy- treatment intensification, deintensification, or additional testing to detect early metastatic disease. Larger studies and clinical validation are needed. Research Sponsor: None.

11 **Poster Session**

Personalized oncogenomic analysis of metastatic squamous cell carcinoma of the anus: Utilizing whole-genome sequencing to guide clinical decision-making. First Author: Renata D'Alpino Peixoto, BC Cancer Agency, Vancouver, BC, Canada

Background: Metastatic squamous cell carcinoma of the anus (SCCA) is associated with significant morbidity and mortality. Due to its relative rarity, there is limited evidence to support systemic therapy regimens informed by genomic data. This study aims to utilize whole-genome and transcriptome sequencing to enhance the understanding of the genetic alterations driving metastatic SCCA and to identify potentially actionable therapeutic targets. **Methods:** This report examines nine cases of metastatic SCCA in patients enrolled in the Personalized Oncogenomics (POG) Program at the BC Cancer Agency. Comprehensive genomic profiling, including whole-genome and transcriptome analysis (WGTA) was conducted on fresh tumor biopsy or formalin fixed paraffin embedded (FFPE) tissue. Tumor genomic alterations were analyzed in detail including: SNVs, indels, copy number alterations, structural variants, mutation signatures, viral presence, tumor mutation burden (TMB), expression outliers, and immune cell scores. The somatic alterations were integrated with existing knowledge of drug-target interactions to identify actionable therapeutic targets. **Results:** Of the nine patients, eight were found to be HPV-positive, with one exhibiting high tumor mutational burden (TMB$>$10mut/Mb). Mutation signatures included seven cases with AID/APOBEC signatures (commonly seen with HPV), and one case with an ID2 DNA replication slippage signature. The PI3K/AKT/mTOR pathway was the most commonly affected signaling pathway (n=4), followed by FGFR amplification and overexpression (n=3), and RAS/RAF alterations (*BRAF* mutation and *KRAS* overexpression, n=2). EGFR amplification was identified in two cases, one of which also exhibited *EGFR* overexpression. Additionally, *FBXW7* mutations were present in two cases. Immunotherapy was recommended for all nine patients: eight based on HPV positivity, either alone or in combination with other alterations, such as high TMB, SWI/SNF complex alterations, and presence of immune infiltrating cells, and the remaining case based on a SWI/SNF complex mutation. Only two patients ultimately received immunotherapy, both of whom derived clinical benefit. One patient, treated with atezolizumab as part of the phase II CAPTIV-8 trial (NCT04273061), demonstrated progression-free survival (PFS) of 10 months (ongoing), while the other, treated with nivolumab, achieved a PFS of 6 months. **Conclusions:** Genomic analysis using WGTA supported the recommendation of immunotherapy for all patients in this cohort. Although only two patients received immunotherapy, both experienced clinical benefit, underscoring the potential of personalized genomic-guided treatment in metastatic SCCA. Research Sponsor: None.

12 **Poster Session**

The role of sequential PET/CT imaging to predict complete response after definitive chemoradiotherapy for locally advanced squamous cell carcinoma of the anal canal. First Author: Marianna Valzano, Azienda Ospedaliero Universitaria Careggi, Florence, Italy

Background: The standard treatment for locally advanced squamous cell carcinoma of the anal canal (SCCAC) is definitive radiotherapy (RT) with concurrent chemotherapy (CRT), with excellent outcomes. Nevertheless, rates of locoregional failure of 10% to 30% have been reported. Due to the narrow therapeutic window between time-dependent response to RT and prompt salvage surgery, the definition of a surveillance strategy after radical CRT is challenging. In particular the contribution of 18-FDG PET/CT is not clearly established. The purpose of this study was to evaluate the effectiveness of follow-up MR and 18-FDG PET/CT to predict response to CRT in patients with SCCAC. **Methods:** We analyzed a cohort of consecutive patients who received definitive RT/CRT from January 2019 to February 2024. After completion of therapy, radiologic (rCR) and metabolic complete response (mCR) was assessed with pelvic MR (RECIST v.1.0) and 18-FDG PET/CT at 3 ad 6 months respectively. Local Control (LC) was defined as time from the end of CRT to local failure or last follow-up. Univariate analysis (UVA) was performed to identify variables associated with outcome using the log rank test. Sensitivity and Specificity analysis was performed to assess the correlation between rCR/mCR and LC. **Results:** A total of 67 patients (median age 67 years, range 44-83) were included. All patients completed RT, and 52 received concurrent CRT. Baseline and surveillance MR and 18-FDG PET/CT were available for 49 patients. After a median follow-up of 28 months, 12 patients experienced disease progression (PD) (10 local; 2 distant), resulting to salvage surgery in 10 cases after biopsy. At 3 years, LC rate was 78%. At UVA, only mCR was correlated with LC (3-year LC 90% versus 59%, p=0,0137; HR 0,01660, IC 95% 0,03-0,84). For prediction of LC, accuracy of mCR at 6 months was 83.7% versus 62.5% for rCR. Specificity/Sensitivity analysis is summarized in the table. **Conclusions:** 18FDG PET/CT at 6 months may identify patients with a higher likelihood of developing complete tumor regression following CRT. Further data is needed to confirm the appropriate surveillance strategy for patient undergoing CRT in clinical practice. Research Sponsor: None.

Specificity and sensitivity analysis.

	Sensitivity	Specificity	Positive Predictive Value	Negative Predictive Value
mCR	83.3%	84.0%	92.6%	68.7%
rCR	25.0%	70.0%	14.0%	82%

LBA13 **General Session**

A randomized phase 2 study of an individualized neoantigen-targeting immunotherapy in patients with newly diagnosed metastatic microsatellite stable colorectal cancer (MSS-CRC). First Author: Joel R Hecht, David Geffen School of Medicine at UCLA, Jonsson Comprehensive Cancer Center, Santa Monica, CA

The full, final text of this abstract will be available at meetings.asco.org on the day of presentation and in the online supplement to the February 1, 2025, issue of the *Journal of Clinical Oncology*.

LBA14 **Oral Abstract Session**

Prognostic and predictive role of circulating tumor DNA (ctDNA) in stage III colon cancer treated with celecoxib: Findings from CALGB (Alliance)/SWOG 80702. First Author: Jonathan Andrew Nowak, Brigham and Women's Hospital, Boston, MA

The full, final text of this abstract will be available at meetings.asco.org on the day of presentation and in the online supplement to the February 1, 2025, issue of the *Journal of Clinical Oncology*.

15 **Oral Abstract Session**

Circulating tumor DNA for detection of molecular residual disease (MRD) in patients (pts) with stage II/III colorectal cancer (CRC): Final analysis of the BESPOKE CRC sub-cohort. First Author: Purvi K. Shah, Virginia Cancer Institute, Richmond, VA

Background: BESPOKE CRC, a multicenter, prospective, observational study investigated the clinical utility of ctDNA for detecting MRD-based early recurrence in pts with surgically resected CRC. The primary endpoint was to assess the impact of ctDNA testing on treatment decisions and asymptomatic recurrence rates. The secondary endpoint was to assess the MRD clearance rate, survival of MRD-negative pts, overall survival, and patient-reported outcomes. **Methods:** Complete clinical and laboratory data were available for 1001 pts with stages II–III CRC. Longitudinal ctDNA testing was performed prospectively using a clinically validated, personalized, tumor-informed 16-plex mPCR-NGS assay (Signatera, Natera, Inc.). Plasma time points (n=8,536) were collected during the MRD (2-12 weeks postoperatively) and surveillance (post-adjuvant chemotherapy [ACT] completion/12 weeks postoperatively for pts observed) windows. We evaluated the correlation between ctDNA status and disease-free survival (DFS) as part of exploratory analysis. **Results:** Following curative resection, 62.4% (625/1001) pts received ACT: 25.9% (115/443) stage II, and 91.3% (510/558) stage III.Among pts with ctDNA results available during the MRD window with a median follow-up of 23.15 (range: 2.89-33.45) months, ctDNA-positivity was observed in 8.1% (34/420) of stage II and 24.9% (126/505) of stage III pts. A significant association between ctDNA positivity and inferior DFS was observed in stage II (HR=10.4; $p<0.0001$), and stage III (HR=10.1, $p<0.0001$) pts. 18/24 month DFS estimates for stages II-III combined were: (MRD-negative: 93.0%/91.7%; MRD-positive: 44.4%/41.4%). Analysis of surveillance was performed separately for pts observed (N=368) vs. ACT-treated (N=597). During surveillance, in the observation cohort, 6.8% (22/323) of stage II, and 33.3% of stage III (15/45) pts tested ctDNA-positive correlating with significantly worse DFS (stage II: HR=34.9; $p<0.0001$, stage III: HR=34.1; p=0.0008). Likewise, in the ACT cohort, 10.9% (12/110) of stage II and 21.1% of stage III (103/487) pts turned ctDNA-positive and had significantly worse DFS (stage II: HR=131.4; $p<0.0001$, stage III: HR=54.6; $p<0.0001$). Analysis of primary and secondary endpoints will be presented. **Conclusions:** ctDNA positivity was highly prognostic of DFS within MRD and surveillance windows in a subset of stages II-III pts enrolled in the BESPOKE CRC study. Our results highlight the potential value of ctDNA-based MRD detection for treatment-decision making, and findings relevant to clinical utility will be reported in the conference presentation. Clinical trial information: NCT04264702. Research Sponsor: None.

16 **Oral Abstract Session**

BREAKWATER: Analysis of first-line encorafenib + cetuximab + chemotherapy in BRAF V600E-mutant metastatic colorectal cancer. First Author: Scott Kopetz, The University of Texas MD Anderson Cancer Center, Houston, TX

Background: Encorafenib + cetuximab (EC) is approved for previously treated BRAF V600E-mutant metastatic colorectal cancer (mCRC) based on the BEACON phase 3 study (NCT02928224). Historically, first-line (1L) treatment of BRAF V600E-mutant mCRC with chemotherapy (chemo) regimens has had limited efficacy.BREAKWATER (NCT04607421) is an open-label, global, randomized, phase 3 study evaluating 1L EC with or without chemo vs standard of care (SOC; chemo with or without bevacizumab). Reported here are the primary analysis of objective response rate by blinded independent central review (ORR by BICR; dual primary endpoint [EP]), the first interim analysis of overall survival (OS; key secondary EP), other secondary EPs, and safety for the EC+FOLFOX (oxaliplatin, leucovorin, and 5-FU) vs SOC arms. **Methods:** Eligible patients (pts) had untreated BRAF V600E-mutant mCRC, measurable disease (RECIST 1.1), and ECOG PS 0-1. Pts were randomized 1:1:1 to receive EC, EC+FOLFOX, or SOC; EC arm enrollment was closed after a protocol amendment. Dual primary EPs were ORR (assessed in the first 110 pts randomized to each of the EC+FOLFOX and SOC arms) and progression-free survival by BICR (EC+FOLFOX vs SOC); OS was a key secondary EP (EC+FOLFOX vs SOC), other secondary EPs included response duration and time to response (TTR). **Results:** Four hundred seventy-nine pts were randomized to the EC+FOLFOX and SOC arms (EC+FOLFOX: n=236; SOC: n=243). Baseline demographics and disease characteristics were similar across arms (median age: 61.0 years; male: 50.5%; ECOG PS 0: 54.3%). At data cutoff (Dec 22, 2023), the EC+FOLFOX arm demonstrated a clinically meaningful and statistically significant improvement in confirmed ORR vs the SOC arm, 60.9% vs 40.0%, odds ratio=2.443, one-sided *P*-value=0.0008, meeting this dual primary EP. The response observed with EC+FOLFOX was rapid and durable. OS data were immature but indicated a sustained survival benefit with EC+FOLFOX vs SOC arm. Serious treatment-emergent adverse events (EC+FOLFOX: n=231; SOC: n=228) occurred in 37.7% vs 34.6% of pts in the respective arms. The safety profile was consistent with that known for each agent. **Conclusions:** BREAKWATER demonstrated a substantially improved response rate that was rapid and durable with EC+FOLFOX in BRAF V600E-mutant mCRC with manageable toxicities and no new safety signals. Clinical trial information: NCT04607421. Research Sponsor: Pfizer, Inc.

	EC+FOLFOX n=110	SOC n=110
ORR by BICR % (95% CI)	60.9 (51.6, 69.5)	40.0 (31.3, 49.3)
Odds ratio (95% CI)	2.443 (1.348, 4.380)	
***P*-value[a]**	0.0008	
	n=67	**n=44**
Estimated median response duration by BICR (95% CI), mo	13.9 (8.5, NE)	11.1 (6.7, 12.7)
Pts with a response duration of ≥6 mo, n (%)	46 (68.7)	15 (34.1)
Pts with a response duration of ≥12 mo, n (%)	15 (22.4)	5 (11.4)
Median TTR by BICR (range), weeks	7.1 (5.7-53.7)	7.3 (5.4-48.0)
	n=236	**n=243**
OS (95% CI), mo	NE (19.8, NE)	14.6 (13.4, NE)
Hazard ratio (95% CI)	0.47 (0.318, 0.691)	

[a]One-sided α=0.001.
NE, not estimable.

17 Oral Abstract Session

Final analysis of modified (m)-FOLFOXIRI plus cetuximab versus bevacizumab for *RAS* wild-type and left-sided metastatic colorectal cancer: The DEEPER trial (JACCRO CC-13). First Author: Akihito Tsuji, Department of Clinical Oncology, Faculty of Medicine, Kagawa University, Takamatsu, Japan

Background: The DEEPER trial (NCT02515734), which evaluated m-FOLFOXIRI (irinotecan 150 mg/m², oxaliplatin 85 mg/m², 5-FU 2400 mg/m²) plus cetuximab (cet) vs. bevacizumab (bev) as initial therapy in terms of depth of response (DpR) as the primary endpoint in *RAS* wild-type metastatic colorectal cancer (mCRC), has demonstrated a significantly better DpR in the cet arm (ASCO 2021). Moreover, favorable progression-free survival (PFS) was reported in the cet arm for patients (pts) with *RAS*/*BRAF* wild-type and left-sided tumors (ESMO 2023). Due to the small number of overall survival (OS) events, it was decided to extend the observation period, and the final survival analysis was performed at the cutoff date of August 2024. **Methods:** Survival analysis was pre-planned in the per-protocol set (PPS), which consisted of pts evaluable for the DpR by an external review board. The clinical outcomes were evaluated according to clinical factors including primary tumor sidedness, liver metastasis status, and *BRAF* status using a log-rank test. All statistical tests were two-sided, and *P* values $\leq$ 0.05 were considered significant. **Results:** 321 of 359 enrolled pts were defined as PPS (median age 65 years, 64% male, performance status [PS] 0/1: 91%/9%, left/right primary: 84%/16%). In *RAS* wild-type and left-sided tumors, median PFS and OS were 13.9 months vs. 12.1 months (HR 0.81, 95% CI 0.63-1.05) and 45.3 months vs. 41.9 months (HR 0.85, 95% CI 0.64-1.12) in the cet vs. bev arm, respectively. *BRAF* status was available in 234 (73%) of the 321 pts in the PPS. An exploratory analysis showed that PFS was significantly better in the cet arm compared to the bev arm (median 14.8 months vs. 11.9 months, HR 0.71, 95% CI 0.52-0.97) in 178 pts with *RAS*/*BRAF* wild-type and left-sided tumors. Additionally, OS was 50.2 months vs. 40.2 months (HR 0.74, 95% CI 0.53-1.05). Moreover, according to liver disease status, m-FOLFOXIRI plus cet was associated with longer PFS and OS in pts with *RAS*/*BRAF* wild-type and left-sided mCRC with extra-hepatic metastases (median 15.1 months vs. 11.4 months, HR 0.66, 95% CI 0.46-0.95 and 50.2 months vs. 38.6 months, HR 0.60, 95% CI 0.40-0.90), but not liver-limited disease (median 14.5 months vs. 15.5 months, HR 0.79, 95% CI 0.44-1.42 and 52.2 months vs. 49.9 months, HR 1.17, 95% CI 0.61-2.24). **Conclusions:** The final survival analysis of the DEEPER trial demonstrated favorable PFS and OS with m-FOLFOXIRI plus cet in pts with *RAS*/*BRAF* wild-type and left-sided mCRC. The m-FOLFOXIRI plus cet regimen may be a good option for initial therapy, offering longer survival times in pts with extra-hepatic metastases. Clinical trial information: jRCTs061180022. Research Sponsor: Merck Biopharma Co., Ltd.

18 Rapid Oral Abstract Session

Performance of a blood-based test for colorectal cancer screening adjusted to the US census age and sex distribution. First Author: Aasma Shaukat, NYU Grossman School of Medicine, New York, NY

Background: Despite availability of various screening options, nearly 40% of eligible U.S. adults were not up to date with the colorectal cancer (CRC) screening. Blood-based testing offers a promising complementary approach and may enhance patient adherence among unscreened individuals. In the PREEMPT CRC study, we evaluated clinical performance of an investigational blood-based screening test for detecting molecular signals of advanced colorectal neoplasia (ACN) in an average-risk population. Freenome blood-based CRC screening test met all primary endpoints demonstrating 79.2% sensitivity for CRC, 91.5% specificity for ACN, 90.8% negative predictive value (NPV) for ACN and 15.5% positive predictive value (PPV) for ACN. Sensitivity for advanced precancerous lesions (APLs) was 12.5%. To evaluate how the test would perform in the standard U.S. population, an analysis of test performance weighted to the U.S. census sex and age distribution was prespecified and performed. **Methods:** Between May 2020 and April 2022, the study enrolled participants aged 45-85 at average risk for CRC. Enrolled participants had a blood drawn before the bowel preparation for the standard of care screening colonoscopy (CS). CS and applicable histopathology reports underwent central pathologist review. Blood samples were processed blind to clinical findings. Primary endpoints included sensitivity for CRC, specificity for ACN, NPV for ACN and PPV for ACN. A secondary endpoint was sensitivity for APLs. **Results:** A subset of 32,731 sequentially enrolled participants were included in the clinical validation cohort. Out of those, 27,010 (82.5%) had evaluable blood samples and CS. The median age of the evaluable participants was 57.0 years, and 55.8% were women versus 50.5% in the U.S. Census. Age distribution included 11.0% participants in the 45-49 age group versus 15.3% in the U.S. Census, 33.0% in the 50-54 age group versus 15.6%, and 32.3% in the 55-64 age group versus 32.3%, 20.8% in the 65-74 age group versus 24.3%, 3.0% participants aged 75 or older versus 12.5%. After performance was weighted to match U.S. Census sex and age distributions, sensitivity for CRC was 81.1% (95% CI 71.3%-88.1%), specificity for those without ACN was 90.4% (95% CI 90.0%-90.7%), NPV for those without ACN was 90.5% (95% CI 90.5%-90.7%), and PPV for ACN was 15.5% (95% CI 13.8%-16.2%). Sensitivity for APL was 13.7% (95% CI 12.4%-15.0%). **Conclusions:** PREEMPT CRC is the largest study evaluating a CRC screening blood-based test to date. The study successfully met all primary endpoints, demonstrating acceptable clinical performance of the investigational blood-based test. The U.S. census sex-age adjusted clinical performance estimates showed improvement in CRC and APL sensitivity. This blood-based test may provide a convenient and effective option for CRC screening in the average-risk U.S. population. Clinical trial information: NCT04369053. Research Sponsor: None.

19 Rapid Oral Abstract Session

Single-cycle neoadjuvant pembrolizumab in patients with stage I-III MMR-deficient colon cancer: Final analysis of the RESET-C study. First Author: Camilla Qvortrup, Department of Oncology, Copenhagen University Hospital - Rigshospitalet, Copenhagen, Denmark

Background: Neoadjuvant treatment with immune checkpoint inhibitors has shown remarkable responses in patients with deficient mismatch repair (dMMR) colorectal cancer. However, the optimal choice and duration of treatment have yet to be established. **Methods:** RESET-C (NCT05662527) was an investigator-initiated, phase II, single-arm, multicenter study investigating the efficacy and safety of single-cycle neoadjuvant pembrolizumab in 85 patients with resectable stage I-III dMMR colon cancer. Additional inclusion criteria were $\geq$ 18 years of age, no indication for neoadjuvant therapy, and Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1. After inclusion, patients received one cycle of pembrolizumab 4mg/kg (maximum 400mg) and underwent surgery within three to five weeks. A tumor-response evaluation was done before surgery, including blood samples, a computed tomography scan of the chest and abdomen, and a colonoscopy. The primary endpoint was the pathological complete response (pCR) rate according to Mandard tumor regression grading (Clopper-Pearson method). Secondary endpoints included surgical complications, safety of pembrolizumab, major pathological response (Mandard tumor regression grade 1 or 2), and disease-free survival. **Results:** Between February 2023 and March 2024, 85 patients were included. The median age was 74 years (IQR, 68-79), 72% were women, and 60% had clinical stage III disease. All patients received pembrolizumab, and 84 patients proceeded to surgery. One patient with stage I disease decided not to undergo surgery. A pCR rate of 44% (37/84; 95% CI, 33-55) and a major pathological response rate of 57% (48/84; 95% CI, 46-68) were found. A significantly higher pCR rate was seen in patients with stage I-II (20/33) versus stage III (17/51) disease (61% vs 33%; p=0.02). A total of 41 surgical complications were seen in 31 patients (37%; 95% CI, 27-48). Of these, eight complications were Clavien-Dindo grade 3a or above, including three patients with anastomotic leakages and two deaths within 30 days. The patients who died were aged 80 and 81, had ECOG performance status 1, and clinical stage IIIB and IIIC disease, respectively. Seven out of 85 patients (8%; 95% CI, 3-16) experienced grade 3 adverse events, three of which were treatment-related. No grade 4 or 5 adverse events were registered. Finally, data on the association between endoscopically evaluated clinical complete response and pCR, along with the 1-year disease-free survival rate expected in January 2025, will be reported. **Conclusions:** A single cycle of neoadjuvant pembrolizumab was efficacious and safe in patients with localized dMMR colon cancer. For most patients with clinical stage I-II disease, a single cycle sufficed to achieve pCR. Clinical trial information: NCT05662527. Research Sponsor: Danish Cancer Society; R343-A19803; Eva and Henry Frænkel's Memorial Fund; Vissing Foundation; 532937 AHO/LAX; Helge Peetz and Verner Peetz and wife Vilma Peetz's Scholarship; Merchant L.F. Foght's Fund; 22.340; Medical Doctor Sofus Carl Emil Friis and Wife Olga Doris Friis' Foundation; Axel Muusfeldt's Foundation; 2022-0016 and 2023-0486; NEYE Foundation; Danish Comprehensive Cancer Center; 16; Dagmar Marshalls Foundation; Frimodt Heineke Foundation; Manufacturer Einar Willumsen's Memorial Fund; Timber Merchant Vilhelm Bang's Fund.

20 Rapid Oral Abstract Session

Tumour-free ctDNA detection as a decision tool to support organ preservation in node-negative rectal cancer undergoing neoadjuvant chemotherapy, excision, and observation in the phase II NEO trial (CCTG CO.28). First Author: Jonathan M. Loree, BC Cancer, Vancouver, BC, Canada

Background: In the NEO trial (CCTG CO.28, NCT03259035) patients with node negative (N0) T1-T3 rectal cancer were treated with neoadjuvant CAPOX/FOLFOX and transanal excision surgery (TES) with the goal of organ preservation. Patients without documented response following neoadjuvant chemotherapy were recommended total mesorectal excision (TME). We present a post-hoc assessment of ctDNA detection in this early-stage population and correlate kinetics with response and outcomes. **Methods:** Fifty-eight patients enrolled in CO.28. Whole blood was collected in EDTA tubes and processed for future ctDNA analysis at up to six timepoints: pre-chemotherapy, after neoadjuvant chemotherapy (and pre-TES), yearly during surveillance for 3 years and upon progression. A total of 195 samples were analyzed with the Guardant Reveal™ assay: a tissue-free epigenomic assay leveraging $>$20,000 epigenomic regions for ctDNA detection. **Results:** Of 48 available pre-treatment samples, ctDNA was detected in 22 (46%). Sensitivity by T-stage was 14% (1/7) for T1, 53% (17/32) for T2, 44% (4/9) for T3 cancers. ctDNA detection was lower following neoadjuvant chemotherapy (4/46, 9%, $p<0.001$ vs pre-chemotherapy). Of 41 patients with paired pre- and post-chemo samples, 49% (20/41) had undetectable ctDNA at both timepoints, 44% (18/41) had reduced tumour fraction following chemotherapy, the majority of which completely cleared ctDNA (94%; 17/18) and 7% (3/41) had an increased tumour fraction following chemotherapy (67%, 2/3 changing from negative to positive), all three of whom failed to respond to neoadjuvant chemotherapy and had TME recommended (100%). In contrast, of those that cleared ctDNA following chemotherapy, 35% had TME recommended (6/17, p=0.074). Of five recurrences (3 distant, 2 local), all had negative or reduced ctDNA tumour fraction during neoadjuvant therapy before surgical intervention. Among patients with local relapse who were managed with TES, 1/2 (50%) had detectable ctDNA at the time of local progression. No samples were available for those with distant recurrences (3/5) at time of relapse. **Conclusions:** The Guardant Reveal tissue-free assay was able to detect ctDNA even in this extremely early-stage cohort and identified cancers with inadequate response to therapy and in whom TME was recommended. A tissue-free approach may support timeliness of ctDNA results that would support ctDNA as an additional decision tool with endoscopic and MRI assessments to increase physician and patient comfort with organ preservation. Clinical trial information: 03259035. Research Sponsor: Canadian Cancer Society.

21 Rapid Oral Abstract Session

WW vs TME in patients with rectal cancer with a complete or near-complete response to TNT: Pooled analysis of CAO/ARO/AIO-12 and OPRA trials. First Author: Hannah Williams, Memorial Sloan Kettering Cancer Center, New York, NY

Background: Patients with locally advanced rectal cancer (LARC) and a clinical complete response (cCR) or near-complete response (nCR) to total neoadjuvant therapy (TNT) can be offered watch-and-wait (WW) with the intention of achieving organ preservation. However, one third of patients on WW develop tumor regrowth and require total mesorectal excision (TME). Assessing the safety of WW in patients with a cCR or nCR is challenging due to the difficulty of finding an adequate control group. Here we compare the survival outcomes of patients with LARC treated with TNT and either mandatory TME or selective WW stratified by tumor response. **Methods:** This is a pooled analysis of two multicenter, phase II trials (CAO/ARO/AIO-12 [CAO] and OPRA) that randomized patients with stage II/III rectal cancer to either induction or consolidation TNT. All patients in the CAO trial underwent TME within 6 weeks of TNT. Pathologic tumor regression grade (TRG) was categorized as complete (TRG 4), intermediate (TRG 2,3) or poor (TRG 0,1). Patients in the OPRA trial were restaged by endoscopy and MRI 8±4 weeks after end of TNT. Clinical response was graded as cCR, nCR, or incomplete clinical response (iCR). Patients with a cCR or nCR were offered WW; patients with an iCR were recommended for TME. Survival curves were estimated using the Kaplan-Meier method and the log-rank test. **Results:** The study included 628 patients (*n*=304 CAO; *n*=324 OPRA). Median follow-up was 3.6 (IQR 1.13) and 5.1 (IQR 2.2) years. Patients in the CAO trial were more likely to have cT3/4 and cN positive disease while patients in the OPRA trial had tumors closer to the anal verge. Compliance with TNT and rates of grade 3+ adverse events were similar between studies. A total of 144 (47%) patients in the OPRA trial achieved long term organ preservation. We found no differences in survival between trials. In addition, we found no differences in DFS between studies for patients with an excellent (TRG4 87% [95% CI, 79-97%], cCR 87% [95% CI, 81-93%]) or intermediate (TRG2-3 69% [95% CI, 62-76%], nCR 71% [95% CI, 68-80%]) response. **Conclusions:** This pooled analysis demonstrated that almost half of LARC patients treated with TNT can achieve organ preservation and that a selective WW strategy yields similar survival outcomes compared to mandatory TME. In addition, we found no differences in survival based on paired clinical and pathologic tumor response grades following TNT. This data provides further evidence supporting the safety of WW for patients with LARC and a cCR or nCR to TNT. Research Sponsor: None.

Three-year survival outcomes in the CAO/ARO/AIO-12 and OPRA trials.

	CAO/ARO/AIO-12 (95% CI)	OPRA (95% CI)	*P*
3y DFS	73 (71–81)%	76 (68–78)%	0.3
3y DRFS	82 (78–87)%	82 (78–87)%	0.7
3y LRFS	95 (92–98)%	95 (92–97)%	0.4
3y OS	92 (89–95)%	94 (89–95)%	0.5

DFS: disease-free survival, DRFS: distant recurrence-free survival, LRFS: local recurrence-free survival, OS: overall survival.

LBA22 Poster Session

A randomized, double-blind, phase III study comparing trifluridine/tipiracil (FTD/TPI) versus placebo in patients with molecular residual disease following curative resection of colorectal cancer (CRC): The ALTAIR study. First Author: Hideaki Bando, Department of Gastroenterology and Gastrointestinal Oncology, National Cancer Center Hospital East, Kashiwa, Japan

The full, final text of this abstract will be available at meetings.asco.org on the day of presentation and in the online supplement to the February 1, 2025, issue of the *Journal of Clinical Oncology*.

23 Rapid Oral Abstract Session

Preliminary results from a randomized, open-label, phase 2 study of botensilimab (BOT) with or without balstilimab (BAL) in refractory microsatellite stable metastatic colorectal cancer with no liver metastases (MSS mCRC NLM). First Author: Marwan Fakih, Department of Medical Oncology and Therapeutics Research, City of Hope Comprehensive Cancer Center, Duarte, CA

Background: BOT is an Fc-enhanced, multifunctional anti-CTLA−4 antibody designed to improve Fc gamma receptor-mediated effector functions and extend the reach of I-O to tumor types such as MSS mCRC. Here we present preliminary data from a randomized, open-label, phase 2 study in patients (pts) with MSS mCRC NLM treated with BOT ± BAL (anti-PD−1; NCT05608044). The study aimed to inform dose and contribution of components based on the primary endpoint of objective response rate (ORR) by RECIST 1.1 per investigator, and safety, and was not powered for statistical comparisons between arms. **Methods:** A total of 234 pts (intent-to-treat [ITT]) were randomized to BOT (up to 4 doses) 75 or 150 mg every 6 wks (Q6W), BOT 75 or 150 mg Q6W plus BAL 240 mg Q2W (up to 2 years), or standard of care (SOC; regorafenib or trifluridine/tipiracil). **Results:** Median age was 58 yrs (range 23—90), 50% male, 39% rectal, 44% 3L+, 43% ECOG 1, 58% KRAS mutant, 4% NRAS mutant, 83% prior bev, all MSS and/or pMMR by local testing. Key characteristics were well balanced with some exceptions including median time from diagnosis of metastatic disease to study entry (30 mos across arms; 45 mos SOC) and presence of peritoneal metastases (34% across arms; 42% 75 mg BOT / 240 mg BAL; 27% SOC). As of July 29, 2024, median follow-up was 9.8 mos. Key efficacy and safety data are shown (Table). Image based endpoints by blinded independent review, as well as overall survival will be reported in the future. Grade ≥3 treatment-related adverse events (TRAEs) were highest with SOC followed by BOT + BAL combination, and then BOT monotherapy, with dose dependency. Treatment-related immune-mediated diarrhea/colitis (imDC) was manageable and highest with 150 mg BOT / 240 mg BAL. No new safety signals and no treatment-related deaths occurred. **Conclusions:** The study met the objectives of informing dose and contribution of components. Overall ORR was higher with BOT + BAL vs BOT monotherapy. ORR was highest with 75 mg BOT / 240 mg BAL with less toxicity as compared to 150 mg BOT / 240 mg BAL. Consistent with published data, there were no objective responses in SOC whereas most responses seen with BOT + BAL were ongoing, similar to the durable responses observed in the ph1 study. These responses are differentiated from previous I-O-only combinations and SOC, supporting further investigation of 75 mg BOT / 240 mg BAL vs SOC in a planned global ph3 trial. Clinical trial information: NCT05608044. Research Sponsor: Agenus Inc.

	75 mg BOT/240 mg BAL	150 mg BOT/240 mg BAL	75 mg BOT	150 mg BOT	SOC
ITT (Randomized)	*n=62*	*n=61*	*38*	*n=40*	*n=33*
ORR, %	19%	8%	0%	8%	0%
(95% CI)	(10–31)	(3–18)	(0–10)	(2–20)	(0–11)
Safety (Treated)	*n=62*	*n=60*	*n=37*	*n=39*	*n=21*
Any Grade TRAEs, n (%)	54 (87)	59 (98)	28 (76)	31 (79)	19 (90)
Grade ≥3 TRAEs, n (%)	22 (35)	25 (42)	8 (22)	9 (23)	12 (57)
Any Grade Treatment-related IMDC, n (%)	19 (31)	28 (47)	13 (35)	12 (31)	0 (0)

24 Poster Session

Disparities in stage of diagnosis of colon cancer: Differences by Hispanic background, race, and ethnicity. First Author: Dimitrios N. Varvoglis, University of North Carolina at Chapel Hill, Chapel Hill, NC

Background: Given noted heterogeneity in Hispanic adults, cancer outcomes reported in aggregate may not detect disparities within specific Hispanic subgroups. The objective was to assess whether any potential differences in colon cancer (CC) stage at diagnosis among patients with Hispanic background (HBackground) could be explained by differences in socio-economic determinants of health (SDOH) including income, education and insurance status. **Methods:** The National Cancer Database was queried for patients diagnosed with CC from 2004 to 2021. Two logistic regression models generated adjusted odds ratios of late-stage (III-IV) vs early-stage (I-II) CC at diagnosis, disaggregated either by HBackground or R&E, compared to non-Hispanic White (NHW) patients. Model 1 controlled for patient/facility characteristics; Model 2 controlled for additional SDOH. **Results:** Of the 442,742 CC patients identified, 6% were Hispanics. Compared to NH adults, Hispanic patients were more likely (6% higher odds, $p<0.0001$) to present with late-stage CC. When Hispanic patients were disaggregated by HBackground, some were more likely (Mexicans 16%; Cubans 20% and South/Central Americans 12%) than NHW adults to present with late-stage CC while others (Puerto Ricans and Dominicans) were not. Disaggregation by R&E indicated that only Hispanic-White patients were more likely (12%) to present with late-stage disease than NHW patients. Controlling for SDOH, R&E differences in stage at diagnosis persisted but HBackground differences became insignificant (Table). **Conclusions:** Disaggregation of Hispanic adults by HBackground identifies specific subgroups at increased risk for late-stage CC at diagnosis. Understanding modifiable SDOH may identify opportunities for improving early detection of CC and outcomes in specific Hispanic subgroups. Research Sponsor: None.

Odds of presenting with late-stage colon cancer of "Hispanic" patients disaggregated by race and ethnicity or Hispanic background compared to non-Hispanic Whites.

Disaggregation based on:	Model 1: Adjusted for age, facility type, year of diagnosis, and Charlson-Deyo comorbidity aOR (95% CI)	p-value	Model 2: Adjusted for age, facility type, year of diagnosis, Charlson-Deyo comorbidity, neighborhood income, neighborhood education, and insurance status aOR (95% CI)	p-value
Non-Hispanic White (n=418,046)	(ref)		(ref)	
Hispanic White (n=24,058)	1.12 (1.09, 1.15)	<.0001	1.04 (1.01, 1.07)	0.01
Hispanic Background				
Mexican (n=3,612)	1.16 (1.08, 1.24)	<.0001	1.05 (0.97, 1.13)	0.25
South/Central American (n= 1,734)	1.12 (1.02, 1.24)	0.02	1.01 (0.91, 1.12)	0.83
Cuban (n=937)	1.20 (1.05, 1.36)	0.001	1.11 (0.97, 1.27)	0.13
Puerto Rican (n=1,280)	1.07 (0.95, 1.19)	0.28	1.01 (0.90, 1.14)	0.88
Dominican (n=465)	1.14 (0.94, 1.39)	0.18	1.02 (0.83, 1.25)	0.85

aOR, Adjusted odds ratio; CI, confidence Interval.

25 **Poster Session**

Association of social deprivation and colon cancer mortality: A population-based analysis. First Author: Mosffa Ullah, Texas Tech University Health Sciences Center School of Medicine, Lubbock, TX

Background: Colon cancer mortality is largely influenced by social factors, with higher rates seen in deprived areas. The Social Deprivation Index (SDI) measures factors like poverty, education, and unemployment, which have been linked to poor health outcomes. This study examines the relationship between county level social deprivation and colon cancer mortality in Texas. **Methods:** We conducted a population-based cohort study of colon cancer deaths in Texas. We queried the National Cancer Institute Surveillance, Epidemiology and End Results Program for county level colon cancer deaths in Texas over the most recent available year, 2021. We calculated age-adjusted mortality rates (AAMRs) per 100,000 population using the direct standardization method based on the age group weights from the 2000 standard US population. Confidence intervals for AAMR were derived by estimating the standard error as the AAMR divided by square root of number of mortalities. The SDI by county was linked to mortality rate using Federal Information Processing Standards (FIPS) codes. The significance of the difference between two AAMR was evaluated by determining if the intersection of 95% confidence intervals was empty. The association between SDI and AAMR was tested using both unweighted and weighted Spearman correlation with county population as weight. Results are reported and Spearman correlation and 95% confidence interval (ρ [95% CI]). Sensitivity analysis was performed by considering only the individual components of the SDI. Subgroup analyses included age group, sex, urban counties, rural counties, and race/ethnicity groups. **Results:** A total of 3,800 colon cancer mortalities were identified in 2023. The statewide AAMR was 12.5 (95% CI 12.1 to 12.9). Out of 254 counties, 29 (11.4%) had an SDI $\leq$ 25, 61 (24.0%) had 26 $\leq$ SDI $\leq$ 50, 87 (34.3%) had SDI 51 $\leq$ SDI $\leq$ 75, and 77 (30.3%) had 76 $\leq$ SDI $\leq$ 100. Total population in counties with SDI $\geq$ 76 was 15,106,726 (51.1%) with AAMR 12.959 (12.379 to 13.538). In contrast, the population in counties with SDI $\leq$ 25 was 4,771,473 (16.1%) with significantly lower mortality (AAMR 9.752 [95% CI 8.879 to 10.625]). The population weighted Spearman correlation between county level SDI and AAMR was positive (ρ = 0.339 [95% CI 0.209 – 0.514]). The unweighted Spearman correlation between county level SDI and AAMR was also positive (ρ = 0.178 [95% CI 0.089 – 0.238]). Households with no vehicle, poverty, and unemployment were most correlated with AAMR while renter occupied housing had no association. The correlation of county level SDI and AAMR were similar for rural counties, urban counties, males, females, Whites, Hispanics and Blacks. **Conclusions:** This study reveals a significant positive correlation between higher SDI and increased colon cancer mortality at the county level in Texas. The findings highlight the need for targeted public health interventions in deprived areas to reduce mortality and address disparities. Research Sponsor: None.

26 **Poster Session**

Incidence trends of early-onset colorectal cancer (EOCRC) in the United States, by age, sex, race and ethnicity, and urbanicity (2000-2021). First Author: Erin Van Blarigan, Greater Bay Area Cancer Registry, University of California, San Francisco, San Francisco, CA

Background: Research is needed to understand the extent to which the incidence of early onset colorectal cancer (EOCRC)is rising across population sub-groups, including by age, sex, race/ethnicity, and urbanicity, to inform hypotheses regarding its causes and strategies for screening and prevention. **Methods:** We used data on colorectal cancer cases diagnosed among adults 20-49 years (yr) between 2000-2021 in the delay-adjusted Surveillance, Epidemiology, and End Results (SEER) database (April 2024). We calculated delay- and age-adjusted annual incidence rates (IR) per 100,000, overall and by age, sex, race/ethnicity, and urbanicity. Urbanicity was defined using USDA Rural-Urban Continuum Codes for county; 1-3 were classified as urban and 4-9, 98, and 99 were classified as non-urban. We used NCI's Joinpoint regression models (Joinpoint version 5.2.0.0) to estimate annual percent change (APC) and 95% confidence intervals (CI). We also calculated IRs for the most recent 5-year period (2017-2021) and used incidence rate ratios (IRR) to compare across groups; IRs and IRRs were calculated using SEER*Stat version 8.4.3. **Results:** Analyses included 155,500 cases of EOCRC. Most cases (72%) occurred among adults 40-49 yr; 22% occurred among those 30–39 yr and 6% in people 20-29 yr. Among adults 40-49 yr, the incidence of EOCRC increased 4.4% / yr on average between 2018-2021 (95%CI: 2.5, 5.6). Among adults 30-39 yr, incidence of EOCRC increased 3.0% / yr, on average, starting in 2010 (95%CI: 2.4, 4.5). For adults ages 20-29 yr, there were large increases in incidence from 2000-2016 (2000-2013, APC: 3.5; 95%CI: 2.4, 4.1; 2013-2016, APC: 11.2; 95%CI: 6.8, 13.5) but no change from 2016-2021. For the most recent 5-yr period (2017-2021), females had lower IR compared to males (IRR: 0.90; 95%CI: 0.88, 0.91), but trends over time were similar by sex. The incidence of EOCRC increased across races and ethnicities between 2000-2021, albeit at different rates. In the most recent period (2017-2021), the IR per 100,000 was 22.0 for Non-Hispanic American Indian and Alaskan Native (AIAN), 15.0 for Non-Hispanic White, 14.9 for Non-Hispanic Black, 12.4 for Hispanic or Latino, and 11.2 for Non-Hispanic Asian American or Pacific Islander adults. People in non-urban areas had a higher incidence of EOCRC compared to those in urban areas (IRR: 1.18; 95%CI: 1.14, 1.22), with similar trends over time. **Conclusions:** The incidence of EOCRC is highest among adults 40-49 yr, males, Non-Hispanic AIAN adults, and in non-urban areas. Trend analyses generally showed similar increases in incidence (3-5%/yr) in recent periods across sex, race/ethnicity, and urbanicity. Research Sponsor: National Cancer Institute; Centers for Disease Control and Prevention; California Department of Public Health.

27 **Poster Session**

Comparative analysis of demographics and outcomes in hospitalized patients with young- versus average-onset colorectal cancer in New York state (2017-2022). First Author: Mrinalini Ramesh, University at Buffalo, Buffalo

Background: Recent studies have shown an increasing incidence of colorectal cancer (CRC) among young patients. This study investigates clinical outcomes and healthcare utilization among young ($<$50) versus average onset ($\geq$50) CRC patients admitted to hospitals in New York State (NYS). **Methods:** We performed a retrospective analysis using the Statewide Planning and Research Cooperative System (SPARCS) database from 2017 to 2022. Patients were divided into two groups: young-onset colorectal cancer patients (YOCRC, $<$50 years) and average onset colorectal cancer patients (AOCRC, $\geq$50 years). The study population was further stratified by demographic and clinical characteristics. All associations were compared using the Kruskal-Wallis test, along with multivariate linear and logistic regression in RStudio version 4.4.1, at a significance level of $\leq$0.05. Clinical characteristics, including severity of illness and risk of mortality, were defined using the All Patient Refined (APR) grading system. **Results:** A total of 9,904 patients were identified (948 YOCRC and 8,956 AOCRC) with a primary admitting diagnosis of CRC from 2017 to 2022. In comparing the two age groups, AOCRC patients had a higher proportion of White, and female patients, whereas the YOCRC group included a higher proportion of Black and male patients ($p<0.001$). The median length of stay was greater for AOCRC patients (5 days) compared to YOCRC patients (4 days), $p<0.001$. Both age groups showed an upward trend in total cost of stay in the recent years. AOCRC patients were more likely to have increased severity of illness graded as major-extreme (60% vs 48%), while a higher proportion of YOCRC patients were classified as minor-moderate disease (51% vs 40%), $p<0.001$. Similarly, AOCRC patients experienced a higher risk of mortality compared to YOCRC patients ($p<0.001$). Regarding procedures, AOCRC patients needed more transfusions (13%) than YOCRC patients (9.3%), $p<0.001$. YOCRC patients were also more likely to be discharged home (71% vs 46%), while AOCRC patients often required home health services or skilled nursing ($p<0.001$). Notably, while the risk of mortality was higher for AOCRC patients, it has been decreasing overall from 2017 to 2022 [OR 0.92 (95% CI: 0.84-0.99), p=0.039]. **Conclusions:** AOCRC patients experienced greater severity of disease, higher mortality risk, and required more assistive care at discharge compared to YOCRC patients. The total cost of hospital stays has been rising for both groups. Policies focused on earlier outpatient interventions may help reduce the burden of inpatient care and rising costs. Research Sponsor: None.

28 **Poster Session**

Genetic differences in colorectal cancer across race and ethnicity. First Author: Carrie Sha, NewYork-Presbyterian Hospital and Weill Cornell Medical Center, New York, NY

Background: Colorectal cancer (CRC) is the fourth most common cause of cancer death and cancer-related mortality is expected to increase exponentially. Ethnic minorities, in particular African Americans, have increased mortality from CRC. Although multifactorial, differences in somatic gene mutations could contribute to these racial discrepancies. Mutations in the critical oncogene KRAS are associated with worse prognosis in CRC. Recent literature suggests that in CRC patients, KRAS mutations are more frequently found in African Americans compared to Caucasians. This finding led to our hypothesis that molecular alterations across multiple races/ethnicities could explain differences in patient outcomes. This project will also inform a larger analysis using artificial intelligence learning models to predict race from tumor-specific genetic variants, as a tool to identify genetic drivers of race-specific patient outcome. **Methods:** We examined the molecular alterations in 304 patients diagnosed with CRC who had genetic testing between June 2021 and April 2023 at NYP Queens, NYP Brooklyn, and NYP Cornell. Race and ethnicity, genetic mutations, mortality, tumor location, and age, stage, and the presence of metastasis at diagnosis was collected. Patients self-identified as Non-Hispanic Black (NHB), White (NHW), Asian (NHA), American Indian (NHAI), or Hispanic, Other, or Declined. Median and interquartile range were used to summarize continuous variables, and frequency and proportion were used to summarize categorical variables. The significance of difference across ethnicity groups was tested using Kruskal-Wallis rank sum test for continuous variables and Fisher's exact test for categorical variables. Mutational correlates of race using a multiple-instance learning artificial intelligence approach (i.e. Anaya et al., Nature Biomedical Engineering, 2023), will be presented. This method will predict race from patients' tumor-specific genetic variants, allowing us to identify specific variants and patterns predictive of race, and potential drivers of patient outcome. **Results:** Amongst the 304 CRC patients, we identified 116 NHW, 51 NHA, 40 NHB, 35 Hispanic and 1 NHAI, for a total population of 243. BRAF mutations were almost absent in NHB (2.5%) and NHA (3.9%), compared to NHW (13.8%; p=0.02). KRAS mutations were also more prevalent in underserved populations; 65.7% Hispanic, 57.5% NHB, compared with 44.8% NHW (p=0.14). We further identified substantial differences in KRAS mutations when dividing the Asian population into East Asian (62.9%) and South Asian (37.5%). **Conclusions:** We identified significant differences in key molecular drivers between ethnicities. Our findings suggest that genomic and tumor specific differences across ethnicities could in part explain differences in patient survival across these groups. Mutational correlates of race and survival using natural language processing will be presented. Research Sponsor: None.

29 Poster Session

Adherence to the 2018 World Cancer Research Fund/American Institute for Cancer Research (WCRF/AICR) recommendations and overall mortality among adults with colorectal cancer in a multiethnic cohort study. First Author: Edgar Asiimwe, National Clinician Scholars Program at UCSF, San Francisco, CA

Background: High adherence to AICR recommendations on diet, adiposity, and physical activity is associated with improved survival among adults with colorectal cancer (CRC). However, over 80% of participants in existing studies self-reported as non-Hispanic White (NHW). We studied the association between adherence to AICR recommendations and overall mortality among adults with CRC in a multiethnic cohort. **Methods:** This prospective analysis used data from the Multiethnic Cohort (MEC) study. The MEC is a racially/ ethnically diverse cohort of over 200,000 participants residing in Hawaii and Los Angeles. Lifestyle behaviors in the MEC were assessed by questionnaire at enrolment/Q1 (1993-1996) and periodically over follow-up. We included those with incident CRC before completing Q3 (2003-2008). Follow-up was from Q3 survey completion to December 2019. The main exposure was the standardized AICR score [0-7 points] at Q3. The primary outcome was overall mortality; CRC-specific mortality was secondary. We used multivariable Cox proportional hazards regression to estimate adjusted hazard ratios (HR) and 95% confidence intervals (CIs). **Results:** There were 1127 adults diagnosed with CRC prior to Q3. Of these, 1079 (95.7%) had non-missing data on the AICR score and comprised our study population. Median (IQR) age at diagnosis was 69 (62-75); median (IQR) follow-up was 12.2 (0.1-16.4) years. There were 489 (45.3%) women. By self-reported race/ethnicity, there were: 153 (14.2%) Black/African Americans, 453 (42.0%) Japanese Americans, 79 (7.3%) Latinos, 165 (15.3%) Native Hawaiians, and 229 (21.2%) NHWs. Only 48 (4.4%) adults had high adherence (AICR≥5 points). Those in the highest category of AICR scores (AICR≥4.5) had statistically significant lower risk of overall mortality relative to those in the lowest category (AICR≤2.5) (HR: 0.64, 95% CI, 0.47-0.89). A similar (but non-statistically significant) association was observed with CRC-specific mortality [HR (high vs low): 0.70, 95% CI, 0.31-1.60]. Those who increased their score between Q1&Q3 had lower overall mortality (HR: 0.87, 95% CI,0.79 - 0.96). Similar trends were observed in stratified analysis by race/ethnicity; however, statistical power was limited. **Conclusions:** High adherence to AICR recommendations was associated with lower risk of overall mortality in this multiethnic population of adults with CRC. Efforts to increase adherence to AICR recommendations in this population are needed. Research Sponsor: None.

30 Poster Session

Demographic predictors of multi-target stool DNA test completion in an underserved primary care population: A retrospective analysis. First Author: Arup Ganguly, University of Connecticut Health Center, Farmington, CT

Background: Colorectal cancer screening remains crucial for early detection, with multi-target stool DNA(Exact Sciences, Madison, WI, USA) serving as a non-invasive option. However, adherence to test completion may vary across different demographic groups. This study aims to assess the association between patient demographics and the completion of Cologuard testing in an underserved primary care population in order to further explore the disparities that exist. **Methods:** We conducted a retrospective analysis of 12071 eligible patients, aged 45-75, across Hartford Healthcare Primary Care Clinics across Connecticut, USA, between November 2022 and April 2024 for whom an order for this test was placed. Demographic data including gender, age, English proficiency, and insurance status(public/private vs no insurance) were analyzed. The outcome of interest was whether patients had results available within 120 days of placing the order. Logistic regression was used to examine the relationship between these variables and the likelihood of Cologuard test completion. **Results:** 11704(96.9%) of all patients were insured. Among insured patients, 6945 (59.9%) had results available, while 4759 (41.1%) did not have results at the 120 day mark. Among uninsured patients, 46 (12.5%) had results, while 321 (87.5%) did not. Insured patients were more likely to have test results available compared to uninsured patients (OR = 10.184, 95% CI: 7.460-13.901, $p < 0.01$). 10977(90.9%) patients were English proficient. They were more likely to have results compared to Non-English proficient patients (OR = 3.474, 95% CI: 3.038-3.972, $p < 0.01$). Gender was not a significant predictor of test completion (OR = 0.964, 95%CI: 0.895-1.039, p = 0.341), and older age demonstrated a small, non-significant effect (OR = 1.004, p = 0.094), with a test for non-linearity confirming a linear relationship between age and test completion (p = 0.162). **Conclusions:** Insured status and English proficiency are significant predictors of Cologuard test completion in this primary care cohort and are more likely complete testing. These findings underscore the need for targeted interventions to increase colorectal cancer screening adherence, particularly among this population. Research Sponsor: None.

31 Poster Session

Comprehensive analysis of Native Hawaiians and other Pacific Islanders with early-onset colorectal cancer. First Author: Manasawee Tanariyakul, Department of Medicine, John A. Burns School of Medicine, University of Hawaii, Honolulu, HI

Background: Rates of early-onset colorectal cancers (EOCRC) are increasing in Hawaii across all racial groups. Although previous studies have shown that Native Hawaiians have a higher mortality rate compared to other racial groups, the adjustment for sociodemographic factors have been limited. Our objective is to perform a comprehensive analysis of outcomes among patients with EOCRC in a racially diverse population accounting for tumor factors and patient sociodemographics. **Methods:** Patients diagnosed with EOCRC between 2000 and 2022 in Hawaii. Overall survival of Asians, Whites, and Native Hawaiian or Other Pacific Islanders (NHOPI) were calculated using the Kaplan-Meier method. Cox proportional hazard regression models assessed survival predictors adjusting for clinical and sociodemographic factors. **Results:** We included 379 patients in the final analysis. NHOPI had higher prevalence of underinsured status and higher histopathology grade compared to Whites and Asians. In unadjusted Cox regression, NHOPI race, underinsured status, grade, and stage were prognostic for survival. After adjusting for confounders, underinsured status, grade, and stage remained prognostic, but race was not significantly associated with survival. **Conclusions:** This study concludes that while NHOPI patients with EOCRC demonstrated poorer survival outcomes compared to other racial groups, this disparity was largely explained by the large percentage of underinsured NHOPI patients. Addressing treatment disparities in underinsured or uninsured is essential to improve survival outcomes. Research Sponsor: None.

Univariate and multivariate analysis of survival.

	Univariate		Multivariate	
	HR (95% CI)	p-value	HR (95% CI)	p-value
Insurance				
Private Insurance	-	-	-	-
Underinsured & Uninsured	1.865 (1.331-2.612)	<0.001	1.843 (1.285-2.643)	<0.001
Race				
White	-	-	-	-
Asian	1.179 (0.746-1.865)	0.48	0.936 (0.585-1.496)	0.781
NHOPI	2.005 (1.231-3.265)	0.005	1.051 (0.714-1.547)	0.8
Age at diagnosis	0.996 (0.967-1.025)	0.771	1.002 (0.971-1.034)	0.901
Histopathology grade				
Grade 1&2	-	-	-	-
Grade 3&4	2.785 (1.958-3.963)	<0.001	2.278 (1.571-3.303)	<0.001
Stage				
Stage 1	-	-	-	-
Stage 2	1.104 (0.49-2.487)	0.81	1.028 (0.451-2.342)	0.948
Stage 3	3.255 (1.666-6.359)	<0.001	2.873 (1.447-5.706)	0.003
Stage 4	14.851 (7.499-29.41)	<0.001	12.778 (6.283-25.987)	<0.001
Unknown	3.738 (1.506-9.275)	0.004	3.008 (1.189-7.608)	0.02
Sex				
Male	-	-	-	-
Female	0.812 (0.587-1.123)	0.22	0.852 (0.609-1.194)	0.352
Microsatellite status				
Stable	-	-		
Unstable	0.481 (0.148-1.562)	0.204		
Not done/Unknown	1.08 (0.737-1.582)	0.739		
Time to Treatment				
Less than 1 month	-	-	-	-
More than 1 month	1.221 (0.863-1.727)	0.26	1.123 (0.784-1.609)	0.527
Location of Tumors				
Right	-	-	-	-
Left	1.239 (0.76-2.02)	0.391	1.466 (0.886-2.425)	0.137
Rectal	1.391 (0.86-2.249)	0.178	1.211 (0.738-1.986)	0.449

NHOPI: Native Hawaiians and Other Pacific Islanders; HR: Hazard Ratio; CI: Confidence Interval.

32 Poster Session

Comprehensive analysis of socioeconomic and clinical factors contributing to treatment delay in early-onset colorectal cancer. First Author: Manasawee Tanariyakul, Department of Medicine, John A. Burns School of Medicine, University of Hawaii, Honolulu, HI

Background: The prevalence of early-onset colorectal cancers (EOCRC) is rising globally. Disparities in social determinants of health can delay access to treatment and impacting outcomes. This study aims to analyze factors influencing time from diagnosis to treatment among EOCRC patients in a racially diverse population, considering tumor factors and sociodemographics. **Methods:** We analyzed EOCRC data from patients diagnosed between 2000 and 2022 at Queen's Medical Center in Honolulu, Hawaii. Time from diagnosis to treatment for Asians, Whites, and Native Hawaiian or Other Pacific Islanders (NHOPI) was assessed using the Kaplan-Meier Method. Cox proportional hazards regression models were used to identify predictors, adjusting for clinical and pathological factors. **Results:** A total of 379 patients were analyzed. NHOPI were more like to be underinsured compared to Whites and Asians. In the unadjusted Cox model, underinsured/uninsured status, older age, unknown stage, left sided, and rectal locations were associated with delayed treatment. After adjustment, underinsured/uninsured status, older age, and rectal location remained significant predictors of delayed treatment. **Conclusions:** EOCRC patients with underinsured/uninsured status, age, and rectal tumor location are significantly associated with delayed in cancer treatment. While age was statistically significant, its clinical impact appears minimal. The need for a multidisplinary approach and additional preoperative staging in rectal cancer may explain treatment delays even after adjustment. Improving treatment accessibility and reducing delays, especially for underinsured EOCRC patients, is important. Research Sponsor: None.

Univariate and multivariate analysis of time from diagnosis to treatment.

	Univariate		Multivariate	
	HR (95% CI)	p-value	HR (95% CI)	p-value
Insurance	-	-	-	-
Private insurance	-	-	-	-
Underinsured/Uninsured	0.692 (0.545-0.878)	0.003	0.706 (0.55-0.906)	0.006
Race				
White	-	-	-	-
Asian	0.912 (0.696-1.195)	0.503	0.817 (0.619-1.077)	0.151
NHOPI	0.749 (0.548-1.025)	0.071	0.779 (0.562-1.08)	0.134
Age	0.978 (0.96-0.997)	0.021	0.968 (0.949-0.987)	0.001
Histopathology grade				
Grade 1&2	-	-	-	-
Grade 3&4	1.046 (0.798-1.371)	0.746	1.083 (0.813-1.441)	0.587
Stage				
1	-	-	-	-
2	1.022 (0.718-1.456)	0.902	1.071 (0.744-1.539)	0.713
3	1.227 (0.896-1.682)	0.203	1.387 (0.997-1.931)	0.052
4	0.964 (0.673-1.381)	0.842	1.039 (0.716-1.507)	0.842
Unknown	0.631 (0.406-0.981)	0.041	0.748 (0.473-1.182)	0.213
Sex				
Male	-	-	-	-
Female	1.029 (0.836-1.267)	0.785	1.072 (0.868-1.325)	0.518
Microsatellite status				
Stable	-	-		
Unstable	1.074 (0.616-1.875)	0.8		
Not done/Unknown	1.259 (1.004-1.577)	0.046		
Location of Tumors				
Right	-	-	-	-
Left	0.631 (0.468-0.851)	0.003	0.735 (0.535-1.011)	0.058
Rectal	0.43 (0.317-0.582)	<0.001	0.448 (0.327-0.613)	<0.001

NHOPI: Native Hawaiians and Other Pacific Islanders. HR: Hazard Ratio CI: Confidence Interval.

33 **Poster Session**

Disparities in incidence and mortality of colorectal cancer in Appalachian Kentucky. First Author: Lindsay Austin O'Shaughnessy, University of Indiana School of Medicine, Indianapolis, IN

Background: Since the late 1990s, both the incidence and mortality of invasive colorectal cancer (CRC) have decreased across the United States, including in Kentucky. Despite this, CRC incidence and mortality are above national average in Kentucky. Over the last two decades, a disparity in CRC incidence and mortality has emerged between Appalachian Kentucky (AK), a distinct eastern region, and Non-Appalachian Kentucky (NAK). This disparity is attributed to differences in social and structural determinants of health (SSDH) between these populations. **Methods:** Age-adjusted CRC incidence and mortality rates (2016-2020) for both Kentucky and the U.S. were obtained from the National Cancer Institute's public database. Data from the Kentucky Cancer Registry were used to compare CRC incidence and mortality rates in AK and NAK counties from 1994-2021. Incidence rate ratio analyses across discrete time periods demonstrated statistical differences between these regions. **Results:** Compared to national data (2016-2020), Kentucky's CRC age-adjusted incidence and mortality rates were significantly higher: 45.9 vs. 36.5/100,000, 25.8%, for incidence, and 16.2 vs. 13.1/100,000, 23.7%, for mortality (both $p<0.0001$). From 1995-2003 incidence and 1994-2003 mortality of CRC, were higher in NAK, these rates eventually decreased and equalized between NAK and AK. By 2010-2015 both incidence and mortality rates were higher in AK, 55.8 vs 48/100,000, 16.3%, ($p=<0.0001$) and 20.1 vs 15.8/100,000, 27.2%, ($p=<0.0001$) respectively. The most recent data (2016-2021) shows that these significant disparities persist, with AK's incidence and mortality 51.8 vs 44.1/100,000, 17.5% ($p=<0.0001$) and mortality 19.0 vs 15.1/100,000, 25.8% ($p=<0.0001$) higher, respectively. **Conclusions:** While CRC rates have decreased nationwide and in Kentucky, a pronounced gap in both incidence and mortality has developed between Appalachian and Non-Appalachian Kentucky. Appalachian counties—which are more rural and impoverished—face greater barriers to accessing preventive healthcare, contributing to this disparity. Additionally, AK has higher rates of obesity, poverty, smoking, and lower educational attainment and screening rates . Recent public health efforts, such as promoting stool-based screening and increasing preventive care discussions, aim to reduce these disparities. However, delays in CRC screenings during the COVID-19 pandemic— resulting in approximately 1.7 million missed colonoscopies nationally— may have skewed the most recent data, necessitating further analysis in the coming years. Research Sponsor: None.

34 **Poster Session**

Differences in disease characteristics in young vs older adults with colorectal carcinoma. First Author: Shivani Modi, Albert Einstein Healthcare Network, Philadelphia, PA

Background: Colorectal carcinoma (CRC) exhibits distinct disease characteristics between younger ($\leq$50 years) and older ($>$50 years) adults, which may significantly influence treatment decisions and outcomes. This study aims to investigate differences in disease characteristics of colorectal carcinoma among patients aged $\leq$50 years compared to those aged $>$50 years, utilizing data from the Surveillance, Epidemiology, and End Results (SEER) database. **Methods:** A retrospective analysis was performed using the SEER database covering the years 2000 to 2020. A total of 150,000 diagnosed CRC cases were included. Key variables analyzed comprised stage at diagnosis, tumor grade, histological subtype, and 5-year survival rates. Rates of CRC diagnoses in various age brackets were calculated for three time periods: 2000-2010, 2010-2020, and overall from 2000-2020. **Results:** Within the total cohort, 12% (18,000 cases) were diagnosed in young adults, while 88% (132,000 cases) were in older adults. An analysis of age-specific rates showed: 2000-2010**:** Young adults ($\leq$50 years) represented 10% of CRC cases, with 16% presenting at advanced stages (Stage III or IV). 2010-2020**:** This group increased to 12%, with 65% diagnosed at advanced stages (Stage III or IV), compared to 52% in older adults ($p < 0.001$). In terms of tumor characteristics, poorly differentiated tumors were more prevalent in young adults (28% vs. 18%, $p < 0.001$). The prevalence of microsatellite instability-high (MSI-H) tumors was significantly greater in young adults at 18%, compared to 12% in older adults ($p < 0.05$). The 5-year overall survival rate for young adults was 68%, notably higher than the 48% for older adults ($p < 0.001$). **Conclusions:** This study underscores significant differences in disease characteristics between young and older adults with colorectal carcinoma. Young patients are more likely to present with advanced disease but have better survival outcomes. These findings indicate the necessity for an age specific approach to screening and treatment strategies. Research Sponsor: None.

Differences in stage at diagnosis, tumor grade, presence of MSI-H, and survival outcomes between younger and older adults diagnosed with colorectal carcinoma.

Category	Young Adults (≤50 years)	Older Adults (>50 years)	p-value
Proportion of Total CRC Cases	12% (18,000 cases)	88% (132,000 cases)	
Advanced Stage (Stage III or IV)	65% (2010-2020)	52% (2010-2020)	< 0.001
Poorly Differentiated Tumors	28%	18%	< 0.001
Microsatellite Instability-High (MSI-H)	18%	12%	< 0.05
5-Year Overall Survival Rate	68%	48%	< 0.001

35 **Poster Session**

Disparities and barriers in germline testing among patients with early-onset gastrointestinal cancers. First Author: Gideon T Dosunmu, Univeristy of Chicago, Chicago, IL

Background: Early-onset gastrointestinal cancers (EOGIC) are strongly associated with hereditary predisposition, making germline testing essential for treatment and familial risk management. Disparities in germline testing persist. This study investigates disparities and barriers to germline testing in EOGIC. **Methods:** This IRB-approved retrospective analysis included 864 patients aged 18-49 with GI cancers at the University of Chicago (2018-2023). Data collected included race, gender, age, insurance, family cancer history, and subtype. Outcomes were physician referrals for germline testing and completion. Chi-square tests and logistic regression models assessed predictors of recommendations and completion. **Results:** Of 864 patients, 473 (54.7%) received germline testing recommendations, and 323 (37.4%) completed testing. Completion rates: 30.2% for no insurance, 29.1% for Medicare/Medicaid, and 40.3% for private insurance (p=0.032). Patients with family history of cancer were more likely to be recommended for testing (62.9% vs. 39.5%, $p<0.001$) and complete it (44.0% vs. 23.7%, $p<0.001$). In our cohort, no significant racial disparities were observed (p=0.712 and p=0.358). Females were more likely to receive recommendations (59.1%) and complete testing (41.2%) than males (50.9% referral, 34.0% completion, p=0.016 and p=0.029). Patients under 30 had the highest rates of testing recommendation (66.1%) and completion (46.4%), followed by 30-39 (61.7%, 44.4%) and 40-49 (51.2%, 34.0%), p=0.006 and p=0.009. By cancer subtype, colorectal cancer (CRC) patients had the highest referral (66.1%) and completion (45.9%) rates, while anal cancer had the lowest (13.2% referral, 5.3% completion, OR 0.08, $p<0.0001$). Gastric cancer (52.3% referral, 29.2% completion, OR 0.49, p=0.005) and esophageal cancer (36.9% referral, 20.0% completion, OR 0.31, $p<0.0001$) also had lower testing rates. Liver cancer (35.4% referral, 27.1% completion, OR 0.38, $p<0.0001$) and cholangiocarcinoma (45.5% referral, 22.7% completion, OR 0.28, p=0.016) showed similarly low rates. **Conclusions:** Disparities in germline testing were seen among uninsured patients, males, and those with public insurance. CRC had the highest testing rates, while anal, esophageal, liver, and cholangiocarcinoma had lower rates, likely due to lower suspicion of hereditary predisposition. Targeted interventions are needed to improve testing uptake and completion, especially among uninsured and underrepresented populations. Research Sponsor: Bucksbaum Pilot Grant.

36 **Poster Session**

Temporal trends in rural vs urban colon cancer mortality in Texas: A SEER database analysis from 2012 to 2021. First Author: Ahmed Bashir Sukhera, Texas Tech University Health Sciences Center, Odessa, TX

Background: Colorectal cancer is a major concern for the U.S. healthcare system, being the fourth most common cancer by incidence in the country with 152,810 cases in 2024. Measuring the burden this disease places on patients and health resources may serve to optimize our approach to the detection, management and resource allocation for combating this ailment. **Methods:** Our study is a population-based study cohort of colon cancer deaths in Texas. We queried the National Cancer Institute Surveillance, Epidemiology and End Results Program (SEER) for colon cancer deaths in Texas over the most recent available decade, 2012 through 2021. The National Center for Health Statistics Urban-Rural Classification Scheme was used to classify counties according to the 2013 US Census classification: urban ($\geq$ 50,000) and rural ($<$ 50,000). We calculated age-adjusted mortality rates (AAMRs) per 100,000 population using the direct standardization method based on the age group weights from the 2000 standard US population. Confidence intervals for AAMR were derived by estimating the standard error as the AAMR divided by square root of number of mortalities. Differences between AAMR were tested by comparing the 95% confidence intervals (95% CI) for the individual rates. The annual percent change (APC) in AAMR was tested using negative binomial regression. Subgroup analyses included age group, sex, and race/ethnicity. **Results:** Over the study period, 10.8% of the study population lived in rural areas and 89.2% lived in urban areas. A total of 33,591 colon-cancer deaths were reported during the study period with 5,423 (16.1%) in rural counties and 28,168 (83.9%) in urban counties. At the beginning of the study period the rural AAMR (13.9 [12.7 to 15.1]) was significantly higher than urban AAMR 12.1 [95% CI 11.7 - 12.6]). The rural AAMR rose significantly over the study period (APC 1.1% [95% CI 0.3% - 1.9%]; p = 0.0050) to AAMR 14.9 (95% CI 13.7 to 16.1). In contrast, the urban AAMR did not change over study period (APC -0.3% [95% CI -0.7 to 0.1]; p = 0.1444) and was equal to the initial AAMR 12.1 (95% CI 11.7 to 12.5). On subgroup analyses, males, people ages 25-64 years, Hispanics, Non-Hispanic Blacks, Non-Hispanic Asians, and Non-Hispanic Native Americans in rural areas each had significant and increasing trends in AAMR. **Conclusions:** Our analysis revealed an increase in AAMR in rural counties of Texas, reflecting a concerning trend with regard to colorectal cancer in this region. This may be due to higher poverty rates, lower level of education, more distance from healthcare facilities and difficulties with transportation faced by rural counties. Furthermore, rural areas also report higher rates of obesity and poor dietary habits. Rural areas also face a shortage of Primary Care Physicians (PCP's), with metropolitan counties retaining 1.7 times as many PCP's as rural counties, and 82.5% of counties without a PCP being rural. Research Sponsor: None.

37 Poster Session

Disparities in organ preservation rate for high-risk stage I rectal cancer. First Author: Annmarie Butare, East Carolina University/Brody School of Medicine, Greenville, NC

Background: Guidelines for the treatment of high-risk stage I rectal cancer is a curative oncologic resection, while recommendations for stage II/III rectal cancer include neoadjuvant therapy with the option for organ preservation. The paradoxical nature of treating less advanced stage cancer with more morbid surgery calls into question the use of organ preservation strategies (OP) for less advanced disease. While sphincter preservation is appealing, benefits from OP require commitment to preoperative therapy, which may require more significant time and financial commitments than a single-stage curative treatment. This study aims to evaluate disparities in social determinants of health regarding receipt of OP compared to major oncologic resection for high-risk stage I rectal cancer. **Methods:** The National Cancer Database was queried to analyze all patients with high-risk stage 1 rectal cancer between 2004-2021. High-risk stage 1 disease was defined as either T2 or T1 tumors with lymphovascular invasion, or grade 3 or 4 differentiation. Covariates in patients who underwent OP, defined as chemoradiation therapy without oncologic resection or local excision, were compared to those who underwent major oncologic resection. Additional variables included age, sex, race, great circle distance, insurance status, education, facility type, income, rurality, and comorbidity index. Univariate and multivariate analyses were utilized to compare association of covariates to OP. **Results:** Of the 63,762 patients included, 9,261 (14.5%) underwent OP. Increased rates of OP were seen with increasing age (OR 1.029, $p<0.0001$), female sex (OR 1.147, $p<0.0001$), higher education level (OR 1.179, p=0.0034), rurality (OR 1.273, p=0.0003) and lower income (OR 0.757 for income >74K, $p<0.0001$). Interestingly, OP was more frequently utilized at community cancer centers in comparison to comprehensive cancer centers (OR 0.78, $p<0.0001$) and academic centers (OR 0.842, p=0.0021). Race, insurance status, distance to treatment, and comorbidity index were not statistically significant predictors of OP. Of note, those who underwent OP had a considerable delay in time to first treatment, with median time to treatment of 36 days in comparison to 13 days for oncologic resection. **Conclusions:** Socioeconomic and demographic variables are associated with receipt of OP. While plausible that a treatment strategy that requires frequent office visits, travel expenses, and time off from work may preclude OP, lower income and rurality had increased odds of OP. Future aims should be directed to identifying root causes of disparities in treatment and their clinical consequences. Research Sponsor: None.

38 Poster Session

Colorectal cancer care disparities in an urban diverse population: A tale of two hospitals in the nation's capital. First Author: John Marshall, Ruesch Center for the Cure of Gastrointestinal Cancers, Lombardi Comprehensive Cancer Center, Georgetown University Medical Center, Washington, DC

Background: Colorectal cancer (CRC) disparities by race and ethnicity are well-documented nationally. In the District of Columbia (DC), Black Americans have an 18% higher incidence and 46% higher mortality rate compared to overall rates among all races/ethnicities. These disparities often stem from long-standing inequities in the social determinants of health, which affect access to cancer prevention, early detection, and treatment. MedStar Georgetown serves a large, urban, diverse population at 2 hospitals: the MedStar Georgetown University Hospital (MGUH) and the MedStar Washington Hospital Center (MWHC). MWHC serves a higher proportion of minoritized patients (pts) with multiple social determinant barriers to care, including those without private health insurance. We hypothesized that disparities would be found in CRC pt care and outcomes when comparing MWHC with MGUH. **Methods:** We conducted a retrospective study of all CRC pts treated during the 2021-2023 period at MGUH and MWHC using the COTA real-world database. Pt demographics, insurance status, and selected clinical data, including but not limited to treatment site, disease stage, tumor molecular testing, and pt survival, were abstracted for analysis. Descriptive, bivariate, and multivariate analyses focused on gender, race/ethnicity, insurance status, receipt of tumor testing, and the interval between diagnosis and therapy initiation and explored survival by these factors in addition to the site of care (MGUH versus MWHC). **Results:** Of the 379 CRC pts receiving care at the Lombardi Comprehensive Cancer Center during the study period, 53% were treated at MGUH and 47% at MWHC. CRC pts at MWHC were more likely than CRC pts at MGUH to be Black Americans (65% vs. 28%, $p<0.001$) and have 3 or more comorbidities (16% vs. 8%, p=0.22) but were less likely to have early onset CRC ($p<0.01$) and be on private insurance (36% vs. 23%, p=0.01). Pts treated at advanced stages (stages 3-4) were similar between the two hospitals (64%). In spite of these differences, we did not find any differences between MWHC and MGUH in molecular tumor testing (83% vs. 86%, p=0.41), median time (days) to treatment initiation (28 vs. 26 days, p=0.18), or short-term survival (90% vs. 91%, p=0.95). In analyses comparing care/treatment-related factors with race/ethnicity and insurance status across the 2 hospitals, we did not find any statistically significant differences. **Conclusions:** Contrary to our expectations of finding significant cancer care disparities, we did not observe any differences in molecular tumor testing rates, time to initiation of therapy, or short-term survival by treatment site or race/ethnicity status. Our results suggest that integrated, specialized, guideline-concordant care in comprehensive cancer centers has the potential to eliminate CRC outcome disparities among pts diagnosed at these centers. Research Sponsor: None.

39 Poster Session

Survival associated with metastatic early-onset colorectal cancer among Black patients: A real-world account. First Author: Sepideh Mehravar, Cedar Sinai, West Hollywood, CA

Background: Early-onset colorectal cancer (EOCRC), defined as colorectal cancer (CRC) diagnosed under the age of 50, is a growing health concern. The current study explored the racial differences in EOCRC survival in the United States (US). **Methods:** This retrospective cohort study was conducted using the TriNetX database, a platform that provides access to deidentified medical records on more than 130 million patients across 94 healthcare organizations in the US. The study cohort included patients with metastatic CRC (mCRC), aged 20 to 50, diagnosed between January 1, 2002, and January 1, 2020, and received frontline chemotherapy with oxaliplatin and 5- fluorouracil or capecitabine. The cohort was divided into Black or African American cohort and White cohort. The primary endpoint, 5-year overall survival (OS), was compared between these cohorts after performing 1:1 propensity score matching (PSM) for baseline demographics and comorbid conditions (Table). Kaplan-Meier curves, log-rank chi-square statistics, and hazard ratios (HR) were used to compare the outcome. **Results:** The entire cohort consisted of 2,923 patients with a mean age of 44 ($\pm$6) years; 48.58% were female. The majority were white (56.38%), followed by 15.56% of unknown race, Black/African American (9.16%), Hispanic (9.34%), and Asian (8.96%). After propensity score matching (PSM), there were 262 patients in both the Black and White groups. In the matched population, the Kaplan-Meier survival curves showed that Black patients had significantly lower 5-year OS rates (24.84%) compared to White patients (33.76%) [Hazard Ratio (HR): 1.39, 95% CI (1.06, 1.85), $p<0.05$]. There is no significant difference between Black and White population over age of 50 [HR: 1.06, 95% CI (0.966-1.179), P=0.2]. **Conclusions:** Survival among Black patients with EOCRC appears to be worse than White patients despite adjustment for confounding variables, supporting further research focusing onto this patient population. Research Sponsor: None.

Characteristics of the cohorts before and after matching.

	Black cohort (n = 265) and White cohort (n = 1,632) characteristics before propensity score matching			Black cohort (n = 262) and White cohort (n = 262) characteristics after propensity score matching		
	Black patients, n (%)	White patients, n (%)	Sth diff.	Black Patients, n (%)	White patients, n (%)	Sth diff.
Age (Mean ± SD)	36.2 ± 6.18	36.3 ± 6.01	0.022	36.1 ± 6.17	36.2 ± 6.43	0.013
Female	144 (54.5%)	774 (47.4%)	0.142	143 (54.5%)	143 (54.5%)	<0.0001
Tobacco use	15 (5.6%)	96 (5.8%)	0.0087	15 (5.7%)	11 (4.2%)	0.0703
Hypertensive diseases	67 (25.3%)	214 (13.12%)	0.3147	65 (24.8%)	64 (24.4%)	0.0089
Ischemic heart diseases	10 (3.7%)	40 (2.4%)	0.0769	10 (3.8%)	10 (3.8%)	< 0.0001
Type 2 diabetes mellitus	24 (9%)	89 (5.4%)	0.1403	22 (8.3%)	19 (7.2%)	0.0426
Diseases of arteries, arterioles and capillaries	12 (4.5%)	47 (2.8%)	0.0881	11 (4.1%)	10 (3.8%)	0.0195

40 Poster Session

Impact of insurance status on mortality following surgical treatment of colorectal cancers. First Author: Atulya Aman Khosla, Corewell Health William Beaumont University Hospital, Royal Oak, MI

Background: Healthcare disparities in colorectal cancer, driven by insurance status and socioeconomic factors, lead to delayed diagnoses and poorer surgical outcomes. We sought to examine the impact of insurance status on presentation, treatment, and in-hospital mortality in CRC patients undergoing colectomy using the Nationwide Inpatient Sample (NIS) database. **Methods:** We included patients aged 18-65 years diagnosed with colon cancer and undergoing colectomy, as identified by ICD-9-CM codes, who had private insurance, Medicaid, or no insurance from January 1, 2005, through December 31, 2014. The primary variable was insurance status; patient characteristics such as age, sex, race, income, and tumor type were analyzed. We used the Elixhauser comorbidity index to control for confounding and performed a subset analysis on patients without major comorbidities. The primary outcome was in-hospital postoperative death. Associations between this outcome and insurance status were analyzed using the Cox proportional hazard model, both in the full cohort and in a subset of patients without comorbidities, with models stratified by hospitals to account for clustering effects from variations in access to care. **Results:** The study cohort included 301,304 patients, of whom 238,158 (79.0%) were privately insured, 40,417 (13.4%) on Medicaid, and 22,729 (7.6%) were uninsured. Most patients were White (71.6%), followed by African American (12.6%), Hispanic (8.4%), Asian/Pacific Islander (3.8%), and Native American (0.5%). A total of 55.4% of cases took place in teaching hospitals. In the unadjusted analysis, the mortality rate for privately insured patients was 0.7% (95% CI, 0.6%-0.7%) compared with 2.1% for uninsured patients (95% CI, 1.7%-2.5%) and 1.5% for Medicaid recipients (95% CI, 1.2%-1.8%; p=0.001). After adjusting for patient characteristics and stratifying by hospital in patients with no comorbidity, uninsured patients still had a higher risk of experiencing in-hospital death (HR, 1.60; 95% CI, 1.24-2.07) compared with privately insured patients, while no significant disparity was found in Medicaid recipients (HR, 0.95; 95% CI, 0.75-1.22) (Table). **Conclusions:** Uninsured patients undergoing colectomy for colon cancer experienced the highest in-hospital mortality, a disparity not fully explained by overall health differences. These findings underscore the critical role of insurance coverage in improving surgical outcomes and highlight the need for policy interventions to reduce mortality disparities. Research Sponsor: None.

In-hospital mortality after surgery for colorectal cancer.

	Full sample	Patients With No Comorbidity	Patients With No Comorbidity in Urban Teaching Hospitals
Covariate	Hazard Ratio (95% CI)		
Insurance			
Private	Reference	Reference	Reference
Medicaid	0.95 (0.75-1.22)	1.38 (0.91-2.11)	1.66 (0.91-3.02)
Uninsured	1.60 (1.24-2.07)	2.12 (1.42-3.15)	2.28 (1.25-4.16)

41 Poster Session

Regional variation in trajectories of colon cancer mortality, 2012-2021. First Author: Ahmed Bashir Sukhera, Texas Tech University Health Sciences Center, Odessa, TX

Background: Colorectal cancer (CRC) is the fourth most common cancer by incidence in the United States in 2024, and the second most common cause of cancer mortality in Texas. Rates of advanced CRC at diagnosis and cancer related mortality are higher in rural areas. Here we explore the variation in mortality trends across the eleven public health regions (PHR) of Texas from 2012 to 2021. **Methods:** We conducted a population-based cohort study of colon cancer mortality in Texas over the period 2012 through 2021. We queried the National Cancer Institute Surveillance, Epidemiology and End Results Program (SEER) for county level colon cancer deaths in Texas over the most recent available decade. Counties were aggregated into the eleven public health regions defined by the Texas Department of State Health Services. We calculated age-adjusted mortality rates (AAMRs) per 100,000 population using the direct standardization method based on the age group weights from the 2000 standard US population. Confidence intervals for AAMR were derived by estimating the standard error as the AAMR divided by square root of number of mortalities. The annual percent changes (APC) in AAMR were tested using negative binomial regression. Subgroup analyses included age group, sex, and race/ethnicity. **Results:** A total of 33,591 colon cancer mortalities were included in the study. Total colon cancer mortalities rose from 2,973 in 2012 to 3,800 in 2021. The statewide AAMR was initially 12.4 (95% CI 12.0 to 12.9), and did not change over the study period (APC -0.1% [95% CI -0.5% to 0.2%]; p = 0.5155) ending at (AAMR 12.5 [95% CI 12.1 to 12.9]). Divergent trajectories for AAMR across public health regions were observed. The largest decline in AAMR (APC -3.9% [95% CI -6.3 to -1.6]; p = 0.0013) occurred in region 10, where the AAMR fell from 14.2% [95% CI 8.6 to 19.7]) to AAMR 11.0 [95% CI 6.8 to 15.1]. Region 6 also saw a significant decrease (APC -1.1% [95% CI -1.8% to -0.4%]; p = 0.0029) falling from AAMR 12.2 [95% CI 11.3 to 13.2] to AAMR 11.2 [95% CI 10.4 to 11.9]. The largest significant increasing trend (APC +2.3% [95% CI 0.6% to 4.1%]; p = 0.0089) for AAMR was observed for region 9 rising from AAMR 12.4 (95% CI 9.6 to 15.3) to AAMR 15.4 (95% CI 12.3 to 18.4). **Conclusions:** Our analysis showed a significant and increasing trend in CRC related mortality in PHR 9. This region, colloquially termed West Texas, consists of 30 counties of which over 86% are designated as rural. Rural areas struggle with higher poverty, lower degree of higher education attainment, less access to healthcare and physicians, and a higher frequency of uninsured status. West Texas in particular also has a higher rate of smoking and obesity. However, PHR 10, which shares many similarities with PHR 9 such as rurality (83.3%) and similar socioeconomic and geographic challenges, witnessed the greatest decline in AAMR of the eleven PHR's in the same time frame, suggesting the presence of other factors which may be contributing to this increase. Research Sponsor: None.

42 Poster Session

Trends in early onset colorectal cancer–related mortality among adults in the United States from 1999-2020. First Author: Yousaf Zafar, University of Mississippi Medical Center, Madison, MS

Background: Colorectal cancer (CRC) has witnessed a decrease in incidence and mortality in the United States (US) for the past few decades likely due to increased screening and improvements in treatment . Despite the overall burden of CRC has declined and survival has been made for older adults, there is a simultaneous and alarming increase in CRCs diagnosed in individuals younger than 50 years of age. These Early Onset CRC (EOCRC) which now account for roughly 10% of new CRC cases in the US. The purpose of this study was to assess the trends and regional differences in EOCRC-related mortality among adults in the United States. **Methods:** Death certificates from the CDC WONDER (Centers for Disease Control and Prevention Wide-Ranging OnLine Data for Epidemiologic Research) database were examined from 1999 to 2020 for -related mortality in adults $<$45 years of age. Age-adjusted mortality rates (AAMRs) per 10,000 persons and annual percent change (APC) were calculated and stratified by year, sex, race/ethnicity, and geographic region. **Results:** Between 1999 and 2020, 32,850 EOCRC-related deaths occurred among adults 25-44 years. The AAMR increased from 2.1 in 1999 to 2.6 in 2020. Men had consistently higher AAMR than women from 1999 (AAMR men: 2.0 vs women: 1.6) to 2020 (AAMR men: 2.6 vs women: 2.0). Non Hispanic (NH) African Americans had the highest overall AAMR (3.1), followed NH White (2.2), NH Asians (1.6) and Hispanics (AAMR 1.5) Caucasians have had a significant increase in mortality over the past 2 decades (APC 1.4 (95%CI 1.1-.7)) whereas the Hispanics have shown an APC of 4 (95% CI 2.3-5.8 from 2012-2020). AAMR also varied substantially by region (overall AAMR: Midwest 2.1; South: 2.4; West: 1.9; Northeast: 2.0). States in the top 90th percentile of EOCRC-related AAMR were Mississippi, Arkansas, Oklahoma, South Carolina, Alabama West Virginia, Louisiana ,Kentucky and Tennessee which had approximately double the AAMRs compared with states that fell into the lower 10th percentile. **Conclusions:** EOCRC-related mortality in U.S. adults has been steadily increasing over the past two decades . The highest AAMRs were observed among African American adults and men, and among patients living in the Southern , Midwestern regions and those living in the nonmetropolitan United States. Targeted strategies are needed to prevent and treat EOCRC among older adults to curb increasing levels of EOCRC-related mortality. Research Sponsor: None.

43 Poster Session

Association between visceral adipose tissue radiodensity and overall survival among older adults with colorectal carcinoma. First Author: Apoorva Doshi, University of Alabama at Birmingham, Birmingham, AL

Background: Central obesity is a known risk factor for colorectal carcinoma (CRC); however, the association between visceral adipose tissue (VAT) and survival outcomes remains unclear. This study aims to investigate the relationship between VAT and overall survival (OS) among older adults with CRC. **Methods:** We analyzed data from the Cancer and Aging Resilience Evaluation registry, including adults aged $\geq$ 60 years with newly diagnosed CRC who were seen at the University of Alabama at Birmingham for initial consultation between October 2017 and April 2024. Using single-slice computed tomography staging scans, we calculated visceral adipose tissue radiodensity (VATD; in Hounsfield units, HU) at the third lumbar vertebra (L3) using validated methods. A lower VATD (more negative score) indicates higher visceral adipose tissue. Participants were followed until death or the study cutoff date of July 10, 2024. Kaplan-Meier survival analysis was performed to estimate OS for those with low VAT (VATD $\geq$ median VATD) versus high VAT (VATD $<$ median VATD). Additionally, Cox proportional hazards models were utilized to evaluate the association between VATD and OS, adjusting for age at diagnosis, sex, race/ethnicity, skeletal muscle index (SMI), skeletal muscle density (SMD), and cancer stage. **Results:** A total of 190 patients were included, with a median age at diagnosis of 68 years (IQR: 64-75), 56% male, 72% non-Hispanic White, and 40% diagnosed with Stage IV disease. The median SMI and SMD were 42.06 (IQR: 33.81 - 49.52) cm^2/m^2 and 38.68 HU (IQR: 31.38 - 44.89) respectively. The median VATD was -86.14 HU (IQR: -93.78 – -76.39). Over a median follow-up of 60 months, 96 patients (51%) died. Median OS was 27 months for patients with low VAT and 62 months for those with high VAT (log-rank test, $p < .001$). After adjusting for confounders, VATD remained independently associated with OS (adjusted Hazard Ratio: 1.41 per IQR decrease; 95% CI: 1.06 – 1.86; $p \leq .05$), indicating that each IQR decrease in VATD scores (implying an increase in VAT) is associated with a 41% increased risk of all-cause mortality. **Conclusions:** Our findings demonstrate that increased visceral adipose tissue, indicated by more negative VATD scores, is significantly associated with poorer survival outcomes among older adults with CRC. These results underscore the importance of dietary and weight loss interventions to improve outcomes in this population. Research Sponsor: None.

44 Poster Session

Multi-site blinded validation of a deep learning approach for clinical-grade MSI/dMMR detection in colorectal cancer from H&E-stained pathology images. First Author: Cher Bass, Panakeia Technologies Limited, Cambridge, United Kingdom

Background: Testing for microsatellite instability (MSI) or mismatch repair deficiency (dMMR) is part of the diagnosis and clinical management of patients with colorectal cancer (CRC). Healthcare services recommend MSI or dMMR testing for all CRC patients to guide therapeutic choices and assist in identifying Lynch Syndrome. However, in clinical practice, high costs and the demand for timely test results, combined with the rising prevalence of CRC and a shrinking pathology workforce, present a barrier to universal adoption. This highlights the need for rapid and affordable alternatives. PANProfiler CRC (PPC) is a deep learning-based solution for detecting MSI/dMMR in CRC tumors that only requires whole slide images (WSIs) of haematoxylin and eosin (H&E)-stained tissue to provide test results. Using only WSIs, PPC offers an efficient alternative to standard testing. This study evaluates PPC's performance in a multi-site blinded setting. **Methods:** Blinded validation was performed using 3246 WSIs of H&E-stained CRC specimens. PPC provided outputs as "Stable", "Unstable", or "Indeterminate", with "Unstable" indicating dMMR or MSI-High, and "Stable" indicating proficient mismatch repair or non-MSI-High. "Indeterminate" was returned when PPC did not have a definitive result. PPC was evaluated by comparison to standard MSI/dMMR tests. Validation data spanned three cohorts from two sites (Table). St James's University Hospital (SJUH), Leeds, UK, supplied Cohorts 1 and 2; Cohort 3 was sourced from Wales Cancer Biobank (WCB), UK. Blinded analysis was performed at SJUH. **Results:** Results are given (Table). PPC demonstrated an overall percent agreement of 93.91%, a positive percent agreement of 92.17%, and a negative percent agreement of 94.15%, returning a definitive result for 88.05% of WSIs. **Conclusions:** This real-world, multi-site, blinded validation study demonstrates PPC's remarkable performance, comparable to standard tests for detecting MSI/dMMR in CRC, with high test replacement rates. In the clinical setting, PPC could significantly accelerate testing and enable timely delivery of stratified treatment plans. This accurate and cost-effective diagnostic solution promises to revolutionize MSI/dMMR testing in CRC. Research Sponsor: None.

Blinded validation results of PPC with confidence intervals (CI) at 95%.

Site	Cohort	Sample Size (Unstable; Stable)	Overall Percent Agreement % (CI)	Positive Percent Agreement % (CI)	Negative Percent Agreement % (CI)	Test Replacement Rate %
SJUH	1	488 (78; 410)	92.79 (89.86-95.08)	90.16 (79.81-96.30)	93.24 (90.11-95.62)	85.25
SJUH	2	2704 (318; 2386)	94.31 (93.31-95.21)	92.31 (88.48-95.18)	94.57 (93.52-95.50)	88.42
WCB	3	54 (11; 43)	84.31 (71.41-92.98)	100.00 (71.51-100.00)	80.00 (64.35-90.95)	94.44
	All	3246 (407; 2839)	93.91 (92.97-94.76)	92.17 (88.82-94.78)	94.15 (93.16-95.04)	88.05

45 **Poster Session**

Utility of computed tomography (CT) –based staging of mismatch repair deficient (dMMR) colon cancer. First Author: Samantha Linhares, Yale School of Medicine, Department of Surgery, Division of Colon and Rectal Surgery, New Haven, CT

Background: dMMR tumors are characterized by rapid primary tumor growth with reduced likelihood for nodal and distant metastases compared to MMR proficient (pMMR) tumors. Neoadjuvant immune checkpoint inhibitors (ICIs) have demonstrated promising efficacy for localized colon cancer with high pathologic complete response rates and low relapse rates. This raises the question if surgical treatment could be avoided with accurate radiographic staging while avoiding overtreatment of early stage tumors. This study assesses the utility of radiographic staging by CT for dMMR tumors compared to pathologic staging. **Methods:** Patients with stage I-III colon cancer treated with upfront surgical resection were retrospectively reviewed from a single institution from 2012 to 2023 in two cohorts: 1) dMMR colon cancer and 2) pMMR colon cancer limited to similar sample size as matched controls. Clinical stage was determined based on CT scans prior to surgical resection by 2 independent radiologists blinded to pathologic stage, and the results were correlated to pathologic stage. Clinical characteristics were extracted from the electronic medical record. Statistical analysis was performed using SPSS. **Results:** We identified 80 patients with dMMR colon cancer and 66 with pMMR tumors. CT clinical tumor staging was discordant from pathologic staging for 68% of dMMR tumors and 61% for pMMR tumors. For T4 dMMR tumors, CT had a 74% sensitivity and 44% specificity. T4 tumors were understaged by radiology to less than T4 for 56% of cases. For dMMR nodal staging, CT had 89% sensitivity and 50% specificity (Table 1). For pathologic node positive tumors, 14% were understaged by radiologist CT read to node negative. For pathologic node negative tumors, 50% of tumors were overstaged to node positive by radiologist CT read. There was poor inter-rater reliability for both T stage and N stage for dMMR and pMMR tumors. The weakest agreement between radiologists was for T stage (dMMR kappa = 0.246, mMMR kappa = 0.145). **Conclusions:** Radiologic clinical staging of dMMR and pMMR colon cancer does not correlate well with pathologic staging. There were high rates of understaging by CT evaluation for patients who may be candidates for neoadjuvant treatment (T4, N+). Furthermore, there was poor inter-rater reliability between radiologists, indicating that CT staging alone may not be sufficient. Additional diagnostic modalities for nodal detection may be necessary to accurately clinically stage patients before neoadjuvant ICIs in patients with locally advanced disease. Research Sponsor: None.

Comparing utility of pMMR versus dMMR tumors for T4 and node positive disease.

	pMMR (n = 66)	dMMR (n = 80)
T4 versus T3 or less		
Sensitivity (%)	92	74
Specificity (%)	51	47
PPV (%)	80	77
NPV (%)	82	63
Node positive versus node negative		
Sensitivity (%)	60	50
Specificity (%)	75	89
PPV (%)	81	89
NPV (%)	53	50

Abbreviations: PPV = positive predictive value; NPV = negative predictive value.

46 **Poster Session**

A protein-based blood panel test for the early detection of colorectal cancer and advanced neoplasia. First Author: Christine Chavany, Milagen, Inc., Emeryville, CA

Background: Early detection of colorectal cancer (CRC) is key to survival. CRC screening compliance is still low because of either the invasive procedure (colonoscopy) or the so-called "ick factor" (stool-based tests). Stool-based tests also lack ability to detect advanced adenoma (CRC precursor). A blood-based test could improve adherence to screening guidelines, but no one has so far been recommended due to a lack of sufficient sensitivity and specificity. Here we report the potential clinical utility of a blood-based biomarker panel providing high sensitivity (SE) and specificity (SP) for CRC and advanced adenomas (AA) as a screening triage option addressing positive patients to colonoscopy. **Methods:** A total of 241 retrospective serum samples (179 CRC, 22 adenoma, 40 healthy controls) were obtained from the Cooperative Human Tissue Network (CHTN) biobank and tested with a proprietary sandwich ELISA that detects dual CRC specific biomarkers (BF7 and CC3). A standard curve was generated and biomarkers concentration in the samples were calculated using a 4-parameter logistic method. The median biomarker concentration values in each patient group were compared to determine whether differences were statistically significant ($p < 0.05$). Statistical analysis of two-group and multi-group comparisons was carried out using the non-parametric Mann-Whitney U test and the Kruskal-Wallis test, respectively. Receiver operating characteristic (ROC) curves were calculated to evaluate the diagnostic performance, by determining sensitivity (SE), specificity (SP), area under the curve (AUC), and confidence interval (CI). Statistical analysis, ROC curves and scatter plots were performed with GraphPad Prism 7.0. **Results:** Both biomarker's median concentration difference between CRC versus healthy samples groups were found to be statistically significant ($p<0.05$). Both biomarkers were also elevated in AA as compared to healthy samples, but difference in median concentration was only statistically significant for CC3 ($p<0.0001$). BF7 serum levels in cancer patients were significantly associated with TMN stage (p=0.019), N stage (p=0.039) and M stage (p=0.0006). No significant associations were found between this biomarker levels and other clinical characteristics of the CRC patients including age. Our test discriminated CRC (all stages) from healthy individuals with a SE of 80% and a SP of 90%. Sensitivity for early (n=95) and late-stage CRC (n=80) was 79% and 84%, respectively. Our test was also able to detect 81% of serum from advanced adenoma patients who are at increased risk of CRC. **Conclusions:** This study illustrates the potential clinical utility of our blood-based assay for the early detection of CRC which has a better sensitivity for advanced adenoma than tests in current clinical use. A large prospective cohort study is planned to further validate the clinical performance of this test. Research Sponsor: None.

47 **Poster Session**

The role of artificial intelligence in colorectal cancer and polyp detection: A systematic review. First Author: Sakshi Bai, Henry Ford Jackson Hospital, Jackson, MI

Background: Colorectal cancer (CRC) remains a leading cause of cancer-related mortality worldwide. Early detection and accurate diagnosis are essential for improving survival rates and optimizing therapeutic strategies. Recently, artificial intelligence (AI) technologies, such as machine learning algorithms, convolutional neural networks (CNNs), and computer-assisted diagnostic (CAD) systems, have significantly enhanced traditional diagnostic tools like colonoscopy and histopathology. This systematic review evaluates the current role of AI in CRC management, focusing on diagnostic improvements, predictive modeling, and outcome prediction. **Methods:** A comprehensive literature review was conducted using PubMed, Google Scholar, and MEDLINE using MeSH term to identify studies utilizing AI in CRC diagnosis and management, with a focus on diagnostic imaging, histopathological analysis, and predictive modeling. Studies were evaluated for the accuracy of AI-driven diagnostic systems and the predictive performance of AI models in clinical outcomes. **Results:** Out of 147 studies identified, 37 met the eligibility criteria for this review. AI tools such as CNNs, CAD systems, and EndoBRAIN showed high accuracy in CRC detection and polyp classification. Kudo et al. (2020) reported 98% CRC detection accuracy using EndoBRAIN. Blanes-Vidal et al. (2019) achieved 96.4% accuracy in polyp detection via DCNN. Additionally, Wang et al. (2019, 2020) demonstrated improved Adenoma Detection Rates (ADR) of 34.1% and 29.1%, respectively, using AI over traditional methods. For outcome prediction, AI models, including CNNs, predicted patient prognosis with reasonable accuracy, as highlighted by Skrede et al. (2020) with 76% accuracy in prognosis prediction. **Conclusions:** The systematic review and analysis of AI-assisted diagnostic modalities in colorectal cancer and polyp detection reveal promising results, with most models demonstrating high sensitivity, specificity, and accuracy. CNN-based models and CAD systems, in particular, show strong potential for improving detection rates and clinical outcomes in CRC screening. The studies included in this review demonstrate that AI can enhance diagnostic precision, particularly in identifying polyps and predicting clinical outcomes, with high validation rates. However, more external validation and long-term clinical studies are needed to fully establish the robustness of these AI tools in routine clinical practice. Research Sponsor: None.

48 **Poster Session**

Molecular characteristics and prognostic impact of *GNAS* mutations in colorectal cancer: An international collaborative study between United States and Japan. First Author: Elisabeth Arrondo, Translational Research Support Office, National Cancer Center Hospital East, Kashiwa, Japan

Background: *GNAS* (Guanine nucleotide-binding protein, alpha stimulating) encodes the alpha subunit of the stimulatory G protein, and gain of function mutations in this gene have been previously identified, especially those located at the R201 codon. Although $GNAS^{R201C/H}$ has been associated with mucinous histology and poor response to chemotherapy in appendiceal cancer, the impact of *GNAS* mutations in colorectal cancer (CRC) remains unclear. **Methods:** We conducted a comparative, integrative analysis of *GNAS* mutations in CRC using two cohorts: MD Anderson Cancer Center (MDA) across all stages (N=5,249) and the GALAXY study involving only resectable disease (UMIN000039205, N=2,599). We assessed the correlation between *GNAS* pathogenic mutations and microsatellite stability, CRC sidedness, co-mutations, and survival outcomes. The Chi-squared test was used for categorical variables, and the log-rank test for overall survival (OS) and disease-free survival (DFS). **Results:** In the MDA and GALAXY cohorts, *GNAS* mutations were identified in 107 (2.0%) and 61 (2.4%) patients, respectively. Prevalence of *GNAS* mutations was higher in right sided tumors (MDA: 3.8% vs. 1.3%, GALAXY: 6.2% vs 2.8%, $p<0.01$ for both cohorts) and microsatellite instability-high (MSI-H) tumors (MDA: 6.5% vs. 1.8%, GALAXY: 16.5% vs. 1.4%, $p<0.01$ for both cohorts). There was not an association of *GNAS* mutations with stage in either cohort. *GNAS* was significantly enriched for co-mutation with *KRAS* in the MDA cohort (OR=3.26, $P<0.01$), similar to prior observations in appendix cancer, but was not significantly associated with *KRAS* mutations in the GALAXY cohort (OR=1.35, P=0.3). *GNAS* mutations were mutually exclusive with *TP53* in both cohorts (MDA: OR=0.58, P=0.026; GALAXY: OR=0.21, $P<0.01$). With regards to survival, in the MDA cohort, *GNAS* mutant tumors had a trend towards worse OS in the overall population (20 mo vs. 33 mo, HR=1.27, 95% CI=0.61-1, P=0.085). The survival impact of *GNAS* mutations was greatest in metastatic patients (Stage IV: 19 mo vs. 38 mo, HR=1.49, 95% CI=0.47-0.94, P=0.021 vs. Stage I-III: HR=1.35, 95% CI=0.46-1.2, P=0.22). The GALAXY cohort showed no significant association between *GNAS* mutations and DFS (HR=1.05, 95% CI=0.6-1.8, P=0.9). In both cohorts, *GNAS* mutations had a worse prognosis in left-sided CRC (MDA: HR for OS = 1.45, P = 0.091; GALAXY: HR for DFS=2.15, P=0.09) with no significant effect observed in right-sided CRC (MDA: HR for OS=0.91, P=0.68; GALAXY: HR for DFS=0.65, P=0.3). **Conclusions:** Our findings highlighted the intricate role of *GNAS* mutations in CRC, demonstrating its higher prevalence in MSI-H and right-sided colon, significant association with *KRAS* mutations, and divergent prognostic impacts based on tumor sidedness. Further research is needed to elucidate the mechanisms behind the worsened survival in left-sided *GNAS*-mutated CRC. Research Sponsor: Conquer Cancer Career Development Award; (2022CDA-7604125121); SCRUM-Japan Funds.

49 Poster Session

Landscape of metastatic colorectal cancer (CRC) using comprehensive circulating tumor DNA (ctDNA) next-generation sequencing (NGS) in India: Expanding beyond RAS and RAF. First Author: Viraj Kiran Lavingia, Shalby Hospital, Ahmedabad, India

Background: Traditional testing in mCRC primarily focuses on RAS and RAF mutations. However, a broader range of emerging biomarkers, collectively termed the "Pressing panel" (PP), identifies patients with RAS/RAF wild-type (wt) tumors who may particularly benefit from anti-EGFR therapy. Comprehensive liquid biopsies provide a non-invasive method for identifying traditional and emerging biomarkers. Limited genomic data, including the prevalence of PP alterations (alts), is available for mCRC patients in India tested by ctDNA NGS. **Methods:** We reviewed the results of Guardant360 (Guardant Health, Inc.) ordered for patients with mCRC as part of routine clinical practice in India through July 2024. NGS analyzed plasma samples at a CLIA-certified, CAP-accredited central laboratory. Guardant360 analyzes ctDNA for mutations (mts), insertions and deletions, amplifications (amps), and fusions in up to 83 genes, including PP genes, i.e., KRAS, NRAS, BRAF, ERBB2, PIK3CA, PTEN, AKT1, ALK, ROS1, NTRK, RET, and MSI status. The Indian cohort's prevalence of PP mts was compared to that from other countries in Asia and Middle East (AME). Timing of testing relative to line of therapy was unknown. **Results:** Among 145 samples from India, ctDNA was detected in 92% (n=134). Median age was 59 years, with women comprising 40%. The median turnaround time from blood collection to result was 7 days. The mean mts count per sample was 5, and the median variant allele frequency was 1.5%. Frequent mts were in APC (53%), KRAS (56%), and SMAD4 (13%), while amps of EGFR (16%) and FGFR1 (9%) were common. Clinically relevant fusions were observed in 3% of patients. KRAS, NRAS, and/or BRAF V600E alterations were found in 63% of both India and AME samples. PP alts were identified in 24% and 27% of Indian and AME patients, respectively. Most frequent PP alts included PIK3CA (15%), PTEN (3%), and MET amp (2%). The prevalence of most of the PP alts was similar between the Indian and AME cohorts (Table), with minor differences observed in PIK3CA (15% vs 18%) and MSI-High (3% vs 1%). In patients with RAS/RAF wild-type lacking EGFR-amp, 9.5% (India) and 18% (AME) harbored at least one PP alt. **Conclusions:** This study highlights the role of liquid CGP for characterizing the mutational landscape and identifying PP mts for patients from India with advanced CRC. Research Sponsor: None.

Pressing panel alterations.

PP Alteration	INDIA % (n=134)	AME % (n=720)
ERBB2 Amp	1%	2%
MET Amp	2%	3%
ALK Fusion	0%	0%
ROS1 Fusion	0%	0%
NTRK1 Fusion	1%	0%
RET Fusion	0%	1%
ERBB2 Mut	1%	2%
PIK3CA Mut	15%	18%
PTEN Mut	3%	3%
AKT1 Mut	0%	1%
MSI-High	3%	1%

50 Poster Session

Clinical validation of plasma circulating-tumor DNA assay using highly sensitive Safe-SeqS technology for detecting *RAS* and *BRAF* V600E in metastatic colorectal cancer. First Author: Yu Miyashita, National Cancer Center Hospital East, Kashiwa, Japan

Background: We developed a circulating-tumor DNA (ctDNA) test assay for *RAS/BRAF* mutations using next-generation sequencing-based Safe-SeqS technology. The clinical performance of detecting *RAS* and *BRAF* mutations were evaluated versus approved liquid biopsy and tissue testing, respectively. **Methods:** Metastatic colorectal cancer (mCRC) patients (pts) were enrolled. The concordance of the *RAS* mutational status in ctDNA between Safe-SeqS assay and OncoBEAM RAS CRC kit, which is approved as a companion diagnostic in Japan, was evaluated using archival plasma samples. The primary endpoints were positive percent agreement (PPA) and negative percent agreement (NPA). The concordance of the *BRAF* V600E mutational status between plasma-Safe-SeqS assay and tissue-based testing using MEBGEN RASKET-B kit, which is approved as a companion diagnostic in Japan, was prospectively evaluated. The primary endpoints were sensitivity and specificity of Safe-SeqS assay compared to tissue-based testing. Multivariate analysis using LASSO regression model was performed to evaluate the discordance of *BRAF* V600E mutational status between Safe-SeqS assay and tissue-based testing. **Results:** In 125 eligible plasma samples, the PPA and NPA for *RAS* mutational status in ctDNA were 87.1% (95% confidence interval [CI]: 76.1–94.3) and 95.2% (95% CI: 86.7–99.0), respectively. On the other hand, in 103 eligible pts, the sensitivity and specificity of Safe-SeqS assay for *BRAF* V600E mutational status were 69.2% (95% CI: 48.2–85.7) and 98.7% (95% CI: 93.0–100.0), respectively. Multivariate analysis revealed the baseline longest diameter and the number of lesions in pts with peritoneal and/or lung metastasis were factors associated with discordance. The sensitivity for *BRAF* V600E mutations increased to 81.0% (95% CI: 58.1–94.6, fisher's exact test p=0.02) when the 5 pts with lower tumor burden (peritoneal metastasis with the longest diameter $<$20 mm, and lung metastasis with the longest diameter $<$20 mm and $<$10 lesions, according to the previous report [1]) were excluded. **Conclusions:** The clinical validity of the ctDNA assay for detecting *RAS/BRAF* V600E mutations using Safe-SeqS technology was confirmed in pts with mCRC. Careful attention should be paid for mCRC pts with only peritoneal and/or lung metastases having fewer metastases or smaller diameter lesions. 1. Kagawa Y, et al. Clin Cancer Res 2021. Research Sponsor: None.

51 Poster Session

Communication of cancer biomarker testing results: What patients need. First Author: Julie Saliba Clauer, COLONTOWN/PALTOWN, Crownsville, MD

Background: As colorectal cancer (CRC) biomarker testing becomes more widespread and important for treatment decisions, understanding testing results can improve patient engagement, shared decision-making, and facilitate clinical trial enrollment. Our experience in the online 12,000+ member CRC community COLONTOWN (CT), suggests that communication of biomarker results to patients is not ideal. To learn more, we surveyed CT members on their biomarker testing experience. **Methods:** An online survey posted in CT in September 2024 had questions on testing received (what was tested and how), test results experience (communication before testing and after results, patient understanding and resulting impact), suggested provider statements on biomarker results (statements patients prefer) and disease and respondent details. **Results:** Of 83 respondents, 70% had testing that included at least 3 common CRC biomarkers; the following are results from this group. Until receipt of results, 30% were unaware that biomarker testing was being done, and 56% first reviewed results without a clinician. Only 17% of respondents felt that nothing was lacking in the clinician communication of biomarker testing results. Upon receipt of biomarker test results, respondents' feelings improved (30%,18%), did not change (63%, 55%) or worsened (7%, 27%) about treatment options and prognosis, respectively. Test results made 51% feel more confident in their care, and reduced stress for 25%, but increased stress for 39%, with 1/3 citing sub-par communication of results as at least partly driving the stress increase. The value of clear communication was evident, as the most appealing sample provider statements on results with no actionable biomarkers included confirmation of proposed treatment plan and clarity on non-actionable biomarkers. Not surprisingly, 58% learned the most about their biomarker results from COLONTOWN. **Conclusions:** Our survey highlights the gaps in communication of biomarker testing results and provides insights on what may improve them. The perceived value of biomarker test results for patients seem to go beyond actionability as traditionally defined. Context-specific, nuanced communication of "what the results really mean" and a discussion of its impact on patient experience needs to be part of clinician communication of test results. We plan to develop resources to help, and we implore clinicians and testing companies to consider our findings and make changes in how they share test results with patients. Research Sponsor: None.

52 Poster Session

A method for classifying colorectal cancer and gastric/esophageal cancer using blood-based testing. First Author: Elmira Forouzmand, Guardant Health, Redwood City, CA

Background: Identifying the cancer signal of origin is of great value in the clinical use of blood-based cancer testing, especially in those tests that have potential to assess for more than one cancer type. Thus, an approach that can accurately identify cancer signal of origin in order to guide subsequent diagnostic workup is impactful to the adoption of this new technology. Here, we report feasibility data using the DNA methylation signature of plasma cell-free DNA to differentiate between colorectal cancer (CRC) and gastric/esophageal cancer (GEC) types. **Methods:** We developed an algorithm to distinguish between CRC and GEC based on the DNA methylation signature of plasma cell-free DNA molecules from more than 3,000 differentially methylated regions. First, a regression model (a multi-cancer classifier) was trained to identify the samples with sufficient tumor derived methylation signal using 6,822 cancer samples of 24 cancer types and the samples from 4,423 cancer-free individuals. A second regression model (a CRC/GEC classifier) was trained to distinguish CRC from GEC cases on the samples with positive calls from the multi-cancer classifier. The second model was trained on methylation signals from advanced stage CRC samples, advanced stage GEC samples (including esophageal, gastroesophageal, and gastric cancers), and samples from cancer-free individuals. **Results:** The multi-cancer classifier identified 92% (2,954/3,204) of CRC and GEC samples as positive at 90% target specificity, including 93% (2,086/2,253) of CRC samples and 91% (868/951) of gastric/esophageal cancer samples. The performance of the CRC/GEC classifier is evaluated through 10-fold cross validation on these 3,204 cancer samples, where in each fold, an additional 6,298 samples from cancer-free individuals were also used in training. Among the 2,954 CRC and GEC samples detected by the multi-cancer classifier, 2,698 (91%) were accurately classified by the CRC/GEC classifier. Among the detected CRC samples, 91% (1,907/2,086) were correctly classified as CRC. Similarly, 91% (791/868) of the detected GEC samples were correctly classified. The precision of the CRC/GE prediction was also assessed and 91% (2,698/2,954) of the detected cancer samples were correctly classified. Of the 1,984 detected cancer samples predicted as CRC, 96% (1,907/1,984) were true CRC samples. Of the 970 detected cancer samples classified as GEC, 82% (791/970) were correct. **Conclusions:** This assay with high sensitivity and accurate CRC/GEC classification can effectively guide subsequent diagnostic workups following a positive result, maximizing potential clinical utility. Ongoing work will continue further refinements. Research Sponsor: None.

53 **Poster Session**

Use of molecular residual disease (MRD) testing with measurement of circulating tumor DNA (ctDNA) post-operatively in stage II colon cancer (SIICC) in the community-based setting. First Author: Kiran Foster, Astera Cancer Care, East Brunswick, NJ

Background: 80% of adult oncology patients (pts) are treated in the community-based setting. Data on the implementation of emerging technologies such as ctDNA testing is scant in this population. Use of adjuvant chemotherapy in SIICC pts historically was based on clinical/pathological risk factors however new data presented in mid 2022 utilizes the presence or absence of ctDNA 6-8 weeks post operatively to help risk stratify pts. The absence of ctDNA predicts a high chance of disease-free survival. Use of ctDNA was incorporated into the NCCN Colon Cancer guidelines as of 2023. In a private practice setting, we retrospectively looked at the use of ctDNA assay for adjuvant chemotherapy decision-making in SIICC pts. **Methods:** We conducted a retrospective study to quantify ctDNA testing frequency in SIICC pts in our community based oncology practice. The historical cohort was comprised of pts diagnosed prior to the NCCN guideline inclusion of ctDNA; the current cohort included pts diagnosed after the NCCN guideline change. For all pts, we reviewed clinicopathologic risk factors deemed to be high risk (positive margin, $<$ 12 LN removed, tumor budding, lymphovascular invasion, poorly differentiated features; perforation or obstruction) and used this for stratification. Low-risk pts had no high risk features. Medical record data (office notes, pathology, molecular testing) was evaluated to determine ctDNA results including timing and use of adjuvant chemotherapy. Primary endpoint was the percent (%) of current pts who were low-risk with ctDNA testing sent compared to historical control. Secondary endpoint was the % of ctDNA testing sent within 8 weeks of curative surgery in both cohorts. **Results:** In a private practice setting, we identified low-risk SIICC pts spanning a 36-month period and categorized them into two groups: the historic cohort (A) pts were diagnosed between July 2021-Dec 2022, and the active cohort (B) pts were diagnosed between Jan 2023-July 2024. 26 pts were identified in cohort A and 36 pts were identified in cohort B to have SIICC. Out of 26 pts in cohort A, we identified 13 low risk SIICC pts; for cohort B, out of 36 pts, we identified 25 low-risk SIICC pts. The percentage of pts in each group that had ctDNA testing sent was 79.6% (10/13 pts) in cohort A and 92% (23/25 pts) in cohort B. In cohort A, 80% (8/10 pts) and in cohort B, 90.9% (20/22 pts) had ctDNA sent within 8 weeks of surgery. **Conclusions:** Incorporation of ctDNA testing in low-risk SIICC pts in a community-based setting occurred quickly. As ctDNA testing was implemented into practice, a majority of pts had the testing sent in the appropriate time frame. Future best practices, especially in the community, should incorporate tumor specific clinician oversight with use of updated pathways to ensure any recent changes to standard of care are followed. Research Sponsor: None.

54 **Poster Session**

Shifting the paradigm: Early identification of colorectal cancer with C the Signs clinical decision support. First Author: Seema Dadhania, Imperial College, London, London, United Kingdom

Background: The incidence of colorectal cancer is increasing rapidly, particularly amongst younger people. Despite a screening program being widely available, most patients are diagnosed with cancer through symptomatic presentation. Leveraging electronic medical records (EMRs) through clinical decision support platforms offers potential for identifying high-risk individuals before clinical suspicion develops. This study aims to explore the challenges associated with early colorectal cancer diagnosis and evaluate the effectiveness of the cancer case finding feature of a cancer management platform called C the Signs. **Methods:** A retrospective analysis was conducted using data from the Mayo Data Platform, comprising records of 894,275 patients, including 7,348 diagnosed with colorectal cancer. C the Signs was employed to identify patients at risk of colorectal cancer based on EMR data. Sensitivity and specificity analyses were performed to assess the platform's performance in identifying high-risk patients. Additionally, the study evaluated the temporal discrepancy in colorectal cancer diagnoses between those flagged by C the Signs and those detected by primary care physicians. **Results:** The C the Signs screening platform had a sensitivity of 93.8% and a specificity of 19.7% in identifying patients at risk of colorectal cancer. Notably, 29.4% of patients with colorectal cancer were identified as being at risk up to 5 years earlier by C the Signs compared to diagnoses made by primary care physicians, highlighting the platform's potential for early detection. **Conclusions:** Early identification of high-risk individuals holds promise for improving patient outcomes and reducing the burden of colorectal cancer. Whilst the specificity of the C the Signs platform for colorectal cancer is relatively low, the sensitivity is comparable to the most favorable rates reported for colonoscopy. In addition, if C the Signs was used in conjunction with a relatively cheap screening test such as Faecal Immunochemical Testing, this would significantly drive up the specificity whilst maintaining the sensitivity. Further studies are needed to assess its efficacy in conjunction with tests available in primary care. Research Sponsor: None.

55 **Poster Session**

Long-term follow-up of the randomized trial on open versus laparoscopic surgery for primary tumor resection in patients with non-curable stage IV colon cancer (JCOG1107). First Author: Akio Shiomi, Division of Colon and Rectal Surgery, Shizuoka Cancer Center Hospital, Shizuoka, Japan

Background: The Japan Clinical Oncology Group (JCOG) 1107 (UMIN-CTR: UMIN000009715) was a non-inferiority phase III trial to compare laparoscopic surgery (LAP) with open surgery (OP) for symptomatic, non-curable stage IV colon cancer, using progression-free survival (PFS) as the primary endpoint based on the intention-to-treat principle. The planned primary analysis confirmed the non-inferiority of the LAP over the OP. Additionally, the present study aimed to compare the LAP and OP using long-term 3-year follow-up data from the JCOG1107. **Methods:** Eligibility criteria included pathologically proven adenocarcinoma or adenosquamous carcinoma; primary tumor located at the cecum, ascending, transverse, descending, sigmoid, and rectosigmoid colon; primary tumor with bowel stenosis and/or bleeding; and having at least one to three non-curable factors. The sample size was 194 with a power of 70%, one-sided alpha of 5%, and non-inferiority margin of the hazard ratio (HR) of 1.38. **Results:** In total, 195 patients were randomized (OP 95, LAP 100) between January 2013 and January 2021. OP was performed in 92 and LAP in 98 patients. Postoperative chemotherapy was administered in 82 OP and 86 LAP patients. As the data cut-off date of Feb 2024, 93 (98%) and 100 (100%) PFS event occurred in OP and LAP. The 3-year PFS were 5.3% (95% CI, 2.0-11.0) and 3.0% (0.8-7.8) for OP and LAP arms, respectively (HR: 1.028, 95% confidence interval [CI]: 0.772–1.370; P for non-inferiority = 0.0222). Regarding OS, 87 (92%) and 90 (90%) death occurred in OP and LAP. The 3-year OS were 31.5% (22.5-41.0) and 28.5% (20.0-37.6) for OP and LAP arms, respectively (HR: 1.048; 95% CI: 0.780–1.410). **Conclusions:** Long-term follow-up data supported the non-inferiority of LAP over OP for primary tumor resection in patients with non-curable stage IV colon cancer. Our study findings indicate that LAP is acceptable as a standard treatment for symptomatic, non-curable stage IV colon cancer. Clinical trial information: UMIN000009715. Research Sponsor: Ministry of Health, Labour and Welfare, Japan.

56 **Poster Session**

Survival benefits of asymptomatic primary tumor resection after bevacizumab plus FOLFIRI as first-line therapy for patients with metastatic colorectal cancer with synchronous unresectable metastasis. First Author: Hsiang-Lin Tsai, Division of Colorectal Surgery, Department of Surgery, Kaohsiung Medical University Hospital, Kaohsiung Medical University, Kaohsiung, Taiwan

Background: Metastatic colorectal cancer (mCRC) poses a clinical challenge and requires a combination of systemic therapy and conversion surgery. Although first-line chemotherapy and target therapy are considered the standard treatments for mCRC, the role of primary tumor resection (PTR) in asymptomatic synchronous mCRC with unresectable metastatic lesion after initial therapy remains relatively underexplored. **Methods:** A retrospective observational study was conducted from January 2015 to January 2021, involving 74 patients with synchronous mCRC who received bevacizumab plus FOLFIRI as first-line systemic therapy. All 74 patients had unresectable metastatic lesions confirmed through multidisciplinary team (MDT) discussion. Patients' characteristics, PTR data, and radiotherapy (RT) and overall survival (OS) outcomes were analyzed. The patients were categorized into a "PTR" group and a "No PTR" group and then further stratified into IVA, IVB, and IVC subgroups based on the initial mCRC stage. Additionally, four subgroups - namely "PTR(+)/RT(+)", "PTR(+)/RT(-)", "PTR(-)/RT(+)", and "PTR(-)/RT(-)" were formed to assess the combined effects of PTR and RT. **Results:** The median OS for all the patients was 23.8 months (20.5 - 27.1 months). The "PTR" group exhibited a significantly higher median OS of 25.9 months (21.3 - 30.5 months) compared with 21.4 months (15.8 - 27.1 months) in the "No PTR" group (P = 0.048). Subgroup analyses revealed a trend of improved survival with PTR in patients with stage IVA and IVB; however, the results were not statistically significant (P = 0.116 and 0.493, respectively). A subgroup analysis of PTR and RT combinations revealed no significant difference in median OS rates. **Conclusions:** For asymptomatic mCRC with synchronous unresectable distant metastasis. PTR following first-line therapy with bevacizumab plus FOLFIRI can provide a survival benefit, particularly in stage IVA/IVB patients compared with stage IVC patients. Additionally, RT for primary tumor did not provide an additional OS benefit in mCRC with unresectable metastasis. A prospective randomized trial with a larger sample size is essential to further elucidate the role of PTR in this context. Research Sponsor: None.

57 **Poster Session**

Effectiveness of anti-epidermal growth factor receptor agents in elderly patients (age ≥ 70 years) with *RAS* wild-type metastatic colorectal cancer: Nationwide real-world data. First Author: Chao-Yuan Wang, Division of Colorectal Surgery, Kaohsiung Medical University Chung-Ho Memorial Hospital, Kaohsiung, Taiwan

Background: This study investigated the efficacy and safety of anti-epidermal growth factor receptor (EGFR) agents plus chemotherapy for elderly patients (aged > or = 70 years) with *RAS* wild-type metastatic colorectal cancer (mCRC) from nationwide data in Taiwan. **Methods:** Total 756 patients with *RAS* wild-type mCRC who received first-line therapy with anti-EGFR agents (Cetuximab) plus chemotherapy at nationwide multiple institutions from November 2016 to January 2021. The study explored the prognostic impact between two age groups (≥70 vs <70). Progression-free survival (PFS) and cancer-specific survival (CSS) were compared. Additionally, severe adverse events (SAEs) were also compared between two age groups. **Results:** The median PFS and CSS were comparable between the two age groups (14.0 vs 14.0 months, p = 0.098; 32.0 vs 35.0 months, p = 0.226, respectively). Moreover, subgroup analysis revealed similar PFS and CSS between two age groups in patients with synchronous or metachronous mCRC and left-sided or right-sided tumors (all $p > 0.05$). Regarding hematologic SAEs, a significant higher incidence of grade ≥ 3 anemia was found in elderly patient group (13.3% vs 5.0%, p = 0.003). However, no significant difference was observed in the incidence of non-hematologic SAEs (all $p > 0.05$). **Conclusions:** According to this nationwide real-world data, for patients carrying *RAS* wild-type mCRC treated with cetuximab plus chemotherapy, elderly patients could have comparable benefit as younger ones. Except for grade ≥ 3 anemia, other hematologic SAEs and non-hematologic SAEs revealed no significant difference between two age groups. Research Sponsor: None.

58 **Poster Session**

Real-world evidence on genomic profiles and survival outcomes in patients with HER2-positive metastatic colorectal cancer. First Author: Yuki Matsubara, Department of Gastroenterology and Gastrointestinal Oncology, National Cancer Center Hospital East, Kashiwa, Japan

Background: Patients (pts) with HER2-positive (HER2+) metastatic colorectal cancer (mCRC) represent a small, but increasingly important subgroup due to emerging HER2-targeted therapies. Although HER2 positivity is associated with resistance to anti-EGFR therapy (EGFRi), it is unclear if EGFRi or bevacizumab (BEV) is preferable as first-line (1L) therapy. Here, we offer novel evidence on the genomic profiles of pts with HER2+ mCRC and examine differences in survival between key treatment groups. **Methods:** This study used the US-based deidentified Flatiron Health-Foundation Medicine mCRC clinico-genomic database. The deidentified data originated from approximately 280 US cancer clinics (~800 sites of care). Included pts were 18 years or older, had evidence of stage IV or recurrent mCRC diagnosed between 01 Jan 2012 and 31 Mar 2022, and underwent tissue-based Comprehensive Genomic Profiling testing. HER2+ was defined as an *ERBB2* copy number of ≥5 for diploid tumors. A doublet regimen was defined as fluoropyrimidines plus oxaliplatin or irinotecan. Data cut-off was 31 Mar 2024. Kaplan-Meier methods were used to examine real-world overall survival (rwOS) and progression-free survival (rwPFS) overall, by 1L treatment class and across key subgroups defined by tumor sidedness and genomic profile, indexed to the start of 1L with pts censored at their last EHR-recorded activity. **Results:** Among a cohort of 5545 pts, HER2+ was detected in 171 (3.1%) pts. Median age (years) was 57 for HER2+ vs 61 for HER2-; 69.3% of HER2+ pts were White, compared to 72.1% in HER2-. Among pts with HER2+, 31 (18.1%) had *RAS* mutation (mt), *BRAF* V600E mt, or microsatellite instability-high. *ERBB2* copy number was significantly lower in pts with *RAS* mt or *BRAF* V600E mt compared to those without these mutations ($p < 0.01$, $p = 0.045$, respectively). Among pts who received 1L treatment (n=4748), rwPFS significantly differed between HER2+ (n=144) and HER2-negative (HER2-; n=4604) groups (7.6 vs 8.7 months, unadjusted HR (uHR) 1.20, p=0.04). This difference was consistent in pts with left-sided *RAS/BRAF* V600E wild-type microsatellite stable (MSS) mCRC and treated with 1L doublet plus EGFRi (8.7 vs 12.5 months, uHR 2.18, p=0.02; n=10 HER2+,n=117 HER2-) or 1L doublet plus BEV (8.9 vs 10.5 months, uHR 1.65, p=0.04; n=20 HER2+, n=387 HER2-). No significant rwPFS difference was observed between EGFRi and BEV in the HER2+ subgroup. rwOS showed no significant difference between HER2+ and HER2- groups (overall: 26.1 vs 24.5 months, uHR 0.94, p=0.52; n=144 HER2+, n=4604 HER2-). **Conclusions:** Although about 80% of pts with HER2-positive mCRC are *RAS/BRAF* V600E wild-type and eligible for EGFRi, no survival benefit was observed in EGFRi relative to BEV as first-line combination. While the impact of potential confounders needs to be examined in further analyses, these findings highlight the need for specific treatments for pts with HER2+ mCRC. Research Sponsor: None.

59 **Poster Session**

Understanding the unmet needs in colorectal cancer: Insights from patient and survivor experiences. First Author: Kimberley Lynn Newcomer, Colorectal Cancer Alliance, Washington, DC

Background: Colorectal cancer (CRC) is the fourth most commonly diagnosed cancer and the second leading cause of cancer-related deaths worldwide among men and women combined. Despite improved survival rates, there remains a gap in understanding the specific unmet needs of CRC patients and survivors. These needs include emotional support, information on treatment options, coping strategies, quality of life (QoL), and overall healthcare experience. The stress of a diagnosis is often compounded by the emotional and physical demands of treatment, which extend into survivorship when the disease is not active. **Methods:** The Colorectal Cancer Alliance launched a survey for colorectal cancer patients, survivors, and caregivers. This IRB-approved cross-sectional study, comprising more than 153 questions, aimed to inform on patient profiles, diagnosis experience, resources, QoL, access to care, and treatment experience. Participation in the survey was solicited from the Alliance network of social media, email, and web-based communities, underscoring the collective effort to understand and address the needs of CRC patients and survivors. **Results:** The survey included 283 participants, primarily aged 46-55. Key findings showed that 74% of participants had difficulty finding someone who understood their experience, and 41% reported reduced support from others after treatment ended. Additionally, 54% struggled with fatigue, and 51% coped with ongoing stress. CRC significantly impacted careers (64%), dating lives (51%), desire to have children (43%), relationships with spouses or partners (58%), sex lives (80%), and participation in social activities (65%). Although many felt informed before treatment, 46% reported unmet needs for information on complementary or alternative therapies. **Conclusions:** These results highlight the profound psychosocial and physical challenges faced by CRC patients and survivors, emphasizing the need for healthcare providers to develop tailored treatment and survivorship plans and comprehensive support services. Insights from this survey will guide the Colorectal Cancer Alliance (Alliance) in creating targeted actions, care programs, and support initiatives to address these needs. The Alliance aims to enhance patient care and improve CRC patients' and survivors' QoL, potentially reshaping the future of CRC management and survivorship care. Research Sponsor: None.

60 **Poster Session**

Clinical outcomes of hypofractionated radiotherapy (5x5 Gy) with a simultaneous integrated boost (5x6 Gy) in locally advanced rectal cancer. First Author: Maciej Temnyk, Maria Sklodowska-Curie National Research Institute of Oncology, Warsaw, Poland

Background: Hypofractionated RT (5x5 Gy) with or without subsequent chemotherapy is a standard of care used at the National Institute of Oncology, Warsaw, Poland, as a preoperative treatment for locally advanced rectal cancer. To maximize the chance of achieving a complete response (CR), when surgery is not planned or when there are clinically involved lymph nodes located outside the standard TME field, RT can be augmented with a simultaneous integrated boost (SIB) delivering 5x6 Gy. This report aims to evaluate the clinical outcomes of hypofractionated RT with a SIB in rectal cancer. **Methods:** From Dec 2018 to Sep 2022, 78 consecutive pts with a median age of 67 years (range 36–89 years), received hypofractionated RT utilizing the SIB-VMAT technique. The radiation dose fractionation was 5x5 Gy to the rectum, mesorectum and elective lymph nodes and a SIB to the GTV for a total dose of 30 Gy. **Results:** 30 pts (38.5%) received a SIB for the rectal tumor, of which 15 pts (19.2%) were deemed unfit for surgery, 10 pts (12.8%) refused surgical treatment, 3 pts (3.8%) had an oligometastatic disease, and 2 pts (2.6%) had an advanced primary tumor. 48 pts (61.5%) received a SIB to the clinically involved lymph nodes located outside the standard TME field. All 78 pts completed RT as planned without interruptions or dose modifications and with an acceptable toxicity profile. There was a 26.9% (n = 21) incidence of G2 toxicities (diarrhea, proctitis, skin toxicity, cystitis or lumbosacral plexus neuropathy) and a 3.8% (n = 3) incidence of G3 toxicities (diarrhea or proctitis). After a median follow-up of 917 days (range 33–1927 days), an assessment of response was conducted in 68 cases. In 94.1% (n = 64) there was no progression at the site irradiated to a dose of 30 Gy. After completion of RT, mr-TRG was evaluated in 28 pts. In 46.4% (n = 13) mr-TRG 1-2 (cCR) was achieved. Of the 24 pts qualified for the watch and wait strategy, 17 had follow-up. Of this group, 64.7% (n = 11) remained progression-free, and 11 pts had an adequate evaluation for cCR (imaging and endoscopy), resulting in a 63.6% (n = 7) rate of cCR. 43 pts underwent surgery, of which 97.7% (n = 42) had R0 resection. In 41 cases pathological TRG was assessed according to AJCC or Ryan, of which 41.4% (n = 17) achieved TRG 0-1 (pCR). In 40 cases the operated pts had no initial metastatic spread, among them 67.5% (n = 27) remain disease-free. 67 pts reported disease-related symptoms before RT. After treatment, 89.6% (n = 60) of those pts achieved symptomatic improvement in either rectal bleeding, bowel frequency, or pain control. **Conclusions:** Hypofractionated RT with SIB-VMAT delivering a dose of 30 Gy/25 Gy/5 fractions provides good local control and is a safe and viable option for both young and elderly patients who will not undergo surgery and for patients with involved lymph nodes located outside the standard TME field to maximize the chance of achieving CR. Research Sponsor: None.

61 Poster Session

Investigating the real-world outcomes of single- versus multi-agent preoperative chemoradiotherapy for locally advanced rectal cancer: NCDB analysis from 2004-2020. First Author: Gaurav Sharma, University of South Alabama, Mobile, AL

Background: Fluorouracil-based chemoradiotherapy is the standard initial treatment for locally advanced rectal cancer, demonstrating improved pathological response and reduced local recurrences compared to radiation alone. Fluorouracil-based chemo-radiation therapy has no impact on distant recurrences. Oxaliplatin was tested in addition to fluorouracil-based chemo-radiotherapy in clinical trials to improve pathological complete response (pCR), distant recurrence-free survival, and overall survival. The trial results are conflicting. In this study, we aim to compare single-agent (SA) versus multi-agent (MA) chemotherapy using real-world data from the National Cancer Database (NCDB). **Methods:** The NCDB was used to identify 39,667 patients with rectal cancer who received concurrent chemoradiotherapy between 2004 and 2020. We compared patients who received SA chemotherapy with those who received MA concurrent chemotherapy. The primary outcomes of interest were overall survival and pCR. The groups were compared using univariate analysis and Cox proportional hazard models to adjust for potential confounding factors. **Results:** Of 39,667 patients with rectal cancer, 24,452 received single-agent (SA) chemotherapy and 14,215 received multi-agent (MA) chemotherapy. Patients in the SA group were of advanced age, had a greater number of comorbidities (as calculated by the Charleson-Dayo score), were treated within community facilities, had a higher income status, were covered by government insurance, were situated on the east coast, and had lower educational attainment compared to those in the MA group ($p<0.05$). SA group had higher T4 and N1 disease, however MA group had more N2 disease. Upon survival analysis, patient in MA group had higher survival (41.2 vs 38.9 months, p=0.005), however pCR was significantly higher in SA group (10.4% vs 6.4%, $p<0.001$). **Conclusions:** MA chemoradiotherapy for rectal cancer improves survival, with the SA regimen showing enhanced complete pathological response. Differences in baseline characteristics may explain the outcomes. While MA chemotherapy offers modest overall survival benefits, potential toxicity needs to be considered and confirmed in future studies. Research Sponsor: None.

Comparison between single agent vs multi-agent chemo-radiotherapy group.

Outcome	Single-agent chemotherapy	Multiagent chemo-therapy	Chi Squared p-value
No readmission within 30 days of surgical discharge	22,727 (89.3 %)	12,921 (90.9 %)	<0.001
30-day mortality after surgery	360 (1.5 %)	59 (0.5 %)	<0.001
90-day mortality after surgery	716 (3.0 %)	158 (1.3%)	<0.001
Median duration of surgical admission	6.0	5.0	<0.001
Median survival-vital = deceased (months)	38.9	41.2	0.005
Median survival-vital = alive (months)	66.9	52.0	<0.001
Pathological complete response (pCR)	2,644 (10.4%)	904 (6.4%)	<0.001

62 Poster Session

Real-world (RW) treatment patterns and clinical outcomes of patients treated with sequential regorafenib and trifluridine/tipiracil ± bevacizumab (BEV) for metastatic colorectal cancer (mCRC) in the United States (US) community oncology setting: The SEQRT2 study. First Author: Tanios S. Bekaii-Saab, Department of Hematology and Oncology, Mayo Clinic Arizona, Phoenix, AZ

Background: Both Regorafenib (R) and Trifluridine/Tipiracil (T) ± BEV are widely used treatments for mCRC in later lines. This RW study evaluated optimal sequencing and associated outcomes among patients who initiated R and T in sequential lines of therapy (LOT) between LOT1-LOT6 within community oncology settings in the US. **Methods:** A retrospective study utilizing electronic medical record data from The US Oncology Network examined adult patients with mCRC who initiated R and T ± BEV sequentially between 09/01/2015 and 11/30/2022 (index date), with follow-up until 05/31/2023. Treatment patterns, overall survival (OS) and time to next treatment (TTNT) after sequence were assessed overall, R first (R-T) and T first (T-R). Multivariable Cox regression adjusting for age, sex, ECOG, tumor stage, and prior medication, assessed the association of factors with OS and TTNT. **Results:** In total, 308 mCRC patients initiating R and T (R-T, 156 pts [140 R-T and 16 R-T+BEV]; T-R, 152 pts [136 T-R and 16 T+BEV-R]) met study criteria. Baseline demographic (mean age 63 yrs, males 55%) and clinical characteristics (70% colon cancer, 58% ECOG 0-1) were similar between the cohorts. Most patients initiated index therapies at the third line of therapy (LOT3) (49% R-T, 45% T-R). The median (IQR) sequence duration was 7.4 (5.6), 6.7 (4.4), and 7.0 (4.7) months, respectively in patients with R-T, T-R, and overall. The median (95% CI) OS was 12.8 (11.2, 14.1) months among R-T and 10.2 (8.8, 11.9) months among T-R patients. Compared to R-T, the adjusted HR for the T-R sequence was 1.2 (95% CI 0.9–1.6; p=0.2). The median (95% CI) TTNT was 9.3 (8.4, 10.3) and 8.6 (7.8, 9.4) months, for R-T and T-R, respectively; the adjusted HR was 1.1 (95% CI 0.8–1.4; p=0.62). **Conclusions:** Patients initiating R first appeared to have a longer duration of treatment and survival, although not statistically significant. The one-year median duration of survival in this real-world setting confirms the importance of access to these treatments in patients with mCRC and can inform physicians regarding the choice of medications while sequencing therapies in mCRC. Research Sponsor: Bayer Pharmaceuticals.

Clinical outcomes among R-T and T-R mCRC patients.

Variable	Level	R-T (± BEV) [N[1] = 156]	T (± BEV) -R [N[1] = 152]	Adjusted[3,4] HR[5] (95% CI) [R-T Reference]	Effect
Overall Survival	Months – median (95% CI)	12.8 (11.2, 14.1)	10.2 (8.8, 11.9)	1.20 (0.91, 1.58)	0.20
TTNT[2]	Months – median (95% CI)	9.3 (8.4, 10.3)	8.6 (7.8, 9.4)	1.07 (0.83, 1.37)	0.62

[1]Entire R-T and T-R cohorts are presented since 16 patients in each cohort initiated T+BEV.
[2]Time To Next Treatment.
[3]Overall Survival adjusting for age, gender, stage at diagnosis, ECOG, prior anti-VEGF treatment.
[4]Time To Next Treatment adjusting for age, gender, stage at diagnosis, and line of therapy.
[5]Hazard Ratio.

63 Poster Session

Effects of the delay of surgery in localized rectal cancer. First Author: Thiago do Amaral Miranda, Hospita Sírio-Libanês, Brasília, Distrito Federal, Brazil

Background: The treatment landscape for localized rectal cancer has evolved significantly in recent years, with long-course neoadjuvant chemoradiation (CRT) and total neoadjuvant therapy (TNT) emerging as the primary options. However, the optimal timing of surgery remains an important clinical question, particularly in resource-limited settings where surgical delays can occur. Data from previous studies have shown conflicting results. This retrospective study aimed to assess the impact of delaying surgery beyond the longer interval used in previous studies, focusing on its effects on Pathologic Complete Response (pCR), DFS, and OS. **Methods:** We conducted a retrospective cohort study using electronic medical records from three centers in Brazil between the years 2013-2024. The study included 236 patients with localized rectal cancer treated with neoadjuvant CRT and/or chemotherapy. Patients were divided into two pre-specified groups based on the interval between the end of neoadjuvant therapy and surgery: ≤11 weeks and >11 weeks. The primary endpoint was DFS. Secondary endpoints were pCR rates, and OS. Descriptive statistics were used for population analysis, while the Kaplan-Meier method estimated DFS and OS, with comparisons made using the log-rank test. A Cox proportional-hazards model was used to estimate hazard ratios and 95% confidence intervals. **Results:** The median age of the study population was 62 years, with males comprising 56.4% of the sample. The majority of the sample consisted of stage III patients, making up 72.2%, followed by stage II patients at 25.8%, and stage I patients at 1.9%. Most patients (77.8%) had surgery more than 11 weeks after completing neoadjuvant therapy, with a median time to surgery of 16 weeks. The pCR rates did not significantly differ between the two groups (29% for ≤11 weeks vs. 21.5% for >11 weeks; p = 0.271). After a median follow-up of 30 months, the median DFS was not reached, and there was no significant difference between the groups (HR: 0.86; 95% CI: 0.48-1.5; p = 0.63). Similarly, the median OS was not reached, and no difference was observed between the groups (HR: 0.85; 95% CI: 0.47-1.54; p = 0.88). **Conclusions:** Delaying surgery beyond 11 weeks did not result in greater tumor downstaging in patients with localized rectal cancer. Furthermore, DFS and OS were not significantly impacted by extended intervals between neoadjuvant therapy and surgery. Research Sponsor: None.

64 Poster Session

A real-world study of treatment patterns among patients (pts) with metastatic colorectal cancer (mCRC) receiving third-line (3L) or fourth-line (4L) treatment in the US. First Author: Huichu Li, Takeda Pharmaceuticals America, Inc., Cambridge, MA

Background: There are limited real-world data describing 3L and 4L treatments for pts with mCRC. However, the landscape is evolving with new options becoming available, including fruquintinib, a highly selective oral inhibitor of all 3 VEGF receptors, approved by the US FDA in November 2023. This study aimed to understand pt characteristics and real-world treatment patterns in 3L and 4L to characterize the standard of care, adherence to guidelines, and identify unmet needs among pts with mCRC in the US, prior to recent changes in the landscape. **Methods:** A retrospective study of adults with mCRC who had received ≥3L of therapy between 1/1/2018 and 3/31/2023 in the US-based Flatiron electronic health records database. In two cohorts, defined by initiating 3L and 4L systemic treatment, pt demographics, clinical characteristics, and treatment by line of therapy were summarized. **Results:** Overall, 5401 pts were included in the 3L cohort and 2868 pts in the 4L cohort: median age at baseline was 63 and 62 years (yrs), 56% and 56% were male, 43% and 43% had a *KRAS* variant, and 46% and 49% had liver metastasis, respectively. 96% of pts in the 3L cohort received standard chemotherapy (CT) +/- targeted therapy as 1L treatment (55% CT + anti-vascular endothelial growth factor [VEGF] therapy; 34% CT only; 7% CT + anti-epidermal growth factor receptor [EGFR] therapy). In 2L, 87% of pts had standard CT-containing regimens (53% CT + anti-VEGF therapy; 19% CT alone; 15% CT + anti-EGFR therapy). A similar treatment history was observed for the 4L cohort. Commonly used 3L regimens in the 3L cohort were trifluridine/tipiracil (TAS-102; 11%), regorafenib (rego; 10%), and standard CT-containing regimens (54%; including: 12% FOLFIRI + bevacizumab [bev], 7% FOLFOX + bev, 4% FOLFIRI, 3% FOLFOX, and 15% standard CT + anti-EGFR therapy). Approximately 47% (n=2513) of pts did not receive subsequent treatments after 3L. In the 4L cohort, use of TAS-102 (16%) and rego (15%) increased slightly in 4L, accompanied with a moderate decrease in CT-containing regimens (42%; including 6% FOLFOX + bev, 5% FOLFIRI + bev, 3% FOLFOX, 3% FOLFIRI, and 12% standard CT + anti-EGFR therapy). Approximately 4% of pts in the 3L and 4L cohorts received TAS-102 + bev in 3L and 4L, respectively. Pts receiving standard CT-containing regimens were slightly younger (<65 yrs: 3L 57%, 4L 58%) and had better Eastern Cooperative Oncology Group performance status (ECOG PS; 0-1: 3L 77%, 4L 77%) vs pts receiving TAS-102 (<65 yrs: 3L 54%, 4L 58%; ECOG PS 0-1: 3L 70%, 4L 68%) or rego (<65 yrs: 3L 50%, 4L 53%; ECOG PS 0-1: 3L 68%, 4L 70%). **Conclusions:** Characterization of treatment patterns in US pts from 1/2018 to 3/2023 showed that standard CT +/- targeted therapy was used frequently in 3L and 4L, demonstrating an unmet need for additional therapies. Approval of novel treatments, such as fruquintinib, may help to address this need. Research Sponsor: Takeda Development Center Americas, Inc. (TDCA), Lexington, MA, USA.

65 Poster Session

Personalized frailty risk assessment in long-term survivors of colorectal cancer. First Author: Rebecca Forman, Yale New Haven Hospital, New Haven, CT

Background: Frailty, a pathologic form of aging, is associated with reduced quality of life and increased risk of death. Its incidence increases with age. Frailty is a concern for cancer survivors, especially since this population is surviving longer and increasing in size. There is a need to predict which patients are at risk of frailty to tailor preventative measures, particularly in early-stage colon and rectal cancer survivors, the largest group in gastrointestinal cancer survivors. **Methods:** This was a retrospective cohort study of individuals aged 66 and older in the SEER-Medicare linked database, diagnosed with stage I-III colon or rectal cancer between 2003-2012. Patients included in the study received definitive surgical treatment and survived for at least 5 years after diagnosis. Frailty was assessed using administrative claims codes with the Kim frailty index. Patients already frail at year 5 were excluded from the analysis. An increase in frailty score, indicating the onset of frailty or worsening to moderate or severe frailty, occurring within 5-10 years following cancer diagnosis was the primary outcome. Predictors of frailty were identified using restricted mean survival time (RMST) regression, with results less than 1 indicating a shorter time to frailty, and significant factors were used to develop a clinical prediction model and stratify patients into risk tertiles. **Results:** At 10 years or end of available follow-up, 58% of the patients had developed new onset or worsening frailty. There were no significant differences in RMST by patient race, sex, cancer stage and grade. Receipt of systemic therapy 4-5 years after diagnosis, having an ostomy present in years 4-5, advancing age, comorbidities, and living in an area with a greater proportion of residents below the federal poverty line were all associated with shorter time to frailty; the largest effects were seen with advancing age and comorbidities (Table). Our clinical prediction model incorporated age>85 and having multiple comorbidities for both cohorts and ostomy in years 4-5 for the colon cohort. **Conclusions:** Our population level analysis provided a clinical prediction model for frailty 5-10 years after cancer diagnosis in colon and rectal cancer survivors and may help inform long-term survivorship management. Research Sponsor: American Cancer Society; RSG-21-039-01-CPPB.

RMST for selected variables.

Selected Variables	RMST for Colon Cohort (95% CI)	RMST for Rectal Cohort (95% CI)
Age 85-89 (ref: age 71-74)	0.76 (0.73, 0.79)	0.79 (0.74, 0.84)
Age 90+ (ref: age 71-74)	0.67 (0.64, 0.70)	0.74 (0.66, 0.82)
20-100% area-level poverty (ref: 0-<5%)	0.95 (0.92, 0.98)	0.96 (0.91, 1.02)
Receiving systemic therapy in years 4-5	0.89 (0.85, 0.92)	Not selected
Ostomy in years 4-5	0.80 (0.69, 0.92)	0.97 (0.92, 1.02)
Elixhauser Comorbidity Index (ECI) 1-2 (reference: ECI 0)	0.77 (0.75, 0.79)	0.79 (0.76, 0.82)
ECI 3 (ref: ECI 0)	0.60 (0.57, 0.64)	0.67 (0.59, 0.76)

66 Poster Session

Predictors of late recurrence of colorectal cancer (CRC) among older long-term survivors using claims-based evidence. First Author: Faiza Yasin, Yale School of Medicine, New Haven, CT

Background: There are established features that predict early recurrence of CRC in the first 5 years from diagnosis, but predictors of late recurrence 5-10 years after diagnosis are less well understood. Long-term CRC survivors require multidisciplinary follow-up accounting for both cancer and non-cancer related conditions. Identifying predictors of cancer recurrence after 5 years of disease-free survival may help inform survivorship care. **Methods:** This was a retrospective cohort study of patients aged ≥66 in the SEER-Medicare database with non-metastatic CRC diagnosed between 2003 and 2011. As SEER does not collect recurrence data, we used a validated algorithm to identify treated recurrence in Medicare claims. Patients with early recurrence or a second primary malignancy in years 0-5 were excluded. Demographic and cancer-specific clinical data were collected at time of cancer diagnosis. Comorbidities were identified between 4 -5 years after cancer diagnosis and recurrence was modeled starting at year 5 using restricted mean survival time (RMST) regression. Estimates from RMST models were converted to RMST ratios (ratio <1 indicates shorter time to event). The Fine-Gray model was used to estimate cumulative incidence of CRC recurrence or mortality between 5 and 10 years after cancer diagnosis. **Results:** 20,255 patients with non-metastatic CRC were included: 16,158 with colon cancer (33.4% stage I, 41.7% stage II, 25.0% stage III), and 4,097 with rectal cancer (43.1% stage I, 30.1% stage II, 26.7% stage III). Mean age at cancer diagnosis was 77 years. In those who remained disease free five years after initial diagnosis, late recurrence developed in 4.8% (973/20,255) of the combined CRC cohort, 4.2% (679/16,158) of those with colon cancer, and 7.2% (294/4,097) of those with rectal cancer. In colon cancer, stage III disease was associated with a shorter time to recurrence (RMST ratio = 0.97, CI = 0.96-0.98). In both colon (RMST ratio = 0.98, 95% CI 0.96-0.99) and rectal cancer (RMST ratio = 0.97, 95% CI 0.94-0.99) having ≥3 comorbidities was associated with a shorter RMST, consistent across disease stages. For colon cancer, the cumulative incidence of CRC-specific mortality vs. all-cause mortality in years 5-10 from diagnosis was 3% (95% CI 2.8-3.0) vs. 35% (95% CI 34.0-36.0), respectively. For rectal cancer, during the same period, the corresponding rates were 6% (95% CI 5.4-7.0) vs. 30% (95% CI 28.4-31.0). **Conclusions:** Late CRC recurrence is rare in the SEER-Medicare population. Primary risk factors are advanced stage at diagnosis and multiple co-morbidities. Patients who survive 5 years from their diagnosis have a higher risk of all-cause mortality compared to CRC-specific mortality. Interpretation of data is limited by the validated algorithm used to detect recurrence, as it may underestimate recurrence in patients who did not undergo treatment or workup for recurrent disease. Research Sponsor: American Cancer Society.

67 Poster Session

Survival outcomes for right- versus left-sided colon cancer and rectal cancer in patients receiving regorafenib and/or trifluridine/tipiracil for refractory metastatic colorectal cancer: Findings from the multicenter retrospective ReTrITa study. First Author: Carlo Signorelli, Medical Oncology Unit, Belcolle Hospital, ASL Viterbo, Viterbo, Italy

Background: Several studies show that patients (pts) with colon cancer, particularly pts with right colon cancer (RCC), have lower survival rates than those with rectal cancer. This argues that colorectal cancer should be divided into three clinical entities: RCC, left colon cancer (LCC) and rectal cancer (RC). In refractory metastatic colorectal cancer (mCRC), regorafenib (R) and trifluridine/piracil (T) were found to improve survival. This real-life subgroup survival analysis focused on treatment with R and T, sequential or not, according to the above three primary tumor sites. **Methods:** Clinical data of pts diagnosed with mCRC and treated with R and/or T between 2012 and 2023 were retrospectively collected at 17 Italian cancer centers. The purpose of this analysis was to compare the overall survival (OS) and progression-free survival (PFS) of pts who received R and/or T as third line and beyond, according to their primary tumor site. **Results:** The entire ReTrITA study enrolled 1156 patients. 261 (22.5%) of them received the T/R sequence (T/R), 155 (13.4%) the reverse sequence (R/T), 427 (37%) T, and 313 (27.1%) R. For the purpose of this subgroup analysis we identified 381 RCC pts (32.9%), 531 LCC pts (45.9%) and 244 RC pts (21.2%). When comparing the groups treated with the sequential treatment, we observed a statistically significant OS improvement for pts receiving R prior to T: 17,4 vs. 12,2 months (mos) in RCC pts (HR = 0,67; 95% CI = 0,46-0,99; P = 0,0461); 16,1 vs. 15,1 mos in LCC pts (HR = 0,71; 95% CI = 0,51-1,00; P = 0,0559) and 16,6 vs. 11,4 mos in RC pts (HR = 0,57; 95% CI = 0,36-0,89; P = 0,0153). R/T sequence was also found to provide a considerable PFS benefit: 11,8 vs. 7,8 mos in RCC pts (HR = 0,53; 95% CI = 0,37-0,76; P = 0,0006); 11,7 vs. 8,9 mos in LCC pts (HR = 0,64; 95% CI = 0,47-0,88; P = 0,0071) and 10,7 vs. 8,2 mos in RC pts (HR = 0,63; 95% CI = 0,42-0,96; P = 0,0342). The non-sequential administration of R and T had no statistically significant impact on survival outcomes in the three groups. **Conclusions:** Our real-world subanalysis suggests that the R/T sequence significantly increases survival for pts with primary colon and rectal cancer, even after third-line treatment. Examining the pts groups treated with R and T as monotherapy based on the primary tumour site, however, did not allow us to make any significant conclusions. Treatment decisions should, however, constantly include the patient's characteristics, such as sex, ECOG PS, and the extent of metastatic disease. However, in order to verify our findings, prospective studies are required. Research Sponsor: None.

68 Poster Session

Vascular endothelial growth factor receptor (VEGFR) -tyrosine kinase inhibitors combined with programmed death receptor-1 (PD-1) inhibitors as third- or later-line treatment in patients with microsatellite-stable (MSS) metastatic colorectal cancer (mCRC): A retrospective study. First Author: Xiaoqian Li, Department of Medical Oncology, Shandong Cancer Hospital and Institute, Shandong First Medical University and Shandong Academy of Medical Sciences, Jinan, China

Background: Microsatellite-stable (MSS) metastatic colorectal cancer (mCRC) shows a low response to programmed death receptor-1 (PD-1) inhibitors. Recent studies suggest that combining vascular endothelial growth factor receptor (VEGFR)-tyrosine kinase inhibitors (TKIs) with PD-1 inhibitors may enhance the anti-cancer activity for mCRC patients. Multiple antiangiogenic drugs have been approved by the FDA and NMPA for the treatment of later-line mCRC. Nevertheless, the effects of various anti-angiogenic drugs when used in combination with immunotherapy for mCRC treatment is not yet fully established. **Methods:** We conducted a non-interventional, retrospective study. Data were compiled from patients with mCRC at Shandong Cancer Hospital. Eligible patients must have received a combination of anti-angiogenic drugs and PD-1 inhibitors following at least two lines of standard treatment. The treatment period spanned from August 2019 to April 2024. The primary endpoint was overall survival (OS), and the secondary endpoint was progression-free survival (PFS). **Results:** At the cut-off date (April 29th, 2024), 79 eligible patients were available for analysis. Of these, 49 patients received fruquintinib in combination with PD-1 inhibitors (F group), 30 patients received investigator-selected other anti-angiogenic agents in combination with PD-1 inhibitors (NF group). The baseline characteristics of the two groups were well balanced. The median overall survival (OS) was significantly longer in the F group compared to the NF group (hazard ratio [HR]: 0.36, 95% CI 0.14-0.96, P=0.042). The median OS was not reached (95% CI 21.1-NA) in the F group, whereas it was 19.1 months (95% CI 12.6-NA) in the NF group. Subgroup analyses of OS were generally consistent with benefit in the whole population across nearly all subgroups. In contrast to NF group, individuals in F group demonstrated a significant enhancement in overall survival (OS) when patients had received prior bevacizumab (HR: 0.26, 95% CI 0.07-0.75, P=0.015), had no prior treatment with cetuximab (HR: 0.23, 95% CI 0.26-0.97, P=0.045), or had experienced surgical resection at baseline (HR: 0.34, 95% CI 0.12-0.95, P=0.041). The median PFS between two groups showed no statistically significant difference (HR: 0.65, 95% CI 0.38-1.13, P=0.127), F group 6.1 months (95% CI 4.6-10.3) versus NF group 4.4 months (95% CI, 3.0-7.5). **Conclusions:** Fruquintinib demonstrated improved OS over other anti-angiogenic agents when combined with immunotherapy for third or later-line MSS mCRC treatment. Research Sponsor: None.

69 **Poster Session**

A multi-institutional retrospective study evaluating clinical outcomes and prognostic factors in patients with metastatic appendiceal adenocarcinoma. First Author: Koshiro Fukuda, Department of Gastroenterological Chemotherapy, Cancer Institute Hospital, Japanese Foundation for Cancer Research, Tokyo, Japan

Background: Appendiceal adenocarcinoma (AA), a rare cancer, has been reported to have high incidences of mucinous type histology, peritoneal metastasis, and relatively good prognosis. However, its chemotherapeutic efficacy and prognostic factors is not well known. **Methods:** We collected patients with metastatic AA treated with chemotherapy between February 2005 and March 2023 in two high volume cancer centers in Japan. We retrospectively analyzed the clinical outcomes of the chemotherapy of patients with metastatic AA and the clinical factors associated with treatment efficacy and prognosis by multivariate analysis. **Results:** A total of 66 patients with a median age of 54 years, were enrolled. 40 patients (60.6%) received an oxaliplatin-based regimen in combination with bevacizumab. The response rate was 22.0% and the disease control rate was 87.8%. With median follow-up period was 15.5 months [1.9-103.1], median progression-free survival (PFS) was 10.9 months [95% CI: 7.2-15.2] and median overall survival (OS) was 23.0 months [95% CI: 19.6-35.8]. 77.4% of patients received subsequent chemotherapy. In multivariate analysis, pretreatment carcinoembryonic antigen (CEA) and/or carbohydrate antigen 19-9 (CA19-9) were associated with significantly shorter PFS (HR: 2.93, P = 0.0049). Presence of primary tumor (HR:2.50, P = 0.015), ECOG performance status (PS) $\geq$1 (HR: 2.67, P = 0.0091), moderate to massive ascites (HR: 3.06, P = 0.0043) and high tumor marker levels (HR: 2.46, P = 0.026) were associated with significantly shorter OS. Classifying the patients by number of these prognostic factors for OS into three groups (0, 1-2, and 3-4), the group with prognostic factors (1-2 or 3-4) had a significantly shorter prognosis compared to the group without any these factors (0 vs 1-2: HR: 4.70, P = 0.038, 0 vs 3-4: HR: 17.6, $P < 0.01$). In addition, the group with 3-4 prognostic factors had relatively lower rate of subsequent chemotherapy ratio than the group without these factors (66.7% vs 100%, P = 0.25). **Conclusions:** We confirmed our treatment outcomes were consistent with previous studies and elevation of tumor markers presence of primary tumor, ECOG PS $\geq$1, moderate to massive ascites were significantly associated with survival in patients with metastatic AA. Research Sponsor: None.

70 **Poster Session**

Real world characteristics and outcomes of patients with BRAFV600E-mutant metastatic colorectal cancer in Australia: The COALA project. First Author: Nadia Hitchen, Peter MacCallum Cancer Centre, Melbourne, Australia

Background: BRAFV600E-mutant metastatic colorectal cancer (mCRC) represents a unique molecular subset with poor prognosis and less responsiveness to chemotherapy. The BEACON CRC study demonstrated that encorafenib plus cetuximab (EC) significantly improved overall survival (OS) compared to FOLFIRI plus cetuximab in BRAF-mutant mCRC as 2nd or 3rd line therapy. This combination was reimbursed in Australia from January 2022. **Methods:** We analysed data from the Australian TRACC mCRC clinical registry, encompassing patients (pts) with BRAF-mutant mCRC diagnosed in Australia from 2009 to 2024 at 31 hospitals. We assessed baseline demographics, uptake of EC following reimbursement and efficacy analysis of progression-free survival (PFS) and OS for pts that received EC and those who did not. Pts receiving EC on a clinical trial were excluded. **Results:** Among 2,800 pts with known BRAF mutation status, 365 (13%) were BRAF mutated. The BRAF mutant cohort had a median age of 63 years, with a significantly higher mutation rate in the 20-39 age group than other age groups (age 20-39, 25.5%; age 40-59, 10%; age 60+, 13.2%; P$<$0.001); 54.5% were female, 58% right-sided tumours, 64% with synchronous metastatic disease and 21.7% deficient mismatch repair (dMMR). Overall, 314 (86%) pts with BRAF mutant tumours and 2,079 (85%) BRAF wild-type (WT) tumours received 1st line treatment, and 156 (43%) BRAF mutant and 1,144 (47%) BRAF WT pts received 2nd line therapy. The median OS from diagnosis was 17.3 months for BRAF mutant pts versus 34.7 months for BRAF WT pts ($p < 0.0001$). All dMMR BRAF mutant pts received 1st line pembrolizumab since its reimbursement in August 2021. Among 41 BRAF mutant pts progressing on 1st line therapy since January 2022,(32 pMMR, 9 dMMR), 33 (80%) were subsequently treated with EC, and 4 (10%) did not receive further treatment. OS from the start of 2nd line therapy was 9.6 months (95% CI: 7.2-12.5) for pts who received EC and 7.1 months (95% CI: 6.4-8.6) for those who received other systemic treatment. Median PFS for pts receiving EC was 4.6 months (95% CI: 3.4-5.8) and 3.3 months (95% CI: 2.5-5.3) for other 2nd line treatments. **Conclusions:** mCRC pts younger than 40 have a high rate of BRAF mutation, where 1 in 4 tumours harbour the mutation. BRAF mutant mCRC has a poorer outcome than BRAF WT despite a similar proportion receiving 1st and 2nd line treatment. The majority of eligible BRAF mutant mCRC pts received EC following reimbursement in Australia, with PFS and OS outcomes consistent with the BEACON CRC trial. Research Sponsor: None.

71 **Poster Session**

Aflibercept (AFL) vs. ramucirumab (RAM) -based therapy in patients with advanced colorectal cancer: Efficacy comparison using real-world data evidence from TriNetX platform. First Author: Jesus Rodriguez-Pascual, Instituto Oncológico Vithas, Madrid, Spain

Background: FOLFIRI in combination with Aflibercept (AFL) or Ramucirumab (RAM) prolongs overall survival compared with FOLFIRI alone in patients with advanced colorectal cancer (aCCR) as second-line therapy, however, there are no comparative studies between them. **Methods:** Utilizing the TriNetX Global Collaborative Network, a platform that operates globally based on anonymized and aggregated clinical data, a sample of aCCR patients from 128 healthcare organizations (HCOs) who met the initial criteria was selected. 5-year overall survival (OS) was analyzed between these cohorts using a Kaplan-Meier analysis. Hazard Ratio (HR) and its 95% confidence interval (95% CI) were calculated to evaluate the difference between cohorts. Propensity Score Matching (PSM) was used to balance the cohorts based on age, gender, and race mitigating possible cofounding variables. All statistical analyses were conducted utilizing the TriNetX Analytics function in the online research platform. **Results:** A total of 1046 patients met the study criteria and were included in our study. 523 received AFL and 523r received RAM-based therapy. Patients treated with AFL showed a significant better OS compared to patients treated with RAM, both previous to PSM and before PSM (post-PMS median OS 430 days vs 317 days, HR 0.797, 95%CI 0.683-0.930). **Conclusions:** In this study, based on Real World data, Aflibercept showed superior results to Ramucirumab in advanced colorectal cancer patients treatment. To date this is the largest study comparison between this two therapeutic options. Research Sponsor: None.

72 **Poster Session**

Patient-reported outcomes in physical, cognitive, and sexual health in early-onset gastrointestinal cancers (EOGIC). First Author: Susan Feldt, University of Chicago Medical Center, Chicago, IL

Background: Incidence of early-onset (ages 18-49) gastrointestinal cancers is alarmingly increasing. To assess unmet needs in health-related quality of life (HRQoL), we conducted this pilot study using PRO measures in patients with EOGIC. **Methods:** Patients ages 18-49 with any GI cancer seen at our center during a clinic visit were invited to participate in this cross-sectional study. Validated PRO tools PROMIS 29+2, GP5 from FACT-G, AYA Sexual Health PRO Battery and Adult Neuro-QOL Cognitive Function short form were administered. Participants' PROMIS 29+2 responses were scored to generate a T score, and two-tailed t-tests were used to compare T scores with the average score from a large general US population. The electronic medical record was reviewed to collect demographic, diagnosis and treatment information. **Results:** 39 (85% response rate) patients were enrolled from 03-08/24. Table 1 describes demographics, diagnoses, and treatment. In the PROMIS 29+2, EOGIC patients reported significantly lower physical function and higher anxiety, depression, fatigue and pain interference compared to the general US population: t-score (T-test p-value) 40.4 (p 0.0003), 57.7 (p 0.005), 55.7 (p 0.017), 55.1 (p 0.033) and 58.5 (p 0.0001), respectively. 56% reported being at least somewhat "bothered by side effects of treatment," and this measure has a known correlation with clinician reported adverse events. Regarding sexual health (SH) in prior 30 days, 79% of participants felt less whole or "damaged" due to their disease, 64% were sexually active, 56% had some distress due to change in SH, and 50% were worried about their romantic relationship(s). Patients identified discussing safe sex with low counts, body image, contraception and addressing sexual problems as areas of unmet need during their care. They reported not receiving counseling from their care teams: only 15% discussed safe sex with low counts, 28% contraception, 13% STI prevention, and 18% discussed effect of cancer on body image. No significant change in cognitive functioning over the preceding 7 days was reported in this study. **Conclusions:** PROs are an important tool in clinical and survivorship care of young patients with GI cancers. We discovered that decreased physical function, increased anxiety, depression, fatigue and pain significantly effect HRQoL patients with EOGIC. We also found high unmet need to address sexual health in patients with EOGIC. Our ongoing work aims to better understand and address these needs for our patients. Research Sponsor: Bucksbaum Institute for Clinical Excellence, University of Chicago.

Demographic, diagnostic, and therapeutic characteristics of participants.

Age	At study participation	24-49	median 43
Gender	Female	26	66.7%
Primary cancer	Colorectal	19	48.7%
	Pancreatic	5	12.8%
	Biliary tract cancers	5	12.8%
	Esophageal/Gastric	3	7.7%
	Other (including anal, NET, GIST, appendiceal)	7	17.9%
Stage	Metastatic	22	56.4%
Treatment	On any active treatment	24	61.5%
	On chemotherapy	20	51.2%

73 **Poster Session**

Real-world survival comparison of second-line treatment strategies for patients with RAS/RAF wild-type, right-sided metastatic colorectal cancer. First Author: Nishwant Swami, Hospital of the University of Pennsylvania, Philadelphia, PA

Background: For patients with RAS/RAF wild type metastatic colorectal cancer (mCRC), systemic therapy usually includes fluoropyrimidine-based chemo + anti-VEGF or anti-EGFR targeted therapy. The PARADIGM trial established tumor sidedness as an important predictive biomarker for patients with RAS/RAF wild-type disease; in right-sided tumors, there was no survival difference between first-line chemo + anti-EGFR vs. chemo + anti-VEGF. However, there is no comparative data to guide second-line treatment decisions in this population. We aim to compare the effectiveness of second-line chemo + anti-EGFR versus chemo + anti-VEGF therapy for patients with RAS/RAF wild-type, right-sided mCRC who received first-line chemo + anti-VEGF. **Methods:** We used the nationwide Flatiron Health electronic health record-derived database, comprising de-identified patient-level structured and unstructured data obtained from ~280 cancer clinics and curated via technology-enabled abstraction. Patients ≥18 years old with RAS/RAF wild-type, right-sided mCRC who received first-line therapy consisting of chemotherapy ((FOLFIRI or FOLFOX or CAPEOX) + anti-VEGF and initiated second-line chemotherapy with either anti-EGFR or anti-VEGF targeted therapy between January 2013-May 2024 were included. Multiple imputation with chained equations imputed missing values for sidedness, RAS/RAF status, and propensity score covariates (age, gender, year of diagnosis, synchronous/metachronous disease, MMR/MSI status, ECOG score, CEA level, and first-line therapy duration). Cox proportional hazards modeling with stabilized inverse probability of treatment weighting (IPTW) assessed the association of anti-EGFR vs anti-VEGF treatment with overall survival. **Results:** 4,444 patients received appropriate first-line treatment and received chemo + anti-EGFR or chemo + anti-VEGF in the second line. Across 25 imputations, an average of 444 patients met inclusion criteria: 175 patients received chemo + anti-EGFR and 269 patients received chemo + anti-VEGF. Following IPTW, baseline characteristics were balanced between treatment groups. Patients who received chemotherapy + anti-EGFR had a 24% increased hazard of death when compared with patients who received chemotherapy + anti-VEGF, though this was not statistically significant (HR 1.24, 95% CI 0.96 – 1.61, p=0.097). **Conclusions:** Among patients with RAS/RAF wild-type, right-sided mCRC who received first-line chemotherapy + anti-VEGF, there is some evidence to support continuing anti-VEGF therapy vs. switching to anti-EGFR therapy in the second line, though the result was not statistically significant. Future studies should explore predictive biomarkers for EGFR vs VEGF-directed treatment for this population to determine the patients most likely to benefit from anti-EGFR therapy during their disease course. Research Sponsor: None.

74 **Poster Session**

Physical activity and sedentary behavior patterns after early-onset colorectal cancer diagnosis. First Author: Andreana Natalie Holowatyj, Vanderbilt University Medical Center, Nashville, TN

Background: Sedentary lifestyle and physical inactivity are established risk factors for colorectal cancer diagnosed among adults younger than age 50 years (early-onset CRC). However, patterns of physical activity (PA) and sedentary behavior (SB) in patients after an early-onset CRC diagnosis are less understood. **Methods:** The Reproductive Health After Cancer Diagnosis and Treatment (REACT) Study is a patient-partnered, cross-sectional study in cooperation with 23 community partners and patient advocates that recruited adults with a first primary cancer (diagnosis age: 18-49 years) over an 8-week period (October to December 2021). Patient-reported sociodemographic, cancer and cancer impact, PA and SB questions were prospectively collected via questionnaire. Frequency, duration and intensity of leisure-time, work-related and transportation-related PA were summed as minutes per week (min/wk). Time spent on SB was calculated by hours spent sitting on a typical day. Diagnosis age- and year-adjusted Spearman correlation coefficients were calculated between PA and SB by cancer treatment status. **Results:** A total of 131 patients were diagnosed with a first primary early-onset CRC between 2016 and 2021 (median [IQR] age, 40.0 [35-45] years; 82.4% female) in REACT. One hundred thirteen patients (86.3%) were concerned about their physical fitness or getting enough exercise after cancer diagnosis—of whom 62.8% (n=71 of 113) were somewhat or very concerned about their physical fitness/exercise. Among 80 patients with early-onset CRC and complete PA/SB data (86.3% female), 28.8% were not adherent to PA guidelines for aerobic activity (≥150 min/wk) and 56.3% reported long sedentary time (>6 hr/day). The proportion of patients across PA/SB groups also statistically significantly differed by treatment status among those diagnosed between 2019 and 2021 (p=0.049), but not between 2016 and 2018 (p=0.25). While PA was not significantly correlated with SB for early-onset CRC patients who were not actively undergoing treatment (r=-0.16, *p*=0.32), there was a statistically significant moderate inverse correlation between PA with SB among early-onset CRC patients under active cancer treatment (r=-0.58, *p*=0.004). **Conclusions:** One in every 2 early-onset CRC patients reported long sedentary time and one in every 4 early-onset CRC patients in this study did not meet the Physical Activity Guidelines for Americans. With over 80% of patients expressing concern about their physical activity and exercise after an early-onset CRC diagnosis, and differences in PA/SB patterns observed by treatment status, these findings warrant prospective, longitudinal studies of objectively-measured PA and SB among young patients in order to support tailored strategies (e.g., exercise prescriptions) that improve patient health and outcomes across the CRC continuum. Research Sponsor: American Cancer Society; IRG-19-139-59; National Cancer Institute/U.S. National Institutes of Health; P50 CA236733; National Institutes of Health/Eunice Kennedy Shriver National Institute of Child Health and Human Development; K12 HD043483.

75 **Poster Session**

Patterns of metastatic spread and survival outcomes for stage IV left-sided colon vs. rectal cancers: A real-world analysis. First Author: Neal Bhutiani, The University of Texas MD Anderson Cancer Center, Houston, TX

Background: A growing body of literature has reported differences in biology, metastatic potential, and outcomes among patients with colorectal cancer based on the sidedness of their primary tumor (right vs. left). Left-sided primary tumors have comprised both colon and rectal tumors, despite differences in venous and lymphatic drainage patterns and treatment paradigms. The present work sought to investigate patterns of metastases and survival outcomes among patients with left sided colon primary tumors and rectal primary tumors. We hypothesized that these two disease sites metastasize differently and exhibit different survival patterns based on metastatic site. **Methods:** The National Cancer Database (NCDB) was queried for all patients with a diagnosis of metastatic left-sided colon (splenic flexure, descending colon, rectosigmoid) or rectal adenocarcinoma with known primary site and known metastatic site(s) between 2010 and 2021 using parameters outlined in the NCDB Data Dictionary. Patients with left-sided colon primaries (LP) were compared to patients with rectal primaries (RP) along clinicodemographic parameters. Incidence of metastases to the liver, lungs, bone, brain, and non-regional lymph nodes were also compared. Multivariable survival analyses were performed to analyze differences in overall survival (OS). **Results:** Among 84,958 patients, 51,931 (61.1%) had LP while 33,027 (38.9%) had RP. Patients with RP were less likely to have liver metastases than those with LP (79.4% vs. 90.3%), but more likely to have lung (40.7% vs. 29.2%), bone (8.3% vs. 6.0%), and distant lymph node metastases (11.3% vs. 8.6%) (all p<0.001). Compared to patients with LPs, patients with RP were less likely to undergo surgery to the primary tumor (25.1% vs. 46.3%) or any metastatic site (15.3% vs. 18.8%), more likely to receive chemotherapy for metastatic disease (80.8% vs. 75.2%), and more likely to receive radiation to the primary site (17.7% vs. 4.3%) (all p<0.001). Additionally, patients with RP had longer median OS than those with LP (27.3 months vs. 26.4 months, p<0.001). On multivariable analysis controlling for age, gender, tumor grade, RAS mutation status, MSI status, CEA level, clinical T and N category, and treatment details, patients with RP with liver metastases (HR 0.91, 95% CI 0.89-0.92), lung metastases (HR 0.85, 95% CI 0.82-0.87), and bone metastases (HR 0.89, 95% CI 0.84-0.95) (all p<0.001) demonstrated improved OS compared to those with LP. **Conclusions:** Patterns of metastases, treatment regimens, and, in certain instances, survival outcomes, differ among patients with stage IV left-sided colon cancer and rectal cancer. Future work evaluating the implications of sidedness in colorectal cancer as well as future staging efforts should consider assessing left-sided colon cancer separately from rectal cancer, as these demonstrate distinct clinical behavior and outcomes. Research Sponsor: None.

76 **Poster Session**

Treatment sequences in patients with metastatic colorectal cancer in Japan: Real-world evidence of 1st- to 5th-line treatments. First Author: Yoshinori Kagawa, Department of Gastroenterological Surgery, Osaka International Cancer Institute, Osaka, Japan

Background: The development of later-line drugs for metastatic colorectal cancer (mCRC) has expanded treatment options for systemic chemotherapy in patients with mCRC. However, few studies have investigated patients who received later-line treatment in a real-world setting, and available information on treatment sequences and transition rates from early- to later-line treatments is limited. **Methods:** This was a retrospective study using hospital administrative data collected by MDV (Medical Data Vision, Tokyo, Japan) from April 2008 through February 2023 in Japan. Treatment regimens were derived from standard treatments recommended in the Japanese guidelines. The study population was determined using an algorithm from a similar study previously conducted in Japan. Our analysis focused on patients who started CRC surgery or 1st-line treatment after January 2017 to reflect recent late-line real-world practice. Transition rates from 1st- to 5th-line treatment were calculated, and treatment sequences were summarized using a Sankey diagram. Logistic regression was performed to identify the factors associated with the transition from 2nd- to 3rd-line treatment. **Results:** A total of 722,005 patients with a CRC diagnosis record were identified in the MDV database. Of these, 26,456 patients were included in the study population. The median age was 69 years and 61.4% of patients were male. The rates of transition to subsequent lines from 1st to 5th line ranged from 65.7% to 70.4% (1st to 2nd 68.3%; 2nd to 3rd 70.4%; 3rd to 4th 69.9%; 4th to 5th 65.7%). Among 8528 patients who received 2nd-line treatment, ~70% of patients (6006) continued 3rd-line treatment, and 30% of patients (2522) received best supportive care. Comparing these 2 patient groups, younger age (<65 years, OR: 1.50, 95%CI: 1.35-1.67; p<0.001) and longer treatment durations of 1st (≥180 days, OR: 1.26, 95%CI: 1.14-1.39; p<0.001) and 2nd (≥120 days, OR: 1.67, 95%CI: 1.52-1.84; p<0.001) lines were significant factors for 3rd-line treatment continuation. Prior therapy with oxaliplatin or irinotecan plus molecular targeted drugs was also associated with a higher likelihood of proceeding to 3rd-line treatment (OR: 1.42, 95%CI: 1.28-1.58; p<0.001). **Conclusions:** In this study, ~70% of patients were able to transition to subsequent lines at the time of each line transition. However, 30% of patients were unable to continue treatment in real-world practice. To increase treatment continuation rates and maintain patient quality of life, new treatment options and further research are necessary to meet patients' treatment needs, especially in the later-line treatment setting where unfavorable effects from previous treatments are accumulated. Research Sponsor: Takeda Pharmaceutical Company Limited.

77 Poster Session

Bevacizumab plus mFOLFOX6/FOLFIRI sequential bevacizumab plus capecitabine maintenance in RAS-mutated advanced colorectal cancer: A real-world study of first-line/second-line conversion therapy. First Author: Ying Yan, Department of Oncology, The First Affiliated Hospital of University of Science and Technology of China West District, Hefei, China

Background: Currently, the standard first-line/second-line treatment for advanced colorectal cancer with RAS mutations is chemotherapy combined with bevacizumab. Patients with colorectal cancer confirmed maintenance therapy can effectively improve progression-free survival, improve patient quality of life. Therefore, this study aims to evaluate the efficacy and safety of bevacizumab combined with mFOLFOX6/FOLFIRI sequential bevacizumab combined with capecitabine as first-line/second-line conversion therapy for RAS-mutated advanced colorectal cancer. **Methods:** This is an open-label, single-center study in previously untreated patients with RAS mutant mCRC. Eligible patients received first-line chemotherapy of bevacizumab (5mg/m^2) combined with mFOLFOX6/FOLFIRI for 12 cycles randomly, during which the efficacy was assessed every 4 cycles, then followed by bevacizumab(7.5mg/m^2) in combination with capecitabine maintenance therapy which assessed every 2 cycles until disease progression, and then switched to the second-line therapy FOLFIRI/mFOLFOX6 combined with bevacizumab which was assessed every 4 cycles until disease progression or intolerable toxicity. Imaging studies were performed to assess treatment efficacy according to RECIST1.1 criteria, and the primary endpoint was progression-free survival (PFS), including first-line PFS and second-line PFS after progression, and the secondary endpoints included disease response rate (ORR), disease control rate (DCR), overall survival (OS), and safety. **Results:** This study recruited 78 subjects from January 2021 to May 2024. As of September 10, 2024, a total of 70 subjects were analyzable for the first-line treatment, of which 6 subjects met the surgical criteria after treatment and achieved NED status; 45 patients reached the first-line PFS endpoint, with a median PFS (mPFS) of 249 days (95% CI: 204.4-293.6), an ORR of 38.6%, and a DCR of 95.7% . Twenty patients entered the first-line maintenance therapy, with a mPFS of 358 days (95% CI: 53.8-662.2). For the second-line treatment, a total of 43 subjects were analyzable, of which 35 patients reached the second-line PFS endpoint, with a second-line mPFS of 171 days (95% CI: 89.8-252.1). 41 subjects underwent at least one efficacy evaluation, with ORR of 14.6% and DCR of 75.6%. The most common grade 3 treatment-related adverse events were neutropenia, leukopenia, anemia, hypertension, diarrhea, and thrombocytopenia. There were no treatment-related deaths. No new adverse events occurred during the entire combination treatment period. **Conclusions:** This real world study has demonstrated good tolerability and encouraging clinical efficacy, especially with potential implications for treatment planning and management strategies for advanced colorectal cancer with RAS mutations. Clinical trial information: ChiCTR2100042553. Research Sponsor: None.

78 Poster Session

Comparative efficacy of TAS-102 and chemotherapy rechallenge in the third-line setting and beyond in patients with advanced colorectal cancer: A retrospective study. First Author: Kanan Alshammari, Oncology Department, Ministry of National Guard Health Affairs, Riyadh, Saudi Arabia

Background: The optimal approach to third line therapy for patients with metastatic colorectal cancer remains a subject of debate. Options include Trifluridine/Tipiracil (TAS-102), TAS-102 with bevacizumab, regorafenib, chemotherapy rechallenge with or without an anti-EGFR agent, or fruquintinib. This study aims to evaluate the efficacy of standard of care TAS-102, TAS-102 with bevacizumab, or regorafenib versus chemotherapy rechallenge in patients with metastatic colorectal cancer in the third line setting and beyond. **Methods:** We conducted a retrospective analysis of patients with metastatic colorectal cancer who received either TAS-102 with or without bevacizumab, regorafenib or chemotherapy rechallenge with or without an anti-EGFR agent between 2018 and 2023, at the Ministry of National Guard for Health Affairs in Riyadh, Saudi Arabia. Patients received such agents in the third line setting and beyond. **Results:** A total of 75 patients were included in this study, with 39 patients receiving TAS-102 with or without bevacizumab, or regorafenib, and 36 receiving chemotherapy rechallenge. The median patient age was 58 years (range 28-81), with the majority being males at 55%. Among the chemotherapy rechallenge group, 31% were treated with cetuximab and 6% had panitumumab rechallenge, some of which were guided by cell free DNA liquid biopsy testing. In the other group, 66% received regorafenib, 26% TAS-102 alone, and 8% TAS-102 with bevacizumab. At the time of analysis, 56% of patients had passed away. Median overall survival for patients who received TAS-102 with our without bevacizumab, and regorafenib in the third line setting was found to be 13.1 months. Whereas for patients who received chemotherapy rechallenge, their median overall survival was 19.5 months. **Conclusions:** Patients with metastatic colorectal cancer who received chemotherapy rechallenge in the third line setting survived longer compared with patients who received TAS-102 with or without bevacizumab, or regorafenib. Research Sponsor: None.

79 Poster Session

Real-world clinical outcomes of patients (Pts) with metastatic colorectal cancer (mCRC) who received trifluridine-tipiracil (FTD-TPI) monotherapy or FTD-TPI + bevacizumab (FTD-TPI+bev) combination therapy. First Author: Maliha Nusrat, Memorial Sloan Kettering Cancer Center, New York, NY

Background: FTD-TPI+bev has shown survival benefit when compared to FTD-TPI alone in pts with mCRC and was approved in the U.S. in 2023. However, there is a lack of real-world data on outcomes for pts who received FTD-TPI+bev as compared to pts who received FTD-TPI. **Methods:** This retrospective study used data abstracted from electronic medical records and claims in the ConcertAI RWD360 dataset. Adult pts with a diagnosis of mCRC and exposure to FTD-TPI were included. Pts were categorized as having received FTD-TPI or FTD-TPI+bev based on first exposure to FTD-TPI (index date). Kaplan-Meier analyses were used to describe the overall survival (rwOS), real-world time to discontinuation (rwTTD), and time to next treatment or death (rwTTNTD) from the index date. Multivariate Cox regression analyses were used to control for patient characteristics, including demographics, ECOG performance status (PS), comorbidities, sites of metastatic disease, time from mCRC to index date, lab results, and receipt of prior regorafenib. **Results:** This study included 3,680 pts. The FTD-TPI cohort included 3,151 pts (median age 62 years; male 56.3%; White 63.3%; any comorbidities 40.6%; ECOG PS 0-1 34.8%; 2+ metastatic sites 25.4%; median time from mCRC diagnosis to index date 699 days). The FTD-TPI+bev cohort included 529 pts (median age 60 years; male 53.9%; White 68.1%; any comorbidities 34.4%; ECOG PS 0-1 51.4%; 2+ metastatic sites 25.1%; median time from mCRC diagnosis to index date 609 days). Prior to index date, 18.5% and 6.9% of pts had been exposed to regorafenib in the FTD-TPI and FTD-TPI+bev cohorts, respectively. FTD-TPI+bev significantly increased rwOS (median 9.4 [95% CI 8.0-10.1] vs 6.4 [95% CI 6.1-6.6] months; $p<0.0001$) and significantly decreased the risk of death (HR=0.68; $p<0.0001$) in the adjusted model. Other significant factors for rwOS included 2+ metastatic sites (vs.1 site, HR=1.16; p=0.002) and no receipt of regorafenib prior to index date of FTD-TPI (HR=0.80; $p<0.0001$). FTD-TPI+bev also significantly prolonged rwTTD as compared to FTD-TPI (median 3.5 [95% CI: 3.3-3.8] vs 2.4 [95% CI: 2.3-2.6] months; $p<0.0001$) and decreased risk of discontinuation (HR=0.65; $p<0.0001$) after adjusting for patient demographic and clinical characteristics. Additionally, the FTD-TPI+bev cohort had a significantly longer rwTTNTD as compared to the FTD-TPI cohort (median 5.0 [95% CI: 4.6-5.5] vs 3.9 [95%CI: 3.8-4.1] months; $p<0.0001$) in the adjusted model (HR=0.73; $p<0.0001$). **Conclusions:** This is the largest study to compare the clinical outcomes of pts receiving FTD-TPI and FTD-TPI+bev in the real-world setting. FTD-TPI+bev improved clinical outcomes including rwOS, rwTTD, and rwTTNTD, as compared to FTD-TPI in pts with mCRC. These findings align with the results of the SUNLIGHT trial. Research Sponsor: Taiho Oncology, Inc.

80 Poster Session

Treatment experiences and decision-making among patients with metastatic colorectal cancer: Results of an online US patient survey. First Author: Julie Saliba Clauer, COLONTOWN, Crownsville, MD

Background: Developing treatment plans for patients (pts) with metastatic colorectal cancer (mCRC) is complex. Using an online survey, we aimed to understand the treatment experience of pts with mCRC, including pt involvement in treatment decision-making. **Methods:** Adults (≥18 yrs) with mCRC took part in an online survey conducted in the US by The Harris Poll between Jan 17 and Feb 7, 2024. Of 344 respondents, 90 were recruited from patient advocacy groups (PAGs). The 60-question survey explored pts' experience of mCRC. We report results from questions focused on treatment. Raw data were not weighted and are only representative of the participants. All surveys are subject to sources of error. **Results:** Of 256/344 pts who received mCRC treatment, chemotherapy and surgery were the most common therapies(Table). Among pts who received therapy, 67% reported their ability to take treatment was impacted by non-medical factors such as finances or mental health. Overall, 58% of pts have had to decide between affording treatment or meeting basic needs, e.g. safe housing, food, transportation, especially Black and Hispanic pts (Table). Of all 344 pts, 87% reported reliance on their healthcare provider (HCP) when making treatment decisions, primarily their oncology surgeon (48%) or oncologist (45%), despite 268 (78%) pts carrying out independent research on new treatments or clinical trials. Although 266 pts (77%) were involved in treatment decision making, 231 pts (67%) wished for more power in the process. Overall, 64 pts (19%) wished for more support from PAGs during treatment, while 59 pts (17%) relied on PAGs to help guide treatment decisions. The most common factors affecting pt treatment decisions related to costs or insurance coverage (52%), impacts on physical health and quality of life (each 39%), fear of being a burden on family (35%), and side effects (32%). **Conclusions:** Pts with mCRC rely on their HCPs when making treatment decisions and wish for more power in the process. Personal finance and insurance coverage play dominant roles in treatment for mCRC. PAGs are an important resource for treatment information that are underutilized. Research Sponsor: Takeda Development Center Americas, Inc. (TDCA), Lexington, MA, USA.

Treatment experience of pts with mCRC in an online US survey.

n (%)	All pts N=344	Black pts n=87[c]	Hispanic pts n=75[c]	White pts n=158
Received treatment	256 (74)	52 (60)	60 (80)	129 (82)
Chemotherapy[a]	169 (66)	23 (44)	33 (55)	104 (81)
Surgery[a]	147 (57)	23 (44)	32 (53)	86 (67)
Radiotherapy[a]	101 (39)	25 (48)	15 (25)	52 (40)
Immunotherapy[a]	72 (28)	16 (31)	20 (33)	30 (23)
Targeted therapy[a]	61 (24)	12 (23)	14 (23)	30 (23)
Other[a]	13 (5)	0	0	12 (9)
Stopped treatment due to overall burden[a,b]	92 (36)	28 (54)	33 (55)	27 (21)
Had to decide between treatment or basic needs	198 (58)	60 (69)	52 (69)	69 (44)

[a]Among pts who received treatment.
[b]Financial impact, quality of life, etc.
[c]Base size <100 pts; interpret results directionally.

81 Poster Session

Real-world clinical outcomes of trifluridine-tipiracil monotherapy (FTD-TPI) and FTD-TPI + bevacizumab combination therapy (FTD-TPI+bev) in 639 black patients (pts) with metastatic colorectal cancer (mCRC). First Author: Maliha Nusrat, Memorial Sloan Kettering Cancer Center, New York, NY

Background: In the phase 3 SUNLIGHT trial, FTD-TPI+bev improved overall survival (OS) as compared to FTD-TPI in treatment-refractory mCRC. Out of 492 pts in the study, only 7 (1.4%) pts self-identified as Black; therefore, the potential benefit of FTD-TPI+bev in this pt population was not well-examined. We present the largest analysis to date of real-world clinical outcomes of FTD-TPI and FTD-TPI+bev in Black pts with mCRC. **Methods:** We retrospectively identified Black adult pts with a documented exposure to FTD-TPI and diagnosis of mCRC at or prior to the initiation of FTD-TPI using United States-based electronic medical records and claims in the ConcertAI RWD360 dataset. Pts were categorized as having received FTD-TPI or FTD-TPI+bev. Kaplan-Meier analyses were used to compare the real-world overall survival (rwOS) and time to discontinuation (rwTTD) from the date of first exposure to FTD-TPI (index). Multivariate Cox regression analyses were used to control for pt characteristics including demographics, ECOG performance status (PS), comorbidities, metastatic sites, time from mCRC diagnosis to index, lab results, and prior receipt of regorafenib. **Results:** Out of 639 Black pts included in our study, 551 received FTD-TPI and 88 received FTD-TPI+bev. Index years were 2021-2023 for 26% and 92% pts receiving FTD-TPI and FTD-TPI+bev, respectively. Pts receiving FTD-TPI had median age 61 years; male 51.2%; any comorbidities 43.9%; ECOG PS 0-1 36.7%; median time from mCRC diagnosis to index 632 days. Pts receiving FTD-TPI+bev had median age 59 years; male 50.0%; any comorbidities 42.0%; ECOG PS 0-1 60.3 %; median time from mCRC diagnosis to index 579.5 days. The rwOS was significantly improved with FTD-TPI+bev as compared with FTD-TPI (10.8 vs 6.2 months, respectively; p=0.0001), similar to the SUNLIGHT study (Table). In the adjusted model, pts who received FTD-TPI+bev had a significantly lower risk of death as compared to pts who received FTD-TPI (hazard ratio [HR]=0.45; $p<0.0001$). The rwTTD was also significantly prolonged with FTD-TPI+bev as compared with FTD-TPI (3.8 vs 2.4 months, respectively; $p<0.0001$). In the adjusted model, pts who received FTD-TPI+bev had about half the risk of discontinuation as compared with pts who received FTD-TPI (HR=0.51; $p<0.0001$). **Conclusions:** This is the first study to compare the clinical outcomes with FTD-TPI vs. FTD-TPI+bev in Black pts with mCRC in the real-world setting. Our study confirmed OS improvement with FTD-TPI+bev as compared to FTD-TPI in Black pts with mCRC, as seen in the overall population of the SUNLIGHT trial. Research Sponsor: Taiho Oncology, Inc.

	Black n	Current Study rwOS months (95% CI)	Black n	SUNLIGHT OS months (95% CI)
FTD-TPI	551	6.2 (5.7-6.9)	3	7.5 (6.3-8.6)
FTD-TPI+bev	88	10.8 (7.8-16.0)	4	10.8 (9.4-11.8)
		HR 0.45 (95% CI 0.33-0.63; $p<0.0001$)		HR 0.61 (95% CI 0.49-0.77; $p<0.001$)

82 Poster Session

Clinical-pathological characteristics, follow-up, and survival of patients with early-onset vs late-onset colorectal cancer: Institutional experience in Argentina. First Author: Julian Maquieira, Hospital de Gastroenterologia Carlos Bonorino Udaondo, Ciudad Autonoma De Buenos Aires, Argentina

Background: The incidence of colorectal cancer (CRC) in young adults aged 50 or younger (Early Onset - EO) is increasing worldwide, while in patients over 50 years old (Late Onset - LO), it is declining. The objective is to compare the clinicopathological characteristics, time to diagnosis, and survival between Early Onset (EO) and Late Onset (LO) patients diagnosed with colorectal cancer (CRC). **Methods:** The sample was described using measures of central tendency and dispersion for continuous numerical variables and percentages for categorical variables. Clinical characteristics between Early Onset (EO) and Late Onset (LO) groups were compared using the Student's t-test or Chi-square test. Kaplan-Meier curves and the Log-Rank Test were utilized to analyze mortality time. For multivariable analysis, a Cox regression model was applied. A p-value of less than 0.05 was considered statistically significant. The analysis was performed using RStudio. **Results:** A total of 460 patients were evaluated, with 343 diagnosed with colon adenocarcinoma and 117 with rectal cancer. Patients were grouped into two categories: the LO group with 408 (88.6%) patients and a mean age of 69.3 (8.60 SD) years, and the EO group with 52 (11.3%) patients and a mean age of 41.9 (6.23 SD) years. The EO group presented a higher percentage of patients with a PS of 0 compared to the LO group. In the EO group, there was a predominance of left-sided colon cancer (40.4% vs. 40.0%) and rectal cancer, especially in the middle and lower regions (30% vs. 18.8%), with disease diagnosed at stage III (26.9% vs. 26.5%) and IV (5.8% vs. 4.4%) compared to the LO group. Molecular testing for KRAS, NRAS, and BRAF mutations was more frequent in the EO group (19.2%, 3.8%, and 3.8%, respectively) compared to the LO group, as well as mismatch repair (MMR) status at 9.6% in the EO group vs. 6.9% in the LO group. The median time to diagnosis was 186 days in the LO group and 341 days in the EO group (p=0.21). Follow-up was 858 days in the LO group and 725 days in the EO group (p=0.18); while mortality was 200 (49.0%) in the LO group and 24 (46.2%) in the EO group, with a median survival of 1193 days for the LO group and 1414 days for the EO group, respectively (p=0.7). PS and stage were the most influential prognostic factors for mortality. **Conclusions:** In the comparison, the EO group showed a higher incidence of left-sided colon and rectal cancers, more advanced stages, better PS, and a greater number of molecular alterations, with a longer time to diagnosis and shorter follow-up. Although these differences were not statistically significant, they highlight the need to optimize prevention strategies and genetic counseling. Additional studies with a larger number of EO patients in real-life contexts are needed to generate more evidence that can improve clinical decision-making. Research Sponsor: None.

83 Poster Session

Annual incidence of colorectal cancer among patients with average risk in the United States between 2017 and 2023. First Author: Brad Stieber, Exact Sciences Corporation, Madison, WI

Background: Colorectal cancer (CRC) is the second most common cause of cancer-related mortality and third most common cancer in the United States. In 2021, 141,902 new cases of CRC were reported. With recent increases in early onset of CRC incidence, a shift in patterns amongst various demographics has been observed. Together with an estimated 60% increase in burden of CRC globally, it is important to continuously assess the nationwide incidence rates and monitor the changing patterns. In this study, we examined the annual incidence rates of CRC in average risk adults using a large national claims database and report patterns of incidence based on various demographic characteristics. **Methods:** CRC diagnoses were retrospectively identified from a large national claims database, which covers over 165 million lives and is representative of the U.S. population, for each calendar year during the study period from 2017 to 2023. Individuals aged 45 to 75 years, with an average risk of CRC before each index calendar year were included in the study. Individuals were required to be continuously enrolled in the same health plan for at least 1 year prior to and 1 year following the index year. Annual CRC incidence rates per 100,000 people were calculated for each year from 2017 to 2023 within various subgroups based on baseline demographics. **Results:** Overall, annual CRC incidences remained consistent during the study period with 186.86 (95% CI, 184.87-188.86) cases in 2017 and 183.40 (95% CI, 181.32-185.5) cases in 2023 per 100,000. Among younger adults aged 45-49 years, the annual CRC incidence increased by 66% from 2017 (92.04 cases/100,000) to 2023 (153.12 cases/100,000). Conversely, annual CRC incidence decreased by 25% between 2017 (328.52 cases/100,000) and 2023 (244.93/100,000) in older adults aged 65-75 years. Over the study period, CRC incidence in men was higher than women with an overall rate ratio (RR) of 1.206 (95% CI, 1.122-1.297). African American adults had higher CRC incidence than White Americans with an overall rate ratio (RR) of 1.216 (95% CI, 1.152-1.283). **Conclusions:** Based on this large study of national claims data, CRC incidence rates remained relatively stable in the total population among average-risk patients from 2017 to 2023. The continuing increase in recommended CRC screening utilization showed a decreased incidence rate in the older population, while observing increased incidences among younger patients. Research Sponsor: Exact Sciences Corporation.

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Long-term outcomes of total neoadjuvant treatment (TNT) for locally advanced rectal cancer: Maria Sklodowska Curie National Cancer Research Institute (MSCNCRI) cohort from Polish-II study. First Author: Mariola Winiarek, Maria Sklodowska-Curie National Research Institute of Oncology, Warsaw, Poland

Background: The data on long-term outcomes of TNT for locally advanced rectal cancer are still limited. The Polish-II study randomized 515 cT4/fixed cT3 rectal cancer to preoperative chemoradiation (CRT) with 5fu/oxaliplatin or 5x5 Gy radiotherapy with consolidating chemotherapy (TNT). Despite the early separation of survival curves with benefit for TNT, the outcomes at the longer observation were similar reaching 49% in both arms surviving at 8 years. Here, we present the long-term oncological and functional outcomes of patients recruited to the study in MSCNCRI along with the quality of life in survivors at >10 years of observation. **Methods:** Of 180 patients treated in the study in MSCNCRI 88 and 92 pts were randomized to TNT and CRT arms, respectively. The long-term follow-up was performed according to the standard protocol, FACT-C and EORTC QLQ-C30 scales were used to assess the functional outcomes. **Results:** Median survival in the TNT group and CRT was 115 and 61 months (p=0.191). At the cut-off in March 2023, 53.4% and 61.1% of pts died in the TNT and CRT group (p = 0.375). In the TNT group, 73.9% of deaths were attributed to cancer corresponding to 81.8% in CRT (p = 0.725). Late relapses diagnosed over 5 years from the end of the active treatment were identified in 5 of 94 patients in the follow-up (1 local recurrence in the TNT group, 4 patients with lung metastasis – 3 in TNT with one having also CNS relapse - and 1 in CRT). All late relapses in the TNT group could be salvaged with local therapies. Regarding long-term quality of life, no differences were observed in FACT-C and EORTC QLQ C30 in patients treated with either arm: QLQ C30 – TNT: 50.68 ± 12.74 in the range 34.00- 79.00; CRT: 48.61 ± 11.39 in the range 34.00-70.00 (p = 0.507); FACT-C - TNT: 113.00 in the range 42.00-136.00; CRT: 118.00 in the range 74.00-136.00; p=0.252). No difference in physical, family, and emotional status was observed. At the ultimate observation, stoma was present in 40.5% and 40.6% of TNT and CRT groups, respectively ($p>0.999$). **Conclusions:** In this randomized cohort, after more than 10 years of observation TNT led to numerically longer OS than CRT, although this was not statistically significant. The functional outcomes of the long-term survivors are similar in both cohorts. Late relapses are rare but could be salvaged in most cases. This study further supports the use of TNT in patients with locally advanced rectal cancer. Clinical trial information: NCT00833131. Research Sponsor: None.

85 Poster Session

Management of patients with locally advanced rectal cancer in the total neoadjuvant therapy (TNT) era: Experience from Argentina. First Author: Maria Victoria Faura, Hospital Britanico, Buenos Aires, Argentina

Background: In the last years a paradigm shift has changed the way we manage non metastatic rectal cancer. The varied options within TNT have gradually been adapted in our country. This study aims to depict the strategies implemented. **Methods:** A prospective and retrospective observational study was performed from 2020 to 2024 including stage II-III rectal cancer patients treated at three centers in Buenos Aires, Argentina. Data was extracted from medical records. **Results:** 68 patients were included in the analysis. Median age was 59 years (IQR 48–71). 72% were male. 88% (60) had stage III disease, and 12% (8) stage II. Geriatric assessment was performed in 7 of 19 patients aged over 70 (5 frail). All patients were staged with high resolution MRI and discussed in multidisciplinary tumor board. Mismatch repair was assessed in 72%(49) of patients, only one had MLH1 deficiency. 79%(54) patients received TNT: 22%(12) short course vs 78%(42) long course radiotherapy; 24%(13) induction vs 76%(41) consolidation chemotherapy. Preferred chemotherapy regimen was CAPOX both in induction and consolidation chemotherapy (69.7% and 63.4%, respectively) Despite small numbers non significant differences in risk factors were found between TNT vs non TNT groups (Table). We registered 13 Grade 3-4 toxicity events: 3 in induction chemotherapy, 7 in consolidation and 3 in radiotherapy. Clinical complete response was achieved in 25.9% of TNT (11 consolidation, 3 induction) and 7.1%(1) of non-TNT (p=0.25). Watch and Wait was employed in 9 patients in TNT and 1 in non TNT group. 40 patients underwent surgery in the TNT and 11 in the non-TNT group; complete pathological response was reached in 17.5% (7) and 9% (1), respectively. Overall relapse rate was 19.14% (13): 2.94% (2) local and 16.2% (11) systemic. (TNT: 1.85%(1) local vs 18.5%(10) metastatic; nonTNT: 7.14%(1) both local and metastatic) p=0.4. At a median follow-up of 20 months (IQR 15–32 months), 2-year progression-free survival (PFS) rate was 87% (95% CI: 38–98%) in the non-TNT group and 78% (95% CI: 61–88%) in the TNT group. **Conclusions:** This study highlights the feasibility of worldwide adoption of TNT. We describe the preferred regimens and results of TNT in three high volume centers in Argentina. Long course radiotherapy and consolidation treatment with CAPOX are the predominant strategies. Non operative management is routinely offered to complete responders. Our efficacy results are comparable with published literature. Comprehensive geriatric assessment is recommended but still partially applied. In line with recent findings MMR testing has been widely adopted as an initial workup parameter. Research Sponsor: None.

	TNT %(n=54)	non-TNT %(n=14)	p
T4 disease	26 (14)	21 (3)	0.99
N2	37 (20)	29 (4)	0.75
EMVI positivity	49 (26)	18 (3)	0.16
Circumferential resection margin <1 mm	29.6 (16)	28.57 (4)	0.23
Low-lying tumors	33 (18)	43 (6)	0.54

86 Poster Session

The efficacy of anti-EGFR therapy across different lines of treatment in *RAS* wild-type left-sided metastatic colorectal cancer. First Author: Anita Archwamety, Division of Medical Oncology, Department of Medicine, Faculty of Medicine Siriraj Hospital, Mahidol University, Bangkok, Thailand

Background: Anti-epidermal growth factor receptor monoclonal antibodies (anti-EGFR mAbs) are recommended as a first-line treatment in *RAS wild-type (RASwt*) left-sided metastatic colorectal cancer (mCRC). However, the efficacy of these therapies across different treatment lines remains unclear. This study aims to evaluate the efficacy of anti-EGFR mAb across first, second, third, and later-line treatments. **Methods:** We conducted a retrospective analysis of patients diagnosed with *RASwt* mCRC who received anti-EGFR mAb in all treatment lines at Siriraj Hospital between 2008 and 2023. The impact of anti-EGFR mAb treatment line on progression-free survival (PFS) and overall survival (OS) was determined using the Kaplan-Meier method and compared with the log-rank test. **Results:** Of the 350 patients with *RASwt* mCRC receiving anti-EGFR mAb, 33% (115 of 350) were treated in the first-line, while 16%, 45% and 6% were treated in the second-, third- and later-line settings, respectively. Eighty-six percent of patients had left-sided tumors, and anti-vascular growth factors were used in 34% of patients. The median follow-up time was 30.4 months (mo). Among patients with left-sided tumors, those receiving first-line anti-EGFR mAb treatment exhibited a significantly longer mPFS (11.9 mo, 95% CI 7.4-16.3, $p<0.001$). The mPFS in the second-, third- and later-line settings were 6.1 mo (95% CI 4.6-7.5), 4.9 mo (95% CI 3.8-6.0), and 5.1 mo (95% CI 2.3-7.9), respectively. There were no significant differences in terms of OS across treatment lines (p=0.27). **Conclusions:** Although first-line treatment with anti-EGFR mAb had the longest mPFS in left-sided mCRC, OS did not significantly differ among treatment lines. Therefore, anti-EGFR mAb can be incorporated as a part of therapy at any line of treatment. Research Sponsor: None.

87 Poster Session

Digesting danger: A two-decade study exploring the association between food processing and colon cancer. First Author: Elise Katsnelson, University of Miami Miller School of Medicine, Miami, FL

Background: Dietary patterns play a significant role in cancer prevention and progression, yet the specific association of food processing with colon cancer remains largely under-investigated. The NOVA classification system categorizes foods by their level of processing, providing a unique way to examine this relationship. Colon cancer remains a leading global health concern, and emerging evidence points to diet as potential contributors. This study explores associations between NOVA-classified foods and colon cancer, offering new insights into the role of food processing in cancer development. **Methods:** Survey data from the NHANES database (from 1999 to 2018) was probed for demographic and clinically relevant data, and day one food journals were converted into NOVA classifications. Participants aged 20 and older were included for analysis, including those with and without colon cancer (n=339 and 47615, respectively). Independent two-sample t-tests were conducted to compare the mean proportion of daily energy consumption of various NOVA food categories between the two groups, and a Benjamini-Hochberg correction was applied to adjust for multiple comparisons at $p<0.05$. Sensitivity analyses were performed using total grams and percent total caloric intake as inputs. Subgroup analyses were conducted via a logistic regression model including BMI, age, ethnicity, gender, education, country of birth, and military service. **Results:** Individuals with colon cancer had higher intake of ham and salted meats (NOVA 3, adjusted p=0.028) compared to those without cancer. Conversely, individuals without cancer consumed more legumes (NOVA 1, adjusted p=0.028), oils (NOVA 2, adjusted p=0.023), and "other processed foods" that did not fall into a specific processed food category (NOVA 3, adjusted p=0.038). Subgroups analyses revealed a greater association with colon cancer for age greater than 50 and non-hispanic White ethnicity. **Conclusions:** To our knowledge, this is the only study at this scale that evaluates colon cancer prevalence in the context of dietary behaviors stratified by level of industrial processing. The significant associations between ham and salted meats consumption, age greater than 50, and non-Hispanic White ethnicity with increased cancer prevalence highlight the impact of dietary and demographic patterns on colorectal carcinogenesis. Additional studies are needed to confirm these hypotheses in the prospective setting. Research Sponsor: None.

88 Poster Session

Multi-target stool-DNA test for colorectal cancer screening and effects on health and economic outcomes compared to blood-based tests: Influence of adherence. First Author: Vahab Vahdat, Exact Sciences Corp., Madison, WI

Background: There are an estimated 60 million individuals in the US who are not up-to-date with average-risk colorectal cancer (CRC) screening, as screening rates and adherence have remained stubbornly below the national target of 80%. Efforts are ongoing to improve CRC screening performance and participation, including the development of new blood-based tests. Despite high expectations for these tests, performance remains lower than other guideline-recommended strategies, particularly with respect to advanced precancerous lesion (APL) sensitivity. We conducted a simulation study of estimated clinical and economical outcomes for CRC screening with a blood-based test (at perfect adherence) compared to the multi-target stool-DNA (mt-sDNA) test (at published adherence rates). **Methods:** Utilizing CRC-AIM, a calibrated and validated microsimulation model, CRC screening outcomes were calculated for 1 million average-risk individuals screened between ages 45-75 years with triennial blood-based versus mt-sDNA testing. CRC and APL sensitivity and specificity inputs were derived from two large clinical validation studies for these screening modalities: ECLIPSE (NCT04136002) and DeeP-C (NCT01397747), respectively. To demonstrate the maximum benefits and burdens of blood-based screening, adherence to initial screening and follow-up colonoscopy after a positive blood test was modeled at 100%, while real-world adherence estimates of 65.6% were used for the mt-sDNA test. Outcomes of interest included life years-gained (LYG) and CRC cases missed by the blood-based test compared to the stool-based test, additional treatment costs imposed by theblood-based test, and additional CRC-related deaths per 1 million screened. **Results:** Compared to triennial screening with mt-sDNA at real-world adherence, blood-based screening at perfect adherence resulted in a higher number of incident (n=18,464) and fatal (n=6,483) CRC cases that would been avoided with the mt-sDNA test. Over a lifetime, screening with blood-based tests required 1,276,310 more tests (21% more) and 52,942 additional follow-up colonoscopies (7% more) compared to mt-sDNA screening. This increased testing burden did not generate additional benefits and instead reduced life-years gained by 67,645 per 1 million screened, resulting in an additional $1.6 billion in treatment costs compared to the mt-sDNA strategy. **Conclusions:** Data from this CRC-AIM modeling study show that even with perfect adherence, blood-based screening yields inferior clinical and economic benefits compared to mt-sDNA screening, due to suboptimal APL detection with the former test. Research Sponsor: Exact Sciences Corp.

89 Poster Session

Comparison of screening outcomes associated with advanced precancerous lesion versus colorectal cancer sensitivity for a blood-based test: A simulation study. First Author: Vahab Vahdat, Exact Sciences Corp., Madison, WI

Background: With estimates of over 150,000 incident and 50,000 fatal colorectal cancer (CRC) cases in 2024, efforts are ongoing to improve the national screening performance and engagement in the US. The Centers for Medicare & Medicaid Services (CMS) have established criteria for covering new blood-based CRC screening tests if minimum CRC sensitivity and specificity thresholds are met. However, the current blood-based tests are challenged by their relative inability to detect advanced precancerous lesions (APLs). At present, the importance of APL versus CRC sensitivity on screening outcomes remains undetermined. To address this knowledge gap, we conducted a simulation study to compare the screening effectiveness and outcomes of a hypothetical blood-based test across a range of sensitivities for CRC and APL detection. **Methods:** Outcomes were simulated using the CRC-AIM microsimulation model, which has been previously calibrated and validated. Predicted outcomes were calculated for average-risk individuals screened triennially with a blood-based test between ages 45-75 years, assuming perfect adherence. Base-case screening sensitivity and specificity inputs were based on CMS minimum thresholds (74% CRC sensitivity, 90% specificity) with assumed 10% APL sensitivity. Additional scenarios considered an increase in either APL or CRC sensitivity. Outcomes of interest included life years-gained (LYG), CRC incidence, and mortality reductions. **Results:** Compared to the CMS minimum performance (Table), increasing APL sensitivity by 1% (from 10% to 11%) resulted in a similar increase in LYG as compared to increasing CRC sensitivity by 11% (from 74% to 85%). Additionally, increasing APL sensitivity by 1% resulted in greater reductions in CRC cases and deaths (3% decrease) as compared to increasing CRC sensitivity by 11% (0% and 1% reductions, respectively). **Conclusions:** Data from this study show that increased APL sensitivity is a key determinant of screening-related outcomes, such as LYG, derived from a blood-based strategy, and that similar results would require much larger increases in CRC sensitivity. With the current CMS minimum performance threshold, blood-based tests that are designed to have high APL sensitivity can outperform tests with higher CRC sensitivity and little to no APL sensitivity. Research Sponsor: Exact Sciences Corp.

Comparison of CRC screening health outcomes with CMS proposed minimum threshold blood-test versus hypothetical tests with either increased APL and CRC sensitivity.

APL Sensitivity	CRC Sensitivity	CRC Specificity	Follow-up Colonoscopies	CRC Cases (% reduction)	CRC Deaths (% reduction)	Life-Years Gained (LYG) (% increase)
10%	74%	90%	809	50 (ref)	18 (ref)	220 (ref)
11%	74%	90%	813	48 (-3%)	17 (-3%)	226 (3%)
10%	85%	90%	809	50 (0%)	18 (-1%)	227 (3%)

APL: Advanced precancerous lesions; CRC: Colorectal cancer.

90 Poster Session

Birth cohort effects in appendiceal adenocarcinoma incidence across the United States. First Author: Andreana Natalie Holowatyj, Vanderbilt University Medical Center, Nashville, TN

Background: Appendiceal adenocarcinoma incidence rates are increasing across all age groups, sexes and histological subtypes in the United States. Birth cohort patterns of appendiceal adenocarcinomas can provide us with new, etiologic clues into these rising rates but have not been examined for this rare malignant tumor type. We estimated appendiceal adenocarcinoma incidence rates across birth cohorts in the United States. **Methods:** Using the National Cancer Institute's Surveillance, Epidemiology, and End Results (SEER) Program data from eight population-based cancer registries, we identified a total of 4,858 persons aged 20+ years when diagnosed with a pathologically-confirmed primary appendiceal adenocarcinoma (non-mucinous, mucinous, goblet cell, signet ring cell carcinoma) between 1975 and 2019. Birth cohorts (1900-1904 to 1990-1994) were created using five-year age groups and time periods. The ratio of age-specific incidence rates was estimated in each birth cohort relative to the 1945-1949 birth cohort and reported as incidence rate ratios (IRRs) with 95% confidence intervals (CI). **Results:** Compared to persons born in 1945-1949, appendiceal adenocarcinoma incidence rates more than tripled among persons born in 1980-1984 (IRR 3.41, 95%CI 2.54-4.56) and quadrupled among persons born in 1985-1989 (IRR 4.62, 95%CI 3.12-6.82). For all tumor histological types, age-specific appendiceal adenocarcinoma incidence rates increased across successive birth cohorts after 1945-1949—although to varying degrees reported across histological types (e.g., 1980-1984: mucinous adenocarcinoma: IRR 2.71, 95% CI 1.73-4.26; non-mucinous adenocarcinoma: IRR 2.60, 95% CI 1.46-4.63; goblet cell adenocarcinoma: IRR 4.95, 95% CI 2.77-8.85). **Conclusions:** There are strong, yet distinct birth cohort effects for appendiceal adenocarcinomas across histological subtypes that remain unexplained. These patterns signal that recent exposure changes may be influencing this increasing disease risk for Generation X and Millennials now entering mid-adulthood. With the trend toward more non-operative management of appendicitis, another significant consideration for providers is to keep occult appendiceal tumors in the differential diagnosis of patients who present in this way. The current absence of appendiceal cancer prevention and screening modalities emphasizes the importance of efforts to support earlier detection in a rare malignancy where clinical trials have been historically very limited. Moreover, as similar trends have been reported in other gastrointestinal cancers, this is also suggestive of potential shared exposures contributing to this rising cancer burden across generations. Research Sponsor: Appendix Cancer Pseudomyxoma Peritonei (ACPMP) Research Foundation; National Cancer Institute/U.S. National Institutes of Health; P50 CA236733; National Institutes of Health/Eunice Kennedy Shriver National Institute of Child Health and Human Development; K12 HD043483.

91 Poster Session

Impact of COVID-19 on colorectal cancer screening using Medicare Centers for Medicare and Medicaid Services (CMS) claims data. First Author: Soyun Lim, Seoul National University College of Medicine, Seoul, South Korea

Background: When COVID-19 pandemic started in 2020 in the US, the Centers for Medicare and Medicaid Services (CMS) recommended that all non-urgent procedures, including screening colonoscopies, be delayed until pandemic stabilization. This was done to prevent the spread of the virus via colonoscopy since the presence of the virus in fecal samples has been reported in infected patients. Little is known of the impact of COVID-19 on the number of colorectal cancer (CRC) screening performed in the US. **Methods:** We utilized the Centers for Medicare & Medicaid Services Database which provides Medicare claims data for 2013-2022. We searched "Medicare Physician & Other Practitioners - by Geography and Service" using CPT code for colonoscopy(G0121, G0105), fecal occult blood test (FOBT; G0328, 82270), stool-based gene analysis (81528), sigmoidoscopy (G0104), and barium enema (G0120) and collected the number of CRC screening claims at national level and calculated the year-over-year (YOY) change. **Results:** The number of CRC screening claims in the US from 2013 to 2022 is listed (Table). The total number of CRC screening claims increased an average of 0.5% YOY from 2014 to 2019 then decreased 23.4% in 2020. From 2014 to 2019, the average YOY change in colonoscopy, FOBT, stool-based gene analysis, sigmoidoscopy, and barium enema claims were -0.5%, -6.0%, 58.7%, 8.1%, and -6.0%, respectively whereas in 2020, the rates were -27.5%, -25.0%, -14.4%, -25.3%, and -1.7%, respectively. The total number of CRC screening claims in year 2021 and 2022 increased from 2020, however, was still lower than 2019. **Conclusions:** CRC screening claims decreased in the US in 2020 likely due to COVID-19 and not yet returned to pre-pandemic levels. Research Sponsor: None.

Number of CRC screening claims in the US from 2013 to 2022.

Screening test	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Colonoscopy	759,883	736,400	743,873	765,974	767,063	755,806	736,247	533,972	685,439	707,672
FOBT	1,307,054	1,263,602	1,161,214	1,130,130	1,085,777	996,016	897,993	673,614	646,550	556,950
Stool-Based Gene analysis				123,748	231,765	335,459	482,761	413,224	497,748	536,348
Sigmoidoscopy	2,666	2,316	3,254	3,390	3,835	3,784	3,975	2,968	3,893	3,680
Barium enema	107	93	80	116	128	97	58	57	44	65
Total CRC screening	2,069,710	2,002,411	1,908,421	2,023,358	2,088,568	2,091,162	2,121,034	1,623,835	1,833,674	1,804,715

Data are expressed in numbers (N).
CRC; colorectal cancer; FOBT; fecal occult blood test.

92 Poster Session

Gender imbalance in colon cancer screening rates in Texas: A population-based cohort study. First Author: Jorge Rodriguez Vazquez, Texas Tech University Health Sciences Center, Odessa, TX

Background: Males have a higher risk of acquiring Colorectal Cancer (CRC) than females. In Texas, males are diagnosed with CRC more than females, regardless of race or ethnicity. Texas CRC screening compliance rates are lower than the national average of 72.2%. It is important to understand the gender trends in screening to apply more specific strategies. **Methods:** We used the Texas Outpatient Public Use Data File (TOPUDF), a deidentified and publicly available statewide dataset, to conduct a population-based cohort study of outpatient colon cancer screenings of residents age 45 -74 years old at hospitals and ambulatory surgery centers during 2021. Colon cancer screenings were identified using Current Procedural Terminology (CPT) and Healthcare Common Procedure Coding System (HCPCS) codes. We queried the National Cancer Institute Surveillance, Epidemiology and End Results Program (SEER) for age grouped populations in 2021 stratified by gender, race, age group, and county of residence. We calculated age-adjusted outpatient colon cancer screening rates per 100,000 population (AASRs) using the direct standardization method based on the age group weights from the 2000 standard US population. Confidence intervals for AASR were derived by estimating the standard error as the AASR divided by square root of number of colon cancer screenings. The statistical significance of the difference between female and male AASR were tested by comparing the intersection of 95% confidence intervals (95% CI) for the individual rates. Subgroup analyses included age group, rural *vs* urban residence, and race/ethnicity. **Results:** A total 514,601 outpatient colon cancer screenings were included in the study of which 275, 189 (53.5%) were for female residents. During 2021, 51.1% of state residents age 45-74 years old were female. The overall statewide AASR in 2021 was 6071.5 (95% CI 6054.6 – 6088.4)) with the female AASR (6282.1 [95% CI 6258.1 – 6306.0]) significantly higher than the male AASR (5847.7 [95% CI 5823.9 – 5871.5]). For residents aged ≥ 45 years the statewide AASR was 5580.8 (95% CI 5565.5 – 5596) with the female AASR (5735.2 [95% CI 5715.9 – 5754.4]) significantly higher than the male AASR (5429.8 [95% CI 5410.1 – 5449.6]). Female residents had higher AASR amongst White, Black, and Hispanic race/ethnicity groups with the largest difference observed for Blacks; female AASR 5843.9 (5777.7 - 5910.1) *vs* male AASR 4733.5 (4668.5 - 4798.5). The difference between male and female outpatient screening rates was significant among both rural and urban residents. **Conclusions:** Despite a clear relationship between male gender and CRC, females have a higher rate of colon cancer screening. The Black population has the largest gender difference among race/ethnicity groups. Research Sponsor: None.

93 Poster Session

Cost-effectiveness of blood-based colorectal cancer screening: A simulation model incorporating real-world longitudinal adherence. First Author: Elifnur Yay Donderici, Guardant Health, Palo Alto, CA

Background: Colorectal cancer (CRC) is a predominant cause of cancer-related mortality worldwide. Regular screening improves outcomes. However, despite there being many U.S. Preventive Services Task Force (USPSTF) recommended CRC screening options, many people elect not to use any of them. Most cost-effectiveness models assume perfect (100%) adherence initially and over time (longitudinally), which is implausible as approximately one-third of eligible individuals are not up to date with CRC screening. Our study evaluates the cost-effectiveness of Shield, an FDA-approved blood-based CRC test, while incorporating real-world adherence. **Methods:** The CAN-SCREEN (Colorectal cANcer SCReening Economics and adherENce) model is a validated, discrete-event simulation model designed to evaluate the clinical and economic outcomes of CRC screening strategies under real-world adherence scenarios. We simulated individual lifetime outcomes for a cohort of 10,000 people beginning screening at age 45, running 4,000 trials per cohort. We conducted a cost-effectiveness analysis using the CAN-SCREEN model, and compared the Shield blood-based test administered every three years to no screening. Costs were calculated from a healthcare perspective and adjusted to 2023 US dollars. Shield was considered cost-effective if the incremental cost-effectiveness ratio (ICER) was below $100,000 per quality-adjusted life-year (QALY) gained. The analysis also assessed the maximum price Shield could maintain while remaining cost-effective. **Results:** Compared to no screening, Shield increased the number of QALYs by 154 per 1,000 individuals and raised costs by $7.5 million per 1,000 individuals at its commercial test cost. With an incremental cost of $48,662 per QALY gained, Shield was found to be cost-effective relative to no screening at $100,000 cost per QALY willingness-to-pay threshold. Furthermore, the unit cost of Shield can be as high as $3,241 and $4,942 to be considered cost-effective at $100,000 and $150,000 per QALY gained, respectively. **Conclusions:** This study used the CAN-SCREEN model to compare Shield, an FDA-approved blood-based CRC screening test, against no screening, finding that Shield is cost-effective. With about one-third of the U.S. population not up to date on CRC screening, Shield's noninvasive approach offers a promising, cost-effective solution to increase adherence and reduce CRC mortality. Research Sponsor: None.

Economic outcomes of Shield against no screening using CAN-SCREEN model.

Intervention	Test Unit Cost	QALYs/ Person	Cost ($)/Person	Cost ($)/QALY Gained vs No screening
No				**Intervention**
-	15.269	$6,312.18	-	
Shield	$1,495	15.423	$13,786.72	$48,662
Shield	$3,241	15.423	$21,672.34	*$100,000.00*
Shield	$4,942	15.423	$29,352.42	*$150,000.00*

94 Poster Session

Influence of preoperative Lynch syndrome diagnosis on surgery in patients with colorectal cancer. First Author: Mohammad Ali Abbass, Memorial Sloan Kettering Cancer Center, New York, NY

Background: Lynch Syndrome (LS) can guide surgery for colorectal cancer (CRC), particularly for *MLH1/MSH2* carriers, who may benefit from extended procedures: total colectomy (TC) or total proctocolectomy (TP). We investigated timing of germline genetic testing (GGT) and surgical approach in patients (pts) with LS and CRC. **Methods:** Integrated GGT (Labcorp, formerly Invitae Corp.) and insurance claims (Komodo Healthcare Map) data for adult pts with non-metastatic colon (CC) or rectal (RC) cancer and CRC surgery from 2015-23, ≥ 6 months of claims pre-CRC diagnosis, and GGT for *EPCAM, MLH1, MSH2, MSH6,* and *PMS2*. X^2 and t tests and multivariable logistic regression compared GGT results and surgical approach. **Results:** Of 1616 CRC pts (1553 CC, 63 RC), 15% were LS positive (15% CC, 11% RC). Compared to pts with negative GGT, pts with LS were more likely male, Black, younger at diagnosis and GGT, to have family history of GI/any cancer and to have GGT pre-CRC surgery. Of 129 LS pts with CC and *MLH1/MSH2/EPCAM,* 41 (32%) had GGT pre-surgery and 20/41 (49%) had TC/TP (Table). 5/6 (83%) of the corresponding RC pts had GGT pre-surgery and 3/5 (60%) had TP. 88 CC *MLH1/MSH2/EPCAM* pts had GGT post-surgery and 17/88 (19%) had TC/TP. 1 *MLH1/MSH2/EPCAM* RC patient had GGT post-surgery and 0 had TP. More CC pts had TC/TP than RC pts ($p<0.001$). Compared to pts with negative GGT, pts with LS and CRC were more likely to have TC/TP (21% vs. 6%, $p<0.001$), particularly with GGT pre-surgery (43% vs. 13%, $p<0.001$). RC LS pts with GGT pre-surgery had shorter mean months from GGT to surgery than pts with negative GGT (6 vs 14, p=0.026). CC pts with *MLH1* or *MSH2/EPCAM* and GGT pre-surgery had higher odds of undergoing TC/TP (odds ratio (OR): 6, confidence interval (CI): 3-10; OR: 7, CI: 4-14; OR: 4, CI: 2-6) than pts with negative or post-surgery GGT. **Conclusions:** GGT performed pre-surgery for a new diagnosis of CRC was more likely to result in extended procedures, especially in *MLH1/MSH2/EPCAM* carriers. More data in RC is needed to better understand the influence of GGT on surgical approach. Research Sponsor: None.

CC surgery type and timing in relation to GGT.

	GGT pre-surgery (N=295)				GGT post-surgery (N=1258)			
Pts w/ CC surgery	**Total**	**Partial Colectomy** (N=245) N (% total)	**TC/TP** (N=50) N (% total)	**P-value: partial vs TC/TP^**	**Total**	**Partial Colectomy** (N=1203) N (% total)	**TC/TP** (N=55) N (% total)	**P-value: partial vs TC/TP^**
***MLH1* (N=75)**	24 (32)	13 (54)	11 (46)	**0.001**	51 (68)	42 (82)	9 (18)	**<0.001**
***MSH2/ EPCAM* (N=54)**	17 (32)	8 (47)	9 (53)	**0.006**	37 (69)	29 (78)	8 (22)	**<0.001**
***MSH6* (N=49)**	13 (27)	11 (85)	2 (15)	1	36 (74)	35 (97)	1 (3)	1
***PMS2* (N=51)**	6 (12)	3 (50)	3 (50)	0.252	45 (88)	43 (96)	2 (4)	1
LS positive (N=229)	60 (26)	35 (58)	25 (42)	**<0.001**	169 (74)	149 (88)	20 (12)	**<0.001**
LS VUS (N=61)	6 (10)	6 (100)	0 (0)	1	55 (90)	55 (100)	0 (0)	0.526
LS Negative (N=1263)	229 (18)	204 (89)	25 (11)	**<0.001**	1034 (82)	995 (96)	39 (4)	**0.006**

Note: Excludes pts with RC due to small N.
^Adjusted for multiple testing.

95 Poster Session

Comparison of risk factors, treatment strategy, and outcomes of early- vs. older-onset colorectal cancer within the prospective ColoCare study. First Author: Zachary Gottschalk, Fred Hutchinson Cancer Research Center, Seattle, WA

Background: The incidence of colorectal cancer among young adults aged 20-49 has continually increased in recent decades. Early-onset colorectal cancer (EOCRC) patients are more likely to be diagnosed at a more advanced stage of disease, have hereditary cancer syndromes, and have different molecular pathological features of disease than later-onset patients. Herein we compare the risk factors, treatment approaches, and responses to treatment of early-onset versus later-onset colorectal cancer within. **Methods:** The ColoCare Study is a prospective cohort enrolling individuals diagnosed with primary colorectal cancer at six different cancer centers in the United States and one in Germany. Serial questionnaires assess demographic characteristics and symptoms experienced prior to diagnosis, while clinical characteristics (including staging, pathology, treatments given) are abstracted from medical records. Restricting to patients with available age and stage, patients diagnosed <50 years old were labeled as "early-onset" and ≥50yo as "later-onset." Associations between patient age and characteristics of interest were tested using Fisher's Exact Test for count data. **Results:** Of the3,173 patients included in this analysis, 78% were later-onset (mean: 65 years) and 22% were early-onset (mean: 42 years). Demographic data was comparable between the two groups, with 43% being female. Though similar rates of ever alcohol use were seen between the two groups, early-onset patients reported consuming fewer weekly alcoholic beverages than later-onset patients (mean (standard deviation) of 5.5 (6.6) vs. 8.4 (9.3)). Early-onset patients were less likely to have ever smoked tobacco (38% vs 51%, $p<0.001$) and had lower rates of obesity (27% vs 31%) or overweight (34% vs 40%) than later-onset patients ($p<0.001$). For stage II/III disease, early-onset patients were more likely to have received chemotherapy and/or radiation treatment compared to later-onset patients (87% vs 70%, $p<0.001$), though both groups had similar rates of surgery. Despite this, younger patients did not appear to experience better outcomes, with 21% of younger patients experiencing a recurrence or new metastasis during follow-up versus 20% of later-onset patients. **Conclusions:** The risingincidence of colorectal cancer in early-onset patients does not appear to be explained by traditional risk factors such as alcohol use, tobacco use, or high BMI. Early-onset patients are more likely to receive chemotherapy or radiation, but observed outcomes are notably poorer compared to later-onset patients. Further statistical analysis on specifics of treatments by stage is ongoing. Research Sponsor: None.

96 Poster Session

dMMR/MSI tumour occurrence during follow-up of patients with Lynch syndrome treated with immune checkpoint inhibitors for a metastatic digestive cancer between 2015 and 2024: A retrospective monocentric cohort. First Author: Anna Pellat, Gastroenterology and Digestive Oncology Unit, Cochin Hospital, AP-HP, Paris, France

Background: Immune check point inhibitor(s) (ICIs) transformed the prognosis of patients with metastatic dMMR/MSI cancers especially for patients with Lynch syndrome (LS). The aim of our study was to describe a population of LS carrier patients treated with ICI(s) for metastatic digestive cancer, specifically assessing the risk of developing metachronous cancer. **Methods:** We collected data of 93 patients with LS, treated by ICI(s) for a metastatic digestive tumour. These patients were selected from the "immunoMSI" patient cohort, which included all patients presenting with a MSI and/or dMMR digestive cancer treated with ICI(s) between February 2015 and April 2024 in the medical oncology department of Saint-Antoine Hospital (Paris, France). Our primary outcome was the cumulative risk of metachronous MSI/dMMR cancer; secondary outcomes included cumulative risk and time to de novo colorectal polyps, as well as PFS and OS under ICI(s). We performed a cox model to assess the association of predefined variables with OS and PFS under ICI(s). **Results:** Patients most frequently carried a germline pathogenic variant on the *MSH2* (or *EPCAM*) gene (47%) or on the *MLH1* gene (30%). The majority presented with colorectal cancer (CRC) as their primary cancer (83%), followed by small intestine tumours (6%), stomach and pancreatic cancers (4%) and bile duct cancer (1%). Median age at cancer index was 62 years-old (26-77). A total of 12 out of 93 patients presented metachronous cancers during or after treatment with ICI(s). Ten of these metachronous cancers were dMMR or within the LS spectrum (urothelial n=4, CRC n=2, bilio-pancreatic n=1 and 3 skin cancers including 2 keratoacanthomas). All patients who developed a metachronous cancer had a complete or partial response under ICI(s) for the index cancer. During a median follow-up of 47.8 months, 44 patients had colonoscopies and a total of 17 patients (39,5 %) developed pre neoplastic colorectal polyps. In multivariate analysis, PS ≥ 2 and age ≥ 70 years were associated with poorer OS with HRs of 6.3 (95% CI [1.5-25.9], p=0.01) and 5.5 (95% CI [1.6-18.9], p=0.006) respectively. **Conclusions:** Our results show that ICI(s) for patients with LS treated for an MSI/dMMR metastatic digestive cancer can still develop metachronous cancers and pre neoplastic polyps, which underlines the importance of maintaining dedicated follow-up after ICI(s) treatment in this population. Research Sponsor: None.

97 **Poster Session**

Effectiveness of the Colorectal Cancer Alliance's digital screening quiz and navigation model for promoting CRC screening. First Author: Kimberley Lynn Newcomer, Colorectal Cancer Alliance, Washington, DC

Background: Evaluating the effectiveness of a Colorectal Cancer screening navigation model delivered in a digital platform offered by a patient advocacy organization. Designed as a digital screening quiz tool and live navigation to promote CRC screening, the Colorectal Cancer Alliance's Screening Quiz is a risk stratification tool that connects respondents to live certified cancer care navigators delivering evidence-based interventions to increase CRC screening, education and patient and family support. The framework utilizes small and mass media, one-on-one education and patient navigation services to help address and reduce barriers to care. Research on these critical interventions shows that on-time screening for CRC leads to better outcomes. **Methods:** The Screening Quiz has been promoted through social sharing, community outreach, and media, including influencer campaigns to increase outreach to at-risk communities. Questions assess personal and family history, hereditary syndromes, and demographics to provide a CRC screening recommendation based on current USPSTF guidelines. Respondents overdue for screening are triaged to the Alliance's patient navigation team. This interaction includes phone calls and emails for symptomatic individuals/high risk for CRC, telephone and email for average-risk uninsured, and emails for average-risk insured. At-risk individuals are connected to resources, such as financial assistance, free at-home screening tests, and colonoscopy scheduling services. **Results:** July 2022 to July 2024, the Screening Quiz provided over 50,000 screening recommendations. The median age of respondents was 46; 32% of respondents were between the ages of 45 to 64. Of the respondents, 29% were male and 71% were female. Most were White (79.2%), but nearly half of those who opted into navigation identified as BIPOC, including 16.7% Black/AA, 6.6% Asian, and 2.1% AI/AN. Additionally, 8.4% of respondents identified as ethnically Hispanic/Latino with another race. Most respondents were symptomatic (66.6%) but 6.5% were asymptomatic high-risk. These two groups represented 88% of 1,869 respondents triaged to patient navigation. **Conclusions:** The digital Screening Quiz, combined with live navigation, represents a promising approach for increasing CRC screening rates and addressing care gaps compared to traditional methods. The Alliance's ability to engage at-risk communities through its national network of volunteers and partners, combined with evidence-based interventions, positions it well to continue closing screening gaps. Going forward, the Alliance aims to leverage its relationships with healthcare providers and community organizations further to expand the reach of its CRC screening efforts. Research Sponsor: None.

98 **Poster Session**

Trends in utilization of colorectal cancer screening modalities among patients with average risk in the United States from 2017 to 2023. First Author: Brad Stieber, Exact Sciences Corporation, Madison, WI

Background: Colorectal cancer (CRC) is the second leading cause of cancer mortality in the US. The national CRC screening rate (58%) is still below 80%, which is the goal set by the National Colorectal Cancer Roundtable. National guidelines recommend multiple strategies for CRC screening including colonoscopy, sigmoidoscopy, CT colonography, fecal occult blood test (FOBT), fecal immunochemical tests (FIT), or multi-target stool DNA (mt-sDNA) test. To provide updated insights regarding proportional utilization of different CRC screening strategies, we analyzed modality-specific CRC screening rates across demographic subgroups between 2017 and 2023. **Methods:** A retrospective review was conducted using a large national claims database that covers over 165 million lives and is representative of the US population for the study period from 2017 to 2023. Individuals were included if they were aged 45-75 years, at average CRC risk in the baseline period of 1 year before the index date and continuously enrolled in a health plan for at least 1 year after the index date. The index date was defined as date of the first claim for a CRC screening test. CRC screening uptake by modality and newly screened individuals each calendar year were examined among the entire eligible population and across demographic subgroups. **Results:** Overall rates of CRC screening were relatively stable across the study years (approximately 20%) with the lowest annual proportion of screen eligible patients in 2020 (16.6%) and highest in 2023 (24.4%). For CRC-screened individuals, colonoscopy was the most commonly used screening modality, increasing from 1,943,733 (53%) in 2017 to 2,319,695 (58.7%) in 2023. The mt-sDNA test use increased significantly from 87,769 (2.4%) in 2017 to 807,227 (20.4%) in 2023; while FIT/FOBT use declined from 1,611,730 (44%) in 2017 to 805,150 (20.4%) in 2023. Utilization of colonoscopy and mt-sDNA tests increased notably among the 45–49 age group starting in 2021, while FIT/FOBT use simultaneously declined in this age group. Observed trends were similar to the above findings across additional subgroups defined by gender, race/ethnicity, payer type, and geographic region. **Conclusions:** Data from this large study of national claims data provides updated results for overall and proportional utilization of guideline-endorsed CRC screening strategies. While colonoscopy usage remained stable , the use of the non-invasive mt-sDNA test has increased significantly as FIT/FOBT usage declined over time, reflecting growing interest in this noninvasive, home-based option. Research Sponsor: Exact Sciences Corporation.

99 **Poster Session**

Multi-target stool DNA test adherence in average-risk African Americans from 2017-2024. First Author: William K Johnson, Exact Sciences Corporation, Madison, WI

Background: Colorectal cancer (CRC) stands as the second most common cause of cancer-related mortality in the United States, while being the third most common cancer in the country. African Americans (AA) are disproportionately affected with CRC (incidence and mortality), yet the disease can be prevented through screening and early detection. Data reveals that only 59% of AA between ages 45-75 were up to date with CRC screening as of 2021. Broad availability of the navigation-supported, at home multi-target stool DNA (mt-sDNA) test may help mitigate barriers to screening adherence in this population. We examined multi-target stool-DNA (mt-sDNA) test adherence rates in a nationally representative AA population. **Methods:** Using a national claims database of over 165 million individuals linked with Exact Sciences Laboratories data, mt-sDNA orders for AA patients were retrospectively identified from 2016 to 2024. Average CRC risk individuals aged ≥45 years, and new to mt-sDNA, were included in this analysis. Primary outcome of interest was adherence, defined as mt-sDNA kit return rate within 365-days from shipment. Time to test return was examined as the secondary outcome. Demographics were age, sex, ordering provider, residential geography, payor, and outreach preference. Covariates on adherence were examined using logistic regression. **Results:** Among 434,951 mt-sDNA orders from 2017-2024, the overall adherence rate was 62%, and the average time to test return was 27.6 days. The cohort included higher percentages of women (61.7%), ages 50-64 years (57.8%), located in Southern US (51.4%), and metro areas (90.9%). Most patients were also commercially insured (56.1%) and received their mt-sDNA order from their primary care provider (65.4%). Digital SMS was the most preferred outreach option (47%). Patients ages ≥76 years demonstrated the highest adherence (70.3%), with all age categories near or above 60% (45-49 yrs [59.4%]; 50-64 yrs [60.8%], 65-75 yrs [64.3%]). Across payors, patients with Medicare had the highest adherence rates (66.5%). Across outreach, highest adherence was among those who preferred 'Digital SMS + Email' outreach (64.3%). Men were quicker to return their test than women (26.8 days vs 28 days, $p<0.001$). Those ages ≥76 years and with tests ordered by GIs had the shortest time to test return (21.3 and 22.2 days, respectively). Adjusted analyses revealed multiple factors were associated with increased odds of adherence, including older age (≥76 years; OR=1.54; $p<0.001$), 'Digital SMS + Email' outreach (OR=1.25; $p<0.001$), rural residence (OR=1.23; $p<0.001$), and males (OR=1.05; $p<0.001$). **Conclusions:** Mt-sDNA adherence was higher for AA patients in this large, national cohort as compared to previously reported overall CRC screening adherence data for this population group. Findings suggest mt-sDNA utilization, with its included patient navigation, may help reduce screening disparities among AA patients. Research Sponsor: Exact Sciences Corporation.

100 **Poster Session**

Adherence to the multi-target stool DNA test in the US Hispanic population from 2016-2024. First Author: Mark Camardo, Exact Sciences Corporation, Madison, WI

Background: With colorectal cancer (CRC) as the second most common cause of cancer-related mortality in the US, adherence to CRC screening is a vital component to alleviating the cancer burden. Statistics show that only 52% of Hispanic or Latino individuals ≥45 years were up-to-date with CRC screening in 2021. A guideline-recommended, at-home CRC screening option, such as the multi-target stool DNA test (mt-sDNA), could be an efficient modality for screening average-risk individuals. Using a large national claims database, we examined CRC screening adherence with mt-sDNA in the US Hispanic population. **Methods:** Data was sourced from a database of over 165 million individuals linked with Exact Sciences Laboratories data. Hispanic persons who received mt-sDNA orders and resided in the US were retrospectively identified from 2016 to 2024. The analyses included persons who were ≥45 years of age, new to mt-sDNA and average risk for CRC during the 12-month pre-index period. Primary outcome of interest was adherence, which was defined as the mt-sDNA kit return rate within 365-days from shipment. All patients received a mailed letter along with digital outreach through the accompanying patient navigation program, according to their communication preference. The modes of digital outreach included: SMS, email, or both. Baseline demographics collected included age, sex, ordering provider, residential geography, payor, and outreach preference. Adjusted analyses were used to examine effects of covariates on adherence. **Results:** A total of 447,968 mt-sDNA orders were shipped during the study period and the overall adherence rate was 62.5%. Average time to test return was 28.5 days. Women comprised the majority of the cohort (59.9%), resided mostly in the Southern US (46.9%), aged 50-64 (60.9%), resided in a metropolitan area (91.2%), and received their mt-sDNA order from their primary care provider (62.7%). Across evaluated age groups, adherence was above 60% for all: ages 45-49 [60.5%]; 50-64 [62.6%], 65-75 [62.6%]), and ≥ 76 years [66.8]. By payor, Medicare-insured patients demonstrated the highest adherence (65.5%). Digital SMS was the preferred contact method (51.2%), and those opting for SMS + Email had the highest adherence rate (64.2%). Of provider specialties, individuals having their tests ordered by GIs produced the shortest time to adherence (23.2 days). Regression analyses revealed that older age (≥ 76 years), living in a rural area, and receiving fully or partially digital outreach were associated with increased mt-sDNA adherence. **Conclusions:** This large, national analysis reveals high mt-sDNA adherence amongst Hispanic individuals in the US, despite reports of low national CRC screening rates in this population. These data support the important role of mt-sDNA in facilitating screening engagement, while emphasizing the importance of preference-based navigation in facilitating screening test completion. Research Sponsor: Exact Sciences Corporation.

101 Poster Session

Adherence to multi-target stool DNA test in the US Asian population from 2017-2024. First Author: Mallik Greene, Exact Sciences Corporation, Madison, WI

Background: In 2024, incidence and mortality cases related to colorectal cancer (CRC) in the US are estimated to reach 152,810 and 53,010, respectively. CRC screening can help identify malignant lesions at earlier stages; however, in 2021, only 48% of Asian American persons between the ages of 45-75 were up-to-date. Multi-target stool DNA test (mt-sDNA), a highly sensitive and guideline recommended CRC screening modality, may improve screening adherence in average-risk individuals within this population. Herein, we examined CRC screening adherence with mt-sDNA in Asian Americans using national claims data. **Methods:** A large claims database of over 165 million individuals linked with Exact Sciences Laboratories data, was used to source mt-sDNA orders for Asian Americans from 2017 to 2024. Eligible individuals were aged ≥45 years, new to mt-sDNA and average risk for CRC during the baseline period of 12 months. Primary outcome was adherence to mt-sDNA, defined as return of mt-sDNA kit within 365-days from shipment. The secondary outcome was time to kit return in days. All individuals received a standard mailed letter along with additional digital outreach through the accompanying mt-sDNA patient navigation program, according to their desired mode of communication. Baseline demographic measures included age, sex, ordering provider, residential geography, payor, and outreach preference. Association of adherence with patient characteristics was evaluated using multivariable regression. **Results:** Of the 182,259 eligible mt-sDNA orders shipped during the study period, overall adherence was 68.9%. Average time to test return was 27.1 days. The cohort was comprised mostly of women (56.7%), aged 50-64 years (58.42%), primary care provider (74.7%), and preferred 'Digital SMS' outreach (50.9%). Average adherence for all age groups, 45-49 years, 50-64 years, 65-75 years, and 76+ years, exceeded 65%, at 67.2%, 68.8%, 69.7%, and 72%, respectively. Medicare recipients demonstrated the highest adherence by payor type (71%, $p < 0.0001$), and persons who opted for 'Digital SMS + Email' communication were most adherent (70.9%; $p < 0.0001$). Of all provider specialties, individuals having their tests ordered by gastroenterologists produced the shortest time to adherence (21.8 days). Multivariable regression revealed that age, digital outreach (compared to none), micropolitan residence, and ordering provider were independent predictors of increased mt-sDNA adherence. **Conclusions:** The results of this large cohort and nationally representative analysis demonstrated high mt-sDNA adherence amongst Asian American persons, despite a low percentage of this population being current with CRC screening overall. The use of mt-sDNA, in concordance with its included patient navigation, is a viable modality to encourage test completion through screening engagement – which has the potential to reduce screening barriers in this population. Research Sponsor: Exact Sciences Corporation.

102 Poster Session

Multi-target stool DNA test adherence among average-risk 45- to 49-year-old patients from 2017-2023. First Author: Mallik Greene, Exact Sciences Corporation, Madison, WI

Background: Colorectal cancer (CRC) is the third most common cancer in the United States and the second most common cause of cancer-related mortality. In 2021, the US Preventive Services Task Force (USPSTF) lowered the age of initial CRC screening from 50 to 45 years of age for average risk individuals. At that time, only 20% of adults aged 45-49 years were up-to-date with CRC screening. With its at-home use, accessibility, and patient navigation, the multi-target stool DNA test (mt-sDNA) could help support screening uptake in this population. Here, we analyzed national claims data to investigate mt-sDNA adherence rates among average-risk adults aged 45-50 years. **Methods:** Using a national claims database of over 165 million individuals linked with Exact Sciences Laboratories data, mt-sDNA orders for AA patients were retrospectively identified from 2017 to 2023. Average CRC risk individuals aged ≥45 years, and new to mt-sDNA, were included in this analysis. Primary outcome of interest was adherence, defined as mt-sDNA kit return rate within 365-days from shipment. Test return time was examined as the secondary outcome. Demographics were age, sex, ordering provider, residential geography, payor, and outreach preference. Covariates on adherence were examined using logistic regression. **Results:** A total of 607,388 mt-sDNA orders were prescribed during the study period, with an overall 365-day return rate of 65.4%. Average time to test return was 30.4 days. Women outnumbered men in the cohort (58.6%), and most mt-sDNA orders were prescribed by primary care providers (59.4%). The majority of individuals preferred 'Digital SMS' communication (62%), and nearly 90% of the cohort was Commercially insured (87.6%). Among those in the cohort participants' self-reported race/ethnicity was: White (37.3%), Hispanic or Latino (10.1%), Black or African American (8.8%), unknown (8.8%), Asian or pacific islander (4.6%), and other (3.7%). Adherences for the four identified races/ethnicities were all near, or above 60%. Men demonstrated higher adherence than women (66.7% vs 64.5%; $p<0.0001$). Time to test return was shortest in men (28.9 days) and African American persons (28.5 days). Tests ordered by gastroenterologists yielded the quickest adherence times amongst providers (26 days). After controlling for race, gender, payor class, setting (urban/rural classification), and preferred language, digital outreach and provider type were significantly associated with increased mt-sDNA adherence. **Conclusions:** This large, national analysis reports a substantially higher rate of adherence among individuals ages 45-49 with the mt-sDNA test than the current nationally reported rates. At-home testing, along with standard and preference-based navigation may have contributed to high mt-sDNA completion rates. More research is warranted to explore screening behaviors and the benefits of screening within this younger population. Research Sponsor: Exact Sciences Corporation.

103 Poster Session

Prevalence of extracolonic malignancies in patients with familial adenomatous polyposis. First Author: Avina Rami, Harvard Medical School, Boston, MA

Background: Familial adenomatous polyposis (FAP) is an autosomal-dominant inherited colorectal cancer syndrome resulting from mutations in the adenomatous polyposis coli (*APC*) gene. Patients with FAP have nearly 100% lifetime risk of colorectal cancer, typically managed with prophylactic colectomies. As patient survival has improved, the incidence of extracolonic tumors, including thyroid malignancies and desmoid tumors, has increased. However, limited research exists on the prevalence of extracolonic malignancies in FAP patients. **Methods:** We prospectively enrolled 219 patients with FAP in the Hereditary Gastrointestinal Cancers Cohort at the MD Anderson Cancer Center between December 2015 and July 2024. Inclusion criteria included either a clinical or molecular diagnosis of FAP. Data on demographics, cancer history (including cancers diagnosed both prior to and during the study period), BMI, tobacco use, medications, endoscopy results, thyroid ultrasound findings, and abdominal/pelvic CT results were collected. Screening for thyroid and desmoid tumors was provider-dependent, with additional screenings as deemed appropriate. Descriptive and inferential statistics were used to identify associations between demographic and clinical factors and the prevalence of extracolonic malignancies. **Results:** Among 219 patients, 20.5% developed desmoid tumors, 18.7% developed colorectal cancer, and 6.8% developed well-differentiated thyroid cancer (6.4% papillary thyroid cancer, 0.4% follicular thyroid cancer). Prophylactic colorectal surgery was performed in 151 patients (68.9%). Per chart review, benign thyroid disease and hypothyroidism were present in 21% and 11.4% of patients, respectively. Female sex was significantly associated with hypothyroidism ($p=0.007$) and thyroid cancer ($p=0.009$). BMI > 30 was also linked to increased thyroid cancer risk ($p=0.046$). Desmoid tumors were found in 20.5% of patients, most commonly in the abdominal wall. Female sex, BMI > 30, and history of colorectal surgery were significantly associated with desmoid tumor development ($p=0.004$, 0.0064, 0.0038). **Conclusions:** FAP patients exhibit a higher prevalence of benign thyroid disease, thyroid cancer, and desmoid tumors compared to the general population. Given these findings, adherence to existing surveillance guidelines, such as those outlined by the National Comprehensive Cancer Network (NCCN), is essential. Our findings further support the importance of these established surveillance strategies in the management of FAP patients. Research Sponsor: None.

104 Poster Session

Refining colorectal cancer screening strategies using polygenic risk scores and classical risk factors: A proof-of-concept study in the UK Biobank cohort. First Author: Elle Kim, Johns Hopkins University School of Medicine, Baltimore, MD

Background: While the rising incidence of CRC in younger populations has led to a push for lowering the screening age to 40, current clinical guidelines lack sufficient consideration of individual risk variations, raising concerns about the potential harms of over-diagnosis and false positives. Our proof-of-concept study evaluates the utility of risk stratification for multi-target stool DNA (mt-sDNA) testing by integrating polygenic risk scores and traditional clinical risk factors, addressing gaps in current clinical CRC screening guidelines. **Methods:** Our study utilized the UK Biobank prospective cohort, consisting of 311,544 unrelated individuals of White British ancestry, aged 40-69, excluding those with any prior malignant cancer diagnosis. We developed a sex-specific Cox proportional hazards model to estimate the risk of developing CRC, with the primary outcome being the first incidence of CRC. Three models were assessed: a polygenic risk score (PRS)-only model, a classical risk factors (RF)-only model, and a combined (PRS + RF) model. The traditional risk factors considered were BMI, smoking status, pack years of smoking, and family history of bowel cancer in first-degree non-adoptive relatives. Model performance was evaluated using the area under the curve (AUC), hazard ratios (HR), and the positive and negative predictive values of mt-sDNA across various risk profiles. **Results:** The combined model for early CRC screening using mt-sDNA demonstrated superior risk stratification (Table) and improved PPV compared to the risk factors-only model, with enhanced performance even among younger individuals at lower baseline risk. If risk stratification was not used, the average PPV for the test in the general population (ages 40-80) would be 2.13% for males and 1.44% for females. In contrast, individuals in the top 1% of risk would have significantly higher PPVs: 5.62% for males and 3.51% for females. If we limit testing to only the high-risk group, per 100,000 individuals, approximately 3,490 additional cases for males and 2,070 additional cases for females would be detected compared to the average-risk population without stratification. **Conclusions:** The findings suggest that risk stratification may enhance the effectiveness of screening by identifying high-risk individuals who would derive the greatest benefit. In contrast, broad application of the test across the general population may not achieve an optimal risk-benefit ratio, given the relatively low PPV in average-risk individuals. Research Sponsor: None.

Model performance of the PRS-only, RF-only, and PRS + RF (Combined) models for females and males, respectively.

	Hazard Ratio (95% CI)	C-index (SE)	5-Year AUC
Female			
PRS-Only	1.50 (1.37 - 1.64)	0.62 (0.01)	0.61
RF + PRS	1.49 (1.36 - 1.63)	0.62 (0.01)	0.61
RF Only	1.09 (0.99 - 1.20)	0.53 (0.02)	0.51
Male			
PRS-Only	1.49 (1.38 - 1.61)	0.63 (0.01)	0.69
RF + PRS	1.49 (1.38 - 1.61)	0.64 (0.01)	0.71
RF Only	1.23 (1.15 - 1.31)	0.57 (0.01)	0.57

105 Poster Session

The TAB-EOAO study: Trends in use and benefit of adjuvant chemotherapy between early onset and average onset locally advanced colon cancer. First Author: Carolina Bernabe, Montefiore Einstein Comprehensive Cancer Center, Bronx, NY

Background: Despite the declining incidence rates (IR) in average onset colon cancer (AO-CC), IR for early onset colon cancer (EO-CC), defined as patients (pts) younger than 50-years-old at diagnosis, has been increasing. Treatment for high risk locally advanced colon cancer (CC) is surgical resection followed by adjuvant chemotherapy (AC). Data indicates a more permissive chemotherapy use in EO-CC. There is limited data about treatment and survival trends in EO-CC. Thus, we aimed to assess the use and benefit of AC between in EO vs AO CC in locally advanced disease. **Methods:** CC diagnosed between 2000-2020 were identified in the Surveillance, Epidemiology, and End Results Program (SEER) Database. Eligible pts were adults (≥18), Stage II or III at diagnosis. Stage II disease was further stratified as low risk (pT1-3) and high risk (pT4). Logistic regression methods were used to assess Odds Ratio (OR) of AC between EO-CC and AO-CC; Cox models were used to compare overall survival (OS) and Cancer-specific survival (CSS) between both groups during 2000-2010 and 2011-2020. **Results:** Our study included 136,676 pts (AO-CC= 121,606; EO-CC =15,070). The median age for EO-CC was 44 years versus 71 years for AO-CC population. The use of AC increased over time for both the EO-CC (63% vs 66%) and AO-CC (33% vs 40%). Moreover, pts with AO-CC received less AC than EO-CC in the entire study time (see table). In pts with AO-CC, AC improved CSS (HR=0.69 for 2000-2010, HR=0.50 2011-2020). However, among those with EO-CC, AC improved CSS in 2011-2020 (HR=0.76) but did not improve CSS in 2000-2010 (HR=1.02). **Conclusions:** Over the past 20 years EO-CC has been managed aggressively, especially for stage II low risk patients where current guidelines do not support the use of AC. In this real world data set we show that for EO-CC AC did not improve CSS in the decade 2000- 2010 but did improve CSS in the 2011- 2020 period. Further analysis may clarify stage specific benefits of AC. There clearly remains a need for better risk stratification, including the use of ctDNA, to identify the subpopulations of EO-CC patients that will best benefit from AC. Research Sponsor: None.

Multivariable logistic regression model of chemotherapy by timeframe with interaction term: Age* stage.

	2000-2010		2011-2020	
AO-CC vs EO-CC	**OR**	**95% CI**	**OR**	**95% CI**
Stage II low risk	0.23	(0.21, 0.26)	0.28	(0.24, 0.32)
Stage II high risk	0.28	(0.22, 0.36)	0.28	(0.21, 0.36)
Stage III	0.36	(0.32, 0.4)	0.30	(0.26, 0.34)

106 Poster Session

Frequency of surveillance CT scan ordering in patients with stage II/III colorectal cancer. First Author: William Silverstein, University of Toronto, Toronto, ON, Canada

Background: Patients with stage II/III colorectal cancer (CRC) undergo surveillance CT scans to detect early asymptomatic oligometastatic spread, which can be treated with curative intent. However, randomized clinical trials show no difference in 5-year mortality in those that undergo low intensity surveillance (annual) compared a higher intensity strategy (every 6 months). In Ontario, the Program in Evidence-Based Care (PEBC), Ontario Health (Cancer Care Ontario) guidelines recommend surveillance CT scan for patients with stage II/III CRC every 12 months. We aimed to determine the frequency at which clinicians in our Ontario-based hospital system ordered surveillance CT scans for patients with stage II/III CRC. **Methods:** This single centre retrospective cohort study included all patients that underwent surgical resection for CRC from April 1, 2018 – March 31, 2019 at our academic tertiary care centre in Ontario, Canada. We excluded patients that were followed at an outside hospital, had stage IV disease, had disease recurrence, had a concurrent neoplasm, or did not have CRC. The primary outcome was the proportion of surveillance CT scans completed 10+ months after the previous scan. We also described sociodemographic and medical characteristics of included patients. Follow-up was until August 2023. **Results:** We included 200 patients. The median age was 62 years (IQR: 52-70), 48% (N=96) were female, the rectum was the most commonly involved disease site (43%, N=86), 38% (N=76) had stage III disease, and 54% (N=107) received adjuvant chemotherapy. A total of 1286 scans were ordered during the study period. Of these, 67% (N=860) were ordered as surveillance. Of these, 18% (n=155) were completed 10 months or longer from the previous scan. The median time interval between surveillance scans was 6.6 months (IQR: 5.9-7.9). Approximately 20% (n=137 of 630) of surveillance scans ordered by general surgeons were completed 10 months or longer from the previous scan, whereas only 9% (n=18 of 205) of scans ordered by medical oncologists were completed 10 months or longer from the previous scan. Less than 1% of surveillance scans (n=3) identified new metastatic disease. **Conclusions:** The minority of surveillance CT scans ordered in patients with stage II/III CRC in our hospital were timed in concordance with Ontario-based guidelines. Future studies should seek to standardize ordering habits such that they align with best practices. Research Sponsor: Sunnybrook AFP Association through the Innovation Fund of the Alternative Funding Plan from the Academic Health Sciences Centres of Ontario.

107 Poster Session

Care delivery and biomarker testing for people with colorectal cancer receiving 4R care sequence plans. First Author: Aparna Kalyan, Northwestern University, Chicago, IL

Background: The 4R Oncology model has shown improvements in patient knowledge of care and timing/sequence of key care events (1,2) but has not been studied in colorectal cancer specific. 4R is Right Info/Care/Patient/Time. The 4R Care Sequence is a patient-facing care plan describing the course and timing/sequence of care. Here we report the impact of using 4R for patients with colorectal cancer. **Methods:** We tested the impact of 4R care sequence plans on patient reported biomarker testing and care delivery referrals and receipt of services at a colorectal cancer program via patient surveys. 4R Plans were provided to patients from Sept '23 – Aug '24 (4R cohort). Patient-reported data were analyzed to compare the 4R cohort (N = 19) to a control cohort of patients who received care pre-4R, Sept '22 – Aug '23 (N = 60). Used Fisher's exact test. **Results:** Response rates: 4R cohort 79% (19/24), Control cohort 87% (41/47). Comparing the 4R cohort to the control cohort respectively, patients were male (47%, 54%), White (89%, 76%), average age (57, 52), and reported income less than $30,000 (21%, 24%). The 4R cohort significantly improved care delivery in five metrics, and directionally improved in care delivery metrics compared to the control cohort (table). One biomarker testing metric significantly improved and other biomarker testing metrics directionally improved in the 4R cohort vs. the control cohort (Table). **Conclusions:** The use of 4R Care Sequence plans for patients with colorectal cancer facilitated improvements in care delivery and patient 's awareness of biomarker testing. While there were directional improvements in all metrics, there is opportunity and room for additional improvements. The team will continue to work on efforts to improve these metrics and optimize the use of 4R Care Sequence plans for patients with colorectal cancer. 1. Trosman 2021. 2. Liu 2023. Research Sponsor: Pfizer and Association of Cancer Care Centers.

Referral and receipt of services, biomarker testing.

	4R chort N=19	Control cohort N=41	Pvalue
Did your provider recommend you see a nutritionist/dietitian?	89%	20%	**0.0001**
If Yes, did you see a nutritionist/dietitian ?	*84%*	*20%*	***0.001***
Did your provider recommend you see your primary care physician about staying as healthy as possible during care treatment?	95%	29%	**0.0001**
If Yes, did you see your primary care physician?	*95%*	*67%*	***0.04***
Did your provider recommend you get support for transportation, food, housing, insurance, or other needs?	42%	12%	**0.01**
If Yes, did you get support for these needs?	*88%*	*67%*	*0.4*
Did you provider recommend you get help for emotional distress or worry?	53%	46%	0.4
If Yes, did you get help for emotional distress or worry?	*70%*	*47%*	*0.2*
Did your care providers talk to you about biomarker testing after your cancer diagnosis?	53%	37%	0.2
If yes, how clear was it to you why biomarker testing was or was not recommended for you?	*90%*	*73%*	*0.3*
Did you get biomarker testing?	63%	34%	**0.03**
Did your care providers discuss results of biomarker tests with you?	92%	87%	0.6

108 Poster Session

Real-world genomic testing performance in colorectal cancer (CRC): The MultiTEAM Systems Framework Precision Oncology Reflex Testing (TEAMSPORT). First Author: Haoran Li, University of Kansas Cancer Center, Westwood, KS

Background: Genomic testing plays a pivotal role in personalizing treatment for CRC patients. Despite recommendations from NCCN, adoption remains uneven. Research indicates that only 64.7% of patients with metastatic CRC (mCRC) underwent molecular testing nationwide. TEAMSPORT seeks to address this gap by evaluating the feasibility of reflex testing and examining how pathology and oncology teams integrate genomic testing into routine practice. **Methods:** This quasi-experimental study tracks testing over time. Eligible patients were 18 years or older, with a histological diagnosis of adenocarcinoma, who received treatment at the University of Kansas Health System (TUKHS). Exclusion criteria included patients discharged to hospice after diagnosis, those whose last contact with TUKHS occurred within 30 days of diagnosis, and patients who passed away within 30 days of diagnosis. Testing rates were calculated as the number of tests performed divided by the number of testing opportunities, where "testing opportunities" refer to NCCN-recommended tests based on cancer stage. An intervention to increase adherence to NCCN-recommended testing began in December 2022. **Results:** From January 2022 to December 2023, 564 colorectal cancer (CRC) cases were eligible for genomic testing, resulting in 1,525 testing opportunities. In localized CRC, testing rates steadily improved, from 92% in January-June 2022 to 98% by July-December 2023 for patients diagnosed outside of TUKHS. At TUKHS, the rates remained high, reaching 100% in the first two periods and staying over 90% in subsequent periods. Patients with mCRC saw similar trends, particularly in the control arm where rates went from 61% to 71%. Specific molecular NCCN-recommended tests were assessed. At TUKHS, testing rates were high for microsatellite instability (MSI - 98%) which is caused by deficiency of the DNA mismatch repair (dMMR) system. Lower testing rates were observed for other markers like *KRAS* (70%), *BRAF* (67%) and *HER2* (48%). Testing rates were assessed based on primary insurance. At TUKHS, the testing rates for Medicaid patients reached 94%, private insurance patients at 78%, and Tricare/VA at 100%. Outside of TUKHS, Medicaid patients had lower testing rates at 21%, private insurance patients at 74%, and CMS patients at 66%. Time to test by processing lab will be presented at the meeting. **Conclusions:** Despite the high testing rates for MSI/dMMR, the overall genomic testing in CRC remains imperfect, with lower rates for *BRAF* and *HER2*. This discrepancy may be related to the fact that targeted therapies for *BRAF* and *HER2* are not yet part of first-line treatment for CRC. Causes for the discrepancy in testing rates by primary insurance have yet to be determined. Future research should address these gaps to enhance precision oncology in CRC management. Research Sponsor: National Cancer Institute; R01CA274560.

109 Poster Session

Disparities in prophylactic tetracycline use among patients with metastatic colorectal cancer undergoing treatment with EGFR inhibitors. First Author: Zach Rubnitz, Department of Internal Medicine, University of Utah Health, Salt Lake City, UT

Background: Epidermal growth factor receptor (EGFR) inhibitors have been shown to improve survival in metastatic colorectal cancer and are considered a standard of care option for patients with RAS/RAF wild-type disease. However, anti-EGFR agents are associated with an acneiform rash that can lead to decreased quality of life and early discontinuation of treatment. Prophylactic doxycycline has been shown in a randomized clinical trial to decrease any skin toxicity ≥ grade 2 by 50%. In this study we sought to determine the rate of prophylactic tetracycline use in metastatic colorectal cancer patients receiving anti-EGFR therapy in the United States. **Methods:** The Flatiron Health Electronic Health Record-derived, de-identified database was used to analyze the records of patients who were at least 18 years old, diagnosed with metastatic colorectal cancer, and received treatment with an anti-EGFR agent (cetuximab or panitumumab). Patients were stratified by receipt of prophylactic tetracycline versus not. This was defined by an order for a tetracycline 28 days before to 1 day after initiation of an anti-EGFR agent. Demographic and clinical characteristics were compared via multivariable logistic regression. The change in percentage of patients who received a prophylactic tetracycline over time was assessed and was compared to the rate of anti-emetic prescription which served as a control to ensure drug administration data was accurately captured. **Results:** The records of 6,713 patients who met inclusion criteria and received an anti-EGFR agent between 2013 and 2023 were included in our analysis. A logistic regression model assessing the association between patient characteristics and the receipt of a prophylactic tetracycline suggested that older age, race (black/African American or Other), presence of RAS mutation, and higher line of therapy were all significantly associated with not receiving prophylactic tetracyclines. Other patient characteristics including gender, ECOG status, mismatch repair status, and RAF mutation status did not appear to impact the likelihood of receiving a prophylactic tetracycline. The rate of prophylactic tetracycline use increased from 10.9% in 2013 to 22.3% in 2023 while the rate of prophylactic anti-emetic remained stable at around 82%. **Conclusions:** This real-world analysis suggests the use of preemptive treatment for skin toxicity in patients with metastatic colorectal cancer who are treated with an anti-EGFR in US is suboptimal. Disparities in cancer care are also evidenced by lower use of tetracyclines in elderly and African Americans. Randomized controlled trials support the use of prophylactic tetracycline in these patients; thus, further research into the reason for this practice gap is necessary. Educational efforts should be undertaken to improve tetracycline prescribing rates in this population. Research Sponsor: University of Utah Study Design and Biostatistics Center, with funding in part from the National Center for Research Resources and the National Center for Advancing Translational Sciences, National Institutes of Health; UM1TR004409 (formerly UL1TR002538).

110 Poster Session

Increasing RAS mutation result availability in newly diagnosed patients with metastatic colorectal cancer by their first outpatient medical oncology clinic visit at a large safety net hospital. First Author: Viral M Patel, UT Southwestern Medical Center, Dallas, TX

Background: Rat sarcoma virus mutation testing (RASmt) is critical in guiding systemic therapy selection for patients with metastatic colorectal cancer (mCRC) in the first-line setting. Missing results at the initial medical oncology visit can negatively impact patient outcomes by delaying treatment or leading to inappropriate treatment selection. This quality improvement (QI) project aims to increase the percentage of patients with newly diagnosed mCRC with RASmt results available by their first outpatient medical oncology appointment at a large safety net hospital to 20% over an 8-month period. **Methods:** A multidisciplinary team of key stakeholders including clinicians, a pathologist, and administrators was assembled to determine factors associated with failure to obtain RASmt. A baseline analysis of all patients referred to the gastrointestinal (GI) oncology clinic from June 2022 to April 2023, with manual review of electronic medical records by investigators, showed that only 6% (3/46) of newly diagnosed patients with mCRC had RASmt results available by their first GI oncology appointment. Two Plan-Do-Study-Act (PDSA) cycles were conducted to assess interventions that were implemented based on process mapping and reasons for unavailable test results. To increase the percentage of patients with RASmt results available, starting December 2023, patients initially diagnosed with mCRC as inpatients who were being followed by the oncology consult service had RASmt ordered on the tissue biopsy prior to discharge (PDSA1). Starting April 2024, the medical directorfor outpatient GI oncology clinics ordered RASmt for all patients with mCRC referred to the clinic at the point of referral (PDSA2). New patient referrals were reviewed through July 3, 2024. **Results:** At baseline, the most common reason for unavailable RASmt results was the lack of test orders, with 70% (32/46) of new patients with mCRC not having RASmt ordered prior to their first appointment. Once RASmt was ordered, the average time for results was 13 days. After the first intervention, 4% (1/23) of patients with mCRC had RASmt results available at their first appointment. After the second intervention, 42% (5/12) of patients had RASmt results available. Additionally, median time from diagnosis of mCRC to first appointment decreased from 23 days at baseline to 17 days during PDSA1 to 12 days during PDSA2. **Conclusions:** By ordering RASmt at the point of referral, the percentage of patients who had RASmt available at their first appointment markedly increased from 6% to 42%. This QI initiative additionally demonstrates the importance of multidisciplinary teams to enhance patient care efficiency at a large safety net hospital. Further efforts are ongoing to increase RASmt availability beyond 42% with a focus on reducing the turnaround time of the test. Research Sponsor: None.

111 Poster Session

Impact of unplanned readmission to the same facility within 30 days following robot-assisted and laparoscopic hemicolectomy for early-onset non-metastatic colon cancer. First Author: Rahul Thusay, Creighton University School of Medicine, Omaha, NE

Background: Colon cancer in younger populations has risen in recent years, prompting changes in screening guidelines. Hemicolectomy remains integral for standard treatment for non-metastatic early-onset colon cancer, and many institutions have adopted robot-assisted approaches for this procedure. However, high costs associated with this technology along with mixed data regarding survival benefits and post-surgical complications warrant further investigations on the specific advantages of robot-assisted surgery. This study aims to examine the impact of unplanned readmission to the same facility within 30 days following surgery for patients receiving robot-assisted and laparoscopic hemicolectomy on long-term survival for early-onset non-metastatic colon cancer to see how current paradigms of colon cancer treatment extend to early-onset patients. **Methods:** The National Cancer Database was used to identify patients under the age of 50 receiving a minimally invasive hemicolectomy for primary non-metastatic colon cancer diagnosed between 2016-2020. Patients were stratified based on surgical approach and whether they underwent an unplanned readmission to the same facility within 30 days following surgery, and survival differences were calculated log-rank test and Cox regression analysis that included co-variates related to tumor characteristics, patient demographics, and facility characteristics. **Results:** 5,076 patients were identified, with 577 receiving a robot-assisted approach and 4,499 receiving a laparoscopic approach. Multivariate analysis found no significant difference in long-term survival between patients receiving a laparoscopic versus robot-assisted approach (HR: 0.893, 95% CI: 0.687-1.161, p=0.398). Additionally, multivariate analysis found that unplanned readmission within 30 days did not have a significant impact on long-term survival outcomes for patients receiving a robot-assisted approach (HR: 0.856, 95% CI: 0.257-2.848, p=0.800), but it did find worsened survival outcomes for the laparoscopic approach (HR: 1.535, 95% CI: 1.070-2.203, p=0.020). **Conclusions:** Patients receiving laparoscopic hemicolectomy for early-onset non-metastatic colon cancer appear to have worsened overall long-term survival outcomes when undergoing an unplanned readmission to the same facility within 30 days of surgery when compared to patients receiving a robot-assisted surgical approach. However, it also appears that overall, both surgical approaches have similar overall long-term survival. Further research on how robot-assisted surgery impacts survival and post-surgical complications in patients with early-onset non-metastatic colon cancer is warranted to examine how this advancing technology could improve cancer care. Research Sponsor: None.

112 Poster Session

Nationwide analysis of Medicare expenditure and utilization of EGFR inhibitors from 2018-2022. First Author: Devang Namjoshi, Saint Vincent Hospital, Worcester, MA

Background: EGFR inhibitors are used extensively for treatment of metastatic colorectal cancer ranging from first line drugs in combination with chemotherapy to monotherapy for refractory disease. As we increasingly move towards targeted treatments for cancer, financial implications of these drugs must be taken into consideration. In this study we aim to analyze changes in Medicare spending for Cetuximab and Panitumumab to better understand the financial impact of these therapies. **Methods:** We obtained data from the publicly available Medicare Part D database from Centers for Medicare and Medicaid Services. Data on total spending, total claims, total beneficiaries, average spending per claim, and average spending per beneficiary was extracted from the database. We adjusted the data for 2018 for 2022 using a standard annual inflation rate. Percentage change was calculated using the adjusted data. **Results:** For Cetuximab, we saw a sharp increase in total spending by 38.77%, from $4,912,903.87 in 2018 to $6,811,781.05 in 2022. The number of claims rose by 31.96%, while the total number of beneficiaries saw a modest increase of 2.44%. Average spending per claim increased by 16.04%, and average spending per beneficiary grew by 49.44%. Panitumumab showed a similar rise in total spending by 43.23%, from $1,594,347.66 in 2018 to $2,283,142.83 in 2022. Claims increased by 25.42%, and beneficiaries grew by 47.83%. Average spending per claim increased by 25.97%, and average spending per beneficiary saw a substantial rise of 65.75%. **Conclusions:** Our results indicate a significant increase in the financial burden related to EGFR inhibitors. As we move away from chemotherapy and shift our focus to targeted therapies, it is imperative to explore generic alternatives and direct policy changes to ensure equitable access across various racial and socio-economic groups. These findings highlight the need for further research into healthcare policy reforms to ease the burden on our health system. Research Sponsor: None.

Drug	Year	Total Spending ($)	Total Claims	Total Beneficiaries	Avg Spending/ Claim ($)	Avg Spending/ Beneficiary ($)
Cetuximab	2018	4,912,903.87	551	123	8,072.80	36,163.51
	2022	6,811,781.05	727	126	9,369.71	54,061.75
	% Change	0.3877	0.3196	0.0244	0.1604	0.4944
Panitumumab	2018	1,594,347.66	5,955	184	242.72	30,714.91
	2022	2,283,142.83	7,470	272	305.81	50,736.51
	% Change	0.4323	0.2542	0.4783	0.2597	0.6575

113 Poster Session

Integrating pharmacogenetic testing in the management of gastrointestinal cancers. First Author: Darshana Patil, Datar Cancer Genetics, Nasik, India

Background: Gastrointestinal cancers are among the most prevalent malignancies, often treated with chemotherapeutic agents whose efficacy and toxicity can be influenced by genetic variations. Pharmacogenetic testing for key metabolic genes, including DPYD, UGT1A1, and G6PD, can guide personalized treatment approaches. **Methods:** We analyzed samples from 3,519 individuals for genotyping using semiconductor-based Next-Generation Sequencing (NGS) technology. High-quality genomic DNA was extracted and subjected to target enrichment via high multiplex PCR amplification using an NGS panel targeting variants of key metabolic genes, including DPYD, G6PD, and UGT1A1. **Results:** For the DPYD gene, 96% of patients were normal metabolizers. However, 4% intermediate and 0.17% individuals had poor metabolizer status. Poor metabolizers possess homozygous non-functional alleles, leading to complete dihydropyrimidine dehydrogenase (DPD) deficiency and increased toxicity risk with fluoropyrimidine drugs like 5-fluorouracil (5-FU), Capecitabine, and Tegafur. Patients with certain homozygous or compound heterozygous variants in the DPYD gene are at increased risk for acute early-onset toxicity and serious, including fatal, adverse reactions due to fluorouracil. 5-FU is not recommended for use in patients known to have certain homozygous or compound heterozygous DPYD variants that result in complete absence of DPD activity. Intermediate metabolizers exhibit approximately 50% reduced DPD activity, necessitating a 50% reduction in starting doses followed by careful titration based on toxicity. UGT1A1 analysis revealed 46% normal, 42% intermediate, and 12% poor metabolizers, affecting Irinotecan metabolism and toxicity risks. Poor metabolizers show decreased UGT1A1 activity, resulting in reduced clearance of Irinotecan and increased risk of dose-limiting toxicities. These patients are also at higher risk for neutropenia, diarrhea, and asthenia, requiring starting dose adjustments. Additionally, they may experience elevated hyperbilirubinemia with treatments like Regorafenib. Recognizing G6PD deficiency in GI cancers can prevent serious complications and ensure safer treatment options. G6PD testing indicated that 98% were homozygous wildtype, 2% heterozygous, and 0.31% homozygous/hemizygous deficient. Patients with homozygous/hemizygous deficient genotype may be at increased risk for drug-induced hemolysis, particularly when treated with Dabrafenib. **Conclusions:** These findings highlight the therapeutic significance of incorporating genetic testing into clinical practice to tailor treatments, enhance drug efficacy, and minimize side effects in GI cancer management. By integrating pharmacogenetic insights, the safety and effectiveness of therapies can be significantly improved, ultimately leading to better patient care in GI oncology. Research Sponsor: None.

114 Poster Session

Genetic variants and clinical factors affecting the response to 5-fluorouracil–based treatment in Chilean patients with advanced colorectal cancer. First Author: Leslie Carol Cerpa-Castro, University of Chile, Santiago, Chile

Background: Colorectal cancer (CRC) ranks as the second most common cancer in Chile, affecting both men and women. Late diagnosis results in approximately 25% of patients presenting with metastatic disease, with a five-year survival rate of about 14%. Standard treatment includes tumor resection followed by adjuvant chemotherapy, commonly involving 5-fluorouracil (5-FU) combined with drugs like oxaliplatin or irinotecan. However, patient responses to 5-FU vary greatly due to genetic polymorphisms in enzymes like thymidylate synthase (TYMS) and dihydropyrimidine dehydrogenase (DPD), which affect the drug's pharmacokinetics and pharmacodynamics. Polymorphisms in genes related to oxaliplatin elimination and efficacy, such as glutathione-S-transferases (GSTs) and DNA repair enzymes, also play a role. However, the relationship between genetic variants and treatment outcomes remains unclear, often varying by population and cancer stage. Although predictive models for treatment response and risk exist for various diseases, no comprehensive model combining genetic and clinical variables to predict chemotherapy safety in Chilean CRC patients has been developed. **Objective:** This study aimed to identify relevant genetic variants in the genes *TYMS*, *TYMP*, *DPYD*, *GSTP1*, *MTHFR*,*ERCC2*, *ABCB1*, *ABCC2*, *ABCC4*, and *ABCG2*, which, along with clinical variables, could help create a predictive model for the safety of 5-FU-based chemotherapy in advanced CRC patients. **Methods:** A retrospective nested case-control study was conducted on 82 advanced CRC patients. Sixteen genetic variants were analyzed to assess their influence on adverse reactions (ADRs) and their severity, using logistic regression to identify potential associations. Multivariate models were developed to predict chemotherapy safety. **Results:** Among the 16 variants analyzed in 82 patients, several key findings emerged: The G allele of *GSTP1* (rs1695) was found to be protective against neuropathy (OR = 0.147; p = 0.012) but increased the risk of mucositis (OR = 2.27; p = 0.036). The C allele of *DPYD* (rs1801265) was associated with an elevated risk of neuropathy (OR = 4.50; p = 0.05). The *TYMS* deletion (rs151264360) provided protection against cutaneous (OR = 0.029; $p < 0.0001$) and hematological ADRs (OR = 0.098; p = 0.005). Additionally, *TYMS* deletion was protective against severe toxic ADRs (OR = 0.098; p = 0.005). Two multivariate models were created to predict anemia (p = 0.027) and pain (p = 0.01) development. **Conclusions:** This study represents an initial step toward developing predictive models for ADRs related to 5-FU, such as neuropathy, mucositis, and hematological and skin toxicities. In the future, these findings may contribute to pharmacogenetic-based dose adjustment models aimed at reducing ADRs in Chilean CRC patients undergoing 5-FU-based treatment. Research Sponsor: FONDECYT Regular Project N°1211948.

115 Poster Session

A protocol for pretreatment testing for antibodies to galactose-alpha-1,3-galactose to mitigate the risk of cetuximab hypersensitivity reactions. First Author: Nicholas Della Marta, Northern Sydney Cancer Centre, Royal North Shore Hospital, St Leonards, NSW, Australia

Background: Cetuximab has been shown to improve survival in patients with KRAS wildtype metastatic colorectal cancer. However, high rates of hypersensitivity reactions (HSRs) limit its use, with HSR rates up to 10-20%. A major driver of cetuximab HSR is from pre-formed antibody response to galactose-1,3-alpha-galactose (alpha-gal). Evidence from retrospective studies supports alpha-gal pre-screening in this setting. We are the first to report on the impact of prospective alpha-gal antibody screening on cetuximab HSR. **Methods:** Records were reviewed across three medical oncology centres that have adopted alpha-gal antibody screening measures. Data for patients with metastatic colorectal cancer treated with cetuximab were retrieved. All centres assessed alpha-gal antibody status using the ImmunoCAP ELISA assay. Due to lack of a shared screening approach across study sites, a pre-determined protocol was retrospectively applied to all cases. This protocol allowed cetuximab administration if alpha-gal levels were $\leq$0.1 kUA/L, but prohibited it if alpha-gal levels were $>$0.1 kUA/L in favour of panitumumab administration. Patients were allocated to either the Protocol Applied or Protocol Not Applied cohorts based on protocol requirements being met. The primary outcome between these patient groups was the incidence of cetuximab HSRs. **Results:** Of 254 assessable patients, 39 underwent the pre-treatment screening protocol. Of the Protocol Applied group, 3% (n=1/38) experienced a cetuximab HSR compared to 16% (n=35/215) in the Protocol Not Applied group (Odds ratio (OR) 7.16; 95% CI 1.13–299.61, p=0.02). Patients with alpha-gal antibody titres $>$0.1 kUA/L were more likely to experience a cetuximab HSR (OR 69.71; 95% CI 5.18–4296.81, p=0.0001). **Conclusions:** Pre-treatment screening for alpha-gal antibodies significantly reduces the incidence of cetuximab HSRs. A testing threshold of 0.1 kUA/L is effective in identifying patients at risk. Implementing this protocol can improve the safety of cetuximab therapy in high-risk populations. Research Sponsor: None.

116 Poster Session

Sex differences in toxicities and survival outcomes among patients with stage III colorectal cancer receiving adjuvant fluoropyrimidine monotherapy: A pooled analysis of four randomized controlled trials (JCOG2310A). First Author: Hidekazu Hirano, Department of Gastrointestinal Medical Oncology, National Cancer Center Hospital, Tokyo, Japan

Background: Fluoropyrimidine agents are the key component of adjuvant chemotherapy for pathological stage III colorectal cancer (CRC). Western studies have shown that female sex is associated with favorable survival outcomes after surgery, yet a risk factor for fluoropyrimidine-related adverse events (AEs) during adjuvant chemotherapy. However, it remains unclear whether there are sex differences in the treatment outcomes in this setting among the Japanese population. **Methods:** Patients enrolled in four randomized controlled trials (JCOG0205, JCOG0404, JCOG0910, and JCOG1006) who received adjuvant fluoropyrimidine monotherapy for pathological stage III CRC were analyzed. Incidences of AEs and survival outcomes (relapse-free survival [RFS], and overall survival [OS]) were compared between males and females. Multivariable logistic regression analyses were performed to evaluate associations between sex and the development of AEs. Multivariable Cox regression analyses were performed using Cox proportional hazard models to evaluate associations between sex and survival outcomes. **Results:** A total of 3170 patients (male, n = 1654; female, n = 1516) were included in the analysis. Compared with male patients, female patients were more likely to require dose reduction (male, 20.4%; female: 28.8%, $p < 0.0001$) and had low completion proportions of adjuvant chemotherapy (male, 81.7%; female, 77.0%, p = 0.0011). Female patients experienced higher incidences of any-grade hematological and non-hematological AEs (Table). Female sex was associated with higher incidences of grade 3/4 hematological AEs (odds ratio [OR]: 1.702 [95% CI 1.272–2.278], p = 0.0003) and non-hematological AEs (OR: 1.666 [95% CI 1.373–2.020], $p < 0.0001$). Female sex was identified as a favorable prognostic factor for RFS (hazard ratio [HR]: 0.729 [0.627–0.848], $p < 0.0001$) and OS (HR: 0.796 [0.653–0.970], p = 0.0238). **Conclusions:** Sex differences had implications for the development of AEs and survival outcomes of Japanese patients with pathological stage III CRC who received adjuvant fluoropyrimidine monotherapy. Research Sponsor: National Cancer Center; 2023-J-03.

Incidences of adverse events.

Adverse event	Any grade			Grade 3/4		
	Male % (95% CI)	Female % (95% CI)	p-value	Male % (95% CI)	Female % (95% CI)	p-value
Any hematological toxicity	63.7 (61.3–66.0)	83.9 (82.0–85.8)	$<$0.0001	5.0 (4.0–6.1)	8.9 (7.5–10.4)	$<$0.0001
Any non-hematological toxicity	93.3 (92.0–94.5)	96.1 (95.0–97.0)	0.0006	13.3 (11.7–15.1)	20.8 (18.8–22.9)	$<$0.0001

117 Poster Session

Meta-analysis of phase II/III randomized controlled trials (RCTs) to evaluate the incidence of hand-foot skin reaction (HFSR) or palmar-plantar erythrodysesthesia (PPE) syndrome in patients with gastrointestinal (GI) cancers treated with fruquintinib. First Author: Rishi Kumar Nanda, Touro University Nevada College of Osteopathic Medicine, Las Vegas, NV

Background: Fruquintinib is a selective inhibitor of vascular endothelial growth factor receptors 1, 2, and 3 blocks the angiogenesis associated with tumor proliferation. Recently, the US FDA approved fruquintinib as a third-line treatment for metastatic colorectal cancer (mCRC). As with any other novel cancer therapies, adverse events are always a concern and all patients should be monitored in order to early detect and appropriately treat any adverse events. This meta-analysis aims to assess the risk of HFSR or PPE syndrome in patients with GI cancers treated with fruquintinib. **Methods:** We conducted a comprehensive literature search using MEDLINE, EMBASE, and COCHRANE databases from inception through August 12th, 2024. Phase II/III RCTs utilizing fruquintinib in GI cancers mentioning HFSR or PPE as an adverse effect were included. Mantel-Haenszel method was used to calculate the estimated pooled risk ratio (RR) with 95% confidence interval (CI). Heterogeneity was assessed with Cochran's Q-statistic. Random effects model was applied. **Results:** A total of 1872 patients, of whom 1131 (60%) received fruquintinib, from 3 phase III RCTs (FRESCO, FRESCO-2, FRUTIGA) and 1 phase II RCT were eligible for evaluation of HFSR or PPE incidence. FRESCO, FRESCO-2, and phase II trials involved patients with mCRC, they compared fruquintinib vs placebo. FRUTIGA trial involved patients with gastric or gastroesophageal junction adenocarcinoma, it compared fruquintinib + paclitaxel vs placebo + paclitaxel. The incidence of any-grade and high-grade HFSR or PPE was higher in the fruquintinib group compared to the placebo group, 27.93% vs 3.64% (RR, 7.64; 95% CI: 4.08-14.33; P<0.00001), and 8.5% vs 0% (RR, 26.0; 95% CI: 6.42-105.29; P<0.00001), respectively. In the mCRC subgroup analysis, the higher incidence of any-grade and high-grade HFSR or PPE in the fruquintinib arm was also statistically significant, 29.70% vs 2.55% (RR, 10.27; 95% CI: 5.59-18.88; P<0.00001), and 8.45% vs 0% (RR, 19.34; 95% CI: 3.84-97.36; P=0.0003), respectively. **Conclusions:** This meta-analysis revealed increased incidence of any-grade and high-grade HFSR or PPE in patients with GI cancers including mCRC treated with fruquintinib. In the FRESCO trial, this association was also shown to confer survival benefit in Chinese patients with mCRC. Prompt identification and providing proper supportive care is crucial in managing this adverse event and ultimately, preserving the patients' quality of life. Research Sponsor: None.

118 Poster Session

Meta-analysis of randomized controlled trials (RCTs) to evaluate the incidence of hemorrhage and venous thromboembolism (VTE) events in patients with gastrointestinal (GI) cancers treated with fruquintinib. First Author: Daniel Thomas Jones, Touro University Nevada College of Osteopathic Medicine, Las Vegas, NV

Background: Fruquintinib is a novel cancer therapy that was recently approved by the US FDA as a third-line treatment for metastatic colorectal cancer (mCRC). It works by inhibiting the vascular endothelial growth factor receptors, that leads to blocking of angiogenesis which is essential for the tumor proliferation. Close monitoring is always crucial in order to promptly detect and manage any adverse events. This study aims to assess the risk of hemorrhagic and VTE events in patients with GI cancers treated with fruquintinib. **Methods:** A comprehensive literature search was conducted using MEDLINE, EMBASE, and COCHRANE databases from inception through August 12th, 2024. Phase II/III RCTs utilizing fruquintinib in GI cancers reporting hemorrhage or VTE events as an adverse event were included. Mantel-Haenszel method was used to calculate the estimated pooled risk ratio (RR) with 95% confidence interval (CI). Heterogeneity was assessed with Cochran's Q-statistic. Random effects model was applied. **Results:** A total of 1872 patients were included. Data were pooled from three phase III RCTs (FRESCO, FRESCO-2, FRUTIGA) and one phase II RCT, all reporting VTE events including deep vein thrombosis and pulmonary embolism and hemorrhage. FRESCO, FRESCO-2, and the phase II trials involved patients with mCRC, comparing fruquintinib to placebo. FRUTIGA trial incorporated patients with gastric or gastroesophageal junction adenocarcinoma, comparing fruquintinib + paclitaxel to placebo + paclitaxel. High-grade hemorrhage incidence was higher in the fruquintinib arm compared to the control arm (29.00% vs 21.18%, RR 1.62; 95% CI: 1.26-2.08, P=0.0001). In the mCRC subgroup, high-grade hemorrhage rate was also higher and statistically significant in the fruquintinib arm 20.87% vs 11.22% (RR 1.84; 95% CI: 1.25-2.70; P=0.002). Incidence of any grade or high grade VTE events was neither statistically significant in GI cancer population or mCRC subgroup. Incidence of any grade VTE events in GI cancer group and mCRC subgroup was 2.74% vs 2.15% (RR, 1.23; 95% CI: 0.53-2.88; P=0.63) and 2.68% vs 1.53% (RR, 0.95; 95% CI: 0.09-9.67; P=0.97), respectively. Rate of high grade VTE events in GI cancer cohort was 1.67% vs 0.67% (RR, 1.96; 95% CI: 0.52-7.36; P=0.32), while in mCRC subgroup was 1.79% vs 0.76% (RR, 1.08; 95% CI: 0.06-19.98; P=0.96). **Conclusions:** This meta-analysis demonstrates that patients with GI cancers, including mCRC, treated with fruquintinib have higher incidence of high-grade hemorrhage. These findings underscore the importance of individualized risk assessment, especially for patients predisposed to hemorrhagic complications. No increased risk of VTE events was noted in patients treated with fruquintinib compared to placebo. Research Sponsor: None.

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Meta-analysis of phase II/III randomized controlled trials (RCTs) to determine the incidence of hypertension and proteinuria in patients with gastrointestinal (GI) cancers treated with fruquintinib. First Author: Ramaditya Srinivasmurthy, Touro University Nevada College of Osteopathic Medicine, Las Vegas, NV

Background: Fruquintinib is a selective inhibitor of vascular endothelial growth factor receptors which impedes angiogenesis associated with tumor growth. On November 8th, 2023, the US FDA approved fruquintinib for use as a third-line treatment for metastatic colorectal cancer (mCRC). With the introduction of novel cancer therapies, monitoring for adverse events is critical. This meta-analysis aims to assess the risk of hypertension (HTN) and proteinuria in patients with GI cancers treated with fruquintinib. **Methods:** We conducted a comprehensive search of MEDLINE, EMBASE, and COCHRANE databases from inception through August 12th, 2024, identifying Phase II/III RCTs that employed fruquintinib in GI cancers and reported HTN and proteinuria as adverse effects. The Mantel-Haenszel method was used to calculate pooled risk ratios (RR) with 95% confidence intervals (CI). Heterogeneity was assessed using Cochran's Q-statistic, and a random effects model was applied. **Results:** Our analysis included 1872 patients, 1131 (60%) of whom received fruquintinib, from three Phase III RCTs (FRESCO, FRESCO-2, FRUTIGA) and one Phase II RCT. FRESCO, FRESCO-2, and the phase II trials compared fruquintinib to placebo in patients with mCRC. FRUTIGA trial involved patients with gastric or gastroesophageal junction adenocarcinoma, comparing fruquintinib + paclitaxel to placebo + paclitaxel. The incidence of any-grade HTN was higher in the fruquintinib group, 34.92% vs 7.55% (RR 3.96; 95% CI: 3.05-5.13; P<0.00001); the rate of high-grade HTN was 13.96% vs 1.21% (RR 9.01; 95% CI: 4.67-17.40; P<0.00001). In the mCRC subgroup, incidence of any-grade and high-grade HTN was 43.02% vs 10.45% (RR 3.97; 95% CI: 2.95-5.33; P<0.00001), and 17.28% vs. 1.27% (RR 12.06; 95% CI: 5.19-28.01; P<0.00001), respectively. Incidence of any-grade proteinuria was also found to be statistically significant and higher in the fruquintinib arm in both GI cancer cohort and mCRC subgroup: 27.3% vs 16.19% (RR 1.89; 95% CI: 1.30-2.74; P=0.0008), and 25.22% vs 11.73% (RR 2.29; 95% CI: 1.17-4.51; P=0.02). The rate of high-grade proteinuria was not statistically significant in both cohorts. In GI cancer group, 1.85% vs 0.53% (RR, 2.49; 95% CI: 0.86-7.16; P=0.09), and mCRC subgroup 2.17% vs 0.51% (RR, 2.87; 95% CI: 0.74-11.12; P=0.13). **Conclusions:** This meta-analysis revealed a higher incidence of any-grade and high-grade HTN with fruquintinib in GI cancer patients, with significant association in the mCRC subgroup. Fruquintinib was also noted to be associated with increased risk of any-grade proteinuria, however, incidence of high-grade proteinuria was not different in patients treated with fruquintinib compared to placebo. Prompt identification and early management of these adverse events are crucial. Research Sponsor: None.

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Quality of life following complete clinical response to chemoradiotherapy for rectal cancer: Patient-reported outcomes. First Author: Maeve O'Neill, St James's Hospital, Dublin, Ireland

Background: Complete clinical response (cCR) after neoadjuvant therapy (chemoradiotherapy/total neoadjuvant treatment) for rectal cancer may permit an organ-preserving approach with "watch and wait" (W&W) surveillance. This confers comparable survival, while avoiding the sequelae of pelvic surgery. Data on quality of life (QOL) for this cohort are lacking. **Methods:** Patient-reported outcomes were collated for patients on W&W following cCR in a single hospital network. Validated questionnaires evaluating surgical, radiation and chemotherapy-related QOL were completed, including Low Anterior Resection Syndrome (LARS) score, Functional Assessment of Chemotherapy: Colorectal (FACT-C), EORTC QLQ-CIPN20, EQ-5D-5L and a linear analogue and ordinal scale ranking self-perception of health. **Results:** From 2016-2023, 76 patients were enrolled on W&W. Median follow-up is 37 months (IQR20-63). 12 patients (15.8%) had local regrowth, of whom 11 (92%) had salvage surgery. Five patients (6.5%) developed metastases at median 8 months (IQR3-12) from W&W entry. Of 57 patients (75%) with sustained cCR on surveillance, 44 (77%) responded to structured questionnaires. Two with stomas were excluded from LARS. 74% of respondents had no LARS, 9% minor LARS and 17% major LARS. 43% reported faecal urgency and 33% faecal incontinence. 41% experienced functional impairment due to chronic chemotherapy-related side effects (Table). Median EORTC CIPN20 score was 23 (IQR20-28; possible range 19-76, high score = poor QOL). Median sensory score was 11 (IQR9-15.5, PR 9-36), median motor score 8 (IQR8-11, PR 8-32) and median autonomic score 4 (IQR3-6) for males (PR 3-12) and 2 (IQR2-2.5) for females (PR 2-8). 54% of men reported erectile dysfunction, of whom 62% experienced severe dysfunction (33% of all males). Median FACT-C score was 123.5 (IQR108-131; PR 0-136, high score = better QOL). Median overall health score was 80% (IQR70-95). 36% of patients reported difficulty with mobility, 27% with usual activities and 11% with self-care. 30% declared ongoing physical pain/discomfort, while 23% live with anxiety/depression related to their diagnosis. Overall, 86% of reported problems were mild/moderate and 14% severe/very severe. **Conclusions:** This study highlights self-reported QOL in patients on W&W pathway following neoadjuvant treatment of rectal cancer. It provides important insight to side-effects of treatment escalation in hope of organ preservation, and presents vital information for the counselling of future patients. Research Sponsor: None.

	Patients (n=41)
Upper limb paraesthesia	12 (29%)
Lower limb paraesthesia	17 (41%)
Difficulty with:	
- Walking/standing (foot drop or impaired sensation)	10 (24%)
- Stairs/sit-to-stand (leg weakness)	11 (27%)
- Hot/cold differentiation	4 (10%)
- Holding pen/manipulating small objects/opening jar/bottle	12 (29%)
Hearing impairment	13 (32%)
Blurred vision	4 (10%)

121 Poster Session

Safety and tolerability of different total neoadjuvant therapy strategies in patients with locally advanced rectal cancer. First Author: Shohei Udagawa, Department of Gastroenterological Chemotherapy, Cancer Institute Hospital, Japanese Foundation for Cancer Research, Tokyo, Japan

Background: Total neoadjuvant therapy (TNT) leads to improved local control and organ preservation for locally advanced rectal cancer (LARC). The chemotherapy part of TNT may play a critical role in achieving local control and in improving OS. Few studies have focused on the toxicity associated with the chemotherapy of TNT. Furthermore, the differences in safety and tolerability between radiotherapy-first and chemotherapy-first strategies remain unclear. **Methods:** LARC patients treated with TNT between June 2020 and July 2023 at a single cancer institute were retrospectively enrolled. TNT consisted of induction chemotherapy followed by long-course chemoradiotherapy (INCT-LC-CRT) or long-course chemoradiotherapy/short-course radiotherapy followed by consolidation chemotherapy (LC-CRT-CNCT/SC-RT-CNCT). Concurrent chemotherapy in LC-CRT was capecitabine (825 mg/m^2 oral twice daily). INCT/CNCT consisted of six cycles of CAPOX (capecitabine 1000 mg/m² oral twice daily on days 1–14 and oxaliplatin 130 mg/m² intravenously on day 1 every 21 days). INCT allowed the addition of bevacizumab (7.5 mg/kg intravenously on day 1). We compared adverse events (AEs) and treatment exposure between INCT-LC-CRT, LC-CRT-CNCT and SC-RT-CNCT. **Results:** A total of 93 patients were included. Patients with INCT-LC-CRT and LC-CRT-CNCT/SC-RT-CNCT were 28 (30.1%) and 45 (48.4%)/20 (21.5%), respectively. Patients who were treated with INCT-LC-CRT (53.6%) had more N2 stage than those with LC-CRT-CNCT/SC-RT-CNCT (21.6%). Grade 3–4 AEs during chemotherapy were significantly more frequent in LC-CRT-CNCT/ SC-RT-CNCT group than in INCT-LC-CRT group (50.8% vs. 21.4%; p = 0.02). Neutropenia was the most common Grade 3–4 AEs during chemotherapy in INCT-LC-CRT group (7.4%) and LC-CRT-CNCT/SC-RT-CNCT group (27.7%). Grade 3–4 AEs during chemotherapy were not significantly different between the LC-CRT-CNCT and SC-RT-CNCT groups (36.9% vs. 40.0%; p = 0.42). All patients underwent the full radiation dose. The completion rate of six cycles of chemotherapy was no significant difference between INCT-LC-CRT group and LC-CRT-CNCT/ SC-RT-CNCT group (92.9% vs. 84.6%; p = 0.37). However, LC-CRT-CNCT/SC-RT-CNCT group required significantly more dose modification during chemotherapy than INCT-LC-CRT group (75.4% vs. 46.4%; $p < 0.01$). Dose modification during chemotherapy between LC-CRT-CNCT and SC-RT-CNCT group was no significant difference (modification rate 77.8% vs. 70.0%; p = 0.54). Chemotherapy discontinuations due to AEs in INCT-LC-CRT group and LC-CRT-CNCT/SC-RT-CNCT group were 1(3.6%) and 6 (9.2%), respectively. **Conclusions:** Grade 3–4 AEs during chemotherapy were more frequent and required dose modification of chemotherapy in a radiotherapy-first strategy. Safety and tolerability of chemotherapy part of TNT may differ between radiotherapy-first and chemotherapy-first strategies. Research Sponsor: None.

122 Poster Session

Tumor lysis syndrome in colorectal cancer: A systematic review of clinical characteristics. First Author: Wan Ying Tan, University of Connecticut School of Medicine, Farmington, CT

Background: Tumor lysis syndrome (TLS) is an oncologic emergency resulting from the release of potassium, phosphorus, and nucleic acids from tumor cells. While rare in colorectal cancer (CRC), TLS is linked with high mortality, and literature is limited. This study aims to summarize the clinical characteristics of patients with TLS in CRC. **Methods:** We conducted a systematic review of TLS in CRC through a comprehensive literature search in PubMed and manual searches from January 1st, 1993, up to September 13th, 2024. Keywords included "Tumor Lysis Syndrome," "Colorectal Cancer," and "Colon Cancer." 32 studies were found, identifying 11 spontaneous and 22 therapy-related TLS cases. Quantitative and qualitative data were extracted for descriptive analysis. **Results:** A total of 33 patients (16 males and 15 females) had a mean age of 56.5 $\pm$ 17.8 years at TLS diagnosis. The mean ages for males and females were 58.3 $\pm$ 7.1 and 55.8 $\pm$ 13.7 years, respectively (p=0.59). The primary histology was adenocarcinoma (92%) followed by neuroendocrine (8%) (data unavailable for 9 patients). Among 15 patients with grading information, Grade 2 tumors were most common (53%), followed by Grade 3 (40%) and Grade 1 (7%). Staging data was available for 32 patients. Most patients had Stage IV disease (94%), with frequent metastases to the liver (93%), lungs (30%), and bones (10%). Hyperuricemia is present in all patients (1 patient without data). Acute kidney injury (AKI) was present in all patients (4 patients without data). Hyperphosphatemia was found in 82% of patients (5 patients without data). Hypocalcemia was noted in 52% (10 patients without data), and hyperkalemia was found in 74% (6 patients without data). Out of the 33 patients, 11 experienced spontaneous TLS (33%) and 22 had therapy-related TLS (67%). Duration of TLS onset following therapy was available for 17 patients, including one outlier at 30 days on Capecitabine. The mean duration for therapy-related TLS, excluding the outlier, was 3.8 $\pm$ 2.5 days. Among the 22 patients on therapy, 59% received 5-Fluorouracil (5-FU)-based treatment. 31% of those receiving 5-FU had combination therapy with Bevacizumab. Death occurred in 59% of patients, with data missing for 4 patients. **Conclusions:** TLS in colorectal cancer (CRC) is associated with a mortality rate of up to 59% and shows no gender preference. Identifying predictive factors is crucial for TLS prophylaxis. Patients with TLS in CRC often present with Stage IV disease, liver metastasis, hyperuricemia, and acute kidney injury (AKI). Research Sponsor: None.

123 Poster Session

First-in-human, phase 1a, dose escalation study of BGB-B167, a CEA x 4-1BB bispecific antibody, as monotherapy or combined with tislelizumab (anti-PD-1), in patients with selected advanced or metastatic solid tumors. First Author: Jayesh Desai, Peter MacCallum Cancer Centre, Melbourne, VIC, Australia

Background: BGB-B167 is potentially a first-in-class IgG-based bispecific antibody that targets CEA, a tumor-associated antigen overexpressed in many cancers, and 4-1BB, a key T-cell co-stimulatory receptor expressed on activated CD4+/CD8+ T lymphocytes. Preclinical data suggest BGB-B167 binds to these target proteins with high specificity/affinity, inducing T-cell activation and antitumor activity, with potential for mitigating strong AEs, eg CRS and hepatoxicity. We present results of a first-in-human, phase 1a, open-label, multicenter trial of BGB-B167 given as monotherapy (Part A [A]) or with tislelizumab (TIS) (Part B [B]) (NCT05494762). **Methods:** Eligibility: $\geq$18 y with unresectable/metastatic CRC, CEA+ GC or CEA+ NSCLC previously treated with standard systemic therapy or for whom treatment is unavailable; pts with CRC and known MSI-H/dMMR status must have received prior ICI, if available. Primary objectives: safety/tolerability; MTD and RP2D for BGB-B167 $\pm$ TIS; key secondary objective: antitumor activity (RECIST v1.1). **Results:** As of July 7, 2024, 54 pts were enrolled (31 in A, 23 in B) with BGB-B167 dose ranges of 5–1200 mg IV QW assessed in A and 50–600 mg IV QW in B. In A and B, respectively, 29/31 (93.5%) and 18/23 (78.3%) had CRC, 2/31 (6.5%) and 3/23 (13%) had GC, 0/31 (0%) and 2/23 (8.7%) had NSCLC; 19/31 (61.3%) and 18/23 (78.3%) had liver metastases; 20/31 (64.5%) and 14/23 (60.9%) received $\geq$3 lines of prior therapy. TEAEs occurred in all pts (Table). Most common treatment-related TEAEs were nausea (4/31; 12.9%) and fatigue (3/31; 9.7%) in A and pruritus (5/23; 21.7%) and diarrhea (4/23; 17.4%) in B; there were no treatment-related TEAEs indicative of hepatotoxicity or CRS. There was a single DLT (gr 3 SJS) in a pt who received 600 mg BGB-B167 + TIS (resolved after treatment discontinuation). MTD was not reached. 3 pts (1 with CRC [A]; 1 with CRC and 1 with GC [B]) had confirmed PRs. DoR was $\geq$5.6 mo for the pt in A and $\geq$5.6 mo and $\geq$6.9 mo for the pts in B; all remained on treatment with ongoing responses as of the data cutoff date. DCR (95% CI) was 33.3% (17.3–52.8) in A and 40.9% (20.7–63.6) in B. Serum exposure of BGB-B167 increased dose dependently from 150 to 1200 mg. **Conclusions:** Overall, BGB-B167 $\pm$ TIS was well tolerated and has demonstrated limited antitumor activity in pts with advanced/metastatic CEA+ solid tumors. Clinical trial information: NCT05494762. Research Sponsor: BeiGene, Ltd.

Safety.

	Part A BGB-B167 monotherapy (N=31)	Part B BGB-B167 + TIS (N=23)
Any TEAE	31 (100.0)	23 (100.0)
Any treatment-related TEAE	17 (54.8)	15 (65.2)
Gr ≥3	4 (12.9)	1 (4.3)
Serious	0 (0)	2 (8.7)
Leading to death	0 (0)	0 (0)
Leading to treatment discontinuation	0 (0)	1 (4.3)
Any imAE	2 (6.5)	5 (21.7)
IRRs	4 (12.9)	6 (26.1)

Patients with multiple adverse events (AEs) are counted once. All AEs are listed as n (%).
imAE, immune-mediated AE; IRR, infusion-related reaction; TEAE, treatment-emergent AE.

124 Poster Session

Phase II study of FOLFIRI with low-dose irinotecan plus ramucirumab as second-line treatment in Japanese patients with metastatic colorectal cancer (RINDO study). First Author: Masashi Hattori, National Hospital Organization Nagoya Medical Center, Nagoya, Japan

Background: Phase III trials (ML18147, VELOUR, and RAISE trials) of second-line combination therapy with molecular-targeted agents after first-line treatment with bevacizumab (BEV) for metastatic colorectal cancer (mCRC) demonstrated significant improvements in overall survival (OS). In the RAISE trial (irinotecan (IRI) dose: 180 mg/m²), the relative dose intensity (RDI) of IRI was lower (63.8%) and the incidence rates of adverse events leading to discontinuation of cytotoxic agents was higher (48.6%) in the Japanese population compared to all patients. Based on these results, we conducted a prospective trial to evaluate the efficacy and safety of fluorouracil, levofolinate, and IRI (150 mg/m², standard dose in Japan) (FOLFIRI) plus ramucirumab (RAM) as second-line treatment for mCRC in Japanese patients. **Methods:** On day 1 of each 2-week cycle, patients with unresectable mCRC who were refractory to oxaliplatin and fluoropyrimidine in combination with BEV or anti-epidermal growth factor receptor (EGFR) antibodies as first-line treatment received 8 mg/kg RAM, followed by the FOLFIRI regimen with low-dose IRI (150 mg/m²). The primary endpoint was progression-free survival (PFS), and secondary endpoints were OS, treatment compliance, and safety. We hypothesized an RFS threshold of 4.3 months and expected RFS of 5.7 months based on data from a Japanese subgroup of the RAISE trial. The protocol treatment was considered to be effective if the lower limit of the 95% confidence interval (CI) exceeded the 4.3-month threshold. **Results:** A total of 62 patients were enrolled from 15 institutions between January 2018 and August 2021. The intent-to-treat and safety populations included 61 and 58 patients, respectively. The cutoff date for the primary analysis was December 2023. Median PFS and OS were 5.9 months (95% CI, 4.8-6.9 months) and 17.0 months (95%CI, 12.0-21.0 months), respectively. Median PFS was 5.7 months (95% CI, 4.4-6.8 months) in patients treated with first-line chemotherapy with BEV and 7.4 months (95% CI, 4.6-11.0 months) in those treated with first-line chemotherapy with anti-EGFR antibodies (hazard ratio [HR], 1.17; 95% CI, 0.64-2.12; p = 0.60), and median OS was 19.8 months (95% CI, 10.4-22.4 months) and 17.5 months (95% CI, 11.5-26.1 months), respectively (HR, 0.96; 95% CI, 0.52-1.78; p = 0.91). The objective response rate and disease control rate were 8.2% and 74%, respectively. Median RDI of IRI, 5-fluorouracil, and RAM were 73.8% (range, 40.3-102.4%), 58.5% (range, 22.8-102.4%), and 80.8% (range, 36.1-102.4%), respectively. The observed Grade $\geq$3 adverse events included neutropenia (40%), anemia (1.7%), diarrhea (8.6%), fatigue (6.9%), decreased appetite (10%), hypertension (6.9%), and proteinuria (3.4%). **Conclusions:** FOLFIRI with low-dose IRI plus RAM is a feasible second-line treatment in Japanese patients with mCRC. Clinical trial information: jRCTs041180074. Research Sponsor: None.

LBA125 **Rapid Oral Abstract Session**

Low-dose aspirin to reduce recurrence rate in colorectal cancer patients with PI3K pathway alterations: 3-year results from a randomized placebo-controlled trial. First Author: Anna Martling, Karolinska Institutet, Stockholm, Sweden

The full, final text of this abstract will be available at meetings.asco.org on the day of presentation and in the online supplement to the February 1, 2025, issue of the *Journal of Clinical Oncology*.

126 **Poster Session**

Retrospective analysis of HAIP therapy in metastatic colorectal cancer: Efficacy, safety, and prognostic factors. First Author: Matthew Sawyer, UT Southwestern Medical Center, Dallas, TX

Background: HAIP (hepatic arterial infusion pump) therapy is a promising intervention for metastatic colorectal cancer (mCRC) with liver metastases, addressing significant challenges in management and prognosis. It offers targeted treatment to maximize efficacy while minimizing systemic toxicity. We present a retrospective analysis of patients with metastatic colorectal adenocarcinoma treated with HAIP therapy at an academic center and safety net hospital. **Methods:** Patients with mCRC and hepatic metastasis who received at least one HAIP floxuridine (FUDR) cycle from 2013 to 2023 were included. Baseline data, including demographics, tumor characteristics, and treatment history, were collected pre-HAIP. The main outcome was overall survival (OS) post-pump placement, with secondary outcomes being real-world progression-free survival (rwPFS) and HAIP-related complications. **Results:** In total, 88 patients were evaluated for inclusion, of whom 63 met inclusion criteria and were included for analysis. The median age of included patients was 46 years (interquartile range (IQR) 41-57) and 54% were male. Prior to HAIP, 38 (60%) of patients received one line of systemic therapy, while 18 (29%) had two. The median OS from HAIP placement was 28.0 months (95% CI 21.9-NR) and median rwPFS was 11.2 months (95% CI 7.1-16.1). Baseline bilirubin, KRAS mutant status, and CEA were associated with OS in the multivariate cox model. The most common complication from HAIP placement was hematoma (n=4, 6.3%), followed by pump infection (n=2, 3.2%). Grade 3 toxicities from FUDR infusion were infrequent, involving alkaline phosphatase (n=6), AST or ALT elevation (n=3), bilirubin increase (n=2), and platelet count decrease (n=1). **Conclusions:** HAIP treatment is associated with durable responses and few complications, making it well-tolerated by patients. For individuals with colorectal cancer and hepatic metastases, HAIP chemotherapy using FUDR presents a viable and effective treatment option, potentially improving overall outcomes and quality of life. Further prospective and randomized work may establish the role for HAIP in mCRC with liver metastases. Research Sponsor: None.

127 **Poster Session**

Zanzalintinib (XL092) alone or in combination with atezolizumab in patients (pts) with refractory metastatic colorectal cancer (mCRC): Results from an expansion cohort of the phase 1 STELLAR-001 study. First Author: Elisa Fontana, Sarah Cannon Research Institute, London, United Kingdom

Background: VEGFR tyrosine kinase inhibitors (TKIs) in combination with immune checkpoint inhibitors (ICIs) have demonstrated clinical activity in pts with previously treated mCRC, particularly in pts without liver metastases (LM). Zanzalintinib (zanza) is a novel, oral, multitargeted TKI with activity against VEGFR, MET, and TAM kinases. Here, we present results from the randomized expansion cohort of pts with previously treated non-MSI-H/dMMR mCRC receiving zanza $\pm$ atezolizumab (atezo) in the phase 1 STELLAR-001 study (NCT03845166). **Methods:** Adults (aged $\geq$18 y) with locally advanced or metastatic CRC that was *RAS*-wildtype, who were refractory/intolerant to prior standard of care (5-FU–based treatment $\pm$ targeted therapies) and had an ECOG PS of 0/1, were enrolled. Pts with MSI-high or dMMR CRC, or who had received prior treatment with a PD-L1/PD-1 targeting ICI, regorafenib, and/or TAS-102, were excluded. Pts were randomized 1:1 to receive zanza+atezo (100 mg QD + 1200 mg Q3W) or single-agent zanza (100 mg QD). Objectives were to evaluate median (m)OS, investigator-assessed ORR, and mPFS per RECIST 1.1, and safety. **Results:** As of May 6, 2024, 107 pts were enrolled (zanza+atezo, n=54; zanza, n=53). In the zanza+atezo/zanza groups, median age was 61/60 years, 39%/51% had an ECOG PS of 1, pts had a median 2.5/3.0 (range, 1–5) prior lines of therapy, and 31%/32% did not have LM. With a median follow-up of 15 months across both arms, the mOS was 14.3 and 11.1 months for zanza+atezo and zanza, respectively (HR 0.75, 95% CI 0.45–1.26), confirmed ORR was 7.4% and 1.9% (all partial responses), mPFS was 4.0 and 3.0 months (HR 0.68, 0.44–1.04), and DCR was 59% and 55%. In pts without LM, mOS was not reached and 12.5 months (HR 0.46, 0.15–1.36) with 6-month survival rates of 88% and 65%, confirmed ORR was 18.0% and 5.9%, mPFS was 8.2 and 3.3 months (HR 0.40, 0.16–1.0), and DCR was 76% and 59%. The most common treatment-related adverse events (TRAEs) were diarrhea (zanza+atezo, 52%; zanza, 49%), nausea (54%, 36%), and decreased appetite (41%, 36%). Grade 3/4 TRAEs occurred in 48% of pts with zanza+atezo and 40% with zanza (Grade 4 in 3.7% and 0); a Grade 5 TRAE occurred in 1 pt (2%) in each group. 89% and 94% of pts had discontinued study treatment. Zanza was discontinued due to TRAEs in 11% and 8% of pts; atezo was discontinued due to TRAEs in 9% (zanza+atezo). Biomarker analysis is ongoing and results will be presented. **Conclusions:** Clinical activity was observed with zanza both as a single agent and in combination with atezo in this cohort of heavily pretreated pts with mCRC. Greater activity was seen with zanza+atezo versus zanza alone, particularly in pts without LM. Both treatments were generally well tolerated. Evaluation of zanza+atezo versus regorafenib is ongoing in the phase 3 STELLAR-303 study in non-MSI-H/dMMR mCRC (NCT05425940). Clinical trial information: NCT03845166. Research Sponsor: Exelixis, Inc.

128 **Poster Session**

Phase Ib/II study of fruquintinib plus 5-fluorouracil/leucovorin after progression on fruquintinib monotherapy in metastatic colorectal cancer. First Author: Wenwei Yang, National Cancer Center, National Clinical Research Center for Cancer, Cancer Hospital, Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing, China

Background: Fruquintinib is a standard third-line treatment in metastatic colorectal cancer (mCRC). However, no standard therapeutic regimen is available for mCRC patients who progressed on third-line therapy. Preclinical experiments supported a rationale for combining antiangiogenic therapy and 5-fluorouracil (5FU) in refractory mCRC with evidence for synergy. This open-label phase Ib/II study was designed to determine the safety and efficacy of fruquintinib plus 5-fluorouracil/leucovorin for mCRC patients who progressed on fruquintinib monotherapy. **Methods:** The enrolled patients would receive fruquintinib at a fixed dose of 4mg once daily (three weeks on, one week off) and 5FU at two dose levels (1800mg/m^2 and 2400mg/m^2) CIV 46 hours on day one and day 15, with leucovorin 400mg/m^2 iv 2 hours before 5FU. Treatments were repeated every 28 days. The primary endpoint was determining the recommended phase II dose (R2PD) of 5FU in phase Ib and the six-month OS rate in phase II. Secondary endpoints for phase II included objective response rate (ORR), disease control rate (DCR), progression-free survival (PFS), and safety. Planned enrollment was 39 patients. **Results:** No dose-limiting toxicities occurred in the phase Ib dose-escalation study, and the R2PD of 5FU was 2400mg/m^2. Between April 2020 and June 2024, 24 patients were enrolled in the phase II stage. The median age was 57.5 years old (from 35 to 68), and 58.33% were female. Left-sided colorectal cancer patients accounted for 75%. Sixteen (66.7%) patients harbored KRAS mutation, and one (4%) had BRAF V600E mutation. Regarding the number of lines of systemic therapy, 83.3% received this study regimen as fourth-line therapy and 16.7% as fifth-line or later therapy. As of June 30, 2024, efficacy was assessed in all 24 patients with a DCR of 62.5% (15 stable disease, nine progression disease). The median follow-up time was 16.5 months (95% CI:12.553-20.447). The primary endpoint six-month OS rate was 91.3%, the median PFS was 4.4 months (95% CI: 1.243-7.557), and the median OS was 17.3 months (95% CI: 12.892-21.708). The most common TRAEs were nausea (25%), hand-foot syndrome (16.7%), abnormal liver function (8.3%), and proteinuria (8.3%). One patient experienced grade 3 mucositis. No treatment-related death occurred. **Conclusions:** Fruquintinib plus 5-fluorouracil/leucovorin regimen preliminarily demonstrated promising efficacy with a tolerable safety profile for mCRC patients despite progression on prior fruquintinib monotherapy. Further investigation into this possible later-line regimen in mCRC is warranted. Clinical trial information: ChiCTR2000032640. Research Sponsor: None.

129 **Poster Session**

Tetravalent death receptor 5 (DR5) agonist ozekibart (INBRX-109) + FOLFIRI in 2L+ colorectal adenocarcinoma (CRC): Preliminary results from a phase 1 study. First Author: David Berz, Valkyrie Clinical Trials, Los Angeles, CA

Background: DR5, a proapoptotic receptor, is selectively expressed on tumor vs normal cells, making it an attractive target. DR5 binds to its ligand, TRAIL, resulting in DR5 multimerization and apoptosis; greater DR5 clustering enhances downstream effects. Ozekibart is a next-generation, recombinant, humanized, tetravalent DR5 agonist that is based on a single-domain antibody platform and precisely engineered to balance agonism and safety. Ozekibart previously demonstrated an acceptable safety profile and clinical benefit in chondrosarcoma (1), prompting a phase 2, randomized trial (ChonDRAgon; NCT04950075). An ongoing phase 1 study (NCT03715933) is evaluating ozekibart in various solid tumors, including 2L+ CRC, for which effective therapies are lacking. We present preliminary results of ozekibart + FOLFIRI in 2L+ CRC (data cutoff: Aug 9, 2024). **Methods:** Parts 1 (single-agent dose escalation) and 2 (single-agent dose expansion; recommended phase 2 dose [RP2D] ozekibart 3 mg/kg IV Q3W) of this 3-part, open-label trial are complete. Part 3 (combination dose expansion with chemo) is ongoing in pts with CRC, Ewing sarcoma, or SDH-deficient GIST. Pts with locally advanced or metastatic, unresectable CRC who received oxaliplatin-based chemo were eligible for part 3. Pts with chronic liver disease were ineligible. Pts in the part 3 CRC cohorts (N=13; enrollment complete) received ozekibart 1 mg/kg (n=5) or 3 mg/kg (n=8) IV Q28D (1 cycle) + FOLFIRI Q14D (FOLFIRI could be stopped after ≥6 cycles); in 4/5 pts, ozekibart dose was escalated to 3 mg/kg (RP2D). The primary endpoint is safety; clinical response is an exploratory endpoint. **Results:** In total, 13 pts with CRC (male, 62%) received ozekibart + FOLFIRI. Median age was 60. All pts had metastatic disease; median number of prior lines of therapy was 2 (range, 1-6). At data cutoff, 2 pts remained on treatment. Ozekibart-related adverse events (AEs) were reported in 85% of pts (grade ≥3, 31%) and led to ozekibart interruption in 3 pts and discontinuation in 1 pt. One combo-related AE (neutropenic sepsis) led to death. The most common ozekibart-related AEs were nausea (n=5) and diarrhea, fatigue, and increased alanine aminotransferase (each n=4). Four pts (31%) had responses (all partial). Disease control rate (response + stable disease) was 77% (10/13) and durable (>180 days) in 46% (6/13) of pts. Most pts had tumor shrinkage. Median PFS was 7.85 mo. **Conclusions:** Ozekibart + FOLFIRI showed encouraging preliminary efficacy, including PRs and durable disease control, as 2L+ therapy in pts with CRC. Given these promising data, a new expansion cohort (C4d) is open to validate these findings in a more homogeneous population. Eligible pts will have had 2 or 3 prior lines of systemic therapy (irinotecan allowed but not in the immediately prior line) and must have archival or fresh biopsy tissue. 1. Subbiah. *Clin Cancer Res.* 2023. Clinical trial information: NCT03715933. Research Sponsor: Inhibrx Biosciences, Inc.

130 **Poster Session**

Genetic mutations and cytotoxic chemotherapy response for advanced solid tumors: A detailed analysis using the C-CAT database in Japan. First Author: Naoya Ishibashi, Department of Precision Cancer Medicine, Institute of Science Tokyo, Tokyo, Japan

Background: Chemotherapy regimens are selected based on information about the primary organ or its histological type and are administered based on the results of previous clinical trials. Even now, as cancer genomic research has progressed and precision medicine, which optimizes therapeutic interventions based on tumor genome profiling using next-generation sequencing(NGS), has become standard treatment, cytotoxic anticancer drugs still remain key drugs. We examined the real-world database of genomic and clinical information established by the national central datacenter, the Center for Cancer Genomics and Advanced Therapy (C-CAT) to systematically analyze the effects of mutations across multiple cancers and therapies. **Methods:** This study delineated the association between genetic mutations and the therapeutic efficacy of cytotoxic chemotherapy in advanced solid tumors in a real-world Japanese clinical context. We incorporated data from 15,474 patients enrolled in the C-CAT database between June 2019 and June 2022. We analyzed the overall response and time to next treatment (TNT) for genetic alterations and five categories of cytotoxic anticancer drugs in gastrointestinal cancers. **Results:** In the C-CAT data, gastrointestinal cancers were frequently registered, with colorectal cancer being the most common, followed by pancreatic cancer, esophageal/gastric cancer, and biliary tract cancer. For platinum-based drugs, in colorectal cancer, the presence of gene X was associated with not only a higher response rate but also a longer TNT (hazard ratio 0.82, $p<0.001$). In the case of colorectal cancer and pancreatic cancer with KRAS gene mutations, the ORR was low, and the TNT was also short in the case of pancreatic cancer (hazard ratio 1.33, $p<0.001$). Interestingly, when BRCA2 genes are mutant, TNT tended to be longer in many organs, suggesting that platinum-based drugs are effective across organs in carcinomas with BRCA2 mutations. Additionally, gene X showed differences in efficacy not only for platinum but also for antimetabolites and topoisomerase inhibitors. Moreover, KRAS mutation was observed in almost all pancreatic cancers and was a treatment-resistant mechanism for all platinum drugs and antimetabolites. **Conclusions:** This study demonstrated that the effects of cytotoxic anticancer drugs differ depending on genetic mutations. When determining chemotherapy regimens, considering not only the organ but also genetic mutations may enable more efficient drug selection. Research Sponsor: None.

131 **Poster Session**

Adagrasib (Ada) + cetuximab (Cetux) for *KRAS*^{G12C}-mutated metastatic colorectal cancer (mCRC): Longer follow-up analysis from KRYSTAL-1. First Author: Rona Yaeger, Memorial Sloan Kettering Cancer Center, New York, NY

Background: *KRAS*^{G12C} mutations occur in 3%–4% of CRC cases and are associated with poor prognosis. In the phase 1/2 KRYSTAL-1 study (NCT03785249), at a median follow-up of 11.9 months (mo), Ada (irreversible inhibitor of *KRAS*^{G12C}) in combination with Cetux (anti-EGFR antibody) demonstrated promising clinical activity (objective response rate [ORR] of 34% per blinded independent central review [BICR] and 43% per investigator [INV]) and was well tolerated in patients (pts) with previously treated *KRAS*^{GT2C}-mutated mCRC. Based on these findings, Ada + Cetux was granted accelerated approval in the United States for these pts. Here, we present longer-term follow-up analyses from this study. **Methods:** Adults with previously treated *KRAS*^{G12C}-mutated mCRC and ECOG performance status of 0 or 1 were treated with Ada (600 mg BID) in combination with Cetux (400 mg/m^2 followed by 250 mg/m^2 QW or 500 mg/m^2 Q2W) until disease progression, unacceptable toxicity, withdrawal of consent, or death, in separate phase 1 and phase 2 cohorts. Primary endpoints were safety (phase 1) and ORR per BICR (phase 2). Secondary endpoints were duration of response (DOR), progression-free survival (PFS), overall survival (OS), and safety (phase 2). **Results:** A total of 94 pts received Ada + Cetux. The median number of prior lines of systemic therapy was 3 (range, 1–9); 23 pts (24%) had received ≥ 4 prior lines of systemic therapy. The most frequent sites of metastases at baseline were the lung (71%) and liver (64%). At a median follow-up of 20.4 mo, ORR per INV was 43% (95% CI 32–53) and all were partial responses; median DOR was 5.9 (95% CI 5.5–7.6) mo. Disease control rate per INV was 86% (95% CI 78–92). Median PFS per INV was 6.9 (95% CI 5.9–7.4) mo; 6- and 12-mo PFS rates were 61% and 19%, respectively. Median OS was 16.0 (95% CI 13.3–18.8) mo; 6- and 12-mo OS rates were 88% and 66%, respectively. Any-grade treatment-related adverse events (TRAEs) were reported in 100% of pts, 28% of which were grade 3/4. TRAEs led to discontinuation in 10% of pts. Efficacy analyses by BICR, subgroup analyses, and additional safety results will be presented. **Conclusions:** In heavily pretreated pts with *KRAS*^{G12C}-mutated mCRC, Ada + Cetux continued to demonstrate clinically meaningful activity and tolerable safety with longer follow-up. These updated results are consistent with those from the primary analysis and constitute the longest duration of follow-up for dual *KRAS*^{G12C}/EGFR blockade in this setting. Efficacy of second-line Ada + Cetux vs chemotherapy in *KRAS*^{G12C}-mutated mCRC is being investigated in the phase 3 KRYSTAL-10 study (NCT04793958). Clinical trial information: NCT03785249. Research Sponsor: Mirati Therapeutics, a Bristol Myers Squibb company.

132 **Poster Session**

Adverse events (AEs) –derived biomarker in predicting clinical outcomes in patients with colorectal cancer (CRC) treated with immunotherapy (IC). First Author: Dung-Tsa Chen, H. Lee Moffitt Cancer Center and Research Institute, Tampa, FL

Background: There are insufficient studies evaluating the role of AEs in CRC patients treated with IC. This study aims to transform AEs from traditional toxicity evaluation tools to informative biomarkers for assessing treatment efficacy by utilizing modern analytical strategies. **Methods:** We analyzed a single arm study of 51 mismatch repair proficient, refractory CRC patients (NCT03712943) treated with regorafenib and nivolumab. Data for analysis included CTCAE version 5 AE data, RECIST treatment response, progression-free survival (PFS), and overall survival (OS). Analytic approach leveraged multiple AE parameters to develop a set of innovative AE biomarkers. One AE biomarker of interest was treatment related low-grade early AEs (TR-LG-AE) which was defined as frequency of all TR-LG-AEs occurring between the 1st day and day 30 after the 1st treatment. **Results:** TR-LG-AE was associated with treatment response ($p<0.001$ by ANOVA trend test). Patients with higher AE events tended to be a responder, compared to PD patients with a lower frequency of AE events. The TR-LG-AE also correlated with improved PFS and OS with hazard ratio of 0.83 (p=0.01) and 0.89 (p=0.03), respectively. Accordingly, patients were grouped into high versus low TR-LG-AE based on optimal cutoff of 6 events for further analysis. The dichotomized TR-LG-AE (high vs low) showed significant association with survival outcomes (median PFS: 15.3 vs 3.7 months; $p<0.001$; median OS: 21.9 vs 9 months; p=0.02). Further interaction analysis showed patients with high TR-LG-AE had improved PFS regardless of occurrence of high-grade AE (HG-AE) (p=0.01), suggesting its independence of HG-AE. Individual AE analysis identified patients experienced with early low-grade rash maculo-papular, lipase increased, or hypertension had improved PFS ($p<0.001$). In biomarker analysis, one key mutation biomarker, RAS, did not show significant association with survival outcomes (PFS: p=0.66; OS: p=0.81). In contrast, integration with TR-LG-AE led to remarkable improvement. Specifically, the TR-LG-AE was able to help improve RAS classification by identifying patients with better treatment benefit. Specifically, patients with RAS mutation and high TR-LG-AE experiences had a median PFS of 15.3 months compared to others (median PFS of 3.7 months; p=0.007), and a median OS of 21.9 months compared to others (median OS of 9.4 months; p=0.04). **Conclusions:** The study demonstrated utility of AE in predicting clinical outcomes in CRC, similar to our previous findings in lung cancer. Moreover, it discovered that integration of AE-derived biomarkers with molecular biomarkers could produce a more robust classifier than when considered individually. This analysis is the first of its kind to identify subgroups of CRC patients that benefit from IC treatment. Research Sponsor: National Cancer Institute; 1R21CA286417; Division of Community Health Promotion/ Public Health Research/Biomedical Research Unit; MOAAX.

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Total neoadjuvant therapy with FOLFOX plus bevacizumab and short-course radiotherapy for Ras mutant, high-risk, locally advanced rectal cancer: The TRAINER trial. First Author: Bo Feng, Shanghai Ruijin Hospital Administrator Consulting Co., Ltd., Shanghai, China

Background: Total neoadjuvant therapy (TNT) intensified by the combination of targeted therapy or immunotherapy is warranted to achieve satisfying local and systemic control for high-risk locally advanced rectal cancer (LARC). Our study aimed to explore the efficacy and safety of a FOLFOX-based TNT regimen plus bevacizumab and short-course radiotherapy (SCRT) in Ras-mutant high-risk LARC patients. **Methods:** In this single-arm, single-center, phase II trial, LARC patients with at least one risk factor (cT3c-4, mesorectal fascia positive, cN2, lateral lymph node involvement, and extramural venous invasion positive) were recruited and only Ras-mutant patients were included. Four cycles of FOLFOX plus bevacizumab were arranged as induction chemotherapy followed by SCRT (25Gy/5Fx) and two cycles of FOLFOX as consolidation chemotherapy. Surgery was performed 8 weeks after completion of radiotherapy. After surgery, high-risk stage III patients (T4 or N2) were recommended to receive an extra 6 cycles of FOLFOX as adjuvant chemotherapy, while low-risk stage III patients were scheduled for regular follow-up. Patients with clinical complete response(cCR) were allowed to choose the "Watch & Wait" (W&W) strategy instead of surgery. The primary endpoint was the complete response (CR) rate, defined as the total rate of cCR plus pathological complete response (pCR). Secondary endpoints included adverse events of TNT, postoperative complications, 2-year disease-free survival, and overall survival. **Results:** Between April 2, 2021, and March 12, 2024, 47 of 51 enrolled patients were eligible for treatment. All patients completed the prescribed TNT cycles, and six (12.8%) of them had dose reduction. The CR was achieved in 18 (38.3%) patients, including 14 (29.8%) pCR. No grade IV-V toxicity was observed and the incidence of grade III toxicities was 34.0%. Four patients (8.5%) achieved cCR and chose the W&W strategy. The other 43 patients underwent curative surgery and anal preservation was achieved in 39 (90.7%) patients. Seven patients (16.3%) experienced postoperative complications, including four (9.3%) cases of anastomotic leakage. **Conclusions:** The intensified TNT regimen combined with bevacizumab achieved a satisfying CR rate with acceptable toxicities and complications for Ras-mutant high-risk LARC patients. Long-term survival data of this cohort are needed to validate the efficacy of this regimen further. Clinical trial information: NCT04923620. Research Sponsor: None.

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Nirogacestat in patients with desmoid tumors and mutations of adenomatous polyposis coli (*APC*): Findings from the DeFi trial. First Author: Noah Federman, David Geffen School of Medicine, University of California, Los Angeles, CA

Background: Nirogacestat (niro) is an oral, targeted gamma secretase inhibitor FDA-approved for adults with progressing desmoid tumors (DT) who require systemic treatment. In the phase 3 DeFi trial (NCT03785964), niro demonstrated significant improvement vs placebo (pbo) in progression-free survival (PFS; HR 0.29 [95% CI: 0.15–0.55], $P<.001$) and objective response rate (ORR; 41% vs 8%, $P<.001$). Most DT are sporadic tumors characterized by somatic mutations in the *CTNNB1* gene. About 10%–20% of DT are associated with *APC* mutations, the majority being germline that may cause familial adenomatous polyposis (FAP), an inheritable trait linked to an increased risk of colorectal cancer and may also confer more aggressive DT behavior. **Methods:** In DeFi, adult patients (pts) were randomized to oral niro (150 mg) or pbo twice daily. In pts with evaluable blood and tumor samples, descriptive post hoc analyses assessed effects of niro in pts with germline and/or somatic *APC* mutations, including pts with co-occurring somatic mutations of *APC* and *CTNNB1*. **Results:** Of the 142 pts in DeFi, 29 pts had *APC* mutations (niro=13, pbo=16; 22 somatic, 21 germline, and 14 somatic and germline), including 3 pts (niro=2, pbo=1) with co-occurring somatic mutations of *APC* and *CTNNB1*. Of these 29 pts, 19 (66%) were female, 16 (55%) were aged ≤30 y, 22 (76%) had a family history of FAP, and 22 (76%) were refractory to prior therapy (median of 3 prior lines of therapy). PFS was improved with niro vs pbo (HR 0.21 [95% CI: 0.05–1.00], $P=.016$). Confirmed ORR was 38% (5/13) for niro vs 13% (2/16) for pbo; median time to response with niro was 8.31 months. All 3 pts with co-occurring somatic mutations of *APC* and *CTNNB1* were female, aged 18–56 y, had DT in the upper extremity or abdominal wall, and had no family history of FAP. The 2 pts on niro had received prior systemic therapy and/or surgery and the 1 pt on pbo had no prior therapy. Of the 2 pts on niro, both achieved a partial response (median time to response: 9.9 months). The 1 pt randomized to pbo experienced disease progression in 2.6 months. In pts with *APC* mutations, diarrhea was the most frequently reported adverse event. In niro-treated pts, increased rates of skin events (maculopapular rash, 62%; dermatitis acneiform, 38%) and stomatitis (46%) were reported in pts with *APC* mutations compared with those in the overall DeFi population (32%, 22%, and 29%, respectively). **Conclusions:** Improvement in PFS and ORR was observed with niro vs pbo in pts with DT harboring *APC* mutations. Efficacy and safety of niro in pts with *APC* mutations were generally consistent with findings for the overall DeFi population. Although analyses were limited due to small sample size, these results suggest that niro can provide clinically meaningful benefit to pts with progressing DT and *APC* mutations, including those with co-occurring somatic mutations of *APC* and *CTNNB1*. Clinical trial information: NCT03785964. Research Sponsor: SpringWorks Therapeutics, Inc.

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Phase Ib study of gevokizumab (GEVO) in combination with standard-of-care (SoC) anticancer therapies in patients (pts) with metastatic colorectal cancer (mCRC), metastatic gastroesophageal cancer (mGEC), and metastatic renal cell cancer (mRCC). First Author: Naureen Starling, Gastrointestinal and Lymphoma Unit, The Royal Marsden NHS Foundation Trust, London, United Kingdom

Background: GEVO, a humanized monoclonal antibody, binds to interleukin-1β (IL-1β) and inhibits its activity. We report results from a phase Ib study of GEVO + SoC anticancer therapies in pts with mCRC/mGEC/mRCC. **Methods:** This open-label study enrolled pts aged ≥18 years in cohorts A (first line mCRC, [A]), B (second line [2L] mCRC, [B]), C (2L mGEC, [C]) and D (2L/third line mRCC, [D]). The primary objectives were to determine the pharmacodynamically-active dose (PAD) of GEVO monotherapy in A/B, the safety and tolerability, and the recommended dose for expansion (RDE) of GEVO + SoC in A/B/C/D, and the efficacy of GEVO + SoC at RDE in A/B/C measured by the progression-free survival (PFS) rate. The proof of concept (PoC) criteria for PFS rates were ≥42% at 15 months (m) (lower 80% CI limit ≥29%) in A, ≥48% at ~9 m (lower 80% CI limit ≥37%) in B, ≥51% at 6 m (lower 60% CI limit ≥38%) in C. The relationship between baseline levels of high-sensitivity C-reactive protein (hs-CRP), interleukin-6 (IL-6) and IL-1β, and on-treatment decrease of circulating tumor DNA (ctDNA) levels with clinical response was explored. **Results:** As of March 1, 2023, 71 (A), 62 (B), 26 (C) and 7 pts (D) were treated. The PAD was established as 120 mg GEVO intravenous (IV) every 4 weeks (Q4W) and the RDE as 120 mg GEVO IV Q4W + SoC. There were 2 dose-limiting toxicities: grade 4 decreased neutrophil count (1 pt in C) and grade 3 hyponatremia (1 pt in D). The most frequent grade ≥3 treatment-related adverse events (≥15% of all pts) were neutropenia (A/B: 21.1%/22.6%), decreased neutrophil count (A/B/C: 18.3%/17.7%/15.4%) and hyponatremia (D: 28.6%). The PFS rates were 29.0% (80% CI: 20.4, 38.2), 39.4% (80% CI: 29.8, 48.9), 20.0% (60% CI: 13.3, 27.6) in A/B/C, respectively. In B, median PFS and overall survival (OS) were significantly longer in pts with baseline hs-CRP <10 mg/L (low [L]) than ≥10 mg/L (high [H]), IL-6 <9.58 pg/mL (L) than ≥9.58 pg/mL (H), and IL-1β <0.12 pg/mL (L) than ≥0.12 pg/mL (H). In A + B, a median on-treatment decrease of ctDNA fraction by >72% (H) was associated with significantly longer OS. **Conclusions:** The safety and tolerability of GEVO combinations was acceptable. The primary PFS rates did not meet the PoC criteria. Baseline levels of hs-CRP, IL-6 and IL-1β in pts with 2L mCRC, and on-treatment reduction of ctDNA in pts with 1L + 2L mCRC may have prognostic value. Clinical trial information: NCT03798626. Research Sponsor: Novartis Pharmaceuticals Corporation.

Variables	PFS HR (95% CI)	p-value	OS HR (95% CI)	p-value
A				
hs-CRP L / H	1.74 (0.81, 3.73)	0.155	1.99 (0.66, 5.98)	0.221
IL-6 L / H	1.38 (0.70, 2.72)	0.352	1.73 (0.71, 4.20)	0.225
IL-1β L / H	1.50 (0.74, 3.01)	0.259	0.98 (0.38, 2.52)	0.973
B				
hs-CRP L / H	2.56 (1.27, 5.18)	0.009	3.04 (1.40, 6.60)	0.005
IL-6 L / H	2.92 (1.33, 6.40)	0.007	3.25 (1.43, 7.36)	0.005
IL-1β L / H	2.41 (1.13, 5.13)	0.022	3.60 (1.54, 8.44)	0.003
A + B				
ctDNA decrease L / H	0.65 (0.39, 1.10)	0.108	0.45 (0.24, 0.83)	0.011

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Risk of residual tumor following removal of a malignant colorectal polyp with high-risk features. First Author: Thikhamporn Tawantanakorn, Memorial Sloan Kettering Cancer Center, New York, NY

Background: National guidelines recommend surgery when pathology analysis of a colorectal polyp reveals high-risk features: positive margin/piecemeal resection (PM/PR), lymphovascular/perineural invasion (LVI/PNI), poor differentiation (PD), or tumor budding (TB). However, some patients choose to avoid surgery, with its morbidity and quality-of-life implications, despite the risk of residual disease. **Methods:** This single-center cohort study determined the rates of luminal or lymph node disease and distant metastasis in patients who underwent surgery and patients who underwent observation following removal of a malignant colorectal polyp with high-risk features between 2015 and 2022. **Results:** Of 336 patients who underwent a polypectomy for a malignant polyp in the colon (n = 226) or rectum (n = 110), 312 had a PM/PR, 96 had LVI/PNI, 47 had PD, and 45 had TB; 208 patients (62%) underwent surgery, and 128 (38%) underwent observation. Patients with a malignant polyp in the rectum were more likely to undergo observation than patients with a malignant polyp in the colon (79 [72%] vs. 49 [22%] patients; $P < 0.01$). In the surgery group, 19 patients (9%) had residual luminal disease, 39 patients (19%) had lymph node disease, and 5 patients (2%) had distant metastasis. In the observation group, disease was identified in 14 patients (11%): regrowth in the lumen, 7 patients (9%); regrowth in lymph nodes, 5 patients (6%); distant metastasis, 1 patient (1%); regrowth with distant metastasis, 1 patient (1%). The 12 cases of regrowth were salvaged: resection, 4 patients; local excision, 2 patients; chemoradiotherapy, 6 patients. **Conclusions:** The risk of residual disease following removal of a malignant colorectal polyp with high-risk features is considerable. Patients concerned about the morbidity of surgery and postsurgery quality of life should be counseled about this risk, as well as the availability of effective salvage options. Research Sponsor: None.

High-risk features (HRF)	Surgery (n = 208)	Observation (n = 128)	Luminal/LN disease at surgery (n = 51; 25%)	Luminal/LN regrowth during observation (n = 12; 9.4%)	Distant metastasis (n = 7; 2.1%)
PM/PR (n = 312; 93%)	193	119	49 (25%)	12 (10%)	7 (2.2%)
LVI/PNI (n = 96; 29%)	56	40	16 (29%)	3 (7.5%)	3 (3.1%)
PD (n = 47; 14%)	33	14	11 (33%)	0	1 (2.1%)
TB (n = 45; 13%)	20	25	5 (25%)	1 (4.0%)	1 (2.2%)
1 HRF (n = 215; 64%)	141	74	31 (22%)	8 (11%)	4 (1.9%)
2 HRF (n = 78; 23%)	39	39	11 (28%)	4 (10%)	1 (1.3%)
3 HRF (n = 35; 10%)	25	10	8 (32%)	0	2 (5.7%)
4 HRF (n = 6; 1.8%)	2	4	1 (50%)	0	0

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Pattern and predictors of D3 lymph node station positivity in patients undergoing curative surgery for adenocarcinoma of colon. First Author: Anjaly Mohan, Regional Cancer Centre Trivandrum, Thiruvananthapuram, India

Background: The prognosis and surgical management of colon cancer is intricately related to its lymphatic drainage. The oncological outcome of apical node positivity has been evaluated in different studies with varying results. There is no published regional data on the rate of apical node positivity in colon cancer. An insight into the probability of apical node metastases and the risk factors for the same may help us in prognosticating colon cancer patients better. **Methods:** This was a prospective observational study conducted at the Regional Cancer Centre Thiruvananthapuram from September 2022 to June 2024. Our primary objective was to determine the prevalence of apical lymph node positivity in patients undergoing surgery for colon cancer with D3 lymphadenectomy. The secondary objectives were to find out the factors associated with the risk of apical node metastasis, the incidence of skip metastases to apical nodes, the adequacy of lymph node harvest and the postoperative complications. Patients undergoing primary surgery for adenocarcinoma of the colon were included. Patients with metastatic disease, synchronous colon or rectal cancers, past history of colonic surgery were excluded. All patients underwent D3 lymphadenectomy along with colectomy depending upon the tumour location. Lymphatic tissue corresponding to the D3 station was retrieved and sent separately after grossing the fresh specimen on the side table in the operating room. **Results:** A total of 139 patients were included in the study. Fifty two (37.4%) patients had stage III disease, 38.9% had stage II disease and the rest had stage I colon cancer. The mean nodal harvest was 17.86 and mean apical node harvest was 2.04. Node harvest was adequate in 89.21% patients. The apical node positivity rate was 7.19% (10 patients). Skip metastasis to the apical nodes was found in 2 patients. Lymphovascular invasion, number of positive lymph nodes and young age (≤45 years) had a significant correlation with apical node positivity on multivariate analysis. Clavien Dindo grade 3 and above complications were seen in 5.03%. **Conclusions:** Our study showed an apical node positivity rate of 7.19%, with a skip metastasis rate of 1.4%. This is the first Indian data on apical node positivity rate in colon cancer and its predictors. The prognostic value of apical nodes and the occurrence of skip metastasis to the apical nodes are compelling reasons to advocate D3 dissection, even with relatively low prevalence of apical node positivity. Research Sponsor: None.

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What is the optimal treatment strategy of salvage line treatment after progression on anti-epidermal growth factor receptor monoclonal antibody in patients with tissue *RAS/BRAF* wild-type metastatic colorectal cancer? First Author: Emiko Tange, Department of Gastroenterological Chemotherapy, Cancer Institute Hospital, Japanese Foundation for Cancer Research, Tokyo, Japan

Background: Trifluridine/tipiracil (FTD/TPI) plus bevacizumab (BEV) and anti-epidermal growth factor receptor (EGFR) monoclonal antibody (mAb) re-challenge are treatment options for tissue *RAS/BRAF* wild-type (WT) metastatic colorectal cancer (mCRC) in salvage line, although the optimal treatment sequence of salvage line remains unclear. **Methods:** *RAS/BRAF* WT mCRC patients refractory to anti-EGFR mAb were retrospectively enrolled at a single cancer institute from November 2017 to April 2024. We compared responses, progression-free survival (PFS), and overall survival (OS) between Group A (anti-EGFR mAb re-challenge given first) and B (FTD/TPI plus BEV given first). We compared responses between Group C (treated with anti-EGFR mAb re-challenge at any line) and D (treated with FTD/TPI plus BEV at any line). We explored the clinical factors related to the treatment efficacy using multivariate analysis. **Results:** A total of 89 patients (median age, 63 years) were included. The cohort included 74 patients with a primary lesion in the left-sided colon, while 15 patients had a primary lesion in the right-sided colon. The response rate (RR) was 3.4%. The disease control rate (DCR) was 52.8%. The median PFS (mPFS) and OS (mOS) were 3.5 months (95% confidence interval [CI], 2.8-4.2) and 12.8 months (95% CI, 8.8-15.7) respectively. Two of 18 patients (11.1%) in Group A and 1 of 71 patients (1.4%) in Group B achieved partial response (PR). The RR of Group C was significantly higher than Group D (9.6% vs 1.2%; p = 0.031). Both PFS and OS showed no significant difference between Group A and B (mPFS: 3.3 vs 3.9 months; hazard ratio [HR], 1.08; 95% CI, 0.63-1.84; p = 0.79) (mOS: 12.8 vs 15.0 months; HR, 0.90; 95% CI, 0.49-1.65; p = 0.72). Multivariate analysis identified that primary tumor resection was associated with PFS (HR, 0.44; 95% CI, 0.25–0.80; p = 0.0068) and primary tumor resection (HR, 0.34; 95% CI, 0.18–0.65; p = 0.0010) and serum CA19-9 level (HR, 1.75; 95% CI, 1.02-3.00; p = 0.043) were associated with OS. **Conclusions:** No significant survival difference was observed between patients administered anti-EGFR mAb re-challenge first and those administered FTD/TPI plus BEV first, whereas the RR of anti-EGFR mAb re-challenge is significantly higher than that of FTD/TPI plus BEV. Research Sponsor: None.

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Neoadjuvant chemoradiotherapy with ipilimumab and nivolumab in rectal cancer (CHINOREC): A prospective randomized, open-label, multicenter, phase II clinical trial. First Author: Johannes Laengle, Division of Visceral Surgery, Department of General Surgery, Comprehensive Cancer Center Vienna, Medical University of Vienna, Vienna, Austria

Background: Immune checkpoint inhibitors (ICIs) seem only effective in a primed tumor immune microenvironment (TIME). Neoadjuvant radiotherapy (RT) has been shown to induce an immunogenic cell death (ICD) and thereby restores the susceptibility to ICI. This study evaluates the safety of neoadjuvant chemoradiotherapy (CRT) with concomitant ipilimumab (IPI) and nivolumab (NIVO) in curative rectal cancer (RC) patients. **Methods:** The CHINOREC study (ClinicalTrials.gov identifier NCT04124601) is a prospective, randomized (ratio 50:30), open-label, multicenter, phase II investigator-initiated trial (IIT). Patients with RC received neoadjuvant CRT (50 Gy in 2 Gy fractions with concurrent capecitabine 1650 mg/m^2/d) alone or in combination with IPI (1 mg/kg IV at day 7), following 3 cycles of NIVO (3 mg/kg IV Q2W, starting on day 14). Surgical resection was performed 10-12 weeks post CRT. The primary endpoint was surgical safety and feasibility (Clavien-Dindo Classification) of neoadjuvant CRT with sequential IPI and NIVO following surgical resection. Secondary outcome was complete response rate (clinical and pathological). **Results:** Between June 2020 and November 2023, 145 patients were screened and 80 randomly assigned to CRT (n=30) or CRT+IPI/NIVO (n=50). The primary endpoint (any surgical complication) did not differ between the 2 groups (78% vs. 77%). The reoperation rate (≥Grade IIIb) was comparable low in both groups (8% vs. 7%). No new safety signals were identified. The rate of major pathological response (MPR) and complete response (CR) was also comparable high in both arms (38% vs. 37% and 30% vs. 22%, respectively). **Conclusions:** Neoadjuvant IPI/NIVO can be safely applied concomitant with CRT in patients with RC, as it does not increase the rate of reoperation or surgical complications. The validity of the encouraging CR rate needs to be elucidated over time. Further translational analyses will elucidate mechanistic insight regarding improved fraction, dosing and timing of CRT and ICI. Clinical trial information: NCT04124601. Research Sponsor: None.

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Survival benefits of adjuvant chemotherapy after surgical resection of paraaortic lymph node metastasis from colorectal cancer: A multicenter retrospective study. First Author: Hiroaki Nozawa, Department of Surgical Oncology, The University of Tokyo, Tokyo, Japan

Background: Surgical removal of metastasized paraaortic lymph nodes (PALNs) can prolong the survival of certain patients with colorectal cancer (CRC). However, the role of postoperative chemotherapy in such patients remains unknown. **Methods:** This multicenter retrospective study examined97 patients with PALN metastasis from CRC who underwent surgical resection at 36 centers in Japan between 2010 and 2015. Based on adjuvant chemotherapy (AC) after the lymphadenectomy, the patients were classified into Non-AC and AC groups (27 and 70 patients, respectively). After the exclusion of patients receiving irinotecan, the latter group was further categorized into 5-fluorouracil (5-FU) and oxaliplatin (L-OHP) subgroups (14 and 52 patients, respectively) according to the use of L-OHP. Background characteristics and postoperative survival were compared among the groups. **Results:** Marked differences were not seen in background characteristics, except for neoadjuvant treatment, between the Non-AC and AC groups. The AC group exhibited better recurrence-free survival (RFS, p = 0.009) and overall survival (OS, p = 0.040) than the Non-AC group. However, RFS and OS of the 5-FU group did not differ from those of the L-OHP group (p = 0.73 and p = 0.92, respectively). **Conclusions:** AC contributed to the improved prognosis of patients after the removal of PALN metastasis from CRC, but L-OHP did not offer additional survival benefits. Prospective studies comparing Non-AC with 5-FU- and L-OHP-based AC are needed to confirm these findings. Research Sponsor: None.

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AGITG SPAR: A randomized, placebo-controlled, phase II trial of simvastatin in addition to neoadjuvant chemotherapy and radiation for rectal cancer. First Author: Michael Jameson, Waikato Clinical Campus, University of Auckland, Hamilton, New Zealand

Background: Retrospective clinical studies and preclinical studies demonstrated that statin use during preoperative (chemo)radiation (pCRT) for rectal cancer is associated with improved survival, response, and toxicity. Tumor regression following pCRT has strong prognostic significance and can be assessed using MRI-based tumor regression grading (mrTRG), including with non-operative management. SPAR was designed to prospectively evaluate the benefits of adding simvastatin (SIM) to pCRT on tumor regression and gastrointestinal (GI) adverse events (AE). **Methods:** SPAR is a double-blind randomized phase 2 trial investigating SIM/placebo (PBO) in addition to long-course fluoropyrimidine-based pCRT for rectal adenocarcinoma. Stratification included trial site, clinical T stage (<4 vs 4), clinical N stage (<2 vs 2), either mesorectal fascia involvement (MRFI) or extramural venous invasion (EMVI) on MRI (yes vs no), and total neoadjuvant therapy (TNT): induction vs consolidation chemotherapy vs none. Study treatment was SIM 40mg/PBO daily for 90 days, starting 1 week prior to pCRT; recent statin use was excluded. Pelvic MRI was repeated 6-8 weeks after pCRT to determine mrTRG. An amendment in January 2022 allowed TNT with either induction or consolidation chemotherapy; the timing of post-pCRT MRI remained unchanged. The design was amended to open-label due to PBO supply issues. Primary objective: rate of centrally assessed grade 1-2 mrTRG. Secondary objectives include centrally assessed favorable pathologic TRG (pathTRG), safety and cancer outcomes. Analysis was by intention-to-treat. **Results:** Between April 2018 - November 2023, 135 of 222 planned participants from 17 sites in Australia and New Zealand were randomized (68 SIM; 67 no SIM). Recruitment was hampered by the COVID-19 pandemic and adoption of TNT as standard care before the protocol amendment. Participant characteristics: median age 59 years; 85 (63%) males; 118 (87%) T2-3 disease, 80 (59%) N0-1 disease; 55 (41%) MRFI or EMVI; 25 (19%) had TNT. Rates of grades 1-2 mrTRG with SIM vs no SIM were 38.5% and 29.7% (X^2 = 1.09, p = 0.30), respectively. There was no significant difference in rates of > grade 2 GI and non-GI AE. Four serious AE were recorded, none related to study treatment. Median follow-up was 3.2 years. 3-year local recurrence rates (LRR) were low: 1 (2.2%) and 3 (5.5%) with SIM vs no SIM, respectively (HR: 0.29, 95% CI: 0.03 to 2.67, p = 0.26). 3-year disease-free survival (DFS) with SIM vs no SIM was 84% and 72%, respectively (HR 0.48; 95% CI: 0.21–1.08, p = 0.07). **Conclusions:** The rates of favorable mrTRG, 3-year LRR and DFS were numerically better with the addition of SIM to pCRT, though the differences were not statistically significant. SIM was well tolerated. Interpretation is limited by reduced sample size and statistical power. PathTRG and other key outcomes will be presented at the meeting. Clinical trial information: ACTRN12617001087347. Research Sponsor: Health Research Council of New Zealand; 18/233; Cancer Society of New Zealand; 15.17; Cancer Australia; 1125958; Cancer Council New South Wales; RG 17-07.

LBA143 Oral Abstract Session

First results of nivolumab (NIVO) plus ipilimumab (IPI) vs NIVO monotherapy for microsatellite instability-high/mismatch repair-deficient (MSI-H/dMMR) metastatic colorectal cancer (mCRC) from CheckMate 8HW. First Author: Thierry Andre, Sorbonne Université, and Hôpital Saint Antoine, Assistance Publique Hôpitaux de Paris, Paris, France

The full, final text of this abstract will be available at meetings.asco.org on the day of presentation and in the online supplement to the February 1, 2025, issue of the *Journal of Clinical Oncology*.

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Fruquintinib plus best supportive care for patients with metastatic colorectal cancer: Characterization of patients who had an overall survival of ≥10 months in the FRESCO-2 study. First Author: Stefan Kasper, University Hospital Essen, Essen, Germany

Background: Fruquintinib (F) is a highly selective, oral tyrosine kinase inhibitor of all 3 vascular endothelial growth factor receptors (VEGFRs -1, -2, -3) that is approved by the US FDA and in the EU, UK, and Japan for previously treated metastatic colorectal cancer (mCRC), regardless of biomarker status. FRESCO-2 (NCT04322539) met its primary endpoint demonstrating significantly improved overall survival (OS) with F + best supportive care (BSC) vs placebo (P) + BSC. This subgroup analysis assessed landmark survival of all FRESCO-2 patients (pts), and baseline (BL) characteristics and safety data of pts treated with F who had OS ≥10 months (mos) compared with the intent-to-treat (ITT) population (pop). **Methods:** Pts were randomized 2:1 to receive F 5 mg or matching P, by mouth, once daily, 3 weeks on, 1 week off, + BSC. Pts had received prior chemotherapy, anti-VEGF therapy and, if *RAS* wild type, anti-EGFR therapy; and had prior exposure to trifluridine/tipiracil (TAS-102) and/or regorafenib. The hazard ratio (HR) between the two treatment arms was calculated from a stratified Cox proportional hazard model with treatment group as the only covariate in the model. **Results:** In the ITT pop, OS was improved with F vs P (HR 0.66); OS rates (95% confidence intervals [CIs]) at 6 and 9 mos were 60.4% (55.9–64.9) vs 41.5% (35.0–48.0) and 41.1% (36.4–45.8) vs 28.2% (22.1–34.3), respectively. Progression-free survival (PFS) rates (95% CIs) at 6 and 9 mos were 23.8% (19.7–28.0) vs 1.1% (0–2.6) and 11.3% (8.1–14.6) vs 0.5% (0–1.6), respectively. Of 461 pts who received F, 113 (24.5%) had OS ≥10 mos (range 10.0–18.9) with a median duration of F treatment of 6.3 mos (range 0.7–19.1). Among pts receiving F, a higher proportion (≥10% difference) of pts with OS ≥10 mos had no liver metastases (mets) and an Eastern Cooperative Oncology Group performance status (ECOG PS) of 0 at BL compared with the ITT pop (Table). In pts with OS ≥10 mos, 61.9% experienced a grade ≥3 treatment-emergent adverse event compared with 62.7% in the ITT pop. **Conclusions:** Longer-term OS and PFS rates at 6 and 9 mos were higher with F vs P in the FRESCO-2 ITT pop. The data show a higher proportion of pts with OS ≥10 mos had an absence of liver mets and ECOG PS 0 at BL than the ITT pop. The safety profile of F in pts with OS ≥10 mos was consistent with the ITT pop. Clinical trial information: NCT04322539. Research Sponsor: HUTCHMED.

BL characteristics of pts treated with F with OS ≥10 mos vs ITT pop.

BL characteristic	Pts with OS ≥10 mos (n=113)	ITT pop (n=461)
ECOG PS (0 / 1), %	54.0 / 46.0	42.5 / 57.5
Median time since 1st CRC diagnosis, mos	52.0	47.2
Primary tumor location at 1st diagnosis (colon / rectum / both), %	51.3 / 35.4 / 13.3	60.5 / 31.0 / 8.5
Primary site at 1st diagnosis (colon left / colon right), %	42.5 / 15.9	41.6 / 21.0
Liver mets, %	58.4	73.5
Median duration of mCRC, mos	42.7	37.9
Prior lines of therapy for mCRC (≤3 / >3), %	26.5 / 73.5	27.1 / 72.9
Prior TAS-102, %	46.9	52.1
Prior regorafenib, %	12.4	8.7

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Neoadjuvant therapy in T3 and T4 colon cancer: Is there a promising future? First Author: Paola Pena, Roger Williams Medical Center, Providence, RI

Background: Neoadjuvant chemotherapy (NAC) is increasingly being considered over standard upfront surgery followed by adjuvant chemotherapy for locally advanced colon cancer. Studies have been underpowered to look at the impact of NAC on overall survival (OS) in individual clinical T stages. **Methods:** A nationwide retrospective analysis using the National Cancer Database to evaluate patients with clinical stage T3 and T4 colon adenocarcinomas from 2010 to 2020. Cox proportional hazards model was used to evaluate the impact of NAC approach compared to upfront surgery on OS. Clinical and pathological T stages were compared among those who underwent upfront surgery to evaluate accuracy of clinical T staging. **Results:** Total of 60,557 patients were identified - 2,313 patients undergoing NAC and 58,245 undergoing upfront surgery. In the past decade, use of NAC increased from 1.73% to 12.13%. In our multivariable survival analysis combining clinical T3 and T4 patients, we observed a worse OS in the NAC group (HR 1.16, 95% CI 1.07-1.26). We found no OS benefit for patients undergoing NAC approach for clinical stage T3 (HR 1.13, 95% CI 0.99-1.28) but found a significant OS benefit for patients undergoing NAC approach for clinical stage T4 (HR 0.85, 95% CI 0.77-0.95). We observed that clinical T4 stage was accurately staged in 90.48% with overstaging in 9.52%; clinical T3 stage was accurately staged in 95.24% with overstaging in 2.18% of patients. **Conclusions:** Our analysis demonstrated OS benefit of the NAC approach in T4 colon adenocarcinoma but not in T3. The study highlights the importance of evaluating survival benefit of NAC separately for T3 and T4 colon adenocarcinoma in future studies, and the challenges of accurate clinical staging of colon cancer for NAC planning. Research Sponsor: None.

145 **Poster Session**

Updated analysis of a phase 2 study of fruquintinib plus trifluridine/tipiracil (TAS-102) as third-line treatment in patients with metastatic colorectal adenocarcinoma. First Author: Jianjun Peng, Center of Gastrointestinal Surgery, The First Affiliated Hospital, Sun Yat-sen University, Guangzhou, China

Background: Fruquintinib and TAS-102 are two standard therapies for patients (pts) with previous-treated mCRC worldwide. The study aimed to evaluate the efficacy and safety of fruquintinib plus TAS-102 as third-line treatment for mCRC. We previously reported that fruquintinib plus TAS-102 was well tolerated with encouraging clinical activity in pretreated mCRC. Here, we reported the updated results at the data cutoff of Sep 3, 2024. **Methods:** In this open-label, single-arm, multi-center, phase 2 trial (NCT05004831), pts with mCRC who had failed at least two prior standard treatment regimens were enrolled. Eligible pts received fruquintinib (4mg, qd, d1-21) and TAS-102 (35mg/m^2, bid, d1-5, 8-12) orally every 4 weeks until disease progression or unacceptable toxicity. Primary endpoint was PFS (per RECIST v1.1). Secondary endpoints were OS, ORR, DCR and safety (per NCI-CTCAE v5.0). **Results:** From Mar 2022 to Aug 2023, 50 eligible pts were enrolled. Median age was 60 years (range 39-76) and 58.0% were male. 82% pts were of left-sided colon and rectal cancer, 42.0% pts were *RAS* mutant, 58.0% had liver metastasis and 18.0% had peritoneal metastasis. Prior treatments and proportion were standard chemotherapy 100%, anti-VEGF 88.0% and anti-EGFR 26.0%. As of Sep 3, 2024, ORR was 10.9% (95%CI: 3.6-23.6) and DCR was 73.9% (95%CI: 58.9-85.7). The median PFS was 6.33 months (m) (95% CI: 4.20-8.62). The 6-m, 9-m and 12-m PFS rates were 53.0% (95% CI: 40.2-70.0), 28.3% (95% CI: 17.4-45.9) and 23.1% (95% CI: 13.2-40.5), respectively. At a median follow-up of 17.6m, median OS was 18.4m (95% CI: 12.0-NA). The 6-m, 9-m and 12-m OS rates were 87.0% (95% CI: 77.8-97.3), 66.9% (95% CI: 54.0-82.9) and 64.3% (95% CI: 51.1-80.8), respectively. Median PFS was comparable in liver metastasis (LM) and non-LM pts (6.33m [95%CI: 4.13-8.62] vs. 6.46m [95%CI: 3.74-NA], *P*=0.54). Similar results were observed in peritoneal metastasis (PM) and non-PM pts (6.07m [95%CI: 3.74-NA] vs. 6.33m [95%CI: 4.20-8.27], *P*=0.95). None of *RAS* gene mutation, prior treatment lines $\geq$ 3, age $\geq$ 65 years, metastatic sites $\geq$ 2, peritoneal metastasis and right-sidedness were identified as factors significantly associated with PFS or OS (by univariable Cox proportional-hazards model). The most common treatment-related adverse events (TRAEs) of any grade and grade $\geq$ 3 were mainly hematological toxicities. TRAEs (any grade, grade $\geq$ 3) in $\geq$ 50% of pts including neutrophil count decreased (80.0%, 54.0%), white blood cell count decreased (70.0%, 26.0%), anemia (58.0%, 20.0%) and proteinuria (50.0%, 4%). No treatment-related deaths were observed. **Conclusions:** The updated analysis demonstrated encouraging survival benefits of fruquintinib plus TAS-102 as third-line treatment in patients with mCRC with acceptable toxicities. This regimen could be an alternative therapeutic approach for these patients. Clinical trial information: NCT05004831. Research Sponsor: None.

146 **Poster Session**

Phase I/II study of cabozantinib alone or in combination with panitumumab in patients with *MET*-amplified metastatic colorectal cancer. First Author: Saori Mishima, Department of Gastroenterology and Gastrointestinal Oncology, National Cancer Center Hospital East, Kashiwa, Japan

Background: Despite the introduction of anti-EGFR therapies, resistance remains a significant challenge in metastatic colorectal cancer (mCRC). MET amplification is a recognized resistance mechanism. Cabozantinib, a multi-kinase inhibitor targeting MET, has shown promise in overcoming resistance to anti-EGFR therapy. This study aimed to evaluate the efficacy and safety of cabozantinib alone (Cabo) or in combination with panitumumab (Cabo+Pmab) in patients with MET-amplified mCRC. Patients and **Methods:** The study population included patients with RAS and BRAF wild-type mCRC who had progressed on anti-EGFR therapy and subsequently confirmed MET amplification by circulating tumor DNA. The phase I part aimed to establish the recommended cabozantinib dose in combination with panitumumab by assessing dose-limiting toxicity (DLT) within the first four weeks. The primary endpoint of phase II part was objective response rate (ORR) in patients receiving the study treatment as third-line therapy or later. In phase II part, patients were randomized to Cabo or Cabo+Pmab group. Based on a threshold ORR of 1.6% response rate and an expected efficacy of 30%, the planned sample size was 14 patients with one-sided alpha of 1.25% and power of 80% in each arm, Cabo and Cabo+Pmab. A tumor response of at least 3 out of 14 patients would warrant further investigation of these treatments. **Results:** A total of 35 patients were enrolled (6 in phase I and 29 in phase II, 28 for the primary analysis; 13 in Cabo and 15 in Cabo+Pmab). In phase I, no DLTs were observed. This allowed researchers to set the maximum tolerated dose and recommend a dosage of 60 mg/day of cabozantinib in combination with panitumumab for phase II. Of 28 pts for the primary analysis, ORR was 7.7% (1/13; 95%CI 0.2-36%) in Cabo group and 0% (0/15; 95%CI 0-21.8%) in Cabo+Pmab group, median progression-free survival was 3.1 months in both groups, and cumulative proportion of pts with grade $\geq$3 treatment-related adverse events was 38.5% (including hypertension [23.1%]) in Cabo and 60% (including hypomagnesaemia [20%]) in Cabo+Pmab group. **Conclusions:** The combination of cabozantinib 60 mg plus panitumumab had manageable safety profile. However, this treatment did not demonstrate significant clinical benefit in patients with MET-amplified mCRC. Further investigation is warranted to explore alternative therapeutic strategies for this patient population. Clinical trial information: jRCT1080224637. Research Sponsor: Takeda Pharmaceutical Co., Ltd.

147 **Poster Session**

A multicenter, prospective phase II trial of second-line aflibercept plus FOLFIRI in patients with metastatic colorectal cancer refractory to anti-EGFR agents (HGCSG1801): Updated analyses. First Author: Hiroshi Nakatsumi, Department of Gastroenterology, National Hospital Organization Hokkaido Medical Center, Sapporo, Japan

Background: The HGCSG1801 trial evaluated the treatment outcomes of aflibercept (AFL) plus FOLFIRI for patients(pts) with metastatic colorectal cancer (mCRC) refractory to an oxaliplatin-based regimen combined with an anti-EGFR agent. Here we report results of the updated efficacy, safety, and a biomarker analysis. **Methods:** This was a prospective open-label phase II trial. AFL (4 mg/kg iv) followed by FOLFIRI (irinotecan 180 mg/m^2, leucovorin 200 mg/m^2 iv, bolus 5-fluorouracil [5-FU] 400 mg/m^2, and infusional 5-FU 2400 mg/m^2/46 h) was given every 2 weeks until progression or unacceptable toxicities. The primary endpoint was 6-month progression-free survival (PFS) rate, and the secondary endpoints included overall survival (OS), PFS, overall response rate (ORR), disease control rate (DCR), and adverse events. Angiogenic factors were analyzed in plasma sample using a Luminex multiplex assay using the Cox proportional hazards model. This study was sponsored by the Non-Profit Organization Hokkaido Gastrointestinal Cancer Study Group and supported by a grant from Sanofi. **Results:** Forty-three pts were enrolled between November 2019 and October 2022. The data cut-off date was April 30, 2024. The 6-month PFS rate was 59.0% (90% confidence interval [CI], 45.8–72.1%). Median PFS and OS were 7.3 months (95% CI, 5.5–11.0 months) and 18.8 months (95% CI, 12.9–26.6 months), respectively. The ORR was 20.9% (95% CI, 10.0–36.0%) and DCR was 88.4% (95% CI, 74.9–96.1%). No deaths and no new safety signals with a causal relation to the study treatment were observed. A biomarker analysis using pretreatment plasma samples showed a trend toward better PFS in pts with low VEGF-A (hazard ratio [HR] 0.40 [95% CI, 0.18-0.91]), TSP-2 (HR 0.40 [95% CI, 0.18-0.88]) or IL-8 levels (HR 0.43 [95% CI, 0.20-0.93]). There was also a trend toward better OS in pts with low levels of TSP-2 (HR 0.30 [95% CI, 0.11-0.81]) or TIMP-1 (HR 0.20 [95% CI, 0.06-0.69]). **Conclusions:** These updated data further support the activity and manageable safety profile of AFL plus FOLFIRI for pts with mCRC who failed an oxaliplatin-based regimen combined with an anti-EGFR agent. It was suggested the association between some angiogenic factors in plasma samples and the activity of AFL plus FOLFIRI in mCRC. Clinical trial information: jRCTs011190006. Research Sponsor: Non-Profit Organization Hokkaido Gastrointestinal Cancer Study Group; Sanofi; SGZ-2017-11875.

148 **Poster Session**

Phase II study of organ preservation (OP) in node-negative (NN) low rectal cancer (RC): Updated clinical and patient reported outcomes (PROs). First Author: Vanessa Wookey, Fox Chase Cancer Center, Philadelphia, PA

Background: Total mesorectal excision (TME) is a highly effective treatment for rectal cancer (RC) but is associated with significant morbidity and mortality. Organ preservation (OP) approaches for locally advanced RC have been successful but not well studied for node-negative (NN) low RC. The objective of this investigator-initiated trial (IIT) is to determine the feasibility of performing successful local excision (LE) after neoadjuvant chemotherapy (CTX) in NN low RC. **Methods:** With IRB approval, patients (pts) with clinical stage T1-3, N0 low rectal adenocarcinomas ($<$6 cm from anal verge) were included in this single arm phase II IIT. Pts received 6 cycles of FOLFOX (5-FU bolus 400 mg/m2 and infusion 2400 mg/m2, Leucovorin 400 mg/m2, Oxaliplatin 85 mg/m2). Those with evidence of a response underwent LE (transanal excision) 6-12 weeks later. To target occult nodal metastases and reduce in-bowel recurrences, LE was followed by chemoradiotherapy ([CXRT] capecitabine 825 mg/m2 and long course RT to 54 Gy). The primary endpoint was the proportion of pts with successful LE after neoadjuvant CTX (NCT03548961). **Results:** Nineteen pts with low RC were enrolled; nine were female, and the mean age at diagnosis was 65 years. T stage was as follows: T1- 1(5%), T2 - 11(58%), T3 - 7(36%). Eighteen (95%) pts completed at least 5 cycles of neoadjuvant Ctx and 16/19 (84%) underwent LE. Negative margins were achieved in 79% (15/19), meeting our primary endpoint ($p < 0.001$). Ten (53%) were downstaged, and complete pathological response was achieved in 5/16 (31%) of pts who underwent LE. Four (21%) pts (all T2 stage) did not meet the primary endpoint (3 with inadequate response to CTX and 1 with LE and positive margins). Of these, three were salvaged (2 with CXRT; 1 with TME), and one died of infection (unrelated complication). All 16 who underwent LE completed CXRT. Of the 15 pts with successful OP at a median follow-up of 25.5 months, no local recurrences have been reported and 1 pt with lung nodules below the threshold for metastases at study entry ultimately developed clear lung metastases. Median DFS was not reached. Patient reported outcomes (PRO) assessment revealed a slight decline in physical health scores after neoadjuvant CTX but improved to baseline at follow-up. No decline was seen in mental health scores. Symptom specific QOL (sexual and bowel function) and long-term survival data will be presented. **Conclusions:** Neoadjuvant CTX and LE allows for OP in NN low RC and results in a margin negative LE in over three fourths of pts with durable long term local control and preserved QOL. Ongoing trials are investigating an OP approach in NN RC (NCT03259035). Our approach adds to the mounting evidence and will provide early data of the benefit of adjuvant chemoradiation to target occult nodes in this setting. Clinical trial information: NCT03548961. Research Sponsor: Colon Cancer Coalition and Institutional Funds.

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Optimal cut-off value for lymph node harvesting in colon cancer based on tumor stage and sidedness: A retrospective analysis of a Japanese nationwide cohort. First Author: Kazuma Ito, Division of lower GI surgery, Department of Gastroenterological Surgery, Hyogo Medical University, Nishinomiya, Hyogo, Japan

Background: The presence of fewer than 12 harvested lymph nodes (LN) is currently considered a risk factor in stage II colon cancer (CC). However, this cut-off value does not take into consideration the biological differences associated with tumor sidedness. Moreover, the applicability of this cut-off value to stage III CC is unknown. **Methods:** Patients with pathological stage II/III CC who underwent curative surgical resection were assessed using data from the Japanese Society for Cancer of the Colon and Rectum database (2001–2015). The optimal cut-off value of harvested LN was evaluated by analyzing all cut-off values (i.e., 3–40), taking into account the point estimate of the hazard ratio (HR) and lower limit of the 95% confidence interval (CI), according to the tumor stage (II/III) and sidedness (right/left). A Cox proportional hazards model was used to investigate the impact of the number of harvested LN on overall survival, adjusting for age, sex, T stage, histology, size of tumor, sidedness, adjuvant chemotherapy and extent of colon resection. Multivariate analysis was also conducted. **Results:** From the data, 4249 (left/right; 3248/1001) and 3549 (left/right; 2798/751) patients with stage II and III CC, respectively, were identified. The median number of harvested LN was identical between stage II (18, interquartile range [IQR] (12–15)) and III (18, IQR (13–25)). However, this number was higher in right-sided CC (21, IQR (14–31)) versus left-sided CC (17, IQR (12–24)). In stage II, the conventional cut-off value (i.e., 12 LN) was identified as a prognostic factor in the multivariate analysis (HR 1.45; 95% CI: 1.20–1.75; p<0.001). Nevertheless, the prognostic impact of the number of harvested LN was stronger when the optimal cut-off value was set to 7 (HR 1.88; 95% CI: 1.44–2.44; p<0.001). In stage III, the optimal cut-off value was 12 (HR 1.88; 95% CI: 1.44–2.44; p<0.001). When evaluated based on tumor sidedness, the optimal cut-off value differed, especially for stage III. Data are shown in the table. Similar results were obtained following the exclusion of the extent of bowel resection as variable in Multivariate analysis. **Conclusions:** While the presence of 12 harvested LNs remains a valid prognostic factor, the optimal cut-off value can be further evaluated based on tumor stage and sidedness. Research Sponsor: None.

	Stage II			Stage III		
	All	Left	Right	All	Left	Right
Cut-off value	12	12	12	12	12	12
N (>cut-off/<cut-off)	3209/1040	2371/877	838/163	2814/735	2181/617	633/118
HR (>cut-off/<cut-off) (95% CI)	1.45 (1.20-1.75)	1.49 (1.20-1.84)	1.31 (0.85-2.02)	1.30 (1.09-1.55)	1.27 (1.05-1.54)	1.43 (0.97-2.13)
P value	<0.001	<0.001	0.216	<0.003	0.015	0.073
Cut-off value	7	7	10		20	13
N (>cut-off/<cut off)	3930/319	2977/271	889/112		1200/1598	609/142
HR (>cut-off/<cut-off) (95% CI)	1.88 (1.44-2.44)	2.07 (1.56-2.74)	1.39 (0.84-2.27)		1.35 (1.13-1.60)	1.59 (1.11-2.28)
P value	<0.001	<0.001	0.198		<0.001	0.011

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Short-course radiotherapy-based total neoadjuvant therapy combined with tislelizumab in the treatment of locally advanced rectal cancer (Neo-STAR): Early outcomes of a prospective, randomized phase II trial. First Author: Fengpeng Wu, Department of Radiation Oncology, the Fourth Hospital of Hebei Medical University, Shijiazhuang, China

Background: It has been found that radiotherapy possess synergistic anticancer effect with immune checkpoint inhibitors (ICIs). Therefore, this phase II randomized clinical trial (RCT) (ClinicalTrials.gov identifier: NCT05086627) aimed to evaluate the efficacy and safety of SCRT followed by CAPOX and tislelizumab in patients with LARC. **Methods:** Patients initially diagnosed rectal adenocarcinoma with cT1-2N+M0 or cT3-4NanyM0 were recruited, randomly assigned to receive SCRT (25Gy/5f), followed by four cycles of CAPOX combined with tislelizumab or CAPOX alone, respectively. After total mesorectal excision (TME), two cycles of postoperative adjuvant chemotherapy were given according to the patient's preferences. The pCR rate was set as the primary endpoint, and the major pathological response (MPR) (tumor regression grade (TRG) 0+1), 3-year progression free survival (PFS), 3-year overall survival (OS) and treatment safety were set as the secondary endpoints. **Results:**Between September 2021 and March 2024, 118 patients were randomly assigned and 111 patients started allocated treatment, with 53 in the trial group and 58 in the control group. Ninety-six patients (86.5%) completed SCRT and 4 cycles of CAPOX chemotherapy with/without tislelizumab, including 49 patients (92.5%, 49/53) in the trial group and 47 patients (81.0%, 47/58) in the control group. Eighty-two patients (85.4%) had surgical resection, including 42 patients (85.7%, 42/49) in the trial group and 40 patients (85.1%, 40/47) in the control group. Among 82 patients who underwent surgery, 54.8% (23/42) patients in the trial group achieved pCR compared with 37.5% (15/40) in the control group. During the neoadjuvant treatment period, the incidence of grade 3-4 AEs were 9.4% in the trial group and 10.3% in the control group. **Conclusions:** SCRT-based TNT combined with tislelizumab followed by TME exhibited a favorable pCR rate in LARC. Clinical trial information: NCT05086627. Research Sponsor: Hebei Province Science and Technology Department; 22377733D.

Surgical and postoperative pathological outcomes.

Characteristic	Experimental group (n=42)	Control group (n=40)
R0 resection		
Yes	42 (100%)	40 (100%)
No	0 (0%)	0 (0%)
pCR		
Yes	23 (54.8%)	15 (37.5%)
No	19 (45.2%)	25 (62.5%)
TRG score		
0	23 (54.8%)	15 (37.5%)
1	3 (7.1%)	2 (5%)
2	14 (33.3%)	15 (37.5%)
3	2 (4.8%)	8 (20%)
Pathological T stage		
T0	24 (57.1%)	15 (37.5%)
Tis	1 (2.4%)	0 (0%)
T1	0 (0%)	1 (2.5%)
T2	9 (21.4%)	6 (15%)
T3	6 (14.3%)	10 (25%)
T4	2 (4.8%)	8 (20%)
Pathological N stage		
N0	39 (92.9%)	36 (90%)
N1	3 (7.5%)	3 (7.5%)
N2	0 (0%)	1 (2.5%)
Type of surgery		
Anterior resection	36 (85.7%)	24 (60%)
Abdominoperineal resection	6 (14.3%)	13 (32.5%)
Hartmann procedure	0 (0%)	3 (7.5%)
Anus preservation		
Yes	36 (85.7%)	27 (67.5%)
No	6 (14.3%)	13 (32.5%)

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Fruquintinib combined with FOLFIRI/mFOLFOX6 as second-line treatment in patients with RAS-mutant metastatic colorectal cancer (mCRC): A multicenter, open-label, phase II study. First Author: Yun Xu, Department of Colorectal Surgery, Fudan University Shanghai Cancer Center, Shanghai, China

Background: Anti-angiogenic drugs combined with chemotherapy could bring good clinical benefit to patients (pts) with mCRC, but the efficacy in pts with RAS mutant mCRC was limited. Fruquintinib is a highly selective oral inhibitor of vascular endothelial growth factor receptors (VEGFRs -1, -2, and -3) and was approved for previously treated mCRC. Our study was to explore the efficacy and safety of fruquintinib combined with FOLFIRI/mFOLFOX6 in the second line treatment of RAS-mutant mCRC (NCT05634590). **Methods:** Eligible pts with histologically confirmed RAS-mutant mCRC who had received first-line standard chemotherapy regimens were included in the study, and pts received Fru (4 mg, p.o. qd, 3 weeks on/1 week off, q4w) plus FOLFIRI/mFOLFOX6 (d1-15, q4w) until disease progression, unacceptable toxicities, or withdrew consent. The primary endpoint was progression-free survival (PFS). Secondary endpoints included objective response rate (ORR), disease control rate (DCR), overall survival (OS)and safety. **Results:** As of Aug 30, 2024, 25 eligible pts were enrolled and received at least one time treatment. Baseline characteristics included median age (66.0 [range: 35-73]), female/male (56.0% / 44.0%), ECOG PS 1(68.0%), left-sided and right-sided colon (72.0% / 28.0%), liver metastasis (64.0%), prior anti-VEGF therapies (76.0%). 14 pts had received at least one tumor assessment. Five pts achieved partial response (PR), 9 pts achieved stable disease (SD), and the ORR was 35.7% (5/14) (95% confidence interval [CI]: 12.8-64.9), the DCR was 100.0% (14/14) (95% CI: 76.8-100.0). Compared to liver metastases, the ORR was higher without liver metastases (42.9% [95%CI 9.9-81.6]). The median PFS was 6.41 months (95% CI: 6.37 - NA), and median OS was not reached yet. In the subgroup analysis, Median PFS was also longer without liver metastases pts than with liver metastases (8.84 mo vs 6.41 mo; HR=1.230; 95%CI 0.2462-6.148; p=0.7775). Additionally, Grade ≥3 treatment-emergent adverse events (TEAEs) included platelet count decreased (18.2%), hypertension (9.1%) and neutrophil count decreased (9.1%). **Conclusions:** Fruquintinib combined with FOLFIRI/mFOLFOX6 as second-line in patients with RAS-mutant mCRC achieved a promising clinical benefit, with a manageable safety profile. This trial is ongoing and more updated clinical data will be presented in the future. Clinical trial information: NCT05634590. Research Sponsor: None.

152 Poster Session

A retrospective observational study to evaluate the efficacy of trifluridine/tipiracil ± bevacizumab in metastatic colorectal cancer with MSI-high/deficient MMR. First Author: Rin Inamoto, Department of Gastroenterology, Saitama Cancer Center, Kitaadachi-Gun, Saitama, Japan

Background: MSI-high (MSI-H)/deficient MMR (dMMR) metastatic colorectal cancer (mCRC) have reported to be resistant to various cytotoxic agents including fluorouracil. On the other hand, preclinical study showed that trifluridine was effective to fluorouracil-refractory dMMR CRC cell lines. Previous studies on trifluridine/tipiracil (FTD/TPI) ± bevacizumab (BEV) for mCRC as later line treatment showed an objective response rate (ORR) of 1.1-5.6% and disease control rate (DCR) of 44-76.6%. However, there are no reports on the efficacy of FTD/TPI± bevacizumab in MSI-H/dMMR mCRC patients. **Methods:** We retrospectively evaluated the efficacy and safety of FTD/TPI ± BEV as second- or later-line treatment which patients with MSI-H/dMMR mCRC received between June 2012 and January 2023 at the 10 institutions. The primary endpoint was investigator-assessed objective response rate (ORR). The secondary endpoints were progression-free survival (PFS), overall survival (OS), disease control rate (DCR), and adverse event rates. We also compared the FTD/TPI + BEV (FTB group) with FTD/TPI (FT group). **Results:** A total of 18 patients (FTB/FT, 9/9) were included. Patient characteristics were as follows (FTB/FT): median age, 70 (70/70) years; ECOG PS of 0/1, 7 (5/2)/11(4/7); primary tumor location of right side/left side, 12(6/6)/6(3/3); *BRAF* V600E mutant/wild-type, 6(3/3)/11(6/5); number of previous treatment lines of 1/≥2, 2(1/1)/16(8/8); number of metastatic sites of 1-2/≥3, 8(4/4)/10(5/5); prior use of anti-PD-1 therapy, 10(3/7). Efficacies in the whole population were as follows: ORR, 16.7% (95%CI, 3.6-41.4); DCR, 72.2%; median (m) PFS, 5.5 months; mOS, 11.8 months. Efficacies in each group (FTB vs. FT) were as follows: ORR (22.2% vs. 11.1%) and DCR (88.9% vs. 55.6%) were favored in FTB group than in FT group, while the median PFS were similar between two groups (5.6 months vs. 5.2 months). OS in FTB group was longer than in FT group (median, 18.9 months vs. 7.1 months; hazard ratio, 0.22; p value=0.03). Of 11 (FTB/FT, 7/4) patients who received subsequent treatment, 6 in FTB group and 1 in FT group received anti-PD-1 therapy. The most common grade 3 or more adverse events in each group (FTB vs. FT) were neutropenia (77.8% vs.77.8%), anemia (22.2% vs. 33.3%) and febrile neutropenia (0% vs. 22.2%). **Conclusions:** FTD/TPI ± BEV showed a promising efficacy and favorable safety. Although this was retrospective study with a small sample size, FTD/TPI ± BEV therapy may have better efficacies for MSI-H/dMMR mCRC than those for MSS/proficient MMR mCRC in previous prospective studies. Next generation sequencing as biomarker analysis using pretreated tissue samples is ongoing. Research Sponsor: TAIHO PHARMACEUTICAL.

153 Poster Session

Cetuximab every second week in combination with daily encorafenib in patients with BRAF V600E mutated metastatic colorectal cancer: Interim analysis from NEW BEACON. First Author: Martina Eriksen, Department of Oncology, Copenhagen University Hospital - Rigshospitalet, Copenhagen, Denmark

Background: Weekly administered cetuximab in combination with the BRAF inhibitor encorafenib has improved survival in pre-treated patients (pts) with BRAF V600E mutated (BRAFmut) metastatic colorectal cancer (mCRC), with a median progression free survival (PFS) of 4.3 months in the BEACON study. It is now standard-of-care in the second line treatment. However, a regimen with cetuximab administered every second week may demonstrate comparable efficacy and is more convenient for the pts. Thus, we initiated the NEW BEACON study investigating cetuximab given every second week together with daily encorafenib. Details on the rationale and aims of the study have previously been published (doi: 10.1186/s12885-022-10420-x.). **Methods:** NEW BEACON is an open-label, single-arm, multicenter, phase II study including pts with pre-treated BRAFmut mCRC. The primary end point is 2 months PFS rate. We report here the result of a pre-specified interim analysis after treatment of the first 19 included pts. We collected fresh tumor tissues for comprehensive molecular profiling in order to explore features associated with response and resistance. Baseline genomic variants will be reported. **Results:** From February 2021 to February 2024 20 pts were included in the study. 1 patient never started study treatment due to rapid disease progression. At the data cut-off date 19 pts had initiated study treatment. The 2 months PFS rate was 89% (17/19 pts). We had tumor tissue for molecular profiling available from 12 pts. Pathogenic variants in TP53 were the most frequently detected co-mutations at baseline in 83% of pts (10/12 pts). Available preliminary safety, efficacy, and survival data and more extensive genomic data will be presented at the meeting. **Conclusions:** The preliminary PFS data from pts with BRAFmut mCRC treated with cetuximab every second week together with daily encorafenib is promising demonstrating a 2 months PFS rate of 89%. This is similar to what was reported from the BEACON study (2 months PFS rate of 84%). Loss-of-function mutations in tumor suppressor genes were frequently detected as co-mutations at baseline. Clinical trial information: EudraCT: 2020-003283-10. Research Sponsor: Kræftens Bekæmpelse (The Danish Cancer Society); Pierre Fabre.

155 Poster Session

Phase II study of trifluridine/tipiracil (TAS-102) plus irinotecan in combination with bevacizumab as a second-line therapy for patients with metastatic colorectal cancer. First Author: Yongkun Sun, National Cancer Center Cancer Hospital, Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing, China

Background: Metastatic colorectal cancer (mCRC) patients have poor prognoses with limited response to second-line chemotherapy. Therefore, it is urgent to refine and design an optimal second-line treatment regimen to improve the response rate and prolong the survival of patients with mCRC. This study assessed the safety and efficacy of the trifluridine/tipiracil (TAS-102), irinotecan, and bevacizumab regimen in second-line settings for mCRC patients. **Methods:** This was a multicenter, single-arm, phase II trial. The mCRC patients who are refractory to standard first-line treatment, including fluoropyrimidine and oxaliplatin, are eligible. Based on the recommended phase II dose established in phase I, enrolled patients received TAS-102 (30 mg/m2 twice daily on days 1-5) and irinotecan (150 mg/m2 on day 1) combined with bevacizumab (5 mg/kg on day 1) every two weeks. The primary endpoint was the objective response rate (ORR). **Results:** Between October 1, 2023, and August 31, 2024, 60 patients were enrolled, and all of them were evaluated for efficacy. As of August 31, 2024, efficacy was assessed in all 60 participants with an ORR of 18.3% (2 had complete response, CR; 9 had partial response, PR; 39 had stable disease, SD). Among the 39 patients with SD, 29 patients experienced tumor shrinkage. The disease control rate (DCR) was 83.3%. The median follow-up was 8.6 months (95% confidence interval [CI], 6.743-10.457). The six-month progression-free survival (PFS) was 59.6%. The median PFS was 6.9 months (95% CI 5.016-8.784), whereas the median overall survival (OS) was not reached. The 1-year OS rate was 92%. At the time of analysis, four patients have died, and 56 patients are still alive. All patients were evaluable for safety assessment. The most common treatment-related adverse events (TRAE) were nausea (100%), neutropenia (86.7%), and anemia (83.3%). The most frequent grade 3/4 TRAE were neutropenia (48.3%), febrile neutropenia (8.3%), nausea (3.3%), and anemia (3.3%). No treatment-related death occurred. **Conclusions:** The regimen of irinotecan, TAS-102, plus bevacizumab preliminarily demonstrated promising efficacy with tolerable toxicity for mCRC patients as second-line treatment. This regimen warrants further exploration in refractory mCRC patients. Clinical trial information: NCT06202001. Research Sponsor: None.

156 Poster Session

Long-course chemoradiation vs short-course radiation in total neoadjuvant therapy for rectal cancer: Propensity score case-matched analysis. First Author: Kentaro Ochiai, The University of Texas MD Anderson Cancer Center, Houston, TX

Background: Short-course radiotherapy (Short-TNT) is not widely used in the United States as part of total neoadjuvant therapy (TNT) for rectal cancer, due to concerns for lower tumor regression, increased local recurrence, and poorer quality of surgical specimens compared to long-course chemoradiotherapy (Long-TNT). However, there have been few studies that directly compared outcomes of Long-TNT versus Short-TNT. This study aimed to evaluate outcomes of Short-TNT versus Long-TNT. **Methods:** Consecutive patients with stage II-III rectal cancer who underwent TNT at an NCI-designated comprehensive cancer center from 2019 to 2022 were identified. Outcomes after Short-TNT and Long-TNT were compared using 1:1 propensity score matching. **Results:** Among 318 patients, 170 (85 Short-TNT, 85 Long-TNT) were eligible for the matched analysis (Table). The median follow-up was 31 months. The two groups were similar in tumor distance from anal verge (5cm vs. 6cm), clinical stage (stage III: 90.6% vs. 89.4%), radiologic high-risk features, and treatment sequence (induction chemotherapy: 30.5% vs. 28.2%). The Long-TNT was associate with a higher proportion of nonoperative management (38.8% vs. 24.7%, p=0.03). A local regrowth rate with long-TNT was not lower than short-TNT (39.4% vs. 28.6%, p=0.42). There were no differences in rates of clinical complete plus near-complete response (52.9% vs. 52.9%, p=1.00), overall complete response (pathologic plus sustained clinical complete response; 32.2% vs. 29.9%, p=0.74), and surgical outcomes including retrieved lymph nodes number (19 vs. 22, p=0.27), sphincter preservation (46.2% vs. 62.5%, p=0.10), distal margin (3cm vs. 3.5cm, p=0.77), CRM+ (83.8% vs. 3.1%, p=0.42) and postoperative complications (overall: 34.6% vs. 25.0%, p=0.14; ≥Grade3: 5.8% vs. 9.4%, p=0.60). **Conclusions:** Short-TNT had a comparable overall complete response rate with equivalent surgical outcomes to Long-TNT. Short-TNT should be revisited in the US as a viable alternative TNT strategy both for nonoperative management and for treatment of locally advanced rectal cancer. Research Sponsor: None.

Propensity score case-matched analysis.

Outcomes	Total (n=170)	Long-TNT (n=85)	Short-TNT (n=85)	p value
Clinical complete/near complete response, number (%)	90 (52.9)	45 (52.9)	45 (52.9)	1.00
Non-operative management, number (%)	54 (31.8)	33 (38.8)	21 (24.7)	0.03
Overall complete response, number (%)*	54 (31.0)	26 (30.6)	28 (32.9)	0.74
Surgical outcomes	Surgery (n=116)	Surgery after Long-TNT (n=52)	Surgery after Short-TNT (n=64)	
Sphincter-preserving surgery, number (%)	64 (55.2)	24 (46.2)	40 (62.5)	0.10
Circumferential resection margin +, number (%)	4 (3.4)	2 (3.8)	2 (3.1)	0.42
Distal margin, cm (range)	3.4 (0.1-12)	3 (0.2-6.5)	3.5 (0.1-12)	0.77
Major complications (≥Grade3), number (%)	9 (7.8)	3 (5.8)	6 (9.4)	0.60

*Pathologic + sustained clinical complete response.

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Prognosis of patients with liver single-organ metastases: Updated survival results of BBCAPX-II. First Author: Ying Yuan, Department of Medical Oncology, Key Laboratory of Cancer Prevention and Intervention, Ministry of Education, The Second Affiliated Hospital, Zhejiang University School of Medicine, Hangzhou, China

Background: Although immunotherapy has changed the treatment strategy for many cancers with great success, patients with microsatellite stable (MSS) and RAS-mutant metastatic colorectal cancer (mCRC) especially with liver metastases have a low response rate to immunotherapy. Sintilimab plus bevacizumab and CapeOX (BBCAPX) has proved its efficacy and safety in above unresectable mCRC patients (1). Here, we report subgroup analysis of updated survival results of this single arm, open-label, phase 2 trial. **Methods:** Eligible patients were histologically confirmed unresectable metastatic colorectal adenocarcinoma by multidisciplinary team, and had RAS gene mutation and confirmed MSS status. All patients received treatment with sintilimab plus bevacizumab, oxaliplatin and capecitabine of each 21-day cycle. ORR, DCR, PFS, OS, and subgroup analyses based on metastasis site were performed. **Results:** From April 2021 to December 2021, 25 patients were enrolled. The ORR was 84% and the DCR was 100%. Six (24%) patients including 5 patients with liver single organ metastases underwent surgical treatment and unexpectedly achieved no evidence of disease status. At the data cut-off day (September 13, 2024), patients with liver single organ metastases presented better prognosis. Median PFS was 25.3 months (95% CI, 4.8-NA) in the patients with liver single organ metastases and 11.5 months (95% CI, 4.83-24.3) in the patients with other metastasis. Median OS was not reached in the patients with liver single organ metastases and 35.5 months (95% CI, 9.36-NA) in the patients with other metastasis. The OS rate at 24 months was 72% (95% CI, 56.4-91.9) in FAS. Median OS was not reached in FAS and 35.5 months (95% CI, 11.4-NA) in PPS. 32% patients had at least one grade 3 or 4 TRAEs. No grade 5 adverse events occurred during the study. **Conclusions:** This study provides a highly promising regimen for patients with RAS-mutant, MSS, unresectable mCRC, especially those with liver single organ metastases. Furthermore, we are launching a phase III, randomized, open-label, multicentric clinical trial (NCT05171660) to further analyze the effects, safety, and prognostic biomarkers of this regimen. 1. Xuefeng Fang et al. ASCO 2022. Clinical trial information: NCT04194359. Research Sponsor: National Natural Science Foundation of China; 82373415; National Natural Science Foundation of China; 82403806.

PFS and OS results of BBCAPX-II study.

	Median PFS	6m PFS rate (%)	12m PFS rate (%)	Median OS	12m OS rate (%)	24m OS rate (%)
Full analysis set (FAS), n = 25	17.9 (95% CI, 8.84-27)	84 (70.8-99.7)	56 (39.6-79.3)	NA	76 (61-94.7)	72 (56.4-91.9)
Per-protocol set (PPS), n = 19	9.79 (95% CI, 6.44-27.9)	78.9 (62.6-99.6)	42.1 (24.9-71.3)	35.5 (95% CI, 11.4-NA)	68.4 (50.4-92.9)	63.2 (44.8-89)
Classified by metastatic organs (FAS, n = 25)						
Liver single organ metastases, n = 10	25.3 (95% CI, 4.8-NA)	90 (73.2-100)	70 (46.7-100)	NA (11.4-NA)	90 (73.2-100)	80 (58.7-100)
Other metastases, n = 15	11.5 (95% CI, 4.83-24.3)	80 (62.1-100)	46.7 (27.2-80.2)	35.5 (9.36-NA)	66.7 (46.6-95.3)	66.7 (46.6-95.3)

158 Poster Session

Preoperative botensilimab (BOT) with or without balstilimab (BAL) for patients with resectable locally advanced pMMR or dMMR colon cancer: Results from the UNICORN trial by GONO. First Author: Filippo Ghelardi, Department of Medical Oncology, Fondazione IRCCS Istituto Nazionale dei Tumori, Milano, Italy

Background: Standard treatment for patients (pts) with localized colon cancer (CC) is surgery and adjuvant chemotherapy where indicated. Neoadjuvant strategies allow for earlier exposure to systemic therapy, potentially leading to tumor downstaging, micro-metastases eradication, improved survival, and may pave the way for non-operative management (NOM). BOT, a novel Fc-enhanced multifunctional anti-CTLA-4 antibody plus BAL, an anti-PD-1 antibody, have demonstrated activity in proficient mismatch repair (pMMR) CC pts with durable responses in metastatic CC and unprecedented rates of pathological complete response (pCR) in the localized setting. **Methods:** UNICORN is a window-of-opportunity, multicohort, umbrella platform phase II trial enrolling non-metastatic, radiologically staged rT3-4 N0-2, resectable CC pts to be treated with a short course preoperative targeted treatment according to a prespecified molecular profile assessed by immunohistochemistry and next-generation sequencing on tumor biopsy. Pts with pMMR or deficient (dMMR) tumors received intravenous (IV) BOT at 1 mg/kg on day 1 (cohorts 4 or 6) or IV BOT 1 mg/kg on day 1 and BAL 3 mg/kg on days 1 and 15 (cohorts 5 or 7, enrolled after 4 and 6) and underwent radical surgery on day 35 $\pm$ 5. Pathological response (pR), major response (pMR) and pCR rates were defined as $\leq$ 50%, $\leq$10% and 0% residual viable tumor. The primary endpoint was centrally assessed pMR rate in each cohort. According to a Fleming 1-stage design, choosing β=80% and α=5%, 14 pts were enrolled in each cohort and the treatment was judged promising if $\geq$5/14 pMRs were observed. **Results:** All 56 enrolled pts completed preoperative treatment and underwent surgery. Timely surgery was performed in most pts (98%), except 1 who underwent surgery with a delay $<$ 4 weeks due to treatment-related hyperthyroidism. Among pts with pMMR disease, activity was limited for BOT monotherapy (pMR 0%, pR 43%). With BOT + BAL, pCR was 29%, pMR 36% and pR 71%. Among those with dMMR status, BOT led to pCR, pMR and pR in 29%, 36% and 64% pts, respectively. Notably, BOT + BAL led to pCR and pMR in 93% and 100% pts. Adverse events (AEs) of any grade occurred in 28/56 pts (50%), and 22 (39%) were deemed immune-related. Serious AEs occurred in 9 pts (16%) and were treatment-related in 3 pts (5%). **Conclusions:** The primary endpoint was met in all cohorts except for BOT monotherapy in pts with pMMR status. Despite limited sample size, the activity of 1 neoadjuvant cycle with BOT + BAL favorably compares to ipilimumab/nivolumab in both pMMR and dMMR subgroups. pCR rate was remarkable in pMMR and the highest ever reported in dMMR pts, paving the way to NOM studies irrespective of MMR status. Clinical trial information: EU-CT 2022-501308-90-00. Research Sponsor: Fondazione GONO; AGENUS.

Molecular status	n°	Treatment	pCR, n (%)	pMR, n (%)	pR, n (%)
pMMR	14	BOT	0 (0)	0 (0)	6 (43)
	14	BOT/BAL	4 (29)	5 (36)	10 (71)
dMMR	14	BOT	4 (29)	5 (36)	9 (64)
	14	BOT/BAL	13 (93)	14 (100)	14 (100)

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The crossover of the bevacizumab and panitumumab survival curves in the PARADIGM study: A clue to time-related effects of bevacizumab on the risk of tumor progression. First Author: Dan Aderka, Tel Aviv Sourasky Medical Center, Tel Aviv, Israel

Background: The PARADIGM study in CRC patients demonstrated a cross over of the bevacizumab survival curve below the panitumumab survival curve at 28 months, followed by a steady decrease of the bevacizumab survival curve below the panitumumab survival curve up to 11% by 5 years. Similar differences were seen in the FIRE3, CALGB-80405 and PEAK trials starting at 20 months, suggesting an unsuspected time-related effect of bevacizumab on patient's survival starting at about 2 years after its 1st administration. **Methods:** Review of time related effects of bevacizumab in 1st and 2nd line trials conducted in CRC, breast, ovary and renal cancer showed a significant PFS but no OS survival benefit, with a detrimental effect on survival in the adjuvant setting (AVANT, NSABP-08 studies). Preclinical data on tumor biology, intratumor-heterogeneity, clonal evolution and the cross talk between bevacizumab and the diverse tumor microenvironments, was correlated to the clinical data, in an attempt to explain the time-related effects of the biological. **Results:** In the NSABP-08 trial, bevacizumab reduced relapse for 6 months, effect diminished and lost at 12 months. At 24 months after bevacizumab's 1st administration, the relapse of the bevacizumab treated population increased steadily over the control. Identical kinetics were shown in AVANT study and metastatic ovarian patients suggesting that these time-related effects of bevacizumab are universal. The biologic explanation of the clinical data is that short-term treatment with bevacizumab (plus chemotherapy) induces initial tumor stasis and shrinkage, followed by selection and emergence of more aggressive and resistant tumor clones, characterized by an accelerated growth, invasiveness and metastasis shifting the overall tumor growth kinetics to a steeper angle. Bevacizumab appears to be effective for the time of its administration, followed by a slow rebound after its discontinuation. The initial survival benefit will be lost at about 20-24 months after its first administration, as clearly demonstrated in the PARADIGM, FIRE-3, CALGB 80405 and the PEAK studies. **Conclusions:** As bevacizumab benefit persist for about 2 years after its 1st administration, it explains the "survival benefit" obtained in studies with with median survival of 20 months but not in studies with survivals $>$30 mo. (PARADIGM, FIRE-3, CALGB 80405 and the PEAK). This phenomenon also explains the crossover of the survival curves in PARADIGM study and the benefits of bevacizumab obtained preferentially in patients with short survival (rt. sided tumors), high tumor burden, and 2nd and 3rd line CRC. Research Sponsor: None.

160 Poster Session

Outcomes of simultaneous, adaptive, MRI-guided liver SBRT in patients with heavily pretreated colorectal cancer and four or more liver metastases. First Author: Mustafa M Basree, University of Wisconsin Hospitals and Clinics, Madison, WI

Background: About one-third of colorectal cancer (CRC) patients (pts) develop metastases, most commonly in the liver (CRLM). Ablative treatment of multiple liver lesions is difficult and often not done with conventional CT-based radiation therapy, but feasible with MRI-guided adaptive radiation. This study reviews our experience treating pts with high burden CRLM using MRI-guided stereotactic body radiation therapy (MRgRT). **Methods:** This study identified pts with 4 or more CRLM treated with MRgRT between 7/2019 and 4/2024, all of whom had progressive disease during or after standard-of-care therapy. Ablative radiation was delivered to all sites of disease with no concurrent systemic therapy. All with Child-Pugh score 5-6 (class A) liver function. Local control was defined as no progression in any treated lesion on a per-patient basis. Descriptive statistics summarized the data. A Cox proportional hazards model was used to identify predictors of survival, while the Kaplan-Meier method was used to estimate survival functions. Statistical analysis was performed using SPSS. **Results:** Seventeen pts with 110 lesions were included, with a median follow-up of 10.7 months (range, 1.7–36.9). The majority (n=15) received systemic therapy in the year before MRgRT. The median number of prior lines of systemic therapy was 2 (range, 1–5). Eight pts (47.1%) had nodal and/or lung progression in addition to CRLM, and 4 pts (23.5%) had at least one prior course of MRgRT. A median of 6 liver lesions (range, 4–11) were prescribed a median dose of 60 Gy (range, 60–80), each delivered in 5 fractions. Median total gross tumor volume per pt was 23.8 cm^3 (range, 0.97–60.1). Seventy-four fractions were available for radiation delivery analysis, of which 55 (74.3%) were adapted. At least one target was modified in 51 (92.7%) of the adapted fractions. Acute toxicity occurred in 7 pts, all with self-limited fatigue and nausea. Progression in Child-Pugh classification from A to B occurred in three pts 3-6 months post-MRgRT. Late toxicity occurred in 1 patient with mild, possibly radiation-related peripheral biliary dilation. Two pts developed malignant biliary strictures. Per-patient local control was 50.8% and 34.8% at 1 and 2 years, respectively, while liver control was 28.7% at 2 years. Six-month systemic therapy-free survival was 31.9%. Two-year overall survival (OS) was 43.9%, with a median of 17.1 months (range, 8.6–25.7). The size of the largest tumor correlated with OS (HR 1.23, p=0.014). **Conclusions:** In this small series of CRC pts with 4+ CRLM, MRgRT is well-tolerated and shows promise as astrategy to consolidate disease and delay the immediate need for systemic therapy changes. Patient selection is critical to identify those most likely to benefit, though the overall impact on survival and disease trajectory remains uncertain. Research Sponsor: None.

161 Poster Session

Risk of peritoneal recurrence following laparoscopic versus open surgery for stage II or III colorectal cancer (JCOG2310A): An integrated analysis of three phase III randomized controlled trials. First Author: Marie Hanaoka, Department of Gastrointestinal Surgery, Science Tokyo, Tokyo, Japan

Background: Peritoneal recurrence in colorectal cancer (CRC) has long been a concern. Although studies have examined the higher risk of peritoneal recurrence following laparoscopic (LAP) compared with open (OP) surgery, the issue remains unresolved, largely due to the heterogeneity of existing data and the limited number of observed peritoneal recurrence events, preventing any definitive conclusions. This study aimed to evaluate whether LAP for pStage II/III CRC presents a potential risk factor for peritoneal recurrence compared with OP. **Methods:** Data from three phase III trials (JCOG0404, JCOG0910, JCOG1006) were extracted from an ancillary research database (JCOG2310A). From this cohort, patients with pStage II or III CRC who underwent curative surgery were included, and the incidence of peritoneal recurrence was compared between LAP and OP. A multivariate analysis was conducted to adjust covariates between the two groups. The primary outcome was the cumulative incidence of peritoneal recurrence, and the secondary outcome was recurrence in specific high-risk subgroups identified through a univariate analysis. **Results:** From 3061 patients (OP: 1734; LAP: 1327), pT4 (25% vs. 17%), pStage III (65% vs. 83%), and right-sided tumor (63% vs. 70%) were different between OP and LAP, respectively. With a median follow-up for peritoneal recurrence of 6.2 years, the 5-year cumulative incidence of peritoneal recurrence was 2.8% in the OP group and 2.9% in the LAP group (hazard ratio (HR): 0.940, 95% confidence interval (CI): 0.621-1.425 in the univariate analysis, and HR: 1.086, 95% CI: 0.683-1.727 in the multivariate analysis. The univariate analysis identified pT4, pN2, pStage III, right-sided tumor, poorly differentiated (por/sig/muc), and ECOG-PS 1 as risk factors for peritoneal recurrence. No significant difference was observed in the subgroups (see table). An exploratory propensity score matching analysis with 1066 patients in each group also showed no difference between the groups (the 5-year cumulative incidence of peritoneal recurrence was 2.8% in the OP group and 3.0% in the LAP group [HR: 1.002, 95% CI: 0.625-1.607]). **Conclusions:** This study revealed that LAP was not a risk factor regarding peritoneal recurrence following curative surgery for Stage II/III CRC. Research Sponsor: None.

Subgroup analysis of cumulative incidence of peritoneal recurrence.

Outcome Measures	Subgroups					
	pT4	pN2	pStage III	Right-sided	Poorly differentiated	ECOG PS 1
N (OP vs. LAP)	428 vs. 229	274 vs. 184	1133 vs. 1105	635 vs. 403	122 vs. 66	56 vs. 25
Event (OP vs. LAP)	31 vs. 23	13 vs. 10	45 vs. 32	28 vs. 18	10 vs. 2	7 vs. 2
5-year peritoneal recurrence rate (OP vs. LAP)(%)	6.8 vs. 10.1	4.8 vs. 5.4	3.6 vs. 2.9	4.1 vs. 4.5	7.5 vs. 3.0	9.1 vs. 8.0
HR (95% CI)	1.403 (0.819-2.402)	1.144 (0.503-2.603)	0.725 (0.461-1.139)	1.014 (0.562-1.829)	0.372 (0.081-1.699)	0.609 (0.130-2.862)

162 **Poster Session**

Nelmastobart (hSTC810) combined with capecitabine therapy in metastatic colorectal cancer with resistance or intolerance to oxaliplatin and irinotecan-based chemotherapy: A phase 1b clinical trial. First Author: Soohyeon Lee, Korea University Anam Hospital, Korea University College of Medicine, Seoul, South Korea

Background: Treatment options for patients with metastatic colorectal cancer (mCRC) who experienced disease progression after receiving oxaliplatin and irinotecan-based chemotherapy are limited. Nelmastobart (hSTC810), a humanized anti-BTN1A1 (butyrophilin subfamily 1 member A1) monoclonal antibody of the IgG4 isotype, showed acceptable safety profile and the disease control rate was 39.3% in the first-in-human study, suggesting its potential efficacy. This phase Ib study was designed to evaluate the safety and explore synergistic anti-tumor activity of nelmastobart in combination capecitabine. The study aims to determine the maximal tolerated dose (MTD) and recommended phase II dose (RP2D) of this combination therapy in patients with mCRC who are refractory to standard therapy. **Methods:** Dose escalation phases were conducted in patients with mCRC who were administered orally twice daily capecitabine (850mg, 1000mg, 1250mg orally, days 1-14) and nelmastobart (400mg, 800mg intravenously every 3 weeks). Standard "3 + 3" dose escalation was used to define the MTD. In this phase 1b, 12 participants enrolled between February 5 2024 and June 11 2024 were 18 years of age or older with mCRC previously treated oxaliplatin and irinotecan in the metastatic setting. **Results:** The first 3 patients at the dose level 0 (nelmastobart 400mg, capecitabine 1,000 mg/m^2 twice daily) and level 1 (nelmastobart increased to 800 mg, capecitabine 1,000mg/m^2 twice daily) had no dose-limiting toxicities (DLTs). In dose level 2 (nelmastobart 400mg, capecitabine 1,250mg/m^2 twice daily), 2 of 3 had DLTs (grade 3 hand-foot syndrome), which were attributed to capecitabine. When dose level 2 was expanded to 3 additional patients, 2 more patients experienced DLTs (grade 3 mucositis and hand-foot syndrome). Consequently, the combination of nelmastobart 800 mg once every three weeks and capecitabine 1,000 mg/m^2 twice daily (2 weeks on 1 week off) was established as the MTD/RP2D in patients with mCRC. The most common AEs were hand-foot syndrome (n=4), nausea (n=3) and stomatitis (n=2) which were related with capecitabine. Four patients experienced grade 1 fatigue, which was considered to be related to nelmastobart. Two patients (16.7%) out of 12 showed partial response and 8 patients have maintained stable disease for more than 4 months among 9 patients who had a median follow-up of more than 4 months (as of August 30). **Conclusions:** The combination of hSTC810 and capecitabine is well tolerated at the MTD, without unexpected AEs and shows promising antitumor activity in patients with heavily treated mCRC. The phase 2 study is currently in progress. Clinical trial information: NCT0599054. Research Sponsor: STCube.

163 **Poster Session**

Immunotherapy outcomes in dMMR/MSI-H and/or TMB-H metastatic colorectal cancer (CRC) with peritoneal metastases (PM). First Author: Osama M Mosalem, Mayo Clinic in Florida, Jacksonville, FL

Background: CRC with PM has a poor prognosis and limited treatment options. Immunotherapy has shown promise in improving outcomes for patients with microsatellite instability-high (MSI-H), mismatch repair-deficient (dMMR), and high tumor mutational burden (TMB-H) tumors. However the peritoneal niche has a unique microenvironment, which can influence the effectiveness of immune check point inhibitors (ICIs). We assessed the impact of ICI on progression-free survival (PFS) and overall survival (OS) in CRC patients (pts) and PM, considering key clinical and molecular characteristics. **Methods:** A retrospective review aiming to identify CRC pts with MSI-H/dMMR and PM treated with immunotherapy between 2015 and 2023 was performed. Data collected included pt demographics, tumor characteristics, and treatment details. Responses were evaluated using RECIST v1.1 criteria. **Results:** 37 pts were included, 9 (24%) had Lynch syndrome. The median age of diagnosis was 61 years and most pts were females (65%). Most pts had MSI-H/dMMR tumors, while one pt was treated based on ultra-hypermutated phenotype (TMB 202) and POLE mutation. Among those with available TMB data, the median TMB was 52 mutations/Mb (0-202). Common mutations were *PMS2* (51%), *MLH1* (38%), *MSH6* (16.2%), and *MSH2* (16.2%). Right-sided tumors were in 54%, mucinous histology in 62%, and ascites in 54%. Pts received a median of 11 cycles of ICI (2-52). Pembrolizumab was the most frequently used ICI (78%), while Nivolumab was used either alone (11%) or in combination with Ipilimumab (11%). Median PFS in our cohort was 7.8 months and median OS was 29.9 months. The disease control rate (DCR) was 81%, with an objective response rate (ORR) of 51.3% (9 complete responses, 12 partial responses). At the time of data collection, 30% of pts were off systemic therapy and 5-FU-based chemotherapy combined with a biologic agent was the most common subsequent treatment in 32% who had PD. Univariate analysis showed no significant differences in PFS or OS based on tumor sidedness, isolated PM, liver metastases, mucinous vs. non mucinous histology or *BRAF*/*RAS* status (WT vs mutant). Malignant ascites was associated with worse OS(HR 0.3; p= 0.03). **Conclusions:** Treatment with ICIs in pts with MSI-H/dMMR and/or TMB-H CRC and PM improved disease control in CRC pts with PM, however, as compared to historical controls for patients with MSI-H/dMMR disease treated with ICI, this group showed less favorable outcomes. Malignant ascites was associated with poorer overall survival. Further research of the immune response and tumor microenvironment in the peritoneal cavity is warranted. Research Sponsor: None.

164 **Poster Session**

Encorafenib and cetuximab with/without binimetinib therapies for patients with *BRAF*-mutated metastatic colorectal cancer with prognostic factors: The BEETS trial (JACCRO CC-18). First Author: Yu Sunakawa, Department of Clinical Oncology, St. Marianna University School of Medicine, Kawasaki, Japan

Background: BRAF inhibitor combination therapies, triplet (encorafenib, binimetinib and cetuximab) and doublet (encorafenib and cetuximab), have been approved for *BRAF*-mutated metastatic colorectal cancer (mCRC) in Japan. The treatment with triplet is recommended for patients (pts) with clinical prognostic factors according to the subgroup analysis of the BEACON CRC trial. However, it is unclear whether the treatment strategy is useful. We therefore conducted a study on the use of combination therapy with triplet or doublet based on observational study that accumulated real world data. **Methods:** The BEETS trial was a multicenter observational study to prospectively evaluate the efficacy and safety of BRAF inhibitor combination therapy as a 2nd- or 3rd-line treatment in pts with *BRAF*-mutated mCRC in clinical practice [UMIN000045530]. Two hundred pts were assigned to either triplet or doublet therapy based on the physician's choice. The primary endpoint was overall survival (OS). The secondary endpoints included overall response rate (ORR), disease control rate, progression-free survival (PFS), and safety. The exploratory analysis was performed to evaluate the efficacy of triplet and doublet therapies in all pts and pts with clinical prognostic factors (PS1, 3 or more metastatic sites, elevated CRP value, or primary tumor present) using inverse probability weighting (IPW) analysis based on propensity scores to account for potential cofounding factors. **Results:** Among 203 pts enrolled from Oct 2021 to Nov 2023, 195 pts were evaluable (median age 67 years old, 52% male, ECOG PS 0/1/2 55%/39%/6%, no. of prior regimens 1/2 85%/15%, right-sided primary 61%, MSI-high 7%, triplet regimen 57%). Median OS was 13.2 months (95%CI 11.4-17.4) with 1y-survival rate of 53%, and median PFS was 4.9 months. Comparing the triplet and doublet using IPW, the median OS and PFS were comparable between the two treatments (triplet vs. doublet; 16.0 months vs. 13.6 months [HR 0.88, 95%CI 0.53-1.48] and 5.7 months vs. 4.8 months [HR 0.87, 95%CI 0.59-1.28]). In 139 pts with any prognostic factors, median OS and 1y-survival rate were 13.2 months vs. 9.6 months [HR 0.80, 95%CI 0.45-1.43] and 54% vs. 42%, median PFS and ORR were 5.3 months vs. 4.2 months [HR 0.75, 95%CI 0.48-1.19] and 34% vs. 42%, in the triplet vs. doublet therapy, respectively. **Conclusions:** Our study demonstrated that BRAF inhibitor doublet therapy is a reasonable treatment for *BRAF*-mutated mCRC. However, triplet therapy may be a clinically useful for pts with clinical prognostic factors. Clinical trial information: UMIN000045530. Research Sponsor: Ono Pharmaceutical Co. ltd., Bristol-Myers Squibb.

166 **Poster Session**

Efficacy and safety of fruquintinib alternating with bevacizumab plus capecitabine as maintenance therapy after first-line treatment in metastatic colorectal cancer (mCRC): A multicenter, open-label, phase II study. First Author: Wangjun Liao, Department of Oncology, Nanfang Hospital, Southern Medical University, Guangzhou, China

Background: Maintenance therapy with bevacizumab (Bev) plus capecitabine (Cap) is widely recommended to unresectable mCRC patients (pts). Fruquintinib (Fru) is a highly selective TKI that inhibits vascular endothelial growth factor receptor (VEGFR)-1,2,3. This study is to compare the therapeutic potential of alternating treatment with fruquintinib and bevacizumab plus capecitabine as maintenance therapy for mCRC (NCT05659290). **Methods:** Eligible mCRC pts aged 18-75 years with stable disease or better after induction treatment with chemotherapy in combination with Bev, ECOG PS 0-2, adequate bone marrow, liver, and renal function were enrolled. Forty patients were included (20 in phase IIa, 40 in phase IIb). In phase IIa, pts were orally administered with Fru (5 mg, qd, d1-14, q3w) alternating with Bev (7.5 mg/kg, iv.gtt, d1, q3w) plus Cap (850 mg/m^2, orally, twice daily, d1-14, q3w). In phase IIb, pts were randomly assigned (1:1) to either maintenance treatment with Fru alternating with Bev plus Cap or Bev plus Cap. The primary endpoint was progression-free survival (PFS). Secondary endpoints included objective response rate (ORR), disease control rate (DCR) and safety. **Results:** At cutoff date of Sep 5, 2024, In phase IIa, 20 pts (14 males and 6 females) were enrolled with the median age was 59.0 years (range 27-75), ECOG PS 1 (95.0%), liver metastasis (55.0%), left-sided colon and rectal primary (70.0%). 11 pts had received at least one tumor assessment. the DCR was 100.0% (11/11) and the mPFS was immature, but 4 pts showed the PFS of $\geq$ 8 months (8.3, 8.6, 9.2, 13.4 m, respectively). The most common treatment-emergent adverse events (TEAEs) were proteinuria (60.0%), hypoalbuminemia (40.0%), hypertension (35.0%); The most common grade $\geq$ 3 adverse events were hypertension (10.0%), proteinuria (5.0%), and platelet count decreased (5.0%). After fully considering about patient's tolerance and safety, the phase IIb of Fru was adjusted from 5mg to 3mg, these results provided further evidence that 3mg can ensure the safety and tolerance. **Conclusions:** Fruquintinib alternating with bevacizumab plus capecitabine as maintenance therapy after first-line treatment in mCRC showed preliminary anti-tumor activity and manageable toxicity. The phase IIb is ongoing and warrants further exploration in mCRC. Clinical trial information: NCT05659290. Research Sponsor: None.

167 **Poster Session**

Updated survival results of ZL-IRIAN: Irinotecan plus anlotinib or in combination with penpulimab as second-line treatment of metastatic colorectal cancer. First Author: Chenchen Wang, Department of Medical Oncology, Fudan University Shanghai Cancer Center, Shanghai, China

Background: The irinotecan based chemotherapy in combination with bevacizumab had become the standard second-line treatment for metastatic colorectal cancer (mCRC), but the efficacy was limited. Anlotinib, an oral anti-angiogenic tyrosine kinase inhibitor, had initially shown promising clinical effects when combined with chemotherapy in the first-line treatment of mCRC. Whether anlotinib could bring clinical benefit in the second line treatment warrants further exploration. The PD-1 antibody might have partial effects when combined with anti-angiogenic agents with or without chemotherapy in first- and third-line treatment for proficient mismatch repair (pMMR)/microsatellite stable (MSS) mCRC. Our study was to explore the rationality of anlotinib plus chemotherapy or further combined with a novel anti-PD-1 IgG1 antibody—penpulimab in the second line treatment of mCRC. **Methods:** This study was an open label, non-randomized, multi-cohorts, phase II prospective clinical trial that included pts with pMMR/MSS mCRC who were eligible for the second-line systemic treatment. Two cohorts were established as follows: Cohort A consisted of anlotinib(10mg daily, d1-10, q2w) + irinotecan (180mg/m d6, q2w) (n = 23); Cohort B consisted of anlotinib (8mg daily, d1-10, q2w)+irinotecan (180mg/m d6, q2w) + penpulimab (200mg, d6, q2w) (n = 23). The primary endpoint was ORR. Secondary endpoints included progress free survival (PFS), overall survival (OS), duration of response (DoR), disease control rate (DCR) and safety. The ORR was reported at 2024 ASCO, and PFS and OS have been updated in this report. **Results:** At the cutoff date on August 2024, a total of 23 Pts were enrolled in A cohort, and 23 pts were enrolled B cohort but 21 pts received at least one dose of treatment. The median PFS was 7.87 months (95% CI 5.52-10.22) in A cohort, compared to 7.4 months (95% CI 3.17-11.63) in B cohort, p=0.141. The 6-month, 12-month and 18-month PFS rates in A cohort and B cohort were 69.3% vs 80.7%, 27.5% vs 40.3% and 5.5% vs 22.4%, respectively. The median OS of A cohort was 21.73 months (95% CI 9.43-34.03), compared to 19.17 months (95% CI 11.44-26.90 months) in B cohort, p=0.987. The 12-month and 24-month OS rates in A cohort and B cohort were 64.7% vs 73.1% and 20.2% vs 54.8%, respectively. Safety profile indicated that 65.2% (15/23) and 66.7% (14/21) experienced grade≥3 adverse events in A cohort and B cohort, respectively. Common grade ≥ 3 adverse reactions (≥ 20%) mainly include hypertension (4.35% vs 14.29%), diarrhea (13.04% vs 9.52%), neutrophilopenia (39.13% vs 33.33%), and leukopenia (26.09% vs 9.52%). **Conclusions:** There was no statistically significant difference in overall efficacy between the two cohorts, but the long-term PFS and OS rates were higher in group B cohort, indicating that immunotherapy may bring long-term survival benefits to certain patients. Clinical trial information: NCT05229003. Research Sponsor: None.

168 **Poster Session**

Dual versus single immune checkpoint blockade for patients with metastatic colorectal cancer harboring microsatellite instability: The Saint-Antoine immunoMSI score. First Author: Léa Jehanno, Department of Medical Oncology, Saint-Antoine Hospital, AP-HP, Paris, France

Background: Anti-PD1 monotherapy (mono) and anti-PD1 plus anti-CTLA4 therapies (combo) are validated for metastatic colorectal cancer (mCRC) harboring microsatellite instability and/or mismatch repair deficiency (MSI/dMMR). We aimed at assessing the efficacy of combined anti-CTLA4 and anti-PD1 blockade versus anti-PD1 monotherapy and identifying which subgroups benefit more from the combo. **Methods:** All patients (pts) with MSI/dMMR mCRC treated by anti-PD1 ± anti-CTLA4 included in the prospective monocenter immunoMSI cohort were analyzed. The choice between monotherapy or combination therapy varied according to open studies and regulatory approvals, without taking pt characteristics into account. Datalock date was April 23, 2024. Main endpoints were progression-free survival (PFS) per iRECIST and overall survival (OS) estimated with the Kaplan-Meier method. Univariate Cox proportional hazards models were used to estimate hazard ratios with CI. The interactions between studied subgroups and the treatment group were modelized by adding in a single Cox model the studied subgroups, the treatment group parameters, and their interaction term. **Results:** Among 210 pts, 97 were treated combo and 113 with mono. Baseline characteristics were similar in both groups. 25% and 17% were in first line, respectively. Median follow-up was 4.4 years: 4.1 years for combo and 5.4 years for mono. Combo was associated with better PFS (HR=0.48; 95%CI 0.31-0.76) and OS (HR=0.49; 0.29-0.84). Best observed response rates were: 78% of objective responses, 17% of stable diseases and 5% disease progressions with combo versus 60%, 25% and 15% with mono. Subgroup analyses showed significant differential effect of the treatment group for gender (PFS HR=0.21 vs 0.87 for females and males, P-interaction: 0.0067), liver metastases (PFS HR=0.30 vs 0.79 for pts without liver mets and those with liver mets, P-interaction: 0.0479) and sidedness (PFS HR=0.27 vs 0.64 for left-sided and right-sided mCRC, P-interaction=0.0922), but not *RAS*/*RAF* mutations. We computed a predictive score based on gender (female: 2 points), no liver met (1 point) and left side (1 point). PFS HR was 0.1 (0.03-0.33) for the group ≥3 points (N=67), 0.66 (0.36-1.23) for the group with 1-2 points (N=113) and 1.54 (0.50-4.70) for the group with 0 point (N=30). **Conclusions:** Combined CTLA4 and PD1 blockade is associated with improved PFS and OS compared to anti-PD1 alone in MSI/dMMR mCRC. The subgroup deriving the highest benefit of the combo seems to be female pts with left sided tumor and no liver met. Data from the randomization between nivolumab versus nivolumab + ipilumumab in the CheckMate-8HW phase III study are expected. Research Sponsor: None.

169 **Poster Session**

Seven-year efficacy and safety in a randomized phase III trial investigating duration of adjuvant oxaliplatin-based therapy (3 vs. 6 months) for patients with high-risk stage II colon cancer: ACHIEVE-2 trial. First Author: Eiji Sunami, Faculty of Medicine, Kyorin University, Tokyo, Japan

Background: ACHIEVE-2 was conducted as one of four phase III trials for patients with high-risk stage II colon cancer (CC) investigating the duration of adjuvant (adj) oxaliplatin-based therapy in a prospective pooled analysis, IDEA collaboration. The results of the 7-year follow-up are presented here. **Methods:** From Feb 2014 to Jan 2017, 525 Asian patients with high-risk stage II CC (T4, inadequate nodal harvest, poorly differentiated, obstruction, perforation, or vascular invasion) were randomly assigned to 3- or 6-month mFOLFOX6/CAPOX treatment after curative surgery. The cutoff date for the data was Jan 2024. **Results:** Of the 525 randomized patients, 11 were not treated. Among the 514 participants (255 in the 3-month arm; 259 in the 6-montharm), 432 (84%) received CAPOX and 184 (36%) presented with T4 as a high-risk factor for recurrence. The baseline characteristics of the patients in each arm were well balanced between the treatment arms and two therapy regimens. The 7-year disease-free survival (DFS) rates were 83.4% and 81.8% in the 3- and 6-month arms, respectively (hazard ratio [HR]= 1.00 [95% CI, 0.66-1.53]). The 7-year overall survival (OS) rates were 88.4% and 88.8% in the 3- and 6-month arms, respectively (HR= 1.16 [95% CI, 0.70-1.93]). With CAPOX, the HR for DFS and OS of the 3-month arm compared with the 6-month arm were 0.97 (95% CI, 0.61-1.52) and 1.09 (95% CI, 0.64-1.89), respectively. Multivariate analysis of the six high-risk factors for recurrence showed that T4 and inadequate nodal harvest were independent risk factors for both DFS and OS. The rates of any grade of peripheral sensory neuropathy (PSN) lasting up to 96 months in the 3- vs. 6-month arms were 7.9% vs. 25.0% (*P=0.0189*). With CAPOX, the incidence of PSN lasting up to 96 months in the 3- vs. 6-month arms were 9.4% vs. 27.5%. **Conclusions:** The incidence of long-lasting PSN was significantly lower at 3 months than at 6 months of therapy. Three months of CAPOX therapy may be appropriate for high-risk stage II CC, as shorter treatment did not worsen the outcome, even after a long follow-up. Clinical trial information: UMIN000013036. Research Sponsor: Japanese Foundation for Multidisciplinary Treatment of Cancer under the contract with Yakult Honsha.

170 **Poster Session**

Updated efficacy and subgroup analysis of first-line serplulimab plus bevacizumab and XELOX versus placebo plus bevacizumab and XELOX in metastatic colorectal cancer: A phase 2/3 study. First Author: Rui-Hua Xu, Department of Medical Oncology, Sun Yat-sen University Cancer Center, State Key Laboratory of Oncology in South China, Guangdong Provincial Clinical Research Center for Cancer, Guangzhou, China

Background: This is a randomized, double-blind, multicenter phase 2/3 study comparing the efficacy and safety of serplulimab (a novel anti-PD-1 antibody) plus bevacizumab and XELOX chemotherapy vs. placebo plus bevacizumab and XELOX as first-line treatment for metastatic colorectal cancer (mCRC). Our previous presentation at the 2024 ASCO Meeting showed the progression-free survival results. Here we present the updated efficacy and safety findings together with subgroup analysis results after a median follow-up of 31.0 months. **Methods:** A total of 114 patients with mCRC and no prior systemic therapy were randomized 1:1 (serplulimab arm, n = 57; placebo arm, n = 57) to receive intravenous (IV) serplulimab (300 mg) plus bevacizumab (7.5 mg/kg) and XELOX (IV oxaliplatin [130 mg/m^2] and oral capecitabine [1000 mg/m^2]) (group A) or placebo plus bevacizumab and XELOX (group B) once every 3 weeks. Stratification factors were PD-L1 expression level, ECOG PS score, and primary tumor site. The primary endpoint was independent radiological review committee (IRRC)-assessed PFS per RECIST 1.1. Secondary endpoints included other efficacy endpoints, safety, pharmacokinetics, biomarker explorations, and quality-of-life assessments. **Results:** In the phase 2 part, by the data cutoff of June 30, 2024, sustained improvements in PFS (16.6 vs. 10.7 months, stratified HR 0.66, 95% CI 0.37–1.19) and DOR (17.7 vs. 11.3 months, stratified HR 0.45, 95% CI 0.20–0.98) were observed for patients in group A compared to group B in the modified intent-to-treat population (n = 112; two patients in group A did not receive any intended study treatment). 32/55 (58.2%) patients in group A and 35/57 (61.4%) in group B have died. Median overall survival (OS) was 25.6 months in group A and 21.2 months in group B (stratified HR 0.86, 95% CI 0.53–1.42). A trend of PFS and OS benefits was similarly observed for the patients with a microsatellite stable (MSS) status (PFS: 16.8 vs. 10.1 months, stratified HR 0.65, 95% CI 0.33–1.29; OS: 23.5 vs. 20.2 months, stratified HR 0.79, 95% CI 0.45–1.38). 39 (70.9%) patients in group A and 34 (59.6%) in group B had grade ≥3 treatment-related adverse events. Serplulimab/placebo-related serious TRAEs occurred in 13 (23.6%) patients in group A and 14 (24.6%) patients in group B. **Conclusions:** With a follow-up duration of 31.0 months, improved survival benefits demonstrated with the addition of serplulimab were maintained in the first-line treatment of mCRC patients including those with MSS status alongside a manageable safety profile. The phase 3 part of this study is currently ongoing to further evaluate serplulimab plus bevacizumab and XELOX as a first-line treatment option in mCRC. Clinical trial information: NCT04547166. Research Sponsor: None.

171 Poster Session

Overall survival with fruquintinib versus placebo after adjusting for subsequent anticancer therapy in patients with refractory metastatic colorectal cancer in the FRESCO-2 study. First Author: Sara Lonardi, Veneto Institute of Oncology IOV-IRCCS, Padua, Italy

Background: Fruquintinib is a highly selective, oral inhibitor of all 3 vascular endothelial growth factor receptors (VEGFRs -1, -2, and -3) that is approved by the US FDA and in the EU, UK, and Japan for previously treated metastatic colorectal cancer (mCRC), regardless of biomarker status. The phase 3 FRESCO-2 study (NCT04322539) met its primary endpoint demonstrating significantly improved overall survival with fruquintinib + best supportive care (BSC) vs placebo + BSC. As patients may have received subsequent anticancer therapy after progression, understanding the impact on overall survival is important. This analysis examined overall survival data from FRESCO-2 after adjusting for subsequent anticancer therapy. **Methods:** In FRESCO-2, patients were randomized 2:1 to receive fruquintinib (5 mg) or matching placebo by mouth, once daily, 3 weeks on, 1 week off, + BSC. Eligible patients had received prior chemotherapy, anti-VEGF therapy, anti-EGFR therapies (if indicated), and TAS-102 and/or regorafenib. We assessed the impact of subsequent anticancer therapy on overall survival by excluding patients who received subsequent therapies, censoring these patients, and determining the causal hazard ratio using inverse probability of censoring weights (IPCW) and marginal structural models (MSM) to adjust for the potential bias introduced by subsequent treatments. **Results:** In FRESCO-2, a lower proportion of patients in the fruquintinib vs placebo arm received subsequent anticancer therapy (29.4% vs 34.3%). Among these patients, the most common subsequent anticancer therapies in the fruquintinib vs placebo arms were fluorouracil (7.7% vs 9.6%) and regorafenib (7.5% vs 7.8%). The overall survival benefit seen with fruquintinib vs placebo in the intent-to-treat population (hazard ratio [HR]: 0.66; 95% confidence intervals [CI]: 0.55–0.80) was improved when patients who had received subsequent anticancer therapy were excluded (HR: 0.45; 95% CI: 0.36–0.57) or censored (HR: 0.63; 95% CI: 0.50–0.79). In addition, the overall survival benefit seen with fruquintinib vs placebo was improved after adjusting for subsequent anticancer therapy using IPCW (HR: 0.43; 95% CI: 0.33–0.55) and MSM (HR: 0.48; 95% CI: 0.38–0.60). Patients receiving fruquintinib had similar rates of grade ≥3 adverse events, whether or not they received subsequent anticancer therapy (51.5% and 67.4%, respectively). **Conclusions:** Consistent with the primary analysis of the FRESCO-2 intent-to-treat population, fruquintinib improved overall survival vs placebo after adjusting for the impact of subsequent anticancer therapy. Furthermore, these analyses are robust with consistent results reported using IPCW and MSM. These findings support fruquintinib as a new treatment option for patients with previously treated mCRC. Clinical trial information: NCT04322539. Research Sponsor: HUTCHMED.

172 Poster Session

Survival analysis using neoadjuvant rectal score in patients with locally advanced rectal cancer who underwent surgery after neoadjuvant chemoradiotherapy: An ambispective study. First Author: Mohammed Sameer, Regional Cancer Centre Trivandrum, Thiruvananthapuram, India

Background: Colorectal cancer, the third most common malignancy and a major cause of cancer deaths, is treated using a standard protocol of neoadjuvant chemoradiotherapy (NACTRT) or short-course radiotherapy (SCRT) followed by total mesorectal excision (TME) for locally advanced rectal cancers (LARC). Long-term follow-up is essential to determine study endpoints, necessitating surrogate markers for short-term outcomes. This study aims to stratify patients into risk groups based on their Neoadjuvant Rectal Cancer (NAR) score and evaluate its predictive value for oncological outcomes, including overall survival, disease-free survival (DFS), locoregional relapse-free interval (LRFI), and distant metastasis-free interval (DMFI), while also correlating clinicopathological characteristics with these outcomes. **Methods:** This ambispective observational study at the Regional Cancer Centre, Thiruvananthapuram, involved 459 patients with biopsy-proven, locally advanced rectal adenocarcinoma (T3/T4 and/or node positive) who underwent long-course neoadjuvant chemoradiotherapy (NACTRT) followed by curative surgery (Anterior Resection or Abdomino-Perineal Resection) between 2010 and 2015, with a minimum follow-up of three years. Exclusion criteria included previous malignancies, synchronous colorectal cancers, oligometastatic disease, and loss to follow-up. The study aimed to determine the predictive value of the Neoadjuvant Rectal Cancer (NAR) score for oncological outcomes. Clinicopathological details were collected from case records, imaging, and operative notes, and postoperative histopathological records were used to calculate the NAR score. Patients were categorized into three groups based on their NAR scores and followed prospectively for recurrence or death. Data were analyzed using frequency, percentages, mean, standard deviation, Chi-square test, Kaplan-Meier method, log-rank test, and Cox regression analysis, with a significance level set at $p < 0.05$. **Results:** 459 patients were included in the study, with 41.4% females, and 44% being less than 60 years. Most of the patients had lesion 5-10 cm from anal verge (46%). 77% of patients had T3 disease prior to the start of NATCTRT. 51.9% had underwent APR, and 80.4% had open surgery. 11.1% had PCR attained. 27% had disease recurrence of which most were distant metastasis. NAR was a significant predictor of overall survival, disease free survival, distant metastasis free interval and locoregional failure free interval. Other significant predictors of overall survival were pre-NACTRT T stage, pN stage, PLNR, CRM status and PCR, but only NAR was found to be significant on multivariate analysis. **Conclusions:** NAR score is a significant predictor of OS, DFS,LRFI and DMFI and can be used as a short term surrogate for survival data in anticipated trials. Research Sponsor: None.

173 Poster Session

A cross-sectional analysis from a real-world cohort of patients with microsatellite instability colorectal cancer (MSI-CRC) with localized disease from the Spanish RETUD registry. First Author: Ana Fernandez Fernandez Montes, Department of Medical Oncology, Complejo Hospitalario Universitario de Ourense, Ourense, Spain

Background: Microsatellite instability (MSI) localized colorectal cancer (CRC) constitutes a subgroup of patients (pts) with different clinical and molecular characteristics and prognosis compared to microsatellite stable (MSS) disease. Here, we present our real world data regarding the management and clinical outcomes of a cohort of MSI-h localized CRC patients included in the Spanish Group of Treatment of Digestive Tumors (TTD) Registry (RETUD). **Methods:** RETUD is a national, multicenter registry for gastrointestinal tumors from the Spanish TTD Group. In this cross-sectional analysis we evaluated a real-world cohort of MSI-h localized CRC pts diagnosed from 1st January 2017 to 29th April 2024. Baseline characteristics and neo/adjuvant treatments are descriptively presented. Disease-free survival (DFS) and overall survival (OS) analyzed by Kaplan-Meier method are presented along with 95% confidence intervals (CI). **Results:** Five hundred and sixty-one (561) evaluable pts out of 679 MSI-CRC included in RETUD had a localized disease as initial diagnosis. Pts had a median age of 73.1 years and were predominantly Caucasian (97.7%) and female (52.6%). Fifty (12.4%) pts had Lynch Syndrome. Main tumor baseline characteristics are described (Table). Local-locoregional therapeutic procedures were: primary tumor resection in 531 (99.1%) pts and radiotherapy in 18 (3.2%) pts. A total of 198 (35.3%) pts received systemic therapy, mostly chemotherapy (n=165 83.3%) but 18 (9.1%) were treated with pembrolizumab due to unresectable disease. At database cut-off, 123 (21.9%) patients died, mostly due to not related intercurrent illness, and 96 (17.1%) pts developed metastasis. After a median follow up period of 28.5 months, the median (95% CI) OS was not reached and DFS was 78.2 (55.4-NA) months. The impact of adjuvant systemic therapy (chemotherapy vs. non-chemotherapy) for both stage II and stage III pts is under evaluation. **Conclusions:** Our work provides valuable real-world data from a cohort of MSI-h localized CRC pts treated under clinical practice conditions in Spain, reiterating the good disease prognosis exhibited by this subset of pts in contrast to MSS pts. Research Sponsor: Bayer; MDS Pharma Services; Roche.

Main tumor characteristics.	
Stage at initial diagnosis, n (%)	
I/II n (%)	55 (9.8) / 258 (46.0)
III, n (%)	248 (44.2)
Location of primary tumor[a], n (%)	
right colon /left colon /rectum	447 (78.4) / 82 (14.4) / 41 (7.2)
Type histological[b], n (%)	
Intestinal	271 (48.9)
Mucinous (colloid) adenocarcinoma (>50% mucinous)	115 (20.8)
Signet ring cell carcinoma (>50% signet ring)	14 (2.5)
Medullary carcinoma	18 (3.2)

[a]Pts with more than 1 primary tumor; the percentage may be over 100%. [b]Unknown histology in 132 (23.8%) pts and missing data in 7 pts.

174 Poster Session

Exploratory study of TAS-102 combined with intermittent administration of fruquintinib in the treatment of third-line metastatic colorectal cancer. First Author: Jiayu Niu, The First Affiliated Hospital of University of Science and Technology of China West District, Hefei, China

Background: Fruquintinib and TAS-102 are two globally recognized standard therapies for mCRC patients who have previously undergone standard chemotherapy. A phase II exploratory study suggested that the combination of TAS-102 and fruquintinib could prolong progression-free survival (PFS) and overall survival (OS) in these patients. However, the benefits are sometimes limited due to tolerability issues associated with continuous dosing. This study aimed to investigate the efficacy and safety of intermittent administration of TAS-102 combined with fruquintinib as a third-line treatment for mCRC patients. **Methods:** This open-label, single-arm, single-center, exploratory clinical trial enrolled mCRC patients who had failed at least two standard treatment regimens. Eligible patients received oral fruquintinib (3 mg, once daily, on days 1-5 and 8-12) and TAS-102 (35 mg/m², twice daily, on days 1-5) every two weeks until disease progression or unacceptable toxicity occurred. The primary endpoint was objective response rate (ORR). Secondary endpoints included OS, PFS , DCR, and safety. **Results:** From August 2023 to September 2024, 26 eligible patients were enrolled. The median age of the patients was 58 years (range: 19-77), with 61.5% being female. RAS mutations were present in 57.7% of patients, 57.7% had left-sided colon or rectal cancer, 57.7% had liver metastases, and 76.9% had multiple metastases. As of September 10, 2024, 22 patients had undergone at least one tumor assessment, and 12 patients were still on treatment. Among them, 18.2% (4/22) achieved partial response , and 50% (11/22) had stable disease as the best response. Median PFS (mPFS) was 135 days (95% CI: 60.52-209.48). mPFS in RAS-mutant and RAS wild-type patients was 161 days (95% CI: 63.37-258.63) and 135 days (95% CI: 60.20-209.80), respectively, with ORRs of 25% and 12.5%. mPFS in patients with liver metastases (LM) and non-LM was 135 days (95% CI: 45.42-224.58) and 161 days (95% CI: 87.84-234.16), respectively, with ORRs of 14.3% and 25%. mPFS in patients with multiple metastases was 135 days (95% CI: 71.86-198.14), with an ORR of 17.6%. Median OS was not yet mature. Treatment-related adverse events (TRAEs) were generally manageable and tolerable. The most common hematological TRAEs (any grade, grades 3-4) included leukopenia (76.9%, 11.5%), neutropenia (73.1%, 11.5%), and anemia (53.8%, 7.7%). Non-hematological TRAEs were mostly grades 1-2, including anorexia (46.2%) and fatigue (19.2%). One case of grade 3 hypertension was reported. No treatment-related deaths were observed. **Conclusions:** These preliminary results suggest that intermittent administration of TAS-102 in combination with fruquinitinib as a third-line treatment for patients with mCRC reduces hematological toxicity and is well tolerated, with encouraging clinical results. Clinical studies are ongoing. Clinical trial information: ChiCTR2300078241. Research Sponsor: None.

175 Poster Session

A phase 1 dose-escalation study of GCC19CART: A novel CAR T-cell therapy for metastatic colorectal cancer in the United States. First Author: Bridget P. Keenan, University of California, San Francisco, San Francisco, CA

Background: GCC19CART, the first clinical candidate from the CoupledCAR solid tumor platform, pairs solid tumor CAR T-cells with CD19-targeting CAR T-cells to enhance proliferation, activation, and persistence against solid tumors. GCC19CART targets guanylate cyclase-C (GCC), a known mucosal marker. Building on the promising results from a clinical trial in China, a phase 1 study was initiated in the US for patients with metastatic colorectal cancer (mCRC) refractory to standard chemotherapy. As of August 14, 2024, a total of 9 subjects have been treated. **Methods:** Subjects were initially screened for GCC expression, however screening was discontinued after >95% of subjects with mCRC demonstrated GCC expression by immunohistochemistry (95/98). Eligible subjects underwent leukapheresis, lymphodepleting chemotherapy (fludarabine 30mg/m^2 and cyclophosphamide 300mg/m^2 on day-3), and then infusion of a single dose GCC19CART. Objective response was assessed by RECIST v1.1. **Results:** Four subjects have been enrolled at dose level 1 (1x10^6 cells/kg) and 5 subjects at dose level 2 (2x10^6 cells/kg). The most common adverse events were cytokine release syndrome (CRS) in 9/9 subjects (Grade 1: 6/9 [66.7%] and Grade 2: 3/9 [33.3%]) and diarrhea in 8/9 subjects (Grade 1: 3/9 [33.3%], Grade 2: 3/9 [33.3%], Grade 3: 2/9 [22.2%]). Immune effector cell-associated neurotoxicity syndrome (ICANS) occurred in 2/9 subjects (Grade 2: 1/9 [11.1%] or Grade 3: 1/9 [11.1%]) and resolved rapidly with corticosteroid and dasatinib treatment. Three out of 9 subjects had a mild tremor (Grade 1) that resolved with corticosteroids. One subject experienced a dose-limiting toxicity (DLT) at dose level 2, presenting with Grade 3 diarrhea, Grade 4 enterocolitis and Grade 5 sepsis, and died from fungal sepsis 48 days post-infusion. At dose level 1, the overall response rate (ORR) was 50.0% (2/4). Two subjects demonstrated a partial response (PR) with one confirmed PR, while 1 additional subject had partial metabolic response on PET/CT with stable disease (SD) and 1 patient had progressive disease. At dose level 2, the ORR was 80% (4/5), including 1 subject with a pathological complete response, and 3 with a confirmed partial response, of which 1 had a complete metabolic response on PETCT. **Conclusions:** Preliminary results indicate that GCC19CART exhibits significant clinical activity in patients with refractory mCRC. Adverse events were related to GCC19CART mechanism and in line with what has been observed with other CAR T-cell products. Strategies to further optimize the management of treatment-related diarrhea are currently being implemented. The trial is ongoing and updated data will be presented. Clinical trial information: NCT05319314. Research Sponsor: None.

176 Poster Session

A phase Ib study of immunotherapy with ex-vivo pre-activated and expanded CBNK cells in combination with cetuximab, in patients with colorectal cancer (CRC) with minimal residual disease (MRD): Final report. First Author: Maria Pia Morelli, The University of Texas MD Anderson Cancer Center, Houston, TX

Background: Colorectal cancer (CRC) is the second most common type of cancer. Even following successful curative treatment, patients often relapse, likely because of undetected micro-metastatic foci. Tumor-informed ctDNA testing has helped to identify those patients before they developed radiographic evidence of disease, introducing the concept of minimal residual disease (MRD) in solid tumors. No standard-of-care treatment options are available for MRD patients. Therefore, we have designed a phase IB trial of the combination of *in vitro* pre-activated and pre-expanded cord blood natural killer (pre-A+E CBNK) cells with cetuximab. We hypothesized that we could achieve CAR-like efficacy without genetic engineering, by pre-activating NK cells with inflammatory cytokines and by using antibodies to redirect NK cell specificity and antibody-dependent cellular cytotoxicity. The highly immunosuppressive CRC tumor microenvironment is understudied, and novel NK-cell therapy could be highly effective in MRD clearance. **Methods:** 15 patients with CRC-MRD were enrolled in this trial and completed treatment. All the patients received immunodepleting treatment prior to an infusion of pre-A+E CBNK cells, up to a dose of 1x10^8/kg. Patients were monitored inpatient for 24 hours after treatment for infusion-related reaction, and were then seen for toxicity follow-up in the outpatient clinic twice/week for 4 weeks (DLT window). Samples for correlative analysis were collected at different times. **Results:** 8/15 patients enrolled were female and 7 were male, age range 26-69. They all had no measurable radiographic evidence of disease and a positive ctDNA Signatera test at baseline. 11/15 were previously treated for metastatic disease; 13 patients had a history of metastases (including 7 liver, 2 lymph nodal, 2 peritoneal). All of them received 5-FU-based chemotherapy. 7/15 were RAS/RAF WT, 4/10 were KRAS mut, 1/10 had BRAF mutation, 1/10 had HER2 amplification. They were all pMMR. No DLT was observed, only 1 patient had G2 CRS requiring tocilizumab. 9/15 patients treated so far had ctDNA clearance at some point. One patient had negative ctDNA at 3 and 6 months, 4/15 were negative at d28 and 7/15 at d14. 4/15 patients have not met the 90-day time point yet. 10/12 patients analyzed up to the primary endpoint had decreased ctDNA over baseline CBNK cells in blood peaked 3 hours post-infusion and persisted at 3 days. Phenotype analysis of CBNK pre-post, donor NK and other immune cells is ongoing. **Conclusions:** Cetuximab in combination with pre-A+E CBNK is safe and showed promising results for the treatment of MRD-CRC. Clinical trial information: NCT050468. Research Sponsor: MD Anderson Moonshot Program, K12 Paul Calabrese and ASCO CDA Conquer.

177 Poster Session

OP-TNT: Phase II study of total neoadjuvant chemoradiotherapy followed by local resection for early distal rectal carcinoma. First Author: Zhang Xian, Department of Radiation Oncology, Zhongshan Hospital, Fudan University, Shanghai, China

Background: Surgery is standard of care for pMMR early distal rectal cancer (RC), which can be partially exempted by totally neoadjuvant chemoradiotherapy (TNT) followed by watch and wait. But long-term data showed that 1/3 patients had local growth(mostly in the rectal wall)in two years. This study investigates the efficacy and safety of TNT followed by local excision(LE) for early distal rectal carcinoma (cT1-3N0M0) in order to preserve organs. **Methods:** In this single-arm, phase 2 prospective trial, patients were eligible if they were aged 18 years or older, with a performance status of 0-1,newly diagnosed, biopsy proven pMMR/MSS RC which located ≤5cm from the anal verge and staged as cT1-3N0M0. Patients were assigned to receive TNT: 6 courses of CapOx(capecitabine 1000 g/m2 PO BID D1-14, oxaliplatin 130 mg/m2 IV QD D1,Q3W), combined with concurrent long-course radiotherapy (LCRT)(45 Gy/25 Fx to the pelvis with an SIB of 50 Gy/25 Fx to the tumor bed). Evaluation was performed 12 weeks after completion of LCRT. Patients who achieved a clinical complete response (cCR) or near clinical complete response (ncCR) were assigned to LE, whereas those with >ypT1, positive margin, or neurovascular invasion after LE were recommended for total mesorectal excision (TME). The primary endpoint is the 3-year organ-preserving rate. Secondary endpoints include the pathological complete response (pCR)(encompassing response after LE and TME), adverse effects rate, 3y-disease-free survival (DFS)and 3y-overall survival (OS)rate. **Results:** From October 2022 to August 2024, 52 patients had completed therapy. Median age at evaluation and follow-up duration were 58.3 years and 11 months. Of the patients 78.8% (41/52) were defined as MRI stage T3, and 76.9%(40/52) located ≤3cm from the anal verge. Of 52 evaluable patients ,82.7% (43/52) achieved cCR/ncCR and had LE (6 via endoscopic resection and 37 through trans-anal minimally invasive surgery). Among these patients, 25.6% (11/43) were found to have residual tumors after LE, of which 6 were radical-operated while 5 refused this intervention. Nine patients who did not achieve cCR or ncCR received TME resection, and 6 of them were confirmed as pCR. The total pCR rate was 73.1% (38/52). Of all patients, 3 underwent abdominoperineal resection, and sphincter-preserving rate was 94.2%(49/52). No grade 3 or 4 adverse events were reported, and there were no treatment-related deaths. The main adverse event after local resection was the 1-3 months anterior resection syndrome. **Conclusions:** The results of interim analysis showed that tumor residue was found in patients with cCR/ncCR after TNT, similar to the regrowth rate previously reported. LE strategy may become prevention. Long-term outcomes are being followed up. Clinical trial information: NCT05563922. Research Sponsor: None.

178 Poster Session

A cross-sectional analysis from a real-world cohort of patients with microsatellite instability colorectal cancer (MSI-CRC) with metastatic disease from the Spanish RETUD registry. First Author: Cristina Gravalos Castro, Medical Oncology Department. 12 de Octubre University Hospital, UCM, Madrid, Spain, Madrid, Spain

Background: Survival of microsatellite instability (MSI) metastatic colorectal cancer (mCRC) patients (pts) has remarkably increased with immune checkpoint inhibitors (ICI). Here, we present our real-world data regarding the management and clinical outcomes of a cohort of 214 MSI-mCRC patients included in the Spanish Group of Treatment of Digestive Tumors TTD Registry (RETUD). **Methods:** RETUD is a national, multicenter registry for gastrointestinal tumors from the Spanish TTD Group. In this cross-sectional analysis we evaluated a real-world cohort of MSI-mCRC pts diagnosed from 1st January 2017 to 29th April 2024. Baseline characteristics, treatments and treatment response are descriptively presented. Tumoral response was evaluated according to RECIST 1.1 criteria. First line progression-free survival (PFS) and overall survival (OS) analyzed by Kaplan-Meier method are presented along with 95% confidence intervals (CI). **Results:** Among 679 MSI-CRC pts included in RETUD, a total of 214 mCRC pts were evaluable. Pts' age at diagnosis was 69.6 (26-96) years, and they were predominantly Caucasian (97.7%) and female (53.7%). Eastern Cooperative Oncology Group (ECOG) performance status was 0-1 for 168 (78.5%) pts. Seventeen (10.7%) pts presented Lynch syndrome. Main tumor biological characteristics are described at Table 1 and the molecular profile (when available) was: KRAS mutation (24.3%), NRAS mutation (3.5%) and BRAF^{v600E} mutation (53.0%). Surgical resections: 158 (73.8%) pts for primary tumor and 42 (19.6%) pts for metastasis. First line systemic treatment was administered to 187 (97.1%) pts: 92 (49.2%) pembrolizumab, 83 (44.4%) chemotherapy (CT), 4 (2.1%) other ICIs, 8 (4.3%) not reported. With a median follow up period of 17.4 months, the median (95% CI) OS and first-line PFS of the total mCRC population were 32.6 (22-72.4) and 11.1 (8.2-17.9) months (m), respectively. In immunotherapy (IT) pts mOS was not achieved (22.0-NA) and PFS was 26.5 m (12.9-NA) while the mOS was 28.9 m (16.3-69.3) and PFS 7.5 (5.4-9.3) m in CT pts. The overall response rate (ORR) was 54.2% in IT pts vs. 37.3% in CT pts. **Conclusions:** The introduction of IT has changed the evolution of MSI-H mCRC management, offering a beneficial impact in the OS and PFS survival in contrast to other conventional therapies in a real-world context. Research Sponsor: Bayer; MDS Pharma Services; Roche.

Tumor characteristics at initial diagnosis of CRC.

Stage at initial diagnosis of CRC	
I/II, n (%)	4 (1.9) / 26 (12.1)
III/IV, n (%)	66 (30.8) / 118 (55.1)
Location of primary tumor[a], n (%)	
right colon /left colon /rectum	160 (74.8) /37 (17.3) /19 (8.9)
Main metastatic location[b], n (%)	
Liver	84 (39.3)
Peritoneal	76 (35.5)
Lymph	71 (33.2)
Lungs	41 (19.2)

[a]Pts with more than 1 primary tumor; the percentage may be over 100%. Data missing for 4 (1.9%) pts. [b]Pts with more than 1 metastatic site; the percentage may be over 100%.

179 **Poster Session**

Neoadjuvant immunotherapy for locally advanced/metastatic mismatch repair deficient colorectal cancer: A two-year institutional experience. First Author: Maeve O'Neill, St James's Hospital, Dublin, Ireland

Background: MMRd is a tumour agnostic biomarker predictive of response to immune checkpoint inhibitors (ICIs). Given the poor response to 5-FU based chemotherapy, ICIs are now being explored in early MSI-H CRC. The outcomes on this strategy are limited. **Methods:** We retrospectively evaluated early outcomes for patients receiving neoadjuvant ICIs for unresectable locally advanced/oligometastatic MMRd colorectal cancer (CRC), where intention of treatment was downstaging to permit curative resection. Following discussion at regional MDT, suitable patients were administered a PD-1 inhibitor 6-weekly until disease progression or improvement rendering suitability for surgical resection (to max of 2 years). **Results:** From October 2022-September 2024, ten patients with MMRd CRC were commenced on neoadjuvant ICIs. These comprised six right, one left sided colonic and three rectal tumours. Median age at diagnosis was 59 (IQR54–68). Four patients had family history of CRC. One patient had clinical stage II, seven stage III and two patients oligometastatic stage IV disease. All were patients with initially unresectable disease, with potential for curative resection pending response to ICI. Histologically, two tumours were poorly differentiated, two mucinous phenotype and six moderately differentiated adenocarcinomas. BRAF mutation was identified in three patients. None had KRAS mutation. Three rectal cancers received radiotherapy in addition to immunotherapy. Five required a defunctioning stoma either prior to or on ICI. Median follow-up was 12.5 months (IQR7-17). Objective response rate by RECIST 1.1 was 100%, with significant downstaging in 9 of 10 patients. The remaining patient is early in treatment course with no reimaging. To date, complete clinical response (cCR) has been observed in three patients (33%). All 10 patients are alive. Grade 3-4 treatment-related adverse events occurred in two patients (20%) who developed treatment-related strictures (one also had tumour perforation). This was successfully managed with defunctioning ileostomy and antibiotics, with subsequent cCR. **Conclusions:** We report impressive early outcomes of immunotherapy for locally advanced/metastatic MMRd CRC, with excellent downstaging and high predicted cCR. This highlights the importance of screening all CRC for MSI-H/MMRd. In the setting of initially unresectable CRC, an ICI-first approach can render patients eligible for surgery, however deep responses can result in local effects including strictures and/or perforation. Research Sponsor: None.

180 **Poster Session**

A phase 1 trial of folinic acid, fluorouracil, oxaliplatin, bevacizumab, botensilimab, balstilimab (FOLFOX-3B) in microsatellite stable metastatic colorectal cancer. First Author: Marwan Fakih, Department of Medical Oncology and Therapeutics Research, City of Hope Comprehensive Cancer Center, Duarte, CA

Background: Botensilimab (BOT) is a novel Fc fragment engineered CTLA4 inhibitor designed to boost innate and adaptive immune response. The combination of BOT and the PD-1 antibody balstilimab (BAL) has been associated with durable and deep responses in microsatellite stable (MSS) metastatic colorectal cancer (mCRC). Given the potential synergistic activity between checkpoint inhibitors and oxaliplatin and VEGF(R) inhibitors, we performed a phase 1 clinical trial of folinic acid, fluorouracil, oxaliplatin, bevacizumab, botensilimab, and balstilimab (FOLFOX-3B). **Methods:** We enrolled ECOG performance status 0-1 patients (pts) with measurable MSS mCRC, up to 2 prior lines of therapy, and without progression following prior oxaliplatin. FOLFOX was given at a fixed dose, every 2 weeks, across the study: folinic acid at 400 mg/m^2 x 2 hours (hr), oxaliplatin at 85 mg/m^2 x 2 hr, and fluorouracil at 2400 mg/m2 x 46 hrs. BOT was administered at 2 dose levels (DL) of 25 mg and 75 mg IV Q6 weeks x 2. BAL was administered at a fixed dose of 240 mg IV Q2 weeks. The study followed a 3 x 3 escalation design with up to 9 patients per DL. A dose level was deemed safe if $<$ 1 out of 6 pts or $<$ 2 out of 9 pts had a DLT (expansion to 9pts only if 2 DLTs encountered in the 1st 6 pts). The DLT period consisted of 6 weeks. Pts should have received all intended BOT/BAL and FOLFOX doses in the 1st 6 weeks to be deemed DLT-evaluable. **Results:** 14 pts were enrolled on study: 9/14 with liver metastatic disease, 5/14 *RAS*-mutated, all TMB low. 7 were treated on DL1 and 7 on DL2. 6/7 pts at DL1 had received 1-2 prior lines of therapy, 4 of whom included oxaliplatin. 6/7 of pts at DL2 had 1-2 prior systemic therapy, all of whom received prior oxaliplatin. No DLT were noted at DL1 or DL2. 1 pt at each DL was not DLT-evaluable as they were not able to receive all intended therapy due to treatment interruptions. Grade (G)2 and above immune related toxicities included: one pt at DL1 with transient G3 AST/ALT elevation and transient G2 colitis (resolved with steroids and infliximab), one pt at DL1 with G2 hypothyroidism and one patient at DL2 with G2 hyperthyroidism. At DL1, 4/7 had a partial response (PR) - including 2/4 with prior oxaliplatin exposure. At DL2, 6/7 pts had a PR – including 5/6 with prior oxaliplatin exposure. 3 patients proceeded to surgical resection or ablation and are currently followed with observation only. In the combined cohort of DL1 and DL2, the median progression free survival (PFS) is 8 months and remains unreached for DL2. Overall survival outcome has not matured at the time of this analysis. **Conclusions:** FOLFOX-3B with BOT at 75 mg IV Q6 weeks x 2 in combination with BAL 240 mg Q2 weeks is well-tolerated and is associated with promising clinical activity with 5/6 PR and prolonged PFS. The study was amended to explore additional DL that explore higher doses of BOT (150 mg) or longer duration of BOT 75mg (4 doses). Clinical trial information: NCT05627635. Research Sponsor: Agenus.

181 **Poster Session**

The role of stereotactic radiotherapy in brain metastases from gastrointestinal cancer. First Author: Michele Aquilano, IFCA, Radiation Oncology Department, Florence, Italy

Background: The incidence of brain metastases from Gastrointestinal primary tumors (GI-BMs) is approximately of 6%. Appropriate management of these patients, who often presents at advanced disease stage, is poorly defined. Stereotactic Radiotherapy (SRT) could be proposed to improve symptoms and extend disease control. The aim of this study is to assess the impact of SRT in this population. **Methods:** Data from consecutive patients treated for GI-BMs with SRT from March 2015 to March 2024 were retrospectively collected. Dose was expressed as Equivalent Dose in 2Gy Fractions (EQD2). Intra-Cranial Control (IC) and Overall Survival (OS) were calculated from date of SRT to event (intracranial relapse and death respectively) or last follow-up. **Results:** Fifty-four patients (median age 68 years, range 46-84) accounting for 98 GI-BMs were included. Primary tumors were colorectal and gastro-oesophageal adenocarcinoma in 37 (68.5%) and 17 (31.5%) patients respectively. Eighteen (33%) patients presented with a single GI-BM. Extra-cranial disease was present in 41 (76%) patients, including unresected/relapsing primary tumor in 9 (17%) cases. Detection of BMs occurred at diagnosis in 8 (15%) patients, as first site of relapse in 27 (50%) or as progression of known metastatic disease in 19 (35%) patients. Before SRT, surgery and/or WBRT were performed in 10 (19%) and 5 (9%) patients respectively. For each SRT course a median of 1 (range 1-5) GI-BMs was treated. Dose regimens consisted of 12-30 Gy in 1-5 fractions, to a median EQD2 of 50 Gy_{10} (range 22-68). A second SRT course for out-of-field intracranial progression was performed in 7 patients (13%). First- and further chemotherapy lines were administered in 39 (72%) and 15 (28%) patients respectively. IC rate was 88% at 6 months and 70% at 1 year. At univariate analysis (UVA) no variables were correlated with IC. Median OS was 11 months (IC 95% 9-16), 6-months and 1-year OS rate was respectively 76% and 48%. Only systemic treatment beyond first line was correlated with OS at UVA (4 versus 15 months, p=0.02). Two (4%) patients developed mild symptomatic radionecrosis. **Conclusions:** Onset of GI-BMs can occur at any stage of disease. Despite poor overall outcome, selected patients may benefit from SRT to improve IC, particularly in association with first-line systemic treatment. Multiple SRT courses can be delivered in case of further intracranial recurrence. Symptomatic radionecrosis may occur in less than 5% of cases. Research Sponsor: None.

182 **Poster Session**

Phase I/II trial of encorafenib, cetuximab, and nivolumab in microsatellite stable *BRAF*^{V600E} metastatic colorectal cancer following progression on prior BRAF+EGFR targeted therapies. First Author: Kelsey Pan, The University of Texas MD Anderson Cancer Center, Houston, TX

Background: A *BRAF*^{V600E} mutation, present in approximately 8-10% of all colorectal cancers (CRC), is a poor prognostic biomarker for patients with metastatic CRC. The BRAF^{V600E} inhibitor encorafenib (E) and the anti-EGFR antibody cetuximab (C) are approved for refractory, *BRAF*^{V600E} mCRC based upon on overall response rate (ORR) of 20% and median progression-free survival (PFS) of 4.2 months. A single-arm trial adding the anti-PD1 antibody nivolumab (N) to E+C in the same (BRAF-inhibitor naïve) population showed promising efficacy, with an ORR of 50% and median PFS of 7.4 months. We evaluated the efficacy of E+C+N in patients with microsatellite stable (MSS), *BRAF*^{V600E} mCRC who had progressed on prior E+C. **Methods:** Patients with MSS, *BRAF*^{V600E} mCRC with documented prior progression on E+C were eligible for treatment with E (300 mg PO daily), C (500 mg/m^2 q14 days), and N (480 mg IV q28days) were included in the study. Primary endpoint was best overall response by RECIST 1.1. Secondary endpoints were PFS, overall survival (OS), and toxicity (by CTCAE v5). Circulating tumor DNA assessment of molecular alterations in the MAPK pathway in association with response were performed by NGS. Descriptive statistics were used to summarize outcomes. **Results:** Among the 12 participants, none achieved a radiographic response, and 5 experienced stable disease. Disease control rate was 41.7% (95% CI, 13.8-69.6). Median PFS was 3.7 months (95% CI, 1.7-5.7), and median OS was 8.3 months (NE-NE). Four patients (16.7%) had concomitant MAPK-activating alterations at baseline, and *APC*, *TP53*, and *PIK3CA* mutations were present in 25.0%, 66.7%, and 33.3% of patients, respectively. The presence of MAPK-activating mutation was not associated with disease progression (odds ratio 1.3; 95% CI 0.11-16; p=0.82). The most common grade 1-2 treatment-related adverse events (AEs) were anemia (N=5), arthralgia (N=5), rash (N=4), headache (N=4), and fatigue (N=3); 1 of the 12 patients experienced a grade 3 small bowel obstruction. **Conclusions:** In this pilot study, addition of N to E+C did not appear to overcome resistance to prior BRAF + EGFR targeted therapies in patients with MSS, *BRAF*^{V600E} mCRC. While the addition of immunotherapy has demonstrated promising signal in early-phase clinical trials for this population, the benefit appears unlikely when offered sequentially after progression on BRAF combination therapies. Clinical trial information: NCT04017650. Research Sponsor: None.

183 **Poster Session**

Safety, tolerability, and preliminary efficacy of tinodasertib as a monotherapy or in combination with pembrolizumab or irinotecan in metastatic colorectal cancer: Interim results from a phase II open-label, dose-finding, run-in and cohort expansion study. First Author: Marco Matos, Oncology, Pindara Private Hospital, Benowa, Australia

Background: MNK inhibition has been shown to downregulate phosphorylation at serine 209 of eIF4E, a potent regulatory point of CAP-mediated translation of 5′ polyadenylated mRNA associated with growth factor and pro-oncogenic signals in cells. Elevated levels of eIF4E phosphorylation are observed in a broad range of tumors. MNK1/2 knock out mice are healthy and viable, as cellular house-keeping mRNA translation occurs readily using the IRES mechanism, whereas mRNA with more complex and longer 5′ untranslated ends such as those involved in growth factor and pro-oncogenic signaling, are MNK activated. Cells from MNK knockout animals become relatively resistant to subsequent oncogenic transformation. In a program of SAD and MAD Phase I studies conducted in normal healthy volunteers, Tinodasertib was found to be safe and well tolerated with no dose-limiting toxicity (DLT). The objectives of this ongoing Phase 2 study are to evaluate safety, and preliminary efficacy of Tinodasertib as monotherapy and in combination with either pembrolizumab or irinotecan in patients with advanced colorectal cancer (CRC). **Methods:** Patients had to have advanced CRC and previously received ≥2 lines of therapy. The dose escalation phase of the study was open to all patients with CRC. As of 23 July 2024, 22 patients were dosed in a modified 3X3 dose escalation design with Tinodasertib from 20 to 80 mg on alternate days. Out of 22 patients, 12 received monotherapy, 4 were treated with Tinodasertib and Irinotecan and 6 were treated with Tinodasertib and pembrolizumab. Majority (19) had MSS CRC and 3 had unknown status. Nine patients had KRAS-mut CRC. **Results:** No DLTs were observed and MTD was not reached. Grade 3 treatment-related adverse events (TRAEs) were observed in 2 (9%) patients and were related to irinotecan while no Grade 3 AEs were attributed to Tinodasertib by either the investigator or the sponsor. Most common TRAEs were related to gastrointestinal system organ class. There were no Grade 4-5 TRAEs. Of the 22 patients, 18 were evaluable for RECIST 1.1 response. No patient had an objective response to treatment, 12 had stable disease with disease control rate of 67% and a progression free survival of 2.99 months. Patients remained on therapy for up to 28 weeks. Overall survival (OS) at 52 weeks was 52% (CI 29 to 93). **Conclusions:** Tinodasertib either as monotherapy or combined with irinotecan or pembrolizumab, was well tolerated with no DLTs at the dose levels evaluated. Prolonged median time to progression and OS compared to historical controls were observed even during the dose escalation phase for the study population as a whole and for each of the monotherapy and combination arms. Enrolment in the dose escalation phase continues. Clinical trial information: NCT05462236. Research Sponsor: AUM Biosciences Pte. Ltd.

184 **Poster Session**

Rechallenge with epidermal growth factor receptor inhibitors for metastatic colorectal cancer: A systematic review and meta-analysis. First Author: Amy H. Huang, UConn School of Medicine, Farmington, CT

Background: Clonal evolution is a mechanism of treatment resistance against epidermal growth factor receptor inhibitors (EGFRi) in metastatic colorectal cancer (mCRC). Upon EGFRi discontinuation, EGFRi-resistant subclones decay with time, which allows for EGFRi rechallenge. We conducted a meta-analysis to investigate the efficacy of EGFRi rechallenge for patients with mCRC. **Methods:** Clinical trials and cohort studies investigating the efficacy of EGFRi rechallenge were extracted from PubMed and Embase from inception to January 31, 2024. Hazard ratios (HR) of overall survival (OS) and progression-free survival (PFS) were aggregated using random-effects models. **Results:** Among 179 studies initially retrieved from the databases, 23 studies (665 patients) were included in the final analysis. There were 11 single-arm trials, 5 randomized controlled trials (including 3 post-hoc analyses), and 7 observational studies. At least 444 patients (66.8%) had left-sided mCRC. In 19 studies, EGFRi rechallenge was given only for patients previously progressing on 5-fluoropyrimadine, oxaliplatin and irinotecan. RAS/RAF mutation status was assessed with circulating tumor DNA (ctDNA) or tissue-based sequencing right before EGFRi rechallenge in 13 studies. Ten clinical trials only enrolled patients with RAS/RAF-wild type mCRC for EGFRi rechallenge. The median EGFRi-free interval before EGFRi rechallenge was ≧4 months in 13 studies (≥10 months in 9 studies) and between 2 to 4 months in 3 studies. Thirteen studies reported an objective response rate (ORR) up to 70%. Eight out of the 13 studies (61.5%) reported an ORR >10% (7 studies with ORR >20%), which is numerically higher than that achieved by FDA-approved third-line agents for mCRC (ORR <10%). EGFRi rechallenge was associated with significantly longer pooled OS (HR 0.62; 95%CI, 0.42-0.92; I^2=51.9%) and pooled PFS (HR 0.55; 95%CI, 0.38-0.79; I^2=5.5%) than non-EGFRi systemic therapy. Patients without detectable RAS/RAF mutations by ctDNA right before EGFRi rechallenge had a significantly longer OS (HR, 0.39; 95%CI, 0.23-0.66; I2=32%) and PFS (HR, 0.41; 95%CI, 0.29-0.59; I2=0%) than those with detectable RAS/RAF mutations by ctDNA. Acneiform rash was the most commonly reported non-hematologic adverse event (up to 38.9%). No treatment-related deaths were reported. **Conclusions:** EGFRi rechallenge is associated with significantly longer OS and PFS than conventional systemic therapy and a clinically meaningful ORR in heavily treated patients with mCRC, particularly for patients with RAS/RAF-wild type mCRC tested right before EGFRi rechallenge. Research Sponsor: None.

185 **Poster Session**

Neoadjuvant PD-1 blockade in mismatch repair-deficient or microsatellite instability-high, locally advanced and resectable colorectal cancer: A retrospective real-world analysis. First Author: Yugui Lian, The First Affiliated Hospital, Zhengzhou University, Zhengzhou, Henan, China

Background: Neoadjuvant PD-1 blockade therapy has shown encouraging efficacy in patients with mismatch repair- deficient (dMMR)/microsatellite instability-high (MSI-H) locally advanced colorectal cancer. However, data from the real-world setting are scarce. **Methods:** We retrospectively collected clinical data of patients with dMMR/MSI-H locally advanced colorectal cancer diagnosed between 2022-2024 that received neoadjuvant anti-PD-1 monotherapy at the First Affiliated Hospital of Zhengzhou University. Pathological complete response, clinical complete response, R0 resection rate, and safety were analyzed. All statistical analyses were conducted using SPSS Statistics, version 22.0. **Results:** A total of 45 patients were enrolled. As of the data cutoff (Sep 15, 2024), the median follow-up was 14.0 months (range: 4.0-31.4). The median age was 53 years (range: 29.0–78.0) and 25 (55.6%) of 45 patients were male. 32 (71.1%) patients had clinical stage III disease and 39 (86.7%) patients had colon cancer. A higher proportion of patients (53.3%) had clinical T4 stage. 37 patients underwent the surgery of whom 29(78.9%) had a pathological complete response. The R0 resection was 100%. The median treatment duration was 4 cycles (range: 2-10). Among the 18 patients who received equal to or less than 4 cycles, the pCR rate was 77.8%, while for the remaining 17 patients who had more than 4 cycles of treatment, the pCR rate was 84.2%. Eight other patients had a clinical complete response and chose the watch and wait strategy. Treatment-related adverse events ≥3-4grade were observed in 3 patients. All patients survived without any disease recurrence. **Conclusions:** In this retrospective analysis, anti-PD-1 monotherapy is effective and tolerable for patients with dMMR/MSI-H locally advanced colorectal cancer. This study shows that pCR rate increases is associated with a longer treatment cycles probably. Further analysis of larger real-world datasets with longer follow-up will be essential to validate these results. Research Sponsor: None.

187 **Poster Session**

Immunotherapy for MSI-H colorectal cancer with isolated peritoneal metastases. First Author: Daniel Aryeh Metzger, NewYork Presbyterian-Cornell, New York, NY

Background: NCCN guidelines recommend immunotherapy for MSI-H colon cancer (CC) with isolated peritoneal metastases (iPM); however, limited data support its benefit. This study aimed to evaluate the impact of immunotherapy in this population. **Methods:** Patients diagnosed with MSI-H CC with iPM from 2016-2019 were selected from the NCDB. Patients who received immunotherapy were compared to those who did not. Overall survival was assessed using Kaplan-Meier analysis and multivariate Cox proportional hazard regression. The trend of immunotherapy utilization from 2016-2020 was analyzed using the Cochran-Armitage test. **Results:** We analyzed a cohort of 444 patients, with 161 (36%) receiving immunotherapy. Patients in the immunotherapy group were, on average, younger (63 vs. 71 years; P<0.001), with balanced gender distribution between groups. Among those who received immunotherapy, 70% were treated as a first-line agent, and 30% after chemotherapy. Immunotherapy was more commonly administered in the adjuvant setting (66%) compared to neoadjuvant (7%) or without surgery (27%). Immunotherapy was associated with significantly longer median survival (33.6 vs. 14.9 months; P<0.001). In multivariate analysis, immunotherapy was associated with improved overall survival, whether given as a first-line agent (HR 0.54, 95% CI 0.39-0.74; P<0.001) or second-line after chemotherapy (HR 0.59, 95% CI 0.37-0.94; P=0.03). Additionally, primary tumor resection (HR 0.44, 95% CI 0.32-0.60; P<0.001), but not metastatectomy (HR 0.96, 95% CI 0.71-1.29; P=0.8), was associated with prolonged survival. Patients who received both immunotherapy and primary tumor resection (n=118) had a median survival of 43.8 months. From 2016 to 2020, the utilization of immunotherapy in this cohort increased from 29% to 37% (χ^2=0.23, P=0.63). **Conclusions:** Utilization of immunotherapy remains low in MSI-H CRC with iPM and did not change significantly during the study period. Immunotherapy is associated with improved overall survival and should be considered for appropriate candidates. The combination of primary tumor resection and immunotherapy yielded the best overall survival. Research Sponsor: None.

188 Poster Session

Optimal duration of neoadjuvant chemotherapy prior to CRS±HIPEC for colorectal cancer: An assessment of survival and postoperative outcomes. First Author: Frances Jenkins Bennett, Emory University, Atlanta, GA

Background: Cytoreductive surgery (CRS) with or without heated intraperitoneal chemotherapy (HIPEC) represents a viable treatment option for select patients with colorectal cancer peritoneal metastases. Given high recurrence rates after surgery, it is common practice to administer neoadjuvant chemotherapy (NAC). The optimal duration, however, when considering oncologic value and postoperative outcomes, is not known. **Methods:** A single institution database (2009-2024) of colorectal cancer patients that underwent CRS±HIPEC was reviewed. Patients with colorectal cancer undergoing CRS±HIPEC for curative intent (completeness of cytoreduction 0 or 1) with known NAC duration were included. Analysis was stratified by NAC duration of 0-3 months or greater than 3 months. Co-primary outcomes were recurrence free survival (RFS) and overall survival (OS). Secondary outcomes included clinically significant complications, defined as Grade 3 or greater adverse events based on Common Terminology Criteria for Adverse Events Version 5.0. **Results:** From 2009-2024, 108 patients underwent CRS±HIPEC for colorectal cancer. Final analysis included 84 patients that underwent curative intent surgery and had a known duration of NAC (> 3 months NAC: 55 patients, 0-3 months NAC: 29 patients). Median peritoneal cancer index (PCI) score was similar between the two cohorts (> 3 months: median PCI 11 versus (vs) 0-3 months: median PCI 12; p = 0.42). Recurrence-free survival was 9 months for the entire cohort and not significantly different based on duration of NAC (> 3 months: 8 months vs 0-3 months: 15 months, p = 0.14). Overall survival was also similar between both cohorts (> 3 months: 26 months vs 0-3 months: 37 months; p = 0.11). On univariate analysis, > 3 months of NAC was associated with an increased rate of complications of any severity (> 3 months: 83.6% vs 0-3 months: 55.2%; p = 0.01), clinically significant complications (> 3 months: 54.5% vs 0-3 months: 31.0%; p = 0.07), and increased median length of stay (> 3 months: 10 days vs 0-3 months: 8 days; p = 0.01). When controlling for other perioperative variables on multivariable analysis, > 3 months of NAC trended towards an increased risk of clinically significant complications (HR 1.76, 95% CI 0.59-5.30; p = 0.32). **Conclusions:** Even with a similar disease burden, as measured by PCI, an extended duration of neoadjuvant chemotherapy prior to CRS±HIPEC does not appear to be associated with any improved recurrence-free or overall survival. Conversely, postoperative outcomes are worse with higher complication rates and increased length of stay. Thus, despite high recurrence rates after CRS±HIPEC for this disease, it still appears that in well selected patients, an extended duration of neoadjuvant chemotherapy may not be the best treatment strategy. Research Sponsor: None.

189 Poster Session

TAS102 in combination with oxaliplatin (TASOX) for refractory metastatic colorectal cancer. First Author: Sharon Li, Rutgers Cancer Institute of New Jersey, New Brunswick, NJ

Background: TAS102 (trifluridine/tipiracil hydrochloride) is approved for use alone or with bevacizumab in late line metastatic colorectal cancer (mCRC), with overall survival benefit in phase 3 trials (RECOURSE and SUNLIGHT). Oxaliplatin is frequently reintroduced after progression, resolution of limiting neuropathy, or disease recurrence post adjuvant therapy. In preclinical studies, oxaliplatin combined with TAS102 has shown synergism. We hypothesized TASOX may be an option for patients who have progressed/recurred after FOLFOX. **Methods:** We assessed safety and efficacy of TAS102 with oxaliplatin, enrolling patients with metastatic CRC progressed on ≥2 lines of therapy including 5FU, oxaliplatin and irinotecan. Patients with recurrence during or within 6 months of adjuvant chemotherapy were allowed. Exclusion: Prior TAS102 exposure, functional impairing peripheral neuropathy, ≥Grade 3 hypersensitivity to oxaliplatin, or uncontrollable grade 1-2 hypersensitivity to oxaliplatin. Treatment continued until disease progression or unacceptable toxicity. Patients received oxaliplatin 85 mg/m2 and TAS-102 35 mg/m2 bid days 1-5 every two weeks, and bevacizumab per MD choice. The primary endpoint was overall response rate (ORR) by RECIST (by independent radiologists). Secondary endpoints included disease control rate (DCR), safety and tolerability. **Results:** 54 patients enrolled; 53 received treatment on study and 48 had ≥1 disease assessment (median age 59, 58 % male, tumor RAS mutated 69%). Median time on study was 4 months (8 cycles; range 1-37 cycles). ORR was 6% (n=3), with average change in target lesions -45% (range -35 to -67%). 71% (n=34) had SD defined on study, with average change in target lesions -3% (range -29% to +20%). In those treated with bevacizumab (59%; n=28), 79% (n=22) had disease control vs. 75% (n=15). The most common reason for study discontinuation was progressive disease (23%, n=11). 68% (n=36) had a treatment-related adverse event (TRAE) ≥Grade 3 at any point during therapy. The most common G3 TRAEs were anemia 19% (n=10) and leukopenia 28% (n=15) at any time. Two (4%) developed grade 3 neuropathy. **Conclusions:** TASOX is effective as a 3rd line therapy for patients with mCRC. In our study, overall disease control rate was 77% with up to 19 months on study. Anemia and leukopenia are common but manageable. Bevacizumab did not seem to have as great an effect in our trial as observed in SUNLIGHT. Future studies would warrant TASOX in combination to bevacizumab in randomized trials in candidates for oxaliplatin retreatment. Clinical trial information: NCT02848079. Research Sponsor: None.

190 Poster Session

QL1203 vs placebo plus mFOLFOX6 as first-line therapy in RAS wild-type, metastatic colorectal cancer (mCRC): Interim analysis (IA) of a multicenter, randomized, double-blinded, parallel, phase 3 trial. First Author: Weijian Guo, Department of Medical Oncology, Fudan University Shanghai Cancer Center, Shangai, China

Background: Panitumumab plus FOLFOX or FOLFIRI has been approved as the first-line therapy for RAS wild-type mCRC. QL1203 is a panitumumab biosimilar showing similarity to the originator panitumumab (Vectibix, Amgen) in preclinical studies and phase 1 clinical pharmacokinetic study. Here we present results from interim analysis of a phase 3 trial (NCT04233151) investigating QL1203 vs placebo plus mFOLFOX6 as first-line therapy in Chinese patients (pts) with RAS wild-type mCRC. **Methods:** Pts with treatment-naïve, RAS and BRAF wild-type, mCRC unsuitable for radical resection and local therapy were enrolled and randomly assigned (2:1) to receive QL1203 (6 mg/kg) plus mFOLFOX6 once every 2 weeks (Q2W) or placebo plus mFOLFOX6 Q2W, stratified by primary tumor site, liver metastases, and previous neoadjuvant/adjuvant therapy. The primary endpoint was PFS assessed by a Blinded Independ Review Committee (BICR) per RECIST v1.1. Key secondary endpoints include PFS by investigator (INV), OS, ORR, DoR, and safety. This planned IA was conducted at 338 (81.25%) of 416 PFS events occurred. **Results:** From Jan 8, 2020 to Jun 12, 2023, 641 pts were randomized (QL1203/placebo, n=426/215). 551 (86.0%) pts had left-sided tumors (QL1203/placebo, 85.9%/86.0%). 433 (67.6%) pts had liver metastases (QL1203/placebo, 68.1%/66.5%). As of data cutoff of this IA (Mar 22, 2024), median follow-up was 23.8 months. Median PFS by BIRC was 11.20 months for QL1203 and 8.34 months for placebo (hazard ratio [HR], 0.61 [97.42% confidence interval [CI], 0.47-0.79]; stratified one-sided log-rank p<0.0001). Other efficacy results are presented (Table). Incidence of grade≥ 3 adverse events (AEs) related to study drug was 59.9% for QL1203 and 32.9% for placebo. The most common grade≥3 AE related to QL1203 was neutrophil count decreased (20.3%). **Conclusions:** QL1203 demonstrated significantly improved PFS versus placebo and higher ORR. Clinical trial information: NCT04233151. Research Sponsor: None.

	QL1203 (n=426)	Placebo (n=215)
PFS by BIRC		
mPFS, months (95% CI)	11.20 (9.89-13.11)	8.34 (7.26-8.54)
HR (97.42% CI)	0.61 (0.47-0.79)	
PFS by INV		
mPFS, months (95% CI)	10.91 (9.69-11.30)	8.41 (7.39-9.07)
HR (95% CI)	0.74 (0.60-0.91)	
mOS, months (95% CI)	27.66 (24.74-32.79)	24.54 (20.17-27.79)
HR (95% CI)	0.82 (0.64-1.06)	
ORR by BIRC, n (%)	68.31%	47.91%
95% CI	63.66-72.70	41.07-54.81
ORR by INV, n (%)	64.08%	49.30%
95% CI	59.33-68.65	42.44-56.19

191 Poster Session

Efficacy and safety of glecirasib (JAB-21822) monotherapy and in combination with cetuximab in patients with KRAS G12C-mutated advanced colorectal cancer. First Author: Jian Li, Beijing Cancer Hospital, Beijing, China

Background: Glecirasib (JAB-21822), a highly selective, covalent and oral KRAS G12C inhibitor, has demonstrated a favorable safety profile and promising efficacy result as a monotherapy in NSCLC. We previously reported preliminary data of glecirasib monotherapy (NCT05009329) and in combination with cetuximab (NCT05194995) in KRAS G12C-mutated advanced Colorectal Cancer (CRC); here, we present the updated efficacy results. **Methods:** Patients (pts) with advanced CRC harboring KRAS G12C mutation were enrolled and treated with either single agent glecirasib (800mg QD) or glecirasib (800mg QD) combined with cetuximab. The primary endpoint was overall response rate (ORR). The secondary endpoints included duration of response (DoR), disease control rate (DCR), progression-free survival (PFS), overall survival (OS) and safety. **Results:** As of June 30, 2024, 44 pts received glecirasib (median follow up was 21.9 months). The median age was 55.5 years with female (38.6%) /male (61.4%), ECOG PS 0 (25.0%) /1 (75.0.%), and 79.5% having received ≥ 2 prior lines of therapy. The confirmed ORR and DCR were 22.7% (10/44) and 86.4% (38/44), respectively. Median DoR was 4.4months (95%CI: 4.2, 9.7), median PFS was 5.6 months (95%CI: 4.1, 7.0), and median OS was 16.0 months (95%CI: 8.8, 26.3). Any grade treatment-related adverse events (TRAEs) occurred in 41 (93.2%) pts, the most common ones (>20%) being anemia, blood bilirubin increased, hypertriglyceridemia and white blood cell count decreased. Grade 3-4 TRAEs occurred in 9 (20.5%) pts. No grade 5 TRAEs occurred. No patient discontinued treatment due to TRAEs. Among the47 pts treated with combination regimen (median follow up was 18.7 months), the median age was 54.8 years with female (53.2%)/ male (46.8%), ECOG PS 0 (44.7%) /1 (55.3%), and 74.5% having received ≥ 2 prior lines of therapy. The confirmed ORR and DCR were 50% (23/46) and 87.0% (40/46), respectively. Median DoR was 5.1 months (95%CI: 4.1, 6.9), median PFS was 6.9 months (95%CI: 5.4-6.9), and median OS was 19.3 months (95%CI: 13.1, NE). Any grade TRAEs occurred in 46 (97.9%) pts, the most common ones (>20%) being rash, skin fissures, blood bilirubin increased, ALT increased, AST increased, anemia and proteinuria. Grade 3-4 TRAEs occurred in 9 (19.1%) pts. No grade 5 TRAEs occurred. 2(4.3%) pts discontinued treatment due to TRAEs. **Conclusions:** Glecirasib in combination with cetuximab demonstrated better efficacy compared with glecirasib monotherapy in advanced KRAS G12C mutated advanced CRC, while maintaining a favorable safety profile. Clinical trial information: NCT05009329, NCT05194995. Research Sponsor: None.

192 Poster Session

Short-course radiotherapy (SCRT) followed by fruquintinib plus adebrelimab and CAPOX in the total neoadjuvant therapy of locally advanced rectal cancer (LARC): A multicenter, single-arm, open-label, phase II study (UNION TNT). First Author: Zhenyu Lin, Cancer Center, Union Hospital, Tongji Medical College, Huazhong University of Science and Technology, Wuhan, China

Background: We have demonstrated that neoadjuvant SCRT followed by PD-1 inhibitor plus CAPOX for LARC patients (pts) showed a higher pCR rate and a well-tolerated safety profile in phase III UNION study. Fruquintinib is a potent and highly selective VEGFR inhibitor, which had been approved for previously treated mCRC pts. This study aimed to investigate the efficacy and safety of SCRT followed by fruquintinib plus adebrelimab and CAPOX as a total neoadjuvant therapy for high-risk LARC pts. **Methods:** In this prospective, open-label, multi-centers, single-arm phase 2 study (NCT06234007), pts diagnosed with newly diagnosed and high risk LARC were recruited. Pts received SCRT (25Gy/5f) followed by six cycles of fruquintinib (4mg, qd, po, q3w,for the initial 6 pts in the safety run-in period, dosage would be adjusted based on dose-limited toxicities in the 1st cycle) combined with adebrelimab (1200mg, iv, d1, q3w) and CAPOX (q3w). Primary endpoint was CR rate (including clinical CR & pathologic CR). Secondary endpoints included 3-year EFS rate, OS, R0 resection rate and safety. **Results:** As of 30 Aug 2024, 45 pts were enrolled (median age 59 years [range: 24-75], 26 males, 23 clinical T4 stage and 20 clinical N2 stage, 36 with EMVI positive, 31 with MRF positive, 22 tumor located ≤5cm from the anal verge). Up to 20 Sep 2024, all pts completed SCRT, 73.33% (33/45), 57.78% (26/45) and 31.11% (14/45) pts completed two, four and six cycles of combination therapy, respectively. At data cut-off, of 33 pts included in the efficacy analysis, 28 pts underwent MRI examination and mrTRG results were available, the remaining 5 pts had early surgery due to AEs/serious AEs. 35.7% (10/28) and 46.43% (13/28) pts were classified as mrTRG1 and mrTRG2, respectively. Of 19 pts who underwent surgery, R0 rate was achieved in all 19 pts (100%) and pCR was achieved in 12 pts (63.16%). Moreover, 1 pt achieved cCR during neoadjuvant therapy, thus the treatment strategy provided a CR rate of 65% (1 cCR plus 12 pCR divided by 20). Grade 3/4 adverse effects (AEs) occurred in 16 pts (47.06%), and most commonly were lymphocyte count decreased (n=6, 18.18%), diarrhea (n=5, 15.15%) and AST increased (n=2, 6.06%). 4 serious AEs reported with intestinal perforation and all achieved pCR after surgery therapy. No treatment related death was reported. **Conclusions:** SCRT followed by fruquintinib combined with adebrelimab and CAPOX as a total neoadjuvant therapy showed promising CR rate and favorable pCR for high-risk LARC pts, and the safety profile was generally manageable. This treatment strategy might be a potential option as neoadjuvant therapy for high-risk LARC pts. Updated data will be presented in the future. Clinical trial information: NCT06234007. Research Sponsor: None.

193 Poster Session

Update of phase 1b/2 study of muzastotug (ADG126, an anti-CTLA-4 SAFEbody) in combination with pembrolizumab in advanced/metastatic MSS CRC. First Author: Daneng Li, City of Hope National Comprehensive Cancer Center, Duarte, CA

Background: ADG126 is an anti-CTLA-4 IgG1 masked antibody with cleavable masking peptides that is preferentially activated in the tumor microenvironment, which in turn binds to a unique epitope to block CTLA-4 function, prime T cells and deplete Tregs. We previously reported that repeated dosing of the IO doublet therapy ADG126 plus pembrolizumab (Pembro) was well-tolerated with promising clinical efficacy in 3L MSS CRC patients (pts) free of liver metastasis (NCT05405595)[1,2]. Here we update clinical safety and efficacy of the study as result of continued dose optimization for the IO doublet therapy. **Methods:** This is a Phase 1b/2, open-label, multicenter dose escalation (DE) and expansion (EXP) study. Primary endpoints were safety and tolerability. Secondary endpoints were PK, ADA, ORR, DCR, DOR, PFS and OS. **Results:** A total of 72 pts were dosed with ADG126/Pembro (Table). 27.8% pts had ≥ 3 prior therapies and 6.9% pts had prior IO therapies. There was no Grade 4/5 TRAE, and MTD was not reached. Grade 3 TRAEs were 5.9% (1/17), 16.2% (6/37), 33% (4/12) for 10 mg/kg Q6W, 10 mg/kg Q3W and 20 mg/kg loading dose (LD)[&] cohorts, respectively. Repeated dosing of ADG126 at 20 mg/kg (Q3W)/Pembro was not well-tolerated. In MSS CRC EXP cohorts, ADG126 of 10 mg/kg Q3W/Pembro treatment resulted in 4 confirmed PRs (n=24 EE), 17% ORR (95% CI: 5-37%), 75% DCR, 4.7-mon mPFS and OS rate of 88% (6 mons) and 75% (9 mons). Subgroup analysis revealed that in pts without liver and peritoneal metastasis (NLPM), ADG126 of 10 mg/kg Q6W/Pembro (10 EE) showed 70% SD and ~ 100% OS (6 and 9 mons); ADG126 of 10 mg/kg Q3W/Pembro (17 EE) showed 24% ORR (4 PR (95% CI: 7-50%)), 88% DCR, 47% CBR, 8.5-mon mPFS and OS rate of 89% (6 mons) and 83% (9 mons). Detailed safety and efficacy results, including those from ADG126 20 mg/kg LD cohort, will be reported. **Conclusions:** Extensive effort on ADG126 dose optimization, aided by PK-informed dosing strategy, suggested that ADG126 10 mg/kg Q6W and Q3W are appropriate dose/regimen for the IO doublet; this is supported by the well-balanced safety and efficacy (ORR, PFS and OS) profile observed in MSS CRC. ADG126 20 mg/kg Q3W appears to have reached safety ceiling for the IO doublet. The totality of the data further verifies that ADG126 possesses potential best-in-class therapeutic index, and provides basis for continued development of the IO doublet in MSS CRC and for additional exploration of the IO doublet in combination with SOCs/other anti-cancer agents in broader patient populations. Clinical trial information: NCT05405595. Research Sponsor: None.

Patient number for safety and efficacy evaluation.

	ADG126 Dose/Pembro*				
Evaluated for	**10 mg/kg Q6W**	**10 mg/kg Q3W**	**20 mg/kg Q3W**	**20 mg/kg LD[&]**	**Overall**
Safety (all pts of DE & EXP)	17	37	6	12	72
Efficacy (MSS CRC NLM, EXP)	10	24	/	5	39

*200mg, Q3W, IV; [&]20 mg/kg LD: 20 mg/kg x1 cycle followed by 10 mg/kg Q3W. NLM: liver metastasis-free; DE: Dose Escalation; EXP: Dose Expansion; EE: efficacy evaluable. Data cut date: Aug 30, 2024.

194 Poster Session

Efficacy and safety of anlotinib as a later-line treatment for advanced colorectal cancer. First Author: Yang Song, Tangdu Hospital, Air Force Medical University, Xi 'an, China

Background: Colorectal cancer (CRC) is increasingly prevalent in China, with rising incidence and mortality rates. Often, early-stage CRC presents with no symptoms, leading to most patients being diagnosed at an advanced stage, where prognosis is poor. First- and second-line treatments for advanced CRC typically include chemotherapy combined with cetuximab or bevacizumab. However, third-line therapies such as regorafenib, furosemide, and TAS102 have shown limited efficacy. Anlotinib, a multi-targeted tyrosine kinase inhibitor (TKI), has demonstrated improved progression-free survival (PFS) in refractory metastatic CRC (mCRC) in a phase III trial (NCT02332499), showing both efficacy and manageable toxicity. This report provides updated results from extended follow-up. **Methods:** This study enrolled 73 patients with advanced colorectal cancer who had undergone at least two prior standard treatments and were treated with oral anlotinib hydrochloride (8-12 mg daily, days 1-14 of a 21-day cycle) between November 2021 and March 2024 at Tangdu Hospital. Patients were treated until disease progression or intolerable adverse events. Efficacy was assessed using RECIST version 1.1, and adverse reactions were evaluated using CTCAE version 5.0. The primary endpoints were PFS, disease control rate (DCR), and adverse events (AEs), while secondary endpoints included overall survival (OS). **Results:** Of the 73 patients included, 38 had an ECOG performance status (PS) of 0-1, and 35 had a PS of 2. Clinical staging identified 18 patients with stage IIIB/IIIC and 55 with stage IV disease. Patients received anlotinib doses of 8 mg (n=11), 10 mg (n=44), or 12 mg (n=18), with 48 patients also receiving chemotherapy and 25 receiving anlotinib alone. The median PFS was 4.0 months (95% CI: 3.4-4.6), and the median OS was 7.0 months (95% CI: 5.7-8.3). Subgroup analysis indicated that patients under 60 years of age, those without liver metastases, and those with ECOG PS 0-1 derived greater benefit from anlotinib. The most common treatment-related adverse event was hypertension (19.2%). Other notable adverse events included elevated AST (11.0%), proteinuria (6.8%), and jaundice (6.8%). Most adverse reactions were grade 1-2, although 1 patient experienced grade ≥3 gastrointestinal bleeding. **Conclusions:** Anlotinib showed promising efficacy and a favorable safety profile as a later-line treatment for advanced colorectal cancer. These findings suggest that anlotinib may be a viable clinical option for advanced CRC, though larger-scale clinical trials are necessary to confirm these results and further define its role in treatment. Research Sponsor: None.

195 Poster Session

Liposomal irinotecan + 5-fluorouracil + leucovorin + bevacizumab as second-line therapy in metastatic colorectal cancer (IRIS): A multi-center, single-arm, prospective, phase II study. First Author: Xuhua Hu, The Fourth Hospital of Hebei Medical University, Shijiazhuang, China

Background: In patients with advanced colorectal cancer (CRC), the recommended second-line treatment following first-line therapy with oxaliplatin-based regimens is irinotecan-based therapy. Liposomal irinotecan, a novel formulation of traditional irinotecan, has demonstrated potential to enhance efficacy while minimizing toxicity. This study aims to investigate the efficacy and safety of liposomal irinotecan in combination with 5-FU/LV and bevacizumab as a second-line treatment option for metastatic CRC. **Methods:** This is a multi-center, single-arm, prospective, phase II study. Patients with metastatic CRC who received oxaliplatin-based chemotherapy as first-line treatment were enrolled to receive liposomal irinotecan + 5-fluorouracil + leucovorin + bevacizumab regimen until disease progression and/or unacceptable toxicity. The primary endpoint is objective response rate (ORR), secondary endpoints include disease control rate (DCR), progression-free survival (PFS), overall survival (OS), and safety. **Results:** A total of 50 patients were enrolled from Jan 2024 to Jul 2024 across 6 sites in China. The median age was 56.5 years (range: 30.0-76.0), with 58.0% male and 36.0% ECOG 0. Among the patients, 38.0% had left-sided colon cancer, 24.0% had right-sided colon cancer, and 38.0% had rectal cancer. As of Sept 14, 2024, 39 patients had at least one tumor assessment and 19 patients were still on treatment. ORR and DCR were 20.5% (8/39, 95% CI: 9.3%-36.5%) and 84.6% (33/39, 95% CI: 69.5%-94.1%), respectively. Eleven patients had disease progression and 5 patients died, and the median PFS and OS were not reached. The relative dose intensity of liposomal irinotecan and 5-fluorouracil were 92.8% (range: 23.6%-109.6%) and 89.5% (range: 13.3%-116.8%), respectively. During treatment, 43 (86.0%) patients had at least one adverse event (AE) and 24 (48.0%) had grade 3-4 AE. Most common (≥ 10%) grade 3-4 AEs were neutropenia (20.0%), leukocytopenia (16.0%) and diarrhea (10.0%). No unexpected toxicities observed in this study. **Conclusions:** These preliminary results reveled that liposomal irinotecan + 5-fluorouracil + leucovorin + bevacizumab after oxaliplatin-based treatment for metastatic CRC was well tolerated with encouraging clinical activity, which warrants further follow up. Clinical trial information: NCT06184698. Research Sponsor: None.

197 Poster Session

Amivantamab with or without chemotherapy in right-sided metastatic colorectal cancer: Updated results from OrigAMI-1, an open-label, phase 1b/2 study. First Author: Kanwal Pratap Singh Raghav, The University of Texas MD Anderson Cancer Center, Houston, TX

Background: Amivantamab (ami) is an EGFR-MET bispecific antibody with immune cell-directing activity and is FDA approved in *EGFR*-mutated advanced non-small cell lung cancer. High MET expression is observed in ~68% of patients (pts) with metastatic colorectal cancer (mCRC). Additionally, MET amplification occurs in up to 23% of EGFR-resistant mCRC and is implicated in driving resistance to EGFR-targeting antibodies (EGFRi). Compared to left (L)-sided disease, right (R)-sided disease is less responsive to EGFRi and associated with poorer outcomes. We present longer follow-up data among pts with R-sided mCRC. **Methods:** OrigAMI-1 (NCT05379595) is assessing ami as monotherapy and combined with chemotherapy in mCRC. All pts were wild-type for *KRAS*, *NRAS*, *BRAF*, and *EGFR* ectodomain by central ctDNA testing, without *ERBB2/HER2* amplification. One ami monotherapy cohort (Cohort C) enrolled only pts with R-sided disease (all pts must have 2-3 prior lines; prior EGFRi allowed). The ami plus chemotherapy cohorts (Cohorts D [with FOLFOX] and E [with FOLFIRI) enrolled both L- and R-sided disease (1 prior line max; prior EGFRi use was exclusionary). Primary tumor locations of cecum, ascending colon, hepatic flexure, and transverse colon were considered R-sided. Response was assessed by the investigator per RECIST v1.1. **Results:** As of 26-Aug-2024, 23 pts with R-sided mCRC received ami monotherapy (median follow-up of 8.1 months [mo]). Median number of prior lines was 2, and 43% had prior EGFRi. There were 7 pts with R-sided disease who received ami plus FOLFOX or FOLFIRI (median follow-up of 6.5 mo); all 7 had received 1 prior line. Among pts receiving ami monotherapy, objective response rate (ORR) was 22% (5/23) and disease control rate (DCR) was 78% (18/23), with 1 achieving a complete response. Median duration of response (DoR) is 7.4 mo; response and treatment are ongoing for 3 of the 5 responders. In pts receiving ami plus FOLFOX or FOLFIRI, ORR was 43% (3/7) and DCR was 86% (6/7). Median DoR is 5.8 mo, with all 3 responders on treatment, of which 2 are ongoing response. Additional details are in the Table. Biomarker data will be presented at the meeting. Safety profile among R-sided disease was consistent with prior reports, with the most common ami-related grade 3+ AEs being rash and hypoalbuminemia (2 pts each). **Conclusions:** Ami monotherapy or combined with FOLFOX or FOLFIRI demonstrated durable antitumor activity in R-sided mCRC. Clinical trial information: NCT05379595. Research Sponsor: J&J Innovative Medicine.

Efficacy.

	Ami monotherapy (n=23)	Ami + FOLFOX or FOLFIRI (n=7)
Median follow-up, mo (range)	8.1 (0.6–17.5)	6.5 (3.2–8.8)
Median number prior lines	2	1
Prior EGFRi, n (%)	10 (43)	0
ORR, % (95% CI)	22 (8–44)	43 (10–82)
Median DoR, mo (95% CI)	7.4 (3.7–NE)	5.8 (NE–NE)[a]
DCR, % (95% CI)	78 (56–93)	86 (42–100)
Median progression-free survival, mo (95% CI)	3.7 (3.4–5.5)	7.4 (1.8–NE)

[a]All 3 responders remain on treatment; 2 of 3 responders are ongoing response.

198 Poster Session

Preliminary safety and efficacy of upfront FOLFOXIRI and bevacizumab with cadonilimab in patients with proficient mismatch repair/microsatellite stable (pMMR/MSS) metastatic colorectal cancer: A multicenter phase II study (SYLT-026). First Author: Rongbo Lin, Gastrointestinal Medical Oncology, Fujian Cancer Hospital, Fuzhou, China

Background: The AtezoTRIBE phase II trial reported that adding Atezolizumab to upfront FOLFOXIRI plus Bevacizumab significantly improved overall survival (OS) in patients with metastatic colorectal cancer (mCRC). However, the improvement in OS was modest within the pMMR/MSS subgroup (HR, 0.80 [80% CI, 0.63–1.02]; P = 0.117) [1]. Cadonilimab, a humanized bispecific antibody targeting both PD-1 and CTLA-4, may offer enhanced therapeutic benefits. Therefore, this study aims to investigate whether the addition of Cadonilimab to upfront FOLFOXIRI plus Bevacizumab improves efficacy while maintaining safety in patients with pMMR/MSS mCRC. **Methods:** This multicenter, single-arm phase II study enrolled treatment-naive patients with pMMR/MSS mCRC, aged 18–75, who had at least one measurable lesion according to RECIST 1.1, an ECOG performance status (PS) of 0–2, and adequate organ function. Patients with dMMR/MSI-H tumors or prior immunotherapy were excluded. Eligible participants received FOLFOXIRI (irinotecan 165 mg/m², oxaliplatin 85 mg/m², leucovorin 200 mg/m², and fluorouracil 2400 mg/m² as a 48-hour infusion), along with Bevacizumab (5 mg/kg), and Cadonilimab (6 mg/kg). Treatment was administered for up to twelve 14-day cycles, followed by maintenance therapy with fluoropyrimidine and Bevacizumab plus Cadonilimab for up to 52 weeks, or until disease progression, unacceptable toxicity, or withdrawal of consent. Patients were equally enrolled, with 10 having liver metastases and 10 without. The primary endpoint was objective response rate (ORR) per RECIST 1.1. **Results:** As of August 30, 2024, 20 patients were enrolled, with a median age of 65.5 years (range: 33–74). Of these, 40% had an ECOG PS of 1 or 2, 55% had three or more metastatic organs, and 7 patients (35%) had RAS or BRAF mutations. The median follow-up was 5.0 months. Fifteen patients underwent at least two imaging assessments, revealing a confirmed ORR of 100% (15/15) and a disease control rate (DCR) of 100%. The median progression-free survival (PFS) and OS data were not yet mature. All 20 patients experienced treatment-emergent adverse events, most of which were grade 1 or 2. One patient (5%) discontinued treatment due to immune-mediated colitis. The most common grade 3/4 adverse events included neutropenia (55%), leukopenia (15%), infusion reactions (5%), dermatitis (5%), colitis (5%), anemia (5%), elevated ALT (5%), and diarrhea (5%). No new safety signals were identified. **Conclusions:** In conclusion, Cadonilimab in combination with FOLFOXIRI and Bevacizumab demonstrated promising efficacy and a manageable safety profile as a first-line treatment for patients with pMMR/MSS mCRC. Further investigation in larger trials is warranted. 1. Antoniotti, et al, J Clin Oncol, 2024. Clinical trial information: NCT05839470. Research Sponsor: None.

199 Poster Session

Avutometinib plus cetuximab after chemotherapy in patients with *KRAS*-mutated metastatic colorectal cancer: Results of phase 1 dose escalation. First Author: Rushabh Gujarathi, Department of Medicine, Section of Hematology and Oncology, University of Chicago, Chicago, IL

Background: Anti-EGFR antibodies like cetuximab are ineffective in *KRAS* mutated (mt) colorectal cancer (CRC) due to constitutive activation of downstream pathways. Avutometinib is a first in class, oral, novel, dual RAF/MEK inhibitor. In patient-derived xenograft models of *KRAS*mt CRC, the combination of an anti-EGFR antibody with avutometinib conferred stronger tumor growth inhibition than either agent alone. We present results from the dose escalation phase (1b) of an ongoing phase 1b/2 trial evaluating the combination avutometinib plus cetuximab in *KRAS*mt metastatic (m) CRC. **Methods:** This is a single-center, open-label, phase 1b/2 study evaluating the safety and combined tolerability of avutometinib and cetuximab. Patients received avutometinib 2.4mg (at DL0) or 3.2mg (at DL1) orally twice a week for 3 weeks out of a 4-week cycle + cetuximab 500mg/m^2 intravenously q2week. Patients with disease progression on, or intolerance to 5-FU, oxaliplatin, and irinotecan, and measurable disease per RECIST v1.1 were enrolled. The primary endpoint for phase 1b was to establish the maximum tolerated dose (MTD) and hence determine the recommended phase 2 dose (RP2D). Dose limiting toxicity (DLT) period was 28 days. A 3+3 dose escalation design was used with dose level (DLs) 0 (2.4mg avutometinib) as the starting level and one dose escalation to DL+1 (3.2mg avutometinib). **Results:** We report safety results of phase 1b. 10 patients, 7 at DL0 and 3 at DL+1, were enrolled. Median age was 52.5 yrs (range 41 – 75). Any grade (G) treatment related adverse events (TRAEs) were seen in all 10 patients. Most common TRAEs were acneiform rash (100%), chills (30%), diarrhea (30%), and fatigue (30%). G3 TRAEs at DL0 were seen in 4/7 (57.1%) patients and at DL+1 in 2/3 (66.7%) patients. No G4 TRAEs were seen. G3 TRAEs seen at DL0 are noted in the table. One patient from the initial three enrolled was not evaluable due to underdosing. Another patient was later found to not meet eligibility criteria, but they did clear the DLT period. Hence, 6/7 patients at DL0 were evaluated for AE assessment, of which 1 had a DLT. 3 additional patients were enrolled in DL+1. 2/3 patients in DL+1 had grade 3 toxicities (both DLTs) of rash. Hence DL0 was established as the MTD/RP2D. **Conclusions:** This is the first trial evaluating the combination avutometinib plus cetuximab in *KRAS*mt mCRC. RP2D for avutometinib was established at 2.4mg orally twice a week for 3 weeks out of a 4-week cycle when used in combination with q2week 500mg/m^2 cetuximab for *KRAS*mt mCRC. Enrollment continues currently for the dose expansion phase for efficacy assessment. Clinical trial information: NCT05200442. Research Sponsor: Verastem Oncology (funding and drug), Eli Lilly and Company (drug only).

G3 TRAEs at DL0 attributed to both drugs unless specified.

G3 TRAE	# of patients
Acneiform rash	3/7 (42.9%)
Hand-Foot syndrome	1/7 (14.3%)
Extremity edema	1/7 (14.3%)
Cellulitis	1/7 (14.3%)
Sepsis	1/7 (14.3%)
Anemia	1/7 (14.3%)
Creatine phosphokinase elevation	1/7 (14.3% – avutometinib only)

201 Poster Session

Neoadjuvant chemotherapy with FOLFOXIRI for high-risk relapsed locally advanced colon cancer: A single-arm phase II trial. First Author: Yuwen Zhou, Colorectal Cancer Center, West China Hospital, Sichuan University, Chengdu, China

Background: To evaluate the safety and efficacy of neoadjuvant chemotherapy (NAC) with three drug-intensity chemotherapy (oxaliplatin, irinotecan, and fluorouracil [FOLFOXIRI]) in patients with high-risk factors for recurrence and metastasis of locally advanced colon cancer (hr-LACC). **Methods:** A single-arm, open-label, phase II trial was conducted at West China Hospital of Sichuan University from July 2020 to Oct 2023. Patients with T4a/b or (and) N2/fused lymph nodes) and no distant metastases were considered as the hr-LACC patients. Eligible enrolled patients received four cycles of NAC with FOLFOXIRI followed by surgery and adjuvant chemotherapy with capecitabine and oxaliplatin (CAPOX) for two to five cycles or single-agent capecitabine for five cycles or observation. The primary endpoint was the rate of TRG grade 0-2 according to the criteria of NCCN guideline. **Results:** Of 72 patients enrolled (median age 53 [18-74] years,65 were T4a/b [90.2%], 66 were N2 [91.6%]), 58 patients (80.6%) completed four cycles of FOLFOXIRI. Three patients (4.2%) achieved clinical complete response. Twenty-eight patients (39.4%) obtained clinical partial response. The most common adverse event (AE) was nausea (86.1%, 62/72); grade 3-4 AEs occurred in 37 of 72 patients (51.4%).R0 resection was achieved in 64 of 64 patients, and the TRG (0-2) rate was 87.5% (56/64). The pathologic complete response rate was 7.8% (5/64). The proportions of pT0-2 and pN0 were 21.9% and 68.8%, respectively. Only three (4.7%) patients had prolonged hospital stay due to perioperative complications. No patient died within 30 days and 90 days after surgery. The 2-year tumor recurrence rate was 17.2%. **Conclusions:** For patients with hr-LACC, NAC with FOLFOXIRI regimen was safe and effective. Longer follow-up and larger clinical studies are needed to validate this innovative regimen. Clinical trial information: NCT05018182. Research Sponsor: Department of Science and Technology of Sichuan Province; 2024NSFSC1902; the Nation-Sponsored Postdoctor Researcher Program; GZC20231836.

202 **Poster Session**

Immune checkpoint inhibitors in digestive tumors with mismatch repair deficiency or microsatellite instability (dMMR/MSI-H): Effectiveness, safety, and prognostic factors in real-world practice. First Author: Nieves Martinez Lago, Medical Oncology Department, Complexo Hospitalario Universitario de Ferrol, Ferrol, Spain

Background: ICIs have shown durable responses in dMMR/MSI-H digestive tumors in phase II-III clinical trials. The aim of this study is to evaluate the efficacy, safety and prognostic factors in the context of real-world practice. **Methods:** We conducted a retrospective multicenter observational study of patients with dMMR/MSI-H digestive tumors treated with ICIs in routine clinical practice across seven university hospitals in northwest Spain. We analyzed clinicopathological characteristics, treatment response data, efficacy, and adverse events. **Results:** A total of 122 patients treated with ICIs between November 2015 and February 2024 were included. The median age was 70.4 years (range 29-89), and 52.4% were male. The most frequent origin was colorectal (54.9%; 67.2% located in the right colon and 37.3% BRAF V600mt), followed by gastroesophageal adenocarcinoma (34.4%). 16.4% had >3 metastatic sites (including 30.3% liver metastases and 44.3% peritoneal metastases). Baseline performance status (ECOG PS) was 0/1/2 in 21.7%/58.7%/19.8%, respectively. 39.3% had received one prior line of treatment. 92.6% received pembrolizumab monotherapy. The objective response rate (ORR) was 77.7% (including 31.3% complete response, 46.4% partial response, 13.4% stable disease, and 8.9% progressive disease), and the disease control rate (DCR) was 90.1%. Median overall survival (OS) was 53.2 months (95% CI: 42.6 – 63.7 months), and progression-free survival (PFS) was 46 months (95% CI: 30.6 – 61.4 months). The most common grade 3-4 treatment-related adverse events included nephritis (4.0%), hepatitis (3.3%), asthenia (2.5%), and pneumonitis (1.6%). 81.1% of patients discontinued treatment (including 24.6% due to progression, 21.3% after completing 2 years of treatment, and 14.8% due to toxicity). Potential prognostic factors identified in univariate analysis included ECOG PS, primary tumor resection, number of metastatic sites, and best treatment response. In multivariate analysis, a significant association was confirmed between ECOG PS, the presence of liver metastases, and best response to treatment with ICIs and overall survival. **Conclusions:** Our series confirms the efficacy and safety of ICIs in dMMR/MSI-H digestive tumors in routine clinical practice. The identified prognostic factors may be useful for identifying patients who could benefit the most from ICI treatment. Research Sponsor: None.

203 **Poster Session**

Real-world outcomes of targeted therapies and local interventions in BRAF V600E–mutated metastatic colorectal cancer: A single-center retrospective study. First Author: Ziqin Lin, The Sixth Affiliated Hospital, Sun Yat-sen University, Guangzhou, China

Background: Metastatic colorectal cancer (mCRC) patients with BRAF V600E mutations, especially those with microsatellite stable (MSS) tumors, have poor prognoses. Chemotherapy has limited efficacy in this subgroup, especially for second-line or subsequent-line treatments. Current research is increasingly focusing on the integration of targeted therapies and local interventions, but real-world evidence remains scarce, particularly from Asian cohorts. This study leverages real-world data to evaluate the impact of novel therapeutic combinations, including triplet and doublet targeted therapies, alongside local treatments in this high-risk patient population. **Methods:** This retrospective study includes MSS-type mCRC patients with BRAF V600E mutations treated at the Sixth Affiliated Hospital of Sun Yat-sen University between April 2014 and May 2024. Patients receiving targeted therapies were stratified into three groups: Triplet group (BRAF, EGFR, and MEK inhibitors, including Dabrafenib, Cetuximab and Trametinib), Doublet group (BRAF and EGFR inhibitors, including Dabrafenib and Cetuximab or Encorafenib and Cetuximab), and Doublet target-chemotherapy group (BRAF and EGFR inhibitors combined with chemotherapy, including Vemurafenib, Irinotecan, and Cetuximab). Primary endpoint was progression-free survival (PFS). Secondary endpoints included overall survival (OS) and objective response rate (ORR). **Results:** Among 239 enrolled patients, 58 received targeted therapy and 68 underwent conventional chemotherapy as second-line treatment. 63.5% (80/126) of these patients underwent local interventions, including primary tumor resection, metastasectomy, CRS±HIPEC, and liver ablation. Targeted therapy demonstrated significantly improved mPFS compared to conventional chemotherapy (5.5 vs. 4.2 months, HR = 0.44; 95% CI: 0.29-0.68, $p < 0.001$). OS did not differ significantly between the two treatment arms (12.7 vs. 12.1 months, HR = 0.80; 95% CI: 0.50-1.25, p = 0.330). In the triplet group, ORR, mPFS, and mOS were 21%, 5.8 months and 18.0 months, respectively, showing a trend towards enhanced efficacy compared to the other targeted therapy groups, albeit without statistical significance. Patients who received targeted therapy (n=35) or chemotherapy (n=45) in combination with local interventions demonstrated significantly prolonged OS compared to those without local therapy (mOS: 22.3 vs. 9.6 months; HR = 0.19; 95% CI: 0.09-0.42; $p < 0.001$ and 14.3 vs. 10.9 months; HR = 0.53; 95% CI: 0.29-0.98; p = 0.042, respectively). **Conclusions:** Optimal treatment for BRAF V600E mCRC remains under investigation. This study suggests that combining targeted therapies with selective local interventions may significantly improve patient survival in this challenging subgroup, warranting further prospective evaluation. Research Sponsor: None.

204 **Poster Session**

Efficacy and safety of trifluridine/tipiracil and bevacizumab (FTD/TPI+BEV) in patients with metastatic colorectal cancer (mCRC) in real-world practice. First Author: Nieves Martinez Lago, Medical Oncology Department, Complexo Hospitalario Universitario de Ferrol, Ferrol, Spain

Background: In Sunlight trial, the combination of FTD/TPI + BEV showed a significant improvement in Overall Survival (OS) and Progression-Free Survival (PFS), along with better Objective Response Rate (ORR) and Disease Control Rate (DCR), compared to FTD/TPI alone, in patients with mCRC progressing after two chemotherapy regimens. The BeTAS trial aimed to evaluate the effectiveness and safety of the FTD/TPI + BEV combination in real-world practice. **Methods:** This was a retrospective, observational, multicenter study of mCRC patients treated with FTD/TPI+BEV in routine clinical practice across 11 Spanish university hospitals. It included patients with mCRC who were refractory or intolerant to standard therapies. **Results:** A total of 208 patients treated with FTD/TPI+BEV between July 2019 and December 2023 were included. The median age was 69 years (range 33 to 88 years), with 59.8% being male. 15.3% had an ECOG performance status of 2, 55.9% had RAS mutations, 30.3% had three or more metastatic sites, and 81.2% had liver metastases. Additionally, 19.8% had a time from diagnosis of first metastasis of less than 18 months, and 55.3% were categorized in Tabernero's Unfavorable Prognostic Subgroup. Most patients (81.3%) received FTD/TPI+BEV as third-line therapy, with 90.5% having previously received anti-VEGF therapy. The median number of treatment cycles was 4 (range 1 to 31 cycles). The ORR was 4.6%, and the DCR was 50%. The median PFS was 4.9 months (95% CI, 4.0-5.7 months), and the median OS was 11.1 months (95% CI, 9.5-12.8 months). The most common grade 3-4 toxicities included neutropenia (41.8%), asthenia (7.2%), and hepatic toxicity (5.5%). No treatment-related deaths were reported. **Conclusions:** Our series confirms the efficacy and safety of FTD/TPI + BEV in routine clinical practice, even in a population with poor prognosis and an extensive history of previous treatments. Research Sponsor: None.

205 **Poster Session**

The importance of treatment handling and compliance on overall response rate in a phase III study of metastatic colorectal cancer: Post-hoc per protocol analyses of the AGENT trial. First Author: Aldina Pivodic, APNC Sweden, Mölndal, Sweden

Background: To study the compliance in handling the investigational medicinal product (IMP) and its impact on outcomes in patients with metastatic colorectal cancer participating in the AGENT trial (NCT03750786). **Methods:** In a randomized, multicenter, multinational, phase III study, patients were randomized to receive either arfolitixorin (n=245) or leucovorin (n=245) with 5-fluorouracil (5-FU), oxaliplatin and bevacizumab as a first-line treatment. Logistic regression was applied for overall response rate (ORR) and Cox regression for all-cause death, and all-cause death or progression, adjusted for randomization strata. **Results:** Missing or incorrect IMP handling was reported for 28 (11%) patients in the arfolitixorin and 23 (9%) in the leucovorin arm, <80% compliance for duration of bolus 5-FU 2-4min for 101 (42%) and 97 (41%), respectively. Considering the time between the bolus 5-FU and arfolitixorin, non-compliance, defined by <80% doses given within 25-35min PP window, was reported for 92 (38%) in arfolitixorin patients. Adherence to the timing between the first and second arfolitixorin dose (30-60min) was high. In total, 91 (37%) patients in the arfolitixorin and 134 (55%) in the leucovorin arm fulfilled PP definitions (cPP population). The bolus 5-FU dose was equally likely to be reduced in both the arfolitixorin and leucovorin arms across regions (ranging from 22% to 39%), except for Japan where 55% of patients in the arfolitixorin arm had their bolus 5-FU dose reduced, compared to 28% in the leucovorin arm. In the cPP population, the ORR was achieved by 54 (59%) in the arfolitixorin and 69 (51%) in the leucovorin arm, adjusted odds ratio (aOR) 1.40 (95% CI 0.81-2.45), p=0.23. Excluding Japan, the ORR was 41 (62%) in the arfolitixorin and 49 (46%) in the leucovorin arm, aOR 2.11 (95% CI 1.09-4.06), p=0.026. Corresponding aOR for Europe was 1.50 (95% CI 0.71-3.15), p=0.29, and for North America aOR 7.95 (95% CI 1.45-43.62), p=0.017. In Japan, the ORR had better outcome in the leucovorin arm (71.4%) compared to the arfolitixorin arm (52.0%), aOR 0.42 (95% CI 0.13-1.31), p=0.14. However, the relationship was numerically opposite for patients without bolus 5-FU dose reduced. All-cause death during the complete follow-up was reported for 37 (41%) in the arfolitixorin and 61 (46%) in the leucovorin arm. **Conclusions:** In the AGENT trial, 46% of the included patients met per protocol compliance for treatment handling. In this cPP population, there was a non-significant, but clinically relevant, increase in the overall response rate in the arfolitixorin compared to the leucovorin arm. Notably, in all regions excluding Japan, the overall response rate was higher for arfolitixorin, with statistical significance. Adherence to the treatment protocol is crucial for the outcomes of patients with metastatic colorectal cancer. Clinical trial information: NCT03750786. Research Sponsor: Isofol Medical.

206 Poster Session

Phase 1/2 study of XTX101, a tumor-activated, Fc-enhanced anti-CTLA-4 monoclonal antibody, in combination with atezolizumab in patients with advanced solid tumors and in MSS CRC. First Author: Joel R Hecht, UCLA Jonsson Comprehensive Cancer Center, Santa Monica, CA

Background: XTX101 is an investigational tumor-activated, Fc-enhanced, high affinity binding aCTLA-4 designed to block CTLA-4 and deplete regulatory T cells upon activation in the tumor by proteases. XTX101 also contains mutations designed to enhance Fcγ receptor binding. Fc enhancement augments FcγR co-engagement on antigen presenting cells and has been linked with efficacy of aCTLA-4 combinations in patients (pts) with "cold tumors", including microsatellite stable (MSS) colorectal cancer (CRC). Parts 1A/1B of the study evaluated XTX101 monotherapy in pts with advanced solid tumors. XTX101 was generally well tolerated with limited peripheral XTX101 activation (~13%) compared to evidence of XTX101 activation in the tumor microenvironment (73-96%, in n=2 on-treatment tumor biopsies). A durable partial response (PR) and resolution of hepatic metastases was observed in a pt with PD-L1 negative NSCLC at the monotherapy recommended phase 2 dose (RP2D) of 150 mg once every six weeks (Q6W) [1]. **Methods:** Part 1C evaluated the safety and feasibility of XTX101 in combination with atezolizumab in pts with advanced solid tumors using 3+3 dose escalation to establish the RP2D. Phase 2 is enrolling pts with MSS CRC with at least 1 prior regimen for metastatic CRC. Initial safety and anti-tumor activity data from Phase 2 in ~20 pts will be presented. **Results:** As of July 15, 2024, 15 pts were enrolled in Part 1C: MSS CRC (n=12), esophageal cancer, NSCLC, and ampullary carcinoma (n=1 each). Median age 69 and 3 prior lines of therapy (1-12). Across all XTX101 dose levels (75 mg to 150 mg), treatment-related adverse events (TRAEs) of any grade with $>$10% incidence included infusion reactions (n=4), fatigue and diarrhea (n=2 each), no G4 or G5 TRAEs were reported. In Part 1C, 2 pts experienced a dose limiting toxicity: 1 pt had G3 colitis, and 1 pt had G3 liver enzyme elevation (both were at the XTX101 150 mg dose level and resolved with immunosuppression). Two unconfirmed PRs were reported: in a pt with MSS CRC, including resolution of a liver target lesion, and in a pt with ampullary carcinoma (accompanied by a Ca 19-9 decrease from 700 to 41). In addition, a decrease in intra-tumoral Tregs was observed in paired tumor biopsies from a pt. The combination RP2D was established as XTX101 100 mg Q6W and atezolizumab 1200 mg once every three weeks (Q3W), and enrollment in the Phase 2 part of the study for pts with MSS CRC is ongoing at the RP2D. **Conclusions:** In Part 1C dose escalation, the combination of XTX101, a tumor-activated, Fc-enhanced aCTLA-4, and atezolizumab was generally well tolerated in pts with advanced solid tumors and demonstrated initial evidence of anti-tumor activity in cold tumors. These data support the initiation of an ongoing Phase 2 study evaluating the combination in pts with MSS CRC with and without liver metastases. 1. Davar ESMO IO 2023. Clinical trial information: NCT04896697. Research Sponsor: Xilio Development, Inc.

207 Poster Session

Neoadjuvant botensilimab (BOT) plus balstilimab (BAL) in resectable mismatch repair proficient (pMMR) and deficient (dMMR) colorectal cancer (CRC): NEST clinical trial update. First Author: Erika Hissong, Weill Cornell Medical College, New York, NY

Background: Checkpoint inhibitors (CPI) have been transformative for localized dMMR CRC (microsatellite instability-high [MSI-H]), but not for localized pMMR CRC (microsatellite stable [MSS]). The combination of BOT, a novel Fc-enhanced multifunctional CTLA-4 antibody, and BAL, an anti-PD-1 antibody, has demonstrated significant activity in metastatic CRC. We examined this novel dual CPI therapy in patients with MSI-H and MSS localized CRC. **Methods:** This is a single arm trial of neoadjuvant BOT 75 mg/m^2 (day 1) and BAL 240 mg/m^2 every 2 weeks x2 (NEST1) or x4 (NEST2) in patients with localized CRC who were candidates for surgery. The primary endpoint was major pathologic response (MPR), defined as $\geq$90% pathologic tumor regression. The 95% confidence interval (CI) was computed based on exact method for binomial distribution. **Results:** NEST1 (n=12: 9 MSS / 3 MSI-H) accrued from 3/21/23 to 9/25/23 and NEST2 (n=14: 13 MSS / 1 MSI-H; 2 MSS patients had synchronous CRC) accrued from 2/15/24 to 5/6/24; 2 patients in NEST1 were not evaluable (1 with rectal cancer not resected [ypT0N0 on local excision] and 1 patient with occult lung metastasis). In total, 7 MSS CRC tumors were evaluable from NEST1 and 15 from NEST2; and 3 MSI-H tumors were evaluable from NEST1 and 1 from NEST2, leaving a final evaluable study population of 24 patients (20 MSS / 4 MSI-H) and 26 tumors (22 MSS / 4 MSI-H). The Table provides pathologic response rates across NEST1, NEST2, and MSI-H. With a median follow-up of 13.1 months (NEST1) and 4.8 months (NEST2), no patients have recurred and ctDNA testing remains negative. No grade 4 toxicities were observed and the only treatment-related grade 3 toxicity was diarrhea/colitis in 15% and fever in 4%. Treatment-related grade 2 diarrhea/colitis occurred in 8%, grade 2 fever in 15%, and grade 2 fatigue in 4% of patients. Diarrhea/colitis resolved in all cases with steroids $\pm$ anti-TNFα therapy. All patients underwent laparoscopic resection, with 1 conversion to open. No surgeries were delayed due to treatment-related adverse events. Median length of stay was 2 days (range 1-14). Translational data including pathologic evaluation of tumor infiltrate and response will be presented. **Conclusions:** Neoadjuvant BOT/BAL is safe, with no delays to surgery, and effective. We observed high MPR rates in both MSS and MSI-H CRC with no recurrences to date. The MPR and pCR rate improved with extended time to surgery. Clinical trial information: NCT05571293. Research Sponsor: Agenus, Inc.

Pathologic response to neoadjuvant BOT/BAL in localized CRC.

Pathologic Response	NEST 1 n=7 MSS tumors	NEST 2 n=15 MSS tumors	MSI-H (3 NEST1, 1 NEST2)
100% (CR%, 95%CI)	1 (14%, 0.4-58%)	6* (40%, 16-68%)	3** (75%, 19-99%)
≥ 90% (MPR%, 95%CI)	2 (29%, 4-71%)	7 (47%, 21-73%)	4^ (100%, 40-100%)
≥ 50%	4 (57%)	9 (60%)	4 (100%)
Median days to Surgery (range)	29 (21-37)	57 (45-104)	46 (34-78)

*2 with carcinoma *in situ*, no tumor bed. ^MPR was in NEST1. **1 was rectal.

208 Poster Session

Outcomes in management of locally advanced rectal cancer with total neoadjuvant therapy in an underserved population. First Author: Arup Ganguly, University of Connecticut Health Center, Farmington, CT

Background: The use of total neoadjuvant therapy is now standard of care and is favored when treating locally advanced rectal cancer but its treatment is seldom studied in an underserved population. **Methods:** A retrospective study was conducted on patients with Stage II and III locally advanced colorectal cancer treated with either total neoadjuvant therapy (TNT) followed by resection or conventional therapy (CT) with chemoradiation followed by surgery, with or without adjuvant treatment afterwards. Patients completed treatment between 2015 and 2023 at the Hartford Hospital Helen and Harry Gray Cancer Center at Hartford, Connecticut. Data collected included demographics, type of treatment, tumor location (high vs mid vs low) and pathological response to treatment spread out among the two groups. Propensity score matching was used to minimize baseline differences between the two groups, adjusted for age, gender, ethnicity and tumor location and Fisher's exact test was performed to then check for significance. **Results:** Out of the 133 patients analyzed, 90 underwent TNT, and 43 received CT. The average age at diagnosis was 57.5 years (SD: 11.15). In the TNT group, 79 patients (87.7%) achieved either a partial or complete response (PCR), while 11 patients (12.3%) had No response (NR). Tumor locations in the TNT group were distributed as follows: 13.4% (n=12) had high tumors, 34.4% (n=31) had mid tumors, and 52.2% (n=47) had low tumors. In the CT group, 31 patients (72.9%) achieved PCR, while 12 patients (27.9%) had NR. Tumor locations in the CT group were distributed as follows: 9.3% (n=4) had high tumors, 51.2% (n=22) had mid tumors, and 39.5% (n=17) had low tumors. After propensity score matching and adjusting for demographic covariates, the TNT group demonstrated a higher proportion of responders compared to the CT group (p=0.048). Furthermore, 40% (n=36) of patients in the TNT group achieved a complete response, compared to 32% (n=14) in the CT group. Partial response rates were 39.5% (n=17) in the TNT group and 47.8% (n=43) in the CT group. **Conclusions:** TNT demonstrated superior pathological response rates compared to CT, which supports its growing role in the management of locally advanced rectal cancer even in this underserved population. Research Sponsor: None.

209 Poster Session

Efficacy of plogosertib (CYC149462), a novel orally bioavailable PLK1 inhibitor, on patient-derived models of colorectal cancer. First Author: Mohammad Zokaasadi, Department of Medicine, Division of Medical Oncology, Duke University Medical Center, Durham, NC

Background: Colorectal cancer (CRC) is the third most common cancer in adults in the United States. Given the relative rapid cell division in cancer cells compared to most normal cells, drugs targeting mitosis and other cell cycle checkpoints are of interest in CRC treatment. One multi-tasking protein in cell division is polo-like kinase 1 (plk1). Plk1 is a serine threonine kinase that is highly expressed during mitosis and involved in a variety of functions, including spindle assembly, chromosome segregation, G2/M transition, and cytokinesis. Encouragingly, its expression is also significantly higher in CRC compared to normal mucosa, which makes it an attractive cancer-specific target in CRC. In this study, we sought to evaluate the efficacy of plogosertib (CYC149462), a novel plk1 inhibitor, in a series of preclinical models of advanced CRC. **Methods:** Fourteen CRC patient-derived organoids (PDOs) were generated from patients who underwent biopsy or resection for their primary or metastatic CRC under an IRB approved protocol at Duke University. Diagnosis of CRC is confirmed in PDOs using H&E and IHC. Subsequently, PDOs were treated with plogosertib in a range of concentrations from 256pM to 100μM. Drug screens were performed for standard of care agents 5FU, oxaliplatin, and SN38. Viability was assessed with Cell-Titer Glo (CTG) 72 hours after treatment. Using a matched PDX, mice were treated with vehicle or 40 mg/kg daily of plogosertib via oral gavage for 2 weeks, 5 days per week. Cell cycle analysis was performed by flow cytometry, and localization of DNA during mitosis was done by fluorescence staining. **Results:** CRC PDOs were more sensitive to plogosertib (CYC149462) than 5FU, and oxaliplatin, with significantly lower IC_{50} values (518.86$\pm$377.47nM for plogosertib vs. 38.87$\pm$45.63μM for 5FU, and 37.78$\pm$39.61μM for oxaliplatin respectively (ANOVA, $P < 0.05$). Subsequently, we validated our findings *in vivo* using a matched PDX in which plogosertib treatment led to significant tumor growth inhibition compared to vehicle (t-test, $p < 0.05$) without serious adverse effects. Furthermore, we showed that plogosertib induces dysfunction in alignment of chromosomes and subsequent G2/M cell cycle arrest ($p < 0.05$). **Conclusions:** Together, our study shows that pharmacological inhibition of Plk1 using plogosertib represents a promising therapeutic approach for advanced CRC. Research Sponsor: None.

LBA210 **Poster Session**

Short-course radiation(SCRT) followed by 6 cycles of cadonilimab plus mFOLFOX6 as neoadjuvant therapy for patients with locally advanced rectal cancer (LARC): A multicenter, single arm, phase II trial (NeoCaCRT). First Author: Wan He, Shenzhen People's Hospital, Shenzhen, China

The full, final text of this abstract will be available at meetings.asco.org on the day of presentation and in the online supplement to the February 1, 2025, issue of the *Journal of Clinical Oncology*.

211 **Poster Session**

Sensitizing K-Ras G12D mutant colorectal patient-derived organoids via ferroptosis pathway. First Author: Dechokyab De, Columbia University Medical Center, New York, NY

Background: Colorectal cancer remains the third most fatal form of cancer for males and females combined, with K-Ras mutations present in approximately 40% of all cases. These mutations drive cancer progression and confer resistance to conventional therapies, underscoring the need for novel treatment strategies. Ferroptosis, a regulated non-apoptotic form of cell death characterized as iron-dependent lipid peroxidation, has emerged as a target for cancer therapy. A study published in 2023 found that K-Ras mutant pancreatic ductal adenocarcinoma and non-small cell lung cancer cells lack ferroptosis induced lipid peroxidation and concurrently upregulated ferroptosis suppressing protein (FSP-1). It is unknown if K-Ras mutant colorectal tumors also evade ferroptosis-induced cell death as a method of cancer progression and therapy resistance. This study aims to measure the effects of inhibiting K-Ras G12D, the most common K-Ras mutation found in CRC, on ferroptosis through a patient derived colorectal cancer organoid (PDO) model. We hypothesize that blocking K-Ras G12D using a MRTX-1133, a K-Ras G12D specific inhibitor, can induce ferroptosis and synergistically work with ferroptosis inducers. **Methods:** As a pilot study, two patient derived organoid lines were established. Surveillance screening and whole exome sequencing was performed to verify K-Ras mutation. CRC9T has wild-type and CRC30T has mutant K-Ras. Additionally, Columbia Pathology Core confirmed tissues used to generate organoids were moderately differentiated colorectal adenocarcinoma. CTG-3D assays were performed to measure IC50 using ferroptosis inducers RSL3, iFSP, and K-Ras inhibitor MRTX-1133. Organoids were cultured in standard gut media for 72 hours, treated, and collected after 72 hours. After treatment, organoids were imaged and stained with BODIPY-C11, a lipid peroxidation fluorescence dye. Flow cytometry was performed, and results were analyzed using FlowJo to show ferroptosis-induced lipid peroxidation in stained organoids. **Results:** Inhibiting K-Ras G12D and inducing ferroptosis showed a synergistic effect on ferroptosis-induced lipid peroxidation. CRC9T was more sensitive (IC50 RLS3: 2.96 uM, iFSP 3.46 uM) to ferroptosis inducers than CRC30T (IC50: RSL3: 5.68 uM, iFSP: 8.6 uM). Blocking K-Ras in was only efficacious in CRC30T (IC50: MRTX1133: 9.73 uM). Cells from the mutant line showed greatest lipid peroxidation when combining ferroptosis inducers with K-Ras G12D inhibitors in flow cytometry analysis. **Conclusions:** These findings demonstrate a synergistic effect when blocking K-Ras G12D and inducing ferroptosis and align with previous studies that showed a connection between ferroptosis suppression and oncogenic KRAS expression. We plan to validate these findings in other patient derived organoid model systems and to assess the effects of pan-KRAS inhibition and ferroptosis inducer on non-K-Ras G12D PDOs. Research Sponsor: Herbert Irving Comprehensive Cancer Center Support Grant; P30CA013696 (JG, DD).

212 **Poster Session**

Oncomine panel testing in the clinical management of colorectal cancer: An Irish experience. First Author: Mary O Reilly, St Vincent's University Hospital, Dublin, Ireland

Background: Early onset colorectal cancer (EOCRC) is defined as colorectal cancer (CRC) in those under the age of 50. It has been hypothesized that the molecular signatures of EOCRC could differ from those of late onset CRC (LOCRC). Next Generation Sequencing (NGS) using the Oncomine Precision Assay is a useful tool to guide treatment decisions based on the presence or absence of oncogene drivers. While not all of these mutations currently influence the clinical management of CRC patients, ongoing research on the function of these genes and the development of new drugs could offer insights into patient prognosis, treatment choices and prediction of therapy resistance. **Methods:** Of $>$1500 CRC identified between 2010 and 2024, we investigated the clinical and genetic characteristics of 38 EOCRC and 41 LOCRC cases. All cases were microsatellite stable and staged I-III. NGS was performed with the Ion Torrent Genexus integrated platform using the Oncomine 45-gene Precision Assay panel. DNA hotspot mutations and copy number variations (CNVs) were assessed, followed by the classification of clinically relevant mutations using the genetic variant OncoKB database. **Results:** There were 38 EOCRC and 41 LOCRC cases. The age range was 27 to 49 and 50 to 91, with median ages of 44 and 72, in the EOCRC and LOCRC groups, respectively. Based on the stratification of NGS mutation levels as per the OncoKB classification, all patients in both groups (100%) had level 1 actionable mutations. NRAS wild type (WT) status was present in 100% of patients in both groups. However, differences were observed in KRAS status: 84% (32/38) of EOCRC were KRAS WT, while only 65% (27/41) of LOCRC were KRAS WT. While 3 % (n=1) of the EOCRC cohort had an ERBB2 amplification, none were detected in the LOCRC group. BRAF V600E mutations were present in 11% (4/38) of EOCRC and 12% (5/41) of LOCRC. Level 4 mutations were detected in 18% (n=7) of EOCRC, all of which were PIK3CA mutations, while these were seen in 24% (n=10) of the LOCRC group, including 9 PIK3CA mutations and 1 FGFR1 mutation. Regarding concurrent mutations, 2 patients in the EOCRC group had two mutations each: one with concurrent KRAS and PIK3CA mutations, and another with concurrent JAK2 and KRAS mutations. The LOCRC group had a higher incidence of concurrent mutations, with 8 patients affected: 6 had concurrent KRAS and PIK3CA mutations, and 2 had concurrent BRAF and PIK3CA mutations. **Conclusions:** Our study demonstrates that EOCRC have higher KRAS WT status and fewer concurrent mutations compared to LOCRC, who exhibit more frequent KRAS non-G12C oncogenic mutations. It also underscores the importance of routine NGS mutation panel analysis in CRC management. Additionally, identifying specific mutations in patients that relapse can guide targeted therapies, potentially improving outcomes and offering personalized treatment options. Research Sponsor: None.

213 **Poster Session**

Genomic profiling of matched tumor samples in patients with rectal cancer undergoing total neoadjuvant therapy. First Author: Daniel B Rosen, Harvard Radiation Oncology Program, Boston, MA

Background: Total neoadjuvant therapy (TNT) is increasingly used for locally advanced rectal cancer. However, which patients benefit most from TNT remains to be elucidated. In this study we sequenced a TNT cohort with long-term follow-up to identify genomic correlates of TNT efficacy. **Methods:** Patients with locally advanced rectal cancer who underwent TNT followed by surgery (Oct 2016-June 2020) were identified at our institution. MSS patients with paired tissues (normal, biopsy, and surgery) with successful DNA extraction and sequencing were analyzed. Retrospective chart review collected patient demographics and clinical outcomes. Genomic analyses were performed with the R maftools package. Descriptive statistics were calculated with Fisher's exact, Wilcoxon rank sum, and Wilcoxon rank sum exact tests. Univariate survival analyses (UVA) were calculated with Cox proportional hazards regressions. All statistical analyses were performed using R (v4.3.2). **Results:** 53 patients met inclusion criteria and 32 patients with paired tissues (64 samples) were obtained and successfully sequenced. All patients received chemotherapy, most commonly FOLFOX (81.3%) as part of TNT and a 5-FU based chemoradiation therapy (RT) regimen. Almost all patients received 50.4 Gy (93.8%). Median followup was 57 months (IQR: 47-68). There were 24 pathological responders (R) [complete/partial] and 8 nonresponders (NR) [stable disease/progression]. R vs NR were not statistically different across demographic or treatment characteristics. Local recurrences were infrequent (9.4% 3/32). No covariates were prognostic on UVA survival analyses. However, NR was associated with worse distant metastases free survival (DMFS) (HR = 4.11, CI 1.01-16.8, p = 0.048) on UVA. 5-year DMFS was 83% for R and 50% for NR, and the median DMFS was not reached for R, 41 mo for NR. Tumor mutation burden (TMB) did not change with treatment (pre = 4.5 vs post = 6.2 mutations/Mb; r, 1.7-10.3, p=0.59) and was not different on biopsy between R and NR (4.1 vs 5.9, p=0.09). Pre-treatment, the most mutated gene was *APC* (81%), [majority nonsense/frameshift (88%)], followed by *TP53* (69%), [missense (59%)]. When comparing frequent mutations (n $\geq$ 5), *PIEZO1* was more mutated in NR (63%, 5/8) than R (4% 1/24 patients) (p = 0.023 [multiple testing corrected]). Putative radioresistant *KRAS-TP53* co-mutations were uncommon with similar representation in both groups (R 12.5%, 3/24, NR 12.5%, 1/8). **Conclusions:** Pathologic response positively correlated with DMFS in this cohort with long-term followup. *PIEZO1* mutations were negatively prognostic for pathologic NR to TNT. This uncovers a potential new role in chemoRT resistance in addition to a previously reported association with metastatic spread. Further evaluation of *PIEZO1* as a prognostic biomarker may help personalize appropriate treatment for patients with locally advanced rectal cancer. Research Sponsor: None.

214 **Poster Session**

Plasma shallow whole-genome sequencing profiling in patients with *RAS*^{WT} metastatic colorectal cancer (mCRC) undergoing first-line anti-EGFR therapy. First Author: Valentino Martelli, Hospital del Mar Research Institute, Barcelona, Spain

Background: Acquired resistance (AR) to first-line (1L) chemotherapy (CT) + anti-EGFR inhibitors is a major challenge in the management of *RAS*^{WT} mCRC patients (pts). While AR in later lines has been explained with point mutations in the *EGFR* pathway, its real mechanisms in 1L remain poorly understood. Tumor molecular biology can be dynamically studied through circulating tumor DNA (ctDNA) analysis, a real-time, non-invasive approach to detect AR. In this context, we conducted the multicenter prospective PLATFORM-B study, which enrolled 100 *RAS*^{WT} mCRC pts receiving 1L CT + cetuximab across 15 Spanish Medical Oncology Departments, alongside 30 *RAS*^{MUT} pts (control) treated with 1L CT + bevacizumab. Peripheral blood samples have been collected at baseline (BL), week 8, and disease progression (PD). Next-generation sequencing analysis (Ion S5 system) of ctDNA at PD revealed AR mutations in only 10% of pts, suggesting that alterations in the EGFR pathway may only partially explain AR in 1L. Therefore, we conducted shallow whole-genome sequencing (shWGS) to further characterize genomic profiles possibly related with novel mechanisms of AR. **Methods:** Baseline and PD samples with sufficient remaining plasma were considered for this analysis. shWGS was performed with a coverage ranging 0.5X to 2.0X and analyzed using ichorCNA pipeline to evaluate various genomic characteristics such as copy number alterations (CNA), tumor fraction (TF), subclonal fraction (SF), and ploidy. Molecular results were correlated to clinical-pathological features and outcomes. **Results:** Overall, 35 mCRC pts (27 *RAS*^{WT} and 8 *RAS*^{MUT}) were included in the study. At BL (N = 26), median TF (mTF) was 0.19, median ploidy 2, median SC fraction 0.5, median SC genome fraction 0.21, and median subclonal CNA fraction 0.41. At PD (N = 33), mTF was 0.08, median ploidy 2, median SC fraction 0.5, median SC genome fraction 0.17, and median subclonal CNA fraction 0.38. Of all the features analyzed, only the TF showed potential prognostic value. At BL, pts with TF levels above the median had poorer overall survival (OS) compared to those with lower levels, approaching statistical significance (P = .055). At PD, higher TF levels were statistically significantly associated with worse OS (P = .02). No meaningful differences between the *RAS*^{WT} and *RAS*^{MUT} groups were observed. Specific variations in CNA, TF, SF, or ploidy under treatment are being correlated to clinical data to seek for predictive value. **Conclusions:** This preliminary analysis of the PLATFORM-B using shWGS showed the prognostic value of TF. Further analyses are undergoing to better understand the molecular landscape of acquired resistance to 1L anti-EGFR-based therapy in *RAS*^{WT} mCRC pts. Research Sponsor: None.

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Risk factors for relapse based on the molecular profiling of Japanese patients with stage II colorectal cancer. First Author: Shunsuke Kasai, Shizuoka Cancer Center, Shizuoka, Japan

Background: The association between the molecular profiles and prognosis of Stage II colorectal cancer (CRC) remains unclear. This study aimed to evaluate the long-term outcomes of Stage II CRC based on the molecular profiling and to examine the risk factors for relapse. **Methods:** We retrospectively enrolled Japanese patients with pStage II CRC without pre- and/or postoperative adjuvant therapy and whose surgically resected specimens were evaluated using gene expression and whole exome analyses from January 2014 to December 2018. We evaluated the 5-year overall survival (OS) and relapse-free survival (RFS), and examined the risk factors for RFS. **Results:** We evaluated 322 patients with pStage II CRC, including 126 (39.1%) with right colon cancer, 133 (41.3%) with left colon cancer and 63 (19.6%) with rectal cancer. There were 87 patients (27.0%) with pT4, 175 (54.3%) with venous invasion, 120 (37.3%) with lymphatic invasion, and 68 (21.1%) with perineural invasion. Whole exome analyses revealed that there were 48 patients (14.9%) with MSI-High, and the presence of mutations in key genes for colorectal cancer development was as follows: *APC*, 245 (76.1%); *TP53*, 208 (64.6%); *KRAS*, 134 (6.5%); *PIK3CA*, 64 (19.9%); *FBXW7*, 27 (8.4%); and *BRAF*, 21 (6.5%). The 5-year OS and RFS rates were 98.0% and 90.7%, respectively. According to the consensus molecular subtype (CMS) classification based on gene expression, 76 patients (23.6%) had CMS4 and a significantly lower RFS than the other patients (5-year RFS: 83.8% vs. 92.9%, p=0.017). The independent risk factors for RFS were perineural invasion (Hazard ratio [HR] 5.316, $p<0.001$) and CMS4 (HR 2.399, p=0.020). **Conclusions:** Molecular profiling of Stage II CRC to assess the risk factors for relapse may contribute to the indication and drug selection for postoperative adjuvant chemotherapy and lead to personalized treatment to improve prognosis. Research Sponsor: Shizuoka Prefecture.

216 **Poster Session**

Patient body fat and AI ensemble technique and the influence on false-positive rate in AI second observer for colorectal cancer detection. First Author: Neal Kucera, Univesity of Nebraska College of Medicine, Omaha, NE

Background: Colorectal cancer (CRC) is the second leading cause of cancer-related deaths. We previously found that delayed diagnosis due to lack of radiological identification results in significantly worse outcome for patients. We had developed a rudimentary, AI second observer which demonstrated potential for detecting CRC on routine CT abdomen/pelvis (CTAP). However, the AI algorithm detected many false positives. In this study, we analyzed the data using TCIA as test cases and evaluated whether patient peritoneal fat content influenced the false positive rate. This could serve as a guide for future training of AI second observer to minimize false positive detection. **Methods:** 2D U-Net convolutional neural network (CNN) containing 31 million trainable parameters was trained with 58 CRC CT images from Banner MD Anderson (AZ) and MD Anderson Cancer Center (TX) (51 used for training and 7 for validation) and 59 normal CT scans from Banner MD Anderson Cancer Center. 18 of the 25 CRC cases from public domain data (The Cancer Genome Atlas) were used to evaluate the performance of the models (5 had no identifiable cancer and 2 were rejected for having no contrast). The CRC was segmented using ITK-SNAP open-source software (v. 3.8). To apply the deep ensemble approach, five CNN models were trained independently with random initialization using the same U-Net architect and the same training data. Given a testing CT scan, each of the five trained CNN models was applied to produce tumor segmentation for the testing CT scan. The tumor segmentation results produced by the trained CNN models were then fused using a simple majority voting rule (up to 2 voters) to produce consensus tumor segmentation results. The segmentation was analyzed for the number and location of false positives per case. The peritoneal fat content was classified at the level of aortic bifurcation by the distance of fat between adjacent small bowel loops ($\leq$ or $>$ 1 cm). Chi-square test was performed testing fat volume and number of voters as the intervention. **Results:** Our results showed that the higher volume of peritoneal fat ($>$ 1 cm, N=6) decreases the rate of false positive compared with low volume ($\leq$ 1 cm, N=12, p=0.013). When comparing between having one voter and two voter ensemble using low fat volume data, two voter ensemble also decreased the number of false positives but not statistically significant (p=0.286). **Conclusions:** Our results show that AI-based second observer generates more false positives when patients have lower peritoneal fat volume; this implies that future training may require higher percentage of cases with low peritoneal fat to improve second observer precision. Our analysis also showed that increasing the number of voter in the ensemble also decrease the number of false positives per case. Research Sponsor: None.

217 **Poster Session**

Development of new therapeutic strategies to enhance the effectiveness of irinotecan in colorectal cancer treatment. First Author: Koji Ando, Department of Surgery and Science, Graduate School of Medical Sciences, Kyushu University, Fukuoka, Japan

Background: Chemotherapy plays an important role in the treatment of colorectal cancer. Irinotecan, a topoisomerase I inhibitor, is used in therapies such as FOLFIRI and FOLFOXIRI. However, many patients become resistant to irinotecan, and the detailed mechanisms of resistance are not fully understood. The aim is to develop new therapeutic strategies by elucidating the mechanisms of irinotecan resistance. **Methods:** We used in vitro approach to demonstrate the mechanism of topoI degradation. Immunohistochemistry staining was used to explore the biomarker for irinotecan, Finally, in vivo approach was used to develop a new strategy for irinotecan treatment. **Results:** Mechanism of topoisomerase I (topoI) degradation: We investigated the molecular mechanism of topoI degradation in detail. After irinotecan administration, double-strand breaks occur in the DNA, activating DNA-PK. This activated DNA-PK phosphorylates the S10 residue of topoI, and BRCA1-BARD1 ubiquitinates the phosphorylated topoI. Eventually, topoI is degraded by the proteasome. Colorectal cancer cell lines where topoI degradation occurred after irinotecan administration showed irinotecan resistance. Development of a biomarker for irinotecan sensitivity: We created a monoclonal antibody that labels the phosphorylated S10 residue of topoI (topoI-pS10). We examined topoI-pS10 expression in colorectal cancer cases with a history of irinotecan treatment. The results showed a sensitivity of 87.5%, accuracy of 70%, positive predictive value of 70.7%, and negative predictive value of 87%, indicating that topoI-pS10 expression could serve as a biomarker for irinotecan sensitivity.Development of new therapeutic strategies: The degradation of topoI, which causes irinotecan resistance, is carried out via the ubiquitin-proteasome pathway. It is expected that combining irinotecan with a proteasome inhibitor, which blocks this pathway, will enhance irinotecan's effectiveness. When irinotecan was combined with a proteasome inhibitor in irinotecan-resistant colorectal cancer cell lines, cell death occurred at a higher rate compared to when they were not combined. Similar results were observed in experiments using mice. **Conclusions:** Based on the mechanism of topoI degradation, we developed a biomarker for irinotecan sensitivity and a new therapeutic strategy. This study suggests the potential for further improvements in irinotecan therapy. Research Sponsor: None.

218 Poster Session

Diverse stromal phenotypes at spatial resolution in colorectal cancer peritoneal metastasis. First Author: Joseph J Zhao, Department of Haematology-Oncology, National University Cancer Institute, Singapore, Singapore

Background: Peritoneal metastases (PM) in colorectal cancer (CRC) portend a poor prognosis. We sought to elucidate spatially resolved molecular features facilitating transcoelomic dissemination and progression. **Methods:** 50 PT, 116 PM and 37 primary normal samples from 96 patients were retrieved and profiled with digital spatial profiling (GeoMx DSP, Nanostring Technologies). Through unsupervised clustering, we characterized the microenvironment of tumor-stroma compartments and studied the roles of stromal phenotypes in promulgating tumorigenesis. These findings were orthogonally validated through spatial proteomics (multiplex IHC [mIHC] - COMET, Lunaphore Technologies) of 10 paired PT-PM samples retrieved from 6 patients. **Results:** A median of 37 region of interests (ROIs) (IQR: 32-38) were selected from a total of 9 tissue microarray (TMA) slides. A total of 269 ROIs (181 tumor, 88 stroma) from PM samples, 45 ROIs (34 tumor, 11 stroma) from PT samples and 28 (all stroma) ROIs from primary normal samples were retrieved and profiled. Through consensus clustering of stroma ROIs, we identify a fibro-collagenous and immune infiltrated stromal phenotype (stromal cluster [SC] 2) characterized by increased cancer associated fibroblasts (CAFs), memory B cells, M2 macrophages and T-cell exhaustion. Patients with SC2 stroma had poorer survival (p=0.036). SC2 stroma was also observed to support adjacent tumor compartments with enriched oncogenic pathways such as TGF-beta, TNF-alpha, hypoxia and JAK-STAT. Through discriminatory gene expression profiles, we developed a 20-gene composite bidirectional signature of SC2. The prognostic significance of the SC2-signature was externally validated through several cohorts including TCGA and MSI-H only cohorts. Next, by inspecting mIHC retrieved immune cell type densities across ROIs, we confirmed a similar phenomenon in which two distinct stromal clusters were found – one being fibroblast infiltrated but T-cell depleted (SC2-like, n=67 ROI) and another which is T-cell infiltrated but fibroblast depleted (SC1-like, n=104 ROI). Diverging spatial distributions of T-cells and fibroblasts between SC1 and SC2-like stroma was appreciated as well. Between both PT and PM, we find close clustering of T cells in SC1-like stromal compartments and fibroblasts in SC2-like stromal compartments. **Conclusions:** We describe SC2, a protumorigenic stromal phenotype characterized by increased CAFs, T cell exhaustion and is associated with poor prognosis in CRC PM. Research Sponsor: National University Health System; NUHSRO/2024/008/RO5+6/Seed-Sep23/01; National University Hospital; JRA/Sep23/002; Yong Loo Lin School of Medicine, National University of Singapore; National Medical Research Council; MOH-CIRG21jun-0005; National Medical Research Council; MOH-CIRG24jan-0044; National Medical Research Council; MOH-CSAINV22jul-0005; National Medical Research Council; MOH-OFLCG18May-0003 and MOH-OFLCG23-may0031; National Medical Research Council; NMRC/CIRG23Jul-0035.

220 Poster Session

Longitudinal ctDNA measurements for treatment response monitoring in patients with metastatic colorectal cancer undergoing systemic therapy: The ORCA trial. First Author: Denise van Steijn, Netherlands Cancer Institute, Amsterdam, Netherlands

Background: Precise monitoring of treatment response may support and improve treatment decision-making in patients with metastatic colorectal cancer (mCRC). The PLCRC-ORCA study aimed to explore the clinical applicability of personalized ctDNA testing for treatment response monitoring in patients with mCRC. **Methods:** The prospective, observational ORCA study is a substudy of the Prospective Dutch Colo-Rectal Cancer cohort (PLCRC). After informed consent, blood samples were collected before initiating first line of treatment (baseline) and subsequently every 8-9 weeks during routine clinical care. ctDNA levels were analyzed using a clinically validated, personalized, tumor-informed 16-plex PCR NGS assay (Signatera™, Natera, Inc). **Results:** The majority of patients (45/49) had blood samples collected at both baseline and the first on-treatment time point. At baseline, 43/45 (95.6%) patients were ctDNA-positive with a median concentration of 63.5 mean tumor molecules per mL (MTM/mL; range 0.08 to 16,200). In a multivariate linear regression model including metastatic site at baseline and resection status of the primary tumor, ctDNA levels at baseline were lower in patients with metastatic disease limited to the lung ($n = 4$; $p < 0.001$) or peritoneum ($n = 6$; $p = 0.002$) compared to patients with liver-limited metastases. Moreover, primary tumor resection was independently associated with lower levels of ctDNA at baseline ($p = 0.05$). At the first measurement on-treatment, 13/45 patients (28.9%) were ctDNA-negative and 32/45 (71.1%) patients were ctDNA-positive with ctDNA levels ranging from 0.04 to 318 MTM/mL. Reduction in ctDNA levels relative to baseline was observed in 42/45 (85.7%) patients while two patients exhibited increased ctDNA levels. One patient had undetectable ctDNA levels at baseline and on treatment. Among the 30 patients with detectable but reduced ctDNA levels on treatment, the median ctDNA levels decreased 55-fold (95% CI: 26–600). **Conclusions:** The levels of ctDNA in patients with mCRC are influenced by the metastatic site and are lowest among patients with metastases limited to the lung and peritoneum. Our data suggest that clinical applications of ctDNA testing for treatment response monitoring should take the baseline ctDNA levels into account when interpreting ctDNA dynamics. Correlation of ctDNA status with radiological treatment response is underway and will be reported during the conference. Research Sponsor: None.

221 Poster Session

The use of liquid biopsy comprehensive genomic profiling to identify the molecular landscape of patients with metastatic colorectal cancer who are candidates for anti-EGFR rechallange therapy: Findings from the CAVE-2 GOIM trial. First Author: Stefania Napolitano, Medical Oncology, Department of Precision Medicine, Università degli Studi della Campania "Luigi Vanvitelli", Naples, Italy

Background: In the era of precision medicine, biomarker guided therapies are necessary for patients with chemorefractory metastatic colorectal cancer (mCRC). Circulating tumor DNA (ctDNA) comprehensive genomic profiling (CGP) may contribute to the development of more effective treatments. **Methods:** CAVE-2 GOIM trial (ClinicalTrials.gov ID: NCT05291156) is the first randomized phase II study to assess the efficacy of avelumab plus cetuximab compared to cetuximab alone as rechallenge strategy in pre-treated plasma ctDNA *RAS, BRAF* WT mCRC patients. Before being enrolled in the trial, patients were selected by baseline plasma ctDNA analysis by Foundation One Liquid CDx (F1L CDx) 324 genes to identify ctDNA *RAS/BRAF* WT cases. **Results:** As of July 31th 2024, 229 (out of approximately 310) patients have been screened. A total of 1562 pathogenic variants (PV) were detected, with *TP53* (86.0%), *APC* (84.2%), *DNMT3A* (30.5%) representing the most common altered genes. In these anti-EGFR pretreated patients different acquired mechanisms of resistance were observed. In particular, 72/229 (31.4%) patients had *RAS* or $BRAF^{V600E}$ PV (102 *KRAS*, 21 *NRAS* and 8 $BRAF^{V600E}$ mutations, respectively). Additionally, in these 72 patients, multiple $RAS/BRAF^{V600E}$ gene alterations (median of 1 PV, range 1-7) were observed in 41.6% cases. *MAPK, RTK*, and *PI3K/AKT/mTOR* PV were detected in 43.2%, 24.5% and 24.5% patients, respectively. Of note, the variant allele frequency of resistant PV was lower than other gene alterations, suggesting a high tumor heterogeneity as a consequence of the development and/or expansion of EGFR-inhibitor resistant cancer cell sub-clones. All patients were assessed for tumor mutational burden (TMB). Median TMB was 6 (IQR 4-10) mut/Mb, with 29.6% cases showing a TMB $\geq$10 mut/Mb. Finally, a total of 258 actionable genomic alterations were detected in 52,4% of patients according to the ESMO Scale for Clinical Actionability of Molecular Targets (ESCAT). Eighty-one tier I alterations were detected in 32.3% patients, which included TMB high (29,7%), $BRAF^{V600E}$ (3,5%), $KRAS^{G12C}$ (6,5%), and *RET* fusions (0,4%). **Conclusions:** These findings support liquid biopsy-based GCP for individualized therapeutic choices in chemorefractory mCRC. Up-dated final analyses of the entire screened patient population of the CAVE-2 GOIM trial will be presented. Clinical trial information: NCT05291156. Research Sponsor: CAVE-2 GOIM clinical trial is partially supported by a research grant from Merck KGaA Italy.

222 Poster Session

Can overall survival (OS) benefit be predicted from improvements in progression-free survival (PFS) for previously untreated metastatic colorectal cancer (mCRC)? First Author: Jeanine Roodhart, Department of Medical Oncology, University Medical Center, Utrecht, Netherlands

Background: With improvements in treatment of mCRC, OS maintains its gold standard efficacy measure but takes longer to mature than intermediate endpoints. This study analyzed the association between treatment effects on PFS and OS using aggregate-level data from RCTs in previously untreated mCRC patients. **Methods:** A systematic literature review identified RCTs in previously untreated mCRC patients published from 2010–2021 reporting hazard ratios on PFS (HR_{PFS}) and OS (HR_{OS}). All treatments in comparison to chemotherapy alone or chemotherapy with anti-VEGF (bevacizumab) or anti-EGFR (cetuximab) targeted therapy were considered. Correlation between HR_{PFS} and HR_{OS} was evaluated using bivariate random-effects meta-analysis (BRMA) and weighted linear regression (WLR). Predictive performance of the surrogacy equations from WLR was assessed using leave-one-out cross-validation (LOOCV). Surrogate threshold effects (STE), defined as the minimum PFS benefit that would translate into a statistically significant OS benefit with 95% probability, were also derived to gauge the practical utility of the models. Primary analysis consisted of all included trials. Sensitivity analyses omitted trials that (I) had anti-EGFR medications, (II) had anti-VEGF medications, (III) violated proportional hazards assumptions, and (IV) permitted treatment crossover. **Results:** In the primary analysis 47 trials were included. The estimated correlation between PFS and OS was 0.67 (95% CI: 0.48–0.80) using BRMA and 0.70 (95% CI: 0.48–0.84) using WLR. The surrogacy equation derived from WLR was $\log(HR_{OS}) = -0.03 + 0.56 \log(HR_{PFS})$ with a statistically insignificant intercept and significant slope. Estimated STEs corresponding to sample sizes of 200 and 300 patients were 0.55 and 0.62, respectively. Observed HR_{OS}'s were within their 95% prediction intervals predicted from HR_{PFS} for 93.6% of studies in LOOCV. Sensitivity analyses produced moderate correlations with >90% coverage in LOOCV (Table). **Conclusions:** Moderate correlations were found between HR_{PFS} and HR_{OS} using both modeling approaches, highlighting the stability of the findings. Cross-validations of surrogacy equations indicated promising predictive value of PFS benefit for OS benefit in previously untreated mCRC. Research Sponsor: Bristol Myers Squibb.

Analysis Set	# of Studies	Correlation (95% CI)		STE		LOOCV Coverage Rate
		BRMA	WLR	N = 200	N = 300	
Sensitivity Analysis I	34	0.75 (0.57, 0.86)	0.78 (0.55, 0.90)	0.55	0.61	94.1%
Sensitivity Analysis II	14	0.44 (-0.08, 0.77)	0.62 (-0.05, 0.90)	0.44	0.52	92.9%
Sensitivity Analysis III	36	0.61 (0.36, 0.77)	0.59 (0.26, 0.80)	0.46	0.55	94.4%
Sensitivity Analysis IV	43	0.68 (0.48, 0.81)	0.67 (0.41, 0.83)	0.52	0.60	93.0%

223 Poster Session

Alterations in gut microbiome in age- and diet-induced colorectal cancer (CRC) progression in a preclinical model. First Author: Pooja Mittal, Division of Medical Oncology, Norris Comprehensive Cancer Center, Keck School of Medicine, University of Southern California, Los Angeles, CA

Background: CRC is the third leading cause of cancer related deaths worldwide with age and diet among the strongest risk factors. Emerging evidence suggests gut microbiome plays important role in CRC and is highly impacted by the exposome primarily the food intake. Since diet and age plays an important role in gut microbiome diversity, the present study aimed to investigate the alterations in the gut microbiome in young versus old mice harboring CRC allografts and fed with different diets. **Methods:** Two cohorts of C57BL/6 including young (n = 9, age = 6 weeks) and old (n = 9, age = 20-24 months) mice were implanted with $1x10^6$ MSI (microsatellite instable) CRC MC38 tumor cells. Mice in both cohorts were randomized into three different diet groups: normal diet (ND; standard chow), high-fat (HF) and calorie-restricted (CR: 30% reduction in total calories). Mice were housed individually under standard laboratory conditions. Mice fecal samples were collected and subjected to DNA isolation (ZymoBIOMICS-96 MagBead DNA Kit), followed by metagenomic shotgun and whole genome sequencing (ZymoBIOMICSe and Illumina NovaSeq, respectively). $P \leq 0.05$ were considered as statistically significant. **Results:** Fecal microbiome analysis showed that old microbiome has higher alpha diversity compared to the young mice. No statistically significant differences in were found in microbiome diversities between the diet groups. Pairwise permanova results showed statistically significant differences between microbiomes of old vs. young groups (*P*: 0.001), CR vs. HF groups (*P*: 0.012) and HF vs. ND groups (*P*: 0.021). ANCOM-BC analysis determined the differentiating features and microbial functional pathways in the young mice compared to the old mice group and in different diet groups with relative abundance differences of larger than $\log_{10}2$. *Bacteroides thetaiotaomicron* (*P*: 1.43E-39) and *Parabacteroides goldsteinii* (*P*: 2.20E-23) were enriched in young and old groups, respectively. *Akkermansia muciniphila* (*P*: 0.008) was significantly enriched in ND compared to CR group. *Lactococcus lactis* (*P*: 9.94E-160) and *Lachnospiraceae bacterium A4* (*P*: 4.95E-10) were significantly enriched in HF compared to ND diet groups and vice-versa, respectively. Among the functional pathways, CMP-legionaminate biosynthesis I (*P*: 2.38E-13) and L-arginine biosynthesis IV (archaebacteria) (*P*: 4.77E-11) were the most significantly enriched in young and old groups, respectively. **Conclusions:** Our findings highlight the differential diversity and microbial features of gut microbiome in age- and diet-induced CRC progression. Collectively, alteration in diet and age of the host lead to changes in gut microbiota which has a direct impact on activation of specific signaling pathways, metabolism and local and systemic immune responses, which may in turn affect chemosensitivity and outcome in CRC Research Sponsor: National Cancer Institute; P30CA014089; Ming Hsieh research fund; National Cancer Institute; U2CCA252971; Gloria Borges WunderGlo Foundation; Dhont Family Foundation; Gene Gregg Pancreas Research Fund; San Pedro Peninsula Cancer Guild; Daniel Butler Research Fund; V foundation for cancer research; Victoria and Philip Wilson Research Fund; Fong research project.

224 Poster Session

Impact of expression of metabolic genes on patient (pts) outcomes in metastatic colorectal cancer (mCRC): Data from CALGB/SWOG 80405 (Alliance). First Author: Francesca Battaglin, Division of Medical Oncology, Norris Comprehensive Cancer Center, Keck School of Medicine, University of Southern California, Los Angeles, CA

Background: Metabolic pathways are reprogrammed during CRC development, leading to increased glycolysis, glutaminolysis, and fatty acid synthesis. Understanding the impact of metabolic reprogramming on treatment outcomes is paramount. Hence, we investigated whether the tumor gene expression of 4 selected major metabolic genes (*PKM, SLC2A1*, *SLC16A1*, and *CAV1*) could affect treatment response in pts enrolled in the CALGB/SWOG 80405 trial. **Methods:** 433 mCRC pts treated with either bevacizumab (bev, n = 226) or cetuximab (cet, n = 207) in combination with first-line chemotherapy were analyzed. RNA was isolated from FFPE tumor samples and sequenced on the HiSeq 2500 (Illumina). Overall survival (OS) and progression-free survival (PFS) were compared between groups of pts categorized by tertiles of gene expression (high [H], medium [M] and low [L]). Logrank *P*-values provide a non-parametric unadjusted assessment of differences. Likelihood ratio tests (LRT) were computed from multivariate Cox proportional hazards models, adjusting for age, sex, ECOG, tumor side, number of metastatic sites, *KRAS*, CMS, and treatment. **Results:** *PKM* gene expression was associated with cet treatment outcomes with *PKM*-low tumors showing significantly longer PFS and OS (median PFS: 14.3 vs 9.8 vs 8.1 months, L vs M vs H, $P < 0.0001$, LRT $P = 0.011$; median OS: 45.9 vs 31.2 vs 20.9 months, $P < 0.0001$, LRT $P = 0.12$). Low *SLC2A1* was also associated with longer OS in cet-treated pts (32.4 vs 35.8 vs 25.2 months, $P = 0.02$, LRT $P < 0.0001$). Similar results were observed in bev-treated pts where low *SLC2A1* expressing tumors had longer OS (37.4 vs 26.1 vs 29 months, $P = 0.037$, LRT $P = 0.069$) and a non-significant trend for longer PFS (13.1 vs 11 vs 9.5 months, $P = 0.076$). *CAV1* low gene expression was associated with longer OS in cet-treated pts (37.4 vs 34 vs 21.5 months, $P = 0.0048$, LRT $P = 0.013$), while no significant associations were found in bev-treated (interaction LRT $P = 0.0009$). No significant results were observed for *SLC16A1*. **Conclusions:** The intratumoral expression of metabolic genes was prognostic and predictive in mCRC pts treated with first-line therapy. GLUT1 (encoded by *SLC2A1*) and PKM2 (encoded by *PKM*) promote aerobic glycolysis of cancer cells, known as Warburg effect, supporting rapid cell proliferation and survival. CAV1 has been shown to be involved in mitochondrial bioenergetics and fatty acid metabolism and to play either a tumor suppressor or oncogenic role depending on the cancer type and stage. Lower expression of *SLC2A1* and *PKM* was associated with improved survival and increased benefit from cet treatment in our cohort. Low *CAV1* expression was also associated with increase survival suggesting a cancer-promoting role in mCRC. Our results suggest that targeting cancer cell metabolism through novel inhibitors of the aforementioned pathways may be a promising therapeutic strategy. Research Sponsor: U.S. National Institutes of Health; P30CA014089, U10CA180821, U10CA180882, U10CA180820 (ECOG-ACRIN); U10CA180888 (SWOG); Pfizer, Genentech; https://acknowledgments.alliancefound.org.

225 Poster Session

Impact of education on clinical confidence utilizing HER2-targeted therapies in colorectal cancer. First Author: Karine Cohen-Solal, PlatformQ Health, Needham, MA

Background: Novel targeted therapies entered the treatment landscape in patients with HER2-positive metastatic colorectal cancer (mCRC). To ensure that these therapies are used effectively, clinicians must assess tumor HER2 status and recognize how HER2-directed treatment modalities fit into the current paradigm. To further enable clinicians to integrate HER2-targeted therapies, an educational program was designed in 2023. **Methods:** A one-hour online, video-based program was designed for clinicians and their teams and hosted on MedLive from February 2023 for one year. Knowledge and competence questions were administered pre-, and immediate post-activity. Additional questions assessed attitudes, barriers, and intended practice changes related to HER-positive mCRC. **Results:** Over 4,000 participants engaged in the educational activity, 67% of whom were physicians, and 63% noting their specialty as oncology. Approximately 68% identified as treaters seeing 14 patients with CRC per week with 78% of those patients have HER2-positive CRC. All pre-post questions showed significant improvements in knowledge and competence related to HER2 amplification testing methodologies, trial eligibility criteria, identification of patients who would benefit from anti-HER2 targeted regimens, and adverse events associated with anti-HER2 agents. Following the activity, 51% were more likely to incorporate testing for HER2 expression level or amplification prior to the selection of targeted therapy for their patients. The greatest challenges faced when managing patients with HER2-positive mCRC were affordability of therapy for patients (50%), adherence to treatment schedules (25%) and patient anxiety about treatment efficacy (24%). The most common barriers to patient enrollment in clinical trials identified by clinicians were lack of trials at their institutions (35%), lack of trials in their geographic region (21%), and patient lack of interest (20%). Qualitative insights on intended practice changes were shared by learners post-activity. **Conclusions:** The outcomes of this educational activity demonstrate the effectiveness of education in improving knowledge and confidence from testing to adverse event management to better integrate new regimens into practice. Encouraging practice changes were also seen from the write-in responses from learners. Additional needs were also identified to further solidify HER2 testing as a prerequisite for anti-HER2 targeted therapies, selecting eligible patients, and timely management of adverse events. Research Sponsor: Seagen Inc.

226 Poster Session

Enhancing a low-cost, minimally invasive screening test for dihydropyrimidine dehydrogenase mutations linked to 5-fluorouracil sensitivity by integrating computational mutation predictions. First Author: John M Janiga, Molecular Diagnostics Laboratory, Fralin Biomedical Research Institute at Virginia Tech, Roanoke, VA

Background: 5-Fluorouracil (5-FU), a common chemotherapy for solid tumors, is metabolized primarily by dihydropyrimidine dehydrogenase (DPD), encoded by the *DPYD* gene. With > 200 known variants, individuals with nonfunctional *DPYD* alleles exhibit impaired 5-FU metabolism and are at risk of severe toxicity. Approximately 35% of the population has partial DPD deficiency, and 0.2% completely lack enzymatic activity. Despite this, *DPYD* genotyping is not standard practice. This study evaluates the use of non-invasive saliva samples for *DPYD* genotyping and combines AI-based molecular prediction modeling to search for novel variants within *DPYD* coding region. Saliva samples were collected from fifty-six healthy individuals and from gastrointestinal (GI) cancer patients including a family cohort of nineteen individuals and three unrelated patients undergoing chemotherapy. **Methods:** gDNA was sequenced for nine *DPYD* variants reported to reduced or abrogate DPD activity: c.1905+1G > A, c.1679T > G, c.2846A > T, c.1129-5923C > G, c.1236G > A, c.299_302delTCAT, c.703C > T, c.2983G > T, and c.557A > G. cBioPortal was utilized to search the TCGA database for studies involving colon cancer patients with novel DPYD mutations exhibiting a total FIS score of at least 2 or identified through 3D modeling of DPD using PyMol based on their proximity to or the presence of polar contacts within the active site. Putative pathogenic mutations were analyzed using AlphaMissense and ChimeraX to assign a RSMD score, assessing their potential negative impact on DPD function. **Results:** Computational studies identified three mutations in DPYD from TCGA colon cancer patients with an unknown impact on DPD and a FIS greater than 4 (c.198G > C/T, c.2161G > A, c.2185G > A), suggesting potential disruption of its function. Four mutations (c.424T > G, c.427T > G, c.2460G > C, c.2909C > A) were identified through protein modeling using PyMol, but only two had a FIS greater than 2. Of these seven identified mutations, four (c.198G > C/T, c.2161G > A, c.2185G > A, c.2909C > A) also scored highly using AlphaMissense and ChimeraX. *In silico* prediction models from VarSome listed c.2185G > A as Pathogenic Moderate and c.2909C > A as Pathogenic Very Strong. Sequencing of saliva samples indicated mutations c.1129-5923C > G and c.1236G > A were present in four and three volunteers, respectively, and c.1905+1G > A in one case. Mutations identified in GI cancer patients were c.1129-5923C > G and c.1236G > A. **Conclusions:** By combining computational modeling with the analysis of naturally occurring mutations in colon cancer patients, we have successfully identified potentially pathogenic mutations in the *DPYD* gene. This information can enhance existing clinical diagnostic tests, providing a more comprehensive assessment of DPYD mutations, including novel variants. Research Sponsor: Fralin Biomedical Research Institute; Virginia Commonwealth Clinical Research Network.

227 **Poster Session**

Spatial transcriptomics and lethality-associated stromal remodeling of colorectal cancer. First Author: Colin Wood, University of Glasgow, Glasgow, United Kingdom

Background: Colorectal cancer (CRC) is surgically resectable and eminently screenable yet remains a lethal entity with high affinity to metastasise synchronously and metachronously. It has been proposed that cellular and extracellular components of the microenvironment contribute to metastatic potential. Dissociative profiling techniques such as bulk transcriptomic and single-cell RNA sequencing have contributed insight but limited therapeutic progress. Spatial transcriptomic (ST) assessment allows molecular profiling of tissue while preserving tissue architecture. Here we employ ST approaches to interrogate tumor compartments of primary resectable and matched synchronous CRC demonstrating stromal signatures with distinct collagen expression associated with outcome. **Methods:** 25 patients with primary resectable CRC and 4 patients with matched primary CRC and liver metastasis (CRLM) underwent single-cell spatial transcriptomics using the Nanostring CosMx Spatial Molecular Imager (SMI, 1000plex gene panel) (Discovery cohort). 71681 epithelial and 88806 stromal cells with intact spatial resolution were analysed. The spatial single-cell signatures were reconstituted in 89 patients using the Nanostring GeoMx Digital Spatial Profiler (Validation cohort). 3 GeoMx compartments were analysed: Epithelial (PanCK+); Stroma (aSMA+); Immune (PanCK-aSMA-). **Results:** CosMx demonstrated 2 distinct collagen signatures: COL1A1,COL1A2,COL3A1 associated with normal fibroblasts; COL9A2 associated with cancer-related stroma and lethal subtypes of epithelial cell. In the GeoMx validation cohort, 42 patients expressed the COL1A1 signature in the aSMA compartment and 47 did not (5 year recurrence free survival: 0.88 vs 0.49, $p < 0.005$) with different morphological patterns of each distinct aSMA group. In patients with favourable prognosis, chemokine high epithelial subtypes expressed CXCL5 and recruited COL9A2-, IL6+ neutrophils to the surrounding microenvironment. In contrast, epithelial cells in patients with poor prognosis expressed CXCL8 and recruited COL9A2+ neutrophils which were frequently in contact with SPP1+ macrophages. **Conclusions:** We have used ST approaches to interrogate cellular compartments of CRC demonstrating stromal subtypes that impact outcome. These insights could be used in the clinical setting to quantify COL9A2 expression in aSMA+ cells to prognosticate patients. Targeting of tumor collagen has been proposed to augment existing anti-cancer therapies and this work has demonstrated COL9A2 as a potential target in need of further investigation. Research Sponsor: None.

228 **Poster Session**

Unbiased transcriptomic analysis of primary early-onset vs late-onset colorectal cancer. First Author: Alessandro La Ferlita, The Ohio State University, Columbus, OH

Background: Colorectal cancer is the 3rd most diagnosed and 2nd leading cause of cancer death. Despite the decline in incidence late-onset colorectal cancer (LOCRC), there is an increase in early onset of colorectal cancer (EORC). Clinicopathologic examinations from single centers revealed left-sided and rectal enrichment, poorly differentiated, and higher stage at presentation. Genomic studies examining the potential germline mutations/variants for EOCRC demonstrated mixed results. Transcriptomic analysis of the EOCRC compared to LOCRC remains limited. We hypothesized that EOCRC present with enrichment in aggressive molecular signatures consisting of CMS4 and iCMS3 signatures. To understand the mechanism behind EOCRC, we performed an unbiased RNA-Seq analysis in age groups of 32-82 in 95 patients. **Methods:** We utilized RNA-Seq data from the Orien Network, a collaborative between 19 NCI/NCCN cancer centers. We identified 95 patients diagnosed with colorectal cancer ranging in age from 32-82 and undergone RNA-sequencing of their primary tumors. Clinicopathologic assessment were analogously performed. We collected unnormalized raw-count RNA-Seq data from these patients. To account for differences in sequencing depth across samples, raw counts were scaled using the Reads Per Million (RPM) formula. The scaled reads in RPM were then utilized to assess the Consensus Molecular Subtypes (CMS) classification through two distinct approaches: the R package CMScaller and a deconvolution analysis using the R package SpatialDecon with the iCMS gene signature. Differences in age at the sample collection across patients with different CMS classifications were assessed by performing two-tailed unpaired T-tests. Additionally, differential expression analysis between EOCRC and LOCRC patients, defined by the upper and lower quartiles of the patients' ages at the sample collection, was conducted using the LIMMA R package. **Results:** Clinicopathologic assessment of the primary tumors revealed left-sided tumors with poorly differentiated phenotype. Differential gene expression analysis between EOCRC and LOCRC demonstrated no significant differences. Stratification of transcriptomic data by decades also did not reveal any significant differences. Furthermore, there were no differences in CMS or iCMS classifications between different age groups. **Conclusions:** The RNA-Seq analysis comparing EOCRC and LOCRC did not identify any differential gene expression patterns. There was no significant differences were observed in CMS or iCMS classifications. These findings suggest that EOCRC and LOCRC share similar tumor biology based on molecular phenotypes. This implies that the initiating events for the disease may occur earlier in life while triggering a same disease process. As such, the findings underscores the importance of considering preventive measures, including earlier colonoscopy screenings. Research Sponsor: None.

229 **Poster Session**

Phase II study of cabozantinib and nivolumab in refractory metastatic microsatellite stable (MSS) colorectal cancer (CRC). First Author: Alexis Diane Leal, Division of Medical Oncology, Department of Medicine, University of Colorado Anschutz School of Medicine, Aurora, CO

Background: Current data suggest that combining tyrosine kinase inhibitors with immune checkpoint inhibitors may be a promising treatment strategy in patients with metastatic MSS CRC. This prospective, open-label, single-arm, phase II study (NCT04963283) evaluated the efficacy of cabozantinib in combination with nivolumab in patients with metastatic refractory MSS CRC. **Methods:** Patients with metastatic/unresectable MSS CRC who were refractory to chemotherapy in the 3rd line setting and beyond were eligible. Patients were treated with cabozantinib 40 mg orally daily and nivolumab 480 mg IV every 28 days. Tumor assessments were obtained every 8 weeks for the first 16 weeks, and then every 12 weeks thereafter. Response was assessed via RECIST v1.1. The primary endpoint was 16-week disease control rate (16-wk DCR); secondary endpoints were objective response rate (ORR), progression free survival (PFS), overall survival (OS), safety and tolerability, and correlative studies (to be presented at a future date). **Results:** Between June 2021 and January 2024, 49 pts were enrolled; 63% were male; median age 55 [50-64]. Median prior treatment regimens were 4 [2-14]. Most patients (84%) had not received prior trifluridine/tipiracil; only 1 patient had received prior regorafenib. The 16-wk DCR was 40% (19/47) (95% CI 0.264 – 0.557). The 16-wk ORR was 8.5% (4/47) (95% CI 0.024 – 0.204). mPFS was 3.4 months (95% CI 1.8-5.0) and mOS was 10.9 months (95% CI 4.7-13.8). Thirty nine percent of patients experienced a serious adverse event (SAE), with 4% Gr 1, 37% Gr 2, 35% Gr 3 and 4% Gr 4; there were no Gr 5 treatment related adverse events. The most common AEs were diarrhea (53%), fatigue (43%), hypothyroidism (31%), weight loss (29%), nausea (29%), and hypertension (27%). Two patients remain on study at time of abstract submission. **Conclusions:** The combination of cabozantinib and nivolumabwas well tolerated and provided modest benefit in a population of patients with metastatic, chemotherapy refractory, MSS CRC. Clinical trial information: NCT04963283. Research Sponsor: None.

230 **Poster Session**

Preclinical analysis and clinical validation to identify biomarkers for trifluridine/tipiracil (FTD/TPI) efficacy with or without bevacizumab in patients with metastatic colorectal cancer. First Author: Mitsukuni Suenaga, Department of Clinical Oncology, Tokyo Medical and Dental University, Tokyo, Japan

Background: We previously identified candidate biomarkers for FTD/TPI efficacy through preclinical and translational studies in patients with metastatic colorectal cancer (mCRC). Recent studies reported that FTD/TPI plus bevacizumab (BEV) improved survival compared to FTD/TPI alone. This study evaluated the clinical significance of these biomarkers in patients with refractory mCRC receiving either FTD/TPI plus BEV or FTD/TPI alone. **Methods:** Candidate biomarkers were selected based on our transcriptomic and cell biological analyses. We validated these biomarkers in mCRC patients treated with FTD/TPI plus BEV (cohort A) or FTD/TPI alone (cohort B). Blood samples were collected at baseline (BL), before the second cycle (2nd), and progressive disease (PD), with serum biomarker levels measured using ELISA. Change patterns were classified as 'increased' or 'decreased' from BL. To address the unbalanced distribution of baseline characteristics between cohorts, propensity score matching (PSM) was performed. **Results:** A total of 112 patients were included, with 34 in each cohort after PSM. Cohort A showed significantly longer overall survival (OS) (14.4 vs 6.2 months, HR 0.53, 95%CI: 0.30-0.93) and higher disease control (DC) (58.8 vs 34.4%, P=0.047) compared to cohort B. In univariate analysis, increased IL-8 or AREG at 2nd and increased IFI16 at PD were associated with poorer clinical outcomes. Grade $\geq$3 neutropenia (NEU) was linked to better survival in both cohorts. Multivariable analysis identified decreased AREG and Grade $\geq$3 NEU as better markers for progression-free survival and OS. OS was significantly longer in cohort A compared to B (12.8 vs 5.8 months, HR 0.43, 95%CI: 0.21-0.89) when IL-8 increased at 2nd. Cohort A also had longer OS compared to cohort B when EREG (12.8 vs 6.2 months, HR0.27, 95%CI: 0.11-0,65) or AREG (14.2 vs 9.7 months, HR 0.40, 95%CI: 0.17-0.95) decreased at 2nd. There was no significant interaction between the regimen and biomarkers (IL-8, EREG, and AREG) for OS. **Conclusions:** Our data suggest that AREG may serve as a prognostic marker for FTD/TPI efficacy. Changes in IL-8, EREG and AREG potentially contribute to the extended survival benefit of combining BEV with FTD/TPI. Research Sponsor: JSPS KAKENHI.

231 **Poster Session**

Proto-oncogene HRAS transcript level and overall survival in stage II and III colorectal cancer. First Author: Donghyun Kim, University of Iowa Holden Comprehensive Cancer Center, Iowa City, IA

Background: Mutational landscape is prognostic in colorectal cancer (CRC). Driver mutations in the RAS proto-oncogenes such as KRAS and NRAS portend poor survival outcomes, whereas HRAS mutations are extremely rare and their prognostic value remains uncertain. The prognostic implication of RAS gene activity at the transcript level is also obscure. Here, we retrospectively review stage II and III CRC tumor RNA-Seq data and demonstrate that high transcript levels of HRAS are associated with superior overall survival (OS). **Methods:** Tumor RNA-Seq of 734 stage II and III CRC cases and their associated clinical data were retrieved from the ORIEN AVATAR database (Aster Insights). Transcript level was determined by batch-corrected TPM (transcripts per million) of each gene, and cohort median was used as a reference to determine "high" versus "low" in subgroup analyses. The p-value and hazard ratio (HR) derived from Kaplan-Meier survival analysis are log-rank. **Results:** The 4-year HR for death in patients with "very high" HRAS transcript level (top 5%) was 0.15 (95% CI 0.07-0.34, p=0.03) compared to the rest, and this superior OS was independent from TNM stage or perioperative 5-FU treatment history. In subgroup analyses, "high" HRAS expression was associated with superior OS only in patients with concomitant "low" KRAS expression in the absence of pathologic KRAS mutations in codons 12, 13 and 61 – the 5-year HR in "low" KRAS group was 0.42 (95% CI 0.18-0.96, p=0.03) compared to "high" KRAS group. NRAS transcript level did not influence OS. When further sub-grouped based on primary disease location, this superior OS was present only in the right-sided primary disease group with 5-year HR of "low" KRAS group at 0.17 (95% CI 0.03-0.99, p=0.06) compared "high" KRAS group. **Conclusions:** Contrary to the notion that RAS family genes are proto-oncogenic, we demonstrate that high HRAS transcript levels are associated with superior OS in stage II and III CRC. The high HRAS-associated OS benefit was most pronounced in patients with right-sided primary expressing low KRAS transcript levels in the absence of pathologic KRAS mutations. We speculate that when expressed at high levels, HRAS may counteract proto-oncogene KRAS but not oncogene KRAS at the transcript level. The potential of HRAS as a prognostic biomarker should be explored further. Research Sponsor: None.

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Performance of a blood-based screening test for the early detection of colorectal cancer. First Author: Yoshiaki Nakamura, Department of Gastroenterology and Gastrointestinal Oncology, National Cancer Center Hospital East, Kashiwa, Japan

Background: While the clinical benefit of highly sensitive screening tools for colorectal cancer (CRC) has been established, adherence rates remain low. The convenience and familiarity of blood-based testing may reduce disease burden through greater access and compliance to regular and recommended asymptomatic screening. We report on preliminary findings from a minimally invasive blood-based early cancer detection (ECD) screening test for CRC. **Methods:** Plasma samples from healthy individuals and CRC patients sourced from biobanks were included. A targeted panel composed of differentially methylated genomic regions was developed. Using an independent cohort of plasma samples, a machine-learning model was trained to classify patient samples into cancer-free and CRC based on observed differential methylation patterns in circulating cell-free DNA (cfDNA). The assay performance was calculated based on concordance with clinical diagnosis (CRC, cancer-free). **Results:** The test set included 440 individuals: 127 with a confirmed CRC diagnosis (47% stage I/II, mean age 65±12 years, male 46%) and 313 without cancer (mean age 56±10 years, male 41%). Overall sensitivity was 95% (95% CI: 88-100%), and specificity was 91% (95% CI: 84-98%). Among patients with stage I disease, sensitivity was 92% overall, and 88% in screen-detected individuals. Sensitivity was 100% (n=14/14) in microsatellite-high tumors and 95% (n=107/113) in microsatellite-stable tumors. Multivariate analysis demonstrated associations between methylation signal and tumor size, cancer stage, and sex. Compared with a tumor-informed, variant-based circulating tumor DNA (ctDNA) test (Signatera), we observed a PPA (positive percent agreement) of 100%. Moreover, the differentially methylated allele fraction (DMAF) across target regions strongly correlated with variant allele frequencies (VAFs) from the tumor-informed test (Spearman's correlation coefficient, 0.85). **Conclusions:** Using epigenetic markers specific for CRC, we developed a sensitive and specific blood-based test to detect cfDNA from early-stage and asymptomatic disease. The methylation signal strongly correlated with ctDNA VAFs from a clinically validated tumor-informed test and increased with cancer stage and disease burden. This test could serve as a means to increase patient adherence rates for CRC screening, thereby improving patient outcomes. Research Sponsor: None.

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PK/PD analyses of circulating tumor DNA (ctDNA) in patients with metastatic colorectal cancer (mCRC) treated with telisotuzumab adizutecan (ABBV-400). First Author: Ibrahim Abdelgawad, AbbVie Inc., North Chicago, IL

Background: Telisotuzumab adizutecan (ABBV-400) is an antibody drug conjugate (ADC) consisting of a c-Met targeting antibody telisotuzumab conjugated to a potent topoisomerase 1 inhibitor payload adizutecan. The ongoing first in human phase 1 trial (Study M21-404; NCT05029882) shows that ABBV-400 has encouraging radiographic response and ctDNA molecular response (MR) as monotherapy in patients with advanced solid tumors, including mCRC. Herein, correlation between ABBV-400 exposures and % change in ctDNA from baseline and PK/PD analyses using MR are presented for mCRC patients. Impact of c-MET expression and *MET* amplification (*MET*-amp) status on exposures was also evaluated. **Methods:** Analyses utilized preliminary data in mCRC patients (N=122) from study M21-404 where subjects were treated with ABBV-400 (1.6 – 6.0 mg/kg Q3W). Serial PK samples were collected in cycles 1 (C1) and 3 (C3) and sparse samples across other cycles. C1 exposure metrics (C_{max}, C_{avg}, and C_{trough}) for both ABBV-400 conjugate and unconjugated payload were obtained by non-compartmental PK analysis. Baseline and C3 day 1 plasma samples were collected and analyzed using the Guardant INFINITY assay for biomarker analysis. ctDNA (circulating tumor fraction [cTF]) was estimated based on 1) variant allele frequency of somatic mutations in a 74 gene panel and 2) methylation signals across targeted regions of the Guardant INFINITY methylation panel. MR was defined as a 50% decrease in cTF from baseline. PK/PD correlations with % change in cTF at C3 and MR at C3 were evaluated. c-Met protein expression was assessed by IHC (VENTANA MET SP44 RxDx Assay) and *MET*-amp status was based on available local testing results and retrospective ctDNA testing. **Results:** Higher exposures of ABBV-400 conjugate and payload (primarily C_{avg} for both analytes and C_{max} for conjugate) correlated with greater decrease in % change in cTF at C3 from baseline, as assessed by the 74 gene panel (N =66) or methylation panel (N=74). PK/PD analysis showed that higher exposure was strongly correlated with higher probability of MR (n=66-74, nominal $p < 0.05$). ABBV-400 conjugate and payload exposures were comparable across c-Met expression H-score (range 9 – 295), cut-offs (≥90% with 1+ intensity, ≥50% with 2+, ≥10% with 3+, ≥25% with 3+, and ≥50% with 3+), and *MET*-amp status. **Conclusions:** Exposure-response analyses have previously shown correlation between ABBV-400 conjugate exposures and probability of efficacy (objective response rate) as well as safety in mCRC subjects. The current PK/PD analyses demonstrate that higher ABBV-400 conjugate and payload exposures are also correlated with liquid biopsy data, i.e., ctDNA and MR based on ctDNA. c-MET expression and MET-amp did not have impact on ABBV-400 exposures. Clinical trial information: NCT05029882. Research Sponsor: AbbVie.

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A novel immune-based score and its immune contexture exploration in colorectal cancer. First Author: Qian Wu, Laboratory for Digestive Oncology, KU Leuven, Leuven, Belgium

Background: The majority of currently available biomarkers in colorectal cancer (CRC) perform poorly with respect to prognostication and prediction of response. Immunoscore (IS) has been proposed as a signature of host immune cancer response reflecting a favorable prognosis. Next, through AtezoTRIBE, 'Immunoscore immune checkpoint' (IS-IC) was introduced, indicating its potential predictive value for immune checkpoint blockade (ICB) in pMMR metastatic CRC. IS-IC is computed through quantitative and spatial variables related to density and proximity of CD8 cells and programmed cell death-ligand 1 (PD-L1) cells in the tumor core (TC). There are little confirmatory studies. In this work we test the application of IS-IC through multiplex immunofluorescence (mIF) on our own cohort of patients. Next we explore immune contexture differences through subgroup analysis. **Methods:** 15 prospectively collected, treatment naïve, primary CRC samples (9 dMMR/MSI-H, 6 pMMR/MSS) where analyzed with mIF. Cell density, proximity between cells, cell clustering and spatial distribution of selected mIF biomarkers (CD3, CD20, CD11c, CD163, CD15, CD4, CD8, Foxp3, CD56, Granzyme B, PD-L1, programmed cell death protein 1 (PD-1), pan-cytokeratin (CK) and Ki67) were analyzed both at the invasive margin (IM) and TC using digital pathology. All slides subjected to digital image analysis met predefined quality control criteria. **Results:** Serial mIF showed PD-L1 was enriched in stromal cells (CK^-) within IM (p=0.0148) and confirmed its co-localization with macrophages ($CD163^+$) (p=0.0092). Hence, both TC and IM were included to quantify the densities of CD8, PD-L1 and the proximity between them, as a suggested novel IS-IC_{TC+IM}. During exploration of the immune context we found that both IS-IC and IS-IC_{TC+IM} high groups exhibited significantly increased infiltration of PD-1^+ cells (p=0.0074 and p=0.0058, respectively) and $CD8^+$PD-1^+ cells (p=0.0101 and p=0.0062, respectively), compared to the corresponding low groups. Whereas only the IS-IC_{TC+IM} high group showed increased natural killer (NK) cells ($CD56^+$) (p=0.0192), found mainly at IM (p=0.0167) in contrast to TC (p=0.0824). The novel IS-IC_{TC+IM} scoring system captures the immunosuppressive and immune responsive interaction among PD-1^+ cells, PD-$L1^+$ cells, NK cells and $CD8^+$ cells in TC and IM from the cell numbers and cell-cell proximity. A validation cohort is underway. **Conclusions:** In this study we confirm the ease of replicating IS-IC and the presence of ICB sensitive stroma in the high score group. This study further explored the immune context in both TC and IM, emphasizing the importance of regional differences. Additionally it was found that NK cells were only increased at IM. Altogether we suggests adding dual quantification in TC and IM as a novel IS-IC_{TC+IM}, which may be a more sufficient reflection of the tumor micro environment. Research Sponsor: None.

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A comprehensive molecular and clinical study of patients with young-onset CRC. First Author: Elham Nasrollahi, University of Pittsburgh Medical Center (UPMC), Harrisburg, PA

Background: Young-onset colorectal cancer (YO-CRC), defined as colorectal cancer diagnosed in individuals under 50 years of age, has emerged as a distinct clinical entity, often presenting at advanced stages. Despite the increasing incidence, molecular and clinical underpinnings of YO-CRC remain underexplored. This study aims to characterize the clinical and molecular features of YO-CRC and to evaluate their impact on OS. **Methods:** We retrospectively reviewed 110 patients diagnosed with YO-CRCfrom our institution's molecular database. All patients underwent next-generation sequencing. Demographic, clinical, and molecular data, including age, gender, race, tumor location, cancer stage, and mutation status (KRAS, NRAS, BRAF, POLE, ERBB-2/HER2, microsatellite status), were collected by reviewing electronic medical records. For OS analysis, we focused on patients diagnosed with de novo Stage 4. Cox proportional hazards regression and Kaplan-Meier survival analysis were utilized to assess the association of these factors with OS, with statistical significance determined by a p-value threshold of <0.05. **Results:** Among 110 patients, N=44 (40%) presented with local disease (stage 1-3), while N=66 (60%) patients presented with de novo metastatic disease at the time of diagnosis. The median age at diagnosis was 44.5 years. The cohort consisted of 64% males and 36% females, with 84% of patients identified as White. Most tumors were left-sided (77%), including the distal colon/sigmoid (44%) and rectum (33%). KRAS and BRAF mutations were present 36% and 5.5%, respectively. ERBB-2/HER2 amplification and microsatellite instability were observed in 4.5% and 6.4% respectively. Tumor mutation burden (TMB) was <10 in 57% of patients, with 14% having TMB >20. CNV analysis revealed that 14% of patients had copy gains, 12% had concurrent gains/losses, and 31% had copy losses. Among the 66 Stage 4 patients, 44% had died by the time of analysis, with a median overall survival (OS) of 43.6 months (95% CI, 28.7 - not reached). KRAS mutation was found to be significantly associated with worse survival outcomes. Cox regression analysis reveals the prognostic significance of KRAS status, with a hazard ratio (HR) of 3.52 (95% CI: 1.59-7.76, P = 0.002 $<$ 0.05), indicating a significantly higher risk of death for KRAS-mutant YO-CRC patients. **Conclusions:** KRAS mutations are a key prognostic factor in YO-CRC, highlighting the need for therapeutic need to improve outcomes in this high-risk group. Research Sponsor: None.

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Role of cfDNA and DNA integrity index (DII) in colon cancer in response to surgery. First Author: Vipul Goyal, Regional Cancer Centre, Thiruvananthapuram, Thiruvananthapuram, India

Background: Colorectal cancer diagnosis requires new biomarkers for improved detection and monitoring. cfDNA has shown potential in detecting minimal residual disease, evaluating treatment response, and providing prognostic information. Recent studies highlight cfDNA as a promising molecular marker for early disease detection and progression monitoring. This study aims to evaluate the role of cfDNA and DII in assessing surgical response in CRC patients. The primary objective is to assess cfDNA levels pre- and post-curative resection. Secondary objectives include identifying factors influencing baseline cfDNA levels, evaluating factors affecting DII, and comparing cfDNA and CEA levels in response to surgery. Statistical analyses included descriptive statistics, ROC curve, Student T tests and Fisher's exact test. **Methods:** The study was designed as a prospective investigation conducted over a period from September 2022 to April 2024 at the Regional Cancer Centre. Forty pathologically proven non metastatic colonic carcinoma planned for primary surgery were included and patients with sepsis and chronic diseases were excluded from the study. Blood samples (10 ml) were collected pre-surgery and at four weeks post-surgery. Plasma cfDNA was extracted, quantified, and analyzed via quantitative real-time PCR using human β-actin as a reference. The DII was calculated as the ratio of 394 bp to 99 bp amplicons of β-actin. **Results:** The mean age was 62.15 years (range 39-85), with 55% over 60 years. High-risk features included poor differentiation (4.2%), preoperative CEA >5 ng/ml (17.5%), LVI (20.8%), and PNI (6.3%). Surgical approaches included 10 laparoscopic, 29 open, and 1 converted from laparoscopic to open due to excessive bleeding. Overall stage grouping showed 18 patients in Stage III, and 11 each in Stage I and II. Median pre- and postoperative cfDNA levels were 24.2 ng/μl and 29.6 ng/μl (p=0.83), and median DNA integrity index levels were 1.1 and 1.0 (p=0.95). Preoperative cfDNA levels averaged 24.2 ng/μl and postoperative levels 29.6 ng/μl (p = 0.83); DII levels were 1.1 preoperatively and 1.0 postoperatively (p = 0.95). Postoperative cfDNA levels decreased in 21 patients and increased in 19, while DII decreased in 20 and increased in 20. Significant associations were noted for cfDNA changes with surgery type and high-risk features. ROC analysis revealed cfDNA cut-off at 275 ng/μl, DII at 1.5, and CEA at 4 μg/L. Sensitivities for cfDNA and DII were 66.67% and 100%, respectively, with combined sensitivity reaching 100%. **Conclusions:** Persistent high cfDNA levels post-surgery suggest potential minimal residual disease despite radical resection. The residual disease is a harbinger of systemic failure. Monitoring cfDNA and DII can identify patients at risk for recurrence, guiding intensified adjuvant therapy. Further studies with extended follow-up are recommended to validate these findings. Clinical trial information: CTRI/2023/10/058468. Research Sponsor: None.

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Clinical relevance of tumor fraction assessment from circulating tumor DNA in metastatic colorectal cancer. First Author: Van K. Morris II, The University of Texas MD Anderson Cancer Center, Houston, TX

Background: Both tissue (TBx) and liquid (LBx) biopsies are indicated for detecting actionable alterations in metastatic colorectal cancer (mCRC). LBx is easier to obtain and often has faster turnaround time, but its informative results depend on the ctDNA tumor fraction (TF) content. We aimed to (1) evaluate the concordance between LBx and TBx in CRC; and (2) determine baseline lab and clinical factors associated with TF. **Methods:** CRC patients (pts) with both tissue and liquid Foundation Medicine comprehensive genomic profiling within an interval of up to 90 days between samples collection were analyzed. Positive percent agreement (PPA) and negative predictive value (NPV) for RAS (KRAS/NRAS) and BRAFV600E detection were calculated with tissue as reference. Clinical data was obtained by the US-wide de-identified Flatiron Health-Foundation Medicine real-world clinicogenomic CRC database, from ~280 cancer clinics (~800 sites of care) between 1/2011-12/2023. mCRC pts with LBx collected up to 60 days prior to therapy (Tx) start were analyzed for association between TF and baseline factors using logistic regression with TF$\geq$1% as the binary outcome. Baseline factors include age, ECOG, race/ethnicity, line of therapy, practice type, metachronous vs. synchronous disease, tumor side, albumin, alkaline phosphatase (AP), serum creatinine, hemoglobin, lactate dehydrogenase (LDH), neutrophil-to-lymphocyte ratio, opioid and steroid pre-Tx. Foundation Medicine's ctDNA TF is a composite algorithm prioritizing aneuploidy at higher levels to avoid germline signal and prioritizing variant allele frequency of canonical alterations at lower levels to maximize dynamic range. **Results:** A total of 402 pts had TBx and LBx (268 [67%] TF$\geq$1% and 134 [33%] TF$<$1%). Without accounting for TF, the overall PPA for RAS and BRAFV600E was 77% and 83%, and the NPV was 65% and 99%. In LBx with TF$\geq$1%, the PPA for RAS and BRAFV600E was 99% and 100%, and NPV was 98% and 100%. In contrast, for TF$<$1% samples, the PPA for RAS and BRAFV600E was 37% and 50%, and NPV was 57% and 97%. Of the 633 pts with LBx collected prior to Tx, 474 (75%) had TF$\geq$1%. TF$\geq$1% was independently associated with ECOG 1+ (odds ratio [OR]: 1.58, p=0.038), community oncology care site (OR: 2.58, p=0.002), elevated levels of AP (OR: 3.00, p$<$0.001) and LDH (OR: 3.97,p=0.002). TF$\geq$1% was less likely to occur in right-side tumors (OR: 0.53, p=0.012) and metachronous disease (OR: 0.48, p$<$0.001). **Conclusions:** Approximately 70% of LBx samples from CRC pts have TF$\geq$1%, with near-perfect concordance for RAS/BRAFV600E. For LBx with TF$<$1%, interpretation of negative results can be challenging due to low tumor shedding, especially for RAS mutations, where 43% may be false negatives and a subsequent TBx will provide more confidence in the negative results. Clinical factors such as right-sided tumors and metachronous disease may help anticipate low TF and guide decisions to pursue TBx. Research Sponsor: Foundation Medicine, Inc.

238 **Poster Session**

Frequency and impact of a genome-wide homologous recombination deficiency signature (HRDsig+) on the genomic landscape of advanced colorectal cancer (CRC). First Author: Faiza Yasin, Yale School of Medicine, New Haven, CT

Background: There have been significant advancements in novel therapies that target the DNA damage response pathway, and utilizing scar-based measures of homologous recombination deficiency (HRD) beyond genomic alterations (GA) in *BRCA1/2* may provide opportunities for additional biomarker-driven therapies for patients with advanced CRC. **Methods:** We identified 47,028 cases of advanced CRC for comprehensive genomic profiling (CGP) to examine all classes of GA and measure HRDsig+ status. HRDsig status was calculated using a broad set of genome-wide copy number features (PMID 37769224) and was correlated with microsatellite status, tumor mutation burden (TMB), genomic ancestry, trinucleotide mutational signatures, and PD-L1 IHC using the Dako 22C3 tumor proportional score (TPS) system. Categorical variables were compared using the Fisher exact test with the Benjamini-Hochberg adjustment to correct for false discovery rate. **Results:** A total of 796 (1.7%) advanced CRC tumors were HRDsig+ with a median age of 62 years for HRDsig+ and 63 years for HRDsig-. The HRDsig- cohort was more likely to be male compared to HRDsig+ (56.1% vs 50.4%; P=.004). Median GA per tumor was similar (range 5-6) in both groups. Genomic ancestry revealed significantly more African ancestry in the HRDsig- group (12.6% vs 8.8%; P=.002) and more European ancestry in the HRDsig+ group (75.8% vs 71.6%; P=.02). MSI-high status was more frequent in HRDsig- group as compared to the HRDsig+ group (5.5% vs 1.6%; P$<$.0001). Median TMB was higher in the HRDsig+ cohort (5.0 vs 3.6; P$<$.0001), as was frequency of TMB $\geq$ 10 mutations/Mb (16.2% vs 8.5%; P$<$.0001). Trinucleotide mutational signature distribution was similar in both cohorts with the exception that a tobacco signature was more frequent in the HRDsig+ group (1.5% vs 0.2%; P$<$.0001). Both PD-L1 low (1-49% TPS) expression (19.5% vs 13.1%; P=.005) and PD-L1 high ($\geq$50% TPS) expression (4.4% vs 1.5%, P=.0021) were higher in HRDsig+ cases. GA in genes associated with HRD were *BRCA1* (13.6% vs 1.1%; P$<$.0001), *BRCA2* (8.4% vs 2.8%; P$<$.0001), *ATM* (8.0% vs 5.3%; P=.0003), and *RAD21* (6.7% vs 3.2%; P$<$.0001), which were more frequent in the HRDsig+ group. In the HRDsig- group, 83.7%/88.8% of *BRCA1/BRCA2* mutated CRC were not bi-allelic, likely non-driver GA. *ERBB2* GA were similar in both groups (5.3% vs 4.4%; P=0.31). *KRAS* GA were more frequent in the HRDsig+ group (49.1% vs 31.7%; P$<$.0001). **Conclusions:** This analysis is the largest assessment of HRD status in advanced CRC. HRDsig+ was 1.7% and is associated with co-occurring actionable GA compared to HRDsig- tumors and higher levels of PD-L1 TPS, which may provide opportunities to target these tumors with immunostimulatory therapies. Given the advances in novel DNA repair targeted therapies these results may provide insight for CRC clinical development for DNA repair combinations. Research Sponsor: None.

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_TP53_ Y220C mutations in colorectal carcinoma (CRC): A genomic landscape study. First Author: Deevyashali Parekh, SUNY Upstate Medical University, Syracuse, NY

Background: Inactivating genomic alteration (GA) of the TP53 tumor suppressor gene leads to inactivation of p53 protein and is currently the most frequently associated GA across all cancers. Colorectal cancer is a common and lethal subtype of cancer, in the top 5 for incidence, prevalence and cancer-related deaths worldwide. TP53 is the second most frequent GA identified in clinically advanced CRC. It has been postulated that GA in TP53 not only affects the molecular biology of malignant cells but also impacts the tumor microenvironment in CRC. This GA is currently untargetable by approved drugs however, recently a novel approach designed to revert to wild type functioning of TP53 inactivated by Y220C base substitution has gained significant clinical interest. **Methods:** 67,301 cases of clinically advanced CRC underwent hybrid capture based comprehensive genomic profiling to assess all classes of GA. Cases were sequenced to a mean coverage depth of 650X with microsatellite instability (MSI) status and tumor mutation burden (TMB) determined from the sequencing data and PD-L1 expression measured by immunohistochemistry using the DAKO tumor proportional staining system (low positive set at 1-49% staining and high positive set at ≥50% staining). **Results:** 422 (0.6%) of the CRC featured the *TP53* Y220C GA (Y220C+). When compared with the CRC that were *TP53* Y220C negative (Y220C-), the Y220C+ cases were of similar age, gender and number of GA per tumor. The Y220C- cases has significantly higher frequencies of MSI status (6.1% vs 2.2%; p=.0007) and TMB levels greater >10 mutations/Mb (8.7% vs 4.7%; p=.004). The Y220C+ featured lower frequencies of GA in *PIK3CA* (12.9% vs 18.6%; p=.002), *BRAF* (8.8% vs 10.1%; NS) and *PTEN* (7.4% vs 8.3%, NS) and a higher frequency of *ERBB2* amplification (5.0% vs 2.7%; p=.004). GA in *APC*, *KRAS*, *NRAS*, *BRCA2*, *EGFR* and *MET* were similar in both Y220C+ and Y220C- CRC. PD-L1 low expression was similarly infrequent (12.1% to 16.7% range) in both groups. **Conclusions:** The potentially targetable *TP53* Y220C short variant mutation is uncommon in clinically advanced CRC but is associated with unique genomic and biomarker characteristics. Further research might elucidate subpopulations or CRC subtypes with a greater frequency of this mutation who might benefit from a novel targeted therapy. Research Sponsor: None.

	CRC *TP53* Y220C+ (422 Cases)	CRC *TP53* Y220C- (66,879 Cases)	P Value
Gender	41.2% F/ 58.8% M	44.4% F/55.6% M	NS
Median Age (range)	62 (23-89+)	61 (8-89+)	NS
GA/tumor	6.1	6.5	NS
***TP53* non-Y220C GA**	16.1%	75.7%	<.0001
PIK3CA	12.9%	18.6%	.002
ERBB2	6.2% (5.0% amp)	5.1% (2.7% amp)	.004 (amp)
MSI High	2.2%	6.1%	.0007
TMB > 10 mut/Mb	4.7%	8.7%	.004
PD-L1 Low (1-49%TPS)/ PD-L1 High (>50% TPS)	16.7% (132 cases)/2.3%	12.1% (20,759 cases)/1.4%	NS

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Utilization of _SDC2_ gene in conjunction with _SEPTIN9_ gene methylation analysis for the diagnosis and efficacy assessment of colorectal cancer. First Author: Rongjia Xie, Department of Oncology, Shanghai Tenth People's Hospital, Tongji University School of Medicine, Shanghai, China

Background: SDC2 gene encodes the transmembrane proteoglycan molecule Syndecan-2. Syndecan-2 protein affects the proliferation, migration and invasion of colorectal cancer cells by participating in the regulation of cell adhesion, tissue differentiation and angiogenesis. Plasma methylated SEPTIN9 gene has been shown to be a sensitive and specific biomarker for the detection of CRC, which is involved in apoptosis, pseudopod projection, tumor cell migration and invasion. The aim of this study is to validate the value of SDC2 and S9 gene methylation detection in the diagnosis and efficacy evaluation of CRC. **Methods:** A total of 102 patients were enrolled in the study. The methylation levels of fecal SDC2 gene and blood SEPTIN9 gene were detected by fluorescence PCR. The clinical diagnostic accuracy of SDC2 kit was evaluated by patients with benign gastrointestinal lesions, gastrointestinal tumors and healthy subjects. Combined detection of SDC2 and S9 to improve the detection rate of CRC. SDC2m levels were analyzed in 26 patients with partial remission or stable disease after palliative treatment and 32 patients with complete remission after radical surgery. **Results:** The detection rate of SDC2 in patients was 0.0% for UC and CD, 66.7% for AA and 60.0% for NA, 25.0% for HOP. For 8 healthy subjects with negative colonoscopies, the true negative rate was 100%. The detection rates of GC and EC were 50.0% and 0.0%, respectively. Data from 72 patients with CRC were evaluated, with a sensitivity of 89.1% (41/46,95%CI 0.798-0.985) for patients with untreated CRC. The sensitivity of SDC2 was 46.2% (12/26,95%CI 0.256-0.667) in patients with partial response or stable disease after palliative treatment. The difference was statistically significant ($P < 0.001$). The sensitivity of S9 for CRC detection was 88.2% (15/17, 95%CI 0.712-1.053). By two tests, 94.1%(16/17, 95%CI 0.816-1.066) of CRC cases could be detected.The sensitivity was 89.1% (41/46 95%CI 0.798-0.985) in untreated CRC and 46.2% (12/26 95%CI 0.256-0.667) in patients with PR and SD after chemotherapy/radiotherapy/targeted/immunotherapy. The difference rate of SDC2m between two groups was statistically significant ($P < 0.001$). Among 46 untreated patients with CRC, 32 patients with positive SDC2m underwent radical surgery, of which 30 CR patients turned SDC2m test negative after surgery. We found reduced SDC2m in stool from patients with clinical benefit (CR+PR+SD). **Conclusions:** Detection of SDC2m in untreated CRC patients has high sensitivity and has the potential to become a non-invasive diagnostic tool for CRC. Combination of S9 and fecal SDC2 could improve the detection rate of SDC2 in non-dominant population. The methylation level of SDC2 may be helpful for postoperative follow-up of CRC after radical surgery, prediction of recurrence, and monitoring the efficacy of a series of palliative treatments for CRC. Research Sponsor: Shanghai Shenkang Hospital Development Group Medical Enterprise Integration Innovation Synergy Project (SHDC2022CRT009); Preclinical Study of New Traditional Chinese Medicine by Shanghai Science and Technology Commission (21S21901500).

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Assessment of pathologic response in the colorectal cancer cohort of the IMHOTEP phase II trial of neoadjuvant pembrolizumab in dMMR/MSI tumors. First Author: Frederic Bibeau, Pathology Department, Besançon University Hospital, Franche-Comté University, Besançon, France

Background: Neoadjuvant treatment with immune check point inhibitors has shown promising results in localized deficient mismatch repair/microsatellite instability (dMMR/MSI) colorectal cancer (CRC), notably in terms of pathologic response. However, pathologic response has rarely been precisely assessed on both primary tumor and its corresponding lymph nodes (LN) as an end-point. We performed this analysis on the CRC samples selected from the IMHOTEP trial, a multicenter, single-arm study evaluating peri-operative Pembrolizumab in 4 different cohorts of patients with localized resectable dMMR/MSI tumors. **Methods:** We analyzed pathologic response on the 56 surgical specimens available from the CRC cohort. Pathologic response was assessed on primary tumor according to the percentage of Residual Viable Tumor (RVT) score, encompassing 4 groups : 0% = pathologic Complete Response (pCR) ; ≤10% = Major Pathologic Response (MPR) ; >10% < 50% = partial Pathologic Response (pPR) ; > 50% = lack of response (noPR). Pathologic response was assessed on lymph node (LN) according to 4 groups : negative LN without sign of sterilization (yN0 reg-), negative LN with sterilization (yN0 reg+) ; metastatic LN without tumor regression (yN+ reg-) ; metastatic LN with tumor regression (yN+ reg+).The type of regression, including colloid, fibrotic and necrotic features was reported, as well as the presence of tertiary lymphoid structures (TLS). pCR and MPR corresponded to pathologic responder patients and pPR and noPR to pathologic non responder patients. **Results:** In thisCRC cohort pCR, MPR, pPR and noPR were observed in n= 33 (58.9%), 7 (12.5%), 5 (8.9%) and 11 (19.6 %) cases and yN0 reg-, yN0 reg+, yN+ reg-, yN+ reg+ in 41 (73.2%) , 7 (12.5%), 7 (12.5%), and 1 (1.8 %) cases. There was a significant association between pathologic responders (pCR + MPR) and negative LN status encompassing sterilized LN (p=0.04). Pathologic responders were more frequently associated with classical (Lieberkuhnian) adenocarcinoma than with other CRC special subtypes (mucinous, signet ring-cell, poorly cohesive subtypes)($p<0.01$). Colloid response was the most frequent pattern of regression, observed in 42 (75%) cases, and predominant (>80% of pathologic response areas) in 14 (25%) cases. TLS were seen in 46 (82%) cases, without any significant association with the RVT score. However TLS were observed in all CRC cases with MPR. **Conclusions:** These preliminary results show pembrolizumab efficacy as a neoadjuvant monotherapy in terms of pathologic response on both primary CRC and corresponding LN, highlighting the need for a new and more accurate pathologic score. It also illustrates the potential impact of the histological subtypes. Complementary data will be available after updated pathologic reviewing of additional specimens from CRC patients and will also be correlated with outcome. Clinical trial information: NCT04795661. Research Sponsor: MSD.

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A comprehensive study of clinical and molecular features in patients with metastatic colorectal cancer with and without liver mets. First Author: Tara Magge, UPMC Hillman Cancer Center, Pittsburgh, PA

Background: Colorectal cancer (CRC) most frequently metastasizes to the liver, and patients (pts) with liver metastasis (mets) at the time of metastatic diagnosis tend to have poorer outcomes compared to pts with non-liver mets. However, the reason for this discrepancy is not well-defined. We thus aimed to identify any distinct molecular or clinical characteristics that could contribute to the poor outcomes associated with liver mets of CRC. **Methods:** Pts diagnosed with metastatic CRC from 2014-2022 were identified using the institutional molecular database of CRC. Demographic, clinical, and molecular data were collected by reviewing electronic medical records. Time to next treatment was defined as the time between 1st and 2nd line treatment. We utilized Fisher's exact tests for categorical variables and both Kaplan-Meier analysis and multivariate Cox proportional hazards models to analyze survival data, with a p-value threshold of <0.05 for statistical significance. **Results:** We identified a total of 217 pts, including pts with liver mets (N=146) and non-liver mets (N=71) at time of diagnosis. Pts with non-liver mets were more likely to have rectal primary than those pts with liver mets (40% vs. 15%, p=0.0001), indicating a distinct pattern of recurrence. Common driver mutations such as KRAS, BRAF, and TP53 were observed at equal frequencies in the liver and non-liver cohorts. In pts without liver mets, age had a significant effect on OS ($p < 0.05$, HR 1.1). Pts without liver mets with colon primary had worse OS compared to rectum primary (HR= 0.48, 95% CI: 0.24–0.96, p = 0.038). For pts without liver mets, KRAS was not a predictor of outcomes, while BRAF was associated with poorer OS ($p<0.001$, HR 7.09, 95%CI: 2.31-21.8). Contrarily, for pts with liver mets, KRAS mutation was significantly associated with worse survival (mOS 40.4 vs 61.7 mos, HR 2.04, 95% CI: 1.26-3.30, p=0.004). BRAF mutations were also associated with worse OS for this cohort (p=0.039 and HR 2.86, 95%CI: 1.06-7.77). Median time to the next treatment was significantly shorter in pts with liver mets (13.6 mos) than in pts with non-liver mets (22.6 mos) (HR 1.82, 95% CI 1.28-2.59; $p<0.001$). OS was longer for pts with non-liver mets compared to those with liver mets (mOS 52.9 mos vs 46.7 mos) though it did not reach statistical significance, likely due to the small sample size. **Conclusions:** Liver metastasis of CRC is associated with systemic therapy resistance with a shorter time on treatment, indicating treatment resistance associated with liver mets is not limited to immunotherapy. Given similar molecular characteristics in pts with and without liver mets, treatment resistance may be related to the inherent tumor microenvironment of the liver. In this study, pts with distinct molecular features had poorer outcomes, highlighting the importance of developing targeted therapeutics in this population. Research Sponsor: None.

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Impact of baseline alterations in HER2-pathway oncogenic drivers on response to anti-HER2 therapy in patients (pts) with HER2-amplified advanced colorectal cancer (aCRC). First Author: Celine Hoyek, Division of Hematology and Oncology, Mayo Clinic Arizona, Phoenix, AZ

Background: HER2-amplified aCRC is associated with poor prognosis despite new anti-HER2 regimens. Insight on biomarkers that predict response to these therapies is limited. This study aims to evaluate whether baseline alterations in oncogenic drivers along the HER2 pathway predict response to anti-HER2 therapy in HER2-amplified aCRC. **Methods:** In this international multi-center study involving Mayo Clinic, Duke Cancer Institute and National Cancer Center Hospital East (Japan), we collected baseline genomic alterations from tissue or blood samples using next-generation sequencing (NGS) prior to anti-HER2 therapy initiation. We defined PIK3CA, RAS, RAF, MAP2K, and MET as oncogenic drivers of resistance within the HER2 pathway. Overall survival (OS) was evaluated with the Kaplan-Meier method and a log-rank test was used to compare median OS (mOS) between pts with and without alterations in the above genes. A chi-square test (significance level 0.05) was performed to compare overall response rate (ORR) and disease control rate (DCR). **Results:** We identified 39 pts with HER2-amplified aCRC and baseline NGS. Median age was 57 (range 30-77), 54% were male. Baseline alterations in HER2 pathway oncogenic drivers were identified in 17 pts (43.5%). 18 (46%) received monoclonal antibodies (mAbs) plus tyrosine kinase inhibitors, 12 (30%) received dual mAbs and 7 (18%) received trastuzumab deruxtecan. The mOS was 41.9 months (mos) (95%CI 23.7-60.1) for other alterations vs 24.2 mos (95%CI 14.1-34.3) in pts with HER2 pathway alterations (p = 0.1). ORR was 47% vs 41% (p = 0.7) and DCR 82% vs 73% (p = 0.4), for pts with HER2 pathway alterations and without HER2 pathway alterations, respectively. 4 pts receiving trastuzumab/tucatinib had complete response, none of which had HER2 pathway alterations. **Conclusions:** Pts with baseline alterations in oncogenic drivers of the HER2 pathway had clinically worse outcomes compared to pts without such alterations despite receiving HER2-targeted therapy, though results lacked statistical significance. Larger studies are needed to confirm the lack of benefit from anti-HER2 therapy in this subgroup. Research Sponsor: None.

244 **Poster Session**

Prognostic implications and therapeutic response in GNAS-mutated colorectal adenocarcinoma: A retrospective cohort analysis. First Author: Faran Polani, Inova Schar Cancer Institute, Fairfax, VA

Background: GNAS, a key regulator of the G-protein signaling pathway, is frequently mutated in colorectal adenocarcinomas and other malignancies. Despite its role in tumorigenesis, GNAS mutations remain understudied, with no established targeted therapies. Exploring the impact of GNAS alterations on therapeutic response and outcomes in colorectal cancer (CRC) is crucial for developing effective treatment strategies. **Methods:** We reviewed 531 CRC cases from our institutional Molecular Tumor Board database, identifying 8 patients with GNAS mutations. The prevalence ofGNASmutations in our cohort was 1.5%, which aligns with reported rates of 2-5% in various CRC cohorts. Comprehensive clinicopathologic data, including tumor histology, staging, treatment outcomes, and response rates, were analyzed. A detailed evaluation was conducted on stage IV patients to assess their response, progression-free survival (PFS), and overall survival (OS) in comparison to historical stage IV CRC benchmarks. **Results:** The cohort included 8 patients with a median age of 61 years (range: 33-73 years), with balanced gender representation (50% male, 50% female). The racial distribution was 50% Black or African American, 25% White, and 12.5% Asian or Pacific Islander. The majority of patients (62.5%) were diagnosed at stage IV, with primary tumor sites predominantly in the sigmoid colon and cecum. Among the 5 stage IV patients, while no patients achieved a complete or partial response, 3 exhibited stable disease, resulting in a 60% disease control rate (including stable disease). The median progression-free survival (mPFS) was 9 months, reflecting the duration of disease control before progression, while the median overall survival (mOS) was 15 months. These outcomes are notably lower than historical benchmarks for stage IV CRC, where response rates typically range from 40-50% with mOS of 20-24 months, highlighting the potentially adverse prognostic impact of GNAS mutations. **Conclusions:** Patients with GNAS-mutated colorectal adenocarcinoma appear to represent a unique subset with poorer therapeutic outcomes, even when achieving stable disease. The observed disease control in the absence of objective responses underscores the complexity of treating GNAS-mutated tumors and highlights the urgent need for targeted therapies tailored to this molecular subtype. Further research into the biological mechanisms underlying GNAS mutations and their impact on CRC treatment resistance is warranted to improve patient outcomes. Research Sponsor: None.

245 **Poster Session**

Association between ctDNA levels, timing of blood collection, and overall survival in metastatic colorectal cancer. First Author: Akinori Kobayashi, Japan National Cancer Center Hospital East, Kashiwa, Japan

Background: While circulating tumor DNA (ctDNA) detection is influenced by various factors, the impact of blood collection timing remains understudied. This study investigated the relationship between ctDNA detection, and blood collection timing and prognosis in patients with metastatic colorectal cancer (mCRC) from the GOZILA study. **Methods:** GOZILA is a nationwide plasma genomic profiling study using Guardant360 CDx for advanced solid tumors. This study evaluated the association of ctDNA levels with blood collection timing and prognosis in patients with mCRC. ctDNA levels were determined by the maximum variant allele frequency. Blood collection timing was categorized into five groups: naive (before initiation of 1st line treatment), after treatment initiation (days 1-14 of treatment), during treatment (between "after treatment initiation" and "before progression disease [PD]"), before PD (from 28 days before PD to PD), and after PD (after disease progression but before the next line of treatment). Overall survival (OS) was compared across treatment lines using ctDNA level cutoff of 10. **Results:** We analyzed 2,398 blood samples from 1,994 patients with mCRC enrolled in the GOZILA between February 2018 and July 2023. ctDNA levels were significantly lower during treatment compared to the naive (median, 13.2 vs. 5.7, $p < 0.001$). No significant differences were observed between naive and other blood collection timing. Compared to blood collection before 1st line treatment, ctDNA levels significantly decreased before 2nd line (median, 13.5 vs. 8.3, p = 0.02) and 3rd line treatments (median, 13.5 vs. 7.6, p = 0.01). Patients with ctDNA levels$<$10 demonstrated significantly better OS across all treatment lines, including 1st line (median, NA vs. 23.9 months, HR 0.49, 95% CI, 0.34-0.72), 2nd line (median, 22.1 vs. 10.3 months, HR 0.42, 95% CI, 0.33-0.55), and later line treatments (FTD/TPI: median, 12.7 vs. 5.7 months, HR 0.51, 95% CI, 0.26-0.99; FTD/TPI+Bev: median, 9.3 vs. 6.5 months, HR 0.55, 95% CI, 0.37-0.82; Regorafenib: median, 13.5 vs. 7.4 months, HR 0.45, 95% CI, 0.26-0.80). **Conclusions:** In mCRC, blood collection during treatment is suboptimal due to decreased ctDNA levels. Patients with ctDNA levels $<$10 showed significantly better OS across all treatment lines. Optimal timing of blood collection can enhance ctDNA levels, enabling more accurate gene mutation assessment and stratifying outcomes. Research Sponsor: None.

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Effects of targeting CTMP on immunogenicity of MSI-L colon adenocarcinoma cells by metabolism and cell cycle modulation. First Author: Cheng Gu, Department of Oncology, Shanghai Tenth People's Hospital, Tongji University School of Medicine, Shanghai, China

Background: Regimen of metastatic colon cancer (mCOAD) is moving towards combination of immune checkpoint inhibitors (ICBs) and targeted therapy, however ICBs are rarely effective in MSI-L metastatic colon cancers, limiting options for backline treatments, new targeted therapy is on demand to improve immunogenicity of MSI-L mCOAD. As targeting fatty acid metabolism or cell cycle has shown potential in improving the efficacy of ICBs, our team focused on CTMP, a thioesterase (ACOT) family member, mainly modulates fatty acid β oxidation, meanwhile it binds to phosphorylation sites of AKT thus intervenes cell cycle, currently its role on immunogenicity of COAD is not clear. **Methods:** Correlation between CTMP and OS in MSI-L colon cancer patients was assessed by KM plotter in GEO database.11 Clinical samples of MSI-L colon cancer, who received a third-line treatment regimen of anti-PD-1 and fruquintinib, were collected then applied in immunofluorescence and western blot analyses to evaluate CTMP and MHC-I expression. In vitro, Lentiviral transfection of targeted genes were stably established,then applied in RNA-seq and metabolic assays. Proteomic analysis and co-IP were applied to explore CTMP-modulating pathway. In vitro, IFNγ was intratumorally injected. **Results:** KM curve showed CTMP was negatively correlated with OS in MSI-L COAD in GEO database. In clinical samples, lower CTMP expression and higher MHC-I expression were observed in partial response (PR) than progressive disease (PD) group. In vitro, CTMP-OE reduced MHC-I, increased AKT phosphorylation and immune checkpoints (PD-L1, IDO-1), promoted clone formation. Seahorse assay showed increased capacity of glycolysis and oxidative phosphorylation in CTMP-OE than vector. RNA-seq showed CTMP-OE reduced MHC-I augmentation under IFN-γ treatment, meanwhile glucose metabolism genes, ACOT family ,immune checkpoints were also up-regulated, indicating ACOT family related metabolism promotion and immunogenicity suppression of COAD. Proteomic analysis and co-IP assay indicated CTMP interacts REV7, a key mitotic regulatory protein. Enzyme activity assay showed REV7 negatively regulates CDK1. Furthermore, REV7-OE alone is capable to induce MHC I augmentation and G2 arrest, accordingly CTMP-KD enhanced REV7's negative modulation on CDK1,contributing to G2 arrest, that also augmented MHC I . In homologous mice tumor-bearing model, tumor shrinkage was observed in CTMP-KD/ REV7-OE group. CTMP-KD/ REV7-OE increases MHC I expression,CD8+ T cell infiltration under IFN γ intra-tumoral injection. **Conclusions:** In summary, our studies suggested that CTMP regulates the metabolism of colorectal cancer and intervenes MHC-I molecules expression, and targeting CTMP is potential in suppressing tumor cell cycle and promoting immunogenicity through its modulation on REV7. Research Sponsor: Shanghai Shenkang Hospital Development Group Medical Enterprises Interation Innovation Collaborative Special Project; SHD2022CRT009; Shanghai Municipal Science and Technology Commission of the preclinical research of new Chinese medicine; 21S901500.

247 **Poster Session**

Landscape of *MET* genomic alterations in colorectal cancer (CRC). First Author: Faiza Yasin, Yale School of Medicine, New Haven, CT

Background: MET signaling is implicated in the tumorigenesis of many solid tumors, including CRC. MET-targeted therapies are approved in non-small cell lung cancer with novel agents in clinical trials for tumors with *MET* exon 14 splice site mutations, *MET* amplifications, and recently MET expression. There is emerging therapeutic interest in targeting MET in CRC. **Methods:** Tumor samples from 50,500 cases of clinically advanced CRC were analyzed by hybrid capture-based comprehensive genomic profiling (CGP) that evaluated all classes of genomic alterations (GA). MSI-high status, tumor mutational burden (TMB), genomic ancestry, trinucleotide mutational signatures, and homologous recombination deficiency signature (HRDsig) were assessed for patients with activating *MET* GA. PD-L1 expression was determined by IHC (Dako 22C3 with TPS scoring system). Results were compared using the Fisher exact test with the Benjamini-Hochberg adjustment. **Results:** A total of 502 (1.0%) cases of activating GA of *MET* (METmut) were identified, including 418 cases (0.83%) with *MET* amplification (MET-amp), 10 (<0.1%) rearrangements, and 78 (0.1%) short variant (SV) mutations with 19 (<0.1%) cases of *MET* exon 14 splice site mutations. When compared with *MET* wild-type (METwt) CRC, the METmut CRC were of similar age (median 62 years), and numerically more likely to be male (61.4% vs 56.0%; P=.05). There were no differences in genomic ancestry (72% European ancestry for both groups). The METmut CRC featured more cases with a POLE signature (3.6% vs 0.4%; P=<0.0001) and more GA per tumor (7 vs 5; P<0.0001). MSI-high status was similar in both groups, with 4.2% in METmut and 6.0% in METwt (P=0.28). TMB levels were numerically higher in the METmut cases (TMB ≥ 10 mutations/Mb at 12.2% vs 9.0%; P=0.058). Low level PD-L1 expression (1-49% TPS) was similar in both groups (13.9% in METmut vs 13.3% in METwt), and while not statistically significant, high level PD-L1 expression (>50% TPS) was higher in the METmut group (4.3% vs 1.6%; P=0.056). The frequencies of HRDsig+ CRC were rare and similar in both groups with 1.5% in METmut and 1.7% in METwt. METmut CRC had lower frequencies of GA in *KRAS* (34.7% vs 48.8%; P<0.0001), *NRAS* (1.8% vs 4.2%; P=0.011), *APC* (72.5% vs 78.2%; P=0.009) and *PIK3CA* (12.2% vs 18.9%; P=0.0004), and higher frequencies of GA in *CDK6* (10.2% vs 0.6%; P<0.0001), *EGFR* (5.2% vs 0.2%; P=0.0002), *ERBB2* (9.6% vs 5.2%; P=0.0004), *HGF* (8.0% vs 0.8%; P<0.0001), *POLE* (4.2% vs 0.5%; P=0.005) and *TP53* (84.1% vs 75.9%; P<0.0001). In METmut CRC, ERBB2 alterations were 4.1% amplifications, 2.9% sequence mutations, 0.9% with more than one mutation. *BRAF* V600E was seen in 5.9% in METmut CRC and 8.4% of METwt. **Conclusions:** CGP reveals significant differences in the genomic landscapes of METmut CRC and METwt CRC, which may guide selection of potential rational drug combinations with novel MET-targeted therapies. Research Sponsor: None.

LBA248 **Poster Session**

Co-formulated favezelimab plus pembrolizumab versus standard-of-care in previously treated, PD-L1-positive metastatic colorectal cancer: The phase 3, randomized KEYFORM-007 study. First Author: Neil Howard Segal, Memorial Sloan Kettering Cancer Center, New York, NY

The full, final text of this abstract will be available at meetings.asco.org on the day of presentation and in the online supplement to the February 1, 2025, issue of the *Journal of Clinical Oncology*.

249 **Poster Session**

BTN1A1 expression in colon cancer: Implications for immunotherapy and patient stratification. First Author: Stephen S. Yoo, STCube Pharmaceuticals Inc, Rockville, MD

Background: Butyrophilin 1A1 (BTN1A1) is a promising immune checkpoint protein offering a novel option for immunotherapy. We discovered that BTN1A1 and PD-L1 expression are mutually exclusive through the JAK/STAT pathway in various tumors. Recently, Nelmastobart, an antibody targeting BTN1A1, successfully completed a Phase 1 clinical trial at three sites in the United States and two in the Republic of Korea, spanning 98 weeks. This abstract focuses on the results from the Phase 1 trial in patients with colorectal cancer (CRC), emphasizing biomarker diagnostics and therapeutics derived from translational research. **Methods:** We conducted a first-in-human, multicenter, open-label Phase I clinical trial to evaluate the safety, tolerability, pharmacokinetics, and preliminary efficacy of hSTC810 monotherapy in CRC patients. Extensive analyses of BTN1A1 expression were performed using patient specimens through immunohistochemistry and multi-immunofluorescence techniques to identify critical biomarkers and accurately assess nelmastobart's anticancer activity. We then analyzed progression-free survival (PFS), overall survival (OS), objective response rate (ORR), and safety alongside translational research findings using patient tissue samples. **Results:** A total of 47 patients were enrolled in the study. The median PFS across all cohorts was 1.9 months, while the median Kaplan-Meier estimate for OS was 12.8 months. Among the 16 CRC patients enrolled, tissue samples were obtained and analyzed using IHC in 12 patients and OPAL staining in 7 patients. The objective response rate (ORR) for CRC was 2.3%. The median progression-free survival (PFS) and overall survival (OS) were 3.6 months and 11 months, respectively. Notably, PFS, OS, and ORR strongly correlated with BTN1A1 expression levels in CRC patients. High levels of BTN1A1 expression in colorectal tumor tissues correlated with clinical outcomes, identifying BTN1A1 as a potential biomarker for predicting patient responses to targeted therapies. The mutual exclusivity of BTN1A1 and PD-L1 expression suggests that BTN1A1 could serve as a diagnostic marker to identify patients who may be unresponsive to PD-1/PD-L1 blockade therapy and might benefit from BTN1A1 blockade therapy. **Conclusions:** This Phase 1 clinical trial demonstrated that antibody-mediated BTN1A1 blockade effectively suppresses tumor growth in patients with BTN1A1-positive CRC. These findings suggest that BTN1A1-targeted therapies could provide a viable treatment option for patients with colon cancers that are refractory or resistant to anti-PD-1/PD-L1 therapies. Clinical trial information: NCT05231746. Research Sponsor: None.

250 **Poster Session**

VEGF-A as a prognostic indicator in advanced gastric and colorectal cancers and correlation with immune suppression involving myeloid-derived suppressor cells (MDSC). First Author: Masahiko Shibata, Fukushima Medical University, Fukushima, Japan

Background: VEGF-A is important molecule should be controlled in cancer treatment since hypoxic and immunosuppressive conditions of tumor microenvironment are induced by VEGF-A. Myeloid-derived suppressor cells (MDSC), activated by chronic inflammation and VEGF-A, play critical roles in immunosuppression in patients with cancer. **Methods:** Serum levels of VEGF-A were measured by ELISA and circulating levels of MDSC were measured by Flow Cytometry (CD11b+, CD14-, CD33+). Patients with gastric and colorectal cancers were divided into two groups, high VEGF group (serum VEGF-A>330pg/ml) and lo group (≤330pg/ml). **Results:** Prognosis of stages III and IV gastric cancer with high VEGF-A was significantly worse than in low VEGF-A group, while difference was not significant in stages I and II. In colorectal cancer, the results were exactly same as found in gastric cancer. The levels of MDSC were significantly correlated with the levels of VEGF-A in patients with gastric and colorectal cancers. The levels of VEGF-A were also correlated with inflammatory markers including CRP and NLR (neutrophil to lymphocyte ratio), and inversely did to stimulation index of lymphocyte proliferation (SI: parameter of cellular immune reaction) and nutritional parameters such as albumin or rapid turnover protein. **Conclusions:** The observation in the present study suggest that VEGF-A is closely associated with inflammation-associated disease progression especially in advanced stages, and serve as useful marker of immunosuppression involving MDSC and prognosis in patients with gastric and colorectal cancers. Research Sponsor: None.

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Analysis of tumor microenvironmental features of primary and synchronous liver metastases from patients with colorectal cancer using a deep learning algorithm. First Author: Oluwadunni Eunice Emiloju, Department of Hematology/Oncology, Mayo Clinic Rochester, Rochester, MN

Background: The development of colorectal liver metastases (CRLM) is associated with poor prognosis, and recent data suggest that metastasis to the liver is associated with resistance to immunotherapy. We characterized the microenvironment of primary colorectal carcinomas (CRCs) relative to their synchronous CRLM using a validated segmentation algorithm that quantifies 15 distinct morphologic tumor features. **Methods:** Adult CRC patients with synchronous CRLM (N=57) at Mayo Clinic were identified from the electronic health record using Epic Slicer Dicer and from an internal database. Tumor H&E sections were digitized and reviewed for quality control (RKP). QuantCRC (Aiforia, Inc) was applied to digitized images to extract 15 GI pathologist pre-defined morphological features. Tumor features were compared between primaries and CRLM using the Kruskal–Wallis test. The project was approved by the Mayo Clinic Institutional Review Board. **Results:** The study included 57 patients (median age 59 years [IQR 50, 73], 51% female) with CRC primaries and synchronous CRLM. Among primaries, 20 (35%) were right-sided and 37 (65%) were left-sided. QuantCRC identified 6 of 15 morphological features that differed significantly between primaries and their CRLM, including reduced stroma, more high-grade and necrosis, and a higher tumor: stromal ratio (TSR) in CRLM (Table). The increase in TIL density in CRLM vs the primary tumor was of borderline significance (p =0.053). Among patients with left-sided primary tumors, their CRLM had significantly higher TSR, percent high-grade, percent necrosis, and TIL density compared to the primary (all p values $\leq$ 0.02). Among right sided primaries, CRLM had a significantly reduced percent mature stroma (p=0.034) whereas percent necrosis was increased (p =0.01). **Conclusions:** Using deep learning, we identified tumor morphological features that differed significantly between primary CRC and their synchronous CRLM. This included higher TSR in CRLMs compared to primaries that is associated with epithelial-mesenchymal transition and has been shown to contribute to treatment resistance. Analysis of tumor morphological features with patient prognosis is ongoing. Research Sponsor: None.

Deep learning-derived morphological features of primary CRC and CRLM.

Morphological feature (Median)	Primary tumor	Liver metastasis	P-value
Tumor-Stroma Ratio	0.78	1.50	**0.008**
% High-grade	23.08	31.72	**0.005**
TIL Count	44.36	60.20	0.053
% Necrosis	6.35	19.32	**<0.001**
%Signet Ring Cells	0.25	0.13	0.088
TB/PDC	0.89	1.13	0.484
% Stroma	54.89	40.98	**<0.001**
% Immature Stroma	46.76	34.13	**<0.001**
% Mature Stroma	6.27	5.08	**0.030**

252 Poster Session

Effect of homologous recombination deficiency on survival in patients with metastatic colorectal cancer with KRAS mutations. First Author: Minsoo Kim, Korea University Anam Hospital, Seoul, South Korea

Background: Metastatic colorectal cancer (mCRC) patients exhibit intertumoral genetic heterogeneity, making it difficult to apply precision medicine. In the molecular genetic profile of mCRC, there are relatively few other gene alterations that significantly impact treatment decisions beyond KRAS and MSI status. We aim to investigate the prognostic impact of Homologous recombination deficiency (HRD)-related genetic alterations in mCRC patients, focusing on how mutation status influences on patient outcomes. **Methods:** The analysis included 364 mCRC patients who underwent comprehensive genomic profiling at Korea University Anam Hospital from Sep. 2016 to Dec. 2020. HRD was defined by the presence of pathogenic variants in genes including ATM, ATR, BARD1, BRCA1/2, BRIP1, CDK12, CHEK2, FANCA, MRE11A, PALB2, RAD50/51. Overall survival (OS) and progression-free survival (PFS) were evaluated using the Kaplan-Meier method. Multivariate Cox proportional hazards regression analyses were performed to assess the independent impact of HRD status. **Results:** Of the total cohort, 181 of patients (49.78%) harbored KRAS mutations. HRD gene mutations were identified in 38 of patients (10.4%) of patients, predominantly in ATM, BRCA 1/2, ARID1A genes. KRAS-HRD co-mutations were present in 18 of patients (4.7%). There is no significant difference in HRD status between the KRAS mutant and wild-type groups (p=0.73). While first-line treatment PFS was independent of KRAS mutational status (wt 10.8 vs. mut 9.6 months, p=0.31), it was significantly associated with HRD gene mutation status (16.2 vs. 10.6 months, p=0.001). OS was similar with KRAS mutations (34.7 vs. 32.9 months, p=0.72) and HRD mutations (34.7 vs. 39.1 months, p=0.73). However, patients with both KRAS and HRD mutations had significantly worse OS compared to those with either KRAS and HRD mutation (21.3 vs. 36.7 vs. 65.3 months, p=0.034, $<$0.001). A multivariate analysis was performed to evaluate factors affecting OS in mCRC patients. Among the variables, metastasectomy, which 44% of patients underwent, had a significantly low HR (=0.28) and would contributed to more favorable outcomes when evaluating OS. Neither KRAS nor HRD mutation alone was a significant prognostic factor, but their co-mutation was significantly associated with poor survival (HR=2.39, p=0.048). **Conclusions:** Poor OS in patients with KRAS-HRD co-mutations, despite metastasectomy, highlights the need for additional treatment strategies beyond surgery. Future studies with larger sample sizes and analyses stratified by metastasectomy status are necessary to strengthen these findings. Research Sponsor: None.

Variables	HR (95% CI)	p-value
Metastasis site		
Liver	2.16 (1.60 – 2.92)	$<$ 0.001
Peritoneum	1.52 (1.01 – 2.11)	0.013
Bone	1.69 (0.96 – 2.97)	0.067
Metastasectomy	0.28 (0.47 – 0.74)	$<$ 0.001
KRAS	1.03 (0.77 – 1.37)	0.826
HRD	0.75 (0.40 – 1.41)	0.367
Co-mutation	2.39 (1.01 – 5.66)	0.048

253 Poster Session

Real-world analysis of the impact of timing of biomarker testing on first-line treatment choice in patients with metastatic colorectal cancer in British Columbia. First Author: Tae Hoon Lee, BC Cancer - Vancouver, Vancouver, BC, Canada

Background: The treatment landscape for metastatic colorectal cancer (mCRC) has evolved significantly, shifting from chemotherapy alone to more personalized approaches that include immune checkpoint inhibitors and targeted therapies. Treatment decisions in mCRC are driven by molecular characteristics, and current guidelines recommend a minimum standard of biomarker testing, such as mismatch repair (MMR) status, and RAS and BRAF mutations, at the time of metastatic disease diagnosis to guide treatment selection. This study aimed to assess the real-world implementation of biomarker testing in a contemporary cohort of mCRC patients in British Columbia (BC) and its impact on treatment decisions by medical oncologists. **Methods:** We conducted a retrospective chart review of patients who received first-line systemic therapy for mCRC in BC between January 2022 and December 2023. Patient demographics, tumour characteristics, and treatment details were obtained from the BC Cancer provincial pharmacy database. Medical oncology documentation and pathology reports were reviewed to determine the availability of biomarkers, including MMR status, and RAS and BRAF mutations (OncoPanel), at the time of initial medical oncology consultation, and to assess how these results influenced treatment plan. **Results:** To date, 71 eligible patients have been identified, with data collection ongoing. Of these, 82% had proficient MMR status, 17% had deficient MMR status, and 1% had incomplete data. RAS mutations were present in 55% of cases, while BRAF mutations were found in 21% (Table). At the time of the initial medical oncology consultation, MMR status was unavailable in 14 patients (20%), two of whom were later found to have deficient MMR status. OncoPanel results were unavailable for 63 patients (88%) at the time of the initial consultation, and in 16 of these cases (25%), the systemic therapy plan was modified once the results were received. **Conclusions:** Even in contemporary practice with publicly funded biomarker testing, the results for RAS and BRAF mutations were not available at the time of initial medical oncology consultation for most patients. Biomarker testing is an essential tool for guiding therapeutic decisions in mCRC. This subsequently led to treatment plan changes in one-quarter of cases after results were received. These findings highlight the importance of timely biomarker testing to ensure optimal and efficient treatment decision-making for all mCRC patients. Research Sponsor: None.

Metastatic colorectal cancer biomarker testing.

	Proficient/WT	Deficient/Mutated	Incomplete
MMR	58	12	1
RAS	29	39	3
BRAF	53	15	3

254 Poster Session

The prognostic impact of *SLFN11* gene expression in metastatic colorectal cancer (mCRC). First Author: Hiroyuki Arai, Norris Comprehensive Cancer Center, Keck School of Medicine, University of Southern California, Los Angeles, CA

Background: SLFN11 was recently identified as a dominant determinant of response to DNA-damaging anticancer agents. Several studies showed SLFN11 expression led to better sensitivity to topoisomerase inhibitors and PARP inhibitors. However, clinical evidence in mCRC has been lacking. **Methods:** Two independent cohorts (CALGB/SWOG 80405 trial and a real-world patient [Caris CODEai] cohort) were included in this study. Patients were stratified into *SLFN11*-high and -low groups based on the median *SLFN11* gene expression levels. Survival outcomes were compared between *SLFN11*-high and *SLFN11*-low in irinotecan-treated and oxaliplatin-treated patients. Median time was estimated using Kaplan-Meier method and Cox models were used for covariate adjustment. Endpoints of interest were progression-free survival (PFS) and overall survival (OS). **Results:** In the CALGB/SWOG 80405 trial cohort, 433 patients were included (110 treated with FOLFIRI-based chemotherapy and 323 treated with FOLFOX-based chemotherapy). In patients treated with FOLFIRI-based chemotherapy, *SLFN11*-low patients exhibited numerically better PFS (median PFS, 13.4 vs 11.2 months, log-rank p = 0.03, adjusted hazard ratio [HR] = 0.86, 95% confidence interval [CI] 0.55-1.39, p = 0.50) and OS (median OS, 39.6 vs 30.3 months, log-rank p = 0.02, adjusted HR = 0.73, 95% CI 0.46-1.16, p = 0.19) compared to *SLFN11*-high patients. In patients treated with FOLFOX-based chemotherapy, *SLFN11*-low patients exhibited numerically better PFS (median PFS 11.2 vs 9.8 months, log-rank p = 0.03, adjusted HR = 0.83, 95% CI 0.65-1.05, p = 0.12) and OS (31.1 vs 26.1 months, log-rank p = 0.07, adjusted HR = 0.84, 95% CI 0.65-1.08, p = 0.16) compared to *SLFN11*-high patients. No significant interaction was observed between treatment and *SLFN11* expression in terms of PFS (p = 0.69) and OS (p = 0.44). In the Caris CODEai cohort, a better OS in *SLFN11*-low patients than in *SLFN11*-high patients was observed in irinotecan-treated patients (N = 2906) (median OS, 25.5 vs 21.7 months, HR = 0.90, 95% CI 0.82-1.00, p = 0.048). However, no significant difference in OS between *SLFN11*-low and SLFN11-high was observed in oxaliplatin-treated patients (N = 4428) (median OS, 39.5 vs 35.8 months, HR = 0.98, 95% CI 0.90-1.06, p = 0.56). **Conclusions:** Low *SLFN11* expression showed a potential for better prognosis in mCRC patients receiving irinotecan- and oxaliplatin-containing chemotherapies. These findings are not consistently aligned with preclinical data and clinical evidence in other cancer types previously reported. Further investigation is warranted to better understand the biological role and clinical implication of SLFN11 in mCRC. Research Sponsor: National Cancer Institute; P30CA014089, U10CA180821, U10CA180882, U10CA180820, UG1CA233277, and U10CA180888; Pfizer and Genentech; Gloria Borges WunderGlo Foundation; Dhont Family Foundation; Victoria and Philip Wilson Research Fund; San Pedro Peninsula Cancer Guild; Daniel Butler Research Fund.

255 Poster Session

Impact of *RAS* and *BRAF* heterogeneity on the efficacy of EGFR blockade in patients with metastatic colorectal cancer. First Author: Nobuhisa Matsuhashi, Department of Gastroenterological and Pediatric Surgery, Gifu University Graduate School of Medicine, Gifu, Japan

Background: Epidermal growth factor receptor (EGFR) blockade can effectively shrink tumors in metastatic colorectal cancer (CRC) patients without *RAS* nor *BRAF* mutations. However, some patients without RAS or BRAF mutations in their primary tumors may develop these mutations in metastatic tumors (heterogeneity). Circulating tumor DNA (ctDNA) can reflect this heterogeneity in metastatic tumors. Here, we evaluated the efficacy of EGFR blockade in patients who had no *RAS* or *BRAF* mutations in their primary tumors but did have them in ctDNA. **Methods:** We prospectively enrolled 100 patients with confirmed metastatic CRC without *RAS* or *BRAF* mutations in their primary tumors. Patients were treated with first-line systemic chemotherapy including EGFR blockade. We obtained ctDNA from each patient before they started chemotherapy. *RAS, BRAF (V600E),* and *PIK3CA* mutations were detected using digital PCR. **Results:** One of the 100 cases were excluded due to protocol violation. In the ctDNA obtained before starting chemotherapy, *RAS, BRAF,* and *PIK3CA* mutations were detected in 12, 4, and 6 patients, respectively. No patients had both *RAS* and *BRAF* mutations in their ctDNA; however, one patient had both *BRAF* and *PIK3CA* mutations and one had both *RAS* and *PIK3CA* mutations. Eighty-nine patients had measurable tumor lesions. Among these, 3 experienced a complete response (CR), 71 had a partial response (PR), 13 had stable disease (SD), and 2 had progressive disease (PD). The response rates for patients with *RAS* or *BRAF* mutations and for patients with neither mutation were 81% and 83%, respectively (P=0.73). Median progression-free survival (PFS) for patients with *RAS* or *BRAF* mutations and those without mutations were 251 days and 216.5 days, respectively (P=0.82). The presence or absence of PIK3CA mutations did not affect the response rate or PFS. **Conclusions:** Previously, we reported that the incidence of *RAS* and *BRAF* heterogeneity were 10% (1) and 4% (2). In the present study, *RAS* and *BRAF* heterogeneity were 12% and 4%, respectively. The heterogeneity of *RAS* and *BRAF* mutations had no effect on the response rate and PFS of EGFR blockade as first line chemotherapy. The effect of heterogeneity on the overall survival is of great interest; however, about half of the patients are still alive. 1. Yamada T, et al. Cancer Science 2016. 2. Ueda K, et al. Eur J Surg Oncol 2022. Clinical trial information: UMIN000031177. Research Sponsor: None.

256 Poster Session

Prognostic value of circulating tumor DNA for progression-free survival in patients with locally advanced dMMR/MSI-H colorectal cancer managed without surgery. First Author: Michael Brian LaPelusa, The University of Texas MD Anderson Cancer Center, Houston, TX

Background: Immunotherapy has shown the ability to induce a high rate of pathologic and clinical complete response in patients with deficient mismatch repair (dMMR)/ microsatellite instability-high (MSI-H) colorectal cancer (CRC). The role of circulating tumor DNA (ctDNA) as a prognostic biomarker for survival in patients who receive non-operative management (NOM) is unknown. Among patients enrolled in a phase II trial (NCT04082572) that evaluated the efficacy of pembrolizumab in patients with locally advanced dMMR/MSI-H solid tumors, we evaluated the impact of ctDNA on progression-free survival (PFS) among 12 patients with locally advanced dMMR/MSI-H CRC who received NOM. **Methods:** ctDNA testing was performed in the CLIA-certified laboratory at MD Anderson using a 70-gene liquid biopsy panel (LBP-70), which used digital sequencing of cell-free DNA. Formalin-fixed paraffin-embedded tumor samples and germline peripheral blood mononuclear cells were profiled using next-generation sequencing. Mutations in ctDNA were considered tumor-specific if they were confirmed with matched tumor tissue profiling. The lower limit of detection of the LBP-70 assay was a mutational variant allele frequency of $<$0.3%. "ctDNA clearance" and "ctDNA(-)" status were defined as the absence of any detectable tumor-specific mutations in ctDNA. Two-year progression-free survival (PFS) was estimated using Kaplan-Meier techniques and compared with log-rank tests. **Results:** The median follow-up between the first cycle of pembrolizumab and last radiographic assessment was approximately three years (36.7 mo; range (range 1.2 – 51.0 mo). Forty-nine LBP-70 assays were performed among the 12 patients over the course of treatment with pembrolizumab. The median number of cycles was 16 (range 1 – 16 cycles). Six patients were ctDNA(-) prior to pembrolizumab, all of whom remained ctDNA(-) after. None of these six patients experienced disease progression (PD) at their last follow-up. The other six patients were ctDNA(+) prior to pembrolizumab (median baseline VAF 0.4%, range 0.3%–4.0%). Of these six patients, three experienced ctDNA clearance while receiving pembrolizumab. None of the three patients who cleared ctDNA experienced PD as of their last follow-up. Of the three patients who did not clear ctDNA, two experienced PD in their primary tumor (at two and six months after their first cycle, respectively), while the third patient remains without PD at their last follow-up. The two-year PFS rate was 100% among the nine patients who were ctDNA(-) after pembrolizumab compared to 33% among the three patients who were ctDNA(+) after pembrolizumab (p = 0.03). **Conclusions:** ctDNA may be a valuable tool for assessing treatment response and improving the ability to tailor organ-sparing, curative strategies to patients with locally advanced dMMR/MSI-H CRC. Research Sponsor: None.

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EORTC1527/JCOG1609INT/ESSO02: Diffusion-weighted magnetic resonance imaging (DW-MRI) assessment of initially unresectable liver metastasis to improve surgical planning (DREAM)—Primary analysis. First Author: Kozo Kataoka, Division of Lower GI Surgery, Department of Gastroenterological Surgery, Hyogo Medical University, Nishinomiya, Japan

Background: Disappearing liver metastases (DLMs) diagnosed on post-chemotherapy (Cx) computed tomography (CT) is a favorable prognostic factor in patients (pts) with colorectal liver metastases (CRLM). However, the optimal treatment of DLMs - whether they should be resected or left behind - is controversial. This is a prospective, multi-centred, international study examining the added value of MRI (DWI, T1/T2 and contrast-enhanced) to that of CT alone in accurate assessment of the viability of DLMs. **Methods:** Pts with initially unresectable CRLM downstaged to a planned liver resection after Cx were enrolled, based on the obligatory decision of a multidisciplinary team. Pts were imaged by both CT and MRI at baseline and presurgical timepoints. DLMs were defined as disappeared lesions diagnosed by CT alone, while confirmed DLMs (cDLMs) were defined as lesions that disappeared on both CT and MRI. cDLMs were either resected or followed-up for 2 years if not resected. All imaging scans were collected centrally for quality assurance. The primary endpoint was the negative predictive value (NPV) of MRI and CT in confirming the status of cDLMs using either pathological complete response or the absence of a local recurrence at the site of cDLMs during the 2 year follow up. The study was aimed at excluding a NPV $\leq$0.85 with a 1-sided alfa of 5% and power of 90% under the alternative that the NPV $\geq$0.95. The planned sample size was 92 evaluable (resected or left behind) cDLMs, assuming a within-patient correlation between cDLMs of 0.2 and an average number of 2 cDLMs per pt. **Results:** 233 pts were registered at 22 participating centres, and 112 were enrolled for analysis. A median of 8 cycles of Cx was delivered, and pts had a median of 7 CRLMs at baseline. A total of 203 cDLMs were identified while the number of DLMs diagnosed by CT was 296. Of these, 152 cDLMs and 227 DLMs, respectively, were evaluable according to the imaging protocol. Intraoperative ultrasound was performed for 195 cDLMs and, of these, 59 (30.2%) were still detected. The rate of R0/R1 resection was 95.5%. The NPV of evaluable cDLMs either resected or left behind was 62.5 % (95/152, 95% CI:50.8-74.2), which was lower than the prespecified threshold. The NPV of DLMs was 52.9% (119/227, 95% CI:42.7, 63.0). The NPV of resected cDLMs, and those left behind were 56.8% (50/88, 95% CI: 44.2, 69.5) and 70.3% (45/64, 95% CI: 48.6 -92.0), respectively. For DLMs, the NPV of resected DLMs and those left behind were 45.6% (72/158, 95% CI: 35.4-55.7) and 69.6% (48/69, 95% CI: 47.7-91.5), respectively. **Conclusions:** For pts with initially unresectable CRLM, cDLMs diagnosed by both CT and MRI do not correspond to tumor viability, even after highly effective chemotherapy. Ongoing survival analysis (to be presented at the annual meeting) may impact the treatment strategy of pts with DLMs. Clinical trial information: NCT02781935. Research Sponsor: Krebsforschung Schweiz; Ligue nationale contre le cancer; Anticancer Fund; Kom Op Tegen Kanker.

258 Poster Session

Genomic alterations in circulating tumor DNA (ctDNA) and response to telisotuzumab adizutecan (ABBV-400) treatment in patients (pts) with colorectal cancer (CRC). First Author: Michael Cecchini, Yale School of Medicine, New Haven, CT

Background: Telisotuzumab adizutecan is an antibody-drug conjugate comprising the c-Met–targeting antibody telisotuzumab conjugated to a topoisomerase 1 inhibitor payload, adizutecan. A phase 1 trial (NCT05029882) in advanced solid tumors reported higher response rates in pts with CRC and high c-Met expression. We present an analysis of responses based on genomic alterations in ctDNA. **Methods:** Pts with CRC that is refractory to standard treatment were enrolled. Telisotuzumab adizutecan was given IV Q3W. ctDNA was isolated from baseline (BL) and cycle 3, day 1 plasma samples and analyzed with the GuardantINFINITY assay. Circulating tumor fraction (cTF) was estimated on the basis of variant allele frequency of somatic mutations in a 74-gene panel and methylation signals across targeted regions of the GuardantINFINITY methylation panel. Molecular response (MR) was defined as 50% decrease in cTF from BL. Radiographic response (RR) was assessed by RECIST v1.1. **Results:** Overall, 113 pts had both BL ctDNA and RR data. Confirmed ORR was 18% (20/113). The most prevalent gene alterations included *TP53* (75%), *APC* (68%), and *KRAS* (66%) mut; *PTPRT* (56%), *ASXL1* (53%), and *FLT1* (52%) amp; *SMAD4* (44%) and *TP53* (43%) deletions; and *MET* (4%), *BRAF*, *FGFR3,* and *NRG1* (all 3%) fusions. The Table shows RRs in pts with specific CRC biomarkers, and MR using the 2 panels with corresponding clinical outcomes. RRs were seen in pts with positive plasma samples at BL for TMB, *RAS, BRAF,* and *HER2*. MRs were detected in 64% and 65% of pts using the 74-gene and methylation panel, respectively. ORR was higher in pts with MR (35%) than without MR (8%) using the 74-gene, and 35% vs 0% using the methylation panel, respectively. Median PFS (mPFS) was longer in pts with MR. **Conclusions:** Telisotuzumab adizutecan has promising efficacy. RRs were seen in CRC pts with heterogeneous genomic profiles, including pts positive for actionable biomarkers in the metastatic setting. More pts with MR experienced RRs and benefited from treatment. Clinical trial information: NCT05029882. Research Sponsor: AbbVie Inc; n/a.

	Pts with ctDNA and radiographic response data (N=113)				
Radiographic cPR, n/N (%)			20/113 (18)		
Genomic alteration	**TMB[a] high (≥20 mut/Mb)**	***KRAS* mut**	***KRAS* G12C mut**	***BRAF* mut[b]**	***HER2* amp**
cPR, n/N (%)	11/48 (23)	11/73 (15)	1/5 (20)	4/14 (29)	2/9 (22)
Pos	9/51 (18)	9/40 (23)	19/108 (18)	16/99 (16)	18/104 (17)
Neg					
MR, n/N (%)	**74-gene panel** 42/66 (64)		**Methylation panel** 48/74 (65)		
ORR, n/N (%)	**MR pos** 15/42 (36)	**MR neg** 2/24 (8)	**MR pos** 17/48 (35)	**MR neg** 0/26 (0)	
mPFS, mo (95% CI)	6.1 (5.3, 6.9)	3.5 (2.6, 4.3)	5.9 (5.3, 6.8)	3.5 (2.7, 4.1)	
Events (n/N)	28/42	21/24	31/48	23/26	
MR in pts with SD, n/N (%)	**74-gene panel** 27/45 (60)		**Methylation panel** 31/53 (58)		
	MR pos	**MR neg**	**MR pos**	**MR neg**	
mPFS, mo (95% CI)	5.3 (4.5, 5.9)	3.9 (2.8, 4.3)	5.3 (4.5, 5.9)	4 (2.8, 4.4)	
Events (n/N)	21/27	16/18	23/31	19/22	

[a]14 pts not evaluable for TMB. Threshold determined by Guardant.
[b]1 pt had V600E mutation.
mo, months; neg, negative; pos, positive.

259 Poster Session

Bragg peak proton irradiation and ATM inhibition for rectal cancer. First Author: Cameron Matthew Callaghan, Mayo Clinic Rochester, Rochester, MN

Background: Rectal cancer patients with a complete response (CR) after chemotherapy and radiotherapy may opt for non-operative management. Currently only 20-40% of patients reach a CR. Escalating the dose of radiation may increase the risk of toxicity. ATM inhibitors (ATMi) are potent radiosensitizers in colorectal cancer (CRC). Proton irradiation exhibits an entrance portion of the beam (ENT) where dose is deposited sparsely and a Bragg Peak (BP) with virtually no 'exit dose' thereafter. ATMi renders cells uniquely hypersensitive to proton BP, and BP+ATMi may increase the response of Mismatch Repair proficient (pMMR) rectal cancer to immunotherapy. **Methods:** Tumor efficacy and immunogenicity was assessed using clonogenic survival, γH2AX foci, micronuclei, and cytoplasmic cGas assays after indicated IR modalities +/-ATMi. Normal tissue toxicity was assessed via murine models of non-tumor bearing BALB/c after whole-abdominal irradiation (WAI) after indicated IR modalities +/- ATMi via acute weight loss, FITC/Dextran assay, and Radiation Injury Score (RIS, Scale 0-18) in jejunal and rectal samples at 4 days post-IR. **Results:** ATMi decreased clonogenic survival across all types of radiation, but the effect was most pronounced with BP protons. BP+ATMi also significantly increased unresolved γH2AX foci, micronuclei, and cytoplasmic cGas compared to ENT+ATMi. Normal tissue toxicity studies demonstrated that BP+ATMi did not significantly increase acute weight loss, RIS in jejunal or rectal tissue; but did increase FITC/Dextran intestinal permeability. **Conclusions:** Inhibition of ATM preferentially radiosensitized BP protons compared to x-rays or ENT protons in CRC but not in normal tissue based on weight loss and histopathological evaluation. Research Sponsor: None.

Differential tumor efficacy and normal tissue toxicity after x-ray, entrance and Bragg Peak Protons +/- ATM inhibitor.

	A	B	C	D	E	F	G	H	I	J
DMSO	1	1	1	2.9	1.5	8764	96.4	10	0.3	0.2
ATMi	0.873	0.842	0.723	8.2	3.0	16300	98.5	5.8	0.7	0.8
x-ray	0.615	0.751	0.109				82.4	6.2	4.8	1.0
x-ray+ATMi	0.123	0.344	0.011				80.0	14.4	5.2	1.5
ENT protons	0.293	0.515	0.080	10.2	5.8	10429	78.5	6	2.9	1.5
ENT+ATMi	0.091	0.066	0.013	15.1	7.2	12274	73.0	23.8	5.2	3.8
BP protons	0.213	0.340	0.032	12.0	12.6	7192	75.6	24.6	4.1	3.7
BP+ATMi	0.074	0.038	0.004	22.7	16.2	43677	73.6	310.4	5.3	4.3

Clonogenic survival of pMMR colorectal cancer cell lines after IR (4 Gy) +/- 10nM AZD1390 (ATM inhibitor) ofA) HT29, B) CT26, and C) SW480 cell lines. D) γH2AX foci at 24 hours post-IR in HT29. E) Micronuclei, and F) cytoplasmic cGas integrated density per cell via immunofluorescence in SW480 72 hours post-IR. WAI (10Gy/1 fraction) was delivered to non-tumor bearing BALB/c mice which were followed for G) acute weight loss (% from baseline), and H) FITC/Dextran permeability assay. Tissue samples were graded for Radiation Injury Score in I) jejunal, and J) rectal samples at 4 days post-IR (n=5 per cohort, 3 samples per mouse). Figures are means of triplicate data.

260 Poster Session

A comparison of real-world outcomes by mismatch repair status in patients with stage II/III rectal cancer in the Netherlands. First Author: Renee A Lunenberg, Department of Medical Oncology, University Medical Center Utrecht, Utrecht, Netherlands

Background: A subset of rectal tumours (~3–5%) have mismatch repair deficiency (dMMR); the remaining are classified as MMR proficient (pMMR). A recent trial for stage II/III dMMR rectal cancer (RC) showed that 6 months of neoadjuvant treatment with dostarlimab, a programmed cell death protein 1 inhibitor, induced a 100% clinical complete response rate and allowed for organ preservation. Using real-world data from the Netherlands, a dMMR patient cohort was compared to a matched pMMR cohort to assess if clinical outcomes differed, even after matching for baseline characteristics. **Methods:** Utilizing data from the Netherlands Cancer Registry, this retrospective analysis assessed patients with stage II/III RC treated from 2015–2022 with known MMR status. The dMMR cohort was matched 1:2 to a cohort of pMMR patients on age, year of diagnosis, tumour clinical (cT) stage, and node clinical (cN) stage. Event-free survival (EFS; defined as first occurrence of locoregional failure, distant metastasis, a new primary colorectal tumour, or death from any cause) and overall survival (OS) were compared using Kaplan-Meier methodology and a univariable Cox model. **Results:** Among the 7939 patients included, 184 (2.3%) had dMMR tumours. After matching and a loss of 8 dMMR patients due to ineligibility, 176 dMMR patients were compared to 363 pMMR patients. Matching successfully removed differences in age, cT, and cN stage; however, dMMR patients continued to have more BRAF mutants (p=0.014), more poorly differentiated tumours ($p<0.001$) and more variation in histology (p=0.01). dMMR patients had a statistically significant lower risk of experiencing an event (hazard ratio [HR] 0.54 [95% confidence interval (CI) 0.38–0.77], $p<0.001$), with a 3-year EFS of 79.6% (95% CI 73.7–86.1) compared to pMMR patients (64.7% [95% CI 59.8–70.1]). For OS, there was no significant difference between the dMMR and pMMR cohorts (HR 0.73 [95% CI 0.47–1.16], p=0.181); however, within the small group of patients with an event, OS at 1 year for dMMR patients was 68.4% (95% CI 55.1–84.9) compared to 93.5% (95% CI 89.5–97.7) for pMMR patients (p=0.001). **Conclusions:** dMMR RC is rare, and tumour characteristics differ from those of patients with pMMR tumours (1). After controlling for clinical stage and age, this study suggests dMMR patients have improved EFS compared to pMMR patients. There is no significant difference in OS between groups but dMMR patients with an event have worse OS than pMMR patients, highlighting an additional need to prevent recurrence of disease within dMMR patients.1. Lunenberg, R et al. Poster presented at ESMO Congress (Presentation 569P), 13-17 Sept 2024, Barcelona, Spain. Research Sponsor: GSK; 219473.

261 Poster Session

Leveraging large-scale PDO-based assays to optimize antibody-drug conjugate efficacy in CRC. First Author: Lélia Polit, Orakl Oncology, Le Kremlin-Bicêtre, France

Background: Clinical trials are one of the major barriers in contemporary drug development. Functional assays based on patient-derived organoids (PDO) are a promising new tool for derisking clinical development of new therapies, but their use has been limited due to small cohort sizes and the absence of systematic validation studies. Using the largest cohort of matched PDOs, longitudinal clinical data, transcriptomic and WES data in CRC, we explore the use of large PDOs collections to characterize ADC efficacy and affinity through systematic mapping of the relationship between target antigen affinity and payload efficacy. **Methods:** We have assembled a collection of 85 PDOs from colorectal cancer (CRC), generated from primary tumour samples and metastatic biopsies. Patients ranged from 32 to 89 years old and had experienced an average of 1.54 lines of treatment prior to the PDO-generating tissue sampling. Using gene expression levels of well known target antigens such as CEACAM5, ERBB2, MSLN and TROP2 as a proxy for protein concentration, we explore the relationship between the target antigen affinity and the efficacy of a subset of common payloads including topoisomerase and microtubule inhibitors. This allows us to identify the relationship between two of the three main vectors of ADC efficacy in CRC. **Results:** We show that topoismerase inhibitor and microtubule inhibitor efficacy relates to the RNA expression level of multiple target antigens commonly used in active clinical trials in CRC. These data match the target antigen and payload couples currently under development or approved in the clinic. We are developing a tool to optimize clinical trial design. This approach will allow us to match patients more accurately with the appropriate payload, ensuring a higher likelihood of response. **Conclusions:** We report that large-scale PDO-based assays can be a useful tool to anticipate clinical readouts. This work suggests that well-characterized PDO collections could also help redefine patient populations, increasing the likelihood of identifying those more likely to respond to a given ADC, and thus improving the success rates of clinical studies for new treatments in CRC. Research Sponsor: Fondation Jean-Jacques et Felicia Lopez-Loreta pour l'Excellence Académique; Banque Publique d'Investissement.

262 Poster Session

Highly accurate detection of early-stage colorectal cancer using tumor and immune extracellular vesicles biomarkers. First Author: Todd A. Hembrough, Nexosome Oncology, Durham, NC

Background: Colorectal cancer (CRC) is one of the most common cancers worldwide, and early detection is critical for successful treatment and improved survival rates. However, precancerous and early-stage CRC present significant diagnostic challenges due to the small size of lesions and the very low expression of tumor-specific biomarkers in the bloodstream. Current genomics-based diagnostic methods struggle to detect these early-stage and precancerous lesions, making it imperative to develop more sensitive and specific approaches. Circulating extracellular vesicles (EVs) are emerging as a promising solution for early-stage CRC detection. Since EVs are produced by tumor cells as well as tumor microenvironment and host immune response cells, EVs offer the means to detect and monitor small early-stage tumors through both direct detection of tumor-associated biomarkers as well as tumor specific host response and tumor microenvironment biomarkers. **Methods:** To identify novel early-stage CRC specific biomarkers, we performed proteomics analysis on EVs purified from patient plasma using size exclusion chromatography and a proprietary buffer system that enhances EV and corona protein recovery. TrueDiscovery Data-independent acquisition (DIA) mass spectrometry (MS) analysis was conducted on 24 pre-cancer/stage 0 and 25 stage 1 CRC patients and 75 normal patient plasma samples. An in-house developed machine learning pipeline was used to identify differentially expressed proteins and model candidate multiplexes to identify those with extremely high diagnostic accuracy ($>$ 0.99). **Results:** An average of ~2,500 proteins were identified per sample and included in bioinformatics analysis. Comparative analysis between control patient EVs and either pre-cancerous/Stage 0, or Stage 1 colon cancer EVs using an in-house Machine Learning pipeline identified 336 and 493 differentially expressed proteins for each group (FDR adjusted p-value $<$ 0.001). Of these, the best 17 pre/stage 0 proteins and 53 stage 1 proteins were trained and tested using Support Vector Machine to assess all potential 3-plexes in order to identify those with mean diagnostic accuracy $>$ 99%. This yielded 5 3-plexes in precancer/stage 0 and 34 3-plexes in stage 1 with near perfect diagnostic accuracy. These plexes are currently being assessed in single analyte and multiplex immunoassays with the goal of translating candidate 3-plexes to the clinical. **Conclusions:** Key biomarkers from CRC patient EVs indicate the potential to detect and diagnose early-stage and low tumor burden cancers using our methods. Interestingly, the most accurate 3-plexes consist of proteins from immune, inflammatory and metabolic processes, suggesting that for the detection of early stage cancer it may be critical to include both tumor and host specific biomarkers. Research Sponsor: None.

263 **Poster Session**

Correlation of mid-chemoradiation ctDNA results and clinical complete response to total neoadjuvant therapy (TNT) for locally advanced rectal adenocarcinoma. First Author: Seth Felder, H. Lee Moffitt Cancer Center, Tampa, FL

Background: The total neoadjuvant therapy (TNT) paradigm for locally advanced rectal cancer (LARC) is evolving with intensification and de-escalation neoadjuvant therapies. Chemoradiation (CRT) followed by consolidation chemotherapy offers the highest rate of organ preservation. Circulating tumor DNA (ctDNA) response during TNT may serve as a biomarker that allows for therapy personalization to better select candidates for rectal organ preservation (Watch-and-Wait, WW). **Methods:** We enrolled patients with mismatch repair proficient LARC eligible for TNT on a prospective protocol (NCT05108428). Patients underwent CRT to 50.4 Gy with a mid-treatment evaluation (27 Gy) on a MRI-linac followed by 8 cycles of consolidation FOLFOX. A sequential MRI-guided adaptive radiotherapy boost to 59.4 Gy was delivered if rectal tumor volume reduced $>$ 30% at 27 Gy. Standard restaging with diagnostic MRI and endoscopy occurred after TNT to evaluate organ preservation candidacy. ctDNA was obtained at diagnosis, mid-CRT (27 Gy), completion of TNT, and regular intervals in surveillance. ctDNA analysis was performed using a clinically validated, personalized, tumor-informed assay (Signatera, Natera, Inc.). Descriptive statistics for clinical response for mid-treatment assessment are reported. Additional clinical outcomes will be reported as follow up matures. **Results:** A total of 20 patients were enrolled (70% male, 70% cN+) with average age of 58 (range: 30-86) and 40% with threatened or involved radial margin. Average follow up is 12 months (range: 4-24 months) from date of completion of TNT. Average initial tumor volume was 26cc (range: 8.6cc- 66.8cc) with an average tumor reduction of 68.5% at 27 Gy. Volumetric change based on mid treatment MRI $>$30% did not predict WW eligibility (2 patients with $<$30% volumetric change maintains clinical complete responses). All patients demonstrated positive ctDNA at diagnosis (average 7.08 MTM/mL, range: 0.08 MTM/mL -77.96 MTM/mL). 25% of patients had elevated CEA at diagnosis. Conversion to negative ctDNA at mid-CRT occurred in 53% of patients. 57% of patients with a positive ctDNA at this timepoint underwent formal oncologic mesorectal excision with adenocarcinoma confirmed in the specimens. 25% of patients with negative ctDNA at this timepoint underwent a local excision, with no histologic invasive cancer in both specimens. We observed a 100% rectal organ preservation rate when early conversion to negative ctDNA at the 27 Gy of CRT timepoint. **Conclusions:** ctDNA clearance during CRT may be a good early predictor for patients that may be a candidate for a WW protocol. This biomarker may also guide consolidation therapy escalation (FOLFIRINOX) or de-escalation (chemotherapy omission) for select patients. Longer follow up is needed to correlate with clinical outcomes and future studies with more patients is needed. Clinical trial information: NCT05108428. Research Sponsor: Natera, Viewray, Moffitt.

264 **Poster Session**

KRAS status in lung oligometastases treated with SBRT. First Author: Mauro Loi, Radiation Oncology Unit, Azienda Ospedaliera Universitaria Careggi, University of Florence, Florence, Italy

Background: Stereotactic Body Radiotherapy (SBRT) to lung metastases (LM) from colorectal cancer (CRC) has been applied to improve disease control in in the oligometastatic (OM) and oligoprogressive (OP) setting. The aim of this study is to assess the impact of KRAS mutational (KRASmut) status in the response to SBRT in LM. **Methods:** Data from a consecutive cohort of OM/OP CRC patients treated at our Institution with SBRT for LM from May 2019 to March 2024 were retrospectively reviewed. Patients (age, KRASmut in primary tumor, OM/OP disease, chemotherapy lines prior to SBRT) and treatment-related variables (dose regimen and Biologically Effective Dose: BED, toxicity) were collected. Local Control (LC) was defined as time from SBRT to local failure or last radiologic follow-up. **Results:** Sixty-two patients (median age: 74 years, range 49-85), accounting for 103 LM were included. LM were treated in the OM and OP setting in 65(63%) and 38(37%) cases, respectively. SBRT was delivered before chemotherapy (n=49,48%) or following first (n=31,30%) or further chemotherapy lines (n=23,22%). KRASmut was present ab initio in 49% of cases (n=50). Dose regimens included 48-60 Gy in 3-8 fractions, resulting in a median BED of 115.5 (range 76.8-151.2) Gy_{10} : a BED$>$100Gy_{10} was delivered in 92% (n=95) of treatment courses. Median follow-up was 16 (range 2-45) months. Median LC was not reached; 1- and 3 years LC rates were 85% and 67% respectively. At multivariate analysis (MVA), only a BED$\leq$100Gy_{10} (p=0.029, HR 2.99, IC95% 1.1- 9.5) and prior second or further chemotherapy line (p=0.031, HR 3.2, IC95% 1.1- 8.1) were independently correlated with impaired LC. No grade $>$2 toxicity was observed. Focusing on patients receiving no chemotherapy or first-line treatment before SBRT, 1- and 3 years LC rates were 92% and 77% respectively (median: NR). At MVA, while BED$\leq$100Gy_{10} (p=0.036, HR 6.96, IC95% 1.9-25.7) was associated with poorer LC, lack of KRASmut (p=0.037, HR 0.26, IC95% 0.07- 0.9) was independently correlated with improved LC. **Conclusions:** SBRT in LM from OM/OP CRC results in high LC rates, particularly with intensive regimens in non-heavily pretreated patients. No impact of KRASmut was observed. Due to possible occurrence of secondary mutations in the pretreated population, a subset analysis in the chemotherapy-naive/first-line subset suggests an impact of KRASmut in radioresistance that should be further explored in a dose-escalation perspective. Re-biopsy or liquid biopsy may be useful to identify de-novo mutations. Research Sponsor: None.

265 **Poster Session**

A retrospective study of colorectal cancer CT imaging features involving microsatellite instability. First Author: Scott McLaughlin, AT Still University, Kirksville, MO

Background: This study aimed to identify correlations between CT imaging characteristics accessible to the radiologist with microsatellite instability (MSI) status among colorectal cancer. Noninvasive identification of MSI status can decrease time to treatment if pathology can be bypassed. **Methods:** 109 colorectal cancer patients were identified retrospectively from 2011 to 2018 with an average age of 61 years (58 male, 51 female) and had MSI pathological assessment. These subjects all had CT abdomen and pelvis obtained at the time of initial diagnosis. Imaging features of both the primary and metastatic lesions were assessed. Primary lesion features included: location, size, stage, attenuation compared to the liver, growth pattern, tumor margin, primary mass area, and the presence/absence of mesenteric infiltration. Metastatic lesion features included: size, attenuation relative to liver, and enhancement pattern. Patient age, sex, and tumor staging were obtained from patient medical records. Statistical analysis was performed using various methods, including chi-square, Mann-Whitney and Kruskal-Wallis tests. **Results:** 5 tumor characteristics displayed a statistically significant relationship with MSI status. MSI-H lesions were more likely to be clinical Stage 2, and MSI-L lesions were more likely to be Clinical Stage 3 (p=0.012). Tumors with distant metastasis were more likely MSI-L versus regional metastasis were more likely MSI-H ($p<0.001$). Primary mass area was larger in the MSI-H than MSI-L group ($p<0.001$). Tumors in the left colon are more likely to be MSI-L versus CRC tumors in the right colon are more likely to be MSI-H ($p<0.001$). MSI-H tumors had a lower primary tumor Hounsfield Unit standard deviation (SD) than MSI-L tumors (p=0.002). The remaining features were not statistically significant. For primary lesion, these include mesenteric infiltration (p=0.189), tumor margin (p=1.000), enhancement pattern (p=0.498), growth pattern (p=0.127), tumor CT density (p=0.162), and liver density (p=0.105). Enhancement of metastasis also had no difference (p=0.376). **Conclusions:** Multiple CT imaging characteristics were predictive of MSI classification in CRC tumors, holding potential clinical relevance that can help guide individualized therapy. While these results were statistically significant, more research must be done to validate these findings and to derive useful nomograms. Research Sponsor: None.

266 **Poster Session**

Development of a methylation-based, tissue-agnostic test for the detection of molecular residual disease by circulating tumor DNA. First Author: John Paul Y.C. Shen, The University of Texas MD Anderson Cancer Center, Houston, TX

Background: Clinical validation studies support tumor-informed molecular residual disease (MRD) as a prognostic biomarker for disease recurrence across multiple solid tumor types. However, these tests are not always feasible due to the occasional lack of tumor tissue. Here, we discuss the design of a test for tissue-agnostic MRD detection and its application to a cohort of patients with colorectal cancer (CRC). **Methods:** A targeted panel composed of differentially methylated regions was developed. A machine-learning model was trained on differential methylation patterns in order to classify plasma samples as MRD-positive or MRD-negative. Performance of the independently trained classifier was assessed in a cohort of 247 patients enrolled in the Bespoke CRC trial (NCT04264702). These patients had MRD results available using a tumor-informed circulating tumor DNA (ctDNA) assay (Signatera), of whom 163 were persistently MRD-negative without clinical progression and 84 had MRD-positive results. Tissue-agnostic MRD results were compared to the tumor-informed results by calculating the percent positive agreement (PPA). Additionally, the differentially methylated allele fraction (DMAF) from the tissue-agnostic test was compared with variant allele frequencies (VAFs) from the tumor-informed test. **Results:** In the Bespoke CRC clinical cohort (72% non-Hispanic White, 54% male, mean age 61.4$\pm$12.3 years), 71 (28%) patients had stage II CRC, and 147 (60%) had stage III CRC. Overall, PPA was 86% (95% CI: 70-100%) and specificity was 97% (95% CI: 93-100%). When categorizing based on tumor-informed MRD VAF levels, PPA was 97% for VAF $>$0.2%, 100% for VAF 0.1-0.2%, 89% for VAF 0.04-0.1%, and 68% for VAF $<$0.04%. The DMAFs strongly correlated with the VAFs from the tumor-informed test and the correlation was independent of disease stage, histology, age, and sex. **Conclusions:** This is the first study of its kind demonstrating high concordance between a tissue-agnostic MRD test and a clinically validated tumor-informed ctDNA assay. These findings demonstrate that in cases where tissue is not available or of inadequate quality, a methylation-based tissue-agnostic assay may serve as a potential alternative for MRD detection. Research Sponsor: None.

267 **Poster Session**

Prognostic factors for thermo-ablation of liver metastases in colon cancer: A Maltese perspective. First Author: Malcolm Buhagiar, Sir Anthony Mamo Oncology Centre, Msida, Malta

Background: Malta is a small Mediterranean archipelago with an annual colon cancer incidence of 300 patients per year and a cumulative 5-year overall survival of 58%. Despite major improvements in therapeutic strategies, colon cancer remains the third leading cause of cancer deaths locally with an annual mortality rate of circa 110 patients per year. Percutaneous thermal ablation of colorectal liver metastases involves exposure of metastatic deposits to temperature extremes using radiofrequency, microwave or cryotherapy. The role of this case series was to examine the prognostic value of clinico-pathological variables to provide insight into patient selection for ablative therapy. **Methods:** This retrospective analysis of 66 adult patients with colorectal liver oligometastatic disease who underwent thermal ablation between January 2014 and January 2022, reviews demographic and pathological criteria including tumour grade, presence of vascular invasion, mismatch repair and k-RAS & BRAF mutational status, CEA & Ca19.9 assay at time of treatment. Other parameters include metachronous versus synchronous disease, hepatic burden and concurrent chemotherapy. All these covariables were analysed using the Cox Regression Hazard model to assess their impact on progression free- (PFS) and overall survival (OS). **Results:** For this patient cohort, the mean PFS and OS were 13.1 and 33.8 months. Our data indicates that most individuals experience progression within the first 10 months, with a gradual decline in PFS for the remaining individuals. A Ca19.9 level within normal limits (p=0.004, HR 0.157-0.700 95% CI) was related to an improved PFS but not OS. The presence of synchronous metastases (p=0.004, HR 0.168-0.715 95% CI) and ablation of one lesion (p=0.013 HR 0.191 (0.052-0.702 95 CI) were favourable prognostic markers for improved OS. k-RAS wt status was a prognostic adverse biomarker for PFS (p=0.002, HR 1.529, 6.686 95% CI). There was a paucity of data for BRAF and MMR status so these were excluded from the final analysis. No statistical survival benefit was noted for patients treated with chemotherapy, and other variables including gender, age, primary tumour sidedness, LVSI or CEA value. **Conclusions:** There are two surprising trends here- the paucity of survival benefit for the use of chemotherapy in conjunction with ablation and the worse outcome for k-ras wt tumours. We postulate that the absence of BRAF sub-stratification at the time of patient accrual, impacted statistical hazard. This case series suggests that the prognosis and survival of colorectal liver oligometastatic disease is multifactorial and complex, highlighting the need for further genomic and molecular insights. Research Sponsor: None.

268 **Poster Session**

Landscape of novel gene fusions in gastrointestinal tumors. First Author: Aysegul Balyimez, Exact Sciences, Madison, WI

Background: Gene fusions are recognized as critical drivers of cancer and targets for precision therapies. The 2022 ASCO provisional clinical opinion on tumor genomic testing includes recommendations for fusion testing and recognizes the strengths of RNA-based approaches. Targeted DNA/RNA sequencing panels can be used to detect well characterized, clinically actionable fusions, but their ability to detect rare novel fusions (i.e., those with uncharacterized fusion partners) can be limited by assay design. Whole-exome, whole-transcriptome sequencing (WES/WTS) may improve the identification of actionable gene fusions by detecting both known and novel fusions. This work describes clinically actionable novel fusions detected in gastrointestinal (GI) cancers. **Methods:** The study included samples from GI cancers that were analyzed using the OncoExTra test between May 2018 and May 2024. The assay uses tumor-normal WES/WTS to identify somatic alterations, including gene fusions with known and unknown partners and breakpoints.Clinical data were queried foractionable fusions with novel fusion partners that were unique in the entire laboratory database and have not, to our knowledge, been previously reported in the existing scientific literature or public databases. Actionable fusions were defined as those associated with FDA-approved targeted therapies or those that made patients eligible for a clinical trial. **Results:** Tumor samples from 2,776 patients with GI cancers were evaluated with a total of 75 actionable fusions identified in samples from 74 patients. Of these, 25.3% (n=19) were novel, and 74.7% (n=56) were previously described or recurrent fusions. Novel fusions were detected in small bowel (2 of 81), colorectal (4 of 1,621), gastric/gastroesophageal junction (GEJ) (2 of 259), pancreatic (7 of 466) and biliary (4 of 131) cancers. Thirteen (63.2%) of the novel fusions involved genes associated with FDA-approved therapies, including FGFR2 (n=6), BRAF (n=3), MET (n=2), FGFR1 (n=1) and NTRK2 (n=1). Of the fusions associated with FDA-approved therapies, 20% (8/40) of recurrent fusions and 38.5% (5/13) of novel fusions were detected only by RNA sequencing. No fusions were identified in appendiceal or esophageal cancers. **Conclusions:** Gene fusions with novel fusion partners represented one fourth of all actionable fusions detected in GI tumors. Use of a comprehensive WES/WTS testing approach that detects fusions agnostic of the fusion partner may affect clinical care in GI cancers by increasing the number of patients eligible for targeted therapies. Research Sponsor: Exact Sciences.

269 **Poster Session**

Impact of sirtuin genes expression on first-line treatment outcomes in patients (pts) with metastatic colorectal cancer (mCRC) enrolled in CALGB/SWOG 80405 (Alliance). First Author: Pooja Mittal, Division of Medical Oncology, Norris Comprehensive Cancer Center, Keck School of Medicine, University of Southern California, Los Angeles, CA

Background: Sirtuins (SIRT1-7) are class III histone deacetylases with different roles in CRC as tumor suppressive and oncogenic, mainly influencing proliferation, migration, invasion, DNA damage and chemoresistance. In the present study, we explored the association between tumor gene expression of sirtuins and first-line treatment outcomes in mCRC pts. **Methods:** We evaluated the impact of the tumor gene expression of sirtuin genes (*SIRT1-7*) on treatment outcomes (overall survival: OS; progression free survival: PFS) in 433 pts enrolled in the randomized, phase III CALGB/SWOG 80405 (NCT00265850) trial. RNA extracted from FFPE tumor samples was sequenced using the HiSeq 2500 (Illumina). Subgroup analyses were based on treatment (bevacizumab [bev] (N = 226) vs. cetuximab [cet] (N = 207)). For each gene, expression was divided into tertiles of high, medium and low. The associations between tumor gene expression and outcomes (PFS and OS) were assessed using log-rank test in univariate analysis. Multivariable Cox proportional hazards regression model was used to compute hazard ratios (HR) and 95% confidence intervals while adjusting for baseline characteristics. Treatment-gene interactions were analyzed using the likelihood ratio test. **Results:** Of the seven genes analyzed, *SIRT3*^{high} was associated with both longer PFS (HR: 0.59 [95% CI: 0.46-0.74], $P < 0.001$) and OS (HR:0.64 [95% CI: 0.50-0.83], $P = 0.002$) overall. Treatment-gene interaction testing found significant interactions for *SIRT1* and *SIRT5* with PFS (P values: 0.021 and 0.016, respectively) and OS (P values: 0.038 and 0.014, respectively) with targeted therapies (bev vs. cet). Multivariate analysis showed *SIRT1*^{high} associated with longer PFS (HR: 0.62 [95% CI: 0.42-0.89], $P = 0.014$) and OS (HR: 0.59 [95% CI: 0.40-0.89], $P = 0.036$) in cet-treated pts; in contrast with shorter observed PFS (HR: 1.38 [95% CI: 0.96-1.99], $P = 0.22$) and OS (HR: 1.31 [95% CI: 0.90-1.93], $P = 0.23$) in bev-treated pts. *SIRT5*^{high} showed similar results and associated with longer PFS (HR: 0.64 [95% CI: 0.45-0.91], $P = 0.039$) and OS (HR: 0.57 [95% CI: 0.39-0.83], $P = 0.013$) in cet-treated pts, with no similar associations in bev-treated pts. **Conclusions:** Our results highlight a potential role for *SIRT3* as prognostic and, *SIRT1* and *SIRT5* as predictive biomarkers for first-line treatment in mCRC, with differential effects based on targeted agents. Significant interaction favored cet treatment in *SIRT1*^{high} and *SIRT5*^{high} tumors. However, previous reports suggest an opposite role of *SIRT5* that has been linked to chemotherapy and/or cet resistance in CRC via the inactivation of succinate dehydrogenase complex subunit A. Therefore, further studies are warranted to dissect the mechanism behind the interaction of *SIRT5* with cetuximab activity and establish the potential of targeting *SIRT1/5* in CRC. Research Sponsor: National Cancer Institute; P30CA014089; National Cancer Institute; U10CA180821; National Cancer Institute; U10CA180882; SWOG; U10CA180888; National Cancer Institute; UG1CA180830; Genetech.

270 **Poster Session**

Prediction of immunotherapy response in deficient mismatch repair colorectal cancers using an immune-enhanced exome and transcriptome platform. First Author: Frank A. Sinicrope, Departments of Medicine and Oncology, Mayo Clinic Rochester, Rochester, MN

Background: While immune checkpoint blockade (ICB) is used to treat metastatic deficient mismatch repair (dMMR) colorectal cancers (CRCs), such tumors are heterogeneous and resistance to ICB is frequent. To date, biomarkers that can consistently predict patient response to ICB are lacking. We determined whether biomarkers that integrate tumor and immune-related molecular mechanisms could predict ICB response. **Methods:** Consecutive patients with metastatic dMMR CRCs (N=32) who had been treated with anti-PD-1 therapy were profiled using a validated immune-enhanced exome and transcriptome platform (ImmunoID NeXT, Personalis) using tumor and paired normal tissue. This platform provides antigen-specificity of T or B cells, indicated by the TCR or BCR respectively, neoantigen prediction, tumor mutation burden (TMB), immune profile, HLA allele-specific loss of heterogeneity (LOH) and a Neoantigen Presentation Score (NEOPS) that combines a neoantigen prediction tool with mechanisms of immune evasion. Tumor variables were analyzed in relationship to objective tumor response, and a Cox proportional hazards model was fit to predict progression-free survival (PFS) or overall survival (OS). **Results:** In CRCs, TCR and BCR repertoire diversity were each significantly increased in tumors from responders versus those with stable or progressive disease after anti-PD-1 treatment (p=0.015 and p=0.048, respectively). Neoantigen burden score, TMB, and HLA allele-specific LOH (reflects loss of antigen presentation machinery) were each unable to distinguish responders from nonresponders. An immune checkpoint gene expression signature was increased in nonresponders relative to responders (P= 0.005). Neoantigen presentation score (NEOPS) was prognostic when adjusted for tumor grade only for OS (p=0.048). **Conclusions:** The T-cell and B-cell receptor (TCR/BCR) repertoire, as well as NEOPS, were each significantly predictive of favorable response to anti-PD-1 therapy in dMMR tumors. In addition, an immune checkpoint gene expression signature was predictive of resistance to ICB. Research Sponsor: None.

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Differential gut microbiota profiles among normal colorectal mucosa, adenomatous and hyperplastic polyps in healthy volunteers: ColoMAR-1 study. First Author: Maria Dolores Martos-Morillo, Hospital Vithas Madrid La Milagrosa, Madrid, Spain

Background: Increasing evidence indicates that the interaction between intestinal microbiota and host cells plays an important role in colorectal cancer oncogenesis through immune, inflammatory and/or metabolic alterations. However, there are no clear evidence in the very early adenoma-carcinoma sequence at which microbiota species influences future steps to colorectal carcinoma. This study aims do find a point in the adenoma-carcinoma sequence ad witch microbiota species suggest the risk of a future tumor to develop less invasive forms of early diagnosis. **Methods:** In this cross-sectional study, healthy volunteers above 45 years old were selected to participate. Exclusion criteria includes previous colonoscopy performed in the last 5 years. Stool sample were collected before a colonoscopy with biopsy. Samples of normal mucosa were systematically obtained, and all polyps found were removed. In fecal samples, microbiota species were identified using GoodGut test: *Bacteroides fragilis, Subdoligranulm variabile, Ruminococcus, Roseburia, Coprococcus*, eubacteria, *Faecalibacterium prausnitzii* and *Gemella morbillorum*. Number of genomic copies/μL were analyzed using the Wilcoxon test. **Results:** From February to June 2024, 29 healthy volunteers between 45 and 69 years were recruited. 62% were female. Adenomatous polyps or hyperplastic polyps were found in 9 volunteers. No case of colorectal cancer was found. For species *S. variabile, Ruminococcus, Roseburia, Coprococcus* and *F. prausnitzii*, higher counts were found in stool samples from volunteers with adenomatous or hyperplastic polyps than in those from volunteer with a normal colonoscopy, both in the total count and after correction for bacterial load ($p<0.001$). For *B. fragilis*, bacterial count was higher in samples from volunteers with adenomatous polyps, but not in the samples from people with hyperplastic polyps compared with those samples from volunteers with a normal colonoscopy. *G. morbillorum* counts were not related to any type of polyps. None of this counts met the laboratory-designated threshold related to colorectal cancer. **Conclusions:** The presence of *B. fragilis* could be related to the differential risk of malignancy of adenomatous compared to hyperplastic polyps. Research Sponsor: None.

Bacterial count for each sample and specie.

Bacteria specie*	Normal mucosa	Adenoma	Hyperplastic
B. fragilis	22.13	22.3	20.42
S.variable	6.06	7.29	7.53
Ruminococcus, Roseburia, Coprococcus	6.08	7.29	7.47
F. prausnitzii	4.7	6.28	7.09
G.morbillorum	20.91	20.54	20.84

*Number of genomic copies/μL.

272 Poster Session

Biological and therapeutic impact of tumor mutational burden in microsatellite instability/mismatch repair-deficient (MSI/dMMR) metastatic colorectal cancer (mCRC) treated with immune checkpoint inhibitors (ICI). First Author: Sarah Blanchet Deverly, Department of Medical Oncology, Saint-Antoine Hospital, AP-HP, Paris, France

Background: Microsatellite instability (MSI) is linked to hypermutability and response to ICI(s), especially in colorectal cancer. The predictive impact of tumor mutational burden (TMB) in ICI-treated MSI/dMMR metastatic colorectal cancer remains controversial, with some studies suggesting low TMB as a potential biomarker of resistance to ICI(s). We aimed to investigate the biological relevance of TMB in MSI/dMMR mCRC and its impact on survival outcomes in ICI(s)-treated patients (pts). **Methods:** MSI/dMMR mCRC pts treated with ICI(s) from the NIPICOL trial (NCT03350126) and the immunoMSI prospective monocentric cohort with available data from whole exome sequencing (WES) and RNA-sequencing (RNA-seq) were included. All cases were centrally assessed for MSI and dMMR status. TMB was calculated as described (Dupain et al., BMC, 2024). The search for a cutpoint able to optimize the difference in survival between 2 groups of patients with low and high-TMB was applied. The main endpoint was progression free survival (PFS) by iRECIST criteria. The cohort (NIPICOL, immunoMSI) was included as a covariate in the analysis of differential gene expression and pathway enrichment, with various tumor mutational burden (TMB) thresholds evaluated, specifically 10 and 30 mutations per megabase (mut/Mb). **Results:** A total of 99 pts were analyzed, with 4 (4%) excluded after central MSI/dMMR status review. Of these, 52 received anti-PD1 plus anti-CTLA4 therapy, while 43 received anti-PD1 alone. Median follow-up was 25 months. Median TMB was 51.8 mut/Mb (IQR: 35.9-74.9). TMB was not predictive of patient PFS, whatever the TMB threshold applied, including cutpoints already reported in the literature, the 2-year PFS rates being 80% versus 80% for TMB $\leq$ 10 and $>$ 10 mut/Mb, and 79% versus 82% for TMB $\leq$ 30 and $>$ 30 mut/Mb, respectively. Multivariate analysis including treatment type confirmed no association between TMB and PFS (HR 1.00, 95% CI [0.97–1.04]). TMB levels were not linked with specific signalling pathways related to immune checkpoints or antigen presentation as evaluated by RNA-seq with Gene Set Enrichment Analysis (GSEA) at a TMB threshold of 10 or 30 mut/Mb. However pathways involved in cell proliferation (e.g., "cell cycle", "DNA replication") were increased in TMB high (adjusted p-value <0.05). RNA-seq signature quantification and deconvolution analysis revealed similar cell populations (e.g., fibroblasts, B cells, T cells) between high and low TMB groups at a TMB threshold of 10 and 30 mut/Mb. **Conclusions:** TMB has no survival impact for patients with ICI(s)-treated dMMR/MSI mCRC and does not discriminate two biologically distinct subpopulations. Other biomarkers should be developed for personalized medicine amongst these patients. Research Sponsor: GERCOR; Ligue nationale contre le cancer; BMS; MSD AVENIR.

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Serial monitoring of circulating tumor DNA as a marker of response to immune checkpoint blockade in MSI-H gastrointestinal tumors. First Author: Victoria Higbie, Hematology/Oncology, The University of Texas MD Anderson Cancer Center, Houston, TX

Background: Microsatellite instability- high (MSI-H) tumors have high rates of response to immune checkpoint blockade (ICB) with prolonged durability of response. However, predicting who will have a response is difficult and there are limitations to radiographic evaluation of response. As circulating tumor DNA (ctDNA) offers a noninvasive approach to monitoring response and therapeutic efficacy, we sought to correlate ctDNA change with response to ICB in MSI-H gastrointestinal (GI) tumors. **Methods:** We conducted a retrospective study of MSI-H GI patients receiving ICB who underwent serial ctDNA sequencing. Tissue-informed ctDNA monitoring utilized comprehensive genomic profiling results of pre-treatment tumor tissue from a clinical trial assay based on FoundationOne CDx, a variant selection algorithm to exclude non-tumor variants, and multiplex PCR of up to 16 patient-specific variants to detect and quantify ctDNA. ctDNA changes were correlated with radiographic response. **Results:** A total of 20 patients with MSI-H tumors were evaluated including 17 colon, 2 small bowel, and 1 pancreatic. For 12 patients, it was the first exposure to ICB whereas 8 had previously progressed and were being rechallenged with ICB. Response was seen in 9 patients with a best overall response (BOR) of complete response (CR) in 4 and partial response (PR) in 5 while 5 showed stable disease (SD) and progression (PD) in 6. All patients had baseline ctDNA samples and at least 1 serial ctDNA after ICB initiation (median number of samples was 3). The mean time from baseline to first on-treatment sample was 30 days (range 13-64 days). 4 patients had undetectable ctDNA levels at baseline and remained undetectable on treatment. Of the 16 that had detectable ctDNA at baseline, 11 had a decrease in ctDNA levels at 1st on-treatment sample and 5 had an increase. The mean percent change of ctDNA at 1st on-treatment for those who had a response (CR + PR) was a 76% decrease while the mean percent change for those with SD was a 7% increase and for those with PD a 72% increase (p=0.05). Of the 6 patients with BOR of PD, 5 had an increase in ctDNA and 1 had undetectable levels at baseline. Of the 5 patients with BOR of SD, 2 had a decrease, 1 had an increase, and 2 were undetectable at baseline. Duration of SD did not correlate with initial ctDNA response. **Conclusions:** Early ctDNA change correlated with response in MSI-H GI tumors suggesting that serial ctDNA monitoring may provide early indication of response even prior to radiographic evaluation. Research Sponsor: None.

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Investigating the clinical and molecular characteristics of class II and III BRAF mutations and their response to anti-EGFR therapy in MSS CRC: A comprehensive analysis. First Author: Ibrahim Halil Sahin, University of Pittsburgh Medical Center (UPMC), Pittsburgh, PA

Background: *BRAF* mutations (mts) are a heterogeneous group of molecular alterations seen in colorectal cancer (CRC). Class I *BRAF* mts (V600) are known to be associated with aggressive biology to CRC, but the knowledge of clinical characteristics of class II and III *BRAF* mts is limited. In this large cohort analysis, we evaluated the clinical and molecular features of class II and III *BRAF* mts and performed comparative analyses for their impact on survival outcomes. **Methods:** A total of 24402 MSS CRC samples were profiled by NGS (592-gene, NextSeq; WES, WTS NovaSeq) (Caris Life Sciences, Phoenix, AZ). *BRAF* mts were detected by NGS and classified using published literature (1). MAPK pathway activity score (MPAS) was calculated using RNA expression data. Real-world overall survival (OS) was obtained from insurance claims and calculated from tissue collection to last contact, while post-treatment survival from first treatment to last contact. KM estimates were calculated for defined patient cohorts. Significance was determined as $P<0.05$. **Results:** A total of 1270, 134, and 327 patients with class I, II, and III *BRAF* mts were identified. *BRAF* mts overall and class I *BRAF* mts were significantly less common among younger pts (age$<$50) compared to patients $\geq$ 50 (4.7% vs. 7.4 %; $P<0.001$ and 3.2% vs. 5.7%, respectively, $P<0.001$). Class I *BRAF* mts were enriched with consensus molecular subtype 1 (CMS1) (class I, II, and III: 46% vs. 14% vs. 18% respectively) while class II and III mts had more CMS2 subtype compared to class I (2%, 30% and 29%, $p<0.05$). Higher MPAS scores were noted among patients with class I *BRAF* mts than those with class II and III mts (1.73 versus 0.38 vs 0.79). Class I *BRAF* mts and KRAS mts were nearly mutually exclusive (0.5%), while KRAS mts were relatively common among class II and III mts (12.8% and 27.4%, respectively). Median OS for patients with wild-type *BRAF*, class I, II, and III *BRAF* mts were significantly different (29.8, 17.3, 21.8, and 24.6 months, respectively, $p<0.01$). Similar OS was seen among patients who did not receive anti-EGFR therapy (28.3, 15.2, 20.7- and 21.9 months $P<0.001$). Patients with class III mts had significantly better OS compared to patients with class I mts (HR=1.28 CI: 1.09-1.48 p=0.002), and while no significant difference was noted for patients with class II vs class I mts, numerically more favorable outcomes were noted (mOS 21.8 vs 17.3 months P=0.133). Among treated with anti-EGFR, patients with class II and III *BRAF* mts had numerically better post-anti-EGFR survival compared to class I mts; however, this was not statistically significant (21.2 vs 14.2 vs. 10.4 respectively P=0.3). **Conclusions:** Class II and III *BRAF* mts represent a distinct biological subgroup of MSS CRC with distinct prognoses. Class II and III *BRAF* mts are associated with improved survival outcomes compared to patients with class I mts. 1. Sahin et al. JCO OP 2021. Research Sponsor: None.

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Identifying the optimal post-surgical timing of molecular residual disease (MRD) detection in colorectal cancer (CRC) using an ultra-sensitive assay: Interim results from the VICTORI study. First Author: Joao Paulo Solar Vasconcelos, BC Cancer - Vancouver, University of British Columbia, Vancouver, BC, Canada

Background: While detection of MRD using ctDNA is prognostic for recurrence in CRC, some patients still recur prior to MRD detection. VICTORI is prospectively investigating NeXT Personal, an ultra-sensitive NGS-based MRD assay, to profile patients with resected CRC. **Methods:** Patients with CRC treated with curative intent (all stages) are tested for MRD using NeXT Personal, a bespoke assay with up to ~1,800 tumor-informed single nucleotide variants (SNVs) identified from whole-genome sequencing. Plasma is collected prior to surgery, every 2 weeks post-surgery up to week 8 (MRD landmark window), and every 3 months for up to 3 years (surveillance). We present preliminary results on 397 samples from the first 62 patients. **Results:** A total of 62 patients (N=36 rectal [58%], N=26 colon [42%]; N=46 stage I-III [74%], N=16 stage IV [26%]) were included in our analysis. Baseline pre-surgical sensitivity (treatment naive) was 93.5% [N=29/31]. Pre-surgical positivity rate in patients who had received neoadjuvant therapy and had residual cancer at the time of surgery was 67% [N=16/24]. 60 patients were evaluable for clinical outcomes. At a median follow-up of 355 days, 15 patients (25%) had a recurrence. Of these, all patients were ctDNA-positive prior to recurrence (100%, 14/14; 1 pt excluded due to lack of samples prior to recurrence). ctDNA detection preceded clinical relapse by a median of 194 days [range: 5-397]); ctDNA for 78.6% (N=11/14) were first detected in the MRD landmark window. All landmark-positive recurrences occurred within one year of surgery. The 14 recurrent cancers with samples were first detected at a median ctDNA concentration of 28.7 parts per million (PPM) (range 2.4-111,120), with 64.3% (9/14) of those detections in the ultra-low range of <100 PPM. MRD detection at week 4 and week 8 had the greatest reduction in RFS (HR 12.86 [2.74-60.28], p=0.0012 week 4; HR 16.14 [3.52-74.09], p=0.0004 week 8), with weeks 4, 6 and 8 having similar higher prevalence of ctDNA detection (36.0% [18/50], 35.3% [18/51], 38.5% [20/52] respectively). Week 2 detection rate was 17.0% (8/46). cfDNA concentration was highest at week 2 (4.63ng/ml vs. 2.22 at baseline, p=0.00028) and 4 (3.68 vs. 2.22, p=0.0081), returning to baseline levels at week 6 (2.28 vs. 2.22, p=0.67) and 8 (2.48 vs. 2.22, p=0.64). **Conclusions:** In this interim report after a median ~1 year follow-up, NeXT Personal detected MRD for all patients prior to disease recurrence. Most initial MRD detection was in the ultra-sensitive range <100ppm and detection at 4-8 weeks after surgery was highly prognostic for recurrence. Research Sponsor: Personalis, Inc.

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Clinical and molecular characterization of *FAP* and *SPP1* in colorectal cancer (CRC), CALGB (Alliance)/SWOG 80405 and real-world data. First Author: Karam Ashouri, Division of Medical Oncology, Norris Comprehensive Cancer Center, Keck School of Medicine, University of Southern California, Los Angeles, CA

Background: *FAP* and *SPP1* contribute to immune modulation in the CRC tumor microenvironment (TME). FAP is expressed by cancer associated fibroblasts, aiding in tissue remodeling and tumor invasion. SPP1 is an integrin-binding protein expressed by tumor associated macrophages that promotes tumor growth, adhesion, and metastasis. We present a clinical and molecular characterization of *FAP* and *SPP1* in CRC. **Methods:** We analyzed 24,257 CRC samples tested at Caris Life Sciences (Phoenix, AZ) with WTS (Illumina NovaSeq), NextGen DNA sequencing (NextSeq), and PD-L1 expression (SP142, positive ≥ 2+, 5%). RNA deconvolution analysis estimated cell infiltration in the TME. Data from the phase 3 CALGB/SWOG 80405 trial (NCT00265850) on 433 metastatic CRC patients treated with bevacizumab (Bev, n = 226) or cetuximab (Cet, n = 207) in combination with first-line chemotherapy were also evaluated. RNA isolated from FFPE tumor samples were sequenced with HiSeq 2500 (Illumina). Overall survival (OS) and progression-free survival (PFS) were compared using Cox regression in categorical gene expression tertiles (high (T3), medium (T2), and low (T1)) for *FAP* and *SPP1*. **Results:** *FAP*-T3 and *SPP1*-T3 had increased PD-L1 positivity (q<0.05), while *SPP1*-T3 also demonstrated increased MSI-H (8.4% vs 5.3%) and TMB-H(>10 mt/mb, 15.0% vs 10.1%) status relative *SPP1*-T1 (all q<0.001). T3 of both genes of correlated with increased M1/M2 macrophages, NK cells and T cell inflamed score (q<0.001); dendritic cells and neutrophils were increased in *SPP1*-T3 and *FAP*-T1 tumors (q<0.05). *FAP*-T3 and *SPP1*-T3 had increased pathway activation of epithelial-mesenchymal transition (EMT), inflammatory response, TNF-a signaling, angiogenesis and *KRAS* signaling (all $q < .005$). In Caris cohorts, *FAP*-T3 demonstrated worse OS in Cet/panitumumab treated CRC (T3: 23.6 vs T1: 21.0 months [mo], $P = .005$; HR 0.85, 95% CI [0.77-0.95]), but *SPP1* did not correlate with OS. In 80405, *FAP*-T3 showed shorter PFS (T3: 9.5 vs T2: 11.5 vs T1: 12.6 mo, T3 vs T1 (reference) adjusted HR 1.27 [1.07-1.51]) and OS (25.2 vs 29.4 vs 35.5 mo, adjusted HR 1.31 [1.10-1.57]). Similarly, *SPP1*-T3 demonstrated worse PFS (9.0 vs 12.7 vs 14.0 mo, adjusted HR 1.29 [1.12-1.48]) and OS (20.9 vs 34.0 vs 36.3 mo, HR 1.24 [1.07-1.44], $P < 0.001$). Treatment interaction tests noted *FAP*-T1 CRC benefited from Cet over Bev with respect to PFS ($P = 0.003$) and OS ($P = 0.044$), while SPP1-T1 tumors demonstrated a PFS benefit with Cet ($P = 0.009$). **Conclusions:** Our results indicate that increased *FAP* and *SPP1* expression is associated with immune cell infiltration, EMT, and inflammatory signaling. Additionally, their expression may be prognostic and predictive of targeted therapy. These data support the evaluation of *FAP* and *SPP1* as predictive markers and therapeutic targets in CRC. Research Sponsor: P30CA014089, UG1CA180830 [HJL]; UG1CA180830 and U10CA180888 (SWOG).

277 Poster Session

Acquired gene alterations potentially related to resistance to anti-HER2 therapy in HER2-positive metastatic colorectal cancer (mCRC). First Author: Celine Hoyek, Division of Hematology and Oncology, Mayo Clinic Arizona, Phoenix, AZ

Background: While HER2-targeted therapies benefit patients (pts) with advanced HER2-positive mCRC, the development of resistance remains inevitable. This study aims to explore the landscape and frequency of genetic alterations putatively associated with acquired resistance to anti-HER2 therapy in HER2-positive mCRC. **Methods:** In this retrospective-multicenter international study involving Mayo Clinic, Duke Cancer Center and National Cancer Center Hospital East (Japan), we descriptively compared comprehensive genomic analyses from tissue or blood samples of pts with HER2-positive CRC pre-treatment and post-treatment with HER2-targeting agents. **Results:** A total of 36 pts were included in this study. 29 (80%) had HER2-positive mCRC as confirmed by immunohistochemistry (IHC) and fluorescent in situ hybridization (ISH), the remaining pts had HER2 amplification detected on blood or tissue . 34/36 (94%) pts were administered their first anti-HER2 agent after progressing on ≥2 prior lines of treatment. The most common anti-HER2 regimen administered was trastuzumab/pertuzumab (50%, 18/36), followed by trastuzumab-tyrosine kinase inhibitors (TKIs) (tucatinib or neratinib) (30%, 11/36) and trastuzumab deruxtecan (14%, 5/36). Putative genomic mechanisms of acquired resistance were detected in 72% (26/36), while the remaining pts lacked new genomic alterations that might explain resistance to anti-HER2 therapy, suggesting alternative escape mechanisms. Acquired alterations in downstream or alternate bypass pathways were identified in 72% (26/36) of pts; most commonly in the RAS/RAF pathway and in EGFR (34.6%, 9/26 each), followed by 27% (7/26) of pts with *MET* point mutations (mts) and amplification; activating muts in PI3K/AKT and in 27% (7/26), most commonly CDK12 (60%, 4/7). HER2 receptor alterations were detected in 3 pts (8.3%), all of whom received trastuzumab-tucatinib and included *HER2 L755S, HER2 V777L, HER2 D769Y, HER2 G727A* and *HER2 S310F,* which are known to cause resistance to , however, information about tucatinib resistance is not yet available. Other altered pathways included AR, RB, NOTCH and HRD. Two pts lost HER2 expression on blood and tissue NGS post anti-HER2 therapy. **Conclusions:** The emergence of resistance might be in part related to acquired alterations that constitutively activate signaling pathways parallel or downstream of HER2, bypassing HER2 inhibition. Additional histologic studies, deep mutational scanning, larger populations and correlative analysis are needed to validate these findings. Research Sponsor: None.

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Comprehensive landscape of FGFR variations and prognosis revelance in colorectal cancer from circulating tumor DNA and tissue gene analyses in 2083 patients. First Author: Xiaoshuang Lyu, The Sixth Affiliated Hospital, Sun Yat-sen University, Guangzhou, China

Background: FGFR genomic alterations have been identified, and several FGFR inhibitors are approved for various malignancies, including bladder and cholangiocarcinoma. However, limited studies have shown FGFR alterations and their prognostic impact in colorectal cancer (CRC). **Methods:** This retrospective analysis involved tissue and blood samples from a single-institute cohort (The Sixth Affiliated Hospital of Sun Yat-sen University) of stage I-IV CRC patients from 2014 to 2024. FGFR genomic alterations were analyzed using tissue NGS and plasma ctDNA. Disease-free survival (DFS) for stages I-III and progression-free survival (PFS) for stage IV were compared between FGFR-altered and negative CRC patients, matched in a 1:2 ratio. Validation of genomic alterations and their prognostic impact was conducted using cBioPortal and TCGA data. **Results:** FGFR alterations were found in 7.89% of 608 tissue samples and 10.95% of 2083 ctDNA samples. The most frequent alteration was FGFR1 amplification in all cohorts, along with FGFR2 p.R165W and FGFR3 p.A634T mutations in ctDNA. FGFR alterations correlated with younger age, left-sided colon tumors, adenocarcinomas, poor differentiation, advanced TNM staging, and multi-organ metastasis. Prognostic analysis showed a higher proportion of FGFR alterations in stage IV patients (66.23% vs. 59.14%), lower rates of no evidence of disease (NED) (6.62% vs. 17.68%), and significantly shorter PFS (median 17 vs. 20 months, HR 2.184, p=0.007). In stages I-III, FGFR-altered patients also had a lower NED rate (79.22% vs. 82.19%) and significantly shorter DFS (median 13 vs. 29 months, HR 2.833, p=0.002). Among frequent alterations, FGFR1 amplification was significantly associated with poor prognosis (PFS 17 vs. 20 months, HR 2.221, p=0.014; DFS 13 vs. 29 months, HR 2.964, p=0.002), while others showed no significant correlation. In stage IV patients receiving bevacizumab, FGFR1 alteration was linked to significantly shorter PFS (p=0.0021). KEGG analysis indicated FGFR1 was associated with angiogenesis pathways. FGFR alteration profiles and prognostic outcomes were consistent across tissue samples, ctDNA analyses, and databases. **Conclusions:** Our study outlines the genomic landscape and prognostic significance of FGFR in CRC, highlighting FGFR1 amplification as the dominant alteration. FGFR alterations were associated with poorer outcomes, as reflected in lower NED rates and shorter DFS and PFS. FGFR1 amplification was notably linked to reduced DFS and PFS, and it correlated with shorter PFS in patients receiving bevacizumab, indicating a potential link to resistance against bevacizumab. Research Sponsor: None.

	ctDNA n=2083	Tissue gene n=608	cBioPortal Database n=6998
FGFR Total	228/10.95%	48/7.89%	528/8.0%
FGFR1	76/3.65%	29/4.77%	264/4.0%
FGFR2	78/3.74%	10/1.64%	125/1.9%
FGFR3	104/4.99%	10/1.64%	139/2.1%

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Clinical-pathologic characteristics in young patients with metastatic colorectal cancer: Experience from a specialized cancer institute in Peru. First Author: Eder Christian Veramendi Cabana, Instituto Nacional de Enfermedades Neoplásicas (INEN), Lima, Peru

Background: Colorectal cancer (CRC) represents the third cause of malignant neoplasm in the world and the second in terms of mortality. CRC is a major cause of cancer-related morbidity and mortality in Peru. Diagnosis at younger ages represents a challenge. In this study, we evaluated prognostic factors in patients who were metastatic at diagnosis or progressed to metastatic disease during follow-up. **Methods:** A descriptive cross-sectional study, between 2013 and 2022, 66 patients under 40 years old and stage IV at diagnosis were analyzed. Parameters evaluated were age, gender, comorbidity, performance status and stage of the disease at the diagnosis, localization, surgery undergone, systemic treatment regimen, response to first-line treatment, incidence of K-RAS, sites and number of metastases, expression of tumor predictors (CEA levels), and survival times. A multivariate analysis was conducted with factors that were considered statistically significant in the univariate analysis. **Results:** Median age was 35 years [29.25-37] and the female/male ratio was 34/32. From 66 patients at stage IV disease, 30 patients (45.5%) were right-side, 36 (54.5%) were left-side colon cancer. At the end of follow-up, 51 patients had died, and 15 survived. Overall survival (OS) was significantly better in patients older than 35 years old ($p<0,05$), those who underwent primary tumor surgery ($p<0,001$), negative perineural invasion ($p<0.0045$). The treatment most common was CAPOX with a median OS of 17 months ($p<0.013$). The multivariate analysis shows that patients who did not undergo primary surgery had worse disease-specific OS (hazard ratio, CI 2.21.95% [1.15-4.25], $p< 0.018$) and there was no statistical difference between chemotherapy regimens and treatment line. **Conclusions:** In young patients with metastatic CRC, Primary tumor surgery and negative perineural invasion are significant factors influencing overall survival. Patients older than 35 years and those who underwent surgery showed better survival. These findings reflect the importance of early colorectal cancer screening and improve the risk assessment. Research Sponsor: None.

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Challenges in patient selection for surgical options in V600E BRAF–mutated metastatic colorectal cancer. First Author: Giulia Maddalena, Department of Gastrointestinal Medical Oncology, Division of Cancer Medicine, The University of Texas MD Anderson Cancer Center, Houston, TX

Background: ^{V600E} BRAF mutated metastatic colorectal cancer (mCRC), which represents almost 10% of mCRC, is associated with a decreased disease-free interval and increased risk of recurrence compared to other mCRC molecular subtypes; however, data show wide outcome heterogeneity within this subgroup. Surgery is known to be associated with improved outcomes in patients with mCRC, particularly in those with liver-only disease. While many patients with ^{V600E}BRAF have very aggressive disease and never have the chance to undergo surgery, it is not clear how to navigate the clinical heterogeneity and choose patients that might benefit the most from loco-regional treatment. **Methods:** We retrospectively evaluated^{V600E} BRAF mutated mCRC patients from two independent high-volume institutions (MDACC and IOV IRCCS) who underwent metastectomy from Oct 2009 to Oct 2023. Patients with MSI-H disease and those who had R2 surgery were excluded. Clinicopathological features, such as sex, age, histology, radicality and site of surgery, were collected from institutional datasets. Relapse free survival (RFS) was defined as the time from surgery to first relapse or progression. Overall survival (OS) was defined as the time from surgery to the date of death or last follow up. Survival was estimated with the Kaplan Meier method, and the Cox proportional Hazard model was adopted for survival comparison. **Results:** We collected data from 71^{V600E} BRAF mutated mCRC patients (N=37 MDACC, N=34 IOV IRCCS); 49% were female, 61% had right-sided primary tumor and 62% had synchronous metastatic disease. Median age at metastatic disease diagnosis was 62 years old (IQR: 61-28). For 52% of patients, surgery was performed on liver-only disease; 35% involved the peritoneum. There were no emergency interventions. Median RFS was 7.9 months (95% CI: 5.4 – 10.5) and median OS was 39 months (95% CI: 28.8 – 51.7). No clinicopathological features, including sex, sidedness, histology, differentiation, presentation (synchronous, oligometastatic), or site of intervention, were associated with RFS. However, radicality (R1) was associated with worse RFS (HR=3.5 [95% CI: 1.8 – 6.5], p=0.0003) and OS (HR=2.7 [95%CI: 1.3 – 5.6], p=0.0122). **Conclusions:** In this retrospective evaluation of patients with ^{V600E} BRAF mutated mCRC who underwent metastectomy, only surgical radicality, observed in the post-operative setting, was associated with recurrence and survival outcomes. Therefore, while surgery should be evaluated whenever radical resection is feasible, the identification of patients with ^{V600E} BRAF mutated mCRC who might benefit the most from metastasectomy during pre-operative prognosis evaluation remains challenging. Research Sponsor: None.

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Role of *RAS/BRAF* and *PIK3CA* mutations in tissue and plasma for prognostic assessment in metastatic colorectal cancer (mCRC). First Author: Eduardo Terán-Brage Sr., Vall d'Hebron University Hospital, Barcelona, Spain

Background: Mutations in *RAS/BRAF* (*RAS/BRAFm*) and *PIK3CA* (*PIK3CAm*) genes have been described as mechanisms of resistance to anti-EGFR agents in mCRC. However, the utility of liquid biopsy in their baseline detection remains unknown. We aim to analyse oncological outcomes according to *RAS/BRAF* and *PIK3CA* mutational status in tissue & plasma in mCRC. **Methods:** We conducted aretrospective study in patients (pts) with mCRC from 2017 to 2024. *RAS/BRAF* and *PIK3CA* mutational status assessment was performed at baseline using Idylla (Biocartis) and Cobas PIK3CA kits, respectively. Cox models were used to analyse overall survival (OS) in *RAS/BRAFm* and *PIK3CAm* vs wild-type (WT) population. **Results:** We analysed 404 paired samples (tissue & plasma) from 202p with mCRC. *RAS/BRAFm* was described in 45% (91p) in tissue and in 43.6% (88p) in plasma. *PIK3CA* was tested in 84p, with *PIK3CAm* detection in 19% (16p) in tissue and 11.9% (10p) in plasma. Liver involvement was more frequent in *RAS/BRAFm* vs WT (80.6% vs 63.9%; p=0.02), and in *PIK3CAm* vs WT population (94.4% vs 63.6%; p=0.01). Concordance study (tissue & plasma) for *RAS/BRAF* and *PIK3CA* are presented (Table). Subjects with concordant results had worse OS in *RAS/BRAFm* (18.1m vs 51.1m, HR=2.60; $p<0.001$) regardless of liver involvement, and showed the same trend in *PIK3CAm* (22.6m vs 28.1m, p=0.81). Patients with discordant results and plasma *RAS/BRAFm* or plasma *PIK3CAm* showed worse OS (23.7 m vs 51.1m; p = 0.006 and 16.4 m vs 42.5 m; p=0.18, respectively) and a lower disease control rate with anti-EGFR therapy (p = 0.04 and p=0.42, respectively). *PIK3CAm* were evenly distributed in *RAS/BRAF* WT vs mutated tumors (50% vs 51.5%; p=0.92) and had no independent prognostic impact in OS after stratifying by *RAS/BRAF* mutational status; *RAS/BRAF* WT (p=0,93) and *RAS/BRAFm* (p=0,91). **Conclusions:** Baseline determination of *RAS/BRAFm* and *PIK3CAm* by liquid biopsy in mCRC provide valuable prognostic and predictive information, extending its utility beyond tissue-based analysis. Research Sponsor: None.

Correlation study of *RAS/BRAF* and *PIK3CA* mutational status in tissue and plasma in mCRC.

***RAS/BRAF* tissue**					***PIK3CA* tissue**			
***RAS/BRAF* Plasma**	**Mutated n (%)**	**WT n (%)**	**Total n (%)**		***PIK3CA* Plasma**	**Mutated n (%)**	**WT n (%)**	**Total n (%)**
Mutated n (%)	72 (80)	15 (13.4)	87 (43.1)		Mutated n (%)	7 (43.8)	2 (2.9)	9 (10.7)
WT n (%)	18 (20)	97 (86.6)	115 (56.9)		WT n (%)	9 (56.2)	66 (97.1)	75 (89.3)
Total n (%)	90 (100)	112 (100)	202 (100)		Total n (%)	16 (100)	68 (100)	84 (100)
	Positive agreement: 72/90: 80%					Positive agreement: 7/16: 43.8%		
	Negative agreement: 97/112: 86.6%					Negative agreement: 66/68: 97.1%		
	Overall agreement: 169/202: 83.6%					Overall agreement: 73/84: 86.9%		
	KI*: 0.67					KI: 0.49		

KI: Kappa Index.

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Investigating survival outcomes in patients with early- vs late-onset colorectal cancer with liver metastases. First Author: Judy Li, Department of Surgery, Division of Surgical Oncology, Icahn School of Medicine at Mount Sinai, New York, NY

Background: Approximately 50% of patients with colorectal cancer develop liver metastases (CRLM). The incidence of early onset colorectal cancer (EOCRC), defined as <50 years of age at time of diagnosis, is increasing. In this study, we aim to explore the prognosis of early vs late onset colorectal cancer patients with CRLM who receive curative-intent treatment. **Methods:** A retrospective study was conducted to assess clinical outcomes in patients with CRLM presenting with EOCRC compared to those with late onset colorectal cancer (LOCRC). All patients underwent surgical resection and/or ablation between 2009 and 2024 at a tertiary care center. Groups were defined by patient age at the time of diagnosis of the primary tumor $<$ 50 years versus $\geq$ 50 years. Demographics, tumor characteristics, and treatment modalities were compared between groups using Chi-square, independent samples t-test, and median test. Survival outcomes were assessed using Kaplan-Meier. **Results:** 342 patients met inclusion criteria. There were 85 (24.9%) EOCRC patients, with a median age at diagnosis of 44 years versus 63 years in the LOCRC group. EOCRC patients had fewer major medical comorbidities. EOCRC patients had a smaller proportion of right-sided (15.9% vs 29.6%, P=0.049) tumors. Rates of synchronous metastases (74.1% in EOCRC vs 72.4%, P=0.781), median size (28cm in EOCRC vs 25cm, P=0.421) median number of metastases (2 in EOCRC vs 2, P=0.385), and initial CEA levels (339.18 + 1276.48 ng/ml vs 328.87 + 2520.59 ng/ml, P=0.482) were similar between groups. However, EOCRC patients had higher rates of node-positive disease (75.6% vs 58.8%, P=0.010) and Fong Clinical Risk Scores 3-5 (57.0% vs 39.7%, P=0.009). Both groups underwent similar operative management with no differences in morbidity, but more EOCRC patients received neoadjuvant therapy (82.4% vs 66.1%, P=0.017). There were no significant differences in *RAS/RAF* (37.6% in EOCRC vs 34.2%, P=0.669) or *TP53* (25.9% vs 34.6%, P=0.98) mutational status or microsatellite instability (stable in 74.1% in EOCRC vs 69.3%, P=0.523). Both intra- (46.5% in EOCRC vs 50.6%, P=0.535) and extra-hepatic (47.7% in EOCRC vs 38.1%, P=0.129) recurrences occurred at similar rates, with no significant differences in post-recurrence treatments received. Patients with CRLM and EOCRC had significantly worse overall survival, with a median time of 80 months versus 116 months (P=0.042). **Conclusions:** Despite similar clinical, pathologic, and genetic features, fewer medical comorbidities, and receiving similar medical and surgical treatments, EOCRC patients had worse median overall survival than LOCRC patients. High risk Fong Clinical Risk Scores were more common in the EOCRC group, who also received neoadjuvant therapy at higher rates. Additional research is needed to understand the differences in survival. Research Sponsor: None.

283 Poster Session

Sex differences in the colorectal tumor immune microenvironment. First Author: Elizabeth Seitz, Case Western Reserve University School of Medicine, Cleveland, OH

Background: Males have increased risk and mortality rates of colorectal cancer (CRC) compared to females, but the reasons for these disparities are not well understood. Immune responses differ by sex, and a robust T cell response is an important prognostic indicator in CRC. We hypothesized that females exhibit a stronger CRC-associated T cell response, contributing to reduced mortality in females compared to males. **Methods:** The study cohort included 2,751 incident CRC patients (1,337 female; 1,414 male) from a population-based case-control study. T cell receptor (TCR) abundance and clonality were quantified using immunoSEQ, and a gastrointestinal pathologist scored tumor-infiltrating lymphocyte counts per high-powered field (TILs/hpf). We used logistic regression to estimate odds ratios (ORs) and 95% confidence intervals (CIs) for the associations between T cell metrics and sex, adjusting for age, anatomic site, and microsatellite instability (MSI) status. Cox proportional hazards regression was used to estimate hazard ratios (HRs) and 95% confidence intervals for the association between sex and 5-year all-cause and CRC-specific survival, adjusting for known prognostic indicators. We also conducted a formal test for interaction between sex and T cell features in relation to survival outcomes. **Results:** There were no associations between sex and TCR abundance (OR=1.05, 95% CI=0.98-1.13, p=0.14), TCR clonality (OR=0.99, 95% CI=0.91-1.06, p=0.72), or TILs/hpf (OR=1.05, 95% CI=0.88-1.25, p=0.60). We observed no differences in overall or disease-specific five-year survival between males and females. Further, the associations between T cell responses and survival did not significantly differ by sex. **Conclusions:** In a well-powered analysis, no robust sex differences were observed in the CRC immune microenvironment or in the relationship between T cell infiltration and CRC outcomes. The high burden of CRC and the emergence of immunotherapy as an important therapeutic approach suggest a need for further research investigating immunological sex differences in the setting of treatment response. Research Sponsor: James and Ruth Levitan Cancer Research Endowment from the Case Comprehensive Cancer Center; National Cancer Institute at National Institutes of Health; R01CA197350, R01CA081488, P30CA014089, U19CA148107, & P01CA196569; Gift from Daniel and Maryann Fong.

Summary of associations between the colorectal tumor T cell response and sex.

T cell Features	Odds Ratio (Female vs. Male)	Lower 95% CI	Upper 95% CI	P Value
TCR Abundance	1.05	0.98	1.13	0.14
TCR Clonality	0.99	0.91	1.06	0.72
TILs/hpf* <2	1.05	0.88	1.25	0.60

*A total of 99 patients (45 female; 54 male) with missing TILs/hpf data were excluded from the regression analysis.

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Circulating tumor DNA for predicting complete response to total neoadjuvant therapy in locally advanced rectal cancer: ENSEMBLE-2. First Author: Jun Watanabe, Department of Colorectal Surgery, Kansai Medical University, Osaka, Japan

Background: Total neoadjuvant therapy (TNT) has dramatically shifted the paradigm in the treatment of locally advanced rectal cancer (LARC), prolonging survival with high rates of pathologic complete response (pCR) and introducing non-operative management (NOM). However, no predictive biomarkers of TNT efficacy, the regrowth of NOM, or prognosis have been developed. Circulating tumor DNA (ctDNA) is a minimally invasive biomarker used to detect molecular residual disease (MRD) and predict recurrence in colon cancer after curative resection. The effectiveness of ctDNA MRD status as a predictive biomarker for TNT was evaluated in the Phase II TNT study, ENSEMBLE-2 (jRCTs071210143), conducted in Japan. **Methods:** Patients with LARC undergoing TNT were enrolled in ENSEMBLE-2. Protocol treatment was defined as total mesorectal excision (TME) following long course chemoradiotherapy (LCCRT: 50.4Gy, capecitabine) plus four cycles of CAPOX. NOM was allowed if a clinical complete response (cCR) was achieved in the evaluation after TNT. ctDNA MRD was measured by Signatera (Natera, Inc.) at the following time points: baseline, after LCCRT, after TNT, post operative 4w, 12w, 24, 36 and 48w in the GALAXY trial (UMIN000039205). **Results:** A total of 28 patients were enrolled in the study. Treatment was discontinued at the patient's request in one case. After completing TNT, TME and NOM were performed in 21 (77.8%) and 6 (21.4%) patients, respectively. ctDNA positivity rates were 96.4. % (27/28) at baseline, 14.8% (4/27) after LCCRT, and 34.6 % (9/26) after TNT. Post-LCCRT ctDNA status was not significantly associated with cCR + near CR (nCR) vs. incomplete clinical response (iCR), pCR vs. non-pCR (p=0.065 and p=0.539, respectively). In contrast, ctDNA status after TNT was significantly associated with cCR + nCR vs. iCR (p=0.028), as well as pCR vs. non-pCR (p=0.038). **Conclusions:** Our study indicates that post TNT ctDNA status may be a predictive biomarker for TNT response in LARC patients. ctDNA negativity upon completion of neoadjuvant treatment may indicate a favorable response. Clinical trial information: jRCTs071210143. Research Sponsor: None.

Correlation with ctDNA, clinical response, treatment after TNT and pathological response.

	After LCCRT	p value	After TNT	p value
Clinical Response	cCR + nCR 18 / 1	0.065	cCR + nCR 15 / 4	0.028
cCR + nCR vs. iCR (ctDNA -/+)	iCR 5 / 3		iCR 2 / 5	
Treatment after TNT	TME 17 / 3	0.438	TME 11 / 9	0.063
NOM vs. TME (ctDNA -/+)	NOM 6 / 0		NOM 6 / 0	
Pathological response after TME	pCR 5 / 0	0.539	pCR 5 / 0	0.038
pCR vs. non pCR (ctDNA -/+)	non pCR 12 / 3		non pCR 6 / 9	

Fisher's exact test was used to statistically examine the correlation between ctDNA MRD status and the factors at each timing.

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Evaluation of CSF1R as a potential prognostic biomarker in colorectal cancer (Alliance). First Author: Sandra Algaze, University of Southern California Norris Comprehensive Cancer Center, Los Angeles, CA

Background: Immune checkpoint inhibitors (ICIs) have shown limited efficacy in colorectal cancer (CRC) outside of mismatch repair (MMR) deficient/microsatellite instability-high (MSI-H) tumors. Most CRC cases are MMR proficient /microsatellite stable and demonstrate resistance to ICIs, attributed to the tumor microenvironment. Colony Stimulating Factor 1 Receptor (CSF1R), a regulator of tumor-associated macrophages, has emerged as a potential target due to its association with poor prognosis in CRC. **Methods:** Data from 433 metastatic CRC (mCRC) patients in the CALGB/SWOG 80405 trial were analyzed. CSF1R RNA was extracted from formalin-fixed, paraffin-embedded tumor samples and sequenced. Patients were divided into tertiles of CSF1R expression (high, medium, low). Progression-free survival (PFS) and overall survival (OS) were compared using Kaplan-Meier curves, log-rank tests, and Cox proportional hazards models. Subgroup analyses were conducted based on liver metastases, treatment type (Bevacizumab, Cetuximab, FOLFOX, FOLFIRI), and MSI status. **Results:** Low CSF1R expression (CSF1R-L) was associated with significantly better survival outcomes. Median PFS was 12.9 months for CSF1R-L, compared to 11.2 and 9.2 months for CSF1R-M and CSF1R-H (p = 0.003). Median OS was 35.8 months for CSF1R-L versus 32.5 and 24.4 months for CSF1R-M and CSF1R-H (p = 0.00086). Stratification by liver metastases showed that CSF1R-L remained a strong prognostic factor only in patients with liver metastases. There were no significant survival differences based on CSF1R expression in bevacizumab-treated patients. However, in cetuximab-treated patients, CSF1R-L tumors had significantly longer PFS (12.9 vs 8.8 months, p = 0.037) and OS (35.8 vs 21.4 months, p = 0.002) than CSF1R-H tumors. Similarly, in FOLFOX-treated patients, CSF1R-L tumors showed longer PFS and OS (p = 0.018 and p = 0.013, respectively), and in FOLFIRI-treated patients, OS was significantly longer for CSF1R-L tumors (42.2 vs 24.7 months, p = 0.022) than CSF1R-H tumors. **Conclusions:** Our data suggests that CSF1R expression may serve as a prognostic biomarker in CRC, with lower expression potentially linked to improved survival, particularly in patients with liver metastases. These findings warrant further investigation of CSF1R as a possible prognostic and predictive biomarker, as well as a potential therapeutic target in CRC. Clinical trial information: NCT00265850. Research Sponsor: None.

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Safety and biomarker assessment of ST316, a novel peptide antagonist of ß-catenin, in patients with advanced solid tumors. First Author: Anthony B. El-Khoueiry, Norris Comprehensive Cancer Center, Keck School of Medicine, University of Southern California, Los Angeles, CA

Background: Constitutive activation of ß-catenin, resulting in its nuclear accumulation and deregulated transcriptional activity, is a key event in colorectal cancer (CRC). Additionally, pathway mutations correlate with immune exclusion in CRC and across different tumor types. Disruption of the interaction of ß-catenin and its co-activator BCL9 is sufficient to suppress oncogenic ß-catenin transcriptional activity, without impacting its homeostatic functions in normal tissue. ST316 is a first-in-class, cell-penetrating peptide antagonist of the ß-catenin and BCL9 interaction. Preclinical evaluation revealed significant bioavailability and potent activity in CRC models with no impact on ß-catenin homeostatic functions such as intestinal stem cell survival or bone morphology. **Methods:** A phase 1-2 (P1-2) escalation-expansion study has enrolled patients (pts) with advanced solid tumors likely to harbor abnormalities in the Wnt/ß-catenin pathway, to assess the safety, pharmacokinetics (PK), biomarker and preliminary activity of ST316, and to recommend a P2 dose (RP2D). Paired biopsies were collected when feasible and assessed for drug penetration and pharmacodynamic biomarkers. Peripheral blood (PB) collected pre- and post-treatment was assessed by flow cytometry for polymorphonuclear myeloid-derived suppressor cells (PMN-MDSCs). **Results:** As of Sep.24, 2024; 23 pts with CRC (N=14), NSCLC (N=4), PDAC (N=2), or Breast and Ovarian cancer (N=1, each), were treated in 6 cohorts (0.5-12mg/kg) with IV ST316, once weekly. Treatment related AEs were observed in 17/23 pts and were mostly low grade, except for 2 G3 AEs that resolved: Fatigue (1) and ALT/AST elevation (1). No dose-limiting toxicities (DLTs) were observed. Single agent ST316 showed disease stabilization in 4 pts. PK analysis showed dose-proportional increases in Cmax and AUC for doses up to 8 mg/kg. IHC analysis showed substantial tumor penetration of ST316 in all analyzed samples and evidence of a treatment-induced redistribution of β-catenin in a subset of pts. PB analysis indicated high baseline levels of the ß-catenin-driven immunosuppressive PMN-MDSCs , followed by a significant decrease after ST316 exposure ($p<0.005$). The RP2D selected for combination cohorts in CRC is 8mg/kg; additional PD assessments are ongoing. **Conclusions:** Monotherapy ST316 was well-tolerated with dose-proportional PK and substantial tumor uptake. Pharmacodynamic analyses demonstrate mechanistic evidence of antagonism of ß-catenin signaling, including redistribution of ß-catenin subcellular localization and significant depletion of the ß-catenin-driven immunosuppressive PMN-MDSC population. Based on these data, ST316 advanced into P2 in CRC pts in combination with SoC in 2nd and 3rd line. Clinical trial information: NCT05848739. Research Sponsor: None.

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A multimodal deep learning approach for the prognostic stratification of patients with colorectal cancer (CRC). First Author: Nuha Shaker, Department of Pathology, University of Pittsburgh, Pittsburgh, PA

Background: Traditional prognostic systems like the AJCC TNM staging for colorectal cancer (CRC) often fall short in predicting long-term patient outcomes. These systems often rely on limited pathologic features, leading to generalized approaches to treatment despite diverse tumor heterogeneity, such as the complex interaction between the tumor-host microenvironment. There is therefore a pressing need for more accurate, scalable tools to enhance decision-making and predict outcomes for patients with CRC. **Methods:** Tumor sections from 191 CRC patients were comprehensively stained for CD8 to examine CD8-positive tumor-infiltrating lymphocytes (TILs). To account for tumor heterogeneity, multiple areas of the tumor were evaluated to include intertumoral and peritumoral areas. Slides were digitized and quantitatively analyzed using a cloud-based platform (SpatialX Diagnostics, Inc. USA). Additionally, clinicopathologic features including patient demographics, 8th edition AJCC staging, lymphovascular invasion (LVI), perineural invasion (PNI), mismatch repair (MMR) protein status, tumor location, presence/development of distant metastases, and patient follow-up were collected. Merging CD8 AI findings with clinicopathologic features, a complete data set was available for 175 patients. Cases were randomly assigned into 70% training and 30% validation. In the training set, unsupervised artificial intelligence (AI) models integrated clinicopathologic data and CD8-positive TIL counts using vision transformer models. The final model combined spatially resolved pathologic features with clinical variables to statistically predict overall survival (OS) and distant metastasis risk (DMR). **Results:** In the validation cohort, the model stratified patients into two groups based on a deep learning-derived risk score, with statistically significant differences between low-risk and high-risk groups for OS ($p<0.02$) and DMR prediction ($p<0.0001$), For OS prediction, the model selected tumor site, PNI, MMR status, epithelial and stromal CD8-positive TILs, and 5 deep vision features as the top 10 clinically significant features. For DMR prediction, the age, T- and N-stage, PNI, and 6 deep vision-derived features as key predictors. **Conclusions:** As captured by vision transformers, the distribution and spatial arrangement of CD8-positive TILs are critical factors in patient stratification for OS and DMR. AI-driven models like the one herein offer the potential for more personalized management strategies, advancing CRC management and improving patient outcomes beyond traditional prognostic staging systems and parameters. Research Sponsor: None.

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Deep learning–powered analysis of tumor-infiltrating lymphocytes (TILs) in colorectal cancer. First Author: Elio Adib, Brigham and Women's Hospital, Boston, MA

Background: TILs are a promising biomarker in various cancers, including colorectal cancer (CRC). Advances in deep learning enable objective and efficient TIL assessment. This study applies a novel deep learning model to quantify TIL density in a large cohort of CRC tumors, examining its relationship with clinicopathologic factors and survival. **Methods:** We applied a deep learning model (Deep-ICP) to H&E-stained images from CRC patients treated at Dana-Farber Cancer Center to quantify TIL density (immune cells/mm²). Kruskal-Wallis and Wilcoxon tests were used to compare categories. Overall survival (OS) was calculated from the date of diagnosis to the date of death or last follow-up. Cox proportional hazards models analyzed the association between TIL density tertiles (lower, middle, upper) and OS, adjusting for age, sex, stage, biopsy site, tumor mutational burden (TMB, mutations/megabase), and mismatch repair (MMR) status. **Results:** Of 1,477 tumor specimens tested, 1,039 (70%) were primary and 438 were metastatic. The most common sites of metastasis were liver (n=205), lung (n=86), and soft tissue (n=35). The median age at diagnosis was 56 (range 19–91) years and 706 were female (48%). Most tumors (1,214/1,300; 93%) were MMR proficient (MMRp); 86 were MMR deficient (MMRd). Median TIL density was 664 (interquartile range; 469–934) cells/mm² in primary tumors, 478 (358–685) cells/mm² in liver metastases, 415 (251–645) cells/mm2 in soft tissue ($p<0.001$). MMRd tumors had higher median TIL density than MMRp (881 vs. 620 cells/mm²; $p<0.001$). After adjusting for covariates, patients with highest TIL tertile had significantly longer OS than patients with lowest TIL tertile (median OS: 75.4 [95%CI 67.4-87] months; 100.5 [83.9-116] months and 99.1 [92.4-157] months for lower, middle and upper TIL tertiles respectively; Table). In the MMRp subset, higher TIL density was still predictive of improved OS (p adj=0.010). **Conclusions:** This study demonstrates the feasibility of deep learning powered TIL quantification in CRC, potentially informing differences in tumor microenvironment and prognostication. Future prospective studies are needed to validate the results and explore their therapeutic implications in personalized CRC management. Research Sponsor: None.

Variable	Hazard Ratio (95% CI)	P-value
TIL upper tertile (vs. lower)	0.70 (0.50-0.99)	0.04
TIL middle tertile (vs. lower)	1.04 (0.77-1.39)	0.80
Age (continuous)	1.02 (1.00-1.03)	0.008
Male (vs. female)	1.13 (0.86-1.47)	0.38
Stage II (vs. stage I)	4.8 (0.59-40)	0.14
Stage III (vs. stage I)	9.4 (1.20-73)	0.03
Stage IVA (vs. stage I)	23 (2.94-178)	0.003
Stage IVB (vs. stage I)	46 (5.6-369)	0.0003
Liver specimen (vs. primary)	1.39 (1.00-1.92)	0.050
Lung specimen (vs. primary)	0.78 (0.47-1.29)	0.33
Lymph node specimen (vs. primary)	0.77 (0.27-2.17)	0.62
Other metastasis (vs. primary)	1.23 (0.78-1.93)	0.37
TMB≥10 (vs. <10)	1.17 (0.82-1.68)	0.38
MMRd (vs. MMRp)	0.65 (0.29-1.46)	0.30

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The prevalence and prognostic significance of *KRAS G12C* mutation in metastatic colorectal cancer. First Author: Joseph Lutz, Washington University in St. Louis, Siteman Cancer Center, St. Louis, MO

Background: Kristen rat sarcoma virus (*KRAS*) mutations are reported in 40-45% of colorectal cancer (CRC), and they are associated with poor prognosis. *KRAS G12C* mutation is reported in 3-4% of CRC patients and may be associated with worse prognosis compared to *RAS non-G12C* mutations. Here, we explored the prevalence and prognostic significance of *KRAS G12C* mutation compared to *RAS non-G12C* and *BRAF V600E* in a cohort of CRC patients from a tertiary cancer center in the United States (US). **Methods:** The cBioCancer Genomics Portal was used to obtain genomic data. The prevalence of *KRAS G12C* mutation, *RAS non-G12C* (*KRAS* G12D, *KRAS* G13D, *KRAS* G12V, *KRAS* A146T, *KRAS* G12A, *KRAS* G12S, *KRAS* G13C, *KRAS* V14I, *KRAS* Q61H, *KRAS* G12R, *KRAS* Q61R, *KRAS* K117N, *NRAS Q61K*, *NRAS Q61R*, *NRAS Q61L*, *NRAS Q61H*, *NRAS G13D*, *NRAS G13C NRAS G13R NRAS G12D*, *NRAS G12C*, *NRAS G12V*, *NRAS G12V* and *NRAS G60V*) mutations and *BRAF V600E* mutations was calculated. The median overall survival (mOS) was explored in CRC (MSK, Cancer Cell 2018 cohort). Patients in this cohort were not treated with *KRAS G12C* targeted therapy. Survival analysis was performed using the Kaplan-Meier method with log-rank test comparisons. **Results:** In this cohort of patients with stage IV CRC (N=1134), the prevalence of *KRAS G12C* mutation in patients with stage IV CRC was 3.1%. In patients with *RAS non-G12C* mutations, *BRAF V600E* mutation, and *RAS/BRAF V600E* wild-type (WT), the prevalence was 41.7%, 7.5%, and 47.7% respectively. The mOS in *KRAS G12C* mutation, *RAS non-G12C* mutations, *BRAF V600E* mutation, and *RAS/BRAF V600E* WT was 65 months (m), 55m, 19m and 105m respectively. Compared to *RAS non-G12C*, the mOS of CRC patients with *KRAS G12C* was (65m vs 55m, p=0.54) . Compared to patients with *BRAF V600E* mutation, the mOS was better in patients with *KRAS G12C* (65m vs. 19m, $p<0.0001$) and in patients with RAS non-G12C (55m vs. 19m, $p<0.0001$). **Conclusions:** The prevalence of *KRAS G12C* mutation in this cohort was relatively similar to other reports at 3.1%. Consistent with several other reports, there was no statistically significant difference in the mOS between patients with *KRAS G12C* mutation and patients with *RAS non-G12C* mutations. Patients with BRAF V600E mutations had worse mOS compared to patients with *KRAS G12C mutation* and *RAS non-G12C* mutations. The reported high mOS in this cohort is possibly due to selection bias (patients able to afford and receive treatment at this tertiary center). Research Sponsor: None.

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Pan-cancer microsatellite instability testing using a multiplex assay including long mononucleotide repeats. First Author: Dustin A. Deming, University of Wisconsin Carbone Cancer Center, Madison, WI

Background: Microsatellite instability (MSI)/mismatch repair deficiency (dMMR) testing is a critical component of molecular testing for gastrointestinal cancers, among others. These analyses are important for the identification of patients with lynch syndrome and eligibility for treatment with immunotherapy. The current standard of care testing can identify most cases, however improved identification of MSI-H status is needed for non-colorectal cancers and for cancers with alterations outside MLH1/PMS2. A recent analysis indicated the potential for improved MSI-H detection with a multiplex assay including long mononucleotide repeats (LMR). **Methods:** Patients were selected based on MSI status, MMR gene variants, or the presence of a high tumor mutation burden and consented for participation as part of an IRB-approved protocol (UW15068). Formalin-fixed tissue slides were annotated by a pathologist for tumor and normal tissue. Slides were scraped and DNA isolated. The LMR MSI Analysis System (Promega, Madison, WI) and the OncoMate MSI Dx Analysis System (Promega) were performed on cancer DNA and normal tissue DNA when available per the manufacturer's protocols. Instability of microsatellites and the MSI score were reported and correlated with clinical MMR immunohistochemistry (IHC) and commercial next-generation sequencing (NGS). **Results:** A total of 96 cancer samples were tested across 94 patients with over 20 different cancer types, including lung, colon, rectal, gynecologic, and esophagogastric cancers. Tumors from 20 patients were known to have microsatellite instability based on clinical NGS testing. 27 cases were found to be MSI-H based on the OncoMate assay. All of these cases were also MSI-H per the LMR assay and an additional 3 cases were found to be MSI-H using only the LMR assay. Of these 3 cases all were microsatellite stable by NGS. This number includes 1 colorectal and 2 esophagogastric cancer cases. One of the cases was found to have loss of MSH6 by IHC with an MSH6 mutation (MSH6 L1061fs). One of the cases was MMR intact by IHC, though did have a MSH6 exon 7 splice acceptor alteration that failed NGS calling quality control. The last case did not have MMR IHC performed, no detected MMR gene alterations, and a tumor mutation burden of 6.3 mutations per megabase. **Conclusions:** In this limited dataset, the utility of various methods for determination of MSI, dMMR, and TMB including PCR, IHC, and NGS were demonstrated using a patient cohort spanning solid tumor types. Promising results were observed with the LMR MSI Analysis System. These findings need to be further validated in a larger dataset and correlated with the response to immunotherapeutics. Research Sponsor: Promega.

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5-hydroxymethylcytosine as a liquid biopsy biomarker in colorectal cancer. First Author: Christopher Maher, Washington University, St. Louis, MO

Background: 5-hydroxymethylcytosine (5hmC) is an epigenetic modification which regulates gene expression and is associated with active transcription. The optimization of 5hmC sequencing in cell-free DNA (cfDNA) could therefore enable assessment of gene activity through a liquid biopsy. We aimed to investigate the 5hmC landscape of colorectal cancer (CRC) in plasma-derived cfDNA to evaluate the potential of 5hmC modifications as a liquid biopsy-based biomarker of CRC. **Methods:** Genome-wide 5hmC modifications were analyzed with a low-input whole-genome 5hmC sequencing method based on selective chemical labeling in cfDNA from 72 CRC patients and 70 healthy samples. Differential 5hmC analysis between CRC and healthy samples was conducted using DESeq2 (padj $<$ 0.05 and log2FC $<=$ -1 and $>=$1). Low-pass whole genome sequencing (LP-WGS) was performed on all cfDNA samples. Tumor fraction was estimated using ichorCNA to classify samples into high ct-fraction ($>=$0.1) and low ct-fraction ($<$0.1) groups. Nucleosome profiling was performed on LP-WGS to reveal chromatin accessibility patterns. Elastic Net Regression (ENR) was applied to identify changes in 5hmC levels for classifying samples as cancerous or healthy. **Results:** Differential cfDNA 5hmC levels of 139 genes distinguished high ct-fraction CRC (n=16) and healthy samples (n=70). Although the remaining CRC samples (n=56) had similar ct-fractions as the healthy samples, as estimated by ichorCNA, we found a differentially hydroxymethylated gene signature could discriminate between these low ct-fraction CRC and healthy samples. Nucleosome profiling analysis with LP-WGS in cfDNA was used to identify active transcriptional drivers of the genes with increased 5hmC levels. The analysis revealed increased chromatin accessibility near the binding sites for the transcription factor caudal-related homeobox transcription factor 2 (*CDX2*), a master transcriptional regulator of intestinal cell fate and cancer, in CRC patients relative to healthy samples. Corroborating this, *CDX2* and its downstream targets had increased 5hmC levels in cfDNA in CRC relative to healthy samples, consistent with colorectal-derived tumor DNA in CRC plasma samples. **Conclusions:** Genes with differential 5hmC levels in cfDNA can discriminate CRC and healthy samples. Our 5hmC signature identified CRC samples that did not harbor copy number gains or losses; which is typically used to estimate ct-fraction and routinely included in targeted ctDNA sequencing assays. Integration of 5hmC and LP-WGS can be used to identify gene activation in disease. In the context of CRC, CDX2 displayed increased chromosome accessibility in the promoters of its target genes which in turn displayed increased 5hmC levels. Further studies are aimed at optimizing and validating 5hmC-based biomarkers throughout patient treatment. Research Sponsor: St. Louis Community Foundation; ClearNote Healthy.

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The role and molecular mechanism of GDF5 in regulating cell migration and angiogenesis via the Hippo-YAP signaling pathway in colorectal cancer. First Author: Peijie Lei, The Affiliated Hospital of Qingdao University, Qingdao, China

Background: Distant metastasis is a leading cause of poor prognosis and mortality in colorectal cancer (CRC) patients, with angiogenesis playing a crucial role in tumor progression. The TGF-β superfamily has been shown to significantly influence tumor angiogenesis, yet the biological function and molecular mechanism of Growth Differentiation Factor 5 (GDF5), a member of this superfamily, remain unexplored in CRC. Given that the efficacy of current anti-angiogenic therapies is limited to a subset of patients, identifying new therapeutic targets involved in tumor angiogenesis is of great clinical importance. GDF5, with its potential role in angiogenesis regulation, presents a promising candidate to bridge this therapeutic gap. **Methods:** In this study, we analyzed GDF5 expression in CRC tissue samples using both the TCGA dataset and samples from our center, revealing significantly lower GDF5 expression in tumors compared to adjacent normal tissues. Moreover, GDF5 downregulation correlated with lymph node metastasis and disease recurrence. Functional assays in vitro demonstrated that GDF5 overexpression reduced tumor cell migration and invasion, while its knockdown produced the opposite effect. Co-culture assays showed that GDF5 knockdown significantly enhanced endothelial cell migration and angiogenesis, whereas overexpression diminished these effects. Additionally, F-actin cytoskeletal staining indicated that GDF5 overexpression led to cytoskeletal remodeling in endothelial cells. Mechanistically, we found that GDF5 overexpression in tumor cells downregulated the epithelial-mesenchymal transition (EMT) marker Slug and upregulated E-cadherin, while downregulating N-cadherin. In endothelial cells, GDF5 inhibited the Hippo-YAP signaling pathway, increasing YAP phosphorylation and reducing nuclear YAP levels. Conversely, GDF5 knockdown suppressed YAP phosphorylation and promoted its nuclear translocation. **Results:** Our findings demonstrate that GDF5 functions as a tumor suppressor in CRC by regulating cell migration and angiogenesis. This study reveals that GDF5 exerts its anti-cancer effects by modulating both EMT in tumor cells and cytoskeletal changes in endothelial cells, leading to the inhibition of the Hippo-YAP signaling pathway and reduced angiogenic potential. These results identify GDF5 as a promising new target for anti-angiogenic therapy in CRC, offering potential avenues for improving treatment strategies in patients resistant to current therapies. **Conclusions:** GDF5 acts as a tumor suppressor in colorectal cancer by inhibiting cell migration and angiogenesis through modulation of the Hippo-YAP signaling pathway, making it a promising therapeutic target for anti-angiogenic treatment in patients resistant to current therapies. Research Sponsor: Shandong Province Natural Science Fundation; ZR2020MH216.

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Retrospective study of fluoropyrimidine chemotherapy dosing and toxicity in patients with screen-detected deleterious *DPYD* gene variants. First Author: Chanbormey Leatheng, Dartmouth-Hitchcock Medical Center, Lebanon, NH

Background: Fluoropyrimidine chemotherapy agents are commonly used to treat solid tumors. Fluoropyrimidines are degraded by the enzyme dihydropyridine dehydrogenase (DPD), encoded by the *DPYD* gene. Deleterious variants in *DPYD* leads to functional DPD deficiency and increased toxicity risk. **Methods:** We conducted a retrospective study of patients with screen-detected deleterious *DPYD* variants who were treated with fluoropyrimidine chemotherapy between 01/01/15-04/15/23. Outcomes of interest were initial fluoropyrimidine dosing, dose adjustments (cycles 2-6), and early toxicities. **Results:** We identified 12 patients with heterozygous *DPYD* variants, including *2A (n=5), c.2846A$>$T (3), c.1129-5923C$>$G (2) and *13 (2). Initial chemotherapy regimens included FOLFOX (7), mFOLFIRINOX (3) and CAPEOX (1). Of 11 patients receiving FOLFOX/mFOLFIRINOX, nine began treatment with a 50-60% reduction of the 5-FU infusion dose, one received a 25% reduction, and one received a standard dose (due to treatment initiation prior to return of the genotype result). Seven patients tolerated the initial 5-FU infusion dose without further dose reductions, including three who tolerated limited dose-escalation. Four patients required further dose reduction from the cycle 1 dose, due to toxicity. The patient who received a standard 5-FU infusion dose in cycle 1 experienced grade 3 toxicity, but tolerated subsequent treatment after 50% dose reduction. The patient treated with CAPEOX tolerated 4 planned cycles with a 50% capecitabine dose reduction. **Conclusions:** Most patients who screened positive for deleterious *DPYD* gene variants tolerated fluoropyrimidine chemotherapy with a 50% dose reduction. However, 3 of 12 patients experienced toxicity requiring treatment discontinuation or dose reduction to less than 50% of the standard 5-FU dose. Research Sponsor: None.

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UGT1A1 mutation rates in a minority-rich comprehensive cancer center: A retrospective pharmacogenomic study. First Author: Christiana Mo, Montefiore Einstein Comprehensive Cancer Center, Bronx, NY

Background: Irinotecan is a widely used and effective treatment for various gastrointestinal malignancies. Its toxic metabolite, SN-38, is inactivated by UDP glucuronosyltransferase-1A1 (UGT1A1). However, variability in individual clearance of irinotecan and SN-38 can cause unpredictable toxicities. Mutations in the *UGT1A1* gene reduces enzyme activity, resulting in dose-limiting neutropenia and diarrhea, hospitalizations, treatment delays, and dose reductions. The best characterized germline variant is a polymorphism in the UGT1A1*28 allele, but data in underrepresented groups remains particularly lacking. **Methods:** We conducted a retrospective cohort study of patients who were treated at our minority-rich academic medical center. A total of 63 patients with gastrointestinal malignancies were tested for the UGT1A1*28 polymorphism. Data for irinotecan treatment course, related toxicities, and dose reductions was available for 41 patients. Analysis was performed using Fisher's exact test, one-way ANOVA, and Welch's two-sample test. **Results:** 73% (n = 46) of patients were identified with the UGT1A1*28 polymorphism, including 63% (n = 29) with heterozygous genotype and 36.9% (n = 17) with homozygous genotype. There were significantly more patients of UGT1A1*28 polymorphisms carriers in the African American and Hispanic population as compared to the white population (p = 0.001, 0.01 respectively). The polymorphism frequency was 0.66 in the African American group [95% CI (0.49 – 0.81)], 0.48 in the Hispanic group [95% CI (0.34 – 0.62)], and 0.06 in the white group [95% CI (0.001 – 0.30)]. The frequency was significantly lower in the white population when compared to all other races (p-value = 0.0002), to African American (p- value $<$ 0.001), and to Hispanic (p-value = 0.003) populations. There were higher rates of overall toxicities of any grade in the heterozygotes (76%) and homozygotes (64%) when compared to the wild-type group (56%). 55% (n = 6) of the homozygous and 48% (n = 10) of the heterozygous patients developed G1-G4 neutropenia, as compared to the 22% (n = 2) of the wild-type patients. **Conclusions:** Ethnic differences are known to exist in UGT1A1*28 polymorphism, with the prevalence being significantly higher in African and Caucasian populations. Historical data based on large-scale genotyping report allele frequencies of 0.42 in African/African American, 0.37 in Hispanic American, and 0.29 in Europeans. Our local population has a higher rate overall in both the African American and Hispanic populations with significantly higher rates than the white population. Patients with the UGT1A1*28 polymorphism experienced more toxicities and neutropenia of any grade. *UGT1A1* genotyping is not routinely performed in patients initiating irinotecan treatment, however it may prove to be beneficial in certain demographics with higher rates of UGT1A1*28 polymorphisms. Research Sponsor: None.

295 **Poster Session**

Real-world treatment patterns, healthcare resource utilization (HCRU), and costs in newly diagnosed patients with rectal cancer (RC) in the United States (US). First Author: Laura Moore-Schiltz, Inovalon, Bowie, MD

Background: RC constitutes 30% of colorectal cancer cases, the fourth most common cancer in the US, and is associated with a substantial economic burden. This retrospective claims analysis examined patient (pt) characteristics, all-cause and RC-related HCRU, and costs in pts who received surgery, radiotherapy (RT), or systemic therapy (ST) for RC. **Methods:** Medicare Fee-for-Service and MORE^2 Registry claims data were used to select pts with RC (RC cohort) and matched, non-RC controls (non-RC cohort). Pts with RC were required to have a diagnosis of RC in the primary position of ≥1 inpatient claim or in any position of ≥2 outpatient claims between Jan 1, 2016, and Dec 1, 2021, and ≥1 claim for an RC surgery, RT, or ST after diagnosis. Pts (aged ≥18 y) also needed to have continuous enrollment for 12 months preceding and minimum 30 days following the treatment index date (earliest RC treatment claim). Non-RC controls were matched to pts with RC on birth year, sex, payer type, and Deyo-Charlson Comorbidity Index (DCI). All-cause HCRU and costs (per pt per month [PPPM]) were compared for the RC and non-RC cohorts. **Results:** A total of 20,953 pts were included in each cohort (median age, 70 y; 52.5% male; median DCI, 1.0 in both cohorts). Median follow-up time was 24 months. In the RC cohort, 88.5% of pts underwent ≥1 surgery after RC diagnosis and 51.3% had ≥4 surgeries during follow-up. Colonoscopy and biopsies (59.4%), total mesorectal excision (47.7%), and proctectomy (45.2%) were the most frequent surgeries/procedures. RT was used as neoadjuvant and adjuvant therapies in 21% and 15.7% of pts, respectively; 5-fluorouracil was the most common neoadjuvant (12.7%) and adjuvant (20.9%) ST. Among nonsurgical pts (11.5%), 72.8% were treated with RT and 49.9% with 5-fluorouracil; a small proportion received immunotherapy (pembrolizumab, 1.2%; nivolumab, 0.7%). All-cause HCRU was significantly greater ($P<0.001$) in the RC cohort vs the non-RC cohort (proportion of pts with ≥1 claim for physician office/clinic visit, 81.3% vs 73.6%; emergency room visits, 68.7% vs 45.3%; hospitalization, 71.1% vs 25.9%). The mean all-cause costs PPPM were $6043 for the RC cohort vs $1538 for the non-RC cohort ($P<0.001$); mean RC-related costs PPPM were $3119. Outpatient and inpatient services were the main cost drivers in the RC cohort, accounting for 57.1% and 36.6% of all-cause expenditure, respectively, and 58.1% and 40.5% of RC-related expenditure. **Conclusions:** Most pts newly diagnosed with RC received standard of care surgical treatment, often with neoadjuvant or adjuvant RT or ST. All-cause HCRU was significantly greater for pts with RC vs those without RC; all-cause costs were almost 4 times higher, with about half of these costs related to RC and driven by outpatient services and hospitalizations. These results show a high economic burden and a need for additional treatments for pts with RC. Research Sponsor: None.

296 **Poster Session**

Effect of changes in body composition and sociodemographic factors on colon cancer survival. First Author: Ignacio Catala Vilaplana, University of British Columbia, Vancouver, BC, Canada

Background: Muscle mass, muscle quality, and adipose tissue volumes have been shown to affect colon cancer survival. Further research is required to understand how change in body composition during treatment impacts colon cancer outcomes, and how covariables such mental health and socioeconomic status affect this relationship. The aim of this project was to examine how changes in body composition during chemotherapy for colon cancer impact the duration of disease-free survival (DFS) and how sociodemographic covariables (age, sex, social isolation, depression, anxiety, income, community size) influence this relationship. **Methods:** Computed tomography (CT) scans from the time of diagnosis and the end of chemotherapy were obtained from individuals treated for stage III colon cancer with oxaliplatin at BC Cancer between 2012 and 2016. Whole body tissue volumes were estimated based on CT images at the level of the third lumbar vertebra, analysed using *DAFS Express* (Voronoi Health Analytics Inc., Vancouver BC). Muscle quantity was measured as skeletal muscle index (SMI), and muscle quality was measured as skeletal muscle density (SMD) and skeletal muscle gauge (SMG). Quantity of visceral adipose tissue (VAT), subcutaneous adipose tissue (SAT), intermuscular adipose tissue (IMAT), and total adipose tissue (TAT) were also measured. Social isolation, anxiety, and depression symptoms were measured using the validated Psychosocial Screen for Cancer–Revised. Neighbourhood income and community population were estimated using postal codes. Cox proportional hazard models were calculated for the effect of body composition variables on DFS, and interactions with sociodemographic variables. **Results:** Significant reductions in SMI, SMD, SMG, VAT, IMAT and TAT were observed between diagnosis and post-chemotherapy (n=282). Improved survival was associated with higher SMD and lower IMAT at diagnosis and post-chemotherapy, lower VAT at diagnosis, and an increase in VAT or TAT during chemotherapy. Body composition had a more significant effect on survival in individuals who were older, female, or lived in smaller communities. **Conclusions:** Significant reductions in SMI, SMD, SMG, VAT, IMAT and TAT were observed between diagnosis and post-chemotherapy (n=282). Improved survival was associated with higher SMD and lower IMAT at diagnosis and post-chemotherapy, lower VAT at diagnosis, and an increase in VAT or TAT during chemotherapy. Body composition had a more significant effect on survival in individuals who were older, female, or lived in smaller communities. Research Sponsor: None.

297 **Poster Session**

Impact of COVID-19 pandemic on critically ill patients with colon cancer: A population-level study. First Author: Jorge Rodriguez Vazquez, Texas Tech University Health Sciences Center, Odessa, TX

Background: The COVID-19 pandemic strained healthcare systems and disrupted pre-pandemic levels and trends in the short-term mortality of colon cancer hospitalizations. Colon cancer is the fourth most diagnosed cancer in the US and Texas. The impact on short term mortality in Texas in the years since the COVID-19 pandemic have not been explored. **Methods:** We used publicly available, deidentified, and state-wide data to conduct a population-base cohort study of hospitalizations aged ≥ 18 years admitted to intensive care units at acute care hospitals in Texas during 2016-2022 with a diagnosis of colon cancer. Hospitalizations transferred to another acute care hospital were excluded from the study. Colon cancer was identified using ICD-10-CM codes C18x and C20 selected from Clinical Classification Software Refined category NEO015: Gastrointestinal cancers - colorectal. Short-term mortality was defined as in-hospital mortality or discharge to a hospice. The response variables are pre-pandemic time trend, level change, change in time trend, and final time trend. The primary analysis method was interrupted time series analysis (ITSA) based on multilevel-multivariable logistic regression with results reported as average marginal effects per quarter (AME) and 95% confidence intervals (95% CI) in changes to the absolute probability of mortality. The matched pairs t-test was applied to ITSA and counterfactual predicted probabilities to further elucidate the impact of the COVID-19 pandemic. Sensitivity analyses was performed for hospitalizations aged ≥ 65 years and for hospitalizations without a diagnosis of COVID-19. **Results:** A total of 49,007 hospitalizations were included in the study of which 21,995 (44.9%) occurred after the beginning of the pandemic. Unadjusted short-term mortality was lower in the pre-pandemic period compared to the post-pandemic period (18.1% *vs* 20.1%). On adjusted analysis, the post-pandemic period remained associated with higher mortality rates. The pre-pandemic short-term mortality time trend was negative (AME -0.19%/quarter, 95% CI [-0.28% to -0.10%], p = 0.0001). The change in time trend was positive (AME 0.39%/quarter, 95% CI [0.23% to 0.55%], $p < 0.0001$) and the change in level was also positive (AME 1.25%, 95% CI [0.0.04% to 2.47%], p = 0.0436). The final time trend was significant and positive (AME 0.20%/quarter, 95% CI [0.07% to 0.33%], p = 0.0023). The point estimate and 95% CI for the difference between ITSA predicted and counterfactual predicted probabilities of mortality was 3.45% (3.41% to 3.48%). Similar results were observed on sensitivity analyses. **Conclusions:** Since the onset of the pandemic there was a 2% increase in the absolute probability of short-term mortality and a change from negative to positive time trend in short-term mortality. The average impact of the COVID-19 pandemic on the absolute probability of short-term mortality is estimated at over 3%. Research Sponsor: None.

298 **Poster Session**

Development and validation of a deep learning–based pathomics signature for prognosis and chemotherapy benefits in colorectal cancer: A retrospective multicenter cohort study. First Author: Shenghan Lou, Harbin Medical University Cancer Hospital, Harbin, China

Background: The current TNM staging system fails to provide adequate information for prognosis and adjuvant chemotherapy benefits in colorectal cancer (CRC). Pathomics, an emerging field, shows promise in improving prognosis estimation and decision-making. In this study, we developed and validated a pathomics signature (PS_{CRC}) that directly analyzes hematoxylin and eosin–stained slides using deep learning to predict outcomes. **Methods:** A total of 883 whole slide images from two cohorts, Harbin Medical University Cancer Hospital and The Cancer Genome Atlas (TCGA), were retrospectively analyzed. An interpretable, multi-instance deep learning model was proposed to establish the PS_{CRC}. Shapley additive explanations were employed to interpret the model's decisions, and gradient-weighted class activation mapping was applied to visualise the pathological phenotypes of the PS_{CRC}. The transcriptomics data from TCGA cohort was used to explore the potential pathogenesis underlying the PS_{CRC}. **Results:** The PS_{CRC} was identified as an independent prognostic factor associated with both overall survival and disease-free survival. Incorporating the PS_{CRC} into the TNM stage model resulted in a significant improvement in prognosis estimation, as evidenced by a notable increase in net reclassification improvement and integrated discrimination improvement. Moreover, among stage II and III CRC patients with low levels of PS_{CRC}, satisfactory benefits from chemotherapy were observed. Notably, the main underlying features of PS_{CRC} include tumor cell infiltration, adipocyte accumulation, fibrous tissue deposition, and stromal infiltration. Transcriptome analysis further support the relevance of PS_{CRC} to tumor progression and immune suppression. **Conclusions:** Our finds highlight the significant potential of histopathology images-based deep learning in predicting prognosis and assess therapeutic response of CRC. The PS_{CRC} could serve as an effective tool in clinical decision for CRC management, providing insights into the underlying pathogenic mechanisms. However, prospective studies are still necessary for further validation. Research Sponsor: None.

299 Poster Session

Man versus machine: Comparative analysis of ChatGPT's colon cancer guidance and expert opinion. First Author: Sujata Ojha, Dell Medical School at The University of Texas at Austin, Austin, TX

Background: AI Chatbots, such as ChatGPT, are increasingly being utilized across various aspects of society, including healthcare. Despite this growing usage, their role within the field of oncology remains underdeveloped. The objective of this study is to evaluate ChatGPT's ability to accurately respond to patient inquiries regarding colon cancer by comparing its responses to assessments from expert clinical oncologists. **Methods:** Ten comprehensive questions were compiled by reviewing commonly asked questions from reputable sources, including the American Society of Colon & Rectal Surgeons, Mount Sinai, the National Cancer Institute, Mayo Clinic, and the American Cancer Society. The questions were categorized into two categories based on their content: General Oncology Characteristics (covering symptoms, screening, and prevention) and for Diagnosis & Treatment. These questions were then entered into ChatGPT, with prompts designed to simulate patient inquiries. The AI-generated responses were subsequently evaluated by oncology experts using a 5-point Likert scale to assess their accuracy and relevance, with scores reflecting the experts' level of agreement with the answers. **Results:** On a five-point Likert scale, with 1 representing "strongly disagree" and 5 representing "strongly agree," the mean score was 4.72. ANOVA analysis was performed and there was no statistically significant difference in the mean score across all raters (p = 0.221). However, ratings between the two categories were statistically significant (p=0.034). **Conclusions:** This study demonstrates that ChatGPT can provide accurate and relevant responses to patient inquiries about colon cancer, as assessed by medical oncology experts. With an average rating of 4.72 on a 5-point Likert scale, ChatGPT's responses closely align with expert opinion, particularly for general characteristics category which included symptoms, prevention, and screening. However, responses related to diagnosis and treatment showed a statistically significant difference between expert opinions and AI, indicating that the experts agreed less with this component of AI's responses. These findings highlight the potential of AI chatbots in supplementing patient education in oncology, though further research is necessary to explore its limitations and expand its clinical utilities. Research Sponsor: None.

300 Poster Session

Risk factors for mortality and financial burdens in patients with colon cancer. First Author: Zaid Zahid, Hamilton Medical Center, Dalton, GA

Background: Colon cancer remains one of the most common cancers of the digestive system and continues to rank high among the leading causes of cancer-related deaths, despite the increased prevalence of screening. In this study, we aimed to investigate risk factors for mortality and health care costs in hospitalized patients diagnosed with colon cancer. **Methods:** Patients admitted to the hospital in 2019 and 2020 were identified using the Nationwide Inpatient Sample database and categorized into two groups based on the presence of colon cancer (with relevant ICD-10 codes). To account for confounding factors, multivariate regression analysis was applied to calculate mortality. Statistical analyses were conducted to assess mortality rates, length of stay, and hospital charges. **Results:** A total of 169,805 patients were admitted with a primary diagnosis of colon cancer, with a mean age of 67.6 years. The majority were White (69.9%), and 50.2% were male. Age over 60 (OR: 1.67, $p < 0.001$) and sepsis (OR: 16.01, $p < 0.001$) were associated with increased in-hospital mortality among colon cancer patients, while higher household income (76th to 100th percentile, OR: 0.71, p = 0.011) and treatment at an urban teaching facility (OR: 0.54, $p < 0.001$) were linked to a lower risk of in-hospital mortality. Female gender (OR: 0.88, p = 0.080) and having inflammatory bowel disease (OR: 0.67, p = 0.055) did not significantly affect in-hospital mortality, nor did Black race (OR: 1.1, p = 0.348) or Hispanic race (OR: 1.15, p = 0.287) compared to White patients. Sepsis (+11.8 days, $p<0.001$; +$192,699, $p< 0.001$), age over 60 (+0.99 days, $p<0.001$; +$6,134, $p<0.001$), and care at urban teaching hospitals (+0.54 days, $p< 0.001$; +$32,230, $p< 0.001$) were associated with longer hospital stays and higher total charges. Conversely, female gender (-0.21 days, p=0.001; -$5,529, $p<0.001$) and higher income (-0.69 days, $p<0.001$; -$1,565, p=0.466) were linked to shorter hospital stays and lower total charges. **Conclusions:** Early recognition of sepsis in colon cancer patients could potentially reduce in-hospital mortality rates, as well as significantly decrease costs and length of stay. The higher mortality rates observed in patients with sepsis may be attributed to the immunocompromised state associated with cancer. Implementing targeted interventions to address the higher risks in older patients and those with sepsis could improve outcomes and reduce the overall burden of colon cancer-related hospitalizations. Research Sponsor: None.

Impact of patient demographics, socioeconomic factors, and clinical conditions on mortality, length of hospital stay, and total hospital charges.

Data	Mortality	Length of Hospital stay	Total Charge
Age>60	1.67 ($p< 0.001$)	0.99 days, $p<0.001$	$6428, $p< 0.001$
Race (Hispanic compared to White race)	1.15 (p=0.287)	0.38 days, p= 0.004	$14,127, $p<0.001$
Race (Black compared to White race)	1.1 (p=0.348)	1.4 days, $p< 0.001$	$5370, p=0.002
Female gender	0.88 (p=0.08)	-0.21 days, p= 0.001	-$5529, $p< 0.001$
High house hold income	0.71 (p= 0.011)	-0.69 days, $p<0.001$	-$1565, p= 0.466
Treatment at urban teaching facility	0.54 ($p<0.001$	0.54 days, $p<0.001$	$32230, $p< 0.001$
Sepsis	16.01 ($p<0.001$)	11.8 days, $p<0.001$	$192699, $p< 0.001$
IBD	1.05 (p=0.888)	0.67 days, p=0.055	$5348, p=0.293

301 Poster Session

Exploring the link between liver cirrhosis and liver metastasis in colorectal cancer. First Author: Chaula Desai, Brooklyn Hospital Center, Brooklyn, NY

Background: Colorectal cancer (CRC) is the third most common cancer and the second most common cause of cancer mortality in the world. Although it is now associated with improved outcomes due to advancements in treatment modalities, CRC continues to be the most common source of liver metastasis. Previous studies have shown cirrhotic livers to be at reduced risk of metastatic disease. Our study aims to investigate the association between colorectal cancer and liver metastasis in the presence of concomitant liver cirrhosis. **Methods:** The National Inpatient Sample (NIS) was queried to identify all hospitalizations with colorectal cancer utilizing ICD-10 codes C18.x, C19.x, and C20.x from 2016 to 2020 and were further classified based on the presence or absence of cirrhosis. Demographic and clinical data were analyzed using chi-squared tests, independent sample t-tests, and binary logistic regression (adjusted for age, gender, and Charlson comorbidity index or CCI). The primary outcome studied was liver metastasis. Secondary outcomes were hepatic encephalopathy, portal hypertension, and spontaneous bacterial peritonitis. Statistical significance is indicated by a p-value less than 0.05. **Results:** A total of 255,599 hospitalizations with colorectal cancer were identified. Out of these, 5,370 (2.1%) had liver cirrhosis and 250,229 (97.9%) did not. Hospitalizations with CRC and concomitant cirrhosis had higher comorbidities as per the Charlson Comorbidity Index (10.5 vs 8.1, p-value<0.001) and higher mortality rates (8.5% vs 4.2%, p-value<0.001). Higher rates of liver metastasis (26.5 vs 23.4, $p <0.001$) and portal hypertension (26.1% vs 0.3%, $p<0.001$) were observed in the cirrhosis subgroup. This subgroup also has a significantly higher risk of SBP (2.8% vs 0.2%, $p <0.001$). **Conclusions:** This is a large retrospective study analyzing the impact of cirrhosis on liver metastasis in patients with colorectal cancer. Patients with concomitant cirrhosis consistently have increased liver metastasis and worse outcomes. The results of this study imply that diligent CRC screening is of utmost importance in patients with cirrhosis. Patients with CRC and coexisting cirrhosis would benefit from routine surveillance such as carcinoembryonic antigen (CEA) screening or CT scans to allow early detection of liver metastases which could potentially improve outcomes. Research Sponsor: None.

	Colorectal Cancer Patients		
	No Cirrhosis (N=250,229)	Cirrhosis (N=5,370)	p-value
Age (in years)	65.79 ± 13.8	65.04 ± 10.9	<.001
Sex (Female)	47.3%	36.8%	<.001
Charlson Comorbidity Index	8.1 ± 3.6	10.5 ± 3.7	<.001
Mortality	4.2%	8.5%	<.001
Length of Stay	6.6 ± 7.2	7.4 ± 7.2	<.001
Liver Metastasis	23.4%	26.5%	<.001
Hepatic Encephalopathy	0.0%	0.2%	<.001
Portal hypertension	0.3%	26.1%	<.001
Spontaneous Bacterial Peritonitis (SBP)	0.2%	2.8%	<.001

302 Poster Session

Circulating tumor DNA as a marker of recurrence risk in stage III colorectal cancer: The α-CORRECT study. First Author: Robert E. Schoen, Department of Medicine, Division of Gastroenterology, Hepatology and Nutrition, and Department of Epidemiology, University of Pittsburgh, Pittsburgh, PA

Background: Patients with stage III CRC undergo surgery and adjuvant chemotherapy. Evaluation of MRD using a highly sensitive and specific, tumor-informed ctDNA assay may help predict which patients will have a recurrence. **Methods:** Patients with stage III CRC were enrolled after surgery in an observational study across 19 Pennsylvanian hospitals between 2016-2020. Tumor tissue and serial blood samples (every 3 months for years 1-3, biannually for years 4-5) were collected. Following whole exome sequencing (WES) of the tumor and selection of 50 – 200 somatic variants, a personalized MRD assay was used to measure ctDNA status of each sample. Three timepoints were used to analyze the association of ctDNA with recurrence: the single sample post-surgical (PS) and post-definitive therapy (PDT) timepoints, and the surveillance period, which included the PDT and subsequent samples. Primary endpoint was the association of ctDNA status during surveillance with recurrence-free survival (RFS). **Results:** A total of 124 patients with 1029 ctDNA results were analyzed. Almost all patients (96.0%) received adjuvant chemotherapy, primarily with an oxaliplatin-based regimen. On average, patients had 10 (SD = 4.5) blood draws and the median follow-up was 4.8 years. WES identified 172 (range 50 – 200) somatic variants per patient on average. Patients who were ctDNA positive during surveillance had a significantly higher likelihood of recurrence than patients who remained ctDNA negative (HR 49.6, 95% CI: 16.6 – 148.3; ctDNA status as time-dependent variable). At all timepoints, ctDNA detection was strongly associated with recurrence (Table). In multivariable analyses, ctDNA was strongly prognostic for recurrence but CEA was not. The median lead time between positive ctDNA result and a clinical diagnosis of recurrence was 10.4 months. **Conclusions:** In this cohort of patients with stage III CRC with long-term follow-up, a tumor-informed ctDNA assay using up to 200 variants was strongly prognostic for recurrence at all timepoints. Research Sponsor: National Cancer Institute; U01CA152753; Exact Sciences Corporation.

Hazard ratios (HR), sensitivity, and specificity, with 95% confidence intervals (95% CI), for the association of ctDNA status with recurrence-free survival (RFS) at the surveillance, postsurgical (PS), and post-definitive therapy (PDT) timepoints (three-year RFS for the PS and PDT timepoints are also shown).

Timepoint	Statistic	Result
Surveillance (N = 111)	HR	49.6**
	(95% CI)	(16.6 – 148.3)
	Sensitivity	20/22 = 90.9%
	(95% CI)	(72.2 – 97.5%)
	Specificity	82/87 = 94.3%
	(95% CI)	(87.2 – 97.5%)
PS (N = 88)	HR	9.6**
	(95% CI)	(3.2 – 29.5)
	Sensitivity	14/18 = 77.8%
	(95% CI)	(54.8 – 91.0%)
	Specificity	53/66 = 80.3%
	(95% CI)	(69.2 – 88.1%)
	RFS at 3 yrs ctDNA(+), ctDNA(-)	54.5%, 96.1%
PDT (N = 97)	HR	16.7**
	(95% CI)	(6.9 – 40.3)
	Sensitivity	10/21 = 47.6%
	(95% CI)	(28.3 – 67.6%)
	Specificity	75/76 = 98.7%
	(95% CI)	(92.9 – 99.8%)
	RFS at 3 yrs ctDNA(+), ctDNA(-)	18.2%, 90.0%

*$P< 0.001$, **$P<0.0001$.

303 Poster Session

Systematic literature review (SLR) of prognostic factors (PFs) in locally advanced rectal cancer (LARC) and mismatch repair deficient (dMMR)/ microsatellite instability-high (MSI-H) rectal cancer (RC). First Author: Jaume Capdevila, Vall d'Hebron University Hospital, Barcelona, Spain

Background: Innovative immunotherapies arebeing developed to improve clinical outcomes in patients (pts) withdMMR/MSI-H LARC who respond poorly to the current standard of care. A comprehensive overview of PFs in dMMR/MSI-H LARC may provide insight into the natural history of this disease to inform clinical study design and pt risk stratification. **Methods:** Systematic searches of bibliographic databases (EMBASE, MEDLINE, Cochrane; 2013–2023) and congress websites (2021–2023) were conducted to identify PFs for tumor response and survival outcomes in adult pts with LARC (SLR1) and dMMR/MSI-H RC (SLR2). Given the expected high volume of evidence, SLR1 was limited to large observational studies (≥1000 pts) and clinical trials (≥100 pts); SLR2 had no such limits. **Results:** 95 of 2307 publications met the inclusion criteria for SLR1. Of the PFs reported in ≥20 studies, older age, more lymph node involvement or fewer nodes assessed, higher T-stage or larger tumor size, higher histologic grade, and more comorbidities were most frequently significantly associated with poorer patient prognosis (Table). 3 of 289 publications met the inclusion criteria for SLR2. Neoadjuvant chemoradiation therapy (CRT) compared to chemotherapy (CT) was associated with statistically significant worse disease-free survival (DFS) and distant metastasis-free survival (DMFS), while T- or N-stage, age, and sex did not significantly impact these outcomes. Low vs middle tumor localization was also associated with worse DMFS. Stage III and IV disease were associated with worse RC-specific survival, while surgical procedure (segmental vs extended) showed no difference in overall survival (OS). The addition of neoadjuvant CRT or adjuvant CRT/CT to surgery alone had no impact on OS but was associated with worse DFS. **Conclusions:** Evidence regarding PFs in dMMR/MSI-H RC is scarce and may limit its interpretability, highlighting the need for further research that may be informed by the PFs identified from SLR1. The results from these SLRs may be helpful for optimizing patient management and treatment decision making, and to further contextualize RC clinical trials. Research Sponsor: GSK; 221776.

PFs reported in ≥20 studies with ≥75% of studies reporting significant results in pts with LARC.

PF	Studies reporting PF (n)	Studies reporting significant* impact of PF on prognosis (%)	Poorer prognosis group	Top 2 most frequently reported outcomes
Age	59	81.4	Older	OS, DFS
Lymph node involvement/ assessment	55	78.2	Yes or higher N stage	OS, DFS
T-stage or tumor size	53	79.2	Higher or larger	OS, DFS
Histologic grade	37	75.7	Higher	OS, pathological complete response (pCR)
Comorbidities	23	87.0	More	OS, pCR

*p-value <0.05 was considered significant as reported in the identified studies.

304 Poster Session

Impact of synchronous vs metachronous primary tumor resection on prognosis of BRAF-V600E mutant metastatic colorectal cancer (mCRC). First Author: Arwa Isameldin Ahmed, Ottawa Hospital and Cancer Center-General Campus, Ottawa, ON, Canada

Background: The goal of this study is to evaluate if resection of the primary tumor in synchronous versus metachronous metastatic disease predicts survival of BRAF-V600E mCRC in the era prior to combination BRAF/EGFR inhibitors. **Methods:** This is a retrospective study of patients treated at the Ottawa Hospital Cancer Centre with BRAF-V600E mutation mCRC who did not receive combination BRAF/anti-EGFR therapy, diagnosed with colorectal cancer prior to March 31, 2021. Demographic, clinical, pathologic and cancer characteristics were abstracted from electronic medical records. Outcomes of interest included median duration of therapy and overall survival (OS). **Results:** 71 patients were included with a median age of 70 years:, 37 (52%) were females, 49 (69%)had right-sided tumors and 19 (27%) were mismatch repair deficient (dMMR). The primary tumor was unresected in 22 (30%) and resected in 49 (70%), completely resected in 44 (62%), and partially resected in 5 (7%) patients. Overall, median OS was 12.9 months 95% CI (8.4, 19) with 17 months 95% CI (11, —) mOS in patients with resected primary and 6.3 95% CI (4.8, 15) months in patients with unresected primary, HR 2.48 95% CI (1.42, 4.32), p=0.001. In univariate analysis: signet ring cell (p=0.007), good performance status (p<0.001), distant lymph nodes (p<0.001), peritoneal metastasis (p=0.008), metasetectomy (p=0.002), Charlston Comorbidity Index (p=0.003) were significant prognostic factors. Synchronous versus metachronous primary tumour was not a significant prognostic factor. **Conclusions:** Real world data confirms that patients with BRAF-V600E mutant mCRC have poor clinical outcomes. Our analysis suggests that resection of the primary tumor did not seem to impact on survival of BRAF-V600E mutant mCRC patients. This suggests that prognosis is dictated by the systemic disease and not complications of the primary tumour itself. Research Sponsor: None.

Of the patients with unresected primary tumour (n=21), 6 were synchronous.

Metastatic status	Primary Tumor Resected	Primary Tumor unresected	Total
Synchronous	23	22	45
Metachronous	26	0	26
Total	49	22	71

305 Poster Session

Impact of inpatient palliative care on end-of-life care among patients with early-onset colorectal cancer. First Author: Suriya Baskar, Brooklyn Hospital Center, Brooklyn, NY

Background: Palliative care has proven benefits in elderly patients with advanced cancer; however, the objective benefits of palliative care in younger patients with cancer remain understudied. We sought to examine the impact of inpatient palliative care consultation on end-of-life (EOL) care among hospitalized patients with early-onset colorectal cancer (EO-CRC). **Methods:** This study analyzed all admissions in the National Inpatient Sample (NIS) between 2016 and 2020 for patients younger than 50 years old with a diagnosis of CRC and death during the admission. Patients with CRC were identified utilizing ICD-10 codes C18.x, C19.x, and C20.x. Patient cohorts were made based on if they received inpatient palliative care consultation (ICD-10 code Z515). We assessed these encounters for patient demographics, Charlson comorbidity index (CCI), utilization of mechanical ventilation, vasopressors or blood transfusion, and presence of a Do Not Resuscitate (DNR) order. **Results:** Among 4,060 patients with EOCRC, 2,425 (59.7%) received inpatient palliative care consultation. There was no significant difference in age or length of stay between groups. The cohort of patients who received palliative care consultation had more female patients (44.3% vs 39.8%, p=0.004) and had higher comorbidity burden measured by the CCI (8.6 vs 8.3, p<.001). Patients receiving inpatient palliative care consultation at EOL were significantly more likely to have a DNR order (83.3% vs 44.6%, p<.001) in place and less likely to receive aggressive interventions, including mechanical ventilation (20.4% vs 43.4%, p<.001), blood transfusion (14.8% vs 22.6%, p<.001), vasopressor use (6.2% vs 8.6%, p=.004), and surgical intervention (3.0% vs 7.3%, p<.001). There was no difference in chemotherapy use between cohorts (2.8% vs 2.9%, p=0.8). Admissions including palliative care consultation were also associated with significantly lower average hospital charges ($99,387 vs $131,993, p<.001). **Conclusions:** Inpatient palliative care consultation at EOL among patients with EOCRC was associated with lesser use of aggressive interventions and higher rates of DNR code status. We also noted significantly lower costs of hospitalization among patients receiving inpatient palliative care consultation at EOL. These results underscore the importance of integration of inpatient palliative care consultation among patients with EOCRC at EOL. Research Sponsor: None.

	No Palliative N=1635	Palliative N=2425	p-value
Age (in years)	41.9 ± 6.1	42.1 ± 6.5	.28
CCI	8.3 ± 2.6	8.6 ± 2.3	**<.001**
Total Charges	131993 ± 206404	99387 ± 151864	**<.001**
DNR	730 (44.6%)	2020 (83.3%)	**<.001**
Blood transfusion	370 (22.6%)	360 (14.8%)	**<.001**
Mechanical ventilation	710 (43.4%)	495 (20.4%)	**<.001**
Vasopressor	140 (8.6%)	150 (6.2%)	**.004**
Chemotherapy	45 (2.8%)	70 (2.9%)	.800
Surgery	125 (7.3%)	75 (3.0%)	**<.001**

TPS306 Trials in Progress Poster Session

Trial in progress for a colorectal cancer detection blood test. First Author: Sarah Sawyer, Natera, Inc., Austin, TX

Background: Blood-based biomarkers for cancer screening have garnered much anticipation in recent years. Current colorectal cancer screening methodologies such as colonoscopy and stool-based tests are plagued with low compliance rates. While scientific advancement calls for clinical trial recruitment from diverse backgrounds to ensure health equity, traditional clinical trial design has led to the underrepresentation of such populations, resulting in a gap in our understanding of disease conditions, preventative measures, and appropriate treatment across populations. Here, we describe an ongoing, decentralized, prospective sample collection clinical study, PROCEED-CRC, to develop a blood-based CRC screening test in an average-risk population. **Methods:** The primary endpoint of the study is to achieve the successful enrollment of up to 1500 evaluable patients. Key eligibility criteria include participants aged ≥40 years, planning to undergo an asymptomatic screening colonoscopy, willing to provide a blood sample within 120 days prior to a colonoscopy procedure, and an informed consent prior to participation in the study. Exclusion criteria include individuals who have had a prior malignancy, have undergone diagnostic colonoscopy in the past 9 years, or any other recent CRC screening tests, precancerous findings on recent colonoscopy, or are at high risk for CRC based on germline carrier status. Primary analysis would include summarizing the number and percentage of participants classified to have CRC, advanced adenoma, or non-advanced neoplasia. In order to reach a diverse study cohort that is representative of this average-risk US population aged over 45 years, we implemented several non-traditional recruitment outreach efforts, such as engaging a pharmacy provider to whom individuals may have previously consented to receive messages/ notifications. Additional outreach efforts include digital advertisements and referral letters from healthcare professionals and community advocacy groups. Through these efforts, interested participants will have access to a link, a web address, or QR code to be screened for the study. After informed consent and enrollment, participants may schedule a blood draw via a mobile phlebotomy center based on their availability. Participants are considered evaluable if they meet the inclusion criteria, provide informed consent, provide an adequate blood draw, complete the screening colonoscopy within 120 days of the blood draw, and have medical records and pathology reports provided. The data from this study is expected to be completed in May 2025. Research Sponsor: None.

TPS307 Trials in Progress Poster Session

Colorectal cancer legal and administrative burden support (COLLABS): A pilot clinical trial. First Author: Quan Phung, University of Minnesota, Minneapolis, MN

Background: Patients with colorectal cancer often have health-related social needs (e.g. food insecurity, housing instability, employment loss). Failure to adequately address these is associated with financial hardship, worse quality-of-life, psychosocial distress, lower rates of treatment receipt, and worse survival. Cancer clinical care teams often lack the resources, knowledge, and processes to resolve these issues, many of which are actually legal in nature. Engagement with legal teams—medical-legal partnerships—can screen for, prevent, and resolve health-related social needs, as well as address related legal issues such as insurance denials, employment rights and protections, disability benefits, debt management, and estate planning. Cancer Legal Care (CLC) is a nonprofit providing free legal services to persons affected by cancer in MN. Past CLC clients have indicated a preference for proactive connection to CLC to prevent crises and overcome barriers to access such as concerns regarding perceived costs of legal services. **Methods:** This is a single-arm, pilot study to evaluate the feasibility, acceptability, and preliminary efficacy of proactive and free legal care support for patients with colorectal cancer at the University of Minnesota. The enrollment goal is 20 patients and the trial duration is 6 months. Inclusion criteria include English-speaking adults with advanced stage colorectal cancer and estimated life expectancy of >6 months. There is no requirement for baseline health-related social need(s) or time from diagnosis. After participants complete baseline surveys, we will connect them to a CLC attorney for an initial "legal care checkup"— the timing and method of connection (in-person, video visit, or phone call) is based on patient preference. Patients can bring up self-identified or active issues themselves, while CLC will also screen for needs. CLC will provide any required legal services and engage with the cancer care team as needed. CLC will follow-up with participants at 3 and 6 months, though services may continue after the study period off trial. Participants will complete surveys at baseline, 3 months, and 6 months. Surveys will assess patient-reported outcomes on financial toxicity, quality-of-life, comfort levels with accessing cancer care, status of health-related social and legal needs, perceived stress, life mastery and engagement, relationship with CLC contact, and acceptability of the intervention. The primary outcomes are feasibility (rates of initial engagement, continued engagement, and intervention completion; feasible if >50% on each metric) and acceptability. We will conduct exit qualitative interviews. We will collect sociodemographics, rates of treatment adherence/ completion, legal services provided, outcome of legal issues, and monetary savings or recovery for patients. Enrollment opened August 2024 and we have enrolled 15/20 participants. Clinical trial information: NCT06475664. Research Sponsor: Minnesota Colorectal Cancer Research Foundation.

TPS308 Trials in Progress Poster Session

Phase 2 randomized study evaluating safety, efficacy, and optimal dose of ABBV-400 in combination with fluorouracil, folinic acid, and bevacizumab in previously treated patients with metastatic colorectal cancer. First Author: Kanwal Pratap Singh Raghav, The University of Texas MD Anderson Cancer Center, Houston, TX

Background: Colorectal cancer (CRC) is the third most common cancer. In the metastatic (m) setting, the 5-year relative overall survival is approximately 15%. Conventional treatment comprises fluorouracil (5-FU)–based chemotherapy. Recently, targeted therapies have been studied for specific molecular subtypes. c-Met overexpression frequently occurs in a variety of tumors, including CRC. ABBV-400 is a c-Met–directed antibody-drug conjugate composed of the monoclonal antibody telisotuzumab conjugated to a potent topoisomerase 1 inhibitor payload. Preliminary data from the first-in-human study of ABBV-400 in patients with advanced solid tumors indicate encouraging efficacy of ABBV-400 monotherapy in patients with third-line or later mCRC. This phase 2 randomized study evaluates the safety, efficacy, and optimal dose of ABBV-400 in combination with 5-FU, folinic acid (FA), and bevacizumab (bev) in patients with mCRC with progression after first line (1L) treatment. **Methods:** Global, open-label, phase 2 randomized controlled study (NCT06107413). Eligible patients (≥18 years) have confirmed unresectable mCRC and measurable disease per RECIST v1.1, are microsatellite stable or mismatch repair proficient, *BRAF* V600E wild type, and have progression after 1L combination chemotherapy ± an anti-vascular endothelial growth factor or anti-epidermal growth factor receptor antibody. Primary objectives are (a) optimize ABBV-400 dose in combination with 5-FU, FA, and bev; (b) evaluate the efficacy of the combination, using objective response and progression-free survival as dual primary endpoints; (c) evaluate the safety and tolerability of the combination. Approximately 206 patients planned for enrollment in 2 stages: safety lead-in dose escalation (stage 1; n=30) and dose optimization (stage 2; n=176). In stage 1, patients receive escalating doses of ABBV-400 either every 2 weeks (Q2W; 0.8–2.4 mg/kg) or every 4 weeks (Q4W; 1.6–3.0 mg/kg) in combination with Q2W 5-FU (2400 mg/m^2 infusion), FA (200 mg/m^2), and bev (5 mg/kg) in 28-day cycles. Dose escalation of ABBV-400 uses a Bayesian optimal interval design, with target toxicity rate of 30%. Dose-limiting toxicities (DLT) are assessed during cycle 1, with ≥6 DLT evaluable patients required to declare a dose safe for the dose-optimization stage. In stage 2, patients are randomized to up to 4 ABBV-400 dose cohorts (2 with Q2W and 2 with Q4W ABBV-400 schedule; all in combination with Q2W 5-FU, FA, and bev) and a comparator cohort (irinotecan [180 mg/m^2] + 5-FU [400 mg/m^2 bolus and 2400 mg/m^2 infusion] + FA [200 mg/m^2] + bev [5 mg/kg]; all Q2W). Patients are treated until progression, unacceptable toxicity, or other discontinuation criteria are met. Enrollment was initiated in November 2023, with 3 patients enrolled as of January 4, 2024. Clinical trial information: NCT06107413. Research Sponsor: None.

TPS309 Trials in Progress Poster Session

Alliance A022101/NRG-GI009: A pragmatic randomized phase III trial evaluating total ablative therapy for patients with limited metastatic colorectal cancer—Evaluating radiation, ablation, and surgery (ERASur). First Author: Paul Bernard Romesser, Memorial Sloan Kettering Cancer Center, New York, NY

Background: Patients with oligometastatic colorectal cancer (CRC) often experience long-term progression-free survival and overall survival (OS) following aggressive local treatment of isolated metastases, especially in the liver. However, the effectiveness of local ablative therapies, such as microwave ablation (MWA) and stereotactic body radiation therapy (SBRT), in patients with inoperable limited metastatic CRC or those with additional metastases outside the liver or lungs, remains uncertain. Despite the historical use of local therapy for oligometastatic CRC, often influenced by provider bias rather than strong evidence, questions persist about the potential benefits of applying this approach to patients with more extensive disease. This trial aims to address this clinical challenge by evaluating the safety and efficacy of adding total ablative therapy (TAT) of all disease sites to standard systemic treatment in patients with limited metastatic CRC. **Methods:** A022101/NRG-GI009 is a randomized phase III study within the National Clinical Trials Network, set to enroll 364 patients with newly diagnosed metastatic CRC (BRAF wild-type, microsatellite stable). Patients with liver-only metastatic disease are excluded. Pre-registration during first-line systemic therapy induction is encouraged. Eligible lesions must be treatable with surgical resection, MWA, and/or SBRT. Patients without significant radiographic progression after 12-39 weeks of first-line systemic therapy, with 4 or fewer disease sites post-induction chemotherapy, will be randomized 1:1. Randomization will be stratified by the number of metastatic organ sites (1-2 vs. 3-4), timing of metastatic disease diagnosis (de novo vs. secondary), and the presence of metastatic disease outside the liver/lungs in at least one site. Patients in Arm 1 will receive TAT, targeting all metastatic sites with a combination of surgical resection, MWA, and/or SBRT, followed by standard systemic therapy. Patients in Arm 2 will continue with standard systemic therapy alone. The choice of systemic therapy is at the discretion of the medical oncologist and may include continued therapy, maintenance chemotherapy, or observation. The primary endpoint is OS, with secondary endpoints including event-free survival, treatment-related toxicities, and local recurrence. Exploratory biomarker analyses will also be conducted. The study requires 346 evaluable patients across the two arms to demonstrate an OS improvement with a hazard ratio of 0.7, providing 80% power with a one-sided alpha of 5%. The trial employs a group sequential design with two interim analyses (at 25% and 50% of events) to assess futility. The trial began in January 2023, and patient recruitment is ongoing. Clinical trial information: NCT05673148. Research Sponsor: National Cancer Institute.

TPS310 Trials in Progress Poster Session

NRG-GI008: Colon adjuvant chemotherapy based on evaluation of residual disease (CIRCULATE-North America). First Author: Christopher Hanyoung Lieu, Division of Medical Oncology, Department of Medicine, University of Colorado Anschutz Medical Campus, Aurora, CO

Background: Currently, there are no biomarkers validated prospectively in randomized studies for resected colon cancer (CC) to determine need for adjuvant chemotherapy (AC). However, circulating tumor DNA (ctDNA) represents a highly specific and sensitive approach (especially with serial monitoring) for identifying minimal/molecular residual disease (MRD) post-surgery in CC patients (pts), and may outperform traditional clinical and pathological features in prognosticating risk for recurrence. CC pts who do not have detectable ctDNA (ctDNA-) are at a much lower risk of recurrence and may be spared the toxicities associated with AC. Furthermore, for CC pts with detectable ctDNA (ctDNA+) who are at a very high risk of recurrence, the optimal AC regimen has not been established. We hypothesize that for pts whose CC has been resected, ctDNA status may be used to risk-stratify for making decisions about AC. **Methods:** In this prospective phase II/III trial, up to 1,912 pts with resected stage IIB, IIC, and III CC will be enrolled. Based on the post-operative ctDNA status using personalized and tumor-informed assay (Signatera, bespoke assay), those who are ctDNA- (Cohort A) will be randomized to immediate AC with fluoropyrimidine (FP)+oxaliplatin (Ox) for 3-6 mos per established guidelines *v* serial ctDNA monitoring. Patients who are ctDNA+ post-operatively, or with serial monitoring (Cohort B), will be randomized to FP+Ox *v* more intensive AC with addition of irinotecan (I) for 6 mos. One cycle of chemotherapy is allowed while awaiting ctDNA testing results for cohort assignment. The primary endpoints for Cohort A are time to ctDNA+ status (phase II) and disease-free survival (DFS) (phase III) in the immediate *v* delayed AC arms. The primary endpoint for Cohort B is DFS in the FP+Ox *v* FP+Ox+I arms for both phase II and phase III portions of the trial. Secondary endpoints include prevalence of detectable ctDNA post-operatively, time-to-event outcomes (overall survival and time to recurrence) by ctDNA status, and the assessment of compliance to adjuvant therapy. Biospecimens including archival tumor tissue, as well as post-operative plus serial matched/normal blood samples, will be collected for exploratory correlative research. Active enrollment across the NCTN started in June 2022 with CCTG sites joining in August 2023. Current accrual (as of 9-20-2024): 500/1,912. Clinical trial information: NCT05174169. Research Sponsor: National Cancer Institute; U10CA180868; National Cancer Institute; U10CA180822; National Cancer Institute; UG1CA189867; Natera, Inc.

TPS311 Trials in Progress Poster Session

Colorectal cancer metastatic dMMR immunotherapy (COMMIT) study: A randomized phase III study of atezolizumab (atezo) monotherapy versus mFOLFOX6/bevacizumab/atezo in the first-line treatment of patients with deficient DNA mismatch repair (dMMR) or microsatellite instability-high (MSI-H) metastatic colorectal cancer 9NRG-GI004/SWOG-S1610). First Author: Michael J. Overman, The University of Texas MD Anderson Cancer Center, Houston, TX

Background: Approximately 45% of dMMR/MSI-H metastatic colorectal cancer (mCRC) in the immunotherapy arm progressed at 12 mos (KEYNOTE 177). We hypothesize that dMMR/MSI-H mCRC patients (pts) may be more effectively treated with the combination of PD-1/PD-L1 (PD-1) pathway blockade and mFOLFOX6/bevacizumab (bev) rather than with anti-PD-L1 therapy (atezo) alone. Preclinical work demonstrated synergistic effects between anti-PD-1/anti-VEGF as well as between oxaliplatin/anti-PD-1 in murine CRC models, and phase II data showed activity of anti-PD-1/anti-VEGF in chemotherapy refractory colon cancer. Within the Atezo-TRIBE 8-pt dMMR CRC subgroup treated with FOLFOXIRI+bev+atezo, median PFS was not reached, with the first progression event at ~16 mos. Additionally, in other solid tumor malignancies, anti-PD-1 plus anti-VEGFr (i.e., HCC and RCC) as well as anti-PD-1 plus chemotherapy (i.e., gastroesophageal and lung cancers) combinations are standard first-line treatments. **Methods:** This two-arm prospective phase III open-label trial randomizes (1:1) mCRC dMMR/MSI-H to atezo monotherapy *v* mFOLFOX6/bev+atezo combination. Key inclusion criteria have been simplified on recent amendments to better mirror clinical practice for pts receiving mFOLFOX6/bev+atezo: One cycle of FOLFOX or CAPOX, with or without bev (or biosimilar) prior to enrollment allowed, dMMR tumor determined by local CLIA-certified IHC assay (MLH1/MSH2/MSH6/PMS2) or MSI-H by local CLIA-certified PCR or NGS panel; pts with total bilirubin ≤4.0 x ULN; duration of therapy for up to two years for both arms; imaging frequency on post-treatment follow-up has been reduced; as has measurable disease per RECIST. Primary endpoint is PFS. Assuming the atezo monotherapy control arm has a 48% PFS at 24 mos as assessed by site investigator, we have 80% power to detect a hazard ratio of 0.6 (equivalent to 64.4% PFS at 24 mos) with alpha 0.025 one-sided. Stratification factors include BRAFV600E status, metastatic site, and prior adjuvant CRC therapy. Secondary endpoints include overall survival, objective response rate, safety profile, disease control rate, and duration of response. Archived tumor tissue and blood samples will be collected for correlative studies. Harmonization of translational analyses is planned between GI004 (COMMIT) and A021502 (ATOMIC). Sample size has been modified with the accrual goal of 120 pts randomized between the two immunotherapy arms needed for study completion. Enrollment actively continues at U.S. sites. Current accrual (as of 9-20-2024): 100/120. Clinical trial information: NCT02997228. Research Sponsor: National Cancer Institute; U10 CA180868; National Cancer Institute; U10 CA180822; National Cancer Institute; U10 CA180888; National Cancer Institute; UG1 CA189867; National Cancer Institute; U24 CA196067; Genentech, Inc.

TPS312 Trials in Progress Poster Session

Total neoadjuvant therapy followed by total mesorectal excision and selective lateral lymph node dissection for locoregionally advanced low rectal cancer: A phase III randomized controlled trial (JCOG2207). First Author: Masayoshi Yasui, Osaka International Cancer Institute, Osaka, Japan

Background: For locally advanced low rectal cancer (RC), total neoadjuvant therapy (TNT) followed by total mesorectal excision (TME) confers a survival advantage compared with neoadjuvant chemoradiotherapy (nCRT) in the West, even though the local control rate was similar between TNT and nCRT. The Japan Clinical Oncology Group (JCOG) Colorectal Cancer Study Group has been developing therapies for RC based on recurrence risk according to presence/absence of clinical lateral lymph node metastasis (cLLNM). Our previous study in Japan showed that the standard of care (SOC) for RC is TME with lateral lymph node dissection (LLND) plus adjuvant chemotherapy (CTx). However, the prognosis for cStage III low RC remains poor regardless of cLLNM status. Therefore, new treatment strategies are needed for this population. In cLLNM-positive cStage III low RC patients, previous research suggests that adding nCRT to LLND would reduce local recurrence and improve survival. TNT and LLND may further improve outcomes. In cLLNM-negative cStage III low RC, several reports have suggested that nCRT as an alternative addition to LLND has comparable efficacy in terms of local control. Thus, in these populations, if TNT can replace LLND, adverse events due to LLND can be avoided while improving prognosis. We hypothesize that TNT + TME + selective LLND would be the most promising strategy for cLLNM-positive and -negative cStage III low RC. **Methods:** Eligibility criteria include low rectal adenocarcinoma or adenosquamous carcinoma, cT2-cT4 tumor depth, clinically positive lymph node metastasis, no distant metastasis, no history of pelvic irradiation or rectal surgery, age 18-75 years, and sufficient organ function. Eligible patients are randomized (1:1) with adjustment factors of institution, sex, and cLLNM status. The experimental treatment arm receives TNT, short-course radiotherapy (25 Gy/5 fractions to whole pelvis), and consolidation chemotherapy consisting of 6 courses of CAPOX (oxaliplatin 130 mg/m2 on day 1 and capecitabine 2000 mg/m2/day on days 1-14). After completing TNT, TME + selective LLND are performed for cLLNM-positive cases or TME alone for cLLNM-negative cases. The SOC arm receives TME + LLND followed by adjuvant chemotherapy with 8 cycles of CAPOX (same as above). The primary endpoint is overall survival (OS) in the intention-to-treat population. To detect an increase in 5-year OS from 76% to 84%, corresponding to a target hazard ratio of 0.64, 408 patients (108 events) would achieve 75% power at a one-sided α of 0.05, an expected accrual period of 4 years, and a follow-up period of 5 years. The total sample size was set at 420 patients to account for patients lost to follow-up. Accrual in JCOG2207 began in October 2023. Clinical trial information: s031230415. Research Sponsor: National Cancer Hospital Japan.

TPS313 Trials in Progress Poster Session

Combining low-dose regorafenib with pembrolizumab for patients with MSI-H colorectal cancer: REGPEM-CRC-01. First Author: Ibrahim Halil Sahin, University of Pittsburgh Medical Center (UPMC), Pittsburgh, PA

Background: Currently, pembrolizumab is one of the front-line therapies for patients with MSI-H CRC. However, approximately 40% of patients who received pembrolizumab experienced disease progression early in the course of disease (KEYNOTE 177). Therefore, there is still an unmet need to enhance the efficacy of checkpoint inhibitors in MSI-H CRC. MSI-H CRC has a higher level of expression of VEGF in blood compared to patients compared to its MSS counterpart (Hansen et al. Colorectal Dis. 2011). Consistently, exploratory analysis of CALBG-80405 and PARADIGM trial showed that patients with MSI-H CRC were more likely to benefit from anti-VEGF therapy than anti-EGFR therapy regardless of the side of the tumor. NSABP C-08 also suggested that anti-VEGF therapy may have biological activity even in as adjuvant therapy for patients with MSI-H colon cancer. Regorafenib is a potent VEGF inhibitor, with preclinical evidence showing its immune modulatory effect in the tumor microenvironment. In this trial, we hypothesize that adding low-dose regorafenib to pembrolizumab may induce synergistic activity beyond their independent clinical efficacy and create deep and durable responses for patients with MSI-H CRC. **Methods:** In the lead arm of this prospective randomized study, 22 patients will be enrolled through Hoosier Cancer Research Network (HCRN-GI23-643). Patients with treatment naïve MSI-H CRC will be enrolled in this front-line trial. One cycle of pembrolizumab and up to 3 cycles of chemotherapy prior to determination of MMR-D/MSI-H is allowed. Patients will receive regorafenib 60 mg daily in combination with pembrolizumab 200mg IV in cycle 1, followed by regorafenib 90mg in subsequent cycles to improve treatment tolerance. The primary outcome that will be measured is ORR, defined as the percentage of partial or complete response to the treatment within 12 months. ORR will be measured using RECIST 1.1. criteria. A formal one-sided hypothesis test will be conducted for futility, assuming that we will reject the null hypothesis of a target ORR only if we have strong evidence. In this study, we assume a null hypothesis that ORR is 0.60, which would reflect significant clinical improvement over the current standard of ORR = 0.43 from KEYNOTE 177. The alternative hypothesis is that ORR is less than 0.60. For the lead-in phase of the study, the emphasis is on controlling Type I error to be small, approximately 0.05. The test statistic will be the number of ORRs in the 22 patients, which we assume to follow a binomial distribution. Clinical trial information: NCT06006923. Research Sponsor: None.

TPS314 Trials in Progress Poster Session

A phase II, randomized study of JMT101 in combination with irinotecan and SG001 versus regorafenib in patients with metastatic colorectal adenocarcinoma (mCRC). First Author: Jianmin Xu, Department of Colorectal Surgery and Colorectal Cancer Center, Zhongshan Hospital, Shanghai, China

Background: Epidermal growth factor receptor (EGFR) inhibitorcombined with-chemotherapy based on5-FU with oxaliplatin and irinotecanis the first-line treatment of wild-type RAS mCRC.JMT101 is a humanized monoclonal IgG1 antibody targeting EGFR. Previous trials showed that JMT101 monotherapy or in combination with chemotherapy had demonstrated a promising anti-tumor activity in advanced solid tumors with good safety profiles. EGFR inhibitor might have a synergetic effect with immunocheckpoint inhibitor (ICI) by activating innate and adaptive immune response. SG001 is a fully humanized and high-affinity immunoglobulin G4 monoclonal antibody that targets programmed death-1 (PD-1), which also had encouraging efficacy in advanced solid tumors. Therefore, EGFR antibody in combination with chemotherapy plus ICI is worthy exploration in patients with mCRC. **Methods:** This is a phase II, multicenter, open-label, two-part study. Part I is a safety run-in, followed by part II randomized treatment. Eligible patients are histologically confirmed mCRC without MSI-H/dMMR, wild-type RAS and BRAF, who progressed on at least 2 prior systemic therapy with chemotherapy based on 5-FU with oxaliplatin and irinotecan, with or without cetuximab and bevacizumab (both parts); no prior treatment with regorafenib, fruquintinib, or TAS-102 (part II). Safety run-in data will be analyzed after 3-6 patients have received JMT101 6 mg/kg plus irinotecan 180 mg/m^2 plus SG001 240 mg Q2W. Upon dose confirmation, 90 patients with mCRC will be randomized into three arms (1:1:1) to receive JMT101 plus irinotecan plus SG001, or JMT101 plus irinotecan, or regorafenib monotherapy. Primary endpoints include incidence of dose limiting toxicities (DLT) and adverse events (part I) and objective response rate (ORR) as per RECIST v1.1 by investigator (part II). Secondary endpoints include JMT101 and SG001 pharmacokinetic profile, antidrug antibodies (ADA) (both parts), disease control rate (DCR), duration of response (DOR), progression free survival (PFS), and overall survival (OS), as well as safety (part II). Prespecified goal for safety run-in part was met; randomized treatment part accrual began in February 2024. Clinical trial information: NCT06089330. Research Sponsor: Shanghai JMT-Bio Technology Co., Ltd.

TPS315 Trials in Progress Poster Session

Chemoradiotherapy (CRT) followed by sequential tislelizumab (TIS) + CAPOX and TIS monotherapy for organ preservation in locally advanced low rectal cancer. First Author: Wentao Tang, Department of Colorectal Surgery and Colorectal Cancer Center, Zhongshan Hospital, Shanghai, China

Background: CRT followed by total mesorectal excision (TME) is the standard care for patients with locally advanced low rectal cancer (≤ 5cm from the anal margin). However, TME may lead to permanent colostomy and severely impact quality of life. Organ preservation is, therefore, a critical unmet need for these patients. The proportion of patients achieving clinical complete response (cCR) after CRT is relatively low, with the literature reporting around 20%. Total neoadjuvant therapy (TNT) has been investigated to increase cCR rate. Several clinical trials demonstrated that PD-1 inhibitor combined with TNT could result in significate improvement in pathological complete response rate (pCR rate, 39.8% vs 15.3%, $p < 0.001$ in UNION study) or complete response rate (pCR+cCR rate, 44.8% vs 26.9%, p = 0.031 in a randomized phase 2 study) compared with TNT alone in rectal cancer patients. The RELIEVE-01 study (NCT06390982) was designed to evaluate the efficacy in organ preservation with CRT followed by sequential TIS + CAPOX and TIS monotherapy in patients with low rectal cancer. **Methods:** This multicenter, single-arm, phase 2 study will enroll 46 pts with histopathologically confirmed rectal adenocarcinoma (≤ 5cm from the anal margin), pMMR/MSI-L/MSS, cT1-3N1M0/T2-3N0M0, and ECOG PS ≤1. Pts will initially receive 6 weeks of CRT (50.4 Gy/28F, capecitabine 825 mg/m^2, bid, d1-5 each week), followed by 4 cycles of TIS + CAPOX (TIS, 200 mg, IV, d1; capecitabine 1000 mg/m^2, bid, d1-14; oxaliplatin 130 mg/m^2, d1) in a 21-day cycle. Then, clinical response will be assessed to determine the subsequent treatment. Patients with cCR will receive TIS + CAPOX (4 cycles) and TIS (up to 9 cycles), and then be managed using the W&W strategy. Patients with non-cCR will undergo TME. Patients with near-cCR will be given local excision, and those achieving pCR will receive the same treatment as cCR patients, and for those with non-pCR after local excision, TME will be performed. The primary endpoint is rate of CR (cCR plus pCR post local resection). Secondary endpoints include organ-preserving rate, event-free survival rate and OS rate at 1, 2 and 3 years, respectively. Recruitment is ongoing. Clinical trial information: NCT06390982. Research Sponsor: None.

TPS316 Trials in Progress Poster Session

Cetuximab and vemurafenib plus liposomal irinotecan (II), leucovorin and fluorouracil for BRAF^{V600E}-mutated advanced colorectal cancer (IMPROVEMENT2): A multicenter, open-label, randomized controlled trial. First Author: Zhan Wang, Department of Medical Oncology, Changzheng Hospital, Naval Medical University, Shanghai, China

Background: Current first-line treatment regimen for advanced colorectal cancer with BRAF^{V600E} gene mutation is chemotherapy combined with anti-angiogenic drug, but the efficacy is somewhat limited. Theoretically, targeting BRAF^{V600E} mutation is an effective treatment strategy and has achieved success in lung cancer and melanoma. However, the objective response rate (ORR) is below 5% in colorectal cancer, suggesting that BRAF^{V600E} mutation in colorectal cancer may have different feedback regulation mechanisms compared to other cancer types. Combining chemotherapy (such as 5-fluorouracil plus irinotecan) with BRAF/EGFR targeted therapy may potentially achieve good efficacy. Subsequently, we conducted the IMPROVEMENT (NCT03727763) study, which showed an objective response rate (ORR) of 85% and a disease control rate of 100% with this regimen, significantly higher than those with chemotherapy combined with anti-angiogenic drug. The recent BREAKWATER study showed similar trends. To further validate the efficacy and safety of this regimen, we designed a randomized controlled trial. **Methods:** IMPROVEMENT2 (NCT06603376) is a multicenter, open-label, randomized controlled trial in patients with histologically or cytologically confirmed advanced colorectal cancer. Adult patients who are previously untreated in the metastatic setting, have a confirmed BRAF^{V600E} mutation, measurable disease per Response Evaluation Criteria In Solid Tumors (RECIST) version 1.1, and Eastern Cooperative Oncology Group performance status of 0-1 are eligible. Patients who received previous treatment in the adjuvant setting but experienced metastasis or recurrence within 12 months are permitted. A total of 75 eligible patients will be randomly assigned (2:1) to receive liposomal irinotecan(II) 60 mg/m^2, leucovorin 400 mg/m^2, fluorouracil 2800 mg/m^2, and cetuximab 500 mg/m^2 on day 1 of each 14-day cycle plus oral vemurafenib 720 mg twice daily, or liposomal irinotecan(II) 60 mg/m^2, leucovorin 400 mg/m^2, fluorouracil 2800 mg/m^2, and bevacizumab 5 mg/kg on day 1 of each 14-day cycle. Treatment will be continued until disease progression or unacceptable toxicity. The primary endpoint is ORR per RECIST 1.1 by blinded independent central review (BICR). Key secondary endpoints are depth of response, progression-free survival per RECIST 1.1 by BICR, overall survival, and safety. The study is actively enrolling patients. Clinical trial information: NCT06603376. Research Sponsor: None.

TPS317 Trials in Progress Poster Session

A phase 2, multicenter, open-label study of AlloStim in-situ cancer vaccine immunotherapy as 3L therapy in MSS/pMMR metastatic colorectal cancer. First Author: Wirote Lausoontornsiri, Mirror Biologics, Inc, Wesley Chapel, FL

Background: Neoantigen therapeutic cancer vaccines have had limited clinical benefit in metastatic tumor settings. Tumor immunoediting selects tumor clones over time that are non-immunogenic. In addition, the metastatic tumor microenvironment (TME) becomes profoundly immunosuppressive. Microsatellite stable (MSS) and mismatch repair proficient (pMMR) metastatic colorectal cancer (mCRC) tumors do not accumulate mutations, resulting in low mutational burden (TMB) and a lack of neoantigen targets. Consequently, therapeutic vaccine approaches for mCRC have had difficulty in demonstrating clinical effectiveness. Novel approaches that boost the immune response in low TMB tumors may benefit mCRC patients. AlloStim is an allogeneic, non-genetically manipulated, living Th1-like cell immunotherapy. The Th1-like cells are ex-vivo differentiated and expanded from healthy blood donors and activated prior to infusion. These cells are designed to be rejected upon intradermal (ID) injection. Repeated ID injections elicit high titers of allo-specific Th1 memory cells in circulation. In allo-primed patients, intravenous (IV) infusion of AlloStim causes a rejection response that releases cytokines, such as IL-12, which non-specifically activate circulating memory T cells and NK cells. Activated host memory T-cells and NK cells extravasate and traffic to metastatic tumor sites. This creates the conditions for "in-situ vaccination". Activated NK cells cause disruption of the cell membrane and release of chaperoned neoantigens into the tumor microenvironment, a process known as immunological cell death (ICD). The release of neoantigens into the microenvironment in the context of inflammation mediated by infiltrating Th1 memory cells creates conditions for *in-situ* vaccination. *In situ* vaccination represents an alternative immunotherapy approach in which a therapeutic cancer vaccine is generated *in vivo* without the need to previously identify and isolate tumor neoantigens. **Methods:** To investigate this novel in-situ vaccine mechanism, a phase 2, multicenter, single-arm clinical trial in 3L MSS/pMMR mCRC subjects that have been previously treated with oxaliplatin-containing and irinotecan-containing chemotherapy regimens, anti-EGFR (RAS wt and left-sided) and anti-VEGF was initiated. Patients were administered weekly AlloStim in three 28-day cycles consisting of 4 ID injections and a combination ID and IV infusion on the final week in each cycle to create self-amplifying waves of allo-specific and tumor-specific host immune cells in circulation which traffic to tumors upon IV infusion. The primary endpoint is overall survival. 21 evaluable patients have been enrolled and are currently being followed for maturity of the survival data. Clinical trial information: NCT04444622. Research Sponsor: Mirror Biologics, Inc.

TPS318 Trials in Progress Poster Session

A phase 2, multicenter, open-label study of AlloStim combined with anti-PD-L1 immunotherapy as 4L therapy in patients with MSS/pMMR metastatic colorectal cancer. First Author: Wirote Lausoontornsiri, Mirror Biologics, Inc, Wesley Chapel, FL

Background: Microsatellite stable (MSS)/mismatch repair proficient (pMMR) metastatic colorectal cancer (mCRC) are immunologically 'cold' tumors that do not respond to immunotherapy. Cold tumors represent a formidable hurdle for checkpoint blockade immunotherapy approaches. Therefore, there is a need for novel strategies that could reprogram mCRC cold tumors to become inflamed hot tumors in a manner that might potentially convert low tumor mutational burden (TMB) tumors into checkpoint blockade responsive tumors. AlloStim is an experimental allogeneic, non-genetically manipulated, Th1-like activated living cell immunotherapy derived from healthy unrelated blood donors. The mechanism of action is designed to convert immunologically cold tumors into inflamed hot tumors. A case study of a single 3L MSS/pMMR mCRC patient primed with AlloStim followed by combination anti-PD-1/anti-CTLA4 checkpoint blockade resulted in a rare objective tumor response (Hirschfeld, et al, Translational Medicine Communications (2024) 9:15). The AlloStim priming results in an abundance of infiltrating interferon-gamma producing Th1 cells. Interferon-gamma has been shown to increase expression of PD-L1 in the TME. Therefore, it is hypothesized that the combination of AlloStim with an anti-PD-L1 checkpoint inhibitor might optimally evoke anti-tumor immune responses in this immunotherapy-refractory MSS/pMMR mCRC population. **Methods:** The study is a phase 2, multicenter, single-arm clinical trial of MSS/pMMR mCRC subjects in 4L that have been previously treated with an oxaliplatin-containing and irinotecan-containing chemotherapy regimen, anti-EGFR (RAS wt and left-sided), anti-VEGF and have progressed on 3L TAS-102 +/- bevacizumab or regorafenib or fruquinitinib. The study is evaluating weekly AlloStim priming from day 0 to day 49 followed by combination of AlloStim IV infusion (q2w) followed a week later by anti-PD-L1 (avelumab 800mg/q2w) from days 63 to day 98. Objective tumor response and disease control rate are the primary end-points. Overall survival is the secondary endpoint. Clinically stable patients at day 112 can continue in an expansion phase with another round of combination AlloStim and avelumab from day 119 to day 154. CT scans are conducted at baseline, day 56 after AlloStim priming, day 112 after combination of alternating q2w infusions of AlloStim and avelumab and day 168 in all patients. CT scans will be analyzed by independent central readers using RECIST 1.1. Up to 50 patients will be enrolled with an interim analysis performed to assess efficacy after 20 patients become evaluable. The study is open and has not yet enrolled any patients at time of submission. Clinical trial information: NCT06557278. Research Sponsor: Mirror Biologics, Inc.

TPS319 Trials in Progress Poster Session

A phase 2 study of intravenous pembrolizumab and intratumorally injected autologous dendritic cells (DCs) in refractory colorectal cancer (CRC). First Author: Sarbajit Mukherjee, Roswell Park Comprehensive Cancer Center, Buffalo, NY

Background: Immune checkpoint inhibitors (ICIs) have shown limited benefit in CRC outside the microsatellite instability high (MSI-H) population. Deficient cytotoxic T lymphocyte (CTL) trafficking to the tumor microenvironment (TME) and excess pro-tumorigenic cytokines and immunosuppressive cells may contribute to primary ICI resistance in microsatellite stable (MSS) CRC. Selective enrichment of T cells with specific chemokine receptors may enhance antitumor immunity. Studies highlight the role of IL-12 in improving progression-free survival (PFS) and overall survival (OS) in patients on DC therapies. Our group developed ST-aDC1s, optimized for high production of high levels of IL-12 and CTL-attracting chemokines when injected intratumorally. In preclinical models, ST-aDC1s combined with anti-PD-1/PD-L1 therapy showed therapeutic synergy, suggesting ST-aDC1s as a promising alternative to traditional DC vaccines for inducing CTL entry into tumors. **Methods:** We designed an open-label, non-randomized, phase 2 study of intratumoral autologous ST-αDC1 combined with pembrolizumab in recurrent/metastatic unresectable MSS CRC. Eligible patients must have prior treatment with, or contraindication to fluoropyrimidine, irinotecan, oxaliplatin, anti-VEGF, and anti-EGFR therapy (if RAS wild type), be anti-PD-1/PD-L1 therapy naïve, have ECOG PS $\leq$1, and have at least two target lesions, one amendable to biopsy and injection. Leukapheresis will collect monocytes for ST-αDC1 generation. Patients will receive intratumoral ST-αDC1s (6 million) on days 1 and 8 with biopsies, and an optional dose on day 50 before a third pembrolizumab dose. Pembrolizumab 200 mg IV will start on day 8, given every 3 weeks for 4 doses during the study period. Pembrolizumab will continue as conventional care afterwards. CT scans will occur after 2 and 4 doses of pembrolizumab. A safety lead-in cohort will include six patients, with up to 17 patients enrolled. The primary endpoint is objective response rate (ORR) at 12 weeks per RECIST 1.1. Secondary endpoints are safety, PFS, OS, and ORR per iRECIST. Correlative analysis will assess baseline levels and post-treatment changes in infiltrating T cells, CD4/CD8 ratios, FoxP3+ Tregs, and Teff-attracting and Treg-favoring chemokines from biopsy specimens, and immune cell subsets and cytokine levels from peripheral blood. With an expected ORR under 5% for standard treatment, 17 patients are required to show at least a 25% ORR with 81.2% power and a one-sided type I error rate of 0.05. A Simon two-stage Optimal design will enroll 9 patients in stage 1, and if $\geq$1 response occurs, 8 more patients will be added in stage 2. If $\geq$3 responses are observed, the treatment will be deemed promising for further study. Clinical trial information: NCT05518032. Research Sponsor: Roswell Park Comprehensive Cancer Center internal funding.

TPS320 Trials in Progress Poster Session

BXQ-350, a novel sphingolipid metabolism modulator, in combination with mFOLFOX7 and bevacizumab in newly diagnosed metastatic colorectal cancer (mCRC) patients: A phase 1b/2 randomized study. First Author: Daniel Blake Flora, St Elizabeth Healthcare, Edgewood, KY

Background: Ceramides and sphingosine-1-phosphate (S1P) are key bioactive signaling molecules. Ceramides are proapoptotic and mitigate chemoresistance. Conversely, S1P promotes cancer cell proliferation, activates multiple oncogenic pathways, and stimulates immuno-suppressor cell populations promoting a pro-tumoral microenvironment. Several studies in colorectal cancer patients have shown high levels of ceramides are associated with improved survival, while high S1P levels are associated with a poor prognosis. Hence, modulation of sphingolipid metabolism could be a promising therapeutic approach. BXQ-350 is a nanovesicle of Saposin C, an allosteric activator of sphingolipid metabolism, that lowers systemic S1P and increases C18 ceramide. BXQ-350 was investigated in a Phase 1 dose-escalation safety study in cancer patients with advanced solid malignancies (NCT02859857). BXQ-350 was safe and well-tolerated (no DLT, no MTD). Also, 13 patients (~17.8% of evaluable patients) had a clinical benefit up to cycle 6 and beyond (PR, SD). Among patients with PFS $>$ 6 months, there were 4 recurrent CRC patients: 1 patient had a PFS of ~12 months, 2 of ~18 months, and 1 is still on study after 7 years suggesting potential long-term benefit. Furthermore, there were signs that BXQ-350 may alleviate symptoms of CIPN as 4 out of 10 patients with chronic CIPN at time of enrollment experienced an improvement of their symptoms. **Methods:** BXQ-350 is being investigated in a Phase 1b/2 study in combination with mFOLFOX7 and Bevacizumab in newly diagnosed mCRC patients (NCT05322590) to assess the efficacy and safety of BXQ-350. Design of the Phase 1b (open label study): 1) A safety dose escalation part to establish the RP2D: patients will initially receive 1.8 mg/kg BXQ-350 in combination with mFOLFOX7 and Bevacizumab. If safe (no MTD), dose of BXQ-350 will be increased to 2.4 mg/kg and 9 additional patients will be entered at this dose level. If safe, then this dose will be the RP2D and 21 additional patients will be enrolled, completing a 30 patient expansion cohort 2) Efficacy will then be evaluated for all patients entered at the RP2D. Primary objectives of the Phase 1b are to assess safety, identify RP2D, and assess preliminary efficacy of BXQ-350 in this combination. A secondary objective is to determine if BXQ-350 decreases CIPN. Enrollment in the expansion cohort of 30 patients is nearly complete, with 26 patients enrolled as of September 2024. Clinical trial information: NCT05322590. Research Sponsor: Bexion Pharmaceuticals.

TPS321 Trials in Progress Poster Session

Phase 1/2 study of the HER3-directed antibody-drug conjugate patritumab deruxtecan (MK-1022) as monotherapy in patients with gastrointestinal cancers. First Author: Do-Youn Oh, Seoul National University Hospital, Seoul, South Korea

Background: A significant medical need exists for more effective and tolerable therapeutic options for patients with advanced gastrointestinal cancers (colorectal cancer [CRC], biliary tract cancer [BTC], and hepatocellular carcinoma [HCC]). Human epidermal growth factor receptor (HER3) expression is high in CRC and BTC, and low-to-intermediate in HCC. Patritumab deruxtecan (HER3-DXd; U3-1402/MK-1022) is a HER3-directed antibody-drug conjugate (ADC) comprising a fully human anti-HER3 immunoglobulin G1 monoclonal antibody (patritumab), which includes a plasma-stable, selectively cleavable linker, and a potent topoisomerase I inhibitor payload (DXd; released payload) that leverages the clinically validated deruxtecan (a derivative of exatecan) technology. DXd is cell membrane permeable and enables a bystander antitumor effect after DXd ADC internalization and linker cleavage, resulting in elimination of both target and neighboring tumor cells. Patritumab deruxtecan has shown antitumor activity in phase 1 and 2 studies of EGFR-mutated advanced non–small cell lung cancer and advanced breast cancer and in preclinical studies of CRC. This nonrandomized, open-label, phase 1/2 study (NCT06596694) will examine the safety and efficacy of patritumab deruxtecan in select gastrointestinal cancers. **Methods:** Approximately 130 adults (aged $\geq$18 years) will be allocated to 1 of 3 cohorts based on their disease. Patients in cohort 1 (CRC; n = 40) must have previously treated unresectable or metastatic colorectal adenocarcinoma. Patients in cohort 2 (BTC; n = 40) must have previously treated locally advanced or metastatic BTC. Patients in cohort 3 (HCC; n $\leq$50) must have previously treated advanced HCC. Prior topoisomerase I inhibitor therapy is not allowed. Patients in cohorts 1 and 2 will receive patritumab deruxtecan intravenously (IV) as monotherapy. Cohort 3 includes a dose-escalation phase to determine the recommended phase 2 dose of patritumab deruxtecan monotherapy and an efficacy phase. In the efficacy phase, patients will receive the recommended phase 2 dose of patritumab deruxtecan. The primary objectives are to evaluate safety and tolerability of patritumab deruxtecan and to evaluate the confirmed objective response rate per RECIST v1.1 assessed by blinded independent central review (BICR) for each cohort. Secondary objectives are to evaluate duration of response and progression-free survival per RECIST v1.1 by BICR and overall survival, and to characterize the pharmacokinetics of patritumab deruxtecan. Clinical trial information: NCT06596694. Research Sponsor: Funding for this study was provided by Daiichi Sankyo Company, Limited and Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA.

TPS322 Trials in Progress Poster Session

Phase 1/2 study of FOG-001, a first-in-class direct β-catenin:TCF inhibitor, in patients with colorectal cancer, hepatocellular carcinoma, and other locally advanced or metastatic solid tumors. First Author: Kyriakos P. Papadopoulos, South Texas Accelerated Research Therapeutics (START), Clinical Research Department, San Antonio, TX

Background: Wnt/B-catenin pathway activation, which is frequently present in most colorectal cancers and a variety of other solid tumors, is associated with poor prognosis and resistance to immunotherapy. FOG-001 is a Helicon peptide that competitively inhibits the interaction between β-catenin and the T-cell factor (TCF) family of transcription factors. Experiments in patient-derived-xenograft (PDX) models of colorectal cancer (CRC) and hepatocellular carcinoma (HCC) have demonstrated that FOG-001 can inhibit tumor growth and promote tumor regression when administered as monotherapy or in combination with immune checkpoint inhibitors or standard of care therapies, including bevacizumab and 5-FU. **Methods:** This first-in-human, Phase 1/2, multicenter, open-label, dose escalation (Part 1) and dose-expansion (Part 2) study evaluates the safety/tolerability, pharmacokinetics (PK), pharmacodynamics (PD), and anti-tumor effects of FOG-001 as monotherapy and in combination with other anti-cancer therapies. Eligible patients must have received at least one prior systemic anti-cancer therapy and either progressed on, not responded to, or be unfit for available therapies. In Part 1, FOG-001 is administered intravenously, either weekly or every other week, at escalating dose levels evaluated sequentially in a standard 3+3 design as monotherapy in patients with: (1a) microsatellite stable (MSS) CRC or any solid tumor with a documented Wnt/b-catenin pathway activating mutation (WPAM) and (1c) HCC with WPAM. Part 1b will evaluate PD effects in cohorts of approximately six patients with MSS CRC. Upon selection of the preliminary recommended Phase 2 dose from Part 1a for weekly dosing, combination dose escalation (1d) evaluates the safety/tolerability, PK and anti-tumor effects of FOG-001 in CRC patients combined with FOLFOX+bevacizumab, anti-PD-1/PD-L1 (also includes other solid tumors with known WPAM), or trifluridine/tipiracil+bevacizumab. Part 2 dose expansion evaluates FOG-001 monotherapy in MSS CRC, HCC, and other solid tumors with documented WPAM. Combination dose expansion evaluates the combinations initially evaluated in Part 1d. Primary endpoints are safety/tolerability and preliminary anti-tumor effects (objective response rate). Secondary endpoints are PK, PD, and other anti-tumor responses (e.g., best objective response, duration of response, and progression free survival). Enrollment in Parts 1 and 2 is ongoing in the USA. Clinical trial information: NCT05919264. Research Sponsor: Fog Pharmaceuticals.

ABSTRACT WITHDRAWN

TPS324 **Trials in Progress Poster Session**

Phase II single-arm, single-center clinical trial of all-trans-retinoic acid, bevacizumab, and atezolizumab in refractory microsatellite stable colorectal cancer. First Author: Syed Mohammad Ali Kazmi, UT Southwestern Medical Center, Dallas, TX

Background: A critical clinical challenge in microsatellite stable (MSS) metastatic colorectal (mCRC) patients is to identify strategies to overcome lack of response to immune checkpoint inhibitors. An immunosuppressive tumor microenvironment comprising myeloid-derived suppressor cells (MDSC), endothelial cells, regulatory T cells, tumor associated macrophages, and cancer associated fibroblasts promotes immune evasion and resistance to these agents. MDSCs are bone marrow derived myeloid cells that suppress T-cell function and promote tumor growth. Among all cancers, mCRC patients have one of the highest frequency of MDSC (CD11b+, CD33+, CD14+ HLA-DRneg) in blood (1). It was demonstrated that altering the MDSC% in the tumor microenvironment by ATRA and anti-VEGF therapy (bevacizumab) can enhance the effects of immune checkpoint inhibitors (2-4). The hypothesis of the current clinical trial is that the combination of ATRA, bevacizumab and atezolizumab will lead to a decrease in MDSC population in tumor microenvironment leading to a clinically meaningful improvement in response rates among refractory MSS mCRC patients. **Methods:** This is a single-arm, open-label, phase 2 clinical trial combining ATRA, bevacizumab, and atezolizumab in refractory MSS mCRC patients. It will enroll a total of 21 patients over 24 months at UT Southwestern Medical Center that are MSS by PCR or NGS testing or are proficient in immunohistochemical expression of all four mismatch repair enzymes (MLH1, MSH2, MSH6, PMS2). ATRA will be administered orally at 45 mg/m2/day in 2 divided doses on days 1-7 and repeated every 14 days; atezolizumab will be given intravenously on day 1 at 840 mg dose every 14 days, and bevacizumab will be administered intravenously on day 1 at 10 mg/kg dose every 14 days. The first six patients enrolled on this study will contribute to the safety lead-in phase of this study. The primary outcome is to assess the overall response rate by RECIST v1.1. Secondary outcomes include assessment of disease control rate and frequency of adverse events using CTCAE v5.0. Exploratory outcomes include assessment of PFS and OS and collecting blood and tissue samples at defined timelines to study the changes in the MDSC population among responders and non-responders (NCT05999812). 1. Kobayashi, M. et. al., Clin Cancer Res, 2019. 2. Mirza, N., et al., Cancer Res, 2006. 3. Tobin, R.P., et al., Int Immunopharmacol, 2018. 4. Tobin, R.P., et. al., Clin Can Res, 2023. Clinical trial information: NCT05999812. Research Sponsor: Eugene Frenkel Award; BMS Foundation.

TPS325 **Trials in Progress Poster Session**

Randomized study to assess colonic microbiome changes in response to energy drink consumption (ROSANNA Trial). First Author: Tim Jang, University of Florida, Gainesville, FL

Background: Colorectal cancer (CRC), the second leading cause of cancer deaths in the US, is increasingly diagnosed in individuals under 50. This rise in early-onset CRC (eoCRC) led the U.S. Preventative Services Task Force to recommend a reduction in the age to begin colorectal cancer screening in average-risk persons, from 50 to 45. The etiology of eoCRC is multifaceted, with one postulated theory pointing to alterations in the young adult colonic microbiome. Hydrogen sulfide (H_2S) producing bacteria, like *Bilophila wadsworthia*, *Fusobacterium nucleatum* and *Atopobium parvulum*, are typically minor gut microbiota components but are overrepresented in CRC cases, linked to inflammation, and may promote a pro-carcinogenic environment. These bacteria preferentially use taurine, an essential amino acid, as a primary energy source. Energy drinks represent one of the largest dietary sources (6-16x normal daily intake) of taurine in contemporary diets. Our hypothesis is that high taurine levels in energy drinks could exacerbate CRC risk by promoting preferential growth and metabolic activities of already present H_2S-producing bacteria, contributing to the rise of eoCRC. **Methods:** This randomized open-label trial is actively enrolling up to 60 young adults (18-40 years) without personal or family history of CRC, other major GI disease/distress, and baseline infrequent energy drink consumption. Group 1 (Arm A; n=30), will consist of subjects consuming at least one of two protocol-specified energy drinks daily for 4 weeks. Group 2 (Arm B; n=30), will consist of usual dietary intake. Enrollments are randomized 1:1 and stratified, by gender, age (18-25 vs. 26-40), and baseline energy drink consumption (rare/none vs. 1-2 per week). Every subject will complete: 1) a daily dietary and fitness log, 2) a weekly wellness log, 3) collection of stool/saliva/skin/urine at 3 sequential time points (baseline, after 2 weeks, after 4 weeks), and 4) blood collection at the same time points. The primary endpoint is the change in H_2S-metabolizing bacterial communities between baseline and 1 month of energy drink exposure within Arm A and also compared to those in Arm B using microbiome analysis (diversity, taxonomy, sulfur-enriched gene pathways). Several secondary and exploratory assessments include changes related to subject demographic and dietary differences, concordance of microbiome changes with dietary logs and in other biospecimens beyond stool, H_2S-production measurements, and analyses of subject metabolic and immune function. Trial opened in February 2024 with active enrollment ongoing. Updated data on trial status and any amendment modifications will be presented at the meeting. Clinical trial information: NCT06137248. Research Sponsor: McJunkin Family Charitable Trust; National Cancer Institute.

TPS326 **Trials in Progress Poster Session**

A phase 3 study of first-line sotorasib, panitumumab, and FOLFIRI versus FOLFIRI with or without bevacizumab-awwb for patients with *KRAS* G12C–mutated metastatic colorectal cancer (CodeBreaK 301). First Author: Tae Won Kim, Department of Oncology, Asan Medical Center, University of Ulsan College of Medicine, Seoul, Korea, Republic of

Background: CodeBreaK 300 (NCT05198934) evaluated sotorasib, a selective inhibitor that irreversibly binds to the Kirsten rat sarcoma (*KRAS*) G12C mutant protein, in combination with panitumumab, an anti–epidermal growth factor receptor inhibitor, and showed that 960 mg sotorasib + panitumumab provided a statistically and clinically significant improvement in progression-free survival compared with trifluridine/tipiracil or regorafenib in patients with chemorefractory metastatic colorectal cancer (mCRC) (1). In CodeBreaK 101 (NCT04185883) subprotocol H expansion cohort 2F, the addition of folinic acid, fluorouracil, and irinotecan (FOLFIRI) to this combination in the first-line setting demonstrated a manageable safety profile and promising response rate of 78% in patients with *KRAS* G12C–mutated mCRC (2). The current study aims to evaluate whether the combination of sotorasib, panitumumab, and FOLFIRI is superior to FOLFIRI with or without bevacizumab-awwb, a VEGF inhibitor, in the first-line setting for mCRC. **Methods:** CodeBreaK 301 (NCT06252649) is a phase 3, multicenter, randomized, open-label, active-controlled study of sotorasib, panitumumab, and FOLFIRI versus FOLFIRI with or without bevacizumab-awwb as first-line therapy for patients with *KRAS* G12C–mutated mCRC. Key eligibility criteria include mCRC with *KRAS* G12C mutation confirmed by central molecular testing, and no prior therapy for metastatic disease. Approximately 450 patients will be randomized 1:1 to receive either sotorasib, panitumumab, and FOLFIRI, or FOLFIRI with or without bevacizumab-awwb. The primary endpoint is progression-free survival assessed by a blinded independent central review of disease response per Response Evaluation Criteria in Solid Tumors Version 1.1 (RECIST v1.1). The key secondary endpoint is overall survival, and additional secondary endpoints include objective response rate and depth of response. Global enrollment is ongoing. 1. Fakih, NEJM 2023. 2. Siena, ESMO 2024. Clinical trial information: NCT06252649. Research Sponsor: Amgen Inc.

327 **Oral Abstract Session**

Intraperitoneal and intravenous paclitaxel plus S-1 versus intravenous paclitaxel plus S-1 in gastric cancer patients with peritoneal metastasis: Results from the multicenter, randomized, phase 3 DRAGON-01 trial. First Author: Chao Yan, Ruijin Hospital, Shanghai Jiao Tong University School of Medicine, Shanghai, China

Background: The optimal chemotherapeutic regimen for gastric cancer with peritoneal metastasis remains undefined. We evaluated the efficacy and safety of intraperitoneal and intravenous paclitaxel plus S-1 (NIPS Group) versus paclitaxel plus S-1 (PS Group) in gastric cancer patients with peritoneal metastasis. **Methods:** In this multicenter, randomized, controlled, phase 3 trial, patients were recruited from 9 cancer centers in China. Eligible patients were aged 18~75 years with an Eastern Cooperative Oncology Group performance status of 0~1, histologically confirmed gastric adenocarcinoma, peritoneal metastases from gastric cancer requiring definitive diagnosis by laparoscopy, without gastric outflow tract obstruction and intestinal obstruction, no prior treatment with chemotherapy, radiation therapy, targeted therapy or immunotherapy. Eligible patients were randomly assigned (2:1) to receive either intravenous paclitaxel at 50 mg/m² and intraperitoneal paclitaxel at 20 mg/m² on days 1 and 8, along with oral S-1 at a dose of 80 mg/m² on days 1~14, or intravenous paclitaxel at 70 mg/m² on days 1 and 8, along with oral S-1 at 80 mg/m² on days 1~14. The primary endpoint was overall survival in the intention-to-treat population. Safety was assessed in all participants. A sample size of 238 patients was calculated to provide 80% power with a one-sided alpha of 0.1, accounting for a 10% dropout rate. Survival was analyzed using Kaplan-Meier curves and compared with the log-rank test. Cox regression was used for multivariate analysis. A *P*-value < 0.05 was considered statistically significant. **Results:** From May 10, 2017, to March 9, 2022, 246 patients were screened and 222 were included in the modified intention-to-treat population, of whom 148 patients were assigned to the NIPS Group and 74 to the PS group. As of data cutoff (March 9, 2024), the median survival time was 19.4 months (95% CI, 17.1~22.9), in the NIPS group and 13.9 months (95% CI, 10.3~16.1) in the PS group (HR = 0.66; 95% CI 0.49 - 0.88; *P* = 0.005). The 1-year and 2-year overall survival rates were 69.6% and 37.2% in the NIPS group, compared to 54.1% and 20.3% in the PS group. The most common grade 3~4 adverse events were leukopenia (21.7% in the NIPS group, and 24.7% in the PS group) and neutropenia (19.9% in the NIPS group, and 23.4% in the PS group). No treatment-related deaths were reported. **Conclusions:** Intraperitoneal and intravenous paclitaxel plus S-1 significantly improved the overall survival compared to intravenous paclitaxel plus S-1 in gastric cancer patients with peritoneal metastasis, with manageable toxicity. Clinical trial information: ChiCTR-IIR-16009802. Research Sponsor: Shanghai Jiao Tong University School of Medicine Multicenter Clinical Research Project; DLY201602.

328 **Oral Abstract Session**

Outcomes of patients with peritoneal-limited metastatic gastric cancer (PLGC) undergoing bidirectional (BD) chemotherapy cytoreductive surgery (CRS) with heated intraperitoneal chemotherapy (HIPEC). First Author: Evelyn Yi Ting Wong, Division of Medical Oncology, National Cancer Centre Singapore, Singapore, Singapore

Background: The prognosis of PLGC patients (pts) is poor with standard systemic therapy. PLGC pts who undergo CRS after BD chemotherapy (chemo) demonstrate prolonged survival. We retrospectively analysed the outcomes of pts who underwent BD chemo followed by CRS and HIPEC in a single institution, stratifying by peritoneal carcinomatosis index (PCI) and chemo regimen. **Methods:** PLGC pts who received BD chemo between January 2016 to September 2023 were enrolled and consented for data collection. BD chemo was either 3 weekly paclitaxel (PTX) (50mg/m2), S-1 (80mg/m2) and IP PTX (20mg/m2) or capecitabine (2g/m2), oxaliplatin (100mg/m2) (CAPOX) and IP PTX (40mg/m2). HIPEC utilised cisplatin 50mg/m2 at 40 for 60 minutes. Trastuzumab and nivolumab were given selectively. Pts without radiological progression after 24 weeks of BD chemo, with PCI7 and potential for complete resection underwent CRS including D2 gastrectomy and HIPEC. Post-resection, BD chemo was stopped. Survival was analysed by Kaplan Meier curves, adverse events (AEs) by CTCAE v5.0 and response by independent radiological review. **Results:** 34 pts were enrolled with median follow-up of 26.9 months (m). Median (range) age was 51 (17 – 79) and median PCI 12 (0-29). 52.9% were female, 8.8% had Cy1 only disease and 38% had prior gastrectomy. 3 out of 6 pts with evaluable disease had partial response. Median progression-free survival and overall survival (mOS) was 9.1m and 16.5m respectively. 7 pts (20.6%) underwent CRS with median (range) PCI of 1 (0-15.0). mOS in CRS vs non-CRS pts was 23.8m vs 14.6m (p=0.4); PCI <7 vs PCI ≥ 7 was 23.8m vs 14.6m (p=0.63); and CAPOX vs S1/PTX systemic chemotherapy was 23.9m vs 14.6m (p=0.29) respectively. Peritoneal cytology negativity increased from 25.8% to 86.4% after BD chemo. Peritoneal lesions biopsied post-BD chemo showed no tumour cells in 64% of cases. There was no treatment-related mortality. 1 pt had IP port removal due to complications. The commonest grade 3 AE was neutropenia (14.7%). Biomarker analysis is currently being performed. **Conclusions:** BD chemo CRS HIPEC was well tolerated with longer survival in patients undergoing CRS. Evaluation of tumour using PCI post-BD chemo may be sub-optimal. Research Sponsor: None.

LBA329 **Oral Abstract Session**

Preliminary results from the multicenter, randomized phase III trial (SCIENCE): Comparing chemotherapy plus sintilimab and chemoradiotherapy plus sintilimab versus chemoradiotherapy for neoadjuvant treatment in resectable locally advanced esophageal squamous cell carcinoma. First Author: Xuefeng Leng, Department of Thoracic Surgery, Sichuan Cancer Hospital & Institute, Sichuan Cancer Center, School of Medicine, University of Electronic Science and Technology of China, Chengdu, China

The full, final text of this abstract will be available at meetings.asco.org on the day of presentation and in the online supplement to the February 1, 2025, issue of the *Journal of Clinical Oncology*.

LBA331 **Oral Abstract Session**

EORTC-1203 GITC "INNOVATION": Integration of trastuzumab (T), with or without pertuzumab (P), into perioperative chemotherapy of HER-2 positive stomach cancer: Overall survival results. First Author: Anna Dorothea Wagner, Lausanne University Hospital and University of Lausanne, Lausanne, Switzerland

The full, final text of this abstract will be available at meetings.asco.org on the day of presentation and in the online supplement to the February 1, 2025, issue of the *Journal of Clinical Oncology*.

332 Rapid Oral Abstract Session

Final analysis of the randomized phase 2 part of the ASPEN-06 study: A phase 2/3 study of evorpacept (ALX148), a CD47 myeloid checkpoint inhibitor, in patients with HER2-overexpressing gastric/gastroesophageal cancer (GC). First Author: Kohei Shitara, National Cancer Center Hospital East, Kashiwa, Japan

Background: Evorpacept (Evo) is a high affinity, CD47-blocker with an inactive Fc region designed to safely magnify anticancer antibody dependent cellular phagocytosis. Evorpacept is being evaluated across cancers, in combination with anticancer antibodies and checkpoint inhibitors. **Methods:** The ASPEN-06 randomized phase 2 trial evaluates Evo in combination with standard trastuzumab (T), ramucirumab (R) and paclitaxel (P) for the treatment of patients with HER2+ GC. Patients with 2nd or 3rd line HER2+ advanced or metastatic GC that has progressed on or after prior anti-HER2 therapy were randomized to Evo (30 mg/kg Q2W) plus TRP or TRP. HER2 status was determined in the most recent GC tissue sample. Primary study objectives were to compare confirmed ORR of Evo-TRP to an assumed ORR of RP (=30%) with one-sided alpha error of 0.025 and to identify a clinically meaningful contribution of Evo to TRP in ORR (delta>10%). **Results:** Among the entire randomized population (N=127) the ORR was 40.3% (Evo-TRP) and 26.6% (TRP), respectively. The difference between Evo-TRP's ORR compared to historical RP's ORR was not statistically significant (p=0.095), but Evo-TRP's ORR demonstrated a meaningful delta of 13.7% over TRP's ORR (p=0.028 in an exploratory analysis). Median DOR for Evo-TRP and TRP was 15.7 and 7.6 months, respectively. In a prespecified population with HER2+ disease in fresh tumor tissues after prior anti-HER2 treatment (N=48), Evo-TRP's ORR (54.8%) compared favorably to historical RP's ORR (p=0.030) with a delta of 31.7% over TRP's ORR of 23.1% (p=0.004 in an exploratory analysis). Evo-TRP was well tolerated with a safety profile consistent with prior experience. Updated results including PFS will be presented at the meeting. **Conclusions:** The addition of Evo to TRP showed promising activities in HER2+ GC. The magnitude of response was greatest in tumors identified as HER2+ using fresh biopsies emphasizing the importance of biopsies post anti HER2 therapy. Consistent with Evo's mechanism of enhancing antibody dependent phagocytosis, these data support its ongoing investigation as an adjunct to anti-HER2 gastric cancer therapy. Clinical trial information: NCT05002127. Research Sponsor: ALX Oncology.

333 Rapid Oral Abstract Session

Efficacy and safety of albumin-bound docetaxel vs. docetaxel in patients with previously treated advanced gastric or gastroesophageal junction adenocarcinoma: A multicenter, randomized, phase II study. First Author: Zhiqiang Wang, Sun Yat-sen University Cancer Center, Guangzhou, China

Background: Docetaxel has demonstrated promising activity in gastric cancer, both as monotherapy and in combination with other agents. Albumin-bound docetaxel (HB1801) is a new kind of taxane and has many advantages compared with docetaxel. Previous phase I study has demonstrated that HB1801 showed preliminary efficacy in patients with gastric cancer. **Methods:** In this phase II study, eligible patients were aged 18-75 years with histologically confirmed gastric or gastroesophageal junction (G/GEJ) adenocarcinoma, who progressed on at least first line of combined chemotherapy of platinum and fluorouracil. Patients were randomized (1:1) to receive HB1801 (100 mg/m^2) or docetaxel (Taxotere, 75 mg/m^2) administered every 3 weeks by intravenous infusion. Stratified factors included previous treatment with immunotherapy (yes or no) and ECOG PS (0 or 1). Primary endpoint was progression free survival (PFS). **Results:** As of June 25, 2024, 128 patients were randomized to the HB1801 group (*n*=65) or the docetaxel group (*n*=63). The two groups were comparable on baseline characteristics. Overall, the median age was 59.0 (range 27-75) years, 106 (82.8%) patients had ECOG PS of 1. 76 (59.4%) patients had received previous treatment with immunotherapy. 76 (59.4%) patients had ≥ 2 metastatic sites. Median PFS was 4.0 months (95%CI 2.8-4.2) with HB1801 versus 2.7 months (95%CI 1.8-3.0) with docetaxel, HR = 0.81(95%CI 0.53, 1.21). Objective response rate and disease control rate were 21.5% (1CR and 13 PRs, 95%CI 12.3-33.5) and 56.9% (20 SDs, 95%CI 44.0-69.2) in the HB1801 group, compared with 14.3% (1 CR and 8 PRs, 95%CI 6.8-25.4) and 42.9% (17 SDs, 95%CI 30.5-56.0) in the docetaxel group. Median overall survival was 11.3 months with HB1801 versus 7.8 months with docetaxel (HR = 0.59 [95%CI 0.35, 1.01], p=0.025). Treatment-related adverse events (TRAEs) occurred in 93.8% of patients receiving HB1801, and 93.7% of patients receiving docetaxel. Alopecia (46.2% vs. 39.7%), fatigue (43.1% vs 23.8%), anemia (40.0% vs 39.7%), hypoalbuminemia (29.2% vs 12.7%), decreased appetite (21.5% vs 12.7%), peripheral edema (16.9% vs 12.7%) and nausea (15.4% vs 19.0%) were the most common TRAEs in the HB1801 group and the docetaxel group. 20 patients (30.8%) in the HB1801 group and 19 patients (30.2%) in the docetaxel group experienced ≥grade 3 TRAEs, with the most common being anemia and fatigue. 3 patients in the docetaxel group experienced treatment emergent adverse events (TEAEs) leading to death, which one was considered to be treatment-related. No patients in the HB1801 group had TEAEs leading to death. No new safety signal was observed in HB1801 group. **Conclusions:** Compared with Taxotere, HB1801 results in 41% reduction in risk of death with a manageable safety profile in patients with previously treated advanced G/GEJ cancer. Clinical trial information: NCT05705635. Research Sponsor: CSPC ZhongQi Pharmaceutical Technology Co., Ltd.

334 Rapid Oral Abstract Session

Conversion effects of PD-1 inhibitor camrelizumab (Cam) combined with Nab-POF regimen in patients (pts) with initially unresectable locally advanced or limited metastatic gastric or gastroesophageal junction adenocarcinoma: FDZL-001 trial. First Author: Qirong Geng, Department of Medical Oncology, Fudan University Shanghai Cancer Center, Shanghai, China; Department of Oncology, Shanghai Medical College, Fudan University, Shanghai, China

Background: For initially unresectable locally advanced or metastatic gastric or gastroesophageal junction adenocarcinoma, the prognosis is poor and chemotherapy is the main treatment option. The AIO-FLOT3 Trial suggested that conversion therapy might improve outcome in pts with limited metastatic gastric cancer (GC). We conducted this prospective, single-arm, phase II trial to improve conversion efficacy of PD-1 inhibitor Cam combined with Nab-POF regimen (nab-paclitaxel , oxaliplatin, fluorouracil) in pts with initially unresectable locally advanced or limited metastatic gastric or gastroesophageal junction adenocarcinoma. **Methods:** Eligible pts received Cam 200mg, nab-PTX 125 mg/m^2, oxaliplatin 85 mg/m^2 or with trastuzumab if Her2 positive, each was an IV followed by fluorouracil 2400 mg/m^2 as a 48-h continuous IV on day 1, every 2 weeks. Patients were assessed for tumor response and surgical feasibility every 3 cycles, and if confirmed feasible for surgical resection by both surgical and medical experts after 6 cycles, they would undergo surgery within 3-6 weeks. The primary endpoint was R0 resection rate. **Results:** In total, 53 pts were enrolled up to 01/2024. 52 pts were evaluable. 50% are younger than 65 years of age. Male was 73.1%. The metastatic sites included liver 46.2% (24), retroperitoneal lymph nodes 40.4% (21), Krukenberg 5.8% (3), and locally advanced unresectable tumors 21.2% (11). 4 pts (7.7%) were dMMR/MSI-H and 10 (19.2%) were Her2 positive. ORR was 88.5% (46/52) and DCR was 98.1% (51/52). 45(86.5%) pts were assessed as resectable. 3 pts (6.8%) refused surgery and 3 (6.8%) were cCR under observation. The R0 resection rate was 75.0% (39/52). CR rate was 23.1% (cCR, 5.8%; pCR,17.3%). The mPFS was not reached (95% CI = 17.0-NE), and the mOS was not reached (95% CI = 25.4-NE), either. The 3-year PFS and OS rate were 56.9% (95% CI = 45.7%-86.3%) and 62.8% (95% CI = 41.6%-78.0%), respectively. The 3 cCR pts have survived utill the last follow-up date of 07/2025 without tumors for 17, 29, and 34 months, respectively. Pts tolerated treatment well, although all pts experienced treatment-related adverse events (AE). Grade 3/4 AEs were 42.3% (22/52) and neutrophil decrease was the most common at 36.5%. 10 patients (19.2%) experienced immune related AEs (irAEs), and 2 of them experienced grade 3 irAEs. **Conclusions:** The PD-1 inhibitor Cam combined with the Nab-POF regimen safely induced a high conversion rate with very high R0 resection and high 3-year PFS and OS rates, preliminarily demonstrating promising effects, providing a new conversion drug treatment option for pts with initially unresectable locally advanced or limited metastatic advanced GC. These findings warrant further prospective, randomized controlled investigations. Clinical trial information: NCT04510064. Research Sponsor: None.

335 Rapid Oral Abstract Session

Effect of SHR-1701 on chemotherapy (chemo)-induced myelosuppression: Data from a phase 3 study in HER2-negative gastric/gastroesophageal junction adenocarcinoma (G/GEJA). First Author: Zhi Peng, Beijing Cancer Hospital, Beijing, China

Background: Blocking TGF-β signaling has the potential to facilitate the recovery from chemo-induced myelosuppression. SHR-1701 is a bifunctional agent targeting PD-L1 and TGF-β. **Methods:** In the multicenter, randomized, double-blind phase 3 study (NCT04950322), SHR-1701 plus CAPOX demonstrated a statistically significant and clinically meaningful benefit in OS compared with placebo plus CAPOX in patients (pts) with previously untreated, unresectable locally advanced or metastatic HER2-negative G/GEJA (ESMO 2024, LBA60). Here, we reported the effect of SHR-1701 on chemo-associated myelosuppression. **Results:** In total, 364 and 366 pts in the SHR-1701 and placebo arm received assigned treatment. As of May 20, 2024, with a KM estimated median follow-up of 13.6 mo, all study medications showed similar exposure duration between the two arms (Table). Occurrence of treatment-related adverse events was generally comparable (any grade, 97.8% in SHR-1701 arm vs 98.4% in placebo arm; grade ≥3, 62.6% vs 59.0%). SHR-1701 reduced the incidence of any grade decreased platelet count, decreased neutrophil count, and decreased white blood cell count by 8.7%, 10.1%, and 11.2%, and for grade ≥3, by 9.1%, 4.3%, and 1.6%, respectively (Table). Compared with the placebo arm, less pts in the SHR-1701 arm used platelet growth factors (30.2% vs 42.9%) and white blood cell growth factors (32.7% vs 39.6%). **Conclusions:** SHR-1701 showed the capacity to suppress chemo-associated myelosuppression. Clinical trial information: NCT04950322. Research Sponsor: Jiangsu Hengrui Pharmaceuticals.

Safety.

	SHR-1701 plus CAPOX (N=364)		Placebo plus CAPOX (N=366)		Treatment difference (SHR-1701 vs placebo)	
Treatment exposure (day), median (IQR)						
SHR-1701/placebo	137.0 (65.5-251.5)		127.5 (62.0-190.0)		NA	
Capecitabine	122.0 (83.0-138.0)		122.0 (73.5-135.0)		NA	
Oxaliplatin	107.0 (66.0-123.0)		108.5 (64.0-121.0)		NA	
Treatment-related hematological toxicities, n (%) or %	**Any grade**	**Grade ≥3**	**Any grade**	**Grade ≥3**	**Any grade**	**Grade ≥3**
Platelet count decreased	217 (59.6%)	69 (19.0%)	250 (68.3%)	103 (28.1%)	-8.7%	-9.1%
Neutrophil count decreased	163 (44.8%)	44 (12.1%)	201 (54.9%)	60 (16.4%)	-10.1%	-4.3%
White blood cell count decreased	147 (40.4%)	13 (3.6%)	189 (51.6%)	19 (5.2%)	-11.2%	-1.6%

336 Poster Session

Effects of neighborhood deprivation index on survival in gastric cancers. First Author: Sawyer Bawek, Department of Internal Medicine, University at Buffalo, Buffalo, NY

Background: Previous studies have shown that living in disadvantaged neighborhoods resulted in poor outcomes in several malignancies. Neighborhood Deprivation Index (NDI) identifies key variables from 13 measures, including the following dimensions of socioeconomic (SES) status: wealth and income, education, occupation, and housing conditions. In this study, we aimed to investigate the effect of NDI among patients with gastric cancer (GC). **Methods:** We conducted a retrospective analysis using the Surveillance, Epidemiology, and End Results (SEER) database from 1996-2015. NDI was divided into two groups: less disadvantaged areas (NDI <60) and highly disadvantaged areas (NDI >60), respectively. Demographic and clinical characteristics were summarized by NDI groups. All associations were compared using Kruskal-Wallis and Chi-Square tests, respectively. Kaplan Meier curves were summarized by the NDI group for overall survival (OS) and disease-specific survival (DSS). All associations were compared using a log-rank test. Cox multivariate regression was performed to measure the association between NDI and OS/DSS. All analyses were conducted in RStudio v4.2.3 at a significance level of ≤ 0.05. **Results:** A total of 19,205 GC (11,137 patients with NDI<60 and 8,068 NDI>60) patients were analyzed. Disadvantaged areas (NDI>60) had a higher proportion of Black and Hispanic, single, uninsured, rural patients with later disease stage as compared to NDI <60. Median OS was 10.0 months for patients with NDI <60 versus 9.0 months for those with NDI >60 (p=0.0007). Median DSS was 12.0 months for NDI <60 and 10.0 months for NDI >60 (p=0.010). Multivariate analysis revealed that factors associated with worse OS included NDI >60 [HR 1.044, p=0.009], age (>60) [HR 1.28, p<0.001], male sex [HR 1.082, p<0.001], single marital status [HR 1.21, p<0.001], uninsured status [HR 1.13], p <0.001], advanced disease stage (III/IV) [HR 2.05, p<0.001], and higher grade [HR 1.32, p<0.001]. On the other hand, non-Hispanic Black [HR 0.98] and Hispanic [HR 0.94] race and urban location [HR 0.91, p<0.001] were associated with improved OS. Factors associated with inferior DSS included NDI >60 [HR 1.037, p=0.038], age (p<0.001), male sex [HR 1.05, p=0.005], single marital status [HR 1.15, p<0.001], advanced disease stage (III/IV) [HR 2.375, p<0.001], and advanced grade (III/IV) [HR 1.40, p<0.001). In contrast, Hispanic [HR 0.94], non-Hispanic Black [0.96], and urban location [HR 0.90, p<0.001] were associated with improved DSS. **Conclusions:** Patients from areas with higher NDI had worse OS and DSS for GC. Tailored healthcare policies that aim to improve healthcare delivery to areas with high deprivation may help improve GC outcomes. Research Sponsor: National Cancer Institute; P30CA016056.

337 Poster Session

Impact of race on end-of-life care among patients with esophageal cancer. First Author: Suriya Baskar, Brooklyn Hospital Center, Brooklyn, NY

Background: Esophageal cancer is an aggressive malignancy with rising incidence globally. Patients with advanced esophageal cancer suffer from a heavy burden of a wide array of symptoms including dysphagia and cachexia/anorexia, necessitating the need for focused symptom management and palliative care. Studies focused on other cancer types have shown that Black patients are more likely to receive aggressive care at end-of-life (EOL) and suffer from a decline in quality of life. We sought to investigate racial differences in EOL care among patients with advanced esophageal cancer in the US. **Methods:** The National Inpatient Sample (NIS) was queried to identify all hospitalizations with esophageal cancer utilizing ICD-10 codes C15x from 2016 to 2020. Hospitalizations of White and Black patients with documented inpatient mortality events were then extracted. Demographic and clinical data were analyzed using chi squared tests, independent sample t-tests, and binary logistic regression (adjusted for age, gender, and Charlson comorbidity index or CCI). Adjusted Odds ratios (aOR) are presented with 95% Confidence intervals (CI). **Results:** A total of 15,130 patients with esophageal cancer were included, of which 13,345 (88.2%) were White and 1785 (11.8%) were Black. Hospitalizations with White patients recorded higher age, more males, and higher comorbidities per CCI. Hospitalizations with Black patients had longer lengths of stay on average, higher total hospital charges, and significantly less likely to have a Do Not Resuscitate (DNR) code status (aOR 0.81, 0.74-0.9 95% CI). Rates of inpatient palliative care consultation at EOL were significantly lower among Black patients (aOR=0.53, 0.48-0.58 95% CI). Black patients were more likely to receive aggressive intervention at EOL demonstrated by rates of blood transfusion (aOR 1.9, 1.6-2.1 95% CI), mechanical ventilation (aOR 1.7, 1.5-1.8 95% CI), and vasopressor usage (aOR 1.4, 1.2-1.6 95% CI). **Conclusions:** We present a large retrospective analysis demonstrating significant racial disparities in EOL care amongst hospitalized patients with esophageal cancer. Black patients had longer lengths of stay, lower rates of palliative care consultation and DNR code status, and higher rates of aggressive interventions at EOL. These findings may have implications for improving health care decision making for EOL care amongst minority patients with esophageal cancer. Research Sponsor: None.

	White N=13345	Black N=1785	p-value
Age (in years)	68.24±10.67	65.12±10.36	**<.001**
CCI	9.4±3.3	8.6±3.9	**<.001**
Mean length of stay (days)	7.5±11.3	10.4±22.8	**<.001**
Total Cost (In thousands of dollars)	108.6±231.0	141.9±388.5	**<.001**
Palliative	60.9%	45.1%	**<.001**
Adjusted OR		**.53 (.48 - .58)**	
DNR	63.8%	58.0%	**<.001**
Adjusted OR		**.81 (.74 - .90)**	
Blood transfusion	13.0%	23.0%	**<.001**
Adjusted OR		**1.9 (1.6-2.1)**	
Mechanical ventilation	31.4%	45.9%	**<.001**
Adjusted OR		**1.7 (1.5-1.8)**	
Vasopressor	7.8%	10.6%	**<.001**
Adjusted OR		**1.4 (1.2-1.6)**	

338 Poster Session

The influence of depression on esophageal cancer outcomes: Using the National Inpatient Sample (NIS) 2021. First Author: Kalaivani Babu, Allegheny General Hospital, Pittsburgh, PA

Background: Esophageal cancer (EC) remains a challenging malignancy with high morbidity and mortality rates. Depression is prevalent among cancer patients and has been shown to affect various clinical outcomes. This study aims to evaluate the impact of depression on mortality, length of stay (LOS), and hospitalization charges in patients with esophageal cancer using data from the NIS. **Methods:** We conducted a retrospective, population-based study using the NIS data to analyze patients diagnosed with esophageal cancer. Depression was the primary exposure variable using the appropriate ICD-10 code. We applied survey-weighted descriptive statistics, logistic regression, and linear regression models to assess outcomes, adjusting for demographic and clinical factors, including age, sex, race, comorbidities (Charlson Comorbidity Index), hospital teaching status, and regional characteristics. **Results:** A total of 40,855 patients with esophageal cancer were included, with 11.4% (n = 4,855) having a concurrent diagnosis of depression. The unadjusted mortality rate for the cohort was 9.18%, with depressed patients showing a lower mortality rate (6.45% vs. 9.53%, p = 0.0018). After adjusting for covariates, depression was associated with significantly lower in-hospital mortality (adjusted OR 0.63, 95% CI 0.48–0.83, p = 0.001). While the unadjusted mean length of stay (LOS) was slightly higher in depressed patients (7.66 vs. 7.25 days, p = 0.132), adjusted models showed no significant difference (β = 0.42 days, p = 0.132). Depression was associated with lower unadjusted total hospitalization charges ($92,321 vs. $104,156, p = 0.023), and in adjusted analyses, depressed patients incurred significantly lower costs (β = -$10,327, 95% CI -$19,233 to -$1,421, p = 0.023). Significant predictors of increased hospitalization charges included teaching hospital status (β = $34,029, p $<$ 0.0001) and large hospital size (β = $38,481, p $<$ 0.0001). **Conclusions:** Depression, while common among patients with esophageal cancer, was unexpectedly associated with lower in-hospital mortality, despite a longer length of stay and higher comorbidity burden. This suggests that depressed patients may receive more comprehensive or prolonged care, which could contribute to their reduced mortality risk. Additionally, despite extended hospital stays, these patients incurred significantly lower hospitalization costs, indicating possible differences in resource utilization or treatment intensity. These findings highlight the complex relationship between mental health and cancer outcomes, emphasizing the need for further research to explore the mechanisms behind these associations and to better tailor care for esophageal cancer patients with depression. Research Sponsor: None.

339 Poster Session

Incidence trends, demographic disparities, and survival outcomes of gastrointestinal malignancies in young adults: A focus on stomach and pancreatic cancers. First Author: Shivani Modi, Albert Einstein Healthcare Network, Philadelphia, PA

Background: The incidence of gastrointestinal (GI) cancers has been rising among younger populations. While prior research has highlighted the increasing incidence of colorectal cancer (CRC) in adolescents and young adults (AYA), defined as individuals aged 15-39, there is less focus on non-CRC GI malignancies, such as stomach and pancreatic cancers, within this age group. This study aims to further elucidate the incidence, mortality, and demographic disparities in the AYA population for stomach and pancreatic cancers compared to older populations (>50 years old). **Methods:** Data was extracted from the SEER database spanning from 2000 to 2020. Demographic, clinical, and secondary cancer characteristics were summarized across the sample, stratified by cancer type. Overall survival (OS) and disease-specific survival (DSS) were analyzed using the Kaplan-Meier method, with comparisons performed via the log-rank test. Univariate and multivariate analyses were conducted to identify predictors of survival outcomes. **Results:** We identified 19,983 AYA patients diagnosed with non-CRC GI malignancies (including esophageal, stomach, small intestine, anal, hepatobiliary, and pancreatic cancers) between 2000 and 2020. Among these patients, the majority were male (58%), White (47%), and resided in metropolitan areas (90%). The median age at diagnosis was 38 years. Pancreatic cancer was 20% more common in the AYA group compared to those over 50 years old (12%), while stomach cancer was also more prevalent in the AYA group (27% vs. 17%). Survival outcomes were significantly better in the AYA group, with a median OS of 24 months compared to 9 months in the older population, and a median DSS of 52 months versus 13 months (p<0.001). Predictors of worse OS and DSS included male sex, African American race, income below $45,000/year, lack of surgical intervention, and delayed initiation of treatment. **Conclusions:** The incidence of stomach and pancreatic cancers is notably higher in the AYA population compared to older adults, with AYA patients demonstrating better overall and disease-specific survival outcomes. However, certain demographic factors, such as male sex, African American race, and lower income, are associated with worse survival, underscoring the need for targeted interventions to address these disparities. Early diagnosis and timely treatment are critical to improving outcomes in this younger population. Research Sponsor: None.

340 Poster Session

Predicting mortality in upper gastrointestinal tract cancer using long short-term memory neural networks. First Author: Manaswitha Thota, Virginia Commonwealth University Health, Richmond, VA

Background: Upper gastrointestinal (GI) tract cancers represent a significant global health burden, characterized by high morbidity and mortality rates, with an increasing incidence among aging populations and varying outcomes based on gender and ethnicity. Early prediction of mortality rates in upper gastrointestinal cancer can significantly enhance treatment strategies and patient outcomes. This study describes current trends in upper gastrointestinal cancer and proposes a predictive model, developed using Long Short-Term Memory (LSTM) neural networks to project future trends using demographic data. **Methods:** We extracted mortality data from the Center for Disease Control and Prevention (CDC) Wide-Ranging Online Data for Epidemiological Research (WONDER) Database spanning 1999, normalized it and encoded categorical variables such as age, sex, ethnicity, and race numerically. Using Studio R to analyze mortality trends, we developed an LSTM model known for its ability to capture and utilize long term data patterns. The model, trained with a sequence length of 20, predicted future data points based on the previous 20. We then assessed its accuracy and generalization with a test set. Ultimately, our model is able to utilize demographic variables to forecast annual mortality and survival rates. **Results:** The mortality rates for upper GI tract cancers are notably higher among elderly individuals, particularly those aged 80 and above, with males and certain ethnic groups, such as Hispanics. The accuracy of the trained model was evaluated by calculating the Mean Squared Error (MSE), a measure of the difference between the predicted and actual value, with zero indicating no difference between the two. Our LSTM model achieved a training MSE of 0.00085 and a validation MSE of 0.0016, indicating good reliability and accuracy. **Conclusions:** Our analysis highlighted clear mortality trends and demonstrated the model's effectiveness in improving mortality predictions in upper gastrointestinal cancer. However, limitations due to the dataset's lack of clinical variables (such as comorbidities, cancer stage, and prior treatments) and its reliance on a single dataset may affect external validity. Future research is needed to determine if the model can be applied for clinical prognosis or strategic planning. We ultimately created and validated an LSTM neural network model with the CDC WONDER dataset, showing promising results. Future development will aim to include clinical characteristics to create a more personalized predictive tool. Research Sponsor: None.

341 Poster Session

Safety and efficacy of first-line nivolumab plus chemotherapy for HER2-negative advanced gastric cancer in real-world patients: Updated analysis. First Author: Keitaro Shimozaki, Department of Gastroenterological Chemotherapy, Cancer Institute Hospital, Japanese Foundation for Cancer Research, Koto-Ku, Japan

Background: Based on the results of CheckMate-649 and ATTRACTION-4, nivolumab plus chemotherapy is the new standard of care as first-line treatment for patients with HER2-negative advanced gastric cancer (AGC) regardless of PD-L1 combined positive score (CPS) status in Japan. This study aimed to evaluate the safety and efficacy of this regimen in real-world patients with HER2-negative AGC with a longer follow-up. **Methods:** In this single-institutional retrospective study, patients with HER2-negative AGC who were treated with first-line nivolumab plus chemotherapy between September 2021 and April 2024 were consecutively enrolled and evaluated. Early tumor shrinkage (ETS) was defined as ≥20% tumor reduction at 8 ± 2 weeks. **Results:** Of 151 patients, median age was 66 years (range, 27–83); 58% were male; 81% had diffuse-type histology; the peritoneal metastasis was seen in 60%; PD-L1 CPS of <1, 1≤/<5, 5≤ was 30%/41%/29%; deficient mismatch repair and/or high microsatellite instability was 7%. At a median follow-up of 14.2 months, the median overall survival (OS) and progression-free survival (PFS) were 21.7 months (95% confidence interval [CI] 14.4–26.2) and 7.7 months (6.4–9.6), respectively. The median OS according to CPS subgroups (<1, 1≤/<5, 5≤) were 21.7 / not reached (NR) /25.0 months, respectively (log-rank P = 0.55). In patients with measurable lesions at baseline according to RECIST ver 1.1, the objective response rate and disease control rate were 62% and 87%, respectively, with a complete response rate of 10%. In the exploratory analysis by ETS, both median PFS and OS were significantly longer in the ETS group compared to the non-ETS group (PFS, NR vs. 6.2 months; hazard ratio [HR] 0.35; 95% CI 0.17–0.68; P = 0.001) (OS, 25.0 vs. 11.7 months; HR 0.44; 95% CI 0.20–0.93; P = 0.03). The immune-related adverse events (irAEs) of any grade were observed in 26% (grade 3–4 irAEs 12%), whereas no treatment-related death was documented. **Conclusions:** The updated analysis suggests that first-line nivolumab plus chemotherapy provides benefit in real-world patients with HER2-negative AGC regardless of PD-L1 status, and suggests that the magnitude of ETS may predict a significant impact on the duration of survival. Research Sponsor: None.

342 Poster Session

Real world data and biomarker analysis of the efficacy and safety of PD-1 inhibitor combined with chemotherapy in first-line treatment of advanced gastric cancer. First Author: Xiaoting Ma, Department of Medical Oncology, National Cancer Center, Cancer Hospital, Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing, China

Background: With the publication of several clinical research results, the efficacy of ICI combined with chemotherapy has gradually been recognized in HER-2 negative advanced GC field. We analyzed the efficacy of 1st-line PD-1 inhibitor combined with chemotherapy in advanced GC patients using real-world data and conducted exploratory analysis. We aim to identify factors that can predict treatment efficacy, and provide a basis for clinical practice. **Methods:** We prospectively collected data from GC patients between November 2020 and April 2024. The clinical staging was stage IV. All patients were HER2 negative and pMMR. All patients received SOX regimen chemotherapy combined with PD-1 inhibitor. Blood samples were collected for exploratory analysis. **Results:** A total of 190 GC patients with histological diagnosis of adenocarcinoma were included from November 1, 2020 to April 1, 2024. As of June 1, 2024, the median follow-up time was 11.75 (95% CI 10.65-13.00) months. Among them, 174 had measurable lesions. 94 patients achieved PR, and 55 patients achieved SD. The ORR and DCR were 61.9% (95% CI 48.8-73.4%) and 96.8% (95% CI 89.0-99.6%), respectively. The survival analysis results indicated that the mPFS and mOS were 9 (95% CI 7.9-14.2) and 27 (95% CI 20- not reached) months. We conducted risk factor analysis on patients' age, gender, CPS, HER2 expression, EGFR expression and EBER status. Age, gender, CPS, EBER status and EGFR expression are not risk factors affecting survival. Specifically, the expression level of HER2 was significantly correlated with therapeutic efficacy. The results indicated that the HER2 expression level was significantly correlated with ORR. In terms of survival analysis, the PFS of treated patients with HER2 '2+' was significantly longer than that of treated patients with HER2 '1+', and there was a trend of prolonged OS, but the statistical difference was not reached. We collected baseline blood samples from 62 patients and performed ctDNA testing. In terms of PFS, the patients with SNV in SMARCA4 and RASA1 was significantly shorter than that of wild-type patients with SMARCA4 (p = 0.02) and RASA1 (p = 0.043). The patients with CNV in EXT1 was significantly shorter than those with wild-type in EXT1 (p = 0.00046). In terms of OS, patients with SNV in BRCA2, TP53, and STK11 had significantly shorter OS than those with wild-type BRCA2 (p = 0.033), TP53 (p = 0.048), and STK11 (p = 0.0083). The patients with CNV in MYC and EXT1 had shorter OS than those with wild-type MYC ($p < 0.0001$) and EXT1 ($p < 0.0001$) patients. **Conclusions:** Real-world data showed that 1st-line treatment with PD-1 inhibitor combined with SOX had shown good efficacy in advanced GC patients. The HER2 expression level in HER2 negative GC may become a direction for further exploration. The EXT1 mutation can serve as a potential predictor, and further investigation is warranted. Clinical trial information: NCT06034964. Research Sponsor: CAMS Innovation Fund for Medical Sciences (CIFMS); 2022-I2M-C&T-B-066.

343 Poster Session

Retreatment with immune checkpoint inhibitors (ICIs) for patients with advanced gastric cancer: A retrospective, real-world study. First Author: Yong-Xu Jia, Department of Medical Oncology, The First Affiliated Hospital, Zhengzhou University, Zhengzhou, Henan, China

Background: Immunotherapy combined with chemotherapy has become the standard first-line treatment for advanced gastric cancer and is recommended by multiple guidelines and consensuses. After the first line immunotherapy, there is still controversy over whether immune inhibitors can continue to be used across lines after progression. **Methods:** This is a single-center, retrospective, observational, real-world study. We collected clinical information of patients with advanced gastric cancer hospitalized at the First Affiliated Hospital of Zhengzhou University from January 2018 to July 2024. All patients received combination therapy with ICIs inhibitors for at least 2 cycles in the first-line and at least 2 cycles of ICI inhibitor-based therapy in the second-line after first-line progression. The efficacy and safety of treatment were evaluated, including overall survival (OS), progression-free survival (PFS), objective response rate (ORR), disease control rate (DCR), the progression pattern of retreatment with immune checkpoint inhibitor, grade 3-5 treatment-related adverse events (TRAEs), and immune-related adverse events (irAEs). Cox regression model was used to investigate the influence of multiple factors on survival. **Results:** We included 217 patients with a median age of 59.0 (18-84) years, of which 70.0% were male. The median OS and PFS were 9.8 (95% CI 8.8-10.8) months and 4.6 (95% CI 4.0-5.3) months. The ORR and DCR were 22.7% (95% CI 17.1-29.0) and 56.2% (95% CI 49.0-63.1). For the second-line treatment regimens, there were 87 (40.1%) patients and 130 (59.9%) patients received immunotherapy combined with chemotherapy (I+C) and immunotherapy combined with chemotherapy and anti-angiogenic drugs(I+C+A) respectively. The median OS was 9.3 (95% CI 8.5-10.1) months and 11.5 (95% CI 9.3-13.7) months, p=0.032. The median PFS was 4.2 (95% CI 3.1-5.3) months and 5.2 (95% CI 4.3-6.1) months, p=0.082. Cox regression model analysis showed that ECOG 0-1, PFS≥6 months in the first-line treatment, second-line regimen (I+C+A), and second-line medication > 3 cycles were independent prognostic factors for OS in the second-line treatment. The incidence of grade ≥3 TRAEs was 20.7%, and the incidence of irAEs was 6.5%. No treatment-related deaths occurred. **Conclusions:** This real-world study results showed that retreatment with immune checkpoint inhibitors had promising clinical benefit for patients with advanced gastric cancer and manageable safety. Research Sponsor: None.

Efficacy of different groups.

		n	mOS (m)	mPFS (m)	ORR (%)	DCR (%)
ITT		217	9.8	4.6	22.7	56.2
ECOG						
	0-1	180	9.8	4.7	19.5	56.8
	2	37	8.6	4.0	38.2	52.9
1-Line PFS						
	< 3m	61	5.5	2.2	1.7	17.2
	≥3m	156	11.7	6.0	31.0	71.7
	< 6m	136	7.4	3.2	2.4	38.7
	≥6m	81	15.1	8.0	54.4	83.5
Second-line treatment						
	I+C	87	9.3	4.2	16.9	54.2
	I+C+A	130	11.5	5.2	26.7	57.5
Second-line treatment cycle						
	≤3	97	7.4	2.9	4.8	31.0
	> 3	120	13.1	6.2	35.3	73.9

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Nimotuzumab combined with chemotherapy plus immunotherapy as first-line treatment for advanced esophageal squamous cell carcinoma. First Author: Quanli Han, The First Medical Center of Chinese PLA General Hospital, Beijing, China

Background: Esophageal squamous cell carcinoma (ESCC) is the primary type of esophageal cancer with poor prognosis in China. Chemotherapy plus immunotherapy as first-line standard treatment has limited benefit in advanced ESCC patients. Nimotuzumab, a humanized anti-EGFR antibody, exhibited favorable efficacy in advanced ESCC with good safety profile. We explored the efficacy and safety of nimotuzumab as first-line therapy for patients with advanced ESCC. **Methods:** In this retrospective study, patients were aged ≥ 18 years with histologically diagnosed advanced ESCC, ECOG of 0-3, given nimotuzumab (400 mg, Q3W) combined with chemotherapy (paclitaxel: 300-400mg and platinum: 400-500mg or tegafur: 60mg/bid, Q3W) plus immunotherapy (PD-(L)1: 200-240 mg, Q3W). They assigned into two groups: metastasis group and locally advanced group. The primary endpoint was overall survival (OS). Secondary endpoints were progression-free survival (PFS), objective response rate (ORR) and safety. **Results:** As of July 2024, 58 patients were enrolled (metastasis group: n=20, locally advanced group: n=38). The median age was 59.5 years (range: 44-80). The median follow-up was 32.59 months (95% CI: 23.0, 36.7). The ORR was 74.1% (95% CI: 0.6, 0.8). Survival analysis showed that the 1-year OS was 79.16% (95% CI: 64.5, 88.3) and 2-year OS was 57.38% (95% CI: 41.3, 70.5), 1-year PFS and 2-year PFS was 64.41% (95% CI: 49.4, 76.0) and 32.80% (95% CI: 19.7, 46.5), respectively. The median OS (mOS) was 31.01months, the median PFS (mPFS) was 17.84 months. Nimotuzumab plus chemotherapy and immunotherapy displayed a favorable survival outcomes both in metastasis group and locally advanced group with OS (1-year OS: 77.76% and 80.37%; 2-year OS: 62.21% and 55.91%) and PFS (1-year PFS: 53.46% and 69.99%; 2-year PFS: 25.06% and 36.66%), the mOS was not reached and 31.01 months, mPFS was 18.27 months and 16.69 months, respectively. There were 74.1% patients occurred grade 1-2 adverse events (AEs), and 13.8% occurred grade 3-5 AEs. The most common AEs included nausea (60.3%), myelosuppression (51.7%), leukopenia (41.4%), thrombocytopenia (29.3%), and neutropenia (24.1%). No serious treatment-related AEs or death recorded. **Conclusions:** This study demonstrated a favorable survival profile and acceptable toxicity in first-line treatment of advanced ESCC patients with nimotuzumab regimen. Further prospective study needed to certify in order to benefit more patients. Research Sponsor: None.

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Real-world treatment patterns and outcomes for patients with neoadjuvant treatment between 2019-2022 for gastric or gastroesophageal junction cancer (GC/GEJC) in the US community setting. First Author: David Cosgrove, Compass Oncology, Vancouver, WA

Background: Treatment options for patients with early-stage GC/GEJC are still limited and the prognosis remains poor, despite recent therapy approvals. With the latest adoption of FLOT use in the US, it is important to understand patients' characteristics, treatment patterns and outcomes in the US community oncology setting. **Methods:** This retrospective observational cohort study used electronic health records data from 278 adult patients with stage II-IVA GC/GEJC adenocarcinoma initiating neoadjuvant treatment (index) within The US Oncology Network between 1/1/2019 and 12/31/2022, based on Food and Drug Administration approval of FLOT in 2019. Patients were followed through 9/30/2023, last patient record or death, whichever occurred first. Descriptive analyses were conducted to evaluate patient and treatment characteristics. Kaplan-Meier was used to assess overall survival (OS) and event-free survival (EFS) from index. Adjusted Cox proportional hazard models were used to analyze the association of surgery as a time-dependent variable with OS and EFS. **Results:** Median (interquartile range [IQR]) age at index was 67.5 (58.2, 74.2) years; most patients were male (68.0%), and half were Caucasian (51.1%). Stage III was the most common initial stage (50.0%), followed by IIB (29.1%), and II (not otherwise specified)/IIA (19.8%). Most patients had an ECOG (0-1) 65.5%. Median (IQR) follow-up from index was 15.6 (8.6, 27.8) months. The most common neoadjuvant regimens were FLOT (51.1%), doublet (28.4%), and CROSS (18.0%). Median OS (months) (95% confidence interval [CI]) for patients with surgery was not reached (NR) (47.7, NR) and those without was 16.1 (8.6, 35.1). Patients with surgery had a longer median EFS (95% CI) (months) than those without 38.3 (25.0, NR) vs 5.6 (4.5, 6.5) (log rank p-value <0.0001). Adjusted hazard ratios (95% CI) for OS and EFS indicated that patients who received surgery were less likely to experience an event than those without, 0.252 (0.139, 0.456) and 0.268 (0.169, 0.426), respectively ($p<0.0001$). **Conclusions:** Study findings showed a high proportion of patients receiving FLOT since its approval in US, with a directional OS benefit similar to the FLOT4 trial. Also, receipt of surgery was a strong predictor for OS and EFS. While the results are promising, most patients who did not receive surgery progressed within 6 months and died a little over a year from index. However, given the small portion of patients who did not receive surgery, further research is needed to understand if there is an unmet need for this group. Research Sponsor: None.

Variable	Doublet	CROSS	FLOT
N	79	50	142
Surgery Rate (%)	60.8%	62.0%	71.8%
GC Rate (%)	79.8%	<20%	85.9%
OS, median (95% CI), months	26.0 (20.9, NR)	37.7 (27.8, NR)	52.1 (47.7, NR)
EFS, median (95% CI), months	10.8 (8.4, 23.2)	20.9 (11.3, 26.3)	37.7 (18.0, NR)

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Prognostic analysis of pathological complete response after neoadjuvant therapy for locally advanced gastric cancer: A national, multicenter, retrospective study. First Author: Jie Chen, Fudan University Shanghai Cancer Center, Shanghai, China

Background: Several research findings indicate that some patients with locally advanced gastric cancer (LAGC) who undergo neoadjuvant therapy followed by curative surgery can achieve pathological complete response (pCR). Whether pCR is equivalent to cure, serves as a good prognostic indicator for long-term survival, and whether patients achieving pCR require subsequent adjuvant therapy are still unknown. Therefore, we conducted a nationwide multicenter retrospective study to investigate the long-term efficacy of patients with LAGC who achieved pCR after neoadjuvant therapy. **Methods:** Clinical and pathological data were collected from 351 patients with locally advanced gastric cancer who achieved pCR after neoadjuvant therapy between January 2018 and October 2023 at 14 medical centers in China. These patients were matched 1:1 with non-pCR patients based on age, gender, clinical T stage, and N stage. This study evaluated general clinical data, neoadjuvant treatment regimens, cycles, surgical outcomes, and follow-up information including postoperative adjuvant therapy, recurrence, metastasis, and survival status up to November 2023. **Results:** Negative Serum tumor markers, non-signet ring cell carcinoma and neoadjuvant treatment regimens were closely associated with pCR. Patients achieving pCR were more likely to reach ypN0 status after neoadjuvant therapy compared to non-pCR patients. Survival analysis showed significantly higher overall survival (OS) and disease-free survival (DFS) in pCR patients compared to matched non-pCR patients. Moreover, OS and DFS were significantly higher in ypN0 patients within the pCR group compared to non-pCR patients. However, there was no significant difference in OS and DFS between ypN+ patients in the pCR group and non-pCR patients. Additionally, postoperative adjuvant chemotherapy did not significantly impact OS and DFS in the pCR group. **Conclusions:** pCR is an independent prognostic factor for OS and DFS. However, survival benefits from pCR are limited to patients achieving ypT0N0 status, while ypT0N+ patients do not benefit from pCR. Furthermore, adjuvant chemotherapy may not improve the prognosis of patients achieving pCR after neoadjuvant therapy. Research Sponsor: None.

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Immune checkpoint inhibitors (ICIs) in deficient mismatch repair (dMMR)/ microsatellite instability (MSI) non-colorectal cancers: Real world Indian data and the use of alternative dosing. First Author: Dishant Vaddoriya, Shalby Hospitals Ahmedabad, Ahmedabad, India

Background: Immune checkpoint inhibitors, predominantly pembrolizumab and dostarlimab, have been approved in deficient mismatch repair (dMMR)/ microsatellite instability (MSI) non-colorectal cancers. There is also an emerging data to suggest significant clinical activity for low-dose ICIs (LD-ICIs) in these cancers. **Methods:** A multi-institutional retrospective analysis in Indian patients with dMMR/MSI-H advanced/ metastatic non-colorectal carcinomas treated with ICIs (irrespective of dose or schedule) between 2017 and 2024 was conducted. The primary objective of the study was to evaluate 12-month Overall survival (OS) for the whole cohort as well as LD-ICI and standard dose ICI (SD-ICI) cohorts, while other secondary objectives included assessment of overall response rates (ORR) (assessed by RECIST v1.1 as per local radiological reporting), 12-month progression free survival (PFS) and incidence of immune related adverse events (IRAEs). **Results:** A total of 51 patients were available for analysis. The most common tumour types were gastrointestinal (64%, commonly gastric cancer, oesophageal cancer and cholangiocarcinoma) and endometrial cancers (23%). Eighty-four percent of patients received ICI monotherapy, while the remaining 16% received varying combinations of ICIs with chemotherapy and tyrosine kinase inhibitors. Nivolumab was used in 24 patients (47%) while pembrolizumab was used in 17 patients (33%). With a median follow-up of 14 months, 12-month OS was 72% and 12-month PFS was 62% in the entire cohort. The corresponding 12-months OS and 12-month PFS for those receiving SD-ICIs (n=29) was 69% and 62%, while it was 72% and 64%, respectively for the LD-ICIs cohort (n=22). The ORR was 61%, with 10 patients (20%) achieving complete response (CR) and 21 (41%) achieving partial response (PR). The most common IRAEs (all grades) were fatigue (62%), thyroid disturbances (37%), and pruritus (23%). **Conclusions:** The current study from India highlights the importance of treating non-colorectal dMMR/MSI-H cancers with ICIs and shows outcomes commensurate with published trial data. Within the confines of a small retrospective study with a short follow-up, low dose ICIs should be explored in these patients in scenarios where standard dosing schedules are not feasible due to logistic reasons. Key Words – Low dose immunotherapy; non-colorectal dMMR/MSI-H cancers; Overall survival. Research Sponsor: None.

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Real-world biomarker testing and first-line treatment patterns among locally advanced, unresectable or metastatic gastric and gastroesophageal junction adenocarcinoma patients in the US. First Author: Lia Tzala, Center for Observational Research, Amgen Inc., Thousand Oaks, CA

Background: Advanced-stage gastric cancer and gastroesophageal junction adenocarcinoma (advG/GEJC) are leading causes of cancer-related morbidity and mortality. Chemotherapy is the traditional first-line (1L) therapy for patients in the advanced setting, with targeted and immunotherapy considered based on biomarker expression. This study aimed to describe real-world HER2 and PD-L1 testing practices and 1L treatment patterns among advG/GEJC patients. **Methods:** A retrospective cohort study was conducted among adult patients diagnosed with advG/GEJC between 01 Jan 2017 and 30 Jun 2023 usingthe US-based Flatiron Health electronic health record-derived de-identified database. HER2 and PD-L1 testing practices were evaluated prior to and up to 1-month post-1L treatment initiation or 6 months post-advG/GEJC diagnosis date in untreated patients. 1L treatment regimens were described overall and stratified by HER2 and PD-L1 expression. **Results:** Among 4,256 advG/GEJC patients, the average age was 67 years with the majority male (68%), non-Hispanic white (44%), and with metastatic disease (64%). Nearly three quarters (72%) of patients included received a HER2 test and 55% had a HER2 test with a valid result within the timeframe mentioned above. More than a quarter of patients (27%) received a PD-L1 test and 25% had a PD-L1 test with a valid result. PD-L1 testing before the approval of nivolumab in 1L (i.e. before 2021) showed testing at 19% (490/2,629; 17% with a valid result) and from 2021 onward at 40% (643/1,627; 37% with a valid result). Across the full cohort, half of the patients received chemotherapy alone (52%) and one quarter received no treatment (25%). Similar treatment patterns were observed in the subset of 1,117 patients who did not receive a HER2 or PD-L1 test; 54% received chemotherapy alone and 36% no treatment. Of the 340 patients who were tested for HER2 and were HER2 positive, half (52%) received trastuzumab combination therapy, 23% received chemotherapy alone, and 18% had no treatment. Of the 301 patients who received a PD-L1 test with a combined positive score (CPS) $\geq$5 from 2021 onward, 41% received immunotherapy, 34% received chemotherapy alone and 18% had no treatment. Prior to 2021, most patients with a PD-L1 test and CPS $>$5 used chemotherapy alone (57%; 132/233), 13% (31/233) received PD-L1 targeted therapies and 19% (43/233) received no treatment. **Conclusions:** This study describes a potential subset of US advG/GEJC patients who do not receive a HER2 or PD-L1 test and/or do not receive eligible targeted or immunotherapy as part of routine clinical practice. As precision medicine options continue to evolve, these data suggest an opportunity to further understand the drivers for biomarker testing as well as the factors impacting the use of biomarker-informed treatment options. Research Sponsor: Amgen.

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The association of antibiotic use and survival among patients receiving first line immunotherapy-based treatment for advanced gastroesophageal cancers. First Author: Nicole Baranda Balmaceda, Division of Medical Oncology, Department of Medicine, University of Colorado Anschutz School of Medicine, Aurora, CO

Background: The advent of immune checkpoint inhibition (ICI) has improved the survival for patients with gastroesophageal cancer (GEC). It is vital to understand the relationship between the tumor microenvironment and the ability of the immune system to prevent and eliminate cancer cells. Antibiotics (abx), frequently used in patients on cancer-directed therapy, can alter gut microbiota leading to dysbiosis and altered immune responses. Some research shows its use is associated with decreased ICI efficacy in RCC, NSCLC, and melanoma. The goal of our study is to investigate the association of abx timing and survival with 1L ICI-based treatments for advanced GEC. **Methods:** The retrospective, multicenter analysis used the Flatiron Health deidentified electronic health record-derived database of real world (rw) patients with advanced esophageal, GEJ, and gastric cancers (EC/GEJC/GC) who received ICI +/- chemotherapy (chemo) as 1L therapy. Abx exposure was defined by antibiotic use from 42 days before to 42 days after starting ICI. The cox proportion hazards models were used to compare rw overall survival (OS) and progression-free survival (PFS) across exposure groups. **Results:** A total of 990 patients were included for the main group analysis with comparison of survival and abx usage and timing: 876 did not receive abx, 58 received abx within 42 days after starting ICI, and 56 received abx within 42 days prior to starting ICI. The median age was 66 with 44.4% EC, 33.5% GC, and 22% GEJC. The majority of patients had adenocarcinoma (87.9%). The unadjusted OS HR for abx (within 42 days) before ICI was 0.99 (0.68-1.46, p=0.971) and (within 42 days) after ICI was 1.56 (1.11-2.19, p=0.011), suggesting worse OS with abx use after ICI initiation. HR trended towards worse PFS for those receiving abx after 1L ICI [1.36 (0.97-1.90 p= 0.073)]. In subgroup analysis, the negative effect of starting abx after 1L ICI over abx before ICI was more prominent in those receiving ICI monotherapy compared to those receiving chemo/ICI. **Conclusions:** Our study identifies a poorer survival association of abx administration after 1L ICI-based therapy for advanced GEC, primarily driven by ICI monotherapy. Research Sponsor: None.

Main group comparison between abx usage and timing: OS	**Abx Use/Timing**	**n (%)**	**HR**
	No	876 (88.5)	
	Yes, after 1L IO	58 (5.9)	1.56 (1.11-2.19, p=0.011)
	Yes, before 1L IO	56 (5.7)	0.99 (0.68-1.46, p=0.971)
Subgroup comparison between abx usage and timing by treatment type			
Chemo + IO: OS	Yes, after 1L IO	44 (6.9)	1.32 (0.88-1.98,p=0.177)
	Yes, before 1L IO	38 (6.0)	1.01 (0.62-1.65, p=0.964)
IO monotherapy: OS	Yes, after 1L IO	12 (4.9)	2.64 (1.38-5.06, p=0.003)
	Yes, before 1L IO	12 (4.9)	1.09 (0.53-2.23, p=0.810)

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Prognostic and biological divergence of upper gastrointestinal adenocarcinomas. First Author: Arun Dang, Kaiser Permanente Santa Clara, Santa Clara, CA

Background: Esophageal (EAC), gastroesophageal junction (GEJ) or gastric (GAC) adenocarcinoma share similar etiologies and treatment approaches. However, it is not clear if they possess similar or different prognostic and biological characteristics. We aimed to examine overall survival (OS) in associations of the common genomic alterations with these three anatomically closely linked adenocarcinomas. **Methods:** Eligible patients had recurrent or de novo metastatic GAC, GEJAC or EAC whose tumors had undergone next generation sequencing (NGS) performed from November 2017 to December 2023. We used Cox regression modeling to examine the association between GAC, GEJAC, EAC and OS, adjusting for demographics, performance status, Charlson comorbidity index, receipt of chemotherapy, and HER2 overexpression or amplification (HER2 positive), *p53* (mutp53*)*, *KRAS* (mutKRAS), *CDKN2A*, *PIK3CA* co-mutations and *MYC* amplification. **Results:** Of 875 total eligible patients, 173 had EAC, 276 had GEJAC and 426 had GAC. GEJAC had substantially better OS than EAC (HR = 0.68, [95% CI, 0.54-0.86]), and modestly better OS than GAC (HR = 0.85, [95% CI, 0.67-1.09]). HER2 positivity was associated with substantially better OS among EAC (HR = 0.62; [95% CI, 0.40-0.96]) and GEJAC (HR = 0.59; [95% CI, 0.38-0.87]) but not among GAC (HR = 0.98; [95% CI, 0.78-1.23]) patients. In addition, p53 gain-of-function versus non-gain-of-function mutations were associated with substantially worse OS among GAC (HR = 1.41; [95% CI, 0.98-2.01]) but not among EAC or GEJAC. MutKRAS was associated with substantially worse OS among EAC (HR = 1.81; [95% CI, 1.10-2.99]) but not among GEJAC or GAC. Surprisingly, *MYC* amplification was associated with dramatically better OS among EAC (HR = 0.19; [95% CI, 0.08-0.42]) but substantially worse OS among GEJAC (HR = 1.98, 95% CI, 1.14-3.40]) and GAC (HR = 1.75; [95% CI, 1.10-2.80]). **Conclusions:** GEJAC, EAC and GAC possess substantially different OS that appears differentially associated with HER2 status, mutp53, mutKRAS and *MYC* amplification. These results suggest distinct prognostic and biological characteristics of these three anatomically closely linked adenocarcinomas that could have important implications in clinical practice and on further investigations on their biological and topological mechanisms. Research Sponsor: None.

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Results of a multicenter retrospective cohort study evaluating the safety and efficacy of FTD/TPI plus ramucirumab in advanced gastric cancer (HGCSG2302). First Author: Ayumu Yoshikawa, Department of Gastroenterology, Japanese Red Cross Kitami Hospital, Kitami, Japan

Background: In advanced gastric cancer, trifluridine/tipiracil (FTD/TPI) is one of the standard treatments for later-line chemotherapy. Ramucirumab (RAM), an anti-VEGFR-2 monoclonal antibody, has shown promising efficacy in combination with FTD/TPI in phase II studies conducted in Japan, suggesting that this combination could improve treatment outcomes. However, there remains a paucity of real-world clinical data comparing the efficacy and safety of FTD/TPI monotherapy and FTD/TPI plus RAM combination therapy in general clinical practice. **Methods:** This retrospective analysis to evaluate safety and efficacy included patients with gastric cancer who were received FTD/TPI plus RAM combination therapy or FTD/TPI monotherapy between August 2019 and March 2023 at 21 institutions. **Results:** 164 patients were analyzed, 38 patients in the RAM combination group and 126 in the FTD/TPI monotherapy group. The distribution of treatment lines (third/fourth/fifth) was 10/13/15 in the RAM group and 26/54/46 in the monotherapy group, respectively. RAM had been previously administered to 34 patients (89.5%) in the RAM group. Objective response rate and disease control rate were 11.1% / 55.6% (OR 2.0, 95% C.I. 0.84-4.77, p=0.126) in the RAM group and 5.5% /38.5% (OR 2.0, 95% C.I. 0.84-4.77, p=0.126) in the monotherapy group, respectively. Median progression-free survival (PFS) was 3.5 months in the RAM group and 2.1 months in the monotherapy group (HR 0.69, 95% C.I. 0.49-1.02, p=0.06), respectively. Median overall survival (OS) was 8.4 months in the RAM group and 5.8 months in the monotherapy group (HR 0.82, 95% C.I. 0.54-1.25, p=0.350), respectively. Grade 3 neutropenia was reported in 48.6%/52.4% (p=0.707), and there were no significant differences in other adverse event profiles between the two groups. Additionally, in the RAM group, a comparison between patients receiving standard dosing (2 weeks on/2 weeks off; n=7) and those on biweekly dosing (1 week on/1 week off; n=24) showed no notable differences in efficacy or safety. **Conclusions:** FTD/TPI plus RAM combination therapy showed a trend toward prolonged PFS compared to FTD/TPI monotherapy, but no significant difference in OS. Research Sponsor: None.

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Impact of prior nivolumab use on the efficacy of second-line taxane-based chemotherapy in advanced gastric or esophagogastric junction adenocarcinoma. First Author: Momoko Sano, Department of Gastrointestinal Medical Oncology, National Cancer Center Hospital, Tokyo, Japan

Background: Taxane-based chemotherapy remains the standard second-line treatment after first-line treatment including anti-PD-1 antibody therapy for patients with advanced gastric or esophagogastric junction adenocarcinoma (GA/EGA). The CheckMate 649 trial suggested that prior use of nivolumab in the first-line setting was associated with longer progression-free survival (PFS) with subsequent treatment. However, there have been few reports on the difference in efficacy of taxane-based chemotherapy as second-line treatment with or without prior nivolumab use in the real-world setting. **Methods:** We retrospectively reviewed the medical records of patients with advanced GA/EGA who received solvent-based paclitaxel or nanoparticle albumin–bound (nab)-paclitaxel chemotherapy regimens as second-line treatment after first-line platinum-based chemotherapy between January 2018 and August 2024. We divided the patients into two groups: those who received nivolumab with the first-line chemotherapy (Nivo group) and those who did not (non-Nivo group). PFS and overall survival (OS) were estimated using the Kaplan–Meier method and compared between the two groups using the log-rank test. Hazard ratios (HRs) and 95% confidence intervals (95%CIs) were estimated by univariate and multivariate Cox regression analysis. **Results:** A total of 115 patients were included: 27 patients in the Nivo group and 88 in the non-Nivo group. Patient backgrounds did not differ significantly in terms of age (median 67/67 years), sex (male 59%/57%), performance status (PS; ≥2 11%/15%), site of primary tumor (EGA 11%/9%), number of metastatic sites (≥3 33%/18%), and ramucirumab use (yes 82%/82%). For the taxane regimens, nab-paclitaxel-based treatment was significantly more common in the non-Nivo group (11.1% /55.7%, p<0.001). There was a tendency toward longer PFS in the Nivo group compared with the non-Nivo group (median 4.2 months [95%CI 2.8-6.0] vs. 2.3 months [95%CI 2.0-3.0], HR 0.65 [95%CI 0.40-1.03]; p=0.067.) The objective response rate was numerically higher in the Nivo group compared with the non-Nivo group (29.4% vs. 17.6%, p=0.315). OS was not significantly different between the Nivo and non-Nivo groups (median 9.6 months [95%CI 4.9-NA] vs. 6.6 months [95%CI 5.5-10.8], HR 0.92 [95%CI 0.52-1.64]; p=0.785). In the non-Nivo group, 64 patients (73%) received subsequent chemotherapy, and 63 patients (72%) received nivolumab. Multivariate analysis identified prior use of nivolumab was a favorable prognostic factor for PFS (HR 0.55 [95%CI 0.31-0.97]; p=0.039). However, prior use of nivolumab did not show a significant benefit for OS (HR 1.09 [95% CI: 0.56-2.13]; p = 0.802). **Conclusions:** Prior use of nivolumab in first-line platinum-based chemotherapy had a positive impact on PFS with taxane-based second-line chemotherapy in advanced GA/EGA. Research Sponsor: None.

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Real-world biomarker testing patterns, first line treatment and drivers of treatment choice in patients with advanced/metastatic gastric/gastroesophageal junction adenocarcinoma in the United States and Europe. First Author: Florian Lordick, University of Leipzig Medical Center, Comprehensive Cancer Center Central Germany, Department of Medicine, Leipzig, Germany

Background: For first line (1L) treatment (tx) in HER2- gastric/gastroesophageal junction (G/GEJ) adenocarcinoma (AC), clinical guidelines recommended platinum-fluoropyrimidine (PF) chemotherapy (CT) in patients (pts) with a PD-L1 combined positive score (CPS) <5 and PF CT in combination with immunotherapy (IO) for CPS ≥5. There is little contemporary evidence into IO tx uptake in the real world. This real world study examined biomarker testing, 1L tx characteristics, and reasons for 1L tx choice in pts with HER2- G/GEJ AC to identify unmet needs and opportunities for optimal clinical practice. **Methods:** Data were derived from the Adelphi G/GEJ Cancer Disease Specific Programme: a systematic approach involving multi-national, multi-faceted data sources including cross-sectional surveys capturing data from physicians and their consulting pts. Physicians reported pt demographics, clinical characteristics, tx received and reasons for tx decisions, via chart review. Data were collected across US and Europe (EU: France, Germany, Spain, the UK) between Oct 2022 to Apr 2023. All pts were alive at data collection and analyses were descriptive. **Results:** Overall, 243 treating physicians (US n=60/EU n=183) provided data on 642 HER2- pts (US n=83/EU n=559) receiving 1L tx at data collection. For pts from US/EU, median age was 66/67 years, 84%/79% had an ECOG of 0-1 and 52%/75% had metastatic disease at data collection. In the US/EU, 87%/80% of pts were tested for PD-L1 status respectively, of which 63%/69% had a known CPS. For US/EU, 67%/45% of pts (n=30/n=139) had a CPS ≥5. Differences in biomarker testing rates were observed in US v EU pts respectively for MSI/dMMR (66% v 58%), NTRK-gene fusion (31% v 4%) and FGFR2 (27% v 3%). Of US/EU physicians, 70%/85% (n=60/n=183), reported the main reason for conducting biomarker testing was 'to inform tx decisions'. Cost related concern (50%/39%) and time constraints (23%/15%) were the main perceived barriers to testing. Among pts with PD-L1 CPS <5, 27% of US pts and 2% of EU pts received IO + CT; in pts with CPS ≥5, 63% of US pts and 76% of EU pts received IO + CT. Whilst progression free survival/overall survival benefit were the most common reasons selected by physicians for 1L tx choice across the US/EU (75%/74% and 60%/61%), differences across regions were reported for tx choice based on maintaining and improving pt's QoL (4%/28%) and pt's performance status (2%/24%). **Conclusions:** PD-L1 testing was not performed by 19% of physicians and approximately one third of pts with CPS ≥5 across the US/EU were not receiving IO + CT in 1L. As informing 1L tx choice was a key driver for biomarker testing, there may be scope for improvement in 1L tx by increasing testing rates. Further studies are needed to address barriers to biomarker testing and uptake of IO in pts who can benefit from 1L IO tx. Research Sponsor: Gilead Sciences.

354 **Poster Session**

Q-TWiST analysis of pembrolizumab plus trastuzumab and chemotherapy for patients with metastatic HER2-positive gastric or gastroesophageal junction (G/GEJ) adenocarcinoma in the KEYNOTE-811 trial. First Author: Sun Young Rha, Division of Medical Oncology, Department of Internal Medicine, Yonsei Cancer Center, Yonsei University College of Medicine, Seoul, South Korea

Background: In the phase 3 KEYNOTE-811 trial (NCT03615326), first-line pembrolizumab plus trastuzumab and chemotherapy (pembrolizumab group) statistically significantly improved PFS and OS versus placebo plus trastuzumab and chemotherapy (placebo group), with manageable safety, in patients with metastatic HER2-positive G/GEJ adenocarcinoma, while maintaining HRQoL, specifically in patients with PD-L1 combined positive score (CPS) ≥1 tumors. The objective of this analysis was to assess quality-adjusted time without symptoms of disease progression or toxicity (Q-TWiST) for patients treated in the KEYNOTE-811 trial. **Methods:** Q-TWiST is a measure of survival weighted by health states utility values. The analysis categorized survival time into 3 health states: time with grade ≥3 adverse events (toxicity [TOX]) and time without symptoms or toxicities (TWiST)—both before disease progression or death—and relapse (REL), defined as time from disease progression to death. Q-TWiST was calculated as the restricted mean survival time spent in each state, with a weighted health state utility for that state. Average utility weights were derived based on pooled EQ-5D-5L scores using the US mapping algorithm and standardized base case utility weights. Relative gains in Q-TWiST of ≥10% and ≥15% were "clinically important" and "clearly clinically important," respectively, as reported in the literature. Treatment difference 95% CIs were generated using the nonparametric bootstrapping method. Data are reported for patients with PD-L1 CPS ≥1 tumors. The data cutoff date was March 20, 2024. **Results:** At the maximum follow-up of 63 months, patients in the pembrolizumab group had a 2.36-month (95% CI, −0.24 to 4.56) longer restricted mean time in TOX (6.98 vs 4.63 months), a 3.40-month (95% CI, −0.44 to 7.45) longer restricted mean time in TWiST (12.03 vs 8.62 months), and a 1.45-month (95% CI, −5.84 to 3.09) shorter restricted mean time in REL (8.87 vs 10.32 months) versus the placebo group. For the restricted mean Q-TWIST at month 63, based on the US mapping algorithm, the difference between the pembrolizumab and placebo groups favored the pembrolizumab group by 3.90 months (95% CI, 1.30 to 6.72), representing a 16.57% relative Q-TWiST gain. Based on the standardized base case utility weights, the difference also favored the pembrolizumab group by 3.86 months (95% CI, 1.47 to 6.67), representing a 16.38% relative Q-TWiST gain. **Conclusions:** Pembrolizumab plus trastuzumab and chemotherapy produced a clearly clinically important improvement in quality-adjusted survival time based on Q-TWiST analyses compared with placebo plus trastuzumab and chemotherapy in patients with previously untreated metastatic HER2-positive G/GEJ adenocarcinoma with PD-L1 CPS ≥1. Clinical trial information: NCT03615326. Research Sponsor: Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA.

355 **Poster Session**

Focusing on clinical trial ineligibility: Nivolumab plus chemotherapy for patients with advanced gastric cancer. First Author: Iori Motoo, Third Department of Internal Medicine, University of Toyama, Toyama, Japan

Background: Nivolumab demonstrates promising efficacy when combined with chemotherapy as first-line treatment in patients with advanced gastric cancer (AGC). However, a large proportion of patients are perceived as ineligible for clinical trials in clinical practice. The aim of this study is to analyze the treatment outcomes and safety among patients deemed ineligible in clinical trials for AGC. **Methods:** A retrospective examination was conducted involving 187 patients who received treatment for HER2-negative unresectable or recurrent AGC between December 2014 and October 2023 at six institutions. Eleven patients were excluded due to massive ascites or poor performance status (PS). Ineligible patients were defined as meeting any of the seven criteria: age 75 years or over, PS 2, bone marrow dysfunction, hepatic dysfunction, renal dysfunction, serious complications, and extended bone metastasis. Of the 104 patients meeting these criteria, individuals were divided into the ICI group (nivolumab plus chemotherapy) and the non-ICI group (chemotherapy only). A comparative assessment of survival outcomes and tumor responses between the two groups was performed. Additionally, the incidence of immune-related adverse events (irAE) in the ICI group was assessed. **Results:** The ICI group comprised 57 patients, while the non-ICI group comprised 47 patients. The median follow-up for survival analysis was 11.6 months. In the ICI group, response rate (RR), median progression-free survival (mPFS), and median overall survival (mOS) were 48.7% among patients with measurable lesions, 7.4 months (95%CI, 5.3-10.0), and 15.1 months (95%CI, 13.1-21.0). In the non-ICI group, RR, mPFS, and mOS were 33.3%, 6.4 months (95%CI, 4.1-7.8), and 11.8 months (95%CI, 10.1-14.6). The OS in the ICI group was significantly better than those in the non-ICI group (log-rank p = 0.0436). Furthermore, irAEs were observed in eight patients (14.0%), including hypothyroidism (n=3), arthritis (n=2), hypopituitarism (n=2), and dermatitis (n=1). No patients discontinued treatment due to intolerable irAEs. **Conclusions:** The combination of nivolumab and chemotherapy showed a favorable outcome even in AGC patients with ineligible group. Research Sponsor: None.

356 **Poster Session**

Prognostic impact of postoperative complications after neoadjuvant therapy followed by esophagectomy for esophageal cancer: An exploratory analysis of phase III trial JCOG1109. First Author: Eisuke Booka, Department of Surgery, Hamamatsu University School of Medicine, Hamamatsu, Shizuoka, Japan

Background: We previously reported that postoperative complications after esophagectomy were significantly associated with poor long-term prognosis in the prospective randomized trial JCOG9907 (Ann Surg, 2017). JCOG1109 evaluated the efficacy and safety of neoadjuvant docetaxel plus cisplatin plus 5-FU (DCF), radiation with cisplatin plus 5-fluorouracil (CF-RT) compared with cisplatin plus 5-fluorouracil (CF) for locally advanced esophageal squamous cell carcinoma, and demonstrated the superiority of DCF compared with CF in improving overall survival (OS). In this exploratory analysis, we investigated the association between postoperative complications and prognosis in JCOG1109, where more intensive preoperative therapies and thoracoscopic esophagectomy (TE) were introduced. **Methods:** Patients with potentially resectable advanced thoracic esophageal cancer were randomly assigned to CF, DCF, or CF-RT therapies and followed by open esophagectomy (OE) or TE with regional lymphadenectomy in JCOG1109. The impacts of postoperative complications ($\geq$ Grade 2) on long-term prognosis including OS and progression-free survival (PFS) were investigated in each of the three preoperative therapies. **Results:** Between December 2012 and July 2018, 601 patients were randomized (CF/DCF/CF-RT; 199/202/200). Of 589 eligible patients, 541 patients underwent esophagectomy (183/181/177). Any postoperative complications, pneumonia, anastomotic leakage, recurrent laryngeal nerve paralysis, or infectious complications had no significant impact on OS in each of the three arms. When divided into OE or TE, the impact of any postoperative complications on OS was attenuated by introducing TE in each of three arms; from neoadjuvant CF followed by OE group [hazard ratio (HR) 1.557, 95% confidence interval (CI) 0.881-2.752] to TE group [HR 0.802, 95% CI 0.424-1.515], from neoadjuvant DCF followed by OE group [HR 1.151, 95% CI 0.575-2.306] to TE group [HR 0.703, 95% CI 0.3363-1.471], and from neoadjuvant CF-RT followed by OE group [HR 1.548, 95% CI 0.840-2.852] to TE group [HR 1.186, 95% CI 0.627-2.245]. **Conclusions:** JCOG1109 showed no association between postoperative complications and long-term prognosis. The prognostic impact of postoperative complications might be attenuated by the introduction of minimally invasive esophagectomy (open to thoracoscopy) even after more intensified preoperative therapy (CF to DCF or CF to CF-RT). Clinical trial information: CRB3180009. Research Sponsor: None.

357 **Poster Session**

Prognosis of signet ring cell (SRC) gastric carcinoma (GC) compared with other histological subtypes. First Author: Rafael Torosian, N.N. Blokhin National Medical Research Center of Oncology, Moscow, Russian Federation

Background: Signet ring cell (SRC) gastric cancer (GC) has its unique clinicopathological features with poor tissue differentiation, high invasiveness, diffuse growth pattern and poor prognosis. Survival rates and clinicopathological characteristics of patients with SRC GC regardless of the stage of the disease has not been well studied. **Methods:** The retrospective analysis was undertaken of 993 patients with diagnosed gastric cancer. These patients received surgical resection in period from January 2013 to December 2018 in N.N. Blokhin National Medical Research Center of Oncology. We compared clinical and pathological features as well as prognostic factors between these groups. **Results:** For early SRC carcinoma, the median survival rates weren't reached; 3- and 5-year survival rates were 100% and 89.0 %; for non-SRC carcinoma 91.4 and 85.3 % (HR=0.73; 95 % CI 0.22–2.42, p = 0.6) respectively. For locally advanced/metastatic SRC carcinoma, the median survival rates, 3- and 5-year survival rates were 38 months, 53.0 and 38.4 %; for non-SRC carcinoma 51.1 months, 59.2 and 48.0 % (HR=1.2; 95 % CI 0.91–1.54, p = 0.2). At univariate analysis SRC morphology was found as independent risk factor for overall survival (HR=1.19; CI 95% 0.94 – 1.50; $p<0.0001$). Multivariate analysis did not reveal SRC morphology to be significant and independent risk factor for overall survival (HR=1.00; CI 95% 0.72 – 1.38; p=0.98). **Conclusions:** Overall survival rates in patients with early (mucosal and submucosal tumor regardless of lymph nodes status) SRC gastric cancer did not show any significant differences in comparison with other histological types of GC. Thus, the prognosis of early SRC GC is equivalent or better than that of other histological subtypes of GC. Overall survival rates of locally advanced/metastatic SRC gastric cancer are worse than non-SRC cancer, but the differences are not statistically significant. Research Sponsor: None.

358 **Poster Session**

Postoperative exercise in gastric cancer patients undergoing gastrectomy: A randomized controlled trial. First Author: Hyoung-Il Kim, Department of Surgery, Yonsei Cancer Center, Yonsei University College of Medicine, Seoul, South Korea

Background: Faster physical recovery after gastrectomy may lead to better physical and psychological health in patients with gastric cancer. Post-surgical exercise may prevent physical dysfunction and improve recovery, but its effects on physical function are not fully explored. **Methods:** In this trial, 52 gastric cancer patients (Stage 1-3) were randomly assigned to early postoperative exercise rehabilitation or usual care (1:1 ratio). Interventions Inpatient Supervised Exercise sessions were conducted for 15 minutes once a day on Postoperative Day (POD) 1 and 2 in the ward. Home-based Exercise involved 30-minute remote exercise sessions at home once a week following discharge, with additional recommendations for daily 30-minute sessions of low to moderate-intensity aerobic exercises and bodyweight exercises. Moreover, the exercise group was provided with an exercise log and instructional videos to record the duration of bodyweight exercises and aerobic exercises. Measurements were taken three times (baseline, POD3, and the first outpatient visit 1 month after surgery). The primary outcome was a Short Physical Performance Battery (SPPB), with secondary outcomes including grip strength, 30-second sit-to-stand (STS), body composition, C-reactive protein (CRP), and quality of life (QOL). **Results:** A total of 46 (88%) participants (mean [SD] age, 60.2 [12.24] years; 26 [57%] male) completed the trial. Compliance with intervention in-patient exercise was 96% and remote exercise was 79%. The significant difference in SPPB scores of 11.4$\pm$1.3 in the exercise group compared with 10.7$\pm$1.43 usual care group (mean difference, -1.09 score; 95%CI, -1.7 to -0.4; p=$<$.001). Moreover, there were significant differences between groups in the STS exercise group 14.3$\pm$2.8 usual care group 13.6$\pm$4.5 (mean difference, 1.86; 95%CI, -3.3 to -0.4; p=.028), and CRP in POD3 exercise group 66.9$\pm$40.73 usual care group 103.7$\pm$54.25 (mean difference, -22.54 score; 95%CI, -46.5 to 1.4; p=.016). However, there was no significant difference in grip strength, body composition, and QOL. **Conclusions:** Postoperative exercise presents as a beneficial strategy to enhance physical recovery in patients undergoing gastrectomy, improving key physical function markers and reducing inflammation. Clinical trial information: NCT06260293. Research Sponsor: None.

359 **Poster Session**

Impact of combined resection of pancreas on long-term survival in patients with gastric cancer. First Author: Taku Hattori, Shizuoka Cancer Center, Nagaizumi, Sunto-gun, Japan

Background: In patients with advanced gastric cancer invading adjacent organs, extended multivisceral resection is required to achieve R0 resection. However, the survival benefit of combined resection of pancreas remains controversial. We investigated the safety and efficacy of combined pancreatic resection in gastric cancer. **Methods:** This study included 64 patients who underwent gastrectomy with pancreatectomy (pancreaticoduodenectomy (PD) or distal pancreatectomy with splenectomy (PS)) for primary gastric cancer with invasion to the pancreas by primary tumor or lymph node metastasis and invasion to the duodenum. Patients who underwent R2 resection were excluded. Short- and long-term outcomes were assessed. **Results:** PD was performed in 18 patients (28.1%) and PS in 46 patients (71.9%). Macroscopic invasion to the pancreas by primary tumor was observed in 47 patients (73.4%), by lymph node in 13 patients (20.3%), and the invasion to the duodenum by primary lesion was in 4 patients (6.3%). Pathologically, pancreatic invasion was confirmed in 27 out of 47 patients (57.4%). Postoperative intra-abdominal infectious complications (PIIC) of Clavien-Dindo (CD) Grade III or higher were observed in 33 patients (51.6%). The most common postoperative complication was pancreatic fistula, which was observed in 31 patients (48.4%). There was no mortality within 30 days after surgery. Univariate analysis revealed that invasion the duodenum was highly associated with PIIC (p=0.038). Twelve patients (18%) resulted in R1 resection due to positive peritoneal cytology in 11 patients and positive surgical margin in one patient. The 5-year overall survival (OS) rate was 42.9%, which was 53.6% in R0 resection and 0% in R1 resection. Multivariate analysis revealed R1 resection was an independent prognostic factor for OS (HR:5.315, $p<0.001$). The 5-year recurrence-free survival (RFS) rate was 44.5%. The 5-year RFS rate was significantly worse in patients with PIIC (28.7%) than in patients without it (58.7%). Multivariate analysis revealed that PIIC was an independent prognostic factor for RFS (HR: 2.345, p=0.036). **Conclusions:** Pancreatic resection for gastric cancer is considered safe and effective when R0 resection is possible. However, surgeon should pay particular attention to avoid the postoperative complications. Research Sponsor: None.

360 Poster Session

Association of early tumor shrinkage with survival in patients with HER2-positive advanced gastric cancer treated with trastuzumab deruxtecan. First Author: Koshiro Fukuda, Department of Gastroenterological Chemotherapy, Cancer Institute Hospital, Japanese Foundation for Cancer Research, Tokyo, Japan

Background: Trastuzumab deruxtecan (T-DXd), an antibody-drug conjugate composed of an anti-human epidermal growth factor receptor 2 (HER2) antibody and a cytotoxic topoisomerase I inhibitor, has been approved as third or later-line treatment for unresectable HER2-positive advanced gastric cancer (AGC) based on the results of phase II DESTINY-Gastric 01 trial. However, there are few studies on real-world clinical outcomes and clinicopathological factors associated with efficacy of T-DXd in patients with HER2-positive AGC. **Methods:** This single-center retrospective study enrolled patients with AGC treated with T-DXd as third- or later-line chemotherapy between March 2018 and December 2023. Early tumor shrinkage (ETS) was defined as the relative change in the sum of longest diameters of RECIST target lesions at 8 $\pm$ 2 weeks compared with baseline. Prognostic outcomes were assessed using the log-rank test and Cox proportional hazards regression model. **Results:** A total of 57 patients were enrolled. The median age was 66 years (range, 31–82); 68% were male; 79% had HER2 3+; 23% had pulmonary metastasis; 67% were treated with T-DXd in the third-line setting. At a median follow-up of 27.6 months, the median progression-free survival (PFS) was 4.5 months (95% confidence interval [CI], 3.8–5.5), and the median overall survival (OS) was 7.3 months (95% CI, 4.9–9.4). Among 40 patients with target lesions, the overall response rate was 35.0% and the disease control rate was 90.0%. Patients with ETS had notably longer PFS (median PFS (months) 8.6 vs 4.1, $P < 0.01$) and OS (median OS (months) 9.1 vs 6.1, $P < 0.01$) than those without ETS. In multivariate analysis, ETS was associated with significantly longer PFS (hazard ratio [HR], 0.23; 95% CI, 0.08–0.63; P = 0.004) and OS (HR, 0.27; 95% CI, 0.12–0.65; P = 0.004), whereas pulmonary metastasis was associated with significantly shorter OS (HR 2.46, 95% CI, 1.07–5.65; P = 0.03). **Conclusions:** This study showed that achieving an ETS was significantly associated with a favorable prognosis in patients with HER2-positive AGC treated with T-DXd. Research Sponsor: None.

361 Poster Session

Preliminary results of the ALTER-E009 study: Efficacy and safety of anlotinib combined with radiotherapy in patients with locally advanced or metastatic esophageal squamous cell carcinoma (ESCC)—A multicenter, multi-cohort retrospective exploratory study. First Author: Xin Wang, Cancer Hospital Chinese Academy of Medical Sciences, Beijing, China

Background: Therapeutic options for metastatic esophageal squamous cell carcinoma (ESCC) primarily involve immunotherapy combined with chemotherapy. For locally advanced ESCC, concurrent chemoradiotherapy (CRT) or radiotherapy is the standard, though outcomes remain suboptimal. Targeted therapies, particularly anti-angiogenic agents, hold promise to enhance CRT efficacy. Anlotinib, a multi-target anti-angiogenic agent inhibiting VEGFR, PDGFR, FGFR, and c-KIT, has shown potential to enhance radiotherapy by alleviating tumor hypoxia. This multicenter, retrospective study evaluates the efficacy and safety of anlotinib combined with radiotherapy in locally advanced or metastatic ESCC. **Methods:** This study screened patients (pts) with locally advanced or metastatic ESCC treated with anlotinib (8-12 mg, p.o., qd, d1-14, q3w) and radiotherapy from June 2018 to June 2024. Two cohorts were included: Cohort 1 (metastatic ESCC) and Cohort 2 (locally advanced ESCC). Cohort 2 was divided into an observational group (OG, anlotinib + radiotherapy $\pm$ chemotherapy) and a control group (CG, radiotherapy $\pm$ chemotherapy). Radiotherapy doses for primary lesions were 45-60 Gy (1.8-2.2 Gy per fraction), while metastatic lesions received either standard fractionation ($\geq$45 Gy) or stereotactic body radiotherapy (SBRT, $\geq$3 Gy per fraction, total dose $\geq$30 Gy). Chemotherapy regimens were not restricted. Cohort 1 aimed to enroll 50 pts, while Cohort 2 included 100 in the OG and 200 in the CG. The primary endpoint was overall survival (OS), with secondary endpoints including objective response rate (ORR), disease control rate (DCR), progression-free survival (PFS), duration of response (DoR), and safety. **Results:** By September 10, 2024, 128 pts had been enrolled: 34 in Cohort 1 and 94 in the OG of Cohort 2. In Cohort 1, all 34 pts (100%) were stage IVB, with a median OS of 24.0 months (95% CI: 12.44-35.60) and a median PFS of 9.2 months (95% CI: 7.60-10.80). The 24-month OS and PFS rates were 54.0% (95% CI: 27.20-74.73) and 30.8% (95% CI: 10.21-54.53), respectively. ORR was 52.9% (95% CI: 35.1-70.2), with 1 complete response (CR) and 17 partial responses (PR), DCR was 88.2% (95% CI: 72.5-96.7). In Cohort 2, 50 pts (53.2%) were stage III, and 28 (29.8%) were stage IVA. The median OS is not yet mature, but the median PFS was 20.1 months. The 36-month OS and PFS rates were 61.9% (95% CI: 46.12-74.34) and 48.4% (95% CI: 29.58-64.87). ORR was 46.8% (95% CI: 36.4-57.4), with 6 CRs and 38 PRs, DCR was 95.7% (95% CI: 89.5-98.8). Safety data collection is ongoing. **Conclusions:** Anlotinib combined with radiotherapy exhibited promising efficacy in locally advanced or metastatic ESCC, though further data are needed to confirm these preliminary findings. Research Sponsor: None.

362 Poster Session

Gastric cancer patient insights: Patient-reported outcomes and real-world perspectives from the gastric cancer patient community. First Author: Martha Raymond, GI Cancers Alliance, Dennisport, MA

Background: According to the National Cancer Institute Surveillance, Epidemiology, and End Results Program, an estimated 26,890 patients in the United States will be diagnosed with gastric (stomach) cancer in 2024. Estimated deaths from the disease in 2024 will reach 10,880, with a five-year relative survival rate of 36%. Understanding the need for increased gastric cancer education and awareness, the GI Cancers Alliance created a 12-month series of engagement opportunities to better understand and listen to the gastric cancer patient voice and perspectives. **Methods:** Our 12-month patient-reported outcomes (PRO) research (September 2023 – August 2024) included an online survey, in person and virtual focus groups, and personalized follow-up interviews. 651 gastric cancer patients and survivors participated in these engagement opportunities, with 89% of participants asking for follow-up personalized interviews. One of our program goals was to reach and interact with patients who often experience barriers in receiving quality cancer care. Our PRO research provided an opportunity to amplify the patient voice and lived experience by engaging patients in medically underserved communities, rural and urban populations, diverse ethnic communities, and patients receiving care in community-based and federally funded oncology centers, and charity (indigent) care. **Results:** Overarching results from our study suggest that while advances have been made, health equity, racial disparities and barriers in cancer care remain a challenge. 651 patient participants (43% male, 54% female, 3% non-binary) identified their race/ethnicity as: 24% Black/African American, 22% Hispanic, 21% Multiracial, Non-Hispanic/White 19%, Asian/Pacific Islander 7%, Native American 2%, and 5% Other. 69% of patients reported that their clinician did not suggest biomarker testing, nor discuss the role of biomarkers to help make informed treatment decisions. 68% of patients were concerned that they lacked knowledge and options to make informed decisions regarding their treatment. 64% of patients reported they suffered from mental health concerns ranging from severe depression, feelings of hopelessness and anxiety; 63% of patients struggled to afford nutritious food to meet their changing dietary needs, especially post-surgery. **Conclusions:** Our PRO research underscores the need for increased education and outreach for gastric cancer patient communities experiencing barriers to high quality cancer prevention, early detection, and treatment. By partnering with our 100+ advocacy partner organizations, we will continue to amplify the gastric cancer patient voice by establishing new programming, initiatives, and enhanced outreach addressing the challenges and barriers to quality cancer care for all patients. Research Sponsor: GI Cancers Alliance; Amgen; Astellas; Astra-Zeneca.

363 Poster Session

Real-world validation of the Khorana risk score in gastric cancer patients receiving immune checkpoint inhibitors. First Author: Junmin Song, Department of Medicine, Jacobi Medical Center, Albert Einstein College of Medicine, Bronx, NY

Background: Thrombosis is a common cancer-related complication, particularly in gastric cancer. Studies show that immune checkpoint inhibitors (ICIs) increase the risk of venous thromboembolism (VTE) and arterial thrombosis. The Khorana risk score is widely used to assess thrombosis risk before starting chemotherapy. This study aimed to validate the Khorana score in gastric cancer patients receiving ICIs using a large global database. **Methods:** We used the TriNetX Global Collaborative Network, which contains de-identified data from over 120 million patients. We included patients diagnosed with gastric cancer who received ICIs between January 2014 and July 2023. The Khorana risk score was calculated based on lab values and BMI within two weeks before ICI initiation. Patients were divided into two cohorts: elevated risk (score 3-6) and intermediate risk (score 2). Cox-proportional hazard analyses were conducted over a one-year follow-up. The primary outcome was VTE (deep vein thrombosis [DVT] and pulmonary embolism [PE]). Secondary outcomes were arterial thrombosis, including myocardial infarction (AMI) and ischemic stroke, and all-cause mortality. **Results:** The study included 551 patients, with 362 in the intermediate risk and 189 in the elevated risk cohorts. The mean age was 64 in both cohorts, with the elevated group consisting of more males (76% vs. 63%) and a higher proportion of Black patients (11% vs. 6%). Pembrolizumab (49%) and nivolumab (47%) were the most used ICIs. Patients with elevated risk had a significantly higher risk of VTE (HR 1.74, 95% CI 1.18-2.58), DVT (HR 1.82, 95% CI 1.07-3.10), and PE (HR 1.74, 95% CI 1.07-2.84). However, no significant associations were found with arterial thrombosis (HR 1.28, 95% CI 0.74-2.20), including cerebral infarction (HR 1.05, 95% CI 0.51-2.16), or AMI (HR 1.46, 95% CI 0.70-3.05). The elevated risk cohort also had significantly higher mortality (HR 1.44, 95% CI 1.11-1.87). **Conclusions:** An elevated Khorana risk score is associated with a higher risk of VTE, but not arterial thrombosis, in gastric cancer patients treated with ICIs. Further studies are needed to determine how the Khorana score can be applied to predict ICI-related thrombosis and guide prophylactic anticoagulation. Research Sponsor: None.

	Khorana risk score 3-6		Khorana risk score 2			
Outcomes	**At risk patients**	**Cases**	**At risk patients**	**Cases**	**Hazard ratio (95% CI)**	**P-value (Log-rank)**
Venous thromboembolism	189	45	362	58	1.74 (1.18-2.58)	0.005
Deep vein thrombosis	189	25	362	30	1.82 (1.07-3.10)	0.025
Pulmonary embolism	189	29	362	36	1.74 (1.07-2.84)	0.025
Arterial thrombosis	189	21	362	34	1.28 (0.74-2.20)	0.375
Cerebral infarction	189	11	362	22	1.05 (0.51-2.16)	0.901
Acute myocardial infarction	189	12	362	17	1.46 (0.70-3.05)	0.317
All-cause mortality	189	93	362	145	1.44 (1.11-1.87)	0.006

364 **Poster Session**

Correlation between PD-L1 CPS and clinical outcomes in patients treated with first-line nivolumab plus chemotherapy for advanced gastric or gastroesophageal junction cancer: A multi-center retrospective study in Japan.
First Author: Misa Onishi, Department of Gastroenterology, Kanagawa Cancer Center, Yokohama, Japan

Background: The CheckMate 649 trial showed the superiority of first-line nivolumab plus chemotherapy (Nivo-CT) compared with chemotherapy alone for advanced gastric and gastroesophageal junction (GEJ) cancer with a programmed cell death ligand 1 (PD-L1) combined positive score (CPS) >5, while survival benefit was poor in PD-L1 CPS <1 and <5. However, there were few studies on correlation between PD-L1 CPS and efficacy of first-line Nivo-CT for advanced gastric or GEJ cancer in real world data. **Methods:** This multi-center retrospective study included patients with histologically confirmed advanced gastric or GEJ cancer who were started on first-line Nivo-CT between December 2021 and December 2023. PD-L1 CPS was assessed before the start of Nivo-CT by classified into three groups at each center: <1, 1~5 and >5, using the Dako PD-L1 immunohistochemistry 28-8 pharmDx assay. The efficacy was evaluated by overall survival (OS), progression-free survival (PFS) and objective response rate (ORR). **Results:** A total of 271 patients were included (median age, 70 [range 18–85] years; male/female, 64/36%; performance status (PS) 0/1/2, 47/49/4%; primary tumor site gastric/GEJ, 90/10%; status unresectable/recurrent, 81/19%; histology intestinal/diffuse, 30/70%). PD-L1 CPS status were evaluated 234 (86%) patients and PD-L1 CPS <1, 1~5 and >5 was found in 35 (15%), 76 (33%) and 123 (53%) patients, respectively. Median follow-up time was 11.3 (range 1-30) months. The median PFS was 9.7 (95%CI, 5.9-16.3), 8.8 (95%CI, 7.2-13.0) and 6.3 (95%CI, 5.3-9.0) months in the patient with PD-L1 CPS <1, 1~5 and >5, respectively. The median OS was 18.7 (95%CI, 17.4-NA), 23.4 (95%CI, 17.7-NA) and 18.3 (95%CI, 13.3-22.5) months, and the ORR were 30.8, 66.7 and 61.3% (p = 0.075) in the patient with PD-L1 CPS <1, 1~5 and >5, respectively. Liver metastases were found in 43 (35.0%) of the patients with PD-L1 CPS >5, significantly more than in the patients with PD-L1 CPS <5 (p < 0.001). Multivariate analysis indicated that PD-L1 CPS status was not associated with prognosis (PD-L1 CPS ≥5 vs. <1, HR 1.18 [95%CI, 0.75-1.86], p = 0.48). In the overall population, grade ≥3 adverse events were observed in 102 (37.6%) patients and grade ≥3 immune-related adverse events were observed in 26 (9.6%) patients. **Conclusions:** There were no significant differences in prognosis and response between PD-L1 CPS in first-line Nivo-CT for advanced gastric or GEJ cancer in real world data. Research Sponsor: None.

365 **Poster Session**

Impact of tislelizumab + chemotherapy versus placebo + chemotherapy on patient-reported symptoms and disease progression by programmed death-ligand 1 expression in gastroesophageal adenocarcinoma: A post hoc analysis of the RATIONALE-305 trial. First Author: Marcia Cruz-Correa, University of Puerto Rico, School of Medicine, San Juan, PR

Background: The relationship between patient-reported outcome (PRO)-based symptom scores, recurrent PRO-based symptomatic deterioration events (RDEs), and terminal events such as progression-free survival (PFS) are rarely examined in the oncology therapeutic domain. Thus, we applied a 3-component joint model (JM) framework aiming to illuminate more clinically interpretable associations between PRO-based treatment effects, RDEs, and PFS among subgroups of patients with programmed death-ligand 1 (PD-L1) expression of ≥1%, ≥5%, and ≥10%. **Methods:** The final analytic sample included 378 patients in the tislelizumab + chemotherapy arm (T+C) vs 401 in the placebo + chemotherapy arm (P+C) for the PD-L1 ≥1% subgroup, 238 in the T+C arm vs 237 in the P+C for the PD-L1 ≥5% subgroup, and 118 in the T+C arm vs 125 in the P+C arm for the PD-L1 ≥10% subgroup. Symptom domain scores from the EORTC QLQ-C30 and QLQ-STO22 symptom were modeled. Change from baseline (CFBL) in each domain was analyzed every cycle up to cycle 6, then every other cycle thereafter (up to cycle 25). The joint model was applied to all PD-L1 subgroups and comprised three components: 1) linear mixed model (LMM) predicting CFBL symptom scores, 2) Cox proportional hazard (CPH) model for disease progression (PFS as terminal event), and 3) frailty (random effects for RDEs) CPH model for time to PRO-based RDEs. Osoba's 10-point threshold was used to define RDEs. **Results:** In the LMM, significant treatment efficacy for the T+C arm compared with the P+C arm was observed for the GHS/QoL (P=0.0080) and dietary restriction scores (P=0.0475) in the PD-L1 ≥5% subgroup. In the PD-L1 ≥1% subgroup, compared with the P+C arm, the T+C arm was associated with less worsening in pain/discomfort (P=0.0376), upper gastrointestinal symptoms (P=0.0088), and dietary restrictions (P=0.0352). In the CPH model, the average hazard ratio indicated that compared with the P+C arm, the T+C arm was associated with a reduction in risk of disease progression across all PRO domains and PD-L1 subgroups. Lastly, the PRO-based RDE frailty predictions were strongly associated with PFS risk in the PD-L1 ≥1% and ≥5% subgroups for GHS/QoL, and in the PD-L1 ≥1% subgroup for physical functioning, fatigue, dysphagia-odynophagia, pain/discomfort, and dietary restrictions, as well as for pain/discomfort and dietary restrictions in the PD-L1 ≥5% subgroup. **Conclusions:** Through the 3-component JM, PRO-based effects were detected and demonstrated strong associations between PRO-based RDEs, and investigator-assessed PFS among varying PD-L1 expression levels. This framework may be a promising alternative for analyzing and interpreting PRO data in the context of PFS. Research Sponsor: BeiGene, Ltd.

366 **Poster Session**

Impact of tislelizumab + chemotherapy versus placebo + chemotherapy on patient-reported symptoms and overall survival (OS) by programmed death-ligand 1 (PD-L1) expression in advanced or metastatic esophageal squamous cell carcinoma (ESCC): A post hoc analysis of the RATIONALE-306 trial. First Author: Harry H. Yoon, Mayo Clinic Rochester, Rochester, MN

Background: While improved survival has been previously demonstrated, the impact of immunotherapy on HRQoL in ESCC has not been well examined. Traditional PRO-based analyses in oncology trials, such as time to deterioration (TTD) and mixed models for repeated measures (MMRMs), are limited by discounting recurrent PRO events. Thus, we applied a 3-component joint model (JM) framework to define more clinically interpretable associations between patient-reported symptoms, treatment effects, and OS among subgroups of patients with ESCC from RATIONALE-306, which met its primary endpoint, with PD-L1 expression of ≥1%, ≥5%, and ≥10%. **Methods:** The final analytic sample included 226 patients in the tislelizumab + chemotherapy arm (T+C), 242 in the placebo + chemotherapy arm (P+C) for PD-L1 ≥1%, 113 in the T+C arm and 103 in the P+C arm for PD-L1 ≥5%, and 168 in the T+C arm and 178 in the P+C arm for PD-L1 ≥10%.From EORTC QLQ-C30 and OES18, 7 keysymptom domains were modeled (GHS, physical functioning, fatigue, dysphagia, pain, reflux, dietary restrictions). PRO data were collected at baseline and at every treatment cycle (up to 6 cycles), then every other cycle, and at safety follow-up, and change from baseline (CFBL) was analyzed. The joint model comprised three components: 1) linear mixed model predicting CFBL symptom scores; 2) Cox proportional hazard model (CPH) for time to OS; and 3) frailty (random effects for recurrent deterioration events [RDEs]) CPH model for time to PRO-based RDEs. Osoba (1998) 10-point threshold was used to define RDEs. **Results:** Adjusted completion rates were >90% in the ITT population for both arms. Significant T+C treatment effects were observed for physical functioning in PD-L1 ≥5% (P=0.0476), and pain in PD-L1 ≥1% (P=0.0028) and PD-L1 ≥5% (P=0.0149) subgroups, but not in PD-L1 ≥10%. For other 5 symptoms (ie, GHS, fatigue, reflux, dysphagia, dietary restrictions), there were no statistically significant differences between treatment arms. However, T+C was associated with significant reductions in the risk of death across all 7 key symptoms and PD-L1 subgroups. As one example, with respect to interaction between pain and OS, T+C was associated with a 22% (HR, 0.78 [95% CI, 0.652-0.931]), 33% (HR, 0.67 [95% CI, 0.515-0.860]), and 47% (HR, 0.50 [95% CI, 0.344-0.720]) reduction in the risk of death in PD-L1 ≥1%, ≥5%, and ≥10%, respectively compared with P+C. **Conclusions:** In this analysis, the addition of tislelizumab to chemotherapy was associated with significantly less deterioration in multiple PRO symptoms including physical functioning and pain, alongside a lower risk of death, after adjusting for recurring deterioration events using a frailty model across multiple PD-L1 expression subgroups. Research Sponsor: BeiGene, Ltd.

367 **Poster Session**

Anlotinib combined with penpulimab and nab-paclitaxel as first-line treatment for advanced esophageal squamous cell carcinoma (ESCC): A single-arm, open-label phase II clinical trial. First Author: Zhiwei Chang, Department of Medical Oncology, The First Affiliated Hospital, Zhengzhou University, Zhengzhou, HeNan, China

Background: Recently, PD-1 blockades combined with dual chemotherapy regimens in first-line setting exhibited encouraging efficacy for patients with ESCC. However, the safety profile of conventional dual chemotherapy remained unsatisfactory. Therefore, PD-1 blockades combined with anti-angiogenic tyrosine kinase inhibitors (TKIs) and single chemotherapy regimen might offer a promising strategy. Anlotinib, a novel multitarget TKI primarily targeting VEGFR1-3, demonstrated promising therapeutic activity as first-line combination therapy or second-line monotherapy for ESCC patients in China clinically. Therefore, this study was designed to explore the efficacy and safety of anlotinib combined with penpulimab (PD-1 blockade) and nab-paclitaxel as first-line therapy in advanced ESCC. **Methods:** Patients with previously untreated metastatic or locally advanced ESCC were recruited and treated with anlotinib (12mg, po, d1~14, q3w) and penpulimab (200mg, iv, d1, q3w) plus nab-paclitaxel (220mg/m^2, iv, d1, q3w) until disease progression or unacceptable toxicity. The tumor response was assessed according to RECIST 1.1 using CT scans every two cycles. Adverse events were recorded by severity in accordance with the NCI CTC AE Version 5.0. The predefined sample size was 30. Primary endpoint was PFS and secondary endpoints included safety, ORR, DCR and OS. **Results:** From Jul 2022 to Sep 2024, a total of 30 patients were enrolled, with 29 patients who had received first tumor response were included in this analysis. The best overall response indicated that there were 1 CR (3.4%), 23 PR (79.3%), 2 SD (6.9%) and 3 NE (10.3%). Therefore, the preliminary ORR was 82.8% (95%CI: 64.2%-94.2%), DCR was 89.7% (95%CI: 72.6%-97.8%). The primary median PFS of the 29 patients was 10.84 months (95%CI: 7.57-14.11). Additionally, safety profile exhibited that the regimen was tolerable. The most common treatment-emergent adverse events among the 29 patients with the incidence >20% were anemia (62%), leukopenia (31%), peripheral neurotoxicity (28%) and rash (21%). Common grade ≥3 treatment-emergent adverse events were leukopenia (7%), glutamic-pyruvic transaminase was elevated (3%), glutamic oxalacetic transaminase increased (3%), hypertension (3%), diarrhea (3%) and rash (3%). **Conclusions:** The combination of anlotinib plus penpulimab and nab-paclitaxel as first-line therapy for advanced ESCC demonstrated promising efficacy and manageable safety profile. And the conclusions needed to be confirmed in subsequent trials. Clinical trial information: ChiCTR2400089133. Research Sponsor: None.

368 Poster Session

Safety and efficacy of perioperative chemotherapy combined with PD-1 inhibitor versus chemotherapy with D2 gastrectomy plus PAND in gastric cancer with PALD metastasis: A single-center retrospective cohort study.

First Author: Yian Du, Department of Gastric Surgery, Zhejiang Cancer Hospital, Hangzhou, Zhejiang, China

Background: Gastric cancer with para-aortic lymph node (PALD) metastasis is associated with a poor prognosis. When the metastasis is limited to the PALD, preoperative systemic therapy combined with D2 gastrectomy plus para-aortic lymph node dissection (PAND) may enhance survival. The benefit of perioperative chemotherapy combined with immune checkpoint inhibitors for these patients is unknown. **Methods:** A total of 133 patients with gastric cancer and limited No.16a2/b1 lymph node metastasis underwent perioperative systemic therapy combined with D2 gastrectomy plus PAND. Patients receiving chemotherapy combined with a PD-1 inhibitor were assigned to Group A (n=62), while those receiving chemotherapy alone were assigned to Group B (n=71). Pathological response, safety, efficacy, and survival outcomes were compared between the two groups. **Results:** Adverse events occurred at comparable rates between the two groups. No treatment-related deaths occurred in either group. In Group A, all patients (100%) achieved R0 resection, compared to 70 patients (98.6%) in Group B. Group A showed significantly better pathological outcomes than Group B, with higher rates of pathological complete response (pCR) (37.1% vs. 4.2%, P < 0.001), and major pathological response (MPR) (58.1% vs. 22.5%, P < 0.001). Group A showed a higher 2-year overall survival (OS) rate compared to Group B (90.3% vs. 62.9%, P = 0.011). The median overall survival (mOS) in Group B was 35.0 months, while follow-up data for Group A are currently insufficient to assess mOS. **Conclusions:** The addition of a PD-1 inhibitor to chemotherapy significantly improves the pathological response rate and prognosis in gastric cancer patients with PALD. Notably, the combination did not increase adverse events. These findings suggest that combining PD-1 inhibitor with chemotherapy could be a promising strategy for improving outcomes in this patient population, though further prospective studies are necessary to confirm long-term survival benefits. Research Sponsor: None.

Postoperative outcomes and pathological response.

Variable	Group A (n=62)	%	Group A (n=71)	%	P Value
Vascular tumor embolus					0.005
Negative	43	69.4%	32	45.1%	
Positive	19	30.6%	39	54.9%	
Nerve infiltration					0.004
Negative	44	71.0%	33	46.5%	
Positive	18	29.0%	38	53.5%	
No.16a2/b1					0.057
Negative	53	85.5%	51	71.8%	
Positive	9	14.5%	20	28.2%	
Tumor resection					0.970
Distal	26	41.9%	30	42.3%	
Total	36	58.1%	41	57.7%	
Serum CEA (U/mL)					0.805
Normal	56	90.3%	65	91.5%	
>5	6	9.7%	6	8.5%	
Serum CA125 (U/mL)					0.280
Normal	30	48.4%	41	57.7%	
>35	32	51.6%	30	42.3%	
Serum CA199 (U/mL)					0.783
Normal	56	90.3%	65	91.5%	
>37	6	9.7%	6	8.5%	
pT-stage					<0.001
≤T2	37	59.7%	19	26.8%	
> T2	25	40.3%	52	73.2%	
pN-stage					0.001
N0/1	42	67.7%	27	38.0%	
N2/3	20	32.3%	44	62.0%	
TRG					< 0.001
0	23	37.1%	3	4.2%	
1	13	21.0%	13	18.3%	
2	22	35.5%	47	66.2%	
3	4	6.5%	8	11.3%	
pCR	23	37.1%	3	4.2%	< 0.001
MPR	36	58.1%	16	22.5%	< 0.001

369 Poster Session

Epidemiological characteristics of gastric and pancreatic cancers in Latin America: GASPAR (LACOG 0222) study.

First Author: Renata D'Alpino Peixoto, Centro Paulista de Oncologia (CPO) - Oncoclínicas & Co, São Paulo, Brazil

Background: Gastric (GC) and pancreatic cancer (PC) have poor prognosis representing major public health challenges, particularly in Latin America. Real-world data are limited. GASPAR (LACOG 0222) study addresses this gap by creating a multicenter database to analyze epidemiological, clinical, and pathological data, treatments, and outcomes for patients (pts) with GC and PC. **Methods:** This retrospective and prospective observational cohort study enrolled pts from 7 research sites in 3 LATAM countries with metastatic/locally advanced not amenable to curative intent GC (Cohort A) and PC (Cohort B) diagnosed from January 2019. Data were collected from medical records. (NCT05924789). **Results:** 201 pts were included: 122 with GC (Cohort A) and 79 with PC (Cohort B). 150 (75%) from Brazil, 29 (14%) from Argentina, and 22 (11%) from Colombia. Median age at diagnosis was 69 years (IQR 57-78), 70 (35%) had history of tobacco use, and 153 (76%) were covered by private healthcare (Table). In Cohort A (GC), HER2 status was assessed in 105 (86%) pts, with 12 (11%) HER2-positive; PD-L1 was tested in 87 (71%): 55 (63%) had ≥1% expression and 28% had ≥10%; all 88 (72%) patients who had MMR tested had MMR proficient. First-line treatment was given to 107 (88%) pts: 78 (73%) chemotherapy (CT) alone (FOLFOX in 43 [40%]; FLOT in 16 [15%]); 23 (21%) CT + immunotherapy; and 7 (7%) CT + anti-HER2. Only 59 (48%) pts received second-line treatment and 31 (25%) third-line. With median follow-up (mFUP) of 29 months (CI95% 18.2-50.4), the median progression-free survival (mPFS) at first-line was 5.6 months (CI95% 4.2-6.3) and the median overall survival (mOS) was 11.2 months (CI95% 9.2-14.1). In Cohort B (PC), 67 (85%) pts received first-line treatment: 41 (61%) FOLFIRINOX and 14 (21%) gemcitabine-nab-paclitaxel. 27 (34%) pts received second-line treatment and 10 (13%) third-line. With mFUP of 21 months (CI95% 15.6-NR), the mPFS at first-line was 5.4 months (CI95%3.9-6.4) and the mOS was 10.2 months (CI95% 8.6-13.1). **Conclusions:** The GASPAR study provides valuable real-world data on GC and PC in Latin America. Notably, the mPFS in both cohorts was slightly lower than historical controls for first-line treatments, while the mOS was similar, suggesting earlier disease progression despite comparable overall survival. Improved therapeutic strategies and healthcare access are needed to enhance outcomes for these populations. Clinical trial information: NCT05924789. Research Sponsor: Funding: Astellas Pharma Inc. Sponsor: Latin American Cooperative Oncology Group (LACOG).

Clinical characteristics.

Characteristic	GC (Cohort A) N=122	PC (Cohort B) N=79
Median age (IQR), *yr*	66 (51–78)	71.0 (61–79)
Male – n. (%)	58 (56)	40 (51)
Race – n. (%)		
White	67 (55)	71 (90)
American Indian or Alaska Native	15 (12)	6 (8)
Black	7 (6)	0
Other	7 (6)	0
Not reported	26 (21)	2 (2)
Disease status at study entry		
Locally advanced disease	4 (3)	14 (18)
Metastatic disease	118 (97)	65 (82)
ECOG status in first-line advanced disease – n. (%)		
0-1	75 (70)	45 (67)
≥ 2	13 (12)	3 (4.5)
Not reported	19 (17.8)	19 (28.4)

370 Poster Session

Patterns of immune checkpoint inhibitor utilization and PD-L1 testing in advanced gastroesophageal cancers using real-world data.

First Author: Jasmeet Kaur, Fox Chase Cancer Center, Temple University Hospital, Philadelphia, PA

Background: Globally, gastroesophageal cancer (GEC) is a leading cause of morbidity and mortality, with poor 5-year overall survival (OS). The addition of immune checkpoint inhibitors (ICIs) such as nivolumab and pembrolizumab to chemotherapy has resulted in significant improvement in OS of those patients (pts) with metastatic GEC (mGEC), most notably in those with tumors exhibiting high PD-L1 CPS scores. However, the pattern of ICI utilization in associated PD-L1 testing since the adoption of this treatment (tx) has yet to be elucidated. We evaluated the patterns of ICI utilization and PD-L1/CPS testing in the first-line tx of mGEC pre and post FDA approval of ICI use in this setting. **Methods:** Pts ≥ 18 years of age, with recurrent/metastatic squamous or adenocarcinoma GEC diagnosed after 2011 were included from the de-identified nationwide Flatiron Health, electronic health record-derived database. We presented the proportion of pts receiving ICI therapy, stratified by line of tx, as well as the proportion of patients who received PD-L1 testing at 6-month intervals. Pts demographics and clinical characteristics, including age, gender, race, ECOG score, documented insurance, and stage at diagnosis (dx) were described and compared using T or Wilcoxon rank-sum tests for continuous variables, and chi-square or Fisher's exact tests for categorical data. We compared tx rates and PDL-1 testing across four time points: time 1: Jan-June 2020, prior to ASCO data presentation; time 2: July-Dec 2020, ASCO data presentation; time 3: Jan- June 2021, FDA label change; time 4: July 2021 to 6 m after FDA label change using Chi-square test. **Results:** A total of 9,573 pts were included with 1,593 (16.6%) receiving ICI-based tx. Among these, 62.3% were White, 7% Black. Descriptive statistics indicated that pts with ICI- tx tend to be younger (median age 65y vs. 66.7; (p <0.001). Within the total cohort, 74% pt were dx before FDA approval of ICI tx, while 26% pts were dx after FDA approval. PD-L1 testing was conducted in 4,065 (42.5%) pts. A total of 1,268 (13%) pts received first-line ICI-based tx. Of these, 154 pts (12.1%) were treated before FDA approval, while 1,114 pts (87.9%) received tx following FDA approval. The rate of first-line ICI tx significantly increased at each subsequent time point Time 2: 11.2% (p <0.0001), Time 3: 32.9% (p <0.0001), Time 4: 37.4% (p <0.0001). PD-L1 testing rates also increased over time, particularly after FDA label change. Testing rates at Time 1 was 68%. Comparatively, testing rates at subsequent time points were: Time 2 (67.2%; p 0.88), Time 3 (71.5%; p 0.324) and Time 4 (77.4%; p=0.004). **Conclusions:** The approval of ICIs for first-line tx of mGEC led to a significant increase in both ICI utilization and PD-L1/CPS testing rates. Despite these advancements, ongoing disparities in tx access and testing highlight the need for further research and efforts to ensure equitable care. Research Sponsor: None.

371 Poster Session

Comparison of long-term outcomes for neoadjuvant chemoradiation (NA-CRT) followed by surgery versus (V) definitive chemoradiation (D-CRT) in localized gastroesophageal cancer (GEC): A single-center analysis.

First Author: Ronan Andrew McLaughlin, Division of Medical Oncology and Hematology, Princess Margaret Cancer Centre – University Health Network, University of Toronto, Toronto, ON, Canada

Background: The therapeutic approach in locally advanced GEC is evolving. NA-CRT followed by surgery has been recommended based on the *CROSS* trial. The *SANO* trial reports active surveillance may be an alternative in clinically complete response after D-CRT in both squamous cell carcinoma (SCC) and adenocarcinoma (AC). SCC and AC in GEC are different diseases with differences in tumor location, etiology, and biology. Despite the evolving landscape, there still may be a role for both *CROSS* and *SANO* approaches. In our large observational cohort study, we sought to compare the long-term outcomes in localized GEC across all histological subtypes treated with NA-CRT+Surgery V D-CR+Surveillance. **Methods:** A retrospective analysis comparing NA-CRT + surgery V D-CRT in localized GEC between 2007-2023 treated at the Princess Margaret Cancer Centre was completed. Baseline characteristics including age, gender, race, performance status and Charlson Comorbidity index (CCI) were reviewed. The primary and secondary endpoints were overall survival (OS) and disease-free survival (DFS), respectively and to compare these outcomes in both histologies. Cox proportional hazards analysis and Kaplan-Meier methods were employed. Outcomes were adjusted for baseline characteristics. **Results:** There were 529 patients (pts) included, 321 (61%) received NA-CRT+Surgery and 208 (39%) D-CR+Surveillance. Median age was 64yrs, 75% were male and 66% had N1+ disease. The proportion of pts with SCC treated with NA-CRT+Surgery and D-CR+Surveillance was 25% and 63%, respectively. There was a statistically significant difference in median OS and DFS between pts who received NA-CRT+surgery and D-CRT+surveillance (Table). Multivariate regression analysis shows a significant association between OS and age (adjusted HR 1.02, p=0.005), male sex (HR 1.55, p=0.003) but no significant association between OS and CCI (HR 1.1 p=0.36). In our subgroup analyses, there was no statistically significant difference in OS (p=0.33) and DFS (p=0.18) between SCC and AC. However, this was significant in D-CRT (OS p<0.0001, DFS p<0.001) favoring SCC. **Conclusions:** In contrast to the *SANO* trial, we found that NA-CRT+surgery was associated with a significantly better OS and DFS compared to D-CRT when subtypes were combined. However, in the D-CRT subgroup, SCC was associated with improved OS and DFS compared to AC. Thus, we continue to recommend current approaches in SCC. Awaited updates from *SANO* will inform whether active surveillance may also be appropriate in pts with AC subtype. Research Sponsor: None.

	Median (95% CI) NA-CRT Surgery	Median (95% CI) D-CRT	Log-rank test p-value
OS (mo)	28.2 (15.1, 57.6)	15.6 (7.7, 33.7)	P=<0.0001
DFS (mo)	17.9 (9.6, 40.8)	9.4 (5.0, 20.7)	P=<0.0001
pathCR (%)	14	-	
R0 resection (%)	89	-	

373 Poster Session

Efficacy and safety of intraperitoneal paclitaxel based regimens in patients with gastric cancer and peritoneal metastasis: A systematic review and meta-analysis. First Author: Muhammad Bakhtiar, King Edward Medical University/ Mayo Hospital, Lahore, Lahore, Pakistan

Background: Peritoneal metastasis (PM) occurs in about 5% - 20% of patients with gastric cancer (GC) at the time of diagnosis, and prognosis remains poor with limited treatment options. There are conflicting results on the efficacy of intra-peritoneal (IP) paclitaxel in patients with gastric cancer with PM. We performed a meta-analysis to examine the current evidence on the effectiveness of IP paclitaxel-based regimens in this patient population. **Methods:** A comprehensive search was conducted across Pubmed, EMBASE, Cochrane library, and Clinicaltrials.gov, resulting in the initial identification of 253 articles. Following screening, 35 articles, comprising only clinical trials (CT) and Randomized Controlled Trials (RCTs), were selected for detailed analysis. Screening and data extraction were conducted by two reviewers concurrently, with quality assessment performed using RoB (Risk of Bias) 2.0 tool and NIH quality assessment questionnaire. Proportional meta-analysis was done to estimate the pooled efficacy of IP paclitaxel based on the Median Survival Time (MST) and 1-year Overall Survival (OS), and tumor response was assessed using RECIST criteria. Subgroup analysis was conducted for patients who underwent conversion gastrectomy. Safety was assessed through the analysis of adverse events. **Results:** A total of 855 patients were included in the evaluation of MST, revealing an overall pooled MST of 20.21 months for GC patients with PM treated with IP paclitaxel-based regimens. Additionally, 1-year OS data was available for 785 patients, revealing an overall cumulative 1-year OS of 71% (95% CI: 66-76). Furthermore, among the 332 patients who underwent conversion gastrectomy, the pooled overall MST was 27 months. For 168 participants who underwent conversion gastrectomy after receiving therapy with IP PAC-based regimens, the 1-year OS was 84% (95% CI: 77-89). The most common adverse events (AE) observed were alopecia (75%) and anemia (60%). Overall, most studies exhibited low RoB, with only a few demonstrating poor quality. **Conclusions:** IP paclitaxel-based regimens demonstrate both safety and encouraging efficacy in the treatment of GC patients with PM. Subsequent conversion gastrectomy in eligible patients can further improve the survival outcomes. Research Sponsor: None.

374 Poster Session

The incidence and risk of cardiovascular events among gastroesophageal cancer patients receiving immune-checkpoint inhibitors. First Author: Furkan Bahar, Mount Auburn Hospital/Harvard Medical School, Cambridge, MA

Background: Immune checkpoint inhibitors (ICIs) have been associated with an increased risk of cardiovascular events. However, these data were mostly derived from patients with lung cancer or melanoma. To date, there is limited data describing the incidence and risk of cardiovascular adverse events associated with the use of ICIs among patients with gastroesophageal cancers. **Methods:** We performed a propensity score-matched cohort study using the TriNetX Analytics Network database, which comprises de-identified data from over 120 participating institutions and 101 million individuals. We defined the ICI cohort as patients who received any of the ICIs and chemotherapy. We defined the control cohort as patients who received only chemotherapy but not an ICI. We matched 2 cohorts based on predetermined variables including age, race, gastroesophageal cancer-directed therapy, cardiovascular agents, underlying comorbidities, and surgical history. The outcomes were myocarditis, pericarditis, myocardial infarction, heart failure, cerebral ischemia/ infarction, atrial fibrillation, conduction disorders, and venous thromboembolism (VTE). All outcomes were captured within 1-year of the start of ICI therapy and identified using the International Classification of Diseases (ICD)-10 codes. **Results:** We identified 2005 patients who received ICI and chemotherapy (ICI cohort) and 14048 patients who received chemotherapy only (control cohort). We matched 952 patients in 2 cohorts. The mean age was 62.8 ± 12.7 and 62.8 ± 12.3 for the ICI and chemotherapy cohorts, respectively. We observed a significant increase in cardiovascular events in the ICI cohort compared to the chemotherapy cohort. In particular, the risk of myocarditis (4 cases in the ICI cohort and no cases in the chemotherapy cohort), pericarditis (Hazard ratio (HR), 2.53 [95% CI: 1.52-4.19]), conduction disorders (HR, 1.64 [95% CI: 1.04-2.59]), and VTE (HR, 1.56 [95% CI: 1.17-2.10]) were higher in the ICI cohort than the chemotherapy cohort. **Conclusions:** The use of ICI was associated with an increased risk of cardiovascular events, particularly myocarditis, pericarditis, conduction disorders, and VTE among patients with gastroesophageal cancer. Further studies are needed to identify strategies to stratify patients at high risk of developing ICI-associated cardiovascular events. Research Sponsor: None.

	ICI + Chemotherapy (ICI cohort)		Chemotherapy only (Control cohort)		Hazard ratio
Outcomes	Cases	At risk patients	Cases	At risk patients	(95% CI)
Myocarditis	4	952	0	952	-
Pericarditis	52	952	21	952	2.53 (1.52-4.19)
Myocardial infarction	18	952	12	952	1.49 (0.72-3.11)
Cerebral Ischemia/Infarction	22	952	17	952	1.30 (0.69-2.44)
Atrial fibrillation	56	952	48	952	1.16 (0.79-1.71)
Conduction disorder	49	952	30	952	1.64 (1.04-2.59)
Venous thromboembolism	112	952	73	952	1.56 (1.16-2.10)
Heart Failure	118	952	114	952	1.04 (0.80-1.34)

375 Poster Session

Envafolimab in combination with lenvatinib and albumin paclitaxel for previously treated advanced GEJ adenocarcinoma: Latest analysis of a phase II, two-cohort study. First Author: Xiaoying Zhao, Department of Medical Oncology, Fudan University Shanghai Cancer Center; Department of Oncology, Shanghai Medical College, Fudan University, Shangai, China

Background: The combination of first-line chemotherapy and PD-1 has emerged as a standard treatment, enhancing the efficacy in advanced gastric/gastroesophageal junction (GEJ) cancer. The efficacy of the combination of second-line chemotherapy and PD-1/PD-L1 remains unclear. This study aims to observe the efficacy of Envafolimab (PD-L1 inhibitor) plus tyrosine kinase inhibitors (TKIs) and second-line chemotherapy in the treatment of patients (pts) with advanced GEJ adenocarcinoma. **Methods:** Eligible pts with HER2-negative, microsatellite stable (MSS), advanced GEJ adenocarcinoma were enrolled in this study. Pts who never received PD-1 or PD-L1 inhibitors before would assign into Group A. Pts in Group B have failed first-line PD-1 antibody combined chemotherapy, with optimal efficacy as or above stable disease (SD). Pts in both groups received Envafolimab (200mg, hypodermic injection, days 1, 15, Q4W) combined with Lenvatinib (8mg/12mg, po, once a day) and albumin paclitaxel (100mg/m2, IV, days 1, 8 and 15, Q4W, up to 6 cycles) until disease progression, unacceptable toxicity, or pts refusal to continue treatment. The primary endpoint was objective response rate(ORR) per iRECISTv1.1, and the secondary endpoints included disease control rate (DCR), progression free survival (PFS), overall survival (OS) and safety. **Results:** From Oct 2022 to Apr 2024, 32 pts were enrolled. Both groups enrolled 16 pts equally, and 15 pts of each group were evaluable for efficacy analysis. The average ages were 60.07 ± 11.73 y and 50.13 ± 12.18 y in group A and B respectively. In group A, 7 pts had been tested for PD-L1 status, of which 2 (13.3%) was positive and 5 (33.3%) were negative. PD-L1 status was negative in 5(33.3%) pts, positive in 9 (60.0%), and untested in 1(6.7%) in group B. As of Sep 14, 2024, The ORR was 60.0% (9/15) in group A and 46.7% (7/15) in group B. The DCR was 100.0% (15/15) in both groups. After a median follow-up of 17.0 mo (IQR: 8.0-18.8) in group A, the mPFS was 8.2 mo (95% CI = 6.1-10.4), and the mOS was 14.8 mo(95% CI = 7.4-22.2). With median follow-up of 9.0 mo (IQR: 6.1-16.2) in group B, the mPFS was 5.9 mo (95% CI= 3.8-8.1) and the mOS was 11.5 mo (95% CI = 3.1-19.9). Overall incidences of adverse events (AEs) of any grade was 100% (30/30), and 80.0% (24/30) pts had ≥ grade 3 AEs during treatment. Most treatment-related adverse events (TRAEs) of grade ≥3 included decreased neutrophil count (73.3%), decreased leukocyte count (63.33%), Anemia (10%), febrile neutropenia (6.67%). **Conclusions:** In pts of advanced gastric cancer who have previously received PD-1 inhibitors or not, this second-line combination therapy demonstrates promising preliminary effects. Pts who have not previously received ICIs might benefit more from this combination therapy. Clinical trial information: NCT06030934. Research Sponsor: None.

376 Poster Session

Q-TWiST analysis of pembrolizumab or placebo plus chemotherapy for patients with advanced HER2-negative gastric or gastroesophageal junction (G/GEJ) cancer in KEYNOTE-859. First Author: Lucjan Wyrwicz, Maria Sklodowska-Curie National Research Institute of Oncology, Warsaw, Poland

Background: In the phase 3 KEYNOTE-859 study (NCT03675737; N = 1579), patients (pts) with locally advanced or metastatic HER2-negative G/GEJ cancer treated with first-line pembrolizumab (pembro) plus chemotherapy (chemo) had a significant and clinically meaningful improvement in OS with manageable toxicity vs placebo (pbo) plus chemo while maintaining HRQoL in the intention-to-treat and prespecified PD-L1 combined positive score (CPS) populations. We report data from an analysis of quality-adjusted time without symptoms of disease progression or toxicity (Q-TWiST). **Methods:** Q-TWiST is a measure of survival weighted by health states utility values. The analysis categorized survival time into 3 health states: time with grade ≥3 AEs (toxicity [TOX]) and time without symptoms or toxicities (TWiST)—both before disease progression or death—and relapse (REL), defined as time from disease progression to death. Q-TWiST was calculated as the restricted mean survival time (RMST) spent in each state, with a weighted health state utility for that state. Average utility weights were derived based on pooled EQ-5D-5L scores using the US mapping algorithm and standardized base case utility weights. Relative gains in Q-TWiST of ≥10% and ≥15% were "clinically important" and "clearly clinically important," respectively, as reported in the literature. Treatment difference 95% CIs were generated using the nonparametric bootstrapping method. Data are reported for all randomly assigned pts and pts with PD-L1 CPS ≥1 and CPS ≥10 tumors. The data cutoff date was August 22, 2023. **Results:** At the maximum follow-up of 56 mo for pembro plus chemo vs pbo plus chemo, RMST was 5.16 vs 2.86 mo for TOX, 7.77 vs 5.87 mo for TWiST, and 6.85 vs 7.12 mo for REL. Between-treatment differences are shown in the table. The difference in restricted mean Q-TWiST and relative gain favored pembro plus chemo. **Conclusions:** A clearly clinically important Q-TWiST gain was observed for pembro plus chemo vs pbo plus chemo in all pts with advanced HER2-negative G/GEJ cancer and pts with PD-L1 CPS ≥1 and CPS ≥10 tumors. Clinical trial information: NCT03675737. Research Sponsor: Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA.

	All pts N = 1579	PD-L1 CPS ≥1 n = 1235	PD-L1 CPS ≥10 n = 553
RMST in TOX, mo (95% CI)	2.30 (1.28 to 3.44)	2.65 (1.46 to 3.99)	4.28 (2.08 to 6.41)
RMST in TWiST, mo (95% CI)	1.90 (−0.02 to 3.79)	2.28 (0.07 to 4.49)	3.72 (0.43 to 7.19)
RMST in REL, mo (95% CI)	−0.28 (−2.43 to 2.01)	−0.22 (−2.88 to 2.39)	−0.45 (−4.52 to 3.71)
Restricted mean Q-TWiST, mo (95% CI)[a]	3.31 (2.10 to 4.59)	3.98 (2.60 to 5.43)	6.45 (4.26 to 8.74)
Relative Q-TWiST gain, % (95% CI)[a]	20.90 (12.49 to 30.56)	25.34 (16.04 to 36.26)	38.05 (23.21 to 56.59)
Restricted mean Q-TWiST, mo (95% CI)[b]	2.91 (1.58 to 4.14)	3.50 (2.02 to 4.88)	5.63 (3.31 to 7.99)
Relative Q-TWiST gain, % (95% CI)[b]	18.38 (9.78 to 27.46)	22.27 (12.65 to 32.52)	33.19 (18.76 to 49.97)

[a]Based on US mapping algorithm.
[b]Based on standardized base case utility weights.

377 Poster Session

FOLFIRI-ramucirumab (FOLFIRI-Ram) versus ramucirumab-paclitaxel (Ram-Pac) in the second line (2L) for patients with advanced upper gastrointestinal (UGI) cancer: A real-world propensity-score matched analysis of survival. First Author: Timothy J. Brown, UT Southwestern Medical Center, Dallas, TX

Background: The approach to 2L treatment of advanced UGI (gastric, esophageal, and gastroesophageal junction) cancers is guided primarily by the results of the RAINBOW trial, a global randomized phase III trial that demonstrated the benefit of Ram-Pac over paclitaxel. However, the onset of peripheral neuropathy limits its use in the palliative-intent setting. The RAMIRIS trial demonstrated the feasibility of 2L FOLFIRI-Ram compared to Ram-Pac, however it failed to meet the prespecified endpoint of $\geq$ 65% 6-month overall survival (OS). Despite this, FOLFIRI-Ram has emerged as an alternative 2L regimen and now endorsed in major society guidelines. Yet, comparative outcomes between FOLFIRI-Ram and Ram-Pac therapy in the real world are lacking. **Methods:** This study used the nationwide Flatiron Health electronic health record-derived deidentified database. During the study period (January 2011-June 2024), longitudinal patient-level data were derived from approximately 280 cancer clinics. Patients were eligible for inclusion if they had advanced unresectable or metastatic UGI cancer and received 2L treatment with either FOLFIRI-Ram or Ram-Pac. Demographics and lab values at the time of 2L initiation were extracted. Given anticipated imbalances in sample size, patients were matched 1:6 (FOLFIRI-Ram:Ram-Pac) for inclusion into the analytic dataset using a greedy match based on a logit model to predict propensity scores from key clinical and laboratory characteristics. The primary endpoint was OS from the start of 2L therapy, analyzed using the Kaplan-Meier method, log-rank test, and Cox proportional hazards model. A hybrid approach was used to construct a multivariate Cox model. **Results:** A total of 15,908 patients were included in the database, of whom 5189 received a 2L treatment. Of these, 631 patients received 2L Ram-Pac and 40 received 2L FOLFIRI-Ram. After matching, the final sample size was 240 patients who received Ram-Pac and 40 who received FOLFIRI-Ram. Baseline clinical and laboratory characteristics were well-balanced. The median OS from initiation of 2L FOLFIRI-Ram was 9.7 months (95% CI 6.9-12.3) and 7.6 months with 2L Ram-Pac (95% CI 6.3-9.2 months). The hazard ratio (HR) for death with FOLFIRI-Ram was 0.77 (95% CI 0.52-1.15, log-rank P=0.20) in reference to Ram-Pac. The adjusted HR death for FOLFIRI-Ram after adjustment for baseline albumin and white blood cell count was 0.87 (95% CI 0.58-1.30, p=0.49) in the final multivariate model. **Conclusions:** This real-world propensity-score matched analysis of patients with advanced UGI cancers receiving FOLFIRI-Ram or Ram-Pac supports the ongoing use of FOLFIRI-Ram in the 2L as an alternative to Ram-Pac. These data are consistent with results from the RAMIRIS study, however further prospective study is needed. Research Sponsor: Flatiron Health.

378 Poster Session

Association between adjuvant component of perioperative chemotherapy and survival outcomes in resected gastric cancer (GC). First Author: Aravind Sreeram, Johns Hopkins University School of Medicine, Baltimore, MD

Background: Perioperative chemotherapy is the standard of care for resectable GC but many patients do not proceed to or complete adjuvant chemotherapy (AC). It remains unclear whether AC following neoadjuvant chemotherapy (NAC) improves survival. **Methods:** Patients with GC who received NAC and underwent curative-intent gastrectomy from 2006-2022 were identified from an institutional database. Patients with mismatch repair-deficient/microsatellite instability-high tumors, GEJ tumors, M1 disease, neoadjuvant radiotherapy, total neoadjuvant intent treatment, incomplete NAC ($<$66% of planned cycles), and incomplete data were excluded. A lymph node ratio (LNR) risk cutoff was determined using maximally selected rank statistics. Kaplan-Meier analysis with log-rank tests were used to compare overall survival (OS) and disease-free survival (DFS). Cox proportional hazards regression models were used to identify independent predictors of OS and DFS. All analyses were performed in R statistical software. **Results:** Of 340 patients included, 264 (78%) received AC. Most common reasons for incomplete AC were postoperative complications (20%), poor performance status (16%), minimal pathologic response to NAC (14%), patient choice (14%), and NAC toxicity (9%). Median follow-up was 6.4 years. In the overall cohort, there was no significant difference in OS (median 10 vs. 8.3 years; p=0.14) or DFS (7.0 vs. 7.5 years; p=0.12) between AC and non-AC groups. In node-negative (ypN-) patients (n=170), there was no association between receipt of AC and OS (12 vs. 10 years; p=0.6) or DFS (12 vs. 10 years; p=0.8). On univariable analysis, lack of AC receipt was not associated with shorter OS (p=0.3) or DFS (p=0.4). In the node-positive (ypN+) cohort (n=170), AC receipt was associated with significantly longer DFS (4.8 vs. 2.7 years; p=0.031) but not OS (5.5 vs. 2.8 years; p=0.071). However, on multivariate analysis, AC only showed a non-significant trend towards longer DFS (HR 0.77 [95% CI 0.49-1.19]; p=0.20). High-risk patients based on LNR (LNR$>$0.0684; n=112) who received AC (n=87; 78%) had longer OS (p=0.033) and DFS (p=0.014). No such differences were observed in the low risk LNR subgroup (LNR$<$0.0684; n=227). **Conclusions:** Following NAC, AC was not associated with improved survival in patients with ypN- GC. The association between AC and survival in ypN+ GC remains unclear but LNR may be used to identify patients with ypN+ disease who may benefit from AC. These observations require prospective validation but may have implications for clinical care and future trial design. Research Sponsor: None.

379 Poster Session

Incidence of mediastinal lymph node positivity in patients undergoing robotic transthoracic esophagectomy. First Author: Kenneth Lee Meredith, Jel-Jellison Cancer Institute, Florida State University College of Medicine, Sarasota, FL

Background: Esophageal cancer is increasing in incidence worldwide. Robotic approaches have quickly become mainstream due to the improved visualization and possible increase in nodal harvest. There remains significant debate on the utility of mediastinal lymph node dissection in esophageal cancer patients. We sought to analyze the incidence of mediastinal lymph node positivity and to evaluate its impact on survival in patients with esophageal cancer undergoing robotic assisted trans-thoracic esophagectomy (RATE). **Methods:** Utilizing a prospectively maintained esophageal database, we identified patients who were diagnosed with EC and underwent RATE between 2009-2024. We then identified patients with positive lymph nodes and stratified by location (celiac or mediastinal). Baseline univariate comparisons were made for continuous variables using both the Mann-Whitney U and Kruskal Wallis tests. Pearson's Chi-square test was used to compare categorical variables. Survival analysis was performed via Kaplain-Meier method. **Results:** We identified 201 patients who underwent RATE with a median age of 70 (41-91). There were 160 (79.6%) males and 41 (20.4%) females. The mean BMI was 26.5 $\pm$ 5.5 and mean ASA score was 3.1 $\pm$ 0.6. Adenocarcinoma was the predominate histology at 162(80.6%). Locally advanced T3/T4 tumors were identified in 53 (26.4%) patients and 63(31.3%) had clinical node + disease. Neoadjuvant therapy was utilized in 159(79.1%) patients. The median lymph nodes resected 18 (14 – 22), and length of hospitalization 8 (7 – 11) days. The 30 and 90-day mortality were 2.2% and 2.6% respectively. R0 resections was performed in 200 (99.5%) of patients and the pathologic complete response rates were 42.1%. The incidence of mediastinal lymph node positivewas 6.3% vs 76.3% abdominal. There were 17.6% of patients that had both mediastinal and abdominal lymph nodes positive. The median survival was 23 months for abdominal LN+, 19.5 months for mediastinal LN+ p=0.23. Multivariate analysis revealed that age, neoadjuvant radiation, nodal positivity were predictors of survival. **Conclusions:** The incidence of mediastinal positive lymph nodes are much higher than previously reported. Perhaps the improved visualization and manipulation of tissue with robotic approaches have enhanced our ability to resect these nodes. Mediastinal lymph node positivity is a poor prognostic factor for survival in patients undergoing robotic esophagectomy. Given these findings, we would continue to support aggressive nodal resection in patients diagnosed with esophageal cancer. Research Sponsor: None.

380 Poster Session

The prevalence of Lynch syndrome and associated clinicopathologic features in patients with mismatch repair deficient gastric cancer. First Author: Sun Mi Lee, The University of Texas MD Anderson Cancer Center, Houston, TX

Background: Approximately 2-4% of patients with primary colorectal cancer are estimated to be associated with Lynch syndrome. However, the prevalence of Lynch syndrome and associated clinicopathologic features in patients with mismatch repair-deficient (dMMR) gastric cancer (GC) is not well understood. In this study, we investigated the frequency of dMMR GCs in unselected GCs and the prevalence of Lynch syndrome, as well as associated clinicopathologic features and molecular alterations in dMMR GCs. **Methods:** A total of 847 cases of unselected GCs from 2010 to 2012 were collected at Jeju National University Hospital. Immunohistochemistry for mismatch repair proteins (MMR), including MLH1, MSH2, MSH6, and PMS2, and MSI analysis were performed on representative tumor sections of surgically resected specimens. Follow-up genetic counseling and genetic and/or epigenetic alterations of the MMR genes were also investigated. **Results:** Loss of one or more MMR proteins and microsatellite instability-high were observed in 72 patients; 67 patients showed loss of MLH1/PMS2, 2 patients showed loss of PMS2 alone, and 3 patients with MSH2/MSH6. Genetic counseling demonstrated that ten patients had a family history of LS-associated cancers and nine patients had a history of metasynchronous cancers. Of the 72 patients, 65 revealed hypermethylation of the promoter region of MLH1. The subsequent genetic germline tests demonstrated 2 patients with MLH1 mutation, 3 with MSH2 mutation, and 2 with PMS2 mutation. **Conclusions:** dMMR tumors were identified in 72 (8.5%) of 847 unselected primary GCs. This study estimated that 0.8% of patients with unselected GC and 9.7% of patients with dMMR GC fulfilled the revised Bethesda criteria and are associated with Lynch syndrome. The detection of dMMR GCs and necessary genetic counseling/tests based on revised Bethesda criteria would help find Lynch syndrome in patients with GC. Research Sponsor: None.

Clinical and immunophenotypical features of patients with dMMR GC.

Age	
≤ 50 years	5 (6.9)
> 50 years	67 (93.1)
Other malignancy history except GAC	
No	63 (87.5)
Yes	9 (12.5)
History of metachronous cancers	
Esophageal cancer	1 (1.4)
Maxillary sinus cancer	1 (1.4)
Papillary thyroid cancer	1 (1.4)
Colorectal cancer	1 (1.4)
Pancreatic cancer	1 (1.4)
Breast cancer	1 (1.4)
Breast cancer and malignant GIST	1 (1.4)
Multiple HCCs	2 (2.8)
History of Family with LS related cancers	
No	62 (86.1)
Yes	10 (13.9)
Fulfillment of the rBG	
No	63 (87.5)
Yes	9 (12.5)
MMR proteins	
Loss of MLH1 and PMS2	67 (93.1)
Loss of PMS2	2 (2.8)
Loss of MSH2 and MSH6	3 (4.2)
Germline mutation test	
MLH1 mutations	2 (2.8)
MSH2 mutations	3 (4.2)
PMS2 mutations	2 (2.8)

381 **Poster Session**

Effect of adjuvant radiotherapy after endoscopic resection of early esophagus cancer to prevent new intraluminal mucosal recurrence. First Author: Hirofumi Fukushima, Department of Gastroenterology, Juntendo University School of Medicine, Tokyo-to, Japan

Background: Endoscopic resection (ER) is effective for treating T1a early esophageal squamous cell carcinoma (ESCC). Occasional distant metastasis and recurrence are inevitable invasion in depth of muscularis mucosa(MM). Additional surgical resection is an effective means for cure and prevent recurrence, but it imposes a heavy physical burden. Chemoradiation therapy (CRT) is a comparable alternative to surgery. Not all patients will want those additional treatment. Another problem is the occurrence of new esophagus intraluminal mucosal recurrence. After endoscopic treatment, there is an increased risk of scarring and deformity. But there are few reports of the effectiveness of additional RT in suppressing new lesions. This study was designed to verify the efficacy of not only curability but also suppressive effect the occurrence of new lesions additional treatment of T1a(MM) ESCC. **Methods:**This study was a retrospective study. We reviewed 71 consecutive patients who underwent ER followed by clinical stage I (T1a: MM) ESCC. We compared patients in the radiotherapy group or non-radiotherapy group after ER. Patients of radiotherapy group received radiation within 3 months after ER. All patients underwent regular follow-up. Recurrence, recurrence-free survival, cancer-specific survival, overall survival, and complications were evaluated. **Results:** There were 34 cases of radiotherapy group and 37 cases of non-radiotherapy group. 3 patients (8.9 %) in the radiotherapy group experienced intraluminal mucosal recurrence, and 2 patients (6.6 %) experienced local recurrence. On the other hand, in the non-radiotherapy group, 9 patients (24.3 %) experienced intraluminal mucosal recurrence, and 7 patients (18.9 %) experienced local recurrence (intraluminal mucosal recurrence: $p < 0.05$, local recurrence p=0.14). The recurrent group received chemotherapy. The 3-year cumulative recurrence-free survival was 97 % in the radiotherapy group and 83.7 % in the non-radiotherapy group ($p < 0.05$). No severe radiation toxicities were recorded. **Conclusions:** The radiotherapy after ESD reduced the risk of lymph nodes recurrence not only local lymph node recurrence but also new intraluminal mucosal recurrence, and safe and effective therapeutic strategy. Research Sponsor: None.

382 **Poster Session**

Optical coherence tomography and elastography for the visualization of architecture and stiffness differences in gastric cancer and dysplasia. First Author: Alana Gonzales, Wyant College of Optical Sciences, The University of Arizona, Tucson, AZ

Background: Individuals at high risk for gastric cancer may be recommended to undergo risk-reducing gastrectomy, which leads to life-long morbidity. Surveillance with endoscopy can be an appropriate and effective option for avoiding unnecessary surgery. Gastric cancer is commonly characterized by changes in tissue architecture and increased tissue stiffness, but visualization of these alterations is challenging for white light endoscopy methods. We utilized optical coherence tomography (OCT), a non-destructive imaging modality that uses reflected near-infrared light to generate cross-sectional images of tissue, to visualize microstructure and relative tissue stiffness. **Methods:** We utilized two modes of OCT for visualization of *ex vivo* tissue samples collected from patients undergoing gastrectomy for gastric cancer, endoscopic mucosal resection, or endoscopic biopsy. Images were collected with a benchtop OCT system (Thorlabs Telesto TEL221). Structural OCT was used to visualize changes in tissue architecture including thickened mucosa and irregular put structure of the mucosa. Optical coherence elastography (OCE) was used to extract relative measures of tissue stiffness using an optical palpation method. A phantom of known mechanical properties was placed on the tissue and an axial force was applied. The resulting deformation (strain) of the phantom provided relative measurements of tissue stiffness. Classification of normal and cancerous tissue was confirmed by histology. **Results:** Structural OCT imaged architectural features of tissue with high resolution ($<$10 um) and depth of imaging up to 1.5 mm. Strain maps generated with OCE data showed that normal gastric tissue had relatively consistent stiffness, whereas stomach cancer and dysplasia exhibited increased tissue stiffness compared to normal. **Conclusions:** Structural OCT and OCE captured changes in tissue architecture and increased stiffness in patient samples of gastric cancer and dysplasia. These tissue features are classic characteristics of cancer, making this technique a useful screening tool for several cancer types. Future work aims to implement endoscopic OCT and OCE to detect cancer at early stages *in vivo*. Research Sponsor: University of Arizona International Research Grant.

383 **Poster Session**

Young onset gastrointestinal cancer: A needs analysis. First Author: Nicola Keohane, St James's Hospital, Dublin, Ireland

Background: The global incidence of Young Onset (YO) cancers diagnosed in adults under 50 years, is increasing for unknown reasons. YO Gastrointestinal (GI) cancers, which account for approximately 27% of newly diagnosed cancer cases and 37% of cancer deaths present distinct challenges in areas including sexual health, finance, career and family. We assessed the holistic needs of these patients through a mixed methods approach. **Methods:** A single institution needs analysis study was undertaken in YO GI patients diagnosed 2014-2024 through an anonymous survey and Focus Group Discussions (FGD). The surveys asked about their experiences of sexual health and function, psychosocial concerns, financial concerns, career developments and when in their cancer journey would they prefer to speak to a health care professional about these. FGD's used nominal group technique to deepen the analysis. **Results:** 88 participants responded via survey and 10 participated in a FGD. The average age at diagnosis was 43 years. 80% of participants reported they would have benefitted from a conversation about fertility, sexual health/ function (82%) and financial supports (98%) at diagnosis. 93% of patients were out of work during cancer treatment with average length of time $>$ 1 year. **Conclusions:** This needs analysis highlights the importance of specialised clinical pathways for young onset GI cancer patients focusing on these unique and complex needs. Financial supports, conversations regarding sexual health/function, fertility preservation and psychosocial support are critical areas requiring structured intervention. Research Sponsor: None.

2014-2024 young onset SJH.

	UGI (N=233)	Colorectal (N=275)
Clinical Stage at diagnosis	Oeso Stage 4 = 39% Gastric stage 4 = 26%	Stage 3= 46% Stage 4 = 20%
Treatment intent %	Radical =54% Palliative =38% Radical/Palliative = 8%	Radical =75% Palliative 25%
Mortality rate	66%	25%
GI Survey results (N=236)	Support offered Y/N	Preferred timing of support
Sexual health/ function	No=85% Yes =15%	82% At diagnosis
Financial support	No=95% Yes= 5%	98% At diagnosis
Fertility preservation	No = 65% Yes=35%	80% At diagnosis

384 **Poster Session**

Clinical utility of nurses' telephone follow-up program (Tsukiji Call) for resectable locally advanced esophageal squamous cell carcinoma patients received neoadjuvant DCF therapy. First Author: Ai Tanaka, Department of Nursing, National Cancer Center Hospital, Tokyo, Japan

Background: The standard neoadjuvant treatment in Japan for resectable locally advanced esophageal squamous cell carcinoma (ESCC) is docetaxel (DTX) + cisplatin (CDDP) + 5-fluorouracil (5-FU) (DCF) based on the results of the JCOG1109 study. Neoadjuvant DCF therapy is administrated triweekly; in the first week in a hospital, and the remaining two weeks at hom. After the administration of neoadjuvant DCF therapy, patients might suffer from febrile neutropenia or gastrointestinal symptoms such as nausea, vomiting, constipation, diarrhea, and stomatitis even during remaining two weeks at home. Therefore, nurses in our hospital teach self-care at home for these patients who received neoadjuvant DCF therapy. However, some patients experienced emergency admission due to worsening adverse events. To prevent these events, nurses in our hospital conducted a telephone follow-up program (Tsukiji Call) for patients at home. **Methods:** This retrospective analysis included locally advanced ESCC patients who received neoadjuvant DCF therapy in our hospital between Feb 2023 and Jul 2024. Three courses of DCF (DTX [day 1]: 70 mg/m2, CDDP [day 1]: 70 mg/m2, 5-FU [continuous infusion on days 1-5]: 750 mg/m2, every three weeks) were planned as neoadjuvant treatment. On days 8-14 of the first course of neoadjuvant DCF therapy, Tsukiji Call was conducted by nurses in our hospital to check fever and nausea, vomiting, constipation, diarrhea, and stomatitis based on the CTCAE ver. 5.0 for patients at home before the first outpatient clinic. Eligible patients were divided into the Tsukiji Call and non-Tsukiji Call groups. We used Fisher's exact test to evaluate the frequency of grade 2 (Gr.2) or more adverse events (AEs) between two groups at the first outpatient clinic. **Results:** This study identified 90 patients (Tsukiji Call: 20, non-Tsukiji Call: 70). The median age (range) and gender male/female, PS 0/1/2, cStage I/II/III/IVA/IVB were 66.5 years (45-76 years) and 15 (75%)/5(25%), 19(95%)/1(5%)/0, 1(5%)/4(20%)/11(55%)/1(5%)/3(15%) in Tsukiji Call group, and were 64 years (46-79 years) and 49(70%)/21(30%), 67(95%)/2(2%)/1(1%), 1(1%)/13(18%)/50(71%)/0/6(8%) in non-Tsukiji Call group, respectively. In the first outpatient clinic, the frequency of Gr.2 or more AEs on fever, nausea, vomiting, constipation, diarrhea, stomatitis and any events between in Tsukiji Call and non-Tsukiji Call groups were 0/0(p = 1), 2(10%)/1(1%) (p = 0.123), 0/0 (p = 1), 0/0 (p = 1), 0/4(5%) (p = 0.572), and 0/10(14%) (p = 0.109), 2(10%)/15(21%) (p = 0.342), respectively. One patient in each group was emergently admitted within 7 days after discharge. **Conclusions:** Non-hematologic toxicities during DCF therapy may be reduced by Tsukiji Call by nurses in the hospital, which seemed to be in preventing the worsening of adverse events. Research Sponsor: None.

385 **Poster Session**

Precision medicine and multidisciplinary care in gastric and gastroesophageal junction (G/GEJ) cancers: Challenges and practice gaps in community cancer clinics. First Author: Matthew Strickland, Massachusetts General Hospital, Harvard Medical School, Boston, MA

Background: The advent of biomarker-driven therapeutic strategies brings increasing precision to the treatment of G/GEJ cancer, placing biomarker testing at the forefront of patient management. The management of evolving logistics for testing and related decision-making can be challenging, especially in community clinics. As part of a multicenter quality improvement initiative, we evaluated current gaps in biomarker testing and practice patterns including multidisciplinary care for G/GEJ cancers in community oncology centers. **Methods:** Between 5/2024 and 8/2024, oncology team members (N = 60) from 5 US-based community oncology practices completed surveys assessing attitudes, knowledge, competence, practice patterns, challenges, and gaps regarding biomarker testing and multidisciplinary care in G/GEJ cancers. **Results:** Despite practice guidelines, many patients with G/GEJ cancer did not receive testing for recommended biomarkers. While most providers report that biomarkers HER2 (67%) and PD-L1 (62%) are routinely reported in pathology reports, less than half said testing for MSI/MMR, TMB, and BRAF p.V600E were reported, and less than 25% reported analysis of *NTRK1/2/3*, *RET*, CLDN.18.2, and *FGFR2*. Biomarkers most frequently requested if not included in the original pathology work-up were PD-L1 (48%), TMB (32%), and BRAF p.V600E (32%). Top challenges reported by providers in integrating biomarker testing into practice were *prioritizing testing for limited tissue* (38%), *determining whether to begin treatment prior to receipt of molecular testing results* (35%), and *optimizing specimen collection for molecular testing* (27%). Only 23% of providers reported feeling very or extremely comfortable (4/5 on 5-point Likert scale) with interpreting and applying biomarker testing results to treatment decision-making. Additional challenges included turnaround time for biomarker testing, uncertainty interpreting biomarker results, and inconsistent or infrequent multidisciplinary tumor boards. Indeed, some providers (12%) reported not participating in multidisciplinary tumor boards at all, while others reported participating weekly (16%), once or twice a month (21%), every other month (12%), or less frequently (23%). Providers believed *improving test ordering process* (47%), *decreasing testing turnaround time* (42%), and *standardizing panels for biomarker testing* (40%) would be most impactful in implementation of biomarker testing into clinical workflows. **Conclusions:** Although most oncology teams reported use of some form of biomarker testing in treatment of G/GEJ cancer, testing was inconsistent and not as comprehensive as currently recommended. These data will be leveraged to develop multidisciplinary action plans with participating clinics to improve biomarker testing for treatment selection in G/GEJ cancers. Research Sponsor: Merck Sharp & Dohme LLC.

386 **Poster Session**

Development and validation of machine learning model for mortality after curative gastrectomy for gastric cancer. First Author: Ki Bum Park, St Vincents Hospital/The Catholic University of Korea, Gyeonggi-Do, South Korea

Background: The risk for postoperative complications can significantly impact perioperative planning. The aim of this study is to develop and validate a model based on various machine learning techniques to predict postoperative mortality (PM) after curative radical gastrectomy for gastric cancer. **Methods:** Data from a nationwide survey led by the Korean Gastric Cancer Association was used for the development set, which included 12,722 patients who underwent gastrectomy in 2019 across 68 institutions in South Korea. The validation set comprised 4,887 patients from 8 institutions between 2009 and 2017. PM was defined as all-cause death within 30 days after surgery or in-hospital death beyond 30 days. Five machine learning models (CatBoost, Gradient Boosting, Light GBM, Random Forest, XGBoost) were evaluated using perioperative clinical characteristics. Model performance was assessed using the area under the receiver operation characteristic curve (AUROC) with 5-fold cross-validation. **Results:** The PM rates were 0.83% and 1.04% in the development and validation set, respectively. In the development set, the Random Forest model demonstrated the highest discrimination capacity with an AUROC of 0.772, followed by Gradient Boosting at 0.767, CatBoost at 0.742, XGBoost at 0.732, and LightGBM at 0.717. However, in the validation set, CaBoost showed the highest AUROC at 0.747, while the AUROC for Random Forest decreased to 0.688. **Conclusions:** The CatBoost model demonstrated robust and consistent performance in predicting PM risk across both development and validation datasets. To further enhance the performance of the model and ensure stable clinical application, future research should focus on developing methods for quantitative measurement of perioperative factors such as surgical techniques. Research Sponsor: Korean Gastric Cancer Association; KGCA2022IC5.

387 **Poster Session**

Outcome of radiation related esophageal stricture from endoscopic dilation. First Author: Yinghong Wang, The University of Texas MD Anderson Cancer Center, Houston, TX

Background: Esophageal strictures present a significant challenge, particularly in cases of radiation-induced strictures. Esophageal dilation is a commonly used intervention aimed at improving swallowing function and nutritional status. However, radiation-induced strictures are amongst the most refractory to dilation. Understanding the success and limitations of this procedure is crucial for optimizing patient outcomes in the cancer population. **Methods:** This was a single-center retrospective chart review of 736 patients who received radiation as well as an EGD between 2014 and 2024. Patients without a radiation-induced stricture that was dilated were excluded. All continuous data are summarized as medians with interquartile ranges (IQR). Independent t-test was used for comparison among different groups. **Results:** 192 were eligible for this study. Our population was mostly male (58%) with a median age of 66 (IQR 67-74). The leading cancer diagnosis were esophageal (37%), head and neck (30.2%), and lung (17.1%). Radiation targets were head and neck (50.5%), lung (45.3%), both (1.6%), or other (1.6%). Patients were receiving active chemotherapy in 9.9% of cases. Mortality by the end of the study period was 51%. The median radiation dose was 5040 cGy (IQR 5040-6600) in a median of 28 fractions (IQR 26-33). Common indications for dilation were dysphagia (91.1%), weight loss (21.9%), and tube feed dependency (17.2%). Strictures were most often dilated 2 to 4 times (39.6%) to a median dilation size of 14 mm (IQR 12-15). Of note, 18.2% of patients needed 8 or more dilations. Dilation was associated with weight gain in 46.4% of patients, PEG tube removal in 3.1%, both forms of improvement in 9.9%, and no change in 40.6%. Complications included perforation (3.6%), aspiration pneumonia (3.6%), chest pain (0.5%), and hospitalization (0.5%). No significant differences were found in maximum dilation size, number of dilations, initial luminal diameter of the stricture, and initial length of the stricture between those who had clinical improvement as compared to those who did not. **Conclusions:** Our study reported 59.4% clinical improvement from esophageal dilation for radiation-induced strictures with nearly 20% of patients needing 8 or more dilations. The stricture dilation parameters did not appear to affect the success rate. This is the largest known single study of esophageal dilation for radiation-induced strictures. Research Sponsor: None.

388 **Poster Session**

The impact of nutritional status on recurrence-free survival (RFS) in patients with locally advanced oesophagogastric adenocarcinoma (LA-OGA) treated with FLOT therapy. First Author: Keiji Sugiyama, Gastrointestinal Unit, Department of Medicine, The Royal Marsden NHS Foundation Trust, London and Surrey, United Kingdom

Background: Malnutrition and weight loss were reported as poor prognostic factors in operable OGA. This study examines the relationship between oncological outcomes and nutritional status in LA-OGA patients under the current standard of care, perioperative FLOT therapy. **Methods:** From 2017 to 2023, 423 patients who had radical resection at Royal Marsden Hospital were screened. Inclusion criteria included LA-OGA (cT2$\leq$ and Nany or Tany and N+), at least one cycle of neoadjuvant chemotherapy (NAC), and ECOG PS 0–2. Nutritional status was assessed by body weight and prognostic nutritional index (PNI), with significant weight loss defined as a 5% loss within six months before diagnosis or during NAC. PNI was calculated as serum albumin (g/L) + 0.005 $\times$ lymphocyte count /μL, with categories of severe ($<$45), mild to moderate (45–49.9), and normal ($\geq$50). The primary endpoint was 3-year recurrence-free survival (3y-RFS). Restricted mean survival time (RMST) at 36 months was calculated to analyze survival time. Multivariate analyses were performed to identify prognostic factors for RFS. The pathological response was assessed using tumour regression grade (TRG) (Mandard criteria). **Results:** Of the 423 patients, 210 met the inclusion criteria. Primary tumour sites were the oesophagus (52.4%), the oesophagogastric junction (18.1%), and the stomach (29.5%). The median follow-up was 26.5 months. Weight loss at baseline and after NAC, as well as PNI, did not significantly impact RFS (p=0.13, p=0.91, p=0.69, p=0.45). However, a PNI decrease from baseline was associated with significantly shorter 3y-RFS (46% vs. 69%, $p<0.001$), and the RMST difference was 5.46 months ($p<0.001$). Multivariable analyses showed ypN positivity ($p<0.001$), R1 resection ($p<0.001$), and PNI decrease (p=0.03) as negative prognostic factors for RFS. Pathological response did not differ regardless of PNI change (28.2% vs. 30.4%, p=0.75). **Conclusions:** A decrease of PNI is a significant poor prognostic factor for RFS in LA-OGA patients undergoing FLOT, suggesting that maintaining nutritional status during NAC could enhance survival outcomes. Research Sponsor: None.

389 **Poster Session**

Assessment of nutritional status in elderly patients with gastroesophageal cancer: A post hoc analysis. First Author: Jasmeet Kaur, Fox Chase Cancer Center, Temple University Hospital, Philadelphia, PA

Background: Elderly patients with gastroesophageal cancer (GEC) frequently suffer from malnutrition due to tumor location and treatment-related toxicity. Malnutrition is associated with increased morbidity and mortality in this population. This study investigates the relationship between body mass index (BMI), Mini Nutritional Assessment (MNA) scores, unintentional weight loss (UWL), and chemotherapy toxicity while identifying key factors influencing nutritional status. **Methods:** This post hoc analysis utilized data from a prospective, multicenter study of patients aged 65 and older with all stages of GEC undergoing chemotherapy. Nutritional assessments were conducted using validated tools, focusing on BMI, MNA, and UWL over three months post enrollment. We defined malnourished patients as those with abnormal BMI (<18.5 or >30), low MNA (0-11), and significant UWL (≥10%). We analyzed associations between nutritional status and geriatric assessment comorbidities, depression score, functional status, cognition, hospitalization, treatment toxicity, and social support using Fisher's exact tests. **Results:** Eighty-two patients were enrolled, with a median age of 73 years (range 65-91 years). The cohort was predominantly male (74%) and primarily presented with stage III-IV disease (82%), most receiving first-line therapy (79%). Primary sites included gastric (32%), esophageal (43%), and gastroesophageal junction (26%). Notably, 34% of patients had abnormal BMI, with 2 patients classified as underweight (BMI <18.5) and 19 as obese (BMI >30). Approximately 69% reported UWL, of 10% or greater. MNA scores indicated malnutrition in about 66% of patients (severe (0-7): 15 patients; moderate (8-11): 37 patients), while 31% scored within the normal range (12-14).Hospital admissions were significantly higher in patients with low MNA (33% vs. 3%; p=0.002) and UWL (29% vs. 8%; p=0.047). No significant correlations were observed between nutritional status and treatment type, treatment toxicity, or other geriatric assessment variables, including functional status and social support. **Conclusions:** MNA and UWL are effective indicators of nutritional status in elderly GEC patients, while BMI is not a reliable measure. Malnutrition significantly correlates with increased hospital admissions, highlighting the need for reliable tools to measure malnutrition to implement necessary interventions. Further research is required to explore the broader implications of malnutrition on treatment outcomes and quality of life in this vulnerable population. Research Sponsor: None.

390 **Poster Session**

An exploratory phase II trial of low-dose decitabine and sintilimab combined with chemotherapy in HER2-negative gastric or gastroesophageal junction adenocarcinoma. First Author: Yunyi Du, Department of Oncology, Changzhi People's Hospital Affiliated to Changzhi Medical College, Changzhi, Shanxi, China

Background: Combining epigenetic modulator with PD-1 inhibitor may enhance treatment outcomes in cancer patients. This study aimed to evaluate the efficacy and safety of epigenetic drug decitabine, sintilimab (a PD-1 inhibitor), and chemotherapy in gastric cancer. **Methods:** In this single-arm, exploratory phase II trial, patients with HER2-negative, locally advanced or metastatic gastric or gastroesophageal junction (G/GEJ) adenocarcinoma were enrolled between August 12, 2021, and October 22, 2022. Treatment included low-dose decitabine (10 mg/m² for 5 days), sintilimab (a PD1 inhibitor, 200 mg every 21 days), and XELOX. The primary endpoint was objective response rate (ORR) and second endpoints included disease control rate (DCR), progression-free survival (PFS), overall survival (OS), and safety. The protocol was approved by the ethics committee of the Changzhi Medical College Affiliated Changzhi People's Hospital (No. 2021K002). All patients provided written informed consent form before study participation. The trial is registered with Chinese Clinical Trial, number ChiCTR2100043312. **Results:** Sixteen patients were initially enrolled, with a median follow-up of 14.8 months. Eleven patients exhibited a PD-L1 combined positive score (CPS) < 5. In the per-protocol sets, 11 patients remained eligible for efficacy analysis. In the intention-to-treat analysis of all 16 patients, 9 achieved partial response and 2 had stable disease, resulting in an ORR of 69% and a DCR of 56%. As of the data cutoff on November 4, 2023, the median PFS was 11.1 months (95% CI, 5.1-not reached) and the median OS was 13.2 months (95% CI, 5.1-not reached). Treatment-related adverse events were generally well-tolerated, with most being grade 1-2. Three patients experienced severe AEs, including one fatal case of grade 4 immune-related pneumonia. **Conclusions:** The combination of low-dose decitabine, sintilimab, and chemotherapy appears promising strategy for locally advanced or metastatic G/GEJ adenocarcinoma, especially in patients with negative HER2 and low PD-L1 expression. These preliminary findings support further investigations to confirm the benefits and explore the underlying mechanisms. Clinical trial information: ChiCTR2100043312. Research Sponsor: Shanxi Provincial Health and Wellness Committee Research Project; 2020129.

Objective response and disease response.

	Full analysis set (n=16)	Efficacy analysis set (n=11)
Complete response	0	0
Partial response	9	9
Stable disease	2	2
Progressive disease	0	0
Non-evaluable*	5	NA
Objective response	9 (56%)	9 (81.8%)
Disease control	11 (69%)	11 (100%)

NA=not applicable.
*Included patients without any post-baseline response assessment.

391 **Poster Session**

Is treatment delivery of perioperative FLOT therapy in locally advanced OGA associated with recurrence-free survival? First Author: Keiji Sugiyama, Gastrointestinal Unit, Department of Medicine, The Royal Marsden NHS Foundation Trust, London and Surrey, United Kingdom

Background: Although perioperative FLOT therapy is a standard for locally advanced oesophagogastric adenocarcinoma (LA-OGA), completing all cycles, especially in the adjuvant chemotherapy (ACT) phase, is often challenging. This study examined the survival impact of treatment delivery in LA-OGA patients receiving FLOT therapy. **Methods:** 423 patients who underwent radical resection at The Royal Marsden Hospital from 2017 to 2023 were screened. Major inclusion criteria included patients with LA-OGA (cT2≤ and Nany or Tany and N+), treated with FLOT (at least one cycle of neoadjuvant chemotherapy) and radical resection, with an ECOG performance status of 0–2. Patients were divided into therapy incomplete (Tx Incomp, less than eight cycles of FLOT) and therapy (Tx comp) groups. We performed univariate and multivariate analyses, as well as propensity score matching (PSM) analysis, to evaluate whether treatment delivery was associated with recurrence-free survival (RFS) and overall survival (OS). Kaplan-Meier curves and restricted mean survival times (RMST) up to 36 months were used to evaluate the survival time. The primary endpoint was the 3-year RFS and OS rate. **Results:** Of the screened patients, 210 met the inclusion criteria, with 79 (38%) in the Tx Incomp and 131 (62%) in the Tx comp group, and 50 (24%) patients did not receive ACT. The median follow-up time was 26.5 months. The 3-year RFS and OS rates in the Tx Incomp and Tx comp groups after PSM were 54% vs 59% (p=0.14) (RMST difference: 4.04 months, p=0.09) and 58% vs 78% (p=0.04) (RMST difference: 6.15 months, p<0.001), respectively. Multivariable analysis showed a significant impact of Tx comp on OS (HR 0.53, 95% CI: 0.32-0.88). In the subgroup of ypN-positive patients, a significant difference of RMST in OS rate was observed between those who started ACT and those who did not start ACT (RMST difference: 7.89 months, p=0.01), whereas no significant difference was found in the ypN-negative group (RMST difference: 0.65 months, p=0.75). **Conclusions:** This study suggests that completing FLOT therapy is associated with better OS. The impact of ACT might differ according to ypN stage. Research Sponsor: None.

392 **Poster Session**

First-line (1L) nivolumab (NIVO) plus chemotherapy (chemo) vs chemo in patients (pts) with advanced gastric cancer, gastroesophageal junction cancer, and esophageal adenocarcinoma (GC/GEJC/EAC): 5-year (y) follow-up of Chinese pts from CheckMate 649. First Author: Lin Shen, Beijing Cancer Hospital, Beijing, China

Background: NIVO + chemo demonstrated clinically meaningful survival benefit and an acceptable safety profile vs chemo in previously untreated Chinese pts with advanced GC/GEJC/EAC from CheckMate 649, consistent with the overall study population. 1L NIVO + chemo is currently approved for pts with advanced non-HER2+ GC/GEJC/EAC in China and other countries. We report 5-y results of NIVO + chemo vs chemo in Chinese pts from CheckMate 649. **Methods:** Adults with previously untreated, unresectable advanced or metastatic, non-HER2+ GC/GEJC/EAC were enrolled regardless of programmed death ligand 1 (PD-L1) expression. Randomized pts received NIVO + chemo, NIVO + ipilimumab, or chemo. Dual primary endpoints for NIVO + chemo vs chemo were overall survival (OS) and progression-free survival (PFS) by blinded independent central review (BICR) in pts with PD-L1 combined positive score (CPS) ≥ 5. **Results:** 208 Chinese pts were randomized to NIVO + chemo or chemo. At 61-month (mo) minimum follow-up, NIVO + chemo continued to demonstrate OS and PFS benefit vs chemo in pts with PD-L1 CPS ≥ 5 and all randomized pts (Table). 5-y OS rate was 24% with NIVO + chemo vs 8% with chemo in pts with PD-L1 CPS ≥ 5 and 20% vs 7% in all randomized pts. Objective response rate (ORR) was higher and responses were more durable with NIVO + chemo vs chemo in pts with PD-L1 CPS ≥ 5 and all randomized pts (Table). No new safety signals were identified with longer follow-up. **Conclusions:** NIVO + chemo continued to demonstrate clinically meaningful long-term survival benefit, more durable responses, and acceptable safety vs chemo in Chinese pts after 5 y of follow-up, consistent with earlier reports and with the overall study population of pts with advanced non-HER2+ GC/GEJC/EAC. These results further support NIVO + chemo as a standard 1L treatment option for Chinese pts. Clinical trial information: NCT02872116. Research Sponsor: Bristol Myers Squibb.

	PD-L1 CPS ≥ 5		All randomized	
Efficacy	**NIVO + chemo (n = 75)**	**Chemo (n = 81)**	**NIVO + chemo (n = 99)**	**Chemo (n = 109)**
mOS (95% CI), mo	15.5 (11.9–21.1)	9.6 (8.0–12.1)	14.3 (11.5–16.5)	10.3 (8.1–12.1)
HR (95% CI)	0.57 (0.40–0.82)		0.63 (0.46–0.85)	
mPFS[a] (95% CI), mo	8.5 (6.0–14.0)	4.3 (4.1–6.5)	8.3 (6.2–12.4)	5.6 (4.2–6.8)
HR (95% CI)	0.51 (0.34–0.76)		0.57 (0.41–0.80)	
ORR[a,b] (95% CI), %	68 (56–79)	48 (36–60)	66 (55–76)	45 (35–56)
mDOR[a,c] (95% CI), mo	12.5 (7.2–23.4)	6.9 (3.9–8.5)	12.5 (7.2–17.7)	5.6 (4.4–8.3)

[a]Per BICR.
[b]In pts with measurable target lesions at baseline.
[c]In responders. DOR, duration of response; m, median.

393 Poster Session

A randomized phase II trial of systemic chemotherapy with or without trastuzumab followed by surgery in HER2 positive advanced gastric or esophagogastric junction adenocarcinoma with extensive lymph node metastasis: 3-year follow-up data of the Japan Clinical Oncology Group study JCOG1301C (Trigger study). First Author: Masaki Aizawa, Niigata Cancer Center Hospital, Niigata, Japan

Background: We previously reported better objective and pathological response with the addition of trastuzumab to preoperative chemotherapy in the randomized phase II trial comparing the efficacy of systemic chemotherapy with or without trastuzumab followed by surgery in HER2 positive advanced gastric or esophagogastric junction adenocarcinoma with extensive lymph node metastasis. Here, we report 3-year follow-up data. **Methods:** Eligible patients with HER2-positive gastric cancer having extensive lymph node metastasis were randomized to receive preoperative chemotherapy with S-1/cisplatin (SP: S-1 80–120 mg/body on days 1–14, and cisplatin 60 mg/m2 on day 1, q21 days, 3-4 courses) in arm A or SP plus trastuzumab (first course 8 mg/kg followed by 6 mg/kg, day 1 on each course) in arm B. After gastrectomy, adjuvant chemotherapy with S-1 was added for 1 year in both arms. The primary endpoint was overall survival (OS), and the sample size was planned to be 130 patients in total, expecting a 10% increase in the 3-year OS (70% vs 80%) with a 1-sided alpha of 0.2, a power of 0.75. **Results:** The study was terminated in March 2021 due to slow patient accrual. In total, 46 patients were randomized to either arm A (22 patients) or arm B (24 patients). Patient characteristics were well balanced between the arms. Planned preoperative chemotherapy was completed in 20 patients (90.9%) in arm A and in 23 patients (95.8%) in arm B. Objective response rate tended to be higher in arm B (66.7% [16/24]) than in arm A (36.4% [8/22], p=0.08), and R0 resection was achieved in 20 patients (90.9%) in arm A and 22 patients (91.7%) in arm B. The adjuvant chemotherapy was completed in 54.6% (12/22) in arm A and in 54.2% (13/24) in arm B. With the median observation period of 4.1 years, the 3-year OS was 68.2% in arm A and was 78.9% in arm B with a hazard ratio of 0.698 (95% confidence interval (CI), 0.221-2.209). The 3-year progression-free survival of arm A and arm B was 39.3% and 62.3%, respectively, with a hazard ration of 0.562 (95% CI, 0.229-1.377). **Conclusions:** Preoperative chemotherapy with SP plus trastuzumab was feasible, and tended to improve the objective response rate, OS and progression-free survival. An integrated analysis with other clinical trials evaluating perioperative trastuzumab is being planned. Clinical trial information: s031180006. Research Sponsor: None.

394 Poster Session

Alternative dosing regimens of tislelizumab proposed using modeling and simulation. First Author: Ahsan Rizwan, BeiGene USA, Inc., San Mateo, CA

Background: Tislelizumab (TIS; BGB-A317) is approved for the treatment of multiple solid tumors, administered at 200 mg every 3 weeks (Q3W), and has demonstrated a flat exposure–response relationship across a wide range of doses. We evaluated alternative dosing regimens of TIS at 150 mg every 2 weeks (Q2W), 300 mg every 4 weeks (Q4W), and 400 mg every 6 weeks (Q6W) using a model-based approach with the aim of alleviating patient burden by providing longer dosing intervals and/or treatment flexibility compatible with background chemotherapy to meet the needs of patients and healthcare practitioners. **Methods:** A previously developed population pharmacokinetic (PK) model was used for simulating PK exposure of the alternative regimens, which were selected by exposure-matching to the reference dose of 200 mg Q3W. PK-based criteria (peak concentration [C_{max}] within 25% and trough concentration [C_{trough}] within 20% of the reference dose) were also used. Alternative dosing regimen exposures in the first least common time interval and at steady state were compared with the reference. Deviations from PK-based criteria were bridged using appropriate safety and efficacy references and exposure–response analyses using data from four phase 1, 2, and 3 clinical trials of TIS in patients with solid tumors, including gastric cancer and esophageal squamous cell carcinoma. **Results:** Simulations at steady state shown here demonstrate that the TIS alternative dosing regimens of 150 mg Q2W, 300 mg Q4W, and 400 mg Q6W produce comparable exposures to the 200 mg Q3W reference regimen. Although the simulated C_{max} at 300 mg Q4W and 400 mg Q6W were higher than with the 200 mg Q3W reference dose, these were below the C_{max} of the 5 mg/kg Q3W safety reference (Table). And while the C_{trough} for the 400 mg Q6W dosing regimen was lower than with the 200 mg Q3W reference dose, it was 10.7% higher compared with the 2 mg/kg efficacy reference dose; therefore, it was within the concentration range where a flat exposure–efficacy relationship of TIS has previously been established. **Conclusions:** TIS alternative dosing regimens of 150 mg Q2W, 300 mg Q4W, and 400 mg Q6W are expected to result in similar safety and efficacy profiles as the 200 mg Q3W reference dosing regimen and may be used interchangeably for indications where 200 mg Q3W is approved. Clinical trial information: NCT04716634, NCT05014828, NCT04379635, NCT02407990, NCT04068519, NCT03430843 and NCT03358875. Research Sponsor: BeiGene Ltd.

Steady-state PK exposure metrics for alternative TIS doses vs reference doses.

Parameter	150 mg Q2W vs 200 mg Q3W	300 mg Q4W vs 200 mg Q3W	300 mg Q4W vs 5 mg/kg Q3W[a]	400 mg Q6W vs 200 mg Q3W	400 mg Q6W vs 5 mg/kg Q3W[a]	400 mg Q6W vs 2 mg/kg Q3W[b]
C_{max}	−5.8 (Yes)	31.4 (No)	−18.7 (Yes)	52.2 (No)	−5.8 (Yes)	–
$C_{average}$	12.3 (Yes)	12.6 (Yes)	–	0.4 (Yes)	–	–
C_{trough}	27.2 (Yes)	0.1 (Yes)	–	−28.4 (No)	–	10.7 (Yes)

Data are presented as difference (% = [$GM_{test} - GM_{reference}/GM_{reference}$] × 100) vs the 200 mg Q3W reference, [a]safety reference, or [b]efficacy reference (meets regulatory criteria, Yes/No).
$C_{average}$, average concentration; GM, geometric mean.

395 Poster Session

Does chemotherapy regimen matter for first-line immunochemotherapy in low programmed cell death ligand 1 (PD-L1)-expressing esophageal squamous cell carcinoma (ESCC)? A systemic review and meta-analysis. First Author: Jhe-Cyuan Guo, National Taiwan University Cancer Center, Taipei, Taiwan

Background: Anti-programmed death 1 (PD-1) therapy plus chemotherapy has become standard first-line therapy for patients with high PD-L1-expressing advanced ESCC. The benefit for patients with low PD-L1-expressing ESCC remains debatable. **Methods:** Eligible studies were phase III trials investigating anti-PD-1/PD-L1 therapy plus chemotherapy (immunochemotherapy) vs chemotherapy as first-line therapy of advanced ESCC. Study-level pooled analyses of hazard ratios (HR) for progression-free survival (PFS) and overall survival (OS) with corresponding 95% confidence intervals (CI), were used to compare the different groups in random-effects model. PD-L1 expression levels with tumor proportion score (TPS) ≥ 1% vs < 1%, tumor area positivity (TAP) ≥ 10% vs < 10%, and combined positive score (CPS) ≥ 10 vs < 10 were collectively categorized as high vs low PD-L1 expression. The impact of two different chemotherapy regimens, TP (paclitaxel plus platinum) and PF (platinum pus fluoropyrimidine), stratified by PD-L1 expression levels, on the survival benefit of immunochemotherapy vs chemotherapy was investigated. **Results:** Eight randomized controlled trials including 4733 subjects were enrolled. Immunochemotherapy vs chemotherapy was associated with significantly improved PFS and OS irrespective of PD-L1 expression levels or chemotherapy regimens (Table). The PFS and OS benefits were more pronounced in high PD-L1 group than low PD-L1 group. In patients with high PD-L1 expression, the PFS and OS benefits were not significantly different between two chemotherapy regimen subgroups. However, in patients with low PD-L1 expression, the PFS benefit was more significant in TP subgroup than PF subgroup, and OS benefit trended to be in favor of TP subgroup. **Conclusions:** The addition of anti-PD-1/PD-L1 therapy to TP chemotherapy may associate with more significant survival benefit than to PF chemotherapy in advanced ESCC with low PD-L1 expression. Research Sponsor: None.

	PFS (immunochemotherapy vs chemotherapy)		OS (immunochemotherapy vs chemotherapy)	
Population	HR (95% CI)	P value (subgroup differences)	HR (95% CI)	P value (subgroup differences)
All patients	0.63 (0.57-0.69)		0.69 (0.64-0.74)	
High PD-L1 group	0.54 (0.49-0.60)	P < 0.01	0.60 (0.53-0.67)	P < 0.01
Low PD-L1 group	0.71 (0.62-0.81)		0.78 (0.70-0.87)	
TP group	0.56 (0.50-0.62)	P < 0.01	0.66 (0.59-0.74)	P = 0.32
PF group	0.68 (0.62-0.75)		0.72 (0.65-0.79)	
Among high PD-L1 patients				
TP subgroup	0.56 (0.45-0.69)	P = 0.75	0.60 (0.46-0.78)	P = 0.90
PF subgroup	0.53 (0.45-0.62)		0.59 (0.50-0.69)	
Among low PD-L1 patients				
TP subgroup	0.59 (0.48-0.74)	P = 0.01	0.72 (0.55-0.93)	P = 0.32
PF subgroup	0.82 (0.72-0.94)		0.84 (0.72-0.97)	

396 Poster Session

Perioperative chemotherapy with either S-1 or 5-fluorouracil plus oxaliplatin and docetaxel for locally advanced gastric or gastroesophageal junction adenocarcinoma: A prospective trial and matched comparison. First Author: Chih Chieh Yen, National Cheng Kung University Hospital, Tainan City, Taiwan

Background: Perioperative triplet chemotherapy improves the survival and surgical outcome of patients with locally advanced gastric or gastroesophageal junction (GEJ) adenocarcinoma in addition to curative surgery. 5-fluorouracil (5-FU), docetaxel, oxaliplatin and leukovorin (FLOT) has been suggested as the treatment of choice. S-1, an oral fluoropyrimidine chemotherapeutic, also shows compelling efficacy as one of the triplet backbone and is exempt from a prolonged infusion time. However, a comparison over efficacy and tolerability between S-1 or 5-FU-containing triplet has not been investigated. **Methods:** The study contains two parts. The first part was a prospective single-arm multi-centered phase II trial with eight cycles of perioperative leucovorin, oxaliplatin, docetaxel and S-1 (LOTS) in patients with ≥T3 any N or ≥T2N(+) non-metastatic gastric/GEJ adenocarcinoma. The primary endpoint was the treatment response assessed by pathological tumor regression (TRG). After confirming the result passing the statistical threshold, we compared them with an identical and independent patient population who received perioperative FLOT under a prospective observation. A propensity score matching (PPSM) was used to balance the heterogeneity between the two groups. The efficacy, survival and tolerability were compared in a post-hoc manner. **Results:** From Feb 2017 to Jan 2024 with a comparable median follow-up time (LOTS, 18.7 ms; FLOT, 18.9 ms), a total of 45 and 154 patients were enrolled and received either LOTS or FLOT treatment. After a one-to-two PPSM procedure, the baseline characteristics of patients were comparable (LOTS, n=45; FLOT, n=90). Patients had similar curative resection rates (LOTS, 91%; FLOT, 86%; p=.359) and above nearly complete pathological regression rates (TRG 0-1: LOTS, 23%; FLOT, 27%; p=.669). The 24-ms recurrence-free (RFS) and overall survival (OS) rates were also comparable (LOTS vs. FLOT: 66 vs. 59% in RFS, p=.463; 73 vs. 68% in OS, p=.491). The toxicities were tolerable, except that more diarrhea and nausea events were reported in the LOTS group, whereas neutropenia, thrombocytopenia and mucositis were significantly prominent in the FLOT group. **Conclusions:** Perioperative LOTS shows reasonable therapeutic efficacy and survival as compared with FLOT in patients with locally advanced gastric/GEJ cancer. While the toxicity profile is distinctive from FLOT, it could be a potential alternative regimen with the convenience to save the prolonged infusion time. Clinical trial information: NCT04999332. Research Sponsor: Clinical Trial Facilitation Program, Ministry of Health and Welfare, Taipei, Taiwan; Year 2023; TTY Biopharm, Taipei, Taiwan.

Pathological outcome.

	Prior to LOTS treat-ment (n=45)		After LOTS treatment (n=43)		Prior to FLOT treat-ment (n=90)		After FLOT treatment (n=86)		p in LOTS vs. FLOT
CAP TRG, n (%)									0.669
0	-	-	1	(2.3)	-	-	3	(3.5)	
1	-	-	9	(20.9)	-	-	20	(23.3)	
2	-	-	14	(32.6)	-	-	23	(26.7)	
3	-	-	19	(44.2)	-	-	40	(46.5)	

397 Poster Session

Efficacy of PD-1 inhibitor plus chemotherapy versus chemotherapy in advanced esophageal carcinoma: A meta-analysis of phase III clinical trials. First Author: Zauraiz Anjum, Rochester General Hospital, Rochester, NY

Background: Programmed death cell death protein-1 (PD-1) inhibitors combined with chemotherapy have demonstrated survival benefits in patients with advanced esophageal squamous cell carcinoma (ESCC) and adenocarcinoma (EAC). Our pooled analysis aims to analyze the efficacy of PD-1 inhibitor plus chemotherapy in advanced esophageal carcinoma (EC) in phase III randomized clinical trials (RCTs). **Methods:** We collected data from eligible phase III RCTs searched through PubMed, EMBASE, ClinicalTrials.gov, and meeting abstracts till July 5, 2024. Initial screening revealed 792 articles. We excluded duplicates, review articles and irrelevant studies, and we included eight RCTs that reported objective response rate (ORR), treatment-related adverse events (TRAEs), overall survival (OS) and progression-free survival (PFS). Odds ratio (OR) for ORR and TRAEs, and hazard ratio (HR) for OS and PFS were computed along with a 95% confidence interval (CI) and p-value for pooled analysis, using a random effect model in RevMan v.5.4. We performed a subgroup analysis based on a combined positive score (CPS) for programmed cell death ligand-1 (PDL-1) expression and tumor histology. **Results:** We included eight phase III RCTs (Jupiter-06, Checkmate 648, Keynote 590, ESCORT-1st, Orient-15, Checkmate 649, ESCORT-NEO, and ASTRUM-007), including 4131 patients with 2137 in group A (PD-1 inhibitor + chemotherapy) and 1994 in group B (placebo + chemotherapy, or chemotherapy). PD-1 inhibitors against EC included nivolumab, pembrolizumab, camrelizumab, sintilimab, toripalimab, and serplulimab. A total of 9.9% of patients were diagnosed with EAC, and 90.1% were ESCC. A pooled analysis showed significant achievement in ORR in group A compared to group B with an OR of 2.19 (CI: 1.77-2.69, $p < 0.05$, $I^2 = 60\%$). In terms of OS benefit, group A led to a 29% reduction in death risk compared to group B (HR: 0.71, CI: 0.66-0.76, $p < 0.05$, $I^2 = 0$). Group A also showed a significantly reduced risk of disease progression compared to group B (HR: 0.62, CI: 0.54-0.70, $p < 0.05$, $I^2 = 60\%$). Safety analysis revealed more TRAEs (OR: 1.93, CI: 1.43-2.60, $p < 0.01$, $I^2 = 11\%$) and grade ≥3 AEs (OR: 1.34, CI: 1.08-1.67, $p < 0.05$, $I^2 = 54\%$) in group A compared with group B. Sub-group analysis based on CPS revealed that patients with CPS ≥10 showed more OS (pooled HR: 0.59 vs 0.75) (p = 0.04) and PFS (pooled HR: (0.59 vs 0.64) (p = 0.18) benefits as compared to patients with CPS <10. Survival analysis between ESCC and EAC also exhibited statistically non-significant OS benefits (p = 0.28). **Conclusions:** PD-1 inhibitor plus chemotherapy as a first-line therapy in advanced EC, mainly in patients with CPS ≥10, showed improved antitumor efficacy in terms of superior ORR, OS, and PFS, with a manageable toxicity profile providing a benchmark for future studies. Research Sponsor: None.

398 Poster Session

Nivolumab (NIVO) + chemotherapy (chemo) vs chemo as first-line (1L) treatment for advanced gastric cancer/gastroesophageal junction cancer/esophageal adenocarcinoma (GC/GEJC/EAC): 5-year (y) follow-up results from CheckMate 649. First Author: Yelena Y. Janjigian, Memorial Sloan Kettering Cancer Center and Weill Cornell Medical College, New York, NY

Background: At 4-y follow-up, 1L NIVO + chemo continued to demonstrate clinically meaningful overall survival (OS) and progression-free survival (PFS) benefit vs chemo with acceptable safety in patients (pts) with advanced non-HER2+ GC/GEJC/EAC from CheckMate 649. We report efficacy and safety results of NIVO + chemo vs chemo at 5-y follow-up. **Methods:** Adults with previously untreated, unresectable, advanced or metastatic, non-HER2+ GC/GEJC/EAC were enrolled, regardless of programmed death ligand 1 (PD-L1) expression. Pts were randomized to NIVO (360 mg Q3W or 240 mg Q2W) + chemo (XELOX Q3W or FOLFOX Q2W), NIVO + ipilimumab, or chemo. Primary endpoints for NIVO + chemo vs chemo were OS and PFS by blinded independent central review (BICR) in pts with PD-L1 combined positive score (CPS) ≥ 5. **Results:** Pts were randomized to NIVO + chemo (n = 789) or chemo (n = 792). NIVO + chemo continued to show OS and PFS benefit vs chemo in pts with PD-L1 CPS ≥ 5, pts with PD-L1 CPS ≥ 1, and all randomized pts at 60-month (mo) minimum follow-up (Table). OS rates at 60-mo were higher with NIVO + chemo vs chemo in pts with PD-L1 CPS ≥ 5, pts with PD-L1 CPS ≥ 1, and all randomized pts (Table), and OS benefit with NIVO + chemo continued to be observed in most prespecified subgroups. Objective response rates (ORRs) were higher and responses were more durable with NIVO + chemo vs chemo in pts with PD-L1 CPS ≥ 5, pts with PD-L1 CPS ≥ 1, and all randomized pts (Table). No new safety signals were identified. **Conclusions:** These results represent the first report of 5-y follow-up for anti–PD-1 + chemo combination therapy in GC/GEJC/EAC to our knowledge. NIVO + chemo continued to provide sustained long-term survival vs chemo with an acceptable safety profile after 5 y of follow-up. These data continue to support the use of NIVO + chemo as a standard 1L treatment for advanced GC/GEJC/EAC. Clinical trial information: NCT02872116. Research Sponsor: Bristol Myers Squibb.

	PD-L1 CPS ≥ 5		PD-L1 CPS ≥ 1		All randomized	
Efficacy	**NIVO + chemo (n = 473)**	**Chemo (n = 482)**	**NIVO + chemo (n = 641)**	**Chemo (n = 656)**	**NIVO + chemo (n = 789)**	**Chemo (n = 792)**
mOS (95% CI), mo	14.4 (13.1–16.2)	11.1 (10.1–12.1)	13.8 (12.4–14.8)	11.4 (10.7–12.3)	13.7 (12.4–14.5)	11.6 (10.9–12.5)
HR (95% CI)	0.71 (0.61–0.81)		0.76 (0.67–0.85)		0.79 (0.71–0.88)	
60-mo OS rate (95% CI), %	16 (12–19)	6 (4–9)	13 (11–16)	5 (4–7)	12 (10–14)	6 (4–8)
mPFS[a] (95% CI), mo	8.3 (7.0–9.4)	6.1 (5.6–6.9)	7.5 (7.0–8.5)	6.9 (6.2–7.1)	7.8 (7.1–8.6)	6.9 (6.7–7.2)
HR (95% CI)	0.71 (0.61–0.82)		0.77 (0.68–0.87)		0.79 (0.71–0.89)	
ORR[a,b] (95% CI), %	60 (55–65)	45 (40–50)	60 (55–64)	46 (42–51)	58 (54–62)	46 (42–50)
mDOR[a,c] (95% CI), mo	9.6 (8.3–12.4)	7.0 (5.7–8.0)	8.6 (7.9–10.5)	6.9 (5.8–7.6)	8.5 (7.7–9.9)	6.9 (5.9–7.6)

[a]Per BICR.
[b]In pts with measurable target lesions at baseline.
[c]In all measurable responders.
DOR, duration of response; m, median.

399 Poster Session

Real-world safety of trastuzumab deruxtecan with gastric cancer: All-patient post-marketing surveillance study in Japan. First Author: Hisato Kawakami, Department of Medical Oncology, Kindai University Faculty of Medicine & Department of Clinical Oncology Tohoku University Graduate School of Medicine., Miyagi, Japan

Background: Interstitial lung disease and pneumonitis (ILD/p) is an important identified risk in the Japanese risk management plan of trastuzumab deruxtecan (T-DXd). In Japan, an all-patient post-marketing surveillance (PMS) study was conducted to investigate the incidence of ILD/p and factors associated with its development among patients with gastric cancer treated with T-DXd in the real-world setting. **Methods:** This is an observational, multicenter, PMS study (jRCT2001200001) with an observation period of 12 months that enrolled all patients who initiated T-DXd treatment for HER2-positive unresectable advanced or recurrent gastric cancer between Sep 2020 and Dec 2021. All suspected ILD/p events reported by physicians were adjudicated by an independent ILD adjudication committee; events adjudicated as drug-related ILD/p were summarized. The factors associated with the development of adjudicated drug-related ILD/p were investigated using a Cox proportional hazards model. Effectiveness endpoints included best overall response rate according to the RECIST 1.1 and overall survival, which was estimated using the Kaplan–Meier method. **Results:** The safety analysis set included 1070 patients (77.4% male, 89.9% ECOG performance status ≤1) with median age of 70 years (range: 23-100). Median initial dose and treatment duration of T-DXd were 6.4 mg/kg (range: 3.1-6.4) and 3.9 months (range: 0.7-12.0), respectively.The incidence of any grade, grade ≥3, and grade 5 adjudicated drug-related ILD/p were 9.6% (n=103), 2.8% (n=30) and 1.2% (n=13), respectively. Within 24 weeks from onset, 73.8% of adjudicated drug-related ILD/p had resolved, were resolving, or resolved with sequelae.A multivariate analysis identified that medical history and/or comorbidity of ILD/p [hazard ratio (HR) = 3.372, 95% confidence interval (95% CI): 1.431, 7.840], radiation pneumonitis (HR = 8.523, 95% CI: 2.006, 36.215), COPD or emphysema (HR = 1.962, 95% CI: 1.032, 3.731) and age ≥75 years (HR = 1.648, 95% CI: 1.058, 2.567) were factors of interest associated with the development of ILD/p in patients with gastric cancer. The overall response rate at 12 months post-treatment initiation of T-DXd was 34.2%. Overall survival after 364 days post-treatment initiation of T-DXd was 43.8% (median survival time: 303.0 days). **Conclusions:** The results suggest that the risk of ILD/p in this PMS study does not differ from that in the Destiny Gastric-01 study, and no new safety concerns were identified. Medical history and/or comorbidity of ILD/p, radiation pneumonitis, COPD or emphysema and age ≥75 years were identified as factors of interest associated with the development of ILD/p in patients with gastric cancer. Further investigation into the identified factors of interest may offer insights into the development of ILD/p among T-DXd treated gastric cancer patients in Japan. Clinical trial information: jRCT2001200001. Research Sponsor: Daiichi Sankyo Co., Ltd.

400 Poster Session

Prognostic factors for gastric cancer patients with positive peritoneal cytology who underwent upfront surgery. First Author: Takashi Abe, Shizuoka Cancer Center, Sunto-Gun Nagaizumi-Cho, Japan

Background: Positive peritoneal cytology (CY1) is classified as Stage IV disease, the standard treatment for which is systemic chemotherapy. However, CY1 is often diagnosed after surgery in patients for whom staging laparoscopy is not indicated. In addition, the treatment strategy for patients with CY1 diagnosed during surgery is controversial. The Japanese guidelines recommend upfront surgery followed by adjuvant chemotherapy for patients diagnosed with CY1 during surgery. To identify patients with CY1 who are most likely to benefit from upfront surgery, we investigated prognostic factors in patients with CY1 who underwent upfront surgery. **Methods:** A total of 169 gastric cancer patients diagnosed as P0CY1 during or after surgery who underwent R1 resection other than positive resection margins were included. The diagnosis of CY1 was obtained by staging laparoscopy or during laparotomy in 69 patients and after surgery in 87 patients. Fourteen patients were initially diagnosed as CY0 by staging laparoscopy but were rediagnosed as CY1 after surgery. The clinicopathological factors of overall survival (OS) and progression-free survival (PFS) were investigated. Prognostic factors were identified using Cox regression models. **Results:** The median patient age was 72 years, and there were 108 males. The histological type was undifferentiated type in all of 102 patients. The macroscopic types were 0 in 10) patients 1 in 3, 2 in 27 3 in 84 and 4 in 45). cT grades and cN grades were cT1 in 3, cT2 in 7, cT3 in 12, cT4 in 147, cN0 in 65cN1 in 30), cN2 in 46 and cN3 in 28. The median OS and PFS were 25.1 months and 14.6 months, with 5-year OS rate and PFS rate of 30% and 23%, respectively. A multivariate analysis for OS identified macroscopic type 0-2, Charlson Comorbidity Index<3, and adjuvant chemotherapy as independent prognostic factors. Similarly, macroscopic type 0-2, tumor size <80 mm, and adjuvant chemotherapy were identified as independent prognostic factors for PFS. In a subgroup analysis of patients who received adjuvant chemotherapy, macroscopic type 0-2 was the only independent prognostic factor for PFS. The 5-year PFS rate in patients with macroscopic type 0-2 was 42%. **Conclusions:** Patients with macroscopic type 0-2 who are suitable for adjuvant chemotherapy may have benefit from upfront surgery, even if peritoneal cytology is positive. Research Sponsor: None.

401 **Poster Session**

Safety and efficacy of conversion therapy for metastatic esophageal cancer: Exploratory analysis of JCOG1314. First Author: Satoru Matsuda, Department of Surgery, Keio University School of Medicine, Tokyo, Japan

Background: Patients with esophageal cancer (EC) with distant metastasis were treated with systemic chemotherapy. Recent advances in multimodal treatments have made conversion therapy (CT) a viable option for patients with metastatic EC. JCOG1314 was a randomized phase III trial to confirm the survival benefit of docetaxel plus cisplatin plus 5-fluorouracil (DCF) versus cisplatin plus 5-fluorouracil (CF) as initial treatment for unresectable or recurrent esophageal cancer. The purpose of this study was to evaluate the safety and survival impact of CT in patients enrolled in JCOG1314. **Methods:** CT was defined as surgery or chemoradiotherapy (CRT) aiming at cure after initial treatment for tumors that were initially unresectable due to distant metastasis. Clinicopathologic factors, surgical outcomes, toxicities during CRT, and survival were compared between CT and non-CT groups. CTCAE v4.0-JCOG was used to evaluate postoperative complications and adverse events during CRT. **Results:** Between September 2014 and April 2021, 240 patients were randomized to CF or DCF. Excluding patients with recurrent disease, 154 patients with initially unresectable esophageal cancer due to distant metastasis were selected for this study. Among them, 21 patients received CT which included conversion surgery (n=5) and conversion CRT (n=16). There was no significant difference in the number of metastatic organs, depth of invasion of the primary tumor, lymph node metastasis, and chemotherapy regiment between CT and non-CT groups. The most common M1 factor in CT group was the thoracic lymph node, followed by abdominal lymph node and cervical node. In conversion surgery group, there was no incidence of Grade 3 or higher pneumonia/leakage. Regarding the Grade 3 or higher non-hematological toxicities during conversion CRT, the incidence of appetite loss/ oral mucositis/pneumonia was 2 (13%)/2 (13%)/1 (6%), respectively. The 3-year overall survival (OS) of CT and non-CT groups was 52.4% and 14.3%, respectively. On multivariable analysis, patients with CT showed significantly better OS compared with the non-CT group (HR 0.36, 95% CI 0.19-0.67, $p<0.01$). In patients who underwent CT, there was no significant difference in OS between conversion surgery and CRT groups. **Conclusions:** This study demonstrated that CT was safely performed with favorable prognosis. A prospective study is warranted to investigate whether CT would improve survival in metastatic esophageal cancer. Research Sponsor: None.

402 **Poster Session**

Short- and long-term outcomes of conversion surgery for initially unresectable advanced gastric cancer: A Japanese multicenter retrospective cohort study. First Author: Itaru Yasufuku, Department of Clinical Anatomy Development Studies, Gifu University Graduate School of Medicine, Gifu-City, Japan

Background: The efficacy of conversion surgery in patients with unresectable advanced gastric cancer at the initial diagnosis who converted to be resectable due to remarkable response to chemotherapy is unclear. To evaluate short- and long-term outcomes of conversion surgery for initially unresectable cStage IVB/pStage IV gastric cancer as the basic data for the JCOG2301 trial, which is a randomized controlled trial comparing conversion surgery with continuation of chemotherapy, a multicenter retrospective cohort study was conducted in the Stomach Cancer Study Group of the Japan Clinical Oncology Group. **Methods:** Inclusion criteria were as follows; (1)gastric cancer with at least one of following distant metastasis at the initial diagnosis; (i) peritoneal metastasis (PER) diagnosed by imaging examination or P1b or P1c disease diagnosed by laparoscopy or laparotomy, (ii) multiple liver metastasis (HEP) more than three lesions, and (iii) distant lymph node metastasis (LYM) beyond paraaortic lymph node station 16a2/b1, (2) conversion to be resectable after systemic chemotherapy, (3) surgery aiming at an R0 resection of the primary and remaining lesions between April 2010 and September 2017. Clinical data were retrospectively obtained from electronic patient records of 58 institutions. **Results:** Atotal of 241 patients were included to the study. Initial unresectable factors were PER, HEP, and LYM in 132(55%), 37(15%), and 63(26%) patients, respectively, and 9(4%) patients had two or more factors. Conversion to be resectable obtained during the first and second/later line chemotherapy was in 209 (87%) patients and 32 (13%), respectively. 182(76%), 29(12%), and 30(12%) patients underwent R0, R1, and R2 resection, respectively. The incidence of postoperative complication of Clavien-Dindo Grade II or higher was 24% and no surgery-related death was observed. The median overall survival (mOS) from surgery was 45.4 (95%CI 30.9-61.4) months in R0 resection, 16.6 (11.5-18.5) months in R1, and 20.9 (11.3-35.6) months in R2. In case of R0 resection, the mOS from surgery was 34.4 (95%CI 26.5-59.2) months in PER, 55.9 (21.8-96.7) months in HEP, 54.6 (30.9-98.3) months in LYM, and not reached (19.7-NA) in two or more factors. **Conclusions:** Conversion surgery appears to be safe and a promising treatment strategy for patients who have initially unresectable advanced gastric cancer and presented remarkable response to systemic chemotherapy. Research Sponsor: None.

403 **Poster Session**

Factors and timing of decreased adherence to S-1 in postoperative docetaxel + S-1 therapy for gastric cancer. First Author: Kazuyoshi Kawakami, Department of Pharmacy, Cancer Institute Hospital, Japanese Foundation for Cancer Research, Tokyo, Japan

Background: The combination of docetaxel + S-1 (DS) therapy in postoperative adjuvant treatment for gastric cancer has proven to be effective and has become a standard therapy. Clarification of factors involved in and the timing of S-1 non-adherence is important to minimize non-adherence. **Methods:** Among 92 patients who started DS treatment as postoperative adjuvant chemotherapy for gastric cancer between November 1, 2015 and April 30, 2021 at the Cancer Institute Hospital of the Japanese Foundation for Cancer Research, 90 were eligible for this study. The S-1 adherence rate per cycle was defined as the number of times a patient took S-1 divided by 28. In this study, adherence to S-1 was assessed through pill counts and by asking patients about reasons for non-adherence in interviews at a pharmaceutical outpatient clinic. Elicited reasons for non-adherence were categorized and analyzed. **Results:** Percentage of remaining S-1 was 3.9% in the first cycle, 7.4% in the second cycle, and 2.0% in the seventh cycle. Median relative dose intensity of S-1 was 80.9%. As reasons for non-adherence, patients most commonly cited nausea/vomiting (18.9%), missed dose (17.6%), and diarrhea (9.9%). From the first to the seventh course, the highest percentage of remaining S-1 was in the second course. **Conclusions:** Nausea/vomiting, missed dose, and diarrhea were identified as the most common factors for decreased adherence to S-1 in postoperative DS therapy for gastric cancer. Between the first and seventh courses, S-1 adherence decreased the most during the second course, when docetaxel was started. Using this information, we intend to develop methods to improve S-1 adherence in the future. Research Sponsor: Japan Society for the Promotion of Science KAKENHI.

404 **Poster Session**

The role of immunotherapy for real-world patients with HER2-negative advanced gastric cancer between 2011-2023. First Author: Keitaro Shimozaki, Department of Gastroenterological Chemotherapy, Cancer Institute Hospital, Japanese Foundation for Cancer Research, Koto-Ku, Japan

Background: Although the appearance of immunotherapy has apparently brought benefits for all patients (pts) with advanced gastric cancer (AGC), the magnitude of the benefit in real-world pts with HER2-negative AGC has not been fully investigated. **Methods:** We conducted a retrospective study on pts with HER2-negative AGC who received first-line platinum-based chemotherapy between 2011 and 2023. We evaluated the clinical outcomes across different periods of immunotherapy approval in Japan: Group A (pre-immunotherapy approval): 2011-2017, Group B (approved for third-line treatment or later): 2018-2021, and Group C (approved for first-line treatment): 2022-2023. **Results:** A total of 949 pts were enrolled (A: n = 477; B: n = 344; C: n = 128). Patient's characteristics were comparable except for the proportion of those with age $\geq$75 years (P = 0.002), prior gastrectomy (P = 0.03), and liver metastases ($P < 0.001$). The median overall survival (OS) was 16.2, 15.2, and 21.3 months in group A, B, and C, respectively, with no significant difference between groups (log-rank P = 0.50). Pts who had received first-line immunotherapy plus chemotherapy (n = 173) showed significantly improved OS compared with pts without having received any immunotherapy-containing treatment at 2011-2017 (n = 382) (hazard ratio [HR], 0.78; 95% confidence interval [CI], 0.61-0.99; P = 0.04). In the multivariate analysis, the use of immunotherapy at first-line treatment was not significantly associated with worse OS, whereas the use of immunotherapy at any lines was prognostic (HR, 0.54; 95% CI, 0.41-0.71; $P < 0.0001$). The proportion of pts who had received any second-line treatment was comparable among groups: 76%, 80%, and 71%, respectively. **Conclusions:** Our study suggests that the immunotherapy might partially have an impact on the survival improvement of real-world pts with HER2-negative AGC, which calls for the need for appropriate treatment strategies, including other agents. Research Sponsor: None.

405 **Poster Session**

Investigation of synergistic effects in trials of combination therapies with immune-checkpoint inhibitors in advanced gastric or gastroesophageal junction cancer. First Author: Jun Okui, Department of Surgery, Keio University School of Medicine, Tokyo, Japan

Background: Combinations of immune checkpoint inhibitors (ICI) and chemotherapy (CT) have been approved for gastric cancer. However, there is a hypothesis that this combination may blunt antitumor immune responses because most chemotherapeutic agents also target lymphocytes. The primary objective was to investigate that the ICI and CT does not interfere each other's therapeutic effects in advanced gastric cancer (GC) or gastroesophageal junction cancer (GEJC) patients. **Methods:** The reconstructed individual patient data was electronically extracted from the Kaplan-Meier curve of phase III randomized controlled trials (RCTs). The observed PFS curve of each constituent monotherapies was used to estimate simulated PFS curves expected under a model of independent drug action. If the observed curve demonstrated significantly better PFS than simulated curve, the combination of ICI and CT may have a synergistic effect, implying a superior outcome compared to simply adding the component monotherapy. **Results:** The study included 2,538 unresectable advanced, recurrent, or metastatic GC or GEJC patients from three RCTs comparing pembrolizumab (KEYNOTE-061, KEYNOTE-062 and KEYNOTE-859). In patients with programmed cell death ligand 1 (PD-L1) combined positive score (CPS) of 1 or greater, the 1-year and median PFS of the observed and simulated curves were 28.0% vs. 27.9%, and 6.89 months vs. 6.88 months, respectively. One sample log-rank test showed no significant differences between the observed and simulated curves (p = 0.107). In the subgroups with PD-L1 CPS $\geq$10 or $<$1, the 1-year PFS of the observed and simulated curves was 34.7% vs 32.8%, and 28.5% vs 26.8%. **Conclusions:** The observed PFS of ICT involving pembrolizumab was comparable to the simulated PFS estimated from the data for each monotherapy regardless of the magnitude of PD-L1 CPS. Although it was not clear whether potential synergies existed for ICT, these findings at least suggest that the benefits of ICI and CT are not interfering each other, thereby providing theoretical support for the efficacy of ICT in patients with advanced GC or GEJC. Research Sponsor: None.

406 **Poster Session**

PD-1 inhibitor (sintilimab) and fruquintinib plus SOX as conversion therapy for initially unresectable gastric/gastroesophageal junction adenocarcinoma (GC/GEJC): Updated results from a single-arm, phase 2 clinical trial. First Author: Fei Ma, Surgical Oncology, Henan Cancer Hospital/Affiliated Cancer Hospital of Zhengzhou University, Zhengzhou, China

Background: The evidence of conversion therapy for unresectable GC/GEJC is limited. Previously, preliminary results of the ongoing open-label, phase 2 study (NCT05177068) have showed promising R0 surgical conversion rate (62.1%), R0 resection rate (100%) and acceptable tolerability in unresectable GC/GEJC with sintilimab and fruquintinib plus SOX (ASCO 2024). Here we present updated results with longer follow-up duration, including more enrolled patients. **Methods:** Eligible patients were administered fruquintinib (4mg/d, po, qd, d1-14), sintilimab (200mg, iv, d1), oxaliplatin (130 mg/m^2, iv, d1) and S-1 (40-60mg based on BSA, po, bid, d1-14) every 3 weeks for 3 or 6 cycles. One more cycle of sintilimab plus SOX was given before resection. Patients were assessed for tumor response and surgical feasibility by radiologic imaging & MDT every 3 cycles. Primary endpoint was R0 resection rate. Secondary endpoints included pathological response, ORR, PFS, OS, and safety. **Results:** As of July 30, 2024, 42 pts (37 males/5 females) with a median age of 64 years (range: 43–76), 71% ECOG PS1, 55% GEJC were enrolled. Nine (21.4%) of the 42 pts had distant metastasis and five (11.9%) pts had more than one unresectable factors. The most common unresectable factors were extensive or bulky lymph nodes 54.8% (23), liver metastasis 11.9% (5), peritoneal metastasis 4.8% (2), para-aortic lymph node metastasis 2.4% (1), and local progression 38.1% (16). Of 35 evaluable pts, ORR was 68.6% and DCR was 97.1%. Among the 29 pts who had completed surgical conversion, the R0 resection rate was 100% (29/29) and R0 surgical conversion rate was 82.9% (29/35). 10.3% (3/29) pts achieved pathological complete response (pCR), and 20.7% (6/29) pts reached tumor regression grade (TRG) 0-1. Additionally, the pathological response rate (pRR) according to JCGC ($\geq$ Grade 1b) was 93.1% (27/29). Seven pts (16.7%) had grade 3 treatment-emergent adverse events, including 4 cases of anemia, 2 cases of neutrophil count decreased, and 1 case of each (lymphocyte count decreased, gastrointestinal hemorrhage, diarrhea, and immune-related pneumonitis). No severe surgery-related complication was observed. **Conclusions:** Fruquintinib combined with sintilimab and SOX yielded very high R0 conversion rate and R0 resection rate with a manageable safety profile in unresectable GC/GEJC, representing a potential and feasible conversion therapy regimen for this population. Survival data will continue to be followed up. Clinical trial information: NCT05177068. Research Sponsor: None.

407 **Poster Session**

A phase II study of fruquintinib plus sintilimab as a second-line therapy for advanced gastric and gastroesophageal junction adenocarcinoma (GC/GEJC): Updated results. First Author: Min Jin, Cancer Center, Union Hospital, Tongji Medical College, Huazhong University of Science and Technology, Wuhan, China

Background: Preliminary results of the open-label, single-arm phase II study (NCT05625737) had demonstrated the combination of fruquintinib and sintilimab as a second-line therapy had a promising anti-tumor activity and an acceptable safety profile in pts with advanced GC/GEJC (ASCO-GI 2024. Poster 332). Here, we update the results with a focus on subgroup analyses. **Methods:** Patients (pts) aged 18-75 years who were HER2-negative and had failed first-line standard treatment were enrolled. Eligible pts received 4mg of fruquintinib orally once daily on days 1-14 and 200 mg of sintilimab intravenously on day 1, with treatment repeated every 3 weeks. An optimal Simon two-stage design was employed. The primary endpoint was objective response rate (ORR). Secondary endpoints included disease control rate (DCR), progression-free survival (PFS), overall survival (OS), and safety. **Results:** At data cut-off (August 25, 2024), 29 pts (7 in the first stage; 22 in the second stage) with median age 51 (range 33–73) years old had been enrolled. 41.4% were male, 37.9% had ECOG PS 1, 69% had lymph node metastasis and 72.4% had previously received immunotherapy. Among the 24 efficacy evaluable pts, the ORR was 33.3% (95% CI 15.6–55.3) and DCR was 66.7% (95% CI 44.7–84.4). After a median follow-up of 13.14 months, the median PFS was 5.13 (95% CI: 2.73–7.03) months and OS result was pending maturity. Pts who had lymph node metastasis exhibited higher ORR compared to pts without lymph node metastasis (ORR-47.1 vs 0%). In particular, pts without prior immunotherapies were more likely to achieve higher response rate (37.5 vs 33.3%) and had an obvious trend for longer PFS than pts with prior immunotherapies (9.00 vs 5.13 months). Majority of treatment-emergent adverse events (TEAEs) were grade 1-2 and the most frequently reported ones ($\geq$20%) were white blood cell decreased (33%), hypertriglyceridemia (25%), lymphocyte count decreased (21%), abdominal pain (21%) and decreased appetite (21%). Grade 3/4 TEAEs occurred in only one patient, consisting of increased blood bilirubin levels, aspartate transaminase elevation, and alanine transaminase elevation. **Conclusions:** Fruquintinib combined sintilimab was well tolerated, with encouraging antitumor activity as second-line treatment for advanced GC/GEJC, especially in pts without prior immunotherapies. Further studies in larger cohorts to validate these findings are warranted. Clinical trial information: NCT05625737. Research Sponsor: None.

408 **Poster Session**

Conversion from radical esophagectomy to definitive chemoradiotherapy after neoadjuvant chemotherapy for advanced esophageal squamous cell carcinoma: Treatment options based on chemotherapy response. First Author: Kiyoshi Narita, Department of Gastroenterological Surgery, Aichi Cancer Center Hospital, Nagoya, Japan

Background: Radical esophagectomy after neoadjuvant chemotherapy (NAC) is the established strategy for resectable advanced esophageal cancer. However, some patients convert to definitive chemoradiotherapy (dCRT) after NAC due to reasons such as their wishes or disease progression, and their prognosis remains uncertain. We aimed to investigate the prognosis of patients who converted to dCRT. **Methods:** Patients who underwent NAC for resectable advanced esophageal squamous cell carcinoma between 2006 and 2020 were enrolled retrospectively. The prognostic impact of subsequent treatment after NAC, planned surgery or conversion to dCRT, was compared. **Results:** We analyzed 686 patients, 70 who converted to dCRT (dCRT group) and 616 who underwent surgery (Surg group). The dCRT group had a poorer prognosis than the Surg group, with more advanced tumors and poorer response to NAC. Therefore, further analysis was conducted by categorizing patients as Responders (complete or partial response) and Non-responders (stable or progressive disease) to NAC. Among Responders in the dCRT group, 76.7% achieved a complete response, and the 5-year esophageal preservation survival rate was 66.9%. The 5-year survival rates for Responders were 77.5% and 71.3% in the dCRT and Surg groups. Multivariable analysis showed that dCRT did not worsen prognosis (P = .706, hazard ratio = 1.13; 95% confidence interval, 0.59 to 2.16). Conversely, in Non-responders, dCRT had a significantly poorer prognosis, with 5-year survival rates of 22.3% and 45.1% in the dCRT and the Surg groups ($P < .001$). **Conclusions:** For patients responding to NAC, conversion to dCRT is considered a potential treatment option. The possibility of chemoselection has been suggested in the choice of treatment for esophageal squamous cell carcinoma. Research Sponsor: None.

409 **Poster Session**

Zimberelimab platform study: Safety and efficacy of zimberelimab in combination with futibatinib and chemotherapy in patients with first-line advanced or metastatic esophageal carcinoma. First Author: Shota Fukuoka, Department of Advanved Medical Development, Cancer Institute Hospital, Japanese Foundation for Cancer Research, Tokyo, Japan

Background: Zimberelimab: AB122 (Zim) platform study is a Phase 1 a/b study with multiple cohorts designed to evaluate the safety and efficacy of Zim, an anti-PD-1 antibody. Futibatinib (Futi) is an FGFR inhibitor that covalently binds FGFR1–4; FGFR inhibitors have been reported to modulate the tumor immune microenvironment, including reducing MDSCs and inhibiting CAFs. These immunomodulatory effects are expected to enhance the efficacy of the anti-PD-1 antibodies. The safety and efficacy of the combination of Zim, Futi, and chemotherapy were evaluated in patients (pts) with first-line advanced or metastatic esophageal cancer (EC) in Japan. **Methods:** Key eligibility criteria included age $\geq$ 18 years, histologically confirmed advanced or metastatic EC with measurable lesions by RECIST v1.1 regardless of FGFR and PD-L1 status, previously untreated, adenocarcinoma or squamous cell carcinoma, and ECOG PS 0 or 1. Patients received Zim 360 mg on day 1, Futi 20 mg QD, fluorouracil 800 mg/m2 on days 1–5, and cisplatin 80 mg/m2 on day-1 of a three-week cycle. The primary endpoint was dose-limiting toxicity (DLT). The secondary endpoints included adverse events, antitumor activity, pharmacokinetics, pharmacodynamics, and pharmacogenomics. **Results:** As of May 10, 2024, 43 pts were enrolled. At the data cutoff date for this analysis (July 15, 2024), treatment was ongoing in 23 pts. Safety and efficacy were evaluated in 43 and 41 pts, respectively. The median age of the pts was 61 years (53–70). An ECOG PS score of 0 was observed in 31 pts. DLTs were evaluated in the first three pts, and none were reported. Common treatment-emergent adverse events (TEAEs) were hyperphosphatemia (81.4%), diarrhea (67.4%), nausea (62.8%), decreased appetite (62.8%), constipation (55.8%), anemia (53.5%), stomatitis (51.2%), neutrophil count decreased (37.2%), alanine aminotransferase increased (34.9%), and palmar-plantar erythrodysaesthesia syndrome (30.2%). Grade $\geq$ 3 TEAEs occurred in 31 pts (72.1%), and common events ($\geq$ 30%) were anaemia (30.2%) and neutrophil count decreased (30.2%). The objective response rate was 70.7% (95% CI: 54.5–83.9), and the confirmed overall response rate was 58.5% (95% CI: 42.1–73.7). The disease control rate was 92.7% (95% CI: 80.1–98.5). Median duration of response was 5.6 months (95% CI: 4.1–5.6), and median progression-free survival was 4.9 months (95% CI: 3.4–6.7). **Conclusions:** The combination of Zim, Futi, and chemotherapy has a manageable safety profile without new safety signals and shows promising antitumor activity in pts with advanced or metastatic EC. As more than half of the enrolled pts are still undergoing treatment, these are preliminary data; follow-up is needed to evaluate the viability of this combination appropriately. Clinical trial information: NCT04999761. Research Sponsor: TAIHO PHARMACEUTICAL CO., LTD.

410 **Poster Session**

Neoadjuvant SHR-1701 (a bifunctional anti-PD-L1/TGF-βRII agent) combined with chemoradiotherapy for resectable locally advanced esophageal squamous cell carcinoma (ESCC): A phase II trial. First Author: Chengixn Liu, Department of Radiation Oncology, Shandong Cancer Hospital and Institute, Shandong First Medical University and Shandong Academy of Medical Sciences, Jinan, China

Background: Neoadjuvant chemotherapy or chemoradiotherapy followed by surgery is the standard of care for resectable locally advanced ESCC. SHR-1701, a new bifunctional fusion protein composed of a monoclonal antibody against PD-L1 fused with the extracellular domain of TGF-β receptor II, may enhance antitumor activity in combination with neoadjuvant standard therapies in ESCC patients (pts). The aim of this phase II trial is to determine the safety and efficacy of neoadjuvant chemoradiotherapy plus SHR-1701 followed by esophagectomy in pts with locally advanced resectable ESCC. **Methods:** Pts with resectable thoracic ESCC, diagnosed as clinical stage of cT1b-cT2N+M0 or cT3-cT4aNxM0 per AJCC 8th were eligible. Preoperative therapy included SHR-1701 (30 mg/kg every 3 weeks for 2 cycles), albumin paclitaxel (50 mg/m^2, once a week for 5 weeks), carboplatin (AUC=2, once a week for 5 weeks) and radiotherapy (41.4Gy in 23 fractions). Following esophagectomy, pts received SHR-1701 up to 1 year. The primary endpoint was pathological complete response (pCR) rate in the per-protocol population. Secondary endpoints included R0 resection rate, major pathological response (MPR) rate, disease free survival (DFS) and safety. **Results:** Between Dec. 2021 and Feb. 2023, 43 pts were screened and 41 met the inclusion criteria, among whom the clinical stage I, II, III, and IVa at baseline were 1 (2.4%), 8 (19.5%), 31 (75.6%), and 1 (2.4%), respectively. All 41 pts received neoadjuvant SHR-1701 combined with chemoradiotherapy. As of Mar. 2024, 30 pts underwent surgery, and all achieved R0 resection. There was no in-hospital and postoperative 30-day mortality. 9 (30.0%) pts achieved pCR in both primary tumor and lymph nodes (ypT0N0), and 12 (40.0%) pts had complete pathological response of the primary tumor with residual disease in lymph nodes alone (ypT0N+). Treatment-related adverse events (TRAEs) occurred in all pts, and most of TRAEs were grade 1-2. Notable toxicity included pneumonitis (31.7%) and anastomotic leak (12.2%). **Conclusions:** The addition of SHR-1701 to neoadjuvant chemoradiotherapy in ESCC demonstrated promising efficacy with acceptable toxicity, and might be a promising approach for neoadjuvant treatment. Clinical trial information: ChiCTR2000041562. Research Sponsor: None.

411 **Poster Session**

Bifunctional anti-PD-L1/TGF-βRII agent SHR-1701 plus chemotherapy as first-line treatment for advanced esophageal squamous cell carcinoma (ESCC): Survival results of a phase II trial. First Author: Yi Wang, Qilu Hospital, Shandong University, Jinan, China

Background: Recently, several clinical trials have demonstrated synergistic efficacy by combining immune checkpoint inhibitors (ICIs) with chemotherapy. However, only a subset of patients (pts) derived benefits. Combining ICIs with agents blocking immunosuppressive pathway may expand the clinical benefit of ICIs to more pts. Transforming growth factor β (TGF-β) participates in tumor immune escape. SHR-1701, a novel bifunctional fusion protein consisting of a monoclonal antibody targeting PD-L1 fused with the extracellular domain of TGF-β receptor II, was evaluated in a phase II trial (ChiCTR2000039909) to assess its efficacy and safety when combined with chemotherapy for pts with unresectable locally advanced, recurrent or metastatic ESCC in China. **Methods:** This trial enrolled treatment-naive pts with histologically or cytologically confirmed unresectable locally advanced, recurrent, or metastatic ESCC. Eligible pts received SHR-1701 (30mg/kg, iv, d1, q3w) in combination with up to six cycles of albumin-bound paclitaxel (125mg/m^2, iv, d1, d8, q3w) and cisplatin (75mg/m^2, iv, d1, q3w). For those without progressive disease, maintenance treatment was administered with SHR-1701 monotherapy until disease progression or unacceptable toxicity. The primary endpoint was objective response rate (ORR). Secondary endpoints included progression-free survival (PFS), overall survival (OS), disease control rate (DCR) and safety. **Results:** At the data cut-off, March 1, 2024, 24 pts had received at least one dose of the study treatment and were included in both the intention-to-treat (ITT) analysis and safety analysis sets. 3 and 12 pts achieved confirmed complete and partial responses, respectively. The confirmed ORR was 62.5% and DCR was 87.5%. With a median follow-up of 20.8 months (95% CI, 15.5-NR), the median duration of response was 9.1 months (95% CI, 6.9-NR). The median PFS was 10.8 months (95% CI, 7.8-NR) and the median OS was 26.1 months (95% CI, 8.6-NR). 12-month PFS and OS rates were 41.4% (95% CI, 22.5-76) and 64.8% (95% CI, 47.8-87.9), respectively. Grade 3-4 treatment-related adverse events occurred in 11 (45.8%) pts and no treatment-related death was observed. **Conclusions:** The results highlighted that SHR-1701 plus chemotherapy as first-line therapy maintained long-term, durable survival benefit in pts with advanced ESCC. Clinical trial information: ChiCTR2000039909. Research Sponsor: None.

412 **Poster Session**

Evaluation of event-free survival (EFS) as a surrogate for overall survival (OS) in patients with locally advanced esophageal carcinoma (LA EC) receiving definitive chemoradiotherapy (dCRT). First Author: Manish A. Shah, Department of Hematology & Medical Oncology, Weill Cornell Medicine, New York, NY

Background: Drug development in esophageal cancer is rapidly evolving, and identifying surrogates for patient outcomes to accelerate clinical drug development remains an important unmet medical need. Although early clinical endpoints such as EFS, disease free survival (DFS) are proven to be valid surrogates for overall survival (OS) in early esophageal (Ajani JA *et al.* 2022), gastric (Oba K *et al.* 2013; Wainberg ZA *et al.* 2023), and gastroesophageal junction tumors (Ronellenfitsch U. *et al.* 2013) after disease resection, there are no such surrogacy data evaluating patients who underwent non-operative management. Specifically, there has been no prior systemic evaluation of the predictive value of EFS as a surrogate for OS among LA EC patients treated with dCRT. This study evaluated the trial-level correlation between EFS and OS in unresectable LA EC patients receiving dCRT. **Methods:** A systematic literature review was conducted to identify randomized controlled trials (RCTs) of dCRT interventions in unresectable LA EC (search cutoff date: April 2023). Studies reporting EFS (or similar outcomes such as progression free survival (PFS), defined as time from randomization to local, regional, or distant recurrence, or death) and OS were included. EFS and OS hazard ratios (HRs) of each treatment comparison were extracted. Correlation between the log(HR) of EFS and OS was assessed using weighted linear regression (WLR). The coefficient of determination (R^2) was used to evaluate the strength of the correlation. A good surrogate was defined as an $R^2 \geq 0.65$ (Ciani *et al.* 2017) or an $R^2 \geq 0.75$ (Lassere *et al.* 2008). Leave-one-out cross-validations and sensitivity analyses were conducted to assess the robustness of the WLR model and correlation analyses, respectively. **Results:** The final analysis included 14 between-treatment comparisons from 11 RCTs (total number of patients = 2,812). We found a strong statistical correlation between EFS and OS, with an estimated slope coefficient of 0.91 ($p < 0.001$) and R^2 of 0.80 (95% confidence interval [CI]: 0.38, 0.96). In the leave-one-out cross validation, 71.4% of observed OS HRs fell within the predicted 95% CIs. Sensitivity analyses showed R^2 of 0.75 among studies conducted in China and R^2 of 0.73 after including studies that did not report EFS definition. **Conclusions:** In a pooled analysis of 11 clinical trials corresponding to over 2800 patients, we observed a strong statistical correlation between EFS and OS outcomes. This analysis supports the utility of EFS as a valid surrogate of OS in patients with unresectable LA EC receiving dCRT. Research Sponsor: Merck & Co., Inc.

413 Poster Session

Response characteristics of tislelizumab (TIS) plus chemotherapy (chemo) in first-line (1L) treatment of locally advanced or metastatic esophageal squamous cell carcinoma (ESCC): A post hoc analysis of RATIONALE-306.

First Author: Yi Li, Fifth Medical Center, Chinese PLA General Hospital, Beijing, China

Background: The global phase 3 RATIONALE-306 study (NCT03783442) demonstrated superior OS, PFS, ORR and manageable safety profile of TIS + chemo as 1L treatment of ESCC, which was well maintained at 3 years. This post-hoc analysis aims to explore the response characteristics of responders in TIS + chemo arm from RATIONALE-306. **Methods:** Advanced or metastatic ESCC patients (pts) were randomly assigned (1:1) to receive 1L TIS + chemo or placebo + chemo. Assessments of tumor response were done every 6 weeks for the first 48 weeks, then every 9 weeks thereafter per RECIST v1.1. Time to response (TTR), depth of response (DpR) and time to maximum response (TTMR) were assessed in responders (pts who achieved CR or PR). TTR was defined as the time from randomization to the first occurrence of a CR or PR, DpR as the percentage of maximal tumor reduction from the baseline of target lesion sum of diameters, and TTMR as the time from randomization to the maximum tumor shrinkage. **Results:** As of 24 November 2023, 207 pts (68.3% of the ITT population) in the TIS + chemo arm achieved responses, with a median TTR of 6.1 weeks (range: 3.3 - 101.3). First response was detected in 64.7% of responders within 8 weeks, 22.2% between > 8 and ≤ 14 weeks, and 13.0% after 14 weeks; the median OS was 21.8, 23.1 and 29.0 mos, respectively in pts within the three TTR categories, and the median OS post first response was 21.3, 21.3 and 23.2 mos (Table). DpR > 30% to ≤ 50%, > 50% to ≤ 80%, and > 80% to ≤ 100% was observed in 29.8%, 50.2% and 20.0% of responders, respectively, and the median OS was 16.1, 23.7, and 34.5 mos. The median TTMR was 19 weeks (range: 3.3 - 201.0), with 32.7% of responders achieved maximum response within 14 weeks, 34.6% between > 14 and ≤ 28 weeks, and 32.7% after 28 weeks, respectively. Median OS of pts with the three TTMR categories was 14.3, 17.4, and 39.9 mos, respectively, and median OS post-maximum response was 12.6, 14.3, and 24.8 mos. **Conclusions:** In the TIS + chemo arm, 64.7% of responders achieved response within 8 weeks, while still 35.2% of responders achieved response at later tumor assessments, and the OS benefits were comparable between early responders and late responders. Responders with deeper tumor response and/or longer TTMR tended to have longer OS. Clinical trial information: NCT03783442. Research Sponsor: BeiGene, Ltd.

	Cutoff	Patients, n (%)	OS, median (95% CI), mos	OS from response*, median (95% CI), mos
TTR	> 0 to ≤ 8w	134 (64.7)	21.8 (16.3, 25.0)	21.3 (16.0, 24.5)
	> 8 to ≤ 14 w	46 (22.2)	23.1 (16.3, 26.0)	21.3 (14.4, 24.3)
	> 14w	27 (13.0)	29.0 (20.4, NA)	23.2 (15.9, NA)
DpR	> 30% to ≤ 50%	61 (29.8)	16.1 (13.2, 20.8)	\
	> 50% to ≤ 80%	103 (50.2)	23.7 (19.6, 25.6)	\
	> 80% to ≤ 100%	41 (20.0)	34.5 (23.7, NA)	\
TTMR	> 0 to ≤ 14w	67 (32.7)	14.3 (8.7, 17.6)	12.6 (8.1, 15.4)
	> 14 to ≤ 28w	71 (34.6)	17.4 (15.8, 24.1)	14.3 (11.8, 20.1)
	> 28w	67 (32.7)	39.9 (28.6, NA)	24.8 (17.5, 39.4)

*From first response for TTR, and from maximum response for TTMR.

414 Poster Session

Tislelizumab (TIS) + chemotherapy (chemo) vs placebo (PBO) + chemo as first-line (1L) treatment in gastric/gastroesophageal junction adenocarcinoma (GC/GEJC) patients with/without peritoneal or liver metastases: A post hoc analysis of RATIONALE-305 study.

First Author: Miaozhen Qiu, Department of Medical Oncology, Sun Yat-sen University Cancer Center, State Key Laboratory of Oncology in South China, Collaborative Innovation Center for Cancer Medicine, Sun Yat-sen University, Guangzhou, China

Background: Gastric cancer patients with peritoneal or liver metastases had poor prognosis, and the efficacy of immunotherapy in these patients remains unclear. The global phase 3 RATIONALE-305 study (NCT03777657) demonstrated that tislelizumab combined with chemotherapy could bring survival benefits to 1L treatment of GC/GEJC patients. Here, we assessed the efficacy of TIS + chemo vs PBO + chemo in patients with/without peritoneal or liver metastases in RATIONALE-305. **Methods:** Patients with systemic treatment-naïve GC/GEJC were randomly assigned (1:1) to receive either TIS + chemo or PBO + chemo. Regression analyses were conducted to explore the associations between peritoneal or liver metastases and OS. Relative treatment effect between tislelizumab and placebo was assessed in each subgroup. The Kaplan-Meier method was used to estimate the median OS, and hazard ratios (HRs) for OS were estimated using Cox proportional hazards models. **Results:** Among the 997 randomized patients (TIS + chemo, n=501; PBO + chemo, n=496), 434 (43.5%) had peritoneal metastases (220 in TIS arm; 214 in PBO arm) and 378 (37.9%) had liver metastases (190 in TIS arm; 188 in PBO arm) at baseline. Regression analyses showed peritoneal and liver metastases were significantly associated with shorter OS. Baseline characteristics were balanced between TIS and chemo arms within each subgroup, including PD-L1 expression levels. As of data cut-off on Feb 28, 2024, OS was longer in the TIS arm compared with the PBO arm in patients with peritoneal metastases (HR= 0.78, 95% CI 0.64-0.96) or without (HR = 0.79, 95% CI 0.65-0.95). OS improvement was also observed in patients with liver metastases (HR = 0.77, 95% CI 0.62-0.96) or without (HR = 0.80, 95% CI 0.67-0.95). **Conclusions:** This post hoc analysis demonstrated OS improvement with TIS + chemo vs PBO + chemo in GC/GEJC patients with/without peritoneal or liver metastases. To our knowledge, RATIONALE-305 is the first global pivotal study reporting survival benefits with TIS + chemo as 1L treatment for GC/GEJC, irrespective of peritoneal or liver metastases. Clinical trial information: NCT03777657. Research Sponsor: BeiGene, Ltd.

	With metastases		Without metastases	
	TIS arm	PBO arm	TIS arm	PBO arm
Peritoneum	n=220	n=214	n=281	n=282
Median OS, mo (95% CI)	12.3 (10.6-14.3)	11.8 (10.6-13.0)	17.3 (15.0-20.3)	14.0 (12.6-16.0)
HR (95% CI)	0.78 (0.64-0.96)		0.79 (0.65-0.95)	
Liver	n=190	n=188	n=311	n=308
Median OS, mo (95% CI)	13.9 (11.4-15.6)	12.9 (10.9-14.5)	16.0 (13.6-18.0)	12.9 (11.9-14.4)
HR (95% CI)	0.77 (0.62-0.96)		0.80 (0.67-0.95)	

415 Poster Session

Clinical outcomes and changes in body composition following endoscopic submucosal dissection versus total gastrectomy in patients with proximal early gastric cancer.

First Author: Ga Hee Kim, Institute of Gastroenterology, Department of Internal Medicine, Yonsei Cancer Center, Yonsei University College of Medicine, Seoul, South Korea

Background: Sarcopenia is associated with poor surgical outcomes and prognosis in patients with gastric cancer. However, the relationship between changes in skeletal muscle following endoscopic submucosal dissection (ESD) versus total gastrectomy (TG) in patients with proximal early gastric cancer (EGC) remains unclear. This study aimed to investigate the clinical outcomes and skeletal muscle mass changes between ESD and TG in patients with proximal EGC. **Methods:** We retrospectively reviewed 329 patients who underwent either ESD (n=127) or TG (n=202) for cT1N0M0 gastric cancer in the upper third of the stomach between 2015 and 2019 at a tertiary care center. The skeletal muscle index (SMI), visceral/subcutaneous fat, and abdominal circumference were measured preoperatively, and at 1 and 3 years after surgery, using cross-sectional computed tomography. We compared these parameters between the two groups and assessed both short- and long-term outcomes, as well as prognostic factors. **Results:** No significant differences were observed between the two groups with respect to overall survival (P=0.266), recurrence-free survival (P=0.079), or disease-specific survival (P=0.24). The recurrence rate was four (3.1%) in the ESD group and two (0.9%) in the TG group. Of the four recurrences in the ESD group, two were cured by gastrectomy, and two by ESD. Two recurrences in the TG group died of cancer progression. In the ESD group, all deaths were unrelated to gastric cancer, compared to 83.3% in the TG group (P=0.376). Patients in the ESD group showed more preserved SMI (P<0.001), visceral/subcutaneous fat (P<0.001), and abdominal circumference (P<0.001) at 1 and 3 years after surgery, compared to the TG group. The incidence of sarcopenia was significantly lower in the ESD group at postoperative years 1 (22.2% vs. 34.7%, P=0.02) and years 3 (25.7% vs. 40.6%, P=0.001). Sarcopenia had a significant impact on survival, with 5-year OS rates being lower in patients with preoperative sarcopenia (85.3% vs. 95.6%, P=0.005). Multivariate analysis revealed that the prognostic nutritional index was significantly associated with OS (hazard ratio, 0.86; 95% confidence interval: 0.75–0.99, P=0.03). **Conclusions:** Although OS was comparable between the ESD and TG groups, ESD was associated with better preservation of skeletal muscle and visceral/subcutaneous fat. These results suggest that ESD should be considered for patients with proximal EGC who are at high risk for sarcopenia. Research Sponsor: None.

416 Poster Session

Outcomes of neoadjvuant chemotherapy versus chemoradiation for esophageal adenocarcinoma: A National Cancer Database analysis.

First Author: Ravi Shridhar, AdventHealth Cancer Institute, Orlando, FL

Background: While neoadjuvant chemoradiation (NCRT) has been the acceptable standard of care for esophageal adenocarcinoma (EAC), there has been a paradigm shift from recent trials have shown that neoadjuvant chemotherapy (NAC) is either equivalent or superior to NCRT. We sought to examine the outcomes of NAC versus NCRT in esophageal cancer from the National Cancer Database (NCDB). **Methods:** Utilizing the NCDB, we identified esophagectomy patients with EAC who underwent multiagent NCT versus multiagent NCRT. Overall Survival (OS) was analyzed with the Kaplan-Meier method and multivariate analysis (MVA) was performed to identify predictors of OS. Propensity score matching (PSM) was used to correct for baseline differences between groups. **Results:** After PSM, we identified 1007 patients in each group. Groups were equally balanced in age, gender, T/N stage, grade, and facility volume. There was improved lymphadenectomy and higher use of adjuvant therapy in NCT patients. There were significant improvements in R0 resection and pathologic response associated with NCRT patients. There was a significant improvement in OS in NCT versus NCRT. The median and 5-year OS was 42.7 months and 42% in NCT patients versus 34.2 months and 35% in NCRT patients (p = 0.001). For pathologic complete responders, median and 5-year OS was 101 months and 64% in NCT patients versus 71.2 months and 53% in NCRT patients (p = 0.04). On MVA, improved mortality was associated with female gender, pathologic N0, >10 nodes removed, pathologic partial or complete response, R0, well/moderate grade, NCT, adjuvant chemotherapy, and higher facility volume. There was also no survival benefit to adjuvant chemoradiation. **Conclusions:** Despite higher R0 and pathologic response associated with NCRT, NCT was superior to NCRT for OS in EAC patients. NCRT needs to be re-examined in its role as a treatment recommendation for operable EAC patients. Research Sponsor: None.

Multivariate analysis for overall survival.

Variable	HR	95% CI	*P* value
Age	1.01	1.00 – 1.02	0.004
Sex			
Male	Reference		
Female	0.79	0.64 – 0.98	0.03
Charlson Deyo score			
0	Reference		
1	1.05	0.91 – 1.21	0.54
Pathologic N-stage			
N0	Reference		
N+	1.82	1.58 – 2.10	<0.001
LN removed			
<10	Reference		
>10	0.81	0.68 – 0.97	0.02
Response			
None	Reference		
Partial	0.81	0.69 – 0.94	0.008
Complete	0.61	0.50 – 0.75	<0.001
Margin status			
R0	Reference		
R1	1.24	1.01 – 1.52	0.04
Grade			
Well/moderate	Reference		
Poor	1.35	1.18 – 1.54	<0.001
Preop radiation			
Yes	Reference		
No	0.77	0.67 – 0.88	<0.001
Adjuvant therapy			
None	Reference		
Chemotherapy	0.77	0.65 – 0.92	0.004
Chemoradiation	0.83	0.64 – 1.08	0.17
Facility volume			
Low (≤10/year)	Reference		
Medium (11-19/year)	0.86	0.72 – 1.03	0.10
High (≥20/year)	0.78	0.67 – 0.90	<0.001

417 Poster Session

Comparison between neoadjuvant chemotherapy followed by surgery and definitive chemoradiotherapy for survival in patients with resectable esophageal squamous cell carcinoma (JCOG2305A). First Author: Motoo Nomura, Department of Head and Neck Oncology and Innovative Treatment, Graduate School of Medicine, Kyoto University, Kyoto, Japan

Background: Neoadjuvant triplet chemotherapy followed by surgery (Neo-S) has become the standard treatment for resectable esophageal squamous cell carcinoma (ESCC) based on the results of JCOG1109. However, definitive chemoradiotherapy (CRT) remains an option for organ preservation with curative intent. This analysis aimed to explore subgroups of patients undergoing CRT, including salvage treatment, to identify those with survival outcomes potentially equivalent to Neo-S. **Methods:** Pooled data from two clinical trials of patients with resectable (T1N1-3M0 and T2-3NanyM0) ESCC. JCOG0909 was a single-arm trial that evaluated upfront CRT consisting of 5-FU (1000 mg/m^2 on day1-4, 29-32) and cisplatin (80 mg/m^2 on day1, 29) combined with radiotherapy at a dose of 50.4 Gy, including salvage treatment. If there was no progression at the time of salvage treatment, it was not regarded as an event for progression-free survival (PFS) in JCOG0909. JCOG1109 compared neoadjuvant treatment with doublet chemotherapy, triplet chemotherapy, and doublet chemotherapy with radiotherapy followed by surgery. The patients were enrolled in JCOG0909 between 2010 and 2014 and in JCOG1109 between 2012 and 2018. Subgroup analyses of clinical data, including tumor location, cT, cN, and clinical stage, were conducted on overall survival (OS) and PFS using Cox proportional hazards regression models for patients with JCOG0909 and JCOG1109. When a subgroup with a multivariable hazard ratio (HR) of more than 0.91 was detected, CRT including salvage treatment was considered not to be inferior to Neo-S in that subgroup. **Results:** Ninety-three patients in the CRT group from JCOG0909 and 196 patients in the Neo-S group from the triplet chemotherapy arm of JCOG1109 were included in this analysis. Although there were no significant differences, PFS and OS tend to be better in the Neo-S group than in the CRT group (PFS, HR 0.72, 95% confidence interval [CI] 0.50-1.03; OS, HR 0.89, 95% CI 0.59-1.36). For patients with Ut/Mt or cT1-2 disease, OS in the Neo-S group had a HR of 0.91 or more compared with those in the CRT group (Table). **Conclusions:** Subgroups in which OS after CRT including salvage treatment was not inferior to that of Neo-S were identified. Research Sponsor: None.

Hazard ratios of Neo-S to CRT by subgroups in multivariable Cox regression.

	Progression-free survival		Overall survival	
Subgroups	Hazard ratio	95% confidence interval	Hazard ratio	95% confidence interval
Ut/Mt	0.85	0.56-1.28	1.13	0.70-1.83
Lt	0.45	0.21-0.97	0.44	0.19-1.00
cT1-2	0.88	0.46-1.68	1.45	0.65-3.24
cT3	0.66	0.42-1.02	0.72	0.44-1.16
cStage IB+II	0.79	0.45-1.36	1.06	0.56-2.00
cStage III	0.75	0.45-1.23	0.92	0.52-1.62
Overall	0.72	0.50-1.03	0.89	0.59-1.36

Ut upper thoracic esophagus, Mt middle thoracic esophagus, Lt lower thoracic esophagus.

418 Poster Session

Phase II study of neoadjuvant chemotherapy with fluorouracil, leucovorin, oxaliplatin and docetaxel for resectable esophageal squamous cell carcinoma. First Author: Shun Yamamoto, Department of Head and Neck, Esophageal Medical Oncology, National Cancer Center Hospital, Tokyo, Japan

Background: Based on the JCOG1109 trial, neoadjuvant docetaxel, cisplatin, and fluorouracil (DCF) followed by surgery has become the standard of care for resectable locally advanced esophageal squamous cell carcinoma (ESCC). Although the combination of fluorouracil, leucovorin, oxaliplatin, and docetaxel (FLOT) demonstrated benefits for esophageal adenocarcinoma as a perioperative therapy, its safety and efficacy for locally advanced ESCC have not been evaluated. **Methods:** We conducted a multicenter phase II study of neoadjuvant FLOT therapy for ESCC. Patients with cT1N1-3M0-1 or cT2-3N0-3M0-1 (only supraclavicular lymph node (SCLN) metastasis is included as M1) based on the 8th edition of the UICC TNM staging system were eligible. Neoadjuvant chemotherapy consisted of oxaliplatin (85 mg/m^2), docetaxel (50 mg/m^2), and l-leucovorin (200 mg/m^2) on day 1, and continuous infusion of fluorouracil (2600 mg/m^2/day) for 24 hours. This regimen was repeated every 2 weeks with a maximum of four cycles. The prophylactic antibody and G-SCF were not used mandatory. After completion of neoadjuvant chemotherapy, esophagectomy with extended lymphadenectomy was performed. Adjuvant treatment was prohibited for all patients. The primary endpoint was the pathological response rate (pRR), defined as the Grade 2 (more than two-thirds of the tumor is necrotic or fibrotic) or 3 (no viable tumor cells), based on the Japanese Classification of Esophageal Cancer. The sample size was determined based on an expected pRR of 38%, aiming for the lower bound of the 95% confidence interval to exceed the predetermined threshold of 20%. The expected number of patients to be enrolled was 60, with enrollment to be stopped when 45 patients with negative SCLN were enrolled. **Results:** Fifty-four patients were enrolled between September 2020 and January 2024. Patients with cStage I/II/III/IVB were 3/16/27/8. Of 54 patients, 45 patients were M0 without SCLN metastasis. Excluding 1 patient who did not receive any treatment after enrollment, 53 patients were included in the full analysis set. During chemotherapy, the most common grade 3 or 4 toxicities were neutropenia (73.6%), and leukopenia (22.6%). Febrile neutropenia was observed in 1 patient (1.9%). Finally, 46 patients underwent surgery. No treatment-related deaths were observed and the incidence of operative morbidity was tolerable. The pRR was 43.4% (23/53) (95% CI 29.8-57.7, p=00002). This study met the primary endpoint. The radical resection rate was 83.0% (44/53). The pathological complete response rate was 13.2% (7/53). **Conclusions:** Neoadjuvant FLOT therapy showed a promising pathological response with acceptable toxicities. It was noteworthy that the incidence of febrile neutropenia was relatively lower with compared to neoadjuvant DCF therapy. This regimen might be a treatment option for locally advanced ESCC. Clinical trial information: jRCTs031200094. Research Sponsor: None.

419 Poster Session

Therapeutic value of para-aortic lymph node dissection in gastric cancer with extensive lymph node metastasis: Integrated analysis of three phase II trials (JCOG2212A). First Author: Takeyuki Wada, Department of Gastric Surgery, National Cancer Center Hospital, Tokyo, Japan

Background: Gastric cancer (GC) sometimes forms metastatic bulky nodes around celiac artery or its branches (Bulky N), or metastasizes to para-aortic node (PAN) without any other distant metastasis. These severe nodal metastases are seldom cured by surgery alone. Japan Clinical Oncology Group (JCOG) defined these tumors as an extensive lymph node metastasis (ELM), and has developed multidisciplinary treatment through 3 phase II trials (JCOG0001, 0405, and 1002). JCOG0405 and 1002 reported the efficacy of D2 gastrectomy and PAN dissection (PAND) after neoadjuvant chemotherapy for GC with ELM with favorable survival outcomes. However, it remains unclear whether PAND truly contributed to the survival. The aim of this study was to clarify an optimal lymph node dissection area for GC with ELM. **Methods:** Therapeutic value index (TI) was investigated using an integrated dataset of JCOG0001, 0405, and 1002. Irinotecan and cisplatin, S-1 and cisplatin, or docetaxel, cisplatin and S-1 were used as neoadjuvant chemotherapy in JCOG0001, 0405, or 1002, respectively. Patients were classified into Bulky N group (only bulky N without PAN) and PAN group (PAN regardless of bulky N). TI was calculated by the following formula: proportion of metastasis (%) $\times$ 5-year relapse-free survival (5y-RFS, %) of patients with metastasis in the respective lymph node area / 100. Subgroup analysis by pathological response was also examined using the Japanese Classification of Gastric Carcinoma (JCGC) criteria. Grade $\geq$ 2a in JCGC criteria was defined as major pathological response (MPR). **Results:** A total of 122 patients were analyzed (Bulky N group: 68, PAN group: 54). Proportion of metastasis in perigastric (PG), supra-pancreatic (SP), and PAN areas were 80.9%, 54.4%, and 15.2% in Bulky N group, and 79.6%, 51.9%, and 44.4% in PAN group, respectively. 5y-RFS of patients with metastasis in PG, SP, and PAN areas were 38.2%, 29.7%, and 30% in Bulky N group and 23.3%, 7.1%, and 4.2% in PAN group, respectively. 30.9% of Bulky N group and 27.8% of PAN group showed MPR.TIs in each lymph node area are shown in the table. **Conclusions:** TI of PAN was extremely low in PAN group, especially in recent 2 trials (JCOG0405 and 1002) and when neoadjuvant chemotherapy was effective. These results suggest that advances in chemotherapy may make PAND unnecessary for GC with PAN metastasis in the future. Research Sponsor: The National Cancer Center Research and Development Funds (2023-J-03).

Therapeutic value index in each lymph node area.

		PG area		SP area		PAN area	
Population	N	Bulky N group	PAN group	Bulky N group	PAN group	Bulky N group	PAN group
Overall	122	30.9	18.5	15.0	3.7	5.6	1.9
JCOG0001	36	25.0	12.5	18.2	12.5	4.5	6.3
JCOG0405	42	40.9	25.0	15.4	0.0	3.8	0.0
JCOG1002	44	26.9	16.7	16.2	0.0	4.5	0.0
MPR	36	19.0	33.3	4.8	0.0	4.8	0.0

PG, peri-gastric; SP, supra-pancreatic; PAN, para-aortic node; MPR, major pathological response.

420 Poster Session

First-line (1L) tislelizumab (TIS) plus chemotherapy (CT) vs placebo (PBO) plus CT in advanced/metastatic esophageal squamous cell carcinoma (ESCC): RATIONALE-306 Japanese subgroup analysis with longer follow-up. First Author: Takashi Kojima, Department of Gastroenterology and Gastrointestinal Oncology, National Cancer Center Hospital East, Kashiwa-Shi, Japan

Background: In the global phase 3 RATIONALE-306 study (NCT03783442), TIS + CT showed a significant overall survival (OS) benefit vs PBO + CT as 1L therapy for advanced/metastatic ESCC. After a minimum 3-year follow-up, the hazard ratio (HR) for OS was 0.70 for all randomized patients (pts) and for pts with programmed death-ligand 1 (PD-L1) Tumor Area Positivity (TAP) score $\geq$10%.We report results for the Japanese subgroup. **Methods:** Eligible pts enrolled in Japan were randomized (1:1) to receive intravenous TIS 200 mg or PBO every 3 weeks + investigator-chosen CT (platinum + fluoropyrimidine/paclitaxel) until disease progression or intolerable toxicity. The primary endpoint was OS in the intent-to-treat (ITT) population. Secondary endpoints included progression-free survival (PFS), objective response rate (ORR), safety, and OS in pts with PD-L1 TAP score $\geq$10%. **Results:** Of 649 randomized pts, 66 (10.2%) were Japanese, median age was 67 years, 89.4% were male, and 28.8% had PD-L1 TAP score $\geq$10%. At study entry, Eastern Cooperative Oncology Group performance status was 0 for 77.3% pts and 97.0% had metastatic disease. As of Nov 24, 2023, 78.8% vs 84.8% Japanese pts on TIS + CT vs PBO + CT received post-systemic therapy (ITT: 51.5% vs 57.9%). After a minimum follow-up of 37.9 months, TIS + CT showed improvements vs PBO + CT in median OS (24.5 vs 15.1 months [mo]; HR: 0.75) in all pts and in pts with PD-L1 TAP score $\geq$10% (HR: 0.79), median PFS (HR: 0.77) and ORR (63.6% vs 45.5%) (Table). Treatment-related adverse events (TRAEs) with TIS + CT vs PBO + CT in Japanese pts were 45.5% vs 36.4% for any grade (ITT: 69.8% vs 60.7%); 27.3% vs 6.1% for grade $\geq$3 (ITT: 32.1% vs 20.2%); 24.2% vs 3.0% for serious TRAEs (ITT: 19.8% vs 8.4%); TRAEs led to treatment discontinuation in 3.0% vs 6.1% (ITT: 13.3% vs 6.5%). No TRAEs leading to death were reported in Japanese pts (ITT: 1.5% vs 0.6%). **Conclusions:** After 3 years, TIS + CT continued to demonstrate robust efficacy and a tolerable safety profile in Japanese pts as 1L therapy for advanced/metastatic ESCC in the RATIONALE-306 study, consistent with the overall population. Clinical trial information: NCT03783442. Research Sponsor: BieGene LTD.

Efficacy outcomes.

	Japan TIS + CT (n=33)	Japan PBO + CT (n=33)	Overall TIS + CT (n=326)	Overall PBO + CT (n=323)
Median OS, mo	24.5	15.1	17.2	10.6
(95% CI)	(17.6, 26.9)	(8.0, 22.5)	(15.8, 20.1)	(9.3, 12.1)
HR (95% CI)	0.75 (0.43, 1.30)	-	0.70 (0.59, 0.83)[b]	-
PD-L1 TAP score ≥10%, n (%)	12 (36.4)	7 (21.2)	116 (35.6)	107 (33.1)
Median OS by PD-L1 TAP score ≥10%, mo	25.5	16.8	16.6	10.0
(95% CI)	(10.9, NE)	(0.9, NE)	(15.3, 23.4)	(8.6, 13.3)
HR (95% CI)	0.79 (0.26, 2.36)	-	0.70 (0.52, 0.95)[b]	-
Median PFS[a], mo	6.8	4.5	7.3	5.6
(95% CI)	(4.4, 8.5)	(4.1, 6.7)	(6.9, 8.3)	(4.9, 6.0)
HR (95% CI)	0.77 (0.45, 1.32)	-	0.60 (0.50, 0.72)[b]	-
ORR[a], n (%)	21 (63.6)	15 (45.5)	207 (63.5)	137 (42.4)

[a]Investigator assessed.
[b]Stratified.
CI, confidence interval; NE, not estimable.

421 **Poster Session**

Perioperative adebrelimab combined with chemotherapy for locally advanced resectable esophageal squamous cell carcinoma: A single-arm, open-label, multicenter study. First Author: Liang Dai, Key Laboratory of Carcinogenesis and Translational Research (Ministry of Education), the First Department of Thoracic Surgery, Peking University Cancer Hospital and Institute, Peking University School of Oncology, Beijing, China

Background: Neoadjuvant programmed cell death 1 (PD-1) inhibitors combined with chemotherapy have shown promising results in treating locally advanced esophageal squamous cell carcinoma (LA-ESCC). However, relatively little attention has been focused on programmed cell death ligand 1 (PD-L1) blockades. This study aims to evaluate the efficacy and safety of neoadjuvant adebrelimab (anti-PD-L1 antibody) plus chemotherapy followed by adjuvant adebrelimab for resectable LA-ESCC. **Methods:** In this single-arm, multicenter study, patients (pts) with newly diagnosed, locally advanced resectable thoracic ESCC (cT1b-2, N+ or cT3-4a, N0/+) received 2 cycles of neoadjuvant chemoimmunotherapy with adebrelimab (1200 mg), paclitaxel (175 mg/m^2) and cisplatin (75 mg/m^2) every 3 weeks, followed by esophagectomy and 16 doses of adebrelimab. The primary endpoints were safety and pathological complete response (pCR) rate. Secondary endpoints included surgery completion rate, R0 resection rate, major pathological response (MPR) rate, event-free survival (EFS) and disease-free survival (DFS). The trial was registered at Chinese Clinical Trial Registry, identifier: ChiCTR2300069179. **Results:** Between May 2023 and January 2024, 36 pts met inclusion criteria, with a median age of 65 years (range: 42-73). Of these, 66.7% (24/36) were classified as stage III-IVa. As of August 1, 2024, the median number of treatment cycles for all pts was 7 (interquartile range, 2-12). The neoadjuvant therapy completion rate was 91.7% (33/36). Thirty-two pts underwent minimally invasive esophagectomy, resulting in an overall surgical resection rate of 88.9% and an R0 resection rate of 90.6%. The pCR rate was 15.6% (5/32) and the MPR rate was 31.2% (10/32). The EFS and DFS outcomes were not mature. During the preoperative treatment period, 35 pts (97.2%) experienced any grade of treatment-related adverse events (TRAEs). Grade 3 or 4 TRAEs occurred in 7 pts (19.4%), and there were no grade 5 TRAEs. The most common grade 3/4 adverse events included neutropenia (11.1%), leukopenia (5.6%), diarrhea (5.6%) and hyponatremia (5.6%). Surgical complications of grade 3/4 were reported in 7 pts (21.9%), with anastomotic fistula (6.3%) being the most common. No perioperative deaths occurred. **Conclusions:** Neoadjuvant adebrelimab combined with chemotherapy and adjuvant adebrelimab demonstrated encouraging clinical efficacy and manageable safety in pts with LA-ESCC. Further study is warranted. Clinical trial information: ChiCTR2300069179. Research Sponsor: None.

422 **Poster Session**

A multicenter retrospective study of salvage surgery for patients with residual or recurrent esophageal squamous cell carcinoma after definitive chemoradiotherapy: SURGES study. First Author: Yohei Ozawa, Department of Surgery, Tohoku University Graduate School of Medicine, Sendai, Japan

Background: Salvage surgery is defined as an operation for patients with residual or recurrent tumors after definitive chemoradiotherapy (dCRT) with $>$ 50 Gy. Although salvage surgery has been considered to improve prognosis, it is associated with a high incidence of postoperative morbidity and mortality. A previous meta-analysis by Faiz et al. reported anastomotic leakage, pulmonary disorder rates, and 90 day-mortality rates of 18.6%, 30.2%, and 8.8%, respectively. The aim of this study was to assess the efficacy and safety of salvage surgery in patients with residual or recurrent esophageal squamous cell carcinoma (ESCC) after dCRT and to explore the risk factors for postoperative complications and poor prognosis. **Methods:** This multi-institutional retrospective study enrolled patients treated with salvage surgery after dCRT between January 2017 and December 2021. The preoperative backgrounds of the patients were analyzed to assess their impact on survival or toxicity. **Results:** In total, 148 patients (59 with recurrent disease and 89 with residual disease) underwent salvage surgery at 10 leading hospitals in Japan. Post-dCRT ycT stages were T0/T1/T2/T3/T4: 12/25/30/78/3, and ycN stages were N0/N1/N2: 87/53/8. Among 136 patients who underwent esophagectomy, 108 underwent minimally invasive surgery and 28 underwent open surgery. Twelve patients had metastatic lymph nodes without primary disease. Severe postoperative complications (≧Clavien-Dindo Grade III) occurred in 39 (26.3%) patients and overall postoperative complications (≧Grade II) rate was 46.6%, including 19 (12.8%) anastomotic leakage and 25 (16.9%) pneumonia. Thirty- and 90-day mortality rates were observed in 0 and 5 (3.3%) patients, respectively. Multivariate logistic regression analysis showed that a total radiation dose ≥60 Gy (odds ratio [OR] 2.65, 95% confidence interval [CI] 1.07-6.55) and residual disease (OR 3.04, 95%CI 1.32-7.00) were associated with severe postoperative complications. The 3-year overall survival (OS) and progression-free survival (PFS) rates were 48.9% and 40.0%, respectively (median follow-up time: 42.1 months). Multivariate analysis revealed that patients with a long interval between dCRT and salvage surgery (≥180 days) had significantly longer OS (hazard ratio [HR] 0.60, 95%CI 0.38-0.95, p=0.032), and ycN0 was associated with longer PFS (HR 0.56, 95%CI 0.36-0.87, p = 0.011). The total radiation dose was not an independent prognostic factor for either OS or PFS. **Conclusions:** Salvage surgery is safe and provides substantial long-term outcomes. In cases with a total radiation dose of ≥60 Gy and residual disease, special caution is required because of the higher incidence of postoperative morbidity. A longer interval between dCRT and salvage surgery and ycN0 were prognostic factors. Research Sponsor: None.

423 **Poster Session**

Updated results from the phase Ib/II study of fruquintinib combined with SOX and toripalimab in patients with advanced metastatic gastric/gastroesophageal junction adenocarcinoma (GC/GEJC). First Author: Xiangrui Meng, The First Affiliated Hospital, Zhengzhou University, Zhengzhou, China

Background: The efficacy of first-line treatment in advanced GC/GEJC patients (pts) with negative or low PD-L1 expression still needs to be improved. This phase Ib/II, open-label study (NCT05024812) evaluating fruquintinib (a highly selective VEGFR-1, -2, -3 inhibitor) plus toripalimab (anti-PD-1), and SOX has shown preliminary antitumor activity as first-line therapy in GC/GEJC. Here we update the results with longer follow-up duration and a specific focus on PD-L1 CPS features. **Methods:** The study of phase Ib employed a 3+3 dose escalation design, pts were treated with fruquintinib 3mg/d (dose level; DL1), 4mg/d (DL2), or 5mg/d (DL3) po, d1-14, in combination with fixed dose of toripalimab (240mg, iv, d1), oxaliplatin (130 mg/m^2, iv, d1) and S-1 (40-60mg based on BSA, po, d1-14) every 3 weeks. It had been reported in phase Ib that fruquintinib 5mg/d was defined as the RP2D. In phase II, a further 64 pts would be treated with the same regimen. Primary endpoint of phase II was PFS per RECIST 1.1. Secondary endpoints included ORR, DCR, OS, DOR and safety. **Results:** As of August 15, 2024, 32 pts (9 in phase Ib; 23 in phase II) had been enrolled. Pts characteristics included: median age 62 (range 38–73); 66% male; 88% with ECOG PS 1, and 31% with liver metastasis. 31 pts had PD-L1 CPS available. 40.6% were CPS $<$ 1 and 68.8% were CPS $<$ 5. Of the 30 pts evaluable for tumor response, the ORR was 63.5% (95% CI 43.9–80.1) with 4 pts achieving complete responses and DCR was 96.7% (95% CI 82.8–99.9). After a median follow-up of 10.94 months, the median PFS was 9.33 (95% CI: 5.68–NA) months and OS result was not reached. Pts with CPS $<$ 5 were more likely to achieve higher response rate (65.0 vs 55.6%) and longer PFS than CPS ≥5 (12.68 vs 8.11 months). Similar trends were observed in pts with CPS $<$ 1 and CPS ≥1 (ORR: 75.0 vs 52.9%; PFS: 12.68 vs 8.11 months). Treatment-related adverse events (TRAEs) were mainly grade 1-2 and the most common ones were hypoalbuminemia (50%), neutrophil count decreased (34%), anemia (41%), platelet count decreased (34%) and white blood cell decreased (34%). Grade 4 TRAEs occurred in 2 pts (impaired liver function, hypertriglyceridemia). There were no treatment related deaths in the trial. **Conclusions:** Fruquintinib combined with SOX and toripalimab provided favorable efficacy and manageable toxicity profile as first-line therapy for pts with advanced metastatic GC/GEJC, especially in pts with negative or low PD-L1 expression. More data including the potential predictive response biomarkers would be further analyzed and reported. Clinical trial information: NCT05024812. Research Sponsor: None.

424 **Poster Session**

An organ-preserving strategy in patients with esophageal squamous cell carcinoma treated with immunochemotherapy. First Author: Bo Zhang, Department of Medical Oncology, National Cancer Center, National Clinical Research Center for Cancer, Cancer Hospital, Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing, China

Background: This study evaluated the strategy of dose-dense chemotherapy plus a PD-1 inhibitor with selective organ-preservation in patients with advanced esophageal squamous cell carcinoma (ESCC). **Methods:** Patients with locally advanced or metastatic (confined to the supraclavicular lymph nodes) ESCC were enrolled to receive treatment with a PD-1 inhibitor plus albumin bound-paclitaxel, cisplatin and 5-FU every two weeks. After six cycles of treatment, patients with clinical complete response (cCR) confirmed by computed tomography (CT), endoscopy and biopsy underwent an organ-preserving strategy with continuation of immunochemotherapy. The primary endpoint was cCR rate in the intention-to-treat population. A post hoc analysis was also conducted in patients with advanced ESCC who had a best response of CR after immunochemotherapy from three prospective studies we previously reported. Consecutive patients who had not received esophagectomy prior to immunochemotherapy, and had durable CR as confirmed by imaging, endoscopy with or without biopsy were identified to provide supporting evidence for the organ-preserving strategy. **Results:** A total of twenty patients were enrolled. At data cut-off (September 12, 2024), eight patients fulfilled the criteria of cCR (40%), while six patients actually adopted the organ-preserving strategy. 75% of the patients experienced ≥ grade 3 treatment-related adverse events (TRAEs), most commonly neutropenia and leukopenia. All TRAEs were managed with appropriate medical care. In the post hoc analysis, out of 71 patients who had a best response of CR to immunochemotherapy in three previous trials, sixteen consecutive patients who had not received esophagectomy before immunochemotherapy, and remained CR assessed with CT at data cut-off were identified. Although endoscopy was not mandatory during follow-up, twelve patients underwent endoscopic exams, and none of them had suspected tumor with endoscopic visualization. Biopsies were taken for eight patients, and were all negative for tumor. The median follow-up for these twelve patients reached 42.7 months, suggesting a long-term disease-free survival after immunochemotherapy. **Conclusions:** Dose-dense nab-paclitaxel, cisplatin and 5-FU combined with a PD-1 inhibitor showed encouraging cCR rate and manageable safety, making organ-preservation a possible subsequent approach in patients with advanced ESCC. Clinical trial information: ChiCTR2300072992. Research Sponsor: None.

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The safety and short-term efficacy of induction DCF plus nivolumab therapy in patients with unresectable locally advanced esophageal squamous cell carcinoma. First Author: Momoko Sano, Department of Head and Neck, Esophageal Medical Oncology, National Cancer Center Hospital, Tokyo, Japan

Background: Definitive chemoradiotherapy (dCRT) is the standard treatment for unresectable locally advanced esophageal squamous cell carcinoma (LA-ESCC) based on the results of the JCOG 0303 study. However, the prognosis in this population remains poor with a median overall survival (OS) of 13.1 months, and fistula formation was reported in about 20% of patients who received dCRT. In such a circumstance, a more effective and safer treatment strategy, such as induction chemotherapy followed by conversion surgery or dCRT, was developed. A phase II study of induction chemotherapy consisting of docetaxel (DTX), cisplatin (CDDP), and 5-fluorouracil (5-FU) (DCF) therapy (IC-DCF) showed a conversion rate of 37.5%, promising efficacy with a median OS of 33.8 months and a fistula formation rate of 4.2%. Therefore, we evaluated whether adding nivolumab to IC-DCF (IC-DCF+Nivo) could improve clinical outcomes. **Methods:** We retrospectively reviewed the medical records of patients with unresectable LA-ESCC who received IC-DCF+Nivo between Nov 2023 and Aug 2024. IC-DCF+Nivo (DTX 70mg/m2 day1, CDDP 70mg/m2 day1, 5-FU 750mg/m2 day1-5, Nivo 360 mg/body day1, every 3 weeks) was administered for 3 courses. We administered prophylactic levofloxacin from day 5 to day 15 during each cycle, and used granulocyte colony-stimulating factor (G-CSF) as a treatment. We evaluated adverse events according to the CTCAE ver5.0 during induction chemotherapy as safety, and objective response rate (ORR) of IC-DCF+Nivo according to the RECIST version 1.1., conversion surgery rate, complete resection (R0) rate, pathological complete response (pCR) rate and clinical CR (cCR) rate after dCRT as efficacy. **Results:** The median follow-up period was 5.5 months. We identified 23 eligible LA-ESCC patients, median age (range): 65 (44-76), 83% male, PS 0/1: 57%/44%, T status T3/T4b 44%/57%, clinical stage III/IVA/IVB 22%/57%/22%. Seven patients (30%) had supraclavicular lymph node metastases, and the most common unresectable factors were tracheal invasion (52%), bronchial invasion (22%) and aortic invasion (13%) of primary tumor or metastatic lymph nodes. Eighteen patients (78%) completed the 3 courses of IC-DCF+Nivo. The frequent grade 3-4 hematological adverse events were neutropenia (13/23, 57%) and anemia (3/23, 13%), and 1 patient (4.3%) had grade 3 diarrhea as non-hematological toxicity. Febrile neutropenia occurred in 3 patients (13%), and G-CSF was used in 7 patients (30%). One patient (4.3%) experienced primary tumor perforation. The ORR after 3 cycles of induction chemotherapy was 53%. Fourteen patients (61%) achieved resectability, and 10 patients (44%) underwent surgery. R0 resection rate was 90% (9/10), and the pCR rate was 20% (1/10). No one achieved cCR in dCRT. **Conclusions:** Induction DCF plus nivolumab therapy showed well-tolerated and well conversion rate. Research Sponsor: None.

426 Poster Session

A phase 2 study to assess the safety and efficacy of FLX475 (tivumecirnon) combined with pembrolizumab in patients with advanced or metastatic gastric cancer. First Author: Do-Youn Oh, Seoul National University Hospital, Cancer Research Institute, Seoul National University College of Medicine, Seoul, South Korea

Background: Regulatory T-cells (T_{regs}) suppress anti-tumor immunity by migrating into tumors via CCL17/CCL22 chemokines binding to CCR4. FLX475 (tivumecirnon), a selective CCR4 antagonist, inhibits T_{reg} recruitment, enhancing the effector T-cell (T_{eff}) response in the tumor microenvironment. Preclinical studies demonstrate that FLX475 increases the T_{eff}/T_{reg} ratio and boosts anti-tumor efficacy, both as monotherapy and in combination with checkpoint inhibitors (CPI). Here, we present findings on the safety and efficacy of FLX475 with pembrolizumab in a Phase 2 trial, aimed at improving therapeutic outcomes in advanced or metastatic gastric cancer (GC) patients. **Methods:** This Phase 2, open-label study (NCT04768686) assesses the safety and efficacy of FLX475 combined with pembrolizumab in advanced or metastatic GC patients. Patients received FLX475 100 mg PO QD and pembrolizumab 200 mg IV Q3W. Cohort 1 included CPI-naïve Epstein-Barr Virus (EBV)-negative GC patients who had progressed on at least two prior treatments. Cohort 2 included CPI-naïve EBV-positive GC patients who had received at least one prior treatment. Both cohorts were selected based on high CCR4 ligand expression and T_{reg} enrichment, expected to enhance treatment response. The primary objective of the study was to assess the anti-tumor efficacy of the study treatment, as assessed by the investigator using RECIST 1.1. **Results:** A total of 20 patients were enrolled in the study, with 10 patients in each cohort. The combination of FLX475 and pembrolizumab was well-tolerated across both cohorts. The most common treatment-related adverse events occurred in more than 10% of patients across cohorts were QTc prolongation (35.0%), and pruritus (15.0%), which were manageable and consistent with the known profiles of the individual agents. In Cohort 2, the confirmed objective response rate (ORR) was 60.0% (95% CI: 26.2 to 87.8). The median time to response (mTTR) was 2.7 months, with a median duration of response (mDOR) of 17.3 months. The median progression-free survival (mPFS) for this cohort was 10.4 months, while the median overall survival was not reached. In Cohort 1, no objective responses were observed, and the best response achieved was stable disease. The relationship between treatment outcomes and biomarkers including CCR4 expression levels and the T_{eff}:T_{reg} ratio was confirmed in exploratory analyses. **Conclusions:** The combination of FLX475 with pembrolizumab was well-tolerated and showed promising anti-tumor activity and durable responses in patients with advanced or metastatic EBV-positive GC. Clinical trial information: NCT04768686. Research Sponsor: None.

		Cohort 1 (EBV (-) GC) N=10	Cohort 2 (EBV (+) GC) N=10
ORR	n (%)	0 (0.0)	6 (60.0)
	(95% CI)	(-)	(26.2, 87.8)
mTTR	Months (95% CI)	-	2.7 (1.2, -)
mDOR	Months (95% CI)	-	17.3 (2.7, -)
mPFS	Months (95% CI)	1.4 (1.1, 1.5)	10.4 (1.0, -)

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Single agent olaparib in advanced oesophago-gastric cancer: Primary results from the SOlar clinical phase II single arm trial. First Author: Georgina Keogh, Royal Marsden NHS Foundation Trust, London, United Kingdom

Background: A subset of pts with advanced oesophago-gastric adenocarcinoma (OGA) are characterised by deficient DNA damage repair (DDR) providing a rationale for PARP inhibition (PARPi) in these tumours. Potential activity was observed in the phase 3 gastric cancer trial with Olaparib + paclitaxel, however the efficacy of Olaparib monotherapy and response biomarkers have not been established. Herein we report the primary clinical results from the SOlar trial. **Methods:** SOlar is a prospective, open label, single arm phase II trial using a Simon's two stage optimal design, evaluating the efficacy and safety of Olaparib 300mg twice daily in pts with locally advanced/metastatic OGA who had previously progressed on ≥1 line of therapy in the advanced setting. Pts with measurable disease by RECIST v1.1 amenable to mandatory baseline biopsy were included. The primary end point was disease control rate (DCR) at 8 weeks from Olaparib initiation by RECIST v1.1. In this two-stage optimal design, if ≥5/27 pts had disease control in stage 1, a further 27 pts were to be recruited in stage 2. If ≥12/54 evaluable pts achieved disease control, Olaparib would warrant further investigation. Other key endpoints include; overall response rate, progression free survival and overall survival, and translational biomarker analyses (NCT03829345). **Results:** Between July 2019 - February 2024, 55 pts with locally advanced/metastatic OGA were enrolled. 51 pts received Olaparib and 50 were evaluable for the primary endpoint (5 withdrew). The median age was 61yrs (range: 54-67), 7 (14%) were *HER2*-positive and 2 had a known germline *BRCA* mutation. 19 (38%) pts had received ≥3 lines of therapy and 22 (44%) pts had received prior immunotherapy for OGA. 30 (60%) pts had objective responses to prior platinum chemotherapy (CR + PR). The DCR at 8-weeks was 28% (91% one-sided confidence interval lower bound: 19.4) with 14/50 evaluable pts having stable disease. No objective response was observed. 36/51 (71%) treated pts experienced a treatment-related adverse event (TRAE) of any grade, and 9 (18%) experienced a grade ≥3 TRAE (most common being anaemia, fatigue, vomiting). Two serious adverse events were reported as treatment-related: definitely related grade 1 pneumonitis, and possibly related grade 4 lung infection. **Conclusions:** In SOlar, Olaparib met its primary endpoint with 28% DCR in this heavily pre-treated group. No new safety concerns were observed. Ongoing translational analyses include; homologous repair deficiency scoring, whole exome sequencing, RNA expression, serial ctDNA, and multiplex immunofluorescent assessment of the tumour micro-environment, with a view to identify a subgroup who might benefit from PARPi to support ongoing PARPi combinatorial studies in advanced OGA. Clinical trial information: NCT03829345. Research Sponsor: AstraZeneca (supplied Olaparib).

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Pembrolizumab plus chemotherapy versus placebo plus chemotherapy in patients with advanced HER2-negative gastric or gastroesophageal junction (G/GEJ) adenocarcinoma enrolled in the Republic of Korea: KEYNOTE-859. First Author: Sun Young Rha, Division of Medical Oncology, Department of Internal Medicine, Yonsei Cancer Center, Yonsei University College of Medicine, Seoul, South Korea

Background: After a median follow-up of 41.6 mo, data from the global phase 3 KEYNOTE-859 study (NCT03675737; N = 1579) continued to show that use of pembrolizumab (pembro) plus chemotherapy (chemo) provided a significant and clinically meaningful improvement in overall survival (OS), with manageable safety, versus placebo (pbo) plus chemo for patients (pts) with locally advanced or metastatic HER2-negative G/GEJ adenocarcinoma. This post hoc analysis evaluated the safety and efficacy of pembro plus chemo in pts enrolled in KEYNOTE-859 in the Republic of Korea. **Methods:** Eligible pts aged ≥18 years with locally advanced unresectable or metastatic HER2-negative G/GEJ adenocarcinoma, measurable disease per RECIST v1.1, and an ECOG performance status (PS) of 0 or 1 were randomly assigned 1:1 to receive pembro 200 mg or pbo IV Q3W for ≤35 cycles. All pts received investigator's choice of chemo (FP or CAPOX). The primary end point was OS. Secondary end points included progression-free survival (PFS), objective response rate (ORR), and duration of response (DOR) per RECIST v1.1 by blinded independent central review, and safety. Efficacy end points were evaluated in all randomly assigned pts (intention to treat). **Results:** A total of 150 pts (66, pembro plus chemo; 84, pbo plus chemo) from the Republic of Korea were enrolled. The median time from randomization to database cutoff (August 22, 2023) was 48.1 mo (range, 32.7-56.8). Baseline characteristics were generally well balanced between treatment groups. Most pts were male (71.3%) with an ECOG PS of 1 (66.7%); 43.3% of pts had ≥3 metastases. The median OS was 19.5 mo (95% CI, 14.2-26.2) for pembro plus chemo versus 15.2 mo (95% CI, 11.7-19.8) for pbo plus chemo (HR, 0.80; 95% CI, 0.55-1.15). Additional efficacy data are reported in the Table. Treatment-related adverse events occurred in 63 pts (95.5%) in the pembro plus chemo group and 78 pts (92.9%) in the pbo plus chemo group; 39 (59.1%) and 31 (36.9%) pts, respectively, were grade 3 or 4. No treatment-related deaths occurred. **Conclusions:** Consistent with the global results, clinical outcomes favored pembro plus chemo versus pbo plus chemo with manageable safety in pts from the Republic of Korea with HER2-negative G/GEJ adenocarcinoma enrolled in KEYNOTE-859. Clinical trial information: NCT03675737. Research Sponsor: Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA.

	Pembro + chemo n = 66	Pbo + chemo n = 84
Median PFS (95% CI), mo[a]	11.0 (6.9-17.6)	7.2 (5.6-8.9)
PFS HR (95% CI)[b]	0.74 (0.48-1.13)	
ORR, % (95% CI)[c]	53.0 (40.3-65.4)	46.4 (35.5-57.6)
Median DOR (range), mo	15.0 (1.4+ to 50.8+)	6.9 (1.4+ to 44.3+)
Responses lasting ≥24 mo,[a] %	42.6	23.6

[a]Based on Kaplan-Meier estimates.
[b]Based on the unstratified Cox regression model with the Efron method of tie handling with treatment as a covariate.
[c]Based on the stratified Miettinen and Nurminen method with strata weighting by sample size.

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A phase I/II clinical trial of preoperative docetaxel/oxaliplatin/S-1 (DOS) combination therapy for esophagogastric junction adenocarcinoma. First Author: Koshi Kumagai, Department of Upper Gastrointestinal Surgery, Kitasato University School of Medicine, Sagamihara-Shi, Kanagawa, Japan

Background: Esophagogastric junction (EGJ) cancer is difficult to cure with surgery alone, necessitating adjuvant chemotherapy. However, administering aggressive adjuvant chemotherapy after invasive surgery is difficult. Neoadjuvant chemotherapy, on the other hand, is expected to enhance compliance, increase the rate of curative resection, and reduce recurrence. **Methods:** We conducted a single center Phase I/II study to evaluate the safety and efficacy of neoadjuvant chemotherapy using docetaxel, oxaliplatin, and S-1 (DOS), followed by adjuvant chemotherapy with S-1 in patients with cT3-4aN0-3M0 adenocarcinoma of the EGJ. In Phase I, we estimated the maximum tolerated dose and dose-limiting toxicity of neoadjuvant DOS therapy and determined the recommended dose (RD). In Phase II, we assessed the safety and efficacy of the RD established in Phase I. The primary endpoint was the curative resection rate, with secondary endpoints including the 3-year overall survival rate, 3-year recurrence-free survival rate, treatment completion rate, response rate to neoadjuvant chemotherapy, pathological response, and adverse events. Neoadjuvant DOS chemotherapy was administered in three 3-week cycles. The starting dose of docetaxel was 50 mg/m², with plans to escalate it to 60 mg/m². Docetaxel and oxaliplatin (100 mg/m²) were administered on day 1, while S-1 (120 mg/body) was given orally on days 1-14. Postoperative adjuvant chemotherapy consisted of S-1 (120 mg/body), administered for 4 weeks followed by a 2-week rest period, repeated for 8 cycles. **Results:** Based on the 6 patients enrolled in Phase I, the RD of docetaxel was determined to be 60 mg/m². A total of 40 eligible patients, including 3 from Phase I, were enrolled in the study. Two patients discontinued protocol treatment during neoadjuvant chemotherapy due to adverse events and 1 patient discontinued due to positive peritoneal cytology. The R0 resection rate was 87.5% (35/40). Surgical procedures included esophagectomy in 18 patients, total gastrectomy in 15, and proximal gastrectomy in 4. Postoperative complications of Grade III or higher, as per the Clavien-Dindo classification, occurred in 15.0% of patients, with no surgery-related deaths. The response rates to neoadjuvant chemotherapy were 12.5% for PR, 2.5% for PD, and 85.0% for non-CR/non-PD. Pathological response rates were: Grade III in 8.1%, Grade II in 35.1% and Grade I in 56.8%. During neoadjuvant chemotherapy, Grade 3 or higher adverse events were observed, including neutropenia (82.5%), nausea (5%) and diarrhea (5%). The neoadjuvant chemotherapy completion rate was 92.5%, although 59.5% of patients required dose reductions. **Conclusions:** Neoadjuvant DOS therapy for esophagogastric junction cancer shows promise, achieving a high R0 resection rate and substantial pathological efficacy, with acceptable levels of adverse events. Clinical trial information: UMIN000022210. Research Sponsor: None.

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Perioperative chemoimmunotherapy for patients with gastric or gastro-esophageal junction cancer: A systematic review and meta-analysis. First Author: Reo Omori, Osaka City General Hospital, Osaka, Japan

Background: Immune checkpoint inhibitors (ICIs) targeting the PD-1/PD-L1 axis, in combination with chemotherapy, are known to improve survival in advanced/metastatic gastric (GC) or gastro-esophageal junction cancer (GEJC). Therefore, several trials incorporated ICIs in the neoadjuvant/perioperative setting with promising preliminary results. The purpose of this meta-analysis was to determine the safety and efficacy of neoadjuvant chemoimmunotherapy for resectable GC or GEJC. **Methods:** PubMed/MEDLINE and Embase were systematically searched for randomized control trials (RCTs) investigating the efficacy of neoadjuvant anti PD-1/PD-L1 ICIs for locally advanced GC or GEJC up to 07/27/2024. Outcomes of interest were surgical outcomes such as pathological complete response (pCR) rate and major pathological response (MPR) rate. Risk ratio (RR) with 95% confidence intervals (CI) were pooled using a fixed model meta-analysis. Statistical analysis was performed with Review Manager 5.4.1. I^2 statistics was used to assess heterogeneity. The Cochrane risk of bias assessment tool was used to assess the risk of bias. All analyses were conducted at a significance level of 0.05. **Results:** Seven RCTs comprising 2,911 GC or GEJC patients were included. Among these, 1,454 patients received neoadjuvant chemoimmunotherapy and 1,457 patients received chemotherapy, respectively. Neoadjuvant chemoimmunotherapy significantly improved pCR rate (RR 2.94; 95% CI 2.34-3.70; $P<0.00001$; I^2=33%), and MPR rate (RR 1.47; 95% CI 1.25-1.72; $P<0.00001$; I^2=30%) compared to chemotherapy alone. Chemoimmunotherapy was associated with slightly higher incidence of grade 3 or worse treatment-related adverse events than chemotherapy (RR 1.07; 95% CI 1.01-1.14; p=0.03; I^2=0%). **Conclusions:** This meta-analysis showed that adding ICIs to chemotherapy significantly improved the pCR and MPR rates and was feasible for patients with locally advanced GC or GEJC. Results of survival outcomes in the included trials and ongoing trials are awaited to further determine the efficacy and utility of chemoimmunotherapy and identify patient populations that benefit from this strategy. Research Sponsor: None.

Included studies.

Author	Year	Trial#	Design	Cancer; Stage	Name of IO	Target	Chemo regimen	Extracted data		
pCR	**MPR**	**TRAE**								
Ding X	2023	NCT04982939	P2	G/GEJ; Stage2-3	Sintilimab	PD-1	SOX	○	○	
Janjigian YY	2023	NCT04592913	P3	G/GEJ; ≥T2Nany/T0-4 N+	Durvalumab	PD-L1	FLOT	○		○
Li C	2023	NCT04208347	P3	G/GEJ; T3-4aN+	Camrelizumab	PD-1	SOX(+apatinib)	○	○	
Lin JX	2024	NCT04195828	P2	G; T2-4N+	Camrelizumab	PD-1	S-1+nabPTX	○	○	○
Lorenzen S	2024	NCT03421288	P2	G/GEJ; ≥T2 and/or N1	Atezolizumab	PD-L1	FLOT	○	○	
Peng Z	2024	NCT04661150	P2	G/GEJ HER2+; resectable	Atezolizumab	PD-L1	CAPOX+Tmab	○		
Shitara K	2024	NCT03221426	P3	G/GEJ; ≥T3N+	Pembrolizumab	PD-1	FLOT or XP/FP	○		○
Yuan SQ	2024	NCT04250948	P2	G/GEJ; T3-4N+	Toripalimab	PD-1	SOX/CAPOX	○	○	○

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FG-M108 plus capecitabine and oxaliplatin (CAPOX) for first-line treatment of CLDN18.2+/HER2- advanced gastric/gastroesophageal junction adenocarcinoma (GC): Update results of a phase I/II trial to determine RP2D. First Author: Jifang Gong, Beijing Cancer Hospital, Beijing, China

Background: There exists unmet need for previously untreated pts with CLDN18.2+/HER2- advanced GC. FG-M108, an mAb specifically targeting CLDN18.2 and exhibiting enhanced ADCC activity, has the potential to benefit these pts. **Methods:** This phase I/II trial evaluated safety, pharmacokinetics and preliminary antitumor activity of FG-M108 plus CAPOX in previously untreated pts with CLDN18.2+/HER2- advanced GC, where FG-M108 was administered intravenously at 300 mg/m^2 or 600 mg/m^2 Q3W. Here we compare results of FG-M108 at 300mg/m^2(Cohort A) with 600mg/m^2(Cohort B) to establish RP2D. **Results:** As of August 28, 2024, 63 pts with CLDN18.2+ (IHC 1/2/3+≥10%) GC were treated, 70% of whom had CLDN18.2 Medium-High expression (CMH, IHC 2/3+≥40%). Baseline characteristics were comparable between Cohorts A & B. Increasing the FG-M108 dosage did not result in a higher incidence or severity of treatment-related adverse events (TRAE) overall. However, nausea and vomiting, identified as CLDN18.2 on-target toxicities, were significantly more frequent and severe in Cohort B compared to Cohort A.When 300 or 600 mg/m² of FG-M108 was administered intravenously Q3W, serum trough concentration exceeded the target drug concentration predicted by in vitro ADCC activity. The 300 mg/m² dose is expected to provide sufficient drug exposure as the effective dose.Moreover, in Cohort A, pts with CMH expression had a confirmed ORR of 78% (unconfirmed ORR=81%) and a median PFS of 11.0 months (95% CI 6.9-12.7) , significantly better than those in Cohort B. Based on safety, PK and efficacy trends, the RP2D was determined to be FG-M108 at 300mg/m^2 plus CAPOX. **Conclusions:** FG-M108 plus CAPOX had a manageable safety profile and promising antitumor activity in GC, with RP2D at 300mg/m^2. A phase III trial is ongoing to confirm the efficacy of FG-M108 at 300mg/m^2 plus CAPOX as first-line treatment (NCT06177041). Clinical trial information: NCT04894825. Research Sponsor: None.

	Cohort A n=52	Cohort B n=11
CMH %	71	64
Primary lesion-Stomach %	92	100
Metastatic organs ≥3 %	27	9
ORR/DCR with CMH %	81/97	57/86
PFS/DOR with CMH months	11.0/9.9	5.5/4.2
≥G3 TRAE %	37	18
TRAE-Nausea % / ≥G3 Nausea %	38/2	64/9
TRAE-Vomiting % / ≥G3 Vomiting %	23/2	73/9

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The safety and short-term efficacy of neoadjuvant chemotherapy in elderly patients with resectable locally advanced esophageal squamous cell carcinoma. First Author: Hiroshi Imazeki, Department of Head and Neck, Esophageal Medical Oncology, National Cancer Center Hospital, Tokyo, Japan

Background: The standard treatment for resectable locally advanced esophageal squamous cell carcinoma (LA-ESCC) is neoadjuvant chemotherapy followed by surgery. However, the efficacy and safety of neoadjuvant chemotherapy for patients aged 76 or older have not been established because the JCOG1109 trial, which established the current standard of care, was limited to patients younger than 76 years. In our hospital, even patients aged 76 or older are treated with docetaxel, cisplatin, and 5-FU (DCF) or 5-FU / leucovorin, oxaliplatin, and docetaxel (FLOT), if the patient's general condition and organ function are maintained, but there have been few reports on the efficacy and safety of these therapies for elderly population. **Methods:** The subjects were 76 years old or older resectable LA-ESCC patients diagnosed as cT1N1-3M0, cT2-3N0-3M0, and cT1-3N0-3M1 (UICC-TNM 8th), who received at least one cycle of neoadjuvant DCF or FLOT in our institution from 2014 to 2023. Adverse events (CTCAE ver. 5.0) and histopathological responses were evaluated. **Results:** Patients who received neoadjuvant DCF (D group) and FLOT (F group) were 19 and 25, respectively. Patients background in the D / F group was as follows; median age 76 / 78 (years), male 84% / 64%, PS1 37% / 64%, clinical, T3 74% / 84%, clinical N1 32% / 52%, N2 47% / 32%, clinical stage III 53% / 64%, stage IVB 21% / 28%. Of the 19 patients in the D Group, 17 completed 3 cycles of treatment (completion rate 89%), 18 underwent surgery, including 2 patients who discontinued during the treatment (1 with disease progression and 1 with adverse event), and 1 underwent CRT. Of the 25 patients in the F group, 16 completed 4 cycles of treatment (completion rate 64%), 22 underwent surgery, including 8 patients who discontinued during the treatment (1 with disease progression, 6 with adverse events, and 1 with other), 2 underwent CRT, and 1 was placed on best supportive care. In the D and F groups, hematological toxicities were more common in the F group, especially grade 3 or more neutropenia was 26% / 96%, but febrile neutropenia was similar at 5% and 12%, respectively. Nonhematological toxicities were more common in the F group for all grades (nausea 42% / 64%, anorexia 63% / 80%, and fatigue 52% / 88%, while grade 3 or more toxicity was not significantly different from the D group (nausea 0% / 4%, anorexia 0% / 4%, and fatigue 5% / 4%). In the 18 patients of the D group and 22 patients of the F group who underwent surgery, R0 resection was performed in 94% / 95%, and pathological complete response (pCR) was 11% / 23%, respectively. **Conclusions:** Neoadjuvant triplet chemotherapy was well tolerated and resulted in high R0 resection rates and pathological responses even in patients aged 76 or older. Further investigation with long-term follow up is warranted to elucidate the clinical benefit of neoadjuvant chemotherapy for elderly LA-ESCC patients. Research Sponsor: None.

434 **Poster Session**

Non-cancer-related deaths in patients with resectable locally advanced esophageal squamous cell carcinoma: A supplementary analysis of JCOG1109. First Author: Hiroshi Imazeki, Department of Head and Neck, Esophageal Medical Oncology, National Cancer Center Hospital, Tokyo, Japan

Background: Neoadjuvant docetaxel, cisplatin, and 5-FU (DCF) has become the standard treatment for resectable locally advanced esophageal squamous cell carcinoma (LA-ESCC) in Japan, based on the result of JCOG1109. In this study, more non-cancer-related deaths were observed in the cisplatin, and 5-FU (CF) plus radiotherapy (CF-RT) arm compared to the CF and DCF arm. However, the details of these non-cancer-related deaths remain unclear. We aimed to evaluate the influence of neoadjuvant therapies on non-cancer-related deaths using the data from JCOG1109. **Methods:** We analyzed patients' characteristics, classification, time of onset, risk factors associated with onset, and cumulative incidences in patients with non-cancer-related deaths. Cumulative incidences of non-cancer-related deaths in each arm were estimated, considering cancer-related deaths as a competing risk. Risk factors were explored using multivariable Fine-Gray model with Firth penalization. **Results:** Among 601 enrolled patients, 164 in the CF arm, 169 in the DCF arm, and 173 in the CF-RT arm were completed the surgery. Of those, non-cancer-related deaths were observed in 19 (12%), 12 (7%), and 32 (18%) patients, respectively. Patient characteristics of non-cancer-related deaths in the CF/DCF/CF-RT arm were as follows: male, 89/92/88%; median age, 69/69/69 years old; ECOG PS 0, 89/100/97%; lower thoracic esophagus, 21/25/19%; previous history of ischemic disease of heart, 5/0/0%; brain, 0/0/0%; other malignancies, 5/8/0%; comorbidity of hypertension, 26/25/47%; COPD, 0/8/3%; thoracoscopic approach, 37/42/38%. The median follow-up time was 5.7 years (range, 0.3-10.4 years). The causes of non-cancer-related deaths in the CF/DCF/CF-RT arm were pulmonary, 4 (21%)/1 (8%)/10 (31%); cardiovascular, 1 (6%)/0/2 (6%); infectious, 2 (11%)/1 (8%)/3 (9%); other malignancy, 2 (11%)/3 (25%)/5 (16%); treatment-related, 0/1 (8%)/1 (3%), and the median time from surgery to onset of non-cancer-related death was 3.3/3.8/4.3 years, respectively. Cumulative incidences of non-cancer-related deaths after 3, 6, 9 years in the CF/DCF/CF-RT arm were 4.9/3.0/7.0%, 10.7/6.1/15.6%, 12.4/8.8/26.1%, respectively. In multivariable analyses, $\geq$ 65 years old (hazard ratio [HR]: 3.243, 95% confidential interval [CI]: 1.797–5.852, vs. $<$ 65 years old), ECOG PS 0 (HR: 3.579, 95% CI: 1.148–11.162, vs. PS 1), and serum CRP $\geq$ 1.0 mg/dl (HR: 2.238, 95% CI: 1.003–4.995, vs. $<$1.0 mg/dl) were significantly associated with the onset of non-cancer-related deaths. Whereas the CF-RT arm was not significantly associated (HR: 1.600, 95% CI: 0.885–2.892, P = 0.12, vs. CF). **Conclusions:** The incidence of non-cancer-related deaths was relatively higher in the CF-RT arm, and pulmonary-related complications tended to be a particularly common cause of death. Research Sponsor: None.

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NRG-GI007: Secondary endpoints of safety assessment and clinical complete response (cCR) of chemoradiation with endoscopically injected oncolytic virus OBP-301 in medically inoperable esophageal cancer. First Author: Geoffrey Yuyat Ku, Memorial Sloan Kettering Cancer Center, New York, NY

Background: Definitive chemoradiation (CRT) is a standard-of-care for patients (Pts) with medically inoperable esophageal cancer (EC). The NRG/RTOG 0436 study of cisplatin/paclitaxel and RT (+/-cetuximab) as definitive therapy showed that 58% of control arm Pts have locally persistent disease. OBP-301 is a conditionally-restricted, replication-competent adenovirus derived from human adenovirus type 5 that adds a human Telomerase Reverse Transcriptase gene promoter, replicating only in tumor cells to cause lysis.It may cause immunogenic cell death, enhance RT increasing radiosensitization by blocking DNA repair and improve local control. **Methods:** In NRG-GI007, a phase 1 study (NCT04391049), OBP-301 was added to weekly carboplatin/paclitaxel and RT (50.4 Gy/28 fxs) for medically inoperable EC Pts with pathologically proven adenocarcinoma or squamous cell carcinoma (SCC) of the esophagus or Siewert Type I/II gastroesophageal junction. Pts receive 1-2mL of intratumoral OBP-301 1×10^{12} virus particles/ml via endoscopy 3 days prior to and then at days 12 and 26 of CRT. The primary endpoint was protocol-defined dose-limiting toxicity, already reported. Secondary endpoints reported here are treatment-related (definitely or probably related to any protocol treatment) AEs - all and grade 4+ non-hematologic, and cCR in the primary tumor, defined as negative EGD/biopsy 6-8 weeks post-CRT. Per the protocol design, no statistical testing is done for these endpoints. **Results:** Initially, 6 evaluable (eligible and started protocol treatment) Pts were enrolled from June 2020 to April 2023. As initial dose level was deemed safe, an additional 9 Pts were enrolled from August 2023 to July 2024, totaling 15 evaluable Pts for secondary endpoints. Median age was 74 years, 9 Pts (60%) with ECOG PS 1. 11 Pts (73%) had adenocarcinoma and 4 had SCC; 11 Pts had N+ disease. 14 Pts (93%) received all planned OBP-301 injections and received 50.4 Gy RT. The most common treatment-related grade 3/4 AEs were decreased neutrophil (6 Pts; 40%) and lymphocyte counts (5 Pts; 33%), expected with CRT. No grade $\geq$3 OBP-301-related non-hematologic AEs were reported. 2 Pts died prior to restaging neitherdefinitely or probably related to OBP-301: 1 Pt developed grade 5 respiratory failure 6 weeks after CRT (suspected RT pneumonitis), while a 2nd Pt had an unrecognized tumor invasion of the bronchus at baseline, which worsened during week 3 of CRT, resulting in a grade 5 tracheo-esophageal fistula. 2 Pts have not yet reached time for restaging. All 11 Pts who have been restaged had a cCR, for a preliminary cCR rate of 100% (95% CI: 74.1%, 100%). **Conclusions:** OBP-301 + CRT is safe and the preliminary cCR rate compares favorably to the historical control from NRG/RTOG 0436. Updated safety and cCR data will be presented. A randomized study is being planned. Clinical trial information: NCT04391049. Research Sponsor: NRG Oncology SDMC; U10CA180822; NRG Oncology Operations; U10CA180868; IROC; U24CA180803; Oncolys BioPharma.

436 **Poster Session**

Transarterial infusion chemotherapy and embolization (TAICE) as a pre-operative therapy for patients with locally advanced gastric cancer (LAGC): A prospective, single-arm, pilot study. First Author: Zheng Liu, Department of Gastrointestinal Surgery, Zhongshan Hospital of Fudan University, Shanghai, Shanghai, China

Background: Interventional therapy is a local intensive therapy for tumor control, often utilized in palliative treatment for solid organ tumors. Yet, its application in tumors of hollow viscera still lacks evidence. Our previous retrospective study revealed transarterial infusion chemotherapy and embolization (TAICE) not only effectively managed tumor-related hemorrhage but also exhibited a notable anti-tumor efficacy in gastric cancer. This prospective study aims to further explore the feasibility and safety of TAICE as a pre-operative therapy for locally advanced gastric cancer (LAGC). **Methods:** Patients diagnosed with locally advanced adenocarcinoma of stomach and gastrointestinal junction (GEJ) with no distant metastasis (cT3-4N+M0) were enrolled. They were given 4 cycles of pre-operative treatments: 2 cycles of TAICE-SOX (Oxaliplatin [85mg/m^2 transarterial infusion on Day 1] and S-1 [40/50/60mg po bid on Day 1-14]) and 2 cycles of SOX (Oxaliplatin [130mg/m^2 IV on Day 1] and S-1 [same as above]), alternately. Then, they received D2 radical gastrectomy, and 4 cycles of SOX post-operatively. In brief, the TAICE procedure was summarized into 4 steps: 1) Celiac arteriography, 2) Super-selection of tumor feeding artery (TFA), 3) Infusion chemotherapy and embolization of TFA, 4) Celiac re-arteriography to ensure the success of embolization. The primary endpoint was pathological complete response (pCR) rate. Other endpoints included major pathological response (MPR) rate, overall survival (OS), progression-free survival (PFS), treatment-related adverse events (TRAEs). **Results:** Between December 2020 and June 2023, twenty patients were enrolled and all of them received TAICE. Among them, 18 (90.0%) patients received 2 cycles of TAICE. The pCR and MPR rate were 55.0% and 70.0%, respectively. The 1-year and 3-year OS rate were 100.0% and 82.6%. The 1-year and 3-year PFS rate were 90.0% and 67.3%. There was no difference in OS between the groups of high and low tumor residual rate with the threshold of 50%. Patients achieving low tumor residual rate had significantly longer PFS (NR vs. 12.17 months, p=0.018) than those with high rate after TAICE. Moreover, we divided TFA into two subtypes: dispersive and branching, depending on the vascular pattern presented in the angiography at the first cycle TAICE. Patients with dispersive TFA had higher risk of recurrence after TAICE (NR vs. 15.17 months, p=0.0264). All patients had TRAEs (vomiting, nausea, abdominal pain and fever) of grade 1 or 2 after the first cycle of TAICE. Only one patient experienced nausea of grade 3 after the second cycle of TAICE. The most common TRAEs were vomiting and nausea. **Conclusions:** TAICE is a promising pre-operative therapy for patients with LAGC for it displaying the satisfactory oncological effectiveness and safety profile. Clinical trial information: NCT05396326. Research Sponsor: Natural Science Foundation of Shanghai (23ZR1410700); Natural Science Foundation of Shanghai (23Y11902100); Natural Science Foundation of Fujian Province (2023J011692).

437 **Poster Session**

Chemoimmunotherapy plus radiotherapy in advanced esophageal squamous-cell carcinoma. First Author: Keke Nie, Qingdao Central Hospital, University of Health and Rehabilitation Sciences (Qingdao Central Hospital), Qingdao, China

Background: Chemotherapy plus Immune checkpoint inhibitors (ICIs) have improved survival outcomes of patients with advanced or metastatic esophageal squamous-cell carcinoma in both first- and second-line settings. However, the benefit of additional radiotherapy for the residual esophageal lesion after chemo-immunotherapy remains unclear. **Methods:** We conducted a one-arm, three-center, open label study involving patients who were treatment naïve, radiological and histological confirmed advanced or metastatic squamous-cell esophageal carcinoma. Enrolled patients were expected the survival time was three months or longer; and ECOG performance status score was 0 or 1. In induction phase, patients received four cycles of TP regimen chemotherapy (docetaxel, 75/mg^2, iv on day 1 and cisplatin, 75mg/m^2, iv on day 1) combined with PD-1 inhibitor (sintilimab, 200mg intravenous infusion on day 1) every 21 days. Patients who finished four cycles of chemo-immunotherapy with stable disease (SD) or partial response (PR) were initiated 50Gy/25f irradiation for residual tumors. 3-4weeks after radiotherapy, maintenance sintilimab therapy was administered every 21 days for another 31 cycles or until disease progression or intolerable toxicity. **Results:** Total of 39 patients were enrolled in this study, 30 of them could be evaluated in efficiency and toxicities. All of them were male and were squamous-cell carcinoma in histology. 12/30 (40.0%) patients were in stage III, and 18/30 (60.0%) were in stage IV. 29/30 (96.7%) patients completed four cycles chemo-immunotherapy, the complete response (CR) rate was 6.7% (2/30), the partial response (PR) rate was 53.3% (16/30) and disease control rate (DCR) was 86.7% (26/30). The median depth of response (DpR) was 34.5% for chemoimmunotherapy and was 64.0% after radiotherapy. The progression-free survival (PFS) was 16.4 months and the overall survival (OS) was not reached. The most common grade 3 or worse adverse event was neutropenia and occurred in 3 (10.0%) patients. **Conclusions:** Four cycles sintilimab plus cisplatin and docetaxel followed by radiotherapy for residual tumors and maintained sintilimab had high response rate, prolonged PFS and tolerable toxicities as first line treatment in patients with advanced or metastatic esophageal squamous-cell carcinoma. Longer OS is promising and more patients enrolled in this study is needed in future. Clinical trial information: NCT06138028. Research Sponsor: None.

Clinicopathological features and patient factors.

Factors	No. of patients (n=30) (%)
Gender	
Male	30 (100.0)
Female	0 (0)
Age (year)	
< 65	27 (43.3)
≥ 65	17 (56.7)
PD-L1 (CPS)	
< 5	9 (30.0)
≥ 5	21 (70.0)
Clinical Stage	
III	12 (40.0)
IV	18 (60.0)
Metastatic Pattern	
Regional lymph node	12 (40.0)
Cervical & Supraclavicular	4 (13.3)
Abdomen	5 (16.7)
Lung	6 (20.0)

438 Poster Session

Efficacy of pembrolizumab in advanced esophageal carcinoma: A systematic review and meta-analysis. First Author: Helai Hussaini, West Anaheim Medical Center, Anaheim, CA

Background: Pembrolizumab, a monoclonal immunoglobulin G4 antibody is a programmed cell death protein 1 inhibitor, approved for treatment of advanced (metastatic or recurrent) esophageal squamous cell carcinoma (ESCC) and esophageal adenocarcinoma (EAC). We performed a pooled analysis to assess the efficacy of pembrolizumab in esophageal carcinoma (EC) and to compare it with chemotherapy alone. **Methods:** We conducted a systemic literature search, using PubMed, EMBASE, and ClinicalTrials.gov, till July 30, 2024. Initial search yielded 412 articles. We excluded duplicates and irrelevant studies and included 4 clinical trials (CT) that reported pembrolizumab's efficacy in EC. We estimated pooled objective response rate (ORR), partial response (PR), and hazard ratio (HR) for overall survival (OS) and progression-free survival (PFS), along with 95% confidence interval (CI) and p-value in RevMan v.5.4. Subgroup analysis, based on combined positive score (CPS) for programmed cell death ligand-1 (PDL-1) expression and tumor histology type was also performed. **Results:** We included four CTs in our study; their characteristics are shown in the table. The estimated pooled ORR and PR of pembrolizumab in all patients was 21% (CI: 0.10-0.41) ($p < 0.001$) and 19% (CI: 0.08-0.37) ($p < 0.001$), respectively. Pooled median PFS was 3.05 mo (CI: 0.92-5.17) (p = 0.004), and pooled median OS was 8.05 mo (CI: 5.21-10.88) ($p < 0.001$). Pembrolizumab use in EC showed a reduced risk of disease progression vs chemotherapy alone [HR: 0.85 (CI: 0.75-0.96) ($p < 0.05$)] and a 21% reduction in mortality [HR: 0.79 (CI: 0.71-0.88) ($p < 0.001$)]. Safety analysis revealed less grade ≥3 adverse events [OR: 0.56 (CI: 0.35-0.90), $p < 0.05$)] with pembrolizumab compared with chemotherapy alone. On subgroup analysis between ESSC and EAC, pembrolizumab exhibited statistically non-significant responses with ORR (OR: 1.29, CI: 0.89-1.88, p = 0.18), OS (pooled HR: 0.94, CI: 0.76-1.15, p = 0.53). Pembrolizumab in patients with CPS ≥10 vs CPS <10 also showed statistically non-significant responses with ORR [OR: 1.70, CI: 0.99-2.91, p = 0.06), OS (pooled HR: 0.63 vs 1.00) (p = 0.05) and PFS (pooled HR: 0.62 vs 0.83, p = 0.08). **Conclusions:** Pembrolizumab alone or combined with chemotherapy compared with chemotherapy alone exhibited favorable responses in terms of ORR, OS and PFS with a manageable toxicity profile, setting a benchmark for future studies. Research Sponsor: None.

Characteristics of included studies.

Trial ID, phase	Histology	Line	Number of patients	Regimen	Median follow-up (mo)	ORR (%)	Median PFS (mo)	Median OS (mo)
KEYNOTE-590, III	ESCC, EAC	1st	749	Pembro + chemo vs chemo	58.8	45 vs 29.3	6.3 vs 5.8	12.3 vs 9.8
KEYNOTE-181, III	ESCC, EAC	2nd	628	Pembro vs chemo	7.1	13.1 vs 6.7	2.1 vs 3.4	9.3 vs 6.9
KEYNOTE-180, II	ESCC, EAC	3rd or later	121	Pembro	5.8	9.9	2.0	5.8
KEYNOTE-028, 1b	ESCC, EAC	3rd or later	21	Pembro	7	30	1.8	7

Pembro: pembrolizumab, mo: month.

439 Poster Session

Correlation between immune-related adverse events and clinical outcomes in patients treated with first-line nivolumab plus chemotherapy for advanced gastric or gastroesophageal junction cancer: A multi-center retrospective study in Japan. First Author: Rin Inamoto, Department of Gastroenterology, Saitama Cancer Center, Kitaadachi-Gun, Saitama, Japan

Background: The CheckMate 649 trial and ATTRACTION-4 trial showed the efficacy of first-line nivolumab plus chemotherapy (Nivo-CT) for advanced gastric or gastroesophageal junction (GEJ) cancer. Recent studies have shown that immune-related adverse events (irAEs) caused by immune checkpoint inhibitors were associated with survival benefit in patients with solid tumors. However, there were few data on correlation between irAEs and efficacy of first-line Nivo-CT for advanced gastric or GEJ cancer. **Methods:** This multi-center retrospective study included patients with histologically confirmed advanced gastric or GEJ adenocarcinoma who were started on first-line Nivo-CT between December 2021 and December 2023. We divided the patients into two groups according to occurrence of irAEs; those with irAEs (irAE group) or those without (non-irAE group) and assessed the efficacy in both groups. Efficacy was evaluated by overall survival (OS), progression-free survival (PFS) and objective response rate (ORR). Landmark analysis at 2 months after initiating Nivo-CT was performed to adjust effects of early progression or death, in which patients who had events up to 2 months were excluded. **Results:** A total of 271 patients were included (median age, 70 [range 18–85] years; male/female, 64/36%; performance status (PS) 0/1/2, 47/49/4%; primary tumor site gastric/GEJ, 90/10%; status unresectable/recurrent, 81/19%; histology intestinal/diffuse, 30/70%; PD-L1 CPS <1/1~5/>5, 15/33/53%). Median follow-up time was 11.3 (range 1-30) months, and 80 (30%) patients developed irAE. In the landmark analysis, the median PFS was 11.1 months in the irAE group and 7.2 months in non-irAE group (HR 0.70 [95%CI 0.50-0.97], p = 0.034), and the median OS was 22.5 months in the irAE group and 17.9 months in non-irAE group (HR 0.77 [95%CI, 0.50-1.17], p = 0.219). The ORR were 67.4% in the irAE group and 66.7% in non-irAE group (p = 0.934). Multivariate analysis indicated that PS ≥1 (HR 1.66 [95%CI, 1.20-2.30], p = 0.02), elevated ALP (HR 1.61 [95%CI, 1.13-2.27], p = 0.008), and absence of irAEs (HR 1.55 [95% CI, 1.08-2.22], p = 0.018) were associated with poor prognosis. The most frequent irAEs were pneumonitis (n = 16), and grade ≥3 irAEs were observed in 26 (9.6%) patients: pneumonitis (n = 6), hepatitis/hypophysitis (n = 5). **Conclusions:** Development of irAEs was associated with improved outcomes in advanced gastric or GEJ cancer patients receiving first-line Nivo-CT. Research Sponsor: None.

440 Poster Session

HLX22 plus trastuzumab and XELOX for first-line treatment of HER2-positive locally advanced or metastatic gastric/gastroesophageal junction cancer (G/GEJC): Updated results with additional patients. First Author: Jin Li, Shanghai GoBroad Cancer Hospital, Shanghai, China

Background: Approximately 12–23% G/GEJC patients have human epidermal growth factor receptor 2 (HER2)-positive disease. Despite survival benefit from combination therapy with trastuzumab and chemotherapy, the prognosis remains unsatisfactory; more effective treatments are needed. This phase 2 study is evaluating the combination of HLX22 (an anti-HER2 monoclonal antibody targeting a different epitope than trastuzumab), trastuzumab, and XELOX chemotherapy as first-line treatment for patients with advanced/metastatic G/GEJC. Following the most recent report of 17 to 18 patients in each group at ASCO 2024 Annual Meeting, herein we present the updated efficacy and safety results with 31 patients in each group. **Methods:** Patients with locally advanced or metastatic HER2-positive G/GEJC and no prior systemic antitumor therapy were enrolled. Herein reported are results from Stage 2 of the study. Eligible patients were randomized to receive either HLX22 + trastuzumab + XELOX or placebo + trastuzumab + XELOX in 3-week cycles. Primary endpoints were independent radiology review committee (IRRC)-assessed progression-free survival (PFS) and objective response rate (ORR) per Response Evaluation Criteria in Solid Tumors (RECIST) v1.1. Secondary endpoints included other efficacy and safety endpoints. **Results:** As of June 30, 2024, 62 patients were randomized to the respective groups (31 vs 31), of whom 51 (82.3%) patients were male. Median follow-up duration was 20.3 and 24.0 months for the respective groups. The efficacy results are shown in the table. Treatment-emergent adverse events (TEAEs) were reported in 30 (96.8%) and 31 (100%) patients, and HLX22- or placebo-related TEAEs of grade 3 or higher were reported in 9 (29.0%) and 6 (19.4%) patients in the respective groups. One patient (3.2%) in the placebo + trastuzumab + XELOX group had a grade 5 HLX22-/placebo-related TEAE. **Conclusions:** With a manageable safety profile, the addition of HLX22 to first-line treatment with trastuzumab plus XELOX conferred survival benefit for HER2-positive G/GEJC patients. Clinical trial information: NCT04908813. Research Sponsor: Shanghai Henlius Biotech, Inc.

Updated efficacy, IRRC-assessed.

	HLX22 + trastuzumab + XELOX (n=31)	placebo + trastuzumab + XELOX (n=31)
Median PFS, months (95% CI)	NR (23.49, NE)	8.3 (5.7, 12.7)
HR (95% CI)	0.2 (0.06, 0.45)	-
12-month PFS rate (95% CI)	73.8 (50.3, 87.4)	34.2 (12.0, 58.1)
24-month PFS rate (95% CI)	61.5 (30.4, 82.0)	24.0 (10.3, 27.3)
Confirmed ORR, % (95% CI)	87.1 (70.2, 96.4)	80.6 (62.5, 92.5)
Week 48 ORR, % (95% CI)	38.7 (21.8, 57.8)	9.7 (2.0, 25.8)
Median OS, months (95% CI)	NR (17.6, NE)	22.0 (10.6, NE)
HR (95% CI)	0.5 (0.20, 1.21)	-
Median DOR, months (95% CI)	NR (22.1, NE)	9.7 (4.6, 20.0)
HR (95% CI)	0.1 (0.04, 0.41)	-

CI, confidence interval. NE, not evaluable. NR, not reached.

441 Poster Session

Phase II preoperative pembrolizumab for MSI high or MSS/PD-L1 gastric cancer followed by surgery and adjuvant therapy with pembrolizumab (NCT03257163): Results of a multicenter study. First Author: Timothy Kennedy, Rutgers Cancer Institute of New Jersey, New Brunswick, NJ

Background: Data suggest that patients with tumors with microsatellite instability (MSI-H), EBV expression, or positive PD-L1 expression can benefit from immunotherapy. This trial evaluated PD-1 immunotherapy in this subset of patients with operable gastric cancer. **Methods:** We present results of a phase 2 multi-institutional clinical trial (NCT03257163). Patients with cT2-T4, N0-N3, M0 gastric adenocarcinoma with MSI-H, PDL1 CPS > 1%, or EBV+ were included. Patients received 2 cycles preop pembrolizumab 200 mg IV q3 weeks followed by surgery. Pembrolizumab was given for 16 additional cycles postoperatively, concurrently with adjuvant chemoradiation (45 Gy) and 5 cycles of capecitabine (825 mg/m2 days 1-14 during cycles 3, 6 and 7; 625 mg/m2 BID days 1-14 during cycles 4 and 5 with radiation). Patients were defined as evaluable if they received > one cycle of postop pembrolizumab. Primary endpoint is DFS, powered to detect 3-yr DFS of 70% with 89% power. **Results:** All enrolled patients received two cycles of pembrolizumab preoperatively (n=45). Eight patients did not undergo surgery (1 due to frailty, 2 with distant progression on preoperative imaging, 2 with peritoneal disease on surgical exploration and 3 with locally advanced, unresectable tumors). Six patients did not receive adjuvant therapy (2 due to surgical complications, 2 for failure to thrive, 1 refusal and 1 pending adjuvant therapy). 31 patients received adjuvant therapy and are included in the predetermined, evaluable analytic cohort. These 31 patients had a median age of 67 years (range 44 – 85) with 22.6% Hispanic, 29.0% Asian, 25.8% Black and 25.8% White. Median follow-up was 31.1 months (range 3.45-63.8). In these patients, 14 (45.2%) had MSI-H, 2 (6.5%) EBV+, and 29 (93.5%) had PDL1 expression >1% tumors. Most tumors were advanced stage with 81% ≥ cT3 and 58% cN+. In the 31 evaluable patients, the 3-year DFS was 79.4%; 3-year OS was 78.9%. For MSI-H patients, 3-year DFS was 81.8%. For patients with MSS/PD-L1+ tumors, 3-year DFS was 77.8%. For the 45-patient cohort, 3-year DFS was 70%; 3-year OS was 65%. Three patients had pCR, two with MSI-H tumors and one with PD-L1 CPS of 4%. Downstaging occurred in 62.2% of patients, and 40% of surgical specimens had pT0/T1 tumors consistent with significant pathologic responses. **Conclusions:** A biomarker-driven preoperative treatment strategy for MSI-H, EBV+, and MSS/PDL1+ operable gastric adenocarcinoma patients is promising and warrants additional evaluation. This treatment strategy is novel and effective, and it incorporates immune checkpoint immunotherapy, low dose chemotherapy, and radiation thus avoiding more intensive combination chemotherapy regimens for patients with these biomarkers. Clinical trial information: NCT03257163. Research Sponsor: Merck.

442 **Poster Session**

Paclitaxel oral solution versus paclitaxel injection as a second-line therapy in advanced gastric cancer: A randomized, open-label, non-inferiority phase 3 trial. First Author: Jin Li, Shanghai GoBroad Cancer Hospital, Shanghai, China

Background: Paclitaxel injection (paclitaxel IV) is category 1 preferred regimen in the second-line therapy of gastric cancer, while it has shortcomings including vehicle-led safety risk, long time injection, premedication or frequent hospital visit. Paclitaxel oral solution (Liporaxel), the world's first successfully developed oral formulation, is an alternative. This study aimed to establish non-inferiority in efficacy and comparable safety profile of paclitaxel oral solution versus paclitaxel IV, as a second-line monotherapy in gastric cancer. **Methods:** This is a randomized, open-label, non-inferiority phase 3 trial conducted at 53 centers in China. Patients with unresectable or recurrent or metastatic gastric cancer progressed after fluoropyrimidine- or fluoropyrimidine plus platinum-based first-line therapy were randomly assigned at 1:1 ratio (stratified by gastrectomy, ECOG PS and prior chemotherapy) to receive paclitaxel oral solution (200mg/m^2 twice daily on days 1, 8, 15 of a 28-day cycle) or paclitaxel IV (175mg/m^2 on day 1 of a 21-day cycle). Co-primary endpoints were progression-free survival (PFS) assessed by blind independent review committee (BIRC) and overall survival (OS), with non-inferiority margin of hazard ratio (HR) of 1.18 for PFS and 1.16 for OS. **Results:** From April 22, 2019, to January 31, 2022 (data cut-off), 536 patients were randomized to receive either paclitaxel oral solution (n=268) or paclitaxel IV (n=268), with a median follow-up of 13.4 vs. 12.6 months. PFS met non-inferiority criteria, with BIRC-assessed median PFS of 3.02 months (95% confidence interval [CI]: 2.69, 3.71) in the oral group vs. 2.89 months (95% CI: 2.53, 3.48) in the IV group (HR=0.894, 95% CI: 0.719, 1.112, p=0.311). OS (cut-off on February 15, 2023, for OS primary analysis) demonstrated superiority for paclitaxel oral solution, with median OS of 9.13 months (95% CI: 7.72, 10.97) vs. 6.54 months (95% CI: 5.75, 7.26), showing 2.59-month improvement (HR=0.770, 95.5% CI: 0.635, 0.934, p=0.006). For the treatment-related adverse events (TRAEs), the oral group showed lower incidences of neuropathy (22.3% vs. 38.7% all grade), alopecia, fatigue, musculoskeletal and connective tissue disorders, and no hypersensitivity occurred without premedication. The most common ≥Grade 3 TRAEs were neutrophil count decreased (47.9% in oral vs. 54.5% in IV), white blood cell count decreased (41.5% vs. 35.3%) and anemia (16.6% vs. 10.9%). Grade 5 TRAEs were rare with comparable rates across two groups (four [1.5%] vs. three [1.1%]). **Conclusions:** Paclitaxel oral solution demonstrated non-inferiority in PFS and superiority in OS compared to paclitaxel IV, with clinically manageable and favorable safety profile, supporting paclitaxel oral solution as a second-line treatment option for gastric cancer. Clinical trial information: CTR20190050. Research Sponsor: None.

443 **Poster Session**

Neoadjuvant adebrelimab plus chemotherapy for resectable locally advanced esophageal squamous cell carcinoma: A phase 2 trial. First Author: Zhigang Li, Department of Thoracic Surgery, Shanghai Chest Hospital, Shanghai Jiao Tong University School of Medicine, Shanghai, China

Background: Several clinical trials have demonstrated the enhanced pathological response with the addition of immune checkpoint inhibitor to neoadjuvant chemotherapy in patients with resectable esophageal squamous cell carcinoma (ESCC), including the phase 3 ESCORT-NEO trial. Adebrelimab, an anti-programmed cell death-ligand 1 antibody, has shown potential as neoadjuvant monotherapy in the phase 1b NATION-1907 trial of resectable ESCC, with a 2-year overall survival rate of 92% in 30 patients. This study evaluated the efficacy and safety of adebrelimab plus chemotherapy as neoadjuvant therapy in patients with resectable locally advanced ESCC. **Methods:** In this single-center phase 2 trial (NCT06178211), patients aged 18-75 years with stage II-III (cT2N1-2M0 or cT3N0-2M0) ESCC received adebrelimab (1200 mg, day 1) plus albumin-bound paclitaxel (100 mg/m^2, days 1, 8, and 15) and carboplatin (area under the curve of 5 mg/mL/min, day 1) every 3 weeks for 2 cycles. Esophagectomy was performed 4-6 weeks after neoadjuvant therapy. The primary endpoint was pathological complete response (pCR; ypT0N0) rate. Secondary endpoints were clinical complete response (cCR) rate (defined as no residual tumor after neoadjuvant therapy, based on computed tomography, positron emission tomography-computed tomography [PET-CT], endoscopic biopsy, and endoscopic ultrasound-guided fine-needle aspiration [EUS-FNA]), margin-free (R0) resection rate, major pathological response (MPR; ≤10% residual tumor cells) rate, event-free survival, overall survival, and safety. **Results:** Between January 2024 and May 2024, a total of 36 patients were enrolled. Most patients (75.0%) had clinical stage III ESCC. Thirty-five patients completed 2 cycles of neoadjuvant therapy, while one patient refused the last dose of albumin-bound paclitaxel. Two patients refused surgery while 34 underwent esophagectomy. The pCR rate was 38.9% (95% confidence interval [CI], 23.1-56.5) in all patients and 41.2% (95% CI, 24.6-59.3) in the surgery set; the R0 resection rate was 94.4% (95% CI, 81.3-99.3) and 100% (95% CI, 89.7-100), and the MPR rate was 63.9% (95% CI, 46.2-79.2) and 67.6% (95% CI, 49.5-82.6), respectively. The cCR rate was 47.2% (95% CI, 30.4-64.5) in all patients. Of 36 patients, grade ≥3 treatment-emergent adverse events during neoadjuvant therapy included decreased neutrophil count (2 [5.6%]), decreased white blood cell (2 [5.6%]), decreased platelet count (1 [2.8%]), impaired hearing (1 [2.8%]), hyponatremia (1 [2.8%]), immune-mediated hepatitis (1 [2.8%]), intestinal obstruction (1 [2.8%]), and esophageal ulcer (1 [2.8%]). Surgical complications occurred in 17 (50%) of 34 patients. **Conclusions:** Neoadjuvant adebrelimab plus chemotherapy shows promising pathological response and favorable safety profile in patients with resectable locally advanced ESCC. Survival follow-up is ongoing. Clinical trial information: NCT06178211. Research Sponsor: Jiangsu Hengrui Pharmaceuticals Co., Ltd.

444 **Poster Session**

Preliminary results of benmelstobart plus anlotinib combined with SOX in the first-line treatment of advanced gastric or gastroesophageal junction (G/GEJ) adenocarcinoma with low PD-L1 expression: A single-arm, multicenter phase II clinical trial. First Author: Yong-Xu Jia, Department of Medical Oncology, The First Affiliated Hospital, Zhengzhou University, Zhengzhou, China

Background: First-line chemotherapy for advanced HER2-negative gastric/gastroesophageal (G/GEJ) adenocarcinoma has limited efficacy. PD-1 inhibitors plus chemotherapy show promise but need improvement, especially for patients (pts) with low PD-L1 expression. Anlotinib, a multi-targeted TKI for tumor angiogenesis/proliferation, is approved in China. This study aims to evaluate the efficacy and safety of benmelstobart a PD-L1 blockade) in combination with anlotinib and SOX (S-1 plus oxaliplatin) as a first-line regimen for advanced G/GEJ adenocarcinoma in patients with low PD-L1 expression, Preliminary findings will be reported promptly. **Methods:** Pts with HER2-negative, unresectable, locally advanced, or metastatic G/GEJ adenocarcinomas and a PD-L1 Combined Positive Score (CPS) < 5, who had not received prior systemic therapy were included. They received benmelstobart (1200mg, iv, d1, q3w) combined with anlotinib (10mg, po, d1~14, q3w), oxaliplatin (130mg/m^2, d1, iv, q3w) and S-1 (40mg, po, bid, d1~14, q3w) for 6 cycles as initial therapy. Maintenance therapy with benmelstobart (1200mg, iv, d1, q3w) plus anlotinib (10mg, po, d1~14, q3w) followed for non-progressive diseaseuntil PD or unacceptable toxicity occurred. Tumor responses were evaluated by RECIST 1.1 criteria. The target sample size was 37, with ORR as the primary endpoint, and safety, DCR, DoR, PFS, and 1-year OS rate as secondary endpoints. **Results:** From Jun 2023 to Sep 2024, 28 pts were enrolled and 27 were available for efficacy and safety evaluation. The patients' characteristics were as follows; median age (range): 59 (38-77) years, Male/Female: 19 (70%)/8 (30%), ECOG PS 0/1: 2 (7%)/25 (93%), Surgical history yes/no: 6 (22%)/21 (78%), CPS: < 1/ ≥1: 4 (15%)/23 (85%), respectively. In the response assessment, 19 PR (70.4%), 6 SD (22.2%), and 2 NE (7.4%) were observed, yielding an ORR of 70.4% (95% CI: 49.8-86.2) and DCR of 92.6% (95% CI: 75.7-99.1). As of September 17, 2024, 6 of 27 patients discontinued (3 due to PD, 3 voluntary), leaving 21 ongoing. The longest DoT was 15.08 months, and the median PFS was not reached. Common TEAEs ≥10% included thrombocytopenia (37%), anemia (33%), leukopenia (22%), HFS (19%), fatigue (15%), liver function abnormalities (15%), and appetite loss (11%). Grade ≥3 TEAEs were thrombocytopenia (4%) and anemia (4%). **Conclusions:** Preliminary findings indicate that the combination of benmelstobart and anlotinib with the SOX regimen as a first-line treatment for advanced G/GEJ adenocarcinoma with low PD-L1 expression demonstrates promising therapeutic efficacy and tolerable adverse events. Further validation of these results in a larger, consecutive patient population is warranted. Clinical trial information: ChiCTR2400085396. Research Sponsor: None.

445 **Poster Session**

Fruquintinib plus camrelizumab combined with paclitaxel liposome and nedaplatin as first-line treatment for advanced esophageal squamous cell carcinoma (ESCC): A single-arm, phase II clinical trial. First Author: Yanhong Gu, The First Affiliated Hospital with Nanjing Medical University, Nanjing, Jiangsu, China

Background: Camrelizumab in combination with chemotherapy has become a standard of care for the first-line treatment of advanced ESCC. Previous studies have indicated synergistic effect of fruquintinib in combination with chemotherapy or immunotherapy. Hence, we conducted a phase II trial to evaluate the efficacy and safety of fruquintinib plus camrelizumab, paclitaxel liposome, and nedaplatin as a first-line therapy for advanced ESCC. **Methods:** This is a single-arm, phase II study consisting of a dose-finding and a dose-expansion phase. A total of 33~36 patients aged 18~80 years with previously untreated advanced ESCC and an ECOG PS of 0-2 were planned to be enrolled. The dose-finding phase followed a 3+3 approach in order to determine the recommended phase II dose (RP2D) of fruquintinib for dose expansion. Patients received fruquintinib (3 mg, 4 mg, and 5 mg once daily on days 1-14, starting at 4 mg) orally, plus intravenous camrelizumab 200 mg d1, paclitaxel liposome 135 mg/m^2 d1, and nedaplatin 70 mg/m^2 d1, 21 days as one cycle. A maximum of 6 cycles was conducted, followed by maintenance therapy with fruquintinib in combination with camrelizumab. The primary endpoint was objective response rate (ORR, RECIST 1.1). Secondary endpoints included RP2D, disease control rate (DCR), progression-free survival (PFS), and adverse events (AEs). **Results:** As of August 24, 2024, 11 patients (9 males) were enrolled with a median age of 66 years (range 50-76). All patients had distant metastases, and the median number of sites of metastasis was 2 (range 1-5). In the dose-finding phase, 3 patients were treated with 4 mg of fruquintinib, and 6 patients were treated with 5 mg of fruquintinib. No dose-limiting toxicities (DLTs) were observed. Thus, the RP2D of fruquintinib was established as 5 mg once daily on days 1-14, 21 days as one cycle. Of the 7 evaluable patients, 6 achieved partial response, and 1 had stable disease. The ORR was 85.7% (95% CI 42.1%-99.6%), and the DCR was 100.0% (95% CI: 59.0%-100.0%). The median PFS was not reached. The most common treatment-related adverse events (TRAEs) were anemia (7/11), hypercholesterolemia (6/11), hypoalbuminemia (5/11), oral mucositis (5/11), hematuria (4/11), hypertension (4/11), fatigue (4/11), constipation (3/11), and neutropenia (3/11). Grade 3 TRAEs were identified in 2 patients, including oral mucositis (2/11), and anemia (1/11). No treatment-related serious adverse events (SAEs) or deaths occurred. **Conclusions:** The combination of fruquintinib, camrelizumab, paclitaxel liposome, and nedaplatin demonstrated significant efficacy and manageable toxicity profiles as a first-line treatment for advanced ESCC, suggesting a potential new treatment strategy. Clinical trial information: NCT06010212. Research Sponsor: None.

446 **Poster Session**

Prediction of relapse-free survival in stage III gastric cancer patients treated with postoperative adjuvant chemotherapy: Random survival forest analysis from the START-2 trial. First Author: Hiroki Sato, OITA University Faculty of Medicine, Yufu, Japan

Background: The START-2 trial demonstrated that docetaxel plus S-1 (DS) was superior to S-1 alone (S-1) in terms of recurrence-free survival (RFS) and overall survival (OS) as the adjuvant chemotherapy for stage III gastric cancer (GC) patients (pts) (J Clin Oncol 2019, Gastric Cancer 2022). We conducted the START-2 AR study as a retrospective analysis using data from the START-2 trial to develop a prognostic model for stage III GC pts treated with postoperative chemotherapy. **Methods:** Among 912 pts in the START-2 trial, 599 pts signed the consent form to participate in the START-2 AR study. There were 542 pts (265 pts in DS group and 277 pts in S-1 group) without missing values of prognostic candidates. Pts were randomly divided into a training set and a validation set with a ratio of 7:3. Prognostic model for RFS was developed using random survival forest (RSF) with the training set. The RSF model was applied to the validation set to assess its performance, evaluated using Harrell's C-index. We calculated tertiles of risk scores from the RSF model to divide the training set into low-, middle-, and high-risk subgroups. These cut-off values were used to classify the validation set. A Cox proportional hazard model was applied to compare the prognosis among the risk groups. **Results:** There were no statistically significant differences in patient characteristicsbetween the 599 pts in the START-2 AR study and the 912 pts in the STRAT-2 trial. RFS was comparable between the two groups. The RSF model for the training set (n = 380) yielded a C-index of 0.873, identifying the number of metastatic lymph nodes, serum albumin, CEA, tumor diameter, the primary tumor location, platelet count, age as prognostic factors in order of importance. In the validation set (n = 162), the C-index was 0.648. Both the low-risk group (n = 35) (hazard ratio [HR]: 0.34, 95% confidence interval [CI]: 0.17 – 0.68, $p < 0.01$) and the middle-risk group (n = 74) (HR: 0.55, 95% CI: 0.33 – 0.89, p = 0.02) in the validation set had significantly better RFS compared to the high-risk group (n = 53). Among the 76 pts in the low-risk group treated with DS, 44 were categorized into the low-risk group when assuming they had been treated with S-1 alone. The analysis of 599 pts after multiple imputation yielded results consistent with those from the complete data. **Conclusions:** We developed the RSF-based predictive model for stage III GC pts, which demonstrated good performance. The number of metastatic lymph modes was identified as the most important prognostic factor. Risk stratification based on this model effectively differentiated prognostic outcomes. Future external validation of the model is required. Clinical trial information: UMIN000011438. Research Sponsor: Taiho Pharmaceutical Co., LTD.; Japan Clinical Cancer Research Organization.

447 **Poster Session**

Results of a phase 2 trial of ramucirumab plus docetaxel as second-line treatment for patients with advanced gastric cancer (HGCSG 1903). First Author: Kentaro Sawada, Department of Medical Oncology, Kushiro Rosai Hospital, Kushiro, Japan

Background: Ramucirumab (RAM) has shown efficacy in combination with paclitaxel (PTX) or irinotecan in the second-line treatment of advanced gastric cancer (AGC). However, the efficacy and safety regarding the combination therapy of RAM and docetaxel (DTX) have not been reported. We hypothesized that RAM plus DTX for patients with AGC could be as effective as RAM plus PTX and could reduce the incidence of neuropathy. **Methods:** HGCSG 1903 was a non-randomized, single arm, phase 2 trial carried out at 20 institutes in Japan. AGC patients with refractory or intolerance to primary chemotherapy were eligible. RAM (8 mg/kg on day1 and 15) plus DTX (60 mg/m^2 on day1) combination therapy administered in 28-day cycle were continued until disease progression or emergence of adverse events requiring discontinuation. The primary endpoint was response rate (RR) with threshold value of 10.7% and an expected value of 27.9%. A total of 35 cases are planned for registration. The secondary endpoints were overall survival (OS), progression-free survival (PFS), and safety. This trial was registered with Japan Registry of Clinical Trials, number jRCTs011200010. **Results:** Between Nov 2020 and Dec 2023, 36 patients were enrolled. Median age was 70 (range, 36–82); female/male were 7/29; ECOG PS 0/1, 16/20, respectively. One patient did not initiate study treatment at the onset of bowel obstruction after enrollment. In the efficacy and safety analysis set of 35 patients, RR was 25.7% (95%C.I., 12.5–43.3%) and disease control rate was 74.3% (95%C.I., 56.7–87.5%). Median OS and PFS were 11.9 months (95%C.I., 3.0–20.7) and 3.1 months (95%C.I., 2.1–4.2), respectively. Grade 3 or higher adverse events that occurred in more than 5% of patients were neutropenia (68.6%), leucopenia (57.1%), febrile neutropenia (17.1%), hypertension (14.3%), anorexia (11.4%), anemia (8.6%), and fatigue (5.7%). Grade 4 neutropenia occurred in 60.0% of patients. The protocol was amended to allow primary G-CSF prophylaxis during study period. The number of patients who received primary G-CSF prophylaxis after the amendment was 7/13 (53.8%). Neuropathy occurred in 65.7% of all grades, with no incidence of grade 3 or higher. No death and new safety signals with a causal relation to study treatment were observed. **Conclusions:** The primary endpoint of response rate was met in HGCSG 1903. Although neutropenia and febrile neutropenia may be more frequent, they were manageable. RAM plus DTX might be considered as an option for second-line treatment of patients with AGC. Clinical trial information: jRCTs011200010. Research Sponsor: Nipro Corporation.

448 **Poster Session**

Updated results from the trastuzumab deruxtecan (T-DXd) 5.4 mg/kg triplet combination of DESTINY-Gastric03 (DG-03): First-line (1L) T-DXd with fluoropyrimidine (FP) and pembrolizumab in advanced/metastatic HER2-positive (HER2+) esophageal adenocarcinoma, gastric cancer (GC), or gastroesophageal junction adenocarcinoma (GEJA). First Author: Yelena Y. Janjigian, Memorial Sloan Kettering Cancer Center, New York, NY

Background: T-DXd is a HER2-directed antibody-drug conjugate; T-DXd 6.4 mg/kg monotherapy is approved for patients with metastatic HER2+ GC/GEJA who have received a prior trastuzumab-based regimen. Preliminary results from DG-03 Part 2 showed feasibility and promising antitumor activity with 1L T-DXd 6.4 mg/kg (arm D) or 5.4 mg/kg (arm F) with FP and pembrolizumab in patients with esophageal adenocarcinoma/GC/GEJA. T-DXd 6.4 mg/kg and FP with pembrolizumab was associated with higher than expected toxicity; alternatively, early safety data from T-DXd 5.4 mg/kg triplet combination showed a manageable safety profile. We report updated results for arm F from DG-03 Part 2 with an approximate time-matched analysis with arm D. **Methods:** DG-03 (NCT04379596) is a Phase 1b/2 multicenter, open-label, dose-escalation (Part 1) and -expansion (Parts 2, 3, and 4) study. In Part 2, patients with HER2+ (immunohistochemistry [IHC] 3+ or IHC 2+/in situ hybridization–positive by local testing) esophageal adenocarcinoma/GC/GEJA, irrespective of programmed cell death ligand 1 status, and no prior treatment for metastatic disease were enrolled. In arm D, patients received T-DXd 6.4 mg/kg intravenous (IV) infusion every 3 weeks (Q3W) and FP (5-fluorouracil [5-FU] 600 mg/m^2 continuous IV or capecitabine [cape] 1000 mg/m^2 twice daily [BID]) with pembrolizumab 200 mg IV Q3W; in arm F, patients received T-DXd 5.4 mg/kg IV Q3W and FP (5-FU 600 mg/m^2 continuous IV or cape 750 mg/m^2 BID) with pembrolizumab 200 mg IV Q3W.Primary endpoint was confirmed objective response rate by investigator assessment (INV) per RECIST 1.1. Secondary endpoints included duration of response by INV and progression-free survival by INV. Safety and tolerability were also assessed. **Results:** A time matched analysis was performed to compare the efficacy of arm D (n=43), data cut-off (DCO) October 27, 2022, and arm F (n=32), DCO May 6, 2024, given the limited follow up of the 5.4mg/kg. The median duration of follow up for the two cohorts was 4.1 months and 4.6 months for arm D and arm F, respectively. Objective response rate for arm D was 41.9% and 59.4% for arm F. Median PFS was 6.4 months (95% CI: 5.0, NC) for arm D and 5.8 months (95% CI: 5.6, NE) for arm F. At DCO, the median overall survival in both arms was not reached. Updated time matched safety and efficacy data will be provided at the time of presentation. **Conclusion:** The time matched analysis shows that lowering the dose of T-DXd to 5.4 mg/kg from 6.4 mg/kg and lowering the starting dose of capecitabine, did not result in decrease in efficacy of the combination of T-DXd + FP + pembrolizumab. Funding: This study is sponsored by AstraZeneca in collaboration with Daiichi Sankyo. In March 2019, AstraZeneca entered into a global development and commercialization collaboration agreement with Daiichi Sankyo for trastuzumab deruxtecan (T-DXd; DS-8201). Editorial acknowledgment: Under guidance of the authors and in accordance with Good Publication Practice, medical writing and editorial support was provided by Carmen Grimaldos, PhD, of Helios Medical Communications, part of Helios Global Group, and was funded by AstraZeneca. Clinical trial information: NCT04379596. Research Sponsor: AstraZeneca in collaboration with Daiichi Sankyo.

449 **Poster Session**

Clinical results of a phase 1 trial evaluating a HER2 directed CAR-T product selective for the tumor microenvironment (TME) in patients with GI and other solid tumor malignancies. First Author: Yuhong Zhou, Department of Medical Oncology, Zhongshan Hospital, Fudan University, Shanghai, China

Background: A major limitation of CAR-T in solid tumors has been the inability to safely target antigens shared between healthy and malignant tissue. One approach to improving selectivity is to regulate CAR affinity through the unique metabolism of the tumor microenvironment (TME). This first in human trial evaluated an autologous (41BB) CAR-T product encoding a tumor metabolism regulated HER2 CAR. **Methods:** This Phase 1, dose escalation, basket trial evaluated the safety, tolerability, pharmacokinetics and efficacy in 12 patients (pts) with HER2 IHC 3+ / FISH + Stage IV r/r solid tumor malignancies (gastric, esophageal, colorectal, breast). They had failed multiple lines of therapy (66.7% patients had failed 4~10 lines of therapies previously). Dose cohorts of 3E5 (n=3), 1E6 (n=3), or 1E7 (n=6) CAR-T cells/kg were evaluated with a 28-day in-patient DLT observation period. Cell product was manufactured from whole blood, with a 12-day closed process and release on day 17. Lymphodepletion regimen was 500 mg/m^2 Cy and 30 mg/kg^2 Flu X 3 prior to infusion of CAR-T. **Results:** Data cutoff 09/16/24. All 12 patients were evaluable for safety and efficacy. No DLTs, ICANS or SAEs were observed. For HER2 positive normal tissues of special concern (e.g. cardiomyocytes), longitudinal functional, biochemical and clinical cardiovascular tests showed no signs of OTOT. One patient experienced Grade 2 CRS and recovered. No AE's lead to study discontinuation. In one patient, there were two Grade 3 AEs attributed to product (neutropenia and hypoxia); both which resolved. With a median follow-up of 302 days for cohort 3, 12-month survival rate in this cohort is 83.3%, compared to the 12-month survival rate of 33.3% for cohorts 1 and 2 combined. Encouraging signals of efficacy were observed with an ongoing (at 24 wks) deep PR (100% reduction SLD) in a GC patient. The median peak copy number of CAR-T cells in peripheral blood (n=12) was 1,818 copies/μg DNA (range 403 – 10,466). **Conclusions:** The tumor metabolism regulated HER2 CAR-T product was generally well tolerated, with no DLTs, and evidence of radiologic objective response in patients with gastroesophageal cancer. Combined, the safety and activity observed in this study suggests that a previously challenging solid tumor antigen may be safely targeted by a CAR with affinity regulated by the metabolism of the TME. Enrollment for the study was complete as of 28-Feb-2024 (NCT04511871). Clinical trial information: NCT04511871. Research Sponsor: Shanghai Perhum Therapeutics, LTD; EXUMA Biotech, Hong Kong.

450 **Poster Session**

Tislelizumab (Tisle) combined with POFI (irinotecan, paclitaxel, oxaliplatin and 5-FU/levoleucovorin) as first-line treatment of advanced gastric/gastroesophageal junction adenocarcinoma (AGC): OS analysis results of the SYLT-023. First Author: Liyu Su, Fujian Medical University Cancer Hospital & Fujian Cancer Hospital, Fuzhou, China

Background: Tisle is an anti-PD-1 antibody. The combination of Tisle + XELOX/FP is a first-line treatment of AGC in China. Both irinotecan and paclitaxel have also shown antitumor activity in AGC. In this phase I/II study, we explore the safety, tolerability, and efficacy of Tisle + POFI as first-line treatment of HER-2 negative, pMMR AGC. **Methods:** In a phase I dose finding study using a standard 3+3 design, subjects received four escalating dose levels (dl) of irinotecan/paclitaxel (mg/m^2): 135/45 (dl #1), 150/45 (dl #2), 135/67.5 (dl #3), and 135/90 (dl #4) in combination with tisle 200 mg plus oxaliplatin 85 mg/m^2, levoleucovorin 200 mg/m^2, and 5-FU 2400 mg/m^2 for 46 hours every 2 weeks. Primary endpoints were safety, tolerability, and RP2D. **Results:** Fifteen subjects with treatment naïve AGC were enrolled (3 each in dl #1, dl #2, and dl #3 and 6 in dl #4). The median age was 65 years (range 36-72); 80% were male. Two subjects (13.3%) were diagnosed with GE junction cancer, 11 (73.3%) had undifferentiated disease, and 5 (33.3%) had liver metastasis. PD-L1 CPS scores were: 5 (n=5); 1 (n=1); and 0 (n=9). All subjects were evaluated for DLT. One DLT (grade 4 neutropenia) occurred within 28 days in dl #4. No maximum tolerated dose was reached; the RP2D was dl #4. As of 18th Sep 2024, confirmed objective response rate in 13 subjects with measurable disease was 100% (1 CR, 12 PR) per RECIST 1.1. Of the 2 subjects with non-measurable disease, one was a CR and the other was non-CR/non-PD. The median PFS was 10.51 (95% CI: 7.44, 13.58) months (versus 6.9 [5.7-7.2] months for ITT population for RATIONALE 305, ESMO 2023), the median DOR was 7.39 (95% CI: 5.34, 9.44) months, and the median OS was 14.75 (95% CI: 5.48, 24.02) months. The OS showed no significant differences between CPS $\geq$ 1 and CPS $<$ 1. All subjects (100%) had AEs. Seven (n=7, 46.67%) of the 15 subjects reported grade $\geq$3 AEs, including neutropenia (n=7; 46.67%), leukopenia (n=3; 20%), anemia (n=2; 13.33%). **Conclusions:** Tisle+POFI was well tolerated and showed preliminary antitumor activity in AGC. A phase II study is ongoing. Clinical trial information: NCT05319639. Research Sponsor: None.

451 **Poster Session**

Neoadjuvant immunotherapy and nonoperative management of dMMR gastroesophageal adenocarcinoma (GEA): A retrospective case series. First Author: Oudai Sahvan, Mayo Clinic Comprehensive Cancer Center, Phoenix, AZ

Background: Perioperative chemotherapy with surgery is the standard of care for patients (pts) with resectable gastric or gastroesophageal junction adenocarcinoma (GEA). However, recent studies have shown more favorable responses after neoadjuvant immunotherapy (NIO) in patients with mismatch repair deficient (dMMR) cancers. Herein, we present the outcomes of 10 cases with dMMR GEA. **Methods:** We retrospectively reviewed patients with GEA at Mayo Clinic Arizona who met the following criteria: have dMMR disease by IHC, nonmetastatic operable disease, and received NIO +/- chemotherapy. Patient demographics, disease characteristics, and clinical outcomes were extracted. The primary endpoint was clinical complete response (cCR) in patients treated with non-operative management (NOM) defined as complete radiological response and no residual disease on endoscopic ultrasound (EUS) and biopsy. **Results:** 10 pts with dMMR GEA with a median age of 70 (43-85) were identified. 6 pts had T2N0, 1 had T4N0, 3 had N+ disease. 8 pts received NIO monotherapy upfront, 1 received NIO after progressing on chemotherapy, and 1 concurrently with chemotherapy. No grade 3/4 toxicities were reported, however, 2 pts discontinued therapy due to arthralgia and colitis. After NIO, 9/10 pts had significant or complete resolution of FGD avidity on PET/CT, while 1/10 had stable disease. 3/10 pts underwent surgical resection, with 2 achieving pathological complete response (pCR) and the third showing T1N0 disease with no recurrence 1 year after surgery. Notably, one of the pts with pCR passed away due to surgical complications. 7/10 patients had NOM, of whom, 4 had cCR and 3 had radiological response only, as no EUS data was available. 4/7 NOM pts had follow up data at 6 months with no evidence of progression, of whom one pt remained progression-free at 33 months. ctDNA testing was performed in 6/10 pts: 2 had undetectable ctDNA at baseline, while 3 became MRD-negative after NIO, consistent with radiological and/or pathological response, and 1 had persistent ctDNA both before and after NIO, despite cCR. **Conclusions:** This study demonstrates exceptional and prolonged response, along with good tolerability to NIO in pts with dMMR GEA, supporting the feasibility of non-operative management in such patients. Research Sponsor: None.

452 **Poster Session**

Phase II study of neoadjuvant NALIRIFOX followed by chemoradiation with paclitaxel and carboplatin in locally advanced gastroesophageal cancer. First Author: Jennifer Yon-Li Wo, Massachusetts General Hospital, Boston, MA

Background: We performed a single-arm phase II study of neoadjuvant NALIRIFOX and chemoradiation (CRT) with concurrent carboplatin/taxol (C/T) followed by surgery in patients with locally advanced gastroesophageal (GE) cancer. **Methods:** Patients were enrolled on an NCI sponsored, prospective, single arm study (NCT04656041). Key eligibility criteria included: histologically confirmed T3/4 or lymph node (LN) positive GE junction or esophageal cancer, ECOG PS $\leq$1, age 18+, and life expectancy $>$ 3 months. Exclusion criteria included: metastatic disease, prior chemotherapy or RT, or prior targeted therapy for diagnosis. Extensive LN disease beyond the surgical field (supraclavicular or para-aortic) was permitted if deemed feasible to be encompassed within a RT field. Laparoscopy was not required. Pts were treated with neoadjuvant NALIRIFOX x 4-8 cycles pending patient tolerance and physician discretion, restaging, CRT (50.4 Gy in 28 fractions) with concurrent C/T, restaging, followed by surgical resection. Dose reductions were at discretion of the treating physician. The primary study endpoint was the pathologic complete response rate following neoadjuvant treatment with NALIRIFOX and CRT with concurrent C/T. Major pathologic response (mPR) was defined as Tumor Regression Grade 0 and 1. Secondary endpoints included: 1) acute toxicity; 2) clinical response per RECIST criteria; 3) PFS and 4) OS. **Results:** From June 2021 to October 2023, 40 patients were enrolled. Median age was 64 (range: 47-82), and 31 patients were male (78%). All patients started NALIRIFOX, and the median number of NALIRIFOX received was 5 (range: 1-8). Reasons for discontinuation of NALIRIFOX were: physician discretion (n=29), subject withdrew from treatment (n=2), progressive disease (n=1), death (n=1). Rates of grade 3+ toxicity for overall, gastrointestinal, and hematologic attributed to NALIRIFOX were 55%, 35%, and 13% respectively. 35 patients started chemoRT (88%) and 33 patients completed chemoRT (83%). Rates of grade 3+ toxicity for overall, gastrointestinal, and hematologic attributed to chemoRT were 86%, 11%, and 9% respectively. Of 33 who completed chemoRT, two patients declined surgery (including 1 with clinical CR), and 31 (78%) patients went for surgical exploration. No patients were found with intraoperative metastases. Therefore, 31 (78%) patients underwent surgical resection. In total, 9 patients had pCR (29% in resected cohort, 23% in ITT cohort) and 15 had a major PR (48% in resected cohort, 38% in ITT cohort). In total, 10 (25% in ITT) patients had pCR or clinically CR at 1 year. **Conclusions:** Neoadjuvant NALIRIFOX followed by CRT is feasible with acceptable rates of treatment completion and grade 3+ toxicity. In our phase II trial, the rate of pCR is promising and further studies incorporating this regimen should be considered. Clinical trial information: NCT04656041. Research Sponsor: None.

453 **Poster Session**

Non-metastatic gastric cancer (GC) in patients aged 80 years or older: Immediate and long-term outcomes of combined treatment. First Author: Nikolay Semenov, SBIH Moscow Clinical Scientific and Practical Center named after A.S. Loginov of DHM, Moscow, Russian Federation

Background: Comorbidities and increased risk of chemotherapy-induced toxicity in elderly patients presents significant challenges in management of GC. Here we presents a single-center experience in treatment patients aged 80 years or older with non-metastatic GS, reporting both immediate and long-term outcomes of surgical treatment (ST). **Methods:** This retrospective analysis includes 61 patients (pts) diagnosed with non-metastatic GC or cardio-esophageal junction (CEJ) cancer. The primary endpoint was relapse-free survival (RFS); secondary endpoints included overall survival (OS) and the incidence of surgical complications (SC). **Results:** Among the 61 pts (median age 82.7 years, range 80.0-89.2) 39.3% were men. ECOG PS was 0-1 in 49 pts (80.6%) and 2 – in 12 pts (19.4%). The primary tumor site was the stomach in 57 pts (93.4%) and the CEJ in 4 pts (6.6%). Tumor stage had been defined as cT_{1-2} in 33.3% pts, cT_{3-4} in 57.1%; cN_0 – in 36.1%, stage remained unknown in 6.3%. Radical surgery was performed in pts 78.7% (n=48), 29 pts (47.5%) of them underwent ST only without perioperative chemotherapy (CT). Neoadjuvant CT (NACT) received 23 pts (37.7%): platinum-based doublet regimen – 82.6% (n=19), while 17.4% (n=4) received capecitabine only. ST was performed in 47.8% (n=11) pts received NACT, other 12 pts didn't undergo ST due to progression of disease (8.3%), comorbidities (33.3%), or patient preference (16.7%) in 1, 4 and 2 cases, respectively. Adjuvant CT was administered to 9 pts (14.3%). Surgical interventions included gastrectomy in 16.6% pts (n=9), proximal or distal resection – in 8.3% (n=4) and 74.1% (n=35), respectively. R0-resection was achieved in 42 pts (88.6%). SC has 20.8% (n=10) pts: classified as Clavien-Dindo grade (G) II – 4.4%, G IIIa – 8.9%, G IIIb - 8.9%; these included anastomotic leakage (6.7%), abscess formation (2.2%), cardiac events and thrombosis (4.4%), and eventration (2.2%). SC G IV or V not reported. The median duration of postoperative hospitalization was 8 days. In the overall cohort, the 3-y RFS was 53% (95% CI 28.9-53.9), 3-y OS – 67% (95%CI 27.9-75.2). Among the pts with stage IA (n=15), the 3-y RFS was 72% (95%CI 49.1-85.1), with a maximum follow-up period of 80 months and no cancer-related deaths. In patients with $cT_{2-4}N_{any}$ 3-y RFS was 70% in the overall cohort and 74% in the surgery-only group (HR 1.09; 95%CI 0.45-2.65, p=1). **Conclusions:** Our findings demonstrate that over 65% of patients aged 80 years or older survive for at least 3 years following surgery for GC, with 53% living without evidence of metastases. Additionally, the rate of SC in this elderly cohort is comparable to that observed in younger populations. The role of perioperative or neoadjuvant CT in this age group remains a topic of ongoing research and discussion. It is critical to manage such population at referral oncological centers with expertise in treating elderly cancer pts. Research Sponsor: None.

454 **Poster Session**

Updated efficacy results of ASKB589 combined with CAPOX and PD-1 inhibitor as first-line treatment for metastatic gastric/gastro-esophageal junction (G/GEJ) adenocarcinoma from a phase Ib/II study. First Author: Zhi Peng, Beijing Cancer Hospital, Beijing, China

Background: ASKB589 is a non-fucosylated fully human IgG1 monoclonal antibody that binds CLDN18.2 with high affinity, specificity, and improved ADCC and CDC activities. Previous findings indicated that ASKB589 exhibited a well-tolerated safety profile at doses up to 20 mg/kg and demonstrated promising antitumor effects. A dose of 6 mg/kg has been selected for further evaluation. Here, we present the updated data from dose expansion part in NCT05632939 study. **Methods:** This phase 1b/2 study was consisted of dose-escalation part and dose-expansion part. The expansion part aimed to assess the safety and efficacy of ASKB589 combined with CAPOX plus a PD-1 inhibitor as the initial treatment for advanced G/GEJ cancer with CLDN18.2 positive, regardless of PD-L1 expression. CLDN18.2 and PD-L1 expression were evaluated through IHC analysis using clone DS-3 and clone E1L3N at a central lab. All pts were given ASKB589 intravenously at a dosage of 6 mg/kg. Responses were assessed by RECIST 1.1 every 6 weeks. Adverse events were graded using CTCAE v5.0. **Results:** As of July 29, 2024, 49 evaluable pts with CLDN18.2 M/H expressions (≥2+ membrane staining intensity in ≥40% of tumor cells) regardless of PD-L1 CPS score. Forty pts (81.6%) achieved a partial or complete response, with a confirmed objective response rate (cORR) of 73.5% (95%CI: 58.9, 85.1). The median duration of response was 11.14 months (6.93, NE). Three subjects underwent surgical treatment as a result of tumor reduction following treatment. All 49 pts achieved SD or better with a disease control rate of 100%. cORR was 76.2% and 74.1% in the CPS<1 and CPS<5 groups, respectively. Of all treated pts (N=50), PFS rate at 9 months was 58.1% (95% CI: 42.2, 71.0) and OS rate at 12 months was 77.1% (95% CI: 61.2, 87.2). Mature PFS and OS data will be reported at the time of the conference. The safety profile of the triplet is generally consistent with the safety data of ASB589 plus CAPOX combination in first-line G/GEJ cancer pts presented previously, which was mainly characterized by manageable on-target-off-tumor effects, including hypoalbuminemia, nausea, and vomiting. **Conclusions:** The combination of ASKB589 with CAPOX and a PD-1 inhibitor as the initial treatment for pts with G/GEJ cancer was well tolerated with no unexpected safety signals. It has shown a high confirmed ORR, 100% disease control, deep and durable antitumor activity, regardless of PD-L1 expression levels. A phase 3 study of the triple regimen for the 1L G/GEJ cancer is actively enrolling pts in China. Clinical trial information: NCT05632939. Research Sponsor: None.

cORR based on CLDN18.2 and PD-L1 expression.

n/N (%)	PD-L1CPS<1	PD-L1CPS 1-5	PD-L1 CPS≥5	Total
CLDN18.2 M&HcORR	16/21(76.2%)	7/10 (70.0%)	13/18 (72.2%)	36/49 (73.5%)
CLDN18.2 Moderate cORR	2/3(66.7%)	0/1(0.0%)	6/9(66.7%)	8/13(61.5%)
CLDN18.2 High cORR	14/18(77.8%)	7/9(77.8%)	7/9(77.8%)	28/36(77.8%)

455 **Poster Session**

Individual patient data (IPD) pooled analysis on the optimal therapeutic management of patients with microsatellite instability-high (MSI) resectable gastroesophageal adenocarcinoma (GEA). First Author: Alessandra Raimondi, Department of Medical Oncology, Fondazione IRCCS Istituto Nazionale dei Tumori, Milan, Italy

Background: Patients with resectable MSI/dMMR GEA showed improved survival and modest if any benefit from chemotherapy. Preoperative treatment with immune checkpoint inhibition (ICI) showed high rate of major-complete pathologic response in single arm trials possibly allowing the design of chemotherapy/surgery-free approaches. **Methods:** This was a multinational IPD analysis including patients with resectable GEA with MSI/dMMR status enrolled in INFINITY and NEONIPIGA phase II trials, with dual CTLA-4/PD-(L)1 ICI followed by surgery +/- adjuvant ICI; PROSECCO retrospective study, with perioperative FLOT chemotherapy and surgery, and the dataset of our previous IPD analysis on MAGIC, CLASSIC, ARTIST and ITACA-S randomized trials of patients treated with surgery alone or plus older perioperative/adjuvant chemo(radio)therapy regimen. Primary endpoint was the evaluation of rates of pathologic complete response (pCR) defined as TRG1a Becker and major-complete pathologic response (pCR/MPR) defined as TRG1a/b Becker according to preoperative treatment schedule in patients who underwent surgery. Univariable and multivariable analyses were conducted using a random effects logistic model adjusted with propensity score. Secondary endpoints were event-free survival (EFS) and overall survival (OS) according to the therapeutic strategy in the overall study population. Multivariable mixed-effects Cox models weighted with propensity score were performed. **Results:** The IPD included 197 patients. Of these, 49 received ICI +/- surgery, 27 FLOT chemotherapy plus surgery, 33 surgery alone and 88 older chemo(radio)therapy regimens plus surgery. In the 69 patients resected after neoadjuvant ICI or FLOT standard of care treatment, ICI demonstrated a higher rate of pathologic response compared to chemotherapy (pCR 61.9% vs 3.7%, OR 54.8 p=0.002; pCR/MPR 78.6% vs 10%, OR 39.3 p<0.001). In ITT population, no significant difference in OS and EFS was shown in patients treated with ICI, FLOT plus surgery, old chemo(radio)therapy plus surgery or surgery alone. **Conclusions:** In resectable MSI/dMMR GEA, upfront ICI showed comparable survival outcomes to surgery alone, with limitations of study design and sample size. The impact on survival of ICI versus surgery alone should be investigated prospectively to avoid overtreatment or identify specific risk categories with benefit. The high rate of major-complete pathologic response may allow to study or perform organ sparing surgery procedures or non-operative management to reduce surgical morbidity/mortality and improve quality of life. Research Sponsor: None.

	OS			EFS		
	HR	2.5-97.5% CI	p	HR	2.5-97.5% CI	p
Ref: Surgery only	-	-	-	-	-	-
Chemo+surgery	1.16	0.27-4.98	0.84	1.10	0.39-3.07	0.85
FLOT+surgery	1.10	0.19-6.22	0.92	2.38	0.90-6.27	0.08
ICI +/- surgery	1.97	0.43-8.96	0.38	1.39	0.51-3.77	0.52

456 **Poster Session**

Safety and preliminary efficacy of ATG-022 in patients with advanced/metastatic gastric cancer (CLINCH). First Author: Jinfeng Ma, Shanxi Cancer Hospital, Taiyuan, China

Background: ATG-022 is an antibody-drug conjugate (ADC) targeting Claudin 18.2 (CLDN 18.2) tumor antigen broadly expressed in gastric and other solid tumors. ATG-022 consists of a humanized monoclonal antibody that binds to CLDN 18.2 with high affinity of sub-nM grade and a conjugated linker-payload VC-MMAE, demonstrating activity across a wide range of CLDN18.2 expression levels, including both high and low/ultra-low expression levels. **Methods:** The CLINCH study included dose escalation phase and dose expansion phase. In the dose escalation, patients(pts) with advanced solid tumors regardless of CLDN 18.2 expression received ATG-022 once every three weeks (0.3-3.0 mg/kg Q3W) to evaluate the safety, tolerability, and pharmacokinetics. CLDN 18.2-positive patients were enrolled in dose expansion to receive ATG-022 at recommended phase 2 doses (RP2D) to evaluate the efficacy and safety. The primary endpoints included dose-limiting toxicity (DLT) and adverse events (AEs). The efficacy endpoints included objective response rate (ORR) and disease control rate (DCR), evaluated per RECIST1.1 criteria. **Results:** As of Aug 21, 2024, 16 pts with advanced solid tumors, including 7 gastric cancer (GC), were treated with ATG-022 from 0.3 to 3.0 mg/kg in dose escalation, with 1 DLT of grade 3 nausea found at 3 mg/kg among 6 pts, and 21 GC pts were enrolled in dose expansion to receive RP2D of 2.4 mg/kg Q3W. Among all 37 pts, median age was 61 years. Baseline ECOG were 0 (9 pts) and 1 (28 pts); 33 (89.2%) pts had stage IV disease; 26 (70.3%) pts had received ≥ 2 prior lines of systemic therapy and 14 (37.8%) pts had prior anti-PD-1/L1 therapy. At 2.4 mg/kg in expansion, 3 (14.3%) pts had ≥ 1 Serious TRAEs; 8 (38.1%) pts had grade ≥ 3 TRAEs; the most common TRAEs included neutrophil count decreased (47.6%), nausea (38.1%) and white blood cell count decreased (33.3 %). In dose escalation, 7 GC pts were enrolled, including 3 CLDN 18.2-positive pts. One PR and 1 CR was observed in pts with GC at the dosage of 1.8 mg/kg and 2.4 mg/kg, respectively. Among 12 GC pts who had at least 1 tumor evaluation in dose expansion, 5 PR and 7 SD were observed, resulting in an ORR of 41.7%, and a DCR of 100%. One CR and 1 PR were observed among 3 GC pts receiving 2.4 mg/kg and with CLDN 18.2 expression < 2+, 5%. The Cmax of both total antibody and ADC drug is increased as dose increased from dose 0.3 – 3.0 mg/kg. No pronounced exposure accumulation was found after multiple dose treatment. Preliminary data showed Dose-Normalized AUC of MMAE for ATG-022 is comparable with other vcMMAE ADC drugs. **Conclusions:** ATG-022 demonstrated a manageable safety profile, comparable PK properties in current dosing levels, and encouraging preliminary antitumor effects in GC pts from high CLDN 18.2 expression to low CLDN 18.2 expression, suggesting further clinical investigation in pts with variable CLDN 18.2 expression. The enrollment of GC and other solid tumors are ongoing. Clinical trial information: NCT05718895. Research Sponsor: Antengene.

457 **Poster Session**

Robot-assisted esophagectomy with extended mediastinal lymph node dissection: The learning curve and surgical outcomes. First Author: Hirofumi Kawakubo, Department of Surgery, Keio University School of Medicine, Tokyo, Japan

Background: We have performed thoracoscopic surgery (TMIE) and robot-assisted surgery (RAMIE) for esophageal cancer with extensive mediastinal lymph node dissection, including resection of the sympathetic nerve layers and the thoracic duct. We studied the learning curve, short-term results, and mid-to-long-term outcomes of RAMIE for esophageal cancer at our institution. **Methods:** We compared surgical times at different stages of the procedure. For mid to long-term outcomes, we analyzed 360 cases of TMIE/RAMIE performed between January 2009 and June 2022, comparing the incidence of postoperative complications, recurrence-free survival, and patterns of recurrence between RAMIE and TMIE. Recurrent laryngeal nerve palsy (C-D I) was evaluated using postoperative endoscopy or contrast swallowing tests to assess even mild paralysis. **Results:** The average console time in the thoracic procedure was 234.4 minutes, with Group A (294.5 minutes, range: 211-397 minutes) showing significantly longer times compared to Group B (190.5 minutes, range: 130-257 minutes). Blood loss was almost 0ml in 10 cases (50%) in Group A and 25 cases (62.5%) in Group B. Postoperative complications included pneumonia in 5 cases (8.3%) and recurrent laryngeal nerve palsy (C-D III) in 5 cases (8.3%), with no significant difference between the groups. However, surgical time for lymph node dissection around the recurrent laryngeal nerve was significantly shorter in Group B compared to Group A. Long-term analysis included 326 TMIE and 34 RAMIE cases. Mild recurrent laryngeal nerve palsy (>C-D I) was observed in 92 cases (28%) of TMIE and 7 cases (21%) of RAMIE, showing a lower tendency in the RAMIE group (p=0.338). Pneumonia (>C-D II) occurred in 57 cases (18%) in the TMIE group and 3 cases (9%) in the RAMIE group, with a relatively lower incidence in the RAMIE group (p=0.178). For patients with cT1N0M0 who did not receive preoperative chemotherapy, the 3-year recurrence-free survival rates were 95% for TMIE and 93% for RAMIE. **Conclusions:** RAMIE demonstrated a significant reduction in operative time due to the standardization of the procedure, particularly in the dissection of recurrent laryngeal nerve lymph nodes and other upper mediastinal maneuvers. RAMIE also showed a potential reduction in the incidence of recurrent laryngeal nerve palsy and pneumonia compared to TMIE. In terms of long-term outcomes, RAMIE maintained local control similar to TMIE, with further improvements in safety expected. Research Sponsor: None.

458 **Poster Session**

Impact of time to surgery on outcomes in patients with advanced esophageal squamous cell carcinoma following neoadjuvant therapy: An exploratory analysis of phase III trial JCOG1109. First Author: Ryosuke Kita, Department of Surgery, Kyoto University Graduate School of Medicine, Kyoto, Japan

Background: JCOG1109, a multicenter three-arm phase III trial, evaluated three neoadjuvant treatment regimens for advanced esophageal squamous cell carcinoma (ESCC), cisplatin and 5-fluorouracil (CF), docetaxel, cisplatin, and 5-fluorouracil (DCF), and cisplatin, 5-fluorouracil, and radiation therapy (CF-RT). The trial demonstrated the superiority of neoadjuvant DCF therapy in improving overall survival (OS). This exploratory analysis investigates the impact of the interval time from completion of neoadjuvant therapy to surgery (time to surgery, TTS) on both short- and long-term outcomes across different neoadjuvant therapy regimens. **Methods:** This analysis included patients from JCOG1109 who underwent surgery after neoadjuvant therapy. Patients were categorized into four TTS subgroups within each arm, based on quartiles of the overall cohort. Short-term outcomes such as perioperative complications, as well as long-term outcomes including OS and progression-free survival (PFS) were evaluated across these TTS subgroups. **Results:** Of the 601 patients enrolled in JCOG1109, 546 patients proceeded to surgery following neoadjuvant therapy. The median TTS was 35 days (range: 16-81) for the CF arm, 38 days (range: 17-109) for the DCF arm, and 41 days (range: 14-98) for the CF-RT arm. Baseline characteristics were well-balanced across all TTS subgroups in each treatment arm. Short-term outcomes revealed that operative time and overall proportions of postoperative complications were comparable. However, in the CF-RT arm, a longer TTS was associated with increased blood loss (200 ml, 210 ml, 300 ml, 370 ml; p-value for trend test = 0.010) and tended to increase the risk of anastomotic leakage (6%, 7%, 14%, 18%; p-value for trend test = 0.073). No significant differences in OS or PFS were observed across TTS subgroups in any treatment arm, with hazard ratios indicating no clear trend toward improved or worsened long-term outcomes. **Conclusions:** The timing of surgery following neoadjuvant therapy does not appear to affect long-term prognosis in patients with advanced ESCC, irrespective of the neoadjuvant therapy regimen. However, for patients undergoing neoadjuvant CF-RT, earlier surgery may be advantageous in reducing the risk of increased surgical complexity and anastomotic leakage. Research Sponsor: None.

459 **Poster Session**

Anlotinib plus benmelstobart as adjuvant therapy in patients with esophageal squamous cell carcinoma: A phase II clinical trial (ALTER-E005). First Author: Changying Guo, Jiangxi Provincial Tumor Hospital, Nanchang, China

Background: Surgery represents a primary radical treatment for esophageal cancer, However, the risk of recurrence remains noteworthy after surgical resection, underscoring the need for more effective and tolerable postoperative regimens for esophageal squamous cell carcinoma (ESCC). Benmelstobart, a novel PD-L1 blockade, demonstrated promising antitumor activity when combined with anlotinib, an antiangiogenic agent, in the treatment of various tumors including advanced ESCC (NCT05038813) and biliary tract cancer (NCT03996408). Therefore, ALTER-E005 study aimed to evaluate the efficacy and safety of combining anlotinib with benmelstobart as adjuvant therapy in patients (pts) with ESCC. **Methods:** This was a single-arm, multi-center phase II clinical trial. Thirty pts (≥18 years) with histologically confirmed T1-2N1-3M0 or T3-4NanyM0 ESCC who had undergone radical (R0) resection with no recurrence 6 to 12 weeks after surgery and an ECOG PS ≤ 1 were eligible for enrollment. Enrolled pts received oral anlotinib (12 mg, days 1-14) and intravenous benmelstobart (1200 mg, day 1) every 3 weeks for up to 16 cycles or until disease recurrence. The primary endpoint was disease recurrence free (DFS). while secondary endpoints included safety, 1-year DFS rate, 3-year DFS rate, 1- year overall survival (OS) rate, and 3-year OS rate. **Results:** As of the data cut-off date (September 20, 2024), a total of 30 patients were enrolled with a median age of 67. Among the 30 surgically resected patients with ESCC, 26 (86.7%) had an ECOG score of 1, and 25 (83.3%) were male. According to the eighth edition of the AJCC Cancer Staging Manual, 9 (30.0%) were classified as stage II, and 21 (70.0%) were classified as stage III. As of the data cut-off date, the median follow-up time was 8.9 months (95% CI: 7.9-12.3). Four pts experienced disease recurrence, and the median duration of DFS has not yet been reached, with 6-month and 1-year DFS rates of 95.7% (95% CI, 72.9%-99.4%), 84.0% (95% CI, 57.7%-94.6%) respectively. Currently, 13 pts are still undergoing therapy. Out of the total 30 patients, 9 (30%) experienced grade 3 treatment-emergent adverse events (TEAEs). No grade 4 or higher TEAEs were observed. The grade 3 TEAEs included hand-foot syndrome (10.0%), hypertension (3.3%), abnormal liver function (3.3%), immune-related hepatitis (3.3%), pneumonia (3.3%), hyperglycemia (3.3%) Lymphocyte count decreased (3.3%), acute cholecystitis (3.3%) and syncope (3.3%). **Conclusions:** This study highlighted the promising efficacy and safety preliminarily of combining anlotinib with benmelstobart as adjuvant therapy in patients with ESCC who underwent radical (R0) resection and showed no recurrence 6 to 12 weeks after surgery. Clinical trial information: NCT05252078. Research Sponsor: None.

460 **Poster Session**

Clinical study of radiotherapy combined with tegafur bolus synchronization in the treatment of inoperable elderly middle and advanced esophageal cancer. First Author: Xixi Zhao, Department of Radiation Oncology, The Second Affiliated Hospital Of Xi'an Jiaotong University (Xibei Hospital), Xi'an, China

Background: For patients with inoperable oesophageal cancer, simultaneous radiotherapy is the standard treatment option. However, elderly patients are often unable to tolerate CCRT, so a safe, effective and tolerable treatment option is needed. Tegafur, a derivative of 5-FU, has shown efficacy and safety in the treatment of oesophageal cancer. This clinical study evaluated the efficacy and toxicity of radiotherapy combined with a Tegafur bolus in elderly patients with oesophageal cancer. **Methods:** In this study, 62 patients with unresectable intermediate and advanced ESCC (stage IIA to IVA) between March 2022 and February 2024 were screened and divided into two groups: 32 in the experimental group (simultaneous radiotherapy and chemotherapy) and 30 in the control group (radiotherapy alone). The total radiotherapy dose was 50.4-64 Gy/5.6-6.4 wGy (1.8-2.0 Gy per fraction) in all cases, and the chemotherapy regimen: tegafur bolus (0.5 g, qd/bid, d1-14,q3w) was given from the first day of radiotherapy. The primary endpoint was near-term efficacy, and secondary endpoints included objective remission rate (ORR), disease control rate (DCR), T-cell subpopulation indices, and safety. **Results:** The CR, PR, SD and PD of radiotherapy combined with tegafur bolus synchronized treatment were 1 (3.1%), 17 (53.2%), 13 (40.6%) and 1 case (3.1%), respectively. The incidence of CR, PR, SD, and PD in the radiotherapy group was 0 (zero), 8 (26.7%), 19 (63.3%), and 3 (10%), respectively. The ORR of the simultaneous radiotherapy group and the control group were 56.3% and 26.7%(P=0.018), and the DCR of the two groups was 96.9% and 90% of the disease control rate of the patients in both groups (P=0.346). The results of univariate analysis showed that the influencing factor of ORR was the treatment regimen (P=0.018), and multivariate logistic regression analysis of variables with $P<0.2$ showed that the recent efficacy was correlated with the treatment regimen (P=0.039), and that the patients with radiotherapy synchronized with tigafluoroembolization chemotherapy had a better recent efficacy. After treatment, the levels of $CD3^+$, $CD4^+$, $CD8^+$, $CD4^+/CD8^+$were higher in the synchronized radiotherapy group than before treatment,$CD3^+$ and $CD8^+$ were higher in the control group than before treatment, and $CD4^+$ and $CD4^+/CD8^+$ were lower than before treatment. In terms of adverse reactions, the incidence of acute hematological toxicity (P=1), nausea and vomiting (P=1), radiation esophagitis (P=0.197), radiation pneumonitis (P=0.238), and esophageal fistula (P=0.613) were slightly increased in the simultaneous radiotherapy group compared with the control group. **Conclusions:** Radiotherapy synchronized with Tegafur suppositories is effective, with relatively mild side effects, convenient and safe for the treatment of inoperable advanced elderly patients with oesophageal cancer. Clinical trial information: ChiCTR2300073845. Research Sponsor: None.

461 **Poster Session**

Fruquintinib combined with PD-1 inhibitors and chemotherapy in the first-line treatment of HER2-negative advanced gastric or gastroesophageal junction adenocarcinoma (FDZL-FIX): A single-arm, open-label phase 2 study. First Author: Chenchen Wang, Department of Medical Oncology, Fudan University Shanghai Cancer Center, Shanghai, China

Background: For unresectable locally advanced or metastatic HER2-negative gastric cancer (GC), the combination of chemotherapy with immunotherapy, specifically PD-1 inhibitors, has emerged as a new standard of care in first-line treatment of advanced GC, but the patients' outcomes remain poor. Early clinical studies investigating the combination of immunotherapy with anti-angiogenic drugs have demonstrated favorable efficacy. Therefore, it is worth exploring whether the addition of fruquintinib, a potent VEGFR-1,2,3 inhibitor, to the combination of PD-1 inhibitor and chemotherapy can further enhance the survival benefits for patients with unresectable locally advanced or metastatic HER2-negative GC. **Methods:** This is an open-label, single-center clinical study. The enrolled patients were all no less than 18 years old with HER2 negative locally advanced unresectable or metastatic gastric or gastrooesophageal junction adenocarcinoma, without any systemic anticancer treatment for advanced disease. Eligible patients received 6 cycles of combined first line treatment with fuquinitinib (4mg daily, d1-14, q3w, oral) combined with PD-1 inhibitor (sintilimab 200mg or nivolumab 360mg intravenously q3w) and chemotherapy (XELOX or SOX) regimen. The following maintenance treatment was fuquinitinib combined with PD-1 inhibitor and S-1/capecitabine until disease progression or intolerable toxicity. The primary endpoint was progress free survival (PFS). Secondary endpoints included objective response rate (ORR), overall survival (OS), duration of response (DoR), disease control rate (DCR) and safety. **Results:** At the cutoff date on Sep 18, 2024, a total of 27 patients were enrolled and received at least one dose of treatment. 23 patients had received the response evaluation, with an ORR of 60.9% (14/23; 95%CI 38.5-80.3) and DCR of 100% (23/23 95%CI 85.2-100.0). In the intention to treatment (ITT) population, the median PFS was 9.79 months (95% CI 5.29-NA). Safety profile indicated that 22.2% (6/27) experienced grade≥3 adverse events. The most common grade ≥ 3 adverse reactions mainly include neutropenia (7.4%, 2/27) and thrombocytopenia (7.4%%, 2/27). **Conclusions:** The combination of fruquintinib and chemotherapy with PD-1 blockade in the first-line treatment for unresectable locally advanced or metastatic HER2-negative GC had shown promising efficacy and acceptable safety profile. Enrollment in the trials is still ongoing. Clinical trial information: NCT06158919. Research Sponsor: None.

462 Poster Session

Surgical treatment results for oligo-recurrence of gastric cancer. First Author: Youjin JIN Jang, Korea University Medical College, Seoul, South Korea

Background: Despite advancements in diverse treatments, approximately 40-50% of advance gastric cancer patients experience recurrence even after curative resection surgery. The condition where there is a limited recurrence after the control of the primary tumor site is referred to as oligo-recurrence. However, there is scarce reporting on oligo-recurrence after gastric cancer surgery. This study compiled cases of patients who underwent surgery for recurrent gastric cancer after R0 surgery. Since there are no specific treatment guidelines for recurrence after gastric cancer surgery, this research could be beneficial in providing insights and assistance. **Methods:** This retrospective study includes data from 25 patients who underwent surgery for suspected oligo-recurrence of gastric cancer among those who had undergone surgery from June 2008 to May 2018. If a solitary mass suspected of recurrence is identified during surveillance, a PET-CT is performed. Patients with suspicion of peritoneal dissemination, carefully excluded through a multidisciplinary approach, are selectively omitted. All patients undergo excision surgery for both treatment and tissue examination purposes. **Results:** Out of 25 cases, there were 6 cases of paraaortic lymph node enlargement, 9 cases liver nodules, 8 case lung nodules, and 1 case tumor bed mass. In the postoperative tissue analysis, 4 cases of paraaortic lymph node enlargement, 5 cases of the liver nodules, and 1case tumor bed mass were confirmed to be recurrences of gastric cancer. All of lung nodule were diagnosed with lung cancer. All 10 cases underwent chemotherapy after metastatectomy. Among them, 3 cases experienced recurrence and died an average of 12.3 months after surgery. Seven cases did not experience recurrence and have survived with an average survival time of 102 months to date. **Conclusions:** The metastatectomy is safe and effective for patients with oligo-recurrence from GC and can improve their prognosis. For an accurate diagnosis, oligometastectomy is essential. However, our results need to be confirmed by more randomized controlled clinical studies. Research Sponsor: None.

Treatment result after metastatectomy.

Case	Diagnosis	Chemotherapy aftermetastatectomy	Recurence after metastatectomy	DFS after metastatectomy	OS after metastatectomy	OS after gastrectomy	Status
1	LN metastasis 1/1	15cycles FOLFOX		147	147	156.4	ALIVE
2	LN metastasis 1/1	37cycles TAS-118	liver metastasis	22	40.6	74.9	DEATH
3	LN metastasis 1/3	12cycles FOLFOX		98.1	98.1	125.5	ALIVE
4	LN metastasis 1/1	35cycles FOLFOX		50.9	50.9	71.3	ALIVE
5	Liver metastasis	ONO tiral	Liver, bone mets.	3.73	27.9	64.6	DEATH
6	Liver metastasis	12cycles FOLFOX		68.3	68.3	122.2	ALIVE
7	Liver metastasis	12cycles FOLFOX		96	96	105.8	ALIVE
8	Liver metastasis	82cycles FOLFOX		55	55	77.4	ALIVE
9	Liver metastasis	12cycles FOLFOX		94.4	94.4	103.9	ALIVE
10	Tumor bed	12cycles FOLFOX		9	25.5	24.4	DEATH

463 Poster Session

Efficacy, safety and DNA methylation analysis of cadonilimab combined with taxane and cisplatin as first-line treatment for advanced esophageal squamous cell carcinoma (ESCC): Updated results from an open-label, multicenter phase II trial (AK104-IIT-014). First Author: Wang Qu, Department of Medical Oncology, National Cancer Center, National Clinical Research Center for Cancer, Cancer Hospital, Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing, China

Background: Immune checkpoint inhibitors combined with chemotherapy had become the first-line standard treatment for advanced ESCC and cadonilimab, as a bispecific antibody simultaneously targeting PD-1 and CTLA-4, may further boost anti-tumor activity with a satisfied safety profile. Here, we present the updated data for the safety and efficacy of cadonilimab combination therapy as the first-line treatment in advanced ESCC. The correlation between DNA methylation and clinical response was also investigated. **Methods:** Treatment-naïve patients (pts) with unresectable locally advanced or metastatic ESCC were enrolled. Cadonilimab (10mg/kg, iv, d1, q3w) combined with paclitaxel or nab-paclitaxel (175 mg/m^2, iv, d1, q3w) and cisplatin (65 to 75 mg/m^2, iv, d1, q3w) were administrated for up to 6 cycles, then Cadonilimab (10mg/kg, iv, d1, q3w) monotherapy continued as maintenance until progressive disease or unacceptable toxicity, with a maximum of 24 months. The primary endpoint was objective response rate (ORR). Secondary endpoints were progression-free survival (PFS), overall survival (OS), disease control rate (DCR) and safety. Plasma cell-free DNA samples were collected before and after treatment and prepared for methylation level sequencing. **Results:** As of September 20, 2024, 43 pts were enrolled with a median age of 61 years (range 44-75), 81.4% were male, 39.5% had PD-L1 CPS≥10 and 95.3% had metastatic diseases. All pts were evaluable for safety and efficacy. The ORR was 81.4% (95%CI: 66.1%-91.1%) and DCR was 97.7% (95%CI: 86.2%-99.9%). The median PFS (mPFS) was 7.05 months (mo) (95%CI: 5.86-8.24) and OS analysis was immature. In the PD-L1 CPS≥10 pts, the ORR was 100.0% (95%CI: 77.1%-100.0%) and mPFS was 7.05 mo (95%CI: 5.19-8.91). In the PD-L1 CPS < 10 pts, the ORR was 77.3% (95%CI: 54.2%-91.3%) and mPFS was 7.05 mo (95%CI: 5.87-8.24). Six hyper-methylated CpG sites, EPTIN9、PKNOX2、DLEU7、SOX7、CNRIP1 and LINC00554 might be the candidate biomarkers as the mean pretreatment methylation levels were significantly higher in the PR pts than those in the non-PR pts ($p=7.8\times10^{-7}$). Grade 3-4 treatment-related adverse events (TRAEs) were reported in 44.2% (19/43) pts, mainly including neutropenia (25.6%), leukopenia (9.3%) and hyponatremia (7.0%). The infusion-related reactions (IRR) occurred in 14.0% (6/43) and grade ≥3 IRR occurred in 4.7% (2/43) pts. 7 pts discontinued cadonilimab administration due to TRAEs. **Conclusions:** The updated results suggested that bispecific antibody cadonilimab combined with taxane and cisplatin as first-line treatment continued to show encouraging anti-tumor activity and manageable safety in pts with advanced ESCC. DNA methylation level might be a potential biomarker for guiding patient outcomes. Clinical trial information: NCT05522894. Research Sponsor: None.

464 Poster Session

Pembrolizumab or placebo plus chemotherapy for advanced HER2-negative gastric or gastroesophageal junction (G/GEJ) adenocarcinoma: An updated analysis of KEYNOTE-859 for patients enrolled in Asia. First Author: Chia Jui Yen, National Cheng Kung University Hospital, Tainan, Taiwan

Background: After a median follow-up of 41.6 months, data from the global phase 3 KEYNOTE-859 study (NCT03675737; N = 1579) continued to show that use of pembrolizumab (pembro) plus chemotherapy (chemo) improved overall survival (OS), progression-free survival (PFS), and objective response rate (ORR), with manageable safety, versus placebo (pbo) plus chemo for patients (pts) with advanced HER2-negative G/GEJ adenocarcinoma. Here, we report updated results from the subgroup analysis of pts enrolled in Asia after an additional 11 months of follow-up from the first interim analysis. **Methods:** Eligible pts aged ≥18 years with locally advanced unresectable or metastatic HER2-negative G/GEJ adenocarcinoma, an ECOG performance status (PS) of 0 or 1, and measurable disease per RECIST v1.1 were randomly assigned 1:1 to receive pembro 200 mg or pbo IV Q3W for ≤35 cycles; all pts received investigator's choice of chemo (FP or CAPOX). The primary end point was OS. Secondary end points included PFS, ORR, and DOR per RECIST v1.1 by blinded independent central review, and safety. Efficacy end points were evaluated in all randomly assigned pts (intention to treat). **Results:** A total of 525 pts (263, pembro plus chemo; 262, pbo plus chemo) were enrolled in KEYNOTE-859 in Asia. The median time from randomization to database cutoff (August 22, 2023) was 39.2 months (range, 26.0-56.8). The median OS was 17.3 months (95% CI, 14.8-19.5) for pembro plus chemo versus 13.0 months (95% CI, 11.8-14.4) for pbo plus chemo (HR, 0.75; 95% CI, 0.62-0.91). The median PFS was 8.4 months (95% CI, 7.1-9.6) for pembro plus chemo versus 5.8 months (95% CI, 5.6-6.9) for pbo plus chemo (HR, 0.72; 95% CI, 0.58-0.89). The ORR was 61.2% (95% CI, 55.0-67.1; 36, complete response [CR]; 125, partial response [PR]) for pembro plus chemo and 48.5% (95% CI, 42.3-54.7; 24, CR; 103, PR;) for pbo plus chemo. The median DOR was 10.0 months (range, 1.2+ to 50.8+) for pembro plus chemo and 5.8 months (range, 1.3+ to 44.3+) for pbo plus chemo; 28.9% and 23.3% of pts, respectively, had a response lasting ≥24 months. Grade 3-5 treatment-related adverse events (AEs) occurred in 155 pts (59.2%) in the pembro plus chemo group and 119 pts (45.4%) in the pbo plus chemo group. Treatment-related AEs led to death in 1 pt (0.4%) in the pembro plus chemo group (unknown cause) and 2 pts (0.8%) in the pbo plus chemo group (cerebral hemorrhage and abnormal liver function). Immune-mediated AEs and infusion reactions occurred in 85 pts (32.4%) in the pembro plus chemo group and 35 pts (13.4%) in the pbo plus chemo group. **Conclusions:** The addition of pembro to chemo improved OS, PFS, and ORR in pts with advanced HER2-negative G/GEJ adenocarcinoma from Asia enrolled in KEYNOTE-859, with no new safety signals. These results further support first-line pembro plus chemo as a treatment option for this population. Clinical trial information: NCT03675737. Research Sponsor: Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA.

465 Poster Session

Exploring the role of bispecific and CAR-T cell therapies in gastric cancer: A systematic review and meta-analysis. First Author: Moazzam Shahzad, H. Lee Moffitt Cancer Center and Research Institute, University of South Florida, Tampa, FL

Background: Gastric cancer (GC) is associated with significant mortality due to the limited efficacy of current conventional treatments, necessitating the need for novel therapies. In this systematic review and meta-analysis, we aim to explore the potential role and outcomes of bi-specific antibodies and chimeric antigen receptor T-cell (CAR-T) immunotherapy in treating GC. **Methods:** Using PRISMA guidelines, searches were conducted on PubMed, Cochrane, and Clinicaltrial.gov for 'Gastric cancers', 'bispecific antibodies', and 'CAR-T therapy' as of March 30, 2024. Out of 35 studies, 9 were selected for pooled analysis in R (v4.3.3) using the Der Simonian-Laird Estimator, calculating inter-study variance and extracting data with 95% CI. **Results:** 138 gastric cancer (GC) patients from 2 phase I (22.22%), 3 phase Ib (33.33%), 3 phase II (33.33%), and 1 case report (11.11%). Median age was 57 (28-77) years, and 76% (31/41) were male. Bispecific antibodies were MCLA-128 (25/138), catumaxomab (15/138), AK104 PD-1/CTLA-4 (16/138), ABL 001 DLL4VEGFA (19/138), and KN026 (25/138) while CART therapy was CT041 (38/138). The pooled overall response (OR), partial response (PR), and complete response (CR) for bispecific antibodies was 37% (95% CI, 0.12-0.71, I2=76%, n=100, p <0.01), 21% (95% CI,0.05-0.55, I2=73%, n=85, p<0.01), and 11%(95% CI, 0.02-0.38, I2=63%, n=85, p=0.03) respectively, while the pooled OR, PR, and CR for CART was 56% (95% CI, 0.40-0.71, I2=0%, n=38, p=0.41), 53%(95% CI, 0.36-0.69, I2=0%, n=38, p=0.38,), and 12% (95% CI, 0.02-0.42, I2=38%, n=38 p=0.2), respectively. The pooled disease control rate was 75% (95%CI,0.46-0.92, I2=79%, n=80, p < 0.01) for bispecific antibodies and 27% (95% CI,0.16-0.40, I2=0%, n=65, p=0.37) had the stable disease while pooled incidence of progressive disease was 18% (95% CI,0.09-0.31, I2=0%, n=55, p=0.53). The median progressive free survival was 6.74 months, with a median overall survival of 14.75 months. Common adverse effects included anemia, diarrhea, fatigue, and cytokine release syndrome grades 1 or 2. **Conclusions:** This analysis shows promising results in GC patients using bi-specific antibodies and CAR-T cell therapy. However, further clinical trials are needed to fully explore their potential. Research Sponsor: None.

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Phase II study of neoadjuvant FLOT and chemoradiation for trimodality therapy of esophageal/GEJ adenocarcinoma. First Author: Jeffrey R. Olsen, University of Colorado School of Medicine, Aurora, CO

Background: Recent data support use of perioperative FLOT (5-FU/leucovorin/oxaliplatin/docetaxel) chemotherapy over pre-operative chemoradiation (CRT) for patients with resectable esophageal/GEJ adenocarcinoma with limited studies evaluating combined FLOT/CRT. This prospective phase II study (NCT04028167) evaluated a neoadjuvant approach of FLOT followed by CRT for resectable esophageal/GEJ adenocarcinoma (EAC). **Methods:** Operable patients with cT1-2 N1-2 or cT3-4N any4 Nany non-metastatic EAC were eligible. Neoadjuvant treatment consisted of FLOT x 3 cycles followed by restaging PET and chemoradiation (41.4 Gy / 23 fractions to the primary site and regional lymphatics with weekly carboplatin/paclitaxel). Adjuvant nivolumab for non-pathologic complete response (pCR) was permitted. The primary endpoint was pCR rate among patients receiving the study regimen and undergoing resection, with secondary endpoints including disease free survival (DFS), overall survival (OS), toxicity, and correlative studies. DFS/OS was measured using the Kaplan Meier (KM) method. Interim results are presented after a planned safety analysis. **Results:** Since 4/2020, 23 patients with median age of 62 (range 40-73) have enrolled. cT3-4 and cN+ disease was present in 20/23 (87%) and 16/23 (70%) of patients, respectively. FLOT was completed by 21/23 patients (reasons for discontinuation included allergic reaction and G5 sepsis/endocarditis). All 21 patients initiating chemoradiation completed planned treatment and 17/21 underwent resection (2 refused surgery after achieving cCR, 1 died of cardiac arrest, 1 patient is awaiting surgery). A pCR was observed in 5/17 resected patients (29%), with pCR or near pCR observed in 9/17 (53%). The complete response rate (pCR + cCR for patients refusing surgery) was 7/19 (37%). Adjuvant nivolumab was administered in 9 patients. The median follow-up among all patients was 20 months with 2-year DFS and OS estimates of 61% and 73%. The regimen was tolerated with G3+ hematologic, G3+ gastrointestinal, and G3+ other toxicity observed in 39%, 13%, and 22% of patients, respectively. **Conclusions:** In a locoregionally advanced population, neoadjuvant sequential FLOT followed by CRT was well tolerated with encouraging DFS/OS and a high rate of complete response compared to prior studies. This hybrid platform may enhance both locoregional and distant control and is of interest for expanded evaluation. Clinical trial information: NCT04028167. Research Sponsor: American Cancer Society; Cancer League of Colorado.

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Identification of candidate biomarkers for gastric cancer by bioinformatics analysis of pooled microarray gene expression datasets in Gene Expression Omnibus. First Author: Fen Wang, Silver Bamboo Health Science, Fairfax, VA

Background: Gastric cancer (GC) remains one of the most prevalent malignant diseases worldwide. The high mortality rate is largely due to the limited treatment options and the challenges in diagnosing the cancer at an early stage. Understanding the genetic differences driving the proliferation of GC is crucial for advancing diagnostic methods and developing targeted treatments. This study aims to identify biomarkers involved in the pathogenesis of GC through integrated bioinformatics analysis. **Methods:** Extracted from the Geo Expression Omnibus, three DataSets (GSE118916, GSE79973, and GSE13861) were analyzed for differentially expressed genes (DEGs) using the R statistical language. The three data sets were then compared for common DEGs, which would serve as the focus of this study. These DEGs were functionally analyzed for pathways by Gene Ontology (GO) and the Kyoto Encyclopedia of Genes and Genomes (KEGG). A protein–protein interaction (PPI) network was constructed to screen out candidate genes. **Results:** The results revealed that 123 DEGs of the three datasets containing 26 upregulated genes and 97 downregulated genes were ascertained in our study. The GO and KEGG enrichment analysis results showed that the functions of DEGs were mainly involved in the retinol metabolic process, xenobiotic metabolic process, collagen type I trimer, potassium: proton exchanging ATPase complex, benzaldehyde dehydrogenase (NAD+) activity, alcohol dehydrogenase (NADP+) activity, Collecting duct acid secretion, and Tyrosine metabolism. Through the PPI analysis network, the core genes with the highest degree of 7 nodes were selected: COL1A1, COL1A2, COL11A1, THBS2, ATP4A, COL10A1, and CXCL8. **Conclusions:** The 7 genes identified in this study may play a vital regulatory role in the occurrence and development of GC and may also be potential therapeutic targets. Further mechanistic studies of the pathogenesis and treatment of GC may be able to identify new targets using these shared pathways. These findings could contribute to a better understanding of the origins of gastric cancer and facilitate the development of early interventions. The bioinformatics analysis conducted in this study offers new insights and directions for further research on gastric cancer. Research Sponsor: None.

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Relationship in gene amplification between *ERBB2* and other oncogenes: Implications for the therapeutic efficacy of trastuzumab (Tmab)-based chemotherapy (CTx) in HER2 positive gastric cancer. First Author: Shigenori Kadowaki, Department of Clinical Oncology, Aichi Cancer Center Hospital, Nagoya, Japan

Background: This study aimed to elucidate the molecular characteristics, focusing on the co-amplification of *ERBB2* and other oncogenes, and to investigate their impact on treatment efficacy in HER2 positive GC. **Methods:** We conducted a phase 2 clinical trial (UMIN 00017602) evaluating the efficacy of S-1, oxaliplatin, and Tmab. Next-generation sequencing was performed using the TS-170 (Illumina). **Results:** 32 patients were enrolled, of whom 6 (19%) were IHC2+/FISH+. The median progression-free survival (PFS) and overall survival (OS) were 11.8 and 28.8 months. *ERBB2* amplification with a cut-off value of 2.84 was present in 84.4% with a median copy number (CN) of 9.3 (range 2.4-76.6). Amplification of oncogenes other than *ERBB2*, such as *KRAS, FGFR2, MET,* and *CCNE1*, was observed in 87.5%, and the median CN of the highest CNs among oncogenes other than *ERBB2* was 7.8 (range 2.4-22.9). Eventually co-amplification was observed in 75%, and CNs of *ERBB2* and other oncogenes showed an inverse correlation (r=0.345, p=0.058). In all tumors having *ERBB2* CN $\geq$ 7.25 obtained by the ROC analysis, CN of *ERBB2* was the highest among oncogenes (*ERBB2* dominant). Conversely, in 92% of the tumors having *ERBB2* CNs $<$ 7.25, CNs of oncogenes other than *ERBB2* was higher than ERBB2 (*ERBB2* non-dominant) ($p<0.001$). Moreover, the highest CN of the oncogenes other than *ERBB2* was higher in tumors with *ERBB2* CNs $<$ 7.25 than that with *ERBB2* CNs $\geq$ 7.25 (median 12.6 vs 6.4 p=0.018). Patients with *ERBB2* non-dominant tumors showed significantly shorter PFS (median 6.9 vs 17.0 months, HR 6.40, p=0.001) and OS (median 14.8 vs 35.5 months, HR 2.72, p=0.016) and numerically lower response rate (63.6 vs 90.0%, p=0.151) compared with those with *ERBB2* dominant tumors. **Conclusions:** Most patients had co-amplification of *ERBB2* and other oncogenes, and the dominance of oncogene amplification in the tumor was associated with treatment efficacy of Tmab-based CTs in HER2 positive GC. These results suggest that the dominance of oncogene CNs within the tumor may determine tumor drivenness and influence treatment efficacy. Research Sponsor: None.

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A liquid biopsy analysis of exosomal miRNA to detect lymph node metastasis in early gastric cancer. First Author: Masaaki Nishi, Department of Surgery, Tokushima University, Tokushima, Japan

Background: Additional surgical resection is required to achieve a complete cure in patients with early gastric cancer (EGC) due to the potential risk for lymph node metastasis (LNM) after pathological analysis. However, LNM is estimated to occur in approximately 10% of patients with high-risk EGC. In this study, we investigated a blood-based liquid biopsy assay of exosomal miRNA for the noninvasive detection of LNM in patients with high-risk EGC. **Methods:** Two genome-wide miRNA expression profiling datasets (GSE164174 and TCGA) were analyzed to prioritize biomarkers in pretreatment plasma samples from clinical training and validation cohorts of GC patients. An integrated exosomal miRNA panel was developed and a risk stratification model combining the miRNA panel with clinical risk factors was established. **Results:** Using comprehensive expression profiling of public datasets, we identified a transcriptomic panel of four miRNAs (miR-34b, miR-130a, miR-375, and miR-627) that robustly identified patients with LNM [area under the curve (AUC) = 0.86, 95% confidence interval (CI) = 0.77-0.92]. We assessed panel performance in a training cohort (AUC=0.86, 95% CI=0.67-0.96) and validated it in an independent validation cohort (AUC=0.84, 95% CI=0.64-0.95). Our risk stratification model was more accurate than the panel and was an independent predictor of LNM identification (AUC=0.97). **Conclusions:** A novel noninvasive, liquid biopsy-based method for patients with EGC may predict those traditionally classified as high-risk patients with LNM who are unlikely to benefit from surgical resection. Research Sponsor: None.

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Quantitative amplification ratio for established and emerging biomarkers in gastroesophageal (GE) cancer. First Author: Matthew Strickland, Massachusetts General Hospital, Harvard Medical School, Boston, MA

Background: Gene copy number (CN) amplifications (amps) and/or protein overexpression in *ERBB2*, *EGFR*, *FGFR2*, and *MET* represent established and emerging clinical targets in GE. Various methodologies are used to report gene amp or overexpression with some data suggesting FISH ratio may predict therapy response. Quantitative methods for NGS CN and relationship to ISH/IHC remains largely unresolved. We studied the landscape of targetable CN amps in GE to understand correlations of quantitative CN amp ratio and outcomes. **Methods:** GE tissue (N = 20,468) and liquid (N = 1,760) samples underwent NGS-based comprehensive genomic profiling (CGP). A genome-wide CN model for each sample was generated, segmented, and combined with variant allele frequencies of heterozygous SNPs to estimate the sample purity, ploidy, and CN for each segment. Genes were considered amplified at CN $\geq$ ploidy + 3 for *ERBB2* and + 4 for other genes. Subanalysis using amp ratio (ratio of the gene CN to the sample ploidy) $\geq$ 3 was performed. ctDNA tumor fraction (TF) was quantified using a composite algorithm (PMID: 38990098). The Flatiron Health-Foundation Medicine clinico genomic database, a nationwide (US-based) de-identified EHR-derived database linked to CGP data, was queried to assess external HER2 FISH or IHC and describe real-world (rw) progression-free survival (rwPFS) and overall survival (rwOS) of GE patients (pts) with *ERBB2* amp on trastuzumab-based regimens. **Results:** $\geq$1 gene amps were detected in 15,341 (75%) GE tissue cases, and in 67% of cases using an amp ratio $\geq$3 threshold. The rates of amps in current clinical targets using each threshold were *ERBB2*: 17%/12%, *KRAS*: 13%/12%, *EGFR:* 8.7%/7.2%, *MET:* 5.2%/4.0%, *FGFR2:* 3.1%/2.7% of tissue samples. The median amp ratio and percent of amps with ratio $\geq$ 3 were *ERBB2*: 7.1/73%, *KRAS*: 14/92%, *EGFR*: 13/82%, *MET*: 6.3/76%, and *FGFR2*: 29/88%. Amp ratio $\geq$ 3 was associated with focal amplicons vs amp ratio $<$ 3 (median amplicon size 0.95 vs. 3.8 Mbp, p $<$ 0.001). Of 1,382 pts with HER2 NGS and IHC results, concordance with IHC3+ was 68% and increasing *ERBB2* amp ratio was associated with IHC 3+ status. Of 604 pts with HER2 NGS and FISH, concordance was 80% (118 positive, 366 negative). Higher *ERBB2* amp ratio was also associated with longer rwOS and rwPFS as a continuous variable on first-line trastuzumab-based regimens. $\geq$1 gene amps were detected in 597 (34%) liquid samples (median TF 18%) and 32% with the amp ratio $\geq$3 threshold, increasing to 79% and 73% with TF $\geq$ 20% (n=355). **Conclusions:** Gene amps are common established or emerging GE targets and *ERBB2* amp ratio is associated with outcomes for pts on trastuzumab-based regimens. Amp detection in liquid correlates with higher TF and amp ratio. As the number of therapies targeting gene amps and overexpression grows, quantification of the degree of amplification and harmonization across methodologies may be an important tool to select pts for optimal outcomes. Research Sponsor: Foundation Medicine.

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Role of intraperitoneal paclitaxel in eosinophil activation and recruitment in the peritoneal cavity and anti-tumor effects against peritoneal metastasis from gastric cancer. First Author: Misaki Matsumiya, Jichi Medical University, Shimotsuke, Japan

Background: Recent studies indicate that the efficacy of chemotherapy is significantly influenced by the tumor immune microenvironment. Peritoneal cavity contains many immune cells, however, the relationship between immune response within the peritoneal cavity and tumor response remains poorly understood. **Methods:** Single cell suspensions were obtained from ascites or peritoneal lavages from 41 patients with PM from GC. In 28 patients, cells were collected both before and after 1-3 courses of IP chemotherapy. These samples were immunestained with mAbs targeting to specific lymphoid or myeloid subsets, and their proportions in CD45(+) leukocytes were analyzed using flowcytometry. Eosinophils were purified by magnetic cell sorting for subsequent RNA-Seq analysis. **Results:** Among 41 patients with PM, tumor leukocyte ratio (TLR) calculated as CD326(+) tumor cells divided by CD45(+) leukocytes varied from 0.002% to 58.1%. TLR was not correlated with any ratios of immune cells, however, the ratios of CD8(+) T cells, NK cells and CD16(-) CD193(+) eosinophils tended to decrease with the increase in TLR. Conversely, the ratios of CD19(+) B cell, CD14(+) macrophages and CD16(+) neutrophils increased with elevated TLR. Mean survival time of these patients was 17.7 months. None of the leukocyte subsets showed significant association with patient outcome. In 28 patients, the ratios of lymphocytes mostly decreased with significant difference in CD4(+) T cells and CD19(+) B cells after IP treatment. In contrast, the ratio of CD11b(+) myeloid cells increased in peritoneal cavity after IP treatment. Among the myeloid cells, the ratio of CD16(-) CD193(+) eosinophils significantly increased. The change was particularly prominent in in post-treatment negative peritoneal lavage cytology (CY0) patients (M=0.47% vs M=10.0%, p$<$0.0001), while the difference was not significant in post-treatment positive peritoneal lavage cytology (CY1) patients (M=0.93% vs M=0.97%, p=0.84). Patients with eosinophil ratio of $\geq$2% after IP chemotherapy in peritoneal cavity had significantly longer overall survival compared to those with lower eosinophil ratios (17.3 mo vs 26.7 mo, p=0.034; HR =0.23[0.06-0.89]). The peritoneal eosinophils after IP treatment exhibited elevated levels of CD11b and CD63 expression compared to circulating eosinophils (p$<$0.01), while SSC-A levels tended to be lower. Gene ontology pathways revealed enrichment of cytokine-mediated signaling pathway, extracellular matrix organization and response to interferon-gamma in peritoneal eosinophils (p$<$0.0001). **Conclusions:** IP-PTX induces eosinophil activation and recruitment in the peritoneal cavity, while also affecting other types of immune cells. These effects may potentially contribute to the anti-tumor response against peritoneal metastases. Research Sponsor: None.

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Unraveling the role of interferon-stimulated neutrophils: A promising biomarker for immunotherapy efficacy and prognosis in gastric adenocarcinoma. First Author: Hongda Pan, Fudan University Shanghai Cancer Center, Shanghai, China

Background: Despite significant advancements in immunotherapy, the identification of effective predictive biomarkers for treatment response in gastric adenocarcinoma (GAC) remains a critical challenge. Discovering such biomarkers could enhance patient stratification and improve therapeutic outcomes. **Methods:** We conducted a comprehensive analysis of 18 specimens obtained from 10 GAC patients, both prior to and following immunotherapy. Utilizing single-cell RNA sequencing, we compared the cellular composition and functional phenotypes of these specimens, with particular emphasis on tumor-infiltrating immune cells and the identification of previously uncharacterized neutrophil subclusters. **Results:** Our analysis identified eight distinct neutrophil subclusters, including Neu01_S100A12, Neu02_VEGF, Neu03_CCL3_CCL4, Neu04_IL1β, Neu05_CXCR4, Neu06_IFN-Stimulated, Neu07_Proliferative, and Neu08_B-cell_related. Notably, the presence of IFN-Stimulated neutrophils was positively correlated with the efficacy of immunotherapy in GAC, a finding that was further validated through analyses of public datasets and in-house cohort studies employing multiplex immunohistochemistry. Additionally, IFN-Stimulated neutrophils served as prognostic indicators across multiple datasets. Investigations revealed that these cells engage in significant cellular communication with other immune components within the tumor microenvironment, including CD8+ T cells, macrophages, and B cells, thereby promoting a pro-inflammatory and anti-tumor phenotype. Importantly, pan-cancer analyses demonstrated the consistent prognostic and immunotherapy efficacy-predicting potential of IFN-Stimulated neutrophils across various cancer types. **Conclusions:** IFN-Stimulated neutrophils represent a promising biomarker for predicting immunotherapy response and patient prognosis in GAC. Our findings offer novel insights into the underlying mechanisms of immunotherapy and highlight potential avenues for personalized treatment strategies in the management of GAC. Research Sponsor: National Nature Science Foundation of China.

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The prognostic value of HER2 gene dosage and heterogeneity at a single-cell resolution in gastroesophageal adenocarcinoma. First Author: Yashika Parashar, Dana-Farber Cancer Institute, Boston, MA

Background: HER2 gene dosage (i.e. level of amplification) and heterogeneity are key prognostic variables for HER2+ gastroesophageal adenocarcinoma (GEA), yet no standardized approach to test these in routine clinical use exists. Here we modify the analytical pipeline for HER2 FISH to quantitatively estimate these prognostic variables and correlate them with clinical outcomes. **Methods:** Patients with metastatic HER2+ GEA who received trastuzumab-based therapy between 2011-2024 at Dana-Farber Cancer Institute and had available pre-treatment tissue for FISH analysis were included. HER2 FISH was conducted in a CLIA approved clinical pathology laboratory. Standard HER2/CEP17 ratio (an average of 50 cells) and individual cell HER2/CEP17 ratio were calculated. Kaplan-Meier method was used to estimate progression free (PFS) on trastuzumab-based therapy and overall survival (OS). Survival outcomes were adjusted for age, gender, tumor site, grade, prior resection, number of metastatic sites, and HER2 IHC using cox proportional hazard regression models. **Results:** A total of 77 patients were included. Median age was 60 years, 88% were male, and 92% had HER2 3+ tumors. Individual cells with HER2/CEP17 ratio $\geq$ 2.0 were defined as amplified (same cutoff as current bulk standard) and those with HER2/CEP17 ratio $\geq$ 4.95 (population median) were labeled highly HER2 amplified providing objective metrics to assess heterogeneity and gene dosage. Using these metrics we defined three subgroups: 1. Heterogenous (Het, n = 15): $<$75% cells HER2+; 2. Homogenous, low gene dosage (HL, n = 30): $>$75% cells HER2+, $<$50% cells with high HER2 amplification; 3. Homogenous, high gene dosage (HH, n = 32): $>$75% cells HER2+, $>$50% cells with high HER2 amplification. Cutoffs for both metrics were determined using data distribution patterns. The subgroups were prognostically relevant and PFS and OS were significantly lower for heterogenous tumors and numerically lower for homogenous tumors with low gene dosage after multivariable adjustment (Table). **Conclusions:** HER2 gene dosage and heterogeneity significantly impact outcomes in HER2+ GEA. More importantly by deconvoluting clinical HER2 FISH data at single cell level, we demonstrate an objective metric with prognostic value that could be adopted for routine clinical use and for patient stratification. Further insights into differing biology of these molecular subtypes are needed. Research Sponsor: Dana Farber Cancer Institute.

Outcomes of patients by HER2 heterogeneity and gene dosage. Hazards ratio (HR) and p values are adjusted using multivariable cox proportional hazard regression model.

		PFS			OS		
	Patients, No.	Median, mo	HR (95%CI)	*P* Value	Median, mo	HR (95%CI)	*P* Value
HH	32	16.2	1		16.4	1	
HL	30	12.1	1.70 (0.79-3.66)	0.173	15.7	1.71 (0.93-3.14)	0.085
Het	15	6.7	3.12 (1.35-7.2)	0.008	12.4	2.72 (1.28-5.75)	0.009

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Blood TCTP as a biomarker associated with immunosuppressive features and resistance to immunotherapy in metastatic gastric cancer. First Author: Hyung-Don Kim, University of Ulsan College of Medicine - Asan Medical Center (AMC), Seoul, South Korea

Background: Cancer biomarkers are essential for determining the prognosis of the disease and/or predicting its response to specific treatments. However, established biomarkers representing specific myeloid cell populations and representative biomarkers in gastric cancer remain unavailable. Therefore, this study aimed to explore the prognostic and immunological relevance of plasma translationally controlled tumor protein (TCTP) in patients with advanced gastric cancer treated with immune checkpoint inhibitor (ICI) and cytotoxic chemotherapy. **Methods:** Plasma samples were prospectively collected from the cohorts of patients with gastric cancer who were treated with 1st line fluoropyrimidine plus platinum chemotherapy (n = 143, cohort 1) and 3rd line nivolumab (n = 165, cohort 2). Plasma TCTP levels were quantified using ELISA, and multiplex proteomic analysis (Olink) was conducted to assess expression levels of immune-related proteins. An external single-cell RNA sequencing (scRNA-seq) dataset was employed to validate the findings. **Results:** Patients with high plasma TCTP levels (TCTP-high group) exhibited poor survival outcomes with 1st line chemotherapy compared to those with low levels (TCTP-low group) in cohort 1. In the TCTP-high group, proteins associated with immunosuppressive myeloid cells, angiogenesis, and immune exclusion of T/NK cell function were upregulated, whereas proteins involved in T-cell activation/exhaustion were significantly upregulated in the TCTP-low group. The scRNA-seq analyses of myeloid cells revealed that TCTP pathway-associated genes (i.e., *TLR2*, *CXCL1*, and *CXCL2*) were specifically expressed in immunosuppressive *APOE*$^{+}$ macrophages, *THBS1*$^{+}$macrophages, and *CD1C*$^{+}$conventional type 2 dendritic cells. In the TCTP-high group, patients treated with nivolumab (cohort 2) also experienced poor survival outcomes. **Conclusions:** Plasma TCTP is a readily measurable prognostic biomarker, reflecting immunosuppressive signals of myeloid cells in patients with gastric cancer treated with ICI and chemotherapy. Research Sponsor: This research was supported by the Korea Drug Development Fund, funded by the Ministry of Science and ICT, Ministry of Trade, Industry, and Energy, and Ministry of Health and Welfare, Republic of Korea).

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Effect of tRF-28-ZJ47D112Z4DZ on the pathogenesis of gastric cancer through targeting cyclin D1/CDK4 and its early diagnostic value. First Author: Xiuchong Yu, The First Affiliated Hospital of Ningbo University, Ningbo, Zhejiang, China

Background: Gastric cancer exhibits high incidence and mortality rates worldwide. Current research on gastric cancer aims to discover stable and effective early diagnostic markers, as well as uncover the disease's pathogenesis. tRNA-derived fragment (tRF) is a newly discovered regulatory non-coding RNA, which plays an important role in tumor diagnosis and treatment. **Methods:** This study, through transcriptome sequencing and MINTBase v2.0 database, screened out a new molecule tRF-28-ZJ47D112Z4DZ (tRF-28) related to gastric cancer. An absolute quantitative method was established to quantify the level of tRF-28 in plasma. ROC curve, survival curve, Cox regression model were constructed, and clinicopathological factors were collected to analyze the diagnostic value of tRF-28. "Overexpression" and "knockdown" tRF-28 gastric cancer cell models and xenograft tumor models were established to study the effects of tRF-28 on cell proliferation. Using bioinformatics, qRT-PCR, Western blot, dual luciferase reporter gene, fluorescence in situ hybridization, RNA binding protein immunoprecipitation assay, rescue experiment and immunohistochemistry to study the regulatory role of tRF-28 by binding to the 3'UTR of CCND1. **Results:** This study clarified the tRF-28 levels in the plasma of healthy individuals, and gastric benign and malignant lesions, providing a new candidate biomarker for the early diagnosis of gastric cancer. Through upregulating and downregulating the level of tRF-28 in gastric cancer cells, it was revealed that tRF-28 inhibits the proliferation of gastric cancer cells in vivo and in vitro by regulating key proteins such as Cyclin D1, CDK4, Rb, and p-Rb. In studying specific molecular mechanisms, tRF-28 and CCND1 were co-located in the cytoplasm. tRF-28 combines with Ago2 protein to form a complex and plays a role in silencing the target gene CCND1 in the cytoplasm. The rescue experiments revealed that silencing the target gene CCND1 can reverse the carcinogenesis effect of tRF-28 inhibitors, clarifying the biological functions of tRF-28 by regulating CCND1. **Conclusions:** Based on a series of research findings, we elucidated that tRF-28 form a complex with Ago2 protein, binding to the 3' UTR of CCND1 to downregulate Cyclin D1, CDK4, and other critical proteins, thereby inhibiting the proliferation of gastric cancer cells. Our study aims to offer new insights into identifying potential early diagnostic markers and therapeutic targets for gastric cancer. Research Sponsor: National Natural Science Foundation of China; 82302611.

476 Poster Session

Predictive role of circulating tumor DNA for locally advanced esophageal squamous cell carcinoma before surgery. First Author: Ryota Kobayashi, Keio University School of Medicine, Shinjuku-Ku, Tokyo, Japan

Background: A precise preoperative tumor monitoring method that reflects tumor burden during neoadjuvant treatment is required to guide individualized perioperative treatment strategies for esophageal squamous cell carcinoma (ESCC). This study examined the clinical significance of preoperative circulating tumor DNA (ctDNA) in the plasma of patients undergoing neoadjuvant chemotherapy (NAC) followed by esophagectomy. **Methods:** Plasma samples were collected longitudinally for ctDNA analysis as well as genomic DNA from primary lesions from patients with histologically confirmed ESCC who received neoadjuvant chemotherapy (NAC) followed by subtotal esophagectomy. Next-generation sequencing was used to identify mutations in both the plasma and primary tumors. We evaluated the relation ship between ctDNA alterations and recurrence in patients with locally advanced ESCC. **Results:** Pretreatment samples from 25 patients (100%) showed the same mutations in both ctDNA and primary tumors; therefore, they were classified as ctDNA-positive before treatment. In the cohort of 25 patients analyzed, those who tested positive for ctDNA after NAC had a significantly higher risk of recurrence; the 36-month recurrence-free survival rates were 92% for ctDNA-negative patients and 8% for ctDNA-positive patients ($p < 0.001$). **Conclusions:** Preoperative ctDNA status may be a promising prognostic biomarker that can be assessed before surgery in patients with ESCC who received NAC. Expanded cohort validation will allow for more personalized multidisciplinary treatment approaches for ESCC tailored to ctDNA analysis. Research Sponsor: None.

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Phase 1b study of futibatinib plus pembrolizumab in patients with esophageal carcinoma: Evaluating the immunological effects. First Author: Takashi Kojima, Department of Gastroenterology and Gastrointestinal Oncology, National Cancer Center Hospital East, Kashiwa, Japan

Background: Combination therapy using FGFR inhibitors and Immune checkpoint inhibitors (ICIs) are being explored as a novel strategy for esophageal cancer (EC) patients (pts). We reported the tolerability and encouraging antitumor activity of futibatinib (futi) plus pembrolizumab (pem) in cohort (CO) A (≥2nd line, ICIs naïve) and COB (≥2nd line, ICIs refractory) and futi plus pem with chemotherapy in COD (1st line, ICIs naïve) with EC pts at Ooki et al ASCO 2024. The purpose of this Cohort E (COE) was to evaluate the effects of futi as a single agent or futi in combination with pem on tumor immune microenvironment by detecting immune cells and immune-related gene expression. **Methods:** In COE of the study, advanced or metastatic EC pts with at least one prior therapy with a fluorouracil and platinum-based drug received a single agent of futi the first 7 days, followed by futi 20 mg once daily (continuous dosing) plus pem 200 mg/every 3 weeks (up to a maximum of 35 cycles). Regardless of FGFR overexpression level and ICIs naïve or refractory pts were include on the study. Tumor samples were collected by biopsy during the screening period, 7 days after the administration of futi, and 21 days after the combination administration of futi and pem. Expression of 18 biomarkers including CD3, CD4, CD8 were analyzed by multiplex IHC (NeoGenomics). Gene expression related to immunity in tumor microenvironment were evaluated by nCounter system (NanoString). **Results:** As of March 10, 2024, 18 pts (ICIs naïve, 3 pts; ICIs refractory, 15 pts) were enrolled. As a result of multiplex IHC, an increase of CD3^{+}CD8^{+} T cells was observed in the samples 7 days after a single treatment of futi comparing with baseline samples and a trend towards greater increased CD3^{+}CD8^{+} cells was observed 21 days after combination treatment. An increasing CD8A gene expression was observed by nCounter analysis after a single treatment of futi comparing with baseline and a trend towards greater increase in CD8A gene expression was observed after combination treatment. Confirmed partial responses were observed in 3 pts (2 pts were ICIs naïve and 2 pts were FGFR positive). ORR was 16.7% (3/18; 1 pt had no target lesion, 95% CI: 3.6, 41.4). Most common TRAEs were hyperphosphatemia (77.8%), diarrhea (33.3%), ALT increased, and AST increased (22.2%). **Conclusions:** The preliminary multiplex IHC and nCounter results suggest that futi promoted CD8^{+} T cells infiltration into EC tumor microenvironments by single treatment and following combination treatment with pem at tolerable dosage. Research Sponsor: Taiho pharmaceutical co., LTD. Merck & Co., Inc.

Summary of CD3^{+}CD8^{+} T cell and CD8A gene expression after futibatinib and combination treatment (% change relative to the baseline).

	N	Mean (S.D)	95%CI
Cell:CD3^{+}CD8^{+}T cell			
C1D7 (futi)	15	176.4 (88.8)	[127.2, 225.5]
C2D21 (Combination)	12	306.3 (229.8)	[160.3, 452.3]
Gene:CD8A			
C1D7 (futi)	16	139.4 (54.0)	[110.6, 168.2]
C2D21 (Combination)	14	245.9 (190.8)	[135.8, 356.1]

478 **Poster Session**

Concordance between human epidermal growth factor receptor 2 (HER2) status by immunohistochemistry (IHC) and next-generation sequencing (NGS) in tissue and blood samples from gastric and gastroesophageal junction cancers (GC/GEJC). First Author: Steven Brad Maron, Memorial Sloan Kettering Cancer Center, New York, NY

Background: An estimated 18% of patients (pts) with GC/GEJC have HER2-positive (HER2+; IHC 3+ or 2+/in situ hybridization−positive [ISH+]) tumors. Although IHC/ISH are the standard testing methods for determining HER2 positivity, NGS can identify *HER2* (*ERBB2*) amplification alongside other biomarkers. This study assessed the concordance of detecting *HER2* amplification by tissue/plasma NGS with IHC/ISH HER2 status in pts with GC/GEJC. **Methods:** Pts were retrospectively identified in the Tempus database. The primary analysis assessed tissue and plasma NGS *HER2* copy number (CN) concordance with a HER2+ (IHC 3+ or 2+/ISH+) or HER2− (IHC 0, 1+, or 2+/ISH−) status; NGS samples were collected in ≤30 days of the IHC sample collection date. *HER2* amplification was defined as tissue CN ≥8 or plasma log2(CN) ≥0.5; additional amplifications were captured by manual inspection. A secondary analysis assessed the correlation between paired tissue and plasma *HER2* log2(CN) in pts with plasma and tissue NGS samples collected ≤30 days apart. The secondary analysis included pts irrespective of IHC sample collection date. **Results:** Of 1578 pts with GC/GEJC and a curated IHC score evaluated in the primary analysis, 1507 and 305 pts underwent tissue and plasma NGS testing, respectively. Among pts with tissue and plasma NGS results, 902 (59%) and 201 (66%) had primary tumors of the stomach, respectively; all remaining pts had primary tumors of the cardia.According to IHC/ISH, 229 (15%) and 40 (13%) pts were HER2+ among those with tissue and plasma NGS testing, respectively. Of the pts identified as *HER2* amplified by tissue and plasma NGS,95% and 92% were HER2+ by IHC/ISH, respectively. Positive percent agreement (PPA) of *HER2* amplification by tissue and plasma NGS with IHC/ISH HER2+ status was 70% and 58%, respectively. Concordance rates by PPA, negative PA (NPA), and overall PA (OPA) are summarized in Table. In the secondary analysis, a positive correlation in NGS *HER2* log2(CN) was observed in pts with paired tissue and plasma NGS (mutant variant allele frequency >5%; R=0.85; P<2.2e−16; n=307). **Conclusions:** Based on PPA, we observed moderate concordance between NGS *HER2* amplification detection (plasma or tissue) and standard HER2 IHC assessment. NGS is not sufficient to detect all pts with HER2+ GC/GEJC. Further testing is warranted to determine if NGS testing could complement IHC/ISH in identifying pts who may benefit from HER2-directed therapies. Research Sponsor: AstraZeneca in collaboration with Daiichi Sankyo.

	IHC 3+ or 2+/ISH+ (HER2+), n	IHC 0, 1+, or 2+/ISH− (HER2−), n	Total
Total	**229**	**1120**	
Tissue *HER2* amplified, n	160	8	**168**
Tissue *HER2* not amplified, n	69	1112	**1181**
	PPA: 69.9%	**NPA: 99.3%**	**OPA: 94.3%**
	IHC 3+ or 2+/ISH+ (HER2+), n	**IHC 0, 1+, or 2+/ISH− (HER2−), n**	**Total**
Total	**40**	**233**	
Plasma *HER2* amplified, n	23	2	**25**
Plasma *HER2* not amplified, n	17	231	**248**
	PPA: 57.5%	**NPA: 99.1%**	**OPA: 93.0%**

479 **Poster Session**

Concordance between PD-L1 assays 28-8 and SP263 and their respective scoring algorithms in procured upper gastrointestinal adenocarcinoma samples. First Author: Thinh Pham, Gilead Sciences, Inc, Foster City, CA

Background: PD-L1 is being studied as a predictive immunohistochemical (IHC) biomarker for gastroesophageal junction (GEJ), gastric (GC), and esophageal (EAC) adenocarcinomas, and anti-PD-(L)1 therapies are used to treat them. Here, we compare the VENTANA PD-L1 (SP263) IHC assay and its tumor area positivity (TAP) algorithm with the clinically validated Agilent PD-L1 IHC 28-8 pharmDX combined positive score (CPS) algorithm in GEJ, GC, and EAC samples. The degree of concordance between SP263 TAP ≥ 5% and 28-8 CPS ≥ 5 is important for interpreting real-world settings and emerging clinical trial data. **Methods:** GEJ, GC, and EAC tumor samples procured from company-sponsored clinical trials (NCT02545504, NCT02862535, NCT02864381) and external vendors with informed consent were analyzed 4 ways: SP263 TAP, SP263 CPS, 28-8 TAP, and 28-8 CPS. Readouts were evaluated by a single pathologist and were analyzed with Pearson correlation coefficients. **Results:** 286 patient samples were analyzed (GEJ, n = 106; GC, n = 88; EAC, n = 92). The agreement among the SP263 TAP ≥ 5% and 28-8 CPS ≥ 5 prevalence is shown in the Table. The overall correlation coefficient between 28-8 CPS and SP263 TAP was high (R = 0.95; $P < 2.2 \times 10^{-16}$). In addition, when applying the same scoring algorithm to both assays (SP263 TAP vs 28-8 TAP or SP263 CPS vs 28-8 CPS), the correlation coefficients remained high (R = 0.97; $P < 2.2 \times 10^{-16}$). When both algorithms were applied to the same IHC assay (SP263 TAP vs SP263 CPS or 28-8 TAP vs 28-8 CPS), the correlation coefficients were also high (R = 0.98; $P < 2.2 \times 10^{-16}$). **Conclusions:** In this controlled experiment the PD-L1 assays 28-8 and SP263 concordance was high. The scoring algorithms for GEJ, GC, and EAC samples were observed as highly correlated; minor differences were likely driven by the distinct PD-L1 monoclonal antibody clones utilized in the IHC assays. These data suggest that the 2 assays are comparable for evaluating PD-L1 expression at the TAP ≥ 5% and CPS ≥ 5 cutoffs. Clinical trial information: NCT02545504, NCT02862535, NCT02864381. Research Sponsor: Gilead Sciences, Inc.

Concordance of SP263 TAP ≥ 5% with 28-8 CPS ≥ 5.

Positive percentage agreement: 0.96 **Negative percentage agreement: 0.95** **Overall percentage agreement: 0.95**	**28-8 CPS ≥ 5: Positive**	**28-8 CPS ≥ 5: Negative**
SP263 TAP ≥ 5%: Positive	116	9
SP263 TAP ≥ 5%: Negative	5	156

480 **Poster Session**

Genomic landscape study of esophago-gastric adenocarcinoma (EGAC) in young patients under the age of 30 years. First Author: Myungwoo Nam, Department of Hematology/Oncology, SUNY Upstate Medical University, Syracuse, NY

Background: Despite recent advances in treatment of EGAC, prognosis continues to be poor with reported 5-year survival of less than 30%. EGAC is more frequently diagnosed at an old age with the median age at diagnosis around 65 years. However, approximately 10% of EGAC are diagnosed at a younger age of less than 50 years and those patients present with more advanced stages resulting in worse prognosis. Our study investigated the genomic landscape of EGAC diagnosed under the age of 30 years to better understand the characteristics of EGAC occurring in very young population. **Methods:** 8,001 cases of clinically advanced EGAC who underwent comprehensive genomic profiling (CGP) from 2012 to 2024 were included in the study. Patients aged less than 30 years were classified as 'EGAC-young' and patients aged 50 and older were classified as 'EGAC-old'. All classes of genomic alterations (GA), microsatellite instability high (MSI-high) status, tumor mutation burden (TMB), genomic ancestry, genomic signature, and homologous recombination defect signature (HRDSig) were determined from the sequencing data. PD-L1 was measured by IHC using the Dako 22C3 tumor proportional score (TPS) system. Comparisons utilized the Fisher Exact method with the Benjamini-Hochberg adjustment to reduce the false discovery rate. **Results:** Out of 8,001 cases, 7389 (92.4%) were classified as 'EGAC-old' and 26 (0.3%) as 'EGAC-young'. Both groups had a higher frequency of male patients but there was no significant difference between the two groups (86.4% vs 76.9%; NS). Median GA per tumor was 6 in both groups. The 'EGAC-old' featured a significantly higher frequency of EUR ancestry (92.0% vs 73.1%; P=0.026). There were no significant differences in clinically important GA between both groups, including similar ERBB2 GA (21.2% vs 30.8%; NS) and FGFR3 GA (1.2% vs 7.7%; NS). There was a trend of more frequent GA of AKT2, CCNE1, and KEAP1 noted in 'EGAC-young' (0.9% vs 11.5%, 8.5% vs 26.9%, 1.3% vs 11.5%, respectively; P=0.071). MSI-high status was 3.1% in the 'EGAC-old' but not detected in the 'EGAC-young' (NS). The median TMB was higher in the 'EGAC-old' vs 'EGAC young' (3.8 mutations/Mb vs 1.9 mutations/Mb; P=.003). The frequencies of a positive HRDSig were similar in both groups (5.6% vs 8.0%; NS). There were no significant differences in COSMIC trinucleotide signatures. PD-L1 expression was identified in greater than 20% of the 'EGAC-old' but was not measured in the 'EGAC-young'. **Conclusions:** Clinically advanced EGAC in young patients under the age of 30 years features a genomic landscape that differs from EGAC identified in older patients, including lower TMB and a trend of increased GA frequency of genes related to poor prognosis in other types of cancer. Further studies are warranted to utilize CGP findings in identifying potential implications for treatment and prognostication in these rare cases of EGAC occurring in very young patients. Research Sponsor: None.

481 **Poster Session**

Relationship between PD-L1 expression and the number of biopsy specimens in advanced gastric cancer. First Author: Taro Mizuno, Department of Clinical Oncology, Aichi Cancer Center Hospital, Nagoya, Japan

Background: Programmed cell death ligand 1 (PD-L1) expression shows spatial heterogeneity in gastric cancer. Compared with single biopsies, multiple biopsies may provide reliable PD-L1 expression results. However, the optimal number of biopsy specimens remains unclear. This study aimed to assess the relationship between PD-L1 expression and the number of biopsy specimens in advanced gastric cancer. **Methods:** We retrospectively analyzed patients with advanced gastric cancer who received first-line chemotherapy at a single institution between December 2021 and June 2024. Those in whom PD-L1 expression was evaluated using tumor-containing biopsy specimens were included. PD-L1 expression was measured using the Dako PD-L1 IHC 28-8 pharmDx assay. PD-L1 positivity was defined as a combined positive score of ≥5. The association between PD-L1 positivity and the number of biopsy specimens was tested using the chi-square or Fisher's exact test. **Results:** Of the 183 patients screened, 110 were included. The patient characteristics were as follows: median age, 71 (range: 34–87) years; sex (male/female), 75/35; primary site (gastric/gastroesophageal junction), 102/8; and histological type (diffuse/intestinal/other), 59/34/17. The PD-L1 positivity prevalence was 71.8%. In 97 patients, human epidermal growth factor receptor 2 (HER2) expression was detected in the same specimens as PD-L1, with a HER2-positive rate of 14.4%. The mean numbers of biopsy and tumor-containing biopsy specimens were 5.14 (range: 2–10) and 4.25 (range: 1–10), respectively. The PD-L1 positivity prevalence was significantly higher when the number of biopsy specimens was ≥5 compared with ≤4 (77.5% vs 56.7%, *P*=0.03; Table). The same trend was observed according to the histological type (diffuse type, 75.0% vs. 46.7%, *P*=0.04; intestinal type, 79.2% vs. 60.0%, *P*=0.39). In HER2-negative cases, PD-L1 positivity was significantly more prevalent when the number of biopsy specimens was ≥5 compared with ≤4 (83.6% vs 54.5%, *P*=0.006; Table). However, there was no such difference in HER2-positive cases (54.4% vs 66.7%, *P*=1). **Conclusions:** The data suggest that collecting at least five biopsy specimens increases the likelihood of identifying PD-L1 positivity in advanced gastric cancer, particularly in HER2-negative cases. This approach may improve PD-L1 evaluation accuracy and lead to better treatment decisions. Research Sponsor: None.

Relationship between the number of biopsy specimens and PD-L1 positivity.

Number of biopsy specimens	PD-L1 positivity (%)	PD-L1 negativity (%)	*P* value
Overall			0.03
≥5	77.5	22.5	
≤4	56.7	43.3	
HER2-negative			0.006
≥5	83.6	16.4	
≤4	54.5	45.5	

482 Poster Session

The expand indication of HER2 targeted therapies by monitoring of circulating tumor cells. First Author: Koichi Suzuki, Jichi Medical University, Saitama Medical Center, Saitama, Japan

Background: Despite success in the HER2 targeted therapies for metastatic gastric cancer patients, the conventional assessment of HER2 status in tumor tissue specimens is still under debate. **Methods:** On-chip sorting system was used for the isolation of circulating tumor cells (CTCs). The fluorescence signal intensity > 50 is set to be a threshold value for distinguishing between positive and negative HER2 on CTCs. Epithelial mesenchymal transition (EMT) was assessed by the vimentin and cytokeratin. DNA was extracted from CTCs and applied to the cancer gene panel. **Results:** In 27 patients with metastatic gastric cancer, 13 patients showed HER2 positive in tumor tissues (Group A) who were treated with cytotoxic agents and HER2-targeted therapy while 14 patients exhibited negative and treated with cytotoxic agents. In these 14 patients, 8 patients showed HER2 positive (Group B) and 6 patients displayed HER2 negative (Group C) on CTCs. HER2 expression on CTCs was associated with EMT ($p<0.001$). Increased expression of EMT was seen in Group A and B, indicating they are likely to have metastasis. Group B exhibited worse PFS than others (13.8 M in A, 7.0 M in B and not reached in C, p=0.012). In addition, genomic analysis identified cancer gene mutations, some of which were involved in mechanisms underlying EMT and were shared between Group A and B. Considering HER2 expression on CTCs in Group B, an additional HER2-targeted therapy may be beneficial. **Conclusions:** Our study suggests that monitoring of circulating tumor cells leads to the expand indication of the HER2 targeted therapies in patients with metastatic gastric cancer. Research Sponsor: None.

483 Poster Session

Use of nucleotyping and tumor stroma ratio to predict prognosis in patients with resected intestinal-type gastric cancer. First Author: Dan Sha, Department of Minimally Invasive Treatment of Cancer, Shandong Provincial Hospital Affiliated to Shandong First Medical University, Jinan, China

Background: Heterogeneity exists in recurrence and survival among patients with intestinal-type gastric cancer (IGC). Nucleotyping is the alterations in chromatin organization. We sought to identify the prognostic utility of nucleotyping, tumor stroma ratio (TSR) and DNA ploidy in these patients. **Methods:** Radical resection of gastric carcinoma (> 15 LNs removed) was performed in 283 patients (stage I, n=73; stage II, n=102; stage III, n=108) at our university hospital between 2010 and 2018. Automated image cytometry tools were used to quantitatively measure nucleotyping, TSR and DNA ploidy. Cutpoints of nucleotyping and TSR were 0.044 and 50%, respectively. Associations of these features with clinicopathological characteristics, time-to-recurrence (TTR) and cancer specific survival (CSS) were analyzed by univariate and multivariate analysis. **Results:** Among 283 patients, chromatin heterogeneity, high TSR and non-diploid were significantly associated with larger tumor size, higher T and N stage, and poorer TTR and CSS at 5 years. In a Cox multivariable analysis, nucleotyping and TSR were independently associated with TTR and CSS ($P < 0.05$) (Table). The top three contributors to TTR were N stage (33.9%), nucleotying (29.7%) and TSR (25.5%). The top three contributors to CSS were N stage (35.6%), nucleotying (29.7%) and TSR (18.0%). As a combined variable, the percent relative contribution of nucleotyping and TSR ranked first for TTR and CSS (56.0%, 50.0%), ahead of N stage (33.4%, 34.6%) and TILs (8.4%, 10.6%). **Conclusions:** Nucleotyping and TSR were each significantly and independently associated with TTR and CSS, and their combined variable was more strongly prognostic than either alone. Research Sponsor: None.

Cox multivariable analysis of TTR and CSS.

Features	TTR HR (95%CI)	TTR P value	CSS HR (95%CI)	CSS P value
Tumor size (>3cm vs ≤3cm)	1.10 (0.48-2.54)	0.83	0.77 (0.35-1.70)	0.56
T stage (T3-T4 vs T1-T2)	1.07 (0.46-2.49)	0.87	1.41 (0.53-3.76)	0.49
N stage (N2N3 vs N0N1)	2.84 (1.39-5.78)	<0.01	7.69 (3.37-17.55)	<0.01
Adjuvant Therapy (with vs without)	1.32 (0.72-2.40)	0.37	0.85 (0.48-1.52)	0.59
TB (high vs low)	1.29 (0.62-2.69)	0.50	1.61 (0.85-3.07)	0.15
TILs (low vs high)	1.73 (0.96-3.13)	0.07	1.90 (1.06-3.40)	0.03
Nucleotyping (Chromatin heterogeneous vs Chromatin homogeneous)	2.19 (1.21-3.99)	0.01	3.44 (1.96-6.03)	<0.01
TSR (high vs low)	2.12 (1.16-3.82)	0.01	2.51 (1.43-4.41)	<0.01
DNA ploidy (Non-diploid vs Diploid)	1.20 (0.55-2.63)	0.65	0.98 (0.44-2.15)	0.95

1 TTR, time-to-recurrence; 2 CSS, cancer specific survival.

484 Poster Session

Histopathological growth patterns of gastric cancer liver metastasis and their association with patient prognosis. First Author: Dan Sha, Department of Minimally Invasive Treatment of Cancer, Shandong Provincial Hospital Affiliated to Shandong First Medical University, Jinan, China

Background: The histopathological growth patterns (HGPs) of liver metastasis reflect the complicated and varied interactions between tumor cells and the host microenvironment. We studied HGPs of gastric cancer liver metastasis (GCLM) and sought to predict them using radiomic and clinical variables. **Methods:** Hepatectomy was performed in 62 patients with GCLM at our university hospital between 2011 and 2021. HGPs were evaluated according to international consensus guidelines, and were classified as either desmoplastic HGP (dHGP) or non-dHGP. CD4, CD8, VEGF and CD31 were analyzed by immunohistochemistry. Eight hundred and fifty-one radiomics features were extracted from CT portal venous phase images and the optimal radiomics signature was determined by univariate analysis and LASSO regression. Multivariable regression analyses and ROC curves were used to assess the predictive performance of HGPs with a radiomic and clinical combined model. **Results:** Among 62 patients, 14 were categorized as dHGP, and 48 as non-dHGP. Non-dHGP vs. d-HGP was associated with higher tumor N stage and CEA level, and with decreased $CD8^+$ T cells, and lower VEGF and CD31 expression. HGP status was independently associated with patient cancer-specific survival (CSS) ($P < 0.05$) multivariately.Analysis of percent relative contribution revealed that HGP ranked first for CSS (35.9%), ahead of T (28.9%) and N (28.9%) stage. A model with combined radiomic and clinical features demonstrated robust performance with area under the curve (AUC) values of 0.993 and 0.933 in the training and validation sets, respectively. **Conclusions:** Non-invasive determination of radiomic and clinical features was shown to predict HGPs types. Compared to dHGP, GCLM with non-dHGP exhibited significantly lower immune cell infiltration and decreased angiogenesis. Among features examined in GCLM, HGP was the most robust for prognostication. Research Sponsor: None.

Univariate and multivariate analysis of prognostic factors for cancer-specific survival.

	Univariate analysis		Multivariate rergression analysis	
Characteristics	HR (95%CI)	P	HR (95%CI)	P
Gender (Women vs Men)	1.13(0.61-2.09)	0.71		
Age (>60years vs ≤60years)	1.08(0.60-1.70)	0.98		
HGP (non-desmoplastic HGP vs desmoplastic HGP)	3.63(1.78-7.41)	0.00	3.48(1.37-8.88)	0.01
T stage (T3-T4 vs T1-T2)	2.03(1.08-3.80)	0.03	2.29(1.15-4.58)	0.02
N stage (N1-N3 vs N0)	3.46(1.66-7.23)	0.00	2.54(1.17-5.56)	0.02
CEA (>5ng/ml vs ≤5ng/ml)	2.37(1.25-4.48)	0.01	1.56(0.70-3.48)	0.27

485 Poster Session

A study of the homologous recombination deficiency signature (HRDSig) status of advanced esophageal squamous cell carcinoma (AESCC) by utilizing comprehensive genomic profiling (CGP). First Author: Zakee Mohamed Jiffry, SUNY Upstate Medical Univerity, Syracuse, NY

Background: The role of poly-ADP-ribose polymerase (PARP) inhibitors is currently under investigation as a potential therapeutic option for AESCC. Homologous recombination deficiency signature status (HRDSig) positivity has been shown to predict biallelic loss of BRCA1/2 which may simultaneously portend a worse prognosis and a sensitivity to poly-ADP-ribose polymerase (PARP) inhibitors. In this study, we attempt to characterize the genomic alterations present in AESCC based on HRDSig status utilizing comprehensive genomic profiling (CGP) techniques. **Methods:** 2,029 cases of AESCC underwent comprehensive genomic profiling with an examination of all genomic alterations (GA). MSI high status, tumor mutation burden (TMB) levels, genomic ancestry and genomic trinucleotide signatures were determined from the sequencing data. HRDSig status was calculated using a broad set of genome-wide copy number features (PMID 37769224). Results were compared using the Fisher exact system with the Benjamini-Hochberg adjustment to correct for false discovery. **Results:** 154 (7.6%) of the 2029 AESCC cases featured a positive HRDSig status (HRDSig+). The age (66-67) and gender (58%-63% male) distribution were similar in the HRDSig+ and HRDSig- AECSS cases as was the frequency of GA/tumor (9 for both). The median TMB was higher in the HRDSig+ (6.3 vs 3.8; $P<.0001$) as was frequency of TMB > 10 mutations/Mb (19.5% vs 10.0%; P=.004). An APOBEC trinucleotide signature was also more frequent in the HRDSig+ AESCC (11.0% vs 5.4%; P=.05). As anticipated, GA in genes associated with HRD including *BRCA1* (10.4% vs 1.4%; $P<.0001$) and *BRCA2* (14.3% vs 2.0%; $P<.0001$) was present. MTOR pathway activating mutations were also more frequent in the HRDSig+ AESCC group including *PTEN* (13.0% vs 7.3%; P=.06). **Conclusions:** With a 7.6% frequency, HRDSig+ status is a relatively rare event in AESCC. CGP with determination of HRDSig status in AESCC may prove useful in tailoring PARP inhibitor-based treatment regimens and may potentially uncover other genomic alterations that can aid in designing targeted therapy combinations in future. Research Sponsor: None.

Characteristics of comprehensive genomic profiling of clinically advanced esophageal squamous cell carcinoma by HRDSig status.

	AESCC HRDSig- (n=1875)	AESCC HRDSig+ (n=154)	P-value
Pathogenic genomic alterations			
BRCA1	1.4% (26)	10.4% (16)	0.00
BRCA2	2.0% (36)	14.3% (22)	0.00
PTEN	7.3% (137)	13.0% (20)	0.06
COSMIC trinucleotide signature			
APOBEC	5.4% (101)	11.0% (17)	0.05
Tumor mutational burden (TMB)			
Median TMB (range) (IQR)	3.8 (0-85) (2.5-6.3)	6.3 (0-61) (3.8-8.8)	0.00
TMB≥10 mut/Mb	10.0% (188)	19.5% (30)	0.00

486 **Poster Session**

Prognostic value of tumor regression grade (TRG) in patients with gastric/gastroesophageal junction cancer treated by peri-operative chemotherapy and surgery: A single-center experience (Institut Mutualiste Montsouris, Paris, France). First Author: Antoine Lefèvre, Department of Medical Oncology; Institut Mutualiste Montsouris, Paris, France

Background: Management of locally advanced gastric and gastroesophageal junction (GOJ) cancer relies on peri-operative chemotherapy and radical surgery. Response to neoadjuvant chemotherapy can be evaluated by Mandard Tumor Regression Grade (TRG) performed on the removed specimens. The aim of this study was to investigate the prognostic value of TRG, in terms of post-surgical disease-free survival (DFS) and overall survival (OS). **Methods:** Retrospective analysis of all consecutive patients (Jan 2010 to Dec 2021) who underwent oncological gastrectomy for gastric or GOJ adenocarcinoma after neoadjuvant chemotherapy was performed. TRG was evaluated according to Mandard classification, ranging from 1 (complete pathological response) to 5 (absence of response), TRG 1-2 being considered as "good responders" and TRG 3-5 as "bad responders". Univariate and multivariate analysis were performed for DFS and OS. **Results:** One hundred and ninety-nine patients were included, 157 treated with FOLFOX and 42 with FOLFOX + Taxanes (either FLOT or FOLFOX + Abraxane). TRG 1-2 was observed in 30% in patients (pts) treated with FOLFOX and 24% in pts treated with FOLFOX + Taxanes. With a median follow-up over 5 years (yrs), median DFS and OS were not reached (DFS rates at 3, 5 and 10yrs of 66.5%, 64.1% and 55.7%, respectively; OS rates at 3, 5 and 10yrs of 75.6%, 66.8% and 60.4%, respectively). In univariate analysis, TRG 1-2 was significantly associated with favorable outcome compared to TRG 3-5 in terms of DFS (HR 2.4, p-value 0.0056) and OS (HR 2.6, p-value 0.0061), as well as ECOG at diagnosis (0 vs. 1), ypT stage (yp T0-2 vs. yp T3-4), ypN stage (ypN- vs. ypN+), tumor localization (antrum + body vs. cardia + esophagus), and lymphatic, vascular and peri nervous embolisms (negative vs. positive) for both DFS and OS. However, in multivariate analysis (selection backward Cox-model), only ypT stage and vascular embolisms for DFS, and ypT and ypN stages for OS were found to be independent prognostic factors. **Conclusions:** Although TRG is a prognostic factor in patients with gastric or GOJ cancer, it does not appear as an independent factor for DFS and OS in this series of 199 pts. These results raise the issue of "good responder" definition, especially for TRG 3. Prognostic impact of TRG needs to be investigated for patients treated with combination of immune checkpoint inhibitors and chemotherapy in neoadjuvant regimens. Research Sponsor: None.

487 **Poster Session**

SOX/FOLFOX+nivolumab followed by conversion surgery for HER2 negative primary gastric or gastroesophageal junction cancer with oligo-metastasis or borderline resectable tumors: Early results of a consecutive 41 cases. First Author: Masashi Nishino, Department of Gastric Surgery, National Cancer Center Hospital, Tokyo, Japan

Background: Gastric or gastroesophageal junction cancer with oligo-metastasis or borderline resectable tumors could be cured by conversion surgery after primary chemotherapy. Because SOX/FOLFOX plus Nivolumab showed high clinical response regardless of CPS score, this nivolumab-based chemotherapy followed by surgery may be promising strategy for these tumors. **Methods:** The study examined the patients who had HER2 negative primary gastric or gastroesophageal junction cancer with synchronous oligo-metastasis or borderline resectable tumors. Oligo-metastasis was defined as peritoneal dissemination localized to the upper abdomen, para-aortic lymph node metastasis, liver metastasis limited up to 3 segments, mediastinal lymph node metastasis, ovarian metastasis, or esophageal intramural metastasis. Borderline resectable tumors were defined as regional lymph node metastasis forming bulky mass or primary tumors invading adjacent organs. The patients with these tumors were treated with primary chemotherapy with SOX/FOLFOX plus Nivolumab for 3 to 22 months, then were proceeded to conversion surgery in case the tumors are responded to chemotherapy and are considered to achieve R0 resection based on the MDT conference. **Results:** During November 2021 and February 2024, 41 consecutive patients were included in the study. Disease status was peritoneal dissemination in 1, para-aortic lymph node metastasis in 26, liver metastasis in 5, mediastinal lymph node metastasis in 1, ovarian metastasis in 1, esophageal intramural metastasis in 2, bulky nodal metastasis in 6, and primary tumors invading pancreas in 1 (distant oligo-metastasis in 34 patients and borderline resectable in 7). CPS score was >5 in 13, 5> and >1 in 23, and >1 in 5 patients. Chemotherapy regimen was SOX+Nivo in 38 and FOLFOX+Nivo in 3. Of 41 patients, 33 (80.5%) (SOX+Nivo: 30, FOLFOX+Nivo: 3) were proceeded to the conversion surgery, and 32 patients achieved R0 resection. 6 patients showed disease progression and 2 patients were judged as unresectable. Major pathological response (disappearance more than 80% of the primary tumor) rate was 54.5% (18/33) and pCR rate was 33.3% (11/33). **Conclusions:** SOX/FOLFOX plus Nivolumab for HER2 negative gastric or gastroesophageal junction cancer with oligo-metastasis or borderline resectable tumors achieved a high R0-conversion surgery rate and pathological response rate. Research Sponsor: None.

488 **Poster Session**

Use of immune repertoire sequencing analysis to detect differences in treatment responses for advanced esophageal squamous cell carcinoma treated with camrelizumab and platinum-based chemotherapy. First Author: Xiaoling Zhang, Department of Oncology, Changzhi People's Hospital Affiliated to Changzhi Medical College, Changzhi, Shanxi, China; Laboratory Animal Center, Shanxi Medical University, Taiyuan, Shanxi, China; Department of Basic Medical Sciences, Sha, Changzhi, Shanxi, China

Background: The study evaluated the efficacy and safety of camrelizumab combined with platinum-based chemotherapy (platinum plus paclitaxel [TP] or fluorouracil [FP] agents) as first-line treatment in patients with advanced esophageal squamous cell carcinoma (ESCC), and explored potential biomarkers for treatment response using immune repertoire (IR) sequencing. **Methods:** In this multi-center, prospective cohort study, advanced ESCC patients were treated with camrelizumab and TP or FP chemotherapy. Primary outcome was 1-year progression free survival (PFS). Secondary outcomes included 1-year overall survival (OS), objective response rate (ORR), disease control rate (DCR) and safety. Patients were categorized based on their treatment responses into the non-ORR and ORR groups. IR sequencing was exploratorily performed on the peripheral blood mononuclear cells from 15 patients in each group. **Results:** From June 2020 to April 2023, 88 out of 98 screened patients were enrolled, with a median follow-up of 15.9 months. The 1-year PFS was 56.8%, OS was 68.2%, ORR was 64.8%, and DCR was 91.1%. Most treatment-related adverse events were grade 1-2, though 12.5% experienced grade ≥3 toxicities. The FP regimen's efficacy was comparable to the TP regimen. The analysis of the complementarity-determining region 3 polypeptide sequences revealed significant differences in amino acid composition between the non-ORR and ORR groups in T-cell receptor beta-chain (TRB) and immunoglobulin heavy chain (IGH) repertoire. Furthermore, the ORR group exhibited a more concentrated distribution of clones within TRB. Two V genes (TRBV29-1 and TRBV4-1) and one J gene (TRBJ1-3) in the TRB region, along with six V genes (IGHV1-45, IGHV3-20, IGHV3-48, IGHV3-49, IGHV4-4, and IGHV5-51) in the IGH region, were differentially expressed between the two groups. **Conclusions:** Camrelizumab with platinum-based chemotherapy is effective and well-tolerated as first-line treatment in advanced ESCC. IR sequencing may provide insights into the underlying mechanisms influencing treatment response. Clinical trial information: ChiCTR2000037942. Research Sponsor: None.

Objective response and disease response.

Efficacy evaluation	Total (N=88)	Camrelizumab plus T (n=69)	Camrelizumab plus FP (n=19)	p
1-year PFS rate	50/88 (56.8%)	39/69 (56.5%)	11/19 (57.9%)	1.000
1-year OS rate	60/88 (68.2%)	47/69 (68.1%)	13/19 (68.4%)	1.000
ORR	57/88 (64.8%)	43/69 (62.3%)	11/19 (57.9%)	0.793
DCR	81/88 (91.1%)	63/69 (91.3%)	15/19 (79.0%)	0.213
ORR group, n (%)	54/88 (61.4%)	43/69 (62.3%)	11/19 (57.89%)	0.793
Non-ORR Group, n (%)	34/88 (38.6%)	26/69 (37.7%)	8/19 (42.1%)	1.000
2-year PFS rate	38/88 (43.2%)	30/69 (43.5%)	8/19 (42.1%)	1.000
2-year OS rate	42/88 (47.7%)	33/69 (47.8%)	9/19 (47.4%)	1.000

*p<0.05, a statistically difference.

489 **Poster Session**

Transcriptomic features of Epstein-Barr virus-associated gastric cancer based on PD-L1 status. First Author: Sun Young Rha, Yonsei Cancer Center, Songdang Institute for Cancer Research, Yonsei University College of Medicine, Seoul, South Korea

Background: Epstein-Barr virus (EBV)-associated gastric cancer (EBVaGC) accounts for 5-10% of all gastric cancers and has a relatively favorable prognosis compared to other subtypes. While the genomic profile of EBVaGC has been previously reported to exhibit extensive CpG island methylation, elevated levels of PD-L1/2, and an absence of TP53 mutations, transcriptomic features remain less explored. The response to immunotherapy of EBVaGC has been various. This study comprehensively investigates the transcriptomic characteristics of EBVaGC in relation to PD-L1 status. **Methods:** RNA sequencing was conducted on formalin-fixed, paraffin-embedded samples from 27 EBV-positive and 12 EBV-negative patients with HER2-negative gastric cancer. Gene expression was quantified using the RSEM package. Predict-IO, a machine learning-based predictive algorithm developed at Auristone Pte Ltd based on initial work done at the Genome Institute of Singapore (GIS), assessed the likelihood of a patient responding to immunotherapy based on transcriptome expression data. PD-L1 immunohistochemistry was performed using the 22C3 pharmDx assay, and status was classified by CPS. **Results:** A total of 200 differentially expressed genes (DEGs; log2FC > 2, FDR < 0.05) were identified in EBV-positive versus negative samples, with 157 genes down-regulated and 43 genes up-regulated. Upregulated genes included immune response related genes such as DMBT1, IDO1, PLA2G2A, and IL13RA2, while downregulated genes were primarily associated with cell adhesion, including LAMA1 and CCN6. In the EBV positive group, immune response pathways were enriched; however, within the PD-L1-Low group (CPS < 1), activated signaling pathways such as MYC, E2F, and MTORC1 were observed. Conversely, the PD-L1-High group (CPS ≥ 5) exhibited upregulated snoRNAs. In the EBV negative group, the PD-L1-Low cohort displayed suppressed immune response and ECM remodeling factors, whereas the PD-L1-High cohort showed enrichment of ECM remodeling factors. The Predict-IO Score, which predicts immunotherapy response, was significantly higher in EBV-positive compared to EBV-negative samples (p = 0.003), and scores were elevated in the PD-L1-High group compared to the PD-L1-Low group. **Conclusions:** In summary, EBVaGC includes subsets of different PD-L1 expression patterns with distinct activated signaling pathways, differences in ECM remodeling factors, and varying Predict-IO Scores. With these information, we could properly select the responsive patients to immunotherapy among EBVaGC. Research Sponsor: None.

490 **Poster Session**

Biomarkers of response in patients with gastric/gastroesophageal junction adenocarcinoma (GEA) treated with telisotuzumab adizutecan. First Author: John H Strickler, Duke University Medical Center, Durham, NC

Background: c-Met protein (MET protein) overexpression (OE) and *MET* amplification (amp) occur in several cancer types, including GEA, and are associated with poor prognosis. The c-Met–directed antibody-drug conjugate telisotuzumab adizutecan (ABBV-400) is being assessed in advanced solid tumors in an ongoing, first-in-human study (NCT05029882). Results showed promising antitumor activity and a tolerable safety profile in GEA (Strickler et al. ESMO 2024, 1439P). Here, we present our analysis on associations between c-Met OE, *MET* amp, and ctDNA-derived molecular response with clinical outcomes. **Methods:** c-Met expression was evaluated retrospectively in FFPE tissue (archival/fresh) collected at study screening using IHC (SP44 Ab clone) or whole transcriptomic RNA sequencing (Caris). Plasma ctDNA collected at baseline (BL) and cycle 3 day 1 (C3) was analyzed with the GuardantINFINITY assay. Molecular response (MR) was assessed using circulating tumor fraction (cTF), estimated on the basis of ctDNA variant allele frequencies in a 74-gene panel (Guardant360 CDx) and methylation signals (GuardantINFINITY methylome platform). MR was defined as a 50% decrease in cTF from BL. *MET* amp status was determined retrospectively by the Guardant Health algorithm. Radiographic response was assessed per RECIST v1.1. **Results:** Clinical data from 42 patients (pts) with advanced GEA were included in this biomarker analysis. There was a negative correlation between c-Met expression and tumor size change from BL (R=-0.59). Of 42 pts, 12 (29%) had tumors with *MET* amp per ctDNA analysis/local site reports, which was higher than the bioinformatic analysis of real-world data (Caris-ConcertAI linked RWD360 and PT360; ~3%). Concordance between c-Met expression per IHC and RNA was observed (R=0.75, $p<10^{-6}$). There were 12 responders; 7 (58%) had *MET* amp, and 5 (42%) did not. Responders had lower ctDNA percentage change from BL vs nonresponders (1.5-fold and 1.6-fold difference based on 74-gene and methylation panel, respectively). Pts with MR had a greater decrease in tumor size from BL vs those without MR. A positive correlation between change in ctDNA levels (BL to C3) and change in tumor size was observed using the 74-gene panel and methylation panel (R=0.73 and 0.69, respectively). Additional biomarker analyses focused on the association between *MET* and other key biomarkers relevant in GEA will be discussed. **Conclusions:** In pts with advanced GEA treated with telisotuzumab adizutecan, c-Met OE/*MET* amp and ctDNA-based MR were associated with enhanced antitumor activity. Compared with real-world data, *MET* amp was enriched in this study population. Phase 2 studies will continue to explore c-Met OE and *MET* amp prevalence and their potential predictive value with other biomarkers in a clinically representative population. Clinical trial information: NCT05029882. Research Sponsor: AbbVie; n/a.

491 **Poster Session**

The Gastric Cancer Registry: A multi-omic cellular and molecular resource for cancer biomarker and therapeutic discovery. First Author: Hanlee P. Ji, Division of Oncology, Department of Medicine, Stanford University School of Medicine, Stanford, CA

Background: The Gastric Cancer Registry (GCR) is a comprehensive clinical and genomic data resource focused on gastric cancer (GC) patients and individuals with a family history of GC or a germline CDH1 mutation. As part of the GCR's ongoing enrollment, patients contribute cancer history information, archival tumor samples and other biospecimens. The cancer samples undergo extensive molecular and cellular analysis that includes genomic sequencing and spatial analysis. We have generated a robust dataset, publicly accessible online through the GCR Genome Explorer (GCR-GE). This genomic, molecular and cellular resource provides a rich data set that accelerates multidisciplinary research aimed at improving detection and treatment strategies for GC. GC disproportionately affects Latin American populations - we have included new cohorts from Latin America. The GCR enables researchers, clinicians and patients to address specific questions about the molecular and cellular basis of GC. **Methods:** GC samples underwent whole genome, whole exome, and bulk RNA sequencing were conducted. The genomic features identified include copy number variation, gene mutations, gene expression, microbiome composition, estimation of tumor-infiltrating immune cells, tumor neoantigens, MSI/MSS status, and HLA subtypes. In addition to genomic data, clinical data was compiled from survey responses and pathology reports and integrated into the GCR-GE. We have incorporated single cell and spatial analysis that includes diverse cell types, single cell gene expression and cellular architecture in native tumor tissues. This data was released to the GCR Genome Explorer website. **Results:** A total of 817 subjects have enrolled in the GCR including 598 diagnosed with GC and 219 at high-risk for GC through family history of GC and/or a CDH1 mutation. Of the 598 with GC, some met multiple criteria including at least 40 with a family history of GC, 10 with a CDH1 mutation, and 18 with both a family history of GC and a CDH1 mutation. The GCR-GE includes data from 257 gastric tumors in addition to external datasets from TCGA. In the latest data release, clinical and genomic data from 78 gastric tumors from patients in Latin America were integrated into the GCR-GE. **Conclusions:** The GCR is a comprehensive database of crucial clinical and genomic data necessary for driving forward GC research. Through global expansion, the GCR has generated a more representative dataset by increasing racial diversity and including underrepresented populations that are at higher risk for GC. Research Sponsor: Gastric Cancer Foundation.

492 **Poster Session**

Claudin 18.2 determination in locally advanced gastric and gastroesophageal adenocarcinoma treated with neoadjuvant chemotherapy. First Author: Lorenzo Gervaso, European Institute of Oncology, IEO IRCCS, Milan, Italy

Background: Gastric and esophago-gastric adenocarcinoma (GAC) represent a major unmet need. For tumors amenable with curative intent, prognosis remains poor with 5-year survival rates below 40%. Several strategies have been added to surgery including peri-operative chemotherapy which represents the standard of care for locally advanced GAC. Claudins is a family of transmembrane junctional proteins which acts in paracellular permeability, transport and communication. Claudin 18.2 (CLDN), a tissue-specific variant, is physiologically expressed in both gastric mucosa and gastric cancer. In metastatic GAC, CLDN positivity settles around 40% of cases and the addition of target therapy zolbetuximab to chemotherapy resulted in a significant improvement in overall survival. However, limited data are present about CLDN in early-stage disease. The aim of our study is to assess rates of expression of CDLN in locally advanced GAC treated with neoadjuvant chemotherapy and correlate it with clinical outcomes. **Methods:** We conducted a multi-omics prospective trial at IEO enrolling patients with locally advanced GAC treated with peri-operative chemotherapy and surgery. Pre-chemotherapy endoscopic samples were selected for CDLN determination. Positive cases were defined as moderate-to-strong membranous staining in $\geq$ 75% of tumor cells by IHC using the VENTANA CLDN18 (43-14A) Assay. CLDN expression was analyzed for correlation with pathologic complete response (pCR), tumor regression grade (TRG), event-free survival (EFS), and overall survival (OS). Survival was calculated using Kaplan–Meier method and log-rank tests. Statistical analyses were performed using SPSS software (version 29.0.2.0). **Results:** We included 48 patients, 67% male, median age 62 years (35-78). 31 tumors originated from gastric body, 14 from the junctional region and 4 from distal esophagus. Most frequent histotypes were diffuse (43%) and tubular (39%). Five cases (11%) presented microsatellite instability. All patients received neoadjuvant chemotherapy, mainly FLOT (92%) and underwent surgery, with 93% of R0 resection. Three patients (6%) achieved pCR and 18 had a TRG 1A or B. Of the total, 20 cases were defined as positive (42%). After a median follow-up of 20 months, 12 patients relapsed and 3 died. No significant association was observed for CDLN positivity and pCR or TRG1 rate, as well as for primary site or histotype. PFS and OS were not statistically different according to claudin status, even though numerically shorter in CDLN positive cases (PFS: 31 vs 33 months, p=.43; OS: 36 vs 39 months, p =.10). **Conclusions:** In our study of locally advanced GAC, rate of CDLN positivity was 40%, similarly to advanced stage. However, no differences in pCR rate, regression grade and survival outcomes have been observed. Larger studies and longer follow-up are required to assess the impact of CDLN in early-stage GAC. Research Sponsor: None.

493 **Poster Session**

The immunogenomic landscape of primary and metastatic gastroesophageal adenocarcinoma. First Author: Xin Wang, Department of Medical Oncology, University of Toronto, Sunnybrook HealthSciences Centre, Toronto, ON, Canada

Background: Metastatic gastroesophageal adenocarcinoma (GEA) is a heterogeneous disease with an overall poor prognosis. The tumour microenvironment (TME) between the primary and metastatic compartment is not well characterized. This study aims to describe the immunogenomic features of matched primary and metastatic GEA and their correlation with clinical outcomes. **Methods:** We performed whole exome sequencing (WES, n=36) and total RNA sequencing (RNA-seq, n=36) on a prospectively collected cohort of metastatic GEA, of which 14 had matched primary and metastatic tissues. We used TME deconvolution (Bagaev et al., 2021) comparing between the primary and metastatic compartments. Using an orthogonal technology, we further integrated immune checkpoint-directed multiplex immunohistochemistry (mIHC) for 22 patients with matched primary and metastatic samples. **Results:** The median age of our cohort was 61 (range 27-89), primarily male (n=28, 77%), and non-Asian (n=32, 89%). All had metastatic GEA at time of diagnosis. Differential expression showed activation of immune pathways such as tumor necrosis factor signalling and interferon-gamma response in the metastatic compartment. Deconvolution of the TME demonstrated 29% of primary tissues having an immune-inflamed TME compared to 61% of metastatic tissues (p=0.032). Exploring immune cell types, metastatic compartment is characterized by higher abundance of naïve and mature B cell populations ($p<0.001$), while primary compartment is enriched with CD4+ T cells ($p<0.01$). In matched cases, only 2 of 14 (14%) had concordant TME subtypes between primary and metastatic samples suggesting high degree of immune divergence. Furthermore, expression of common immune checkpoints aligned more with TME subtypes than within patient compartments suggesting common pathways of immune evasion. M-IHC showed both T-cell and immune checkpoint markers are enriched at the tumor margins compared to the tumor center in both primary and metastatic compartments (CD4; $p<0.0001$, CD8; $p<0.001$, CD68; $p<0.001$). There was a significant increase in the number of CD68+/CD163+/PDL1+ M2-like macrophages in the tumor margins ($p<0.0001$). Consistent with our deconvolution analysis, there are higher CD4+ T cells in the matched primary compared to metastatic samples. Long-term survivors (>16 months) had decreased number of CD68+ macrophages in the tumor center (p=0.019). **Conclusions:** Primary and metastatic GEA have divergent TME. This may partially explain varying responses to chemo-immunotherapy approaches. Therapies aimed at modifying TME may provide personalized treatment options. Research Sponsor: Physician Services Incorporated Foundation.

494 Poster Session

Molecular profiling of a gastroesophageal adenocarcinoma (GEA) multi-cohort using next-generation sequencing (NGS). First Author: Eduardo Terán-Brage Sr., Vall d'Hebron University Hospital, Barcelona, Spain

Background: The mutational landscape of GEA is rapidly evolving with the identification of promising molecular targets. However, histological subtypes are insufficient to capture the molecular heterogeneity in these patients (pts). We aim to assess the histopathological and molecular features in a GEA cohort using NGS techniques, and to evaluate the mutational frequency and its prognostic correlation. **Methods:** We conducted a retrospective study on GEA from 2019 to 2024. Immunohistochemistry (IHC) was used to evaluate HER2 expression, microsatellite instability (MSI), Epstein-Barr virus (EBV) and PD-L1 status. In-house and commercial NGS platforms were used to detect molecular alterations from tumor samples. Cox regression model was used to analyze overall survival (OS) based on the clinicopathological and molecular subgroups. **Results:** Using NGS techniques, we identified the following alterations with potential clinical relevance in the 115-pts included: pathogenic mutations in *PIK3CA* (5.2%), *BRCA1/2* (3.5%) and *POLD1* (0.9%), along with copy number alterations (CNA) in *ERBB2* (10.4%), *EGFR* (8.7%), *MET* (5.2%) and *FGFR2* (4.3%) genes, among others. Additionally, a high tumor mutational burden (≥ 13 Mut/Mb) was observed in 6 pts (5.2%). Clinical characteristics are detailed in the table. We detected a higher rate of alterations in *CDH1* (10.42% vs 4.48%; p=0.22) and *PIK3CA* (8.33% vs 2.99%; p=0.20) genes in early (<50y) vs late-onset (≥50y) GEA. Additionally, we identified a significantly higher mutation frequency in diffuse vs intestinal tumors; *ARID1A* (17.6% vs 2.2%; p=0.01) and *CDH1* (11.8% vs 0%; p=0.02) genes. Also, pts with peritoneal involvement showed a greater prevalence of mutations in *CDH1* (12% vs 0%; p=0.04) and *RHOA* (10% vs 0%; p=0.06) compared to those with liver disease. We observed a significantly worse OS in pts with diffuse vs intestinal tumors (15.7m vs 19.1m; p=0.04) and a trend toward shorter survival in pts with *CDKN2A* mutations (11.97m vs 17.83m; p=0.42). Although not significantly, CNA in *ERBB2* were associated with a better prognosis (22.2m vs 16.2m; p=0.09). **Conclusions:** Our results support the utility of NGS platforms in detecting novel molecular targets with prognostic and clinical impact, beyond *ERBB2*. Additionally, we observed poorer prognosis in diffuse subtype tumors, which exhibited a higher frequency of mutations in *ARID1A* and *CDH1* genes. Research Sponsor: None.

Clinicopathological characteristics of the 115-pts with GEA.

		N=115 (%)
Age	54 [18-83]	
Median years [range]		
Sex	Male	69 (60.0)
Tumor site	Gastric	72 (62.6)
	Gastroesophageal junction	37 (32.2)
	Esophagus	6 (5.2)
Histological subtype	Diffuse or pooly cohesive	51 (44.3)
	Intestinal or tubular	46 (40.0)
	Others	18 (15.7)
Stage at diagnosis	Advanced	74 (66.9)
	Localized	41 (33.1)
Metastatic site	Peritoneum	53 (46.1)
	Liver	34 (29.6)
Molecular profile (IHC)	HER2 (+)	20 (17.4)
	MSI	5 (4.3)
	EBV (+)	2 (1.8)
	PDL1 (CPS) ≥1	35 (30.4)

495 Poster Session

Exploration of novel biomarkers and novel therapies for immune checkpoint inhibitors (ICIs) in genetic variants of esophageal squamous cell carcinoma (ESCC): A study of 1,059 Japanese cases. First Author: Ryuichi Morita, Gastoenterology and Hepatology, Kyoto Prefectural University of Medicine, Kyoto, Japan

Background: In Japan, cancer genome profiling (CGP) tests will be covered by insurance in 2019, and more than 1,000 ESCC cases have been accumulated in the national database. ICIs are currently the key drugs for esophageal cancer, and MSI and TMB status can work as biomarkers for ICIs in solid tumors. Therefore, to improve the prognosis of esophageal squamous cell carcinoma (ESCC), we aim to search for gene variants that may predict the efficacy of ICIs and explore the possibility of new treatment from genetic data accumulated in the Japanese nation-wide C-CAT database. **Methods:** We retrospectively analyzed 1,059 ESCC cases from C-CAT data accumulated from June 2019 to February 2024. We used the registered clinical and genetic information to examine gene variants and carcinogenic pathways that could serve as biomarkers for ICIs. **Results:** Of the 1059, 152 (14.4%) had TMB-H and 3 (0.3%) had MSI-H, with a median TMB of 5.0 Muts/Mb (range: 0-268). *XPO1* and *TP53* were found to be positively and negatively correlated as genes affecting TTF in nivolumab monotherapy (FDR P value < 0.01). TMB has the potential to be a biomarker for ICI, but TMB was significantly higher in the group with variants in the *ARID* family (*ARID2*, *ARID1A* plus *ARID5B*) (median TMB 5.0 vs 8.0 Muts/Mb, P<0.001). In addition, 709 patients (66.9%) had any variants in three pathways (TP53, RTK and cell cycle), which are thought to be involved in the carcinogenesis of ESCC. PIK/Akt/mTOR and epigenetic modulation pathways were also found to be abnormal in many cases, and abnormalities in the epigenetic modification pathway were found to contribute significantly to TMB-H (FDR P value<0.01, Logarithmic value=19.2). **Conclusions:** This is the first case-specific report on the relationship between mutations in the *ARID* family and TMB in ESCC and the signalling pathways leading to carcinogenesis. In the present study, the association between the *ARID* family and TMB, and between abnormalities in the epigenetic modification pathway and TMB, which has been previously reported in other cancer types but not in esophageal cancer, suggests that multiple pathological mutations including these may be new biomarkers for ICI in 1059 Japanese patients. Furthermore, multiple pathways are often involved in ESCC carcinogenesis at the same time, and a separate, combined approach that takes these into account may be necessary in the development of new drugs. Research Sponsor: None.

496 Poster Session

Molecular characteristics of early-onset versus average-onset gastroesophageal adenocarcinoma. First Author: Amit Mahipal, University Hospitals Seidman Cancer Center, Case Western Reserve University, Cleveland, OH

Background: Incidence of early-onset gastroesophageal adenocarcinoma (eoGEA, ages <50 years) has been increasing in the United States since the 1990s, the etiology of which is still unclear, and limited data exists on molecular drivers. This study evaluates somatic and germline profiles in eoGEA compared to average-onset GEA (aoGEA, ages≥ 50 years). **Methods:** This is a retrospective, cross-sectional study utilizing data from de-identified records of GEA (esophageal, gastroesophageal junction and gastric adenocarcinomas) patients who underwent somatic NGS testing via the Tempus xT assay (595-648 gene DNA panel) from 12/2017 to 07/2024. This assay assesses somatic tumor mutations (including SNVs, Indels, CNVs, and select SVs), MSI, TMB, and incidentally detected germline SNVs and small indels in matched normal tissue. Clinical characteristics and immunotherapy markers were compared between eoGEA and aoGEA by Wilcoxon Rank Sum or Chi-squared test. Somatic and germline mutations were compared between two age groups with false discovery rate adjustments. **Results:** We compared the demographic, clinical, and molecular features of a total of 5863 patients with eoGEA (n=785, median age 43) and aoGEA (n=5078, median age 68). Over 80% of patients were diagnosed with stage IV disease. The eoGEA group had a higher proportion of females, non-white, and Hispanic or Latino patients compared to the aoGEA group (p<0.001). Prevalence of MSI-H/ dMMR, and TMB-H was lower in eoGEA vs aoGEA (p<0.001), while PD-L1 expression was similar. Prevalence of somatic *CDH1*, *CDKN2A*, and *TP53* SNVs, and *FGFR2* and *KRAS* CNAs were higher in eoGEA vs aoGEA (q<0.001). In a subset of 3286 patients with tumor/normal match testing (eoGEA=464 and aoGEA=2822), eoGEA also had a higher prevalence of germline CDH1 mutations (q<0.001). **Conclusions:** eoGEA has a unique somatic and germline mutation profile compared to aoGEA; further study evaluating the molecular landscape including epigenetic changes could shed light on underlying mechanisms responsible for the rising incidence of eoGEA. Research Sponsor: None.

	eoGEA (n=785)	aoGEA (n=5078)	p-value/q-value*
Baseline characteristics			
Median age at diagnosis (IQR)	43 (38, 47)	68 (61,74)	<0.001
Gender: male	509 (65%)	3798 (75%)	<0.001
Race (white)	280 (69%)	2347 (80%)	<0.001
Ethnicity (not hispanic or latino)	206 (60%)	1768 (86%	<0.001
Stage IV	495 (85%)	2883 (81%)	0.073
Immunologic markers			
TMB ≥ 10	25 (3.2%)	519(10%)	<0.001
MSI-H	13 (1.7%)	261 (5.1%)	<0.001
dMMR	7 (2.2%)	120 (5.9%)	0.007
Somatic mutations profiles *			
CDH1	126 (16%)	334 (6.6%)	<0.001
TP53	515 (66%)	3764 (74%)	<0.001
CDKN2A	101 (13%)	1011 (20%)	<0.001
KRAS	92 (12%)	931 (18%)	<0.001
NOTCH1	12 (1.5%)	228 (4.5%)	0.002
Germline mutation profiles *			
	eoGEA (N=464)	aoGEA (N= 2,822)	
Overall prevalence	46 (9.9%)	207 (7.3%)	
CDH1	10 (2.2%)	8 (0.3%)	0.001
TP53	3 (0.6%)	0 (0%)	0.039
BRCA2	2 (0.4%)	38 (1.3%)	0.6
BRIP1	4 (0.9%)	10 (0.4%)	0.6

497 Poster Session

Impact of Lauren's subtype in locally-advanced gastric cancer (LAGC) prognosis: A call for change. First Author: Tiago Cordeiro Felismino, A.C. Camargo Cancer Center, São Paulo, Brazil

Background: Patients with LAGC should undergo perioperative chemotherapy and surgery. Lauren's diffuse subtype and signet-ring cell carcinoma have been linked to worse outcomes and chemoresistance in this setting. This study aimed to assess the impact of Lauren's classification on response to neoadjuvant chemotherapy and prognosis in LAGC. **Methods:** This analysis involved adult patients from a prospectively collected gastric cancer database who had histologically confirmed LAGC and received curative-intent perioperative chemotherapy and surgery. Patients were classified into two groups according to Lauren's classification: intestinal and non-intestinal subtypes (including diffuse, signet-ring cell carcinoma and mixed histologies). Clinicopathological and treatment characteristics (age, gender, primary site, HER-2 status, staging laparoscopy, neoadjuvant chemotherapy, surgery and pathological complete response) of the two groups were compared using chi-square tests. Survival curves were obtained using Kaplan-Meier, and log-rank and Cox proportional models used to compare survival between groups. **Results:** Between 2015 and 2022, 174 patients met the inclusion criteria, with a median follow-up of 62.2 months (95% CI: 57.9-66.9). The intestinal subtype comprised 72 patients (41.4%) and the non-intestinal subtype included 102 (58.6%). The median age of diagnosis was significantly higher for the intestinal subtype (63 years) compared to the non-intestinal subtype (56 years; p<0.001). Female patients represented 26.4% of the intestinal subtype and 46.1% of the non-intestinal subtype (p=0.013). The primary gastric site was more prevalent in the non-intestinal subtype (78.4%) compared to the intestinal subtype (51.4%; p<0.001). HER-2 positivity was observed in 12.5% of intestinal and 4.5% of non-intestinal subtypes (p=0.14). Staging laparoscopy was conducted in 67 out of 72 (93%) patients with the intestinal subtype and in 96 out of 102 (94.1%) patients with the non-intestinal subtype, p=1. FLOT and FOLFOX/CAPOX protocols were administered in 50% and 40.2% of intestinal subtype cases, and in 53.9% and 36.2% of non-intestinal subtype cases, respectively (p=0.14). Pathological complete response rates were 8.5% for intestinal and 11.5% for non-intestinal subtypes (p=0.46). Three-year recurrence-free survival rates were 67.1% for intestinal and 64.6% for non-intestinal subtypes (HR=0.99, 95% CI: 0.61-1.61, p=0.96), while five-year overall survival rates were 73.3% and 67.2%, respectively (HR=1.12, 95% CI: 0.63-1.97, p=0.70). **Conclusions:** Our findings demonstrate that the non-intestinal subtype is more common in younger, female patients with primary gastric sites. However, Lauren's subtype did not significantly correlate with pathological staging post-neoadjuvant chemotherapy or prognosis in LAGC. Lauren's subtype should not influence treatment decisions in LAGC. Research Sponsor: None.

498 Poster Session

Surrogacy between pathological complete response and overall survival: An individual patient data analysis of eight RCTs on neoadjuvant treatment plus surgery for esophageal cancer. First Author: Jun Okui, Department of Surgery, Keio University School of Medicine, Tokyo, Japan

Background: Overall survival (OS) is the gold standard endpoint of treatment efficacy but requires an extended follow-up period. This study aimed to determine the validity of pathological complete response (pCR) as a surrogate endpoint for OS. **Methods:** The individual-level surrogacy between categorical outcome and OS was assessed using Kendall correlation coefficient, τ. Because no method has been reported to estimate τ between categorical variables and OS, we proposed the new method using an inverse probability of censoring weighting (IPCW) estimator adjusted for tied data that occur when applied to categorical outcomes. To evaluate the validity of this new method, we conducted simulations to compare its estimation accuracy with existing methods. A systematic review for all phase III RCTs comparing therapies in perioperative settings for resectable advanced esophageal and gastroesophageal junction cancer was performed and individual patient data (IPD) were requested from all included trials. Finally, surrogacy between pCR and OS was assessed using our proposed method. **Results:** Across all scenarios with varying τ values, sample sizes, and censoring proportions, the proposed method showed the lowest bias compared to the existing statistical methods. In total, 23 eligible trials were identified and IPD were available from eight RCTs (JCOG1109, JCOG9907, JCOG9204, FFCD9901, FFCD9102, SAKK75/08, CROSS, and KOK), including 1688 patients who received neoadjuvant therapy. For trial-level analysis, pCR showed a strong correlation with a determination of correlation of 0.65 when limited to neoadjuvant chemotherapy (NAC) trials. For individual-level analysis, in patients who received NAC, the hazard ratio (HR) for OS of pCR was 0.25 (95% CI: 0.13 - 0.47), and τ between OS and pCR was 0.225. While in patients who received neoadjuvant chemoradiotherapy (NACRT), the HR was 0.54 (95% CI: 0.45 - 0.66), and τ was 0.189. Additionally, τ between OS and pathological grade (pGrade), OS and tumor regression grade (TRG) was 0.196 and 0.143, respectively. As a reference, hypothetical data showed that in order to achieve a τ of 0.8 between pCR and OS, it is necessary to be able to clearly identify the prognosis to the extent that an HR of 0.09 (95% CI: 0.08 - 0.11) can be obtained. **Conclusions:** While pCR is sensitive enough to stratify prognosis and exhibit strong trial-level surrogacy, no individual-level surrogacy was demonstrated, making them insufficient to replace OS. It should be noted that there are substantial differences in prognostic stratification and potential use of pCR as primary endpoints in clinical trials. Research Sponsor: None.

499 Poster Session

Development and validation of a prognostic model for advanced gastric and oesophageal carcinoma (AGOC) using individual patient data from two clinical trials. First Author: Sayeda Kamrun Naher, NHMRC Clinical Trials Centre, University of Sydney, Camperdown, NSW, Australia

Background: Estimating survival in advanced gastric and oesophageal carcinoma (AGOC) remains a significant challenge. We developed a prognostic model incorporating readily accessible patient and clinical data and specific patient-reported outcomes (PROs), utilizing individual participant data from two randomized trials. **Methods:** We used data from 2 trials comparing regorafenib to placebo; AGITG INTEGRATE IIa (n=251) for model development and AGITG INTEGRATE (n=152) for validation. Both trials enrolled metastatic or AGOC following failure of 1 or more prior lines of systemic therapy. Candidate variables were chosen from systematic literature review and expert consultation. Significant prognostic factors for inclusion in the multivariable model were identified using univariable Cox proportional hazards models with p-value threshold of 0.1. Multivariable models were developed via Lasso regression. An initial model was created using clinical variables alone, followed by the integration of PROs. C statistics were used to discriminate the model's efficiency. **Results:** Univariable analysis identified 9 clinical variables and 4 PRO domains as significant, including body mass index (BMI)(p=0.08), ECOG PS (0.02), extent of cancer ($p<0.001$), liver involvement (p=0.04), treatment with regorafenib (p=0.005), neutrophil-lymphocyte ratio (NLR) ($p<0.001$), LDH ($p<0.001$), albumin ($p<0.001$), and CA 19-9 (p=0.007). Significant baseline PROs included appetite loss ($p<0.001$), constipation ($p<0.001$), fatigue ($p<0.001$), and pain ($p<0.001$). Gastrectomy violated the Cox model proportional hazards assumption necessitating that the analysis be stratification by gastrectomy. The primary model (M1) incorporated region (Asia vs non-Asia, p=0.25, was included because it was significant in previous analysis (INTEGRATE), ECOG PS status, cancer extent, treatment with regorafenib, NLR, BMI, LDH, CA 19-9, and albumin. The final model (M2), which included PROs, demonstrated superior discriminative ability with the highest c-statistic values in both the gastrectomy and non-gastrectomy strata. The c-statistic increased from 0.695 for M1 to 0.723 for M2, indicating the added prognostic value of including PROs. **Conclusions:** The prognostic model that integrated both clinical data and PROs, proved robust for predicting survival in AGOC. The addition of PROs significantly enhanced prognostic accuracy, highlighting their importance in survival models. Further validation and refinement of the model is ongoing. Research Sponsor: NHMRC CTC University of Sydney.

500 Poster Session

Assessment of clinical value and barriers to use of new investigational targeted agents in the community setting: A study of claudin 18.2 and zolbetuximab. First Author: Griffin M Wright, Aptitude Health, Atlanta, GA

Background: Rapid integration of novel agents in clinical practice can be lifesaving. However, community-based practices may experience extended adoption times, due to practical barriers. REFLECT was developed to assess new drug implementation in the community setting. **Methods:** REFLECT consists of a 20-minute prerecorded presentation, followed by a data-collection event for community oncologists. It was developed to evaluate and assess challenges of implementing novel targeted agents in the community practice setting. In this pilot study, awareness of zolbetuximab (zolbe), a potential first-in-class claudin 18.2 (CLDN18.2) monoclonal antibody and companion diagnostic, was assessed. Between April and August 2024, 53 data-collection events were held representing 26 states across the US. Data from 492 oncologists were recorded and analyzed. **Results:** Nearly half the community oncologists (47%) were not familiar with CLDN18.2 as a therapeutic target in gastric/gastroesophageal junction (G/GEJ) cancer. Awareness of zolbe data was low among community oncologists; 79% indicated they had not seen the results from the phase 3 studies. However, 96% of oncologists reported the data will at least moderately impact their treatment of CLDN18.2+ patients with metastatic (m)G/GEJ cancer. Only 40% of oncologists reported having access to CLDN18.2 testing. Uncertainty regarding availability of CLDN18.2 testing was high among oncologists in the Central US (52%), representing a significant awareness gap for CLDN18.2 testing. Opinions were divided regarding the treatment approach for CLDN18.2+, immunosensitive patients (Table). In a HER2−, CLDN18.2+ patient with mG/GEJ cancer and a combined positive score (CPS) of 10; 58.1% would recommend first-line chemotherapy (chemo) plus immunotherapy (IO) and 40.4% would recommend chemo plus zolbe. **Conclusions:** First-in-class targeted agents often encounter additional challenges and knowledge gaps regarding adoption in the community practice setting, particularly associated with molecular testing. Timely education on patient identification and correct utilization may accelerate adoption of new regimens, and potentially save lives. Data-sharing and -collection platforms including REFLECT can be instrumental in the identification of challenges associated with widespread and rapid implementation of novel agents in the community oncology setting. Research Sponsor: None.

What treatment approach would you recommend in a patient with the following characteristics, if all treatments were available? mGEJ adenocarcinoma, HER2− disease, CLDN18.2+ staining in >75% of tumor cells, CPS 10.

Treatment		Total: N = 492 % (n)
Chemo	**mFOLFOX6**	**1.0 (5)**
	CAPOX	**0 (0)**
Chemo + IO	**mFOLFOX6 + IO**	**52.8 (260)**
	CAPOX + IO	**5.3 (26)**
Chemo + zolbe	***mFOLFOX6 + zolbe***	***33.9 (167)***
	CAPOX + *zolbe*	***6.5 (32)***
	Other	**0.4 (2)**

501 Poster Session

Influence of data review on community physicians' treatment preferences in frontline esophageal cancer (EC). First Author: Robert Bone, Cardinal Health, Dublin, OH

Background: EC remains a global health challenge with a rising incidence and poor prognosis despite advancements in treatment modalities. Standard therapies include surgery, chemotherapy (CT), radiation therapy, and recently, targeted and immunotherapy. The PD-L1 inhibitors pembrolizumab (pembro) and nivolumab (nivo) received FDA approval for frontline treatment in EC in March and May 2021, respectively, based in part on the KEYNOTE-590 (KN590) and CheckMate 648 trials. In this study, we aimed to uncover rationale in physician preference for PD-L1 inhibitors in EC. **Methods:** US-based oncologists convened at two in-person meetings in April 2024 to discuss clinical and real-world data presented at the 2024 ASCO GI Cancers Symposium. Among data discussed included the KN590 5-year update. Survey questions were fielded at the meeting to capture participants' impressions. Demographics were captured via an online survey ahead of the meeting. Responses from participants who manage EC and responded to all survey questions were aggregated. **Results:** 86 participants qualified; and, collectively, are 76% community physicians with 18 years average in clinical practice, see 18 patients median on clinic days, and spend 83% of their time in direct patient care on average. ECOG PS status (50%), tumor histology (52%), and location/stage (48%) were reported as critical factors in treatment decision-making; only 14% said PD-L1 status was a major factor. Participants were queried on a hypothetical patient case of esophageal squamous cell and adenocarcinoma before and after reviewing the KN590 data. Prior to reviewing KN590, physician preferences in the squamous cell 1L setting were split between nivo+CT and pembro+CT; following data review, pembro+CT was favored nearly 3:1. In the adenocarcinoma 1L setting, preference shifted from 60/40 nivo+CT to 60/40 pembro+CT, see Table. The cohort were most impressed with the overall survival (87%) and objective response rates (40%), and 55% would consider pembro+CT for both squamous cell and adenocarcinoma disease while 37% would consider it for PD-L1 CPS>10 squamous cell disease only. **Conclusions:** Our study showed PD-L1 status was not a major factor in clinical decisions. The shift in treatment preference toward pembro+CT for both patient cases following KN590 data review suggests an impact for data review to influence physician behavior. This highlights the importance of educational initiatives to disseminate clinical updates to providers. It is still to be determined the optimal platform and length for these initiatives, and to quantify their effectiveness. Research Sponsor: None.

65-year-old male patient with advanced EC; PD-L1 CPS ≥10; ECOG PS of 1	Squamous cell disease		Adenocarcinoma	
	Before	After	Before	After
Fluoropyrimidine (FP) + platinum (P)-based CT	1%	1%	1%	0%
FP +P-based CT + nivo	55%	27%	60%	36%
FP + P-based CT + pembro	43%	71%	38%	63%
Nivo + ipilimumab	1%	1%	0%	1%
None of the above	0%	0%	0%	0%

502 Poster Session

Impact of inpatient palliative care consultation on end-of-life care among patients with esophageal cancer. First Author: Suriya Baskar, Brooklyn Hospital Center, Brooklyn, NY

Background: Esophageal cancer is an aggressive malignancy with a high global prevalence. Patients often suffer from various debilitating symptoms like dysphagia and cachexia, particularly with more advanced stage disease. Most patients continue to receive aggressive interventions until terminal stages of disease without timely initiation of formal palliative care (PC). We sought to investigate the impact of formal inpatient palliative care consultation on end-of-life (EOL) care among patients with esophageal cancer. **Methods:** The National Inpatient Sample (NIS) was queried to identify all hospitalizations with esophageal cancer utilizing ICD-10 codes C15.x from 2016 to 2020. Hospitalizations with documented inpatient mortality events were extracted and stratified based on reception of PC consultation (ICD-10 code Z51.5). Demographic and clinical data were analyzed using chi-squared tests and independent sample t-tests. A threshold of p-value less than 0.05 was set to determine statistical significance. **Results:** A total of 17,745 patients with esophageal cancer were included, of which 10,370 (58.4%) received PC consultation. Patients that received PC consultation at EOL were more likely to have higher comorbidities per CCI. No significant difference between age and sex was observed between the two cohorts. Patients who received PC consultations at EOL had higher rates of Do Not Resuscitate (DNR) code status (78.1% vs 43.2%, p $<$0.001) and lower rates of aggressive interventions such as chemotherapy (0.9% vs 1.6%, p$<$.001), blood transfusions (12.3% vs 18%, p$<$.001), and mechanical ventilation (28.5% vs 41%, p$<$0.001). No significant difference was observed in the rates of vasopressor use between the two groups (p=0.83). Hospitalizations with PC consultations were associated with significantly lower total hospitalization cost ($97,879 vs $146,128, p$<$.001). **Conclusions:** Inpatient PC consultations among patients with esophageal cancer were associated with lower rates of aggressive interventions and higher rates of DNR code status in addition to significantly lower total hospitalization costs. These findings underscore the importance of inpatient PC consultation at the EOL in helping minimize aggressive interpretation and facilitate a greater focus on comfort and quality of life. Research Sponsor: None.

	No Palliative N=7375	Palliative N=10370	p-value
Age (in years)	67.58 ± 10.9	67.58 ± 10.9	.98
CCI	9.1 ± 3.5	9.4 ± 3.3	**<.001**
Mean length of stay (days)	8.9 ± 14.9	7.5 ± 11.3	**<.001**
Total Charges	146128 ± 321830	97879 ± 195868	**<.001**
DNR	3170 (43.2%)	8100 (78.1%)	**<.001**
Blood transfusion	1320 (18.0%)	1280 (12.3%)	**<.001**
Mechanical ventilation	3015 (41.0%)	2960 (28.5%)	**<.001**
Vasopressor	585 (8.0%)	835 (8.1%)	0.83
Chemotherapy	115 (1.6%)	95 (0.9%)	**<.001**

503 Poster Session

The impact of protein-energy malnutrition on clinical outcomes in hospitalized gastric cancer patients: A population-based analysis. First Author: Rabia Iqbal, Brooklyn Hospital Center, Brooklyn, NY

Background: Protein-energy malnutrition (PEM) is considered an adverse prognostic factor in various clinical conditions. Patients with gastric cancer are at an increased risk of malnutrition because of factors such as loss of appetite due to the tumor, difficulty swallowing, malabsorption, and metabolic changes caused by the cancer. However, the implications of protein-energy malnutrition (PEM) in gastric cancer patients are not well studied. Through this study, we wanted to examine how PEM affects clinical outcomes in hospitalized patients with gastric cancer. **Methods:** We used the National Inpatient Sample (NIS) from 2018 to 2020 to detect patients with a primary diagnosis of gastric cancer and a secondary diagnosis of protein energy malnutrition (PEM). We used the chi-square and student T-test to analyze categorical and continuous variables. Confounders were adjusted using multivariable regression analysis, and the adjusted odds ratio (aOR) was calculated. **Results:** We detected a total of 1673915 patients with the primary diagnosis of gastric cancer using ICD-10 codes. Of them, 401,739 (24%) of the patients had a secondary diagnosis of PEM. The mean age of the patients was 66 years (p-value $<$0.01). Of these, 56% of the patients were male, while the remaining 44% were female (p-value $<$0.001). Based on race, 68% of the patients belonged to the White Race, 12% were African Americans, and 11% were Hispanics (p-value $<$0.001). It was observed that the presence of PEM was associated with higher mortality (aOR 1.6, p-value $<$0.001). There was also higher hospitalization cost ($109522 vs. 75725, p-value $<$0.001) and mean length of stay (LOS) (9.1 days vs. 5.6 days, p-value $<$0.001). PEM in gastric cancer patients was also associated with higher odds of sepsis (aOR 1.6, p-value $<$0.001), gastritis (aOR aOR 1.2, p-value $<$0.001), gastrointestinal (GI) bleeding (aOR 1.1, p-value $<$0.001) and acute kidney injury (AKI) (aOR 1.7, p-value $<$0.001). **Conclusions:** Our study demonstrates that protein-energy malnutrition (PEM) is associated with increased mortality, higher hospitalization costs, and longer stays in hospitalized gastric cancer patients. These findings emphasize the need for regular nutritional assessment and management to improve clinical outcomes in this population. Research Sponsor: None.

Comparison of outcomes in hospitalized gastric cancer patients with and without protein-energy malnutrition (PEM).

Hospital Outcomes	Gastric Cancer Patients with PEM	Gastric Cancer Patients without PEM	Adjusted Odds ratio (aOR)	95% Confidence Interval	P-value
Mortality	9.1% (n=36999)	5.3% (n=68100)	1.6	1.5-1.7	$<$0.001
Length of Stay	9.1	5.6	3.0	2.7-3.3	$<$0.001
Sepsis	10% (n=41915)	6.4% (n= 82035)	1.6	1.5-1.6	$<$0.001
Gastritis	3.9% (n=16174)	3.3% (n=41879)	1.2	1.1-1.31	$<$0.001
GI Bleeding	7.1% (n= 28949)	6% (n= 28949)	1.1	1.0-1.2	$<$0.001
AKI	28% (n=117439)	19% (n=1268134)	1.7	1.7-1.8	$<$0.001

TPS504 Trials in Progress Poster Session

A randomized phase II clinical trial evaluating the safety and efficacy of initial dose adjustment of zolbetuximab plus chemotherapy in patients with HER2-negative and CLDN18.2-positive unresectable advanced or recurrent gastric, gastroesophageal junction cancer or esophageal adenocarcinoma (GENTLE-Z). First Author: Mashiro Okunaka, Department of Pharmacy, National Cancer Center Hospital East, Kashiwa, Japan

Background: Zolbetuximab, a monoclonal antibody targeting CLDN18.2, was recently approved in Japan for patients with HER2-negative/CLDN18.2-positive metastatic gastric or gastroesophageal junction cancer (mGC/GEJC). On-target GI toxicities such as nausea and vomiting are commonly observed during first infusion, occasionally leading to early treatment discontinuation. Effective management of these toxicities is essential for the successful use of zolbetuximab in clinical practice. This study aims to assess whether omitting the loading dose in the first cycle can reduce zolbetuximab-related GI toxicities without compromise efficacy**. Methods:** GENTLE-Z is a prospective, open-label, multi-intuitional, randomized phase II trial designed to compare the safety and efficacy of fixed-dose of zolbetuximab without loading dose vs. the standard zolbetuximab dose in combination with first-line mFOLFOX6 or CAPOX for patients with HER2-negative/CLDN18.2-positive mGC/GEJC. Patients are randomly assigned (1:1) to receive either a fixed dose of zolbetuximab (400 mg/m^2 q2w or 600 mg/m^2 q3w) or standard zolbetuximab regimen (800 mg/m^2 loading dose followed by 400 mg/m^2 q2w or 600 mg/m^2 q3w) with chemotherapy. The primary endpoint is the vomit-free rate during cycle 1, and secondary endpoints include efficacy outcomes such as objective response rate, progression-free survival and overall survival. The sample size is set based on the hypothesis that omitting loading dose will improve the vomit-free rate from 50% to 75%. The planned sample size is 86 patients, with a one-sided alpha error of 5% and power of 75%, over an estimated enrollment period of 1.5 years. Enrollment has been ongoing since September 2024 across 55 sites in Japan. This trial is registered in Japan Registry of Clinical Trials (jRCTs031240347). Clinical trial information: jRCTs031240347. Research Sponsor: None.

TPS505 Trials in Progress Poster Session

Phase 2 trial of defactinib in combination with avutometinib in patients with advanced diffuse-type gastric cancer. First Author: Ryan H. Moy, Columbia University Irving Medical Center – Herbet Irving Pavillion, New York, NY

Background: Diffuse-type gastric cancer (DGC) accounts for approximately one third of gastric cancer diagnoses. DGC is characterized by poor differentiation, discohesive growth, frequent peritoneal spread, chemoresistance, and inferior survival compared to intestinal-type gastric cancer. Moreover, DGC is commonly genomically stable and often lacks clinically actionable targets such as *ERBB2* amplification. Alterations enriched in DGC including loss-of-function *CDH1* mutations and gain-of-function *RHOA* mutations can facilitate increased focal adhesion kinase (FAK) signaling that drives tumor proliferation, invasiveness, and metastasis. Recent preclinical studies demonstrated that the FAK inhibitor defactinib can inhibitor DGC growth *in vitro* and *in vivo*. FAK inhibitor monotherapy leads to compensatory mitogen-activated protein kinase (MAPK) pathway upregulation, which can be overcome with combined FAK and MEK inhibition. In preclinical DGC xenograft models, defactinib plus the RAF/MEK clamp avutometinib led to deep tumor regressions, suggesting synergistic activity of the combination. The goal of this study is to investigate the efficacy and tolerability of avutometinib plus defactinib in patients with advanced DGC refractory to standard therapy. **Methods:** This investigator-initiated, multicenter phase 2 trial will enroll 27 patients with metastatic/unresectable gastric or gastroesophageal junction carcinoma classified as diffuse, poorly cohesive, signet ring cell, or mixed type with progression on at least one line of therapy including platinum/fluorouracil chemotherapy. Patients with known pathogenic *CDH1* and/or *RHOA* alterations regardless of histology are also eligible. Patients must have ECOG PS 0/1 and can have either measurable or evaluable non-measurable disease per RECIST 1.1 criteria. Avutometinib is given 3.2 mg orally twice weekly and defactinib is given 200 mg orally twice a day, both for 3 weeks on/1 week off for each 28-day cycle. The primary endpoint is 6-month PFS rate, with secondary endpoints including ORR, OS, DOR, and tolerability. Exploratory studies will interrogate FAK/MAPK pathway activation, ctDNA dynamics, and the impact of the combination of avutometinib/defactinib on the tumor microenvironment from pre- and on-treatment biopsies. The trial was opened for enrollment in August 2024. Clinical trial information: NCT06487221. Research Sponsor: Gateway for Cancer Research.

TPS506 Trials in Progress Poster Session

Neoadjuvant cadonilimab plus anlotinib followed by surgery for locally advanced esophageal squamous cell carcinoma: A single arm, phase 2 trial. First Author: Wenhan Weng, Peking University People's Hospital, Beijing, China

Background: Esophageal cancer (EC) is the seventh most common cancer in China, where most of patients have locally advanced esophageal squamous cell carcinoma (LA-ESCC) at the time of diagnosis. Neoadjuvant chemotherapy (nCT) or chemoradiotherapy (nCRT) followed by surgery is standard treatment strategy for LA-ESCC in China. Immune checkpoint inhibitors (ICIs) have revolutionized the treatment of ESCC. Programmed death 1 (PD-1) inhibitors are the most commonly used ICIs in treatment of ESCC. Lately, the phase 3 ESCORT-NEO/NCCES01 trial demonstrates that neoadjuvant PD-1 inhibitor plus chemotherapy obtains superior pCR rates compared to chemotherapy alone for LA-ESCC. Except for PD-1 inhibitors, cytotoxic T lymphocyte-associated antigen-4 (CTLA-4) inhibitor is another effective ICI for ESCC, especially combined with PD-1 inhibitor according to the CHECKMATE-648 study. Although neoadjuvant immunotherapy plus chemotherapy has brought earth-shaking changes to the treatment of LA-ESCC, the chemo-based treatment modality still results in a high incidence of adverse events, reduces the quality of life and compliance of patients. Cadonilimab is a dual immune antibody drug comprised of PD-1 antibody and CTLA-4 antibody. Anlotinib is an oral, small molecule anti-angiogenic targeted agent that has been approved for use in advanced or metastatic ESCC in China. Therefore, we launched this trial to assess the efficacy and safety of cadonilimab plus anlotinib, a novel "chemo-free" strategy, followed by surgery for LA-ESCC. **Methods:** Study design: This is a single-arm, phase 2 study (NCT06426797). The primary endpoint of this study is the pCR rate. The secondary endpoints include safety, major pathological response (MPR) rate, objective response rate (ORR), R0 resection rate, and event-free survival (EFS). Study procedures: Cadonilimab will be intravenously administered at day 1 with a dose of 10 mg/Kg every three weeks for three cycles. Anlotinib will be orally administered daily at day 1 to day 14 with a dose of 12mg every three weeks for three cycles. In the end, the subjects will receive subtotal esophagectomy 4 to 6 weeks later after the last dose of cadonilimab. Major inclusion criteria: 1. Age $>$ 18 years. 2. Histopathologically confirmed ESCC: c-stage II–IVA (8th ed. AJCC). 3. Enough tissue for PD-L1 IHC test. 4. Eligible for anlotinib treatment. 5. ECOG performance status $\leq$ 2 points. Statistics: We hypothesized that the expected pCR rate would be 50%. 24 subjects are needed at an α-level of 0.05 (one-sided) and power of 80%. So, 25 subjects in total should be enrolled. The descriptive statistical method will be applied to the presentation of baseline characteristics, toxicities and measures of effectiveness. The Kaplan–Meier method will be used to calculate survival rates. Current enrollment: The study started on August 1st, 2024, and 3 of planned 25 patients have been enrolled. Clinical trial information: NCT06426797. Research Sponsor: Akeso Pharmaceuticals, Inc.

TPS507 Trials in Progress Poster Session

CLARITY-Gastric 01: A randomized phase 3 study of AZD0901, a Claudin18.2 (CLDN18.2)-targeted antibody-drug conjugate, in second- or later-line (2L+) advanced gastric or gastroesophageal junction cancer (GC/GEJC). First Author: Yelena Y. Janjigian, Memorial Sloan Kettering Cancer Center, New York, NY

Background: GC/GEJC is the fifth most commonly diagnosed cancer and the fourth leading cause of cancer-related death worldwide (Sung H, et al. *CA Cancer J Clin* 2021). Prognosis is poor in GC/GEJC; median overall survival (OS) is less than 10 months in the 2L setting and less than 6 months in the 3L+ setting (Wilke H, et al. *Lancet Oncol* 2014; Shitara K, et al. *Lancet Oncol* 2018). New treatment options are needed. CLDN18.2 has emerged as a new therapeutic target in GC/GEJC. AZD0901, a potential first-in-class CLDN18.2-targeted antibody conjugated to monomethyl auristatin E via a protease cleavable linker, has demonstrated clinical activity in a Phase 1 study in advanced GC/GEJC (Xu R-H, et al, *J Clin Oncol* 2023). CLARITY-Gastric 01 (NCT06346392) is a randomized, open-label, sponsor-blinded, global Phase 3 study that will assess the efficacy and safety of AZD0901 versus investigator's (INV) choice of therapy after $\geq$1 prior therapy for advanced GC/GEJC expressing CLDN18.2. **Methods:** Participants (pts) will be randomized 1:1:1 to AZD0901 dose level 1 intravenous (IV) every three weeks (Q3W; Arm 1), AZD0901 dose level 2 IV Q3W (Arm 2), or INV choice of therapy (2L: ramucirumab [ram] + paclitaxel [PTX], PTX, or docetaxel [for pts with contraindication to ram only]; 3L+: irinotecan, TAS-102 [excluding China], or apatinib [China only]; Arm 3). One AZD0901 arm (Arm 1 or Arm 2) will stop enrollment after a pre-planned dose selection decision. Randomization will continue in a 1:1 ratio to the two remaining arms (selected AZD0901 dose arm and Arm 3). Eligible pts must have histologically confirmed, unresectable, locally advanced or metastatic GC/GEJC (non-HER2+, CLDN18.2 expression). Pts must have disease progression on/after $\geq$1 prior regimen (including platinum-fluoropyrimidine) and have an ECOG performance status of 0 or 1. Pts previously treated with CLDN18.2-targeting therapy (other than naked monoclonal antibody) are excluded. Recruitment has begun and pts are planned to be enrolled across 16 countries in Asia, Europe, and North America. Dual primary endpoints are progression-free survival (PFS; intent-to-treat [ITT] pts) and OS (3L+ pts). The secondary endpoints include OS (ITT pts), PFS (3L+ pts), objective response rate and duration of response (ITT and 3L+ pts), and safety/tolerability. Efficacy data will be summarized and analyzed in the ITT population. All efficacy endpoints will assess the selected AZD0901 dose arm versus Arm 3. Safety and tolerability will be assessed in all pts who receive $\geq$1 dose of study treatment. Previously presented at European Society for Medical Oncology - Gastrointestinal Cancers Congress 2024, Final Publication Number: 495TiP, Kohei Shitara et al. Reused with permission. Clinical trial information: NCT06346392. Research Sponsor: AstraZeneca.

TPS508 Trials in Progress Poster Session

Valemetostat and trastuzumab deruxtecan (T-DXd) in previously treated advanced or metastatic human epidermal growth factor receptor 2 (HER2)-positive gastric cancer or gastro-esophageal junction (GEJ) adenocarcinoma. First Author: Kohei Shitara, National Cancer Center Hospital East, Kashiwa, Japan

Background: Valemetostat tosylate (valemetostat), a novel and selective dual inhibitor of enhancer of zeste homolog (EZH)2 and EZH1, has demonstrated clinical efficacy and tolerability in multiple hematologic cancers (200 mg orally [PO] once daily [QD]). EZH2 inhibition by valemetostat is expected to upregulate DNA/RNA helicase Schlafen 11 expression and sensitize tumor cells to DNA damaging agents such as topoisomerase I inhibitor antibody–drug conjugates (ADCs). Topoisomerase inhibitors cause DNA strand breaks, which may synergize with valemetostat. HER2 gene amplification or protein expression has been associated with gastric cancer (GC) and GEJ adenocarcinoma. Patients with HER2+ locally advanced or metastatic gastric adenocarcinoma who progress after first line treatment have poor prognosis and limited therapeutic options. T-DXd is approved globally for HER2+ locally advanced or metastatic GC previously treated with a trastuzumab-based regimen. This study evaluates the clinical activity and safety of valemetostat in combination with T-DXd in patients with GC or GEJ adenocarcinoma as part of a Master Protocol trial evaluating valemetostat in combination with ADCs in solid tumors. **Methods:** This global, phase 1b, multicenter, open-label Master Protocol study (NCT06244485) will enroll ~ 70 patients per sub-protocol in multiple regions, including the US and Japan. The gastric cohort enrolls patients with previously treated advanced or metastatic HER2+ GC or GEJ adenocarcinoma. The dose-escalation (Part 1) will determine the recommended dose for expansion (RDE), combining valemetostat 50–200 mg PO QD with T-DXd 5.4 or 6.4 mg/kg intravenously every 3 weeks. The dose-expansion (Part 2) will assess the efficacy of valemetostat + T-DXd at the RDE. Primary endpoints include safety and tolerability in Part 1 and the objective response rate in Part 2. Secondary endpoints include response duration, progression-free and overall survival, pharmacokinetics, and HER2 expression by immunohistochemistry and in situ hybridization with clinical response. An interim futility analysis will be performed when 20 patients are enrolled with $\geq$ 6 months of follow-up. Clinical trial information: NCT06244485. Research Sponsor: Daiichi Sankyo Inc.

TPS509 Trials in Progress Poster Session

Beamion BCGC-1: A phase Ib/II trial of the HER2-selective tyrosine kinase inhibitor (TKI) zongertinib (BI 1810631) + trastuzumab deruxtecan (T-DXd) or trastuzumab emtansine (T-DM1) for patients with metastatic breast cancer (mBC) and metastatic gastric, gastroesophageal junction, or esophageal adenocarcinoma (mGEAC). First Author: Izuma Nakayama, National Cancer Center Hospital East, Kashiwa, Chiba, Japan

Background: Approximately 15% of mGEAC tumors overexpress HER2. Currently, the HER2-targeted antibody-drug conjugate (ADC), T-DXd, is approved in mGEAC as a second-line or later-line treatment option following the failure of prior trastuzumab-containing therapy. Zongertinib (BI 1810631) is a novel, orally administered HER2-selective TKI that spares EGFR. In a Phase I trial (NCT04886804), zongertinib showed encouraging tolerability and efficacy in patients with HER2 aberration-positive advanced solid tumors, including mGEAC. Beamion BCGC-1 (NCT06324357) is a Phase Ib/II dose escalation/optimization trial investigating zongertinib plus T-DXd or T-DM1 in patients with HER2+ mGEAC or mBC. Here we focus on the mGEAC cohorts, in which patients will receive zongertinib in combination with T-DXd. **Methods:** Across the whole trial, approximately 240 patients will be enrolled from approximately 48 sites in 6 countries. In the mGEAC cohorts, approximately 80 patients with histologically/cytologically confirmed locally advanced/metastatic, unresectable, HER2-positive, pre-treated mGEAC will be enrolled. All patients must have investigator-assessed progression and documented HER2-positive disease (overexpressed and/or amplified; confirmed by immunohistochemistry and *in situ* hybridization). In Phase Ib, approximately 20 patients will receive escalating doses of zongertinib plus a fixed approved dose of T-DXd (6.4 mg/kg, once every 21 days). Dose escalation will be guided by a Bayesian Logistic Regression Model, with overdose control. In Phase Ib, the primary endpoint is the occurrence of dose-limiting toxicities (DLTs) during the maximum tolerated dose evaluation period (first 21-day cycle); secondary endpoints include objective response (RECIST v1.1), DLTs during the on-treatment period and pharmacokinetics. In Phase II, up to 60 patients with mGEAC will be randomized 1:1 to receive one of two zongertinib dose levels (dose levels informed by Phase Ib), plus a fixed approved dose of T-DXd. In Phase II, the primary endpoint is objective response; secondary endpoints include progression-free survival, disease control, treatment-emergent adverse events leading to dose reduction, pharmacokinetics and patient-reported outcomes. Patients will remain on treatment until disease progression, undue toxicity or any other protocol-defined stopping criterion. Enrollment is ongoing. Clinical trial information: NCT06324357. Research Sponsor: Boehringer Ingelheim.

TPS510 Trials in Progress Poster Session

A safety trial of antibiotic fecal microbiota transplantation for esophageal and gastric cancer patients treated with immune checkpoint inhibitors (Biorich2 trial). First Author: Yuri Yoshinami, Department of Gastrointestinal Medical Oncology, National Cancer Center Hospital, Tokyo, Japan

Background: The development of immune checkpoint inhibitors (ICIs) has improved the prognosis of unresectable advanced or recurrent esophageal and gastric cancer. However, the prognosis remains poor. The human intestinal microbiota plays various roles in digestion, immunomodulation, and metabolism and is known to influence tumor suppression. Previous studies reported that intestinal microbiota can impact immune responses and the effectiveness of ICIs. Therefore, this study investigates the safety and efficacy of antibiotic fecal microbiota transplantation (A-FMT) in patients with unresectable advanced or recurrent esophageal or gastric cancer. **Methods:** Eligible patients are diagnosed with unresectable advanced or recurrent esophageal squamous cell carcinoma or gastric adenocarcinoma. The primary endpoint is to examine the dose-limiting toxicity (DLT) of A-FMT. Secondary endpoints include response rate, disease control rate, progression-free survival, overall survival, and biomarkers of efficacy and adverse events. This trial consists of two parts: safety confirmation, which assesses DLT incidence, and an expansion part, which examines efficacy and safety. The safety confirmation part will enroll 3-6 patients in cohort 1 (combination therapy with ICIs and chemotherapy) and cohort 2 (combination of anti-PD-1 and anti-CTLA4 antibodies). The expansion part will enroll untreated patients: 10 for esophageal cancer (ICIs + chemotherapy), 10 for anti-PD-1 + anti-CTLA4 antibodies, and 11 for gastric cancer (ICIs + chemotherapy). After enrollment, patients receive antibacterial agents for 1 week before FMT. Patients then undergo total colonoscopy after bowel cleaning with polyethylene glycol plus ascorbic acid. A 200 mL fecal suspension from a healthy donor is transferred to the patient's cecum, and ICIs with/without chemotherapy are initiated thereafter. Biomarker analysis explores diagnostic or surrogate biomarkers, aiming to identify molecular targets, cell populations, microbial species, and functions involved in efficacy by analyzing the effects of A-FMT and ICIs on the tumor microenvironment and intestinal microbiota using methods such as proteome analysis. This trial is registered in the Japan Registry of Clinical Trials as jRCTs031240170 and was initiated in August 2024, with enrollment ongoing. Clinical trial information: jRCTs031240170. Research Sponsor: Metagen Therapeutics Inc.

TPS511 Trials in Progress Poster Session

KEYMAKER-U06: Phase 1/2 umbrella platform study of TROP2 antibody-drug conjugate sacituzumab tirumotecan plus pembrolizumab plus chemotherapy (substudy 06C) and sacituzumab tirumotecan plus paclitaxel (substudy 06D) in gastroesophageal adenocarcinoma. First Author: Do-Youn Oh, Seoul National University College of Medicine, Seoul, South Korea

Background: Treatment options for patients with gastroesophageal adenocarcinoma (GEA) are limited and prognosis is poor. KEYMAKER-U06 is a multicenter, open-label, phase 1/2, umbrella platform study designed to evaluate investigational agents with or without pembrolizumab and/or chemotherapy for GEA. Substudy 06C (NCT06469944) will be conducted to evaluate sacituzumab tirumotecan(TROP2 antibody-drug conjugate; formerly MK-2870/SKB264) plus pembrolizumab plus investigator's choice of chemotherapy as first-line therapy for patients with advanced or metastatic GEA. Substudy 06D (NCT06445972) will be conducted to evaluate sacituzumab tirumotecan plus paclitaxel as second-line therapy for patients with advanced or metastatic GEA. **Methods:** In both substudies, patients aged ≥18 years with confirmed advanced or metastatic HER2-negative GEA (gastric, gastroesophageal junction, or esophageal adenocarcinoma), measurable disease per RECIST v1.1 by investigator review and verified by blinded independent central review (BICR), and an Eastern Cooperative Oncology Group performance status score of 0 or 1 are eligible; patients in substudy 06C should be treatment naive and patients in substudy 06D should have disease progression on or after 1 prior therapy. Both substudies will have a safety lead-in phase of ≤10 patients to determine the recommended phase 2 dose of sacituzumab tirumotecan when used in combination with pembrolizumab plus chemotherapy (substudy 06C) or paclitaxel (substudy 06D). In the safety lead-in phase, patients will receive sacituzumab tirumotecan 3 mg/kg or 4 mg/kg IV per a Bayesian optimal interval design. In the efficacy phase of substudy 06C, ≅120 patients (including 10 patients in safety lead-in) will be allocated or randomly assigned to receive pembrolizumab 400 mg IV every 6 weeks plus investigator's choice of chemotherapy (arm 1) or sacituzumab tirumotecan IV on days 1, 15, and 29 of each 6-week cycle plus pembrolizumab plus investigator's choice of chemotherapy (arm 2). In the efficacy phase of substudy 06D, ≅80 patients (including 10 patients in safety lead-in) will be allocated or randomly assigned to receive ramucirumab 8 mg/kg IV on days 1 and 15 of each 4-week cycle plus paclitaxel 80 mg/m^2 IV on days 1, 8, and 15 of each 4-week cycle (arm 1) or sacituzumab tirumotecan IV on days 1, 15, and 29 of each 6-week cycle plus paclitaxel (arm 2). Primary outcomes for both substudies are safety and tolerability for the safety lead-in and ORR per RECIST v1.1. by BICR for the efficacy phase (calculated based on pooled data from safety lead-in and efficacy phase); secondary outcomes include DOR and PFS per RECIST v1.1 by BICR, OS, safety, and tolerability. Enrollment is ongoing in both substudies. Clinical trial information: NCT06469944; NCT06445972. Research Sponsor: Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA.

TPS512 Trials in Progress Poster Session

A single arm phase II trial of trastuzumab deruxtecan in patients with gastrooesophageal adenocarcinoma cancer who are ctDNA and HER2 positive: DECIPHER. First Author: Elizabeth Catherine Smyth, Oxford University Hospitals NHS Foundation Trust, Oxford, United Kingdom

Background: Oesophagogastric adenocarcinoma (OGA) is globally prevalent and frequently advanced at presentation. Less than 50% of patients with operable OGA are cured when treated with multimodality therapy. Patients in the UK with operable OGA, are treated with FLOT prior to and following surgery. However, patients who have circulating tumour DNA (ctDNA) in their blood after surgery have worse survival than ctDNA negative patients. Therefore, use of novel treatment approaches aiming to cure this micrometastatic disease are considered a rational and appropriate treatment approach. Trastuzumab deruxtecan (T-DXd), a novel HER2-targeting antibody drug conjugate, is licensed to treat advanced, previously treated HER2 positive OGA. Therefore, there is a need to evaluate T-DXd in resectable HER2 and ctDNA positive OGA, where options are limited. **Methods:** Co-ordinated by the Cancer Research UK Southampton Clinical Trials Unit, DECIPHER is a single arm, multicentre, phase II trial testing the effect of T-DXd on reducing micrometastatic disease burden in HER2 positive OGA patients who are ctDNA positive after chemotherapy and surgery. To be eligible, participants must have been treated with chemotherapy before surgery for at least 6 weeks and must have recovered from surgery with no evidence of metastatic disease on post-surgical imaging. HER2 positive patients are recruited to the tissue testing stage of the trial prior to surgery. Patients who are ctDNA positive after surgery will be treated with T-DXd at a dose of 6.4 mg/kg intravenously every 21 days for a maximum of 8 cycles or until disease recurrence. If required, patients may dose reduce, but no dose re-escalation will be permitted. A clinically validated, personalised, tumour-informed ctDNA assay (signatera, Natera, Inc.) will be utilised to detect ctDNA in patient plasma samples. The primary endpoint is the percentage of patients who are ctDNA negative after 4 cycles of treatment. Secondary endpoints include ctDNA clearance after each cycle, disease free survival, overall survival, quality of life and frequency of adverse events. DECIPHER (NCT05965479) opened in April 2024 and will run in approximately 15 UK secondary care hospitals with the aim of recruiting 25 evaluable patients (recruitment to the tissue testing stage is currently 27 on 23-Sep-2024). Clinical trial information: NCT05965479. Research Sponsor: Astra Zeneca.

TPS513 Trials in Progress Poster Session

A phase I trial of intraperitoneal (IP) mesothelin (MSLN)-targeted CAR T-cell therapy in patients with MSLN-positive esophagogastric cancer (EGC) with peritoneal carcinomatosis. First Author: Joan Choo, Memorial Sloan Kettering Cancer Center, New York, NY

Background: Outcomes for patients with advanced EGC remain poor. The peritoneum is an immune-privileged niche and a common site of metastases occurring in up to 40% of patients with metastatic EGC; this is characterised by poor survival because of resistance to immunotherapy and a lack of measurable disease, which excludes trial participation. MSLN was selected as a target for CAR T cell therapy as it is overexpressed in up to 60% of EGC, is associated with aggressive disease biology and has low levels of expression in normal tissues. In pre-clinical *in vivo* models of peritoneal carcinomatosis, improved efficacy was observed with regional administration of IP CAR T cells, compared with systemic CAR T cells. No significant toxicity was observed at clinically relevant doses of IP administered CAR T-cells. M28z1XXPD1DNR CAR T-cells are a next-generation MSLN-targeted CAR, equipped with a modified CD3z (1XX), and a PD-1 dominant negative receptor (PD1DNR) that provides T-cell intrinsic checkpoint blockade. **Methods:** This is a single-arm phase I study assessing the safety, maximum tolerated dose (MTD) and preliminary efficacy of IP administered M28zXXPD1DNR CAR T cells in patients with MSLN-positive (IHC ≥ 25%) EGC with evidence of peritoneal carcinomatosis (imaging and/or cytology). Patients with concomitant extraperitoneal disease are eligible. Patients must have received at least one prior line of treatment for advanced or metastatic disease and have at least one measurable or evaluable lesion per RECIST 1.1. A minimum of 4 patients and maximum of 18 patients will be treated with 4 escalating doses (with 1 "fallback" dose) of CAR T cells to a maximum of 3 x 10^7 cells/kg administered via peritoneal catheter (Table). Lymphodepletion with fludarabine 30mg/m^2/day and cyclophosphamide 300mg/m^2/day is given for 3 days prior to the CAR T cell infusion. Adverse events, response rates (RECIST 1.1), survival outcomes and immuno-monitoring (immune cell phenotyping and serum/peritoneal fluid cytokine analysis) will be assessed. The trial is currently enrolling patients. Clinical trial information: Will be registered before ASCO GI. Research Sponsor: U.S. Department of Defense; Memorial Sloan Kettering Cancer Center.

Dose-escalation scheme based on the continuous reassessment method.

Dose Level	Dose (cells/kg)
1	1 x 10^6
2 (Starting dose)	3 x 10^6
3	6 x 10^6
4	1 x 10^7
5	3 x 10^7

TPS514 Trials in Progress Poster Session

A phase II study of neratinib (Nera) in combination with chemotherapy/ trastuzumab/pembrolizumab (C/T/P) in HER2 overexpressing gastroesophageal cancers (GECs). First Author: Dae Won Kim, H. Lee Moffitt Cancer Center and Research Institute, Tampa, FL

Background: Human epidermal growth factor receptor 2 (HER2) is overexpressed in about 15-20% of GECs and is tested using immunohistochemistry and/or in-situ hybridization. Per 3rd interim analysis of KN-811 study, addition of P to C/T improved overall response rate (ORR) and overall survival (OS) in patients with HER2 positive tumors. HER2 heterogeneity and activation of other signaling pathways ultimately leading to T resistance limits its potential to have a durable response. Nera is an anilinoqiunoline derivative intracellular oral kinase inhibitor that irreversibly binds to epidermal growth factor receptor (EGFR), HER2 and HER4. In this study, we hypothesize that targeting other oncogenic pathways with Nera may help derive additional therapeutic benefit in HER2 positive GECs when combined with C/T/P. **Methods:** This is an open-label, single arm, multi-institutional Phase II study of Nera in combination with C/T/P in HER2 overexpressing GECs as 1st line treatment. The first 6 pts as a lead-in phase will be evaluated for toxicity. If no dose limiting toxicity (DLT), a total of 30 patients (include the first 6pts in the lead-in phase) will be used to evaluate efficacy in the phase II portion. All pts will receive standard dose mFOLFOX/trastuzumab q2w along with pembrolizumab 400 mg q6w. Nera will be dosed at 240 mg daily. In case of DLTs observed with this dose, a dose of 160 mg/day for Nera will be considered. Estimated sample size is 30-36 pts using Simon 2 stage design. Primary endpoint in ORR. The null hypothesis will be rejected if 26 or more responses are observed in 30 patients. Key inclusion criteria include: HER2 positive GECs (as determined by local testing), treatment naïve for stage IV disease, measurable disease, ECOG 0-1, preserved cardiac function. Uncontrolled brain metastases or impaired organ function that poses risk with treatment are key exclusion criteria. Primary endpoint is ORR and secondary endpoints include safety, duration of response, clinical benefit rate and OS. The trial is registered on clinicaltrials.gov under NCT06109467. Clinical trial information: NCT06109467. Research Sponsor: National Comprehensive Cancer Network (NCCN) through a grant provided by Puma Biotechnology.

TPS515 Trials in Progress Poster Session

A phase II study of agenT-797 (invariant natural killer T-cells), botensilimab (Fc-enhanced CTLA-4 inhibitor) and balstilimab (anti-PD-1) in patients with advanced, refractory gastroesophageal adenocarcinoma. First Author: Samuel Louis Cytryn, Gastrointestinal Oncology Service, Department of Medicine, Memorial Sloan Kettering Cancer Center, New York, NY

Background: Esophagogastric (EG) cancer, the fifth most common cancer worldwide and second leading cause of cancer-related mortality, results in 1.3 million deaths annually. Despite recent advances in frontline management, nearly all patients with metastatic disease experience progression, highlighting the urgent need for novel therapies. Botensilimab (BOT), an Fc-enhanced, multifunctional anti-CTLA-4 antibody, has shown promise in combination with balstilimab (BAL, anti-PD-1) across multiple treatment-refractory cancers. Additionally, agenT-797, an allogeneic off-the-shelf invariant Natural Killer T (iNKT) cell therapy, enhances immune responses, including in settings refractory to immune checkpoint inhibitors (ICI), without requiring HLA matching or lymphodepletion. In a cohort of 82 previously treated patients, agenT-797 demonstrated a favorable safety profile with no severe Cytokine Release Syndrome, Graft-versus-Host Disease, or immune effector cell-associated neurotoxicity syndrome. Notably, in patients with relapsed/refractory gastric cancer, agenT-797 plus anti-PD-1 showed important clinical activity that warrants further exploration. Given their complementary mechanisms of action in modulating anti-tumor immunity, we aimed to evaluate the safety and efficacy of combining BOT, BAL, and agenT-797 with standard ramucirumab (anti-vascular endothelial growth factor receptor-2 antibody) and paclitaxel chemotherapy in patients with advanced EG cancer who have progressed on frontline therapy. **Methods:** This is an investigator-initiated, single-arm phase II trial in adult patients (≥18 years) with esophageal, gastric, or gastro-esophageal junction (GEJ) adenocarcinoma who have received one prior line of therapy, performed at Memorial Sloan Kettering Cancer Center. Eligible patients receive a single infusion of agenT-797 on cycle 1 day 1, BOT 75 mg on day 1 of cycles 1 through 3, BAL 240 mg on days 1 and 15, ramucirumab 8 mg/kg on days 1 and 15, and paclitaxel chemotherapy 80 mg/m^2 on days 1, 8, and 15 of a 28-day cycle. Treatment is continued until disease progression (defined by Response Evaluation Criteria in Solid Tumors [RECIST] version 1.1), unacceptable toxicity, or withdrawal of consent. The primary endpoint is overall response rate. The secondary endpoints are safety and tolerability, disease control rate, duration of response, progression-free survival, and overall survival. The regimen will be considered worthy of further investigation if at least 15 out of 37 patients achieve an objective response. To date, 12 patients have been enrolled and started on treatment. This trial is registered with ClinicalTrials.gov, NCT06251793. Clinical trial information: NCT06251793. Research Sponsor: None.

TPS516 Trials in Progress Poster Session

A phase II double window of opportunity trial evaluating the oral TGF-beta receptor I inhibitor vactosertib in patients undergoing standard of care preoperative therapy for locally advanced esophageal adenocarcinoma. First Author: Kanchi Patell, University Hospitals Seidman Cancer Center, Case Comprehensive Cancer Center, Case Western Reserve University, Cleveland, OH

Background: Esophageal adenocarcinoma (EAC) is the seventh most common cancer globally and ranks sixth in overall mortality. Despite improvements in pre-operative therapy, EAC has a high recurrence rate, especially among patients whose tumors do not achieve pathologic complete response (pCR), thus there is a critical need to enhance pCR rate and improve long-term outcomes. We recently discovered that a subset of EACs are driven by tumor-intrinsic oncogenic TGFβ signaling and have identified two potential biomarkers to predict response to anti-TGFβ therapy. One of these biomarkers is enriched in poorly differentiated EAC. We hypothesize that TGFβ signaling blockade will exert a tumor-inhibitory effect in patients with EAC and propose a double window-of-opportunity trial to test if TGFβ receptor I inhibitor vactosertib can improve pCR and induce tumor metabolic response in patients undergoing neoadjuvant therapy for localized EAC. **Methods:** This phase II double window of opportunity study assesses the efficacy and pharmacodynamics of the oral TGFβ Receptor I inhibitor vactosertib in patients with locally advanced esophageal adenocarcinoma (EAC). Patients will be treated with single-agent vactosertib in two windows of opportunity: the first will be prior to initiation of standard of care neoadjuvant therapy and the second after completion of neoadjuvant therapy prior to surgery. Patients will be eligible if they have poorly differentiated EAC and are appropriate for neoadjuvant therapy followed by resection as per standard of care. The primary objectives are improvement in pCR compared to historical controls and metabolic response (decrease in fluorodeoxyglucose uptake by PET CT) in the primary tumor after the first window. Secondary objectives will evaluate biomarkers of response identified in preclinical studies. Pre- and post-first window of opportunity treatment biopsies will be obtained for correlative and exploratory analyses. Clinical trial information: NCT06044311. Research Sponsor: None.

TPS517 Trials in Progress Poster Session

Preclinical and phase 1/2 data of the CHK1 inhibitor BBI-355 in development for esophageal and gastric cancers (EGC) with *EGFR* or *FGFR2* amplifications. First Author: Steven Brad Maron, Memorial Sloan Kettering Cancer Center, New York, NY

Background: High-copy number amplifications of oncogenes (e.g., *EGFR*, *FGFR2*) frequently occur on extrachromosomal DNA (ecDNA), highly transcribed units of circular non-chromosomal DNA. While targeted therapies have improved survival for patients with oncogene mutations, they have limited activity in oncogene amplified cancers. In Barrett's esophagus-associated esophageal adenocarcinoma (EAC) ecDNA can be found early in the transition from high-grade dysplasia to EAC. In EAC and gastric cancer (EGC), ~7% and ~3% have *EGFR* and *FGFR2* amplifications, respectively, with possibly >50% occurring on ecDNA. Tumor cells with oncogene amplifications, particularly on ecDNA, have increased DNA replication stress and are sensitive to inactivation of checkpoint kinase 1 (CHK1). We developed BBI-355, an oral, potent, and selective small molecule inhibitor of CHK1, which is in Phase 1/2 clinical development. **Methods:** BBI-355-101 is a first-in-human Phase 1/2 study of BBI-355 alone or in combination with targeted therapies for advanced or metastatic solid tumors with oncogene amplification (NCT05827614). Part 1 is dose escalation of BBI-355 (PO Q2D). In Parts 2 and 3, patients with *EGFR* or *FGFR*1-3 amplifications are treated with BBI-355 in combination with the EGFR inhibitor erlotinib (150 mg PO QD) or the FGFR inhibitor futibatinib (20 mg PO QD). While this study is enrolling all tumor types, EGC are some of the most likely to harbor ecDNA amplifications. Amplification-driven cancers rarely contain driver oncogene mutations or fusions, and so are excluded. **Results:** *Preclinical:* Tumors that harbor oncogene amplifications on ecDNA can evade targeted therapeutic pressure via ecDNA-based resistance mechanisms. BBI-355 in combination with agents targeting the amplified oncogenes (*FGFR2* or *EGFR*) prevented resistance to targeted therapies in multiple ecDNA+ gastric cancer CDX and PDX models. *Clinical:* Safety and PK data of BBI-355 alone and in combination with erlotinib or futibatinib will be presented. The most common drug-related TEAEs, SAEs, and DLTs were hematologic (i.e., leukopenia, neutropenia, lymphopenia and thrombocytopenia) which are on target for CHK1 inhibition and generally well managed. Dose escalation is ongoing and the RP2D/MTD has not been determined. PK showed dose-dependent increase in exposure of BBI-355. Target engagement as determined by pCHK1 IHC as a PD biomarker was observed at all dose levels of BBI-355. **Conclusions:** The first ecDNA directed therapy, BBI-355, demonstrated significant synergistic anti-tumor activity in combination with targeted therapies in multiple ecDNA+ oncogene amplified EGC models. BBI-355 alone and in combination with erlotinib/futibatinib was well tolerated. Clinical testing in patients with *EGFR* and *FGFR2* oncogene amplifications is ongoing with a strong rational for EGC. Clinical trial information: NCT05827614. Research Sponsor: None.

518 General Session

Adjuvant cytokine-induced killer cell immunotherapy in hepatocellular carcinoma: Extended follow-up of a randomized controlled trial and post-treatment immune cell profiling. First Author: Jeong-Hoon Lee, Seoul National University College of Medicine, Seoul, South Korea

Background: Notwithstanding the pressing need for adjuvant therapy for hepatocellular carcinoma (HCC), most adjuvant therapies, including atezolizumab/bevacizumab, have failed. Meanwhile, adjuvant immunotherapy utilizing autologous cytokine-induced killer (CIK) cells for HCC improved recurrence-free survival (RFS) in a previously reported randomized controlled trial (RCT) and real-world data. This study aimed to assess the longer-term outcomes of the RCT and elucidate the underlying mechanisms of sustained effects of CIK cell treatment. **Methods:** This study comprised two parts: a long-term follow-up of the preceding RCT and an analysis of immune cells in patients who received adjuvant CIK cell treatment. In the original RCT, 226 patients who underwent curative treatment for stage I or II HCC were randomly allocated to either the CIK (n=114, 16 injections of 6.4×10^9 CIK cells over an 11-month period) or control group (n=112). The follow-up period was extended to 9 years after the enrollment of the last patient. The primary endpoint was recurrence-free survival (RFS). The secondary endpoints included cancer-specific survival (CSS) and overall survival (OS). Parallelly, a prospective study was conducted to investigate post-treatment changes of immune cells in peripheral blood of 7 patients, who received repeated transfer of CIK cells after curative treatment for HCC, using flow cytometry. **Results:** In the extended follow-up of the RCT (median follow-up=115.7 months, interquartile range=74.2–130.5 months), the CIK group sustained a significantly prolonged RFS (median=43.5 vs 27.4 months; hazard ratio [HR] =0.74, 95% confidence interval [CI]=0.55–0.99, *P*=0.045) and CSS (median=unreached; HR=0.49, 95% CI=0.25–0.95, P=0.04) compared to the control group. Adjuvant CIK cell therapy reduced the risk of overall death by 30%, although it did not achieve statistical significance (median=unreached; HR=0.70, 95% CI=0.44–1.11, P=0.1). A preliminary analysis of peripheral blood immune cells revealed that the CIK cell treatment tended to increase the frequencies of $CD8^+$ and $CD4^+$ classical memory cells. **Conclusions:** Adjuvant CIK cell treatment demonstrated significantly enhanced RFS and CSS in the prolonged follow-up of the RCT extending to 9 years in patients who received curative treatment for HCC. The CIK group also demonstrated a consistent trend of improved OS. The increase in memory T cell populations might be linked to the sustained off-treatment anti-tumor efficacy of CIK cell treatment. Research Sponsor: None.

519 Oral Abstract Session

Adjuvant chemotherapy or chemo-radiation in gallbladder cancer: A phase III randomized controlled study (ACCELERATE trial). First Author: Atul Sharma, Department of Medical Oncology, All India Institute of Medical Sciences, New Delhi, India

Background: Role of adjuvant therapy in gallbladder cancer is still evolving and no prospective trial has compared chemotherapy (ChT) alone to combination of chemotherapy and chemo-radiation (CRT). We designed this trial to answer whether adding CRT to ChT improves relapse free survival. **Methods:** In this open-label, multicentric, phase 3, non-inferiority academic trial, operated gallbladder cancers (R0 or R1) patients were randomized to physicians choice of ChT alone (either 6 cycles of mGemOx-Gemcitabine 900 mg/m2 and oxaliplatin 80 mg/m2 IVI days 1 and 8 every 3 weeks or GemCis- Gemcitabine 1000 mg/m2 and cisplatin 25 mg/m2 1 and 8 every 3 weeks) or 3 cycles of physicians choice of ChT (as above) followed by CRT (radiation 45Gyin 25 fractions in 5 weeks with concurrent oral capecitabine in the dose of 825 mg/m2 twice a day on days of radiation and further 2-3 cycles of mGemOx or GemCis. The primary endpoint was the relapse free survival. Planned sample size was 100 subjects in each arm. **Results:** Between April 2018 and January 2021, 137 patients were screened, and 94 eligible patients were randomized, 49 (52.1%) in chemotherapy alone arm (standard-arm1) and 45 (47.9%) in chemotherapy plus chemo-radiation arm (experimental-arm 2). Slow accrual led to premature closure of trial. COVID pandemic might have contributed to that. Baseline characteristics were well balanced (like, sex, presenting symptoms, ECOG PS, duration of symptoms, comorbidities, location of tumour in GB, presence of gallstone disease, and tumour markers were well balance in both groups) except that a greater number of patients had deranged baseline LFTs in arm1 and there was a trend towards higher numbers of stage IIA in arm 1. The median age was 55 years (range 27-73 years). Females constituted 32 in each group. 43 and 45 patients had ECOG PS of 0-1 in arm 1 and arm 2 respectively. One patient in arm 1 was ineligible. All had R0 resection. Incidences of dose reductions and dose delays were similar. A greater number of patients in arm 1 experienced diarrheal episodes (p=0.021) and peripheral neuropathy (0.001). 42 (85.7%) patients in arm 1 and 28 (62.2%) patients in arm completed 5-6 cycles. 18 (36.73%) and 23 (51.11%) died till last follow up. 14 (28.57%) and 20(44.44%) died because of disease progression. One patient in each arm died of toxicity. The primary end point of study, relapse free survival was not estimable in arm 1 and was 34.39 months in arm 2 (p=0.202). Median overall survival was not estimable in arm 1 and was 34.56 months in arm 2 (p=0.123). Mean RFS was 51.96 Vs 43.99 months in arm 1 and arm 2 respectively. **Conclusions:** This trial suggests that addition of CRT to ChT doesn't improve outcome in resected gallbladder cancer compared to ChT alone. A larger trial is needed to address this issue. Clinical trial information: CTRI/2018/04/013218. Research Sponsor: None.

520 Rapid Oral Abstract Session

Nivolumab (NIVO) plus ipilimumab (IPI) vs lenvatinib (LEN) or sorafenib (SOR) as first-line (1L) therapy for unresectable hepatocellular carcinoma (uHCC): CheckMate 9DW expanded analyses. First Author: Masatoshi Kudo, Kindai University Hospital, Osaka, Japan

Background: In the phase 3 CheckMate 9DW study (NCT04039607), 1L NIVO + IPI demonstrated significant overall survival (OS) benefit vs LEN/SOR, higher objective response rate (ORR) with durable responses, and manageable safety in uHCC. We present efficacy by best overall response (BOR) subgroups and baseline characteristics, and additional safety analyses from the preplanned interim analysis. **Methods:** Patients (pts) with previously untreated HCC not eligible for curative surgical or locoregional therapies, Child-Pugh score 5 or 6, and ECOG performance status 0 or 1 were randomized 1:1 to receive NIVO 1 mg/kg + IPI 3 mg/kg Q3W (up to 4 cycles), then NIVO 480 mg Q4W or LEN 8 mg or 12 mg QD or SOR 400 mg BID until disease progression or unacceptable toxicity. NIVO was given for a maximum of 2 years. The primary endpoint was OS; secondary endpoints included ORR and duration of response (DOR) per blinded independent central review (BICR) using RECIST v1.1. **Results:** A total of 668 pts were randomized to NIVO + IPI (n = 335) or LEN/SOR (n = 333). At a median follow-up of 35.2 (range 26.8–48.9) months (mo), median OS (95% CI) was 23.7 (18.8–29.4) mo with NIVO + IPI vs 20.6 (17.5–22.5) mo with LEN/SOR (HR 0.79 [95% CI 0.65–0.96]; *P* = 0.0180). ORR (95% CI) per BICR was significantly higher with NIVO + IPI vs LEN/SOR (36% [31–42] vs 13% [10–17]; $P < 0.0001$); median DOR (95% CI) was 30.4 (21.2–not estimable [NE]) mo vs 12.9 (10.2–31.2) mo. Survival benefit of NIVO + IPI vs LEN/SOR was observed across BOR subgroups at the 24-week landmark timepoint (Table). In subgroup analyses, ORR (95% CI) per BICR was higher with NIVO + IPI vs LEN/SOR across HCC etiologies (uninfected: 35% [26–44] vs 8% [4–15]; HBV infected: 25% [17–34] vs 17% [10–25]; HCV infected: 50% [39–61] vs 16% [9–25]) and in pts with Barcelona Clinic Liver Cancer stage ≤B (33% [23–43] vs 13% [6–21]) or stage C (37% [31–44] vs 14% [10–19]). Safety data are shown in the Table. Additional exploratory analyses will be presented. **Conclusions:** These additional analyses from CheckMate 9DW demonstrate the efficacy and manageable safety of 1L NIVO + IPI in uHCC and further support its use as a potential standard-of-care treatment option in this setting. Clinical trial information: NCT04039607. Research Sponsor: Bristol Myers Squibb.

OS by BOR at week 24 landmark	NIVO + IPI			LEN/SOR		
BOR	CR + PR (n = 101)	SD[a] (n = 105)	PD (n = 47)	CR + PR (n = 28)	SD[a] (n = 212)	PD (n = 31)
Median OS (95% CI), mo	NR (44.4–NE)	30.0 (23.5–37.8)	16.0 (12.0–18.7)	28.3 (20.6–NE)	22.5 (20.5–24.8)	13.5 (8.7–25.3)

All treated pts	NIVO + IPI (n = 332)	LEN/SOR (n = 325)
Any-grade/grade 3–4 TRAEs, n (%)	278 (84)/137 (41)	297 (91)/138 (42)
Hepatobiliary	44 (13)/35 (11)	15 (5)/10 (3)
Cardiovascular	10 (3)/3 (< 1)	138 (42)/39 (12)
Hemorrhagic	2 (< 1)/1 (< 1)	20 (6)/5 (2)

[a]Includes non-CR/non-PD. CR, complete response; NR, not reached; PD, progressive disease; PR, partial response; SD, stable disease; TRAE, treatment-related adverse event.

521 Rapid Oral Abstract Session

A multicenter prospective study to evaluate the efficacy of resection for initially unresectable hepatocellular carcinoma after atezolizumab combined with bevacizumab (the RACB study): Short-term outcomes. First Author: Maho Takayama, Hepato-Biliary-Pancreatic Surgery Division, Department of Surgery, Graduate School of Medicine, The University of Tokyo, Tokyo, Japan

Background: The IMbrave150 study showed that atezolizumab plus bevacizumab (atezo+bev) is superior to sorafenib in progression-free survival and overall survival, making it the recommended first-line systemic therapy for unresectable hepatocellular carcinoma (HCC). The RACB study was scheduled to evaluate the efficacy of atezo+bev in achieving conversion surgery for initially unresectable HCC. This time, we report the short-term outcomes of the RACB study. **Methods:** This prospective, single-arm, multicenter Phase II trial evaluates atezo+bev to achieve conversion surgery in patients with technically or oncologically unresectable HCC. Eligible patients had unresectable HCC without prior systemic chemotherapy and at least one target lesion based on RECIST ver1.1. Patients received atezolizumab (1200 mg) plus bevacizumab (15 mg/kg) every three weeks. If resectable on radiological assessment at 12 weeks, atezolizumab monotherapy was given followed by surgery. The primary endpoint was the progression-free survival. The secondary endpoints included the overall response rate, overall survival, resection rate, curative resection rate, on-protocol resection rate, and ICG retention rate at 15 min after atezo+bev therapy. The study is registered with the Japan Registry of Clinical Trials (jRCTs051210148). **Results:** Between March 2022 and March 2024, 55 patients from 17 centers were enrolled. Four patients were excluded due to ineligible lesions or other reasons. A total of 50 patients were included in the efficacy evaluation set. Macrovascular invasion and extrahepatic metastases were observed in 38 (76.0%) and 6 patients (12.0%), respectively. Two patients (4.0%) had metachronous extrahepatic metastases without intrahepatic tumors. The complete and partial response rates were 0% and 13% based on RECIST, and 2.2% and 26.1% based on mRECIST. The overall resection rate was 48%, corresponding to 24 cases. R0, R1, and R2 resections were performed in 21 (87.5%), 1 (4.2%), and 2 (8.3%) patients, respectively. Complications ≥Grade III according to the Clavien-Dindo classification included 3 bile leakages, 2 infections, 1 anastomotic leakage, and 1 hemoptysis. There were no surgery-related deaths. Of the 51 patients included in the safety analysis set, adverse events occurred in 34 cases (66.7%), with 16 Grade 3 or higher instances with one Grade 4 upper gastrointestinal bleeding, 1 Grade 4 encephalopathy, and 1 unexpected death after 1 cycle of atezo+bev. **Conclusions:** The RACB study's short-term outcomes demonstrated the feasibility of conversion surgery after atezo+bev in patients with initially unresectable HCC. The response rate of atezo+bev, as measured by mRECIST, was similar to IMbrave150, with a 48% resection rate, though careful monitoring of severe adverse events is essential. Clinical trial information: 051210148. Research Sponsor: CHUGAI PHARMACEUTICAL CO., LTD.

LBA522 **Rapid Oral Abstract Session**

Transarterial chemoembolization (TACE) combined with camrelizumab and apatinib versus TACE alone in the treatment of unresectable hepatocellular carcinoma eligible for embolization: A multicenter, open-label, randomized, phase 2 study (CAP-ACE). First Author: Haidong Zhu, Center of Interventional Radiology & Vascular Surgery, Department of Radiology, Zhongda Hospital, Medical School, Southeast University, Nanjing, Jiangsu, China

The full, final text of this abstract will be available at meetings.asco.org on the day of presentation and in the online supplement to the February 1, 2025, issue of the *Journal of Clinical Oncology*.

523 **Poster Session**

Racial disparities in secondary metastasis in patients with hepatocellular carcinoma: A 5-year National Inpatient Sample study. First Author: Sakditad Saowapa, Department of Medicine, Texas Tech University Health Sciences Center, Lubbock, TX

Background: Hepatocellular carcinoma (HCC), a primary liver malignancy, is one of the leading causes of cancer-related mortality worldwide. The incidence and outcomes of HCC vary significantly across racial groups. Secondary metastasis, a key factor in disease progression and prognosis, also appears to differ by race, though these disparities remain poorly understood. This study aims to explore how race influences the metastatic progression of HCC, providing valuable insights into potential racial differences in disease outcomes. **Methods:** A retrospective analysis of National Inpatient Sample (NIS) data from 2016 to 2020 was conducted, identifying adults with HCC using International Classification of Diseases, Tenth Revision (ICD-10) codes, and categorizing them based on the presence of metastasis. The primary outcome was racial differences in organ metastasis. Multivariable logistic regression models were used to adjust for potential confounders and to evaluate the impact of race on all metastatic patterns, with Caucasians as the reference category. Statistical significance was defined as p-values $\leq$ 0.05. **Results:** The study cohort included 236,985 adult patients with HCC, of whom 34,875 had secondary metastasis. The cohort was predominantly male (77.37%), with 22.63% female. The racial distribution was 52.61% Caucasian, 19.08% African American, 15.04% Hispanic, and 13.27% from other racial groups. After adjusting for confounders, African American patients had significantly higher odds of lung (aOR 1.42, $P<0.001$), bone (aOR 1.51, $P<0.001$), and brain metastasis (aOR 1.47, P=0.02), but lower odds of adrenal gland metastasis (aOR 0.74, P=0.003) compared to Caucasians. Asian/Pacific Islander patients demonstrated increased odds of lung (aOR 2.00, $P<0.001$) and brain metastasis (aOR 1.95, P=0.02). In contrast, Hispanic patients had lower odds of abdominal lymph node (aOR 0.69, P=0.008) and bone metastasis (aOR 0.83, P=0.007). **Conclusions:** This study highlights significant racial disparities in the metastatic patterns of hepatocellular carcinoma, emphasizing the need to consider these differences in clinical management and treatment strategies for HCC patients. Further research is required to investigate the underlying causes of these disparities and their impact on long-term survival. Research Sponsor: None.

524 **Poster Session**

Gender and ethnic disparities in hepatocellular carcinoma: Trends in incidence rates across U.S. populations from 2000 to 2021. First Author: Lauren Guerrero Negrón, Universidad Central Del Caribe School of Medicine, Bayamon, PR

Background: Hepatocellular carcinoma (HCC) is a primary malignancy of the liver strongly associated with liver cirrhosis and the etiologies producing cirrhosis. As such, racial and ethnic factors might affect incidence rates of HCC resulting in disparities. While disparities have been reported, the effect of gender within certain populations, particularly Hispanic and non-Hispanic White ancestry, have not been fully explored. This study looks at the changing incidence rates of HCC in the US amongst different populations with a particular focus on the impact of gender on incidence. **Methods:** Incidence rates of HCC between 2000 and 2021 among different sexes within racial and ethnic groups in the United States were analyzed utilizing delay-adjusted data from the National Cancer Institute's Surveillance, Epidemiology, and End Results (SEER) program. Annual percent change (APC) was calculated by using Joinpoint Regression software. **Results:** Overall, HCC incidence rates since 2014 to 2021 show a slight decrease (APC, -0.06). While Hispanic men HCC incidence rates declined from 2012 to 2021 (APC, -0.44), Hispanic women incidence have had a steady increase since 2000 (APC, +2.52). Non-Hispanic Black men (APC, -3.07) rates vastly declined from 2015 to 2021 while non-Hispanic black women interestingly experienced a slight decrease in incidence from 2013 to 2021 (APC, -0.34). Non-Hispanic White men experienced stagnant rates of HCC since 2014 (APC, -0.05) while non-hispanic white women (APC, +3.59) showed increasing trends in HCC incidence since 2003. For comparison, non-Hispanic Asian/Pacific Islander men (APC, -2.19) and women (APC, -2.36), which both experienced a decline since 2007. This analysis was repeated without significant change. Further corroborations are necessary to validate these findings. **Conclusions:** These findings show that while the incidence rates of HCC overall have decreased, this benefit appears to be largely driven in male populations. In contrast, in women, and particularly Hispanic and Non-Hispanic White women, incidence rates have been increasing. This phenomenon likely reflects the changes in etiologies of cirrhosis and HCC. Surveillance strategies and screening might need to be optimized to reflect these trends. Research Sponsor: None.

525 **Poster Session**

Mortality trends and racial disparities in hepatocellular carcinoma associated with metabolic syndrome in the United States and Texas: A 21-year analysis (1999-2020) using CDC WONDER database. First Author: Sidra Naz, The University of Texas MD Anderson Cancer Center, Houston, TX

Background: The prevalence of hepatocellular carcinoma(HCC) and metabolic syndrome (MetS) is increasing globally. This study aims to examine annual trends of mortality in MetS-associated HCC in the United States and its state of Texas, from 1999 to 2020, to evaluate public health initiatives and identify socioeconomic factors. **Methods:** Mortality trends for adults aged $\geq$25 years with MetS-associated HCC were analyzed using CDC WONDER database, identifying deaths through ICD-10 codes C22.9 "Liver Malignant neoplasms" and E88.9 "Metabolic Syndrome". Crude and age-adjusted mortality rates (AAMRs) per 100,000 people were extracted. Annual percent changes (APCs) in AAMRs, with 95% confidence intervals, were determined across various demographic (sex, race/ethnicity, age) and geographic (state, urban-rural, regional) subgroups. **Results:** Between 1999 and 2020, 271,046 documented deaths were attributed to HCC due to metabolic syndrome. The AAMR for MetS-related HCC mortality increased in the US from an adjusted rate (AR) (3.7) in 1999 to (4.5) in 2009 (APC:2.10%; 95% CI: -0.13% to 3.10%), after which it increased to (7.6) in 2018 (APC:5.83%; 95% CI:4.25% to 6.99%) and increased to 9.9 in 2020 (APC:13.19%; 95% CI:8.78% to 15.7%). In Texas, AAMR for MetS-related HCC mortality increased from AR (4.9) in 1999 to (8.9) in 2015 (APC:3.95%; 95% CI:3.11% to 4.69%) after which it increased to (15.3) in 2020 (APC:10.64%; 95% CI:8.18% to 17.33%). Men had consistently higher AAMRs than women (12.4 vs. 8). The AAMR in U.S. men increased from (5.1) in 1999 to (12.4) in 2020. The AAMR in U.S. women increased from (2.7) in 1999 to (8) in 2020. The non-Hispanic (NH) American Indian or Alaska Native population has the greatest AAMR (13.4), followed by the NH Black or African American(AA) (12.7) and the Hispanic or Latino population with AAMR (10.9). The low-risk population was NH White with AAMR (9.6) and NH Asian or Pacific Islander (6.6). The AAMR also varied by region (South:11.5; Midwest:9.9; West:9.4; Northeast:7.4) and non-metropolitan areas had higher AAMR (small metro:12.1; non-core areas:11; micropolitan areas:11.7) than metropolitan areas (large central metropolitan: 8.8; large fringe areas: 8.7). The states in the upper 90th percentile of MetS-related HCC AAMRs were Texas, Louisiana, Arkansas, Michigan, Nevada exhibited an approximately two-fold increase in AAMRs, compared to states falling in the lower 10th percentile Nebraska, Wisconsin, Washington, Maine. **Conclusions:** Mortality rates from hepatocellular carcinoma associated with metabolic syndrome have risen in the United States and Texas over the past two decades. However, significant demographic and geographic disparities in MetS-related HCC mortality continue, emphasizing the need for further investigation and the development of targeted treatment strategies. Research Sponsor: None.

526 **Poster Session**

Racial, gender and socioeconomic disparities in survival outcomes of hepatocellular carcinoma. First Author: Oboseh John Ogedegbe, Trinity Health Ann Arbor, Ypsilanti, MI

Background: Hepatocellular carcinoma (HCC) is the sixth most commonly occurring cancer worldwide, with mortality rates increasing globally, particularly in the Western world. The 5-year survival rate of persons with HCC is less than 20%. It is, therefore, essential to investigate the extent to which social determinants of health influence survival outcomes. In this study, we explored the effect of gender, race/ethnicity and socioeconomic factors on survival outcomes among patients with HCC in the United States. **Methods:** We extracted data on patients with a histologic diagnosis of hepatocellular carcinoma from the Surveillance, Epidemiology and End Result (SEER) Registry from 2000-2021. The primary outcome of interest was the cause-specific survival of these patients. According to the International Classification of Diseases for Oncology, Third Edition (ICD-O-3) system, the histological code of HCC is 8170/3. We excluded patients of missing or unknown race, income, location, and patients outside the years of study. Cox proportional hazard regression (SPSS) was utilized to determine the association between patients' race/ethnicity, gender, annual median income, location and cause-specific survival. **Results:** 80,445 met the criteria and were included in our study. Gender proved to be a predictor of survival; compared to females, males had worse outcomes, with males HR 1.040 (95% CI; 1.020-1.060, p = <0.001). Race was a predictor of survival outcomes. Using American Indian/Alaska Native as our reference, Asian/Pacific Islander HR 0.847 (95% CI; 0.790-0.90, p = <0.001), Black HR 1.019 (95% CI; 0.950-1.093, p = 0.603), White HR 0.955 (95% CI; 0.893-1.023, p = 0.186). We also analyzed the effects of socioeconomic factors on survival. Patients with higher annual median income had better outcomes. >$120000 HR 0.817 (95% CI; 0.783-0.851, p = <0.001) while patients with lower annual median income had worse outcomes < $40000 HR 1.217 (95% CI; 1.134-1.305, p < 0.001). The effect of location on survival was not significant. Compared to non-metropolitan areas, metropolitan areas HR 0.988 (95% CI; 0.971–1.004, p = 0.138). **Conclusions:** Race/ethnicity, income and gender disparities all played a significant role in cause-specific survival in patients with HCC, with improved survival observed in females, Asians and patients with higher annual median income. More efforts are needed to bridge these survival disparities. Research Sponsor: None.

527 **Poster Session**

Ethnic and socioeconomic disparities in cancer-free and all-cause survival in cholangiocarcinoma patients under 60: A population-based study. First Author: Adit Dharia, HCA Healthcare/USF Morsani College of Medicine Oak Hill Hospital, Brooksville, FL

Background: Cholangiocarcinoma in individuals under 60 years of age is rare, and survival outcomes can vary based on ethnic and socioeconomic factors. While previous studies have examined these disparities in other cancers, little is known about their impact on cancer-free and all-cause survival in younger cholangiocarcinoma patients. This study investigates the role of race and income in survival outcomes for this population. **Methods:** Data for individuals under 60 diagnosed with cholangiocarcinoma were extracted from the Surveillance, Epidemiology, and End Results (SEER) 22 registry. Kaplan-Meier curves were constructed to assess overall and cancer-free survival. Median survival times were calculated for each racial and income group, and log-rank tests were used to compare survival differences across the groups. **Results:** A total of 11,214 individuals under 60 years of age diagnosed with cholangiocarcinoma were included. Median all-cause survival was 12.0 months for Asian patients, 11.0 months for White patients, and 9.0 months for both Black and Hispanic patients. Significant differences were observed between Asian vs White ($p < 0.005$), Asian vs Black ($p < 0.005$), Asian vs Hispanic ($p < 0.005$), White vs Black ($p < 0.005$), and White vs Hispanic ($p < 0.005$), with no significant difference between Black and Hispanic patients ($p = 0.859$). For cancer-free survival, the median was 13.0 months for Asians, 12.0 months for Whites, 11.0 months for Blacks, and 10.0 months for Hispanics, with significant differences between Asian vs Black ($p < 0.005$), Asian vs Hispanic ($p < 0.005$), and White vs Black ($p = 0.017$). The difference between Asian vs White was not statistically significant ($p = 0.052$), and no significant difference was observed between White vs Hispanic ($p = 0.418$) or Black vs Hispanic ($p = 0.418$). By income, patients with an income >$75,000 had significantly longer all-cause survival (12.0 months vs 9.0 months, $p < 0.005$) and cancer-free survival (13.0 months vs 11.0 months, $p < 0.005$) compared to those with an income of $0-$75,000. **Conclusions:** Significant disparities in cancer-free and all-cause survival were found based on ethnicity and income. Asian and White patients had longer survival compared to Black and Hispanic patients, and higher-income patients experienced better outcomes. These findings highlight the need for targeted efforts to address healthcare inequalities in cholangiocarcinoma care. Research Sponsor: None.

528 **Poster Session**

Disparities in liver cancer risk and outcomes in the LGBTQ community: A literature review. First Author: Fnu Anamika, Lenox Hill Hospital, New York, NY

Background: Liver cancer continues to be a significant public health concern, particularly among high-risk groups such as individuals with hepatitis infections, alcohol use disorders, and metabolic diseases. While liver cancer disparities have been well-documented among racial and ethnic minority groups, emerging research highlights significant health inequities within the LGBTQ community, including higher rates of risk factors and limited access to screening and care. This literature review aims to assess the extent of liver cancer disparities within the LGBTQ community and identify gaps in prevention, screening, and treatment. **Methods:** We performed a systematic search across PubMed, Scopus, and Google Scholar for peer-reviewed studies published between 2000 and 2024. The inclusion criteria focused on studies addressing liver cancer risk, prevalence, or outcomes in LGBTQ populations. Data on key liver cancer risk factors such as hepatitis B and C prevalence, alcohol use, obesity, and access to healthcare services were extracted and synthesized. Additionally, we reviewed public health reports and clinical guidelines relevant to LGBTQ health disparities. **Results:** The review identified significant disparities in liver cancer incidence and mortality among LGBTQ individuals. Hepatitis B infection rates were about 1.5 times higher, and hepatitis C rates up to 1.7 times higher among LGBTQ populations, particularly among gay and bisexual men. Alcohol use disorder was about 30-40% more prevalent in LGBTQ individuals, which further increases the risk of liver cancer. Screening for liver cancer in LGBTQ populations is underutilized, with rates approximately 20% lower than the general population. There were reported delays in diagnosis in LGBTQ community due to stigma, discrimination, and lack of culturally competent healthcare services. While some interventions targeting LGBTQ health have shown promise, the overall lack of targeted liver cancer prevention and treatment strategies remains a significant barrier. **Conclusions:** LGBTQ individuals face increased liver cancer risk due to a higher prevalence of key risk factors and inadequate healthcare access. Current screening and preventive strategies are insufficient for addressing these disparities. Targeted interventions, including enhanced screening efforts, hepatitis vaccination and treatment programs, and the development of LGBTQ-specific healthcare protocols, are needed to reduce the liver cancer burden in this underserved population. Policymakers and healthcare providers must address the unique needs of LGBTQ individuals to achieve equitable cancer outcomes. Research Sponsor: None.

Prevalence of liver cancer risk factors in LGBTQ vs. general population.

Risk Factor	LGBTQ Population (%)	General Population (%)
Hepatitis B Infection	15	10
Hepatitis C Infection	18	12
Alcohol Use Disorder	25	16
Metabolic Syndrome	22	18
Screening Rate	40	60

529 **Poster Session**

Hepatocellular carcinoma: A 30-year trend analysis of global burden stratified by risk factors, and socio-demographic indexes. First Author: Charmi Bhanushali, Saint Vincent Hospital, Worcester, MA

Background: Hepatocellular carcinoma (HCC) remains a leading cause of cancer-related morbidity and mortality worldwide, driven primarily by chronic infections with Hepatitis B, Hepatitis C, and alcohol consumption. This study systematically examines global trends of HCC from 1991 to 2021, focusing on age-standardized mortality rates (ASMR) and Disability-Adjusted Life Years (DALYs) according to risk factors across Socio-Demographic Index (SDI) regions. By highlighting these patterns, the study provides insights into the interplay of socio-economic disparities and healthcare advancements on HCC trends. **Methods:** We utilized data from the Global Burden of Disease database to extract ASMR, and age standardized DALYs for HCC between 1991 and 2021. The data was stratified by etiology (Hepatitis B, Hepatitis C, alcohol consumption) and stratified across five SDI regions: low, low-middle, middle, high-middle, and high SDI. **Results:** The global HCC ASMR decreased by 4% from 1991 to 2021, with the largest reduction (20.4%) in low SDI regions. Hepatitis B-related HCC ASMR dropped by 16.3%, with reductions across all SDI groups. ASMR of Hepatitis C-related HCC remained stable globally but fell by 15.1% in low SDI regions. Alcohol-related HCC ASMR increased by 10% globally, with the highest rise in high SDI regions (24.5%), and NASH-related HCC ASMR increased by 26.7% globally, with a 42% rise in high SDI regions. Hepatitis B-related HCC DALYs decreased by 22.2%, with the sharpest declines recorded in high (31.4%) and low SDI regions (29%). Hepatitis C-related HCC DALYs saw a 8.8% global reduction, which was consistent across all groups. Alcohol-related HCC DALYs rose by 4.6% globally, with increases of 14.4% in high SDI and 20.5% in low-middle SDI regions. NASH-related HCC DALYs increased by 18.7%, with a 34.7% rise in low-middle SDI regions. **Conclusions:** The study reveals substantial progress in reducing the global burden of Hepatitis B- and C-related HCC, particularly in low SDI regions. This progress likely reflects advances in vaccination, antiviral treatments, and screening programs. However, the rising mortality and disability associated with alcohol- and NASH-related HCC, especially in high SDI regions, emphasize the growing impact of lifestyle-related risk factors. These findings underscore the need for tailored public health strategies to mitigate the increasing effects of these preventable risk factors on global HCC outcomes. Research Sponsor: None.

530 Poster Session

Non-contrast MRI as a surveillance tool for recurrent hepatocellular carcinoma after curative treatment: A prospective multicenter intra-individual comparison trial. First Author: Dong Ho Lee, Department of Radiology, Seoul National University Hospital, Seoul, South Korea

Background: Hepatocellular carcinoma (HCC) often recurs following curative treatment, significantly impacting long-term survival. Contrast-enhanced multiphasic computed tomography (CECT) is widely utilized for HCC recurrence surveillance but carries risks such as radiation exposure and adverse reactions to contrast agents. This prospective multicenter trial aimed to compare the accuracy of non-contrast abbreviated magnetic resonance imaging (NC-MRI) with CECT in detecting late recurrent HCC. **Methods:** This prospective multicenter intra-individual head-to-head comparison trial (study identifier: NCT05690451, KCT0006395) enrolled participants who had undergone curative treatment for HCC and remained recurrence-free for over two years. Participants underwent three rounds of surveillance at 2–6-month intervals using both CECT and NC-MRI. The primary outcome was the detection accuracy of each modality, compared using the McNemar test. Secondary outcomes included detection sensitivity and specificity. **Results:** A total of 203 participants underwent 528 rounds of paired CECT and NC-MRI, detecting recurrent HCC in 22 participants (10.8%). Among these cases, 21 showed intrahepatic recurrent HCC with a median tumor size of 1.3 cm, and one exhibited aortocaval lymph node metastasis. NC-MRI demonstrated a detection accuracy of 96.6% (196/203), significantly higher than CECT's 91.6% (186/203) (P=0.006). NC-MRI also exhibited significantly higher sensitivity in detecting recurrent HCC compared to CECT (77.3% [17/22] vs. 36.4% [8/22]; P=0.012), while specificity did not differ significantly between NC-MRI and CECT (98.9% [179/181] vs. 98.3% [178/181]; P=0.999). **Conclusions:** NC-MRI showed significantly higher sensitivity and accuracy in detecting recurrent HCC among patients who had been disease-free for more than two years after curative treatment, establishing it as the preferred surveillance modality for late recurrence. Clinical trial information: NCT05690451. Research Sponsor: the Ministry of Health & Welfare, Republic of Korea; HA21C0163.

	Contrast enhanced CT		Non-contrast MRI		Difference, % (95% CI)	P-value
	Estimates, % (n/N)	95% CI, %	Estimates, % (n/N)	95% CI, %		
Accuracy	91.6% (186/203)	87.8-95.5	96.6% (196/203)	94.0-99.1	4.9 (1.7, 8.2)	0.006
Sensitivity	36.4% (8/22)	14.5-58.2	77.3% (17/22)	58.3-96.3	40.9 (16.8, 65.0)	0.012
Specificity	98.3% (178/181)	96.5-100	98.9% (179/181)	97.4-100	0.6 (-0.5, 1.6)	0.999

531 Poster Session

MRI imaging and machine learning based radiomics for detection of mixed HCC and CCA tumors: Still a need for liver biopsy? First Author: Yuquan Qian, National Cancer Center, National Clinical Research Center for Cancer, Cancer Hospital, Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing, China

Background: Primary liver cancer (PLC), comprising hepatocellular carcinoma (HCC) and cholangiocarcinoma (CCA), is a leading cause of cancer mortality globally. The combined hepatocellular cholangiocarcinoma (cHCC-CC) subtype may be less common but is relevant to treatment efficacy. We therefore evaluated the diagnostic accuracy of various approaches in distinguishing these liver cancers. **Methods:** Patients diagnosed with HCC, CCA, and cHCC-CC at Beijing University Cancer Hospital and Institute, China were included. Radiologists of varying expertise independently assessed MRI scans, and we measured their diagnostic consistency. Radiomic features were extracted from MRI scans, and machine learning was applied to differentiate the cancer types. **Results:** Standard imaging was insufficient to reliably characterize cHCC-CC. Abdominal imaging experts (AIEs) had a higher mean sensitivity for HCC and CCA, 88% and 84% respectively, while non-experts (NIEs) had a lower sensitivity of 50% for HCC and 38% for CCA (HCC: p=0.03, CCA: p=0.008). Radiomic analysis found 'Sphericity' and 'ClusterShade' as the most relevant features. However, radiomics algorithms were also not sufficient to distinguish cHCC-CC from either HCC or CCA. Regarding sensitivity, the radiomic-based model was not better than radiologists for any of the three classes (p=0.065 for HCC, p=0.426 for CCA, and p=1.0 for cHCC-CC). The random forest algorithm yielded an accuracy of 76% in the test set, since it correctly classified most HCC and CCA, while only one quarter of cHCC-CC tumors. **Conclusions:** Until improved diagnostic tools are available, biopsy of liver cancer remains critical to the detection, diagnosis, and effective treatment of these cancers. Research Sponsor: Sino-German Center for Research Promotion; GZ-1546 and C-0012; State Ministry of BadenWuerttemberg for Sciences, Research and Arts supporting the Clinical Cooperation Unit Healthy Metabolism at the Center for Preventive Medicine and Digital Health; CCU Healthy Metabolism; Foundation for Biomedical Alcohol Research; Baden-Wuerttemberg Center for Digital Early Disease Detection and Prevention; BW- ZDFP; Federal Ministry of Science and Education of Germany; Q-HCC, 01KD2214.

532 Poster Session

First line atezolizumab + bevacizumab (atezo/bev) or durvalumab ± tremelimumab (durva±treme) in unresectable hepatocellular carcinoma (HCC): A real world, multi-institutional retrospective cohort study. First Author: Ioannis Kournoutas, Department of Medicine, Mayo Clinic Rochester, Rochester, MN

Background: HCC accounts for 80% of primary liver cancers, and worldwide, is the third common cause of cancer-related death. Two FDA approved 1st line systemic treatment options include atezo/bev and durva ± treme. Herein we report outcomes of these regimens in the 1st line setting via a real-world, multi-institutional retrospective cohort study. **Methods:** We reviewed clinical outcomes of patients (pts) who initiated 1st line atezo/bev or durva ± treme for unresectable HCC between March 2017 and February 2024, across 5 institutions (Mayo Clinic, University of Alabama, Cedars-Sinai Medical Center, University Hospitals Seidman Cancer Center and University of Michigan). Time to treatment failure (TTF) and overall survival (OS) were analyzed using the Kaplan-Meier method. Best (objective) response was tabulated. Multivariable analyses were performed using Cox models and logistic regression for time-to-event and binary outcomes, respectively, while adjusting for age, sex, race, HCC etiology, Child-Pugh class, and ALBI grade. **Results:** A total of 433 HCC pts were included; atezo/bev (336), durva ± treme (97). Median age at the start of therapy was 68 years; 77% were male, 82% were Caucasian. Etiologies of HCC were viral hepatitis (38%), MASLD (20%), alcohol (10%), non-viral multifactorial (10%), or unknown (22%). Pts receiving durva ± treme were less likely to have cirrhosis (11% vs. 28%, $p<0.01$) but similar in terms of macrovascular invasion (63% vs. 58%, p=0.35) compared to pts receiving atezo/bev. Reasons for treatment discontinuation were similar between the two groups, with the most common being disease progression - atezo/bev (56%) and durva ± treme (47%). The differences in clinical outcomes were not statistically significantly and remained consistent after multivariable adjustment. (Table, atezo/bev as the reference group). Outcomes were significantly different between Child-Pugh classes (CP: A,B,C) respectively, OS: 19.0 m, 6.4 m, 1.9 m ($p<0.001$); TTF: 6.1 m, 2.7 m, 1.3 m ($p<0.001$). **Conclusions:** In this real-world retrospective cohort study of unresectable HCC, the differences in overall survival, objective response, and time to treatment failure were not statistically significant between atezo/bev and durva ± treme in the 1st line setting. Research Sponsor: None.

Outcome	Atezo/bev	Durva±treme	Univariate p-value	Multivariable adjusted hazard ratio (95% CI)	Multivariable adjusted odds ratio (95% CI)
Overall Survival, Median (month)	14.0	14.6	0.77	0.78 (0.54-1.14)	–
Time to Treatment Failure, Median (month)	4.9	4.1	0.71	0.95 (0.72-1.26)	–
Objective response, %	26.3%	21.1%	0.31	–	0.78 (0.43-1.39)

533 Poster Session

Development and validation of the SPEAR scoring model and predicting overall survival in unresectable hepatocellular carcinoma patients treated with TACE, molecular targeted therapies, and immune checkpoint inhibitors. First Author: Wenli Li, Nanfang Hospital, Southern Medical University, Guangzhou, China

Background: To develop and validate a predictive model to identify hepatocellular carcinoma (HCC) patients likely to benefit from combined treatment of transarterial chemoembolization (TACE), molecular targeted therapies (MTTs), and immune checkpoint inhibitors (ICIs). **Methods:** Data from 500 patients with unresectable HCC treated with TACE, MTTs, and ICIs (2019-2023) were analyzed. Patients were divided into a training set (241 from Nanfang Hospital) and a validation set (259 from various hospitals). Key prognostic variables were identified using random forest survival analysis, and the top five were used to construct a Cox proportional hazards model. Model performance was assessed using time-dependent AUC values, calibration curves, clinical decision curves (DCA), and net reclassification improvement (NRI) indices. Risk scores and optimal cut-off values for risk stratification were determined and validated. **Results:** Five key survival variables were identified: serum alpha-fetoprotein (AFP), red cell distribution width coefficient of variation (RDW-CV), aspartate aminotransferase (AST), platelet-to-lymphocyte ratio (PLR), and extrahepatic metastasis (EHM). The Cox model had a C-index of 0.694 in the training and 0.691 in the validation set. The training set's 1-year, 2-year, 2.5-year, and 3-year AUC values were 0.77, 0.73, 0.73, and 0.72, respectively, and 0.74, 0.74, 0.76, and 0.66 in the validation set. The ARAPE model demonstrated a more significant net benefit than existing models. Median overall survival (mOS) was 33.9 months for low-risk, 20.4 months for intermediate-risk, and 14.2 months for high-risk groups in the training set; in the validation set, mOS was 30.8, 18.7, and 10.5 months, respectively. **Conclusions:** The SPEAR model (Serum AFP, PLR, EHM, AST, and RDW-CV) model effectively stratifies risk for HCC patients receiving TACE combined with MTTs and ICIs, aiding in personalized treatment decisions. Research Sponsor: National Key Research and Development Program of China; National Natural Science Foundation of China; 82102879; National Natural Science Foundation of China; Natural Science Foundation of Guangdong Province; Science and Technology Projects in Guangzhou.

534 **Poster Session**

Characterization of the real-world first line (1L) use of single dose tremelimumab regular interval durvalumab (STRIDE), for treatment of unresectable hepatocellular carcinoma (uHCC), using two electronic medical record (EMR) databases. First Author: Aparna Chauhan, AstraZeneca, Gaithersburg, MD

Background: In October 2022, the US FDA approved the STRIDE (Single Dose Tremelimumab Regular Interval Durvalumab) regimen, comprising tremelimumab 300 mg for 1 dose plus durvalumab 1500 mg every 4 weeks, for the 1L treatment of uHCC in adults aged ≥18 years. Thus far there is limited data characterizing STRIDE's 1L use in the real world. This observational study aims to describe patient characteristics, and clinical profiles of a 1L STRIDE cohort. **Methods:** A retrospective cohort study was conducted using EMR data from two community oncology practice networks, Florida Cancer Specialists and American Oncology Network, from January 1, 2018, to March 31, 2024. Adults with HCC who received ≥1 administration of the STRIDE regimen as 1L treatment following diagnosis were included. Patients with evidence of other cancers or systemic therapies pre-index date were excluded, to establish a cancer-naïve cohort at index. Index date was first administration of STRIDE. The analysis described the patient characteristics and clinical profile at index date. Patient demographics, severity indicators (cancer stage, albumin-bilirubin index (ALBI), alpha fetoprotein (AFP), Eastern Cooperative Oncology Group (ECOG)) and lab data were assessed on or closest to index date. **Results:** 79 patients initiated the 1L STRIDE regimen. The median (P25, P75) age at index was 70 (63.2-76.4) years. 72.2% were males, 64.6% were white, and 69.6% were overweight or obese patients. 95% had their first HCC diagnosis between 2022-2024 and the mean time from diagnosis to index date was 3.8 months (SD: 10.2). Among patients with reported cancer stage (n=57), most (82.5%, n=47) were diagnosed with stages IIIA to IVB. 18% had ECOG scores ≥2. 86% had ALBI grades between 2-3; 42.1% had AFP level ≥400 ng/ml; 30.4% had cirrhosis (among those that reported comorbidities) and 24.1% received any loco-regional therapy. **Conclusions:** Our study provides a real-world profile of patients on 1L STRIDE. Compared to the HIMALAYA trial, these findings suggest that patients in this real-world study tend to be older, diagnosed at late-stage, have a poor liver profile, and potentially worse prognosis over time. Further research is required to explore treatment pathways after index date, over a long follow-up, which may provide more insight into the continuity and effectiveness of IL STRIDE in the real-world. Research Sponsor: AstraZeneca.

535 **Poster Session**

The impact of protein-energy malnutrition on outcomes in patients with cholangiocarcinoma admitted with sepsis: Insights from a national database. First Author: Adamsegd Isac Gebremedhen, Marshall University Joan C. Edwards School of Medicine, Department of Internal Medicine, Huntington, WV

Background: Patients with cholangiocarcinoma (CCA) are prone to infection and related complications. Protein-energy malnutrition (PEM) is common among these patients, arising from either the malignancy or its treatments. PEM is linked to worse medical and surgical outcomes and longer hospital stays across various cancers. The impact of PEM on CCA patients with sepsis, however, remains underexplored. This study aims to fill this gap by assessing how PEM affects in-hospital outcomes in CCA patients admitted with sepsis. **Methods:** We queried the Healthcare Cost and Utilization Project National Inpatient Sample and identified adult patients with CCA admitted with the principal diagnosis of sepsis from 2018 to 2021, using the ICD-10 code. We categorized the cohort based on the presence or absence of PEM and applied complex sampling weights for national representativeness. We compared socio-demographic characteristics and comorbidities between CCA patients with sepsis who have PEM and those who do not. The primary outcome studied was all-cause in-hospital mortality. Secondary outcomes examined were acute kidney injury (AKI), renal replacement therapy (RRT), respiratory failure (RF), invasive ventilation (IV), septic shock, vasopressor requirement (VP), mean length of hospital stay (LOS), and mean total hospitalization charges (THC). Multivariate regression models assessed outcome disparities between groups, adjusting for patient characteristics and comorbidities, with statistical significance set at $p < 0.05$. **Results:** We identified a total of 22,680 adults with CCA who were admitted with the principal diagnosis of sepsis from 2018 to 2021. Among these, 7,245 patients (31.9%) had a concurrent diagnosis of PEM. The PEM group had a higher proportion of female (47.3% vs 44.1%, p=0.04) and non-white (41.7% vs 37.2%, p<0.001) patients and patients with Charlson Comorbidity Index > 3 (86.5% vs 83.3%, p<0.001). Patients with PEM had higher odds of all-cause in-hospital mortality with an adjusted odds ratio (aOR) of 1.26 (95% confidence interval (CI) 1.05-1.51). They also had higher odds of AKI with an aOR of 1.32 (95% CI 1.16-1.50), RF with an aOR of 1.30 (95% CI 1.09-1.51), and septic shock with an aOR of 1.22 (95% CI 1.06-1.40). Patients with PEM also had a higher LOS (9.0 vs 6.3 days) with an adjusted incidence rate ratio (aIRR) of 1.41 (95% CI 1.33-1.49) and THC ($115,971 vs $83,176) with aIRR of 1.35 (95% CI 1.25-1.45). **Conclusions:** Our study revealed that CCA patients with PEM admitted with sepsis had significantly higher odds of in-hospital mortality, AKI, RF, and septic shock compared to those without PEM. They also experienced longer hospital stays and higher hospitalization charges. These findings highlight the urgent need to address PEM in CCA patients and tailor inpatient management to reduce complications in this high-risk group. Research Sponsor: None.

536 **Poster Session**

Real-world impact of matched targeted therapy on survival in advanced biliary tract cancer: An international collaborative study. First Author: Binbin Zheng-Lin, Department of Hematology and Oncology, Mayo Clinic Arizona, Phoenix, AZ

Background: The prognosis of advanced BTC remains poor. Although targeted agents have expanded late-line options, many patients (pts) become ineligible for these therapies after progressing on first-line treatment. Herein, we examined the real-world impact of targeted therapy on survival in advanced BTC. **Methods:** In this international collaborative study between the Mayo Clinic (US) and the National Cancer Center East (Japan), we included pts with advanced BTC with comprehensive tumor molecular profiling who had received at least 30 days of systemic therapy. Pts were categorized into 3 groups: 1) non-actionable, 2) pts with alterations who received matched targeted therapy at any point during their treatment (matched group), and 3) those with mutations who did not receive matched therapy (unmatched group). OS was calculated from the start date of first-line therapy until death and compared among three subgroups using Cox regression models. **Results:** Of 932 pts, 366 (39.3%) had an actionable alteration, and 135 of these received matched therapy. 566 (61.7%) did not have an actionable alteration (Table). The most common alterations were somatic *BRCA1/2* (N=106, 11.4%), *HER2* amplification (N=65, 7%), *FGFR2* fusion (N=61, 6.5%), *KRAS G12C/D* (N=46, 4.9%), and *IDH1* (N=38, 4.1%). 78.7% of *FGFR2* fusions were found in Caucasian pts, whereas *BRCA1/2* (97.5%), *HER2* (78.5%), and *KRAS G12C/D* (67.4%) alterations were more prevalent in Asian pts. The median OS was significantly longer in the matched group (21.4 months; 95% CI 18.4-27.9) compared to the unmatched group (14.6 months; 95% CI 12.8-17.6) and the non-actionable group (17.2 months; 95% CI 15.5-19.0). The 36-month OS rate was 33% in the matched group vs. 7% in the unmatched group ($p < 0.0001$). After adjusting for age, gender, prior surgical resection, primary tumor location, and country of origin, matched targeted therapy remained a strong independent predictor for improved OS (HR 0.61; 95% CI 0.48-0.79). **Conclusions:** These real-world findings suggest that matched targeted therapies significantly improve survival in advanced BTC, underscoring the need to incorporate them earlier in the treatment course and address barriers to treatment access. The regional prevalence of actionable alterations should also be considered when developing new targeted therapies. Research Sponsor: None.

Patient baseline characteristics stratified by actionable alterations and targeted therapy.

	Non-actionable (N=566)	Unmatched group (N=231)	Matched group (N=135)
Site, n (%)			
Japan	409 (72%)	166 (72%)	50 (37%)
US	157 (28%)	65 (28%)	85 (63%)
Age at Dx > 65, n (%)	324 (57%)	122 (53%)	54 (40%)
Female, n (%)	235 (42%)	92 (40%)	77 (57%)
Primary tumor location (category), n (%)			
Intrahepatic	217 (43%)	115 (55%)	99 (76%)
Extrahepatic	150 (30%)	39 (19%)	17 (13%)
Gallbladder	142 (28%)	54 (26%)	14 (11%)
Prior surgical resection, n (%)	220 (42%)	74 (34%)	48 (37%)

537 **Poster Session**

Real-world (RW) experience with atezolizumab + bevacizumab (A+B) for the treatment of unresectable HCC (uHCC): A multicenter study. First Author: Maen Abdelrahim, Houston Methodist, Houston, TX

Background: A+B is the preferred first-line (1L) standard of care for uHCC, with emerging RW evidence supporting its use in a broad patient (pt) cohort. We evaluated pt characteristics, clinical and treatment outcomes in pts treated at five U.S. institutions: Mayo Clinic, Houston Methodist, Moffitt Cancer Center, Mays Cancer Center, and University of Arizona. **Methods:** This is a retrospective study of uHCC pts who initiated A+B after January 1, 2019. De-identified pt data were extracted by treating oncologists. Overall survival (OS) and real-world progression-free survival (rwPFS) were assessed using Kaplan-Meier methods for all pts and for a "trial-like" subgroup with characteristics similar to those in the IMbrave150 trial (ECOG performance status [PS] 0-1, Child-Pugh [CP] class A, albumin-bilirubin [ALBI] grade 1-2). HCC-related hospitalizations were evaluated within 1 year of A+B initiation. **Results:** A total of 300 eligible pts were treated with 1L A+B (median age 68 years, 79% male, 79% White, 82% BCLC C, 44% viral etiology,12% ECOG PS >1). Liver function was CP A in 73% (36% A5, 31% A6, 6% not reported), CP B in 26% pts (15% B7, 7% B8, 5% B9), and CP C in 2 (<1%) pts. Compared with CP A, more CP B pts had characteristics of compromised liver function (CP A vs B: cirrhosis: 61% vs 91%; ascites: 7% vs 46%; encephalopathy: 4% vs 17%; esophageal varices: 14% vs 32%), bile duct invasion (2% vs 11%), and ECOG PS>1 (8% vs 22%), all Ps<0.001. Over a median follow-up of 8.7 mos, 244 (81%) pts discontinued A+B, primarily due to disease progression. Toxicity-related treatment discontinuation and hospitalizations due to treatment-related adverse events (TRAE) were similar between the two groups (Table). Median OS (mOS) was 14.4 mos (95% CI: 12.3, 18.2) and median rwPFS was 6.8 mos (95% CI: 5.8, 8.4). In the trial-like subgroup (n=194), mOS was 19.5 mos (95%CI: 14.6, 24.7) and median rwPFS was 8.8 mos (95% CI: 7.6, 12.1). In the CP B subgroup, mOS was 5.6 mos (95% CI: 4.6, 11.3) and median rwPFS was 3.1 mos (95% CI: 2.4, 5.8). **Conclusions:** This study demonstrates the RW outcomes of 1L A+B in a diverse pt cohort. Results from the "trial-like" pts further confirm the reproducible efficacy of A+B in clinical practice. Although pts with CP-B had higher hospitalization rates due to disease progression or symptoms, consistent with their underlying liver complications, they had similar rates of toxicity-related treatment discontinuation and TRAE-related hospitalizations compared to CP-A. Further characterization of safety for A+B in CP B pts within a prospective, controlled trial is warranted. Research Sponsor: F. Hoffman La-Roche Ltd.

N (%)	CP A (n=219)	CP B (n=79)
Treatment discontinuation*	167 (76)	75 (95)
Atezolizumab-related toxicity	20 (12)	13 (17)
Bevacizumab-related toxicity	26 (16)	16 (21)
Hospitalization*	87 (40)	58 (73)
Symptom-related	42 (48)	35 (60)
Progression	8 (9)	11 (19)
TRAE	7 (8)	6 (10)

*P-value comparing CP A vs CP B <0.001.

538 Poster Session

Financial toxicity in patients with newly diagnosed hepatocellular carcinoma. First Author: Seohyuk Lee, Beth Israel Deaconess Medical Center, Boston, MA

Background: Hepatocellular carcinoma (HCC) remains a leading cause of cancer-related deaths and has been associated with significant patient (pt) financial liability, however no study has yet examined pt-reported measures of financial toxicity (FT) in this population. We sought to assess pt-reported FT in newly diagnosed HCC and to identify pt-, disease-, and treatment-related factors associated with FT. **Methods:** Pts with HCC were recruited prior to treatment initiation during their initial visit at a multidisciplinary liver cancer clinic at a tertiary care center between January 2023-May 2024. The Comprehensive Score for Financial Toxicity-Functional Assessment of Chronic Illness Therapy (COST-FACIT), a validated survey for assessing financial distress in pts with cancer (range, 0-44; lower scores reflect worse financial wellness), was completed by participants during their initial visit. Sociodemographic (age, sex, race, ethnicity, primary language, insurance status, employment status, marital status, highest education completed, household income, living arrangement, home zip code), clinical (Barcelona Clinic Liver Cancer stage, performance status, age-adjusted Charlson comorbidity index, albumin-bilirubin score, presence and etiology of chronic liver disease, Child-Pugh grade, MELD-Na score), and planned treatment (surgery, liver-directed therapy, systemic therapy, clinical trial enrollment) characteristics were evaluated by questionnaires or medical record review. The EORTC QLQ-HCC18 was used to assess pt-reported quality of life (QOL). Area Deprivation Indices were calculated using pts' home zip codes. Mean COST-FACIT scores were compared using two-sided t test for binary variables and one-way ANOVA for non-binary variables. **Results:** Among 47 pts enrolled, 35 (74%) completed the COST-FACIT survey. Younger age ($<$65 vs $\geq$65 yrs; mean, 20.3 vs 27.8; p=0.05), non-English primary language (non-English vs English; mean, 16.2 vs 26.7; p=0.04), less completed education (high school or earlier vs college or beyond; mean, 23.3 vs 32.0; p=0.02), and worse self-reported QOL (EORTC QLQ-HCC18 score $\geq$cohort median vs $<$cohort median; mean, 21.4 vs 29.2; p=0.02) were significantly associated with worse FT. Although not statistically significant, Hispanic ethnicity (Hispanic vs not Hispanic; mean, 16.1 vs 26.4; p=0.07), lack of employment (not employed vs retired vs employed; mean, 15.5 vs 29.4 vs 24.3; p=0.07), and no plan for liver-directed therapy (no vs yes; mean, 17.9 vs 26.7; p=0.07) trended towards worse pt-reported FT. **Conclusions:** Among pts with newly diagnosed HCC, significantly higher pt-reported FT is associated with younger age, non-English primary language, less completed education, and worse QOL. Efforts are underway to longitudinally follow our study cohort to assess the course of self-reported FT over 1-year follow-up as well as its associations with treatment response and survival. Research Sponsor: None.

539 Poster Session

Efficacy and safety of durvalumab plus tremelimumab for unresectable hepatocellular carcinoma in Thailand: A real-world multicenter observational study. First Author: Krittiya Korphaisarn, Division of Medical Oncology, Department of Medicine, Faculty of Medicine Siriraj Hospital, Mahidol University, Bangkok, Thailand

Background: Durvalumab plus tremelimumab (Durva/Treme) andatezolizumab plus bevacizumab are preferable first-line therapy for unresectable hepatocellular carcinoma (uHCC). This study aimed to evaluate the efficacy and safety of Durva/Treme in real world setting. **Methods:** Fifty patients with uHCC who enrolled in the expanded access program (EAP) for Durva/Treme as the first-line treatment from 12 centers in Thailand were included. Analysis was assessed for objective response rate (ORR), survival, and adverse events. We also compared data with the HIMALAYA study. **Results:** Median follow-up time was 9.8 months. There were 24 patients alive and 26 patients deceased. The median age was 62 years. Majority of the patients were male (80%). Hepatitis B, C and non-viral cause were identified in 44%, 30%, and 26%, respectively. Ninety-two percent of patients had Child-Pugh A, while 8% had Child-Pugh B(7). Macrovascular invasion was found in 24% of cases. The full data on efficacy and safety were summarized in table below. **Conclusions:** To date,this is the largest real-world data of Durva/Treme in the first-line treatment of uHCC. Our study demonstrated that Durva/Treme was effective but had lower ORR and higher liver toxicities than what reported in the HIMALAYA study. Research Sponsor: None.

	EAP (N=50)		HIMALAYA (N=393)	
Response by RECIST1.1, n (%)	2(4)		12(3.1)	
CR	4(8)		67(17)	
PR	24(48)		157(39.9)	
SD	18(36)		157(39.9)	
PD	2(4)		-	
Not evaluable				
ORR (%)	12		20.1	
DCR (%)	60		60.1	
mPFS (range), mo	5.36(0.2-12.5)		3.78(3.7-5.3)	
mOS (range), mo	NA		16.4 (14.2-19.6)	
Adverse events, n (%)	**Any grade**	**Grade > 3**	**Any grade**	**Grade > 3**
Any AE	37(74)	12(24)	378(97.4)	196(50.5)
Diarrhea	3(6)	1(2)	103(26.5)	17(4.4)
Rash	12(24)	0	87(22.4)	6(1.5)
AST elevation	23(46)	6(12)	48(12.4)	20(5.2)
ALT elevation	21(42)	7(14)	36(9.3)	10(2.6)
Increase bilirubin	8(16)	4(8)	20(5.2)	3(0.8)
Hypothyroidism	7(14)	0	47(12.1)	0
Myocarditis	1(2)	1(2)	-	2(0.5)
Myositis	1(2)	1(2)	-	3(0.8)

540 Poster Session

Prevalence and clinical characteristics of patients with brain metastases at diagnosis with advanced hepatocellular carcinoma in a retrospective registry. First Author: Quincy Harris, Helen Diller Family Comprehensive Cancer Center, University of California, San Francisco, San Francisco, CA

Background: Hepatocellular carcinoma (HCC) is a cancer with poor prognosis and rising incidence. The combination of atezolizumab plus bevacizumab (A+B) prolongs survival as 1st line therapy in advanced HCC (aHCC) but confers a rare risk of serious bleeding. HCC brain metastases (BM) have high hemorrhage rates and contraindicate the A+B regimen. Alternative immunotherapies also prolong survival without increased bleeding risk. We conducted a retrospective review of a large, diverse, cancer center registry to estimate the prevalence of and identify clinical characteristics associated with BM in aHCC. **Methods:** The University of California, San Francisco (UCSF) Helen Diller Family Comprehensive Cancer Center (HDFCCC) registry was queried for new cases of HCC between 2004 and 2022. AJCC stage III or IV were defined as aHCC. Cases with central nervous system (CNS) metastases (including brain as well as skull, face, or orbit as adjacent high-risk sites) at diagnosis were identified. Key clinical covariates including demographics, liver disease etiology, and symptomatology were described. **Results:** Among 4002 new HCC cases, 832 were classified as aHCC. 12/832 (1.4%, 95% CI: 0.8, 2.5) had synchronous BM. Key demographics are displayed in Table. 10/12 (83%) patients with BM were symptomatic, including headache (25%), neurologic deficits (67%), and/or syncope (8%). Bleeding complications and/or hemorrhagic features were reported in 17%. **Conclusions:** BM are rare at diagnosis with aHCC, present in only 1.4% of this registry dataset. The majority (83%) were symptomatic, suggesting that routine CNS imaging may not have clinical utility in asymptomatic patients. A high proportion with BM at diagnosis had viral etiology. Limitations of this retrospective registry analysis include potential for underdiagnosis in asymptomatic patients and incomplete data for clinical covariates. Research Sponsor: None.

Clinical characteristics of aHCC cases with BM at diagnosis between 2004-2022 in UCSF HDFCCC Registry.

	Overall aHCC Cases (N=832)	Cases with BM (n=12)
Median age (range) (yrs.)	61 (8-92)	60 (48-65)
Race/ethnicity (%)		
Asian	29	25
Black	10	8
Caucasian	57	67
Hispanic (any race)	15	17
Gender (%)		
Male	79	75
Female	21	25
Etiology (%)		
Viral hepatitis	29	75
Unknown/negative	71	25
Symptomatic BM (%)	NA	83
Headache		25
Neurologic deficits		67
Syncope		8
Hemorrhagic BM (%)	NA	17

541 Poster Session

Regorafenib combined with immunotherapy and prognosis of unresectable hepatocellular carcinoma. First Author: Tianxiang Li, The Affiliated Hospital Of Qingdao University, Qingdao, Shandong, China

Background: To investigate the prognosis and influencing factors of patients with unresectable hepatocellular carcinoma (uHCC) receiving regorafenib in different treatment modes. **Methods:** A total of 227 patients with uHCC who received at least 2 cycles of regorafenib treatment in the Affiliated Hospital of Qingdao University between June 2018 and August 2022 were retrospectively collected through the Hospital Information System (HIS). 204 patients were finally enrolled by inclusion and exclusion criteria after screening. **Results:** The median follow-up was 25.9 (4.5-69.7) months. The median overall survival (OS) of 204 patients with uHCC was 27.7 (95%CI 21.98-33.42) months. The 1, 2, 3 and 5-year OS were 80.9%, 55.9%, 41.4% and 26.7%, respectively. Univariate analysis showed that combination with immune checkpoint inhibitor (ICI) and metabolic-associated complications were independent factors for survival benefit of regorafenib, while the Eastern Cooperative Oncology Group Performance Status (ECOG-PS) scale = 2, undecreased Alpha-fetoprotein (AFP), macrovascular invasion and extrahepatic metastasis were independent risk factors for prognosis ($P<0.05$). Compared with 60 patients receiving regorafenib alone, 144 patients receiving regorafenib combined with ICI had a significant survival benefit (median OS, 16.9 vs. 31.5 months, respectively, P=0.006). The median OS of 153 patients receiving second-line therapy with regorafenib was 30.3 months. Among 51 patients receiving third or more lines of therapy with regorafenib, compared with 4 patients receiving regorafenib alone, 47 patients receiving regorafenib combined with ICI had a significant survival benefit (median OS, 9.4 vs. 24.1 months, respectively, P=0.000). Among 148 patients receiving local treatment, the median OS of patients with regorafenib combined with ICI (n=115) or without ICI (n=33) was 33.2 and 16.5 months, respectively (P=0.005). **Conclusions:** Regorafenib has a good prognosis in second-line treatment of uHCC. The OS of patients with regorafenib combined with ICI were significantly better than that of patients without ICI. The OS benefit of regorafenib combined with ICI was also equivalent in patients with third or more lines. The benefit of regorafenib combined with ICI was better for patients receiving local treatment. Research Sponsor: None.

542 **Poster Session**

Concurrent transarterial chemoembolization plus atezolizumab and bevacizumab in unresectable hepatocellular carcinoma: Interim analysis from a multicenter real-world study (CHANCE 023). First Author: Rong Yan, Department of Radiology, Zhongda Hospital, Center of Interventional Radiology and Vascular Surgery, Medical School, Southeast University, Nanjing, Jiangsu, China

Background: Combining transarterial chemoembolization (TACE) with systemic therapies such as atezolizumab and bevacizumab (T+A) has shown potential efficacy in patients with unresectable Hepatocellular carcinoma (HCC) This interim analysis aims to evaluate the efficacy and safety of this combination therapy when administered concurrently within a two-month period. **Methods:** This multicenter, real-world study included patients diagnosed with unresectable HCC who received TACE combined with T+A from January 2020 to June 2024, across 37 centers in mainland China. The primary endpoint was overall survival (OS). This trial is registered with ClinicalTrials.gov, NCT06024252. **Results:** Out of a cohort of 335 patients, 293 (259 males) met the inclusion criteria, and 37.5% (112/293) adopted this regimen as second-line therapy. The interim analysis showed a median OS of 29.97 months and a one-year survival rate of 76.50%. The median PFS was 19.23 months, and the objective response rate (ORR) was 41.6%, with 26 complete responses and 96 partial responses. The disease control rate (DCR) was 84.6%. Common adverse events included bone marrow suppression, decreased white blood cell or platelet counts, among others. Serious adverse events occurred in 9.2% of patients (27/293) and were manageable with standard interventions. **Conclusions:** This study is a real-world analysis to highlight the potential benefits of combining TACE with T+A in this patient population. However, longer follow-up is required to better assess both overall survival and the long-term safety profile of this treatment strategy. Clinical trial information: NCT06024252. Research Sponsor: None.

543 **Poster Session**

Organ-specific responses to nivolumab plus ipilimumab in patients with advanced hepatocellular carcinoma: A multicenter, retrospective study. First Author: Youngun Kim, CHA University School of Medicine, Seongnam, South Korea

Background: The Check-Mate 9DW trial demonstrated that combining ipilimumab with nivolumab (Ipi/Nivo) improved survival outcomes compared with sorafenib or lenvatinib in advanced hepatocellular carcinoma (aHCC). Accumulated evidence suggests that immune checkpoint inhibitor (ICI) monotherapy elicits limited intrahepatic responses in patients with aHCC. Here, we aimed to examine the organ-specific objective response rate (OSORR) of Ipi/Nivo compared with that of nivolumab monotherapy (Nivo) and to evaluate the OSORR of Ipi/Nivo based on the prior use of ICI treatment. **Methods:** We conducted a retrospective assessment of 69 Ipi/Nivo-treated patients and 164 nivolumab-treated patients for aHCC at five referral hospitals in Korea. Only patients classified as Child-Pugh Class A were included. Organ-specific response criteria were adopted from RECIST 1.1, according to the indicated primary sites, including the liver, lungs, lymph nodes and other sites of metastasis. **Results:** Baseline characteristics of the Ipi/Nivo and Nivo groups were comparable. However, 65.2% of patients in the Ipi/Nivo group had prior ICI exposure, compared with only 4.3% in the Nivo group. The objective response rate (ORR) of Ipi/Nivo was 29.0%. The Ipi/Nivo combination therapy elicited OSORRs for 204 individual lesions: 18.0% in the liver, 17.7% in the lungs, 30.0% in the lymph nodes, and 12.5% in other sites. Within the Ipi/Nivo group, the ORR was 45.8% for patients without prior ICI exposure and 20.0% for those with prior ICI exposure. Patients without prior ICI exposure demonstrated OSORRs for 72 individual lesions: 29.0% in the liver, 31.3% in the lungs, 33.3% in the lymph nodes, and 23.1% at other sites. Conversely, patients with prior ICI exposure exhibited OSORRs for 132 individual lesions: 11.5% in the liver, 11.4% in the lungs, 27.8% in the lymph nodes, and 7.4% at other sites. Patients who exhibited a response in one organ were more likely to demonstrate responses in other organs irrespective of the organ type, with only two patients exhibiting variable responses across different organs. Furthermore, these responders had improved survival outcomes, including longer progression-free survival and overall survival, compared with non-responders. In terms of nivolumab monotherapy, the ORR was 21.3%. Nivo yielded OSORRs for 305 individual lesions: 13.5% in the liver, 25.3% in the lungs, 39.3% in the lymph nodes, and 18.4% at other sites. **Conclusions:** In contrast to lesions of patients who received Nivo, intrahepatic lesions of Ipi/Nivo-treated patients without prior ICI exposure exhibited favorable responses, comparable with treatment responses of extrahepatic lesions. Conversely, Ipi/Nivo-treated patients with prior ICI exposure had reduced organ-specific responses across all organs when compared with those without prior ICI exposure. Research Sponsor: None.

544 **Poster Session**

Real-world assessment of MDM2 amplification and survival among patients with advanced or metastatic biliary tract cancer in the United States. First Author: Renuka Iyer, Roswell Park Comprehensive Cancer Center, Buffalo, NY

Background: There are few real-world studies on biliary tract cancer (BTC), especially those related to biomarkers for potential novel targeted therapies. The aim of this study was to provide additional information of the prevalence of biomarkers, treatment patterns, and clinical outcomes of patients with BTC in the US, with a focus on patients with mouse double minute 2 homolog (*MDM2*) amplified, tumor protein 53 wild type (TP53 WT) BTC. **Methods:** This was an observational cohort study using deidentified data from the Flatiron Health-Foundation Medicine BTC Clinico-Genomic Database, which included patients with advanced or metastatic BTC who underwent genomic analysis and received antineoplastic therapy in the US between January 1, 2013 and June 30, 2023. Three cohorts were created: 1) 1L+ cohort that included all patients that initiated 1st line (1L) therapy during the study period, 2) 2L+ cohort that included all patients that initiated 2nd line (2L) therapy during the study period, and 3) *MDM2* amp + TP53 WT cohort that included patients in the 1L+ cohort that had *MDM2* amplified ($\geq$ 8 copies), TP53 WT. The prevalence of select biomarkers was assessed, as well as treatment patterns, and clinical outcomes including time to next therapy (TTNT) and overall survival (OS). **Results:** A total of 1,548 patients met the selection criteria for the 1L+ cohort, of whom, 875 initiated 2L therapy (2L+ cohort), and 55 had *MDM2* amplified, TP53 WT BTC (*MDM2* amp + TP53 WT cohort). In the 1L+ cohort, 66 (4.3%) had *MDM2* amplification and 872 (56.3%) had TP53 WT. The most common treatment regimens were cisplatin/gemcitabine (44.5%, 50.7%, 41.8% of patients in the 1L+, 2L+, and *MDM2* amp + TP53 WT cohorts, respectively) at 1L and FOLFOX (12.9%, 22.9%, 16.4% of patients, respectively) at 2L. The median (95% CI) OS from start of 1L was 10.7 (10.0 – 11.3) months for the 1L+ cohort and 11.9 (10.4 – 17.7) months for the *MDM2* amp + TP53 WT cohort. The median OS from the start of 2L was 8.0 (7.2 – 8.8) months for 2L+ cohort. Furthermore, the median TTNT was similar between 1L+ cohort and *MDM2* amp + TP53 WT cohort, 5.3 (5.0 – 5.7) and 4.8 (4.0 – 7.6) months, respectively. The median TTNT from the start of 2L in the 2L+ cohort was 4.3 (3.8 – 4.6) months. We will present co-mutation data in *MDM2* amp + TP53 WT cohort. **Conclusions:** The *MDM2* amplification and TP53 WT status does not appear to be a prognostic biomarker in advanced/metastatic BTC with similar median TTNT and OS in the 1L+ and *MDM2* amp + TP53 WT cohorts. Targeted treatments may help improve OS among these patients and thus further research into these treatments and how they impact survival is warranted. Research Sponsor: Boehringer Ingelheim.

545 **Poster Session**

Real-world insights into immune-related adverse events (irAE) in gastrointestinal (GI) malignancies. First Author: Ashish Manne, The Ohio State University Comprehensive Cancer Center, Columbus, OH

Background: This retrospective review aimed to examine the incidence and baseline characteristics (BLC) that irAE risk in patients with gastrointestinal (GI) cancers undergoing immune checkpoint inhibitor (ICI) therapy at our institution. **Methods:** GI cancer patients who received at least one dose ICI-only between 1/1/2017 and 7/31/2021 at the Ohio State University were included in the study. The BLC including hematological and chemistry labs were extracted with immune-related adverse events (irAE) details. Statistical analyses included descriptive statistics, chi-square test, logistic regression, and matched pairs t-tests were performed. **Results:** Our analysis included 198 patients (14 anal (AI), 65 hepatobiliary (HPB), 46 lower (LGI) and 74 upper (UGI)) with irAEs reported in 37 patients (19%). Median age of irAE-group was 67 years (range: 26 to 88) with 54% males and 86% Caucasians. Most of them (n=36) in had single agent ICI. Pneumonitis was most frequently (n=7) noted followed by dermatitis (n=), colitis (n=5), hypophysitis (n=5), and hepatitis (n=4). Relatively rare ones were hypothyroidism (n=3), nephritis (n=2), encephalitis (n=2), diabetes mellitus (n=1), neutropenia (n=1), and myasthenia gravis (n=1). In irAE-group, 18 (49%) required hospitalization ($\geq$ grade 3) and 5 (14%) had irAE-related deaths. Complete recovery from irAEs was reported in 20 patients (54%), with 13 patients (35%) restarting ICI therapy. The irAE incidence in HPB group was significantly higher than other GI malignancies (HPB vs. LGI vs. AI vs. UGI = 46% vs. 24% vs. 16% vs. 14%, p=0.004). History of asthma (41% vs. 16% (p=0.004) and female gender (26% vs. 15%, p=0.04) increased irAE-risk; presence of liver (9 % 22%, p=0.04) and $>$ 3 lung (8 vs. 21%, p=0.04) metastasis reduced the risk. Compared to non-irAE group, the irAE-group had significantly lower baseline (mean) whole blood cell count (WBC: 6.3 vs. 7.6 K/uL, p=0.02), neutrophil count (NC: 4.2 vs. 6.8 K/uL, p=0.04), and neutrophil-lymphocyte-count (NLR: 5.1 vs. 9.3, p=0.03), and lower albumin (3.7 vs. 3.5 mg/dL, p=0.02). A significant rise in NC (2.1 K/uL, p=0.0008), NLR (by 2.5, p=0.004), WBC (by 2.1 K/uL, p=0.002) from baseline was noted at irAE-incidence. Heterogeneity in the primary tumors and treatments received is a notable limitation of this analysis. **Conclusions:** Overall, the risk of irAEs in patients with GI cancers is relatively low, and certain BLC can predict their incidence. Research Sponsor: None.

546 **Poster Session**

Zanidatamab dose optimization in patients with HER2-positive biliary tract cancer (BTC). First Author: Sheryl Trueman, Jazz Pharmaceuticals, Palo Alto, CA

Background: Zanidatamab (zani) is a dual HER2-targeted bispecific antibody that binds 2 distinct HER2 domains, resulting in receptor crosslinking, clustering, and internalization as well as leading to reductions in cell-surface levels of HER2, phosphorylation of HER family members (EGFR, HER2, and HER3), downstream signaling, and ligand-dependent and -independent proliferation. Zani also induces complement-dependent cytotoxicity as well as antibody-dependent cellular cytotoxicity and phagocytosis. In the phase 2 HERIZON-BTC-01 study (Study 203), zani 20 mg/kg Q2W demonstrated a 41.3% confirmed objective response rate (cORR) and a manageable safety profile in patients (pts) with previously treated HER2+ BTC (IHC3+/2+ and ISH+). The objective of this analysis is to report optimal dose selection for zanidatamab in patients with HER2+ BTC through pharmacokinetic modeling. **Methods:** Source data included the phase 1 ZW25-101 study (Study 101; NCT02892123), Study 203 (NCT04466891), and *in vitro* study for ligand-dependent growth inhibition (LDGI) in HER2-expressing human cancer cell lines. In the LDGI study, cancer cells were cultured in media containing epidermal growth factor and were treated with different zani concentrations. The IC90 value for LDGI was obtained by nonlinear regression using the logistic dose-response model. To assess target saturation, the population pharmacokinetic model was used to simulate clearance at steady state following 5-30 mg/kg Q2W. The clinical utility of zani was evaluated based on the correlation of zani concentrations with both efficacy and safety. **Results:** Study 101 included 192 patients with HER2-expressing solid tumors (dose ranged from 5-30 mg/kg including QW, Q2W, and Q3W). The Ctrough values following the first zani 5 mg/kg QW dose were below the IC90 (25.0 μg/mL) for LDGI, and no confirmed responses were observed at this dose. The Ctrough values following the first zani dose of 10 mg/kg QW or 20 mg/kg Q2W were above IC90 and the 2 dose levels demonstrated a cORR of 25% and 28.6%, respectively. The simulated clearance suggests that the target-mediated elimination pathway was saturated at the dose level of 20 mg/kg Q2W, since clearance was comparable to that of the higher dose of 30 mg/kg Q2W. The efficacy and safety of 20 mg/kg Q2W was further evaluated in 87 pts with HER2-amplified BTC enrolled in Study 203. The clinical utility analysis based on the data from Study 203 demonstrated that the majority of pts had an exposure range on the plateau part of the efficacy curve. **Conclusions:** The zani dose of 20 mg/kg Q2W is expected to maximize the clinical benefit for pts with HER2+ BTC based on: reaching the desired target exposure (IC90 for LDGI), saturating target-mediated elimination pathway, and the clinical utility analysis of safety and efficacy. Higher exposures are not expected to result in higher responses. Clinical trial information: NCT02892123, NCT04466891. Research Sponsor: Jazz Pharmaceuticals.

547 **Poster Session**

Real-world treatment patterns among U.S. adults with unresectable hepatocellular carcinoma (uHCC) eligible for locoregional therapy (LRT). First Author: Shital Kamble, Merck & Co., Inc., Rahway, NJ

Background: Treatment (tx) in uHCC has mostly relied on locoregional liver-directed tx, such as transarterial chemoembolization (TACE) or radioembolization (TARE), but recent pharmacotherapy advances have provided more tx options. This study aims to describe current real-world tx patterns and clinical outcomes of uHCC eligible for LRT. **Methods:** uHCC patients enrolled in TARGET-HCC (an ongoing, 5-year, longitudinal, observational cohort of HCC patients receiving usual care at academic and community sites in the U.S.), who initiated tx between 6/1/2020-6/1/2023 were included and followed until 12/31/2023 (data cutoff). Patients receiving curative HCC tx (ablation, resection, or transplant) or with metastatic HCC, portal vein involvement, or Child-Pugh-C liver disease at diagnosis (dx) were excluded. Patient and disease characteristics, and txs data were abstracted from electronic health records. Index tx was defined as first tx after dx, with a 30-day grace period for combination txs. Subsequent tx was defined as first tx after the grace period; for index systemic therapies, the subsequent tx differed from the index tx. Real-world progression-free survival (rwPFS) was defined as time from index tx to first event: radiological progression (including new lesions, extrahepatic involvement, vascular invasion, or >20% increase in tumor burden from baseline), death, or hospice entry. Real-world time to progression (rwTTP) was defined as time from index tx to first radiological progression. **Results:** A total of 115 patients were included; mean age was 65.1 years and 77% were male. At dx, 41% of patients were BCLC B, 14% were non-metastatic BCLC C, and 38% were BCLC A with lesion >5 cm, rapidly progressive disease, or contraindications to LRT. Tx initiation with TACE/TARE was common, either alone (n=59, 51%) or in combination with systemic tx (n=5, 4%). Among patients initiating systemic tx (n=46 ,40%), 30 received IO-IO (immuno-oncology) combinations, 12 received IO only, and 4 received a TKI only (tyrosine kinase inhibitor). Following index tx, 21 (18%) patients died receiving no subsequent tx and 27 (23%) were censored at data cutoff. Among those who received subsequent tx (n=67, 58%), there was heterogeneity in tx choice: TACE/TARE (n=32), systemic tx (n=16) other LRT (n=10), multimodal therapy (n=5), and transplant (n=4). We observed 60 progression events and 38 deaths corresponding to a median rwPFS and rwTTP of 7.9 (95% CI: 5.0, 13.3) months and 11.3 (95% CI: 6.7, 28.9) months, respectively. **Conclusions:** Our results showed high variability in management of uHCC eligible for LRT with TACE/TARE commonly utilized. Tx choice may be influenced by factors beyond staging criteria. Prognosis for uHCC eligible for LRT remains poor, and further research on adaptive, multimodal tx strategies is needed to inform tx guidelines and improve outcomes. Research Sponsor: Merck.

548 **Poster Session**

Real-world survival differences in advanced biliary tract cancer patients with ctDNA detected IDH1 mutations and FGFR2 fusions receiving first-line gemcitabine-cisplatin with and without immunotherapy. First Author: Richard D. Kim, H. Lee Moffitt Cancer Center and Research Institute, Tampa, FL

Background: Comprehensive genomic profiling (CGP) is recommended for patients (pts) with metastatic BTC (mBTC) given its aggressive, heterogeneous disease profile. Liquid biopsy is a rapid, non-invasive option with high concordance to tissue-based CGP. First-line (1L) therapy in mBTC is gemcitabine-cisplatin (GemCis) +/- durvalumab or pembrolizumab (IO) regardless of CGP results. However, data suggests pts with targetable alterations may have worse outcomes with addition of IO compared to all comers. We examined outcomes for mBTC pts receiving 1L GemCis vs. GemCis+IO following ctDNA-detected *IDH1* mutation (*IDH1*+) or *FGFR2* fusion (*FGFR2*+). **Methods:** Real-world data was sourced from GuardantINFORM, which comprises aggregated commercial payer health claims and de-identified records from pts with clinical ctDNA testing via Guardant360 (G360). Pts with mBTC and >1 treatment claim after G360 results, and IDH+ or FGFR2+ between April 2019 and June 2024 were analyzed. Only pts treated with 1L GemCis +/- IO were included. Outcomes were assessed via real-world time to treatment discontinuation (rwTTD), real-world time to next treatment (rwTTNT), and real-world overall survival (rwOS), all in months with 95% confidence intervals. Log-rank test was used to compare Kaplan-Meier survival curves. **Results:** 412 pts were *IDH1+* and 154 were *FGFR2+*. In the *IDH1*+ cohort, 184 pts were treated with GemCis and 106 pts with GemCis+IO. Pts receiving GemCis demonstrated improved rwOS vs. GemCis+IO [27.2 (23.6-37.8) vs. 16 (13.6-19.9) mo.; HR=0.4 (95CI 0.27-0.59), p<0.001]; there was no difference in rwTTNT [13 (10.8-14.3) vs. 10 (8.4-12.3); HR=0.85(95CI 0.6-1.22), p=0.4] or rwTTD [4.8 (4.1-5.5) vs. 6.6 (4.7-8.1); HR=1.17 (95CI 0.9-1.5), p=0.25]. Similarly, in the *FGFR2*+ cohort, pts receiving GemCis + IO (n=44) had worse rwOS than pts on GemCis (n=70) [NR (15.6-NR) vs. 43 (34.2-2.1); HR=0.37(95CI 0.18-0.74), p=0.005], with no difference in rwTTNT [8.9 (4.89-NR) vs. 13.78 (7.42-NR); HR=0.63 (95CI 0.36-1.1), p=0.1] or rwTTD [5.0 (3.1-7.8) vs. 4.2 (3.5-5.5); HR=1.08 (95CI 0.72-1.6), p=0.7]. Among all mBTC pts receiving GemCis+IO (n=1113), non-*IDH1*+ pts (n=1007) had numerically better rwOS compared to *IDH1*+ pts (n=106) [18.6 (17.2-21.3) vs. 16.0 (13.6-19.);HR=0.76 (95CI 0.6-1.03), p=0.075]. No survival difference was observed between *IDH1+* (n=184) and non-*IDH1*+ (n=2333) pts receiving GemCis [27.2 (23.6-37.8) vs. 26.2 (24.5-28.2); HR=1.08 (95CI 0.88-1.32), p=0.4]. **Conclusions:** Addition of IO to GemCis in *IDH1*+ and *FGFR2*+ mBTC pts resulted in decreased rwOS. This data supports ctDNA CGP to inform clinical decision-making prior to 1L as well as at progression. Further studies in clinical cohorts are needed to confirm these findings. Research Sponsor: None.

549 **Poster Session**

Real-world experience with atezolizumab plus bevacizumab (A+/-B) in Hispanic patients with advanced hepatocellular carcinoma (HCC) at a Hispanic-majority cancer center. First Author: Jay Parekh, UT Health San Antonio, San Antonio, TX

Background: Combination of atezolizumab (A), a programmed death-ligand 1 (PD-L1) inhibitor, and bevacizumab (B), a vascular endothelial growth factor (VEGF) inhibitor, has demonstrated efficacy for first-line treatment for HCC. However, real-world data on the outcomes of this combination therapy in diverse patient populations, particularly in Hispanic U.S. patients, remains limited. This study aims to evaluate the clinical outcomes of patients intended to treat with A+/-B at a Hispanic-majority U.S. cancer center. **Methods:** This retrospective study of patients diagnosed with advanced unresectable HCC who received treatment with A+B at a Hispanic-serving NCI-designated cancer center. Patients were included if they received at least one dose of A, either as monotherapy or in combination with B, between January 2019 and April 2022. Demographic and clinical data, including age, gender, ethnicity, liver disease etiology, Child-Pugh (CP) classification, comorbidities, and treatment details, were collected. The primary endpoints were the overall response rate (ORR) and complete response (CR) rate. Statistical analyses were conducted to evaluate the association between baseline characteristics and clinical outcomes. The median overall survival (mOS) evaluated by Kaplan-Meier Method. **Results:** 38 patients received at least A (84% A+B, 6 patients had B held due to bleeding risk after consent). Median age 65 years (45-90). 32 (84%) patients were male. 22 (58%) patients were Hispanic. 45% had HCV-related liver disease. 32% had CP B with no statistically significant association between CP Class and ethnicity. 53% had concurrent diabetes. 26% patients had obesity. The median time on treatment was 7.5 mo (1- 24 mo) with no statistical difference between Hispanics and non-Hispanics (9 vs 6 mo, p 0.70). CR 37% with ORR of 42%. Using Fisher's exact test, no association was noted between obesity, ethnicity, treatment option or CP class to occurrence of complete or partial response. mOS was 19 mo. There was no statistical difference in mOS between CP A and CP B (19 vs 10 mo, p value 0.6) and no difference between Hispanics (14 mo) vs non-Hispanics (19 mo) (p 0.5), using log rank test. **Conclusions:** This real-world analysis of a Hispanic-majority U.S. cohort of patients with advanced HCC shows 42% ORR and 37% CR. The findings suggest that this regimen is effective across a diverse patient population, including those with comorbid conditions, such as liver dysfunction (CP B 32%), HCV (45%), diabetes (53%, and obesity (26%). Limitations are the small study population and retrospective nature of the study. Evaluation of toxicity data is ongoing. Further prospective studies are needed to identify biomarkers of outcomes in Hispanic patients with HCC, which is ongoing at our cancer center (NCT03894917). Research Sponsor: None.

550 Poster Session

HCC in the Rio Grande Valley: Outcomes in a predominant Latino community. First Author: Sergio Hernandez Llamas, DHR Health, Edinburg, TX

Background: Hepatocellular carcinoma (HCC) has slowly increased in the U.S. in recent years, with a higher incidence among racial and ethnic minorities, particularly Latinos. This retrospective study examines the different etiologies, presentation of diagnosis, management, and outcomes of patients presented in a multidisciplinary-liver tumor board in the Rio Grande Valley, home of 1.3 million Latinos. **Methods:** A retrospective chart review was conducted on patients with HCC diagnosis who were presented at the Liver Tumor Board of DHR Health Transplant Institute between 2017 and 2023. The data collected was extracted from the EHR, including medical history, laboratory results, and radiological studies required for staging, transplant eligibility assessments, clinical characteristics, and treatments. **Results:** A total of 250 cases were reviewed, and 70 patients with complete clinical data and a diagnosis of HCC were included in the study. Of those, 98.5% were identified as White/ Hispanic. The median age at diagnosis was 64.4 $\pm$ 9.2 years, with 72.9% male and 11.4% uninsured. The average BMI was 31.5 kg/m^2, metabolic-associated liver disease (MALD) was present in 28.6% of patients, 25.7% were alcohol-related, and 38.6% presented multiple etiologies. A descriptive analysis of this population is found in the table. Ablative therapy was used in 68 (97.1%) of the patients. The number of patients sent for liver transplant evaluation was 31 (44%), and only 3 (4.2%) patients received a liver transplant. Barriers to transplant included lack of transportation, inadequate social support, underinsurance, and fear of the transplant itself. **Conclusions:** Our data by staging aligns with the national distribution of incidence cases from 2012-2022 per SEER program. The rate of transplantation at our institution is below the national average, highlighting the need to bridge this disparity and address race-specific barriers to transplant within the Latino community. Research Sponsor: None.

Sociodemographic data.

Parameter	Stage IA N=15 (21.4%)	Stage IB N=20 (28.6%)	Stage II N=10 (14.3%)	Stage IIIA N=8 (11.4%)	Stage IIIB N=9 (12.9%)	Stage IVA N=5 (7.1%)	Stage IVB N=3 (4.3%)	All Stages N=70 (100%)
Child Pugh A	7 (46.7%)	8 (40.0%)	5 (50.0%)	4 (50.0%)	2 (22.2%)	2 (40.0%)	1 (33.3%)	29 (41.4%)
Child Pugh B	6 (40.0%)	6 (30.0%)	5 (50.0%)	2 (25.0%)	5 (55.6%)	2 (40.0%)	2 (66.7%)	28 (40.0%)
Child Pugh C	2 (13.3%)	6 (30.0%)	0 (0.0%)	2 (25.0%)	2 (22.2%)	1 (20.0%)	0 (0.0%)	13 (18.6%)
MELD-Na mean, (min-max)	12.8 (7.0-21.0)	13.6 (6.0-23.0)	12.8 (6.0-20.0)	12.3 (5.0-17.0)	13.6 (8.0-26.0)	13.0 (10.0-19.0)	11.3 (7.0-14.0)	13.6 (6.0-26.0)
Therapy*: Ablative (TACE, Y90, Radiation, other)	15	21	10	6	8	6	2	
Surgery (including transplant)**	1	1	2	2	2	0	0	
Systemic	2	6	5	4	6	3	1	
Overall Survival (months)	29.5	26.8	34.7	17.5	24.1	17.3	14.9	25.3

*Several patients received one or more types of therapy.
**All transplant patients received ablation therapy.

551 Poster Session

Nivolumab plus ipilimumab in advanced hepatocellular carcinoma with Child-Pugh B: A multicenter retrospective study. First Author: Jung Sun Kim, Department of Medical Oncology, CHA Bundang Medical Center, CHA University School of Medicine, Seongnam, South Korea

Background: The phase 3 CheckMate 9DW trial proposed that nivolumab plus ipilimumab (Nivo/Ipi) significantly improved the survival of patients with advanced hepatocellular carcinoma (aHCC) compared with sorafenib or lenvatinib as first-line treatment, but it only included patients with Child-Pugh A. Given the limited treatment options for patients with poor liver function, we aimed to assess the efficacy and safety of Nivo/Ipi in aHCC patients, stratified by liver function. **Methods:** This study included patients with aHCC who received Nivo/Ipi treatment at three referral hospitals between March 2020 and September 2023. Patients received nivolumab (1 mg/kg) plus ipilimumab (3 mg/kg) every 3 weeks (four doses), followed by nivolumab (240 mg) every 2 weeks. **Results:** Of 106 patients treated with Nivo/Ipi, 96 evaluable patients were included in the final analysis. Among them, 76 patients had Child-Pugh A, while 20 had Child-Pugh B. The baseline characteristics were generally comparable between the two groups, except for a significantly higher rate of extrahepatic metastasis in patients with Child-Pugh A (86.8% vs. 65.0%, p = 0.043) and a higher incidence of macrovascular invasion in patients with Child-Pugh B (55.0% vs. 27.6%, p = 0.032). The objective response rate (ORR) was 28.9% (22/76, 4 complete responses [CR] and 18 partial responses [PR]) in the Child-Pugh A group and 10.0% (2/20; 0 CR and 2 PR) in the Child-Pugh B group. Disease control rates were 42.1% in the Child-Pugh A group and 25.0% in the Child-Pugh B group. In a median follow-up duration of 29.3 months (95% confidence interval [CI], 28.8-29.8), the median overall survival (OS) was 8.9 months (95% CI, 5.5-17.6) in the Child-Pugh A group and 3.9 months (95% CI, 2.1-4.8) in the Child-Pugh B group ($p < 0.001$). The median progression-free survival (PFS) was 1.6 months (95% CI, 1.2-3.5) in the Child-Pugh A group and 1.2 months (95% CI, 0.8-2.1) in the Child-Pugh B group (p = 0.003). Adverse events of any grade occurred at a comparable rate between the two groups, although immune-related adverse events were significantly more frequent in the Child-Pugh A group. Multivariable analysis indicated that Child-Pugh B was consistently associated with a worse PFS and OS, while prior exposure to immune checkpoint inhibitors (ICIs) was associated with a poorer PFS. Notably, both responders with Child-Pugh B were both ICI naïve. **Conclusions:** While Nivo/Ipi treatment showed reduced efficacy in patients with Child-Pugh B, identifying potential responders beyond the current clinical trial criteria may offer opportunities to improve outcomes in this challenging population. Research Sponsor: None.

552 Poster Session

Outcomes of durvalumab (D) with or without tremelimumab (T) in routine clinical practice according to HIMALAYA trial eligibility: Preliminary results of the international DT-real study. First Author: Ciro Celsa, Department of Surgery and Cancer, Imperial College, Hammersmith Hospital, London, United Kingdom

Background: HIMALAYA showed that D+T and D are effective options for unresectable hepatocellular carcinoma (uHCC). However, data on outcomes according to the adherence to HIMALAYA inclusion criteria in routine clinical practice are lacking. **Methods:** In the context of a prospectively maintained database including 1293 patients (pts) with uHCC treated with immunotherapy, we analysed pts treated with D+T or D across 8 centres in USA, Asia and Europe. Pts who met >1 key exclusion criterion of HIMALAYA (prior systemic therapy, Child-Pugh class B-C, Vp4 thrombosis) were defined HIMALAYA-OUT and compared with HIMALAYA-IN pts for overall survival (OS), progression-free survival (PFS), objective response rate (ORR) and disease control rate (DCR) by RECIST 1.1 and treatment-related adverse events (TRAEs) per CTCAE v.5.0. **Results:** Up to February 2024, 108 pts (mean age 66 years, male sex 81%) started D+T (n=69, 64%) or D (n=39, 36%). 62 pts (57%) were treated in 1° line and 46 (43%) in >2° line. Child-Pugh class was A in 67 pts (62%). Vp4 was present in 17 pts (16%). 31 pts (29%) were HIMALAYA-IN and 19/31 (61%) received D+T. After a median follow-up of 4.3 months (m, 95%CI 3.3-4.9), median OS (mOS) was 11.5 m and 12-m OS rate was 42%. mOS was not reached in HIMALAYA-IN pts (12-m OS rate 62%) and 8.9 m (95%CI 6.0-12.1) in HIMALAYA-OUT pts. Survival hazard ratio (HR) for HIMALAYA IN vs OUT was 0.28 (95%CI 0.09-0.93, p=0.037). Median PFS was 2.6 m (95% CI 2.2-5.2) overall, 4.6 m (95%CI 2.1-8.5) in HIMALAYA-IN and 2.6 m (95%CI 1.9-5.2) in HIMALAYA-OUT pts (HR 0.70, 95%CI 0.38-1.30, p=0.266). ORR and DCR (evaluable in 53 pts, 49%) were 15.1% (95%CI 6.5-29.7%) and 43.4% (95%CI 27.5-65.1) (Table). Any grade TRAEs occurred in 31.5% (95% 21.8-44.0%), grade 3-4 TRAEs in 8.3% (95%CI 3.8-15.8%), TRAEs requiring systemic corticosteroids in 8.3% (95%CI 3.8-15.8%) and discontinuation due to toxicity in 3.7% (95%CI 1.0-9.5%). **Conclusions:** Preliminary observational data from DT-Real study suggest a reproducible efficacy and safety of D+T and D in pts with uHCC fitting the inclusion criteria of HIMALAYA in routine clinical practice. Research Sponsor: None.

		HIMALAYA IN			HIMALAYA OUT	
	HIMALAYA IN (n=31)	D+T (n=19)	D (n=12)	HIMALAYA OUT (n=77)	D+T (n=50)	D (n=27)
mOS (m, 95%CI)	NR	NR	NR	8.9 (6.0-12.2)	11.2 (6.6-13.2)	4.9 (2.6-12.2)
12-m OS (%)	61.8	63	88.9	37.2	37.4	44.4
mPFS (m, 95%CI)	4.6 (2.1-8.4)	8.5 (2.1-8.5)	2.4 (1.6-2.5)	2.6 (1.9-5.2)	2.4 (1.8-6.7)	2.6 (1.8-5.2)
ORR (%,95%CI) (N=53)	23.1 (4.8-67.4)	25.0 (5.1-73.1)	0	12.5 (4.1-29.2)	13.8 (3.8-35.3)	9.1 (0.2-50.6)
DCR (%, 95%CI)	46.2 (16.9-100)	50.0 (18.3-100)	0	42.5 (24.8-68.1)	41.4 (21.4-72.2)	45.4 (14.8-100)
Any grade TRAEs (%,95%CI)	32.3 (15.5-59.3)	31.6 (11.6-68.7)	33.3 (9.1-85.4)	31.2 (20.0-46.4)	28.0 (15.3-47.0)	37.1 (17.8-68.1)
Grade 3-4 TRAEs (%, 95%CI)	9.7 (2.0-28.3)	15.8 (3.3-46.1)	0	7.8 (2.9-17.0)	4.0 (0.5-14.4)	14.8 (4.0-38.0)

553 Poster Session

Comparison of efficacy and safety of PD-1 and PD-L1 antibodies in combination with hepatic arterial infusion chemotherapy and target therapy for unresectable intrahepatic cholangiocarcinoma: A multicenter retrospective study. First Author: Xinhao Xiong, Department of Liver Surgery, Sun Yat-sen University Cancer Center, Guangzhou, China

Background: Immune checkpoint inhibitors (ICIs) in combination with hepatic arterial infusion chemotherapy (HAIC) and target therapy have shown promising antitumor activity in unresectable intrahepatic cholangiocarcinoma (iCCA). The present study aimed to compare the efficacy and safety of programmed cell death protein 1 (PD-1) inhibitors and programmed cell death ligand 1 (PD-L1) inhibitors in this setting. **Methods:** In this multicenter retrospective study, we included patients with iCCA who received ICIs + HAIC + target therapy from June 2018 to June 2023. Propensity score matching (PSM) was performed to reduce confounding factors. Patients were divided into two groups: those treated with PD-L1 antibody + HAIC + target therapy (Group A) and those treated with PD-1 antibody + HAIC + target therapy (Group B). Progression-free survival (PFS), overall survival (OS), and objective response rate (ORR) were compared between the two groups. Adverse events were assessed according to CTCAE v5.0. **Results:** A total of 336 patients met the inclusion criteria and were enrolled for analysis. There were no significant differences in OS, PFS, or ORR between the two groups (median OS: 17.7 vs. 25.3 months, P=0.139; median PFS: 8.3 vs. 10.0 months, P=0.534; ORR: 48.4% vs. 31.1%, P=0.051). Following PSM (ratio = 3, caliper = 0.2), 30 patients in Group A and 83 patients in Group B were included. The ORR, according to the modified RECIST criteria, was significantly higher in Group A compared to Group B (50% vs. 28.9%, P=0.037). However, no significant difference in PFS (median 10.0 vs. 9.0 months, P=0.29) or OS (median 17.6 vs. 22.8 months, P=0.293) was observed between the two groups. Adverse events (AEs) occurred in all 113 patients (100%), wherein grade 3–4 AEs were reported for 12 (10.6%) patients in group A and 45 (39.8%) in group B. There was a tendency for Group A to have a lower incidence of grade 3–4 AEs compared to Group B (40.0% vs. 54.2%, P=0.182),although this difference was not statistically significant. **Conclusions:** This study found that in the treatment combining ICIs with HAIC and target therapy for unresectable iCCA, Group A demonstrated a superior tumor response and a more favorable safety profile. Research Sponsor: None.

554 Poster Session

Effects of disease aetiology on outcomes for unresectable HCC treated with lenvatanib or atezobev: A real world retrospective study. First Author: Nick Dominelli Whiteley, The Beatson West of Scotland Cancer Centre, Glasgow, Scotland, United Kingdom

Background: Hepatocellular Carcinoma (HCC) is the most common primary liver cancer and a leading cause of cancer-death worldwide.[1] Standard of care first line treatment for unresectable HCC is Atezolizumab plus Bevacizumab (AtezoBev)[1] while the multi-kinase inhibitor Lenvatinib[2] is also available in the first line setting. No randomised studies have compared these treatments directly. Real world data suggests similar overall survival (OS) with AtezoBev possibly superior in patients with a viral hepatitis aetiology and Lenvatinib superior in NAFLD.[3,4,5,6,7] **Methods:** We evaluated patients treated with Lenvatinib or AtezoBev in a retrospective cohort of patients who commenced first-line treatment in the West of Scotland from 04/12/2015 to 28/11/2023. 102 received Lenvatinib and 52 AtezoBev. We collected data on progression free survival (PFS), OS, demographics and disease aetiology- classified primarily as viral hepatitis, acholic liver disease (ALD) or NAFLD. **Results:** AtezoBev patients had improved mOS compared with Lenvatinib (13.8 vs 9. 1 months, p = 0.04). No statistically significant differences were observed in mPFS (5.5 vs 3.5 months, p = 0.31). Of the three aetiologies considered, the only statistically significant ($p<0.05$) difference between the treatments was OS in the NAFLD group (24.1 vs 9.1 months, p = 0.04). When considering patient demographic and disease factors, Child-Pugh score was most significant. Patients with Childs=5 liver disease had much longer OS and PFS compared with Childs≥6 disease. Similarly, patients with ECOG PS=0 had improved survival than PS≥1. There were no statistically significant ($p<0.05$) differences between AtezoBev and Lenvatinib. Age, sex and social deprivation (SIMD) score had minimal effects on survival although high levels of social deprivation and male sex were over-represented in our population. **Conclusions:** In contrast to other real world analyses, our patients demonstrated improved mOS with Atezo/Bev vs Lenvatinib.[3] However, mPFS was similar for the two regimes. The difference in OS may be explained by differences in patient selection, with the Lenvatinib group disproportionately representing patients with a poorer performance status (PS>0) and higher Childs-Pugh scores (CP>5). Furthermore, patients in the AtezoBev group who progressed were often offered Lenvatinib second-line, possibly improving mOS, whereas Lenvatinib patients who progressed were offered either Sorafenib, a clinical trial or best supportive care. However, we cannot rule out a direct survival benefit from the use of AtezoBev vs Lenvatinib in the first line setting. The choice of first line therapy for HCC remains complex and must account for patient preferences such as convenience and side effect profile in addition to observed trial outcomes and the increasing body of real world evidence. 1. Finn RS et. al. Atezolizumab plus Bevacizumab in Unresectable Hepatocellular Carcinoma. N. Engl. J. Med. 2020 May 14;382(20):1894-1905. 2. Kudo, Masatoshi et. al. Lenvatinib versus sorafenib in first-line treatment of patients with unresectable hepatocellular carcinoma: a randomised phase 3 non-inferiority trial. Lancet. 2018. Volume 391, Issue 10126, 1163 – 1173. 3. Wang, BC. et. al. Lenvatinib Versus Atezolizumab Plus Bevacizumab in the First-Line Treatment for Unresectable Hepatocellular Carcinoma: A Meta-Analysis of Real-World Studies. Targ. Oncol. 2024. 19, 203–212. 4. Casadei-Gardini A et. al. Atezolizumab plus bevacizumab versus lenvatinib for unresectable hepatocellular carcinoma: a large real-life worldwide population. Eur. J. Cancer. 2023; 180:9-20. 5. Rimini M et. al. Atezolizumab plus bevacizumab versus lenvatinib or sorafenib in non-viral unresectable hepatocellular carcinoma: an international propensity score matching analysis. ESMO Open. 2022 Dec;7(6):100591. 6. Su CW et. al. Similar efficacy and safety between lenvatinib versus atezolizumab plus bevacizumab as the first-line treatment for unresectable hepatocellular carcinoma. Cancer Med. 2023. 12(6):7077-7089. 7. Rimini, M., et al. Survival outcomes from atezolizumab plus bevacizumab versus Lenvatinib in Child Pugh B unresectable hepatocellular carcinoma patients. J. Cancer. Res. Clin. Oncol. 2023. 149, 7565–7577. Research Sponsor: None.

555 Poster Session

Molecular and clinical determinants of targeted therapy (TT) outcomes in biliary tract cancer (BTC): Analysis of a prospectively maintained next generation sequencing (NGS) biorepository. First Author: Darren Cowzer, Memorial Sloan Kettering Cancer Center, New York, NY

Background: BTCs are genomically diverse. TT have established antitumor activity in a subset of pts that harbour actionable alterations; however, most patients (pts) do not respond to or experience acquired resistance on TT. We sought to explore genomic mechanisms of response across the spectrum of BTCs. **Methods:** We leveraged a prospectively maintained cohort of pts with histologically confirmed BTC who underwent molecular profiling using an FDA authorized, targeted, NGS assay. Additional clinicopathologic information was retrospectively extracted with an aim to describe the genomic profile and clinical actionability using OncoKB, to explore outcomes for pts treated with TT, and to nominate innate and acquired genomic mechanisms of resistance. **Results:** N = 1254 pts had molecular profiling: 61.2% intrahepatic (iCCA), 16.8% extrahepatic cholangiocarcinoma (eCCA), 22.1% gallbladder cancer (GBC). MSI_{high} was observed in 2.1% and the median TMB was 3.3 (range: 0-74.6 muts/mb). Among MSS tumors, the most frequently altered genes were *TP53* (36%), *CDKN2A* (21%), *ARID1A* (18%), *KRAS* (17%), and *IDH1/2* (16%). Genomic profiles differed significantly based on anatomic subsite. Oncogenic drivers were mostly clonal except for *ERBB2* amplifications and mutations, which were subclonal in 19%, and 28% of cases respectively. OncoKB level 1/2 events observed in 32.2% (iCCA 40%, eCCA 15%, and GBC 22%). Progression-free survival (PFS) for pts with these oncogenic drivers: *IDH*, *FGFR2*, *ERBB2, BRAF V600E*, MSI_{high}, and TMB_{high} from start of matched TT (N=164) is listed in the table. Clinical factors or co-occurring genomic alterations did not impact outcomes. Profiling of paired pre- and post-progression samples (N=25) nominated pathway reactivation in *FGFR2*, *BRAF*, *NTRK,* and *ERBB2* driven tumors with emergence of recurrent oncogenic *FGFR2*, *RAS*, *MEK*, and *MET* alterations, respectively, as well as off target *MYC*amplification and *CDKN2A* loss. Loss of the oncogenic driver at progression was only observed for *ERBB2* driven tumors. **Conclusions:** Acknowledging the limitations of the study design, matched TT in a real-world, prospectively maintained cohort offers meaningful PFS, most notably in those pts with MSI_{high} BTC treated with immunotherapy. In those with acquired resistance to TT, alterations predicted to reactivate the primary oncogenic pathway were identified. ERBB2 heterogeneity and loss is predicted to be a mechanism of resistance to HER2 targeted therapy. Research Sponsor: None.

Outcomes for BTC treated with matched TT.

Target	Median PFS, months (95% CI)
***IDH1/2* (N = 72)**	4.2 (2.9-6.8)
***FGFR2* (N=37)**	7.4 (6.5-9.4)
***ERBB2* (N =22)**	8.2 (2.95-16.7)
***MSI* $_{high}$ (N = 14)**	NR (10.15-NR)
TMB_{high} (N=12)	2.69 (1.64-NR)
***BRAF* V600E (N=7)**	16.6 (3.7-NR)

556 Poster Session

Clinical and genomic characteristics of hepatocellular carcinoma-cholangiocarcinoma: Insights from real-world data. First Author: Oluseyi Abidoye, Division of Hematology and Oncology, Mayo Clinic Arizona, Phoenix, AZ

Background: Combined hepatocellular carcinoma-cholangiocarcinoma (cHCC-CCA) is a rare and complex liver cancer with features of both hepatocellular carcinoma (HCC) and cholangiocarcinoma (CCA), making its management challenging. This study aims to analyze the clinical characteristics, genomic profiles, and treatment outcomes of cHCC-CCA patients in a real-world setting. **Methods:** We conducted a retrospective analysis of cHCC-CCA patients across three first-line treatment groups: chemotherapy alone(Gemcitabine/Cisplatin, FOLFOX), HCC-directed therapies (atezolizumab/bevacizumab, lenvatinib, nivolumab, durvalumab, sorafenib), and chemoimmunotherapy (Gemcitabine/Cisplatin combined with immunotherapy). This international study, involving Mayo Clinic and the National Cancer Center East (Japan), assessed overall survival (OS) using Kaplan-Meier and Cox regression models. **Results:** A total of 68 patients were included, with a mean age of 66 years; 74% were male and 66% were White Non-Hispanics. Cirrhosis was present in 74% of patients at diagnosis, and Hepatitis B and C were common etiologies in 57%. Prior primary resection and regional therapies (Transarterial chemoembolization (TACE), Transarterial radioembolization (TARE)) were recorded in 39% and 50% of patients, respectively. Advanced/metastatic disease was present in 52% of the cohort. The distribution of first-line treatments was as follows: 47% received chemotherapy alone, 39% received HCC-directed therapy, and 14% received chemoimmunotherapy. Genomic alterations were detected in 44% patients, with *TP53* (28%), *ARID1A* (25%), and *TERT* (22%) being the most frequent. The HCC-directed therapy group demonstrated the best overall response rate (52%) and had a median OS of 15.8 months (mo), compared to 11.8 mo for chemotherapy and 4.7 mo for chemoimmunotherapy (p=0.8). Patients who received regional treatments had a longer median OS of 15.8 mo compared to those who did not (10.2 mo) (p=0.4). **Conclusions:** HCC-directed therapy showed a trend towards better survival outcomes compared to other treatments, though not statistically significant. A considerable proportion of patients had genomic alterations, which may have influenced survival outcomes. Ongoing research with larger cohorts is underway to further elucidate the impact of genomic variations on treatment efficacy and refine therapeutic strategies for this challenging malignancy. Research Sponsor: None.

Baseline characteristics of patients with combined hepatocellular carcinoma-cholangiocarcinoma (cHCC-CCA).

	Total (N=68)
Mean age at diagnosis, years (SD)	66(6.4)
Gender, n (%)	
Male	50 (74%)
Ethnicity, n (%)	
White, Non-Hispanic	45 (66%)
Cirrhosis at diagnosis, n (%)	50 (74%)
Etiology of Liver disease, n (%)	
Viral (Hepatitis B and C)	39 (57%)
History of Primary resection, n (%)	27 (39%)
Regional treatment, n (%)	34 (50%)
Afp, n (%)	
< 200	23 (64%)
ca19_9, n (%)	
< 50	27 (82%)

557 Poster Session

Efficacy and safety of endoscopic retrograde cholangiopancreatography (ERCP) vs percutaneous transhepatic biliary drainage (PTBD) for cholangiocarcinoma biliary obstruction: An updated systematic review and meta analysis. First Author: Alan Ortega-Macias, University of New Mexico, Albuquerque, NM

Background: Perioperative and/or palliative biliary drainage for the management of cholangiocarcinoma biliary obstruction has been part of the backbone of the multidisciplinary approach that this condition requires. For both intrahepatic and perihilar cholangiocarcinoma, ERCP and PTBD are the two main techniques available. There has yet to be consensus favoring one approach over the other. **Methods:** Following the PRISMA guidelines, a systematic review and meta-analysis were performed to determine if there was a difference in efficacy and safety of ERCP compared to PTBD for Cholangiocarcinoma Biliary Obstruction. Primary outcomes were successful biliary drainage, mean number of needed procedures, length of stay and overall adverse events rate. Data were analyzed using R version 4.2.2. The risk of bias was assessed by the Robins-I and RoB 2 tool. The quality of evidence was graded using the GRADE scale and Newcastle-Ottawa guidelines. **Results:** Three randomized clinical trials and two observational studies described the successful biliary drainage rate of ERCP vs PTBD for cholangiocarcinoma associated biliary obstruction, totaling 340 patients. Meta-analysis test for overall effect, showed no statistical difference between groups (RR = 0.88; 95% CI = [0.69; 1.13]; p = 0.23; Fig. 1). Similarly, in terms of the mean number of needed procedures per patient, a total of three randomized clinical trials and two observational studies, totaling 340 patients were included, no statistically significant difference was found (SMD = 0.64; 95% CI = [- 9.36; 10.63]; p = 0.57; Fig. 2). Likewise, one randomized clinical trial and two observational studies showed no statistically significant difference on the length of stay (days) mean between groups (SMD = 0.06; 95% CI = [- 1.57; 1.68]; p = 0.73; Fig. 3). Although, the three clinical trials and three observational studies showed a mild decreased adverse event and death (within 3 months) rate in the ERCP group, after undergoing statistical analysis, meta-analysis test for overall effect, showed no statistical difference (RR = 0.99; 95% CI = [0.55; 1.77]; p = 0.97; Fig. 4 and RR = 0.61; 95% CI = [0.10; 3.87]; p = 0.22; Fig. 5, respectively). The risk of bias was low with moderate-to-high quality of evidence amongst the included studies. **Conclusions:** Our updated systematic review and meta-analysis demonstrated no statistically significant difference in the successful biliary drainage rate, mean number of needed procedures, length of stay, death within 3 months and adverse event rate between ERCP vs PTBD for management of cholangiocarcinoma biliary obstruction. Further randomized studies are needed to confirm these findings. Research Sponsor: None.

558 Poster Session

Risk of proteinuria with atezolizumab plus bevacizumab versus lenvatinib in first-line systemic treatment for hepatocellular carcinoma. First Author: JIwon Yang, Department of Gastroenterology, Liver Center, Asan Medical Center, University of Ulsan College of Medicine, Seoul, South Korea

Background: Proteinuria presents a challenging complication during systemic therapy for hepatocellular carcinoma (HCC). This study aims to identify risk factors for proteinuria in patients with HCC treated with atezolizumab plus bevacizumab (Atezo/Bev) or lenvatinib (LEN) as first-line systemic treatment. **Methods:** A retrospective analysis was conducted on 622 consecutive patients with unresectable HCC who received Atezo/Bev or LEN as first-line systemic treatment between October 2013 and October 2022. Cumulative incidence of proteinuria was estimated using Kaplan–Meier curves and compared using log-rank tests. Risk factors for proteinuria were identified using Cox proportional-hazard models, along with propensity score-matched and subgroup analyses. **Results:** Among 367 patients treated with Atezo/Bev and 255 with LEN, the cumulative incidence of proteinuria at 12 months was 27.5%. In the multivariable analysis, Atezo/Bev treatment (HR, 1.57; 95% CI, 1.03–2.42), diabetes (HR, 1.64; 95% CI, 1.03–2.61), Child–Pugh class B (HR, 3.43; 95% CI, 1.34–8.78), macrovascular invasion (MVI; HR, 1.58; 95% CI, 1.04–2.38), and an estimated glomerular filtration rate ≤60 mL/min/1.73 m^2 (HR, 3.21; 95% CI, 1.84–5.62) were identified as risk factors for proteinuria. A higher risk of proteinuria in Atezo/Bev patients compared with LEN was consistently observed in the PS-matched cohort, particularly pronounced in subgroups with MVI (HR, 2.84; 95% CI, 1.23–6.54) compared with those without MVI (HR, 1.31; 95% CI, 0.69–2.47). **Conclusions:** Patients treated with Atezo/Bev as first-line systemic treatmentfor HCC exhibited a higher risk of proteinuria compared with those with LEN, particularly when accompanied by MVI. Research Sponsor: None.

559 Poster Session

Germline pathogenic variants in a cohort of individuals with biliary tract cancer. First Author: Spring Holter, University Health Network (UHN), Toronto, ON, Canada

Background: Biliary tract cancers (BTC) are rare but have been implicated in hereditary cancer conditions such as Lynch syndrome (LS) and *BRCA1* and *BRCA2*. Data from paired somatic and germline genetic testing on BTC have shown that ~5% may be related to germline pathogenic variants in *BRCA1* and *BRCA2*. Identification of these germline pathogenic variants (PV) often provide patients with an opportunity for personalized treatment including platinum-based chemotherapies, PARP inhibitors and/or immunotherapy. Germline testing also provides crucial data to relatives regarding cancer risk and access to increased cancer surveillance and risk-reducing surgeries. Current guidelines recommend germline genetic testing when the likelihood of identifying a PV is >5%. The contribution of germline PV in BTC has not been described in a diverse patient population. **Methods:** In this prospective clinic-based study, individuals with BTC were referred by their oncologist for genetic counselling and testing. The majority of individuals were enrolled based on participation in the Legresley Biliary Registry, which recruits all individuals with BTC. Blood was sent for germline multi-gene panel testing. **Results:** Germline genetic testing was performed for 139 individuals. Average age of diagnosis was 57.8 years (range 28-84). The majority of individuals were diagnosed with intrahepatic cholangiocarcinoma (CCA) (54.7%), followed by extrahepatic CCA (20.9%) and gallbladder cancer (10.1%). The majority of individuals were male (56.1%). The population was ethnically diverse with 55% European, 24% Asian, 9% Middle Eastern/North African and 6% African. A minority of individuals had a previous cancer (18%) and 24.5% met current provincial eligibility criteria. Germline testing was complete on 136 individuals. 25/136 (14.7%) PV were identified in 20 individuals. 8/136 (5.9%) of the cohort carried PV in genes known to increase the risk for BTC (*BRCA1, BRCA2* and LS genes), 9/136 (6.6%) carried PV in other genes with clinically actionability (e.g. *PALB2, ATM*), and 8/136 (5.9%) carried heterozygous PV in genes for recessive diseases. Variants of uncertain significance were identified in 42/136 (30.9%) and negative results were identified in 74/136 (54.4%). Personal and/or family history was not suggestive of the associated cancer condition in 9/16 (56%) of the PV cohort. Tumor profiling by whole genome sequencing on some individuals with PV found corresponding somatic mutational signatures, consistent with variant pathogenicity. **Conclusions:** Detection rates for PV in a diverse BTC cohort was up to 14.7%, including 5.9% among genes known to increase the risk for BTC. The majority of PV were found in individuals lacking personal and/or family history suggestive of the associated hereditary cancer condition. These results suggest that guidelines should be updated to recommend universal germline genetic testing for individuals with BTC. Research Sponsor: None.

560 Poster Session

End-of-life (EOL) outcomes and healthcare utilization for patients with hepatocellular carcinoma (HCC) who received immune checkpoint inhibition (ICI). First Author: Margaret Wheless, Vanderbilt University Medical Center, Nashville, TN

Background: ICI is standard of care treatment for advanced HCC. EOL outcomes for patients with HCC who receive ICI prior to death are unknown. We examine relationships between EOL outcomes, healthcare utilization, and receipt of ICI for patients with advanced HCC referred to our tertiary center. **Methods:** Patients with advanced HCC evaluated at our center on or after January 1, 2020 and who died by March 29, 2024 were included. Patients were identified using diagnostic codes for liver cancer and HCC, and subsequently verified by manual review of the electronic health record. Demographic data, Child-Pugh (CP) status, and treatment history were collected. Primary EOL outcomes include: documentation of advance directives and goals of care (GOC) conversations, location of death, palliative care referral, hospice referral, days in hospice, and code status. Secondary healthcare utilization outcomes include: systemic therapy receipt, emergency department (ED) visits, hospitalization, and intensive care unit (ICU) admissions within 14, 30, and 90 days of death. Outcomes were stratified by ICI or non-ICI as last therapy received and p-values were derived using Pearson's chi-square test for equality of proportions. **Results:** Of 221 evaluated patients, 71 died and met criteria for analysis. Baseline characteristics include mean age 64.1 years, 70.8% male, 71.8% received systemic treatment (median 1 line, range [1-7]), 54.1% CPA, 37.5% CPB, and 8.3% CPC at the time of last treatment. No statistically significant differences in primary EOL outcomes were detected when stratified by receipt of ICI or non-ICI as last therapy. Median days enrolled in hospice were not statistically significant between groups (24.5 days [non-ICI] vs 10 days [ICI]; p = 0.39). A higher proportion of patients who received ICI as last therapy had increased healthcare utilization across all outcomes (see Table). **Conclusions:** Patients with advanced HCC receiving ICI as their last treatment before death compared to those receiving non-ICI had similar EOL outcomes but higher healthcare utilization. This may be due to increased real-world use of ICI in CPB patients. Further investigation into risk stratification to predict high healthcare utilizers could guide decision-making and conversations around ongoing use of ICI, particularly near the EOL. Research Sponsor: U.S. National Institutes of Health; 2T32CA217834-07.

Healthcare utilization outcomes stratified by receipt of ICI.

Outcome Before Death *n*= 71	90 days	30 days	14 days
Therapy within			
Total:	33/71 (46.5%)	18/71 (25.4%)	6/71 (8.5%)
Last therapy ICI	69.7%	66.7%	33.3%
ED visit			
Total:	29/71 (40.8%)	27/71 (38.0%)	22/71 (30.1%)
Last therapy ICI	58.6%	55.6%	59.1%
Hospitalization			
Total:	32/71 (45.1%)	30/71 (42.3%)	29/71 (40.8%)
Last therapy ICI	59.4%	56.6%	58.6%
ICU admission			
Total:	6/71 (8.5%)	6/71 (8.5%)	6/71 (8.5%)
Last therapy ICI	66.7%	66.7%	66.7%

561 Poster Session

The impact of adherence to NCCN and BCLC treatment guidelines on survival in HCC: A single center study. First Author: Kanan Alshammari, Oncology Department, Ministry of National Guard Health Affairs, Riyadh, Saudi Arabia

Background: Hepatocellular carcinoma (HCC) is a prevalent and lethal disease in Saudi Arabia, and globally. Guidelines such as Barcelona Clinic Liver Cancer (BCLC) and National Comprehensive Cancer Network (NCCN) are used to guide therapy for this disease. Adherence to these guidelines should lead to better outcomes. We aim to examine the adherence to guidelines and its impact on survival in a real world setting. **Methods:** This retrospective cross-sectional study was conducted at the Ministry of National Guard for Health Affairs, Riyadh, Saudi Arabia. Study included all adult patients with HCC who were diagnosed between 2020 and 2023, and presented at our HCC multidisciplinary tumor board (MDT). Data extracted included demographics, clinical and disease characteristics, treatment modalities, and survival status. Chi-square tests, and Kaplan-Meier survival analysis were used to explore relationships between tumor characteristics, adherence to NCCN and BCLC guidelines, and survival outcomes. Adherence to guidelines were assessed by two independent physicians. **Results:** The study included 208 patients with HCC, the majority were males (73%) with a mean age of 73 years. Less than half of the patients (41.7%) had an Eastern Cooperative Oncology Group (ECOG) performance status of 1. Tumor staging showed that 35% of the patients had BCLC stage D. Child pugh class A and B were equally present in 45% of the patients. Moreover, 24% of the patients had distant metastases or vascular invasion (in portal vein) placing them in BCLC stage C. It is noteworthy that Alpha-Fetoprotein (AFP) was abnormal (>8ng/ml) in 58% of the patients, 18% of them had AFP of more than 400ng/ml. Additionally, the prevalence of comorbid conditions were remarkable for liver cirrhosis being present in 85% of patients, while diabetes mellitus and hypertension each present in 63% of patients. Finally, survival analysis showed a median survival of 16.8 months in all patients with HCC (range 0-47.8 months). Adherence to guidelines were high, with 97.6% (n=203) following NCCN guidelines and 70.7% (n=147) following BCLC guidelines. Following both guidelines showed better survival with mean of 37 months (SD=1.841) in patients with child pugh class A (n=53), and mean survival of 24 months (SD=2.251) in patients with child pugh class B (n=74). **Conclusions:** Adherence to NCCN guidelines was high in our MDT presented patients compared to BCLC guidelines adherence which was lower, but with comparable survival outcomes. Research Sponsor: None.

562 Poster Session

A global comparative study on MDT practices in the management of hepatocellular carcinoma in high-income countries (HICs) and low- and middle-income countries (LMICs). First Author: Huang Zhiyong, Tongji Hospital, Tongji Medical College, Huazhong University of Science and Technology, Wuhan, China

Background: As the sixth most common malignancy worldwide, hepatocellular carcinoma (HCC) accounted for 4.7% of all new cancer diagnoses in 2022. Low- and middle-income countries seem to bear a disproportionate burden of the disease, with over 50% of new HCC cases estimated to occur in China and Africa. Research on multidisciplinary team (MDT) practices in these regions is limited. The objective of this study is to compare MDT practices in the management of HCC in HICs and LMICs. **Methods:** Data on MDT practices from hospitals that manage HCC in Italy, Spain, Switzerland, Germany, Denmark, Canada and the USA, representing HICs, and in China, South Africa and Egypt, representing LMICs were collected through virtual semi-structured interviews and workshops. **Results:** Ten key areas of differences were identified (Table 1) that could be categorized as organizational/regulatory or clinical. In HIC institutions, the focus currently is on optimizing MDTs as standard of care approach accessible to all patients with HCC; MDT principles are often integrated into national policies. In LMICs, emphasis is on establishing MDTs and expanding their capacity; patient access to MDT care is not a mandatory requirement, nor regulated in national policy. Of note, LMICs are seeking early technology adoption to overcome some potential barriers to MDT interactions; online formats are more often explored to facilitate participation and AI-assisted diagnostic tools are explored to compensate for absence of radiologists. Absence of screening and surveillance for early detection in HCC is more common in LMICs, and therefore likely that patients more often present with advanced stage disease. Also, with limited diagnostic modalities, diagnostic work-up and staging may be sub-optimal. **Conclusions:** Significant variation exists in HCC MDT practices between HICs and LMICs. The ultimate goal of this study is to develop a format for clinical MDTs that will facilitate interaction between HICs and LMICs. Consideration for the differences highlighted above is crucial for the development of this concept. Research Sponsor: None.

10 key areas highlighting differences identified between HICs and LMICs.

MDT development direction: Standard of Care for all patients vs. specific session for complex cases

National/regional regulatory requirements
Institutionalizing MDT principles
MDT composition (specialists)
Use of digital tools
Reimbursement
Access to therapeutic options
Stage presentation of HCC patients
Use of clinical staging
Patient follow-up after MDT discussion

563 Poster Session

Leading the way: How leading Chinese institutions drive MDT excellence in hepatocellular carcinoma. First Author: Jiaping Li, Department of Interventional Oncology, The First Affiliated Hospital of Sun Yat-sen University, Guangzhou, China

Background: HCC MDTs involve specialists collaborating on patient cases to create personalized treatment plans. Implementing MDTs can be challenging in countries with limited resources and high disease burdens. In China, there were 368,000 new liver cancer cases in 2022 representative of 40% of global cases. Therefore, evaluating MDT practices at leading hospitals in China may offer valuable insights that could help improve HCC MDT practices worldwide. **Methods:** This study selected hospitals with distinct MDT models: one for all new patients and one for complex cases. Hospitals handling over 1,000 HCC cases annually were chosen. Desk research and virtual interviews with MDT members from three leading Chinese hospitals—The First Affiliated Hospital of USTC, The First Affiliated Hospital of SYSU, and Tongji Hospital—were conducted to analyze their MDTs and determine key best practices. **Results:** We identified 6 key best practice in HCC MDTs at these leading Chinese hospitals. 1) Quality Control on MDT process: Leading hospitals ensure the MDT's efficient execution and effective outcomes. E.g., USTC established a quality control team to audit the MDT process. 2) Real-World Evidence: Leading hospitals prioritize the construction of real-world databases linked to MDT-care to inform impact on outcomes. E.g., SYSU's preliminary data shows impact of MDT care on survival rates (From 2006 to 2010, 5,627 patients who underwent radical liver cancer resection with MDT care saw their five-year survival rate reach 60%, a notable improvement over historical averages) 3) Comprehensive Teams: These hospitals typically have robust teams, involving around 10 specialties. E.g., Tongji Hospital's MDTs involve ~10 specialties for holistic care. 4) Patient Journey Coverage: MDTs manage patients from diagnosis to rehabilitation, providing a holistic approach to patient management. 5) Guideline-based treatment decisions: They follow latest guidelines coupled with access to approved therapies to make treatment decisions 6) Training: Young physicians are invited to actively participate in MDT consultations (e.g., through case preparation and presentation, etc.). **Conclusions:** Evaluation of MDT practices at leading China institutes highlights key best practices that could be adopted to enhance care quality for patients with HCC. The ongoing efforts and successes of these institutions provide a model for broader adoption and standardization of MDT practices across the country. Research Sponsor: None.

564 Poster Session

Hepatotoxicity following external radiotherapy in patients with hepatocellular carcinoma treated with or without prior yttrium-90 radioembolization. First Author: Claire Niewiara, University of Cincinnati College of Medicine - Student, Cincinnati, OH

Background: A variety of local therapy options exist to treat Hepatocellular Carcinoma (HCC) including external radiotherapy (EBRT). Prior yttrium-90 (Y90) delivered transarterially is often a contraindication to EBRT in clinical trials and guidelines due to concerns of cumulative hepatotoxicity, though clinical data to suggest higher rates of hepatotoxicity is limited. This study examines hepatotoxicity profiles after EBRT with and without prior Y90 treatment using the ALBI grading system. **Methods:** Patients treated for HCC with EBRT from 2016 to 2024 with sufficient laboratory follow-up were retrospectively reviewed. Patients were classified into three ALBI grades based on ALBI scores: 1 (< -2.6), 2 ($-2.6 < x < -1.39$), and 3 (> -1.39). The total EBRT dose was defined using the EQD2 method ($\alpha/\beta = 10$). The ALBI scores at 3- and 6-months post-EBRT were compared to baseline for patients with or without prior Y90 using paired t-tests. The main RT planning goal was $\geq$700 cc liver- (tumor volume+ Y90 volume) receiving $\leq$15 Gy. **Results:** 62 patients were treated with EBRT with (n=14) or without (n=48) prior Y90. At 6-months post-EBRT, 11 patients with prior Y90 and 39 patients without prior Y90 had appropriate laboratory follow-up. EBRT dose, technique, and baseline liver function are outlined in Table 1. The median time from Y90 to EBRT was 12.3 months (range 4-61 months). In patients with prior Y90, ALBI scores were similar at 3-mo (mean = -2.4 SD = 0.8, vs mean= -2.4 SD = 0.7, p= 0.60) and 6-mo (mean= -2.3 SD= 0.9, vs mean = -2.4 SD = 0.6, p= 0.653) post-EBRT compared to baseline, respectively. In patients without Y90, ALBI scores were clinically similar at 3-mo (mean = -2.2 SD = 0.7, vs mean= -2.3 SD = 0.6, p= 0.15) and 6-mo (mean= -2.1 SD= 0.8, vs mean = -2.3 SD = 0.7, p=0.021) post-EBRT compared to baseline, respectively. The proportion of patients with an increase in ALBI grade at 3- and 6-mo post-EBRT was similar regardless of prior Y90 treatment. In patients treated with Y90, 14.3% and 36.4% had an increase in ALBI grade at 3- and 6-months, respectively. In those treated without prior Y90, 21.3% and 35.9% had an increase in ALBI grade at 3- and 6-months, respectively. **Conclusions:** Patients with HCC treated with EBRT after Y90 appear to have a similar rate of hepatotoxicity defined by the ALBI grading system compared to patients treated without prior Y90. In well-selected patients, prior Y90 should not be an absolute contraindication for EBRT. Research Sponsor: None.

Demographics and ALBI changes post-EBRT.

	EBRT alone (n = 48)	EBRT after Y90 (n = 14)
Median Age Years (range)	67.6 (17.9-84.3)	65.6 (55.1-85.2)
Proton, IMRT/SBRT	22 (46%), 26 (54%)	9 (64%), 5 (36%)
Baseline Child Pugh Score (A, B, C)	34 (70.8%), 10 (20.8%), 4 (8.4%)	11 (78.6%), 3 (21.4%), 0 (0%)
Median EBRT EQD2 (range)	82 Gy (44-99)	82 Gy (67-86)
Baseline ALBI Grade (1, 2, 3)	23 (47.9%), 20 (41.7%), 5 (10.4%)	6 (42.9%), 7 (50%), 1 (7.1%)

565 Poster Session

Clinical outcomes of stereotactic body radiation therapy to the liver. First Author: Lauren Kenney, Dana-Farber Cancer Institute, Boston, MA

Background: Stereotactic body radiation therapy (SBRT) can accurately and effectively target liver lesions with an ablative dose. However, prior studies suggest that SBRT directed at tumors near the central hepatobiliary tract may lead to downstream toxicity including biliary stricture and impaired liver function. This study aims to evaluate clinical outcomes and hepatobiliary toxicities associated with SBRT to the liver. **Methods:** This retrospective study reviews 177 patients who received CT- or MR-guided SBRT to the liver at Dana-Farber Cancer Institute and Brigham and Women's Hospital from 2010 to 2024. Patient demographics, treatment characteristics, and clinical outcomes were extracted from electronic medical records. Local control, regional control, and overall survival were assessed using Kaplan-Meier methods. Toxicity was evaluated by tracking the incidence of biliary stricture requiring intervention and Child-Pugh (CP) scores pre- and post-SBRT. **Results:** Of the 177 patients, 47 (26.6%) had primary disease and 130 (73.4%) had metastatic disease, with a median follow-up of 58.7 months. Among those with primary disease, 9 (19.1%) had post-SBRT local progression (median time to progression not reached, 95% CI: 47.3, NA) and 26 (55.3%) had regional progression (median time to progression of 20.5 months, 95% CI: 6.5, 58.1). Of patients with metastatic disease, 44 (33.8%) had local progression (median time to progression of 40.2 months, 95% CI: 23.0, NA) and 96 (73.8%) had regional progression (median time to progression of 5.9 months, 95% CI: 4.8, 9.3). Median overall survival post-SBRT was 34.6 months for primary disease (95% CI: 20.2, 57.7) and 23.6 months for metastatic disease (95% CI: 19.0, 28.8). 18 patients (10.1%) later developed biliary stricture requiring intervention (12.8% of primary disease and 9.2% of metastatic disease). Of the 119 patients with CP scores available, 59 patients (49.6%) had a maximum post-SBRT CP score increase of 2 or more (55.0% of primary disease, 46.8% of metastatic disease). **Conclusions:** This study demonstrates that SBRT is an effective treatment option for local control in the liver, with better survival and progression outcomes for primary disease compared to metastatic disease. Among potential post-SBRT toxicities, initial analysis indicates frequent increase in CP score and low incidence of biliary stricture requiring intervention. An ongoing analysis will explore whether the dose to central bile ducts correlates with late strictures and toxicities. Research Sponsor: Tufts University School of Medicine.

566 Poster Session

Association between immune-related adverse events and prognosis after immune checkpoint inhibitor in hepatocellular carcinoma: A meta-analysis and systematic review. First Author: Soo Young Hwang, University of Maryland Midtown Campus, Baltimore, MD

Background: Immune checkpoint inhibitors (ICIs) have become the preferred treatment for unresectable hepatocellular carcinoma (HCC). Conflicting data exist regarding the prognostic value of immune-related adverse events (irAEs) in these patients. We aimed to explore potential the association between irAEs and prognosis in HCC patients treated with ICIs. **Methods:** We searched PubMed, Scopus, Web of Science, and CENTRAL databases for articles published from inception to June 2024 using keywords including ICI, HCC, and irAEs. Two reviewers independently screened studies for eligibility and relevant data. A random effects model was used for statistical analysis, and subgroup analysis was performed by the grade of irAEs. **Results:** Of 3,028 studies, 25 (n=4,116 patients) met criteria for inclusion. Atezolizumab plus bevacizumab was the most common treatment regimen (n=10 studies). About half (49.5%) of patients experienced irAEs. irAEs were associated with increased objective response rate (ORR) (pooled OR: 1.72; 95% CI: 1.36 – 2.19, I^2=39%), higher complete response rate (pooled OR: 1.45; 95% CI: 1.21-1.74, I^2=74%), and longer progression-free survival (PFS) (pooled HR: 0.66; 95% CI: 0.52-0.84, I^2=71%). Although irAEs were associated with longer overall survival, this difference was not statistically significant (pooled HR: 0.83; 95% CI: 0.62 – 1.11, I^2=73%). Subgroup analysis showed a survival benefit for patients with grade 1-2 irAEs (pooled HR: 0.50; 95% CI: 0.36-0.67, I^2=0%) but not for those with grade 3-4 irAEs (pooled HR: 0.95; 95% CI: 0.24-3.72, I^2=84%). **Conclusions:** The development of irAEs is associated with a favorable prognosis in HCC, including improved PFS and higher ORR. A survival benefit was noted in patients with mild irAEs, but not in those with severe irAEs. Future studies are needed to determine if a differential survival benefit relates to the continuation vs. discontinuation of ICI therapy. Research Sponsor: None.

567 Poster Session

A multicenter, prospective observational study of chemotherapy in elderly patients with biliary tract cancer. First Author: Kohei Nakachi, Department of Medical Oncology, Tochigi Cancer Center, Utsunomiya, Japan

Background: In addition to gemcitabine (GEM) plus cisplatin (GC), GEM plus S-1 (GS) and GC plus S-1 (GCS) have been used as standard treatments for advanced biliary tract cancer (aBTC) based on results from phase III trials in Japan prior to the introduction of immune checkpoint inhibitors. The aim of this study was to evaluate the efficacy and safety of these chemotherapies in elderly patients (pts) with aBTC. **Methods:** This was a multicenter, prospective observational study (UMIN000045156). Inclusion criteria were: age >=70 years; unresectable or recurrent BTC; and scheduled for chemotherapy with GCS, GC, GS, GEM, or S-1. Data were prospectively collected on treatment tolerability, safety and geriatric assessments (G8, IADL, CCI), and overall survival (OS), progression-free survival (PFS), and objective response rate (ORR) were evaluated. Inverse probability-weighted propensity score analyses were performed for comparisons of GCS versus GC and of GC versus GEM. **Results:** A total of 305 pts were enrolled from 53 Japanese hospitals between August 2021 and January 2023. Of these, 294 received chemotherapy (GCS, n=75; GC, n=131; GS, n=26; GEM, n=52; S-1, n=10). Median follow-up was 11.8 months for all pts. Median age was 76 years (range: 70-89 years). Adverse events are summarized in the Table 1. Rates of pts without G8 deterioration after 3 months were 58.7% for GCS, 55.7% for GC, 34.6% for GS, 42.3% for GEM, and 20% for S-1. Rates of pts without IADL deterioration after 3 months were 82.7% for GCS, 73.3% for GC, 42.3% for GS, 57.7% for GEM, and 40.0% for S-1. With inverse probability-weighted propensity score analyses, median OS was 16.4 months (95% confidence interval [CI], 14.7–20.6 months) in the GCS group and 13.3 months (95%CI, 11.7–21.9 months) in the GC group (hazard ratio [HR] 0.80, 95%CI 0.55–1.17; p=0.258). Median PFS was 11.8 months (95%CI, 9.7–14.5 months) in the GCS group and 7.7 months (95%CI, 6.0–10.2 months) in the GC group (HR 0.55, 95%CI 0.38–0.80; p=0.002). ORR was 31.0% in the GCS group and 15.5% in the GC group (p<0.001). Median OS was 13.3 months (95%CI 11.1–19.4 months) in the GC group and 15.5 months (95%CI 6.4–18.7 months) in the GEM group (HR 0.74, 95%CI 0.42–1.29; p=0.282). Median PFS was 6.9 months (95%CI 6.0–9.6 months) in the GC group and 5.1 months (95%CI 3.0–12.0 months) in the GEM group (HR 0.79, 95%CI 0.42–1.49; p=0.463). ORR was 14.1% in the GC group and 7.5% in the GEM group (p=0.305). **Conclusions:** GCS is an effective therapy compared to GC in elderly pts with aBTC without increasing toxicities or deteriorating activities of daily living. Clinical trial information: UMIN000045156. Research Sponsor: None.

Chemotherapy-related adverse events (grade 3 and 4).

Events, n (%)	GCS (n=75)	GC (n=131)	GS (n=26)	GEM (n=52)	S-1 (n=10)
Neutropenia	20 (26.7)	54 (41.2)	8 (30.8)	10 (19.2)	0
Thrombocytopenia	8 (10.7)	9 (6.9)	2 (7.7)	1 (1.9)	0
Anorexia	2 (2.7)	6 (4.6)	2 (7.7)	3 (5.8)	3 (30.0)
Malaise	0	8 (6.1)	4 (15.4)	4 (7.7)	3 (30.0)
Febrile neutropenia	1 (1.3)	1 (0.8)	2 (7.7)	1 (1.9)	0

568 Poster Session

Prompt initiation of durvalumab and tremelimumab treatment for unresectable hepatocellular carcinoma in patients with chronic active hepatitis B. First Author: Yu-Yun Shao, National Taiwan University Hospital, Taipei City, Taiwan

Background: Chronic hepatitis B virus (HBV) infection is a common etiology of HCC, especially in East Asia. Clinical trials using immune checkpoint inhibitors (ICIs) usually exclude patients with chronic active hepatitis B (serum HBV viral load > 2000 IU/mL) or require the HBV viral load to be below a certain level using anti-HBV medications. This study examines the safety and efficacy of concurrently administering the STRIDE regimen (durvalumab with a single dose of tremelimumab) or durvalumab alone with anti-HBV medications in this patient population. **Methods:** We enrolled patients with advanced HCC, Child-Pugh A liver function reserve, and untreated chronic active hepatitis B. Initially, only patients naïve to ICIs were eligible, and patients received 1500mg of durvalumab intravenously every four weeks for up to two years except for disease progression or occurrence of unacceptable toxicities. After the positive results of the HIMALAYA study, we shifted to the STRIDE (single tremelimumab regular interval durvalumab) regimen, adding one dose of 300mg tremelimumab on the first day to regular durvalumab treatment, and allowed patients who had received prior programmed cell death-1 blockade therapy to be enrolled. Anti-HBV treatment with entecavir was initiated within seven days before starting ICI therapy. The primary endpoint was the rate of HBV reactivation (defined as a ≥2 log increase in serum HBV DNA from baseline) during the first six months. **Results:** Following the protocol design, 30 patients were enrolled. Two patients were female, and the mean age was 66.9 years. Ten patients received durvalumab alone, and 20 received the STRIDE regimen. Macrovascular invasion and extrahepatic spread were present in 16 (53.3%) and 20 (66.7%) patients, respectively; 18 (60%) patients had alpha-fetoprotein level ≥ 400 ng/mL. More than half (60.0%) of the patients had received prior systemic therapy for HCC, including 2 patients who had received PD-1 blockade therapy. Seven patients received the study treatment as the 3rd or later line of systemic therapy. The mean±standard deviation baseline HBV viral load was 771.0±354.1 KIU/mL. No patients experienced HBV reactivation or HBV-associated hepatitis. Hepatitis flare was noted in 8 (26.7%) patients, but none of them were associated with HBV reactivation. The objective tumor response rate was 10% and 25% for the durvalumab treatment alone and the STRIDE regimen, respectively. No unexpected toxicities were identified in this study. The most common treatment-related adverse events were skin rash and liver function test abnormalities. **Conclusions:** For patients with chronic active hepatitis B, ICI therapy could be promptly initiated as long as anti-HBV medications were administered simultaneously. This study was partially supported by AstraZeneca. Clinical trial information: NCT04294498. Research Sponsor: National Science and Technology Council, Taiwan; NSC-112-2314-B-002-267, NSC 113-2314-B-002-001, and 113-2923-B-002-001-MY3; AstraZeneca.

569 Poster Session

Real-world Child-Pugh scores and outcomes of patients with hepatocellular carcinoma at a high-volume center. First Author: Ian Tobal, Ochsner Medical Center, New Orleans, LA

Background: Hepatocellular carcinoma (HCC) is the most common primary liver cancer and most often occurs in the setting of cirrhosis. Patients (pts) with advanced HCC are not eligible for curative intent therapies such as ablation, surgical resection, or liver transplantation. For the past five years, immune checkpoint inhibitor (ICI)-based systemic therapy regimens have been the mainstay of first-line treatment for pts with advanced HCC. Most trials assessing ICI-containing regimens, including IMBRAVE150 and HIMALAYA, excluded pts with Child-Pugh (CP) scores worse than A. Unfortunately, fewer than half of pts maintain a CP A score at the time of HCC diagnosis and referral to medical oncology. Real-world data on CP scores of patients undergoing ICI-based therapy for advanced HCC at the time of medical oncology referral and treatment start are lacking. **Methods:** Pts who initiated treatment with either atezolizumab plus bevacizumab (A+B) or durvalumab plus tremelimumab (D+T) in the Ochsner Health system between 1/1/20 and 12/31/23 were included in this analysis. CP scores were calculated at the time of their initial medical oncology visit and again at the time of their first systemic therapy cycle. Pt demographics and clinical outcomes were analyzed. **Results:** 200 pts were identified in this study, 159 of whom (79.5%) were male. Median age at treatment start was 66.8 years. 117 pts (58.5%) were Caucasian and 63 pts (31.5%) were African American (AA). 152 and 48 pts received A+B and D+T, respectively. At the time of first medical oncology visit, 58 pts (29%) were classified as CP B (n=51) or C (n=7). At the treatment start date, 82 pts (41%) were classified as CP B (n=73) or C (n=9). The median survival across all groups was 10.5 m. Median survival CP A, B, and C pts were 15.4, 7.6, and 1.6 months (m), respectively. 6-month survival rates for CP A, B, and C pts were 75.7%, 52.2%, and 0%, respectively. 12-month survival rates for CP A, B, and C pts were 58.2%, 35.1%, and 0%, respectively. There was no statistically significant difference in CP score at either time point between Caucasian and AA pts. **Conclusions:** Although pts with CP scores worse than A were not included in the pivotal studies that have established recent standard of care therapies for pts with advanced HCC, a significant proportion of pts being treated with ICI-based regimens have at least CP B liver disease, with a poorer prognosis in that subset of pts. Extremely poor survival rates for those with CP C scores argue against use of ICI-based treatment for these pts. In this analysis, no significant difference was observed in CP scores between Caucasian and AA pts. Research Sponsor: None.

570 Poster Session

Adjuvant chemoradiation combined with immunotherapy for patients with high-risk resectable extrahepatic cholangiocarcinoma and gallbladder cancer: A phase II, multicenter, randomized controlled trial (ACCORD trial). First Author: Ming Kuang, Center of Hepato-Pancreato-Biliary Surgery, The First Affiliated Hospital, Sun Yat-sen University, Guangzhou, China

Background: Extrahepatic cholangiocarcinoma (ECC) and gallbladder cancer (GBC), as the majority of biliary tract cancer (BTC), has a markedly high risk of recurrence after surgery. However, adjuvant treatments specifically for resectable ECC and GBC patients are scare and adjuvant chemotherapy alone delivers limited efficacy. Immunotherapy and radiotherapy are potential effective treatments and both of them may synergize with chemotherapy. **Methods:** ACCORD was a multicenter, phase 2, randomized controlled trial to assess the efficacy and safety of chemoradiation with immunotherapy as an adjuvant treatment in resectable ECC/GBC, compared to observation. The primary endpoint was overall survival (OS) and the secondary endpoints included recurrence-free survival (RFS) and safety. Patients in the chemoradiation-anti-PD-1 group received Camrelizumab intravenously every 3 weeks after surgery. Camrelizumab therapy was not terminated until disease progression, unacceptable toxic effects occurred or informed consent withdrawal. After 2 courses of Camrelizumab treatment, chemoradiation was carried out simultaneously. Patients in the observation group received no anticancer treatment unless relapse was detected. **Results:** From March 2020 to June 2022, a total of 93 ECC/GBC patients after curative resection were randomized 1:1 into chemoradiation-anti-PD-1 group (n=46, Camrelizumab + concurrent Capecitabine and radiotherapy) and observation group (n=47). The 1-year, 2-year and 3-year OS rate were 95.7%, 71.4%, and 58.2% in the chemoradiation-anti-PD-1 group, and 80.9%, 52.9%, 30.5% in the observation group (Hazard ratio (HR) 0.43, 95% confidence interval 0.24-0.79; P=0.004). The 1-year, 2-year and 3-year RFS rate were 78.3%, 54.0%, and 40.3% in the chemoradiation-anti-PD-1 group, and 55.3%, 27.0%, 17.2% in the observation group, with a hazard ratio of 0.46 (95% CI, 0.28 to 0.76; $P < 0.001$). In the chemoradiation-anti-PD-1 group, the main adverse effect $\geq$ grade 3 were anemia (7 [15.2%]), dermatitis radiation (5 [10.9%]), and nausea (5 [10.9%]); and 100% of patients completed the whole treatment. **Conclusions:** Chemoradiation combined with immunotherapy as an adjuvant therapy demonstrated superior survival outcomes over observation in resectable ECC/GBC patients with a well-tolerable safety profile, supporting the potential of this combination treatment as effective adjuvant therapy for these high-risk patients. Clinical trial information: NCT04333927. Research Sponsor: National Natural Science Foundation of China; Research and Development Plan for Key Areas in Guangdong Province; Key Research and Development Plan of Guangzhou City.

571 Poster Session

Updated results of the PATH study: Adjuvant therapy with donafenib plus a PD-1 inhibitor for patients with hepatocellular carcinoma at high risk of recurrence after resection. First Author: Yiwen Chen, Department of Hepatobiliary and Pancreatic Surgery, The First Affiliated Hospital, Zhejiang University School of Medicine, Hangzhou, China

Background: There is no universally accepted standard treatment option or strategy for post-surgery adjuvant therapy in hepatocellular carcinoma (HCC). Among BCLC A/B patients (pts) with MVI, adjuvant transarterial chemoembolization has been reported to provide a 1-year recurrence-free survival (RFS) rate of around 60-70%. We conducted a pilot study to explore the utility of donafenib combined with a PD-1 inhibitor as adjuvant therapy for pts at high risk of recurrence. Herein, we present the updated results with a median follow-up of 11.6 months. **Methods:** It was planned to enroll 30 HCC pts who had undergone radical surgery and presented with at least one of the following risk factors: a) Presence of microvascular invasion (MVI); b) Presence of satellite nodule(s); c) More than three tumor nodules; d) Portal vein tumor thrombus. Pts with invasion of the main portal venous system were excluded. Donafenib combined with a PD-1 inhibitor (i.e., toripalimab) was initiated at 4 to 8 weeks post-resection and was to be continued for up to six months, unless recurrence occurred or there was an intolerable adverse reaction. No other adjuvant therapies were permitted before enrollment and during the study, except for anti-virus treatment. The primary endpoint was the cumulative 1-year RFS rate. The secondary endpoints included overall survival, quality of life (QoL) measured with the FACT- Hepatobiliary questionnaire, and safety. **Results:** From June 2020 to November 2023, a total of 30 pts were recruited. 27 pts were included in the efficacy analysis. The median time from surgery to enrollment was 1.4 months. MVI was the most common risk factor (n=25, 92.6%). The majority of the pts had only one risk factor (n=23, 85.2%). As of the data cutoff on August 13, 2024, relapse had occurred in four pts and the median RFS was not reached. The 1-year RFS rate was 81.6% (90% CI, 67.8%-95.4%) in all evaluable pts. QoL showed a mild increase from baseline while on adjuvant therapy. No deaths were observed. In the 30 pts who had received at least one dose, the incidence of any grade and grade 3 treatment-related adverse events (TRAEs) were 90.0% (27/30) and 40.0% (12/30), respectively. No grade 4 or 5 TRAEs were reported. The grade 3 TRAEs occurring in at least two pts were palmar-plantar erythrodysesthesia syndrome (13.3%), rash (13.3%), increased alanine aminotransferase (10.0%), and increased aspartate aminotransferase (6.7%). **Conclusions:** The study's duration exceeded expectations due to slow recruitment. Nonetheless, it showed a promising outcome with a regimen of 6 months of adjuvant therapy with donafenib plus toripalimab in BCLC A/B pts at high risk of recurrence. An improvement in QoL was observed among patients during the adjuvant therapy. No new safety issues were identified. Clinical trial information: NCT04418401. Research Sponsor: None.

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Patient (pt) characteristics, treatment patterns, and outcomes in unresectable hepatocellular carcinoma (uHCC) treated with first-line (1L) systemic therapy in the United States (US). First Author: Nguyen H. Tran, Department of Oncology, Mayo Clinic Rochester, Rochester, MN

Background: Guidelines recommend systemic therapy for uHCC ineligible for locoregional therapy. Until the approval of tremelimumab + durvalumab for uHCC, in Oct 2022, 1L systemic therapies in the US included multikinase inhibitors and atezolizumab (A) + bevacizumab (B). It is crucial to understand factors driving real-world treatment decisions and outcomes in uHCC. Characteristics, treatment patterns, and outcomes were assessed in US pts with uHCC treated with 1L systemic therapy. **Methods:** Data were collected from electronic medical records from US Mayo Clinic sites. Pts with uHCC who initiated an NCCN-recommended 1L systemic therapy between Jun 2020–Oct 2022 (A + B was the only approved 1L immunotherapy (IO)-based regimen) with $\geq$2 follow-up visits, were included. Index was the date of first 1L systemic therapy. Pts were assessed by risk of post-index gastrointestinal (GI) bleeding (pts with Child-Pugh class B or C, pre-index GI bleeding, uncontrolled hypertension [HTN] [$\geq$2 anti-HTN drugs or $\geq$2 vital entries of SBP >140 mmHg or DBP >90 mmHg]), or significant varices and band ligation). Treatment patterns, overall survival (OS), and post-index GI bleeding were assessed. **Results:** A total of 186 pts met the inclusion criteria. Most common etiologies of liver disease were metabolic dysfunction-associated steatohepatitis (42.7%) and hepatitis C virus (40.2%). Pre-index GI bleeding was reported in 29.0% of pts, of which 46.3% occurred within 6 months (mo) pre-index; 62.4% of pts had EGD, of which 72.4% had varices. There were 128 (68.8%) pts with GI bleeding risk and 58 (31.2%) pts without GI bleeding risk. The most common 1L systemic therapies for pts with GI bleeding risk were A + B (72.7%) and A only (9.4%), while for pts without GI bleeding risk, A + B (29.3%), nivolumab + ipilimumab (13.8%), and B + chemotherapy (13.8%) were the most common. Median OS and OS rates at 12, 18, and 24 mo were lower, and post-index GI bleeding rate was higher in pts with vs without GI bleeding risk (Table). For pts treated with A + B, median OS and OS rates at 12, 18, and 24 mo were lower in pts with vs without GI bleeding risk (Table). **Conclusions:** In this network, prior to the approval of other 1L IO-based regimens for uHCC, many pts received A + B, despite risk of post-index GI bleeding. Median OS with A + B was shorter in pts with vs without GI bleeding risk. Data highlight the complexity of uHCC and the unmet need for guidance on characteristics-driven treatment decisions. Research Sponsor: AstraZeneca.

	Full cohort			A + B cohort		
	All pts N=186	GI bleeding risk N=128	No GI bleeding risk N=58	All pts N=110	GI bleeding risk N=93	No GI bleeding risk N=17
Median follow-up, mo	20.6	20.9	20.4	21.3	21.6	20.3
Median OS, mo	14.4	11.4	40.3	13.9	12.8	Not reached
OS rates, %						
12 mo	55.2	48.9	68.9	55.2	52.3	70.6
18 mo	45.3	38.1	61.1	44.1	41.6	57.8
24 mo	39.4	31.8	56.8	37.1	34.6	51.3
Post-index GI bleed, %	17.2	23.4	3.4	17.3	19.4	5.9

573 Poster Session

Clinical outcome after complete response with immunotherapy in hepatocellular carcinoma. First Author: Pin-Jung Chen, Samuel Oschin Comprehensive Cancer Institute, Cedars-Sinai Medical Center, Los Angeles, CA

Background: Immunotherapy has shown great efficacy in hepatocellular carcinoma (HCC). No clear consensus exists on when to stop the immune checkpoint inhibitor (ICI) after complete response (CR), and clinical course after stopping ICI remains unknown. **Methods:** We identified 138 HCC patients who received ICI at Cedars-Sinai Medical Center and Mayo Clinic Rochester from 2016-2023. Modified Response Evaluation Criteria In Solid Tumors (mRECIST) was used to assess treatment response. We further described the clinical course of patients who achieved CR after ICI discontinuation, including cancer recurrence, subsequent treatment, liver transplantation (LT) listing and outcomes. **Results:** The median age at ICI initiation was 68 years; 79% were male; 55.9% were white; 51.5% were BCLC stage C. 9.4% CRs, 25.4% partial responses, 39.9% stable diseases, and 25.4% progressive disease. Hepatitis C etiology was associated with higher likelihood of CR (P=0.005). Among 13 patients who achieved CR, 10 stopped ICI (60% for LT evaluation, 40% for toxicity). After stopping ICI, 40% had recurrence (Table). 3 patients underwent local therapy for recurrent HCC within Milan criteria and 1 patient restarted ICI for recurrence beyond Milan criteria. Among CR patients, 9 patients underwent LT evaluation after stopping ICI and 7 patients were waitlisted. After listing, 3 dropouts were observed, due to recurrent HCC beyond Milan criteria, sepsis leading to death, and personal preference. 4 patients underwent LT. 2 patients underwent local therapy before LT for recurrent HCC (Table). All transplanted patients had the last dose of ICI more than 1 year before LT (12, 21, 24, and 29 months). On explanted livers, 2 patients had no residual viable tumor and 2 patients had residual HCC within Milan criteria. 1 patient developed acute cellular rejection and was treated with immunosuppressants and maintained stable graft function. None had graft loss or evidence of HCC recurrence in the follow-up period at 8, 20, 28, and 51 months. **Conclusions:** Immunotherapy demonstrated excellent effects on a subset of HCC patients, including 9.4% who achieved CR. After stopping ICI, 40% had cancer recurrence but 3 out of 4 were within Milan criteria. 4 underwent successful LTs (2.9%) with excellent post LT outcome. Larger prospective data is needed to establish a protocol for ICI use, especially to determine the optimal timing for ICI discontinuation after CR. Research Sponsor: None.

	BCLC at ICI initiation	ICI cycles before stopping, ICI	Recurrence after last ICI (days)	Within Milan at recurrence	Treatment(s) received after recurrence and outcomes	OS (days)
Patient 1	B	7, Nivolumab	730	No	Nivolumab. Died from GI bleeding	1065
Patient 2	B	64, Atezo/Bev	149	Yes	TACE with CR, followed by LT without recurrence.	1203
Patient 3	A	8, Nivolumab	1393	Yes	Y90 with CR. No recurrence.	1984
Patient 4	C	NA, Pembrolizumab	407	Yes	TACE and MWA with CR, followed by LT without recurrence.	1722

574 Poster Session

Patterns of radiological progression in participants (pts) with embolization-eligible hepatocellular carcinoma (HCC) treated with durvalumab (D) + bevacizumab (B) + transarterial chemoembolization (TACE) and placebos + TACE: EMERALD-1 post hoc analysis. First Author: Bruno Sangro, Liver Unit and HPB Oncology Area, Clínica Universidad de Navarra and CIBEREHD, Pamplona – Madrid, Spain

Background: EMERALD-1 (NCT03778957) met its primary endpoint, demonstrating improved progression-free survival (PFS) in pts with locoregional HCC treated with D + B + TACE versus placebos (PBO) + TACE (stratified Cox proportional hazards hazard ratio [HR], 0.77; 95% confidence interval [CI], 0.61–0.98 [Lencioni R, et al. *J Clin Oncol* 2024;42(suppl 3). Abs LBA432]). This post hoc analysis assessed outcomes by radiological progression pattern in pts treated with D + B + TACE or PBO + TACE. **Methods:** Pts included in this analysis received D (1500 mg) or PBO for D (Q4W) in combination with conventional (c)- or drug-eluting bead (DEB)-TACE (investigator choice, 1–4 TACE procedures within 16 weeks). Subsequently, pts received D (1120 mg) + B (15 mg/kg), or PBO for D + B (Q3W). This study analyzed radiological progression patterns at the time of first progressive disease (PD) as assessed by the investigator per modified Response Evaluation Criteria in Solid Tumors. A new lesion was classified as a new intrahepatic lesion (NIH) or new extrahepatic lesion (NEH); tumor growth of existing intrahepatic lesions (increase of ≥20% of an existing target lesion with at least >5 mm absolute increase or unequivocal PD with a non-target lesion) was classified as intrahepatic growth (IHG; categories were not mutually exclusive). Efficacy was assessed by time to progression (TTP). **Results:** In the D + B + TACE arm, 53.9% of pts had PD, and in the PBO + TACE arm 79.0% of pts had PD. The most common pattern of disease progression across both treatment arms was NIH, occurring in 73 (35.8%) and 107 (52.2%) pts in the D + B + TACE and PBO + TACE arms, respectively, with 31 (15.2%) and 34 (16.6%) pts exhibiting IHG, and 24 (11.8%) and 39 (19.0%) pts exhibiting NEH, respectively. Improved TTP was observed in pts treated with D + B + TACE versus PBO + TACE, regardless of progression pattern (Table). **Conclusions:** Overall, the rate of progression was lower with D + B + TACE compared with PBO + TACE. The pattern of disease progression observed with D + B + TACE and PBO + TACE was similar, with NIH the most common pattern of progression in both treatment arms. Consistent benefit in TTP was observed with D + B + TACE versus PBO + TACE, regardless of progression pattern. Clinical trial information: NCT03778957. Research Sponsor: AstraZeneca.

	NIH		IHG		NEH	
	D + B + TACE (n=73)	PBO + TACE (n=107)	D + B + TACE (n=31)	PBO + TACE (n=34)	D + B + TACE (n=24)	PBO + TACE (n=39)
Median (95% CI) TTP, months	13.7 (10.8–16.5)	8.8 (7.0–10.9)	5.1 (3.0–9.0)	4.3 (2.9–4.8)	6.7 (2.8–16.6)	4.6 (2.9–6.8)
HR (95% CI)*	0.78 (0.57–1.07)		0.61 (0.36–1.03)		0.66 (0.36–1.15)	

*HR and CI were estimated using a Cox proportional hazards model, the CI was calculated using a profile likelihood approach.

575 Poster Session

Outcomes by transarterial chemoembolization (TACE) modality from participants (pts) with embolization-eligible hepatocellular carcinoma (HCC) treated with durvalumab (D) + bevacizumab (B) + TACE and placebos (PBO) + TACE: EMERALD-1 subgroup analysis. First Author: Jeong Heo, Department of Internal Medicine, College of Medicine, Pusan National University and Biomedical Research Institute, Pusan National University Hospital, Busan, South Korea

Background: EMERALD-1 (NCT03778957) met its primary endpoint, showing improved progression-free survival (PFS) in pts with locoregional HCC treated with D + B + TACE versus PBO + TACE (hazard ratio [HR], 0.77; 95% confidence interval [CI], 0.61–0.98). This analysis assessed the impact of TACE modality (conventional TACE [cTACE] or drug-eluting bead [DEB]-TACE) on efficacy and safety outcomes. **Methods:** Pts in this analysis received D (1500 mg) or PBO for D (Q4W) plus cTACE or DEB-TACE (investigator choice; TACE modality was a stratification factor). After completing the last TACE, pts received D (1120 mg) + B (15 mg/kg) or PBO for D + B (Q3W). PFS, time to progression (TTP), and overall response rate (ORR; BICR per RECIST v1.1) with D + B + TACE and PBO + TACE (intent-to-treat population) are reported by TACE modality. Safety was assessed in the safety analysis set (pts received ≥1 dose of study treatment [tx], regardless of randomization). **Results:** Overall, 59.3% of pts received cTACE and 40.7% received DEB-TACE in the D + B + TACE arm; similarly, 58.5% of pts received cTACE and 41.5% received DEB-TACE in the PBO + TACE arm. Most pts received 1 or 2 TACE procedures in both cTACE (60% in D + B + TACE arm; 67.2% in PBO + TACE arm) and DEB-TACE groups (55.6% in D + B + TACE arm; 53.6% in PBO + TACE arm). Baseline characteristics in the cTACE and DEB-TACE groups were similar, with some differences in the relative distribution of BCLC, HAP, and tumor burden (BCLC Score A; 29.0% vs 17.9%: HAP Score A; 36.5% vs 25.0%: tumor burden within up-to-7 criteria; 56.4% vs 37.5%, respectively). Baseline characteristics were generally well balanced across tx arms within the cTACE and DEB-TACE groups. PFS, TTP, and ORR improved with D + B + TACE versus PBO + TACE, regardless of TACE modality (Table). In the D + B + TACE and PBO + TACE arms, max Grade 3–4 adverse events possibly related to study tx were reported by 25/100 (25.0%) and 3/116 (2.6%) pts in the cTACE group, and 16/54 (29.6%) and 9/84 (10.7%) pts in the DEB-TACE group, respectively. **Conclusions:** Overall, pts receiving D + B + TACE had improved PFS, TTP, and ORR versus PBO + TACE regardless of TACE modality. Safety was manageable with both cTACE and DEB-TACE. Clinical trial information: NCT03778957. Research Sponsor: AstraZeneca.

	D + B and cTACE (n=121)	PBO and cTACE (n=120)	D + B and DEB-TACE (n=83)	PBO and DEB-TACE (n=85)
Median (95% CI) PFS, months	19.4 (12.4–22.3)	11.1 (7.2–14.0)	11.1 (7.2–14.2)	6.7 (5.0–7.3)
PFS HR (95% CI)	0.80 (0.59–1.10)		0.74 (0.51–1.06)	
Median (95% CI) TTP, months	22.3 (19.4–27.7)	13.6 (9.2–16.6)	15.0 (11.1–22.4)	6.9 (5.1–11.1)
TTP HR (95% CI)	0.70 (0.49–0.98)		0.55 (0.36–0.83)	
Pts with measurable disease at baseline, n	119	119	83	84
ORR, n (%) pts with response*	62 (52.1)	43 (36.1)	26 (31.3)	17 (20.2)
ORR, odds ratio (95% CI)	1.93 (1.15–3.27)		1.80 (0.89–3.69)	

*Includes confirmed complete or partial response.

576 Poster Session

First-line durvalumab with arterial chemotherapy in major portal invaded hepatocellular carcinoma: The phase 2 DurHope study with biomolecular analyses. First Author: Ming Zhao, Department of Minimally Invasive Interventional Therapy, Liver Cancer Study and Service Group, Sun Yat-sen University Cancer Center, State Key Laboratory of Oncology in South China, Guangzhou, China

Background: Current clinical trials for advanced hepatocellular carcinoma (HCC) significantly underrepresent patients (pts) with Vp4 portal vein tumor thrombus (PVTT), a poor prognostic factor, including only 0–15% despite their real-world prevalence of 24–49%. Previous FOHAIC-1 study indicated the favorable efficacy of arterial FOLFOX chemotherapy in this population. Preclinical evidence suggested that oxaliplatin in the FOLFOX regimen can induce immunogenic cell death. The DurHope study is a biomolecular exploratory, phase 2 trial of first-line durvalumab combined with arterial FOLFOX in pts with Vp4 PVTT. **Methods:** This trial enrolled 30 HCC pts with Vp4 (n = 24, 80.0%) or Vp3 (n = 6, 20.0%) PVTT. Arterial FOLFOX was given on day 1–3, plus durvalumab (1120 mg) intravenously on day 7 (± 2) of each 21-day cycle up to 8 cycles. Maintenance durvalumab (1500 mg) was administered every 28 days thereafter until disease progression, unacceptable toxicity, or death. Primary endpoint was 1-year OS rate. Secondary endpoints included progression-free survival (PFS), objective response rate (ORR) per RECIST v1.1, and safety. Single-cell RNA sequencing (scRNA-seq) was performed on pre-treatment biopsies and peripheral blood mononuclear cells (PBMCs) collected at baseline and during treatment from responders (CR/PR) and nonresponders (SD/PD). **Results:** With a median follow-up of 21.6 months, the 1-year OS rate was 63.3%, median OS was 13.9 months (95% CI, 10.7–NR), and median PFS was 9.0 months (95% CI, 4.5–12.2). ORR was 53.3% (CR: 5/30 [16.7%]; PR: 11/30 [36.7%]), and median duration of response was 9.1 months (> 6 months: 11/16 [68.8%]; > 12 months: 4/16 [25%]). Grade ≥ 3 TRAEs occurred in 43.3% of patients, with no unexpected safety events beyond those of each agent. ScRNA-seq of biopsies revealed enrichment of chemotherapy resistance and immune escape signatures in non-responders, while responders exhibited higher immune cell infiltration and stronger immune-malignant cell interaction. Dynamic scRNA-seq of serial PBMCs demonstrated expanded T cell subsets and increased expression of cytotoxicity-related genes (e.g., GZMB, GNLY, and IFNG) and proliferation marker (MKI67) after treatment, which was more pronounced in responders. This was accompanied by decreased non-classical monocyte frequency and reduced M2-directed monocyte polarization in responders. In vivo mouse models further showed enhanced efficacy of the combination of chemotherapy and PD-L1 blockade, indicating the synergistic effects of this therapeutic approach. **Conclusions:** Durvalumab plus arterial FOLFOX showed promising efficacy and manageable toxicities in advanced HCC with Vp4 PVTT. Biomolecular exploratory analyses revealed enhanced cellular immune responses triggered by treatment, along with novel molecular biomarkers that may help predict therapeutic response. Clinical trial information: NCT04945720. Research Sponsor: AstraZeneca.

577 Poster Session

The effect of TU-100 on hepatocarcinogenesis and colorectal liver metastasis in MASH model. First Author: Shinichiro Yamada, Department of Surgery, Tokushima University, Tokushima, Japan

Background: TU-100 is most commonly used herbal medicine in clinical setting. We have already reported that TU-100 suppressed dysbiosis (J Med Invest, 2013), bacterial translocation (Surg Today, 2017) and hepatic fibrosis (Surgery, 2016). This time we investigate the effect of TU-100 on hepatocellular carcinoma (HCC) and colorectal liver metastasis (CRLM) in MASH model. **Methods:** 1: Effect on HCC: Seven-week-old Tsumura Suzuki obese diabetes mice, a model that shows the spontaneous onset of MASH and HCC, were used. They were divided into a TU-100 treated group and a control group. Mice were sacrificed at 24 and 48 weeks to evaluate hepatic steatosis, fibrosis, carcinogenesis, cytokine expression, and microbiome abundance. 2: Effect on CRLM: Six-week-old male C57BL/6J mice were used. They were divided into the western diet (WD) group and the WD+TU-100 group. MC38 colon cancer cells were injected into the spleen at 16 weeks, and mice were killed at 2 weeks after injection to evaluate hepatic steatosis, fibrosis and mRNA expression in the liver. **Results:** 1: Effect on HCC: At 24 weeks, the TU-100 group showed significantly lower expression of IL6, IL1B, and ACTA2 mRNA in the liver ($P<0.05$). At 48 weeks, the TU-100 group showed a lower rate of MASH than the control group (28% vs 72%; P=0.1). Tumor diameter was significantly smaller in the TU-100 group compared with that in the control group ($P<0.05$). Regarding the intestinal microbiome, the genera Blautia and Ruminococcus were increased in the TU-100 group ($P<0.05$), whereas Dorea and Erysipelotrichaceae were decreased in the TU-100 group ($P<0.05$). 2: Effect on CRLM: Steatosis of the WD+TU-100 group was significantly suppressed compared with the WD group ($p<0.05$). The WD+TU-100 group showed significantly smaller tumor diameter than the WD group ($p<0.05$). In the WD+TU-100 group, expression of SAA1 and TIMP1 mRNA in the liver was significantly lower than in the WD group ($p<0.05$). **Conclusions:** TU-100 regulates the intestinal microbiome and may suppress hepatic steatosis and subsequent hepatocarcinogenesis in the MASH model. Furthermore, TU-100 may have effect on CRLM via suppression of SAA1/TIMP1. Research Sponsor: None.

578 **Poster Session**

Clinical outcomes of camrelizumab + rivoceranib vs sorafenib (CARES-310) as first-line treatment for patients with unresectable hepatocellular carcinoma (uHCC) of non-viral and viral etiology. First Author: Rachna T. Shroff, University of Arizona Cancer Center, Tucson, AZ

Background: CARES-310 (NCT03764293) evaluated the combination of PD-1 inhibitor, camrelizumab (cam), and VEGFR-1-3 inhibitor, rivoceranib (rivo), compared to sorafenib (sor) for the treatment of uHCC. Cam + rivo significantly improved median overall survival (mOS) and median progression-free survival (mPFS) compared to sor (mOS, 23.8 months [mo] [95% CI 20.6, 27.2] vs 15.2 mo [95% CI 13.2, 18.5] hazard ratio [HR] 0.64 [95% CI 0.52, 0.79]; one-sided $p<0.0001$; mPFS, 5.6 mo [95% CI 5.5, 7.4] vs 3.7 mo [95% CI 3.1, 3.7]; HR 0.54 [95% CI 0.44, 0.67]; one-sided $p<0.0001$). The most common ($\geq$10%) grade $\geq$3 treatment-related adverse events in the cam + rivo arm were hypertension (38.6%) and AST increased (20.2%). **Methods:** A post-hoc analysis of CARES-310 was performed, where mOS and mPFS were estimated using the Kaplan-Meier method and compared between the 3 etiology groups of non-viral, hepatitis C virus (HCV), and hepatitis B virus (HBV) using the log-rank test. **Results:** mOS was longer with camrelizumab plus rivoceranib compared with sorafenib in patients with non-viral (HR 0.68 [95% CI 0.39, 1.19]), HCV (HR 0.37 [95% CI 0.162, 0.84]), and HBV etiologies (HR 0.70 [95% CI 0.55, 0.89]) (Table). Similarly, mPFS was longer with camrelizumab plus rivoceranib compared with sorafenib in patients with non-viral (HR 0.55 [95% CI 0.34, 0.91]), HCV (HR 0.50 [95% CI 0.23, 1.06]), and HBV etiologies (HR 0.57 [95% CI 0.45, 0.72]) (Table). **Conclusions:** Cam + rivo in CARES-310 suggested clinically meaningful mOS benefit in non-viral and viral HCC vs sor and provides assurance of clinical benefit for first line treatment to patients with uHCC independent of etiology. Clinical trial information: NCT03764293. Research Sponsor: Elevar Therapeutics; Jiangsu Hengrui Pharmaceuticals.

Etiology Group	Camrelizumab + Rivoceranib (n=272)	Sorafenib (n=271)
Non-viral, n (%)	42 (15.4%)	45 (16.6)
mOS, mo (95% CI)	26.8 (10.3, NR)	15.2 (9.6, 23.2)
mPFS, mo (95% CI)	6.1 (4.1, 13.8)	3.7 (2.4, 5.5)
HCV, n (%)	22 (8.1)	29 (10.7)
mOS, mo (95% CI)	31.7 (10.8, NR)	13.3 (8.0, 21.5)
mPFS, mo (95% CI)	12.7 (3.7, NR)	3.7 (1.8, 9.1)
HBV, n (%)	208 (76.5)	197 (72.7)
mOS, mo (95% CI)	23.0 (18.9, 26.0)	15.6 (13.3, 19.2)
mPFS, mo (95% CI)	5.6 (5.5, 7.3)	3.7 (2.7, 3.7)

HCC, hepatocellular carcinoma; mOS, median overall survival; mPFS, median progression-free survival; mo, months; HR, hazard ratio; HCV, hepatitis C virus; HBV, hepatitis B virus, NR, not reached. Results used Cox proportional hazard model.
Confidence Intervals (CI) used the Brookmeyer and Crowley method.

579 **Poster Session**

First-line iparomlimab and tuvonralimab (QL1706) or iparomlimab (QL1604) + bevacizumab (BEV) for unresectable hepatocellular carcinoma (HCC): Updated results from the phase Ib/II DUBHE-H-106 study. First Author: Feng Bi, Department of Abdominal Oncology, West China Hospital, Sichuan University, Chengdu, China

Background: Novel treatment options for unresectable HCC are needed. Iparomlimab and tuvonralimab are anti-PD-1 and anti-CTLA-4 antibodies, respectively. The DUBHE-H-106 study aims to assess safety and efficacy of first-line QL1706 or QL1604 + BEV for HCC. Preliminary data have been reported on 2023 ASCO Annual Meeting. Here, we report updated results. **Methods:** This study consists of three cohorts. Systemic therapy-naive adult patients (pts) with HCC, $\geq$ one measurable untreated lesion per RECIST v1.1, BCLC stage B–C, Child-Pugh score $\leq$ 7, not amenable to or progression after locoregional therapy, ECOG performance status of 0–1 were eligible. In Cohort A, six pts received QL1706 5 mg/kg + BEV 15 mg/kg Q3W. If $\leq$ two pts had dose-limiting toxicities (DLT), another six pts would be enrolled. If $\leq$ three of twelve pts had DLT, the safe dose of BEV would be determined, and eight more pts would be enrolled. Otherwise, enrollment of another dose group (QL1706 5 mg/kg + BEV 7.5 mg/kg Q3W) would initiate, using the same procedure. If number of DLT exceeded the criteria, further dose reduction of BEV or study termination would be discussed. Then 40–60 pts were randomized 1:1 to Cohort A or B. In Cohort B, pts received QL1604 200 mg + BEV (safe dose) Q3W. Enrollment of Cohort C would initiate according to the preliminary results of Cohort A and B. Pts received QL1706 7.5 mg/kg + BEV (safe dose) Q3W, using the same procedure in Cohort A. If $\leq$ three of twelve pts had DLT, 8–28 more pts would be enrolled. **Results:** Between Jun 2021 and Dec 2023, Cohort A, B, and C included 50, 26, and 40 pts, respectively. Baseline data were balanced in each cohort. All pts were in the safety set. No DLT was reported. Incidences of adverse events (AE) were similar in three cohorts. Efficacy evaluable set included 47, 26, and 37 pts in three cohorts. Numerically, QL1706 + BEV showed better efficacy compared to QL1604 + BEV, and higher response and 12-month progression-free survival (PFS) rates were found in Cohort C vs Cohort A. Detailed results were shown in Table. **Conclusions:** First-line QL1706 or QL1604 + BEV showed acceptable toxicities and promising efficacy for unresectable HCC. QL1706 7.5 mg/kg + BEV 15 mg/kg Q3W may have better anti-tumor activity and were recommended for trials in future. Clinical trial information: NCT05603039. Research Sponsor: Qilu Pharmaceutical Co., Ltd. (Jinan, China).

Endpoints	Cohort A	Cohort B	Cohort C
Treatment-related AE (TRAE)	43 (86%)	25 (96%)	37 (92%)
Grade ≥ 3 TRAE	24 (48%)	14 (54%)	16 (40%)
Serious TRAE	11 (22%)	9 (35%)	10 (25%)
Immune-related AE	27 (54%)	9 (35%)	22 (55%)
TRAE leading to dose interruption	23 (46%)	16 (62%)	16 (40%)
Objective response, n (%; 95% CI)	18 (38%; 25%–54%)	6 (23%; 9%–44%)	16 (43%; 27%–61%)
Disease control, n (%; 95% CI)	35 (74%; 60%–86%)	18 (69%; 48%–86%)	30 (81%; 65%–92%)
Median PFS (95% CI), months	7.0 (3.1–9.6)	5.4 (2.4–11.0)	7.0 (4.2–not evaluable)
12-month PFS rate (95% CI)	26.8 (14.7–40.4)	24.4 (9.9–42.1)	40.9 (24.3–56.9)

580 **Poster Session**

Lenvatinib plus hepatic arterial infusion chemotherapy of oxaliplatin, fluorouracil, and leucovorin versus lenvatinib alone for advanced hepatocellular carcinoma. First Author: Feng Shi, Guangdong Provincial People's Hospital, Guangzhou, China

Background: Lenvatinib stands out as a first-line therapy for individuals with advanced hepatocellular carcinoma (HCC). Concurrently, hepatic arterial infusion chemotherapy comprising oxaliplatin, fluorouracil, and leucovorin (FOLFOX-HAIC) has emerged as a potential option for those with advanced HCC. It is necessary to investigate the efficacy and safety of lenvatinib plus FOLFOX-HAIC (lenvaHAIC) for advanced HCC in real-world situations. **Methods:** In this retrospective analysis, 127 consecutive patients underwent lenvaHAIC, while 184 patients received lenvatinib alone as first-line treatment at six Chinese academic centers between January 2019 and June 2022. Following 1:1 propensity-score matching, we established paired cohorts (113 patients in each group) for evaluating survival. Overall survival (OS), progression-free survival (PFS), objective response rate (ORR) evaluated by RECIST 1.1 and mRECIST criteria, and safety profiles were compared between the two groups. **Results:** The lenvaHAIC group exhibited significantly prolonged median PFS and OS than the lenvatinib group (PFS: 12.3 vs. 6.2 months; OS: 25.6 vs. 12.3 months; $P < .001$ for each). In the propensity-score matched cohorts (113 pairs), both PFS and OS were notably extended in the lenvaHAIC group compared with those in the lenvatinib group ($P < .001$). Multivariate analysis identified lenvaHAIC treatment as an independent factor for improved PFS (hazard ratio [HR] 0.45; $P < .001$) and OS (HR 0.38; $P < .001$). Grade 3-4 adverse events, including nausea, vomiting, diarrhea, thrombocytopenia, and neutropenia, were more prevalent in the lenvaHAIC group. **Conclusions:** LenvaHAIC may be a promising treatment in patients with advanced hepatocellular carcinoma, demonstrating enhanced OS, PFS, and ORR and an acceptable safety profile. Research Sponsor: None.

581 **Poster Session**

Selective internal radiation therapy with yttrium-90 resin microspheres followed by lenvatinib plus PD-1 inhibitor for large or huge advanced-stage hepatocellular carcinoma: A retrospective cohort study from China. First Author: Jingjun Huang, Department of Interventional Radiology, Minimally Invasive and Interventional Cancer Center, The Second Affiliated Hospital of Guangzhou Medical University, Guangzhou, China

Background: Yttrium-90 resin microspheres have been introduced into China in recent years, but the efficacy or safety on Chinese patients are rarely reported. This study aimed to evaluate the efficacy and safety of selective internal radiation therapy (SIRT) with yttrium-90 resin microspheres followed by lenvatinib and PD-1 inhibitor in Chinese patients with large (5.1–10.0 cm in diameter) or huge (>10.0 cm) advanced-stage hepatocellular carcinoma (HCC). **Methods:** Data of large or huge Barcelona Clinic Liver Cancer stage C HCC patients treated with SIRT followed by lenvatinib plus PD-1 inhibitor from November 2022 to October 2023 were prospectively collected and retrospectively analyzed. Key endpoints included overall survival (OS), progression-free survival (PFS), complete response rate (CRR), objective response rate (ORR), disease control rate (DCR) based on modified Response Evaluation Criteria in Solid Tumors, and adverse events (AEs) graded based on Common Terminology Criteria for Adverse Events v5.0. Risk factors affecting PFS were analyzed using univariate and multivariate Cox regression models. **Results:** A total of 30 patients (27 males, 3 females; mean age 54 $\pm$ 11 years) were included. The mean longest diameter of the tumor was 9.6 $\pm$ 4.2 cm (range: 5.2–17.6 cm). Nineteen (63.3%) patients had more than three intrahepatic lesions, 15 (50.0%) had vascular invasion, and 13 (43.3%) had extrahepatic metastasis. Sixteen (53.3%) patients had undergone prior anticancer therapies. Two patients received two SIRT procedures, and the rest had one. Lenvatinib and PD-1 inhibitor treatment was initiated 3–7 days after the initial SIRT procedure. CRR was 20.0%, ORR was 73.3%, and DCR was 90.0%. Median PFS was 8.0 months (95% CI, 5.6–10.4), and median OS was not reached. OS rates at 6, 12, and 18 months were 93%, 74%, and 74%, respectively. Univariate and multivariate analysis indicated that baseline alpha-fetoprotein levels, baseline eosinophil count, and neutrophil-to-lymphocyte ratio at 1 month after SIRT were independent risk factors for PFS. AEs of any grade occurred in 86.7% of patients, with grade 3 AEs in 23.3%, and no grade 4/5 AEs were recorded. **Conclusions:** SIRT followed by lenvatinib and PD-1 inhibitor demonstrates promising efficacy and a favorable safety profile in Chinese patients with large or huge advanced-stage HCC. Research Sponsor: None.

582 **Poster Session**

IMMULAB: A phase II trial of immunotherapy with pembrolizumab in combination with local ablation for patients with early stage hepatocellular carcinoma (HCC)—The IKF-IMMULAB trial. First Author: Arndt Vogel, Hannover Medical School, Hannover, Germany

Background: Ablation of neoplastic tissue through radiofrequency ablation (RFA), microwave ablation (MWA), or brachytherapy is a potentially curative option for patients with early stage hepatocellular carcinoma (HCC); however, recurrence rates are high. While updated results from IMbrave050 do not support the use of adjuvant therapy following resection or local therapy, the potential of peri-interventional systemic treatment as a potentially more effective approach to increase recurrence free and long-term survival remains to be determined. We propose that peri-interventional treatment with pembrolizumab may improve outcomes following local ablative therapies. **Methods:** This single arm phase II trial investigates peri-interventional treatment with pembrolizumab combined with RFA/MWA or brachytherapy, or - as recommended for tumors ≥ 3 cm - by combination with TACE in early stage HCC with maintained liver function. 200 mg of pembrolizumab (q3w) was administered intravenously for 2 cycles followed by radiologic imaging and local therapy. Pembrolizumab was continued for up to 12 months. Primary endpoint was radiological response rate according to RECIST 1.1 prior to local therapy. **Results:** 30 pts (ECOG 0 or 1) were enrolled in 9 centers in Germany with a median age of 70 years (73.3% males, median tumor size 25 mm with minimum of 10 mm and maximum of 72 mm, 60% solitary lesions). All pts received at least 1 dose of study treatment and the median number of cycles was 13. ORR was 13.3% with 6.7% CR and 6.7% PR after two cycles. After a median follow-up of 34.5 months, median time to recurrence (TTR) was 16.43 months, with 8 pts remaining free of recurrence. mOS was not reached, with a 2y OS rate of 76% and a 3y OS rate of 66%. No new safety signs were observed. Subsequent local ablation was performed in 13 pts, and 10 pts started on systemic therapies (5 TKI, 3 ICI and 2 pts received TKI and ICI). **Conclusions:** In the IKF-IMMULAB trial the expected objective response rate of 30%, according to RECIST1.1., prior to local therapy was not reached. However, with recent evidence that radiologic response underestimates pathological response in ICI-treated HCC patients and supported by a 2y OS rate of 76% and a 3y OS rate of 66%, there is evidence for the efficacy of peri-interventional treatment with pembrolizumab without new safety signals. Clinical trial information: NCT03753659. Research Sponsor: MSD SHARP & DOHME GMBH.

583 **Poster Session**

TYRA-430: First reversible FGFR4/3 inhibitor designed to overcome current challenges in FGF19-driven hepatocellular carcinoma treatment. First Author: Mohamed A. Ahmed, Tyra Biosciences, Carlsbad, CA

Background: Hepatocellular carcinoma (HCC) is an aggressive malignancy that accounts for approximately 80-90% of all primary liver cancers and is projected to become the third leading cause of cancer-related mortality by 2030. Previous work demonstrated that human Fibroblast Growth Factor 19 (FGF19) was overexpressed in 20-30% of HCC, thereby exerting an oncogenic effect via signaling through its cognate receptors FGFR4, FGFR3, and the co-receptor Klotho b (KLB). Multiple selective covalent FGFR4 inhibitors have been clinically evaluated in FGF19-overexpressing HCC. However, selective inhibition of FGFR4 alone was insufficient and resulted in low response rates and limited duration, likely due to redundant signaling through FGFR3. Here we describe the first report on the preclinical profile of TYRA-430, a first-in-class reversible FGFR4/3 inhibitor. **Methods:** The preclinical profile of TYRA-430 was assessed *in vitro* in HCC cellular assays and *in vivo* in mouse xenograft models. The *in vitro* potency was assessed by measuring cell viability using Cell Titer-Glo 2.0 Luminescent assay. Percentage of tumor growth inhibition (TGI) was used as a measure to evaluate *in vivo* efficacy of TYRA-430. **Results:** TYRA-430 exhibited potency against Ba/F3 cells dependent on wildtype FGFR3 and FGFR4, as well as the known Cys552 and Val550 gatekeeper (GK) resistance mutations in FGFR4. *In vitro* data showed that TYRA-430 displayed superior potency over the reversible multi-kinase inhibitors sorafenib and lenvatinib, and the covalent FGFR4 inhibitors fisogatinib (BLU-554) and roblitinib (FGF401) in KLB/FGF-19/FGFR3/4 driven models of HCC (Hep3B, HuH-7, and JHH-7 cells). Furthermore, in a human HuH-7 HCC xenograft model in nu/nu mice, TYRA-430 achieved 96% tumor growth inhibition (TGI), compared to 75% TGI for lenvatinib and 86% TGI for FGF401. Additionally, in the GA180 FGF19-driven gastric cancer PDX model, TYRA-430 demonstrated superior efficacy with 93% tumor growth inhibition (TGI), compared to 56% TGI achieved by roblitinib. **Conclusions:** TYRA-430 was active and potent in multiple KLB/FGF-19/FGFR3/4 models of human hepatocellular carcinoma *in vitro* and *in vivo*. Moreover, TYRA-430 displayed potent activity against known FGFR4 resistance mutations compared to covalent FGFR4-specific inhibitors in the clinic. TYRA-430 will be investigated in a Phase 1 clinical trial in hepatocellular carcinoma and other advanced solid tumors. Research Sponsor: Tyra Biosciences.

584 **Poster Session**

Trastuzumab emtansine as second-line therapy in HER2 positive advanced biliary tract cancers: Single arm prospective phase II clinical trial (TAB-3 trial). First Author: Vikas S. Ostwal, Tata Memorial Hospital (HBNI), Mumbai, India

Background: HER2 overexpression or amplification is a therapeutic target of interest in advanced BTC, especially gallbladder cancers (GBC). Limited studies evaluate the role of T-DM1 in these groups of cancers. **Methods:** This study was an investigator-initiated, open-label, single-arm, phase II trial in patients (pts) aged 18 years or older with HER2-positive (defined as IHC 3+ or IHC 2+, and FISH positive), advanced BTCs who had previously received systemic therapy (irrespective of prior additional HER2 targeted therapy). Pts received T-DM1 at 3.6mg/kg q 3 -weekly till disease progression (PD), unacceptable toxicities, or patient choice. The primary end-point of the study was an improvement in 3-month progression-free survival (PFS) from 50% (historical cohort) to 70% in the study arm. Secondary endpoints included overall response rates (ORR), disease control rate (DCR), overall survival (OS), and incidence of grade 3 and 4 treatment-related adverse events (TRAE). **Results:** From July 2023 to July 2024, '52' pts with HER2-positive BTC were screened, and 40 enrolled in the study. A majority of patients had GBC (95%) and most commonly received combination chemotherapy with gemcitabine-cisplatin, with or without nab-Paclitaxel (78%). Six patients (15%) had previously received trastuzumab in combination with chemotherapy. A majority of patients (n=24; 60%) had rapid progression (within 3 months) on prior systemic therapy. With a median follow-up of 7.1 months [95% confidence interval (CI): 5.1-9.1] , 24 patients had disease progression with a 3-month PFS of 51.2% (95% CI: 33.4 - 69) and median PFS of 3.1 months (95% CI: 2.3-3.8); median OS was 7.1 months (95% CI: 5.1-9.1). Partial responses (PR) were seen in 5 patients (12.5 %) and 8 had stable disease (20 %) for a disease control rate of 32.5 %. Patients without rapid disease progression on prior therapy (n=16; 40%) had a 3-month PFS of 70% (95% CI: 44.8 - 95.2) and median PFS of 4.9 months (95% CI: 2.9-6.9). Grade 3 and grade 4 TRAE in the overall cohort were noted in 11 patients (28%), with the commonest being grade 2 fatigue in 6 patients and grade 3 thrombocytopenia in 2 patients. **Conclusions:** T-DM1 was well tolerated in pretreated HER2 positive advanced BTCs, but did not statistically improve PFS in comparison to historical data. Patients who did not have rapid progression on prior systemic therapy appeared to have a more favorable survival in comparison to rapid progressors and the role of T-DM1 in such favorable cohorts can be explored in larger studies. Clinical trial information: CTRI/2023/07/055785. Research Sponsor: ZYDUS PHARMA PRIVATE LIMITED; TRACK (TATA MEMORIAL CENTRE RESEARCH FUNDING COMITTEE).

585 **Poster Session**

Current status of physicians' perception of HCC conversion/downstaging therapy: A nationwide survey in China. First Author: Xinyu Bi, Department of Hepatobiliary Surgery, National Cancer Center, National Clinical Research Center for Cancer, Cancer Hospital, Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing, China

Background: Conversion or downstaging therapy for intermediate and advanced unresectable hepatocellular carcinoma (HCC) has emerged as a significant exploring area of clinical practice and research in recent years. Nonetheless, there exists considerable variability in both the selection criteria for patients undergoing conversion therapy and the regimens selected for conversion across different regions and medical institutions. The present study aims to elucidate the current clinical application of conversion therapy in China, as well as to identify the considerations that physicians take into account when selecting eligible patients for conversion therapy and determining the appropriate conversion modality. **Methods:** Physicians meeting predefined inclusion criteria were invited to complete an online questionnaire from January to July, 2024. The collected data were subsequently pooled and analyzed descriptively. **Results:** A total of 120 valid questionnaires were retrieved, mainly form surgical (69.2%, n=83) and interventional (30.8%, n=37) departments. Generally, the study showed that approximately 51% of stage Ib-IIIa patients will be selected for the treatment goal of conversion or downstaging, and the overall successful rate as meeting criteria for surgical resection after treatment was 36%. Three primary factors were prioritized by physicians for determination of conversion therapy: the type of portal vein tumor thrombus (PVTT) (97%, 116/120), the future liver volume (90%, 108/120), and Child-Pugh classification (90%,108/120). In the specific physician group who selected the following attributes, patients classified as Child-Pugh A (82%, 89/108) or Child-Pugh B7 (66%, 71/108), those with tumor diameter exceeding 5 cm (92%, 98/106) and Eastern Cooperative Oncology Group (ECOG) performance score of 0 to 1 (89%, 76/85) were most frequently selected for conversion therapy. Additionally, study showed that patients presenting with Vp1-2 (72%, 83/116) and Vp3 (71%, 82/116) could also be considered for conversion therapy. Currently, the prevalent treatment approach of conversion therapy involves a combination of local and systemic therapies, which is utilized in approximately 73% of cases. Among systemic treatment methodologies, the combination of Lenvatinib and immunotherapy is the most widely adopted by physicians. Besides, higher ORR was the foremost consideration for 83% (99/120) of physicians, followed by rapid response (68%, 82/120), adherence to guidelines and consensus (63%, 76/120), and lower tumor progression rate (58%, 70/120). **Conclusions:** This study elucidates the current status of conversion therapy for HCC in China. The findings underscore the necessity for optimizing conversion modalities and advancing standardized practices in the application of conversion therapy. Research Sponsor: Beijing Kangwei institute of Biological Health Sciences.

586 **Poster Session**

Phase II study of modulation of sorafenib (SOR)-induced autophagy using hydroxychloroquine (HCQ) in advanced hepatocellular cancer (HCC). First Author: Sukeshi Patel Arora, Mays Cancer Center, UT Health San Antonio, San Antonio, TX

Background: SOR is the first systemic therapy approved for advanced HCC, but has shown only modest improvements in survival. Resistance to SOR in pre-clinical models has been attributed to autophagy induction. Autophagy inhibition with HCQ enhanced SOR-induced cell death and apoptosis in early pre-clinic and clinical studies. Data from the phase I study of SOR plus HCQ in advanced solid tumors at showed clinical safety and efficacy. Therefore, we conducted a prospective study to evaluate efficacy of SOR and HCQ in advanced HCC patients (pts) (NCT03037437) and report the final efficacy analysis. **Methods:** Prospective phase II study of SOR 400 mg po BID + HCQ 400 mg daily in pts with advanced HCC (CP A-B8 cirrhosis). Cohort 1: first-line SOR/HCQ. Cohort 2: add HCQ upon progressing on SOR. CP B pts started at 200 mg BID, with dose escalation as tolerated. Cycle = 4 weeks. Primary endpoint: mTTP. Secondary endpoints: mOS, objective response rate (ORR) by RECIST; AEs (NCI-CTCAEv3.0); PD analysis for markers of autophagy and immunity. Pts evaluable for efficacy if completed C1. Historically, the mTTP for patients treated with sorafenib is 5.5 mo (SHARP trial). We predicted the addition of HCQ will improve mTTP by 50% to 8.2 months. **Results:** N=33 completed C1. Median age 64.1 years SD 8.6 (46- 80). 88% Male; 58% Hispanics. ECOG 0-1: 100%. CP B cirrhosis: 35%. Etiology of cirrhosis: HCV 68%, ETOH 32%, NASH 11%. BCLC B 19%, C 81%. AFP>400: 43%, PVT: 27%, extrahepatic/metastases: 68%, post-transplant: 27%. N=5 (14%) were in cohort 2 (prior SOR). Reason off study: PD (n=21), toxicity (n=5), lost to f/u (n=1), withdrew (n=3). mTTP is 8.0 months (95% CI: 4.0-19.0). mOS 10.0 months (95% CI: 6.0-22.0). ORR (CR+PR): 27%. Best response: CR n=1 (3%), PR n=8 (24%), SD n=14 (42%). 4+ cycles: n=19 (58%). AE on treatment: grade 1 (68%), 2 (24%), 3 (7%), 4 (0.6%), 5 (0.3%). No Gr 4/5 related to SOR or HCQ. Dose interruption (20%), dose reduction (10%). **Conclusions:** SOR/HCQ had a better mTTP (8.0 months) than historical control (mTTP=5.5 months, p=0.03), and the trial met its primary endpoint. SOR/HCQ ORR (27%) was higher than historically SOR alone (ORR 2%, SHARP) in pts with advanced HCC, predominantly BCLC C, with CP A and B cirrhosis (35%), and was tolerable. While immune checkpoint inhibitors (ICIs) are taking the forefront in advanced HCC, SOR/HCQ may still have a role in patients with CP B cirrhosis, post-transplant, or contraindications to ICIs. Further, analysis of predictive markers of response is ongoing. Clinical trial information: NCT03037437. Research Sponsor: Mays Cancer Center, Pilot Award; NIH CA054174.

587 **Poster Session**

A multicenter, phase 2 study of atezolizumab plus bevacizumab (atezo+bev) combination therapy in patients with unresectable hepatocellular carcinoma (HCC) and Child–Pugh class B cirrhosis: CHALLENGE trial final results. First Author: Masafumi Ikeda, Department of Hepatobiliary and Pancreatic Oncology, National Cancer Center Hospital East, Kashiwa, Japan

Background: The Phase 2 CHALLENGE trial (jRCTs031210355) evaluated the safety and efficacy of atezo+bev combination therapy in patients with advanced HCC with Child–Pugh class B cirrhosis. Results of the trial's main analysis demonstrated that atezo+bev was well-tolerated. Here, we present the trial's final safety and efficacy outcomes. **Methods:** This multicenter, open-label, single-arm, Phase 2 study enrolled patients with advanced HCC and a Child–Pugh score of 7 or 8 who had not received prior systemic therapy. Patients were administered atezolizumab 1200 mg plus bevacizumab 15 mg/kg every 3 weeks. The primary endpoint was the frequency of severe adverse events (SAEs). Secondary endpoints included the objective response rate (ORR), progression-free survival (PFS), time to progression (TTP), overall survival (OS), and frequency of adverse events (AEs). **Results:** A total of 31 patients were enrolled between December 2021 and April 2023. Child–Pugh score was 7 points in 25 patients (80.6%) and 8 points in 6 patients (19.4%). Fourteen patients (45.2%) were classified as having Barcelona Clinic Liver Cancer stage C. We set 30 eligible patients as the population for analysis. The final analysis performed with a median follow-up duration of 517days showed that the frequency of SAEs was 30.0% (95% confidence interval [CI] 14.7–49.4%). ORRs according to RECIST 1.1 and modified RECIST were 40.0% and 46.7%, respectively. Median PFS, TTP, and OS were 240 days (95% CI: 176–526 days), 240 days (95% CI: 176–526 days), and 470 days (95% CI: 256–576 days), respectively. Frequent Grade ≥3 SAEs were blood bilirubin increased (n=3), proteinuria (n=3), hypoalbuminemia (n=2), and hypertension (n=2). Grade ≥3 severe liver-related AEs occurred in 7 patients (23.3%), including blood bilirubin increased (n=3) and esophageal varices hemorrhage (n=2). Grade ≥3 severe immune-related AEs occurred in 2 patients – these were mucositis oral (n=1) and malaise (n=1). The mean Child–Pugh score (±standard deviation) at the start of treatment was 7.2±0.4, and was 7.0±1.0, 7.2±0.9, and 7.6±1.7 at the third and fifth cycles and at the withdrawal study, respectively. **Conclusions:** Atezo+bev was well-tolerated with demonstrable anti-tumor effects in patients with unresectable HCC and Child–Pugh class B cirrhosis. Additional studies are warranted to confirm the efficacy results of atezo+bev in this patient group. Clinical trial information: jRCTs031210355. Research Sponsor: None.

588 **Poster Session**

Updated results from ALTER-H004 study: Anlotinib combined with TACE as adjuvant therapy in patients with hepatocellular carcinoma (HCC) at high risk of recurrence after surgery—A single arm, multi-center, phase II clinical trial. First Author: Zheng Wu, Department of Hepatobiliary Surgery, the First Affiliated Hospital of Xi'an Jiaotong University, Xi'an, Shaanxi, China

Background: High recurrence rate after curative resection significantly compromised the long-term survival of patients with HCC patients (pts). TACE as adjuvant therapy was proved to improve clinical outcomes for pts with high recurrence risk factors after hepatectomy. However, the prognosis of TACE reminded unsatisfactory and controversial. Therefore, this study aimed to identify the efficacy and safety of anlotinib as maintenance adjuvant treatment following inductional TACE in pts with HCC. The preliminary results had been reported in 2024 ASCO GI (Poster 513) and the update results were reported here with longer follow-up duration. **Methods:** HCC pts with one of the following high risk factors after hepatectomy: ≥5 cm and <10 cm of tumor diameter, tumor number ≥3, tumor microvascular invasion M1 or M2, portal vein tumor thrombus resection (Cheng's classification I/II) were recruited. Eligible pts received cTACE treatment within 1-2 months after hepatectomy, on the 3-5th day after cTACE, oral anlotinib (12mg, qd, d1-d14, q3w) was given until disease recurrence or unacceptable toxicity. The predefined sample size was 30. The primary endpoint was disease free survival (DFS). Secondary endpoints included 1-year DFS rate, time to recurrence and safety. **Results:** The data cut-off date was April 2024, 29 pts were enrolled and 27 pts received at least once tumor assessment. The median follow-up was 34.1 (IQR, 32.0-40.9) months. Of 27 with assessable efficacy, mDFS was 40.9 (95% CI, 11.4-70.3) months, with 1-, 2-, and 3-year DFS rates of 72.1% (95% CI, 50.2%-85.6%), 63.0% (95% CI, 40.7%-78.9%), and 56.7% (95% CI, 33.6%-74.5%), respectively. 21 of 27 pts (77.8%) experienced treatment-related adverse events (TRAEs). Grade 3 TRAEs occurred in 5 pts (18.5%) included hypertension (7.4%), leukocytopenia (7.4%) and ascites (3.7%), and blood bilirubin increased (3.7%). No grade 4 or 5 TRAEs occurred. **Conclusions:** Anlotinib combined with TACE as adjuvant therapy in HCC pts at high recurrence risk exhibited encouraging efficacy and tolerable safety profile. And the conclusions needed to be confirmed in large scale clinical trials subsequently. Clinical trial information: NCT04213118. Research Sponsor: None.

589 **Poster Session**

Niraparib in patients with *BRCA*-mutated unresectable or recurrent biliary tract, pancreatic and other gastrointestinal cancers: An investigator-initiated phase 2 trial (NIR-B trial). First Author: Yasuyuki Kawamoto, Division of Cancer Center, Hokkaido University Hospital, Sapporo, Japan

Background: It has been reported that there are *BRCA1/2*-mutated patients in biliary tract, pancreatic, and other gastrointestinal cancers. Niraparib is a poly(ADP-ribose) polymerase (PARP) inhibitor, and PARP inhibitors exert their cytotoxicity against cancer cells in the context of homologous recombination deficiency, such as *BRCA* mutations. **Methods:** Main eligibility criteria are unresectable, advanced or recurrent biliary tract cancers (BTC; cohort A), pancreatic cancers (PC; cohort B), and other gastrointestinal cancers (cohort C) with *BRCA1/2* gene mutations identified by germline test or genomic profiling test with either circulating tumor DNA (ctDNA) or tumor tissue, refractory or intolerant to previous treatments, and adequate organ function. Patients with body weight of 77 kg or more and a platelet count of 150,000 /μL or more receive 300 mg of niraparib, and less than 77 kg or having a platelet count less than 150,000 /μL received 200 mg of niraparib, orally once daily, until disease progression or intolerable adverse events occurred. Primary endpoint was the investigator-assessed objective response rate (ORR) in each cohort with a threshold response rate of 10% and an expected response rate of 35%. Key secondary endpoints were progression-free survival (PFS), overall survival (OS), disease control rate (DCR), and safety. Furthermore, pre-treatment ctDNA was collected and analyzed by Guardant360. **Results:** A total of 62 pts, 26 in cohort A, 26 in cohort B, and 10 in cohort C were enrolled between March 2021 and April 2023. Because one patient in cohort C did not receive protocol treatment, 61 pts were identified for primary efficacy and safety analysis. Median number of prior regimens was 1/2/2 (range, 1–2/1–2/1-6). In cohort A/B/C, the confirmed ORR was 15.4/15.4/0% [95% highest posterior density credible interval, 8.4–27.9/8.4–27.9/0.0–25.9]. The DCR, median (m) PFS, mOS were 57.7/53.8/22.2% [95% confidence interval (CI), 36.9–76.6/33.4–73.4/2.8–60.0], 2.7/1.8/1.4 months (95% CI, 1.5–4.1/1.4–9.3/1.0–2.8), 7.8/9.5/7.4 months (95% CI, 5.9–9.8/3.5–NE/2.9–8.6), respectively. In the pts with BRCA mutations by Guardant360 (cohort A/B/C were 18/21/8), ORR was 11.1/19.0/0% (95% CI, 2.0-30.2/6.5–38.3/0.0–28.3). The most common treatment-emergent adverse events were thrombocytopenia (31.1%), anemia (27.9%), neutropenia (14.8%), nausea (36.1%), fatigue (29.5%), anorexia (24.6%). **Conclusions:** Although niraparib had signs of clinical activity in pts with *BRCA*-mutated BTC and PC, the primary endpoint was not achieved statistically. No new safety signal was observed. Alternative approaches, such as evaluation in biomarker-selected patients or in combination with other agents, may demonstrate greater clinical activity of niraparib in this setting. Clinical trial information: jRCT2011200023. Research Sponsor: Japan Agency for Medical Research and Development; Takeda Pharmaceutical Co., Ltd.

590 **Poster Session**

Neoadjuvant immune checkpoint blockade in resectable hepatocellular carcinoma: A systematic review and meta-analyisis. First Author: Luís Felipe Leite da Silva, Universidade Federal Fluminense, Niterói, Brazil

Background: Hepatocellular carcinoma (HCC) presents a significant therapeutic challenge due to its high recurrence rates after curative-intent resection. Neoadjuvant immune checkpoint blockade (ICB) has emerged as a promising strategy to improve tumor control by modulating the immune microenvironment prior to surgery. This systematic review and meta-analysis aim to assess the safety and efficacy of neo-adjuvant ICB in improving clinical outcomes for patients with resectable HCC. **Methods:** A systematic search of PubMed, EMBASE, and Cochrane databases was conducted to identify clinical trials examining neoadjuvant ICB in resectable HCC. Primary endpoints included pathological complete response (pCR), major pathological response (mPR), overall response rate (ORR), and grade 3-4 treatment-related adverse events (TRAE). Prespecified subgroup analyses were conducted for studies combining ICB with tyrosine kinase inhibitors (TKIs) or dual ICB therapies. Pooled proportions were calculated using a random-effects model via the R package "meta." Inter-study heterogeneity was assessed using the I^2 statistic and test for subgroup differences were conducted. **Results:** Out of 1,321 studies screened, 14 clinical trials involving 252 patients with resectable HCC were included. Most studies were phase II trials (57%), and patient median age ranged from 57.5 to 68.0 years, with 88.3% of patients being male. The pooled prevalence of mPR was 32% (95% CI, 24–42%; I^2 = 9%)while pCR was achieved in 24% (95% CI, 17–34%; I^2 = 29%)of patients. The ORR was 27% (95% CI, 18–39%; I^2 = 49%), and severe adverse events (grade 3-4 TRAE) had a pooled prevalence of 24% (95% CI, 17–34%; I^2 = 16%). Subgroup analysis of trials combining ICB with TKIs showed an ORR 34% (95% CI, 20–55%; I^2 = 60%), an mPR rate of 27% (95% CI, 19–40%; I^2 = 0%), and a pCR rate of 20% (95% CI, 14–30%; I^2 = 0%). Dual ICB therapies demonstrated a higher mPR rate of 42% (95% CI, 26–67%; I^2 = 16%) and a pCR rate of 41% (95% CI, 29–59%; I^2 = 29%), but were associated with a higher incidence of severe TRAE (41%; 95% CI, 29–60%; I^2 = 0%). **Conclusions:** Neoadjuvant ICB demonstrates promising efficacy in resectable HCC, particularly when combined with TKIs or dual ICB therapies. While dual ICB enhances pathological response rates, it is associated with higher rates of severe TRAE. These results highlight the need of further studies to optimize combination strategies that balance efficacy and safety for patients with resectable HCC. Research Sponsor: None.

592 **Poster Session**

Long-term outcomes and surgical conversion following immunotherapy in MSI-H biliary tract cancers. First Author: Priyanshi Shah, Mayo Clinic Rochester, Rochester, MN

Background: Deficient mismatch repair/microsatellite instability (MSI-H) in biliary tract cancers (BTC) is rare, estimated to be 1-5%. MSI-H across solid tumors has shown unprecedented durable response to immune checkpoint inhibitors (ICI) regardless of anatomic location. Here we describe the clinical outcomes of unresectable BTC patients with MSI-H treated with ICIs and their conversion to surgery. **Methods:** We conducted a multicenter, retrospective analysis of patients (pts) with advanced BTC who had been identified with MSI-H by molecular analysis performed between 2017 and 2024 at Mayo Clinic and University Hospitals Seidman Cancer Center. The outcomes of interest were overall survival (OS), defined as time from initiation of first-line systemic treatment to death; Time-to-treatment failure (TTF), defined as time from initiation of treatment to cessation due to any cause; and response. All analyses were done descriptively. **Results:** We identified 22 MSI-H pts with BTC, 19 with cholangiocarcinoma and 3 with gallbladder cancer. Median age at diagnosis was 61 (range 26-86), 20 (90.9%) were Caucasian, 9 (40.9%) were male gender, and 5 (22.7%) had etiology as Lynch syndrome. Twelve (54.5%) pts had metastatic disease at diagnosis. Twenty pts (90.9%) received first line systemic therapy, of whom 11 (55%) continued onto second line. Among the 20 pts that received first line therapy, median TTF was 7.6 months (95% CI: 4.8-14.3), objective response was observed in 7 (35%) pts, and disease control was observed in 15 pts (75%). Among the 11 pts that continued onto second line systemic treatment, objective response was observed in 4 pts (36%) and disease control was observed in 6 pts (55%). Among the 20 pts that received first line therapy, 19 (95%) ever received ICI +/- chemotherapy as a part of their treatment plan. Two-year OS for pts who ever received ICI was 63% (95% CI: 41%-96%). In total, 4 pts (20%) were restaged to re-sectable following treatment with an ICI, underwent resection, and currently have no evidence of disease. **Conclusions:** Administration of ICI during treatment plan in pts with MSI-H confirmed durable response and even downstage pts to resectable stage for curative intent. Testing for MSI-H should be performed upfront to guide treatment decision. Research Sponsor: None.

593 **Poster Session**

Surufatinib combined with locoregional therapies and immune checkpoint inhibitor (ICI) for treating unresectable or metastatic intrahepatic cholangiocarcinoma. First Author: Wan Guang Zhang, Hepatic Surgical Center, Tongji Hospital, Tongji Medical College, Huazhong University of Science and Technology, Wuhan, China

Background: Advanced metastatic ICC, characterized by poor survival and limited therapeutic options, may benefit from the combined use of surufatinib—a selective tyrosine kinase inhibitor targeting VEGFR 1, 2, and 3, FGFR1, and CSF-1R—and ICIs, alongside locoregional treatments like TACE, DEB-TACE, and HAIC. **Methods:** Eligible pts who were 18 -75 years old with histologically confirmed unresectable or metastatic intrahepatic cholangiocarcinoma were enrolled. Pts received surufatinib (250mg, orally daily), ICI, and locoregional therapies until surgery, disease progression, death, intolerable toxicity, or withdrawal of consent. The primary endpoint was objective response rate (ORR). Secondary endpoints included progression-free survival (PFS), disease control rate (DCR), overall survival (OS), conversion to surgical resection rate and safety. **Results:** By July 31, 2024, 20 pts were enrolled and 10 pts were evaluated. Median age was 56.4 years (range: 36-69), with a majority being male (70%). 9 pts were pathologically confirmed as adenocarcinoma, 1patient was small bile tract type. Treatment included DEB-TACE (20%), TACE (30%), and HAIC (50%). 50% (5/10) pts achieved partial response (PR), 20% (2/10) pts accepted conversion surgery, 30% (3/10) pts achieved stable disease (SD), and the confirmed ORR was 50%, DCR was 80%. The median PFS (95%CI: 6.9-NA months) had not yet matured, with no significant differences in survival benefits based on age or gender, though TACE showed potential benefits. The most common AEs of all grades were nausea (30%), dizziness (30%), and headache (10%), No grade $\geq$ 3 TEAEs or new safety signals occurred. **Conclusions:** Surufatinib plus immune checkpoint inhibitor, along with locoregional therapies showed preliminary anti-tumor activity and manageable toxicity for the 1L treatment of ICC, providing an additional treatment option for pts with ICC. Clinical trial information: NCT05236699. Research Sponsor: None.

594 **Poster Session**

Phase I/Ib study of afatinib with capecitabine in patients with refractory solid tumors and pancreaticobiliary cancers. First Author: Gentry Teng King, University of Washington Fred Hutchinson Cancer Center, Seattle, WA

Background: The epidermal growth factor receptor (EGFR) and ErbB2/HER2 signaling is overactive in many solid tumors. Afatinib, an irreversible inhibitor of ErbB signaling, downregulates thymidine synthase and synergizes with capecitabine in preclinical models. This phase I trial evaluated the safety, maximum tolerated dose (MTD) and preliminary efficacy of afatinib plus capecitabine in patients (pts) with refractory solid tumors and pancreatic ductal adenocarcinoma (PDA) and biliary tract cancers (BTC). **Methods:** The phase I study had a 3+3 design with capecitabine 1000 mg/m^2 twice daily (BID) on days 1-14 and afatinib 20 mg, 30 mg or 40 mg daily (QD) in 21-day cycles. In phase Ib, 15 pts each with PDA and BTC were treated at MTD. **Results:** 41 pts enrolled from November 2016 to October 2021, who received a median of 2 (range 1-5) prior therapies. No dose-limiting toxicities were observed. The MTD was 40 mg afatinib QD plus 1000 mg/m^2 capecitabine BID. Grade 3 treatment related adverse events were diarrhea (12.2%) and nausea (4.9%). Among 36 response evaluable pts, best response was 1 partial response (PR, 3%) in a pt with *KRAS* wild type (WT)/EGFR amplified BTC, and 8 stable diseases (SD, 22%), with disease control rate (DCR) of 25%. PDA pts (n=20) had DCR of 24%, median progression-free (PFS) and overall survival (OS) of 1.9 months (95% CI 1.05, 2.03) and 3.2 months (95% CI 2.01, 5.82) respectively. BTC pts (n=18) had DCR of 31%, median PFS and OS of 1.9 months (95% CI 1.55, 3.39) and 4.6 months (95% CI 1.88, 6.09), respectively. Eight pts with cancers with EGFR/HER2/HER3 pathway alterations, most $KRAS^{WT}$ (n=6), had DCR of 29%, and median PFS and OS of 2.1 months (95% CI 1.05, 4.14) and 3.5 months (95% CI 1.64, 6.09), respectively. **Conclusions:** Afatinib plus capecitabine is tolerable but does not have significant efficacy in pts with refractory PDA/BTC, including in $KRAS^{WT}$ cancers with EGFR/HER2/HER3 alterations. Clinical trial information: NCT02451553. Research Sponsor: None.

595 **Poster Session**

Efficacy and safety of durvalumab and gemcitabine-based chemotherapy combined with or without lenvatinib in advanced biliary tract cancer. First Author: Qi Ding, The First Affiliated Hospital of University of Science and Technology of China, Hefei, Anhui, China

Background: Immunotherapy in combination with gemcitabine-based chemotherapy, has now become the standard first-line treatment for advanced biliary tract cancer (BTC). Some small-sample studies have shown that immune checkpoint inhibitors (ICIs), when used in combination with lenvatinib and chemotherapy as first-line therapy, exhibit high anti-tumor activity in BTC. Therefore, we conducted a retrospective study to assess the efficacy and safety of durvalumab and gemcitabine-based chemotherapy with or without lenvatinib, in combination for advanced BTC in the real world. **Methods:** Between January 2021 and September 2024, patients with advanced BTC who received either durvalumab combined with or without lenvatinib, along with gemcitabine-based chemotherapy, were enrolled in this study. The study endpoints were overall survival (OS), progression-free survival (PFS), objective response rate (ORR), disease control rate (DCR), and safety. **Results:** Thirty patients with advanced BTC were included in this study. Fourteen (46.7%) patients had received durvalumab combined with lenvatinib plus chemotherapy. For all the patients, the median OS was 9.5 months(95% CI: 4.98-14.02), and the median PFS was 5.9 months(95% CI: 4.56-7.24). ORR was 43.4% and DCR was 76.7%. Patients with lenvatinib treatment had a longer median PFS (7.1 vs. 4.6 months, P = 0.06) compared with the non-lenvatinib treatment group. However, the median OS was shorter in patients with lenvatinib treatment group (9.0 vs. 9.5 months, P = 0.915). ORR and DCR in patients with lenvatinib were 57.1% and 92.9%, respectively. However, in patients without lenvatinib treatment group, ORR and DCR were only 31.2% and 62.5%, respectively. All patients experienced adverse events (AEs), addition of lenvatinib does not increase the risk of AEs. The data is still being updated. **Conclusions:** Durvalumab and gemcitabine-based chemotherapy combined with or without lenvatinib has shown to be effective and safe in routine practice. Regarding lenvatinib, its addition has only improved the DCR, ORR, and PFS, but has not been able to prolong OS. Research Sponsor: None.

596 **Poster Session**

Adjuvant capecitabine in resected biliary tract cancers: Real life data from a multicenter study (PRODIGE83-ACABi-PRONOBIL). First Author: Anthony Turpin, Medical Oncology Department, Lille University Hospital, University of Lille, Lille, France

Background: Biliary tract cancers (BTCs) are rare and heterogeneous tumors with a poor survival even in early-stage patients treated with a curative surgery. Since 2019 the current standard of care in the adjuvant setting is oral chemotherapy (CT) with capecitabine for 6 months, as an option following BILCAP phase III trial results. Nevertheless, outside of this trial, real-world data are lacking. **Methods:** We conducted a French, retrospective, multicenter study from the ACABi-PRONOBIL cohort. Eligible patients had intrahepatic, perihilar, distal cholangiocarcinoma or gallbladder carcinoma treated by surgery with curative intent without prior therapy. Two groups were compared, patients who received capecitabine (Cape group) as adjuvant CT and patients without adjuvant treatment before 2017 (surveillance group). The primary endpoint was overall survival (OS). Secondary endpoints were disease-free survival (DFS) and toxicity. **Results:** 324 patients were included, 200 in the Cape group and 124 in the surveillance group. The median OS was 43.5 months (CI95%= 35.4-48.0) and 44.6 months (CI95% 37.7-54.8) in the Cape and surveillance group, respectively. ECOG performance status ≥ 2, pT3 stage, vascular and node invasion were associated with a poorer OS. The DFS was 15.8 months (CI95%= 14.1-20.3) in the Cape group and 13.3 months (CI95%= 10.1-17.5) in the surveillance group. In inverse probability of treatment weighting (IPTW) analysis we observed a significant statistical association between Cape group and DFS (IPTW HR (CI95%) Cape group vs surveillance group = 0.76 (0.60-0.96), pvalue=0.0224) but not with OS. In the Cape group, 49% of patients had at least one dose reduction, 35% had to discontinue treatment due to toxicity (mainly gastrointestinal,13.3% of grade 3-4; and cutaneous, 27% of grade 3-4). **Conclusions:** In this retrospective analysis, adjuvant capecitabine appears to increase DFS without statistically significant impact on OS in patients undergoing surgery for BTC, confirming in real life the BILCAP study results. Research Sponsor: None.

597 **Poster Session**

Comparative restricted mean survival time (RMST) analysis of survival in advanced hepatocellular carcinoma (aHCC) from pivotal phase III trials: IMbrave-150, ORIENT-32, CARES-310, HIMALAYA, and CM-9DW. First Author: Roberto A. Pazo Cid, Medical Oncology Department, Hospital Universitario Miguel Servet (HUMS), Aragon Health Research Institute (IIS-A), Zaragoza, Spain

Background: In advanced hepatocellular carcinoma (aHCC), the evaluation of overall survival (OS) and progression-free survival (PFS) through restricted mean survival time (RMST) provides a nuanced understanding of treatment efficacy. The RMST analysis serves as a valuable complement to the hazard ratio (HR) analysis, offering insights into the average survival time over a specified period. This study compares RMST analyses of OS at 24 and 36 months, and PFS at 12 and 18 months, across five pivotal phase III trials: IMbrave-150, ORIENT-32, CARES-310, HIMALAYA, and CM-9DW. **Methods:** Data from the experimental arms of the five phase III trials were analyzed. The RMST was calculated using the area under the Kaplan-Meier survival curves up to the specified time points. Specifically, OS was evaluated at 24 and 36 months, and PFS at 12 and 18 months. The RMST provides an estimate of the average time that patients survive (for OS) or remain progression-free (for PFS) within a specific timeframe. Kaplan-Meier survival curves were digitized if not available in raw form, and the area under the curve (AUC) was computed using numerical integration methods. RMST values were extracted from the AUC for the specified periods. This approach accounts for censored data and provides a robust comparison of survival times across different treatment groups. **Results:** The results for OS RMST at 24 and 36 months and PFS RMST at 12 and 18 months are summarized in the table. **Conclusions:** Angiogenesis inhibitor + immune checkpoint inhibitor combinations (Atezolizumab + Bevacizumab; Sintilimab + IBI305) provide the most favorable RMST outcomes in terms of OS and PFS. Research Sponsor: None.

Restricted mean survival time (RMST) comparison of OS and PFS (months).

aHCC 1L Phase III Trials	Treatment Arm	OS 24m	OS 36m	PFS 12m	PFS 18m
IMbrave-150 [1, 2]	Atezolizumab + Bevacizumab	17.50	22.79	9.69	13.29
ORIENT-32 [3]	Sinitilimab + IBI305	17.01	21.09	9.18	12.54
CARES-310 [4]	Camrelizumab + Rivoracenib	16.34	19.75	9.29	12.14
HIMALAYA [5, 6]	Tremelimumab + Durvalumab	15.36	18.76	8.95	12.02
CM-9DW [7, 8]	Nivolumab + Ipilimumab	15.87	20.08	8.74	11.78

[1]Finn et al. N Engl J Med 2020;382:1894-905; [2]Cheng et al. J Hepatol 2022;76:862-873; [3]Ren et al. Lancet Oncol 2021;22:977–90; [4]Qin et al. Lancet 2023;402:1133–46; [5]Abou-Alfa et al. NEJM Evid 2022;1(8):EVIDoa2100070; [6]Sangro et al. Ann Oncol 2024;35:448-457; [7]Galle et al. J Clin Oncol 2024;42(suppl 17):abstr LBA4008; [8]Decaens et al. Ann Oncol 2024;35:S657.

598 **Poster Session**

Durvalumab (DUR) effectiveness in advanced biliary tract cancer (aBTC) patients based on real-world data evidence. First Author: Jesus Rodriguez-Pascual, Instituto Oncológico Vithas, Madrid, Spain

Background: Based on TOPAZ-1 trial, Durvalumab (DUR) plus gemcitabine and cisplatin significantly improved overall survival (OS) in advanced biliary tract cancer (aBTC) patients. Real-World data evidence can evaluate the benefit of this combination outside controlled studies comparing patients treated with standard chemotherapy with or without DUR. **Methods:** Using the TriNetX Global Collaborative Network, a platform that operates globally based on anonymized and aggregated clinical data, a sample of aBTC patients from 128 healthcare organizations (HCOs) who met the initial criteria was selected. 5-year overall survival (OS) was analyzed between these cohorts using a Kaplan-Meier analysis. Hazard Ratio (HR) and its 95% confidence interval (95%CI) were calculated to evaluate the difference between cohorts. Propensity Score Matching (PSM) was used to balance the cohorts based on age, gender, and race mitigating possible cofounding variables. All statistical analyses were conducted utilizing the TriNetX Analytics function in the online research. **Results:** A total of 1099 patients met the study criteria and were included in our study. 399 received DUR plus chemotherapy and 700 received only chemotherapy. Patients treated with DUR showed a significant better OS compared to patients treated with only chemotherapy, both previous to PSM and before PSM (post-PMS median OS 418 days vs 359 days, HR 0.862, 95%CI 0.759-0.979, p=0.022). **Conclusions:** This is the largest World-Real Data study exploring the effectiveness of Durvalumab in aBTC patients outside clinical trials, confirming a similar benefit than TOPAZ-1 study. Research Sponsor: None.

599 Poster Session

Efficacy and safety of HER2-targeted therapies in biliary tract cancer: A systematic review and meta-analysis of early phase clinical trials. First Author: Parth Sharma, University of Missouri - Kansas City, Kansas City, MO

Background: Biliary tract cancers (BTC) are a rare type of gastrointestinal malignancy, often presenting in advanced stages and associated with a dismal prognosis. BTC are frequently resistant to conventional chemotherapy, with novel targeted therapies playing a pivotal role in overcoming this challenge. *HER2* has recently emerged as an important actionable target in patients with BTC. We aim to evaluate the efficacy and safety of *HER2*-directed monoclonal antibodies (MOAs), bispecific antibodies, antibody-drug conjugates (ADCs), and their combinations in BTC. **Methods:** A systematic literature search was conducted on PubMed, Embase, and Cochrane Central Register of Controlled Trials for clinical trials investigating *HER2*-directed MOAs, bispecific antibodies, ADCs, and their combinations in BTC. The study was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) Guidelines. A random effects model was used to pool the outcomes along with 95% confidence intervals (CI). Statistical analyses were performed using program R version 4.4.1. **Results:** We included 12 clinical trials with three Phase I and nine Phase II trials. The pooled ORR (objective/overall response rate) was 35.5% (95% CI: 26.2-45.3, I^2: 69%), and the DCR (disease control rate) was 72.2% (95% CI: 65.3-78.8, I^2: 37%). The median follow-up period for DCR was four months. Trastuzumab, in combination with gemcitabine-cisplatin as first-line therapy, had the highest median progression-free survival of 7 months (95% CI: 6.2-7.8), whereas the combination of trastuzumab and tucatinib had the longest median overall survival 15.5 months (90% CI: 6.5-16.7) reported. The most common treatment-related ≥grade 3 adverse events reported across the cohort were anemia and neutropenia. Three cases of confirmed grade 5 interstitial lung disease were reported in patients who received trastuzumab deruxtecan. Clinical trials included in this study are summarized in the table. **Conclusions:** Advances in precision oncology and the emergence of novel targeted therapies have revolutionized cancer management. *HER2*-directed therapies have shown promising results for patients inflicted with BTC having *HER2* alterations in the early phase clinical trials. Further larger studies are awaited to explore the efficacy of *HER2*-directed therapies in BTC. Research Sponsor: None.

Clinical trials of *HER2*-directed therapies in BTC.

HER2-directed therapy	Number of clinical trials	Total number of BTC participants	ORR, in % (95% CI)	DCR, in % (95% CI)
Trastuzumab + chemo	2	124	43.2 (19.4-68.8)	80 (72.3-86.7)
Trastuzumab + pertuzumab	2	45	20.9 (9.1-35.3)	53.4 (37.8-68.8)
Trastuzumab deruxtecan	4	84	27.2 (6.9-52.6)	71.9 (58.4-84)
Trastuzumab + tucatinib	1	30	46.7 (28.5-65.5)	76.7 (57.7-89.7)
Zanidatamab	2	101	40.5 (30.9-50.4)	67.6 (57.9-76.6)
SHR-A1811	1	16	56.3 (29.9-80.2)	81.3 (54.4-96)

600 Poster Session

Long-term survival outcomes of atezolizumab plus bevacizumab treatment in patients with advanced hepatocellular carcinoma. First Author: Ching-Tso Chen, Department of Oncology, National Taiwan University Hospital Hsinchu Branch, Hsinchu, Taiwan

Background: Immunotherapy combinations such as atezolizumab plus bevacizumab (Atezo-Bev) or STRIDE have been established as standard therapies for patients with advanced hepatocellular carcinoma (HCC). After dual immunotherapy treatment, durable responders and long-term survivors had been reported. However, the longevity of such outcomes following Atezo-Bev treatment remains to be evaluated. **Methods:** We analyzed medical records of four medical centers in Taiwan for patients with Child-Pugh class A liver reserve who received first-line Atezo-Bev treatment for advanced HCC from January 2018 to May 2021, to evaluate their survival outcomes after a long-term follow-up. **Results:** We enrolled 54 patients, predominantly male (90.7%) with the median age of 65 years. The main hepatitis etiology was viral hepatitis (92.6%), including 68.5% chronic hepatitis B. After a median follow-up of 61.9 months, the median overall survival (OS) was 21.0 months (95% confidence interval 10.4 – 31.6 months). The 3-year, 4-year and 5-year OS rate was 36.4%, 25.7% and 25.7%, respectively. For patients with tumor response (N = 19, 34.5%) or disease control (N = 44, 80.0%), the 3-year, 4-year, and 5-year OS rates are listed at Table 1. In univariate analysis, younger age (≤ 70 years), absence of intrahepatic tumor, lack of macrovascular invasion (MVI), and ALBI grade 1 were associated with an OS > 3 years. In multivariate analysis, after adjusting sex, viral hepatitis, MVI, extrahepatic spread, and initial alpha-fetoprotein > 400 ng/ml, the absence of intrahepatic tumor (Odds ratio [OR] = 5.36, p = 0.047) and ALBI grade 1 (vs grade 2, OR = 10.02, p = 0.037) remained independent predictors for OS > 3 years. **Conclusions:** In patients with advanced HCC, Atezo-Bev treatment led to a notable proportion achieving long-term survival, especially in patients with good response to the treatment. Absence of intrahepatic tumors and preserved liver function are predictive of prolonged survival. Research Sponsor: Ministry of Science and Technology, Taiwan; National Science and Technology Council; Ministry of Health and Welfare; National Taiwan University Hospital.

3-year, 4-year and 5-year OS rate of patients who receive Atezo-Bev treatment, stratified by treatment response.

	OS rate (%) by treatment response	
Landmark survival	**Disease control**	**Responder**
3 years	35.6	52.2
4 years	30.3	52.2
5 years	30.3	47.7

Responder: patients who achieved a best response of complete response or partial response after treatment with Atezo-Bev.

Disease control: includes responders and patients who achieved a best response of stable disease following Atezo-Bev treatment.

601 Poster Session

Updated results of ALTER-H006: A phase II study of benmelstobart (PD-L1 inhibitor) plus anlotinib as adjuvant therapy in hepatocellular carcinoma (HCC) with high risk of recurrence after radical resection. First Author: Xianhai Ma0, Hunan Cancer Hospital, The Affiliated Cancer Hospital of Xiangya School of Medicine,Central South University, Changsha, China

Background: Although radical resection offers one of the most favorable prognosis for hepatocellular carcinoma (HCC), the rate of postoperative recurrence remains high, which is the primary cause of mortality in HCC patients. The optimal adjuvant treatment strategy for patients is still under active investigation. Herein we evaluated the efficacy and safety of the anti-angiogenic tyrosine kinase inhibitor, anlotinib, plus benmelstobart, a novel programmed death-ligand 1 (PD-L1) inhibitor as an adjuvant treatment for HCC with high risk of recurrence after radical resection. **Methods:** This study enrolled patients diagnosed with HCC confirmed through histological or cytological examination, whose age was 18-75, with ECOG 0-1, Child-Pugh class A, 4~8 weeks after R0 resection with any of the following high-risk factors for recurrence: a) tumor nodules ≥4; b) portal vein tumor thrombus (PVTT): vp1 or vp2; c) hepatic vein tumor thrombus (HVTT): vv1 or vv2. Enrolled patients received anlotinib (12 mg, p.o., qd, d1-14, q3w) plus benmelstobart (1200 mg, i.v., d1, q3w) until disease recurrence or unacceptable toxicity or up to 18 cycles (~1 year), whichever occurred first. Tumor assessment was performed at the end of the second cycle and every four cycles thereafter, according to RECIST v1.1. The predefined sample size was 37. The primary endpoint was 1-year recurrence-free survival (RFS) rate. Secondary endpoints included RFS, 1-year overall survival (OS) rate, and safety. **Results:** As the cutoff date of September 13,2024, a total of 38 patients (pts) were enrolled and 35 pts included in per-protocol set analysis. The majority of the 38 pts were male (94.7%, n = 36), and the median age was 56.5 years (range: 33-75). 18 pts (47.3%) had CNLC Stage IIb and 20 (52.6%) had CNLC Stage IIIa HCC. Among them, 35 pts received at least once tumor assessment. According to RECIST 1.1, of the 35 pts, 18 had no recurrence, 15 experienced relapse, 1 dropped out and 1 discontinued due to serious adverse events. The 1-year RFS rate was 59.97% (95%CI: 39.12-79.68) and the primary median RFS was 16.89m(95%CI: 6.82-26.96). The 1-year OS rate was 91.47% (95%CI: 69.23-97.86). 34 of 38 pts (89.5%) experienced treatment-related adverse events (TRAEs). The most common grade 3 TRAEs occurred in 17 pts (44.7%) including hypertension (32%) and neutropenia(5.3%). No grade 4 or 5 TRAEs were observed. **Conclusions:** The present study indicated that anlotinib plus benmelstobart as adjuvant treatment for HCC with high risk of recurrence after radical resection exhibited promising efficacy and tolerable safety profile. The conclusion should be validated in more participants included subsequently. Clinical trial information: NCT05111366. Research Sponsor: None.

602 Poster Session

Pembrolizumab and stereotactic radiotherapy combined in subjects with advanced HCC: A phase II study (PEMRAD). First Author: Grainne M. O'Kane, St Vincent's University Hospital, Dublin, Ireland

Background: Combination approaches of immune-checkpoint inhibitors and locoregional therapies have an increasing role in advanced HCC. **Methods:** The PEMRAD phase II trial investigated the efficacy of combination pembrolizumab and stereotactic body radiation (SBRT) following progression on sorafenib. Patients with advanced HCC, ECOG 0-1 and Childs-Pugh A were eligible. Patients received pembrolizumab on day 1 followed by SBRT starting Day 2 and delivered in five fractions. Pembrolizumab was continued every 21 days until progression, unacceptable toxicity or patient withdrawal. The primary endpoint was objective response rate (ORR) as measured by RECISTv 1.1. hypothesizing an increase to 40% with the combination. Tissue and liquid biopsies were collected for correlative analyses including multiplex IHC, longitudinal CyTOF and dynamic cytokine changes. **Results:** The trial was stopped early due to a shift in the treatment landscape. Between March 2018 and March 2023 18 patients of a planned 22 patients were enrolled and treated with combination therapy. Of these 10 (55%) had evidence of macrovascular invasion (MVI) and 15 (83%) had extrahepatic disease; viral hepatitis was the underlying etiology in 50%. The ORR was 41% meeting the proposed endpoint. Median number of cycles of pembrolizumab was 7.5 (2-35) .The median number of lesions treated by SBRT was 3 (1-5) and the median dominant gross tumour volume was 184 cc (IQR 69-285). The median PFS was 5.4 (2.8-9.9) months and median OS 12.6 (5.7-25.8) months. Of the 10 patients with macrovascular invasion 5 (50%) had evidence of a vascular thrombus response by RECIST criteria including 1 complete response. Treatment related adverse events ≥ grade 3 were reported in 5 patients (28%). One death due to myocarditis was attributed to treatment. **Conclusions:** In a poor prognostic group of patients in the second line setting, the combination of SBRT and pembrolizumab demonstrated high ORR and may have a specific role in patients with MVI. Correlative analyses will be presented at the meeting. Clinical trial information: NCT03316872. Research Sponsor: Merck.

603 **Poster Session**

Clinical outcome of radiofrequency ablation in elderly hepatocellular carcinoma patients over 80 years old. First Author: Kenta Takaura, Department of Gastroenterology and Hepatology, Musashino Red Cross Hospital, Musashino-Shi, Tokyo, Japan

Background: The SURF trial showed the efficacy and safety of radiofrequency ablation(RFA). However, patients over 80 years old were excluded from the study. Aims: To evaluate clinical outcomes of RFA for largest hepatocellular carcinoma (HCC) diameter ≤3 cm, and ≤3 HCC nodules in patients over 80 years. **Methods:** Patients who underwent RFA at our institution for initial treatment of largest HCC diameter ≤3 cm, and ≤3 HCC nodules from January 2011 to December 2023. Treatment outcome and prognosis were examined in the elderly group for cases 80 years or older and in the non-elderly group for cases under 80 years. The Cox proportional hazards model was used to analyze the factors associated with treatment outcome and prognosis. **Results:** Of the 518 eligible patients, 136 patients were 80 years or older. Median overall Survival (OS) was 80 (95%CI 60-96) vs. 123 (95%CI 101-NA) months (p=0.021) in the elderly vs. nonelderly group. The median recurrence-free survival was 16 (95%CI 14-22) vs. 26 (95% CI 19-30) months (p=0.023), better in the non-elderly group, Interestingly, for liver disease-related deaths, median OS was 97 (95% CI 80-NA) vs. NR (95% CI NA-NA) months (p=0.62) in the elderly vs. non-elderly group. In the multivariate analysis, factors associated with OS were ALBI grade 2 or 3 (HR 1.67, 95%CI 1.07-2.60), DCP ≥ 40mAU/ml (HR 2.08, 95%CI 1.42-3.05), persistent HCV infection (HCV non-SVR) (HR 5.45, 95%CI 3.07-9.67), nonviral liver disease (HR 4.18, 95%CI 2.31-7.55). HCC recurrence was significantly associated with male (HR 1.51, 95%CI 1.18-1.94), elderly group (HR 1.46,95%CI 1.10-1.95), PT ≥80% (HR 0.69, 95%CI 0.53-0.91), DCP ≥40mAU/ml (HR 1.37, 95%CI 1.07-1.76), and HCV non-SVR (HR 1.73, 95%CI 1.35-2.21). The factors associated with liver disease-related death were ALBI grade 2 or 3 (HR 2.16, 95% CI 1.25-3.73), DCP ≥ 40mAU/ml (HR 2.34, 95% CI 1.48-3.70) and HCV non-SVR (HR 2.22, 95% CI 1.39-3.56). **Conclusions:** In RFA for diameter ≤3 cm, and ≤3 HCC nodules, over 80 years was not a significant factor associated with OS or liver disease-related death. The results support that RFA would be a promising treatment option for HCC patients over 80 years. Research Sponsor: None.

604 **Poster Session**

Mediation of the liver protective effect of the anti-cancer drug candidate namodenoson via adiponectin. First Author: Pnina Fishman, Can-Fite Ltd., Petah Tikva, Israel

Background: Namodenoson (CF102) is a small, orally available, anti-cancer drug candidate, currently in pivotal phase 3 clinical study for the treatment of hepatocellular carcinoma. Adiponectin is a positive cytokine released by adipocytes and endothelial cells, known to induce liver-, cardio-, and neuro-protective effects. We evaluated the potential role of adiponectin as the mediator of the observed namodenoson-induced protective effects. **Methods:** Proteins extracted fromN1S1 tumor-bearing mice and from murine adipocyte culture treated with namodenoson or vehicle were analyzed using Western blot. Apoptosis was determined in the N1S1 tumor-bearing mice using the TUNEL assay. The effect of namodenoson on hepatic ischemia/reperfusion (IR) was evaluated in Wistar rats by temporarily clamping the hepatic vasculature; the effect on hepatic inflammation was evaluated in Con. A-induced hepatitis in mice. The STAM murine in-vivo model was used to study the effect of namodenoson on metabolic dysfunction-associated steatohepatitis (MASH). Adiponectin levels were determined in the serum of patients in the phase 2 trial evaluating namodenoson vs placebo in patients with metabolic dysfunction-associated steatotic liver disease (MASLD)/MASH. **Results:** The anti-cancer effect of namodenoson was attributed to its capability to bind to the A3 adenosine receptor, inducing inhibition of PI3K and subsequent de-regulation of the Wnt/β-catenin and NF-kB signal transduction pathways, resulting in apoptosis of liver tumor cells. At the same time, namodenoson demonstrated a liver protective effect manifested by prevention of hepatic IR injury, suppression of liver inflammation, and induction of anti-fibrotic and anti-steatotic effects. In adipocyte cell culture, namodenoson stimulated adiponectin production. An increase in adiponectin levels was also shown in the murine MASH model (STAM). These in-vitro and in-vivo results were consistent with translational data from the phase 2 study of namodenoson in MASLD/MASH where upregulation of adiponectin was observed with namodenoson (mean change from baseline of 539 ng/mL for namodenoson 12.5 mg BID vs -78 ng/mL for placebo, $P = 0.032$). **Conclusions:** The anti-cancer effect of namodenoson combined with its protective effects, position it as a unique drug inducing a dual effect in patients with advanced liver cancer. Research Sponsor: Can-Fite BioPharma.

605 **Poster Session**

Results from a phase 2 study of triplet blockade of the IL-27, PD-(L)1, and VEGF pathways with casdozokitug (casdozo, CHS-388) in combination with atezolizumab and bevacizumab in patients with unresectable, locally advanced or metastatic hepatocellular carcinoma (uHCC). First Author: Daneng Li, City of Hope National Comprehensive Cancer Center, Duarte, CA

Background: Casdozo is the first in class and only clinical-stage IL-27 targeting antibody, which induces increased serum IFN-γ and NK cell gene activation, leading to reversal of IL-27-mediated immune suppression. A phase (Ph) 1 study demonstrated a favorable safety profile and antitumor activity of casdozo alone and in combination with PD-1 blockade in indications with known high levels of IL-27 pathway activation, including HCC (NCT04374877). This open-label Ph 2 trial examined the potential antitumor activity and safety of casdozo with PD-L1 and VEGF blockade in uHCC (NCT05359861). Updated clinical and biomarker data are presented. **Methods:** Patients (Pts) with untreated uHCC received casdozo 10 mg/kg, atezolizumab (atezo) 1200 mg, and bevacizumab (bev) 15 mg/kg IV Q3W. Primary endpoints were safety and tolerability, with key secondary endpoints of investigator-assessed ORR by RECIST1.1 and mRECIST, PFS, and OS. **Results:** As of Sept 4, 2024, 30 pts received casdozo/atezo/bev. Most pts were male (77%), Asian (67%), had viral etiology (53% HBV, 17% HCV), and poor-prognosis disease assessed by metastatic spread (67%) and/or macrovascular invasion (13%). Median follow-up was 15 months (mo). Median time on therapy was 8 mo. The most common (>20%) adverse events related to any study drug were proteinuria (43%), hypertension (27%), fatigue (23%), and diarrhea (20%); Grade ≥3 events occurred in 67%, with only hypertension (20%), arising in >10% of pts. Treatment-emergent adverse events (TEAEs) resulting in any study drug discontinuation occurred in 30% of pts. No treatment-related deaths were reported. In the 29 RECIST1.1 response-evaluable pts, ORR was 38% with 17% of pts achieving a CR (5 CRs, 6 PRs); mPFS (95% CI) was 8.1 mo (1.9, 13.6); landmark PFS rates at 6, 12 and 18 mo were 58.6%, 36.1% and 27.1%, respectively; mDoR (95% CI) was NR (6.1, -). In the 28 pts evaluable for mRECIST, ORR was 43% (5 CRs, 7 PRs); mPFS (95% CI) was 8.4 mo (2.0, 14.5); mDoR was NR (12.7, -). mOS (95% CI) was 19.9 mo (14.2, -); the 12-mo survival rate (95% CI) was 85% (65, 94). More than 40% of pts remain on study, and 16.7% remain on study treatment. Available archival tissue of responders expressed IL-27+ TAMs by IHC, including tumors of viral and nonviral etiologies. **Conclusions:** Triplet blockade of IL-27, PD-(L)1, and VEGF pathways with casdozo/atezo/bev continues to show a manageable safety profile with promising antitumor activity in uHCC that warrants continued exploration. Toxicity was consistent with the known profiles of the agents, with no new safety signals identified. Biomarker analysis to evaluate immune response and associations of IL-27 pathway and clinical outcomes is ongoing. Additional clinical studies of casdozo in combination with toripalimab are planned. Clinical trial information: NCT05359861. Research Sponsor: Coherus BioSciences; Roche.

606 **Poster Session**

First-in-human phase 1/2 study (MYCHELANGELO I) of first-in-class epigenomic controller OTX-2002 targeting MYC oncogene in patients with hepatocellular carcinoma (HCC) and other solid tumors. First Author: Jonathan Mizrahi, Department of Hematology and Oncology, Ochsner Clinic Foundation, New Orleans, LA

Background: The c-MYC (MYC) oncogene is a master transcription factor of tumor cell and microenvironment regulation; it is often dysregulated in cancer, including HCC. OTX-2002 is a MYC-targeted Epigenomic Controller (MYC-EC), an mRNA drug substance encapsulated in a clinical lipid nanoparticle (LNP), designed to downregulate MYC expression pre-transcriptionally with high specificity and durability, inhibiting tumor cell viability while sparing normal cells. **Methods:** A first-in-human dose-escalation study (NCT05497453) investigates OTX-2002 monotherapy in HCC and other solid tumors. Key outcomes include safety, tolerability, pharmacokinetics (PK) and the recommended dose for expansion. In addition to standard safety measures, LNP-associated safety measures including cytokine profiling and an assessment of the presence of anti-PEG antibodies (ADA) were included. Initial pharmacodynamic activity (PD) was assessed via measurement of DNA methylation at the MYC locus in cell-free DNA [cfDNA] from whole blood. **Results:** As of August 23, 2024, 24 patients were treated at 6 doses (19 HCC, 1 colorectal, 1 cervical, 1 pancreatic, 2 sarcoma). OTX-2002 was generally well-tolerated; most adverse events (89%) were low grade, with infusion-related reactions being the most common (30%). Consistent with known LNP profiles, OTX-2002 adverse events were transient and manageable, including cytokine and ADA expression levels. Plasma PK exposures of OTX-2002 were linear across the first five doses analyzed. Significant, on-target dose-dependent changes in MYC gene methylation were seen in cfDNA for all patients post-dose and were durable for at least 15 days, consistent with the designed durability of the MYC-EC. Updated data will be presented. **Conclusions:** OTX-2002 is well-tolerated and induces rapid and durable epigenomic changes to the MYC locus. These results provide clinical proof of mechanism of the epigenomic controller platform and support continued development of OTX-2002. Clinical trial information: NCT05497453. Research Sponsor: Omega Therapeutics.

607 **Poster Session**

Real-world data of durvalumab plus gemcitabine and cisplatin in patients with advanced biliary tract cancer: A Thailand multicenter observational study. First Author: Suebpong Tanasanvimon, Department of Internal Medicine, Faculty of Medicine, Chulalongkorn University, King Chulalongkorn Memorial Hospital, Thai Red Cross Society, Bangkok, Thailand

Background: Currently, durvalumab plus gemcitabine and cisplatin is a standard of care for patients with advanced biliary tract cancer (BTC). We aimed to evaluate efficacy and tolerability of durvalumab plus gemcitabine and cisplatin in real world experience in Thailand where there is high incidence of liver-fluke related cholangiocarcinoma (CCA). **Methods:** This is an observation study of patients with advanced BTC participating in the early access program of durvalumab since August 2023. Data were collected and analyzed for efficacy and toxicity. **Results:** Fifty patients participated the early access program in 14 centers across Thailand. Median follow-up time was 8.5 months. Median number of treatment cycles was 8.8 (1-16) cycles. Thirty six (72%) patients were alive and 8 (16%) patients were still on treatment. The mean age was 63 years, 60% were male and 92% were ECOG 0-1. There were 68% intrahepatic cholangiocarcinoma (CCA), 28% extrahepatic CCA and 8% gallbladder cancer. There were 29.5% objective response rate and 79.5% disease control rate. Median progression free survival (PFS) was 6.0 (95%CI 4.49-7.50) months. Eight (16%) patients were still on treatment with PFS more than 10 months. Grade 3 or 4 adverse events (AEs) were reported in 50% of patients. Most AEs were related to chemotherapy, there were 10% immune related AEs. **Conclusions:** In this real-world experience, durvalumab plus gemcitabine and cisplatin was efficacious and tolerable in Thai patients with advanced BTC. Research Sponsor: None.

608 **Poster Session**

Tinengotinib in patients with advanced, metastatic cholangiocarcinoma: Overall survival results and biomarker correlative analysis from a phase 2 clinical trial. First Author: Christos Fountzilas, Roswell Park Comprehensive Cancer Center, Buffalo, NY

Background: Tinengotinib, a spectrum-selective multi-kinase inhibitor with unique binding properties to *FGFR*, potently inhibited *FGFR2* fusion/rearrangement and acquired resistant/gatekeeper mutations in pre-clinical models and exhibited antitumor activity in cholangiocarcinoma (CCA) patients (pts) in phase 1/2 trials. Here we present the overall survival (OS) data and biomarker correlative analysis from a phase 2 clinical trial of tinengotinib in CCA. **Methods:** Eligible pts with advanced/metastatic CCA who had received ≥ 1 prior systemic chemotherapy with ECOG PS 0-1 were treated with tinengotinib 10 mg QD across four Cohorts: A1: *FGFR2* fusion(s) with primary progression on previous *FGFR* inhibitor (*FGFR*i); A2: *FGFR2* fusion(s) with progression after prior response to *FGFR*i; B: *FGFR* alteration(s) without *FGFR2* fusion(s); C: *FGFR* wild-type. Efficacy was evaluated per RECIST v1.1 and safety was assessed by CTCAE v5.0. Genomic alterations in circulating tumor DNA (ctDNA) were assessed by Foundation Medicine NGS panel at baseline, C3D1 and EOT. Planned correlative analyses of response (PR+SD>6 months) and PFS in pt subgroups based on ctDNA mutational status were performed. **Results:** 55 eligible pts were enrolled (18 in A1, 11 in A2, 13 in B, 13 in C) with median age 61.0 [range 24-81] years, ECOG PS 0 in 50.9% pts, 56.4% female, and 60.0% with ≥ 3 lines of prior therapy. Among 42 pts with *FGFR* alterations, 78.4% had ≥ 1 prior *FGFR*i, and 97.6% had prior chemotherapy. Median follow up time was 8.6 months (0.4-30.5). The median OS (months) was 17.1 (95%CI 7.5-19.5) in A1; not reached (95%CI 9.6, -) in A2; 18.0 (95%CI 9.6, -) in A1+A2; not reached (95%CI 8.0, -) in B; 6.5 (95%CI 4.8-16.4) in C. The OS rate in A, B and C was 100%, 91.7% and 75% at month 3; 83.7%, 83.3% and 66.7% at month 6, and 65.8%, 55.6% and 23.8% at month 12. Partial response was observed in 26% (10/39) of evaluable FGFR-altered pts (A1+A2+B). The safety profile was consistent with previous reports. The most common any-grade treatment-related AEs were hypertension (63.6%) and diarrhea (47.3%). Baseline ctDNA results were available in 46 pts, 35 in cohorts A+B and 11 in C. A positive correlation was shown in MED12 mutation vs response (P=0.03476). BCOR and ARID1A mutation were found to be negatively associated with PFS (P=0.01366 and P=0.0437). Acquired/secondary mutations at baseline were observed in cohort A, including V564F/I (3/24 and 5/24) and N549K/H (5/24 and 5/24); favorable anti-tumor activity was noted despite these resistant mutations, consistent with preclinical data. **Conclusions:** Tinengotinib has shown promising anti-tumor efficacy in CCA pts with FGFR fusion after prior *FGFR*i and in those with primary FGFR mutations. An ongoing randomized phase III study will investigate tinengotinib vs. chemotherapy in FGFR inhibitor refractory CCA (NCT05948475). Clinical trial information: NCT04919642. Research Sponsor: None.

609 **Poster Session**

Safety and efficacy of combining PD-L1 and CTLA-4 blockade with transarterial chemoembolization in intermediate-stage hepatocellular carcinoma: A phase II study. First Author: Amy K. Kim, Johns Hopkins School of Medicine, Department of Gastroenterology and Hepatology, Baltimore, MD

Background: Transarterial chemoembolization (TACE) is the standard treatment for intermediate-stage hepatocellular carcinoma (HCC) defined as Barcelona Clinic Liver Cancer (BCLC) B, but the benefit of adding systemic therapy remains unclear. While durvalumab and tremelimumab are approved for advanced HCC, the safety and efficacy in combination with TACE for intermediate-stage disease is undefined. Combining these PDL1 and CTLA4 inhibitors with TACE may enhance anti-tumor activity by leveraging TACE's proinflammatory effects while blocking immune evasion and reducing immune suppression. This phase IIb study evaluates the safety and efficacy of this combination in BCLC-B HCC. **Methods:** We enrolled patients with intermediate-stage HCC that are also unresectable and ineligible for transplant. All patients received drug-eluting bead TACE (DEB-TACE) followed by durvalumab (1500 mg Q4W) and tremelimumab (300 mg once). Durvalumab monotherapy was continued until disease progression or unacceptable toxicity, for a maximum of 13 cycles. Primary endpoints include objective response rate (ORR) based on mRECIST. Secondary endpoints were overall survival (OS), and safety/adverse events (AEs). The trial is designed to rule out ORR of 30%. We obtained 3 serial biopsies and plasma for participating patients prior and one week after TACE and after the administration of immunotherapy for correlative studies. **Results:** 21 patients were enrolled. Of these, 20 patients were considered evaluable after receiving at least one dose of TACE and immunotherapy. ORR was 55% (95% CI: 32%-77%, CR 10% (n=2), PR 45% (n=9), meeting the primary endpoint. Five patients had PD (25%) per mRECIST, of which two were unconfirmed on repeat imaging and continued treatment beyond progression. Three patients received curative intent treatment as the result of treatment: surgical resection (n=1) and transplant (n=2). At a median follow-up of 18.4 months, median OS was 28.8 mo (95% CI: 15.1, NA) and mPFS was 6.1mo (95% CI: 3.3, NA). Grade 3 or higher immune related adverse events were rash (n=2), diarrhea, myalgia, nausea, elevated AST/ALT. One patient developed multiple irAEs after first dose (myocarditis, myositis, pericarditis) and was taken off study. **Conclusions:** The combination of DEB-TACE with PD-L1/CTLA-4 blockade is safe and shows promising activity in patients with BCLC-B HCC. The ongoing immune analysis and biomarker studies on serial blood and tissue biopsies will provide further insights into the mechanistic effects of this combination approach and inform future therapeutic strategies. Clinical trial information: NCT03638141. Research Sponsor: AstraZeneca.

610 **Poster Session**

Modified Frailty Index (mFI-5) as a prognostic tool of mortality, complications, and readmissions in patients undergoing hepatic ablation for hepatocellular carcinoma. First Author: Sushrut Ingawale, Quinnipiac University Frank H. Netter, MD, School of Medicine/St. Vincent's Medical Center, Bridgeport, CT

Background: Hepatocellular carcinoma (HCC) is a leading cause of cancer deaths, often diagnosed late, limiting curative options. Transarterial chemoembolization, Y-90 radioablation, radiofrequency ablation, and thermoablation are commonly used for downstaging, transplant bridging, and palliation in unresectable HCC. Frailty, linked to postoperative complications, has not been studied in hepatic ablation patients. This study evaluates the predictive value of the modified frailty index (mFI-5) for mortality, readmissions, complications, redo procedures, and discharges. **Methods:** We utilized the National Surgical Quality Improvement Program (NSQIP) database from 2015 to 2019, querying cases of hepatic ablation for CPT codes: 47370 (closed hepatic ablation, single site) and 47380 (open hepatic ablation, single site). Selected cases were analyzed for frailty using the mFI-5 criteria. A multivariate analysis was performed using mFI-5 frailty scores, stratified into non-frail (0), pre-frail (1), frail (2), and severely frail (3+). The outcomes analyzed included mortality, discharge to a destination other than home, major complications, minor complications, readmission, and reoperation. **Results:** We identified 1921 patients as: 737 non-frail, 701 pre-frail, 442 frail and 41 severely frail. The rates of readmission (5.4%), reoperation (1.4%), major (5.3%) and minor (3.5%) complications were low. Eight patients died following hepatic ablation, and 79 patients were discharged to a destination other than home. In multivariate analysis, most outcomes were not significantly associated with increasing frailty. However, discharge to a destination other than home was significantly more likely in patients with higher frailty scores. In frail patients, the odds ratio (OR) was 2.325 (95% CI 1.228–4.402), and in severely frail patients, this increased to 8.720 (95% CI 3.389–22.435). **Conclusions:** Hepatic ablation is generally well tolerated with low complication rates; however, frailty significantly influences postoperative outcomes. Frail and severely frail patients are more likely to be discharged to care facilities rather than returning home, suggesting that even in the absence of major complications, these patients require a higher level of post-procedural care. This underscores the need for tailored discharge planning and resource allocation based on patient frailty. Research Sponsor: None.

612 Poster Session

Rates and risk factors for recurrence of hepatocellular carcinoma after initial complete remission in response to transcatheter arterial chemoembolization: A systematic review and meta-analysis. First Author: Aishwarya Thakurdesai, University of Louisville School of Medicine, Louisville, KY

Background: Transcatheter arterial chemoembolization (TACE) is used for palliative treatment of patients with inoperable hepatocellular carcinoma (HCC). Despite remarkable initial remission rates, frequent recurrences of HCC after TACE have resulted in questionable survival benefit. Our investigation aimed to study the rates and risk factors for HCC recurrence following remission after TACE. **Methods:** Comprehensive search of 5 databases (EMBASE, Web of Science, PubMed, CINAHL, ClinicalTrials.gov) was conducted to yield 4263 studies, which were screened based on prespecified inclusion and exclusion criteria. Outcomes assessed were total recurrence rate; 6-month, 1, 2, and 3-year cumulative recurrence rates; and 1 and 3-year survival rates for patients with and without recurrence. Meta-analysis was conducted by the random-effects model and heterogeneity was assessed by the I^2 statistic and 95% prediction interval. Systematic review of risk factors for recurrence was performed. **Results:** Twelve studies and 1,152 patients who underwent complete remission in response to TACE were included. The pooled rate of total recurrence was 60% (95% CI 53-67%). The pooled 6-month, 1-year, 2-year, and 3-year cumulative recurrence rates were 26% (95% CI 18-33%), 33% (95% CI 22-43%), 49% (95% CI 34-64%), and 62% (95% CI 53-71%), respectively. The pooled 1-year survival rates for patients with and without recurrence were 94% (95% CI 89-99%) and 99% (95% CI 99-100%), respectively. The pooled 3-year survival rates for patients with and without recurrence were 61% (95% CI 31-90%) and 91% (95% CI 81-100%). Heterogeneity was considerable based on the 95% prediction interval. Predominant risk factors for recurrence were larger tumor size, multinodularity, portal vein thrombosis, and higher AFP levels. **Conclusions:** HCC frequently recurs after initial remission achieved through TACE, with differences in survival rates between patients with and without recurrence. Identifying high-risk patients based on tumor size, multinodularity, portal vein thrombosis, and AFP levels could improve long-term outcomes. Research Sponsor: None.

613 Poster Session

Development and regulatory approval of a new systemic targeted therapy for advanced hepatocellular carcinoma. First Author: Boris Pasche, Barbara Ann Karmanos Cancer Institute, Wayne State University, Detroit, MI

Background: There are no commercially available devices for the treatment of patients with advanced hepatocellular carcinoma who have failed 1st line and 2nd line therapy. We have identified tumor-specific modulation frequencies in patients with advanced hepatocellular carcinoma. We exposed patients to radiofrequency electromagnetic fields, which are amplitude-modulated from 0.1 Hz to 100 kHz and measured changes in pulse pressure. Frequencies eliciting measurable changes in pulse pressure were selected as tumor-specific frequencies. **Methods:** Treatment is administered to patients with advanced hepatocellular carcinoma by means of a battery-operated portable device emitting 27 MHz radiofrequency electromagnetic fields, which are amplitude-modulated at hepatocellular carcinoma frequencies. The device is connected to a coaxial cable ending with a spoon-shaped antenna placed on the anterior part of the patient's tongue during treatment. Treatment is administered three times a day for one hour. **Results:** A total of 69 patients with advanced hepatocellular carcinoma received treatment with the TheraBionic device until progression or death. The median overall survival of patients who received treatment with the TheraBionic device after failing 1st line and 2nd line therapy was 9.15 (95% CI 1.6-34.8) months and the median overall survival of Child-Pugh A patients was 9.5 months, which are comparatively longer than the 7.8 months pooled estimated medians of the placebo arms of 11 randomized 1st line and 2nd line placebo-controlled studies assessing the safety and efficacy of new agents for the treatment of Child-Pugh A advanced HCC (Llovet, Montal et al., J Hepatol 2019). We also analyzed the survival of patients who had received two lines of systemic therapy and did not receive any additional cancer treatment while and after receiving treatment with the TheraBionic device. The median OS of these patients was 9.8 months and the median OS of the Child-Pugh A patients was 17.2 months, which are comparatively longer than the 7.74 months pooled estimated medians of the placebo arms of the above cited 11 randomized studies. To assess unknown device-related adverse events, we assessed patient reported symptoms using the same scale in the SHARP and Asian Pacific Sorafenib studies. The incidence of any grade adverse events among Child-Pugh A who received treatment with the TheraBionic device was not different from the Placebo group of the SHARP and Asian Pacific Sorafenib studies. There were no NCI Grade 2, 3 or 4 toxicities. One patient developed grade 1 mucositis and one patient developed grade 1 fatigue. **Conclusions:** The data presented here provide reasonable assurance of safety and probable benefit in patients with advanced hepatocellular carcinoma who have failed 1st line and 2nd line therapy: https://www.fda.gov/medical-devices/recently-approved-devices/therabionic-p1-h220001. Research Sponsor: None.

614 Poster Session

Evaluating the safety of immune checkpoint inhibitors prior to liver transplant. First Author: Laia Aceituno, Multi-Organ transplant, Toronto General Hospital, University Health Network, Toronto, ON, Canada

Background: Liver transplantation (LT) offers a 5-year survival exceeding 70% for selected patients with hepatocellular carcinoma (HCC). Immune checkpoint inhibitors (ICIs) may be used to downstage or bridge to transplantation. This study aims to evaluate the safety of ICIs prior to LT. **Methods:** Multicenter retrospective study, involving 9 centers, included adults who received ICIs and, subsequent LT, between 2019 and 2023. The ICI cohort was matched by age, sex, liver diseases and transplant date (1:3) with a similar cohort of HCC transplanted patients that did not receive ICIs. **Results:** The ICI cohort included 45 patients (Table part 1). The most common underlying liver diseases were viral. 15.5% of patients had macrovascular invasion and, 62.2% multifocal disease. The most commonly used ICI was nivolumab (75.6%), followed by atezolizumab-bevacizumab (17.8%). The median follow up was 32.5 months (17.4 - 47.6). The median ICI duration was 147 days (129.5-276.5), with a median interval of 58 days (22–193) between the last ICI dose and LT. There were 8 (17.8%) cases of graft rejection, 2 steroid-resistant leading to 1 graft failure. There were 4 (8%) HCC recurrences. During follow-up there were 6 deaths, 2 related to HCC recurrence and 4 non liver related. When compared with 135 non-ICI cohort patients, there were no differences within the groups (Table part 2). Importantly, we found no significant differences in crude rejection rates (p=0.5). **Conclusions:** Our study suggest that ICI treatment prior to LT is safe. ICI rejection primarily occurs in the acute post-transplant period (within 3 months), with a low risk of death due to graft failure. Further clinical trials are needed. Research Sponsor: None.

Part 1: ICI cohort descriptive analysis. N (%); Median (IQR); Part 2: Comparison between cohorts. N (%); Mean (SD).

(part 1)		ICI-cohort n=45	
General data	Male	38 (84.4)	
	Age	61.2 (7.6)	
	Etiology of liver diseases	Hepatitis C 17 (37.8)	
		Hepatitis B 6 (13.3)	
		MASLD 6 (13.3)	
		Viral +Alcohol 5 (11.1)	
		Alcohol 3 (6.7)	
		Autoimmune 3 (6.7)	
		Other 3 (6.7)	
		Combined viral 2 (4.4)	
	Largest tumor (mm)	39 (22)	
	Macrovascular invasion	7 (15.5)	
	Multifocal	28 (62.2)	
	Locoregional therapy	36 (80.0)	
	Number of locoregional therapies	2 (2)	
Immunotherapy	Nivolumab	34 (75.6)	
	Atezolizumab/bevacizumab	8 (17.8)	
	Pembrolizumab	1 (2.2)	
	Nivolumab+pembrolizumab	1 (2.2)	
	PD1 inhibitor clinical trial	1 (2.2)	
	Days of ICI treatment	147 (203)	
Radiological tumor response ICI	No	17 (37.8)	
Transplant	Deceased donor	43 (95.6)	
	Washout period (days)	58 (171)	
	Induction therapy (anti-thymoglobulin)	8 (17.8)	
Outcomes	Rejection	8 (17.8)	
	Biopsy proven	8 (100)	
	Time to rejection (days)	43 (102)	
	Graft failure	2 (4.4)	
	Rejection	1 (2.2)	
	HCC recurrence	4 (8)	
	Deaths	6 (13.3)	
Follow up	Since transplant (months)	32.5 (30.2)	
(part 2)			
	ICI-cohort (n=45)	**Non-ICI cohort (n= 135)**	**P**
Post transplant rejection rate	8 (17.8)	32 (23.7)	0.53
Time to rejection (days)	180.1 (SD 218)	62.2 (SD 62.8)	0.25
Overall deaths	6 (13.3)	10 (7.4)	0.36

615 Poster Session

MDM2 amplification in patients with biliary tract cancer in RETUD registry. First Author: Andrés J. Muñoz Martín, Medical Oncology Department, Hospital General Universitario Gregorio Marañón, Universidad Complutense, Madrid, Spain

Background: Antagonism of Mouse double minute 2 homolog (*MDM2*) can restore p53 activity and represents a novel therapeutic strategy in biliary tract cancer (BTC). We aimed to characterize the epidemiology of total population versus that harbouring *MDM2* amplifications; potential associations of *MDM2* to other genetic alterations and survival outcomes. **Methods:** We evaluated a large real-world cohort of patients (pt) diagnosed with BTC from 1st January 2017 to 31st December 2022 from the Spanish RETUD registry. Next generation sequencing (NGS) test was performed in all patients. Data collected included demographic and clinical characteristics, molecular profile, systemic oncologic treatment and effectiveness (best response and survival outcomes). Progression free survival (PFS) outcomes and best response were calculated for pts who received first line therapy. Descriptive statistic was used to analyse sociodemographic data, tumor characterization, molecular analysis, and best response. PFS and overall survival (OS) were estimated by the Kaplan-Meier method. **Results:** A total of 301 evaluable pt were included in 24 sites. *MDM2* amplifications were reported in 19 pt (6.3%) and 2 pt (10.5%) had TP53 mutations. At primary diagnosis, in the *MDM2* amplified population the tumour was mostly intrahepatic (63.2%); followed by gallbladder cancer (21.0%). Half of the pt had advanced disease at diagnosis (52.6%) and 21.0% resectable disease. No significant differences were observed in clinical characteristics (age, sex, tumor location, stage, ECOG performance status and metastasis location) and most common actionable genomic alterations (BRAF, HER2, IDH, FGFR2) in pt with *MDM2* amplification compared to overall population. Median overall survival (95% Confidence Interval [CI]) in pt with *MDM2* amplification was 18.4 months (12.3-19.9), and 17.8 months (95% CI, 12.3-19.9, p=0.247 in pt without amplification. Median PFS in first-line was 5.3 months (95% CI, 2.7-8.9) in *MDM2* amplified group, as compared with 6.0 months in non-*MDM2* amplified group (95% CI, 5.3-6.8; p=0.4233. The objective response rate in first-line was 21.4% in the *MDM2* amplified group versus 29.6% in non-amplified group. The proportion of pt who received systemic therapy in first, second and third line was similar in both groups. The main first-line regimen in *MDM2* amplified pt was cisplatin/gemcitabine (n=9, 47.4%), followed by clinical trial (n=3, 15.8%), GEMOX (n=2, 10.5%) and gemcitabine (n=2, 10.5%). The median OS and PFS of patients treated with cisplatin/gemcitabine in first line was similar in both groups. **Conclusions:** This analysis provides insights into the characterization of *MDM2* amplified BTC pt. Similar incidence of *MDM2* amplification was observed compared to other cohorts of BTC. No significant differences were observed in clinical characteristics, molecular profile and survival in *MDM2* amplified pt compared to non-amplified pt. Research Sponsor: Boehringer Ingelheim.

616 **Poster Session**

The relationship between the tumor microenvironment of hepatocellular carcinoma and apparent diffusion coefficient. First Author: Yuji Morine, Department of Surgery, Tokushima University, Tokushima, Japan

Background: The tumor microenvironment is critical for the acquisition of tumor malignancy in various cancer types. The objectives of this study were to investigate whether the levels of cancer-associated fibroblasts (CAFs) and tumor-associated macrophages (TAMs) reflect the prognosis of patients with hepatocellular carcinoma (HCC) after hepatectomy (Hx) and to determine whether the apparent diffusion coefficient (ADC) from diffusion-weighted imaging (DWI) reflects CAF and TAM expression. **Methods:** The study cohort comprised 109 patients who underwent initial curative resection for HCC. Alpha smooth muscle actin (αSMA) was selected as a CAF marker and CD204 as a TAM marker. Protein expression was immunohistochemically evaluated in the intratumoral regions of resected specimens. Clinicopathological factors, including the long-term prognosis after Hx, were investigated between αSMA-negative and -positive tumors and between CD204-negative and -positive tumors. The correlation between CAF/TAM marker expression and the calculated minimum ADC using DWI was also evaluated. **Results:** αSMA-positive expression was correlated with tumor number, invasive growth pattern, and advanced stage. CD204-positive expression was correlated with the presence of venous invasion. Both αSMA-positive expression and CD204-positive expression were significant prognostic factors in the univariate analysis of overall survival and disease-free survival. αSMA/CD204 double positivity was associated with an extremely poor prognosis after Hx and was a significant independent prognostic factor for overall survival (p=0.02, hazard ratio: 3.27). Patients with double positivity also showed a significantly higher ADC^{low} rate (83%). **Conclusions:** Expression of both CAF and TAM markers reflected a poor prognosis after Hx. Furthermore, the preoperative ADC could be a clinical surrogate marker in the tumor microenvironment in patients with HCC. Research Sponsor: None.

617 **Poster Session**

Safety, efficacy, and outcomes of 90-yttrium radioembolization in combination with immune checkpoint inhibitors as a definitive first-line treatment approach in non-surgical hepatocellular carcinoma. First Author: Alexandre Grahovac, University of Queensland - Ochsner Clinical School, Jefferson, LA

Background: Immune checkpoint inhibitors (ICI) have emerged as first line therapy for advanced-stage hepatocellular carcinoma (HCC). However, response rates remain modest ($<$20%). Combination treatments offer potential to improve initial response rates while extending duration of response. While early studies of ^{90}Yttrium transarterial radioembolization (^{90}Y)-ICI showed fewer adverse events (AE) and improved objective responses (OR) (30-40%), there were no improvements in time to progression (TTP) and overall survival (OS). This study evaluates safety, efficacy, and outcomes of first line ^{90}Y-ICI while evaluating the role of ICI regimen and treatment sequencing. **Methods:** A retrospective, multi-center study was conducted in BCLC A-C stage HCC patients (ECOG 0-1) receiving first line ^{90}Y-ICI with atezolizumab/bevacizumab (A+B) or tremelimumab/durvalumab (STRIDE). Treatment response was recorded at ^{90}Y follow-up. TTP, progression-free survival (PFS), and OS were primary endpoints, with AE and safety as secondary outcomes. **Results:** The study included 40 patients receiving ^{90}Y-ICI from 2021–2024, with a median follow-up of 12 months. The cohort was predominantly Child-Pugh A (96%) with HCV-related cirrhosis (76%), BCLC-C stage (51%), and a median target HCC size of 7.7 cm (IQR, 6-11 cm). Grade 3-4 AEs occurred in 15 patients (37%), leading to steroid use (12, 29%), treatment delays (13, 31%), or discontinuation (4, 10%). The first cycle target OR rate was 82% (32/39), with a 51% (20/39) complete response (CR) rate. The first cycle overall OR was 59% (23/39), with a 41% (16/39) CR rate. No significant differences were found in adverse events, delays, or discontinuations based on sequencing or regimen. Sequencing had no significant effect on target or overall OR with no downstream effect on TTP (P = 0.968) or PFS (P = 0.906). Similarly, there was no effect of ICI regimen on OR, TTP (P = 0.813) or PFS (P = 0.904), although the data trended toward higher response rates with the STRIDE regimen. Log-rank analysis revealed target CR was associated with improved TTP (P = 0.004) and PFS (P = 0.009), but not OS (P = 0.237). Achieving an overall CR was associated with further improvements in TTP ($P < 0.001$) and PFS ($P < 0.001$). The cohort 2-year OS was 55% with median OS not yet reached. **Conclusions:** First line ^{90}Y-ICI therapy in BCLC A-C stage HCC is an effective treatment strategy with AEs in the expected range of systemic therapy and low rates of treatment discontinuation. Sequencing and ICI regimen did not significantly impact AEs or treatment outcomes. However, patients achieving CR showed significantly improved TTP and PFS. Although follow-up is limited, these findings suggest achieving CR is key to optimal long-term outcomes. We anticipate long-term follow-up will demonstrate an OS benefit linked to first line CR rate. Research Sponsor: None.

618 **Poster Session**

Association between ex vivo pharmacotyping of patient-derived tumor organoids and personalized therapeutic options for patients with biliary tract cancer. First Author: Payel Chatterjee, SEngine Precision Medicine, a Subsidiary of Tempus AI, Bothell, WA

Background: Biliary tract cancers (BTC), including cholangiocarcinomas and gallbladder adenocarcinomas, present significant therapeutic challenges due to limited treatment options and poor prognoses. We report findings from the CLIA-certified PARIS assay, evaluating drug sensitivities of patient-derived tumor organoids (PDTOs) to a panel of oncology drugs. **Methods:** PDTOs were successfully cultured from 27 out of 46 live tumor samples obtained from 43 BTC patients. The majority of patients presented with advanced metastatic disease (60% stage IV, 13% stage III, 10% stage II, 4.4% stage I, 10% unknown) and had exhausted standard therapeutic options. Each PDTO culture underwent testing against an average of 50 drugs, encompassing chemotherapeutic agents and targeted therapies. **Results:** Comparative analysis of PDTO drug sensitivities with patients' prior treatment histories revealed non-responsiveness to $>$80% of drugs associated with clinical progression, including gemcitabine, cisplatin, and targeted therapies for FGFR translocations. In contrast, 100% (26/26) of the samples demonstrated significant sensitivity to one or more targeted agents, particularly inhibitors of EGFR, MEK, ERK, mTOR, PI3K, MDM2, BCL2, and BET families. Despite these shared sensitivities, each individual PDTO culture exhibited unique responses to targeted therapies, highlighting the genetic and phenotypic diversity between BTC patients. Comparison of *ex vivo* drug sensitivities with tumor genomic profiles validated oncogenic drivers such as HER2 amplification, *KRAS* and *PIK3CA* pathogenic mutations, correlating with sensitivities to EGFR/HER2 inhibitors, MEK inhibitors, and PI3K inhibitors, respectively. While mutations in *KRAS*, *BRAF*, *BRCA1*, *BRCA2*, *ERBB2*, *ERBB3*, or *MET* are present in less than 8% of cases, their role as biomarkers in BTCs requires further validation. PDTOs offer a promising tool to expedite this validation process. The PARIS assay results guided treatment decisions for five patients with metastatic disease, two of whom maintained treatment for over 4 weeks. Notably, two patients showed clinical benefit with everolimus (5 weeks) and dasatinib (11 weeks), experiencing symptom relief and reduced ascites. Intriguingly, 5 out of 7 PDTOs with *FGFR* alterations exhibited exceptional PARIS test responses to dasatinib, suggesting potential efficacy linked to FGFR activation. **Conclusions:** In conclusion, our study demonstrates that ex vivo drug testing of PDTO cultures can inform treatment selection and potentially accelerate drug approvals for BTC, leveraging therapies approved for other cancers. Early integration of such assays in patient management, possibly at diagnosis, holds promise for improving outcomes in this challenging disease. Research Sponsor: None.

619 **Poster Session**

The Oncopig hepatocellular carcinoma model: An innovative large animal platform for evaluating treatment effects. First Author: Ian Chau, Gastrointestinal Unit, Department of Medicine, The Royal Marsden NHS Foundation Trust, London and Surrey, United Kingdom

Background: Hepatocellular carcinoma (HCC) is an aggressive disease associated with poor prognosis and limited treatment options. Systemic therapies for advanced HCC include sorafenib, lenvatinib, and atezolizumab plus bevacizumab. However, given the limited efficacy of systemic treatment options, novel treatment approaches including combination of systemic and locoregional therapies (LRTs) is required to improve patient outcomes. These approaches require advanced large animal models to prove effectiveness. Pigs represent an ideal platform that provide both regulatory acceptable and clinically relevant large animal cancer models due to their similar physiology, size, genetics, immunity, and metabolism in relation to humans. The Oncopig HCC model is an ideal tool for testing LRT and other oncolytic based treatments. **Methods:** The Oncopig cancer model develops tumors following induced expression of $KRAS^{G12D}$ and $TP53^{R167H}$ driver mutations utilizing an adenoviral vector encoding Cre recombinase (AdCre). Oncopig HCC cell lines were developed by isolating hepatocytes from Oncopig liver biopsies and transformed in vitro via exposure to AdCre. Transgene expression and hepatocyte cell origin were validated by RT-PCR and arginase-1 staining, respectively. Comparison of in vitro Oncopig, human, and murine HCC cell line phenotypes (proliferation, migration, and chemotherapeutic response) was performed. Oncopig HCC tumor formation was performed via autologous intrahepatic injection of Oncopig HCC cells. Tumor formation was evaluated using ultrasound and CT imaging. **Results:** Oncopig and human HCC cell lines displayed similar cell cycle lengths and migration rates. Oncopig HCC cells were consistently more predictive of human HCC responses than murine cells when treated with standard chemotherapeutic agents (doxorubicin, cisplatin, mitomycin C, and sorafenib). Importantly, consistent with human HCC, Oncopig HCC cells were non-responsive to 5-fluorouracil, while murine HCC cells were responsive. Intrahepatic injection of Oncopig HCC cells resulted in tumors visible on ultrasound and CT scan within 2-4 weeks. Resulting tumors were consistent with human tumors on imaging. Histological analysis of tumor biopsies confirmed the identity of the resulting masses as HCC tumors based on positive arginase-1 and $KRAS^{G12D}$ staining. **Conclusions:** The Oncopig HCC model may be more predictive of human HCC chemotherapeutic responses than currently employed murine models. Tumors develop rapidly (within 2-4 weeks) and are readily identified on ultrasound and CT imaging, making this an ideal model for evaluation of novel therapeutic approaches. Further exploration and development of additional tumor models including further translational characterization will be important for broader utility. Research Sponsor: None.

620 Poster Session

Analysis of gut and mucosal microbiomes in patients with extrahepatic cholangiocarcinoma. First Author: Hyewon Ryu, Division of Hematology and Oncology, Department of Internal Medicine, Chungnam National University Hospital, Daejeon, South Korea

Background: Cholangiocarcinoma is an aggressive malignancy with a poor prognosis. While gut microbial dysbiosis is linked to various diseases, including biliary disorders and cancer, its association with cholangiocarcinoma remains unclear. **Methods:** This study compared the mucosal and gut microbiomes of patients with extrahepatic cholangiocarcinoma (n = 21) to those with cholecystitis (n = 21) and healthy individuals (n = 28). Mucosal microbiomes were collected from cancer lesions and gallbladders, with fecal samples obtained from 31 participants. The 16S rRNA gene (V3 and V4 region) was sequenced using the Illumina MiSeq platform and analyzed with QIIME 2. **Results:** 16S rRNA sequencing revealed Proteobacteria as the dominant taxon in the cancer cohort, with *Klebsiella, Aeromonas, Staphylococcus, Fusobacterium, and Haemophilus* also enriched. In contrast, *Firmicutes* dominated in the cholecystitis and control cohorts. β-diversity analysis highlighted significant differences between cohorts. Elevated levels of bile acids, particularly chenodeoxycholic acid and ursodeoxycholic acid, were observed in the cancer cohort and positively correlated with *Proteobacteria, Aeromonas, Enterobacter, and Enterococcus.* **Conclusions:** These findings suggest that microbiome dysbiosis contributes to cholangiocarcinoma development. Research Sponsor: None.

621 Poster Session

Evaluating immune cell abundance across hepatocellular carcinoma disease stages I-III: Findings from TCGA analysis. First Author: Jaber H. Jaradat, Mutah University, Al-Karak, Jordan

Background: Hepatocellular carcinoma (HCC), the predominant form of primary liver cancer, is the sixth most prevalent cancer globally and a leading cause of cancer-related mortality. The tumor microenvironment (TME) in HCC, characterized by complex interactions between immune and tumor cells, significantly influences tumor progression and immune suppression. Understanding the immune landscape across the different stages of HCC can inform prognosis and therapeutic strategies. **Methods:** We analyzed RNA seq data from 333 HCC patients (disease stages I-III), from The Cancer Genome Atlas (TCGA), using the immune deconvolution algorithms CIBERSORT and xCell. Differential gene expression analysis was performed using the DESeq2 package in R. Immune cell abundance across various HCC stages was imputed, and survival analyses was conducted using Kaplan-Meier curves and log-rank tests. **Results:** Of the 19,514 genes analyzed, 2,774 were up-regulated and 947 were down-regulated in stage II and III in comparison to stage I. CIBERSORT and xCell analyses revealed significant variability in the immune cell composition between HCC stages. Notable differences were observed in the T cells (CD4 and CD8 subsets), B cells, plasma cells, and stromal components. ANOVA and post hoc tests identified significant stage-specific differences in several immune cell types. For the survival analysis, we categorized patients into low and high groups based on median cell abundances. Overall survival showed no statistically significant differences among the various groups for common myeloid and lymphoid progenitor cells. In contrast, disease-free survival demonstrated statistically significant differences across the groups for both cell types. **Conclusions:** This study emphasizes the tumor microenvironment's pivotal role in HCC, revealing that immune cell composition significantly influences disease-free survival, despite limited effects on overall survival. These findings advocate for further exploration into targeted immunotherapies and the development of predictive models that integrate immune landscape characteristics, ultimately aiming to enhance patient outcomes in HCC. Research Sponsor: None.

Cell type	xCell		Cibersort	
	Mean	SD	Mean	SD
T cell CD4 naive	0.00466	0.01994	1.31e-5	1.69e-4
T cell CD8	0.01368	0.03253	0.03161	0.04947
Eosinophil	3.96e-5	3.41e-4	1.67e-4	8.69e-4
Macrophage	0.02323	0.02441	0.02924	0.04345
Macrophage.M1	0.00914	0.01526	0.01806	0.02054
Macrophage.M2	0.02303	0.01697	0.05439	0.04815
B cell memory	0.00196	0.01418	0.00107	0.00733
B cell naive	0.00163	0.01034	0.01319	0.02148
Neutrophil	2.14e-4	0.00137	0.00112	0.00539
Immune score	0.04621	0.06817	-	-

Table presents various cell type abundances for each algorithm used.

622 Poster Session

Association of GPR120 and NGF with perineural invasion in cholangiocarcinoma: Clinical and experimental insights. First Author: Jun He, The First Affiliated Hospital of Soochow University, Suzhou, China

Background: Cholangiocarcinoma (CCA) is a highly malignant biliary tumor characterized by frequent perineural invasion (PNI), which is associated with a poor prognosis; however, the underlying mechanisms remain unclear. G-protein-coupled receptor 120 (GPR120) has been implicated in the development of various tumors, while nerve growth factor (NGF) has a strong correlation with PNI. This study aims to elucidate the role of GPR120 in the production of NGF and to investigate the mechanisms by which PNI is induced in CCA. **Methods:** We conducted a retrospective review of medical records for 386 patients with cholangiocarcinoma (CCA), including both extrahepatic (ECC) and intrahepatic (ICC) types, who underwent curative resection at the Department of General Surgery, First Affiliated Hospital of Soochow University, between January 2015 and December 2021. NGF and GPR120 expression were assessed in 60 paraffin-embedded archived CCA tissue samples via immunohistochemical staining (IHC), with an additional 60 normal bile duct samples serving as controls. All tissues were sourced from the aforementioned 386 patients, none of whom received preoperative treatment. Tumors were staged according to AJCC classification. Western blot analysis was employed to evaluate NGF and GPR120 expression in CCA cell lines (QBC 939, RBE, HUCCT-1, and HCCC-9810). To assess the invasion and migration capabilities of CCA cell lines, QBC 939 and HCCC-9810 were treated with TUG-891 (a GPR120 agonist) and AH7614 (a GPR120 inhibitor). **Results:** Both GPR120 and NGF were found to be upregulated in CCA tissues, correlating with aggressive clinicopathological features. IHC staining revealed cytoplasmic localization of GPR120 and NGF, with significantly higher expression in cancerous tissues compared to adjacent noncancerous samples. Among the 386 patients, PNI was observed in 339 (87.8%). PNI correlated with tumor diameter, CA 19-9 levels, TNM stage, preoperative bilirubin levels, tumor differentiation, and preoperative drainage. Notably, GPR120 expression increased in poorly differentiated tumors and advanced clinical stages, indicating a clinical association with CCA progression. In vitro studies demonstrated that GPR120 promoted invasion and migration in the QBC 939 and HCCC-9810 cell lines. Western blot analysis confirmed elevated GPR120 levels in these lines, which also exhibited higher NGF expression. The GPR120 inhibitor AH7614 significantly reduced invasion in QBC 939 and HCCC-9810 cells, while the GPR120 agonist TUG-891 showed no effect on any of the CCA cell lines tested. **Conclusions:** GPR120 and NGF are implicated in perineural invasion (PNI) in CCA, with a notable correlation between GPR120 and NGF in the context of PNI occurrence. Research Sponsor: None.

623 Poster Session

Evaluation of EpiSwitch in predicting immunotherapy response in hepatocellular carcinoma and gastrointestinal tumors. First Author: Mahmoud Ouf, Lombardi Comprehensive Cancer Center, Georgetown University Medical Center, Washington, DC

Background: The EpiSwitch Checkpoint Inhibitor Response Test (CiRT) is a blood-based assay that analyzes DNA conformations in immune cells to predict responses to immune checkpoint inhibitors (ICIs) targeting PD-L1/PD-1. CiRT has demonstrated superior predictive accuracy compared to traditional biomarkers, such as tumor mutational burden (TMB) and PD-L1 immunohistochemistry (IHC), for predicting responses in urothelial cancer (ESMO 2022). However, its role in hepatocellular carcinoma (HCC) and other gastrointestinal (GI) tumors remains unclear. As immunotherapy becomes more common in these cancers, identifying reliable predictive biomarkers is essential to optimize treatment outcomes. **Methods:** This retrospective study evaluates CiRT-predicted immunotherapy responses and clinical outcomes in patients with HCC and GI tumors, including cholangiocarcinoma, pancreatic adenocarcinoma, and gastric cancer. All patients received one or more lines of immunotherapy. CiRT responses were categorized into high probability (HP) and low probability (LP). Treatment response was assessed per RECIST 1.1 criteria. Sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), overall survival (OS), and progression-free survival (PFS) were analyzed. **Results:** 43 patients (24 HP, 19 LP) were included. HCC patients (n = 33) received varied immunotherapy regimens, such as combinations of bevacizumab with atezolizumab, tremelimumab with durvalumab, and nivolumab with ipilimumab. Non-HCC patients (n = 10) received combinations of immunotherapy and chemotherapy such as durvalumab with cisplatin and gemcitabine and nivolumab with oxaliplatin and capecitabine. Across all patients, CiRT revealed a sensitivity of 73.91%, a specificity of 65%, a PPV of 70.83%, and an NPV of 68.42%. In HCC, sensitivity was 70.59% and specificity was 68.75%. In non-HCC, sensitivity was 83.33% and specificity 50%. HP responders showed better treatment outcomes compared to LP responders, with higher rates of complete response (CR: 12.5% vs 0%), partial response (PR: 29.2% vs 21.1%), and stable disease (SD: 29.2% vs 10.5%), and a lower rate of progressive disease (PD: 29.2% vs 68.4%) (p = 0.0467). Overall, 70.83% of HP achieved CR, PR, or SD vs. 31.58% of LP (p = 0.0098). PFS was significantly better in HP (p = 0.044, Log-rank test; median PFS = 2.0 months for the LP group, mPFS not reached for the HP group). No statistically significant OS difference was noted due to the lack of events in the HP group. **Conclusions:** CiRT HP status was associated with improved response rates to immunotherapy, longer PFS, and better treatment outcomes in HCC and GI tumors. HP responders experienced notably better outcomes, suggesting that CiRT could serve as a promising predictive biomarker for immunotherapy response, addressing a critical unmet need for reliable biomarkers to treat HCC and other GI tumors. Research Sponsor: None.

624 Poster Session

Influence of MRP4 levels on clinical outcomes in biliary tract cancer. First Author: Gerardo F. Arroyo, CeDIT (Centro de Diagnóstico, Investigación y Tratamiento), Salta, Argentina

Background: Biliary tract cancer (BTC) are rare tumors with a poor prognosis.Multidrug resistance protein 4 (MRP4) is a transmembrane efflux transporter that influences on cellular growth and differentiation. Recent studies have linked high MRP4 levels to increased aggressiveness in pancreatic cancer and other tumor types. This study aimed to evaluate the role of MRP4 in BTC and its association with clinical outcomes in a real-world cohort from endemic areas in Argentina. **Methods:** We conducted a retrospective analysis of clinical outcomes and MRP4 expression in paraffin-embedded tissue samples from 78 patients with BTC. Immunohistochemistry staining was performed, and MRP4 expression was scored based on staining intensity and the percentage of positive cells. A score exceeding 100 was classified as high MRP4 levels. Statistical analyses included Chi-square, Kaplan-Meier survival curves, and log-rank tests, with significance set as $p<0.05$ (two-tailed). **Results:** High MRP4 levels were observed in 46 out of the 78 samples (59%). Patients with gallbladder cancer (n=64) and elevated MRP4 levels had significantly poorer overall survival (OS) compared to those with low MRP4 levels (median 11.3 and 14.5 months, high and low, p=0.038). In the broader BTC cohort evaluable for survival (n=73; 64 gallbladder, 7 cholangiocarcinoma, 1 ampulla of Vater, and 1 biliary undefined) patients with elevated MRP4 levels had non-significant shorter OS (median 12.4 and 13.5 months, high and low, p=0.17). Poorly differentiated tumors were more likely to express high MRP4 levels (p=0.06). High MRP4 expression was found in 59% of localized stages (0-II) and in 57% of advanced stages (III, IV and relapsed). **Conclusions:** High MRP4 expression is common in BTC, particularly gallbladder cancer, and could be a marker of poor prognosis. MRP4 levels appear more elevated in poorly differentiated tumors, and are similar in both, early and advanced stages, suggesting its involvement in early tumor progression. Further prospective studies are needed to confirm MRP4 as a biomarker. If MRP4 is a driver gene, targeting with inhibitors could represent a promising therapeutic strategy for these challenging tumors. Research Sponsor: None.

625 Poster Session

Analysis of clinically actionable alterations in baseline tumor versus plasma samples in participants of the TOPAZ-1 study of durvalumab plus gemcitabine and cisplatin in advanced biliary tract cancer. First Author: Do-Youn Oh, Division of Medical Oncology, Department of Internal Medicine, Seoul National University Hospital, and Cancer Research Institute, Seoul National University College of Medicine, Seoul, South Korea

Background: In the TOPAZ-1 study (NCT03875235), durvalumab + gemcitabine and cisplatin (D+GC) significantly improved overall survival (OS) versus placebo + GC (P+GC) in participants (pts) with advanced biliary tract cancer. Updated results have shown a clinically meaningful long-term OS benefit for D+GC versus P+GC at 3 years. This exploratory analysis tested circulating tumor DNA (ctDNA) in plasma samples for blood-based detection of clinically actionable alterations (CAAs) and investigated the potential of this method to guide treatment decisions. **Methods:** Baseline genomic alterations were retrospectively assessed in evaluable tumor (FMI biomarker evaluable population [BEP], n=441) and plasma samples (GH BEP, n=643) using FoundationOne (Foundation Medicine Inc., Cambridge, MA) and Guardant INFINITY (Guardant Health, Redwood City, CA) assays, respectively. Mutation prevalence and association with outcomes were compared in the FMI BEP and GH BEP. Positive and negative percent agreement of CAAs detected in tumor versus ctDNA were assessed in 419 pts with both tumor and plasma samples (FMI-GH BEP). **Results:** The FMI BEP and GH BEP represented 64% and 94% of the TOPAZ-1 final analysis set (685 pts), respectively. The relative prevalence and overall mutational landscape detected in plasma ctDNA was consistent with that observed by tumor profiling, with the notable exception that genes harboring complex alterations (e.g. gene amplification, rearrangements, or homozygous deletions) were less frequently detected in ctDNA (e.g. *ERBB2*, *FGFR2*, and *CKDN2A/2B/MTAP*). The most common alterations (>15% prevalence in both BEPs) observed were in *TP53* (49%/52%), *KRAS* (24%/17%), and *ARID1A* (21%/16%) (in the FMI/GH BEPs, respectively). The relative prevalence of alterations within geographic and anatomic subgroups was similar in tumor versus plasma for most CAAs. The overall percent agreement in CAAs was ≥93%, and negative percent agreement was ≥97%. However, positive percent agreement was notably low for *ERBB2* amplification (52%) and *FGFR2* fusions (47%). OS hazard ratios for D+GC versus P+GC in the GH BEP were <1 for both CAAs and wild-type, except for *ERBB2* amplification, as previously reported in the FMI BEP. **Conclusions:** The overall concordance and relative prevalence of simple mutations (e.g. single nucleotide variants) were similar using the FoundationOne tumor assay and Guardant INFINITY ctDNA assay, suggesting that plasma ctDNA testing has potential utility in clinical practice. However, negative status by ctDNA for the complex alterations found in *FGFR2* and *ERBB2* would require further testing of tumors, based on their low detectability in plasma. Clinical trial information: NCT03875235. Research Sponsor: AstraZeneca.

626 Poster Session

Clinical landscape of cell free DNA alterations in patients with advanced biliary tract cancer treated with targeted therapies. First Author: Rohit Thummalapalli, Memorial Sloan Kettering Cancer Center, New York, NY

Background: Although 40% of patients (pts) with advanced biliary tract cancer (BTC) harbor actionable alterations amenable to targeted therapy (TT), tumor genotyping can be challenging due to limited tissue and other factors. Next generation sequencing (NGS) of cell free DNA (cfDNA) may aid in prognostication, identification of TT opportunities, and characterization of acquired resistance mechanisms to TT. **Methods:** cfDNA was collected prospectively in pts with BTC treated at MSK from 2016-2023. Samples were analyzed using a CLIA-approved, cfDNA targeted NGS assay (MSK-ACCESS). When available, matched tumor samples were analyzed using an FDA-authorized targeted NGS assay (MSK-IMPACT). Objectives included evaluating association of cfDNA genomic alteration variant allele frequency (VAF) with outcomes and description of genomic features identified on cfDNA at resistance to TT. **Results:** N=170 BTC pts (intrahepatic cholangiocarcinoma [CCA], n=121; extrahepatic CCA, n=29, gallbladder cancer, n=20) underwent cfDNA genotyping, comprising 270 samples overall. A majority of pts (146/170, 86%) had locally advanced or metastatic disease. OncoKB level 1/2 alterations were identified in 19% of pts overall. Pts with treatment-naïve locally advanced or metastatic BTC whose baseline detected cfDNA genomic alterations were VAFhigh (defined as > median VAFmax across all samples, n=60) had significantly worse median PFS (4.5 (95% CI 2.2-6.5) vs. 11.8 (95% CI 5.4-18.4) months, (HR 2.7, p=0.002) and OS (14.1 (95% CI 5.4-19.9) vs. 32.0 (95% CI 13.8-39.9) months, HR 3.1, p=0.002)) to first line treatment. Serial cfDNA was available from 28 pts who received TT. Emergent RAS isoform alterations were identified at resistance in 7/25 (28%) pts who received BRAF-, FGFR-, or HER2-directed TT. These included 3/5 pts who received BRAF/MEK TT for BRAF V600E+ BTC (emergent KRAS G12D in 1, NRAS Q61K in 1, polyclonal KRAS G12D, G12V, G13D in 1), 2/13 pts who received FGFR TT for FGFR2 fusion+ BTC (KRAS Q61H in 1, NRAS Q61R in 1), and 2/7 pts who received HER2 TT for ERBB2-amplified BTC (KRAS G12C in 1, KRAS amp in 1). Serial cfDNA also revealed canonical FGFR2 resistance mutations in 3 pts who received FGFR TT, as well as loss of index ERBB2 amp in 2, acquired ERBB2 resistance mutation in 1, acquired MYC amp in 2, and MET amp in 1 pt who received HER2 TT. **Conclusions:** cfDNA NGS in pts with BTC may assist with prognostication in advanced disease and can identify novel patterns of resistance, including frequent emergence of RAS alterations at resistance across multiple TTs. These may identify therapeutic opportunities for KRAS- and NRAS-directed therapeutics in development, and underscore the importance of serial cfDNA profiling. Research Sponsor: None.

627 Poster Session

Positive homologous recombination signature (HRDsig+) and the genomic landscape of intrahepatic cholangiocarcinoma (iCCA). First Author: Nakul Manish Shah, The University of Texas MD Anderson Cancer Center, Houston, TX

Background: Defective DNA repair has not been a major focus in iCCA clinical research. In this comprehensive genomic profiling (CGP) study, we queried whether the presence of a scar-based HRDsig biomarker could identify a subset of patients with unique genomic alterations (GA) and potential for responsiveness to PARP inhibitor-based treatment regimens. **Methods:** 6,271 cases of clinically advanced iCCA underwent hybrid capture based CGP to evaluate all classes of GA. Microsatellite instability (MSI) status, tumor mutation burden (TMB) levels, genomic ancestry and genomic trinucleotide signatures were determined from the sequencing data. HRDsig status was calculated using a broad set of genome-wide copy number features (PMID 37769224). PD-L1 was determined by IHC using the Dako 22C3 tumor proportional score (TPS). Results were compared using the Fisher exact system with the Benjamini-Hochberg procedure. **Results:** 288 (4.6%) of the sequenced iCCA featured an HRDsig+ status. Gender distribution (46-49% male) and median age (67 years) were similar in both HRDsig+ and HRDsig- iCCA. More pathogenic GA per tumor were found in HRDsig+ vs HRDsig- iCCA (5 vs 4; P<0.0001). Genomic ancestry distribution was similar; European ancestry ranged from 73% to 76% in the two cohorts. The APOBEC signature was uncommon but slightly more frequent in the HRDSig+ cases (1.7% vs 0.4%; P=.034). MSI High status was slightly more frequent in the HRDsig- group (1.8% vs 0.0%; P=0.034). Median TMB was higher in the HRDsig+ cases (3.6 vs 1.2 mut/Mb; P<0.0001) as was the frequency of TMB > 10 mut/Mb (9.1% vs 3.5%; P<0.0001). PD-L1 expression was similar with low level (1-49% TPS) ranging from 19% to 23% in the 2 groups. As anticipated, GA in genes associated with HRD including *BRCA1* (6.6% vs 0.8%; P<0.0001), *BRCA2* (25.0% vs 1.3%; P<0.0001), *PALB2* (8.7% vs 0.3%; P<0.0001), and *ATM* (6.3% vs 3.4%; P=0.033) were all more frequent in the HRDsig+ iCCA. In contrast, GA in genes associated with iCCA targeted therapies were less frequent in the HRDsig+ cases including *FGFR2* (8.0% vs 12.5%; P=0.033), *IDH1* (3.5% vs 14.0%; P<0.0001) and *ERBB2* (3.5% vs 5.7%; NS). In the HRDsig- group, 83.3%/74.1% of *BRCA1/BRCA2* mutated iCCA were mono-allelic likely non-driver GA respectively. MTOR pathway activating GA were more frequent in the HRDsig+ cases including *PTEN* (7.3% vs 3.3%; P=0.003) and *NF1* (8.0% vs 2.8%; P<0.0001). *MTAP* deletion, an emerging iCCA target, was more frequent in the HRDsig+ iCCA (23.3% vs 16.9%; P=0.024). **Conclusions:** Nearly 5% of advanced iCCA feature HRDsig positive status which is associated with both significant differences in the frequencies of therapy associated genomic targets and the potential for introducing PARP inhibitors for these patients. Research Sponsor: None.

	HRDsig- iCCA N = 5983	HRDsig+ iCCA N = 288	P-value
ATM	3.4%	6.3%	0.033
BRCA1	0.8% *(83.3% mono-allelic)*	6.6%	<.0001
BRCA2	1.3% *(74.1% mono-allelic)*	25.0%	<.0001
PALB2	0.3%	8.7%	<.0001

628 Poster Session

A global comparative study on real-world clinical and genomic differences in advanced biliary tract cancer. First Author: Oluseyi Abidoye, Division of Hematology and Oncology, Mayo Clinic Arizona, Phoenix, AZ

Background: Biliary tract cancers (BTC) are aggressive malignancies with poor survival outcomes and limited treatment options. This study compared the epidemiological and molecular differences of advanced BTC patients through a global collaboration involving the Mayo Clinic and National Cancer Center Hospital East. **Methods:** We included patients with advanced BTC who underwent comprehensive tumor molecular profiling who received at least 30 days of systemic therapy. Patients were grouped based on specific alterations. Overall survival (OS) was calculated from the start of first-line therapy until death and compared among subgroups using Cox regression models. **Results:** 332 (36%) out of 932 patients had at least one actionable mutation: *somatic BRCA1/2* mutations N=106, *FGFR2* fusions N=61, *HER2* amplifications N=65, *KRAS G12C/D* mutations N=46, *IDH1* mutations N=38, and TMB >10 mut/Mb N=16. *BRCA1/2*-mutated BTC were more commonly found in older Asian males with intrahepatic tumors and often presented with peritoneal and liver metastases upon initial diagnosis. *FGFR2* fusions were more prevalent in younger White females with intrahepatic tumors, while *HER2* amplifications were mainly seen in Asian females with gallbladder tumors. *KRAS* mutations were primarily observed in Asian males, and *IDH1* mutations were common in White males with intrahepatic tumors, frequently presenting with liver metastasis. Survival outcomes varied significantly based on the alteration type: patients with *FGFR2* fusions had the longest median OS (mOS) at 23.3 months (mo) (95%CI 18.4-34.9), followed by those with *IDH1* mutations (19.3 mo; 95% CI 17.4-NE) and TMB >10 mut/Mb (17.8 mo; 95% CI 13.6-NE). *BRCA* mutations had a mOS of 16 mo (95% CI 14.6-19.8), while *HER2* amplifications and *KRAS* mutations had mOS of 15.5 (95% CI 11.9-20.2) and 11.6 mo (95% CI 10.0-20.9), respectively. **Conclusions:** These findings underscore the heterogeneity in clinical and genomic characteristics among BTC patients, highlighting the importance of tailored therapeutic approaches. Research Sponsor: None.

Baseline patient characteristics stratified by mutation status.

	KRAS G12C/D (N=46)	*BRCA 1/2* (N=106)	*IDH1* (N=38)	*HER2* amplification (N=65)	*FGFR2* fusion (N=61)	TMB>10 mut/Mb (N=16)	P-value
Median Age, year	67	68	64	66	52	73	<0.001
Female, n (%)	19 (41)	34 (32)	18(47)	34 (52)	45(74)	6 (38)	<0.001
Race, n (%)							<0.001
Asian	31 (67)	97 (92)	13 (34)	51 (79)	10 (16)	6 (38)	
White	14 (30)	8 (8)	24 (63)	13 (20)	48 (79)	10 (63)	
Primary tumor location, n (%)							<0.001
Intrahepatic	28 (68)	45 (45)	33 (100)	19 (31)	54(95)	10 (67)	
Extrahepatic	9 (22)	27(27)	0 (0)	10 (16)	2 (4)	4 (27)	
Gallbladder	4 (10)	28 (28)	0 (0)	32 (53)	1 (2)	1 (7)	
Sites of Metastasis at diagnosis, n (%)							
Abdominal Wall / Peritoneal	9 (24)	15 (19)	4(13)	7 (12)	6 (10)	4 (27)	0.309
Bone	33 (92)	70 (90)	26 (87)	2 (3)	9(15)	0 (0)	0.212
Liver	20 (49)	52 (61)	28 (82)	45 (73)	45 (76)	6 (38)	0.001
Lung	26 (70)	65 (82)	22 (69)	36 (58)	27(45)	11 (69)	0.006

629 Poster Session

Impact of HER2-positivity on prognosis and targeted therapeutic outcomes in advanced biliary tract cancer. First Author: Choong-kun Lee, Division of Medical Oncology, Department of Internal Medicine, Yonsei Cancer Center, Yonsei University College of Medicine, Seoul, South Korea

Background: HER2 overexpression and amplification have been identified as a promising druggable target in biliary tract cancer (BTC). This study aims to characterize the clinical and molecular outcomes associated with HER2 positivity in BTC and evaluate the prognostic implications of HER2-targeted therapies. **Methods:** A retrospective study was conducted involving patients with advanced BTC diagnosed at Yonsei Cancer Center (YCC) and the University of Texas MD Anderson Cancer Center (MDACC) between 2009 and 2023. Patients were classified as HER2-positive either by immunohistochemistry (IHC 3+ or 2+/ISH+) or by next-generation sequencing (NGS, *ERBB2* amplification). Overall survival (OS) and palliative first-line progression-free survival (PFS) were analyzed using Kaplan-Meier and Cox regression methods. **Results:** A total of 389 advanced BTC patients were initially screened (310 from YCC; 79 from MDACC). Among the 310 BTC patients from YCC with available HER2 IHC results, 78 (25.2%) were HER2-positive and 232 (74.8%) were HER2-negative. HER2 positivity was significantly more common in gallbladder cancer (55.1%) compared to its prevalence in intrahepatic or extrahepatic cholangiocarcinoma (22.8%, p<0.001). Of the 201 patients with available NGS data, 186 (92.5%) were *ERBB2* non-amplified; among these, 27 (14.9%) were HER2-positive by IHC. For survival analysis, 310 patients were eligible: 239 from YCC (both HER2-positive and negative) and 71 from MDACC (all HER2-positive). Kaplan-Meier survival analysis indicated a trend towards shorter OS in HER2-positive patients compared to HER2-negative patients (median OS, 13.7 vs. 17.1 months, p=0.084, HR=1.246, 95% CI=0.97–1.60) regardless of exposure to HER2-targeted therapy, while first-line PFS on chemotherapy only was significantly shorter in the HER2-positive group (5.1 vs. 7.4 months, p<0.001, HR=1.906, 95% CI=1.46-2.48). Among HER2-positive patients, 56.8% received HER2-targeted therapy in the second line setting. HER2-positive patients who did not receive HER2-targeted therapy had a significantly poorer prognosis compared to HER2-negative patients (median OS 8.1 vs. 17.1 months, HR=2.46, 95% CI=1.73-3.48), whereas those receiving HER2-targeted therapy had comparable OS to HER2-negative patients (median OS 18.2 vs. 17.1 months, HR=0.95, 95% CI=0.71-1.27). Multivariable analysis identified a history of surgical treatment, radiation therapy, HER2 positivity, and HER2-targeted therapy as independent prognostic factors for OS. **Conclusions:** HER2 positivity is associated with a poorer prognosis in advanced BTC. However, HER2-targeted therapies improve outcomes, suggesting their potential inclusion in standard care for HER2-positive subgroups. Discrepancies between IHC and NGS in HER2 assessment emphasize the need for optimized testing guidelines in BTC. Research Sponsor: None.

630 Poster Session

Extra-hepatic (ehCCA) and intra-hepatic cholangiocarcinomas (iCCA): A genomic landscape comparison of peri-hilar (phCCA), common bile duct (cbdCCA) and gallbladder (gbCCA) with iCCA. First Author: Parth J. Sampat, SUNY Upstate Medical University, Syracuse, NY

Background: Metastatic CCAs have poor prognosis. Comprehensive genomic profiling (CGP) was used to compare genomic alterations (GA) in phCCA, with cbdCCA, gbCCA and iCCA. Frequencies of GA in therapy targets such as FGFR2, IDH, ERBB2 & BRAF, & Immuno-oncology (IO) drug response predictors (MSI status, TMB & PDL1) were examined. **Methods:** DNA extracted from 110 cases of phCCA, 743 cases of cbdCCA, 2983 cases of gbCCA and 9178 cases of ihCCA underwent hybrid capture based CGP to evaluate classes of genomic alterations (GA), MSI status, TMB levels, genomic ancestry, genomic signature, HRD score and germline status. phCCA cases were confirmed at surgery and pathology as originating from either left, right or common hepatic ducts. cbdCCA cases were confirmed by cbd biopsies or Whipple procedures. PD-L1 expression was determined by IHC using the Dako 22C3 assay using tumor proportional score (TPS) system. **Results:** The pt age distributions were similar; female gender was more frequent in gbCCA. gbCCA featured higher GA/tumor (P=.0001). The frequencies of FGFR2 & IDH1 targetable GA in phCCA, cbdCCA and gbCCA were uncommon especially when compared iCCA (P<.0001). GA in BRAF, CDKN2A, MTAP and PIK3CA in gbCCA were like phCCA and cbdCCA. At 16.5%, the frequency of GA in ERBB2 were highest in gbCCA and in all ehCCA when compared with iCCA (P<.0001). The frequencies of KRAS GA including KRAS G12C were greater in both phCCA and cbdCCA when compared with gbCCA (P=.0015) and iCCA (P.0001). TP53 GA were highest in gbCCA (P=.0007). IO biomarker frequencies were similar in all. The frequencies of HRD positive (gLOH) status and MMR genomic signature were similar in all. **Conclusions:** ehCCA subtypes feature characteristic GA profiles that have potential to influence therapy selection with lower frequencies of FGFR2 & IDH1 when compared with iCCA. In contrast, targetable ERBB2 & KRAS G12C are more frequent in ehCCA than iCCA, and highest in gbCCA & cbdCCA. Research Sponsor: None.

	peri-hilarCCA	cbdCCA	P-value (phCCA vs cbdCCA)	gbCCA	P-value (phCCA vs gbCCA)	ihCCA	P-value (phCCA vs iCCA)
BRAF	3.6%	4.7%	NS	2.4%	NS	5.1%	NS
ERBB2	14.5%	10.6%	NS	16.5%	NS	5.1%	0.002
FGFR2	0.9%	2.0%	NS	1.5%	NS	11.7%	0.001
IDH1	3.6%	0.8%	NS	0.6%	NS	13.9%	0.006
KRAS	29.1%	46.2%	0.015	14.2%	0.0015	19.8%	0.071
TP53	45.5%	63.8%	0.011	65.7%	0.0007	33.9%	0.061
MSI-high	1.8%	1.3%	NS	1.2%	NS	1.8%	NS
TMB≥10 mut/mb	3.6%	4.2%	NS	5.2%	NS	3.6%	NS
PD-L1 low positive	21.1%	31.3%	NS	23.5%	NS	23.7%	NS

631 Poster Session

Exclusive liver invasion as prognostic indicator in T3 gallbladder cancer. First Author: Woohyung Lee, Asan Medical Center, Seoul, South Korea

Background: T3 gallbladder cancer (GBC) contains a diverse range of invasive tumors. However, distinct prognostic outcomes may be observed among T3GBC confined to localized range despite sharing the same T3GBC. There is uncertainty regarding whether there are noticeable prognostic variances between tumors that invade the liver, or serosa only and those that extend to adjacent organs. This study aims to investigate prognostic differences according to the extent of invasion in patient with T3GBC. **Methods:** The patients with T3GBC who underwent curative intent surgery from February 2002 to December 2021 were analyzed. The patients were divided into those who had tumor invaded liver only, serosa only, or extended beyond. Their clinicopathologic characteristics, perioperative details and prognostic data were compared. Survival analyses were carried out using the log-rand test including subgroups considering nodal metastasis, and other T stages. Cox-proportional hazard model including other factors were used to identify the independent risk factors for survival. **Results:** We identified 194 patients with T3GBC. The patients with tumor invaded liver only, serosa only, and adjacent organs were 86, 33, and 35, respectively. The median survival was 23 months. The patients with invasion limited to the liver showed better survival than those who involve adjacent structures (26 months vs 9 months, p < 0.001). In multivariate analysis, it was demonstrated that the concomitant presence of invasion in structures other than the liver serves as independent prognostic factors for survival. (hazard ratio [HR], 6.8, 95% confidence interval [CI] 1.746 – 26.803, p = 0.006) as well as lymphovascular invasion (HR 2.5, 95% CI 1.093 – 4.193, p= 0.026). **Conclusions:** In the T3 gallbladder cancer, exclusive liver invasion is a better prognostic indicator compared to concurrent invasion around adjacent structure. It is crucial to recognize that the extent of invasion in T3GBC patients before surgery. Multicenter studies are needed for the definite results. Research Sponsor: None.

632 **Poster Session**

Exploring Nectin-4 expression in hepatocellular carcinoma. First Author: Chien Huai Chuang, National Taiwan University Cancer Center, Taipei City, Taiwan

Background: Nectin cell adhesion protein 4 (Nectin-4) is overexpressed in various malignancies, contributing to cancer progression and poor prognosis. We aimed to assess Nectin-4 expression in hepatocellular carcinoma (HCC). **Methods:** Patients who underwent primary liver tumor resection for HCC upon initial diagnosis at National Taiwan University Hospital in 2019 were included. Exclusion criteria consisted of prior systemic or loco-regional therapy, or death within three months post-surgery. Immunohistochemical (IHC) staining for Nectin-4 (EPR15613-68, Abcam) was conducted and independently reviewed by two pathologists. Nectin-4 expression patterns (membranous, cytoplasmic, and nuclear) and intensity were scored on a 0–3 scale, with normal eccrine glands (2+) serving as a positive control. Nectin-4 expression was further assessed in primary and paired progressive or metastatic tissue samples from patients with available paired samples. **Results:** Of the 193 patients, 175 had specimens available for IHC analysis. Nectin-4 expression was detected in 22.9% of these patients, with intensity scores of 0 in 77.1%, 1 in 22.3%, and 2 in 0.6%. The majority of staining pattern was cytoplasmic (94.3%), with membranous staining observed in 5.1% of cases. Nectin-4 expression was significantly associated with higher pathological grades (p = 0.005) but showed no correlation with recurrence-free survival (p = 0.771) or overall survival (p = 0.396) after surgery. Importantly, Nectin-4 expression intensity significantly increased in paired progressive or metastatic samples compared to initial HCC specimens (p = 0.046; n = 27). **Conclusions:** Nectin-4 expression is associated with higher pathological grades but not related to clinical outcomes in HCC. Increased Nectin-4 expression in progressive or metastatic disease highlights its potential as a therapeutic target. Research Sponsor: National Science and Technology Council.

633 **Poster Session**

Use of pathological response versus RECIST assessment in predicting relapse-free survival following neoadjuvant immunotherapy for hepatocellular carcinoma. First Author: Antonio D'Alessio, Imperial College, London, United Kingdom

Background: Immune checkpoint inhibitors (ICI) administered prior to liver resection (LR) lead to pathological responses in patients with hepatocellular carcinoma (HCC). However, the relative value of pathological versus radiological response in predicting relapse-free survival (RFS) remains unclear. **Methods:** We pooled patient-level data from 111 patients (pts) with HCC receiving ICI prior to LR as part of 5 phase I/II trials and observational clinical studies conducted in 12 centres in the United States, United Kingdom and Asia, as part of an academic consortium (NeoHCC). Pathological response was measured as the percentage of non-viable tumour in the resected specimen, with major (pMR) and complete pathological response (pCR) corresponding to ≥70% and 100% tumour regression. Radiological overall response rate (ORR) was assessed by RECIST v1.1 and modified RECIST (mRECIST) criteria. We correlated pathological response and ORR with RFS using Cox regression. **Results:** Pts received preoperative ICI between Oct 5, 2017, and Nov 15, 2023, mostly ICI combinations (69%, n=76). Most pts were male (78%, n=87) with viral chronic liver disease (66%, n=73), BCLC stage A HCC (55%, n=61). ORR was 28% per RECIST v1.1 and 32% per mRECIST criteria (available for n=81). Out of the 104 pathologically evaluable pts, pMR and pCR rates were 32% (n=33) and 18% (n=19). Radiological response by RECIST v1.1 showed a significant correlation with pathological response (R^2 = 0.43, p<0.001). When using RECIST v1.1, 74% of pts with ORR achieved pMR vs 14% of those without ORR (n=23/31 vs 10/73, p<0.001). However, 30% of pMR were not predicted by ORR (n=10/33). The discrepancy between pMR and ORR decreased using mRECIST. ORR per mRECIST was 83% in pMR pts (n=19/23), whereas it was 10% in non-pMR (n=5/51). After a median follow-up of 27.2 mos (95%CI 22.3-32.1), median RFS for the whole cohort was 43.6 mos (95%CI 28.3-NE). Achievement of pMR, pCR and ORR was associated with improved RFS (Table). However, reduction in the risk of relapse and/or death was higher in pts achieving pMR or pCR than ORR, regardless of the use of RECIST v1.1 or mRECIST. Conversely, pts without pMR had a lower mRFS than pts without ORR (28.3 mos [95%CI 12.8-43.8] and 32.8 mos [95%CI 13.7-51.9], respectively). **Conclusions:** Whilst radiological responses to neoadjuvant ICI are associated with improved RFS in pts with HCC, achievement of pMR is more accurate than RECIST and mRECIST-based responses in predicting for RFS, with lack of pMR identifying pts at a higher risk of relapse or mortality. Research Sponsor: None.

Cox regression for RFS	HR (95% CI)	p value
pCR	0.19 (0.05-0.78)	0.02
pMR	0.25 (0.10-0.66)	0.005
RECIST 1.1	0.34 (0.13-0.86)	0.02
mRECIST	0.38 (0.13-1.11)	0.08

RFS, relapse-free survival; HR, hazard ratio; CI, confidence interval; pCR, complete pathological response; pMR, major pathological response; mRECIST, modified RECIST.

634 **Poster Session**

Genomic alterations as predictive biomarkers for cancer-associated thrombosis (CAT) in biliary tract cancer (BTC). First Author: Angelo Pirozzi, Division of Hematology and Oncology, Mayo Clinic Arizona, Phoenix, AZ

Background: CAT is the second leading cause of cancer-related death, yet the genetic underpinnings of VTE risk in BTC remain underexplored. The evaluation of CAT risk is based on Khorana score (KS), which is limited to clinical parameters. Incorporation of a genomic signature in CAT risk assessment may allow for individualized risk assessment. **Methods:** We collected data from consecutive patients (pts) with BTC treated at Mayo Clinic Hospitals (Phoenix, Rochester, Jacksonville) between 1998-2024. Genomic profiling was conducted on FFPEs and blood samples through next-generation sequencing (NGS) (NextSeq, Tempus xT CDx, FoundationOne CDx). Association between CAT and clinicpathologic features (age, gender, race, ethnicity, BMI, tumor site (TS), surgery, stage, radiotherapy, systemic therapy, history of thromboembolism (hVTE)) were assessed with multivariate Cox regression (p<0.05). A separate Cox regression adjusting for age, hVTE, radiotherapy, systemic therapy, metastatic disease stratified for TS was used for gene associations. Benjamini–Hochberg method was applied (q<0.1) based on previous publications. Median overall survival (mOS) was calculated from the date of diagnosis to the date of death or last follow-up. **Results:** 740 adult pts were included, median age 62 years (IQR: 56-71), male/female (51%/49%), median BMI 26.6 (IQR: 22.8-31.1) with intrahepatic (496, 67%) and extrahepatic BTC (173, 23%), gallbladder cancer (71, 10%). 244 (33%) had metastatic disease. 216 (29%) experienced CAT. For those with CAT, 31 (14%) scored 1, 9 (4%) scored 2 and 1 (<1%) scored 3 on KS. mOS was 33 months (mos) (95% CI 28.9-38.4) vs. 55 mos (95% CI 25.1-39.1) (p=0.2) for pts with and without CAT, respectively. hVTE was associated with CAT (HR 3.14, 95% CI 1.98-4.98, p<0.001) along with systemic therapy (HR 1.19, 95% CI 1.11-2.89, p<0.01) and metastatic disease (HR 1.65, 95% CI 1.11-2.44, p<0.01). 305 pts (41%) underwent NGS. The most common altered genes were TP53 (126, 41%), KRAS (64, 21%), CDKN2A (63, 21%). The most common altered currently targetable genes were FGFR2 (43, 14%), IDH1 (30, 10%) and HER2/ERBB2 (16, 5%). MYC (4, 1%) (HR 11.60, 95% CI 3.04-44.25, p<0.001, q=0.058) and HER2/ERBB2 (HR 2.45, 95% CI 1.26-4.77, p=0.008, q=0.087) alterations were significantly associated with an increased risk of CAT. MYC alterations were all copy number gains (100%). The most frequent HER2/ERBB2 alterations were copy number gains (10, 62%). FGFR2 fusions (31, 10%) were linked to a lower CAT risk (HR 0.46, 95%CI 0.22-0.93, p=0.032, q=0.075) with FGFR2-BICC1 being the most frequent (5,16%). **Conclusions:** MYC and HER2/ERBB2 alterations are novel predictive biomarkers for an increased risk of CAT in BTC, while FGFR2 fusions confer lower risk. The knowledge of mutational status lends insight into guidance for anticoagulant prophylaxis management for this subgroup and integration with KS. Research Sponsor: None.

635 **Poster Session**

Expanded biomarker analysis of an international, open-label phase 2 study of regorafenib plus pembrolizumab in patients with advanced hepatocellular carcinoma (HCC) previously treated with immune checkpoint inhibitors (ICI). First Author: Anthony B. El-Khoueiry, USC Norris Comprehensive Cancer Center, Los Angeles, CA

Background: The optimal treatment for patients (pts) with advanced HCC who progress on an ICI-containing regimen has not been defined. In pts who progressed on one prior ICI regimen, regorafenib plus pembrolizumab (rego + pembro) showed modest anti-tumor activity (primary completion analysis: overall objective response rate 7.4%, all partial responses [PR]) and an expected safety profile (NCT04696055; El-Khoueiry, ASCO 2024). We conducted additional exploratory analyses of tumor markers to gain insights into the clinical results. **Methods:** Overall, 95 adults with unresectable advanced HCC (C-P Class A, BCLC stage B/C, ECOG PS 0/1) received oral regorafenib 90 mg once daily 3 weeks on/1 week off plus pembrolizumab 400 mg i.v. every 6 weeks. Prior treatment: Cohort 1 = atezolizumab + bevacizumab (n=68); Cohort 2 = any other ICI alone or in combination (n=27). Biomarker analyses in subsets of pts explored potential correlations with clinical benefit (defined as tumor response or stable disease [SD] ≥ 4 months). Immune cell subtypes in blood were analyzed by flow cytometry. Tumor immune markers were assessed by immunohistochemistry. Gene expression profiling in tumor samples was by RNAseq. **Results:** Twenty-one pts had baseline samples for RNAseq, 7 of whom had clinical benefit (n= 3 Cohort 1 [including 1 PR], n=4 Cohort 2). Pts with clinical benefit had differential gene expression at baseline including enriched expression of genes associated with Notch, Hedgehog, fibroblast, TGF-β, and insulin-like growth factor binding protein signaling; in contrast, there was higher expression of genes associated with myeloid derived suppressor cells in pts without clinical benefit. Of note, tumor samples from pts in this study had increased Tregs and T-cell exhaustion signatures compared with pts treated with rego + pembro in a first-line study (NCT03347292).Analyses of available paired tumor samples from 4 patients (n=2 SD, n=2 progressive disease) found reductions in activated fibroblast, TGF-β, and collagen gene expression signatures from baseline to Week 6 Day 1 of treatment. **Conclusions:** Rego + pembro had modest activity in pts who progressed on one prior ICI-based regimen. Exploratory biomarker analyses suggest different gene expression in pts with or without clinical benefit. Clinical benefit was associated with enriched expression of genes in Notch, Hedgehog, fibroblast, and TGF-β pathways. Pre- and post-treatment biopsies showed reductions in fibroblast, TGF-β, and collagen gene expression, which are known features of HCC. In the context of the limitations of small sample size and absence of a control arm, these findings are hypothesis generating and consistent with HCC biology and regorafenib mechanism of action. Clinical trial information: NCT04696055. Research Sponsor: Bayer Healthcare Pharmaceuticals.

636 **Poster Session**

Radiomics-based prediction of HCC response to atezolizumab/bevacizumab. First Author: Isaac Rodriguez, Division of Hepatology, Division of Clinical Bioinformatics, Department of Medicine II, Medical Faculty Mannheim, Heidelberg University, Mannheim, Germany

Background: Advanced hepatocellular carcinoma (HCC) treatment has evolved with the introduction of atezolizumab/bevacizumab, showing improved outcomes over sorafenib. However, the response varies among patients, particularly between viral and non-viral etiologies. This study aimed to develop and evaluate multimodal prediction models combining quantitative imaging and clinical markers to predict treatment response in HCC patients. **Methods:** From March 2020 to May 2023, patients with advanced HCC treated with atezolizumab/bevacizumab were retrospectively identified from six centers in Germany and Austria. Patients underwent baseline contrast-enhanced liver MRI and follow-up imaging to assess therapy response. Machine learning models, including RandomForestClassifier, were developed for radiomics, clinical, and combined datasets. Hyperparameter tuning was performed using RandomizedSearchCV, followed by cross-validation to evaluate model performance. **Results:** The study included 103 patients, with 70 achieving disease control (DC) and 33 experiencing disease progression (PD). Key findings included significant differences in treatment response and progression-free survival between DC and PD groups. The radiomics model, using 14 selected features, achieved 73.1% accuracy and a ROC AUC of 0.635 on the test set. The clinical model, with 4 selected features, achieved 73% accuracy and a ROC AUC of 0.649 on the test set. The combined model showed improved performance with 69% accuracy and a ROC AUC of 0.753 on the test set. Hyperparameter tuning further enhanced the combined model's accuracy to 80.1% and ROC AUC to 0.771 on the test set. **Conclusions:** The hybrid model combining clinical and radiological data outperformed individual models, providing better predictions of response to atezolizumab/bevacizumab in HCC patients. Research Sponsor: None.

637 **Poster Session**

Expression of Trop2, Nectin-4, and FOLR1 cell surface markers in biliary tract cancers: Is it time to repurpose drugs? First Author: Kannan Thanikachalam, Roswell Park Comprehensive Cancer Center, Buffalo, NY

Background: Biliary tract cancers (BTC), including gall bladder cancer, intrahepatic and extrahepatic cholangiocarcinoma are aggressive malignancies, with poor prognosis. Despite advances in frontline setting, and in biomarker selected population (IDH1, FGFR2, Her2, BRAF, HRD), there is a dire need to advance therapy in subsequent lines. **Methods:** We identified BTC patients (pts) at Roswell Park Cancer Center who underwent curative intent surgery between 01/2006-01/2020. After excluding duplicate samples and those with insufficient tissue, we found 78 pts with adequate follow up for analysis. Expression of trophoblast cell surface antigen 2 (Trop2), nectin4, folate receptor alpha (FOLR1) was assessed by immunohistochemistry and correlated with clinical outcomes. H score, a weighted index score that factor in, the % of cells across different intensity was calculated. H scores were classified as negative (0) and positive (>1). Positive H scores were grouped as weak (1-100), intermediate (101-200) and strong (201-300). Descriptive statistics were conducted, and multivariate analyses performed (R studio) to study any correlation between expression of markers and survival outcomes. **Results:** The median age of our cohort was 67.5 years (range 40-87) with 58% male; 24.4%, 30.8%, 20.5%, 7.7% had stages I, II, III, IV disease. Staging information was missing in 16.7% of pts. Of 73 analyzable samples for Trop2, all were positive for Trop2, with a mean H score of 86, SD of 68.9. 56.4% of patients had a weak H score, 30.8% had an intermediate H score and 6.4% had a strong H score for Trop2. Median overall survival (mOS) among low, intermediate and high expressing group of pts with Trop2 was 48.9 months (m), 23.9 m, and 41.9 m with median Progression Free Survival (mPFS) of 37.8m, 15.3 m, and 37.4 m, respectively. Of 75 analyzable samples for FOLR1 and Nectin4, FOLR1 was expressed in 51.3% with a mean H score of 6.97 (SD 21.9). Nectin4 was expressed in 78.2%, with a mean H score of 13.7 (SD 26.3). Pts with Trop2 expression had a mOS of 39.5 m and a mPFS of 28.7 m. There were no significant differences in mOS (48.9 m vs 33.3 m, p=0.14) and mPFS (37.8 m vs 26.2 m, p=0.35) among patients with positive vs negative FOLR1 respectively. There was no difference in mOS (41.6 m vs 38.3 m, p=0.92) and mPFS (32.8 vs 26.4 m, p-value=0.5) in patients with positive vs negative Nectin4. Multivariate analyses showed that pts with intermediate Trop2 (H score=101-200) had a high risk of death (HR 21.44, CI 2.81-163.4, p= 0.003). **Conclusions:** Though no difference was seen in clinical outcomes in pts with strong Trop2 expression, FOLR1, and Nectin4 expression, this could be explained by the small sample size. Trop2 is frequently expressed in BTCs and patients with Intermediate Trop2 H score had lower mOS and mPFS, with a higher risk of mortality. This warrants further investigation into potential use of Antibody Drug conjugates against Trop2 in BTC. Research Sponsor: Roswell Park Start-Up Fund.

638 **Poster Session**

Tumor vascularity as a predictor of FGFR inhibitor response in FGFR2-fused intrahepatic cholangiocarcinoma. First Author: Kelly Meza, Baylor College of Medicine, Houston, TX

Background: Fibroblast growth factor receptor 2 (FGFR2) fusion is a well-established mutation in intrahepatic cholangiocarcinoma (iCCA), accounting for approximately 10% of cases. Current treatments, including pemigatinib, infigratinib, and futibatinib, have shown promising results with response rates of less than 40%. Most patients experience progression within 6-8 months, and reliable biomarkers to predict treatment response are still lacking. This study aimed to assess the clinical outcomes of FGFR inhibitors in patients (pts) with FGFR2-fused iCCA, while investigating tumor vascularity as a potential predictor of treatment efficacy. **Methods:** We retrospectively analyzed data from 134 pts treated at MD Anderson Cancer Center between 2009 and 2024, focusing on 73 pts with comprehensive clinical data. The cohort included pts treated with futibatinib (n=16), infigratinib (n=16), pemigatinib (n=36), and derazantinib (n=5). Tumor vascularity was assessed via Hounsfield units (HU) from CT images, including a mathematical parameter to calculate a ratio of hypervascular tumor rim thickness to tumor diameter. Transcriptomic analysis of 11 pts was performed to explore mRNA expression related to tumor vasculature. Kaplan-Meier analysis for progression-free survival (PFS) and ANOVA for statistical comparisons were applied. **Results:** The median age was 60 years, and 41 pts were female. For FGFR2-fused iCCA, median PFS for first-line gemcitabine-based chemotherapy with or without immunotherapy was 4.07 months [3.0-5.1]. Among FGFR inhibitors, mPFS was 7.07 months [95% CI: 5.10-8.16], with no significant difference in outcomes between FGFR inhibitors. 59 pts had comparable CT images and were grouped into 3 groups based on K-means clustering: Group 1 (n=30) had mPFS of 3.07 months [2.03-4.10]; Group 2 (n=22), mPFS of 9.10 months [8.2-11.2]; Group 3 (n=7), mPFS of 34.03 months [18.2-37.6]. Group 1 had the most hypovascular tumors (the lowest HU in the periphery of tumors), while Group 3 had the most hypervascular tumors (p=0.017). The HU ratio of the hypervascular tumor rim thickness to tumor diameter showed a trend, with the highest ratio in Group 3 and the lowest in Group 1 (p=0.09). Transcriptomic analysis revealed a trend toward higher VEGF/VEGFR gene expression in hypervascular tumors. Eight pts received second-line FGFR inhibitors, with mPFS of 5.0 months [2.1-8.3]. **Conclusions:** Pts with FGFR2-fused iCCA have a shorter PFS on first-line chemotherapy, and PFS outcomes for FGFR inhibitors were consistent with existing data. Tumor vascularity, as assessed by CT imaging, may be a valuable predictor of response to FGFR inhibitors. This study suggests that hypervascular tumors may have better outcomes, warranting further investigation in a larger cohort to confirm these findings and establish tumor vascularity as a predictive biomarker for FGFR inhibitor therapy. Research Sponsor: None.

639 **Poster Session**

Safety and efficacy of anti-HER2 agents in the treatment of biliary tract cancers: A systematic review. First Author: Nikolas Naleid, Department of Internal Medicine, University Hospitals Cleveland Medical Center, Cleveland, OH

Background: Limited treatment options exist for patients with locally advanced or metastatic biliary tract cancers (BTCs). Recently, several clinical trials provided preliminary evidence for human epidermal growth factor receptor-2 (HER2) as a new target for patients with HER2 expressing BTC. We conducted a systematic review and pooled analysis of the safety and efficacy of anti-HER2 agents in patients with advanced BTC. **Methods:** A comprehensive search of PubMed/MEDLINE and EMBASE was performed to identify phase 1, 2, or 3 clinical trials published between January 2019 and March 2024 that evaluated anti-HER2 therapy in locally advanced or metastatic BTC. Primary endpoints included objective response rate (ORR), disease control rate (DCR), and progression-free survival (PFS). Secondary endpoints included the incidence of treatment-related adverse events (TRAEs), rate of treatment discontinuation, and death. Forest plots were generated to summarize the results and demonstrate the effect size estimates for each study, along with their confidence intervals and the pooled estimate from the random-effects model. The random-effects models were configured to use the Restricted Maximum Likelihood method for estimating between-study variance for all analyses. **Results:** After excluding duplicate and irrelevant articles, data extraction and analysis were performed on eight selected clinical trials. The analysis included 368 patients with a median age of 64 (range 49-68), with 55% being female. Patients were treated with several anti-HER2 agents, including zanidatamab, pertuzumab plus trastuzumab, tucatinib plus trastuzumab, trastuzumab deruxtecan, trastuzumab plus chemotherapy, trastuzumab-pkrb plus chemotherapy and neratinib. The pooled ORR and DCR were 34% (95% CI 24-44) and 64% (95% CI 51-77) respectively. The pooled weighted PFS and median OS survival were 4.8 and 9.4 months, respectively. The pooled duration of response was 5.0 months. In our subset analysis, which included only patients who had received ≥1 prior therapy, the pooled ORR was 30% (95% CI 22-39), the pooled DCR was 62% (95% CI 48-75), and the pooled median PFS was 4.4 months. In the study cohort, 82.6% of patients experienced any adverse event, and 32.1% experienced a grade 3-4 adverse event. Only 5.7% of the patients discontinued treatment secondary to TRAEs. **Conclusions:** In patients with HER2 expressing BTCs, anti-HER2 therapies demonstrated preliminary signals of efficacy with manageable toxicities, particularly in the second-line setting. Research Sponsor: None.

640 Poster Session

Economic burden of hepatocellular carcinoma recurrence after curative treatment in Spain. First Author: Maria Varela, Liver Unit, Hospital Universitario Central de Asturias, Universidad de Oviedo, IUOPA, ISPA, FINBA, Oviedo, Spain

Background: The literature on burden of disease studies of hepatocellular carcinoma is scarce. The aim of this analysis was to quantify the economic burden of disease recurrence in very early or early-stage HCC patients who underwent liver resection, ablation or transplantation in Spain. **Methods:** A decision-tree model was developed to estimate the cost of a recurrence after curative treatment with an initial complete response of very early or early-stage primary HCC tumor over a 5-year time horizon. Clinical parameters were collected by a multidisciplinary group of experts who also validated the assumptions and results of the analysis. According to the Barcelona Clinic Liver Cancer (BCLC) recommendations, patients with recurrent HCC were classified into five HCC stages (0, A, B, C, and D) and allocated to different treatment options. A payer perspective was adopted, so only direct costs (€2024) were considered (local/locoregional treatments costs, drug acquisition and administration costs, adverse events management costs, diagnostic and follow-up costs and end-of-life costs). Sensitivity analyses were performed to assess uncertainty. **Results:** A total of 1,250 HCC recurrences were estimated in Spain, resulting in a total cost of €32,817,142 in the first year of recurrence management. After 5-years follow-up, an additional €10,1M was estimated, mainly due to patient surveillance (€2,8M) and 228 new recurrences (€5,7M). The average cost of a relapse was €26,253 euros in the first year of recurrence management, rising to €34,342 euros for 5-years follow-up. A hypothetical patient experiencing a second relapse 5 years after their first relapse would incur costs of €58,758. **Conclusions:** This study is the first in Spain to explore the economic burden of hepatocellular carcinoma. Estimating the cost of recurrence after curative treatment and complete response of a primary HCC tumor is complex and depends on multiple factors, so further studies exploring both efficacy and costs in hepatocellular carcinoma are needed. Research Sponsor: Roche.

641 Poster Session

Alteration of gut microbiota in patients with advanced hepatocellular carcinoma. First Author: Thanakorn Charoenthanadhol, Faculty of Medicine, Khon Kaen University, Muang, Thailand

Background: Hepatocellular carcinoma (HCC) is the most common primary liver cancer and the third leading cause of cancer-related death worldwide. Despite advancements in treatment, such as novel combinations of immunotherapy and targeted therapy or dual immunotherapy, the prognosis remains poor due to a lack of predictive biomarkers for treatment response. Gut microbiota alterations have been proposed as both potential diagnostic biomarker and predictive biomarker. However, data in this field are not well-established and are mostly derived from studies conducted in China and Western countries, which may not fully reflect the situation in Thailand. **Methods:** Fecal samples were collected from pre-treated advanced HCC patients who visited medical oncology outpatient clinic at Srinagarind hospital, Khon Kaen University (HCC group = 27) and analyzed using 16S rRNA sequencing of gut microbiota. Additional sequenced data from a healthy population (Control group = 31) who had no underlying disease and had normal liver ultrasonography were retrieved from a previous study conducted by Khon Kaen University (Cholangiocarcinoma Screening and Care Program, CASCAP). The datasets were compared using the Wilcoxon rank-sum test to analyze the alteration of gut microbiota between HCC group and control group. **Results:** The HCC group exhibited significantly lower biodiversity index ($p < 0.001$) than control group in terms of richness, Shannon diversity index and Simpson's index. The HCC group had significantly higher relative abundance of the phylum Proteobacteria ($p < 0.001$), Firmicutes ($p < 0.001$) and a lower abundance of the phylum Actinobacteria ($p < 0.001$) compared to the control group. Additionally, the HCC group had significantly higher relative abundance of Stenotrophomonas, Granulicatella, Ruminococcus, Blautia, Phascolarctobacterium, Butyricoccus, Streptococcus, Escherichia-Shigella, Flavonifractor, Haemophilus and Lachnospira at the genus level. Further analysis of relative abundance of gut microbiota at the genus level and clinical prognostic parameters showed positive correlation between Streptococcus and 6-month survival status. **Conclusions:** Gut microbiota profile in patients with advanced HCC was significantly altered from healthy control group. This could be the evidence of microbial dysbiosis and may potentially be biomarker for prognosis. Research Sponsor: Faculty of Medicine, Khon Kaen University, Thailand.

642 Poster Session

Total cost of care of immune-oncology treatments for unresectable hepatocellular carcinoma: A Singapore healthcare perspective analysis. First Author: John Fullarton, Violicom Medical Limited, Aldermaston, United Kingdom

Background: Tremelimumab + durvalumab (STRIDE) and atezolizumab + bevacizumab are recommended first-line immune-oncology treatment options for unresectable hepatocellular carcinoma (uHCC). This study aimed to quantify the total economic burden to the Singapore healthcare system of treating patients with uHCC using STRIDE *versus* atezolizumab + bevacizumab. **Methods:** A model was developed to estimate the total cost of care of using STRIDE or atezolizumab + bevacizumab, which followed patients with uHCC eligible for first-line systemic therapy, considering direct medical costs. Costs considered in the model included treatment acquisition and administration, endoscopy, healthcare resource use, and cost for treating grade ≥3 treatment-emergent adverse events (TEAEs). Data on treatment regimens and TEAEs were taken from the HIMALAYA (STRIDE) and IMbrave150 (atezolizumab + bevacizumab) trials. Resource use rates were based on an average of 168 days treatment for both regimens. All costs were obtained from Singaporean sources. **Results:** The total cost of care for STRIDE and atezolizumab + bevacizumab was estimated at SGD$40,277 (USD$30,932) and SGD$43,605 (USD$33,487) *per* patient, respectively. This translates into an average cost savings of SGD$3,328 (USD$2,555]) *per* patient when treated with STRIDE compared to atezolizumab + bevacizumab. The main driver of cost savings was treatment acquisition and administration (-SGD$1,917 [-USD$1,472], -6%) though lower costs were also associated with resource use (-SGD$927 [-USD$712], -8%), endoscopy (-SGD$342 [-USD$263], -77%) and TEAE management (-SGD$141 [-USD$108], -74%). Assuming an incident population of 260 patients eligible for treatment, the total burden was estimated at SGD$10,489,604 (USD$8,055,696) when all patients were treated with STRIDE, and SGD$11,356,273 (USD$8,721,271) when treated with atezolizumab + bevacizumab (Table). The incremental cost offset to the health plan was SGD$866,669 (USD$665,575) when treated with STRIDE. **Conclusions:** This study suggests that STRIDE offers cost savings over atezolizumab + bevacizumab for uHCC treatment in the Singapore healthcare system. Additional real-world evidence studies are warranted to capture the long-term survival benefits and associated treatment costs. Research Sponsor: AstraZeneca.

Annual cost of care based on 260 patients receiving first-line immune-oncology treatment.

Category	STRIDE	Atezolizumab + Bevacizumab	Incremental difference
Endoscopy costs	$26,358	$115,555	-$89,197
Resource Use costs*	$2,607,000	$2,848,307	-$241,307
Adverse events costs	$12,899	$49,716	-$36,817
Treatment acquisition and administration	$7,843,347	$8,342,696	-$499,349
TOTAL	**$10,489,604**	**$11,356,273**	**-$866,669**

All costs in SGD$. *Includes: physician visits; laboratory and radiological tests; and hospitalisation.

643 Poster Session

Obesity and biliary tract cancers: Changing epidemiology. First Author: Layal Al Mahmasani, Memorial Sloan Kettering Cancer Center, New York, NY

Background: The incidence of biliary tract cancers (BTC) continues to rise (Sung et al. 2024). The reasons remain ill-defined. Over 70% of patients are diagnosed without identifiable predisposing factors. Obesity and overweight have emerged as putative contributors. We examined the epidemiologic and molecular characteristics of patients with BTC, with a focus on body mass index (BMI) at presentation and overall survival (OS). **Methods:** A retrospective cohort study of patients diagnosed with BTC at Memorial Sloan Kettering was conducted from January 2022 to December 2023. Patients were divided based on BMI into: obese (BMI≥30 kg/m^2) and non-obese (BMI<30 kg/m^2). We collected patients' baseline demographics, and results of somatic next generation sequencing using the MSK-IMPACT platform and compared between obese and non -obese groups using Fisher's exact test for categorical and Wilcoxon rank sum test for continuous variables. OS was calculated from date of diagnosis, estimated using Kaplan-Meier methods and compared between groups using log rank test. **Results:** A total of 142 patients were identified. The table summarizes the baseline characteristics including age, BMI, sex, stage at diagnosis, family history, history of malignancy, and race. Obesity was significantly associated with family history of malignancy and white race. Among surviving patients (n=93), with a median follow-up of 13.8 months, 6-month, 1-year, and 2-year overall survival (OS) rates were 85% [95%CI: 80%-92%], 71% [95%CI: 64%-79%], and 54% [95%CI: 43%-68%] respectively. OS was not statistically different among obese and non-obese patients. However, when stratifying OS by BMI and gender, obese males were noted to have inferior OS compared to obese females (6-mo OS 75% and 90% respectively), though not statistically significant. Genetic sequencing did not reveal a specific molecular pattern in either group. **Conclusions:** Obesity appears not to be associated with young age, stage, or particular somatic alterations in this limited study. OS was not statistically different among obese and non-obese patients. Males with obesity appeared to have a worse prognosis. This observed gender-based difference needs to be further studied, and if confirmed further research is needed to explore potential factors that may explain the clinical observation. Research Sponsor: None.

Baseline characteristics.

Characteristics	Overall n=142	Obese (BMI≥30 kg/m^2) n=54	Non-Obese (BMI<30 kg/m^2) n=88	p-value
Age	68 (35-91)	69 (35-87)	67 (36-91)	>0.9
BMI (kg/m^2)	28.1 (16.6-48.5)	33.5 (30-48.5)	24.7 (16.6-29.6)	
Sex female/male	79 (56%)/63 (44%)	32 (59%)/22 (41%)	47 (53%)/41 (47%)	0.6
Stage Localized/advanced	21 (15%)/121 (85%)	11 (20%)/ 43 (80%)	10 (11%)/ 78 (89%)	0.3
Family history of cancer	91 (65%)	41 (76%)	50 (57%)	0.030
History of second malignancy	26 (18%)	12 (22%)	14 (16%)	0.4
Race White	99 (70%)	44 (81%)	55 (63%)	0.004
African	12 (8.5%)	4 (7.4%)	8 (9.1%)	
Asian	20 (14%)	1 (1.9%)	19 (22%)	
Other	11 (7.7%)	5 (9.3%)	6 (6.8%)	

644 Poster Session

Utilization of circulating tumor DNA (ctDNA) testing in biliary tract cancers. First Author: Nikolas Naleid, Department of Internal Medicine, University Hospitals Cleveland Medical Center, Cleveland, OH

Background: There is limited data to guide surveillance in biliary tract cancers (BTCs) following resection. Minimal residual disease (MRD) assays utilize circulating tumor DNA (ctDNA) to detect early recurrence. Current data is limited on the use of ctDNA in biliary tract cancers in conjunction with surveillance imaging. In this study, we aim to elucidate the role of ctDNA monitoring in patients with biliary tract cancers who have undergone curative resection. **Methods:** We conducted a retrospective search at a tertiary care and several satellite settings to identify patients with BTCs who underwent curative surgery and subsequently had ctDNA monitoring. We collected demographic variables, disease characteristics, treatment regimens, and follow-up data. The primary endpoint of the analysis is recurrence-free survival (RFS), comparing patients with detectable ctDNA to those with negative ctDNA post-surgery. Kaplan-Meier analysis and Log-Rank tests were performed to evaluate RFS, and Cox proportional hazards(PH) models were employed for both univariate and multivariate analyses. **Results:** We identified 36 pts with a median age of 72 (34-87) yrs at diagnosis. In this study cohort, 22 (59%) were female, 32 (88%) were white, and 3 (8%) were African American. 12 (33%) pts were ctDNA positive at any point during surveillance, while the remainder were ctDNA negative. In the ctDNA-positive group, recurrences developed in 7 out of 12 (58%) pts. In the ctDNA negative group, only 2 out of 24 (8.3%) pts developed recurrences. The median time to recurrence following primary surgery was 10.7 months, while the time to ctDNA positivity following surgery was 7.8 months. The median time from ctDNA positivity to recurrence was 1.1 months. Pts with ctDNA positivity had a significantly lower 1-year RFS of 47.62% (95% CI: 7.5%-81%) compared to those with ctDNA negativity (85%, 95% CI: 60%-95%; p = 0.0008). Multivariate analysis demonstrated that ctDNA positivity at any point was associated with a significantly higher risk of recurrence (HR: 21.8, 95% CI: 2.3–203, $p<0.0001$). **Conclusions:** In pts with BTCs who underwent resection, ctDNA testing had independent prognostic value. Further prospective trials are needed to help define the role of ctDNA in this patient population. Research Sponsor: None.

Baseline demographics and clinical characteristics of patients who were ctDNA positive vs. negative.

Characteristics	Patients with Biliary Tract Cancers who were ctDNA positive	Patients with Biliary Tract Cancers who were ctDNA negative
N	12	24
Age in years, median (range)	72 (62 – 87)	72 (34 - 85)
Sex n (%)		
Male	7 (58)	8 (33)
Female	5 (42)	16 (67)
Race n (%)		
White	12 (100)	20 (83)
Black/African American	0 (0)	3 (13)
Asian	0 (0)	1 (4)
Primary Tumor Location n (%)		
ICC	4 (33)	11 (46)
ECC	4 (33)	5 (21)
GBC	2 (17)	6 (25)
AC	2 (17)	2 (8)
Margin Status n (%)		
Positive	2 (17)	3 (13)
Negative	10 (83)	21 (87)
Clinical Stage		
I	0 (0)	1 (4)
II	2 (17)	12 (50)
III	8 (66)	10 (43)
IV	2 (17)	0 (0)
Unknown	0 (0)	1 (3)

TPS645 Trials in Progress Poster Session

A phase IIa clinical trial of first-line cyclical therapy alternating gemcitabine, cisplatin, and durvalumab with pemigatinib for advanced biliary tract cancers with *FGFR2*-alterations. First Author: Chunjie Li, The Ohio State University Wexner Medical Center, Columbus, OH

Background: Unresectable biliary tract cancer (BTC) has a poor prognosis. Frontline chemo-immunotherapy (C-IO) for advanced BTC leads to ~25% overall survival (OS) at 24-month, but 93-99% of patients reported adverse events (AEs), and 13-19% of patients stopped treatment drugs due to AEs. FGFR kinase inhibitors (FGFRi) are second-line therapy for 10% of BTC with *FGFR2* fusions. However, clinical trials of upfront FGFRi have not accrued well. Cumulative toxicities frequently interrupt treatment and limit feasibility of concurrent C-IO + FGFRi. Over time, side effects from each of therapies have a negative impact on patients' quality of life (QoL). We previously treated two *FGFR2*-altered BTC patients with cyclical chemotherapy (CT) and FGFRi and observed prolonged survival with better QoL. By monitoring serial circulating tumor DNA (ctDNA), we observed development of FGFRi-resistant subclones while on FGFRi and their subsequent clearance after resuming CT. Thus, we have developed a clinical trial to validate our hypothesis that upfront cyclical therapy alternating between C-IO and FGFRi can improve therapeutic efficacy and QoL. Alternating therapy is hypothesized to avoid accumulating toxicities and treat emerging resistance subclones. **Methods:** In this single-center, single-arm, Phase IIa trial, all treatment-naive patients with *FGFR2*-altered advanced BTCs, PS $\leq$ 1, are eligible. Patients will receive cyclical therapy alternating between two therapy-blocks, starting from C-IO block (2 cycles of gemcitabine, cisplatin and durvalumab, 28-day/cycle), then FGFRi block (3 cycles of pemigatinib on day 1-14 of 21-day/cycle). Block-switch continues until progressive disease (PD), and then we will continue alternated block until the next PD. A sample size of 30 is calculated based on an expected 12-month OS rate of $\geq$ 55%. The trial is anticipated to enroll the first patient in early 2025. **Results:** The primary objective is 12-month OS rate. Secondary objectives are overall response rate, progression-free survival, AEs. We will stratify patients to PD-L1 CPS <1 or $\geq$ 1 to assess if cyclical therapy can enhance IO. Exploratory objectives are (1) assessing dynamics of *FGFR2*-ctDNA variants throughout therapy correlating with clinical outcomes, using a custom liquid biopsy (FGFR-Dx) developed at the Ohio State University; (2) assessing QoL scores with EORTC-QLQ-C30 and -BIL21 questionnaires, correlating with clinical outcomes. Results will be compared to historical controls from TOPAZ-1 trial. **Conclusions:** Cyclical therapy will be one of the first to combine alternating C-IO and targeted therapy in cancer. This strategy can be validated in other tumors if it is proven effective in our trial. Cyclical therapy holds great promise to introduce FGFRi earlier into treatment for BTC and improve both efficacy and QoL. Research Sponsor: Incyte Corporation (pharmaceutical company).

TPS646 Trials in Progress Poster Session

A multicenter, open-label, first-in-human study of TYRA-200 in advanced intrahepatic cholangiocarcinoma and other solid tumors with activating *FGFR2* gene alterations (SURF201). First Author: Jordi Rodon Ahnert, The University of Texas MD Anderson Cancer Center, Houston, TX

Background: Approved Fibroblast Growth Factor Receptor (FGFR) inhibitors have demonstrated clinical benefit in locally advanced or metastatic cholangiocarcinoma harboring oncogenic *FGFR2* fusions or other rearrangements. However, the emergence of acquired resistance mutations limits the clinical activity and duration of responses to currently available inhibitors. TYRA-200 is an orally bioavailable FGFR1/2/3 inhibitor designed to specifically address these clinically observed acquired resistance alterations in *FGFR2*. **Methods:** SURF201 is a single arm, multicenter, open-label, first-in-human, dose escalation and expansion study designed to investigate TYRA-200 in patients (pts) with advanced intrahepatic cholangiocarcinoma and other solid tumors with primary activating *FGFR2* gene alterations and on-target acquired known *FGFR2* resistance mutations. The study is being conducted in two parts. Part A dose escalation uses an i3+3 design and is evaluating the safety, tolerability, and the pharmacokinetic profile of TYRA-200. Part B dose expansion will further characterize the safety profile and evaluate the preliminary anti-tumor activity of TYRA-200 by RECIST v1.1 in pts with unresectable locally advanced/metastatic intrahepatic cholangiocarcinoma who have previously received an FGFR inhibitor(s) and have developed acquired *FGFR2* kinase domain resistance mutations. The study is currently planned for approximately four centers in the US and is actively enrolling (NCT06160752). Clinical trial information: NCT06160752. Research Sponsor: Tyra Biosciences, Inc.

TPS647 Trials in Progress Poster Session

KIRROS: A phase II trial in progress of first-line atezolizumab (atezo) with or without bevacizumab (bev) in patients with unresectable hepatocellular carcinoma (HCC) and Child-Pugh B (CPB) cirrhosis. First Author: Amit Singal, Department of Internal Medicine at UT Southwestern Medical Center, Dallas, TX

Background: HCC treatment is more complex for the sizable proportion of patients with CPB cirrhosis. These patients have poor survival due to both the cancer and underlying liver dysfunction. As such, they are typically excluded from pivotal clinical trials for first-line systemic therapy for unresectable HCC. The IMbrave150 trial established atezo + bev as a global standard of care for unresectable HCC, demonstrating a statistically significant improvement in overall survival (OS), progression-free survival (PFS) and confirmed objective response rate (ORR) of atezo + bev vs sorafenib (Finn N Engl J Med 2020; Cheng J Hepatol 2022). Real-world studies have consistently suggested benefit of atezo + bev in patients with HCC and CPB cirrhosis (e.g., Tanaka Hepatol Res 2022, D'Alessio Hepatol [Baltimore, Md] 2022). However, prospective clinical trial data are needed to comprehensively characterize safety and efficacy in this population. Single-agent immunotherapy is often used in patients with HCC and CPB cirrhosis given potential safety concerns of combination therapy. Safety data from real-world studies are limited by ascertainment and measurement biases. Therefore, a prospective study is warranted to address the unmet need for safety and efficacy data for first-line atezo $\pm$ bev in patients with CPB cirrhosis. **Methods:** KIRROS (NCT06096779) is a prospective, open-label, multicohort, multicenter phase II study in patients aged $\geq$18 years who have measurable and confirmed unresectable HCC, and CPB-7 or -8 cirrhosis who have not received prior systemic therapy. Eligible patients will be recruited into 2 cohorts for which endpoints will be non-comparatively assessed: Cohort A (n=60) patients will be treated with atezo 1200 mg intravenous (IV) + bev 15 mg/kg every 3 weeks (Q3W), and Cohort B (n=60) patients, those who do not meet Cohort A inclusion criteria (e.g., main portal vein tumor thrombus and proteinuria), will receive atezo 1200 mg IV Q3W only. Patients will receive treatment until unacceptable toxicity or loss of clinical benefit. The primary objective is to evaluate the safety of the treatment regimens as determined by incidence, nature and severity of adverse events and laboratory abnormalities graded per the National Cancer Institute Common Terminology Criteria for Adverse Events v5.0. Secondary objectives are to estimate efficacy with ORR, duration of response, and PFS per Response Evaluation Criteria in Solid Tumors (RECIST) 1.1 and HCC modified RECIST, as well as OS. Quality-of-life secondary endpoints include patient-reported tolerability of the treatment regimens assessed by PRO-CTCAE and EORTC IL46, and health-related quality of life per EORTC QLQ-C30 and QLQ-HCC18 scores. As of September 2024, the KIRROS study is recruiting patients at 40 sites in the US and Puerto Rico. Clinical trial information: NCT06096779. Research Sponsor: Genentech Inc.

TPS648 Trials in Progress Poster Session

HERIZON-BTC-302: A phase 3 study of zanidatamab with standard-of-care (SOC) therapy vs SOC alone for first-line treatment of human epidermal growth factor receptor 2 (HER2)-positive advanced/metastatic biliary tract cancer (BTC). First Author: James J. Harding, Memorial Sloan Kettering Cancer Center, New York, NY

Background: Zanidatamab is a humanized, IgG1-like, HER2-targeted bispecific antibody that simultaneously binds to 2 non-overlapping domains on HER2. Zanidatamab is being investigated for the treatment of HER2-expressing solid tumors, including BTC. In the phase 2 HERIZON-BTC-01 trial, zanidatamab monotherapy demonstrated promising antitumor activity in 80 patients with previously treated HER2-positive BTC. Confirmed objective response rate (cORR) was 41.3% with rapid and durable responses and a manageable safety profile. This phase 3 trial is assessing zanidatamab + SOC therapy vs SOC alone for first-line treatment of HER2-positive advanced/metastatic BTC. **Methods:** This ongoing, global, phase 3, randomized, open-label trial (NCT06282575) is investigating the efficacy and safety of zanidatamab with cisplatin and gemcitabine (CisGem) vs CisGem alone ± a programmed cell death protein-1/ligand 1 (PD-1/L1) inhibitor (pembrolizumab or durvalumab at physician's discretion if locally approved) as first-line treatment for patients with advanced HER2-positive BTC. Eligibility criteria include ≥18 years of age; locally advanced, unresectable, or metastatic HER2-positive BTC by immunohistochemistry and in situ hybridization assay (IHC 3+ or IHC 2+/ISH+); and Eastern Cooperative Oncology Group performance status ≤1. Patients may have received ≤2 cycles of a gemcitabine-based regimen ± pembrolizumab or durvalumab. Prior HER2-targeted therapy is not allowed except for patients who completed it for breast cancer >5 years prior to BTC diagnosis. Eligible patients will be randomized to receive zanidatamab (flat 2-tiered dosing: 1800 mg intravenous [IV; body weight <70 kg] or 2400 mg IV [body weight ≥70 kg] every 3 weeks) + a standard dose of CisGem ± a PD1/L1 inhibitor or CisGem alone ± a PD1/L1 inhibitor (≤8 cycles). The primary endpoint is progression-free survival (PFS) in patients with IHC 3+ tumors. Secondary/exploratory endpoints include overall survival (IHC 3+ subgroup; overall population), PFS (overall population), cORR, incidence and severity of adverse events, and patient-reported outcomes. The study is currently recruiting patients. Clinical trial information: NCT06282575. Research Sponsor: Jazz Pharmaceuticals.

TPS649 Trials in Progress Poster Session

A multicenter, open-label study investigating RP2 oncolytic immunotherapy in combination with second-line systemic atezolizumab plus bevacizumab in patients with locally advanced unresectable or metastatic hepatocellular carcinoma (HCC). First Author: Kevin Kim, University of Maryland School of Medicine, Baltimore, MD

Background: Despite advances in unresectable HCC treatment, long-term survival rates remain poor. The combination of atezolizumab (Atezo) plus bevacizumab (Bev) is approved as frontline therapy for advanced HCC, but only a minority of patients (pts) respond, and secondary resistance usually occurs within months. HCC has an immune-suppressed tumor microenvironment (TME) mediated by the expression of immune checkpoint signals and angiogenesis pathways, which may contribute to therapeutic resistance. RP2 is an enhanced potency oncolytic herpes simplex virus type 1 (HSV-1) that expresses GM-CSF, a fusogenic glycoprotein (GALV-GP-R−), and an anti−CTLA-4 antibody-like molecule. RP2 showed preliminary clinical activity alone or combined with anti−PD-1 in a phase 1 study in pts with advanced solid tumors. The direct oncolytic effect coupled with immune stimulation by RP2 in the TME is intended to provide systemic antitumor activity and synergize with anti−PD-1/PD-L1 agents, such as Atezo. Preclinical data have demonstrated improved distribution of oncolytic HSV within tumors when administered with Bev, supporting the clinical combination of RP2 with Bev. This study will evaluate the safety and efficacy of RP2 combined with Atezo plus Bev as second-line systemic therapy for unresectable advanced HCC (NCT05733598). **Methods:** This is an open-label, single arm, phase 2 trial. Up to 30 pts will be enrolled and receive RP2 in combination with Atezo plus Bev. Key inclusion criteria include advanced unresectable HCC with ≥1 measurable tumor of ≥1 cm in longest diameter, Child-Pugh class A, an Eastern Cooperative Oncology Group performance status of 0 to 1, and progression on 1 prior systemic treatment, which must have included a PD-1/PD-L1−directed agent. Key exclusion criteria include untreated/incompletely treated esophageal and/or gastric varices with bleeding or at high risk for bleeding and macroscopic invasion of the tumor into any major blood vessel(s) and/or main bile ducts. Pts will receive intratumoral RP2 Q2W for 4 doses, then Q3W for up to 4 doses. Bev will be given at 10 mg/kg Q2W starting with the first dose of RP2, then at 15 mg/kg Q3W starting with cycle 4; Atezo will be given at 840 mg Q2W for cycles 2 and 3, then at 1200 mg Q3W starting with cycle 4. Pts will receive treatment until confirmed progressive disease, loss of clinical benefit, or unacceptable toxicity. The primary endpoint is overall response rate (ORR), defined as the proportion of pts achieving a best overall response of complete response or partial response per Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1 as modified for this study. Secondary endpoints include safety, ORR using HCC-modified RECIST, duration of response, complete response rate, and progression-free survival. Clinical trial information: NCT05733598. Research Sponsor: Replimune, Inc.

TPS650 Trials in Progress Poster Session

AMBER part 2F: Cobolimab in combination with dostarlimab in treatment-naïve patients with locally advanced/metastatic and/or unresectable hepatocellular carcinoma (HCC). First Author: Stephen Lam Chan, Department of Clinical Oncology, Chinese University of Hong Kong, Ma Liu Shui, Hong Kong

Background: The introduction of targeted treatment (tx) and immune checkpoint inhibitors (ICIs) has transformed the 1L advanced/metastatic (A/M) HCC tx landscape.[1] Combinations of ICIs with anti-angiogenic agents or other ICIs have demonstrated synergism in patients (pts) with A/M HCC, and shown improved efficacy versus single agents or tyrosine kinase inhibitors.[1,2] Combinations with anti-T-cell immunoglobulin and mucin-domain containing-3 (TIM-3) and anti-programmed cell death protein-1 (PD-1) ICIs have shown promising anti-tumor activity in pts with solid tumors[3,4] including advanced HCC in US pts.[5] AMBER (NCT02817633) part 2F will evaluate the efficacy and safety of anti-TIM-3 cobolimab plus anti-PD-1 dostarlimab in tx-naïve pts with A/M and/or unresectable HCC. **Methods:** AMBER is a two-part, global, open-label, Phase Ib study assessing cobolimab as monotherapy or in combination with other drugs in pts with advanced solid tumors. For part 2F, ~45 pts are planned for enrollment. Key eligibility includes pts aged ≥18 years, with Eastern Cooperative Oncology Group (ECOG) performance score 0–1, histologically confirmed A/M HCC with measurable disease, Child-Pugh Class A liver function, no prior systemic tx, documented hepatitis B and C test at screening, and a pre-tx biopsy sample. Eligible pts will receive cobolimab (300 mg IV) plus dostarlimab (500 mg IV) every 3 weeks (Q3W) for up to 2 years or until progression, toxicity, discontinuation, or death. The primary endpoint is investigator-assessed objective response rate (ORR) per Response Evaluation Criteria in Solid Tumors version 1.1 (RECIST v1.1); secondary endpoints include investigator-assessed disease control rate (DCR), duration of response (DOR) and progression-free survival (PFS) per RECIST v1.1, and overall survival, alpha-fetoprotein response, and safety. Exploratory endpoints include immune-related (ir)-ORR, irDCR, irDOR and irPFS per irRECIST, and biomarker and pharmacokinetic assessments. Disease assessments will consist of CT/MRI evaluations of the chest, abdomen, and pelvis, Q9W from first dose then Q12W after 1 year of tx. Pts will undergo safety follow-up at 30 and 90 (±7) days after last study tx. Efficacy analysis will be based on the safety population (pts who receive ≥1 cobolimab dose) and will include summary statistics and point estimates with 2-sided 95% confidence intervals. Time-to-event analyses will be performed via Kaplan–Meier methods. After ~25 pts have been followed-up for ≥3 scans, an interim analysis may be performed. References: 1. Gordan JD, et al. *J Clin Oncol*. 2024:42:1830–50; 2. Finn RS, et al. *N Engl J Med*. 2020;382:1894–1905; 3. Curigliano G, et al. *Clin Cancer Res*. 2021;27:3620–29; 4. Davar D, et al. *J ImmunoTher Cancer*. 2023:11(Suppl 1):596; 5. Acoba JD, et al. *J Clin Oncol*. 2023:41:580. Funding: GSK (213348). Clinical trial information: NCT02817633. Research Sponsor: GSK; 213348.

TPS651 Trials in Progress Poster Session

A phase II open-label study of sacituzumab govitecan in patients with previously treated locally advanced, recurrent, or metastatic cholangiocarcinoma (SIGNA). First Author: Anup Kasi, University of Kansas Cancer Center, Westwood, KS

Background: Cholangiocarcinoma (CCA) is a highly aggressive cancer with limited treatment options. Trop-2, a transmembrane calcium signal transducer, plays a key role in cellular self-renewal, proliferation, and transformation, thereby promoting tumorigenesis. Its overexpression has been associated with disease progression and shorter survival in several epithelial tumors, including CCA. Sacituzumab govitecan (SG) is an antibody-drug conjugate composed of a humanized anti-trophoblast cell-surface antigen 2 (Trop-2) monoclonal antibody coupled to SN-38, the active metabolite of the topoisomerase inhibitor, irinotecan. Based on the results from the Phase III ASCENT trial, SG was approved by the FDA for treating relapsed or refractory triple-negative breast cancer and hormone receptor-positive breast cancer, regardless of Trop-2 expression. Given Trop-2's expression in CCA, SG is being investigated as a potential treatment option in this setting. **Methods:** This Phase II, non-randomized, open-label, single-arm study is being conducted at the University of Kansas Cancer Center and its affiliated sites. *Eligibility*: Key inclusion criteria include patients with locally advanced, recurrent, or metastatic CCA who have undergone at least one line of prior therapy, have adequate archival tissue for biomarker evaluation or are willing to undergo a biopsy, and have an ECOG performance status of 0-1. Key exclusion criteria include known homozygosity for the UGT1A1*28 allele, which is associated with irinotecan toxicity, and prior treatment with topoisomerase I inhibitors. *Treatment Plan*: SG is administered at 10 mg/kg intravenously on Days 1 and 8 of every 21-day cycle. Imaging will be performed every 6 weeks. Biopsies will be required both prior to treatment and on-treatment (between Days 15-21 of Cycle 1). *Objectives*: Primary objective: To determine the anti-tumor activity of SG as measured by the overall response rate (ORR) according to RECIST 1.1. Secondary objectives: To assess treatment safety in participants who have received at least one dose of SG, and to evaluate anti-tumor activity based on progression-free survival (PFS), disease control rate (DCR), and overall survival (OS). *Statistical Plan*: The sample size (n=22) was calculated using a modified Simon two-stage design, hypothesizing an improvement in ORR to 25%, compared to a baseline value of 8%. In the first stage, 14 patients will be enrolled. If ≤1 participant demonstrates an ORR, the study will be stopped for futility. If ≥2 participants show ORR, an additional 8 patients will be enrolled (total n=22). The study will be deemed successful if ≥4 participants achieve an ORR. This design provides 80.9% power with an 8.5% type I error rate. Stopping rules include halting the trial if the grade 4-5 treatment-related toxicity rate reaches 10%. Enrollment is ongoing. Clinical trial information: NCT06178588. Research Sponsor: Gilead.

652 **Oral Abstract Session**

A phase III study of combination therapy with everolimus plus lanreotide versus everolimus monotherapy for unresectable or recurrent gastroenteropancreatic neuroendocrine tumor (JCOG1901, STARTER-NET). First Author: Susumu Hijioka, Department of Hepatobiliary and Pancreatic Oncology, National Cancer Center, Chuo-Ku, Japan

Background: There is limited evidence regarding the benefit of adding somatostatin analogs to molecular targeted agents for well-differentiated gastroenteropancreatic neuroendocrine tumors (GEP-NETs). This phase III trial was conducted to compare everolimus plus lanreotide (EVE/LAN) with everolimus monotherapy (EVE) in patients with unresectable or recurrent GEP-NETs in the first-line setting. **Methods:** Patients with grade 1 or grade 2, nonfunctioning GEP-NETs with poor prognostic factors (Ki-67 labeling index (LI) 5-20% or diffuse liver metastases) were randomly assigned (1:1) to EVE (10 mg/day) or EVE/LAN (EVE + LAN 120 mg every 28 days). The primary endpoint was progression-free survival (PFS). The key secondary endpoint was overall survival (OS), and other secondary endpoints were objective response rate (ORR), disease control rate (DCR), and safety. The study was based on the hypothesis of an assumed median PFS of 11.0 months for EVE, expecting a 4-month improvement by EVE/LAN (HR: 0.73). The planned sample size was 250, requiring 195 events overall with a one-sided alpha level of 5%, a power of 70%, an accrual period of 5 years, and a follow-up period of 1.5 years. **Results:** Between April 2020 and June 2024, a total of 178 patients were enrolled, and the planned interim analysis was conducted in 145 patients (72 in the EVE and 73 in the EVE/LAN) in June 2024 with a data cut-off date of Nov 2023. The median PFS was 11.5 months in the EVE arm and 29.7 months in the EVE/LAN arm (HR 0.38 [99.91% CI 0.15–0.96], P = 0.00017 $<$ the prespecified significance level of 0.00046, by the stratified log-rank test). The HR for OS was 0.97 (95% CI: 0.24-3.90). The ORR and DCR were 8.7% (6/69) and 87.0% (60/69) in the EVE arm and 26.8% (19/71) and 91.5% (65/71) in the EVE/LAN arm, respectively. Both hematologic and non-hematologic toxicities tended to be more frequent in the EVE/LAN arm than in the EVE arm. No treatment-related deaths were observed in either arm. Based on the efficacy results, the Data and Safety Monitoring Committee recommended early termination of the study. **Conclusions:** The EVE/LAN provides statistically significant prolongation of PFS compared with EVE monotherapy, and the safety profile of EVE/LAN was manageable. The EVE/LAN might be a new standard treatment in the first-line setting for well-differentiated grade 1/2 GEP-NETs with poor prognostic factors. Clinical trial information: jRCT1031200023. Research Sponsor: National Cancer Center Research and Development Fund; 29-A-3, 2020-J-3, 2023-J-03; Japan Agency for Medical Research and Development (AMED); 22ck0106766h0001, 23ck0106766h0002, 24ck0106766h0003.

653 **Poster Session**

Survival disparities in small intestine carcinoid tumors: A SEER-based analysis. First Author: Adit Dharia, HCA Healthcare/USF Morsani College of Medicine Oak Hill Hospital, Brooksville, FL

Background: Small intestine carcinoid tumors (SI-carcinoids) are rare neuroendocrine neoplasms with variable survival outcomes. This study examines the impact of age, sex, race, income, and geographic location on overall survival (OS) in patients with SI-carcinoid tumors. **Methods:** A retrospective cohort study was conducted using the Surveillance, Epidemiology, and End Results (SEER) registry 22 to identify patients aged 20 and above with SI-carcinoid tumors from 2000 to 2021. Kaplan-Meier survival analysis and log-rank tests were used to assess differences in survival. Cox proportional hazards models estimated hazard ratios (HR) for Overall Survival (OS). **Results:** 20,777 patients were included, with 7,029 deaths recorded. Median OS varied significantly by age, sex, race, income, and tumor site. Patients aged 60-85 had a median OS of 10 months (95% CI: 9.75-10.42), while those over 85 had 2.92 months (95% CI: 2.67-3.33). Females had better OS (13.92 months; 95% CI: 13.42-14.42) than males (12.83 months; 95% CI: 12.42-13.25). Non-Hispanic Black patients had worse OS (12.33 months; 95% CI: 11.50-13.33) compared to Non-Hispanic Whites (13.25 months; 95% CI: 12.92-13.67). Patients earning less than $50,000 had worse OS (11.42 months; 95% CI: 10.67-12.42) than those earning $100,000–$120,000 (15.42 months; 95% CI: 14.25-17.83). Geographic disparities were noted, with nonmetropolitan areas showing worse OS (11.42 months; 95% CI: 10.83-12.92) compared to large metropolitan areas (14.00 months; 95% CI: 13.50-14.42). Cox models confirmed age (HR for 85+ years: 21.99; 95% CI: 17.40-27.81), male sex (HR: 1.12; 95% CI: 1.07-1.17), and Black race (HR: 1.29; 95% CI: 1.18-1.42) were linked to worse OS. Tumor site also affected OS, with ileal tumors showing better survival compared to duodenal tumors (HR: 0.89; 95% CI: 0.84-0.94). **Conclusions:** Age, sex, race, income, and geographic location significantly affect OS in patients with small intestine carcinoid tumors, highlighting the need for tailored interventions for vulnerable groups. Research Sponsor: None.

654 **Poster Session**

Effect of marital status and race on survival of patients with well differentiated neuroendocrine tumors (NETs). First Author: Amy D. Stringer, Lewis Katz School of Medicine, Temple University, Philadelphia, PA

Background: Socioeconomic status factors (SES) such as marital status, income, and race can influence and enhance disparities in cancer incidence and outcomes. Marriage has documented positive impacts on overall survival of numerous cancer types. Consequences of divorce have been less explored. The unique indolent, secretory nature of NETs may be differentially affected by SES factors. This study identifies implications of marital status and sex on 5-year survival rates of different racial groups diagnosed with NETs within the Surveillance, Epidemiology, and End Results (SEER) database between 2000-2020 while controlling broadly for the type of treatment received. **Methods:** Within the SEER data base patients with reported sex, race, marital status and treatment who were diagnosed with well differentiated NETs were selected excluding carcinomas. Relative and disease specific 5-year survival analyses were calculated with 95% confidence intervals and age standardized. Three groupings were analyzed single, separated/divorced/widowed and domestic-partner/married. **Results:** For all races and sexes, relative survival for individuals in a union was higher than those who were single, or separated, divorced/widowed [86.9% (CI 86.1% - 87.7%); 79.3% (77% to 81.3%); and 79.2% (78% to 80.3%) respectively]. When individuals of each marital status group were compared by matching sex and race, there were no significant differences in relative survival. But males of any race who left a union had lower chances of survival than corresponding females [72.4% (69.7% to 74.9%) vs 81.6% (80.3% to 82.8%)]. As expected, all groupings who received surgical treatment had a higher relative survival than those who did not. Those who received surgical treatment while in a partnership had a higher relative survival then those who were separated [94.5% (93.6% to 95.3%) vs 88.7% (87.3% to 89.9%)]. All marital groupings who underwent surgery or did not, had comparable relative survival between races; except, black individuals in a partnership had improved outcomes compared to the white population. More specifically, when accounting for race, sex, marital status and treatment received, black men who did not undergo surgery had a higher relative survival compared to white men [69.9% (62.2% to 76.3%) vs 59.4% (56.6% to 62.1%)]. All the same conclusions were statistically significant for disease specific survival. **Conclusions:** Due to NETs slow growing nature SES factors play a major role in the care of patients living with them. Our findings demonstrate generally, irrespective of race, sex, or surgical treatment, being involved in a union is advantageous, showing a protective effect on patient survival, while parting from a union could be deleterious as it is tied to less favorable outcomes. These findings underscore the importance of considering these factors in treatment and care strategies. Research Sponsor: National Cancer Institute/U.S. National Institutes of Health; P30CA006927.

655 **Poster Session**

Automated radiomics model for preoperative pancreatic neuroendocrine tumor grade prediction. First Author: Pratik Chandra, Department of Surgery, Memorial Sloan Kettering Cancer Center, New York, NY

Background: Pancreatic neuroendocrine tumors (PNETs) have varying biologic behavior based on factors including tumor grade. Many prognostic factors can only be determined post-pancreatectomy; thus, are unable to guide clinical decision making preoperatively. This is particularly relevant in small, non-functional (NF)-PNETs. Imaging analysis with radiomics may be able to elucidate biologic data such as tumor grade, without the need for tissue sampling. Our study aims to create an automatic pipeline from segmentation to radiomics-signature building to preoperatively predict tumor grade. **Methods:** Patients that underwent resection between 2003-21 with adequate preoperative arterial phase CT scans were divided into training and validation cohorts. In the training cohort, the pancreas and tumor region were manually segmented and used to train an auto-segmentation model. The validation cohort then underwent automatic segmentation. A total of 255 radiomics features were extracted from the tumor region. Correlated features were removed, and the minimum redundancy maximum relevance (mRMR) method was used to select the final feature set. These features were used to train a Support Vector Machine (SVM)-based classifier through the training cohort to predict grade, which was dichotomized into grade I versus II/III. The radiomic based prediction model was then evaluated in the automatically segmented validation cohort. **Results:** A total of 186 patients were identified and divided into training (n = 140) and validation (n = 46) cohorts. Median age was 57 (27, 85) and 52% were male. 113 (62%) and 70 (38%) patients were grades I and II/III, respectively. Median tumor size was 24 mm (6, 200) and 43 (26%) patients had positive nodal disease. The auto-segmentation model was able to accurately segment the tumor region in 33 (72%) patients of the validation cohort. In the training cohort (n = 140), the radiomics-signature produced an area under the curve (AUC) of 0.87 (0.81, 0.93). Sensitivity, specificity, positive predictive value (PPV) and negative predictive value (NPV) were 0.94 (0.9, 0.98), 0.74 (0.66, 0.81), 0.72 (0.65, 0.80) and 0.97 (0.94, 1.00) respectively. In the auto-segmented validation cohort (n = 33), the radiomics model produced an AUC of 0.75 (0.61, 0.90). Sensitivity, specificity, PPV and NPV were 0.88 (0.77, 0.99), 0.62 (0.46, 0.79), 0.71 (0.56, 0.87) and 0.83 (0.71, 0.96), respectively. **Conclusions:** The proposed auto-segmentation model was able to automatically identify tumor regions with a high degree of accuracy. Similarly, the subsequent radiomics-signature was also able to strongly discern PNET grade. This automatic pipeline bypasses manual segmentation and could help incorporate a radiomics-signature into preoperative clinical decision-making for PNETs. Research Sponsor: None.

656 Poster Session

Clinicopathological factors related to survival in gastroenteropancreatic neuroendocrine tumors (GEP-NET): 20 years experience in the American British Cowdray Medical Center. First Author: Susy Wuotto Alvarado, The American British Cowdray Medical Center, Mexico, DF, Mexico

Background: Neuroendocrine neoplasms (NEN) are a diverse group of tumors originating from neuroendocrine cells found throughout the body. Their "rarity", heterogeneity, and non-specific presentation contribute to limited reliable epidemiological data, especially in developing countries such as Mexico. This study aims to analyze the clinical and pathological features of gastroenteropancreatic neuroendocrine tumors (GEP-NET) and their association with survival in a third-level center in Mexico. **Methods:** A retrospective cohort study at the American British Cowdray Medical Center, in Mexico City, was conducted. We included all histopathological confirmed GEP-NET from 2004 to 2024. Clinical, biochemical, and radiological features at diagnosis were analyzed. Survival was estimated using Kaplan-Meier and Log-Rank test. **Results:** We identified 110 NENs, of which 89% were GEP-NET, 56.1% were female, and the median age was 57.7 (SD 13.8) years. The clinical stage was metastatic (50%), localized (37.7%), and locally advanced (6.1%). The primary sites were the pancreas (49%), small bowel (23.5%), gastric (10.2%), colon (7.1%), rectum (5.1%), appendix (3.1%), and unknown (2%). ECOG performance status were 0-1 in 96.9%. In the metastatic setting, liver involvement was 96.2%, mainly high hepatic burden (59.6%). Symptoms were abdominal pain (34.6%), diarrhea (16.3%), and bowel obstruction (6.1%). Functioning tumors were 26.8%, and carcinoid syndrome was found in 9.2%. Grades 1, 2, and 3 were 62.9%, 34%, and 3.1% respectively. The median of Ki-67 was 6.4%. At diagnosis, Ga-68 PET/CT scan, serum CgA, and urinary 5-HIIA were performed in 98.2%, 39.7%, and 16.3% of patients. Among metastatic patients, 87.1% received at least one line of systemic treatment, primarily with somatostatin analogs (62.9%). In all population, the median follow-up was 42.5 months and the median overall survival (OS) was 164.9 (SD 12.2) months. Univariate analyses showed that primary site (p=0.02), Ki-67 $<$5% (p=0.02), CgA $<$700 ng/ml (p=0.03), and low hepatic burden (p=0.01) were statistically significantly associated with OS. **Conclusions:** This study provides critical insights into GEP-NET characteristics in a Mexican population, highlighting clinical, biochemical, and radiological factors associated with better survival. Research Sponsor: None.

657 Poster Session

Efficacy and safety of surufatinib in unresectable or metastatic grade 3 pancreatic neuroendocrine tumors (NET G3): Real-world analysis. First Author: Jian Wang, Qilu Hospital, Shandong University, Jinan, China

Background: Currently, there is limited exploration regarding the treatment of NET G3, resulting in restricted treatment options. Therapeutics for NET G1/2 may serve as an alternative. Surufatinib, with its potent targeting of VEGFRs, FGFR1, and CSF-1R, not only exerts inhibitory effects on tumor angiogenesis but also modulates the immune microenvironment. Due to its demonstrated efficacy and manageable safety profile, it has secured approval for the treatment of G1/2 NETs. **Methods:** This is a multi-center real-world study that retrospectively analyzes the clinical data of pts with NET G3 who received surufatinib-based treatment from December 2021 to February 2024, aiming to assess its efficacy and safety. Eligible pts were unresectable or metastatic p-NET with ki-67$>$20%. The primary endpoint was PFS, with secondary endpoints were ORR, DCR, and safety. **Results:** As of September 2024, data from 37 pts were analyzed. The median age was 54 years (range: 32-80), with 59.5% male. The median Ki-67 index was 30% (range: 25-70%), and 94.6% of the cases were non-functional p-NETs, with 89.2% having liver metastases and 35.1% having bone metastases. In terms of prior treatments, 18.9%, 40.5%, and 40.5% of pts had received 0, 1, and $\geq$2 systemic therapies, respectively. Among them, 81.1% had previously undergone chemotherapy, 35.1% had received interventional therapy of liver metastases, and 13.5% with everolimus or anti-angiogenic agents. 48.6% pts had surufatinib-based combination treatment. The median PFS was 9.30 months (mo, 95% CI: 8.02–13.02), with an ORR of 21.6% and a DCR of 81.1%. First-line (1L) treatment showed a higher ORR (42.9% vs. 16.7%) and a trend of better PFS (9.67 vs. 9.04 mo, p=0.7) compared to $\geq$2L treatment. Combination therapy improved ORR (33.3% vs. 10.5%) with a similar PFS (9.30 vs. 9.44 mo, p=0.93) compared to surufatinib monotherapy. Liver metastases had limited impact on PFS (9.93 vs 9.30 mo, p=0.55). Common grade $\geq$3 adverse events included hypertension (8.1%), proteinuria (5.4%), and elevated bilirubin (5.4%). **Conclusions:** Surufatinib has demonstrated initial efficacy and manageable safety in pts with NET G3, providing a new treatment option for these pts. First-line therapy may be a favorable factor for improving PFS. To delve deeper into these findings, additional subgroup analyses, with a particular emphasis on pts with ep-NET G3 will be conducted in the future. Research Sponsor: None.

658 Poster Session

Patient-centered adverse event reporting in clinical trials: A novel approach using ECOG-ACRIN E2211. First Author: Wan Ying Tan, University of Connecticut School of Medicine, Farmington, CT

Background: The Common Terminology Criteria for Adverse Events (CTCAE) classifies adverse events (AEs) in clinical trials but traditionally limits the analysis to a single occurrence per patient and grade 3-4 AEs, overlooking frequent and lower grade AEs. We propose a new approach of AE reporting that captures both the safety profile and patient experience. **Methods:** We analyzed related AEs collected in the ECOG-ACRIN E2211 [Phase II capecitabine + temozolomide (Arm A) vs. temozolomide (Arm B) in advanced pancreatic neuroendocrine neoplasms] by CTCAE v4.0 and limited to hematologic and gastrointestinal (GI) categories for this exploratory analysis. We calculated AE burden, defined as the average occurrence of AE per patient experiencing the AE, measured by total occurrence of AEs, and divided by number of patients who developed the AE. We then compared burden of grade 3-4 AEs vs. grades 1-4 AEs. Sex differences were evaluated using t-tests. **Results:** For hematologic AEs, 72 patients reported 426 occurrences across grades 1-4 and 24 patients reported 39 occurrences for grades 3-4, resulting in AE burdens of 5.9 (grades 1-4) versus 1.6 (grades 3-4). For GI AEs, 108 patients reported 997 grade 1-4 events, and 13 patients reported 22 grade 3-4 events, resulting in AE burdens of 9.23 versus 1.69, respectively. Sex differences in AE burden are tabulated below (Table). For example, in Arm B, anemia burden was 4.8 (males) vs 2.7 (females) [p=0.068] for grades 1-4 and 0 (males) vs 1 (females) for grades 3 and 4. **Conclusions:** This AE reporting approach, accounting for the burden across grade 1-4, offers a different and more patient-centered assessment of AEs in clinical trials and may help better identify sex-based differences. Research Sponsor: None.

Burden of AEs by treatment arm and sex distribution.

Categories of AEs	Burden of Related AEs											
	Arm A (N=65)						Arm B (N=68)					
	Grade 1-Grade 4			Grade 3-Grade 4			Grade 1-Grade 4			Grade 3-Grade 4		
	Male	Female	p-value	Male	Female	p-value	Male	Female	p-value	Male	Female	p-value
Hematologic												
Anemia	2.9	3	0.937	0	0		4.8	2.7	0.068	0	1*	
Neutropenia	1.7	2.2	0.686	1	1*		1.2	2.3	0.064	1	1.1	0.356
Thrombocytopenia	4.3	2.6	0.241	1*	1.2		3.6	3.3	0.781	1	1	**
Gastrointestinal												
Abdominal pain	2.5	2.2	0.835	0	0		3.1	3	0.955	0	1*	
Constipation	4.1	3.2	0.515	1*	0		4.9	3.3	0.169	0	0	
Diarrhea	1.8	3.6	0.158	0	0		2.6	2.5	0.940	1.25	1	0.390
Nausea	4.2	4.9	0.619	0	0		4.4	4.2	0.893	1	1**	**
Vomiting	2.2	1.9	0.703	0	0		1.8	2.8	0.241	1	1	**

*Only 1 patient in each group **No variance within each group.

659 Poster Session

Phase II study of mFOLFIRINOX efficacy as first and subsequent lines of advanced gastrointestinal neuroendocrine tumors G3 and neuroendocrine carcinomas therapy. First Author: Zhulikov Yarik, National Medical Research Center of Oncology named after N.N. Blokhin, Moscow, Russian Federation

Background: Gastrointestinal (GI) neuroendocrine carcinomas (NEC) are rare tumors and account for 3% of all NECs. Despite the differences in genetic characteristics, treatment principles of GI NECs are extrapolated from lung NECs. The standard first-line therapy is platinum and etoposide-containing regimens, which allow to achieve a median PFS of 5-6 months. A small retrospective study demonstrated the potential effectiveness of the mFOLFIRINOX regimen with a median PFS of 5.4 months, despite that the majority of patients (pts) received this regimen as second or subsequent lines. The aim of this study is to evaluate mFOLFIRINOX +/- somatostatin analogues (SA) efficacy in GI NEC subgroup. **Methods:** This prospective, single center phase II study used a two-stage Simon design. The primary endpoint is disease control rate (DCR) $\geq$ 6 months. Statistical hypothesis: investigated therapy improves DCR compared with historical control from 50 to 70%. Secondary endpoints are progression-free survival (PFS), overall survival (OS), objective response. At the first stage, enrollment of 16 pts was planned (α=0.05, power 80%). If a disease control $\geq$ 6 months was achieved in at least 9 of the 16 pts, it was planned to continue enrollment up to 39 pts. Here we present results of the first stage of this study. Inclusion criteria: pts $\geq$18 y.o. with histologically confirmed advanced GI NEC or neuroendocrine tumor (NET) G3 ki67$\geq$55%; ECOG 0-2. Recruitment of pts was carried from 2019 to February 2024. **Results:** The study included 16 pts, 12 male and 4 female. The most common sites of primary tumor are stomach (N=8, 50%) and pancreas (N=5, 31.3%). 10 (62.5%) pts had large cell carcinoma, 1 – small cell carcinoma (6.3%), 5 (31.3%) – NET G3. The median ki-67 was 70% (40-95%). 15 pts (93.7%) had IV stage, 1 (6.3%) – III. The most common site of metastases is liver – 11 (68.8%), more than half of pts have isolated metastases in liver (N=9, 56.3%). ECOG status was evaluated as 0-1 in 15 (93.7%) cases. Majority of pts (N=14, 87.5%) received mFOLFIRINOX as first line therapy. 7 pts (46.7%) had positive expression of SSTR-2A or -5 and received concurrent treatment with SA. ORR was 62.5% (N=10/16), stable disease – 37.5% (N=6/16). DCR $\geq$ 6 months was 93.7% (N=15). With a median follow-up of 13.2 months, median PFS was 10.8 months (95% CI 7.57-14.1). One pts had serious adverse event - myocardial infarction, no grade 5 toxicity was observed. **Conclusions:** Chemotherapy with mFOLFIRINOX showed promising results in GI NECs or NETs G3 ki67$\geq$55% treatment. Primary endpoint was met, enrollment of patients in the study continues. Research Sponsor: None.

660 **Poster Session**

Multicenter, retrospective, observational study of outcomes of treatment for unresectable neuroendocrine tumors G3 of gastroenteropancreatic or unknown primary origin. First Author: Taro Shibuki, Department of Hepatobiliary and Pancreatic Oncology, National Cancer Center Hospital East, Kashiwa, Japan

Background: The 2017 edition of the WHO classification categorized neuroendocrine neoplasms (NEN) as neuroendocrine tumors (NET) G1, G2, G3, and neuroendocrine carcinoma (NEC). Prior to 2017, NET G3 was categorized as NEC in the WHO 2010 classification. Platinum-based chemotherapy, commonly used as first-line treatment for these tumors, shows suboptimal efficacy for NET G3. Additionally, large-scale studies and robust evidence on chemotherapy outcomes for NET G3 are lacking. We investigated chemotherapy regimens for NET G3 and evaluated their outcomes. **Methods:** We included 73 patients (pts) who were clinically and pathologically diagnosed with unresectable NET G3 of gastroenteropancreatic or unknown primary origin and who started chemotherapy between Jan 2011 and Dec 2019. Clinical data were collected retrospectively and analyzed for regimen selected, treatment efficacy, and prognostic factors. A central pathological review was conducted if tissue samples were available. **Results:** The median age of pts was 65 years (range 27-85 years), with 45 being male. Among the pts, 49 had pancreatic NETs, 10 had gastrointestinal NETs, and 14 had NETs of unknown primary origin. The median Ki-67 index was 30% (range 20-70%). Twenty-seven patients (37%) had received prior treatment, such as surgery or local therapy. A central pathological diagnosis was confirmed NET G3 in 42 out of 44 patients (95%). Primary treatment regimens were NET-based (somatostatin analogues, everolimus, sunitinib, streptozocin-based chemotherapy, and capecitabine and temozolomide [CAPTEM]) and NEC-based (platinum-based chemotherapy) in 54% and 44% of patients, respectively. NET-based regimens were selected in 36% and 78% of pts, respectively, before and after publication of the WHO 2017 classification. The overall response rate (ORR) was 16% for NEC-based regimens and 19% for NET-based regimens. However, when focusing on NET-based chemotherapy (streptozocin-based and CAPTEM), the ORR was 54%. Median progression-free survival (PFS) was 118 days (95% confidence interval [CI]: 60-337 days) for NET-based regimens and 229 days (95%CI: 133-426 days) for NEC-based regimens. Median overall survival (OS) was 841 days for NET-based regimens (95%CI: 480-989 days) and 658 days (95%CI: 455-1144 days) for NEC-based regimens. There were no significant differences in PFS or OS between the regimens. In multivariable analysis, hepatic tumor volume $>$25% was a negative predictor of PFS, and NSE $\geq$16.3 ng/mL was a negative predictor of OS. **Conclusions:** Since the introduction of NET G3 in the WHO classification, NET-based regimens have become the preferred treatment choice. Although there were no significant differences in PFS or OS between the regimens, the NEC-based regimens was limited. Pts with high hepatic tumor volume or elevated NSE levels had a worse prognosis. Research Sponsor: None.

661 **Poster Session**

Phase 1b portion of the ACTION-1 phase 1b/3 trial of RYZ101 in gastroenteropancreatic neuroendocrine tumors (GEP-NET) progressing after ^{177}Lu somatostatin analogue (SSA) therapy: Safety and efficacy findings. First Author: Jonathan R. Strosberg, Moffitt Cancer Center and Research Institute, Tampa, FL

Background: RYZ101 (^{225}Ac-DOTATATE) is an alpha-emitting radiopharmaceutical in development for SSTR2+ solid tumors. Alpha-particles have a shorter path length/higher linear energy transfer than beta-particles, causing more frequent double-strand DNA breaks and potentially improving the therapeutic index. ACTION-1 (NCT05477576) is a two-part, global, randomized, controlled, open-label, phase 1b/3 trial of RYZ101 in advanced, well-differentiated SSTR+ GEP-NETs progressing after ^{177}Lu-SSA therapy. Herein, we report updated results from the phase 1b portion of the trial. **Methods:** The phase 1b portion of the ACTION-1 trial had a dose de-escalation/Bayesian optimal interval design with boundaries based on a dose-limiting toxicity (DLT) rate of 25%. Patients received RYZ101 IV every 8 weeks for 4 cycles. Planned dose levels (n=6/level): Level 0 (starting dose) 120 kBq/kg; Level 1 90 kBq/kg; Level 2 60 kBq/kg. DLTs were assessed for 56 days after the first RYZ101 dose. Treatment-emergent adverse events (TEAEs) were graded by NCI-CTCAE v5.0. A Data Review Committee oversaw dose de-escalation decisions/safety data. Tumor response was assessed locally by RECIST v1.1. **Results:** Seventeen patients received at least one dose of RYZ101 at 120 kBq/kg (4 doses: 15 patients; 2 doses: 2 patients; median 8.3 MBq). Baseline patient characteristics: median age 63 years; male (n=11); ECOG PS 0/1 (n=10/7); primary tumor site GI/pancreas (n=12/5). As of 14 December 2023, the most frequent TEAEs were anemia (58.8%), nausea (58.8%), and fatigue (52.9%). Serious adverse events (SAEs) were observed in 6 patients (35.3%, none were treatment-related); grade $\geq$3 TEAEs occurred in 9 patients (5 were treatment-related, 29.4%). No TEAEs led to treatment discontinuation. Four patients had TEAEs leading to dose modification, dose hold, and/or dose delays. There was one grade 5 TEAE of hepatic failure that was deemed unrelated to the study drug. The confirmed objective response rate was 29.4% (n=5; 1 complete response, 4 partial responses). One additional patient had an unconfirmed partial response at the time of data cutoff, which was subsequently confirmed. Seven patients (41.2%) had stable disease and 3 (17.6%) had progressive disease. The median duration of response was not estimable (95% CI 9.26 months, not estimable), and the median progression-free survival was not estimable (95% CI 12.16 months, not estimable). **Conclusions:** RYZ101 was well tolerated, and a fixed dose of 10.2 MBq was declared the recommended phase 3 dose. Initial data suggest promising efficacy and a manageable safety profile. Part 2 (phase 3) of the ACTION-1 trial is enrolling and will compare RYZ101 at 10.2 MBq every 8 weeks for 4 cycles with standard of care in patients with advanced SSTR2+ GEP-NETs progressing after ^{177}Lu-labeled SSAs. Clinical trial information: NCT05477576. Research Sponsor: RayzeBio.

662 **Poster Session**

Efficacy of immune checkpoint inhibitors in patients with advanced pancreatic NETs displaying high TMB and MMR alterations following treatment with alkylating agents. First Author: Jonathan R. Strosberg, H. Lee Moffitt Cancer Center and Research Institute, Tampa, FL

Background: Alkylating-based chemotherapy (ALK) is one of the main treatments used in patients with advanced pancreatic neuroendocrine tumors (PanNETs). Its use has been associated with an increased risk of grade progression and acquisition of a hypermutator phenotype, associated with frequent alterations of the mismatch repair (MMR) genes. This suggests a potential benefit of immune checkpoint inhibitors (ICI) that are otherwise ineffective in ALK-naïve PanNETs, which are mostly hypomutated. We aimed to describe the efficacy of ICI in patients with advanced PanNET pretreated with ALK. **Methods:** We assembled an international retrospective cohort of patients with advanced PanNET who received at least one cycle of ICI, and who had previously received at least 3 months of ALK. Patients with poorly-differentiated neuroendocrine carcinoma were excluded. The primary endpoint was progression-free survival (PFS) and the secondary endpoint was the objective response rate (ORR). The impact of tumor mutational burden (TMB, high being defined by $>$10 mut/Mb) and MMR alterations (mutations or IHC) was explored. **Results:** We included 39 patients (median age 58.7, male sex 62%). All PanNETs were progressive at baseline and presented a median of 3 (IQR, 2-4) metastatic sites. Tumors were predominantly classified as G2 (47%) or G3 (42%), with a median Ki-67 of 18.5% (IQR, 12-35). 23/27 patients with tumor tissue or ctDNA NGS testing had high TMB (median, 35 mut/Mb; IQR, 24-106) and 12/22 had MMR alterations. Patients had previously received an average of 14 cycles (IQR, 7-25) of ALK (temozolomide 78%, streptozotocin 18%, dacarbazine 4%), which had yielded objective response in 72% of cases. In addition, patients had received a median of 4 (IQR, 3-5) other treatments (including platinum drugs in 80% and PRRT in 51%). ICI mainly consisted in an anti-PDL1/PD1 + anti-CTLA4 combination (85%) or a monotherapy (15%), with a median of 3 (IQR, 2-8) cycles. The ORR was 18% and the disease control rate was 46%. Median PFS was 2.8 months (95% CI, 0.62-4.96) and 6-month PFS rate was 31%. 19% of patients had grade 3-4 toxicity. Patients with high TMB had higher ORR (30% vs. 0%, p=0.03) and longer median PFS (4.9 vs. 2.1 months, p=0.004) compared with patients with low/unknown TMB. Patients with altered MMR had higher ORR (42% vs. 7%, p=0.02) and longer median PFS (8.9 vs. 2.5 months, p=0.014) compared with patients with no/unknown MMR alterations. **Conclusions:** While ICI shows limited efficacy in unselected patients with advanced PanNETs pretreated with ALK, it may be an effective treatment option in case of high TMB and MMR alterations. Research Sponsor: None.

663 **Poster Session**

A systematic literature review of studies assessing first-line (1L) treatments in patients with advanced, grade 2/3 (G2/G3) gastroenteropancreatic neuroendocrine tumors (GEP-NETs). First Author: Jaume Capdevila, Medical Oncology Department, Vall Hebron University Hospital, Vall Hebron Institute of Oncology (VHIO), Barcelona, Spain

Background: There is no universally accepted standard of care for newly diagnosed patients with G2/G3 advanced GEP-NETs. The phase 3 NETTER-2 trial showed that [^{177}Lu]Lu-DOTA-TATE plus long-acting octreotide demonstrated a significant improvement in progression-free survival (PFS) vs. high-dose long-acting octreotide alone in this patient population. We conducted a comprehensive review of the published literature to summarize the available evidence on other 1L treatments assessed in advanced, well-differentiated, G2/G3 GEP-NETs. **Methods:** Embase, MEDLINE, and CENTRAL were systematically searched (until Jan 2024) to identify relevant clinical trials and observational studies assessing 1L systemic therapies among adults with advanced, well-differentiated, G2/G3 GEP-NETs. Key conferences from the past three years were also reviewed. Comparators of interest included somatostatin analogues (SSAs; octreotide and lanreotide), targeted therapies (sunitinib and everolimus), and chemotherapies (CAPTEM, STZ+5-FU, FOLFIRI, and platinum-based therapies). Studies with mixed patient populations were also included if $\geq$80% of the study participants matched the eligibility criteria. The 2022 WHO classification system was applied in studies with available Ki-67 index data but lacking grade information. **Results:** A total of 31 studies met the review eligibility criteria (three randomized controlled trials, one open-label extension study, three single-arm trials, and 24 observational studies). None of the included studies entirely matched the NETTER-2 population but provided data only for specific sub-populations. The included studies were categorized either by overall GEP-NET (n=16) or pancreatic-NET only population (n=15) with no study exclusively assessing gastrointestinal-NET patients. Majority of the comparator studies assessed chemotherapies (n=14), followed by SSAs (n=10), targeted therapies (n=3), a combination of targeted therapy and chemotherapy (n=1), and mixed treatments (n=2). Heterogeneity was also observed in terms of patient characteristics (data reported only in 12 studies), sample size ($<$50 patients in 20 studies), and disease grade (G2/G3: nine studies, G2 only: 12 studies, and G3 only: 10 studies). Grading was derived using Ki-67 index in two studies. Majority of the studies reported PFS data (n=22), with nine studies providing data for both PFS and overall survival (OS); Kaplan-Meier curves for PFS and OS were reported for only 13 and four studies, respectively. **Conclusions:** This systematic literature review highlights a substantial evidence gap and an unmet need in 1L treatment of G2/G3 GEP-NETs. Robust quantitative comparison of NETTER-2 data with published evidence on other 1L treatments was largely restricted by low sample sizes and the aforementioned heterogeneity. Research Sponsor: Novartis Pharma AG, Basel, Switzerland.

664 **Poster Session**

Matching adjusted indirect comparison (MAIC) of ^{177}Lu-DOTATATE vs. sunitinib as first-line (1L) treatment for advanced grade 2 (G2) pancreatic neuroendocrine tumors (pNETs). First Author: Raj Srirajaskanthan, King's College Hospital NHS Foundation Trust, London, United Kingdom

Background: The phase 3 NETTER-2 trial recently demonstrated a statistically significant improvement in progression-free survival (PFS) with [^{177}Lu]Lu-DOTA-TATE (^{177}Lu-DOTATATE) plus long-acting octreotide (Oct LAR) vs. high-dose Oct LAR in patients with G2/G3 gastroenteropancreatic neuroendocrine tumors in the 1L setting, across tumor grades and origins (including pNETs). However, no head-to-head comparison of ^{177}Lu-DOTATATE to other treatment options was conducted. Sunitinib is a currently recommended 1L treatment for patients with advanced G2 pNETs. The Raymond 2018 study is the only study that assessed sunitinib in advanced pNETs and provided sub-group data for the 1L setting (80% of the patients had G2 tumors). We aimed to determine the relative efficacy (PFS) of ^{177}Lu-DOTATATE + Oct LAR vs. sunitinib as 1L treatment for advanced G2 pNETs using MAIC. **Methods:** An unanchored MAIC (due to absence of a common comparator) of PFS in patients with advanced G2 pNETs treated in 1L with ^{177}Lu-DOTATATE + Oct LAR vs. sunitinib was conducted using individual patient data from NETTER-2 and aggregated data published in the Raymond 2018 study. Baseline characteristics and PFS data for the sunitinib arm were available for the treatment-naïve subgroup and were considered for the analysis. Baseline characteristics considered in the MAIC included age, sex, race, prior somatostatin analogue use, and number as well as location of involved disease sites. Multiple scenarios with various combinations of covariates were considered with multivariate Cox-regression of NETTER-2 used to identify covariates predictive of PFS. Finally, the combination of covariates with a reasonable effective sample size (ESS) was selected. **Results:** The final ESS was estimated to be 22 for ^{177}Lu-DOTATATE + Oct LAR, representing 43% of the original sample size (n=50), whereas sample size for sunitinib arm was 61 patients. After matching and weighting, patient characteristics were well balanced. MAIC analysis showed a statistically significant PFS benefit with 1L ^{177}Lu-DOTATATE + Oct LAR vs. sunitinib in G2 pNET (PFS hazard ratio: 0.23 [95% CI: 0.12–0.45]). Unadjusted comparison showed similar results (PFS hazard ratio: 0.34 [95% CI: 0.19–0.61]). Results were consistent across various scenarios assessed. **Conclusions:** Using MAIC to adjust for cross trial differences, PFS comparison favored ^{177}Lu-DOTATATE + Oct LAR over sunitinib as 1L treatment of patients with G2 pNET. Research Sponsor: Novartis Pharma AG, Basel, Switzerland.

665 **Poster Session**

Adjuvant therapy with somatostatin analogs for recurrent gastric neuroendocrine tumors type 1. First Author: Alla Anatolievna Markovich, National Medical Research Center of Oncology named after N.N. Blokhin, Moscow, Russian Federation

Background: Neuroendocrine gastric tumors (gNETs) type 1 are associated with chronic autoimmune atrophic gastritis, hypergastrinemia and usually have multifocal lesions and high frequency of recurrence (up to 70% for 2 years). Somatostatin analogs (SA) showed promising data in neoadjuvant setting with 84% of complete responses in a small retrospective trial. The aim of this study was to evaluate the efficacy of adjuvant treatment with SA for recurrent gNETs type 1. **Methods:** This retrospective, single-center study included patients (pts) with recurrent gNETs who received adjuvant treatment with SA. Recruitment of pts was carried out from 2012 to February 2024. **Results:** The study included 35 pts, 33 females and 2 males. Endoscopic mucosal resection (EMR) was performed in 34 cases (97.1%), gastric resection in 1 (2.9%). The median of previous EMR was 2 (1-6) procedures. Most of the pts have multifocal lesions – 28 (80%), the median of lesions is 5. The median tumor size – 6 mm, tumor size $>$10 mm was observed in 2 cases (5.7%), one of them had N+. The median of ki67 was 3,5% (1-10%), grade 1 (G) - 14 (40%) and G2 - 21 (60%). All pts received adjuvant therapy with SA, 33 (94.3%)long-active releasing octreotide30 mg and 2 (5.7%) lanreotide 120 mg every 4 weeks. The time of SA therapy was $<$12 months in 23 cases (65.7%) and $\geq$12 months in 12 (34.3%). The median of follow-up was 30.5 months. The median disease-free survival (DFS) was not reached (95% CI, 67-NR). We compared median DFS after previous EMR (DFS1) without adjuvant SA and DFS after EMR with adjuvant SA (DFS2). The DFS2 was significantly higher – median of DFS1 was 8.8 months (95% CI, 2.7-14.9) not reached (95% CI, 67-NR, p$<$0.01) . Adjuvant treatment with SA $\geq$12 months was associated with improved DFS compared to $<$12 months – not reached (95% CI, 58.1-NR) versus 61 months (95% CI, 26.2-96.2, p=0.05). Adjuvant SA reduced gastrin levels after a year of therapy by $\geq$50% from baseline in 13 cases (37.1%). **Conclusions:** Adjuvant treatment with SA significantly improved DFS and should be considered for recurrent and multifocal gNETs type 1. Duration of therapy $\geq$12 months was significantly associated with improvement in DFS compared with shorter duration of therapy. Research Sponsor: None.

666 **Poster Session**

Efficacy and safety of cabozantinib for advanced gastrointestinal (GI) neuroendocrine tumors (NET) after progression on prior therapy: Subgroup analysis of the phase 3 CABINET trial (Alliance A021602). First Author: Jonathan R. Strosberg, H. Lee Moffitt Cancer Center and Research Institute, Tampa, FL

Background: In the phase 3 CABINET trial (NCT03375320), cabozantinib (CABO) significantly prolonged median progression-free survival (PFS) compared to placebo (PB) in patients with advanced, previously treated, progressive extrapancreatic NET (epNET) [HR 0.38 (95% CI, 0.25-0.59, P$<$0.0001)] and pancreatic NET (pNET) [HR 0.23 (95% CI, 0.12-0.42, P$<$0.0001)] (Chan et al., NEJM, 2024). In this analysis, we focus on outcomes of patients with epNET arising in the GI tract. **Methods:** Patients with locally advanced or metastatic epNET or pNET were randomized 2:1 in separate cohorts to receive CABO 60 mg daily vs PB. The epNET cohort included patients with tumors arising in the GI tract (not including pancreas), lung, unknown primary, and other rare primary sites. In this subgroup analysis of the epNET cohort, patients with GI NET were included. Eligibility included progression by RECIST within 12 months (mo) prior to registration, $\geq$ 1 prior systemic therapy. Primary endpoint: PFS by blinded independent central review (BICR). Secondary endpoints: objective response rate, overall survival, safety. **Results:** 116 of the 203 patients in the epNET cohort had GI NET (CABO, n=70; PB, n=46). Median age was 66 yrs; females: 46%; G1/G2/G3/unknown grade: 35%/57%/6%/2%. ECOG performance status: 0/1-2: 43%/57%. White race: 86%. The most common primary tumor locations were ileum/cecum (54%), small intestine with location not specified (20%); non-cecum colon or rectum (11%), stomach (4%); duodenum (3%); jejunum and nonspecified midgut site (3% each). Prior systemic therapies included somatostatin analogs (SSA) (100%), everolimus (59%); Lu-177 dotatate (77%); temozolomide +/- capecitabine (16%). In patients with GI NET, CABO was associated with improved PFS by BICR compared to PB (stratified HR, 0.50; 95% CI: 0.28-0.88, 1-sided stratified log-rank P=0.007). Median PFS with cabozantinib was 8.5 mo (95% CI, 6.0-16.7) compared to 5.6 mo (95% CI, 3.9-11.0) with PB. CABO demonstrated potential benefit in patients with midgut GI NET and across clinical factors including grade, functional status related to hormone secretion, concurrent SSA use, and prior therapy with Lu-177 dotatate or everolimus. The most frequent grade 3/4 adverse events attributed to therapy with CABO included hypertension (19%), diarrhea (13%), fatigue (10%). **Conclusions:** CABO is associated with improvement in PFS compared to PB in patients with advanced GI NET. Potential benefit was observed across clinical factors including grade, functional status, concurrent SSA, and prior treatment. The safety profile for CABO in patients with GI NET is consistent with that previously reported. Support: U10CA180821, U10CA180882; U10CA180820 (ECOG-ACRIN), U10CA180868 (NRG Oncology), U10CA180888 (SWOG); Exelixis; https://acknowledgments.alliancefound.org. Clinical trial information: NCT03375320. Research Sponsor: National Cancer Institute; U10CA180821, U10CA180882 (Alliance for Clinical Trials in Oncology); U10CA180820 (ECOG-ACRIN), U10CA180868 (NRG Oncology), U10CA180888 (SWOG); Exelixis.

668 **Poster Session**

Interim safety and efficacy data of [^{212}Pb]VMT-α-NET in somatostatin receptor 2 (SSTR2) expressing neuroendocrine tumors (NETs). First Author: Richard L. Wahl, Mallinckrodt Institute of Radiology, Washington University School of Medicine, St. Louis, MO

Background: After the approval of 177Lu DOTATATE, a new generation of radiopharmaceutical therapies (RPT) utilizing alpha-emitting radiometals is being developed as front-line therapies for advanced NETs with the hope of higher clinical efficacy. In this Phase I study, [^{212}Pb]VMT-α-NET, a novel targeted alpha radionuclide therapy (TAT) to SSTR2, is being evaluated in SSTR2-expressing NETs tumors in patients with no prior PRRT treatment. Here we present interim safety and the first efficacy results from dose escalation cohorts 1 and 2. **Methods:** In this first-in-human dose-escalation study, safety, PK, and efficacy of [^{212}Pb]VMT-α-NET were investigated in a population of well-differentiated adult NETs of any grade which progressed after prior standard of care therapy as assessed by RECIST v1.1 (NCT05636618). No patient may have received prior ^{177}Lu-DOTATATE or PRRT. The dose escalation design includes four cohorts, 3 mCi, 5 mCi, 7.5 mCi and 10 mCi based on a Bayesian modified toxicity probability interval (mTPI-2) design. Subjects in cohorts 1 and 2 underwent dosimetry evaluations using the therapeutic surrogate [^{203}Pb]VMT-a-NET. Therapeutic [^{212}Pb]VMT-a-NET was administered for 4 cycles Q8W in combination with reno-protective amino acids. The DLT assessment period is defined as the first 6 weeks of cycle 1. The primary objectives include the evaluation of safety and tolerability, the PK and RP2D of [^{212}Pb]VMT-a-NET as well as ORR as assessed by RECIST 1.1 criteria. **Results:** The current data cut occurred on September 19th, 2024. A total of 9 patients were enrolled (2 in cohort 1 and 7 in cohort 2). Median exposure for cohort 1 was 2.5 mCi (2.5 mCi, 2.5 mCi) and for cohort 2 was 5.1 mCi (4.7 mCi, 5.2 mCi). No DLTs were observed. Two grade 3 AEs occurred in Cohort 2 (diarrhea, syncope). The most frequent related TEAE were alopecia, fatigue, anemia, lymphopenia and nausea (all grade 1 or 2). One subject in Cohort 2 was discontinued due to progressive disease. Landmark PFS at 9 months is 89%. ORR is not yet mature and will be shown at the time of the meeting. Safety Monitoring Committee (SMC) meeting was held on July 17th, 2024, recommended dose escalation to cohort 3 at 7.5 mCi. No nephrotoxicity was reported in either cohort. **Conclusions:** [^{212}Pb]VMT-α-NET is safe up to 185 MBq (5 mCi) dose level, and the SMC supported dose escalate to cohort 3 at 277.5 MBq (7.5 mCi) following a mandated FDA review in fall. Cohort 2 remains open for dose level expansion. Early efficacy results demonstrate encouraging progression-free survival. Clinical trial information: NCT05636618. Research Sponsor: Perspective Therapeutics, Inc.

669 Poster Session

Molecular and clinical profiling of gastroenteropancreatic (GEP) neuroendocrine tumors (NETs): An analysis of the Oncology Research Information Exchange Network database. First Author: Benjamin Edward Ueberroth, Division of Medical Oncology, Department of Medicine, University of Colorado Anschutz School of Medicine, Aurora, CO

Background: The incidence of neuroendocrine tumors (NETs) is approximately 7 per 100,000 persons and rising. By location, tumors in the gastroenteropancreatic (GEP) region, particularly midgut NETs, are most common. The Oncology Research Information Exchange Network (ORIEN) database contains complementary clinical, genomic, and transcriptomic profiling, providing opportunities to identify novel associations between molecular features and clinical outcomes in GEP-NETs. **Methods:** Survival analyses were performed using the log-rank testing, and clinical features were evaluated using Wilcoxon and chi-squared tests. Mutational analyses utilized sample-level enrichments from whole exome sequencing data, and statistical tests were performed using the one-sided Fisher Exact test. Transcriptomic analyses utilized a student's t-test. We reviewed 240 samples from 226 patients, with 124 of pancreatic origin, 90 small bowel, 16 gastric, and 10 colorectal. All specimens were well-differentiated; further grading information was unavailable on the ORIEN platform. A p-value <0.05 was considered significant. **Results:** Patients with metastatic disease were significantly younger at diagnosis (58 vs. 61 years, p=0.0242) and had significantly higher tumor mutational burden (TMB) (0.48 vs. 0.32 mut/Mb, p=0.001). Among the 100 most common alterations across the entire cohort, *BPTF* (p=0.03) and *PRKCA* (p=0.007) mutations were more prevalent in pancreatic NETs than in other primary sites. The most frequently mutated genes identified included *TTN* (27.7%), *MUC16* (25.1%), *CCDC168* (22.1%), *KIR3DL1* (20%), *TGIF2LX* (20%), *MUC17* (19.1%). *TTN* mutation was significantly associated with higher TMB ($p<0.0001$). The most common copy number alteration was *MUC3a* amplification (7q22.1) in 61.5% of samples. This amplification was associated with more prolonged overall survival (OS) with a significance of p=0.0501. Patients with NETs harboring a mutation in *CSNK2A2* (p=0.0149), *WDR74* (p=0.0386), or *NCMAP* (p=0.0248) experienced significantly shorter OS compared to the cohort as a whole. **Conclusions:** We report the first clinical, genomic, and transcriptomic analysis of ORIEN GEP-NET cases. These findings create multiple avenues for further investigation and reinforce the value of multi-institutional consortia such as ORIEN in deepening our knowledge of well-differentiated NETs. Research Sponsor: None.

Most prevalent mutations in GEP-NETs from ORIEN database.

Mutation	Prevalence (Percentage of Samples)
TTN	27.7%
MUC16	25.1%
CCDC168	22.1%
KIR3DL1	20.0%
TGIF2LX	20.0%
MUC17	19.1%

670 Poster Session

Characteristics and outcomes of patients with early-onset versus average-onset small bowel neuroendocrine tumors. First Author: Udhayvir Singh Grewal, University of Iowa Hospitals and Clinics, Iowa City, IA

Background: The global incidence of early-onset gastrointestinal cancers (GI) is increasing. Although patients with more common early-onset GI cancers, e.g. colorectal cancer, often present with more advanced disease and experience relatively worse outcomes; the characteristics and treatment outcomes for less common early-onset small bowel neuroendocrine tumors (SBNETs) have yet to be explored. **Methods:** We used the Surveillance, Epidemiology, and End Results (SEER) database to identify patients with SBNETs (2004-2021), using ICD-0-3 SEER histology codes (8240/3, 8241/3, 8243/3, 8244/3, 8249/3, 8246/3). Patients with poorly differentiated histology were excluded. We used the chi-square test to compare categorical variables between early-onset SBNETs (EO-SBNETs, aged 30-49 years) and average-onset SBNETs (AO-SBNETs, aged $\geq$ 50 years) patients. Overall survival was estimated using the Kaplan-Meier method followed by application of the log-rank test to assess for differences in survival. A multivariable Cox regression analysis included early onset of disease, sex, race/ethnicity, marital status, urban/rural residence, prior treatments, and disease extent as covariates. All tests were conducted with a significance threshold of $p < 0.05$ using RStudio. **Results:** A total of 3,713 patients were included, of which, 925 (25%) were EO-SBNETs. Compared to AO-SBNETs, EO-SBNETs patients were more likely to be females (51.1% vs 46.7%, p=0.02), non-Hispanic Black (20.8% vs 17.6%, p=0.04), Hispanic (13.6% vs 8.8%, $p<0.0001$) and less likely to be non-Hispanic White 60.8% vs 69.9%, $p<0.0001$). There were no differences in stage at diagnosis between the EO-SBNETs and AO-SBNETs, i.e,, localized disease (33.5% vs 31.0%, p=0.15), regional spread (41.9% vs 42.8%, p=0.64) and distant metastatic disease (20.9% vs 21.9%, p=0.50). There were no differences in receipt of surgical resection (90.5% vs 89.1%, p=0.25) or chemotherapy (5% vs 6.3%, p=0.15) comparing EO-SBNETs and AO-SBNETs respectively. In a multivariate cox regression analysis, early onset was associated with a significantly better overall survival (HR=0.49, 95% CI 0.41-0.59, $p<0.0001$). **Conclusions:** With this first of its kind analysis, significantly superior overall outcomes among EO-SBNETs despite no differences in stage and receipt of surgery establishes age at diagnosis as a significant prognostic factor for SBNETs. Further studies investigating the role of factors such as differences in tumor biology and patient preferences in receipt of other therapies besides surgery are needed. Research Sponsor: None.

671 Poster Session

Comparative outcomes of neuroendocrine tumors in the historically underserved population of Bronx, NY: Montefiore Einstein Comprehensive Cancer Center. First Author: Andrew Takchi, Montefiore Einstein Comprehensive Cancer Center, Bronx, NY

Background: Neuroendocrine tumors (NETs) pose different clinical implications based upon age at diagnosis, with regards to early-onset (EO$<$50yrs) versus late-onset (LO$>$=50yrs) disease. While primary-tumor resection is crucial for localized NETs, its benefit in metastatic cases without metastasectomy remains uncertain. This study examines variables associated with survival in EO versus LO cases, including primary tumor resection. **Methods:** A retrospective cohort of 91 patients with gastrointestinal(GI)-NETs were identified as part of an IRB-approved project. Disease-stage (local vs. metastatic), tumor grade (G1-3), 5-year-survival, primary-tumor removal, and race/ethnicity were included in this analysis. **Results:** Among all patients, 31.9% identified as Hispanic, 33% as non-Hispanic Black, 22% as non-Hispanic White, and 13.1% as other. Among all patients, 79 were classified as G1/G2. Among LO patients, 87.5% (n=63) were G1/G2, among which 46% (n=29) were metastatic. Among EO patients, 84% (n=16) were G1/G2, among which 68.8% (n=11) were metastatic. In the LO group with G1/G2 tumors, 5-year survival rate was 48.3% for metastatic disease and 58.8% for localized disease. For EO with G1/G2 tumors, the 5-year survival rate was 72.7% for metastatic disease and 80% for localized disease. In the EO G1/G2 group, 5 of the 11 patients with metastatic disease had primary tumor removal without metastasectomy with 100% 5-year survival rate. In the LO G1/G2 group, 7 of 29 metastatic patients had primary tumor removal without metastasectomy, with 86% 5-year survival rate. Resection of primary tumor (without metastasectomy) in patients with metastatic G1/G2 NETs was associated with improved 5-year survival, independent of age at diagnosis (*p-value*=0.008), as shown in the table. **Conclusions:** This study helps to emphasize the importance of primary tumor removal in improving survival outcomes for patients with metastatic low-grade GI-NETs, regardless of age. Further investigation is warranted to study the role of certain imperative variables such as site of metastases, number of metastases, and patient co-morbidities. Research Sponsor: None.

Survival outcomes based on primary tumor removal in metastatic low-grade GI-NETs.

	No 5-year survival	5-year survival
Primary tumor removal without metastasectomy	1	11
No surgery	12	8

TPS672 Trials in Progress Poster Session

A first-in-human phase I trial with antibody drug conjugate ADCT-701 in neuroendocrine tumors and carcinomas. First Author: Jaydira Del Rivero, Developmental Therapeutics Branch, National Cancer Institute, National Institutes of Health, Bethesda, MD

Background: Neuroendocrine neoplasms (NENs) consist of neuroendocrine tumors (NETs) and neuroendocrine carcinomas (NECs), which are rare malignancies found in various anatomical sites, including the gastrointestinal tract, pancreatic islets, lungs, and adrenal glands. Poorly differentiated NECs are classified as high-grade carcinomas that bear resemblance to small-cell lung cancer (SCLC). Treatment for poorly differentiated NECs typically follows small-cell carcinoma guidelines, utilizing platinum-based regimens; however, these patients experience a high risk of relapse and demonstrate a limited response to additional systemic therapies. Adrenocortical carcinoma (ACC) is another rare malignancy associated with a median survival of approximately 14.5 months post-diagnosis, and advanced disease management options show limited efficacy. Currently, no targeted therapies have demonstrated significant effectiveness for this condition. Preclinical investigations have indicated that Delta-like non-canonical notch ligand 1 (DLK1) is expressed in various neuroendocrine neoplasms, including ACC, SCLC, neuroblastoma, pheochromocytoma, and paraganglioma. ADCT-701, a humanized antibody targeting DLK1, has shown potential in suppressing tumor growth and enhancing survival across multiple cancer models expressing this marker. **Methods:** This Phase I dose escalation study (NCT06041516) aims to enroll adult patients with histologically or cytologically confirmed neuroendocrine neoplasms or malignant ACC with evaluable disease, as defined by the Response Evaluation Criteria in Solid Tumors (RECIST) 1.1. Participants will be part of a dose-finding trial. A 3+3 design is utilized to assess up to ten different dose levels of ADCT-701. Dose escalation continues unless dose-limiting toxicities occur, the maximum tolerated dose (MTD) is reached, or an optimal dose is established. The primary endpoint of the study is to ascertain the recommended phase 2 dose of ADCT-701 in patients with neuroendocrine neoplasms or malignant adrenocortical carcinoma. Secondary endpoints will assess safety, preliminary tumor activity, response rate, overall survival, pharmacokinetics, and the immunogenicity profile of ADCT-701. The trial aims to enroll up to 70 evaluable patients across a maximum of 10 dose levels. For further details, please refer to clinical trial information NCT06041516. Clinical trial information: NCT06041516. Research Sponsor: None.

673 Oral Abstract Session

Pamrevlumab plus nab-paclitaxel/gemcitabine (Pam + GA) as first- and second-line therapy in metastatic pancreatic cancer (mPDAC): Results from Precision Promise (PrP) Bayesian platform trial. First Author: Vincent J. Picozzi, Virginia Mason Hospital and Medical Center, Seattle, WA

Background: Pamrevlumab (Pam) is a fully human recombinant monoclonal antibody against connective tissue growth factor. Early clinical data with Pam plus chemotherapy showed a favorable safety profile and potential efficacy in PDAC. In PrP, Pam + GA was tested as first line (Line 1) and second line (Line 2) therapy for mPDAC vs GA. PrP is a phase 2/3, innovative Bayesian adaptive platform trial sponsored by the Pancreatic Cancer Action Network testing multiple experimental arms efficiently against common controls (1). **Methods:** Randomization is 70% (adaptive amongst experimental arms in stage 1) and 15%:15% amongst two control arms (GA, mFOLFIRINOX). Pam + GA graduates from stage 1 to 2 if the Bayesian predictive power (PP) of eventual success is $\geq$ 35% for one of the arm's signatures [Line 1, Line 2 or Line 1 & 2 (All)]. All participants (pts) are followed for 12 months (mos) after last pt randomized and treated in Pam + GA. Stages 1 and 2 are combined for final analysis. Efficacy is defined by overall survival (OS) hazard ratio (HR, experimental vs control), by a Bayesian statistical model. Superiority (HR $<$ 1) is claimed at final analysis if the Bayesian probability of superiority is $\geq$ 98%. **Results:** Pam + GA entered PrP in Jun 2021. Between 6/2021 - 1/2023, at 23 US sites, 317 pts were treated and are included in the mITT analysis. 213 pts were treated in the Pam + GA arm (102 Line 1; 111 Line 2), 45 in the GA arm (23 Line 1; 22 Line 2), 31 in the mFOLFIRINOX arm and 28 in other experimental arms. All treated pts before and during Pam + GA enrollment were in the Bayesian model, with outcomes adjusted via a time machine to increase the trial's statistical power. Baseline characteristics were balanced across the arms. Pam + GA met criteria to graduate to stage 2 in Sep 2022 in the All signature (Line 1 & 2). At final analysis, Pam + GA did not meet the OS primary endpoint [model estimated HR: 1.18 (95% credible interval, 0.88, 1.56), posterior Pr(HR $<$ 1) = 0.14, below specified $\geq$ 0.98]. No benefit was seen in PFS nor ORR (Table). No new safety signals were seen. **Conclusions:** PrP, the first Bayesian platform trial in mPDAC, performed as designed; Pam + GA did not improve OS versus GA in Line 1 and Line 2 mPDAC. The novel Bayesian design warrants further study in mPDAC to enhance efficiency of drug development. Future designs should explore different strategies to allocate patients to control arm(s) vs experimental arms and accommodate novel agents intended to benefit subsets of mPDAC. 1. Picozzi, ASCO TPS 4188, 2022. Clinical trial information: NCT04229004. Research Sponsor: Pancreatic Cancer Action Network; FibroGen, Inc.

	Line 1			Line 2		
	GA	Pam + GA	Hazard Ratio	GA	Pam + GA	Hazard Ratio
Model Estimated mOS (mos)	11.3	9.7	1.18 (95% CI* 0.88, 1.56)	7.8	6.6	1.18 (95% CI* 0.88, 1.56)
mPFS (mos)	5.3	5.9	0.64 (95% CI^ 0.36, 1.14)	7.0	3.9	1.35 (95% CI^ 0.78, 2.33)
ORR (%)	26.1	35.3		4.5	9.0	

*Credible Interval.
^Confidence Interval.

675 Rapid Oral Abstract Session

LAPIS: Randomized phase 3 trial of chemotherapy (CTX) with and without pamrevlumab (PAM) for locally advanced pancreatic cancer (LAPC). First Author: Vincent J. Picozzi, Virginia Mason Hospital and Medical Center, Seattle, WA

Background: PAM improved response, surgical eligibility, and resection rates when combined with CTX in phase 1/2 LAPC trials (NCT01181245; NCT02210559). LAPIS (NCT03941093) evaluated efficacy and safety of PAM + CTX in unresectable LAPC. **Methods:** LAPIS was a global, double-blind, placebo-controlled phase 3 trial of adults with treatment-naïve, -confirmed unresectable LAPC. Patients received (randomized 1:1) PAM (35 mg/kg Q2W) or placebo (PBO) plus CTX (gemcitabine + nab-paclitaxel [GnP] or FOLFIRINOX [FFX] per standard protocol) for up to six 28-day cycles. based on changes in -9, on FDG-PET, resectability, with final surgical decision made by surgeon. Primary endpoint: overall survival (OS); secondary endpoints: event-free survival (EFS), progression-free survival (PFS), and objective response rate (ORR. Safety (including treatment-emergent adverse events [TEAEs]) was evaluated in patients who received treatment. **Results:** 143 and 141 patients (median [range] age, 65.0 [31–90] yrs; 53.2% male) were randomized to PAM and PBO arms. 142 and 141 received treatment: GnP, 114 (79.7%) and 112 (79.4%); FFX, 28 (19.6%) and 29 (20.6%). PAM (64.8%) and 96 PBO (68.1%) patients completed 6 treatment cycles. . Survival similar for PAM and PBO arms (Table) and did not differ by. For PAM + FFX vs PBO + FFX, respiratory, thoracic and mediastinal disorders (39.3% vs 27.6%) and skin and subcutaneous tissue disorders (53.6% vs 37.9%) occurred $>$10% more with PAM; TEAE frequencies were similar with PAM + GnP and PBO + GnP. One treatment-related death occurred in PAM arm. **Conclusions:** LAPIS contained important LAPC trial innovations: pre/post-CTX FDG-PET imaging, objective criteria for surgical intervention, external surgical review panel, and composite EFS measurement. Addition of PAM to CTX was not associated with additional toxicity but did not improve survival outcomes for unresectable LAPC. Clinical trial information: NCT03941093. Research Sponsor: None.

LAPIS endpoints.

	PAM arm (n=143)	PBO arm (n=141)	
OS, months, median (95% CI)	events=118 (82.5%) 17.25 (15.47, 18.89)	events=112 (79.4%) 17.94 (14.59, 20.34)	HR (95% CI) 1.08 (0.83, 1.41) p=0.55
EFS,[a] months, median (95% CI)	events=99 (69.2%) 5.72 (5.59, 6.01)	events=102 (72.3%) 5.78 (5.62, 6.37)	HR (95% CI) 1.05 (0.78, 1.39)
PFS, months, median (95% CI)	events=48 (33.6%) 9.36 (7.75, 11.79)	events=44 (31.2%) 9.40 (7.69, 10.84)	HR (95% CI) 1.01 (0.65, 1.56)
ORR, n (%)	43 (30.1)	64 (45.4)	OR (95% CI) 0.50 (0.31, 0.82)
Complete response	0	0	
Partial response	43 (30.1)	64 (45.4)	

CI, confidence interval; HR, hazard ratio; OR, odds ratio.
[a]Earliest of 1) failure to achieve local disease-free status at end of treatment and/or after surgery; 2) local or distant recurrence/progression; or 3) death.

676 Rapid Oral Abstract Session

Effect of baseline geriatric and quality of life assessments on treatment outcomes in ECOG-ACRIN EA2186 (GIANT): A randomized phase II study of gemcitabine and nab-paclitaxel compared with 5-fluorouracil, leucovorin, and liposomal irinotecan in older patients with treatment-naïve metastatic pancreatic cancer. First Author: Efrat Dotan, University of Pennsylvania, Ann B. Barshinger Cancer Institute, Lancaster, PA

Background: Data is lacking to guide the care of vulnerable older adults (OA) with newly diagnosed metastatic pancreatic adenocarcinoma (mPDAC). EA2186 trial demonstrated poor outcomes among vulnerable OA with mPDAC treated with dose-reduced chemotherapy. To understand the factors driving treatment outcomes in this patient population, we analyzed the correlation between baseline geriatric and quality of life (QOL) assessments and treatment outcomes. **Methods:** Vulnerable OA $\geq$70 yo with mPDAC, ECOG PS 0-2 were enrolled. Vulnerability was defined by screening geriatric assessment (GA) demonstrating mild abnormalities in function, comorbidities, cognition, or age$\geq$80y. Pts were randomized to Arm A: Gemcitabine (1000mg/m2) + Nab-Paclitaxel (125mg/m2) q14 days or Arm B: 5-Fluorouracil (2400mg/m2 46hr) + Leucovorin (400mg/m2) + Liposomal Irinotecan (50mg/m2) q14 days. GA and QOL evaluations were completed at baseline and 3 time points. Secondary endpoints of the study included evaluating the correlation between baseline GA, QOL and treatment outcomes. Regression models were used to evaluate the associations between baseline GA and QOL factors, survival and grade 3 or higher toxicity, with 80% power to detect doubling in grade 3 toxicity for GA/QOL measures. **Results:** 176 pts (88 per arm) enrolled with median age 77 (range 70-90), 24% ECOG-0, 64% ECOG-1 and 12% ECOG-2. Pts were deemed vulnerable by cognition (46%), age (36%) or comorbidities (31.4%), with 35% meeting vulnerability criteria in $\geq$2 domains. No significant difference was seen in median OS (4.7 vs. 4.4 months; p=0.72) or $\geq$grade 3 toxicity rate (45.6% vs. 58.7%; p=0.10) between arms A and B, respectively. Strong correlation was found between OS and baseline instrumental activities of daily living score (HR 0.84; p=0.02), nutritional scores (HR 0.82; p$<$0.0001), depression scores (HR 1.07; p=0.02), and scores of all QOL measures (HR 0.98; p$<$0.0001). No correlation was found between OS and comorbidity, cognition, and Activities of Daily Living scores. After adjustment for age and PS, only baseline WBC level (OR=0.35; p=0.0054), and depression scores (OR=1.20 per score unit; p=0.021) and to a lesser extent FACT-G score (OR=0.98 per score unit; p=0.061) were found to correlate with rates of $\geq$grade 3 toxicity. **Conclusions:** Baseline GA and QOL factors among vulnerable older adults with mPDAC correlate strongly with survival and treatment tolerance. Supportive care to address these factors may favorably affect outcomes in this patient population. Clinical trial information: NCT04233866. Research Sponsor: National Cancer Institute.

677 Poster Session

Assessing socioeconomic disparities and the likelihood of multimodal treatment utilization in patients with pancreatic cancer using SEER Medicare data. First Author: M. Muska Nataliansyah, Medical College of Wisconsin, Milwaukee, WI

Background: Optimal treatment for pancreatic cancer necessitates multimodal therapies combining surgery, chemotherapy, and radiation. Socioeconomic determinants of health may influence access to these treatments, leading to disparities in outcomes. This study examines the association between socioeconomic status (SES) and the likelihood of receiving multimodal treatment among pancreatic cancer patients using SEER Medicare data. **Methods:** Our study population consists of elderly persons diagnosed with pancreatic cancer from the 2008–2017 SEER-Medicare dataset. A multinomial logistic regression controlling for sociodemographic characteristics, cancer stage, comorbidities, geographic region, and ecological measures of urbanicity and SES was used to assess the likelihood of receiving one of three mutually exclusive treatment categories: no treatment, one modality (surgery, chemotherapy, or radiotherapy), or multiple modalities (different combinations of surgery, chemotherapy, and radiotherapy), with multiple modalities as the reference group. Survival analysis was conducted using Kaplan-Meier curves (KMC) and Cox proportional hazards (CPH) models. **Results:** Adjusted analyses showed that among stages 0–3 patients, living in areas with a high percentage of individuals without a high school education (non-HS) was associated with increased odds of receiving no treatment (OR = 1.26, p $<$ 0.01) and one modality (OR = 1.20, p $<$ 0.01) versus multiple modalities. Low median household income was linked to higher odds of no treatment (OR = 1.30, p $<$ 0.01) and one modality (OR = 1.16, p $<$ 0.05). This pattern persisted in stage 4 patients, where high non-HS percentage increased odds of no treatment (OR = 1.38, p $<$ 0.01), and low median income was associated with higher odds of no treatment (OR = 1.25, p $<$ 0.05). Non-Hispanic Black patients had higher odds of receiving no treatment in both stages 0–3 (OR = 1.26, p $<$ 0.05) and stage 4 (OR = 1.91, p $<$ 0.01) compared to Non-Hispanic Whites. Being married was associated with reduced odds of receiving no treatment for stages 0–3 (OR = 0.53, p $<$ 0.01) and stage 4 (OR = 0.55, p $<$ 0.01). Lower SES and being Non-Hispanic Black are associated with decreased likelihood of receiving multimodal treatment across all stages. Patients receiving multimodal therapies demonstrated improved survival on our KMC analysis, and CPH models indicated lower mortality for those receiving multiple modalities. **Conclusions:** Socioeconomic disparities significantly influence multimodal treatment utilization in pancreatic cancer. Lower SES indicators—high non-HS percentages, low median income—and being Non-Hispanic Black are associated with decreased use of multimodal therapies and poorer survival. Addressing these disparities is essential to ensure equitable access to optimal treatments and improve survival rates. Research Sponsor: American Cancer Society; Medical College of Wisconsin.

678 Poster Session

Social determinants of health and financial burden among patients with pancreatic cancer at a tertiary cancer center. First Author: Khalid Jazieh, Mayo Clinic Rochester, Rochester, MN

Background: Financial toxicity (FT) represents a challenge for cancer patients that impacts treatment decisions, patient satisfaction, and quality of life. Social determinants of health (SDoH) screening may predict who is at risk of FT and could potentially benefit from financial interventions. **Methods:** This is a retrospective study of all cancer patients presenting to Mayo Clinic sites between Sept 2023 – Jun 2024. Patients completed SDoH questionnaires and were given risk scores of Low Risk, Medium Risk, and High Risk for SDoH categories: Financial Strain Risk, Transportation Risk, and Food Insecurity Risk. Patient financial data including active debt amounts, insurance coverage, and financial aid were collected. Patients with pancreatic ductal adenocarcinoma (PDAC) were compared to those with pancreatic neuroendocrine tumors (PNET) and patients with other cancers (Others). Categorical variables were compared with Pearson's Chi-squared test, continuous variables with Kruskal-Wallis rank sum test. P values < 0.05 were considered significant. **Results:** Data was obtained from 88,039 patients, 2,607 of whom had PDAC and 449 had PNET. Compared to PDAC patients, PNET patients and Others were more likely to have Medium/High risk scores for Financial Strain Risk (6% vs 6.5% and 8.5%, respectively) and Food Insecurity (1.7% vs 3.8% and 3.2%) (Table). There was no statistically significant difference in median medical debt and need of financial assistance between the groups. Among PDAC patients, there were differences in distributions among race, primary language spoken, and patient electronic record access depending on Financial Risk categories. African American patients had the greatest proportion of Medium/High risk (12.9%) compared to Asian (4.4%) and White (5.8%) patients ($P<0.001$). Among patients with English as their primary language, 64.9% had Low Risk scores compared to 27.7% with other primary languages ($P<0.001$). Patients with electronic record access were more likely to report High Risk scores (6.9%) compared to those without access (1.6%). **Conclusions:** Per patient responses collected, PDAC patients presenting to our center have lower risk of financial strain and similar medical debt levels compared to the remainder of cancer patients. Among PDAC patients, higher financial strain was experienced by non-whites (7.9% vs 5.8%). Research Sponsor: None.

Diagnosis	PDAC (N=2607)	PNET (N=449)	Other (N=84983)	p value
Current Financial Risk (Grouped)				**< 0.001**
- Low Risk	1662 (63.8%)	304 (67.7%)	59298 (69.8%)	
- Medium/High Risk	157 (6.0%)	29 (6.5%)	7209 (8.5%)	
- Unknown/Missing	788 (30.2%)	116 (25.8%)	18476 (21.7%)	
Food Insecurity Risk				**< 0.001**
- Low Risk	1764 (67.7%)	308 (68.6%)	63235 (74.4%)	
- Medium Risk	45 (1.7%)	17 (3.8%)	2703 (3.2%)	
- Unknown/Missing	798 (30.6%)	124 (27.6%)	19045 (22.4%)	
Median Active Debt in USD	1102.230	1020.010	1141.135	**0.593**

679 Poster Session

Racial disparities in receipt of guideline concordant pancreatic cancer care among older adults in the US. First Author: Joshua Herb, The University of Texas MD Anderson Cancer Center, Houston, TX

Background: Patients racialized as Black experience higher incidence and mortality from pancreatic cancer (PC). The aim of this study was to examine differences in receipt of guideline concordant care (GCC) among older adults with PC racialized as Black, White, or Hispanic ethnicity. **Methods:** Patients 65 years and older with incident PC racialized as White, Black, or Hispanic ethnicity were identified in the SEER-Medicare database from 2004-2019. The primary outcome was receipt of GCC (stage I/II: surgery+chemotherapy+/-radiation; stage III: chemotherapy+/-surgery+/-radiation; stage IV: chemotherapy+/-radiation). Multivariable logistic regression was used to identify factors associated with GCC. Oaxaca-Blinder decomposition was used to examine the contribution of measured and unmeasured variables to racial disparities in receipt of GCC. **Results:** Of 12,772 patients included, 85.5% of patients racialized as White (n=10,915), 7.8% Black (n=992), and 6.8% Hispanic ethnicity (n=865). Patients racialized as Hispanic ethnicity and Black were more often dual-eligible for Medicare/Medicaid than patients racialized as White (38.6%, 26.9%, and 7.5%, respectively). In total, 56.3% received GCC. On adjusted analysis, patients racialized as Black were less likely to receive GCC for early stage (stage I/II) disease compared to patients racialized as White (OR 0.66; 95% CI 0.53-0.83), but not for stage III (OR 0.65; 95% CI 0.41-1.01) or stage IV (OR 0.87; 95% CI 0.66-1.14) disease. No difference in GCC was observed among patients of Hispanic ethnicity when compared to patients racialized as White. Increased age and dual-eligibility were associated with decreased likelihood of GCC across stages. Differences in receipt of GCC among patients racialized as Black and White remain largely unexplained. Of the measured factors, comorbidities, dual-eligibility, and area level poverty contributed most to the disparity in GCC (Table). **Conclusions:** Just over half of an insured population with PC receive GCC. Despite an increased focus on equity in cancer care, Black-White racial disparities in receipt of GCC are prevalent, particularly among patients with early-stage disease. Further studies are critical to identify and address the unmeasured factors driving treatment disparities. Research Sponsor: U.S. National Institutes of Health; T32 CA 009599; MD Anderson Cancer Center; P30 CA 016672.

Contributing factors to racial disparities in guideline concordant care for patients with pancreatic cancer.

	Stage I/II	Stage III	Stage IV
Difference (White-Black)	11.6%	9.4%	4.1%
Explained			
Age	-3.0%	-2.0%	-1.0%
Sex	0.1%	0.8%	0.1%
Dual-Eligibility	2.0%	3.1%	0.8%
Charlson Comorbidity Index	1.3%	0.2%	0.7%
Area Level Poverty	2.2%	-0.8%	0.7%
Hospital/Practice Volume	1.2%	-0.3%	-0.03%
Region	0.4%	0.9%	-0.2%
Year of Diagnosis	-0.1%	0.5%	0.1%
Unexplained	7.5%	7.0%	3.0%

Positive values indicate these factors increased the measured disparity, while negative values indicate these decreased the measured disparity.

680 Poster Session

Association of malnutrition at diagnosis with survival among older adults with pancreatic cancer. First Author: Apoorva Doshi, University of Alabama at Birmingham, Birmingham, AL

Background: Malnutrition is common among older adults newly diagnosed with pancreatic cancer. However, the implications of malnutrition among this vulnerable population are poorly understood. We sought to understand the association between malnourishment at diagnosis and overall survival (OS) among older adults with pancreatic cancer. **Methods:** We included adults aged $\geq$ 60 years with newly diagnosed pancreatic cancer, who were enrolled in the University of Alabama at Birmingham (UAB) Cancer and Aging Resilience Evaluation (CARE) registry between October 2017 and April 2024. To assess nutritional status, we administered an abridged version of the Patient-Generated Subjective Global Assessment (abPG-SGA) to patients at baseline prior to receiving chemotherapy. Using abPGSGA scores, we categorized patients as either normal (score $\leq$ 5) or malnourished (score $\geq$ 6). All participants were followed until death or until the cutoff date of July 10, 2024. We employed Kaplan-Meier survival analysis to estimate overall survival (OS) for both malnourished and non-malnourished groups and compared them using log-rank test. Additionally, we built multivariable Cox proportional hazards models to test the association between malnutrition and OS, while controlling for age at diagnosis, sex, race/ethnicity, and cancer stage. **Results:** A total of 274 patients were included; the median age at diagnosis was 70 (interquartile range, IQR: 65-75) years with 52% males, 75% non-Hispanic White and 52% with Stage IV disease at diagnosis. At baseline, 59% of patients were malnourished, and the median abPGSGA score was 8 (IQR 4-14). Patients who were malnourished had a similar demographic and clinical characteristics as compared to those without malnutrition. Over a median follow up of 36 months, a total of 202 patients (74 %) died. The median overall survival (OS) was significantly lower for malnourished patients at 13 months compared to 21 months for non-malnourished patients (log-rank test, $p < .001$). After adjusting for potential age, sex, race/ethnicity and cancer stage, malnutrition at diagnosis remained independently associated with worse survival outcomes (adjusted HR: 1.762; 95% CI: 1.316–2.358; $p < .001$). **Conclusions:** Our study reports that malnutrition at baseline is significantly associated with worse overall survival in older adults with pancreatic cancer. These findings underscore the importance of early nutritional assessment and intervention to enhance outcomes in this population. Research Sponsor: None.

681 Poster Session

Redefining the use of a first-line FOLFIRINOX-like regimen in older patients with metastatic pancreatic cancer. First Author: Arthur Winer, Inova Schar Cancer Institute, Fairfax, VA

Background: Prospective elderly-specific data for treating pancreatic cancer are scarce. First-line treatment with triplet chemotherapy (e.g., FOLFIRINOX) is often too toxic for older adults (OA). An alternating regimen of FOLFOX and FOLFIRI (aFOLFIRINOX) has demonstrated efficacy and decreased toxicity in patients with colon cancer. This ongoing multicenter phase II trial evaluates the tolerability and efficacy of aFOLFIRINOX in OA with metastatic pancreatic adenocarcinoma (mPDAC). Herein we report the interim safety analysis results. **Methods:** Eligible patients (goal n=37) $\geq$65 years old with measurable untreated mPDAC, ECOG PS 0-2, receive standard doses of FOLFOX alternating with FOLFIRI every 2 weeks for up to 6 months. Patients can continue treatment thereafter at the discretion of their physician. The primary endpoint is rate of treatment-limiting grade $\geq$3 adverse events (TLAE) per CTCAE v5.0 in the first 8 weeks of therapy. These toxicities are defined as treatment-related grade 3 non-hematologic AE persisting $>$ 2 weeks despite best supportive care, grade 3 anemia or thrombocytopenia, grade $>$3 neuropathy, or any grade $>$ 4 AE. The study is powered to detect a 20% improvement in grade $\geq$3 TLAE: null vs alternate: 45% vs 65% of patients without TLAE at 8 weeks. Secondary endpoints include progression free & overall survival (PFS & OS) and objective response rate (ORR). FACT-G and PROMIS-10 quality of life (QOL) surveys and geriatric assessments are collected and will be correlated with treatment outcomes. **Results:** 21 eligible patients were included in the interim safety analysis: 11 male, 10 female; median age:73 (range 65-82); 18 patients had ECOG PS 0-1. 16 (76%) patients did not experience TLAE as defined above. TLAE included neutropenia, thromboembolism, and hypokalemia, and are similar to reported toxicity with FOLFOX and FOLFIRI in this patient population. One patient died of their disease within 8 weeks of treatment initiation and another within 2 weeks due to pulmonary embolism deemed unrelated to treatment. Of the 18 patients evaluable for response, ORR was 11% (partial response: n =2), Disease Control Rate was 89% (stable disease: n =14). 7 patients completed 6 months of treatment on study. **Conclusions:** The interim safety analysis of this clinical trial fulfilled the predefined futility rule without excess treatment-limiting or unexpected toxicity among OA with mPDAC treated with first-line aFOLFIRINOX, suggesting this may be a reasonable treatment option in this patient population. Trial is ongoing for evaluation of efficacy data of this regimen. Clinical trial information: NCT05360732. Research Sponsor: None.

682 Poster Session

Optimizing clinical trial design to mitigate racial disparities in pancreatic cancer clinical trial enrollment. First Author: Tracey Pu, National Cancer Institutes/ National Institutes of Health, Bethesda, MD

Background: Participation in pancreatic ductal adenocarcinoma (PDAC) clinical trials remains dismal among patients from minoritized groups. This study aimed to investigate if clinical trial design, including eligibility criteria, affects representation of minoritized populations. **Methods:** United States Phase I-IV PDAC clinical trials between 2000-2020 were extracted from the ClinicalTrials.gov database. The Surveillance, Epidemiology, and End Results (SEER) database was used to determine cancer incidence linked with years of trial enrollment. Enrollment fraction (EF) was defined as the number of trial participants divided by the incidence of PDAC in each racial and ethnic group. Representation Quotient (RQ) was defined as the proportion of trial participants divided by proportion of PDAC incidence for each subgroup, where underrepresentation was RQ $<$1.0. **Results:** 8451 patients from 228 unique PDAC clinical trials were analyzed. 55.7% (n=128) and 44.3% (n=102) trials reported racial and ethnic demographic data, respectively. Non-Hispanic Black (NHB), Non-Hispanic Asian/Pacific Islander (NHAPI), Non-Hispanic American Indian/Alaskan Native (NHAIAN), and Hispanic patients were underrepresented (Table). NHB patients were more likely to be underrepresented in trials for metastatic PDAC disease (89.4% vs. 73.8% of trials, p=0.035). NHB patients demonstrated an increase in median RQ (0.28 vs. 0.48) in trials from 2000-2010 vs. 2011-2020, while Hispanic patients had a decreased median RQ (0.36 vs. 0.25). Trials with renal insufficiency exclusion criteria without cutoffs or explicit methodology were more likely to accrue NHB patients (RQ 0.88 vs. 0.46, p=0.02), whereas trials that used a creatinine cutoff or the modified Cockcroft-Gault (CG) formula did not significantly affect RQ. Other eligibility criteria including cardiac history, HIV, Hepatitis B/C, and receipt of neoadjuvant therapy did not significantly affect RQ for any racial or ethnic group. Trial location and funding source did not affect RQ for any group. **Conclusions:** Systemic barriers persist in minority enrollment in PDAC clinical trials. Clinical trial design optimizing inclusive eligibility requirements is a first feasible step in increasing minority clinical trial enrollment. Research Sponsor: National Cancer Funding.

Racial and ethnic representation in pancreatic ductal adenocarcinoma clinical trials.

	Enrollment Fraction (Median, IQR)	Representation Quotient (Median, IQR)
NHW	0.28 (0.12-0.62)	1.17 (1.09-1.24)
NHB	0.15 (0-0.33)	0.47 (0-0.87)
NHAPI	0 (0-0.23)	0 (0-0.45)
NHAIAN	0 (0-0)	0 (0-0)
Hispanic	0.07 (0-0.21)	0.26 (0-0.68)

*Abbreviations: Interquartile Range (IQR), Non-Hispanic Black (NHB), Non-Hispanic Asian/Pacific Islander (NHAPI), Non-Hispanic American Indian/Alaskan Native (NHAIAN).

683 Poster Session

Disparities in outcomes of pancreatic cancer hospitalizations: A ten-year trend study. First Author: Ayobami Gbenga Olafimihan, John H. Stroger, Jr. Hospital of Cook County, Chicago, IL

Background: Pancreatic cancer is the third leading cause of cancer mortality in the United States with 51,750 estimated deaths from Pancreatic cancer in 2024. Demographic variations have been reported with Pancreatic cancer. Our aim was to identify trends in demographic-specific disparities in inpatient pancreatic cancer outcomes. **Methods:** This is a retrospective longitudinal trends study using the Nationwide Inpatient Sample (NIS) database (2010-2019). We identified hospitalizations with pancreatic cancer using ICD-9 and ICD-10 codes. We highlight disparities in mortality outcomes stratified by age, gender and race. **Results:** There were 992,550 hospitalizations of pancreatic cancer patients between 2010 and 2019. Over the period, the mean age at hospitalization was similar at 68 years (P = 0.006). A majority of the admissions were elderly patients ($\geq$65 years) identifying as males and Caucasians (P $<$ 0.01). Most patients had Medicare and were in large and urban teaching hospitals (P $<$ 0.001). Over the decade, there was an overall decline in inpatient mortality, from 8.2% in 2010 to 7.1% in 2019 (P trend = 0.004). Amidst the age subgroups, only the elderly patient group experienced a significant decline in mortality rate (8.8% to 7.3%; P trend = 0.004). The young adult population ($<$45 years) had a trend towards increasing mortality rate, but it was not significant (P trend= 0.139). On gender subgroup analysis, only males had a significant decline in mortality rate (P trend= 0.003). On racial subgroup analysis, only Caucasians had a significant decline in mortality trend (P trend= 0.004) during the study period. **Conclusions:** There was an overall improvement in the mortality outcome of pancreatic cancer hospitalizations. On sub-group analysis, significant decline in mortality over the decade were seen only in elderly, male and Caucasian patient groups. Targeted research is needed in understanding the root causes of these disparities in pancreatic cancer outcomes and develop effective intervention to bridge this gap. Research Sponsor: None.

Trends in mortality outcomes of pancreatic cancer hospitalizations by demographic subgroups.

Mortality rate, %	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	P- Trend value
Overall	8.21	7.94	7.85	7.70	7.73	7.68	7.46	7.87	7.47	7.14	0.004
Young (<45 years)	4.45	5.55	5.23	6.08	6.27	5.32	3.63	6.84	6.95	6.35	0.14
Middle-aged (45- 64 years)	7.55	7.24	7.74	7.32	7.03	6.99	7.62	6.80	6.74	6.91	0.06
Elderly (>65 years)	8.81	8.44	8.04	8.00	8.22	8.19	7.54	8.48	7.87	7.29	0.004
Females	7.72	7.32	7.23	7.15	6.99	7.31	6.72	7.38	7.01	6.87	0.08
Males	8.67	8.57	8.44	8.22	8.44	8.04	8.13	8.32	7.90	7.38	0.003
Whites	7.94	7.49	7.48	7.39	7.81	7.29	6.81	7.39	7.19	6.80	0.004
Blacks	9.87	9.40	9.38	9.15	8.35	8.13	9.40	9.01	8.55	8.41	0.07
Hispanics	7.10	8.99	8.02	6.85	6.66	8.80	7.54	7.43	7.10	6.14	0.05

684 Poster Session

Utility of circulating tumor DNA in patients with unresectable pancreatic cancer using a patient-specific panel: ARTEMIS-PC study. First Author: Taro Shibuki, Department of Hepatobiliary and Pancreatic Oncology, National Cancer Center Hospital East, Kashiwa, Japan

Background: Circulating tumor DNA (ctDNA) analysis has proven to be a highly accurate method for assessing treatment efficacy and detecting molecular residual disease (MRD) in cancer patients, outperforming tumor markers and imaging modalities. However, the ctDNA detection rate is reported to be low, and its utility remains unclear in pancreatic cancer (PC). We conducted the prospective "ARTEMIS-PC" study to evaluate the utility of a personalized, tumor-informed MRD assay in patients with unresectable PC. **Methods:** Eligible patients had histopathologically confirmed unresectable PC, were previously untreated, and had disease measurable per Response Evaluation Criteria in Solid Tumors v1.1. A patient-specific tumor-informed assay (Invitae Personalized Cancer Monitoring) was used for the detection of ctDNA. Blood samples were collected at structured pre- and post-systemic therapy timepoints. **Results:** A total of 99 patients were eligible for the ARTEMIS-PC study, and personalized panels were successfully created for 92 patients. The median age of patients was 70 years (range 45-81 years), with 43/92 being female. Thirty patients were stage III, and 62 stage IV. Median size of the primary tumor was 38 mm (range 13-100 mm). MRD positivity at enrollment was 88.0%, with 73.3% in stage III patients, and 95.2% in stage IV (p=0.007). During follow-up, ctDNA clearance was achieved in 33 patients (40.7%). ctDNA clearance was associated with a significantly higher objective response rate (61.5% vs. 17.6%, p=0.001) and disease control rate (100% vs. 64.7%, p=0.002) compared to those who did not achieve clearance. Patients who achieved ctDNA clearance had significantly longer progression-free survival (PFS) than those who did not (9.0 vs. 3.5 months, hazard ratio 0.2, p$<$0.001). The predictive performance of select biomarkers (variant allele frequency [VAF], CEA, CA19-9) was evaluated using area under the curve (AUC) for treatment response and disease control. For disease control, VAF showed superior predictive performance with AUCs of 0.84 at enrollment and 0.97 at both Week 4 and Week 8, compared to CEA (AUCs: 0.65, 0.73, 0.60) and CA19-9 (AUCs: 0.51, 0.57, 0.56). The predictive performance for treatment response was lower than AUC 0.75 for any of the biomarkers. Among patients who were MRD negative before treatment or became negative during follow-up (n=44), 45.5% (n=20) subsequently turned MRD positive. Of these, disease progression was confirmed by CT in 14 patients. The median time from MRD turning positive to CT-confirmed progressive disease was 88.5 days (range: -28 to 385 days). **Conclusions:** ctDNA clearance is a useful predictor of improved objective response, disease control, and PFS, with VAF emerging as the more accurate biomarker for disease control, underscoring its potential utility in treatment monitoring in patients with unresectable PC. Clinical trial information: UMIN000043561. Research Sponsor: None.

685 Poster Session

A systematic review of the impact of artificial intelligence in pancreatic cancer detection. First Author: Jahnavi Ethakota, Henry Ford Allegiance Health System, Jackson, MI

Background: Pancreatic cancer is the fourth leading cause of cancer deaths in the United States, and early detection remains a significant challenge. Screening the general population is not feasible, but the rise of artificial intelligence (AI) has introduced new possibilities for improving early diagnosis and patient outcomes. **Methods:** A systematic literature search was conducted using PubMed, Google Scholar, and MEDLINE using MeSH terms Artificial intelligence or AI, and diagnosis and pancreatic carcinoma or pancreatic adenocarcinoma. Prisma guidelines were adhered to, and a total of 19 studies resulted, 6 of them were retrospective studies, 4 of them were literature reviews, and 4 of them were randomized trials. The rest were duplicates. The inclusion criteria for this study is AI being used in the diagnosis, only in pancreatic cancer, within the last 5 years, only in English, and only retrospective studies were included. **Results:** One study used Digital Imaging Processing (DIP) for analyzing Endoscopic ultrasound (EUS) images from 153 pancreatic cancer patients, yielding a sensitivity of 97.98% and a specificity of 94.32%. Two studies explored Computer Aided Diagnosis (CAD) models applied to PET/CT and EUS images, achieving a sensitivity of 95.23% and specificity of 97.51% in PET/CT scans and 83.3% and 93.3% in EUS images, respectively. Another study used a Faster R-CNN model to analyze CT images from 338 pancreatic cancer patients showed high diagnostic accuracy in much less time. Additionally, two studies utilized Natural Language Processing (NLP) for identifying family histories of pancreatic cancer and detecting pancreatic cysts, with the latter achieving sensitivity and specificity rates of 99.9% and 98.8%. **Conclusions:** The current strategies for early diagnosis of pancreatic cancer focus on serum biomarkers and EUS-guided Fine Needle Aspiration (EUS-FNA) and sensitivity varies but depends on the physician's expertise. AI is showing promise in improving pancreatic cancer diagnosis by enhancing early detection and accuracy. Techniques like deep learning, NLP-based models, Faster R-CNN, and CAD systems analyze medical data and images more effectively than manual methods. AI holds the potential to shape the future of pancreatic cancer diagnosis and improve patient outcomes. Research Sponsor: None.

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Impact of intrathecal targeted drug delivery (ITDD) on oral morphine equivalents and pain scores in advanced pancreatic cancer pain. First Author: Nicole Eye, Avera McKennan Hospital and University Health Center, Sioux Falls, SD

Background: Pancreatic cancer accounts for approximately 3% of all cancers and affects ~64,000 adults in the United States each year. Seventy five percent of patients present with abdominal pain, leading to diagnosis. This is one of the most painful cancers, with pain severity strongly correlating to prognosis. Severe intractable abdominal pain negatively impacts quality of life. Traditionally, opioids have been used as the first line agent for the treatment of severe abdominal pain. However, opioids carry significant risk and often unwarranted side effects. With escalating doses to treat refractory pain there is also potential for unintentional overdose. Intrathecal targeted drug delivery (ITDD) provides pain medications directly to the intrathecal space limiting systemic side effects. Our study aimed to determine the impact ITDD has on pain scores as well as the amount of systemic opioid needed in patients with advanced pancreatic cancer. **Methods:** A retrospective review was conducted on 217 advanced cancer patients that underwent ITDD placement from March 2019 through July 2024. Thirty-five patients with a diagnosis of pancreatic cancer met inclusion criteria. Pain scores and oral morphine equivalent (OME) use was evaluated at baseline (prior to ITDD), at 1 month and 3 months post ITDD implant. A numeric scale of 0-10 was used to determine pain scores, while a comprehensive chart review and patient response was used to determine OME use. **Results:** We found that post ITDD placement, pain scores were reduced on average by 4.55 points at 1month and 4 points at 3-months after implantation. This was statistically significant with a two-paired t-test analysis ($p < 0.05$). Post ITDD placement, there was a 96.7% reduction in OME at 1-month ($p < 0.005$) and 98.1% reduction in OME at 3 months ($p < 0.04$). **Conclusions:** Despite the small sample size of patients, ITDD showed a significant reduction in pain scores and OME use compared to baseline with systemic opioid use only in patients with advanced pancreatic cancer. This reduction appeared to be maintained for 3 months post ITDD placement. Further studies conducted on a larger sample size would be beneficial to demonstrate statistical outcomes of ITDD therapy on pain scores and OME use. Research Sponsor: None.

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Impact of long-term chemotherapy (CTx) on outcomes in pancreatic ductal adenocarcinoma (PDAC): A real-world UK multi-centre study. First Author: Umair Mahmood, Department of Gastrointestinal Oncology, University College London Hospitals NHS Foundation Trust, London, United Kingdom

Background: PDAC is associated with poor outcomes with limited treatment options. Here, we present a multi-institutional review evaluating outcomes following short and long-term CTx with or without treatment breaks in PDAC patients (pts). **Methods:** All consecutive PDAC pts receiving > 3 CTx cycles between 2019 – 2023 at University College London Hospitals and Oxford University Hospitals were included. Treatment response, survival outcomes and predictors of clinical benefit were evaluated. Wilcoxon test, Kaplan-Meier and multivariate Cox regression models were performed. **Results:** Of the 213 screened pts, 127 eligible subjects met the study criteria. 1st, 2nd and 3rd line CTx were received by 127, 40 and 2 pts, respectively. 21 pts had resectable disease (9 remained relapse-free after adjuvant CTx and 8 pts alive at post 2 years follow-up surveillance). 106 pts had unresectable or metastatic disease and were selected for final analyses (12% borderline resectable (BR), 19% locally advanced (LA), 9% localised disease who developed metastases and 60% de novo metastatic pts). 7 pts (7%) pts underwent genetic profiling on pt request or previous clinical trial screening; *KRAS* aberrations (N = 4), actionable *PLAB2* / *BRCA2* mutations (N = 2). BR and LA pts (N = 33) achieved a median PFS1 (Progression Free Survival while on 1st line CTx) of 8.28 (95% Confidence Interval (CI), 5.49 – 15.11) and Overall Survival (OS) of 15.15 (95% CI, 9.46 – 26.41) months (mos). De novo metastatic pts (N = 64) attained a PFS1 of 6.64 (95% CI, 5.98 – 8.18) and OS of 9.30 (95% CI, 8.05 – 12.81) mos. Patient-specific factors in both cohorts comprising of age, gender, performance status, smoking history, comorbidities were not associated with PFS1 and OS. Improved PFS was associated with ≥ 6 cycles of 1st line CTx ($P = < 0.001$), median duration of 1st CTx of ≥ 3.58 mos ($P = < 0.001$) and best response to 1st CTx ($P = 0.007$). A favourable OS was associated with > 1 line of CTx ($P = < 0.001$), ≥ 6 cycles of 1st line CTx ($P = 0.009$), median duration of 1st CTx of ≥ 3.58 mos ($P = < 0.001$) and best response to 1st CTx ($P = 0.002$). Subjects receiving ≥ 6 cycles of 1st CTx were younger than pts tolerating fewer cycles (median age 62.41 vs 72.25 years) ($P = 0.04$). In the localised disease group, improved PFS was associated with ≥ 6 cycles of 1st line CTx ($P = 0.003$), median duration of 1st CTx of ≥ 2.99 mos ($P = 0.008$) and local treatment after commencing 1st line CTx ($P = < 0.001$). These three factors were also associated with OS; $P = 0.02$, $P = 0.02$ and $P = < 0.001$, respectively. Median number and duration of 1st line CTx interruptions were not associated with PFS1 or OS in both cohorts. **Conclusions:** Despite challenges associated with long-term CTx, survival outcomes may be better in pts receiving CTx for longer periods and multiple treatment lines. Genetic profiling currently offers limited value but in carefully selected pts, localised treatment options may be associated with improved survival outcomes. Research Sponsor: None.

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Clinical outcomes of immunotherapy in metastatic pancreatic carcinoma: A systematic review and meta-analysis. First Author: Moazzam Shahzad, H. Lee Moffitt Cancer Center and Research Institute, University of South Florida, Tampa, FL

Background: Metastatic pancreatic carcinoma is one of the most challenging malignancies to treat. Traditional treatment modalities such as chemotherapy and radiation therapy have shown limited efficacy. This meta-analysis aims to investigate immunotherapy as a potential therapeutic option for metastatic pancreatic carcinoma. **Methods:** PubMed, Cochrane, Embase, and Clinicaltrials.gov were searched as per PRISMA guidelines. Data extraction was conducted independently by two reviewers, and any discrepancies were resolved through discussion. Pooled analysis was performed using the 'meta' package by Schwarzer et al. in the R programming language (version 4.16-2). **Results:** 834 patients from the 17 studies reporting immunotherapy outcomes in metastatic pancreatic carcinoma were included in this systematic review and meta-analysis. The studies included (There were) were 4 phase I (23.5%), 2 phase Ib (11.8%), 1 phase Ib/II (5.9%), and 5 phase II (29.4%), 1 retrospective study (5.9%) and 4 prospective studies (23.5%). The median age of the patients was 61 years and 55.6% (398/716) were male. Immunotherapies included were GVAX, tremelimumab, Mesothelin-specific Chimeric Antigen Receptor T cells, durvalumab, pembrolizumab, canerpaturev, Sotigalimab, nivolumab, CO-1.01, allogeneic natural killer cell immunotherapy, Anti-EGFR chimeric antigen receptor-modified T cells, bispecific antibody armed T cells, and clivatuzumab tetraxetan. The median follow-up duration was 17.8 months. The median overall survival was 7.9 months, and the median progressive-free survival was 4.6 months. The pooled overall response rate was calculated at 16% (95% CI 0.09-0.26, I2=70, $p<0.01$, n= 426) and the pooled partial response rate was 13% (95% CI 0.09-0.26, I2= 67%, $p<0.01$, n= 272). The pooled stable disease was calculated to be 28% (95% CI, 0.2-0.39, I2= 59%, $p<0.01$, n=277). The pooled progressive disease was 36% (95% CI, 0.23-0.52, I2=34%, p=0.22, n=68). The most common overall adverse effects were anemia, diarrhea, and thrombocytopenia. **Conclusions:** This meta-analysis demonstrates that novel immunotherapies have promising results on metastatic pancreatic cancer. Future new clinical trials with a follow-up on current early-phase results are needed. Research Sponsor: None.

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A retrospective study of the prognostic value of early induction of S-1 postoperative adjuvant therapy for pancreatic cancer. First Author: Aki Yukutake, Department of Surgery, Nagasaki University Graduate School of Biomedical Sciences, Nagasaki, Japan

Background: The usefulness of oral S-1 administration as adjuvant therapy (AC) after pancreatic cancer surgery has been demonstrated (1). Our study also suggested the importance of maintaining relative dose intensity or postoperative early induction (2). In a large Japanese data set, a comparison of induction before and after the postoperative 10 weeks showed a worse prognosis in the group which started 10 weeks after surgery (3). However, the number of patients of which early induction is possible are gradually increasing with recent improvements of surgical outcomes. In the present study, we retrospectively evaluated the survival benefit of AC which was started within 2 weeks after surgery. **Methods:** 129 patients with invasive pancreatic cancer who underwent radical surgery from January 2018 to May 2023 were enrolled in this retrospective study. The patients were divided into the following groups: 46 (36%) underwent induction within 2 weeks, 56 (43%) within 3-6 weeks, 13 (10%) after 7 weeks, and 14 (10%) without AC (4 with low renal function, 4 with deteriorating physical condition after surgery, 4 with not desired, and 2 with comorbid disease treatment). The background of each group was as follows (Described in the following order; induction within 2 weeks after surgery: 3-6 weeks: after 7 weeks: without AC), age 71: 71: 71: 78 (n.s.), pancreatic head cases 52: 68: 69: 50% (n.s.), BR cases 35: 23: 8: 21% (n.s.), NAC performed 86: 38: 15: 29% ($p < 0.001$), preoperative CEA ≥ 5.0ng/ml 24: 23: 8: 21% (n.s.), preoperative CA19-9 ≥ 37U/ml 48: 52: 54: 64% (n.s.), PV/SMV reconstruction 20: 16: 31: 14% (n.s.), postoperative complications (Clavien Dindo ≤ 3) 0: 16: 31: 43% ($p < 0.001$), Pathological stage 2 or higher 83: 65: 77: 79% (n.s.). **Results:** The need for reduction of S-1 dosage was 87: 89: 92% of the patients (n.s.), and the 6-month completion rate was 70: 71: 61% (n.s.). Median overall survival duration in each group was as follows: not reached: 49.5: 35.5: 48.6 months, and 1-year postoperative recurrence-free survival was 73: 66: 45: 71%, median recurrence free survival time was 34.0: 20.8: 10.3: 12.1 months, and induction within 2 weeks after surgery group significantly improved the recurrence-free survival in comparison with the group of after 7 weeks ($p < 0.05$). No significant factors were detected in multivariate analysis. **Conclusions:** These results suggest that early induction of AC as a result of improvement of surgical outcomes would contribute to postoperative recurrence-free survivals. 1. Uesaka et al, Lancet, 2016. 2. Matsushima et al. Anticancer Res, 2022. 3. Tomimaru et al, J Gastroenterol, 2023. Research Sponsor: None.

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Overall survival (OS) of patients with metastatic pancreatic ductal adenocarcinoma (mPDAC) treated with first-line (1L) FOLFIRINOX (FFX): Bridging the gap between the NAPOLI 3 trial and real-world practice. First Author: Paul Cockrum, Ipsen Biopharmaceuticals, Inc., Crowley, TX

Background: NALIRIFOX (liposomal irinotecan + 5-fluorouracil/leucovorin + oxaliplatin) demonstrated significantly improved OS compared to 1L gemcitabine + nab-paclitaxel in the phase 3 NAPOLI 3 trial (NCT04083235). The median OS (95% CI) in the NALIRIFOX intention-to-treat population (N= 383) was 11.1 (10.0, 12.1) months. However, the relative efficacy of NALIRIFOX versus FFX (irinotecan + 5-fluorouracil/leucovorin + oxaliplatin), another standard of care for 1L treatment of mPDAC, has not been studied. To further contextualize findings from NAPOLI 3 trial, the current study described OS among patients treated with 1L FFX in the real-world setting. **Methods:** Three cohorts of patients with mPDAC treated with 1L FFX were identified from the Flatiron Health database. The all-comer cohort included all adult patients with mPDAC treated with 1L FFX since January 2014. Among them, the trial-aligned cohort included patients with mPDAC treated with 1L FFX between January 1, 2020 and July 31, 2022 (to align with the time period of NAPOLI 3 trial) who met the eligibility criteria of NAPOLI 3. Finally, a subgroup of patients in the trial-aligned cohort who received modified FFX (mFFX) regimen composed the trial-aligned mFFX cohort. OS was assessed using Kaplan-Meier analysis. **Results:** The all-comer cohort consisted of 3,271 patients. The trial-aligned cohort included 219 patients, of whom 154 (70%) were treated with mFFX. The distribution of baseline characteristics was similar across three cohorts (Table). The median age of the trial-aligned cohort was 65 years old, with 46% of patients being female, 67% being White, and 47% having ECOG score of 1. The median OS (95% CI) in the all-comer cohort and trial-aligned cohort was 9.0 (8.5, 9.3) months and 9.1 (7.8, 10.9) months, respectively; the median OS (95% CI) was 8.6 (7.3, 10.5) months in the trial-aligned mFFX cohort. **Conclusions:** NALIRIFOX regimen in the NAPOLI 3 trial showed numerically improved OS compared to FFX, including mFFX, in the real-world setting. Analysis adjusting for baseline characteristics are warranted and will provide further insights for the comparative efficacy of the two regimens. Research Sponsor: Ipsen Corp.

Characteristics of real-world FOLFIRINOX cohorts.

	All-comer cohort (N=3,271)	Trial-aligned cohort (N=219)	Trial-aligned mFFX cohort (N=154)
Baseline characteristics			
Age, years, mean (SD)	63.5 (9.0)	64.6 (8.2)	65.1 (8.5)
Female, n (%)	1,377 (42.1)	100 (45.7)	68 (44.2)
White, n (%)	2,118 (64.8)	146 (66.7)	100 (64.9)
ECOG score = 1, n (%)	1,327 (48.9)*	103 (47.0)	77 (50.0)
Endpoint			
Median OS, months (95% CI)	9.0 (8.5, 9.3)	9.1 (7.8, 10.9)	8.6 (7.3, 10.5)

Abbreviations: CI, confidence interval; ECOG, Eastern Cooperative Oncology Group; OS, overall survival; SD, standard deviation.
*Proportion calculated based on 2,711 patients with ECOG available at baseline.

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Real-world clinical outcomes of locally advanced unresectable and de-novo metastatic pancreatic ductal adenocarcinoma: A multicenter retrospective study from Saudi Arabia. First Author: Mohamed Aseafan, Section of Medical Oncology, Department of Internal Medicine, Security Forces Hospital, Riyadh, Saudi Arabia

Background: Pancreatic ductal adenocarcinoma (PDAC) has dismal clinical outcomes, even with existing treatment regimens. This study evaluates the clinical characteristics and outcomes of patients with locally advanced unresectable or de-novo metastatic PDAC in Saudi Arabia to provide real-world data that can be compared to international benchmarks. **Methods:** This is a retrospective, multicenter study. Patients diagnosed with unresectable locally advanced or de-novo metastatic PDAC between January 2015 and November 2023 were included. Data were collected from 10 oncology centers across Saudi Arabia. Patient characteristics, treatment regimens, and clinical outcomes, including progression-free survival (PFS) and overall survival (OS), were analyzed using univariate and multivariate analyses. **Results:** A total of 350 patients were identified, with a median age of 60 years at time of diagnosis, 63% of patients presenting with multiple metastatic sites, primarily in the liver (66.3%). Among patients tested, 7.1% had deficient mismatch repair (d-MMR), and 5.8% harbored BRCA mutations. FOLFIRINOX was the most common first-line treatment (55.1%), followed by gemcitabine plus nab-paclitaxel (15.1%). The median PFS for first-line treatment was 5.3 months, with FOLFIRINOX achieving the longest PFS (6.5 months). The median OS was 10.34 months for the entire cohort, with better survival outcomes observed in patients receiving FOLFIRINOX (12.3 months). Independent prognostic factors for PFS and OS included performance status, type of first-line regimen, and neutrophil-lymphocyte ratio (NLR). **Conclusions:** This real-world study confirms that clinical outcomes for locally advanced unresectable and metastatic PDAC in Saudi Arabia are consistent with international data, with trend of FOLFIRINOX showing superior outcomes over gemcitabine-based regimens. However, both treatments reflect the persistent poor prognosis of PDAC, underscoring the need for novel therapeutic strategies. Further research is warranted to optimize treatment selection and improve survival outcomes in this population. Research Sponsor: None.

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Real-world dose intensity (DI) and time to adverse events (AEs) in a commercial database during first-line (1L) treatment of metastatic pancreatic adenocarcinoma (mPDAC) with FOLFIRINOX (FFX), FFX without 5FU bolus, and gemcitabine + nab-paclitaxel (GnP). First Author: Syvart Dennen, Genesis Research Group, Hoboken, NJ

Background: Onset of severe AEs during 1L FFX and GnP treatment for mPDAC negatively impacts clinical outcomes, requiring supportive care and/or treatment adjustment. Occasionally they lead to discontinuation. The 5FU bolus may be omitted from FFX (FFXnb) to increase tolerability. We examined DI and time to key AEs (anemia, neutropenia, thrombocytopenia, diarrhea, fatigue, nausea/vomiting, and neuropathy) during 1L FFX, FFXnb, and GnP. **Methods:** This retrospective study utilized Optum Market Clarity claims + EHR linked data. Inclusion criteria were: adults diagnosed with mPDAC between 1/1/2015 and 5/31/2023; initiated 1L FFX, FFXnb, or GnP (index date) within -14 to +90 days; ≥6 pre- and ≥1 month post-index enrollment; ≥1 AE of interest in 1L. Follow-up was index to earliest of death, disenrollment, start of 2L, or last administration + 14 days. DI was defined as cumulative dose divided by follow-up in months. Grade 3/4 hematologic AEs (hAEs) were measured from lab values. Ungraded AEs were sourced from EHR notes and claims diagnoses. FFX was defined as including 5FU bolus + infusion in the 1st cycle; FFXnb as 5FU infusion only. Kaplan-Meier (KM) methods estimated time AE. **Results:** Median follow-up in months for patients (pts) with hAEs was: FFX 4.2; mFFX 3.6; GnP 3.1. Median DI (mg/m2/month) for FFX vs FFXnb pts with hAEs was: 5FU 4567 vs 4581; leucovorin 632 vs 398; oxaliplatin 119 vs 142; irinotecan 253 vs 251. For GnP, DI was: gemcitabine 2230; nab-paclitaxel 269. The table below presents sample size (N) and median (95% CI) time to AE. Median time to anemia was numerically 2–4 weeks shorter for GnP and FFXnb vs FFX. Time to neutropenia was 8–12 days less for FFXnb vs GnP and FFX. Time to thrombocytopenia was 7 weeks less for GnP vs FFX and FFXnb. Time to neuropathy was nearly 4 weeks shorter for FFXnb vs FFX and GnP. **Conclusions:** While neutropenia is higher risk for FFX vs GnP, prophylactic growth factor is standard during FFX, which may result in the similar onset time vs GnP. Early onset of severe thrombocytopenia differentiates the GnP sample. Oxaliplatin DI was higher for FFXnb vs FFX pts, and may be associated with earlier onset of anemia, neutropenia, diarrhea, and neuropathy. Research is merited on interactions between dosing, AE timing, and treatment discontinuation. Research Sponsor: Ipsen Biopharmaceuticals, Inc.

Time to AE.

Grade 3/4 hAEs	FFX N	FFXnb N	GnP N	FFX median days (95% CI)	FFXnb median days (95% CI)	GnP median days (95% CI)
Anemia	47	25	97	64 (38–96)	36 (19–70)	47 (39–56)
Neutropenia	75	53	112	35 (25–47)	23 (14–43)	31.5 (18–42)
Thrombocytopenia	33	22	54	63 (35–79)	65.5 (32–94)	14 (13–34)
Ungraded AEs						
Diarrhea	216	264	297	27.5 (22–36)	17.5 (14–27)	27 (21–36)
Fatigue	211	267	418	34 (23–43)	27 (17–34)	27 (21–30)
Nausea/vomiting	337	403	561	6 (3–8)	9 (7–14)	9 (7–12)
Neuropathy	151	155	190	92 (70–112)	67 (48–79)	92 (80–111)

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Real-world excess costs associated with hematologic adverse events (hAEs) during first-line (1L) treatment of metastatic pancreatic adenocarcinoma (mPDAC) with FOLFIRINOX (FFX), FFX without 5FU bolus, and gemcitabine+nab-paclitaxel (GnP). First Author: Syvart Dennen, Genesis Research Group, Hoboken, NJ

Background: During 1L treatment of mPDAC with FFX and GnP, hAEs can harm patients (pts) and accrue excess costs to the health system, i.e., costs beyond those of a treated pt without the hAE. This study examined excess total cost of care (TCoC) and components (total inpatient; outpatient transfusion, granulocyte colony-stimulating factor [G-CSF], and thrombopoietic growth factor [TGF]) for three key hAEs (anemia, neutropenia, thrombocytopenia) during 1L FFX, FFX without 5FU bolus (FFXnb), and GnP. **Methods:** This retrospective observational study utilized Optum Market Clarity claims + EHR linked data. Inclusion criteria were: adult pts diagnosed with mPDAC between 1/1/2015 and 5/31/2023; initiated 1L FFX, FFXnb, or GnP within -14 to +90 days (index date); ≥6 months pre-index enrollment. hAEs during 1L were detected from lab values and grouped all grades combined. Pts were divided into "hAE groups" and "ref. groups" with and without any of the three hAEs of interest, respectively. Pts were matched 1:1 without replacement based on follow-up time. Costs were measured from the day a lab value indicated an hAE until 30 days later and during the corresponding times for ref. pts, then standardized to per-pt-per-month (PPPM). This study did not adjust for differences in pt characteristics. **Results:** Means and differences (95% CI) in TCoC following hAE detection for each regimen and hAE of interest are presented (Table). PPPM TCoC was greater for all three regimens and hAEs, driven by 1.6–4.9 times higher inpatient costs. Inpatient costs were $5,614 and $5,665 higher for FFX and FFXnb neutropenia pts vs. ref. groups. Use of outpatient G-CSF at any time during 1L was lower for FFX and FFXnb neutropenia pts, at 76% and 63% vs. 83% and 75% in the ref. groups, respectively. Outpatient TGF and transfusions were rare. **Conclusions:** All hAEs showed excess PPPM TCoC across regimens, driven by inpatient costs. FFX and FFXnb neutropenia pts had lower overall utilization of outpatient G-CSF but higher inpatient costs following detection of neutropenia, which may represent an opportunity for improved pt management. Research Sponsor: Ipsen Biopharmaceuticals, Inc.

Excess TCoC of hAEs.

AE (any grade)	FFX, hAE group (N=199, 127, 157)	FFX, ref. group (N=199, 127, 157)	Diff. (95% CI)	FFXnb, hAE group (N=202, 106, 137)	FFXnb, ref. group (N=202, 106, 137)	Diff. (95% CI)	GnP, hAE group (N=406, 268, 270)	GnP, ref. group (N=406, 268, 270)	Diff. (95% CI)
Anemia TCoC, mean	30,561	22,310	8,251 (4,396–12,106)	31,866	23,380	8,487 (4,007–12,966)	32,662	27,697	4,965 (1,320–8,610)
Neutropenia TCoC, mean	30,382	24,952	5,430 (-31–10,891)	26,588	22,571	4,017 (-1,288–9,322)	32,975	31,051	1,924 (-4,629–8,477)
Thrombocytopenia TCoC, mean	30,566	22,535	8,031 (3,895–12,166)	31,434	22,958	8,476 (2,962–13,991)	33,863	30,751	3,113 (-3,310–9,536)

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Costs of generic vs. branded nab-paclitaxel (nP) during first-line (1L) treatment of metastatic pancreatic adenocarcinoma (mPDAC) with gemcitabine + nab-paclitaxel (GnP) in a real-world commercial database. First Author: Syvart Dennen, Genesis Research Group, Hoboken, NJ

Background: In 2013, the FDA approved GnP for 1L treatment of mPDAC. The total cost of care (TCoC) to payers during treatment is driven by the cost of nP, which first faced competition from generics in April 2022. We examined cost per generic vs. branded nP claim and cost per patient (pt) before and after generic entry. **Methods:** This retrospective observational study utilized Optum Market Clarity claims + EHR data. Inclusion criteria were: adult pts diagnosed with mPDAC between January 1, 2015 and May 31, 2023; initiated 1L GnP (index date) -14 to +90 days from diagnosis; 6 months pre-index enrollment. Claims were grouped into branded vs. generic based on NDC. For per-pt-per-month (PPPM) analyses, pts were required to have ≥30 days post-index enrollment and no nP claims missing NDCs. Pts were grouped into 1) branded pts with all nP claims for the branded NDC, and 2) generic pts with all claims for generic NDCs. TCoC, inpatient, outpatient, and chemotherapy drug cost to payers were calculated. The full study period and the period of generic entry beginning April 1, 2022 were examined. **Results:** Costs for branded vs. generic nP claims and GnP pts for 1L treatment initiated April 1, 2022 and after are shown (Table). The mean cost of generic claims was affected by outliers and was more than twice that of branded. The median nP claim cost, less affected by outliers, was still $2,408 greater for generic vs. branded. Costs for branded nP claims were similar before and after generic entry (not shown). 20 1L GnP pts had only branded nP claims, 24 had only generic, and 156 had mixed or were missing ≥1 NDC. PPPM mean TCoC was approximately twice as high for generic vs. branded. Medians were more similar, with generic median $2,037 higher vs. branded. Differences were driven by drug cost; other inpatient and outpatient costs were similar between the two groups (not shown). Costs of branded GnP pts before and after generic entry were similar (not shown). **Conclusions:** Publicly available prices for branded nP are $1,673 vs. $1,460 for generic per 100mg. In real-world data, we found generic nP costs per claim and PPPM tended to be higher than branded. While GnP is accepted by payers and clinicians as a standard 1L regimen for mPDAC, its real-world drug cost is at the level of branded therapy, despite generic availability. Future research should evaluate this finding as more post-generic-entry data becomes available. Research Sponsor: Ipsen Biopharmaceuticals, Inc.

Branded vs generic costs.

	Statistic	Branded	Generic
Claims	**N**	**343**	**308**
Cost per claim	Mean (SD)	4,721 (2,197)	11,761 (40,559)
	Median (IQR)	3,482 (2,945–6,583)	5,890 (3,482–6,964)
Patients	**N**	**20**	**24**
TCoC, PPPM	Mean (SD)	25,410 (9,531)	52,129 (121,166)
	Median (IQR)	22,276 (17,911–32,180)	24,313 (17,231–36,774)
GnP drug cost PPPM	Mean (SD)	11,056 (4,334)	35,910 (118,205)
	Median (IQR)	10,162 (7,658–13,508)	10,554 (7,162–16,797)

695 Poster Session

Real-world use of supportive medications to prevent and manage key adverse events during first-line (1L) treatment of metastatic pancreatic adenocarcinoma (mPDAC) with FOLFIRINOX (FFX), FFX without 5FU bolus, and gemcitabine+nab-paclitaxel (GnP). First Author: Syvart Dennen, Genesis Research Group, Hoboken, NJ

Background: 1L treatment of mPDAC with FFX or GnP typically requires supportive care to alleviate toxic effects. Two of the most common adverse events are neutropenia, which may be treated with granulocyte colony-stimulating factor (G-CSF), and diarrhea, often treated with atropine and/or loperamide. Medications may be administered as prophylaxis (PPx) or to manage symptoms as they occur. Other methods to improve tolerability include modifying regimens with lower doses or omitted components, such as removing the 5FU bolus from FFX (FFXnb). This study examines PPx vs non-PPx G-CSF, atropine, and loperamide during 1L treatment with FFX, FFXnb, and GnP. **Methods:** This retrospective observational study utilized Optum Market Clarity claims + EHR linked data. Inclusion criteria were: adult patients (pts) diagnosed with mPDAC between January 1, 2015 and May 31, 2023; initiated treatment (index date) with 1L FFX, FFXnb, or GnP -14 to +90 days from diagnosis; no evidence of other cancers in 6 months pre-index continuous enrollment; at least one EHR record post-index. G-CSF PPx was defined as administration/fill within 7 days of the start of a treatment, inclusive. PPx with atropine or loperamide was defined as administration/fill the day before or day of a treatment; all other use was categorized "non-PPx". Pts may appear in both PPx and non-PPx categories. **Results:** Rates of overall use, PPx, and non-PPx for each medication of interest are shown (Table). The overall G-CSF use rate was >3.5 times higher for FFX and FFXnb vs GnP. While ≥70% of FFX/FFXnb pts received PPx, non-PPx still occurred in more than 1 in 8 pts. One-fifth of FFX and FFXnb pts received atropine at any time, vs. just 6% of GnP pts. Atropine PPx was less common than non-PPx for FFX/FFXnb and was rare in GnP pts. Similar to the other medications, loperamide was observed more for FFX and FFXnb vs. GnP, but was the least common of the medications, used by <10% of pts in any regimen. PPx and non-PPx use were similar. Use of the three medications was similar between FFX and FFXnb. **Conclusions:** In a RW cohort of 1L mPDAC pts, G-CSF and antidiarrheal use was more common in FFX and FFXnb than GnP pts. Most FFX and FFXnb pts received PPx G-CSF, reflecting clinician efforts to manage the high risk of neutropenia with these regimens. Non-PPx G-CSF still occurred in 14–18% of pts. FFXnb had similar rates of concomitant medications as FFX, indicating that removing the 5FU bolus may have minor impact on the need for some supportive medications. Research Sponsor: Ipsen Biopharmaceuticals, Inc.

Concomitant medication use.

Medication		1L FFX (n=511)	1L FFXnb (n=583)	1L GnP (n=958)
G-CSF	% treated overall	77%	71%	20%
	% PPx	76%	70%	17%
	% non-PPx	18%	14%	9%
Atropine	% treated overall	20%	20%	6%
	% PPx	11%	9%	2%
	% non-PPx	15%	16%	5%
Loperamide	% treated overall	7%	9%	3%
	% PPx	4%	5%	1%
	% non-PPx	4%	6%	2%

696 Poster Session

Real-world microscopic residual disease (MRD) monitoring with circulating tumor DNA (ctDNA) in pancreatic adenocarcinoma (PDAC). First Author: Peter Chi-Win Li, Brooke Army Medical Center, Ft Sam Houston, TX

Background: ctDNA is a powerful tool that can detect MRD with limited but emerging data in patients (pts) with PDAC. We describe the clinical outcomes of pts with PDAC who underwent real-world MRD testing at our institutions. **Methods:** We retrospectively analyzed pts at 2 institutions with ≥1 tumor-informed ctDNA test (Natera, Inc) ordered for MRD testing post-operation (post-op) between June 2020-August 2024. Descriptive statistics were used to characterize clinicopathologic factors, ctDNA positivity (ctDNA+), and relapse rates (RR). Median relapse-free survival (mRFS) and overall survival (mOS) were calculated using Kaplan-Meier methods, comparisons between groups by log-rank test, and hazard ratios (HR) by Cox proportional hazard models. Associations between clinicopathologic factors and outcomes were assessed by Fisher's exact and Chi-square tests. **Results:** Of the 54 pts who underwent MRD testing, 48 pts (88.9%) had successful tests. Failures were due to insufficient tissue. Patients with successful tests were staged as follows: 31.2% stage 1, 50.0% stage 2, and 18.8% stage 3, with 12.5% having R1/2 resections and 46.8% being node-positive. With a median follow-up of 21.0 months (mo), 22 (45.8%) pts had relapsed and 22 (45.8%) were anytime-ctDNA+. Sensitivity and specificity of ctDNA for relapse was 77.3% and 86.4%, respectively. Between anytime- and never-ctDNA+ pts, RR were 77.0% vs. 23.1%, mRFS 14.2 mo vs. not reached (NR) (HR 4.9, 95% CI 1.9-12.9), and mOS 31.7 mo vs. NR (HR 3.3, 95% CI 1.0-10.9), respectively. For 27 pts tested in the MRD window (2-12 weeks post-op), mRFS was 6.6 vs. 25.0 mo (HR 3.1, 95% CI 1.0-9.4) and mOS was 25.5 mo vs. NR (HR 2.5, 95% CI 0.6-10.2) in ctDNA+ vs. negative pts. ctDNA+ preceded radiographic relapse in 54.5% of patients with a median lead time of 166 days (IQR 70-332). In anytime-ctDNA+ pts, there were non-statistically significant trends for improved RFS and OS based on ctDNA decrease and clearance (table). No clinicopathologic factors predicted ctDNA+. Post-op ctDNA+ (p <0.001) and CA 19-9 (p=0.03) were the only factors that had a significant association with RFS. **Conclusions:** Post-op ctDNA is a strong prognostic marker for survival in PDAC, predicting relapse about five months before imaging. ctDNA+ status is independent of clinicopathologic factors and the strongest predictor of recurrence. These findings highlight a window where novel therapies could be tested to eradicate MRD. Longer follow up and prospective studies are needed to validate its predictive role with therapies in the MRD setting to improve outcomes in high-risk pts. Research Sponsor: None.

		RFS events/ total	mRFS (mo)	HR (95% CI)	p	OS events/ total	mOS (mo)	HR (95% CI)	p
ctDNA clearance	Yes	6/9	17.0	0.61 (0.2-1.7)	0.35	2/9	NR	0.37 (0.08-1.8)	0.20
	No	11/13	10.3			8/13	31.7		
ctDNA MTM/mL reduction	≥50%	8/11	15.5	0.93 (0.3-2.9)	0.90	3/11	NR	0.54 (0.1-2.4)	0.42
	<50%	6/8	13.6			4/8	31.7		

697 Poster Session

Changes in peri-operative chemotherapy and radiation use among patients with resected pancreatic ductal adenocarcinoma (PDAC) between 2004-2021 in the US SEER dataset. First Author: Alan Su, Providence Cancer Institute, Portland, OR

Background: PDAC peri-operative management changed over 20 years. Randomized data supports adjuvant and multiagent chemotherapy (CT), while peri-operative chemoradiotherapy (RT) was used selectively in the US. We examined recent trends in treatment with outcomes in surgically resected PDAC patients. **Methods:** Using the Surveillance, Epidemiology, and End Results (SEER) 17 dataset, we identified surgically resected primary M0/MX PDAC cases from 2004-2021 over 3-year periods (P1-P6). Neoadjuvant (NCT), adjuvant (ACT), or multiagent (MACT) CT scheduling was included. χ^2 analysis assessed cohort characteristics over time and multinomial logistic regression assessed odds of CT allocation adjusted by cohort characteristics. Multivariable Cox proportional hazards model assessed treatment-related overall survival (OS) adjusted for cohort characteristics. **Results:** Among 55,293 cases of M0/MX PDAC, 21,904 (40%) completed surgery. Median age was 66 years; stages were T1/T2/T3/T4 (8.0/25/59/6.0%) and N0/N+ (38/54%). RT use declined (P1:P6, 41%->18% p<0.001) while CT use increased (P2:P6, 63%->79%, p<0.001). NACT (P2:P6, 4.1%->23% p<0.001) and MACT (P2:P6, 1.6%->20%, p<0.001) use increased while ACT decreased (P2: P6, 58%->36%, p<0.001). CT is likely to be allocated to young patients (<65yo) with Stage II-III tumors located at the pancreatic head (OR>1.00, p<0.001). Median OS increased over time (P1: P6, 18mo->32mo, p<0.001). Multivariable analysis shows that Adjuvant RT (HR 0.84 [0.81-0.87], p<0.001) and all CT (NCT: HR 0.71 [0.66-0.77], p<0.001; ACT: HR 0.63 [0.60-0.66], p<0.001; MACT: HR 0.65 [0.60-0.71], p<0.001) improved OS. **Conclusions:** This study shows decreased RT and increased CT adoption from 2004-2021. Median OS improved over the years, potentially due to peri-operative advances during this study period. Research Sponsor: None.

Trends in treatment use over 3-year periods.

	RT N (%)	CT N (%)	Neo CT N (%)	MACT N(%)
P1: 2004-2006 N=2661	1087 (41)	0 (0)[a]	0 (0)[a]	0 (0)[a]
P2: 2007-2009 N=3185	1185 (37)	2021 (63)	129 (4.1)	50 (1.6)
P3: 2010-2012 N=3536	1226 (35)	2460 (70)	267 (7.6)	138 (3.9)
P4: 2013-2015 N=3947	1210 (31)	2880 (73)	434 (11)	244 (6.2)
P5: 2016-2018 N=4224	971 (23)	3256 (77)	676 (16)	491 (12)
P6: 2019-2021 N=4351	787 (18)	3430 (79)	1003 (23)	857 (20)

[a]No cohorts reported chemotherapy use during period 1.
RT: Radiation Therapy; CT: Chemotherapy; NCT: Neoadjuvant Chemotherapy; MACT: Multiagent Chemotherapy.

698 **Poster Session**

Investigating communication barriers impacting postoperative decision regret in patients with pancreatic ductal adenocarcinoma. First Author: Judy Li, Department of Surgery, Division of Surgical Oncology, Icahn School of Medicine at Mount Sinai, New York, NY

Background: Patient communication barriers are associated with worse quality of care and clinical outcomes. However there are limited data exploring their impact on shared decision-making (SDM) in the preoperative setting, especially for pancreatic ductal adenocarcinoma (PDAC). This study investigates the impact of patient communication barriers on patient-physician communication surrounding the decision to pursue pancreatectomy. **Methods:** This cross-sectional study investigates the impact of communication barriers on the degree of decisional regret (DR) in patients with PDAC who have undergone curative-intent resection at least 6 months prior to study recruitment. Patients completed validated surveys assessing communication preferences, SDM participation, and DR. Health literacy was assessed by the BRIEF Health Literacy survey. Groups were stratified by presence or absence of DR, assessed by the Decisional Regret Scale (DRS). Characteristics and survey scores were compared by chi-square and independent t-test. Given significant findings, results of this interim analysis are reported below. **Results:** 45 patients met inclusion criteria and completed all questionnaires. 19 (42.2%) patients expressed regret about their decision to pursue surgery with 5 patients expressing moderate to severe regret (DRS $\geq$25). Baseline characteristics were similar between groups. There were no significant differences in neoadjuvant or adjuvant treatments, or operative characteristics. Both groups reported high participation in preoperative SDM (average score 34.47 $\pm$ 9.67 in DR group vs 36.08 $\pm$ 8.88, P=0.567). Despite similar levels of education between groups, those expressing regret had a lower level of health literacy (14.68 $\pm$ 4.10 vs 17.35 $\pm$ 2.07, P=0.016). Fewer patients also identified English as their primary language in the DR group (78.9% vs 100%, P=0.026). While most patients preferred an active role in the final decision making process, fewer patients in the DR group reported this happening in actuality (k= 0.635 vs k=0.934). **Conclusions:** Almost half of patients expressed some regret pursuing surgery for PDAC. Despite similar educational backgrounds, a higher proportion of those expressing regret did not identify English as their primary language and had lower health literacy. While there was high participation in SDM, there was greater discordance between preferred and actual roles in the final decision making process. This identifies an important disparity to address in future preoperative discussions. Research Sponsor: None.

699 **Poster Session**

Assessing impact of postoperative outcomes and quality of life on decisional regret after resection of pancreatic ductal adenocarcinoma. First Author: Judy Li, Department of Surgery, Division of Surgical Oncology, Icahn School of Medicine at Mount Sinai, New York, NY

Background: Surgery is potentially curative for patients with localized pancreatic ductal adenocarcinoma (PDAC). Given the high rate of complications, decline in quality of life (QoL) postoperatively, and high rates of recurrence, clear preoperative communication and expectation setting is critical. This study investigates clinical outcomes that affect postoperative decisional regret (DR) in patients with PDAC. **Methods:** This mixed-methods study investigates the impact of clinical outcomes and QoL on the degree of DR in patients with PDAC who have undergone curative-intent resection at least 6 months prior to study recruitment. Patients completed validated surveys assessing DR and QoL by the European Organization for Research and Treatment of Cancer (EORTC). Groups are stratified by presence or absence of DR, assessed by the Decisional Regret Scale (DRS). Characteristics and survey scores were compared by chi-square, independent t-test, and median test. Narrative responses were thematically analyzed. Given significant findings, results of this interim analysis are reported below. **Results:** 45 patients met inclusion criteria and completed all questionnaires. 19 (42.2%) patients expressed regret about their decision to pursue curative-intent surgery, with 5 patients expressing moderate to severe regret (DRS $\geq$ 25). There was no significant difference in median postoperative timing of survey completion between groups (56 months in DR group vs 34 months, P=0.465). Baseline characteristics, medical treatments, and operative approaches were similar. Both groups had similar recurrence rates at time of survey completion. While 30 and 90 day readmission rates were similar, a greater proportion of the DR group had 30 day complications (42.1% vs 15.4%, P=0.045). Those expressing regret reported poorer physical functional status (mean score 79.30 $\pm$ 18.84 vs 92.31 $\pm$ 10.27, P=0.011) and greater impact on social activities (mean score 67.54 $\pm$ 33.55 vs 84.67 $\pm$ 19.19, P=0.038). Thematic analysis of narrative responses revealed that a majority of patients expressed a strong desire to have a greater focus on potential QoL changes during preoperative discussions. **Conclusions:** Patients expressing postoperative DR experienced more 30 day complications, and still report lasting effects on their current physical status and social activities. This emphasizes the importance of thorough preoperative counseling on surgical risks and potential changes in postoperative QoL so that patients may make the best informed decision. Research Sponsor: None.

700 **Poster Session**

Comparison of demographics, mortality outcomes and resource utilization in early-onset versus later-onset pancreatic cancer: An analysis of the US National Inpatient Sample (NIS). First Author: Sneha Singh, Sunrise Hospital and Medical Center, Las Vegas, NV

Background: Pancreatic cancer is associated with higher mortality, with 5-year relative survival of 12%. Early-onset pancreatic cancer (EOPC) is relatively uncommon and is seen in 5-12% of the cases. Recent studies indicate a rising trend in the incidence of EOPC. In this study, we categorized EOPC for ages less than 50 and later-onset pancreatic cancer (LOPC) for ages above or equal to 50 years. The aim of the study is to understand difference in demographics, mortality, and resource utilization between the two subcategories of patients admitted to acute care hospitals in the United States. **Methods:** This is a retrospective cohort study of adult patients hospitalized with pancreatic cancer as the primary diagnosis at acute care hospitals across the United States in 2021, using the National Inpatient Sample (NIS) database. International Classification of Diseases, 10th Revision (ICD-10) codes were utilized to identify patients with a primary diagnosis of pancreatic cancer. The primary outcome is comparing demographics and in-hospital mortality between EOPC and LOPC. The secondary outcomes are length of stay (LOS) and total hospitalization charges. We performed multivariate logistic regression analysis to assess in-hospital mortality, LOS, and total hospitalization charges after adjusting for potential confounders such as sex, race, Charlson comorbidity index, hospital teaching status, and hospital region. **Results:** Our study included 38,654 patients with pancreatic cancer, of which 2,242 were EOPC and 36,412 were LOPC. The majority of the baseline characteristics were similar in both groups, except that EOPC has a higher Hispanic population (16.06% versus 8.77 %, $p < 0.001$), higher Medicaid as the payer source (28.28 % versus 7.32%, $p < 0.001$), and a more significant number of patients admitted to an urban teaching hospital (89.2 % versus 85.2%, $p < 0.05$) compared to LOPC subpopulation. No statistically significant difference in mortality was found between the two sub-categories (odds ratio 0.97, 95 % CI 0.60-1.5). Charlson index, hospital location, hospital bed size, and hospital region are independent predictors of mortality. There was no statistically significant difference between the two sub-categories for mean length of stay and total hospitalization charges. **Conclusions:** No significant difference was observed between EOPC and LOPC regarding in-hospital mortality, total length of stay, and total hospitalization charges. Our study highlights that EOPC disproportionately affects the Hispanic population and also a greater number of EOPC patients being admitted to urban teaching hospitals. Future research is needed to understand the disproportionate effect on the Hispanic population in the EOPC group and continued research in the domain, as the incidence of early-onset pancreatic cancer is expected to rise. Research Sponsor: None.

701 **Poster Session**

Understanding rates of attrition and barriers to surgical resection in patients undergoing neoadjuvant therapy for localized pancreatic cancer. First Author: Erin M Dickey, Department of Surgery, University of Miami Miller School of Medicine, Miami, FL

Background: Treatment paradigms are shifting toward neoadjuvant therapy (NAT) for resectable (R) and borderline resectable (BR) pancreatic cancer (PC) based on the rationale of treating occult metastatic disease, potentially downstaging patients (pts) to resectability if anatomically unresectable and, most importantly, selection of biology and pts most likely to benefit from potentially morbid surgery. While the benefits of NAT for PC have been well described, what remains underreported are the true attrition rates and underlying reasons for failure to proceed to surgical resection among pts receiving NAT. Herein, we present true intent-to-treat (ITT) attrition data of pts undergoing NAT for localized PC (LPC). **Methods:** Pts with LPC and who underwent $\geq$ 1 cycle NAT from January 2013-2022 were included. Clinicopathologic data were compared between those who did and did not proceed to pancreatectomy. Multivariable models identified factors associated with odds of attrition. Overall survival (OS) was estimated using the Kaplan-Meier method. **Results:** 427 LPC pts received NAT, including 57 (13%) with R, 133 (31%) BR, and 237 (56%) locally advanced (LA) disease. 64% had head tumors with median tumor size of 3.3 cm (IQR 2.7-4.2) and median CA 19-9 of 267 U/mL (IQR: 58-766). 60% of pts received FOLFIRINOX and 28% Gemcitabine-Abraxane for a median of 4.2 months (IQR 2.5-5.7). During NAT, 22% switched regimens and 30% received radiation. Following NAT, 207 (48%) pts underwent attempted resection, and 182 pts (43%) had successful pancreatectomy. The rate of attrition among all LPC pts was 52%, of which 20% had disease progression, 18% remained inoperable, 11% had physiologic decompensation, 2% were lost to follow-up, and 1% declined surgery. Rates of attrition were: 23% for R, 44% for BR, and 73% for LA ($p<0.01$). Higher stage was associated with increased odds of attrition (OR 9.7, 95% CI 4.8 – 19.5 for LA and OR 3.0, 95% CI 1.4 – 6.1 for BR compared to R). The most common reasons for attrition were: metastatic progression (5% in R, 18% in BR, 26% in LA), the primary tumor was deemed inoperable (2% in R, 12% BR, 32% LA), or physiologic decompensation on NAT (7% R, 11% BR, or 12% LA). Pts undergoing pancreatectomy had a median OS of 30 vs 16 months for those not ($p<0.01$). ITT median OS inversely correlated with stage – R: 30, BR: 21 and LA: 18 months ($p<0.01$). There was no difference in OS for those who received NAT and resection – R: 31, BR: 27, and LA: 34 months (p=0.66). **Conclusions:** At a tertiary center with 87% BR and LA pts, 43% of pts proceeded to pancreatectomy following NAT including 27% with LA tumors. Attrition of pts remains a significant issue even in the setting of R, highlighting the importance of NAT in pt selection. Pts who receive NAT and pancreatectomy experience similar OS, regardless of stage, further supporting the NAT approach in choosing patients most likely to benefit from resection. Research Sponsor: U.S. National Institutes of Health; T32CA211034.

702 Poster Session

Evaluation of the impact of lymph node ratio on survival in patients with pancreatic cancer undergoing neoadjuvant therapy. First Author: Maren Thomas, Florida State University, Sarasota, FL

Background: Pancreatic cancer continues to have a dismal prognosis despite improvements in surgical care. Resection status and lymph node status continues to be key factors impacting prognosis. There remains significant debate on the utility of increasing nodal harvest in patients undergoing surgery, while others report an improvement in survival. We sought to evaluate the impact of lymph node positivity to lymph nodes resected in patients with pancreatic cancer undergoing resection. **Methods:** Utilizing the National Cancer Database, we identified patients diagnosed with pancreatic cancer who underwent neoadjuvant chemotherapy (NCT) or neoadjuvant chemoradiation (NCR) followed by resection. We then stratified by nodal status and calculated a LN+/LNR (lymph node resected) ratio. Pearson's Chi-square test was used to compare categorical variables. Survival analysis was estimated by the Kaplan-Meier method and associated log-rank test. Propensity score matching (PSM) was completed for variables of age, sex, tumor size, LN removed, LN positive, pathogenic T- stage and N-stage, grade, 30- and 90-day mortality, R0 resection, facility volume, response, adjuvant therapy, and LNR. Significance was considered at p-value $\leq$0.05. **Results:** We identified 4117 patients with a median age of 65 (25-90). There were 2323 patients who underwent NCT and 1794 who underwent NCR. The median LN resected in the NCT was 18 vs 15 in the NCR, $p<0.001$. However, there were more LN+ in the NCT vs NCR, 2.1 vs 0.9, $p<0.001$. The R0 resection rates were lower in the NCT vs NCR 81% vs 86%, $p<0.001$. Additionally, complete response rates were higher amongst the NCR, 9.2% vs 4.7%, $p<0.001$. Lymph node ratios were consistently higher in the NCT cohorts among all ratio strata, $p<0.001$. Patients with pN+ disease had a median and 5-year survival of 26.4mo and 17% compared to their pN- counterparts at 38.8mo and 36%, $p<0.001$. Patients treated with NCT demonstrating a LNR of 0.01-0.1 had median and 5-year survival of 32.8 and 27%, LNR 0.11-0.4 (25.5mo and 12%), and LNR$>$0.4 (19.1mo and 10%), $p<0.001$. Patients treated with NCR demonstrating a LNR of 0.01-0.1 had median and 5-year survival of 29.1 and 21%, LNR 0.11-0.4 (25.8mo and 17%), and LNR$>$0.4 (21.3mo and 5%), $p<0.001$. PSM confirmed a higher LNR in the NCT cohort compared to NCR, $p<0.001$, however survival was also longer in these patients among the LNR 0.01-0.1, and 0.11-0.4, $p<0.001$. **Conclusions:** Lymph node status in patients with pancreatic cancer continues to be a significant prognostic indicator of survival. Moreover, lymph node ratio may be a more accurate predicter of survival in patients undergoing neoadjuvant chemotherapy and neoadjuvant chemoradiation. Research Sponsor: None.

703 Poster Session

Perioperative versus adjuvant therapy in resectable pancreas cancer. First Author: Tianyi Wang, University of Michigan, Ann Arbor, MI

Background: Pancreatic ductal adenocarcinoma (PDAC) is the third-leading cause of cancer death in the United States with a 5-year survival rate of 13%. Recurrence rates are high following resection and optimization of neoadjuvant and adjuvant therapy remains under investigation, with the hypothesis that neoadjuvant/perioperative therapy may convert borderline resectable tumors to resectable and prevent progression of micrometastases in the postoperative setting. Our study evaluates patient outcomes and models predictive and prognostic markers to assist with clinical decision-making in consideration of neoadjuvant therapy for resectable and borderline resectable pancreatic cancers. **Methods:** We performed a single-institution retrospective analysis of patients who underwent pancreatic resection between January 1, 2016 and December 31, 2020 (n=532). We excluded patients with unresectable disease or a non-PDAC diagnosis. Analysis of the remaining patients (n=175) was performed using automated and manual chart review, basic statistical comparisons with T-tests or Chi-squared tests, Kaplan-Meier Survival analyses, and Cox Proportional Hazard Testing using intention to treat analysis. **Results:** There were no significant differences ($p>0.05$) the presenting demographics between the adjuvant (AJ, n=97) and neoadjuvant/perioperative (NP, n=76) groups. Patients were more likely to have a history of pancreatitis, a risk factor for PDAC development, in the NP group (p=0.0153). Patients in the AJ group included patients who ultimately did not receive the recommended adjuvant therapy for various reasons (n=25) and had overall worse performance status (p=0.004) and received less chemotherapy overall compared to the NP cohort (p=0.0003). Adjuvant therapy was more likely to be gemcitabine with capecitabine (n=37, p=0.0007) in the AJ group and gemcitabine with nab-paclitaxel (n=11, p=0.0018) in the NP group. In the neoadjuvant setting, 62% of patients received FOLFIRINOX and 28% received gemcitabine with nab-paclitaxel. Patients in the NP group were more likely to receive radiation (p=0016), including 6 patients who received neoadjuvant radiation. Median overall survival (33.9 [28.8-43.8] versus 38.2 [28.2-47.8] months) and progression free survival (21.1 [18.1-34.2] versus 16.6 [11.7-23.6] months) were not significantly different between the AJ and NP groups, respectively. **Conclusions:** These results demonstrate similar presenting characteristics between patients with resectable PDAC treated in a period when general practice shifted from adjuvant therapy to a neoadjuvant/perioperative approach. Although patients treated neoadjuvantly received more chemotherapy overall and the chemotherapy regimens differed between the groups, patients had similar outcomes in the two groups. Research Sponsor: None.

704 Poster Session

Outcomes of surveillance in patients with intraductal papillary mucinous neoplasms (IPMNs) who tested negative for high-risk mutations (HRMs) in next-generation sequencing (NGS). First Author: Amudhan Kannan, University of Texas Southwestern, Dallas, TX

Background: HRMs in NGS have demonstrated high sensitivity and specificity for predicting advanced neoplasia in patients with IPMNs. However, the outcomes of surveillance in patients who tested negative for these HRMs have not been investigated in prior studies. This study aimed to identify the predictors of progression in patients with negative NGS results for HRMs. **Methods:** We conducted a retrospective review of prospectively collected data from patients who underwent NGS testing between 2016 and 2023 in our pancreatic cancer prevention program. Patients without IPMNs (i.e., negative for KRAS/GNAS), with HRMs, and those without at least one follow-up imaging post-NGS were excluded from the study. Progression was defined as the development of new or additional worrisome features or high-risk stigmata (per Kyoto guidelines), or advanced neoplasia (HGD/PDAC) following NGS testing. A Cox proportional hazards regression model was used to identify predictors of progression. **Results:** A total of 543 patients underwent NGS testing during the study period, of which 210 patients met the inclusion criteria. The median follow-up duration was 22 months (IQR = 12-36). Of these, 72 patients developed progression following NGS testing. A total of 11 patients underwent surgical resection, and 5 of them had HGD. Factors predictive of progression included a history of smoking (HR = 1.74; 95% CI = 1.06 - 2.86), larger cyst size (HR = 1.09; 95% CI = 1.03 - 1.16), and the presence of a dilated main pancreatic duct (MPD) (HR = 2.56; 95% CI = 1.5 - 4.4) prior to NGS testing. The median time to progression for patients with worrisome features was 33 months, compared to 59 months for those without worrisome features at baseline. Similarly, the median time to progression for patients with cysts $\geq$ 2 cm was 38 months, compared to 57 months for those with cysts $<$ 2 cm. **Conclusions:** Patients who test negative for high-risk mutations in NGS are not exempt from the risk of progression. Any smoking history and presence of any worrisome features before NGS testing showed to be independent predictors of disease progression. For patients with smoking history, worrisome features or cysts $\geq$ 2 cm, frequent surveillance should be recommended following NGS testing. Research Sponsor: None.

Results of Cox proportional hazards regression model for identifying the predictors of progression.

Variables	Hazards ratio (HR)	95% Confidence Interval	P-value
Age	0.99	0.96 – 1.01	0.257
Male sex	1.02	0.62 – 1.7	0.93
White race	1.07	0.63 – 0.83	0.8
Family history of pancreatic cancer	0.6	0.26 – 1.36	0.22
Personal history of other cancers	0.95	0.56 – 1.62	0.86
Diabetes	1.22	0.72 – 2.05	0.46
Any history of smoking	1.74	1.06 – 2.86	**0.03***
Pre-NGS cyst size (cm)	1.09	1.03 – 1.16	**0.006***
Pre-NGS dilated main pancreatic duct (≥5mm)	2.56	1.5 – 4.4	**0.001***
Pre-NGS presence of mural nodules	0.32	0.04 – 2.37	0.26

*Significant p-value.

706 Poster Session

Trajectories of premorbid healthcare contact days in veterans with pancreatic cancer. First Author: Whitney Victoria Johnson, University of Minnesota, Minneapolis, MN

Background: Early detection of pancreatic cancer is a key clinical priority. Although prior work has evaluated machine learning models to identify at-risk patients for pancreatic cancer screening, this is limited by complexity and cost. "Contact days"— days spent receiving healthcare outside the home for any reason or length of time—have previously been used to measure the "time toxicity" of cancer care post-diagnosis. We sought to characterize trajectories of premorbid contact days, alongside standardized risk scores and healthcare costs to explore their use as potential markers for early pancreatic cancer diagnosis. **Methods:** We identified decedents with pancreatic adenocarcinoma diagnosed between 2012 and 2022 who had at least one primary care visit in the Veterans Affairs (VA) system in each of the two years prior to diagnosis using the VA Clinical Cancer Registry (VACCR) and the Corporate Data Warehouse (CDW). We grouped patients into two cohorts: (1) surgical (receipt of curative intent surgery), and (2) non-surgical and collected sociodemographic and clinical characteristics. During a 2-year premorbid period we calculated monthly contact days (24 data points) and plotted them as percentages (e.g., 3 contact days in a month=10%) and visualized their trajectory using cubic smoothing splines. We plotted trajectories for each sociodemographic and clinical subgroup and visually evaluated differences. We similarly plotted monthly Care Assessment Need (CAN) scores, a validated VA risk score identifying patients at high risk of hospitalization/death, and monthly VA healthcare costs. **Results:** We included 8,654 patients (649 surgical - median age 69 years, 74% white, 78% stage 1-2 and 8,005 non-surgical - median age 71 years, 72% white, 21% stage 1-2). For both groups, trajectories of contact days were similar. Baseline contact days (-24 to -12 months) were 7-8% ($\approx$2 days/month). Contact days started to rise on average 9 months prior to diagnosis at an approximate rate of 1% per month, reaching 15-20% at diagnosis ($\approx$5-6 days/month). In the surgical group, patients with a Charlson comorbidity score of $\geq$2, urban residence, and Black race had higher baseline contact days. Those $<$65 years and rural residence had a steeper slope of rise in contact days. CAN score trajectories closely mirrored contact day trajectories in both groups. Cost trajectories were similar in both groups ($\approx$\$1000/month at baseline), rose gradually to $\approx$\$2000/month by -3 months, and then rose sharply to $>$\$6000/month 2 months before diagnosis. **Conclusions:** In this decade-long national study of Veterans diagnosed with pancreatic cancer, we found that contact days and CAN scores, but not costs, start to rise 9 months prior to diagnosis. These data provide the first evidence supporting their further evaluation in pancreatic cancer risk models which may aid early diagnosis. The similar trajectories in patients who go on to receive curative-intent surgery or not may imply they may be markers for pancreas cancer diagnosis but perhaps not outcomes. Research Sponsor: Pancreatic Cancer Action Network.

707 Poster Session

Second-line therapy with Nal-IRI/5-FU/FA after failure of gemcitabine/nab-paclitaxel in advanced pancreatic cancer (PC): Predictive role of 1st-line therapy (PREDICT) and impact of quality of life (QoL). First Author: Meinolf Karthaus, Department of Hematology and Oncology, Munich Hospital Neuperlach, Munich, Germany

Background: Metastatic PC has the lowest survival rate of all malignant diseases and is the fourth most common cancer. QoL remains an unresolved issue in PC for patients (pts) after treatment failure of 1st-line CTx (TTF1). Nanoliposomal irinotecan (Nal-IRI) with 5-FU/folinic acid (FA) increased survival of PC pts from 4.2 months (mo) to 6.1 mo as compared to 5-FU/FA alone, with a rate of toxicities that may impact QoL. PREDICT is an open label, single arm, multicenter Phase IIIb trial (NCT03468335) and examined 2nd-line Nal/IRI/5-FU/FA for PC. Here we report on QoL during 2nd-line Nal-IRI after failure of 1st-line gemcitabine/nab-Paclitaxel (gem/nab-Pac). **Methods:** In this prospective trial, 151 patients with locally advanced or metastatic PC were treated with biweekly Nal-IRI/5-FU/FA (70 mg/m², 2400 mg/m², 400 mg/m²) after failure of gem/nab-pac (TTF1). Primary end point (EP) was the time to treatment failure of 2nd-line therapy (TTF2) and has been reported elsewhere. Secondary EP were QoL and Health related quality of life (HR-QoL) which were evaluated using questionnaires (EORTC QLQC30, QLQ-PAN26, EQ-5D-5L). Evaluation of time to definitive deterioration of QoL (TDD) during 2nd-line therapy, defined as the time from screening/baseline until loss of ≥10 points in the EORTC QLQ-C30 was compared to baseline. **Results:** QoL analyses were performed with all QoL evaluable subjects set (QAS). The QAS included 143 pts, of which 48 pts were included in TTF1 high (TTF of 1st-line ≥213 d) and 49 pts in the TTF1 low (TTF ≤119 d) cohort, with 79 (54.1%) female and 67 (45.9%) male pts. Mean age was 68.2±8.9 years. Overall, a low global health status / QoL mean score was observed at baseline for TTF1 high and low cohorts (about 48 points), which slightly improved during the study for TTF1 high (3.5±23.7) and slightly worsened for TTF1 low cohort (-0.7±22.5). Substantial improvements were observed for both cohorts in emotional functioning scale at certain study time points (TTF1 high: up to 9.1±25.7, TTF1 low: up to 6.6±17.0). A relatively high mean score was observed for cognitive function scale for both cohorts at baseline (TTF1 high: 70.1±30.2; TTF1 low: 72.5±21.4). Pts in the TTF1 low vs. high cohort showed more pronounced worsening of cognitive function. Median TDD was numerically longer (about one month) for TTF1 high (5,3 mo) compared to TTF1 low cohort (3,9 mo). The probability to maintain QoL after six months was similar between the two cohorts. **Conclusions:** QoL showed similar global and HR-QoL scores with only slight changes during the study. Median TDD was numerically longer for high TTF1 pts, but similar after six months between pts still on 2nd line Nal/IRI/5-FU regardless of duration of TTF in 1st-line. Clinical trial information: NCT03468335. Research Sponsor: None.

708 Poster Session

Disease-free survival as a surrogate endpoint for overall survival in adults with early-stage pancreatic cancer: A trial-level surrogacy analysis. First Author: Luís Felipe Leite da Silva, Universidade Federal Fluminense, Niterói, Brazil

Background: In early-stage cancer, overall survival (OS) is the gold-standard endpoint for assessing therapeutic efficacy. However, long follow-up periods make OS challenging and costly to measure in clinical trials. Disease-free survival (DFS), which captures the time to recurrence or death, has been proposed as a surrogate endpoint for OS in early-stage pancreatic cancer (PC). This surrogacy relationship could allow for a more rapid assessment of treatment benefits, but its validity requires robust analysis across clinical trials. Understanding the trial-level correlation between DFS and OS is crucial to optimizing trial design and grant accelerated drug approval. **Methods:** We conducted a systematic search of Embase, PubMed, and Cochrane databases through August 2024, identifying randomized controlled trials (RCTs) that evaluated adjuvant or neoadjuvant systemic therapies in adult patients with PC. Included studies reported both OS and DFS outcomes. Hazard ratios for DFS (HR_DFS) were analyzed as surrogate endpoints for OS (HR_OS). Statistical analyses utilized linear regression models, weighted by trial sample sizes, to determine the strength of the correlation. Surrogacy strength was categorized by the coefficient of determination (R^2), with $R^2 \geq 0.7$ indicating a strong correlation, 0.69–0.5 a moderate correlation, and values < 0.5 indicating weak correlation. **Results:** Out of 3,048 studies screened, 30 RCTs met the inclusion criteria, comprising 5,275 patients in adjuvant therapy trials and 557 patients in neoadjuvant therapy trials. The median follow-up duration ranged from 16 to 53.5 months, with 12 phase II trials included. Overall, DFS showed strong surrogacy for OS in adjuvant trials (R^2 = 0.70; 95% CI, 0.50–0.91). Subgroup analysis indicated strong correlation in phase III trials (R^2 = 0.71; 95% CI, 0.46–0.96) and moderate correlation in phase II trials (R^2 = 0.67; 95% CI, 0.21–1.00). The strongest surrogacy was observed in studies incorporating adjuvant radiation therapy (R^2 = 0.81; 95% CI, 0.52–1.00). In the analysis of neoadjuvant trials, HR-DFS also presented strong correlation with OS (R^2 = 0.90; 95% CI, 0.56–1.00). **Conclusions:** There is a strong trial-level correlation between DFS and OS in both adjuvant and neoadjuvant therapy trials for early-stage pancreatic cancer, particularly in phase III studies and trials incorporating adjuvant radiation therapy. However, the strength of this surrogacy varies depending on the treatment setting and trial design. Research Sponsor: None.

709 Poster Session

A mixed methods evaluation of the Canopy Cancer Collective's implementation of network-wide screening initiative in 2023. First Author: Priya Pathak, Department of Radiation Oncology, Johns Hopkins School of Medicine, Baltimore, MD

Background: The Canopy Cancer Collective, a Learning Health Network (LHN) comprised of 14 pancreatic cancer centers across the U.S., set ambitious goals in 2023 to improve screening rates across four key areas: clinical trial screening (from a baseline of 30% to 100%), assessment of enzyme replacement (37% to 100%), goals of care discussions (57% to 100%), and palliative care screening (30% to 100%). This study aimed to identify the barriers and facilitators influencing the implementation of these screening aims, with the goal of developing informed, actionable recommendations for future implementation efforts. **Methods:** Throughout 2023, data were gathered from monthly assessments completed by site coordinators and quality improvement coaches at each of the 14 centers. These assessments captured insights into planning, implementation, and challenges related to the screening aims. Additionally, a comprehensive site assessment conducted in October 2023 was analyzed. We employed content analysis to identify the key factors influencing the implementation process, with percentage improvements in screening aims documented. **Results:** A total of 128 assessments were reviewed. The most common barriers identified were insufficient personnel and time for screening implementation (n=7), delayed responses and lack of engagement from supporting teams (n=6), unanticipated delays and logistical barriers to the procedures (n=6), recruitment challenges (n=3), and changes in the staff or management (n=5). Facilitators that supported implementation included: identification and testing of patient-centered strategies for implementation of screening (n=11), team expansion (n=8), consistent data collection and sharing through the LHN database (n=7), launching of other initiatives tailored to institutional needs (n=5), and collaboration between teams within or across institutions (n=4). Among the barriers to form ideal interprofessional composition, resource shortages (n=7), provider shortages (n=6), and coordination difficulties (n=5) were included. By January 2024, the network's screening rates had improved significantly: 92% for clinical trial, 87% for enzyme need and palliative care each, and 97% for goals of care. **Conclusions:** While common challenges such as staffing limitations and logistical barriers posed obstacles, facilitators like patient-centered strategies, team growth, and improved collaboration were key to the successful implementation of the screening aims. These findings offer actionable insights for enhancing future implementation efforts, with a focus on addressing resource limitations and fostering team engagement. Moving forward, we plan to assess the outcomes of these screening initiatives to examine their impact on patient care and evaluate the perceptions of site on utilization of implementation strategies to promote its adoption. Research Sponsor: None.

710 Poster Session

Impact of neoadjuvant radiation therapy modalities on post-surgical lymphopenia in operable pancreatic cancer. First Author: Soon Khai Low, Medical College of Wisconsin, Milwaukee, WI

Background: Recent advances in therapeutic vaccines show promise for *KRAS*-mutated pancreatic cancer (PC), with ongoing clinical trials exploring their potential further. Lymphocytes are essential in generating vaccine-induced immune response. Patients who develop lymphopenia following neoadjuvant radiation therapy (RT) are frequently excluded from these vaccine trials. We aimed to assess how different modalities of neoadjuvant RT impact the severity and incidence of lymphopenia. **Methods:** We identified patients with PC who received neoadjuvant therapy including RT followed by surgery at our institution between March 2009 and June 2024. Patients were stratified by the type of RT they received: conventional long-course chemo-RT (LCRT) with 50.4 Gy over 28 fractions (Fr), short-course chemo-RT (SCRT) with 36 Gy over 15 Fr, and stereotactic body RT (SBRT) with 30 Gy over 5 Fr. The absolute lymphocyte count (ALC, 10^3 cells/μL) was compared at various timepoints: baseline, during RT, 6-weeks after surgery (post-op), and every 3-month interval up to 1 year post-op. Moderate lymphopenia was defined as ALC $<$1.0 and severe lymphopenia as ALC $<$0.5. **Results:** Among the 627 patients included in the study, 522 had baseline pre-RT ALC data available for analysis. There was no difference in baseline ALC among the different RT groups (Table). During RT, patients receiving LCRT experienced a higher incidence of severe lymphopenia (86.1%; median ALC (mALC) nadir 0.28) compared to SCRT (41.9%; mALC nadir 0.50) or SBRT (20.0%; mALC nadir 0.72, p$<$0001). No difference in the mALC or rates of lymphopenia was observed beyond 6 weeks post-op. Notably, 53% of the patients experienced lymphopenia for as long as 1 year after surgery. **Conclusions:** LCRT resulted in more severe lymphopenia compared to SCRT or SBRT during RT, but this difference in severity diminished 6 weeks after surgery. Alternatives in treatment sequencing for operable pancreatic cancer can have profound effects on patterns of disease recurrence. When applying multiple treatments in series, toxicities such as lymphopenia may impact receipt of therapy. Techniques to mitigate such toxicities, while maximizing response, will be important to consider. Research Sponsor: None.

Absolute lymphocyte count (ALC, 10^3 cells/μL) at different time point.

Patients with baseline ALC	LCRT (n=454)	SCRT (n=34)	SBRT (n=34)	P-value
Baseline ALC, median (IQR)	1.48 (0.74)	1.27 (0.94)	1.51 (0.74)	0.74
6-week lymphopenia, n (%)				
No	173 (37)	13/27 (48)	18/35 (51)	0.10
Moderate	219 (46)	11/27 (41)	16/35 (46)	
Severe	79 (17)	3/27 (11)	1/35 (3)	
1-year lymphopenia, n (%)				
No	101/217 (47)	3/8 (37)	9/13 (69)	0.054
Moderate	90/217 (41)	5/8 (63)	1/13 (8)	
Severe	26/217 (12)	0/8 (0)	3/13 (23)	

711 Poster Session

A novel 2-2 5-5 model for stratifying liver metastasis burden in patients with pancreatic cancer. First Author: HyeonKi Kim, Seoul National University Bundang Hospital, Seongnam, South Korea

Background: In metastatic pancreatic cancer (MPC), hepatic metastasis is uniformly counted as just 'one organ metastasis' regardless of its metastatic tumor burden. This study aims to develop a novel stratification system for liver metastasis based on tumor burden, and to evaluate its prognostic significance in MPC patients. **Methods:** Patients with MPC treated at Seoul National University Bundang Hospital from 2011 to 2021 were included as the training cohort. Patients with liver-only metastasis were selected for analysis. Liver metastasis burden was categorized by the number and maximum diameter of metastases as measured from diagnostic CT images. Optimal cutoffs for metastasis number and size were determined using Cox proportional hazards regression, adjusted for chemotherapy regimen. An independent cohort from Seoul St. Mary's Hospital and Hallym University Sacred Heart Hospital served as the validation group. The 'H category classification' suggested by the Japanese Society for Cancer of the Colon and Rectum (JSCCR) was also referenced. **Results:** A total of 1,257 patients were included, with 1,084 in the training cohort and 173 in the validation cohort. In the training cohort, data from 311 patients with liver-only metastasis were analyzed to derive the Two-two five-five ('2-2 5-5') criteria, classifying metastasis into Oligo-metastasis (≤2 metastases and ≤2 cm), Meso-metastasis (non-oligo and non-mega), and Mega-metastasis (≥5 metastases or ≥5 cm). Median overall survival (OS) significantly differed between the groups: Oligo (11.4 months), Meso (9.2 months, HR 1.66, 95% CI 1.14–2.43), Mega (5.6 months, HR 2.63, 95% CI 1.94–3.58) in the training cohort; Oligo (5.8 months) vs Mega (1.81 months, HR 1.94, 95% CI 1.16–3.22) in the validation cohort. **Conclusions:** This 2-2 5-5 model effectively stratifies MPC patients with hepatic metastasis into three prognostic groups, demonstrating significant survival differences in both training and validation cohorts. This could rationalize the independent therapeutic strategies by extent of hepatic metastasis. Research Sponsor: None.

Demographic characteristics of patients classified according to "two-two five-five" criteria.

	Oligo-metastasis (n = 88)	Meso-metastasis (n = 59)	Mega-metastasis (n = 164)	Total (n = 311)	P-value*
Age, median (range)	66.0 (33.2-91.0)	67.1(37.2-86.9)	68.7 (40.1-94.6)	67.9 (33.2-94.6)	0.587
Male, n (%)	43 (48)	36 (61)	91 (55)	170 (54)	0.330
Tumor location(Head), n(%)	47 (53)	27 (45)	60 (36)	134 (43)	<0.05
Number of liver metastasis, median (range)	1 (1-2)	3 (1-4)	12 (2-14)	5 (1-14)	<0.05
Maximal size of liver metastasis, cm (IQR)	10.6 (8.1-13.1)	15.9 (10.1-24.2)	21.6 (16.8-33.7)	17.2 (10.9-25.2)	<0.05
Average size of liver metastasis, cm (IQR)	10.3 (8.0-12.1)	13.5 (7.8-19.1)	13.2 (10.5-19.2)	11.9 (9.2-17.1)	<0.05
CA19-9 (U/mL)	462 (103-1,334)	560 (89-1,480)	1,170 (100-9,500)	745 (100-3,700)	<0.05

712 Poster Session

Treatment preferences in advanced pancreatic ductal adenocarcinoma (PDAC) from qualitative patient (PT), caregiver (CG), and oncologist (OC) interviews. First Author: Mark Andrew Lewis, Intermountain Healthcare, Murray, UT

Background: Most treatment regimens for patients with PDAC focus on extending survival while managing symptoms and maintaining quality of life (QoL). Here we report insights from PT, CG, and OC interviews on PDAC treatment preferences. **Methods:** A targeted literature review of 7 quantitative/preference and 12 qualitative/patient-reported outcome studies was used to develop semi-structured interview guides. Single 60-minute interviews of US PTs with stage 4 PDAC, CGs, and OCs with ≥5 years' experience treating PDAC were undertaken virtually by staff from March to April 2024. Interviews had 5 sections: background/clinical experience (OC only); key treatment benefits and risks to avoid (ranked by importance); treatment convenience; and 2 hypothetical treatment preference questions. Participants completed a sociodemographic and clinical questionnaire. Interviews were recorded with permission, transcribed, and analyzed with ATLAS.ti v8+. **Results:** Ten PTs, CGs, and OCs completed interviews (N=30 total), including 4 PT/CG dyads. Mean PT age was 56.8 (standard deviation [SD] 12.15) years; 6 were men and 4 did not have CGs. Mean CG age was 51.1 (SD 14.2) years; 9 were women and 7 cared for their partners. OCs had mean 18.5 (SD 6.9) years' clinical experience and 10 were men. Top-ranked treatment benefits were increased overall survival (OS) (6 PTs [60%], 8 CGs [80%], 7 OCs [70%]); improved QoL (6 PTs [60%], 5 CGs [50%], 7 OCs [70%]); stopping disease progression (5 PTs [50%], 3 CGs [30%], 7 OCs [70%]), and increased life expectancy (7 PTs [70%], 5 CGs [50%], 3 OCs [30%]). PTs' top 3-ranked side effects to avoid included fatigue (n=4 [40%]), nausea (n=4 [40%]), and vomiting (n=3 [30%]). CGs ranked trouble remembering (n=4 [40%]), neutropenia (n=4 [40%]), and neuropathy (n=3 [30%]) in the top 3, while OCs ranked vomiting (n=5 [50%]), neuropathy [n=5 [50%]), and neutropenia (n=4 [40%]) highest. Most PTs/CGs were indifferent to ≥1 infusion/visit and infusion duration. In the first hypothetical treatment preference question, the treatment with higher efficacy but higher side effect risk and longer infusion time was preferred by 80% of PTs/CGs and 70% of OCs. In the second hypothetical treatment question, treatments had the same efficacy; 90% of PTs/CGs/OCs preferred the treatment with shorter infusion time and lower risk of gastrointestinal side effects and hypersensitivity reactions, despite higher risk of severe hypokalemia and severe neutropenia. **Conclusions:** OS and QoL were ranked as similarly important by participants, suggesting that extending survival is insufficient without acceptable QoL. Avoiding side effects was important to all participants, though PTs, CGs, and OCs ranked specific risks with differing importance. More studies are needed to quantify tradeoffs between treatment efficacy and side effects in the care of patients with advanced PDAC. Research Sponsor: Astellas Pharma Inc.

713 Poster Session

Effect of ketorolac on serum GDF-15 and IL-8 in patients with pancreatic cancer with cachexia. First Author: Abrahm Levi, Cedars-Sinai Medical Center, Los Angeles, CA

Background: Cancer cachexia deteriorates the quality of life, chemotherapy tolerance and survivability of pancreatic cancer (PC) patients. Although a prevalent condition affecting PC patients, there are no approved therapies for cachexia and its mechanism of action is poorly understood. Elevated growth differentiation factor 15 (GDF-15), a stress-induced cytokine, has been associated with the pathogenesis of cachexia. Similarly, prior studies have displayed positive correlations between the proinflammatory factor interleukin-8 (IL-8) and PC cachexia. This abstract focuses on the exploratory endpoints (serum GDF-15 and IL-8 levels) of a feasibility trial on ketorolac as a treatment for PC cachexia. **Methods:** Patients took ketorolac 10 mg PO QID for 5 consecutive days, with weights measured on Day 1 (D1) and Day 6 (D6). Serum cytokines were drawn on D1 and D6. IL-8 was measured using a HDF15 assay (Eve Technologies). GDF-15 was measured using a human R-plex GDF-15 assay. Activity (e.g., number of steps) was measured via Fitbit. Differences in weight, activity, GDF-15, and IL-8 were analyzed using two-sample paired t-tests. **Results:** Twenty-nine patients were enrolled, of which 19 were evaluable. Median age was 70, 47.4% (9/19) were female, and 94.7% (18/19) were adherent with ketorolac. Statistically significant increases in both weight per baseline BMI and activity were observed from D1 to D6 (0.0716±0.1079 kg/BMI, p=0.012; 1423±2477 steps, p=0.0221). Of the 19 evaluable patients, 16 patients provided complete D1 and D6 pairs of serum cytokines. From D1 to D6, the mean GDF-15 level decreased by 277.5 pg/mL (p=0.529) and mean IL-8 decreased by 65.13 pg/mL (p=0.103). In all patients who gained weight (n=14), a trend toward reduced IL-8 from D1 to D6 was observed (79.57±39.57 pg/mL, p=0.065). Similarly, for patients who gained ≥ 1-kg from D1 to D6 (n=10), a trend toward reduced GDF-15 was observed (951.2±482.3 pg/mL, p=0.080). Patients with increased activity from D1 to D6 (n=12) had mean reductions of 157.7 pg/mL (p=0.643) and 70.17 pg/mL (p=0.179) in GDF-15 and IL-8, respectively. **Conclusions:** Given the growing literature suggesting associations between inflammatory cytokines and cancer cachexia, it is critical to investigate treatments that may possibly alter cytokine levels. Although our findings suggest that ketorolac aids in weight gain and increased activity, larger sample sizes are needed to determine whether these differences are related to reductions of specific serum cytokines. However, given reductions that trended towards significance for both GDF-15 and IL-8 serum levels within subsets of patients that gained weight on ketorolac, further studies investigating these cytokines are warranted to hopefully provide insight on the mechanistic action of ketorolac in cachexia. Clinical trial information: NCT05336266. Research Sponsor: Yinuoke, Ltd.

714 Poster Session

The value and indications for surgery in pancreatic cancer with synchronous liver metastases. First Author: Ruili Wei, Capital Medical University, China-Japan Friendship School of Clinical Medicine, Beijing, China

Background: Over half of pancreatic cancer patients present with advanced disease and distant metastases, primarily in the liver, at initial diagnosis. Although systemic combination therapies have improved outcomes, the role of surgery remains contentious. This study seeks to pinpoint prognostic factors for patients with synchronous liver metastatic pancreatic cancer (sPDACLM) who receive effective treatments, and to identify patients likely to benefit from surgery. **Methods:** This is a retrospective cohort study. Included patients received first-line chemotherapy (AG/mFOLFIRINOX), with efficacy assessed every 4-6 cycles through imaging and biochemical evaluations. Imaging assessments were conducted according to RECIST 1.1 criteria, while biochemical evaluation focused on the reduction of CA19-9 (calculating cutoff values). Patients with good responses were considered for curative surgery, while those with minor responses decided between surgical treatment or continued maintenance therapy after being fully informed. **Results:** From May 2017 to February 2024, a total of 48 patients with sPDACLM who met the inclusion criteria were recruited for this study. The prognosis of patients with resectable or borderline resectable primary tumors was significantly superior to unresectable cases (18 months vs 11 months, p = 0.0011). Multivariate analysis revealed independent favorable prognostic factors: surgery (HR 0.22; p = 0.026), partial response (PR) per RECIST 1.1 criteria (HR 0.09, p = 0.040), and tumor marker reduction >85% (HR 27.99; p = 0.005). Achieving conversion effectiveness, as defined by imaging and biochemical criteria, was associated with a significantly improved prognosis compared to cases where conversion was ineffective (28 months vs. 12 months, p = 0.0086). Notably, effective surgery patients had a median survival of 49 months and a 5-year survival rate of 33%. Conversely, for patients who did not respond to systemic therapy, surgery did not confer any significant impact on overall survival; however, the corresponding survival curve indicated a trend towards shorter survival compared to those who did not undergo surgery. **Conclusions:** Patients with sPDACLM fulfill both imaging and biochemical criteria, exhibit improved prognoses; additionally, surgery has been shown to significantly prolong their median overall survival times. Research indicates that accurate screening for cases suitable for conversion surgery can significantly improve the survival rates of sPDACLM. Research Sponsor: National key clinical specialty construction project of General Surgery Department (Hepatobiliary Surgery Department); 2021-QTL-004.

Overall survival time and survival rates in patients with synchronous liver metastatic pancreatic cancer undergoing conversion therapy.

	OS	1-year survival rate	2-year survival rate	3-year survival rate	5-year survival rate
Effective group					
Surgery	49 months	100%	88%	66%	33%
Non-surgery	18 months	80%	20%	-	-
Non-effective group					
Surgery	11 months	33%	-	-	-
Non-surgery	12 months	36%	-	-	-

OS, overall survival time.

715 Poster Session

Nivolumab plus ipilimumab (N+I) in patients (pts) with pancreatic cancer (PC) with *BRCA1/2* mutation (mut): Results from the Targeted Agent and Profiling Utilization Registry (TAPUR) study. First Author: Peter Joel Hosein, Sylvester Comprehensive Cancer Center, University of Miami, Miami, FL

Background: TAPUR is a phase II basket study evaluating antitumor activity of commercially available targeted agents in pts with advanced cancers with genomic alterations. Results from a cohort of pts with PC with *BRCA1/2* mut treated with N+I are reported. **Methods:** Eligible pts had measurable disease, ECOG performance status (PS) 0-2, adequate organ function, and no standard treatment (tx) options or prior immune checkpoint inhibitor tx. PD-L1 expression testing was not required. Genomic testing was performed in CLIA-certified, CAP-accredited site-selected labs. Most genomic tests did not distinguish between germline or somatic mut. Pts received I at 3 mg/kg every 3 weeks (wks) for 4 doses with N at 1 mg/kg IV every 3 wks for 4 doses. N alone was then continued at 240 mg every 2 wks or 480 mg every 4 wks until disease progression. Primary endpoint was disease control (DC) per investigator defined as complete (CR) or partial (PR) response per RECIST v. 1.1, or stable disease of at least 16 wks duration (SD16+). CR is based on radiographic assessment. CA 19-9 levels were not collected. Simon 2-stage design tested null DC rate of 15% vs. 35% (power = 0.85; α = 0.10). If $\geq$2 of 10 pts in stage 1 have DC, 18 more pts are enrolled; otherwise, the cohort is closed. If $\geq$7 of 28 pts have DC, the null DC rate is rejected. Secondary endpoints were objective response (OR), progression-free survival (PFS), overall survival (OS), duration of response (DOR) and SD, and safety. DOR is defined as time from pt's first documented OR to progressive disease. **Results:** 32 pts with PC with *BRCA1* (n=8), *BRCA2* (n=22) or both *BRCA1/2* (n=2) were enrolled between October 2017 and March 2023. 4 pts were not evaluable for efficacy. Pts had adenocarcinoma (n=29), acinar cell carcinoma (n=2), and poorly differentiated carcinoma (n=1). Table shows demographics and outcomes. 1 CR, 3 PR and 4 SD16+ were observed for a DC rate of 31% (90% CI, 18 to 100) and OR rate of 14% (95% CI, 4 to 33). The pt with CR had a *BRCA1* mut, the pts with PR had *BRCA1* (n=1) and *BRCA2* (n=2) muts, and the pts with SD16+ had *BRCA1* (n=2) and *BRCA2* (n=2) muts. The null DC rate was rejected (p=0.04). The pt with CR remains on tx with a DOR of 311 wks as of May 2024. 15 pts received prior tx with olaparib and 1 of those pts had SD16+ on this study. 17 pts had $\geq$1 drug-related grade 3-4 adverse event (AE) or serious AE. All were consistent with N+I labels except generalized muscle weakness and lymphopenia. **Conclusions:** N+I shows antitumor activity in pts with PC with *BRCA1/2* mut and warrants further study. Clinical trial information: NCT02693535. Research Sponsor: Bristol-Myers Squibb; Astrazeneca, Bayer, Boehringer Ingelheim, Eli Lilly and Company, Genentech, Merck, Pfizer, Seagen, now a wholly owned subsidiary of Pfizer Inc., Taiho Oncology.

Demographics (N=32) and efficacy outcomes (n=28).

Median (Med) age, years (range)		**66 (37-80)**
ECOG PS, %	0-1	31 (97)
	2	1 (3)
Prior systemic regimens, %	0-2	10 (31)
	$\geq$3	22 (69)
DC rate, % (OR and SD16+) (90% CI)		31 (18, 100)
OR rate, % (95% CI)		14 (4, 33)
Med PFS, wks (95% CI)		9 (7, 16)
Med OS, wks (95% CI)		34 (14, 46)

716 Poster Session

Effect of dose adjustments on overall survival (OS) in patients with metastatic pancreatic ductal adenocarcinoma (mPDAC) treated with NALIRIFOX: A post hoc analysis of NAPOLI 3. First Author: Anjan J Patel, Florida Cancer Specialists, Sarasota, FL

Background: NALIRIFOX is approved by the US Food & Drug Administration and European Medicines Agency for the first-line treatment of adults (no upper age limit) with mPDAC. Approval was based on the results of NAPOLI 3 (NCT04083235), in which NALIRIFOX demonstrated significant improvements in OS (primary outcome) and progression-free survival compared with nab-paclitaxel plus gemcitabine. In this exploratory post-hoc analysis of NAPOLI 3, we examined the impact of NALIRIFOX dose adjustments on OS in patients treated in North America. **Methods:** Patients (N = 770) with confirmed untreated mPDAC were randomized (1:1) to receive liposomal irinotecan 50 mg/m^2 + oxaliplatin 60 mg/m^2 + leucovorin 400 mg/m^2 + 5-fluorouracil 2400 mg/m^2 (NALIRIFOX) on days 1 and 15 of a 28-day cycle or nab-paclitaxel 125 mg/m^2 and gemcitabine 1000 mg/m^2 on days 1, 8 and 15 of a 28-day cycle. In this analysis, liposomal irinotecan dose reductions, delays, and exposure were examined descriptively in patients who received NALIRIFOX at centers in North America. OS was evaluated using Kaplan–Meier methods; no statistical tests were performed. **Results:** Overall, 120 patients were randomized to receive NALIRIFOX at centers in North America (intention-to-treat [ITT] population) and 112 received treatment (safety population). Dose adjustments (dose reductions and dose delays) occurred in a higher proportion of patients presenting with longer OS. Patients with the longest duration of liposomal irinotecan exposure and highest cumulative dose had the longest OS. **Conclusions:** This finding suggests that tolerability-guided dose modification of liposomal irinotecan does not adversely affect efficacy outcomes and suggests a path forward to further optimize the OS of patients with mPDAC receiving NALIRIFOX. Clinical trial information: NCT04083235. Research Sponsor: Ipsen.

717 Poster Session

Impact of *UGT1A1*28* polymorphism on tolerability in patients with metastatic pancreatic ductal adenocarcinoma (mPDAC) treated with NALIRIFOX in NAPOLI 3. First Author: Maen Abdelrahim, Houston Methodist, Houston, TX

Background: SN-38, the active metabolite of irinotecan, is cleared by the liver enzyme uridine diphosphate glucuronosyltransferase 1A1 (UGT1A1) and biliary excretion. *UGT1A1*28* (genotype 7/7) homozygosity is associated with reduced activity of UGT1A1 and a recommendation for dose adjustment of non-liposomal irinotecan, a component of the FOLFIRINOX regimen. In this post-hoc analysis, we analyzed the impact of *UGT1A1*28* homozygosity on the incidence and profile of treatment-emergent adverse events (TEAEs) among patients enrolled in NAPOLI 3 (NCT04083235). **Methods:** Patients (N = 770) with confirmed untreated mPDAC were randomized (1:1) to receive liposomal irinotecan 50 mg/m^2 + oxaliplatin 60 mg/m^2 + leucovorin 400 mg/m^2 + 5-fluorouracil 2400 mg/m^2 (NALIRIFOX) on days 1 and 15 of a 28-day cycle or nab-paclitaxel 125 mg/m^2 and gemcitabine 1000 mg/m^2 (Gem+NabP) on days 1, 8 and 15 of a 28-day cycle. *UGT1A1*28* status was evaluated in all patients before enrollment. All patients, regardless of *UGT1A1*28* status, received the full starting dose of liposomal irinotecan and followed the same dose reduction rules. This exploratory analysis of TEAEs by *UGT1A1*28* status was descriptive; statistical analyses were not performed. **Results:** Overall, 83 patients (11.0%) were homozygous for *UGT1A1*28*, among whom the incidence of TEAEs was similar to patients with non-homozygous status (Table). The most frequently reported TEAEs ($\geq$ 40%) among the homozygous group (vs the overall NAPOLI 3 population) were diarrhea (59.0% vs 70.5%), nausea (56.4% vs 59.5%), vomiting (46.2% vs 39.7%) and anemia (41.0% vs 26.2%) in the NALIRIFOX arm and nausea (45.5% vs 42.7%), fatigue (45.5% vs 37.7%), diarrhea (45.5% vs 36.7%) and anemia (40.9% vs 40.4%) in the Gem+NabP arm. **Conclusions:** The incidence of TEAEs, including TEAEs leading to death, was similar between patients with homozygous and non-homozygous *UGT1A1*28* status, in both treatment arms. The profile of TEAEs was consistent with that of the overall NAPOLI 3 population. *UGT1A1*28* status did not substantially influence the safety profile or subsequent need for dose reductions of liposomal irinotecan in NAPOLI 3. Clinical trial information: NCT04083235. Research Sponsor: Ipsen.

	*UGT1A1*28* homozygous		*UGT1A1*28* non-homozygous	
TEAE, n (%)	**NALIRIFOX (n = 39)**	**Gem+NabP (n = 44)**	**NALIRIFOX (n = 328)**	**Gem+NabP (n = 331)**
Any	39 (100.0)	44 (100.0)	327 (99.7)	328 (99.1)
Related to any drug	39 (100.0)	41 (93.2)	310 (94.5)	308 (93.1)
Grade ≥ 3	31 (79.5)	34 (77.3)	288 (87.8)	288 (87.0)
Serious	24 (61.5)	24 (54.5)	176 (53.7)	168 (50.8)
Leading to treatment discontinuation	15 (38.5)	10 (22.7)	103 (31.4)	100 (30.2)
Leading to dose reduction of any drug	23 (59.0)	14 (31.8)	182 (55.5)	174 (52.6)
Leading to interruption of any drug	1 (2.6)	1 (2.3)	15 (4.6)	3 (0.9)
Leading to death	2 (5.1)	4 (9.1)	20 (6.1)	19 (5.7)

718 Poster Session

Optimal preoperative treatment strategy for conversion surgery in unresectable locally advanced pancreatic cancer: A project study by the Japanese Society of Hepato-Biliary-Pancreatic Surgery. First Author: Satoshi Yasuda, Department of Surgery, Nara Medical University, Kashihara, Japan

Background: Conversion surgery (CS) for unresectable locally advanced pancreatic cancer (UR-LAPC) has been widely accepted when feasible. However, optimal preoperative treatment strategies remain unclear. To determine the most effective preoperative approach for UR-LAPC in the current treatment landscape, we conducted a project study in the Japanese Society of Hepato-Biliary-Pancreatic Surgery. **Methods:** We analyzed 465 UR-LAPC patients who underwent CS following chemotherapy with FOLFIRINOX (FFX), gemcitabine plus nab-paclitaxel (GnP), or modified regimens between 2015 and 2020 across 84 Japanese board-certified training institutions. Overall survival (OS) was the primary endpoint. We used the Kaplan-Meier method for estimating survival analyses, the Cox proportional hazards model for multivariate analyses, and Maximally Selected Rank Statistics to determine the optimal preoperative treatment duration. **Results:** Median OS was 43.8 months, with a 5-year survival rate of 37.2%. The optimal preoperative treatment duration was 6.1 months. Patients receiving $>$6 months of preoperative treatment (n=350) showed significantly better median OS and recurrence-free survival (RFS) compared to $\leq$6 months (n=115) (50.4 vs 29.7 months and 15.6 vs 9.1 months, respectively; both $P<0.001$). FFX-based regimens (n=116) showed better OS and RFS than GnP-based regimens (n=349) (65.0 vs 36.5 months and 19.3 vs 12.4 months, respectively; both $P<0.01$). Multivariate analysis identified preoperative treatment $>$6 months (HR 0.53, 95% CI 0.407-0.697, $P<0.001$) and initial FFX-based regimens (HR 0.63, 95% CI 0.464-0.843, $P=0.002$) as independent prognostic factors. Additional favorable factors included normal CA19-9 and CEA levels, PNI$>$45 before CS, adjuvant chemotherapy, R0 resection, and Evans grade III/IV. **Conclusions:** This large-scale retrospective study has demonstrated that preoperative treatment duration $>$6 months and the use of FFX-based regimens as initial treatment were associated with improved survival outcomes in UR-LAPC patients undergoing CS. These findings provide critical information for considering CS in UR-LAPC patients, especially in current clinical practice. Research Sponsor: None.

719 Poster Session

Intra-arterial gemcitabine versus intravenous gemcitabine: Pharmacokinetic sub-study of the TIGeR-PaC phase 3 clinical trial. First Author: Paula M. Novelli, University of Pittsburgh Medical Center (UPMC), Pittsburgh, PA

Background: Localized, dual balloon catheter-mediated, intra-arterial delivery of gemcitabine (IAG) targeted to tumors/tissue can provide higher local drug potency (1). Furthermore, IAG may result in a decreased systemic drug concentration and associated side effects. In patients with locally advanced pancreatic cancer (LAPC), this approach is currently being tested in the TIGeR-PaC Phase 3 clinical trial. Herein we report the results of a 19-patient pharmacokinetics (PK) sub-study analysis within TIGeR-PaC. **Methods:** PK analyses were performed for a total of 19 participants across 6 TIGeR-PaC study sites; 11 participants received IAG with the RenovoCath dual balloon catheter at 1000 mg/m^2 over 20 minutes and 8 participants received intravenous gemcitabine (IVG) (5 at 1000 mg/m^2, 2 at 800 mg/m^2, 1 at 500 mg/m^2) over 30 minutes. IAG target treatment arteries were either the superior mesenteric artery (SMA) (n=5) or branches of the celiac axis (n=6). At T = -5, 10, 15, 20, 30, 40, 60, and 90 minutes from the onset of infusion, 2 mL of blood was collected in heparinized tubing containing 25 mcg/mL of tetrahydrouridine. Plasma from each sample was frozen and shipped to a reference lab for gemcitabine assays. Maximum plasma drug concentration (C_{max}) and the area under the drug plasma concentration curve (AUC) based on the terminal phase were compared between the two groups; for these dose-dependent parameters, only participants who received 1000 mg/m^2 gemcitabine (IAG, N=11; IVG, N=5) were included in the comparison analysis. **Results:** As shown in the table, AUC was significantly lower with IAG (4.99) compared to IVG (9.97) (P = 0.019). Peak plasma gemcitabine concentrations were also lower with IAG (12.9 mcg/mL) vs. IVG (14.6 mcg/mL) despite a 50% higher drug concentration during IAG vs. IVG (1000 mg/m^2 infused over 20 minutes vs. 30 minutes, respectively). There was no difference in plasma levels between IAG treatment sites (SMA vs. celiac axis). **Conclusions:** In this analysis, localized, dual-balloon catheter-mediated IAG resulted in decreased systemic levels of gemcitabine compared to IVG. Thus, in addition to providing increased local potency, the IAG approach may also be beneficial in decreasing gemcitabine-related systemic side effects. 1. Farsad K, et al. 2024. *JVIR* 35:1043-48 e3. Clinical trial information: NCT03257033. Research Sponsor: RenovoRx.

Effects of treatment mode on mean pharmacokinetic parameters for gemcitabine.

PK parameter	IAG Group (N=11)	IVG Group (N=5)
C_{max} (mcg/mL), Mean (SE)	12.9 (±2.4)	14.6 (±1.2)
AUC (hr · mcg/mL), Mean (SE)	4.99 (±0.96)	9.97 (±1.9)

AUC = area under the curve; C_{max} = maximum serum concentration; IAG = intra-arterial gemcitabine; IVG = intravenous gemcitabine; PK = pharmacokinetic; SE = standard error.

720 Poster Session

Influence of time to first therapy (TTFT) variable on survival outcomes of neoadjuvant chemotherapy versus upfront surgery approach in resectable pancreatic ductal adenocarcinoma. First Author: Qusai Almasad, Roger Williams Medical Center, Providence, RI

Background: Evidence demonstrating overall survival (OS) benefit of neoadjuvant chemotherapy (NAC) followed by surgical resection over upfront surgical resection for resectable pancreatic ductal adenocarcinoma (PDAC) has been mixed. The time to first therapy (TTFT) variable has not been studied as a contributing factor. **Methods:** A nationwide retrospective analysis using the National Cancer Database to evaluate patients with clinical stage T1 and T2 PDACs from 2010 to 2020. Cox proportional hazards model was used to evaluate the impact of NAC followed by definitive surgery compared to upfront surgery on OS with and without TTFT. **Results:** Total of 43,196 patients were included – 9,880 patients with clinical stage T1 and 33,316 patients with T2 PDACs. There were increasing trends in the NAC approach from 2.9% in 2010 to over 25% by 2020 and decreasing trends in the upfront surgery approach from 69.35% in 2010 to 31.84% by 2020. There were significant differences in TTFT according to the treatment choice with upfront surgery group having a significantly shorter TTFT – proportion of those receiving first treatment within the 1st week was 24.32% in the upfront surgery compared to 4.22% in the NAC group. In the adjusted cox regression without the TTFT variable, there was a 25% higher rate of death in the upfront surgery compared to the NAC group (HR 1.25, 95% CI 1.19 – 1.30). When the adjusted regression was performed with addition of a TTFT interaction term, there was survival disadvantage of upfront surgery approach in patients whose TTFT occurred after 1 week, but not in those with TTFT occurring in less than 1 week (HR 1.01, 95% CI 0.86 - 1.17). **Conclusions:** Our study emphasizes the importance of incorporating TTFT variable when comparing NAC versus upfront surgery approach in PDAC. Future studies comparing NAC to upfront surgery in resectable PDAC should consider incorporating the TTFT variable. Research Sponsor: None.

721 Poster Session

Real-world evidence of nanoliposomal irinotecan and fluorouracil with folinic acid in patients with unresectable or recurrent pancreatic cancer: Final results of a multicenter observational study. First Author: Yudai Shinohara, Department of Hematology/Oncology, Japan Community Healthcare Organization Kyushu Hospital, Fukuoka, Japan

Background: Nanoliposomal irinotecan (NAL-IRI) and fluorouracil with folinic acid (NFF) is the standard regimen after gemcitabine-based therapy for unresectable or recurrent pancreatic cancer (urPC). However, we have almost no prospective data on its efficacy and safety in the real-world, so we conducted this study to investigate them both retrospectively and prospectively (NAPOLEON-2 study). We previously reported the retrospective data in ASCO-GI 2023, and here we report the final results of the prospective part. **Methods:** We prospectively collected data of urPC patients treated with NFF who had received at least one previous chemotherapy line in 17 hospitals in Japan from June 2021 to October 2023. The primary endpoint was overall survival (OS). Secondary endpoints were overall response rate (ORR), disease control rate (DCR), progression-free survival (PFS), dose intensity (DI) and adverse events (AEs). In addition, OS and PFS among the therapeutic lines of NFF were also analyzed. **Results:** A total of 150 urPC patients were prospectively enrolled. The median follow-up period was 7.2 months (95% confidence interval [CI], 6.3–8.8); median age, 72 years (range, 45–85), with 81 female patients (54%). The Eastern Cooperative Oncology Group performance status was 0/1/2/3 in 46/93/10/1 patients, respectively. Sixteen patients (11%) had locally advanced disease; 134 (89%) had metastatic disease; 77 (51%) had liver metastasis; and 47 (31%) had peritoneal metastasis. All patients previously received gemcitabine-based therapy. NFF was administered as 2nd/3rd/4th-or-later-line therapy to 87/53/10 patients, respectively. The median OS was 7.8 months (95% CI, 6.6–9.2); median PFS, 3.7 months (95% CI, 2.8–4.9); overall response rate, 11%; and disease control rate, 56%. The relative dose intensity was 72.7% with NAL-IRI and 79.4% with fluorouracil. The initial dose of NAL-IRI was reduced in 84 patients (56%), mainly owing to decreased organ function (17%), followed by age (12%), worsened performance status (10%), or UGT1A1 examination status (5%). Dosage reduction of NAL-IRI during treatment (independent of the initial dose reduction) was performed in 74 patients (49%), mainly owing to neutropenia (22%) and anorexia (11%). Frequent Grade 3/4 adverse events were neutropenia (27%), anorexia (19%), and leukopenia (16%). Grade 5 peritoneal infection was observed in only one patient. The median OS and PFS for NFF in the 2nd-line group, compared with the 3rd-or-later-line group, were 7.4 vs 7.8 months (hazard ratio [HR], 0.97; 95% CI, 0.68–1.39; p=0.88) and 3.3 vs 4.2 months (HR, 1.04; 95% CI, 0.74–1.45; p=0.84), respectively. **Conclusions:** NFF had appropriate efficacy and manageable toxicity profiles, consistent with our previous report. NFF can be a candidate for 2nd-or-later-line regimens in the real-world. Clinical trial information: UMIN000043939. Research Sponsor: None.

722 Poster Session

Safety, efficacy, and on-treatment circulating tumor DNA (ctDNA) changes from a phase 1 study of RMC-6236, a RAS(ON) multi-selective, tri-complex inhibitor, in patients with RAS mutant pancreatic ductal adenocarcinoma (PDAC). First Author: Ignacio Garrido-Laguna, Huntsman Cancer Institute, University of Utah, Salt Lake City, UT

Background: PDAC is the third leading cause of cancer mortality, with limited treatment options. Even with multi-agent chemotherapy, patients with second-line (2L) PDAC have a median progression-free survival (PFS) of ≈2–3.5 months and median overall survival (OS) of ≈6–7 months. RAS mutations occur in >90% of patients with PDAC, mainly KRAS G12X (G12X = non-synonymous mutations in KRAS codon 12 [G12]). RMC-6236 is an oral, RAS(ON), multi-selective, tri-complex inhibitor of the active GTP-bound state of both mutant and wild-type RAS. In a Phase 1 study, RMC-6236 demonstrated efficacy and manageable safety in patients with PDAC harboring KRAS G12X or other RAS mutations (NCT05379985). Data on ctDNA reduction with targeted therapy in PDAC are sparse. We report safety, updated efficacy, and exploratory analyses of early ctDNA reduction with clinically active doses of RMC-6236 in patients with RAS mutant PDAC. **Methods:** Escalating doses of RMC-6236 were administered orally to patients with previously treated RAS mutant PDAC. Additional patients were enrolled for dose optimization and expansion. Patients receiving clinically active doses (160–300 mg QD) of RMC-6236 ≥14 weeks before the July 23, 2024 data cutoff were included. Plasma samples were collected for ctDNA analysis of change in RAS mutant variant allele fraction at baseline (BL; cycle 1, day 1 [C1D1]), and on treatment (C2D1 or C3D1). **Results:** As of July 23, 2024, 127 patients with RAS mutant PDAC had received RMC-6236 160–300 mg QD. The most common (≥10% of patients) any-grade treatment-related adverse events were rash (91%), diarrhea (48%), nausea (43%), vomiting (31%), stomatitis (31%), fatigue (20%), paronychia (13%), mucosal inflammation (13%), decreased appetite (11%), and peripheral edema (10%). The Table shows objective response rate (ORR), PFS, and OS with 2L RMC-6236. Paired plasma samples were tested in 106/127 patients. Of these, 68 patients with RAS mutant allele ctDNA at BL were evaluable for ctDNA response (Table). **Conclusions:** RMC-6236 showed a manageable safety profile and encouraging efficacy in patients with previously treated RAS mutant PDAC, and early and deep reductions in RAS mutant ctDNA. RASolute 302, a global, randomized, Phase 3 trial evaluating RMC-6236 as 2L treatment vs chemotherapy in patients with metastatic PDAC, is ongoing. Clinical trial information: NCT05379985. Research Sponsor: None.

	KRAS G12X	RAS mutant
Efficacy with 2L RMC-6236		
	(n=42)	**(n=57)**
ORR, % (95% CI)	29 (16–45)	25 (14–38)
[confirmed + pending confirmation]		
Median PFS, months (95% CI)	8.5 (5.3–11.7)	7.6 (5.9–11.1)
Median OS, months (95% CI)	14.5 (8.8–not evaluable)	14.5 (8.8–not evaluable)
ctDNA Response with 2L+ RMC-6236		
	(n=56)	**(n=68)**
>50% decrease from BL, n (%)	53 (95)	63 (93)
100% decrease from BL, n (%)	28 (50)	32 (47)

723 Poster Session

Evaluating the relative treatment efficacy of CD40 agonist mitazalimab in combination with mFOLFIRINOX in patients with metastatic pancreatic ductal adenocarcinoma (mPDAC) using unanchored indirect treatment comparisons (ITCs). First Author: Eileen M. O'Reilly, Memorial Sloan Kettering Cancer Center, New York, NY

Background: Mitazalimab, a human CD40 agonistic IgG1antibody has demonstrated encouraging results in combination with mFOLFIRINOX (mFFX) in the phase 2 single arm trial OPTIMIZE-1 in patients with mPDAC. We conducted unanchored ITCs to assess the relative efficacy of mitazalimab + mFFX vs FOLFIRINOX (FFX) / mFFX and NALIRIFOX (NFX), the most effective therapies to date. **Methods:** Using data from OPTIMIZE-1 and published data from five RCTs identified via a literature review, matching-adjusted indirect comparisons (MAIC) and simulated treatment comparisons (STC), two commonly used ITC methods for health technology assessment (1) were conducted to balance effect modifiers and prognostic factors between trials. Analysis outcomes were overall survival (OS), objective response rate (ORR), and progression free survival (PFS). Reconstructed survival data from FFX, mFFX, and NFX studies were pooled together for the comparison with FFX/mFFX (Pool #1) and FFX/mFFX/NFX (Pool #2) pooled studies. **Results:** In the STC analysis mitazalimab + mFFX showed a significantly higher OS versus Pool #1 [hazard ratio (HR): 0.64; 95% CI: 0.46 – 0.87] and #2 [HR: 0.65; 95% CI: 0.47 – 0.87], corresponding to an improvement in median OS by 3.4 and 3.3 months respectively. The OS HRs of study-level pairwise comparisons ranged from 0.57 (95% CI: 0.39 – 0.80) to 0.85 (95% CI: 0.53 – 1.39) with only one non-significant comparison. MAICs showed similar trends, although the comparison with Pool #1 was close to statistical significance (HR: 0.68; 95% CI: 0.45 – 1.02). Results of the PFS and ORR comparisons are shown in the table below. **Conclusions:** Literature based ITCs show a significantly better efficacy of mitazalimab + mFFX versus the current available therapies in patients with mPDAC. These trends will have to be confirmed in a randomized Phase 3 study. 1. Phillippo DM et al, 2019. Clinical trial information: NCT02829099. Research Sponsor: Alligator Bioscience.

ITC results – HR (for OS and PFS) and odds ratios (for ORR) and their 95% CIs.

	STC			MAIC		
Comparator study	**OS**	**PFS**	**ORR**	**OS**	**PFS**	**ORR**
PRODIGE 4 (FFX)	**0.57** (**0.39 – 0.80**)	**0.63** (**0.44 – 0.88**)	1.62 (0.87 – 2.99)	**0.60** (**0.40 – 0.89**)	**0.56** (**0.37 – 0.82**)	1.55 (0.84 – 2.87)
PANOPTIMOX (FFX)	**0.58** (**0.39 – 0.84**)	**0.68** (**0.47 – 0.98**)	1.17 (0.59 – 2.30)	**0.60** (**0.38 – 0.94**)	0.67 (0.44 – 1.02)	1.04 (0.52 – 2.05)
AVENGER500 (FFX)	**0.71** (**0.50 – 0.98**)	1.15 (0.82 – 1.60)	1.35 (0.75 – 2.42)	0.69 (0.46 – 1.03)	1.11 (0.76 – 1.62)	1.30 (0.72 – 2.34)
SWOGS1313 (mFFX)	0.85 (0.53 – 1.39)	0.89 (0.57 – 1.36)	0.90 (0.43 – 1.89)	0.91 (0.55 – 1.50)	0.86 (0.55 – 1.33)	0.88 (0.42 – 1.86)
Pool #1	**0.64** (**0.46 – 0.87**)	0.81 (0.59 – 1.10)	1.33 (0.77 – 2.32)	0.68 (0.45 – 1.02)	0.81 (0.56 – 1.18)	1.20 (0.69 – 2.09)
NAPOLI-3 (NFX)	**0.68** (**0.49 – 0.93**)	0.85 (0.61 – 1.17)	0.97 (0.55 – 1.71)	**0.67** (**0.46 - 0.97**)	0.83 (0.58 – 1.20)	0.92 (0.52 – 1.62)
Pool #2	**0.65** (**0.47 – 0.87**)	0.82 (0.60 – 1.11)	1.17 (0.68 – 2.02)	**0.68** (**0.47 – 0.99**)	0.84 (0.59 – 1.19)	1.11 (0.64 – 1.91)

724 Poster Session

Preliminary safety, antitumor activity, and circulating tumor DNA (ctDNA) changes with RMC-9805, an oral, RAS(ON) G12D-selective tri-complex inhibitor in patients with KRAS G12D pancreatic ductal adenocarcinoma (PDAC) from a phase 1 study in advanced solid tumors. First Author: Alexander I. Spira, Virginia Cancer Specialists Research Institute, Fairfax, VA

Background: RMC-9805 is a potent, oral, RAS(ON) G12D-selective, covalent, tri-complex inhibitor targeting the active, GTP-bound state of oncogenic RAS G12D isoforms. Despite low response rates to current standard of care (SOC), no RAS-targeted therapy is approved for G12D PDAC. **Methods:** In this Phase 1 study (NCT06040541), patients with previously treated, advanced KRAS G12D solid tumors received escalating RMC-9805 doses (150-1200 mg once daily [QD] or 300-600 mg twice daily [BID]). Treatment cycles were every 3 wks. Antitumor activity was assessed every 6 wks the first 24 wks then every 9 wks. Additional patients were enrolled at RMC-9805 doses that cleared the dose-limiting toxicity (DLT) evaluation to further characterize pharmacokinetics, safety, antitumor activity, and biomarkers. Plasma for ctDNA analysis was collected at baseline (BL; cycle 1, day 1 [C1D1] or screening), and on treatment (OT; C2D1 or C3D1). **Results:** As of Sept 2, 2024, 179 patients with KRAS G12D solid tumors (most having PDAC; n=104 [58%]) received 5 escalating dose levels of RMC-9805 (150-1200 mg daily). No DLTs or Grade 4 or 5 treatment-related adverse events (TRAEs) were reported, and the maximum tolerated dose (MTD) was not reached. Among patients who received a candidate recommended phase 2 dose (RP2D) of 1200 mg daily (1200 mg QD [n=60] or 600 mg BID [n=39]), the most common ($\geq$10% of patients) TRAEs were nausea (27%), diarrhea (20%), vomiting (15%), and rash (10%), all of which were Grade 1 or 2 severity. One Grade 3 TRAE (ALT elevation) was observed in a patient with PDAC and a history of ALT elevation, biliary stenting, and liver metastasis. No patients treated with 1200 mg daily discontinued treatment due to TRAEs and 4% of patients had dose reductions due to TRAEs. In patients with PDAC receiving RMC-9805 daily at 1200 mg QD (n=20) or 600 mg BID (n=20) who enrolled at least 14 wks prior to data cutoff, the objective response rate (confirmed response or pending confirmation) was 30%, and the disease control rate was 80%. Of 28 patients who had KRAS G12D mutations detected in ctDNA at BL and were evaluable for OT assessment, 86% had an OT decrease $>$50% from BL of detectable KRAS G12D mutations, and 39% had a 100% OT clearance. Updated data will be presented. **Conclusions:** Oral RMC-9805 showed encouraging initial antitumor activity with early and deep reductions in KRAS G12D ctDNA in patients with KRAS G12D PDAC. Tolerability was favorable relative to SOC chemotherapy for PDAC and manageable. This overall safety profile and antitumor activity support continued evaluation as monotherapy in patients with KRAS G12D PDAC, and in combination with chemotherapy and targeted therapies, including the RAS(ON) multi-selective inhibitor RMC-6236. Clinical trial information: NCT06040541. Research Sponsor: Revolution Medicines.

725 Poster Session

MEK inhibitor-based therapy in metastatic pancreatic adenocarcinoma with *KRAS* G12R alteration. First Author: Elizabeth Auckley, Medical College of Wisconsin, Milwaukee, WI

Background: Pancreatic ductal adenocarcinoma (PDAC) is largely driven by oncogenic alterations in the *KRAS*, *TP53*, *CDKN2A/B*, and *SMAD4* genes. *KRAS* G12R accounts for nearly 20% of *KRAS* alterations in PDAC, is biologically unique, and offers the potential for improved response to MEK inhibitor (MEK-inh)-based therapy. As part of the MCW-Master-PREDICT (NCT05802069) observational study, we evaluated the efficacy of MEK-inh-based therapy matched to the unique molecular alterations in the tumor of each patient as recommended by the molecular tumor board. **Methods:** From March 2022 to December 2023, 8 patients with metastatic PDAC were treated with MEK-inh-based therapy. Molecular profiling was performed by Tempus (648 genes) in 7 patients, and FoundationOne (324 genes) in 1 patient. Description of molecular alterations, matched therapies, and efficacy of treatment are provided. **Results:** The median age of patients starting MEK inhibitor-based therapy was 69 years, with 62% of them being female. The molecular characteristics and matched therapies are summarized (Table). MEK-inh-based therapy was administered to 3 (37.5%) patients as a first or second-line treatment, and to 5 (62.5%) patients as a third to fifth-line treatment for metastatic PDAC. The overall survival and progression free survival (PFS) on MEK-inh-based therapy was 8.4 and 4.9 months, respectively. Four patients had PFS beyond 4 months (4.9, 5.5, 5.6 and 6.9 months). **Conclusions:** Individualized MEK-inh-based therapy demonstrated efficacy in late-line treatment of metastatic PDAC with *KRAS* G12R alterations. Using this approach earlier in the treatment course and in combination with RAS inhibitors or cytotoxic systemic therapies may further enhance outcomes. Research Sponsor: None.

Tumor alterations and matched therapies.

Gene altered	Number of patients (%)	Number of patients matched (%)	Drug family matched
KRAS G12R	8 (100)	8/8 (100)	MEK inhibitor
TP53	7 (87.5)	4/7 (57)	VEGF/VEGR inhibitor*
CDKN2A	4 (50)	4/4 (100)	CDK4/6 inhibitor**
SMAD4	2 (25)	2/2 (100)	MEK inhibitor + (EGFR or Pan-HER inhibitor)***

*PMID: 27466356; **PMID: 33472910; ***PMID: 36127339, PMID: 24625091.

726 Poster Session

MEK inhibitor-based therapy in metastatic pancreatic adenocarcinoma with *KRAS* G12D or *KRAS* G12V alteration. First Author: Stephanie Yohay, Medical College of Wisconsin, Milwaukee, WI

Background: Up to 95% of pancreatic ductal adenocarcinomas (PDAC) are driven by oncogenic alterations in the *KRAS* gene, with *KRAS* G12D (40%) and *KRAS* G12V (30%) mutations being the most common. However, in most cases, *KRAS* alterations are accompanied by alterations in other tumor suppressors such as *TP53*, *CDKN2A/B*, and *SMAD4*. Therefore, targeting *KRAS* alone may not be sufficient for effective treatment of PDAC . As part of the MCW-Master-PREDICT (NCT05802069) observational study, we evaluated the effectiveness of MEK inhibitor (MEK-inh)-based therapy, tailored to the unique molecular alterations of each patient's tumor, in treating metastatic PDAC (mPDAC) with *KRAS* G12D or *KRAS* G12V alterations. **Methods:** From April 2022 to December 2023, 19 patients (*KRAS* G12D=10 (53%), *KRAS* G12V=9 (47%)) were treated with MEK-inh-based therapy. Comprehensive genomic profiling was performed using Tempus xT (648 genes) in 15 patients (79%), FoundationOne (324 genes) in 4 patients (21%), and STRATA (429 genes) in 1 patient. The genomic characteristics, matched therapies, and treatment outcomes are provided (Table). **Results:** Of the 19 patients, 11 (58%) were female. The median age at the initiation of MEK-inh-based therapy was 67 years. Six (32%) patients received MEK-inh-based therapy as first or second-line treatment, while 13 (68%) patients were treated in the third to fifth-line. Among the 19 patients, the median overall survival (mOS) and median progression-free survival (mPFS) from the initiation of MEK-inh-based therapy were 5 and 2 months, respectively. The mOS from the time of MEK-inh-based therapy was 5.1 months for *KRAS* G12D, and 4.7 months for *KRAS* G12V (P=0.87). The mPFS from the time of MEK-inh-based therapy was 2.3 months for *KRAS* G12D and 1.4 months for *KRAS* G12V (P=0.12). Among all 19 patients, 4 (21%) patients had mPFS above 4 months (*KRAS* G12V=1, *KRAS* G12D=3; 4.1, 4.3, 4.7, 9.9 months). **Conclusions:** The efficacy of MEK-inh-based therapy was limited in the late-line treatment of mPDAC for patients with *KRAS* G12D and *KRAS* G12V mutations. The potential role of MEK inhibitors in earlier lines of therapy, and their combination with *KRAS* inhibitors, requires further prospective evaluation. Research Sponsor: None.

Tumor alterations and matched therapies.

Altered gene	KRAS G12V (N=9)	KRAS G12D (N=10)	Patients matched (N=19)	Type of matched therapy
KRAS G12D/V	9 (100%)	10 (100%)	19 (100%)	MEK inhibitor
TP53	7 (78%)	8 (80%)	8 (42%)	VEGF/VEGR inhibitor*
CDKN2A/B	8 (89%)	5 (50%)	9 (47%)	CDK4/6 inhibitor**
SMAD4	3 (33%)	4 (40%)	4 (21%)	MEK inhibitor + (EGFR or pan-HER inhibitor) ***

*PMID: 27466356; **PMID: 33472910; ***PMID: 36127339, PMID: 24625091.
N= number of patients.

727 **Poster Session**

Phase II study of niraparib and dostarlimab for the treatment of germline or somatic homologous recombination repair mutated (HRD) metastatic pancreatic cancer. First Author: Khalid Jazieh, Mayo Clinic Rochester, Rochester, MN

Background: PARP inhibition (PARPi) is a standard maintenance therapy for patients (pts) with germline *BRCA1/2* mutations in pancreatic cancer. Combination of PARPi therapy and immunotherapy has potential to increase efficacy and may offer benefit beyond germline *BRCA1/2*. **Methods:** We performed a multisite single-arm phase 2 study using the highly potent PARPi niraparib with anti-PD1 drug dostarlimab in pts with germline or somatic HRD mutations in *BRCA1/2, PALB2, RAD51C/D,* or *BARD1*. Pts were required to have metastasis, progressed on ≥1 regimen, and prior platinum-exposure unless refused/intolerant. Pts were treated with niraparib 200 mg orally once daily continuously. Dostarlimab 500 mg was administered IV every 3 weeks for 4 cycles, then 1000 mg IV every 6 weeks. The primary endpoint was disease control rate (DCR) at 12 weeks (DCR12) using iRECIST criteria. At least 8 pts with DCR12 in the first 19 eligible pts (41%) would be considered promising for further trials. **Results:** 22 pts were enrolled from Dec 2020 to Oct 2023. Two pts withdrew prior to receiving therapy, and 2 were later found to be ineligible, leaving 18 eligible pts for final analyses. Median age was 63.0 yrs, 39% male, 94% non-Hispanic white, 1 black (6%). 17 pts were platinum exposed, 7 were platinum-refractory, and 5 had previously progressed on PARPi maintenance. Mutation distribution was: *BRCA2* 13/18 (72%), *BRCA1* 4/18 (22%), *BARD1* 2/18 (11%). One pt had both *BRCA1* (somatic) and *BRCA2* (germline) mutations. Germline mutations were identified in 14/18 (78%). DCR12 was achieved in 5/18 (27.8%), so the primary endpoint (>41%) was not met. Pts received a median 2 cycles (range 1-8) of niraparib + dostarlimab. No clinical factors were significantly associated with improved DCR12 (all $p > 0.05$) in subgroup analyses, but we did observe the following statistically nonsignificant associations. Pts who were platinum-refractory had a lower DCR12 compared to those not (14.3% vs. 36.4%). Counterintuitively, PARPi exposed/refractory pts had a higher DCR12 than non-exposed/non-refractory (50 vs 17% exposed/nonexposed and 40% vs 23% refractory/nonrefractory). Pts with germline mutations had a higher DCR12 compared to those without (38.5% vs. 0%). Non-*BRCA2* mutations had a higher DCR12 than those with *BRCA2* mutations (50% vs.17%), with highest DCR12 rates among *BRCA1* patients (67%). Tolerability was in line with similar combinations, with 60% non-hematologic Gr3 or higher, including fatigue (15%), AST increase (10%), nausea/vomiting (10%), sepsis (10%). Two grade 5 events were not attributed to treatment. **Conclusions:** PARPi plus anti-PD1 checkpoint inhibition was not sufficiently active as later line therapy in metastatic pancreatic cancer for the majority of patients with HRD mutations. Additional molecular analyses to elucidate predictive markers of sensitivity/resistance are ongoing. Clinical trial information: NCT04493060. Research Sponsor: GSK.

728 **Poster Session**

AGITG ASCEND: Randomised, double-blind phase II study of certepetide or placebo added to gemcitabine plus nab-paclitaxel in patients with untreated metastatic pancreatic ductal adenocarcinoma: Initial results. First Author: Andrew Peter Dean, St. John of God Subiaco Hospital, Subiaco, Australia

Background: Gemcitabine (GEM) plus nab-paclitaxel (NAB-PAC) is a standard first-line chemotherapy regimen for advanced/metastatic pancreatic ductal adenocarcinoma (PDAC), with median Progression Free Survival (mPFS) and Overall Survival (mOS) of 5.5 & 8.7 in the MPACT trial and 5.6 & 9.2 months (mo) in the NAPOLI 3 trial, respectively. Certepetide (formerly LSTA1 or CEND-1) is a novel cyclic peptide that improves targeted penetration of co-administered drugs into tumor and stroma, leading to potentially increased anti-neoplastic activity. ASCEND is a randomized phase II trial designed to investigate the impact of adding certepetide to GEM/NAB-PAC. (NCT05042128). **Methods:** Eligible participants (pts) with histologically confirmed advanced PDAC and ECOG 0-1 were randomized 2:1 to receive GEM/NAB-PAC plus certepetide (3.2 mg/kg) or placebo (PLA) on days 1, 8, & 15 of a 28-day cycle. Stratification was by age (<65/≥65 years), ECOG (0/1), presence of liver metastasis (Y/N) and trial site. The primary objective was to determine the effect of adding certepetide to GEM/NAB-PAC on PFS. Objective tumor response rate (OTRR), safety and OS were secondary objectives. This non-comparative phase II design targeted a PFS increase of 17% at 6 mo from 47% to 63%. A sample size of 65 patients in the certepetide arm was expected to have 80% power, with 95% confidence to exclude an uninteresting 6-mo PFS rate of 47%. **Results:** 95 pts (66 certepetide, 29 PLA) were enrolled between May 2022 to December 2023. 6-mo PFS in the certepetide and PLA groups was 49.0% (95% CI 36.4%, 60.5%) and 40.8 (95% CI 22.6%, 58.3%) respectively, with mPFS of 5.5 mo in both groups. mOS was 12.42 mo for the certepetide and 9.72 mo for the PLA. An OTRR of 38.3% (certepetide) and 26.9% (PLA) was observed. Of note, 4 complete responses were observed in the certepetide group vs. 0 in PLA group. In subjects with ECOG 0, 6 mo PFS was 68.1% (95% CI 48.7%, 81.4%) in the certepetide compared to 36.4% (95% CI 11.2%, 62.7%) in the PLA. Grade ≥3 toxicities were similar in both groups at 16% (certepetide) and 15% (PLA). **Conclusions:** The addition of certepetide is safe and despite showing no improvement in 6-mo PFS, a possible signal of benefit in OS and OTRR, including the 4 complete responses was observed, warranting further investigation. A further cohort of the ASCEND study evaluating the addition of a second dose of certepetide is ongoing. Clinical trial information: NCT05042128. Research Sponsor: Lisata Therapeutics, Inc.

729 **Poster Session**

Preliminary result of a phase Ib study: Efficacy and safety of FG-M108 plus gemcitabine/nab-paclitaxel as first-line (1L) treatment in patients with Claudin18.2-positive locally advanced unresectable or metastatic (LA/m) pancreatic cancer. First Author: Funan Liu, The First Hospital of China Medical University, Shenyang, China

Background: FG-M108, an ADCC-enhanced anti-CLDN18.2 monoclonal antibody, showed significant efficacy in 1L treatment of gastric cancer. Herein, we report the safety and efficacy results of FG-M108 in 1L treatment of pancreatic cancer (cohort C2&D2). **Methods:** In this open-label, multicenter phase I/II study patients received FG-M108 (cohort C2: 300mg/m^2 or cohort D2: 600mg/m^2 Q3W) plus gemcitabine (1000mg/m^2, d1, d8, Q3W) and nab-paclitaxel (125mg/m^2, d1, d8, Q3W). Eligible patients were those with CLDN18.2 positive (IHC 1+/2+/3+≥10%) previously untreated LA/m pancreatic cancer. The primary endpoint were the incidence of adverse events (AEs) and preliminary clinical efficacy (ORR, DCR, DOR, PFS, and OS). **Results:** As of Sep 5 2024, 50 patients were enrolled and received FG-M108+gemcitabine/nab-paclitaxel treatment (39 patients in cohort C2, 11 patients in cohort D2). The median age was 61 (range 30-72). 47 (94%) patients were with CLDN18.2 moderate-high expression (IHC 2+/3+≥40%). Out of 50 patients, 44 patients had at least one tumor assessment after baseline and included in the efficacy analysis set. The results of ORR and DCR in cohort C2 were 32.4% and 100.0%, while that in cohort D2 were 30.0% and 90.0%, respectively. In cohort C2, the median PFS and median DOR were 6.8 months and 9.8 months, respectively. In the subgroup patients with CLDN18.2 moderate-high expression, the median PFS and median DOR were 7.6 months and 9.8 months, respectively. The median OS has not been reached. FG-M108 related AEs were observed in all patients, with 39 patients (78%) having Grade 3/4 AEs and no FG-M108 related fatal AE was observed. The most common FG-M108 related AEs in cohort C2 & D2 were anemia (56.2% vs 63.6%), nausea (56.4% vs 36.4%), vomiting (48.7% vs 45.5%), and hypoalbuminemia (46.2% vs 54.5%). **Conclusions:** The combined therapy of FG-M108 plus chemotherapy as 1L treatment for patients with CLDN18.2 positive pancreatic cancer was well tolerated with encouraging survival especially in patients with CLDN18.2 moderate-high expression, deserving further investigation. Clinical trial information: NCT04894825. Research Sponsor: None.

730 **Poster Session**

GOBLET study: Results of the safety run-in for first-line metastatic pancreatic ductal adenocarcinoma (PDAC) patients treated with pelareorep + modified FOLFIRINOX +/- atezolizumab. First Author: Thomas Seufferlein, Department of Internal Medicine I, Ulm University Hospital, Ulm, Germany

Background: Immunotherapies are effective in only the small subset of metastatic PDAC patients with MSI-H or dMMR tumors. Pelareorep (pela) is a non-genetically modified, intravenously administered reovirus that selectively infects cancer cells. It stimulates the expansion of pre-existing tumor-infiltrating lymphocyte (TIL) clones, and it makes tumors visible to the immune system by upregulating interferon-induced gene expression including PD-L1, CXCL9, CXCL10, and CXCL11. PD-L1 upregulation also provides a basis for potential synergy between pela and immune checkpoint inhibitors. Pela combined with gemcitabine/nab-paclitaxel/atezolizumab showed promising tumor responses in mPDAC. Here, we are assessing the safety and preliminary efficacy of pela + mFOLFIRINOX +/- atezolizumab in a new cohort in the ongoing GOBLET study. **Methods:** GOBLET is a phase 1/2, open-label, Simon two-stage study in patients with advanced or metastatic GI cancers. In this new GOBLET cohort (Cohort 5), patients with newly diagnosed mPDAC are randomized to receive either pela + mFOLFIRINOX or pela + mFOLFIRINOX + atezolizumab. The primary endpoints are safety and objective response rate (ORR). In Stage 1, 15 evaluable patients will be enrolled in each arm. Both arms include a 3-6 patient safety run-in that must be successfully completed prior to opening the study to full enrollment. If pre-specified ORR success criteria are met, one or both arms may be advanced to Stage 2 during which 17 additional evaluable patients/per arm will be enrolled. **Results:** The 3-patient safety run-in for both arms of Cohorts 5 (6 patients total) have been enrolled, and all patients have completed the required 1-month evaluation period. Cohort 5 safety data have been reviewed by the independent Data Safety Monitoring Board (DSMB). The DSMB identified no safety signals attributable to the combination of pela and mFOLFIRINOX +/- atezolizumab and recommended that enrollment into these arms of Cohort 5 continue without modification. Reported adverse events are consistent with the known toxicities of the components of the treatment regimen. Tumor response results are pending. **Conclusions:** The results of the GOBLET Cohort 5 safety run-in indicate that pela can be given safely in combination with mFOLFIRINOX +/- atezolizumab to newly diagnosed mPDAC patients. Safety and efficacy of these combination therapies will continue to be monitored (Eudra-CT: 2020-003996-16). Clinical trial information: 2020-003996-16. Research Sponsor: None.

731 **Poster Session**

Clinical outcomes of adjuvant chemoradiotherapy in pancreatic adenocarcinoma with R1 resection. First Author: Wahyu Wulaningsih, The Royal Marsden NHS Foundation Trust, London, United Kingdom

Background: The benefit of chemoradiotherapy (CRT) following adjuvant chemotherapy in pancreatic cancer is unclear, though recent findings suggest some disease-free survival benefit over chemotherapy alone. We aimed to assess overall (OS) and progression-free survival (PFS) and patterns of disease recurrence following adjuvant chemoradiotherapy (CRT) in pancreatic adenocarcinoma patients with R1 resection (defined as $<$1mm). **Methods:** We retrospectively collected data of patients with pancreatic adenocarcinoma who underwent adjuvant CRT (50.4-54Gy) following surgery with R1 resection between 2007-2023 in a single centre in the UK. Clinical data was collected from electronic medical records and information on radiotherapy was collected from treatment planning systems (TPS). Toxicity during and after CRT was graded with Common Terminology Criteria for Adverse Events (CTCAE). OS and PFS were measured with Kaplan-Meier curves from the time of surgery to death or disease progression or last follow up. In-field and locoregional recurrence were determined from restaging scans compared against radiotherapy field in TPS. **Results:** Among 33 patients included, 27 (81.8%) died within median follow-up 39 (14.8-111.4) months. All patients received adjuvant chemotherapy prior to CRT. Median OS was 42 (21.6-52.7) months and median PFS was 18.2 (15.5-28.4) months. In all but one patient, the R1 involved the posterior resection margin. Most patients (84.8%) had documented disease progression. In the majority the site of first progression was distant metastatic disease (51.5%), whilst six (18.2%) had local recurrence within the radiotherapy field. Four (12.1%) relapsed locoregionally. No statistically significance difference in OS or PFS was seen with presence of perineural or lymphovascular invasion, though analysis was limited by low number of patients. The majority of patients (72%) were node positive on pathology. Two (6.1%) patients had grade 3-4 toxicity during CRT (neutropenic and non-neutropenic sepsis). **Conclusions:** Our study supported acceptable safety following CRT as an adjuvant treatment in pancreatic cancer patients with R1 resection on surgery. Despite R1 resection, we observed survival outcomes with adjuvant chemotherapy and CRT somewhat above previously reported data. However the benefit of CRT after adjuvant chemotherapy remains unclear. Further evaluation is warranted to clarify which subgroups of patients benefit. Research Sponsor: None.

732 **Poster Session**

Clinical outcomes and molecular characteristics of patients with locoregional ampullary carcinoma. First Author: Cody Eslinger, Department of Hematology and Oncology, Mayo Clinic Arizona, Phoenix, AZ

Background: Ampullary carcinoma (AC) is a rare malignancy arising from the ampulla of Vater which has a more favorable prognosis compared to other pancreatic malignancies. Current guidelines favor surgical resection followed by adjuvant therapy for average risk patients (pts), however the type of regimen as well as other perioperative treatment modalities are still being explored. The objective of this retrospective study aims to provide insight into the management of locoregional disease. **Methods:** Pathology records from Mayo Clinic (AZ, FL, MN) denoting AC between 2010 to 2024 were searched using Mayo Data Explorer and selected for retrospective review. Pt demographics, treatment courses, and next generation sequencing (NGS) data were collected. Statistical analysis was conducted using SAS version 9.04. **Results:** A total of 137 pts (62% male, n = 85) were identified with median age 66 years old. Histologic subtypes were 54% pancreatobiliary (44/81), 40% intestinal (32/81), 6% mixed (5/81), with 56 pts unknown. Stages at diagnosis were 81% resectable (111/137), 11% locally advanced (15/137), and 8% metastatic (11/137). Nearly all (93%, n = 103/111) pts with localized disease had resection with a 97% R0 resection rate (100/103). Majority (68%, n = 75/111) of resectable pts had perioperative chemotherapy with 5-FU or gemcitabine-based regimens (24%, n = 18/75 neoadjuvant vs. 76%, n = 57/75 adjuvant). Perioperative chemoradiation was given in 17% (19/111) of pts. At median follow up of 30.5 months, recurrence rates were 40% (41/103), primarily to the liver (18/41). Median recurrence free survival (RFS) was 29.3 months (95% CI 24.7 – NE). Median overall survival (OS) has not been reached. The 5-year RFS and OS rates were 0.37 (95% CI 0.25 - 0.55) and 0.69 (95% CI 0.58 - 0.82), respectively. Univariate analysis showed no difference in OS with the addition of neoadjuvant or adjuvant chemotherapy or chemoradiation. Somatic NGS testing showed pathogenic mutations in 90% of pts (63/70): 57% KRAS (36/63; 44% G12D, 19% G12V, 8% G12C, 8% G12D, 19% others); 13% homologous recombination (HR) (8/63); 6% *ERRB2*/Her2 amplification (4/63), and 5% mismatch repair (MMR) (3/63). **Conclusions:** AC has a favorable prognosis in resectable pts. However, high recurrence rates indicate the need for better systemic therapies and improved selection of pts who would benefit from these treatments. Future research should focus on refining perioperative strategies to enhance outcomes and reduce recurrence. Research Sponsor: None.

733 **Poster Session**

Cost-effectiveness of NALIRIFOX compared to other first-line treatments for metastatic pancreatic cancer. First Author: Chia Jie Tan, University of Utah College of Pharmacy, Salt Lake City, UT

Background: NALIRIFOX was recently approved for first-line treatment of metastatic pancreatic cancer, where combination chemotherapy remains the main treatment modality. We conducted a cost-effectiveness analysis of NALIRIFOX compared to both FOLFIRINOX and gemcitabine/protein-bound paclitaxel (GP). **Methods:** A partitioned survival model with three health states (progression-free, post-progression and death) was constructed to model the disease course. Using published data from NAPOLI-3, parametric survival curves were generated for NALIRIFOX, from which survival curves for FOLFIRINOX and GP were extrapolated using pooled efficacy data reported by a network meta-analysis of clinical trials. Model inputs were retrieved from the literature or other publicly available resources. Specifically, drug acquisition costs were from average wholesale prices in the Red Book while administration, disease monitoring and supportive care costs were from the CMS Physician Fee Schedule. Utility values to estimate quality-adjusted life years (QALY) were from the CALGB80303 trial. Adverse effects that were considered included anemia, neutropenia, febrile neutropenia, thrombocytopenia, diarrhea, peripheral neuropathy, nausea and vomiting, and fatigue. Adverse effect rates were based on data pooled from clinical trials while costs and disutility were obtained from a secondary analysis of claims data and a clinician survey, respectively. The time horizon was 5 years and utilized a third-party payer perspective. Both costs ($USD 2024) and outcomes were discounted at 3% annually. A probabilistic sensitivity analysis was conducted by varying model parameters based on 95% confidence intervals if known or 10% in both directions otherwise. **Results:** The overall cost associated with NALIRIFOX was $129,314, followed by GP ($104,593) and FOLFIRINOX ($41,528). Among the 3 regimens, NALIRIFOX incurred the highest treatment-related cost ($105,533) but the lowest adverse effects management cost ($16,541). On average, patients on NALIRIFOX gained a total of 1.14 life years (LY) or after adjusting for utility, 0.86 QALY. FOLFIRINOX led to 1.09 LY or 0.82 QALY while GP resulted in 0.99 LY or 0.75 QALY. Due to its higher cost and lower survival benefit, GP was dominated by FOLFIRINOX. Compared to FOLFIRINOX, the incremental cost-effectiveness ratio (ICER) of NALIRIFOX was $1,952,062/QALY. In our sensitivity analysis, the probability of NALIRIFOX being cost-effective was 0% at all conventional ICER thresholds up to $200,000/QALY. **Conclusions:** Due to the high drug acquisition cost, the ICER of NALIRIFOX far exceeds currently acceptable thresholds for economic value and NALIRIFOX is considered not cost-effective. This is despite that in our model among first line treatments for metastatic pancreatic cancer, NALIRIFOX was associated with reduced costs for managing adverse effects. Research Sponsor: None.

734 **Poster Session**

The effect of renin-angiotensin system inhibitors on patients with pancreatic cancer receiving systemic treatment: A retrospective cohort study from TriNetX US collaborative networks. First Author: Cheng-Wei Chou, Taichung Veterans General Hospital, Taichung, Taiwan

Background: This study aimed to assess the impact of renin-angiotensin system inhibitors, including angiotensin receptor blockers (ARBs) and angiotensin-converting enzyme inhibitors (ACEIs), on patients with pancreatic cancer undergoing systemic chemotherapy. **Methods:** We conducted a global, retrospective collaborative network analysis using TriNetX data from January 1, 2014, to December 31, 2023, to explore this question. The study captured data on pancreatic cancer diagnoses, comorbidities, medications, and survival outcomes. Propensity score matching (PSM) was used to create 1:1 matched groups. Hazard ratios (HR) with 95% confidence intervals (95% CI) were calculated to assess survival outcomes, while the Kaplan–Meier method was employed to evaluate survival probabilities. Subgroup analyses were performed based on sex, age, surgery, race, and comorbidities. **Results:** The study included 1,861 patients who were on ARBs/ACEIs and 4,875 patients who were not on these medications at baseline. After 1:1 propensity score matching, both groups were well-balanced with 1,627 patients each. Compared to the non-ARBs/ACEIs group, patients receiving ARBs/ACEIs demonstrated significantly better overall survival (HR: 0.846, 95% CI: 0.753–0.949). Subgroup analyses revealed significantly improved survival outcomes in patients on ARBs/ACEIs, including those without surgery (HR: 0.846, 95% CI: 0.749–0.956), males (HR: 0.762, 95% CI: 0.640–0.908), patients aged 65 years and older (HR: 0.801, 95% CI: 0.709–0.906), and White patients (HR: 0.806, 95% CI: 0.704–0.924). Notably, similar survival benefits were observed in patients with comorbidities such as diabetes, ischemic heart disease, dyslipidemia, and hypertension. **Conclusions:** In the largest TriNetX matched cohort study on pancreatic cancer, the use of ARBs/ACEIs was associated with a reduced risk of overall mortality. However, prospective studies are needed to further evaluate the effects of concomitant ARB/ACEI use in patients with pancreatic cancer undergoing systemic treatment. Research Sponsor: None.

735 **Poster Session**

Neoadjuvant therapy in borderline resectable and resectable pancreatic cancer. First Author: Kriti Dhamija, Allegheny Health Network, Pittsburgh, PA

Background: To improve clinical outcomes in borderline resectable and resectable pancreatic cancer patients, preoperative chemotherapy with or without radiation can be administered. According to NCCN, there is limited evidence to recommend a specific neoadjuvant chemotherapy regimen. The Dutch randomized phase III PREOPANC trial showed that preoperative chemoradiotherapy led to improved disease free survival. We compared overall survival (OS) between two chemotherapy regimens; mFOLFIRINOX (mFOL) and Gemcitabine/nabPaclitaxel (Gem/nabP). **Methods:** We performed a retrospective analysis of biopsy-proven pancreatic cancer patients from 1/1/2019 to 2/1/2024. We included borderline resectable and resectable pancreatic cancer patients who received neoadjuvant chemotherapy and then underwent pancreatic resection. We divided them into two cohorts: those receiving mFOL and those receiving Gem/nabP. Kaplan-Meier and multivariate Cox regression analyses were performed to analyze OS and rates of R_0 resection amongst the two groups. A p-value < 0.05 was deemed statistically significant. **Results:** From January 2019 to January 2024, 85 pancreatic cancer patients received preoperative chemotherapy prior to pancreatic resection. 50 patients received mFOL and 35 patients received Gem/nabP. The mean overall survival (OS) for the mFOL group was 1,137 days, while for the Gem/nabP group it was 876 days. The hazard ratio was estimated as 1.692 (confidence interval 0.901- 3.177, p-value 0.10). After matching for pathological stage and grade, preoperative radiation, postoperative chemotherapy, and presence of perineural and lymphovascular invasion, the hazard ratio was 1.981 (p-value 0.065). The R_0 resection rate was 92.1% for mFOL and 82.2% for Gem/nabP. The odds ratio for having negative margins with mFOL compared to Gem/nabP was 2.005 with a p-value of 0.37. **Conclusions:** Preoperative chemotherapy with mFOL did not demonstrate improved OS compared to Gem/nabP. The probability of achieving negative margins was also similar amongst the two groups. The SWOG S1505 which is a randomized phase II trial comparing perioperative chemotherapy with mFOL versus Gem/nabP in resectable pancreatic cancer patients has also shown similar OS and progression free survival in both groups. Research Sponsor: None.

736 **Poster Session**

XPO1 inhibition as a clinically viable strategy to enhance durability of response to $KRAS^{G12D}$ inhibitor in pancreatic ductal adenocarcinoma. First Author: Mohammed Najeeb Al Hallak, Barbara Ann Karmanos Cancer Institute, Wayne State University, Detroit, MI

Background: The OFF State (GDP bound) $KRASG^{12D}$ inhibitor MRTX1133 is currently being evaluated in pancreatic ductal adenocarcinoma (PDAC) patients. Nevertheless, as with other OFF state $KRAS^{G12C}$ inhibitors sotorasib and adagrasib, the $KRAS^{G12D}$ inhibitor may yield only a modest increase in disease-free survival due to emergence of drug resistance. This underscores the need to identify strategies that can enhance the efficacy of KRAS inhibitors in PDAC. Nuclear protein transport plays an important role in several pro-tumorigenic pathways and has been shown to be a therapeutic vulnerability in KRAS mutant cancers. XPO1 is the major nuclear exporter and plays a pivotal role in shuttling various critical tumor suppressors, genome surveillance proteins, and transcription factors out of the nucleus and is frequently overexpressed in various cancers, including PDAC. In this study, we employed a $KRAS^{G12D}$ inhibitor resistant model and evaluated its sensitivity to nuclear exporter protein inhibitor. **Methods:** We examined the cytotoxic and molecular effects of $KRAS^{G12D}$ inhibitor MRTX1133 in combination with nuclear transport inhibitor eltanexor in $KRAS^{G12D}$ inhibitor resistant models. The preclinical antitumor efficacy of the combination was evaluated in $KRAS^{G12D}$ mutant PDAC cell derived xenograft (CDX) subcutaneous and orthotopic models. **Results:** Eltanexor treatment reversed MRTX1133 resistance *in vitro* yielding suppressed proliferation of $KRAS^{G12D}$ mutant 2D and 3D cultures of PDAC cell lines and patient derived primary tumors cells. High throughput P-kinome analysis showed suppression of a larger repertoire of MAPK substrates in combination treatment compared to single agent group. Eltanexor and MRTX1133 combination led to reduction in protein expression of KRAS downstream effectors and cell cycle markers. Furthermore, a combined administration of eltanexor and MRTX1133 at sub-optimal doses resulted in remarkable tumor growth inhibition in multiple subcutaneous and orthotopic $KRAS^{G12D}$ mutant mice models. Moreover, eltanexor as a maintenance therapy also prevented tumor relapse indicating durability of response in PDAC. **Conclusions:** This is the first study demonstrating that nuclear transport inhibitors can overcome $KRAS^{G12D}$ inhibitor MRTX1133 resistance in PDAC cells. This study provides a rationale for combining MRTX1133 with eltanexor for the treatment of PDAC patients with $KRAS^{G12D}$ mutant tumors. Research Sponsor: U.S. National Institutes of Health; NIH R37CA215427.

737 **Poster Session**

Predictive factors for the efficacy of nal-IRI + 5FU/LV therapy in patients previously treated with conventional irinotecan. First Author: Yasutsugu Asai, Department of Gastroenterology, Kanagawa Cancer Center, Yokohama, Japan

Background: There were some reports that nal-IRI + 5FU/LV (NAPOLI) therapy was effective for patients with unresectable pancreatic cancer who had a history of treatment containing conventional irinotecan (IRI) though the others did not. We aimed to investigate the factors that contribute to the effectiveness of NAPOLI therapy for patients who have already received IRI-containing treatment. **Methods:** We retrospectively enrolled the consecutive patients who had previously treated with IRI-containing treatment and started NAPOLI therapy at our hospital between June 2020 and December 2022. We compared the patient backgrounds of patients who achieved CR, PR, or SD according to RECISTv1.1 (effective group) and those who did not (ineffective group). Continuous variables were divided into two groups by the median of the entire population, and factors contributing to the effectiveness of NAPOLI therapy were determined by logistic regression analysis. **Results:** The subjective were 52 patients. The reason for discontinuation of IRI-containing treatment was refractory and intolerant in 45 and 7 patients, respectively. The median age was 65 years. Twenty-three patients (44%) had ECOG PS of 0, and 41 patients (79%) had metastatic disease. The median albumin, CRP, CEA and CA 19-9 was 3.8 g/dL, 0.3 mg/dL, 8.5 ng/mL and 2082 U/mL, respectively. Twenty-one patients (40%) showed SD and they were allocated in the effective group whereas 31 patients were in the ineffective group. Of the 21 patients in the Effective Group, 16 patients(76%) were refractory, while of the 31 patients in the Ineffective Group, 29 patients (94%) were refractory. The factors contributing to disease control were PS of 0 (odds ratio, 5.79; 95% confidence interval, 1.34–25.0), CA 19-9 $<$1000 U/mL (odds ratio, 5.53; 95% confidence interval, 1.26–24.3), and Alb $>$3.8 g/dL (odds ratio, 5.19; 95% confidence interval, 1.14–23.7). Age, sex, disease stage, serum level of CRP, and CEA did not affect the effectiveness of NAPOLI therapy. **Conclusions:** Even in patients who have already received IRI treatment, the NAPOLI regimen is expected to be effective in patients with PS of 0, Alb $>$3.8 g/dL, and CA 19-9 $<$1000 U/mL. Research Sponsor: None.

738 **Poster Session**

Chemotherapy selection for advanced or metastatic pancreatic cancer using machine learning models trained with multi-center datasets. First Author: Moonho Kim, Department of Hematology and Oncology, University of Ulsan College of Medicine, Gangneung Asan Hospital, Gangneung, South Korea

Background: Several factors must be considered when selecting the chemotherapeutic regimen for advanced or metastatic pancreatic cancer. The decision between FOLFIRINOX and Gemcitabine/Nab-paclitaxel (GnP) as the first-line therapy is challenging, as survival is influenced by the efficacy and toxicity profiles of these treatments, along with individual patient characteristics and therapeutic vulnerabilities. This study aims to facilitate the selection of an appropriate first-line regimen for advanced or metastatic pancreatic cancer by employing machine learning (ML) methods trained on multi-center datasets. **Methods:** Our study is based on a cohort of 191 patients who underwent systemic chemotherapy for advanced or metastatic pancreatic cancer at the Gangneung Asan Hospital and Asan Medical Center in South Korea between 2014 and 2023. The dataset consisted of 17 types of demographic and clinical characteristics, as well as time-course of response and survival outcomes. We divided the cohort into training and test groups (4:1) in a stratified manner. ML models predicting overall survival (OS) were trained on the former group using the CatBoost method. The models utilizing a minimal subset of variables were selected with respect to ROC-AUC during 5-fold cross validation. A patient was classified as having high risk to a therapy when the predicted probability of death following treatment initiation was above the threshold value as determined during training of the respective model. **Results:** The median age of the entire cohort was 62 years, with 64% being male. The ML models achieved ROC-AUCs of 0.81 and 0.82 when predicting early death within 12 months following the initial administration of FOLFIRINOX or GnP, respectively. The predictive accuracies of the models were 0.77 and 0.80 for the unseen datasets from the two treatment groups, respectively. The median OS of the high- versus low-risk groups of FOLFIRINOX predicted by the ML models were significantly different (9 vs 15 months, $P < 0.0001$), with a hazard ratio (HR) of 2.8. Similarly, the median OS of the high- and low-risk groups of GnP were significantly different (9 vs 18 months, $P < 0.01$, HR of 2.5). We observed that 27% of the patients predicted as high risk to FOLFIRINOX therapy were predicted as low risk to GnP and vice versa for 32% of the patients predicted as high risk to GnP therapy. **Conclusions:** We developed ML models to compute the probability of early death following FOLFIRINOX or GnP therapy from routinely collected data of the patients with advanced or metastatic pancreatic cancer. After confirming robust predictive performance with multi-center datasets, we believe that these models may assist clinicians in selecting a first-line regimens, potentially enhancing personalized treatment strategies for this challenging disease. Research Sponsor: Medical Research Promotion Program through the Gangneung Asan Hospital, funded by the Asan Foundation.

739 Poster Session

Surufatinib combined with TAS-102 as a late-line therapy in patients with metastatic pancreatic cancer (mPDAC): Updated results of a single-arm, phase II trial. First Author: Dong-Sheng Zhang, Department of Medical Oncology, Sun Yat-sen University Cancer Center, State Key Laboratory of Oncology in South China, Collaborative Innovation Center for Cancer Medicine, Sun Yat-sen University, Guangzhou, China

Background: Patients (pts) with mPDAC recurrent after FOLFIRINOX and AG chemotherapies have few treatment options. Previous reports of NCT05481463 showed promising clinical benefits of surufatinib (a small-molecule inhibitor of VEGFR1-3, FGFR1 and CSF-1R) plus TAS-102 (combination of trifluridine and tipiracil hydrochloride) in recurrent mPDAC pts. Here, we report the updated results. **Methods:** This is a single-arm, phase II trial (NCT05481463). Eligible pts with histologically confirmed mPDAC, who had progressed after ≥ 2 lines of previous therapies were treated with surufatinib (250mg, qd, po, q4w) and TAS-102 (35 mg/m^2, bid, po, days 1–5 and 8–12, q4w). The primary endpoint was progression-free survival (PFS). Key secondary endpoint was overall survival (OS). Other secondary endpoints included objective response rate (ORR), disease control rate (DCR), and safety. **Results:** Up to Sep 18, 2024, 20 pts were enrolled (median age 58 years, male 70.0%, TNM stage IV 100.0%, ECOG PS 1 100.0%, liver metastasis 50.0%, peritoneum metastasis 50.0%). Median baseline CA19-9 level of the 20 pts was 996.7u/ml, with a range from 2u/ml to 16612u/ml. 40.0% of pts had received 3L prior therapies before enrollment and 70.0% with primary site on pancreatic body and tail. All pts had received at least one post-baseline tumor assessment (evaluable), among which ORR was 15.0% (3 PRs), DCR was 30.0% (3 PRs, 3SDs). 35.0% pts had stable or shrinking tumors. Median PFS was 2.35m (95% CI 1.91-3.94). Median OS was 6.44m (95% CI 3.81-NR). In subgroup analysis, significantly longer PFS&OS were observed in pts without liver metastasis (mPFS: 3.42m vs 2.01m, p=0.049; mOS: 9.92m vs 3.70m, p=0.026), or with ≤ 2 metastatic organs (mPFS: 3.94m vs 1.91m, p=0.005; mOS: 10.09m vs 3.81m, p=0.003). Additionally, OS was significantly prolonged in pts with baseline CA19-9 level ≤ 996.7u/ml, in contrast with those with baseline CA19-9 level more than 996.7u/ml (9.92m vs 3.70m, p=0.008). All pts experienced treatment emergent adverse events (TEAEs). Most common (≥ 10%) TEAEs (any grade) were anemia (59.1%), neutropenia (54.6%), leukopenia (50.0%), and lymphopenia (45.5%). Grade ≥ 3 TEAEs included neutropenia (31.8%), lymphopenia (13.6%), and anemia (9.1%). **Conclusions:** Surufatinib plus TAS-102 showed encouraging survival benefits and promising anti-tumor activity with acceptable toxicity in the late-line treatment of refractory mPDAC, especially in patients without liver metastasis, with ≤ 2 metastatic organs, or with baseline CA19-9 level ≤ 996.7u/ml. The results warrant further investigations in a large cohort. Clinical trial information: NCT05481463. Research Sponsor: None.

740 Poster Session

Chidamide plus envafolimab combined with S-1 as second-line treatment in advanced and metastatic pancreatic cancer (P-henomS/SCOG-P002): A single-arm, exploratory, multicenter, phase 2 trial—Interim report. First Author: Wei Li, Department of Oncology, The First Affiliated Hospital of Soochow University, Suzhou, China

Background: Pancreatic ductal adenocarcinoma (PDAC) is one of the most lethal cancers with poor prognosis worldwide. Chidamide, a subtype-selective histone deacetylase (HDAC) inhibitor, has significant anti-tumor effects and extensive synergistic effects with immunotherapy. Envafolimab is a light-chain deficient PD-L1 antibody. Here we conducted a single-arm, multicenter, prospective phase II clinical study (ChiCTR2200058431) to evaluate the efficacy and safety of Chidamide plus Envafolimab combined with S-1 as second-line treatment in advanced and metastatic pancreatic cancer. **Methods:** Patients with metastatic PDAC receive Envafolimab (400 mg, on day 1), Chidamide (20 mg orally twice weekly, on days 0, 3, 7, and 10), and S-1 (40-60 mg according to body surface area, orally twice daily from day 1 to 14) every 3 weeks until disease progression, unacceptable toxicity, or patient refusal. The primary end points are safety and ORR. The secondary endpoints were PFS, OS, DCR, QoL and nutrition score. **Results:** Recruitment completed with 16 patients as of September 2023 and the data cut-off for analysis was August 2024, 13 were evaluable. Median age was 66 (range 59 - 80), with 5 males and 11 females. After a median follow-up of 8.5 months, the ORR and DCR by RECIST v1.1 were 30.77% and 76.92%, respectively. Median PFS was 5.83 months (95%CI 2.592-9.068) and median OS was not reached. No new safety signals were observed. Grade ≥3 treatment-related adverse events (TRAEs) occurred in 31.25% of patients. The most common TRAEs were anemia (12.5%), decreased platelet count (6.25%) and neutropenia (12.5%). No treatment-related deaths occurred. **Conclusions:** Preliminary data suggest that Chidamide and Envafolimab in combination with S-1 may be an effective second-line treatment with a manageable safety profile for patients with PDAC. Clinical trial information: ChiCTR2200058431. Research Sponsor: None.

Efficacy evaluation.

Efficacy Evaluation	All (N = 13) n(%)
Best efficacy evaluation	
Partial Response (PR)	4 (30.77%)
Stable Disease (SD)	6 (46.15%)
Progressive Disease (PD)	3 (23.08%)
Objective response rate (ORR)	30.77%
95%CI	9.09-61.43
Disease control rate (DCR)	76.92%
95%CI	46.19-94.96
mPFS (95%CI)	5.83 (2.592-9.068)
mOS (95%CI)	- (NR)

741 Poster Session

Interim analysis of feasibility, safety, and tumor control in two first-in-human trials of a novel alpha-emitting radionuclide for pancreatic adenocarcinoma. First Author: Philip Blumenfeld, Sharett Institute of Oncology, Hadassah-Hebrew University Medical Center, Jerusalem, Israel

Background: This study aims to report the initial feasibility, safety, and tumor response of delivering Diffusing Alpha-Emitter Radiation Therapy (Alpha DaRT) to the pancreas. This innovative approach involves the interstitial implantation of a novel alpha-emitting radiation source via endoscopic ultrasound (EUS) for the treatment of pancreatic cancer. **Methods:** The study enrolled patients treated with Alpha DaRT to the pancreas across two protocols (PANC and ALL). The primary objective was to assess the safety of Alpha DaRT delivery to the target lesion via EUS, while the secondary objective was to evaluate initial tumor response using the Response Evaluation Criteria in Solid Tumors version 1.1 at 1 and 3 months post-Alpha DaRT source insertion. Toxicity was evaluated according to the Common Terminology Criteria for Adverse Events version 4.03. The PANC protocol included patients with biopsy-proven adenocarcinoma, either unresectable, recurrent, or metastatic, who were unfit for standard treatment. The ALL protocol allowed for any malignancy with a targetable lesion and permitted concurrent systemic therapy. For this analysis, only patients from the ALL protocol with pancreatic adenocarcinoma treated to the primary tumor were included. Pre-procedure planning involved CT simulation and EUS, and EUS-guided Alpha DaRT placement was confirmed by CT. The median follow-up time was 91 days. **Results:** A total of 12 patients (9 male, 3 female) median age 73 (range: 59-85) were enrolled (4 in PANC, 8 in ALL). Four patients had localized, unresectable disease and eight had metastatic disease. Three patients had a history of prior radiotherapy. Successful placement of Alpha DaRT sources into the targetable pancreatic lesions was achieved in all cases (100%), as confirmed by post-treatment imaging. Grade 1-2 acute toxicities included fever in 2 of 12 patients (16.7%), fatigue/weakness in 2 of 12 patients (16.7%) and abdominal or back pain in 3 of 12 patients (25.0%). One patient was hospitalized for obstructive jaundice (Grade 3) possibly secondary to inflammation seven days post-implantation requiring percutaneous transhepatic cholangiography; all hepatic laboratory values returned back to normal post procedure. All adverse events resolved. No long-term toxicities (> 3 months) have been seen to date. Tumor response data were available for 8 of 12 patients. Local control was 100% for all evaluable patients. Of these, 6 (75.0%) exhibited stable disease, while 2 (25.0%) showed a partial response (> 30% tumor reduction). One patient had a complete metabolic response on PET-CT at both the primary pancreatic lesion and liver metastasis. **Conclusions:** Alpha DaRT delivered via endoscopy is feasible with a favorable safety profile and demonstrates promising tumor responses. Ongoing longer follow-up and larger studies are required to validate these findings. Clinical trial information: NCT05781555 and NCT05657743. Research Sponsor: None.

742 Poster Session

Safety and preliminary efficacy of ivaltinostat, an HDAC inhibitor, plus capecitabine in patients with pretreated metastatic pancreatic adenocarcinoma (mPDAC): Results of a phase Ib study. First Author: Christos Fountzilas, Roswell Park Comprehensive Cancer Center, Buffalo, NY

Background: Front-line chemotherapy for patients (pts) with mPDAC typically consists of mFOLFIRINOX or gemcitabine/nab-paclitaxel given until disease progression or intolerable toxicity. For pts with durable disease control on front-line treatment, maintenance therapy that can effectively delay disease progression while preserving quality of life with minimal cumulative toxicity is highly desirable. Ivaltinostat (ival) is a pan-histone deacetylase inhibitor that increases histone acetylation, suppresses cell proliferation, and promotes apoptosis. Preclinical data demonstrated synergy with 5-FU/capecitabine (cape) in mouse models. We report here results of the Phase Ib portion of a Ib/2 trial evaluating the safety and recommended phase 2 dose (RP2D) of this combination to use in a randomized study of ival + cape vs cape as maintenance therapy. **Methods:** Key eligibility criteria for phase 1b included unresectable or metastatic PDAC; ≥18 years old; ≥1 prior therapy; ECOG PS 0-1; and no known germline BRCA1/2 mutation. Three cohorts were enrolled, receiving ivaltinostat at 60, 125, or 250mg/m^2 iv (weekly on days 1 and 8) in combination with cape (1000mg/m^2 po BID on days 1-14) of a 21-day cycle. 1o objectives: safety, tolerability and RP2D; 2o objectives: pharmacokinetics (PK) of ival + cape. **Results:** A total of 28 patients were enrolled at the 60 (n=6), 125 (n=10) and 250 (n=12) mg/m^2 ival dose levels. Median prior lines of therapy: 2 (range 1-7). Median age: 67 years (range 50-83). Only one DLT (G4 thrombocytopenia) was observed at 250mg/m^2 dose level. Treatment-emergent AE profile was similar across dose levels, with common TEAEs including fatigue (n=15, 53.6%), nausea (n=14, 50.0%), palmar-plantar erythrodysesthesia syndrome (n=12, 42.9%), diarrhea (n=10, 35.7%), and constipation (n=9, 32.1%). G3 AEs included neutropenia (n=6, 21.4%), anemia (n=4, 14.3%), abdominal pain (n=3, 10.7%) and diarrhea (n=3, 10.7%). G4 AEs included 1 (3.6%) event of neutropenia and 1 (3.6%) of thrombocytopenia. No SAEs related to study drugs occurred in any of the 3 dose cohorts. PK analyses indicated dose proportional increases in exposure with increasing doses of ival. Histone H3K27 acetylation for ival showed the similar trend in optical density evaluation. Across all dose levels and prior treatments, median OS and PFS were 9.6 and 3.3 months, respectively. **Conclusions:** The combination of ival and cape was generally well tolerated in this heavily pretreated patient population. Ival 250 mg/m^2 is the dose selected for the ongoing randomized study of ival + cape vs cape in the maintenance setting for mPDAC pts post-FOLFIRINOX, which is expected to finish accrual in 2025. Clinical trial information: NCT05249101. Research Sponsor: CG Pharmaceuticals Inc.

743 Poster Session

A phase 1b dose escalation trial of gemcitabine and nab-paclitaxel in combination with lixumistat in patients with advanced pancreatic cancer. First Author: Shubham Pant, The University of Texas MD Anderson Cancer Center, Houston, TX

Background: Despite recent advances, there are limited effective therapeutic options for pancreatic ductal adenocarcinoma (PDAC). Conventional chemotherapy can be difficult to tolerate and offers modest survival benefits. Lixumistat is a small molecule biguanide that inhibits the first and rate-limiting step of the oxidative phosphorylation pathway, which is critical to the survival of PDAC tumor cells. A first-in-human dose escalation trial of Lixumistat in solid tumors demonstrated a favorable safety profile and a recommended phase 2 dose (RP2D) of 800 mg per day (QD) (1). **Methods:** A single center Phase 1b study is currently enrolling patients with treatment-naive metastatic PDAC to evaluate the safety and tolerability of Lixumistat when combined with standard-of-care gemcitabine (Gem) plus nab- paclitaxel (NP) (NCT05497778). The initial escalation cohort received oral Lixumistat at 400 mg QD; the subsequent escalation cohort received oral Lixumistat at 800 mg QD. All patients also receive fixed doses of intravenous (IV) NP (125 mg/m^2) and IV Gem (1000 mg/m^2) on Days 1, 8, and 15 (28 day cycle). Here we report the preliminary results from the escalation phase of the trial. **Results:** Thirteen patients have been treated: 7 at 400 mg and 6 at 800 mg. There were no grade 4-5 toxicities. Two patients experienced a DLT at the 800 mg dose (grade 3 diarrhea and grade 3 fatigue); no DLTs were observed at the 400 mg dose. The most common Lixumistat related toxicities were Grade 1/2 nausea/vomiting (controlled with anti-emetics), skin rash, fatigue, and diarrhea. Grade 1-3 thrombocytopenia, neutropenia, and peripheral sensory neuropathy events were no more than expected with this combination and managed with Gem/NP dose reduction, omission of day 8, or therapy hold until resolved to Grade 1. Based on toxicities observed at 800 mg, 400 mg is the RP2D. Among 6 response-evaluable patients treated at the RP2D, 2 achieved an objective partial response (ORR = 33%, 95% CI: 4.3%-77.3%); 4 patients had stable disease with tumor shrinkage observed in 3 of these patients (12%, 22% and 23% reduction); and the DCR was 100% (95% CI: 54.1%-100%). Notably, a patient from the 800 mg cohort who was dose reduced to 400 mg after DLT achieved a complete response in their target lesions. Current median follow-up time is 13.2 months, and 2 patients have died. Estimated median PFS is 12.9 months (95% CI: 3.65, NA) and estimated median OS is 18 months (95% CI: 6.42, NA). Four patients remain on active treatment at the time of analysis. **Conclusions:** When combined with Gem and NP; a dose of 400 mg of Lixumistat QD was demonstrated to be safe, tolerable and identified as the RP2D. This trial is currently enrolling patients at the RP2D in the expansion cohort. Updated data on safety and efficacy will be presented at the meeting. 1. Janku 2022. Clinical trial information: NCT05497778. Research Sponsor: This research is supported by The MD Anderson Cancer Center SPORE in Gastrointestinal Cancer Grant P50 CA221707; Immunomet.

744 Poster Session

Galeterone monotherapy in advanced pancreatic ductal adenocarcinoma: Results from a phase two trial. First Author: Aaron Ciner, University of Maryland Greenebaum Comprehensive Cancer Center, Baltimore, MD

Background: Advanced pancreatic ductal adenocarcinoma (PDAC) is a devastating disease with limited treatment options. Dysregulated MAPK and PI3K signaling in PDAC activates the eukaryotic translational initiation complex, promoting cell growth, chemoresistance and metastatic spread (PMID 25593033). Galeterone, a novel steroidal anti-androgen, downregulates critical mediators of this complex including Mnk1/2 and phosphorylated eIF4E, thereby blocking epithelial-to-mesenchymal transition and NF-kB activity. In-vitro, galeterone exhibits anti-tumor activity both alone and in combination with gemcitabine and slows tumor growth in a MiaPaCa-2 xenograft murine model (PMID 28881737). We present results of the monotherapy arm from an ongoing phase two trial evaluating galeterone +/- gemcitabine for patients with advanced PDAC. **Methods:** Patients with locally advanced or metastatic PDAC, an ECOG performance status of 0-2 and adequate organ function, who progressed on two prior lines of systemic therapy were eligible. Galeterone was administered orally at 2550mg daily and gemcitabine intravenously at 1000mg/m^2 weekly for three weeks on 28-day cycle. The primary endpoint was radiographic response rate per RECIST v1.1 and secondary endpoints included progression-free survival (PFS), overall survival (OS) and safety. We used a Simon's optimal two-stage design with a null hypothesis of p_0=0.05 and one-sided alternative of p_1=0.20 with an early stopping rule for futility in each arm. Due to poor accrual and no signal of activity, the galeterone alone arm closed after enrollment of four patients. **Results:** Four patients were evaluable for the primary endpoint. Ages ranged from 43 to 68. Two patients were male and two were female. Two patients were white and two were black; one patient was Hispanic/Latino. All patients had ECOG PS of 1. No patients had a radiographic response; the best response was progressive disease (PD). PFS from cycle 1 day 1 ranged from 0.9 – 1.7 months and overall survival between 1.8 – 3.6 months. The only treatment-related adverse events (TRAEs) were grade 1 dizziness and grade 3 fatigue in one patient each. There were no treatment-related serious adverse events. No patients required dose modification or discontinued due to TRAEs. **Conclusions:** Galeterone showed no clinical activity, but with a favorable safety profile, as a single-agent in heavily pre-treated patients with advanced PDAC. Activation of the Mnk-eIF4E axis mediates chemoresistance in PDAC and galeterone may prove more efficacious with gemcitabine. This combination arm is currently open to enrollment. Clinical trial information: NCT04098081. Research Sponsor: None.

Clinical outcomes with galeterone monotherapy in PDAC.

Pt #	Age	Prior lines of Therapy	Time on Treatment (days)	Best Response	PFS (months)	OS (months)
1	68	4	46	PD	1.5	3.6
2	59	2	54	PD	1.7	3.8
3	55	2	47	PD	1.4	2
4	43	2	35	PD	0.9	1.8

745 Poster Session

SIL-204 siRNA encapsulated in extended release microparticles for the treatment of localized cancer that harbors a KRAS G12x or G13D mutation. First Author: Mitchell Shirvan, Silexion Therapeutics Corp, Grand Cayman, Cayman Islands

Background: In a Phase 2 trial, treatment of non-resectable locally advanced pancreatic cancer (LAPC) patients harboring the KRAS G12D or G12V mutations, with SiG12DLoder combined with chemotherapy resulted in superior median overall survival (OS) of 9.3 months over chemotherapy (not statistically significant) and Objective Response Rate of 55% (35% over the chemotherapy control), when administered once every 3 months (Clinical trial # NCT01676259, ESMO 2023, abstract FPN: 1626). SIL-204 which is a modified version of siRNA siG12D is composed of a 21-base sense and 23-base antisense strands, both chemically modified and linked to a lipid. SIL-204 is encapsulated in biodegradable PLGA microparticles (MP), for direct release and injection into KRAS mutated solid tumors using an endoscopic ultrasound procedure. The strategy of preventing the synthesis of KRAS mutated protein may have an advantage over other approaches targeting KRAS function using small molecules. **Methods:** Methods are presented in results section. **Results:** SIL-204 has an increase stability over siG12D.To enhance stability, base modifications and phosphorothiate bonds at cleavage sites were implemented. SIL-204 exhibited a half-life of more than 48 hours in human serum. SIL-204 has robust KRAS silencing. When tested in Hepa1-6 cells transfected with individual human KRAS mutations, SIL-204 effectively silenced the various G12x mutations found in pancreatic cancer (D,V,R,C) at sub-nanomolar concentration (IC50 range=0.19-0.59) as well as G13D (IC50 =0.37). Addition of a hydrophobic tail to a prototype of SIL-204, SIL-101, increased the silencing of KRAS. Using PANC-1 cells (KRAS G12D mutation), lipid conjugation of the siRNA doubled the siRNA's silencing efficiency in these cells. In addition, release of SIL-204 from the MP was tested in rats and indicates a prolong and sustained release over our intended treatment regimen SIL-204-MP was administered to tumors from human pancreatic tumor cell lines Capan-1 (KRAS G12V mutation and labelled with luciferase) grown in NSG mice and Panc-1 (grown in Athymic Nude mice). Bioluminescence imaging of tumor growth showed that SIL-204-MP significantly reduced Capan-1 tumor growth compared to vehicle controls, $p<0.0005$. Furthermore, histopathological analyses of tumor center slices from Capan-1 and Panc-1 models showed induction of significant tumor necrosis with SIL-204 compared to vehicle controls. **Conclusions:** The preclinical results of SIL-204-MP are promising and as the potential to further enhance the anti-tumor effect demonstrated by siG12DLoder in pancreatic patients. SIL-204 is now in toxicology studies as preparation for clinical trials in non-resectable pancreatic cancer. Research Sponsor: Silexion Therapeutics Corp.

746 Poster Session

Immunotherapy engagement in pancreatic adenocarcinoma: Provisional results of iLSTA study— Durvalumab, LSTA1 (certepetide), gemcitabine, and nab-paclitaxel for locally advanced pancreatic ductal adenocarcinoma. First Author: Andrew Peter Dean, St John of God Subiaco Hospital, Subiaco, Australia

Background: Pancreatic ductal adenocarcinoma (PDAC) is characterised by a dense, extracellular matrix-rich stroma, which creates a physical barrier against drug penetration. Evidence suggests that the iRGD peptide, LSTA-1, can increase the penetration of anti-cancer drugs to tumours through selective targeting of tumour endothelial cell receptors. Additionally, LSTA-1 administration promotes CD8+ immune cell tumour infiltration, potentially enabling immunotherapy as a treatment modality. iLSTA examined safety, tolerability and effect of the combination. **Methods:** Participants diagnosed with locally advanced PDAC were divided into 3 cohorts in a 1:1:4 ratio. Cohort 1 (n=5) received gemcitabine (1000mg/m^2), nab-paclitaxel (125mg/m^2), placebo LSTA-1 (3.2mg/kg) and placebo durvalumab (750mg). Cohort 2 (n=5) received gemcitabine, nab-paclitaxel, active and placebo LSTA-1, and placebo durvalumab. Cohort 3 (n=20) received gemcitabine, nab-paclitaxel, LSTA-1 and durvalumab. Tissue biopsies via endoscopic ultrasound were obtained pre-treatment and between weeks 12 and 16 for analysis of tumour infiltrating lymphocytes (TILs). Patient tumour sizes were assessed every 8 weeks using radiological imaging according to RECIST v1.1, and serum CA19-9 levels were analysed monthly. **Results:** 6/17 patients showed significant RECIST partial response after 2 cycles of treatment (5 patients in cohort 3), with the remaining 11 patients exhibiting stable disease. After 4 cycles of treatment, 10/17 patients demonstrated partial response (9 patients in cohort 3). Of the remaining 7 patients, 6 demonstrated stable disease, and 1 patient (cohort 2) exhibited a RECIST complete response. 13/17 patients who completed 4 treatment cycles showed a decrease in CA19-9 levels. 6 patients demonstrated >90% reduction in CA19-9 (5 in cohort 3), with the remaining 7 patients showing a >50% reduction in CA19-9 levels (5 in cohort 3). 12 patients underwent repeat biopsies 12-16 weeks post-treatment to assess TILs. 10 patients showed immune cell infiltration (5% to 50% stroma infiltration), 2 had no tumour found (Cohort 3). The combination was safe with no unexpected toxicities. **Conclusions:** The preliminary results of this study suggest that the combination of gemcitabine and nab-paclitaxel with LSTA-1 and durvalumab is safe and potentially induces tumour infiltrating lymphocytes and RECIST response in locally advanced pancreatic adenocarcinomas. Clinical trial information: ACTRN12623000223639. Research Sponsor: WARPNINE.

747 **Poster Session**

Telotristat ethyl to promote weight stability in patients with advanced pancreatic cancer. First Author: Olumide B. Gbolahan, Emory University School of Medicine and Winship Cancer Institute, Atlanta, GA

Background: Cachexia a key feature of pancreatic ductal adenocarcinoma (PDAC), affects up to 80% of patients (pts), and is associated with worse outcomes. Serotonin, due to its tumorigenic potential and effect on gut function may contribute to weight loss in PDAC. Telotristat ethyl (TE) inhibits serotonin production and improves gut motility dysfunction. We developed a phase II trial to evaluate the impact of TE on the weight of patients with metastatic PDAC (mPDAC) receiving first-line chemotherapy. **Methods:** Treatment-naïve pts with mPDAC who had lost at least 10% of baseline weight (Group 1) received gemcitabine and nab-paclitaxel (GnP) in combination with TE (250mg administered orally three times a day). Pts with $<$ 10% weight-loss were recruited in Group 2 and received chemotherapy without TE. Serum serotonin was collected monthly during the study. The primary endpoint was weight stability, evaluated as % weight change at 3 months compared to baseline, with a goal of $<$ 5% decrease. A key secondary endpoint was change in serotonin levels at 3 months compared to baseline. The groups were analyzed independently. **Results:** We enrolled 22 pts, 14 in group 1 (planned 40) and 8 in group 2 (planned 40). The median age was 70 years (IQR 62-75), majority were male (n = 15; 68%) and 11 (50%) were Black. Mean weight at treatment start was 77.9 kg (SD 19.16) group 1, and 90.6kg (SD 24.71) group 2. Mean serotonin level at baseline was 165.07 ng/ml (SD 60.49) group 1, and 296.9 ng/ml (SD 166.65) group 2. Mean % weight change over 3 months in group 1 was -2% (90% CI -5.2 to 1.3, P $<$ -5% = 0.0595). Mean change in group 2 was -5.6% (90% CI -9.4 to -1.7%, P $<$ 5% = 0.53). At 3 months, the median change in serotonin level in group 1 was -37.8% (IQR -59.1 to -13, P = 0.064) and -42.6% (IQR-57.8 to -29.9, P = 0.016) for group 2. Adverse events (AE) were consistent with known AE for GnP and TE. The median overall survival was 10.7 months (95% CI 5.4-16.9) group 1 and 12.2 (95% 6.1 to 21.4) group 2, HR 1.41 (95% CI 0.57 to 3.52 P = 0.46). **Conclusions:** Among pts with mPDAC and cachexia, the addition of TE to chemotherapy led to weight stabilization. These pts experienced 2% weight loss on average during the study (the upper limit of the CI just crossed -5% at -5.2%). The average weight loss was 5.6% in those who did not receive TE suggesting a signal of activity. A larger study is warranted to evaluate the role of TE in the management of cachexia in mPDAC. Funded by Lexicon Pharmaceuticals (NCT03910387). Clinical trial information: NCT03910387. Research Sponsor: Lexicon Pharmaceuticals.

748 **Poster Session**

Implication of the novel microtubule targeting agent, AUS_001, in pancreatic cancer cellular signaling. First Author: Marina Koutsioumpa, Australis Pharmaceuticals, Inc., Simi Valley, CA

Background: Pancreatic cancer (PanCa) is considered a major therapeutic challenge due to its poor prognosis, high mortality rate and resistance to most therapies, underlining the need for new treatment approaches. Our previous investigations revealed that the prenylated hydroxy-stilbene, AUS_001, inhibits β-tubulin polymerization via its unique binding to the colchicine site of tubulin. A cell-based profiling platform identified sensitivity of 12 out of 13 PanCa lines to AUS_001 with a median concentration causing 50% growth inhibition of 0.227μM. The goal of the current study was to elucidate the signaling activity and cellular responses following treatment with AUS_001 and explore the prospect of the latter in PanCa treatment. **Methods:** A series of functional and biochemical assays were performed to delineate the effect of AUS_001 on the growth, colony formation ability, invasive capacity and oxidative status of PanCa cells. Tumor growth was evaluated in PANC-1 implanted CB.17 SCID mice and MIA PaCa-2 implanted nude mice upon intraperitoneal administration of AUS_001 once weekly. Metabolic profiling was carried out using the Seahorse XF Analyzer. RNA silencing and adenovirus-mediated overexpression of the Dominant-Negative Mutant of c-Jun were employed to explore the link of c-Jun activation and AUS_001-induced cytotoxicity. **Results:** The potential of AUS_001 to disrupt PanCa cell functions was established *in vitro* and *in vivo*. Viability and compromised membrane integrity were multiplexed and resulted in half maximal effective concentration agreement indicating drug-induced cytotoxicity. AUS_001-treated cells underwent apoptosis followed by secondary necrosis, exerted diminished anchorage-independent growth and metastatic potential *in vitro*. Importantly, tumor growth was significantly reduced in PanCa murine xenograft models by AUS_001 as a monotherapy. Pretreatment with N-acetyl-l-cysteine did not alter cellular sensitivity to AUS_001, suggesting that the time-dependent induction of reactive oxygen species upon AUS_001 exposure is not the primary cause of cytotoxicity. Notably, the oxygen consumption rate, extracellular acidification rate and function of all mitochondrial complexes (I-IV) of live MIA PaCa_2 cells were significantly impaired in both glucose- and galactose-containing media 48h after AUS_001 administration. Integration of transcriptomic and proteomic data originating from AUS_001-challenged cells uncovered the dramatic induction of c-Jun and FosB stress responders. Further studies demonstrated that c-Jun transcriptional activation acts as a partial mediator of AUS_001-induced cytotoxicity. **Conclusions:** Collectively, our findings provide critical insights into the molecular events triggered by AUS_001 in PanCa cells and serve as supporting data for future exploration of AUS_001 as a novel PanCa therapeutic agent. Research Sponsor: Australis Pharmaceuticals, Inc.

749 **Poster Session**

Toxicology and clinical protocol development of DUO-207: An ultrasmall nanomedicine co-delivering gemcitabine and paclitaxel for the treatment of metastatic pancreatic cancer. First Author: Katherine Eichinger, Duo Oncology, Pittsburgh, PA

Background: The long-term survival rates for metastatic pancreatic cancer have remained stubbornly low for decades. Most pancreatic cancers have dense, fibrous stroma tissue surrounding their tumors, which contributes to tumor growth and acts as a barrier to many therapies. Nanomedicines have been developed for use in oncology to reduce toxicity and improve efficacy, but a majority are too large ($>$ 100nm) to effectively penetrate dense stroma. DUO-207 is a novel ultrasmall nanomedicine (~ 20nm) composed of copolymer-conjugated gemcitabine that encapsulates paclitaxel. Numerous preclinical efficacy models have demonstrated DUO-207's ability to penetrate and accumulate in tumor tissue. Here we describe results from toxicology studies performed in canines that assisted in the design of a phase 1b clinical trial using the continual reassessment method (CRM). **Methods:** A single cycle, bridging toxicology study of DUO-207 with a recovery period was performed in canines. Animal equivalent doses of nano-albumin-bound paclitaxel (nAb-PTX) and gemcitabine were administered as the standard of care (SoC) comparator. Blood was analyzed for hematological toxicity, biomarkers of major organ toxicity, and drug toxicokinetics. Toxicology results from DUO-207 were then used in a statistical model to explore plausible scenarios for clinical dose escalation. **Results:** DUO-207 treated canines demonstrated improved clinical observations compared to SoC. As expected, hematologic toxicities occurred in a dose dependent fashion in the DUO-207 group but no hepato- or nephrotoxicity was noted. Based on these data, a phase 1b CRM design has been developed with 90% power to detect the maximum tolerated dose in key scenarios with an average sample size of 11 patients per group. **Conclusions:** In summary, DUO-207 is a novel, ultrasmall nanomedicine that can co-deliver paclitaxel and gemcitabine in a single intravenous infusion to treat stroma-rich cancers. DUO-207 demonstrated dose dependent hematological toxicities but improved clinical observations compared to SoC and lacked signals of renal and hepatoxicity. Toxicokinetic data from canines was used in a statistical model to determine patient numbers for phase 1b dosing using the CRM. Duo Oncology will conduct its clinical trial in 2025. Research Sponsor: None.

750 **Poster Session**

Dose-escalated radiation for locally advanced unresectable pancreatic cancer. First Author: Arjun K Moorthy, Arizona Center for Cancer Care, Scottsdale, AZ

Background: Patients with unresectable locally advanced pancreatic cancer have limited therapeutic options. Historical treatments have included systemic chemotherapy or palliative radiation. New advanced radiotherapy treatment techniques have allowed dose escalated ablative radiotherapy to be delivered for patients with cancer limited to the pancreas. This study retrospectively reviews the long term outcomes of patients receiving dose escalated radiation for unresectable localized pancreatic cancer. **Methods:** Patients with locally advanced unresectable pancreatic underwent neoadjuvant chemotherapy with either gemcitabine or FOLFIRINOX based regimens. When maximum response was achieved with chemotherapy or this was no longer tolerated, patients were reassessed with imaging. Patients with localized unresectable disease then proceeded to definitive dose escalated radiation with either SBRT radiosurgery (40Gy in 5 fractions) or IMRT radiation (65-70Gy in 30-35 fractions) with concurrent 5FU. Surveillance of disease was performed with CT imaging and Ca19-9 biomarker assessment. **Results:** A total of 38 patients (21 male, 17 female) underwent definitive radiation. Radiation technique was determined by treating physician based on patient anatomy, with 7 patients having SBRT and 31 IMRT/5FU. Average patient age at diagnosis was 71. All patients underwent at least 6 months of follow up and mean follow up was 14 months. At time of latest follow up, 30 patients had no evidence of local progression and 12 patients had no evidence of progression at any site. The most common side effect at last follow up was grade 2/3 duodenal ulcer experienced by 7 patients. **Conclusions:** Advances in systemic chemotherapy have increased the population of patients with locally advanced unresectable pancreatic cancer who have persistent pancreatic disease after induction chemotherapy. Definitive dose escalated radiation is a new and effective technique to provide local control in this setting. This approach is well tolerated with manageable toxicity and effective durable local control. Patients continue to experience distant disease failure despite improving local control rates, though at a lower rate than historical controls. Control of primary pancreatic malignancy with dose escalated radiation can potentially improve quality of life and function for these patients for whom additional chemotherapy or surgery is not indicated. Research Sponsor: None.

751 **Poster Session**

Beyond Khorana score 2: Molecular correlates of venous thromboembolism in pancreatic cancer. First Author: Jonathon Jonah Gerstein, University Health Network, Toronto, ON, Canada

Background: Pancreatic ductal adenocarcinoma (PDAC) is associated with a hypercoagulable state leading to thrombosis. Risk models such as the Khorana score automatically classify PDAC as intermediate-high risk, and recent guidelines recommend consideration of thromboprophylaxis. However, little is known about the molecular correlates of PDAC for venous thromboembolism (VTE). **Methods:** We examined clinical and genomic data from the prospective multi-institution COMPASS trial (NCT02750657), which enrolled patients with treatment-naïve metastatic PDAC who underwent a fresh tumor biopsy for real-time whole genome and transcriptome sequencing. Laser capture microdissection was performed. Patients underwent restaging scans at 8-week intervals. Detailed chart review was conducted to focus on VTE risk factors, timing of VTE diagnosis, and anticoagulation. We also compared clinical and molecular factors based on timing of VTE diagnosis. Overall survival was defined as time from VTE to death. **Results:** Of 268 patients enrolled in the COMPASS trial, 166 patients had detailed clinical data available regarding VTE status. 82 patients (49%) developed a VTE, where 16 (20%) had breakthrough clots. Baseline epidemiological variables were similar for those with and without VTE, including no differences between age, sex, BMI, and baseline CA 19-9. No patients were on routine prophylactic anticoagulation. SMAD4 mutations were more frequently seen in the VTE subgroup (57% vs. 40%, p= 0.04), but no differences were seen in other driver genes (KRAS, TP53, CDKN2A), Moffitt subtype, or homologous recombination deficiency status. Patients who developed an early VTE (within 3 months) had a higher baseline CA 19-9 (median 4015 vs. 891, p= 0.02), and none were port-associated. A higher incidence of KRAS wildtype cases (10%) were observed in the early vs. later VTE groups (p= 0.03), but no differences in KRAS allelic status. There were no differences in burden of SNVs, indels, or SVs. Early VTE occurred in 46% of all basal cases, which are typically more aggressive, and only in 37% of classical subtypes. Overall survival was shorter with early VTE (HR 1.74, p= 0.02). **Conclusions:** A higher incidence of SMAD4 alterations were observed among patients with metastatic PDAC who developed VTE. VTE diagnosed earlier were associated with shorter survival, suggesting that early thromboprophylaxis should be considered at the time of diagnosis. Clinical trial information: NCT02750657. Research Sponsor: None.

752 **Poster Session**

Preclinical development of the lipid nanoparticle–based mRNA vaccine with multiple natural T cell epitopes for pancreatic cancer. First Author: Josephine Chang, Hampton Roads Academy, Newport News, VA

Background: Immunotherapy has recently emerged as a promising avenue in cancer treatment; however, its effectiveness on pancreatic ductal adenocarcinoma (PDAC) is hindered by the lack of effector T cells in the tumor microenvironment. One of the strategies to overcome the resistance of PDAC to immunotherapy involves identifying immunogenic tumor-associated and specific antigens for the development of personalized cancer vaccines. The mRNA platform has become an attractive platform for antigen presentation. The delivery of mRNA cancer vaccines via lipid nanoparticles (LNPs) enables intracellular delivery and controlled release of the payload, offering improved vaccine efficacy. However, mRNA vaccines presenting human T cell epitopes often lack a preclinical model for antitumor immune efficacy testing. **Methods:** A syngeneic mouse model of PDACs was established by implanting KPC tumor cells into the livers by hemispleen surgery. mRNA vaccine expressing five 25-mer peptide regions were synthesized. Two mouse CD8+ T cell epitopes were predicted. Their MHC binding was tested by the T2 binding assay. CD8+ T cells specific for these two epitopes were examined by ELISpot assays and were quantified by ImmunoSpot ELISpot Reader to identify changes in IFN-g secretion. **Results:** Using mass spectrometry analysis, we successfully identified natural HLA class I and class II-binding epitopes in the neoplastic tissues of human PDACs. For the design of mRNA vaccine, we chose five 25-mer peptide regions that each contains both class I and class II epitopes punctuated by AAY and GPGPG linkers and followed by an MITD sequence, which all aim to improve the efficiency of antigen presentation to dendritic cells. Using NetMHC, we predicted H2Kb or H2Db-binding epitopes LYPFPLAL and SMVQMTFL, within these five human peptide regions. The T2 binding assay showed a lack of evident binding of both peptides to the T2-Kb or T2-Db cells. However, the ELISpot assay showed that LYPFPLAL, but not SMVQMTFL, can induce the IFN-g expression from CD8+ T cells derived from splenocytes of the KPC tumor mice vaccinated by a peptide vaccine containing both peptides. Therefore, this result confirmed that LYPFPLAL is a mouse CD8+ T cell epitope and can be used to monitor antigen-specific CD8+ T cell response in response to the mRNA vaccine treatment in the syngeneic mouse model of PDACs. **Conclusions:** Antitumor efficacy of our above-designed mRNA vaccine can be tested in the syngeneic mouse hemispleen model of KPC tumors. Mouse LYPFPLAL epitope-specific CD8+ T cell response can serve as a surrogate marker for the antitumor efficacy of this mRNA vaccine in mouse models for PDAC. Research Sponsor: None.

753 **Poster Session**

Effects of tumor treating fields (TTFields) with FOLFIRINOX on BRCA WT pancreatic cancer cells. First Author: Michael Zheng, Novocure Inc., New York, NY

Background: Pancreatic cancer remains one of the most aggressive malignancies, frequently diagnosed at the locally advanced or metastatic stage. In the advanced setting, first-line chemotherapy options include gemcitabine with nab-paclitaxel or FOLFIRINOX, a combination regimen of fluorouracil, oxaliplatin, irinotecan and leucovorin. The FOLFIRINOX regimen is however associated with greater toxicity and is hence used only in patients with good performance status. BRCA-mutated tumors seem to be more sensitive to FOLFIRINOX due to the DNA-damaging, platinum-based component of this treatment regimen. Tumor Treating Fields (TTFields) are electric fields that disrupt cellular processes crucial for cancer cell viability that have shown efficacy in preclinical pancreatic cancer models; and, in various cancer types, have been shown to reduce expression of proteins from the BRCA-dependent DNA repair pathway. The current study examined the potential use of TTFields for sensitization of BRCA wild-type pancreatic cancer cells to treatment with FOLFIRINOX. **Methods:** Human pancreatic BRCA wild-type cancer cells BxPC3 and AsPC1 were treated with TTFields (150 kHz; 0.7 and 1 V/cm RMS, respectively), using the inovitro device. FOLFIRINOX was administered to the cells at increasing concentrations, with or without co-treatment with TTFields. After 72h of treatment, cell count, colony formation, and apoptosis were measured. For mechanistical insight, gene and protein expression were evaluated following 24, 48, or 72h of TTFields treatment. **Results:** TTFields application to the pancreatic cells together with FOLFIRINOX elevated the cytotoxic, clonogenic and apoptotic effects induced by FOLFIRINOX or TTFields alone. RNA sequencing data revealed that TTFields downregulated DNA damage repair, DNA replication, and cell cycle related processes. Specifically, real-time PCR and Western blot analysis demonstrated reduced expression of several central players in the BRCA DNA damage repair pathway. **Conclusions:** TTFields application together with FOLFIRINOX in pancreatic cancer cells lacking background BRCA mutations exhibits potential improvement relative to FOLFIRINOX alone, that may be rationalized based on downregulation of key DNA repair pathways. Future studies should explore the possibility of reducing FOLFIRINOX treatment dose, potentially alleviating FOLFIRINOX-related toxicity. Research Sponsor: Novocure Ltd.

754 **Poster Session**

$CD8^+$ T cell proximity analysis of the tumor microenvironment (TME) in patients with pancreatic ductal adenocarcinoma (PDAC) and associated clinical outcomes. First Author: Matthew Ebia, Cedars-Sinai Medical Center, Los Angeles, CA

Background: Immunosuppressive cells (cancer-associated fibroblasts [CAFs], tumor associated neutrophils [TANs], M2/M1 macrophages) can influence $CD8^+$ T cell infiltration in the PDAC TME. The spatial proximity of these cells to $CD8^+$ T cells and association with clinical outcomes is not well known. We hypothesize the cell distances between $CD8^+$ T cells and these phenotypes: CXC chemokine receptor 4 $(CXCR4)^+$ TANs, CAFs, focal adhesion kinase $(FAK)^+$ tumor cells, FAK^+ M2 and M1 macrophages will correlate with survival outcomes. **Methods:** Twenty-two stage I/II PDAC patients who consented for tissue collection were stratified into two groups: standard responders (n=11) with overall survival (OS) $>$2 years and poor responders (n=11) with OS $\leq$ 2 years. Multiplex IHC was performed on specimens using multiple biomarkers (CD8, CXCR4, CD66b, FAK, FAP, CD68, CSF1R, CD163, EPCAM) to identify cell phenotypes. Comprehensive spatial analysis with HALO software was used to measure mean cell distance between $CD8^+$ T cells and other cells. The mean cell-to-cell distance among evaluable specimens was determined by phenotype. Patients were divided into two groups with average cell distance below or above the mean by phenotype. OS of each subgroup was compared using Kaplan-Meier method. **Results:** Median OS for standard and poor responders was 47.6 months vs 13.1 months ($p<.0001$), respectively. There was a higher % of $CD8^+$ T cells closer (<50 μm) to $CXCR4^+$ TANs in standard responders (56% vs 28%, $p<.0001$). There were no significant differences in $CD8^+$ T cell distance to other phenotypes. Longer survival was seen with average cell distance below the mean when comparing $CD8^+$ T cells to FAK^+ tumor cells (1274 d vs 742 d, p=.306), to $CXCR4^+$ TANs (854 d vs 442 d, p=.947) and to FAP^+FAK^+ CAFs (890.5 d vs 672 d, p=.662). In contrast, longer survival was seen with a distance above the mean comparing $CD8^+$ T cells to M2 (622 d vs 1274 d, p=.732) and to M1 (613 d vs 1397 d, p=.517) macrophages. **Conclusions:** Our study shows clinical outcomes may be correlated with spatial distance among TME immune cells in PDAC. Longer survival was seen when $CD8^+$ T cells were closer to FAK^+ tumor cells, $CXCR4^+$ TANs, and CAFs and further away from M2 and M1 macrophages. Larger cohort analyses may better elucidate the prognostic and biological significance of cell distance among TME phenotypes, which may have potential implications for response to immunotherapy. Research Sponsor: None.

755 **Poster Session**

Machine learning–based liquid biomarker analysis from the NEONAX trial for response prediction to perioperative (PO) and adjuvant (A) gemcitabine/nab-paclitaxel in resectable PDAC (rPDAC). First Author: Thomas Seufferlein, Department of Internal Medicine I, Ulm University Hospital, Ulm, Germany

Background: The phase II NEONAX trial examined PO and A gemcitabine/nab-paclitaxel (G/nP) efficacy for rPDAC pts with a comprehensive biomarker program. This translational study aimed at identifying liquid biomarker for early prediction of G/nP success in each treatment group (PO and A) and for the combined group using machine learning (ML). **Methods:** Pts from both study arms were stratified into groups based on Short- or Long-DFS (n=80 pts total; PO n=42 pts; A n=38 pts). Blood plasma from selected G/nP naïve pts was analyzed by multiplexed ELISA (mELISA, 80-Plex ProcartaPlex Human Immune Response) and mass spectrometry (MS; feature list generation) and clinical data were acquired for each patient. For ML pts were divided into training (80%) and validation (20%) datasets. The Weka-based algorithm WrapperSubsetEval (WSE) with 8 different classifiers was used for feature selection. The best performing (highest accuracy, best ROC-AUC, minimal signature) classifier with the corresponding feature panel was selected by a 10x10-fold cross-validation (CV). Respective panels for all features combined as compared to clinical features and Ca19-9 blood levels were tested for performance via CV and bootstrap aggregating for training and validation datasets (10x with replacement). **Results:** The feature generation process generated 579 features from G/nP-naive pts (mELISA: 80; clinical data: 99; MS: 400). For response prediction of the whole rPDAC group to G/nP (S-/L-DFS), the ML-based flow identified a panel of 8 proteins from MS (SERPINA1, C1QB, KRT1, C4B, UBB, VCAM1, HPD, LYZ) and 3 proteins from mELISA (Galectin3, IL34, CCL4) combined with a logistic regression classifier. The predictive panel with the best performance for determining PO G/nP success was established using a kernel logistic regression classifier and included 2 proteins from mELISA (CXCL2, IL17A), 4 proteins from MS (C3, IGLV3-25, HLA-B, SHBG), and 2 clinical data (WHO grade, Na/Mg blood level ratio). The best performing biomarker for predicting success in A included 2 clinical data (tumor size at staging, hematocrit) and 1 protein from mELISA (IL22) and was established with a random forest classifier. All panels represented minimal feature signatures (rPDAC: 11; PO: 8; A; 3) and showed a high performance with ROC-AUC (training) $>$ 0.90, ROC-AUC (CV) $>$ 0.85, and ROC-AUC (validation) $>$ 0.90. All panels described were superior compared to similar predictive signatures (with corresponding ML classifiers) for all clinical data or Ca19-9 blood levels. **Conclusions:** We show that minimal liquid biomarker signatures for early prediction of G/nP success in the NEONAX trial based on mELISA, MS and clinical data can be established by ML. The study shows the potential of ML for biomarker panel development and the value of mELISA/MS for generation of feature lists to use in ML. Clinical trial information: NCT02047513. Research Sponsor: Celgene/Bristol-Myers Squibb.

756 **Poster Session**

Impact of pancreatic tumour biomechanics on diagnosis and post-operative complications. First Author: Aidan O'Dowling, Conway Institute of Biomolecular and Biomedical Research, Dublin, Ireland

Background: Around 600 people are diagnosed with pancreatic cancer in Ireland each year, with 90% of these having pancreatic ductal adenocarcinoma (PDAC). Five-year survival for PDAC is around 10%. PDAC upregulates stromal collagen formation which contributes to tissue stiffness. Stiffer pancreatic tumours have worse oncological outcomes due to faster progression and increased chemoresistance. Softer pancreata, conversely, are more likely to develop post-operative pancreatic fistulae (POPF), a significant post-pancreatectomy complication. While incorporated in risk scores, softness is currently determined by subjective intra-operative observation. **Methods:** Atomic force microscopy (AFM) is the gold standard technique used to measure biomechanics at a cellular length scale. We obtained fine needle biopsies (FNBs) of pancreatic tumours during diagnostic endoscopic ultrasound. This is the first time tissue biomechanics have been measured from pancreatic biopsies. Separately, we applied this technique to pancreatic tumours and adjacent normal tissue following resection. We also assessed collagen organisation in these pancreatic tumours to understand its contribution to the observed mechanical properties. **Results:** Of the patients with PDAC who underwent FNB, those presenting with metastatic disease had significantly stiffer tumours than those without, 1,490 Pa (SD 2,048.03 Pa) vs 250 Pa (SD 555.06 Pa) ($p<0.0001$). Neuroendocrine tumours exhibited larger variance compared to PDAC ($p < 0.0001$), SD 5,885.97 Pa vs SD 1,797.23 Pa. Regarding tumours which have undergone curative resection, tumour stroma is significantly stiffer than both cancer and benign epithelium. The collagen strands in this stroma are straighter, more numerous, and more packed than epithelial tissues. Pancreata which develop post-operative pancreatic fistulae (POPF) are softer than those which do not 96.42 Pa (SD 691.14 Pa) vs 343.99 Pa (SD 1,678.37 Pa). **Conclusions:** We describe a novel technique to measure the biomechanical properties of pancreatic biopsies. Using this technique, we now have the opportunity to measure these properties for the first time on all pancreatic tumours, including those presenting with unresectable disease. It is possible to distinguish between different types of tumour based on the biomechanical profile of biopsies. Previous ex vivo studies have linked stiffer tumours to a higher metastatic potential, and our results are the first to suggest that this may be true in a human population. In resected tumours, we confirm the anecdotal belief that softer pancreata are associated with POPF. Overall, this research demonstrates that tissue biomechanics has potential diagnostic and prognostic applications in pancreatic cancer. Research Sponsor: Irish Research Council; Breakthrough Cancer Research; St Vincent's Foundation.

757 **Poster Session**

Potential effect of the gut microbiome on tumour response to anti-cancer treatment in patients diagnosed with pancreatic ductal adenocarcinoma. First Author: Andrew Peter Dean, St. John of God Subiaco Hospital, Subiaco, Australia

Background: Chemotherapy is the standard of care for pancreatic ductal adenocarcinoma (PDAC), however, recent research has identified bacteria in the human gut microbiome that may influence patient response to chemotherapy. These bacteria are also present in faeces, and are thought to be representative of the bacteria in the pancreatic ductal system. Preliminary studies have suggested that patients with more advanced PDAC demonstrate lower levels of Firmicutes, and higher levels of Proteobacteria. Additionally, research has shown bacteria in the Firmicutes phylum possess several anti-cancer properties, which may improve patient treatment response, meanwhile, Gammaproteobacteria (Proteobacteria phylum), have been shown to attenuate gemcitabine efficacy and worsen treatment response. **Methods:** This prospective study involved 14 participants with locally advanced PDAC. Participants received combinations of Gemcitabine, Nab-Paclitaxel, and durvalumab (immunotherapy). Participant tumours were assessed every 8 weeks using radiological imaging according to RECIST v1.1, and serum CA19-9 levels were analysed monthly. Gut microbial profiling was completed on stool samples taken pre-treatment using PacBio HiFi full-length 16S rRNA sequencing. **Results:** 16S rRNA sequencing identified that of the 4 patients that died due to disease progression, 3 patients had a low Firmicutes abundance, and a high Proteobacteria abundance. The rother patient demonstrated low levels of Actinobacteriota and Bacteroidota, and a low bacterial diversity. One participant showed a significant increase in CA19-9 (1066.7%), and exhibited a low abundance of Actinobacteriota and Bacteroidota, a high abundance of Proteobacteria, and a low diversity. 5 participants demonstrated a significant treatment response (according to CA19-9 and RECIST analysis), with 4 patients showing a high Firmicutes abundance, high levels of Actinobacteriota or Bacteroidota, and low levels of Proteobacteria. The remaining patient demonstrated a low abundance of Firmicutes and Proteobacteria, yet, displayed the highest abundance of Actinobacteriota and Bacteroidota. **Conclusions:** This exploratory study suggests an association between microbial composition and treatment response, with Firmicutes potentially being associated with an improved treatment response, and Proteobacteria with a worsened response. We believe this study to be one of the first to explore correlation of microbiome with clinical treatment response in PDAC. Research Sponsor: WARPNINE.

758 **Poster Session**

Genomic analysis of paired primary and metastatic pancreatic ductal adenocarcinoma tumors. First Author: Caitlin A McIntyre, UT Health San Antonio, San Antonio, TX

Background: The genomics of pancreatic ductal adenocarcinoma (PDAC) have been increasingly understood allowing for improvements in treatment, yet the clinical relevance of PDAC evolution is less well studied. The aim of this study is to evaluate genomic differences and clinical correlations between primary pancreatic tumors and paired metastases using a targeted sequencing panel. **Methods:** Patients who underwent surgical resection for PDAC with recurrent disease who had targeted genomic sequencing of both the primary tumor and a metastasis were included. Genomic alterations were determined using a panel of cancer-related genes (MSK-IMPACT). OncoKB was used to annotate driver alterations. MATH scores were used to quantify intratumoral heterogeneity. Comparison between paired samples was performed and association with clinical variables was assessed. **Results:** There were 55 patients included; 20% (n=11) received neoadjuvant therapy, 91% (n=50) had adjuvant therapy and 36% (n=20) had adjuvant radiation. The most common sites of metastasis sequenced were lung (36%, n=20), liver (35%, n=19) and local (13%, n=7). There was concordance between the first site of recurrence and site sequenced in 93% (n=51) of cases with a median time between resection and biopsy of recurrence of 25 months. The most commonly altered genes are *KRAS* (96%), *TP53* (83%), *CDKN2A* (40%), *SMAD4* (25%) and *RNF43* (13%). The majority of alterations are shared between the primary tumor and metastasis. Clonal mutations were defined as cancer cell fraction (CCF) $>$0.8. The majority of shared mutations are clonal; 52% of mutations are clonal in both the primary and metastasis, and clonality is preserved in 76% of shared mutations in metastases. The most commonly shared alterations are in driver genes and the majority are clonal- *TP53* (80%), *KRAS* (78%), *CDKN2A* (77%) and *SMAD4* (64%). The proportion of clonal mutations among drivers is higher than variants of unknown significance (p=0.005). There are 31 mutations private to the metastasis in 19 patients, 12 of which are clonal (39%). Mutations private to the metastasis are more common in patients who received adjuvant systemic therapy (p=0.035); however this was not seen when comparing patients who received adjuvant radiation to those who did not (p=0.38). MATH scores are higher in the metastasis as compared to the primary in all patients (p=0.0096), including those who received adjuvant treatment (p=0.002), indicating increased intratumoral heterogeneity in metastases. **Conclusions:** A targeted sequencing panel demonstrates that the majority of driver mutations are clonal and preserved between the primary tumor and metastasis. Additional private mutations and increased intratumoral heterogeneity are noted in metastasis, and our data suggests that this may be associated with exposure to systemic therapy. Research Sponsor: None.

759 Poster Session

MUC5AC levels as predictors of adjuvant therapy outcomes in pancreatic ductal adenocarcinoma: A study of post-surgical gemcitabine-based regimens. First Author: Ashish Manne, The Ohio State University Comprehensive Cancer Center, Columbus, OH

Background: The role of tissue mucin 5 AC (MUC5AC) in PDA is not fully understood. In our previous work we showed that mature MUC5AC (MM) detected in the extracellular (EC) space and in apical region has better prognostic value than perinuclear immature MUC5AC (IM). Our previous research demonstrated that mature MUC5AC (MM), present in the extracellular (EC) space and apical region, offers better prognostic value than perinuclear immature MUC5AC (IM). Preclinical studies suggest that MM may confer chemoresistance to gemcitabine (Gem), commonly used in adjuvant therapy (AT) either alone or with capecitabine (Cap) or nab-paclitaxel (NP) in patients not eligible for FOLFIRINOX. This study investigates MM's effect on outcomes in PDA patients undergoing post-surgical treatment with Gem-only, Gem-NP, and Gem-Cap. **Methods:** Formalin-fixed, paraffin-embedded tissue samples from resected PDA cases (January 2010 - June 2021) were sourced from the Ohio State University biorepository. Immunohistochemistry using monoclonal antibodies 45M1 (for MM) and CLH2 (for IM) was employed to examine cellular MM and IM expression, with H-scores calculated. Statistical analysis of progression-free survival (PFS) and overall survival (OS) was done using the log-rank test. **Results:** We had 49 patients available for analysis. The median age of the group was 66 years (range: 38 to 89) with majority Caucasians (90%) and 53% (n=26) females. AT distribution within this group was, 31 Gem-only, 7 Gem-NP, and 11 Gem and capecitabine (Cap). Median expression levels for MM and IM were both 150, and 82% (n=40) were EC-MM positive. Using the median MM H-score as a threshold ($<$ 150 vs. $\geq$ 150), outcomes were evaluated within each therapy subgroup (Table). **Conclusions:** This preliminary study suggests that PDA patients with low MM expression may benefit from Gem-Cap therapy. However, larger, prospective studies are needed for definitive conclusions. Research Sponsor: IRP, Pelotonia.

Comparing outcomes in the sub-groups.

Sub-group (n)	OS in days (p-value)	PFS in days (p-value)
MM H-score < 150 vs. ≥ 150		
Gem-only (11 vs. 20)	775 vs. 1006 (0.4)	444 vs. 413 (0.4)
Gem NP (4 vs. 3)	1409 vs. 551 (0.8)	391 vs. 418 (0.5)
Gem-Cap (4 vs. 7)	not evaluable (NE) vs. 766 (0.02)	NE vs. 415 (0.01)

760 Poster Session

Spatially resolved protein profiling in pancreatic adenosquamous carcinoma. First Author: Adel Kardosh, Division of Hematology and Oncology, School of Medicine, Oregon Health & Science University Knight Cancer Institute, Portland, OR

Background: Pancreatic adenosquamous carcinoma (PASC) is a rare histopathological type of pancreatic cancer that is more aggressive than pancreatic ductal adenocarcinoma (PDAC). Although PASC has a unique tumor biology and poorer prognosis, there are no guidelines for differentially treating PASC patients, and its rarity makes it significantly understudied. We used Nanostring GeoMX digital spatial profiling (DSP) to provide a targeted proteomic characterization of key tumor-promoting proteins in PASC and PDAC. **Methods:** FFPE tissue from PASC (n=10 primary, n= 4 met) and PDAC (n=21 primary, n=5 met) cases were analyzed using the DSP platform. Sections were stained with an 82-protein cocktail of antibodies conjugated to UV-photocleavable DNA barcodes. A pan-cytokeratin morphology marker was used to select tumor regions of interest (ROI) and inferred surrounding stromal (i.e., pan-CK negative) ROIs. UV excitation of the defined ROIs was used to release the DNA barcodes and were quantified on the NanoString nCounter system. For PASC, squamous enriched ROI were inferred from corresponding p40 IHC. Corresponding clinical and mutational status for all cases were abstracted from medical records. **Results:** TP53 (PDAC, 80%; PASC, 83%) and KRAS (PDAC, 76%; PASC, 100%) were the most prevalent mutations.In comparing the tumor ROIs of PASC and PDAC (all cases), DSP-assessed global expression and activation of EGFR-MAPK signaling is significantly increased in PASC, whereas PI3K-AKT signaling is significantly decreased. Compared to PDAC, EGFR expression and phosphorylation (p-EGFR), as well as p-MEK1, are significantly increased in PASC. PASC showed significantly lower levels of PLCG1 and INPP4B, while the majority of other AKT signaling effectors were unchanged. For DNA damage response signaling, PASC showed significantly higher levels of p-ATM, p-CHK2, and p21, but not p-ATR. Likewise, expression of oncoproteins c-MYC and BCL6 were significantly increased in PASC (vs. PDAC). **Conclusions:** Given that the normal pancreas lacks squamous cells, their histological presence is a uniquely aberrant feature that can be used to guide deep spatially-resolved interrogation of complex cellular and molecular programs. Our preliminary findings show that regions of squamous cell differentiation within PASC have intrinsically higher levels of active EGFR signaling. With emerging RAS-targeted agents poised to alter the treatment landscape, high EGFR signaling in PASC suggest a possible compensatory resistance mechanism that may require additional targeting. Likewise, reliance on ATM signaling highlights this as preferential target for DDR-directed therapies. Building on these results with additional multiomic assays will inform strategies for improving the management of patients with PASC. Research Sponsor: Knight Cancer Institute; Kuni Foundation.

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***TP53* Y220C mutations in pancreatic ductal adenocarcinoma (PDAC): A genomic landscape study.** First Author: Hannah Ruth Robinson, Division of Medical Oncology, Department of Medicine, University of Colorado Anschutz School of Medicine, Aurora, CO

Background: A novel targeted therapy designed to restore wild-type function to the p53 Y220C mutant protein has demonstrated promising activity in an early-phase trial (NCT04585750). While the *TP53* Y220C mutation is found in $<$ 1% of all solid malignancies, its prevalence is enriched in certain cancers including PDAC. Here we report results of comprehensive genomic profiling (CGP) in a cohort of patients with advanced PDAC, with a focus on *TP53* Y220C mutations. **Methods:** A total of 27,377 cases of clinically advanced PDAC underwent hybrid capture-based CGP to assess all classes of genomic alterations (GAs). Cases were sequenced to a mean coverage depth of 650X. Microsatellite instability (MSI) status and tumor mutation burden (TMB) were determined using sequencing data; PD-L1 expression was measured by immunohistochemistry (Dako 22C3, tumor proportion score [TPS]). **Results:** *TP53* Y220C alterations were identified in 488 (1.8%) PDAC patients (Y220C+). In comparison to PDAC patients lacking the *TP53* Y220C alteration (Y220C-), Y220C+ patients were of similar age, but were more likely to be female (52.9% vs 46.8%; p = 0.0008) and harbor *KRAS* mutations (96.7% vs 92.7%; p = 0.007). Meanwhile, alterations in *ATM* were more frequent in the Y220C- group (4.0% vs 1.4%; p = 0.0008). Frequencies of other GAs were similar between Y220C+ and Y220C- groups, including alterations in the homologous recombination defect-associated genes *BRCA1/2* and *RAD21*, as well as *ERBB2*, *BRAF*, *MTAP* and *PIK3CA*. MSI-high status and elevated TMB ($\geq$ 10 mutations/Mb) were extremely uncommon in both groups. PD-L1 expression was similar in the Y220C+ and Y220C- groups, with positive PD-L1 expression (TPS $\geq$ 1) observed in 31.5% and 34.9% of patients, respectively. **Conclusions:** The potentially targetable *TP53* Y220C mutation was identified in close to 2% of clinically advanced PDAC cases. Those with *TP53* Y220C mutations were more likely to be female and harbor *KRAS* mutations, and less likely to harbor *ATM* mutations. However, *TP53* Y220C mutations were not associated with other distinct genomic characteristics. These findings are consistent with previous reports of interactions between *TP53* and *KRAS* alterations in PDAC. Given the emergence of drugs aimed at restoring p53 function in cases with the Y220C mutation, further study of this alteration in PDAC is warranted. Research Sponsor: None.

Age, number of genomic alterations (GA) per tumor, and alteration frequency of select commonly altered genes based on *TP53* Y220C status.

	TP53 Y220C+ (n = 488)	*TP53* Y220C- (n = 26,889)	P Value
Median Age, years (range)	66 (36-89+)	66 (21-89+)	NS
GA/tumor	5.2	5.0	NS
KRAS	96.7%	92.7%	0.007
CDKN2A	60.1%	56.6%	NS
CDKN2B	28.6%	29.5%	NS
MTAP	24.4%	23.8%	NS
ARID1A	7.8%	8.7%	NS
ATM	1.4%	4.0%	0.0008
BRCA2	2.3%	3.1%	NS

NS = non-significant.

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A real-world prospective analysis of the relationship between nanoliposomal irinotecan and fluorouracil with folinic acid and neutropenia in patients with pancreatic cancer (NAPOLEON-2 study, NN-2401). First Author: Tsuyoshi Shirakawa, Clinical Hematology Oncology Treatment Study Group, Fukuoka, Japan

Background: Nanoliposomal irinotecan and fluorouracil with folinic acid (NFF) is a standard regimen after gemcitabine-based chemotherapy (CTx) for unresectable or recurrent pancreatic cancer (urPC) patients. Neutropenia caused by systemic CTx has been reported to be a prognostic marker in several cancers, however, the relationship between neutropenia and efficacy of NFF in urPC patients is unknown. Therefore, we initiated a prospective observational study to investigate whether neutropenia can be a prognostic marker in the real-world (NN-2401), followed by a retrospective one (NN-2301). **Methods:** We collected the data of urPC patients treated with NFF who had received at least one previous CTx in 17 hospitals in Japan from June 2021 to October 2023. The primary endpoint was overall survival (OS). Secondary endpoints included the progression-free survival (PFS) and overall response rate (ORR). To analyze them, we divided urPC patients into three groups by the grade of the most severe neutropenia during NFF (Cutoff A, grade 0 vs 1-4; Cutoff B, grade 0-1 vs 2-4; Cutoff C, grade 0-2 vs 3-4). Besides that, we compared OS among the following three groups receiving NFF $\geq$4 cycles in Cutoff C; one showed neutropenia with NFF reduction after the 4th cycle (Cutoff C High-Late), another showed neutropenia with NFF reduction in the 2nd or 3rd cycle (Cutoff C High-Early), and the other did grade 0-2 neutropenia (Cutoff C Low). **Results:** Among the 150 patients, the number of patients with the most severe neutropenia of grade 0, 1, 2, 3, or 4 was 79, 7, 23, 34, and 7, respectively. The median OS of severe neutropenia group in Cutoff A, B, or C was 9.8, 9.8, and 10.6 months, respectively. Both OS and PFS were significantly better in all severe neutropenia groups; hazard ratio [HR] of OS for Cutoff A, B, or C was 0.58, 0.54, and 0.60, respectively; HR of PFS was 0.61, 0.56, and 0.56, respectively. Only Cutoff C showed a significant difference with an odds ratio of 3.06 (95% confidence interval [CI], 1.06-8.80; p=0.04) in ORR. Multivariate Cox regression analysis showed significant difference of OS in all Cutoffs; adjusted HR was 0.58 for Cutoff A, 0.53 for Cutoff B, and 0.52 for Cutoff C. Logistic regression analysis showed a significant difference in ORR of Cutoff C with an odds ratio of 4.02 (95%CI, 1.29-12.6; p=0.02). In addition, the numbers of patients were 73, 14, and 19 in Cutoff C Low, High-Late, and High-Early, respectively. The median OS of Cutoff C High-Late was significantly better than that of Cutoff C High-Early or Cutoff C Low (15.0 vs 10.4 or 9.1 months, p$<$0.05). **Conclusions:** In NFF treatment, neutropenia showed the usefulness as a prognostic marker for both OS and PFS, and also a predictive one for tumor response in urPC patients. The results are similar to our previous data, so we conclude that neutropenia can be a useful biomarker of NFF. Clinical trial information: UMIN000043939. Research Sponsor: None.

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Relationship of circulating tumor DNA (ctDNA) tumor fraction with prognosis in patients with metastatic pancreatobiliary cancer. First Author: Priyadarshini Pathak, Massachusetts General Hospital, Boston, MA

Background: Carcinomas of the pancreas and biliary tract are among the most lethal cancers. Currently, it is very difficult to predict individual patient outcomes with only general prognostic information available. Emerging evidence shows that baseline ctDNA tumor fraction (TF) is a prognostic factor for many tumor types, including metastatic pancreatic ductal adenocarcinoma (mPDAC). With an additional year of follow-up, we refresh our prior study of the prognostic value of ctDNA TF in mPDAC and extend the analysis to biliary tract cancers (BTC). **Methods:** A genomic dataset comprising patients who underwent ctDNA testing using FoundationOne Liquid or FoundationOne Liquid CDx as part of routine care was used to determine the distribution of ctDNA TF across pancreatobiliary cancer subtypes. A subset of patients was also included in the nationwide (~280 US cancer clinics, ~800 sites of care) de-identified Flatiron Health-Foundation Medicine clinico-genomic database, which was used separately for all other analyses. Clinical data were derived from the electronic health record. Real-world overall survival (rwOS) was evaluated by ctDNA TF while controlling for relevant covariates. In parallel, ctDNA TF cutoffs of ≥1% and ≥10% were evaluated. Exploratory analysis of trichotomized ctDNA TF at 1% and 10% was also performed. **Results:** In 8817 patients with pancreatobiliary carcinoma, 46.1% of PDAC and 41.9% of extra-hepatic cholangiocarcinoma patients had detectable levels of ctDNA TF, while patients with gallbladder carcinomas had the highest levels of ctDNA TF (median 0.8%). 470 patients with mPDAC and 89 patients with BTC were included in the outcomes analysis. High ctDNA TF was associated with clinical features linked to poor prognosis in mPDAC: Stage IV at diagnosis, pretherapy opioid prescriptions, high neutrophil-to-lymphocyte ratio and CA19-9 at baseline, and alterations in *KRAS*, *TP53*, *CDKN2A*, or *SMAD4*. In BTC, high ctDNA TF was associated with Stage IV at diagnosis and cholangiocarcinoma histology. Higher ctDNA TF in mPDAC was associated with significantly reduced rwOS for both cutoffs (1%: HR 1.55 [1.25-1.93], $P < 0.001$; 10%: HR 1.85 [1.42-2.41], $P < 0.001$). When ctDNA TF was trichotomized, higher ctDNA TF groups had reduced rwOS stepwise. Similar trends were observed in patients with BTC, but results were not statistically significant. **Conclusions:** ctDNA from pancreatobiliary carcinomas can be frequently detected and characterized in liquid biopsy while also assessing key genomic biomarkers. ctDNA TF was a prognostic biomarker for mPDAC, but not for BTC in this cohort. Detection of ctDNA associated with other clinical factors linked to poor outcomes. Cohorts given uniform treatment could possibly help further evaluate the ability of ctDNA TF to identify patients with more aggressive disease and inform future studies to personalize therapeutic decision-making. Research Sponsor: None.

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Natural history study of patients with pancreatic acinar cell carcinoma: A 3-year prospective longitudinal analysis. First Author: Krishna Patel, Laboratory of Molecular Biology, NCI, NIH, Bethesda, MD

Background: Pancreatic acinar cell carcinoma (PACC) is a rare exocrine tumor of the pancreas that comprises 0.2-2% of all pancreatic malignancies. There are no prospective randomized studies in this disease and most data about PACC comes from retrospective database analyses and case reports. **Methods:** The Cancer Moonshot-funded My Pediatric and Adult Rare Tumor (MyPART) network's Rare Solid Tumor Natural History study (NCT03739827) is a prospective, single-institution observational study that enrolls patients of all ages with rare solid tumors (<15 cases per 100,000 people per year), including PACC. Patients can participate remotely (field cohort) or visit the NIH for annual evaluations (in-person cohort). All participants complete medical and family histories, patient reported outcomes (PROs) measures, and provide saliva for DNA analysis. Relevant data such as clinical and family histories, surgical and molecular pathology, and imaging results are extracted from patient medical records and entered in the central study database. Archival tumor samples (when available) are analyzed using a 500+ gene panel (TruSight500, Illumina) and discussed in a molecular tumor board. Patients participating in the in-person cohort undergo additional evaluations including clinical examination, imaging studies, genetic counseling, and blood sample collection for clinical and research purposes (standard clinical labs, germline DNA/RNA, immune phenotypes, cytokines) as indicated. **Results:** From 09/2021 until 09/2024, 17 PACC patients (n=13 for the field cohort, n=4 for the in-person cohort) were accrued. The mean age of diagnosis is 64.7 (48-78) years, and the male to female ratio is 14:3. At the time of diagnosis, 6 patients had Stage I or II, 1 had Stage III, and 10 had Stage IV disease. The median follow-up time since accrual is 7.6 months, and 4 patients have died. No patients have been lost to follow-up. Of the 6 patients diagnosed with early-stage disease, 5 had surgery and all received adjuvant chemotherapy, while 1 patient is receiving neoadjuvant chemotherapy. Among the 9 patients diagnosed with advanced disease, 8 received platinum-based regimens as a first-line therapy. Tumor molecular profiling was available for 16 patients. The most common tumor mutations observed were in CDKN2A (29.4%), BRCA2 (29.4%), and SMAD4 (23.5%). Six patients received targeted therapies matched to their tumor molecular profile. **Conclusions:** The demographic profile of our cohort aligns with previous studies, with a mean age of diagnosis in the early-to-mid 60s and a high male to female ratio. Our cohort highlights a shift in first-line treatment to platinum-based therapies over the past decade. Many PACC tumors have targetable mutations found in molecular profiling, and the efficacy of matched therapy warrants further exploration. Clinical trial information: NCT03739827. Research Sponsor: NCI Center for Cancer Research.

765 Poster Session

Clinical and genomic features of the morphological subtypes in advanced pancreatic cancer. First Author: Eugenia Flores-Figueroa, University Health Network, Toronto, ON, Canada

Background: In resected pancreatic cancer (PDA), morphology from routine histopathology slides provides a rapid inexpensive biomarker that predicts overall survival and correlates with transcriptomic subtypes. In this study, we evaluated clinical and genomic associations of morphological subtypes in resected and advanced disease and validated the consistency of subtypes in patient-derived organoids (PDO) and mouse xenografts (PDXs) during *in vitro* and *in vivo* modeling. **Methods:** Our cohort included PDA tumor tissues from 152 resectable (stage I/II) and 228 advanced cases (Stage III/IV). Hematoxylin and eosin-stained slides were blindly reviewed by two pathologists and classified into subtypes based on Kalimuthu (1). Morphological subtypes were correlated with clinical and genomic data from whole-genome and transcriptome sequencing. Histological preparations from PDOs and PDXs obtained from pancreatic resections and metastases were reviewed using the same criteria. **Results:** Morphological subtypes were significantly associated with clinical patterns. Locally advanced PDA exhibited the highest proportion of glandular tumors. Metastatic tumors were enriched for non-glandular morphologies. The non-glandular morphologies were significantly associated with lower survival rates in both resected and advanced tumors. Furthermore, morphological subtypes were significantly associated with unique genomic alterations. Compared to glandular tumors, non-glandular tumors were associated with increased *KRAS* copy number, *KRAS* imbalances, and polyploid genomes. In advanced settings, glandular and non-glandular tumors mostly exhibited classical and basal-like transcriptional subtypes, respectively. Squamous tumors had the highest mutation burden and the highest proportion of Basal A signature (2). Using differential gene expression, we identified transcriptional signatures of the morphological subtypes. PDOs and PDXs maintained the morphological subtypes, although non-glandular tumors had a lower success rate for PDO establishment. **Conclusions:** Morphological subtypes have distinct clinical and genomic associations across all stages of pancreatic cancer, which can be consistently modeled both *in vitro* and *in vivo*. These results demonstrate that morphological subtyping offers a rapid and biologically-relevant classification of PDA that could be used for drug development and stratification to predict therapy selection. 1. Gut; 69:317-328 (2020). 2. Nat Gen; 52:231-240 (2020). Research Sponsor: None.

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Correlation of mesothelin (MSLN) expression measured by RNA sequencing (RNASeq) and immunohistochemistry (IHC) in MSLN-expressing tumors. First Author: Joel R Hecht, UCLA Jonsson Comprehensive Cancer Center, Santa Monica, CA

Background: BASECAMP-1 (NCT04981119) is a pre-screening study to identify patients with tumor-associated human leukocyte antigen (HLA)-A*02 loss of heterozygosity (LOH) for interventional studies, such as EVEREST-2 (NCT06051695), a phase 1/2 study of logic-gated chimeric antigen receptor T-cell (CAR T) therapy for MSLN-expressing cancers. MSLN is a cell surface protein expressed in several cancer types, including mesothelioma (MESO), colorectal (CRC), non-small cell lung (NSCLC), ovarian (OVCA), and pancreatic (PANC) cancer, which can be associated with poor prognosis (1). Longitudinal clinical, genomic, and biomarker data were collected from patients enrolled in BASECAMP-1, providing a large dataset for translational discovery and propensity scoring. The aim of this analysis was to determine whether MSLN expression measured by RNAseq and IHC are correlated among patients with solid tumors. **Methods:** In BASECAMP-1, tumor tissue from patients with germline heterozygous HLA-A*02 is tested for HLA-A*02 LOH using an investigational next generation sequencing device codeveloped with Tempus Labs (Lozac'hmeur, et al. *NPJ Precis Oncol.* 2024) that detects somatic alterations, including HLA LOH, and generates RNAseq data (eg, MSLN expression). Gene-level transcripts per million read (TPM) values were determined and the log2 TPM +1 was displayed in a scatter plot (the mean and standard deviation [SD] are provided in the Table). Patients with HLA-A*02 LOH also submit tissue for IHC testing for exploratory biomarkers (eg, MSLN expression using anti-MSLN clone 5B2). Statistical significance was determined with ANOVA or t-tests. **Results:** As of June 1, 2024, 314 patients had been screened for BASECAMP-1 and had RNAseq results; 34 patients also had MSLN IHC results. MSLN expression by RNAseq was consistently higher in patients with PANC or OVCA vs CRC or NSCLC ($P<0.001$). MSLN expression by RNAseq and IHC were correlated overall ($P<0.001$), particularly among patients with PANC and OVCA. Among the 8 patients with OVCA or PANC who were IHC+ for MSLN, only 1 had an RNAseq value below 6 log2 TPM+1; this patient had mesonephric-like ovarian adenocarcinoma, which may account for the low value. The one MESO sample with both RNAseq and IHC available was IHC+ and had a high RNAseq value. **Conclusions:** This analysis demonstrated high correlation between RNAseq and IHC MSLN expression in tumor tissue. 1. Faust, et al. *Cancers.* 2022. Clinical trial information: NCT04981119. Research Sponsor: A2 Bio.

MSLN expression by RNASeq by tumor type and IHC status.

	Overall (N=314)		IHC data available (n=34)			
			MSLN Positive		MSLN Negative	
Tumor	n	Mean TPM[a] (SD)	n	Mean TPM[a] (SD)	n	Mean TPM[a] (SD)
All	314	5.1 (2.3)	20	6.0 (2.1)	14	2.3 (1.7)
adNSCLC	33	3.8 (2.4)	2	4.7 (1.7)	7	1.5 (1.3)
sqNSCLC	4	3.1 (1.8)	0		0	
MESO	4	4.5 (4.2)	1	5.7	0	
CRC	148	4.4 (2.0)	8	5.1 (2.1)	7	3.1 (1.6)
PANC	100	6.1 (1.8)	3	7.8 (0.6)	0	
OVCA	25	7.2 (2.1)	5	6.6 (2.8)	0	

[a]MSLN log2 transcripts per million+1.
ad, adenocarcinoma; sq, squamous.

767 **Poster Session**

Serum mucin 5AC (MUC5AC) as a predictor of outcomes and microscopic distant metastasis in pancreatic ductal adenocarcinoma (PDA). First Author: Ashish Manne, The Ohio State University Comprehensive Cancer Center, Columbus, OH

Background: Our prior studies demonstrated that tissue extracellular MUC5AC (EM) detection (EM-D) is associated with early recurrence, and serum MUC5AC (sM) levels have prognostic value in resected PDA following neoadjuvant therapy. This study evaluates the predictive value of post-operative sM before adjuvant therapy (AT). **Methods:** Serum samples, collected from January 2010 to June 2021 and sourced from the Ohio State University biorepository, were analyzed using enzyme-linked immunoassays with a human MUC5AC kit (NBP2-76703). Multivariate (MV) Cox regression models quantified progression-free survival (PFS) and overall survival (OS), while Wilcoxon tests compared group differences. **Results:** Nineteen patients had serum samples available post-operative and pre-AT. The cohort's median age was 65 years (range: 48-83), predominantly Caucasian (n=18), and 58% female. Serum was collected a median of 2 weeks (range: 1-16) after surgery, with a mean sM level of 1.1 ng/mL (range: 0.42 to 3.3). Most patients (17/19) received Gemcitabine-based therapy, one received chemoradiation, and another FOLFIRINOX. Among patients, 3 had no recurrence (Nr), 10 experienced distant metastases only (mets), 3 had local recurrence only (Lr-o), and 3 had both Lr and liver recurrence (Liv/Lr). In the MV model, incorporating sM, EM-D, and AT, higher sM levels (HR=2.83, p=0.04) and negative EM-D (HR=6.85, p=0.02) were associated with increased recurrence risk, whereas AT type did not influence recurrence. Regarding OS, sM was significant (HR=2.95, p=0.04) while other factors were not. Elevated sM levels correlated with metastasis risk (mets vs. Liv/Lr vs. Lr-o vs. Nr = 1.54 vs. 0.61 vs. 0.62 vs. 0.92, p=0.01). Patients with locoregional recurrence had significantly lower sM than those with distant metastasis and no recurrence (mets vs. Liv/Lr+Lr-o vs. Nr = 1.54 vs. 0.62 vs. 0.92, p=0.005). In a separate cohort of 10 metastatic patients (mean sM of 1.8 ng/mL, range: 0.43 to 11.2), liver metastases were linked to significantly lower sM than non-liver metastases (0.54 vs. 3, p=0.02), while peritoneal metastases were associated with higher sM levels (3 vs. 0.6, p=0.04). **Conclusions:** This small study suggests that sM can serve as a biomarker for assessing outcomes and recurrence sites post resection before the initial dose of AT. Elevated sM should prompt suspicion of microscopic metastases, particularly non-liver metastases. Research Sponsor: IRP, Pelotonia.

768 **Poster Session**

Predictive methylation signature for gemcitabine sensitivity in pancreatic ductal adenocarcinoma: A pathway to personalized treatment. First Author: Deepak Sherpally, Metropolitan Hospital Center, New York, NY

Background: Gemcitabine (Gem) is a key drug used in chemotherapy regimens for treating pancreatic ductal adenocarcinoma (PDA). Identifying a biomarker to predict patient response to Gem could be crucial in personalizing treatment selection between Gemcitabine and fluorouracil (5-FU)-based therapies, thereby improving patient outcomes. We hypothesize that DNA methylation changes in genes linked to Gem resistance may serve as surrogates for the expression of corresponding proteins, thereby predicting treatment outcomes in patients with PDA receiving this therapy. **Methods:** We curated a 93-gene panel through a literature review spanning from January 1970 to April 2024. The proteins encoded by these genes have demonstrated either preclinical evidence (from cell line or mouse models) or clinical evidence (from tissue-based studies) of affecting the response to Gem in PDA. We evaluated the prognostic significance of methylation at specific CpG (cytosine-phosphate-guanine) sites within these genes using data from The Cancer Genome Atlas (TCGA) database, focusing on patients who had received Gem (106 /184 patients). We utilized elastic net multivariate analysis to analyze survival and created Kaplan-Meier plots based on our gene panel to estimate overall survival (OS). We compared RNA and gene expression levels for the clinically significant gene signatures between normal (NT) and PDA tissues using TNMplot.com, a web-based tool developed from data in the Gene Expression Omnibus of the National Center for Biotechnology Information (NCBI-GEO), TCGA, Therapeutically Applicable Research to Generate Effective Treatments (TARGET), and The Genotype-Tissue Expression (GTEx) repositories. This tool employs Mann-Whitney or Kruskal-Wallis tests to determine statistical significance. **Results:** We identified an 8-CpG site methylation signature across eight genes, distinguishing a high-risk subgroup within the Gem-treated cohort in the TCGA database. Patients exhibiting methylation at specific CpG sites in this signature experienced significantly worse OS, with median survival times of 20.35 months compared to 49.38 months (p=0.0068). Additionally, the RNA expression (fold change: 2.11, p=2.12e-55) and gene expression (fold change: 1.42, p=3.14e-18) of these genes were markedly higher in PDA tissues compared to normal tissues. Notably, the candidate genes in our signature do not overlap with those used to classify PDA into classical versus basal subtypes. In an ongoing study, we are also investigating methylation changes of these candidate genes in the blood (cell-free DNA) of PDA patients. **Conclusions:** We present a signature that can predict Gem sensitivity or resistance in patients with PDA. This signature serves as a critical first step toward personalizing treatment strategies. Research Sponsor: None.

769 **Poster Session**

More isn't always better: Expanding and refining prognostic methylation signatures for pancreatic ductal adenocarcinoma. First Author: Deepak Sherpally, Metropolitan Hospital Center, New York, NY

Background: In our previous study, we developed protein-informed methylation signatures from a chemoresistance gene panel (RGp, n=122) curated from the literature, identifying 28 signatures with strong prognostic value in The Cancer Genome Atlas (TCGA) pancreatic ductal adenocarcinoma (PDA) database. To enhance the clinical utility of this panel, we incorporated methylation changes from genes with known prognostic (PGp) and diagnostic (DGp) significance in PDA to form a composite gene panel (CGp-PDA) and repeated our analysis. We hypothesized that epigenetic changes in these genes reflect protein expression and that deriving signatures from them could help identify patients at risk of treatment failure. **Methods:** We assembled a composite gene panel (CGp-PDA) of 206 genes (122 RGp + 26 PGp + 82 DGp) through a literature review spanning from January 1970 to April 2024. These PGp and DGp genes are unique, as cell-free DNA methylation changes in them have established prognostic and diagnostic value, respectively in PDA. The same cohort of pancreatic ductal adenocarcinoma (PDA) patients (n=184) from our previous study was analyzed. We applied elastic net multivariate analysis to calculate survival outcomes and generated Kaplan-Meier plots using our gene panel, optimizing values for ? and λ over 100 iterations. **Results:** Our analysis identified 20 signatures with cytosines followed by a guanine residue (CpG) site in genes from our panel. The number of CpG sites in these signatures ranged from 3 to 4248. The most effective signature featured 17 CpG sites (22 months [m] vs. 67m, p=0.03), followed by signatures with 3 (16m vs. 49m, p=0.009) and 12 (17m vs. 36m, p=0.02) CpG sites. The 4248 CpG site signature demonstrated the poorest prognostic value among the signatures (17m vs. 21m, p=0.01). Some signatures included multiple CpG sites within a single gene. Notably, KCNH2 reappeared in several signatures, as in the prior study. Expanding our RGp to CGp-PDA did not produce more (20 vs. 28) signatures or improved prognostic value. **Conclusions:** Targeted methylation signatures, supported by robust preclinical or translational evidence of their impact on treatment resistance, can be clinically valuable for informing treatment selection and prognosis. We may need to shift our focus from machine learning-based broad methylation or transcriptomic analyses to more targeted investigations. Research Sponsor: None.

770 **Poster Session**

Investigation of the association of *CTRB2* exon 6 deletion with time to progression and overall survival in pancreatic ductal adenocarcinoma. First Author: Zafar A Siddiqui, Mayo Clinic Rochester, Rochester, MN

Background: Pancreatic ductal adenocarcinoma (PDAC) is associated with poor prognosis and limited treatment options. A 584 bp deletion in *CTRB2*, which impairs chymotrypsin B2 function, has been linked to increased PDAC risk. This study investigates the impact of this deletion on progression and survival outcomes in patients with PDAC. **Methods:** There were 633 patients in whom *CTRB2ex6* deletion was genotyped; 263 patients received chemotherapy and had time to progression (TTP) information. TTP was calculated from diagnosis date until earliest of 1) progression date (event), 2) surgery, death, or next chemotherapy date (competing risk), or 3) date last known to be alive (censor). Overall survival (OS) was calculated for the cohort from diagnosis until date of death (event) or date last known to be alive (censor). TTP was analyzed using cumulative incidence methods accounting for competing risks; OS was analyzed using Kaplan-Meier methods. Cox proportional hazard models adjusted for age and sex were used to test associations and compute hazard ratios (HR) and 95% confidence intervals (CI). **Results:** Median TTP in those with *CTRB2ex6* deletion negative carrier status (n=209) was 1.3 years vs 5.9 years for those with positive carrier status (n=54). Estimated 3-year progression rates were 54.2% for patients with no *CTRB2ex6* deletions, 43.7% for patients with one deletion (n=48), and 33.3% for patients with two *CTRB2ex6* deletions (n=6). TTP was not associated with *CTRB2* deletion carrier status (HR: 0.8, 95% CI: 0.6-1.3). Median OS for *CTRB2ex6* deletion negative carriers (n=514) was 1.0 year vs 0.9 years for positive carriers (n=119); within deletion carriers, heterozygous carriers (n=110) had a median OS of 0.9 years, and homozygous carriers (n=9) had a median OS of 2.6 years. OS was not associated with *CTRB2* deletion carrier status (HR: 1.0, 95% CI: 0.9-1.3). **Conclusions:** Although we did not detect statistically significant differences in TTP and OS by *CTRB2ex6* deletion carrier status, there was an interesting trend toward longer TTP and OS in the small number of patients who carried homozygous *CTRB2ex6* deletions. Larger studies are warranted to validate these findings. Research Sponsor: None.

771 **Poster Session**

p53 pathway genes' mutations as prognostic factors in patients (pts) with metastatic pancreatic ductal adenocarcinoma (mPDAC): A single center study. First Author: Giovanni Trovato, Oncologia Medica, Comprehensive Cancer Center, Fondazione Policlinico Universitario Agostino Gemelli, IRCCS Università Cattolica del Sacro Cuore, Rome, Italy

Background: Mutations (muts) in oncosuppressor genes have been associated with poor prognosis in different solid tumors. This study aims to evaluate the prognostic impact of p53 pathway muts in mPDAC pts. **Methods:** For each mPDAC pt enrolled in this study a FFPE tumor tissue specimen was collected and Next Generation Sequencing (NGS) was performed by TSO500HT assay (DNA [523 genes], RNA [55 genes]). Primary endpoint was overall survival (OS). The Kaplan–Meier method was used to estimate efficacy outcome; log-rank test, Peto-Peto test and Cox-regression model were used to compare the differences, considering a statistically significant p value < 0.05. **Results:** A total of 149 mPDAC pts were enrolled; TP53 was mutated in 115/149 (77.2%) pts, CDKN2A in 28 (24.3%), CDKN2B in 8 (5.3%), ATM in 6 (4%), TP53BP1 and MDM2 in 2 pts each (1.3%) and ATR in 1 (0.7%). No mutations were found in MDM4, CHEK1 and CHEK2 genes. Overall, 121 (81.2%) pts had at least one mutation in a gene of p53 signaling pathway (p53sigmut) and 28 pts (19.8%) had none (p53sigwt). The most frequent TP53 alterations were single nucleotide variations (SNVs): R175 (10 pts-8.7%), R248 (9 pts-7.8%) and R273 (8 pts-7%); 27 TP53 muts (18.1%) occurred in exon 7, 26 (17.4%) in exon 5, 19 (12.8%) in exon 8 and 13 (8.7%) in exon 4. Median OS (mOS) was 15 months (mos) (CI 95%: 12.8-18.2) in TP53 mutated pts vs 24.3 mos (CI 95%: 17.3-Not reached) in TP53 wt pts, p=0.03. The site of TP53 alteration was correlated with pts' survival: mOS was 34.9 mos (CI 95%: 31-Nr) in exon 4-mutated pts, 16.9 mos (CI 95%: 11.2-21.2) in exon 7-mut, 13.8 mos (CI 95%: 11.2-17.8) in exon 5-mut and 10.7 mos (CI 95%: 8.2-Nr) in exon 8-mut, p= 0.018. CDKN2A mutations were also associated with survival: mOS was 17.9 mos (CI 95%: 15.9-24.1) in wt pts vs 13.4 (CI 95%: 10.7-22.2) in mutated pts, p= 0.04. CDKN2B, ATM, TP53BP1, MDM2 and ATR mutations were not associated with OS. Overall, patients with at least one mutation in one of the genes of p53 pathway had a mOS of 15 mos (CI 95%: 12.8-18.2) vs 19.6 months (mos) (CI 95%: 17.3-Not reached) in p53sigwt patients. In terms of comutations, TP53 wt pts had a higher frequency of KRAS mutations: 94% vs 64.5%, $p<0.001$. **Conclusions:** Our study suggests an important prognostic role of p53 signaling muts in mPDAC pts; wt pts resulted in a longer OS than mutated pts, although some mutations seem to be associated with a longer survival. Further studies are needed to investigate these findings, including an in-depth analysis of structural proteogenomic. Research Sponsor: None.

772 **Poster Session**

Uncovering the role of oncogenic *HER2* variants as drivers of pancreatic cancer. First Author: Elishama Kanu, Duke University Medical Center, Durham, NC

Background: The therapeutic resistance of pancreatic ductal adenocarcinoma (PDAC) to conventional cytotoxic therapy underscores the need for advances in targeted molecular therapeutics. Previous studies have reported approximately 2% of PDAC patients with *HER2* gene amplification. While anti-*HER2* cancer therapy has yielded success in other disease sites, it has had minimal reported benefit in PDAC. Multiple *HER2* isoforms exist that are generated either by loss of exon 16 (*d16 HER2*), or through N-terminal truncations (*p95 HER2*). These isoforms are implicated in differential tumor behavior and response to anti-HER2 therapy in the breast but have not been investigated in PDAC. Herein, we hypothesize that these isoforms may underlie the observed resistance to *HER2* therapy in PDAC. **Methods:** To test for the presence of HER2 structural variants within human PDAC, a tissue microarray was constructed of fifty-one primary tumor samples and two metastatic samples. Multiplex immunohistochemistry (mIHC) was performed for staining of the extracellular N-terminal and intracellular C-terminal HER2 domains. We then utilized our previously described *HER2+* cancer rainbow (Crainbow) mouse model in which human wild-type *HER2* (*WT HER2*), *d16 HER2*, and *p95 HER2* were expressed in the same mouse along with fluorescent protein reporters. *HER2* Crainbow mice were crossed with PDX1-Cre recombinase to initiate recombination and expression of each fluorescently labelled *HER2* isoform during pancreas development. PDX1-Cre / *HER2* Crainbow mice were followed for up to one year and sacrificed to lineage trace each isoform and register *HER2* lineages with histopathology. **Results:** In our human data, four of fifty-three patient samples (three primary and one metastatic) had moderate-to-strong *HER2* staining. All samples identified on mIHC as *HER2* positive stained primarily for the *p95* variant of *HER2*, with absence of the N-terminal domain and preservation of the intracellular domain. In our transgenic mice, 65% of the mouse cohort developed pre-malignant low grade pancreatic intraepithelial neoplasia (PanIN) by eight weeks. At one year, 58.3% of mice demonstrated advanced high grade PanIN, and 25% developed invasive disease. Lineage tracing revealed that while early low grade PanIN began as polyclonal lesions, comprised of *WT-*, *d16-*, and *p95 HER2* expressing cells, advanced PanIN and invasive disease were unanimously monoclonal with one dominant isoform arising. *WT HER2* was represented abundantly in low grade disease, but high grade and invasive disease were predominantly driven by *d16-* and *p95-HER2*. **Conclusions:** Our data implicates the oncogenic *d16-* and *p95 HER2* variants, rather than *WT HER2*, as the primary drivers of *HER2+* PDAC tumorigenesis. In breast *p95 HER2* has been described as a known cause of Trastuzumab resistance. Therefore, future efforts will be aimed at characterizing the mechanisms driving therapeutic resistance in *p95-HER2* PDAC tumors. Research Sponsor: None.

773 **Poster Session**

Analysis of serum and urine miRNAs as biomarkers of cancer cachexia in pancreatic cancer. First Author: Mao Okada, Department of Experimental Therapeutics and Department of Hepatobiliary and Pancreatic Oncology, National Cancer Center Hospital, Tokyo, Japan

Background: Cancer cachexia (CC), characterized by anorexia, weight loss and skeletal muscle wasting, is a common complication in patients (pts) with pancreatic cancer (PC). The association of CC with treatment-related toxicity and decreased quality of life has been reported. Although CC is believed to result from systemic inflammation and cancer-induced metabolic changes, the underlying mechanisms remain poorly understood. Early diagnosis of CC is particularly challenging, highlighting the need for reliable biomarkers. We analyzed microRNAs (miRNAs) in serum and urine as potential biomarkers for the early detection of CC in PC. **Methods:** Clinical information, along with serum and urine samples were collected from pts with PC treated at the National Cancer Center Hospital, Tokyo, from March 2022 to March 2024. CC diagnosis according to Fearon et al: (1) $\geq$5% weight loss in 6 months, (2) $\geq$2% weight loss in individuals with a BMI of $<$20 kg/m², or (3) $\geq$2% weight loss in pts with sarcopenia. In this study, pts were considered to have CC if they met the definition of CC at enrollment or within six months. Serum and urine extracellular vesicle (EV) miRNA profiles were sequenced by NGS and compared by means of counts per million (CPM). Differential expression of miRNAs between pts with and without CC was investigated by DEseq2. **Results:** Of the 84 enrolled cases, 73 had sufficient sample volume for NGS assay. After excluding 4 cases with low miRNA read counts and 10 cases with unknown CC status, 59 cases were analyzed. CC was present in 41 pts (69%). Patients' background was as follows (CC/no CC); median age 70 [IQR:62–75]/ 67 [63–69], male 21/9 pts, cStage {I (7/0), II (1/0), III (18/5), IV (15/13)} pts, median BMI (21.7 [IQR:19.6–24.0]/21.8 [20.7–22.4]) kg/m2, respectively. There were 141 common miRNA species between serum and urine samples, 81 miRNAs unique to serum and 33 unique to urine. The correlation between the common miRNAs was moderate (R = 0.60). No significant differences in miRNA expression were observed between pts with and without CC in serum samples. In urine samples, miR-141-5p was significantly downregulated (log2FoldChange = -0.667, padj = 0.0463), while miR-146b-5p was significantly upregulated (log2FoldChange = 1.03, padj = 0.0463) in pts with CC compared to those without. **Conclusions:** The results of this study suggest that urine is a suitable biofluid for detecting CC in patients with PC, as differentially expressed miRNAs were not observed in serum. However, the mechanism underlying the superiority of urine over serum in detecting CC remains unclear and should be addressed in future studies. Research Sponsor: None.

774 **Poster Session**

Personalized ctDNA monitoring to assess the recurrence of early stage pancreatic cancer after surgery. First Author: Juan Du, Department of Oncology, Nanjing Drum Tower Hospital, Affiliated Hospital of Medical School, Nanjing University, Nanjing, China

Background: Up to 80% of patients with resected pancreatic cancer recur within 2 years. Here, we assessed the feasibility and accuracy of a personalized, tumor-informed circulating tumor DNA (ctDNA) test for early warning of recurrence risk during long-term surveillance. **Methods:** This study retrospectively analyzed 43 patients who underwent radical surgery for pancreatic cancer between November 2021 and June 2023 at the Comprehensive Cancer Centre of Drum Tower Hospital, Clinical Cancer Institute of Nanjing University. A personalised panel was constructed to detect ctDNA in plasma based on whole-exon mutation information from tumour tissue. A total of 139 ctDNA samples were obtained from periods including postoperative, during adjuvant therapy, and post-treatment. **Results:** Until the last follow-up time (May 13, 2024), a personalised ctDNA monitoring panel was successfully constructed for 35 (81.4%) stage I-III patients with a median follow-up of 11.82 months (range: 2-17.8 months). A total of 16 patients experienced recurrence within 15.7 months (range: 5.4-30 months), with a median lead time from first positive ctDNA to imaging recurrence of 3.6 months (range: 1-9.3 months). ctDNA positivity was detected in 25.8% (8/31) of ctDNAs at the landmark time point (postoperative prior to adjuvant therapy), with a positive predictive value (PPV) of 75% (6/8) and a negative predictive value (NPV) of 69.6% (16/23). Patients with positive ctDNA at the landmark time point had inferior DFS (6.9 months vs undefined. HR 3.955, P=0.007) and OS (15 months vs undefined. HR 5.599, P=0.001) compared to those with negative ctDNA. When integrating longitudinal time points, PPV and NPV were 68.2% (15/22) and 91.7% (11/12), respectively. For all 16 relapsed patients, the sensitivity of positive ctDNA detected longitudinally was 93.75% (15/16). The ctDNA-MRD status at longitudinal time points had a more significant difference with DFS (undefined vs 12.6 months. HR 5.096, P=0.002) and OS (undefined vs 21 months. HR 6.487, P=0.038). **Conclusions:** Personalized ctDNA surveillance is effective in identifying high recurrence risk after pancreatic surgical resection. Long-term longitudinal warning of arbitrary ctDNA positivity suggests a recurrence probability of up to 93.75%; therefore, full ctDNA dynamic monitoring is essential. And ctDNA monitoring will be further used to guide the decision-making in the clinic to prolong patient survival. Research Sponsor: None.

775 Poster Session

Analysis of lipogenic and metabolic subtypes and survival outcome in patients with pancreatic adenocarcinoma. First Author: Madappa N. Kundranda, The University of Texas MD Anderson Cancer Center, Phoenix, AZ

Background: Delayed diagnosis and rapid progression are major drivers of poor survival outcomes for Pancreatic Ductal Adenocarcinoma (PDAC). In previous studies, molecular profiling in tumor tissue has identified Cholesterogenic, Classical, or Exocrine-Like pathological subtypes associated with improved prognosis. Herein, in a clinical trial called Project Survival, we molecularly and clinically characterized a prospective PDAC biomarker cohort. This multicenter (n=6) clinical trial (NCT 02781012) of PDAC combined with high-fidelity longitudinal phenotypic characterization and multi-omic profiling was utilized to identify biomarkers with diagnostic and therapeutic utility. The study included assessment of plasma, serum, buffy coat, urine, saliva, and tumor tissue samples and followed 269 PDAC subjects for up to 7 years (Median survival was 700 days, with IQR of 822 days). **Methods:** We assessed plasma samples to identify circulating biomarkers using proteomics, metabolomics, and lipidomics profiling. Multi-omic comparison of patients with Overall Survival (OS) above the 75th percentile (improved survival) against those with OS below the 25th percentile (poor survival) was performed. **Results:** We identified distinct panels of proteins, metabolites, and lipids that were associated with patient survival outcomes. Protein biomarkers that were increased in improved survival outcomes were associated with cholesterol metabolism (7/14, p-value = 0.00085) and glycolysis/gluconeogenesis (5/12, p-value = 0.00089), however, protein markers associated with immune surveillance and inflammation were associated with poorer survival. For metabolite biomarkers, 19 metabolites were increased in poorer survivors of which 13/19 were carnitine metabolites, and 42 metabolites were increased in patients with improved survival. Interestingly, only two lipid molecular species of the diacylglyceride family were increased in patients with poorer survival. **Conclusions:** Project Survival, a prospective biomarker-driven clinical trial identified distinct metabolic and molecular subtypes of patients which characterize survival outcome in PDAC. These subtypes go beyond tumor genomic characterization and align with distinct cholesterogenic and glucogenic phenotypes associated with survival outcomes, which can potentially be leveraged for therapeutic intervention strategies. Research Sponsor: None.

776 Poster Session

Validation of PDACai v2.0 in predicting relative benefit from frontline FOLFIRINOX (FFX) and gemcitabine/nab-paclitaxel (GA) for patients (pts) with metastatic pancreatic cancer (mPDAC). First Author: Michael J. Pishvaian, Johns Hopkins University School of Medicine, Washington, DC

Background: Novel biomarkers beyond genomic alterations in DDR pathway genes (e.g. *BRCA1/2*) have the potential to optimize frontline and subsequent treatment decisions for mPDAC. This real-world evidence (RWE) study validates PDACai v2.0, a machine learning framework that leverages clinical NGS data to predict progression-free survival (PFS) on FFX vs GA in individual pts with mPDAC. **Methods:** We analyzed outcomes from 774 pts with mPDAC who underwent genomic profiling via the Know Your Tumor program or were referred to Perthera by treating oncologists. Independent training and validation cohorts receiving either 1st line FFX or GA were split (60:40). PDACai v2.0 integrates an updated set of clinical (age $<$ 63, sex) and lab-agnostic molecular features derived from genomic testing reports (specific *KRAS* variants, *TP53* GOF vs LOF, DDR pathway redefined). Relative benefit scores predicted by FFX or GA models were evenly binned into three PDACai signature categories representing lower, middle, and upper tertiles for each independent cohort. Statistical differences in median PFS were evaluated using ordinal Cox regression. **Results:** Median PFS followed PDACai predicted trends for each therapy's training and validation cohorts. The predictive utility of PDACai was confirmed in the independent validation cohorts when comparing PFS on FFX (p = 0.01637; HR = 0.73 [95% CI: 0.56-0.94]) and GA (p = 0.0007506; HR = 0.62 [95% CI: 0.47-0.82]) across tertiles. **Conclusions:** Using RWE, the PDACai v2.0 signature successfully predicted PFS differences in pts treated with 1st line FFX or GA in mPDAC. Prospective validation efforts and additional data-driven insights into 2nd line PFS on FOLFOX vs 5FU/nal-irinotecan are underway. Research Sponsor: Pancreatic Cancer Action Network.

Summary of actual PFS in months on 1st line therapies in pts assigned to lower, middle, and upper thirds based on relative PDACai v2.0 scores.

Independent Cohort (# Pts)	Lower TertilemPFS [95% CI]	Middle TertilemPFS [95% CI]	Upper TertilemPFS [95% CI]
1st line FFX Training (243)	6.5 [5.5-8.1]	9.3 [6.5-13.3]	15.8 [14.1-N/R]
1st line FFX Validation (165)	9.4 [7-13.3]	8.6 [5.6-11.4]	13.2 [9.9-39.6]
1st line GA Training (218)	5.6 [4.7-6.3]	6.9 [5.2-8.2]	11.2 [8.8-13.4]
1st line GA Validation (148)	5.6 [4.2-8.4]	7.5 [6.5-13.1]	8.5 [7.7-11.5]

777 Poster Session

Frequency and incidence of oncogenic *RAS* mutations in patients with metastatic pancreatic ductal adenocarcinoma: Derived from the real-world evidence database Foundation Medicine Insights. First Author: Shubham Pant, The University of Texas MD Anderson Cancer Center, Houston, TX

Background: Approximately 60,000 patients are diagnosed with pancreatic cancer in the United States annually, with half of all cases diagnosed at the metastatic stage.[1] The most common type, pancreatic ductal adenocarcinoma (PDAC), accounts for ~90% of all pancreatic cancers[2] and is characterized by frequent *KRAS* oncogenic mutations. Newer investigational therapeutics target HRAS and NRAS, in addition to KRAS. To comprehensively characterize the incidence of RAS mutations in patients with PDAC, we analyzed mutational frequency data from the Foundation Medicine Insights (FMI) database. **Methods:** NGS-based comprehensive genomic profiling (CGP) data from 27,298 PDAC tissue samples were accessed from the FMI database (May 7, 2024 data cut). Only 'short variants' in RAS genes (*KRAS*, *NRAS*, and *HRAS)* occurring at amino acid positions G12, G13, or Q61 that had a frequency 0.1% or greater were included. To estimate the number of new US cases of *RAS*-mutant PDAC, pancreatic cancer incidence estimates[3] were adjusted by the derived *RAS* mutational frequency values. All incidence estimates were rounded to the nearest 100. **Results:** Oncogenic *RAS* mutations were identified in 92% of PDAC patients, with nearly all being *KRAS* (91.8%) and the remaining *NRAS* (0.2%). No *HRAS* mutations passed our minimum frequency threshold requirement. Individual *RAS* mutations with frequency $>$1% included *KRAS* G12D (39.9%), *KRAS* G12V (28.6%), *KRAS* G12R (14.6%), *KRAS* Q61H (4.2%), *KRAS* G12C (1.7%), and *KRAS* Q61R (1.4%). Applying these *RAS* mutation rates to reported PDAC incidence, we estimate that ~56,000 new patients with PDAC in USA per year will have tumors harboring an oncogenic *RAS* mutation, including ~51,000 of those with a *RAS* G12 mutation specifically. **Conclusions:** Oncogenic *RAS* mutations are identified from NGS testing in nearly all patients with advanced PDAC. Both the frequency and diversity of RAS mutations in PDAC suggest that PDAC is largely a RAS-driven cancer and that targeting RAS may be useful broadly in this disease. Research Sponsor: None.

Mutational frequency and incidence at G12, G13, and Q61 amino acid positions in *KRAS* and *NRAS* among patients with PDAC.

		RAS G12	*RAS* G13	*RAS* Q61	*RAS* mutant
Mutational Frequency	KRAS	85.7%	0.8%	6.3%	91.8%
	NRAS	0.1%	Below threshold	0.1%	0.2%
Incidence	KRAS	51,300	600	3,700	55,600
	NRAS	100	Below threshold	100	200

778 Poster Session

The effect of cachexia and anti-cancer therapies on GDF-15 levels in patients with pancreatic adenocarcinoma. First Author: Talya Geller, Cedars-Sinai Medical Center, Los Angeles, CA

Background: Growth differentiation factor 15 (GDF15) is a novel biomarker in the setting of pancreatic adenocarcinoma (PDAC). Preliminary studies have found that downstream effects of GDF-15 promote the progression of pancreatic cancer. PDAC has a high prevalence of cachexia which is commonly treated with enzyme supplementation and nutritional support. Platinum-based therapies often used to treat PDAC increase the likelihood of patients experiencing cachexia. The purpose of this study was to observe the GDF-15 levels in respect to weight change of PDAC patients being treated with cachexia and anti-cancer therapies. **Methods:** Serum and Plasma samples were collected from locally advanced and advanced PDAC patients on 5 IRB approved studies from 2014-2024. Samples were analyzed using the R-PLEX Human GDF-15 Antibody Set. The study objective was compared using two-sample paired t-tests and Pearson correlation matrices. Elevated GDF-15 was determined as $>$1500 pg/mL. **Results:** Samples from 73 patients were utilized for this study (63% female; median age: 69; BMI: 23.31$\pm$4.96 kg/m^2). At baseline 71.8% (46/64) had elevated GDF-15, at Cycle 3 76% (25/33), and at Cycle 5 86% (25/29). In the total sample set, there was a slight negative correlation in the change of weight (-1.75 kg, p = 0.013) and GDF-15 (577.5 pg/mL, p = 0.278) from C1 to C3 (r = -0.185, p = 0.366). There was a slight negative correlation (r = -0.44, p = 0.076) in the difference of weight and GDF-15 from C1 to C3 in patients receiving chemotherapy. A strong negative correlation was observed in the change of GDF15 and weight from C1D1 to C1D8 in patients treated with platinum-based therapy (r = -0.530, p = 0.212) and a weak negative correlation (r = -0.421, p = 0.065) for patients with non-platinum-based therapy. There was a strong positive association (r = 0.993, p $<$ 0.01) between the change in weight and GDF-15 from C1 to C5 in patients not receiving chemotherapy. There was no significant decrease in GDF-15 levels across the patients at all time points. **Conclusions:** Despite cachexia and anti-cancer treatment, the GDF-15 levels of PDAC patients continue to remain elevated in consecutive blood draws. Further research in GDF-15 kinetics is warranted. Clinical trial information: NCT02400398, NCT03207724, NCT04098237, NCT05336266, NCT04634539. Research Sponsor: None.

Weight and GDF-15 levels across different chemotherapy regimens.

	Baseline	Cycle 1		Cycle 3		Cycle 5	
	BMI (kg/m2)	Weight (kg)	GDF15 (pg/mL)	Weight (kg)	GDF15 (pg/mL)	Weight (kg)	GDF15 (pg/mL)
Chemotherapy	23.21	63.83	3243.25	63.44	3498.76	60.14	3201.33
No Chemotherapy	23.77	65.85	3113.73	62.32	2949.56	71.05	2944.54
Platinum Based	20.88	59.89	6527.78	51.7	3600.87	58.1	5152.50
Non-platinum-based chemotherapy	23.86	64.89	2442.14	64.67	3487.41	60.39	2957.43

779 Poster Session

Correlation of tumoral collagen width with overall survival and tumoral collagen straightness with distant recurrence-free survival in patients with pancreatic ductal adenocarcinoma. First Author: Alexander Marwaha, Allegheny Health Network Cancer Institute, Pittsburgh, PA

Background: Quantitative measurements of fibrillar collagen characteristics has been shown to correlate with survival in several types of malignancy. The aim of this study was to determine whether these characteristics correlate with survival in pancreatic ductal adenocarcinoma. **Methods:** Patients who underwent pancreatic resection for pancreatic ductal adenocarcinoma at an Allegheny Health Network hospital between 1/1/2019 and 12/31/2023, and for whom archived formalin-fixed paraffin-embedded tumor tissue was available, were included in the study. Clinical and pathologic characteristics were collected from the patient's electronic medical record. Survival data included overall survival (OS), local recurrence-free survival (RFS), and distant RFS. One section of tumor per patient was stained with Picrosirius Red and a representative digital image was obtained with a camera attached to a microscope at 100x magnification. Each image was analyzed with CT-FIRE software to determine averages of collagen fiber width, length, straightness, and angle. Descriptive statistics were initially computed. Log-rank tests were also conducted to compare the OS rate and RFS rate between four different groups (width, length, straightness, angle). Each group was divided into high and low groups, using the mean as cutoff. In the end, multivariate Cox regression was conducted by controlling the confounders including sex, perineural invasion, lymphovascular invasion, and presence of lymph node metastasis to examine the hazard ratios of OS, local RFS, and distant RFS for the four collagen fiber metrics. All statistical analyses were performed via SAS 9.4 under alpha level .05. **Results:** One hundred three patients met the inclusion criteria. Based on the results of log-rank test, OS was significantly different between high and low collagen width groups, p = .0251. The lower width group had higher survival rate than the upper width group. Also, distant recurrence-free survival was significantly different between high and low collagen straightness groups, p = .0053. The lower straightness group had higher distant RFS rate than upper group. The results of multivariate Cox regression were consistent with the Kaplan-Meier curves. Compared to the high collagen width group, the low collagen width group had a 57% lower chance of dying, hazard ratio = 0.43, 95% CI [0.23, 0.78], p = .0056. The low collagen straightness group had a 60% lower chance of distant recurrence than the high collagen straightness group, hazard ratio = 0.40, 95% CI [0.19, 0.85], p =.0163. **Conclusions:** Quantitative collagen fiber metrics could help identify patients with resected pancreatic adenocarcinoma who are at risk of earlier death and distant recurrence. Research Sponsor: None.

780 Poster Session

Mortality and other outcomes of hospitalizations due to pancreatic cancer. First Author: Bugra Zengin, Hamilton Medical Center, Dalton, GA

Background: Pancreatic cancer continues to be a highly fatal malignancy of the digestive system and imposes a significant financial strain on healthcare systems. Although it is often diagnosed incidentally, several risk factors have been identified, including smoking, family history, age, race, gender, and alcohol use. In this study, we aimed to examine inpatient outcomes and the healthcare burden of patients hospitalized with pancreatic cancer. **Methods:** Patients admitted to the hospital in 2019 and 2020 were identified using the Nationwide Inpatient Sample database and categorized into two groups based on the presence of a pancreatic cancer diagnosis (with relevant ICD-10 codes). To account for confounding factors, multivariate regression analysis was applied when calculating mortality. Statistical analyses were conducted to assess mortality rates, length of stay and hospital charges. **Results: :** A total of 74,584 patients met the inclusion criteria, with a mean age of 68.5 years. The majority were white (69%), while black patients made up 13% of the study population, and 49% were female (36,570 patients). Thrombosis (OR: 2.08, $p < 0.001$), age over 60 (OR: 1.22, p = 0.042), and being of black race (compared to white race, OR: 1.45, p = 0.001) were linked to higher mortality rates. Thrombosis was also associated with longer hospital stays (+1.9 days, $p<0.001$) and increased costs (+$16,003, p= 0.001). However, age over 60 (+0.26 days, p = 0.088; +$589, p = 0.833) and black race (+1.01 days, $p< 0.001$; +$1,493.7, p = 0.649) showed trends toward longer stays and higher costs, these were not statistically significant. On the other hand, pancreatic conditions such as chronic pancreatitis, pancreatic cysts, pseudocysts, or exocrine pancreatic insufficiency (OR: 0.589, $p < 0.001$), female gender (OR: 0.78, p = 0.002), and care at urban teaching facilities (OR: 0.37, $p < 0.001$) were associated with lower mortality. While female gender (+0.06 days, p=0.603; -$1699, p=0.448) did not significantly impact hospital length of stay or costs, pancreatic disease (+0.33 days, p = 0.031; +$10,826, $p < 0.001$) and being treated at urban teaching facilities (+1.8 days, $p < 0.001$; +$57,106, $p < 0.001$) were linked to longer stays and higher costs. **Conclusions:** Although some unmodifiable risk factors significantly increase mortality, thrombosis contributes not only to higher mortality but also to increased healthcare costs and longer hospital stays. However, it may be preventable with anticoagulation therapy. The inverse relationship between pancreatic disease and mortality could potentially be explained by more frequent imaging and earlier detection of malignancies in these patients. Research Sponsor: None.

Analysis of factors influencing in-hospital mortality, length of hospital stay, and total charges in hospitalized patients.

Data	Mortality	Length of hospital stay	Total Charge
Age>60	1.22(p=0.042)	+0.26 days, p=0.088	$589, p=0.833
Race (black compared to white race)	1.45 (p=0.001)	+1.01 days, p< 0.001	$1493.7, p=0.649
Female gender	0.78 (p=0.002)	+0.06 days, p=0.603	-$1699, p=0.448
Treatment at urban teaching facility	0.37(p<0.001)	+1.8 days, p<0.001	$57,106, p<0.001
Thrombosis	2.08(p<0.001)	+1.9 days, p< 0.001	$16,003, p= 0.001
Pancreatic conditions	0.589(p<0.001)	+0.33 days, p= 0.031	$10,826, p<0.001

781 Poster Session

Comparison of mutational landscape in primary and metastatic pancreatic tumors. First Author: Jean-Paul De La O, Exact Sciences, Redwood City, CA

Background: Genomic profiling has brought to the forefront the clinical relevance of intratumoral heterogeneity. Oncogenic *KRAS* mutation is the most common driver event in pancreatic ductal adenocarcinoma (PDAC), but little is known about how *KRAS* alteration may cooperate with other genetic lesions to support dissemination and growth of metastatic disease. Therefore, an improved understanding of molecular events affecting the clinical course of *KRAS*-altered PDAC is needed to identify biomarkers of early disease and expand the repertoire of targeted therapies. **Methods:** We retrospectively analyzed tissue samples from PDAC patients using the OncoExTra tumor-normal whole exome and whole transcriptome test between April 2018 and May 2024. The prevalence of actionable somatic alterations was determined overall and by specimen location (primary vs. metastatic). Fisher's exact test with FDR-based correction for multiple testing was used to identify co-alterations with *KRAS* and 19 DNA damage response (DDR) gene alterations. **Results:** PDAC samples from 434 patients (51.2% female; mean age 65.4±10.9 years) were analyzed. *KRAS* alterations were detected in 369 (85.0%) samples: 198 (45.6%) primary and 239 (55.1%) metastatic. $KRAS^{G12D}$ (32.7%) and $KRAS^{G12V}$ (29.5%) were the most common *KRAS* mutations. The frequency of *KRAS* alterations was 85.9% and 84.5% in primary and metastatic samples, respectively. *TP53* alterations occurred in 68.2% samples overall and in 64.6% and 71.1% of primary and metastatic samples, respectively. Examination of co-alterations in *KRAS*-altered versus *KRAS*-WT samples uncovered enrichment of *TP53* alterations in *KRAS*-altered (277 samples, 75.1%) compared to *KRAS*-WT samples (19 samples, 29.2%) ($p<0.001$). Also, *BRAF* mutations and *SND1:BRAF* fusions exclusively occurred in metastatic *KRAS*-WT samples, although the number of events was small (n=4 and n=2, respectively). We also found that *MTAP* was significantly co-altered in metastatic *KRAS*-WT samples (n=6, 16.2%) compared to *KRAS*-altered samples (6 samples, 3.0%) (p=0.004). A total of 21.9% of all PDAC samples had an alteration in at least 1 DDR gene. No associations were observed between *KRAS* alteration status and alteration in DDR genes. However, *TP53* alterations were more frequent in non-DDR-altered samples ($p<0.001$), while *CDKN2B* alterations were enriched in DDR-altered samples (p=0.025). **Conclusions:** We observed expected rates of alterations in known PDAC driver genes and uncovered higher co-occurrence of *KRAS* and *TP53* alteration in metastatic samples. Additionally, *TP53* alterations were less frequent in DDR-altered samples. Our findings suggest that a number of therapy approaches including BRAF, WEE1, PARP, and KRAS inhibitors as well as epigenetic modulators, alone or in combination, could improve clinical outcomes in PDACs with specific mutational profiles. Research Sponsor: Exact Sciences.

782 Poster Session

Exploring predictive factors for detecting KRAS mutations in pancreatic cancer using liquid biopsy. First Author: Takaaki Furukawa, Department of Hepato-Biliary-Pancreatic Medicine, Cancer Institute Hospital of Japanese Foundation for Cancer Research, Tokyo, Japan

Background: Liquid biopsy offers an alternative to tissue specimens for comprehensive genetic profiling. However, circulating tumor DNA is sometimes insufficient to detect genomic alterations in pancreatic cancer (PC). The detection rate of KRAS mutations was reportedly increased in PC with liver metastasis. In contrast, the predictive value of CA19-9 levels, the timing of liquid biopsy, or the type of liquid biopsy for detecting KRAS mutations remains unclear. We investigated whether these factors could be predictive of detecting KRAS mutations in PC. **Methods:** From September 2021 to August 2024, we retrospectively reviewed patients with PC who underwent liquid biopsy. **Results:** A total of 106 patients were enrolled (median age: 64 years, male: 66). Patient characteristics were as follows: tumor status: locally advanced: 10, metastatic: 90, recurrence: 6; metastatic site: liver: 51, lung: 26, lymph nodes: 20, peritoneum: 38, others: 7; number of metastatic sites: 1: 63, 2: 25, ≥3: 8; treatment line at liquid biopsy: 1: 21, 2: 77, 3:8; median CA19-9 level: 1053.4 U/mL; type of liquid biopsy: FoundationOne Liquid CDx (FL): 75 or Guardant360 (G): 31. The overall rate of KRAS mutations was 47%, with a median variant allele frequency of 0.83 [range:0.1-71.0]. The patterns of KRAS mutations were as follows: G12D: 23, G12V: 18, G12R: 4, and others: 14. The detection rate of KRAS mutations varied by the follows factors: metastatic site: liver: 65%, other organs: 44%; number of metastatic sites: ≥2: 47%, <2: 48%; CA19-9 level: ≥2,000 U/mL: 68%, <2,000: 38%; timing of liquid biopsy: 1st line: 42%, ≥2nd line: 67%; types of liquid biopsy: FL: 49%, G: 42%. In multivariate analysis, liver metastasis and CA19-9 level ≥2,000 U/mL were identified as independent factors associated with improved detection of KRAS mutations. In PC with these two factors, the detection rate of KRAS mutation increased to 76%. **Conclusions:** Our findings indicate that liver metastasis and CA19-9 levels above 2,000 U/mL are predictive factors for detecting KRAS mutations in PC through liquid biopsy. Research Sponsor: None.

783 Poster Session

Prognostic implication of body composition changes in patients with borderline resectable/locally advanced pancreatic adenocarcinoma (PDAC) treated with mFOLFIRINOX. First Author: Jong Hyuk Lee, Department of Oncology, Asan Medical Center, University of Ulsan College of Medicine, Seoul, South Korea

Background: Although cachexia is a known prognostic factor in patients with advanced cancer, individual body composition changes during chemotherapy and their association with survival in PDAC warrant further investigation. We aimed to analyze the prognostic implication of comprehensive body composition, including areas and attenuation of muscle mass, body fat area and its distribution, and body mass index (BMI) during mFOLFIRINOX in patients with PDAC. **Methods:** Patients with borderline resectable or locally advanced PDAC, who were not able to undergo curative surgery and received first-line mFOLFIRINOX between January 2017 and December 2020 from Asan Medical Center, Seoul, Korea, were included in this retrospective study if they had paired CT scans of the abdomen at baseline and after 12 weeks of mFOLFIRINOX. Body composition was measured using artificial intelligence software (AID-UTM, iAID Inc.) from the CT images, and their association with overall survival (OS) were analyzed. **Results:** A total of 377 patients were included. Median age was 63 (range, 56–69) and 214 patients (56.8%) were males and 163 patients (43.2%) were females. During the first 12 weeks, significant changes in the body composition occurred as follows (median changes and interquartile range): skeletal muscle area, -5.8% [-11.1%–0.6%]; normal attenuation muscle area/total attenuation muscle area, -2.8% [-8.7%–3.2%]; visceral fat area (VFA), -6.1% [-23.7%–18.6%]; skeletal fat area (SFA), -12.2% [-27.8%–77.1%]; and BMI, -2.2% [-6.8%–1.5%]. Furthermore, changes in the VFA and BMI was associated with tumor response (Kruskal-Wallis test, p = 0.012 for VFA; p = 0.054 for BMI). There was no significant relationship between body composition abnormalities including sarcopenia, myosteatosis, or obesity at baseline and OS. However, patients who experienced the greatest decrease in SFA and BMI after 12 weeks of mFOLFIRINOX had poorer OS (tertile 1 [greatest decrease] vs 3 [smallest decrease]: HR 0.67 [95% CI, 0.52–0.87] for SFA, p = 0.003; 0.68 [95% CI, 0.49–0.83] for BMI, p = 0.001). **Conclusions:** In patients with borderline resectable/locally advanced PDAC, body composition significantly changes during the first 12 weeks. Decrease in the body fat and BMI from baseline were associated with poorer tumor response and OS. Research Sponsor: None.

784 Poster Session

Clinical outcomes of malignant portal vein thrombosis in patients with pancreatic cancer: A retrospective analysis. First Author: Rabia Iqbal, Brooklyn Hospital Center, Brooklyn, NY

Background: Portal vein thrombosis (PVT) includes portal vein blockage with a blood clot. Recently, there has been an increase in the incidence of PVT with solid malignancy. Portal vein thrombosis (PVT) is a potentially high-risk complication associated with pancreatic cancer. A hypercoagulable state is frequently observed in pancreatic cancer, with the incidence of thromboembolic events reported to range between 17% and 57%. We wanted to analyze the prevalence of PVT in patients with pancreatic cancer and its impact on hospital outcomes. **Methods:** We performed a retrospective analysis using the National Inpatient Sample Database (NIS) from 2018 to 2020. Patients with the primary diagnosis of pancreatic cancer and age >18 were diagnosed using ICD-10 codes. They were then stratified based on the presence of portal vein thrombosis. The Chi-square test was used to analyze categorical variables, and the T-test was used to analyze continuous variables. The confounders were adjusted using multivariate regression analysis. We observed the outcomes on mortality, length of stay, and total hospital charges. **Results:** A total of 336894 patients were diagnosed with a primary diagnosis of pancreatic cancer. Of them, 0.04% (n=13475 patients) were diagnosed with metastatic portal vein thrombosis. The mean age of the patients was 64 years (p <0.001). Out of them, 47% of patients were female. In comparison, the remaining 53% were male (p-value: 0.5). Almost 71% f the patients were White, 13% were African American and 0.09% were Hispanics (p-value <0.001). The presence of PVT was found to be associated with increased mortality in pancreatic patients (aOR 1.6, p-value <0.001) as compared to those without PVT. It was also associated with an increase in mean length of stay (LOS)(7.1 vs 6 days, p-value 0.009) and total hospital charges ($86,000 vs $73794, p-value <0.001). It was also observed that the presence of PVT was associated with increased odds of secondary outcomes, including venous thromboembolism (p-value <0.001), stroke (p-value 0.5), and acute kidney injury (p-value <0.001). **Conclusions:** The presence of portal vein thrombosis (PVT) in pancreatic cancer patients is associated with significantly worse hospital outcomes. These findings emphasize the importance of early detection and aggressive management of PVT in this patient population. Research Sponsor: None.

Comparison of hospital outcomes between pancreatic cancer patients with and without portal vein thrombosis.

Hospital Outcomes	Pancreatic cancer patients with PVT	Pancreatic cancer patients without PVT	Adjusted Odds Ratio (aOR)	95% CI	P-value
Mortality	9.4% (n=1310)	7.2% (n=23300)	1.3	0.3-1.5	<0.001
LOS (Days)	7.1	6.0	1.6	1.1-2.3	0.009
Venous thromboembolism (VTE)	11% (n=1645)	4.2% (n=13734)	2.9	2.5-3.3	<0.001
Stroke	0.03% (n=5)	0.02 (n=79)	1.8	0.2-14	0.5
Acute Kidney Injury	21% (n=2970)	20% (n=68499)	1.2	1.09-1.3	<0.001

785 Poster Session

Cohort-level patient-derived organoid assays and prediction of clinical trial outcomes in PDAC. First Author: Gustave Ronteix, Orakl Oncology, Le Kremlin-Bicêtre, France

Background: Clinical trials are one of the major barriers in contemporary drug development. Improving the number of therapeutic solutions for patients whilst limiting the financial burden of healthcare systems will require dramatically improving clinical trial success rates. Functional assays based on patient-derived organoids (PDO) are a promising new tool for derisking clinical development of new therapies, but their use has been limited due to small cohort sizes and the absence of systematic validation studies. **Methods:** We have assembled a collection of 70 demographically relevant PDAC PDOs. Each PDO was screened with a 22-drug panel including standard of care, chemotherapies and targeted therapies. Using a best-in-class patient response prediction model we generate cohort scale estimations of overall response rates (ORR) and progression-free survival (PFS). The estimations are compared with data from published clinical trials to determine the validity of the approach. **Results:** The PDOs exhibit representative treatment histories in terms of number of lines of treatment (mean = 1.60) and treatment types (prior treatment types: folfirinox = 68.6%, gemcitabine = 50%). All the organoids' mutational profile, past patient treatments and subsequent responses to future lines were analyzed and compared to the distributions observed in the clinic. The cohort-level drug efficacy on our PDO collection matches retrospective clinical trial ORRs reported in the literature. Furthermore, our preclinical assay predicts the relative performance of drugs in different clinical trial arms. **Conclusions:** We report that large-scale PDO-based assays predict clinical ORR and PFS with best-in-class accuracy. This work shows that well-characterized PDO collections can be used to improve the success rates of clinical studies for new potential treatments in PDAC. Research Sponsor: Fondation Jean-Jacques et Felicia Lopez-Loreta pour l'Excellence Académique.

786 Poster Session

Updated trends in incidence and mortality of pancreatic cancer. an analysis of the Surveillance, Epidemiology, and End Results (SEER) database. First Author: Olanipekun Lanny Ntukidem, Trinity Health Ann Arbor, Ann Arbor, MI

Background: Pancreatic cancer is the fourth leading cause of cancer deaths in the United States. The prognosis for pancreatic cancer remains poor despite advances in cancer therapy. In this study, we provide updated trends in pancreatic cancer incidence and mortality using the SEER database, with a subset analysis of age and gender. **Methods:** Using the SEER Stat software, we gathered data from the Surveillance, Epidemiology, and End Results (SEER) database (2000–2021). We employed Jointpoint regression to analyze the secular trend of incidence and incidence-based mortality (IBM), which we classified and arranged according to age and sex. **Results:** 235,116 cases met the inclusion criteria for our study. These included diagnosed cases of pancreatic cancer from 2000 to 2001. The total age-adjusted incidence rate was 12.5 per 100,000. 118,791 (50.5%) were males and 116,325 (49.5%) were female. While the annual percentage change (APC) from 2000-2019 was 0.92 ($p < 0.05$), signifying an annual cumulative increase, APC from 2019-2021 was -0.085, signifying an annual cumulative decrease. For males, the APC was 0.94 ($p < 0.05$) from 2000 to 2018 and -0.55 from 2018-2021, and for females, the APC was -0.89 ($p < 0.05$) from 2019-2021. The pancreatic cancer IBM APC from 2000-2022 was 22.60 ($p < 0.05$) and 0.54 ($p < 0.05$) from 2002-2021. For males, IBM APC was 21.10 ($p < 0.05$) from 2000-2002 and 0.58 from 2002-2021. For females, IBM APC was 25.15 from 2000-2002 and 0.42 from 2002-2021. We also studied pancreatic cancer IBM based on age. For 30-49 years, APC from 2000-2002 was 34.65 ($p < 0.05$), and -0.64 ($p < 0.05$) from 2002-2021. For 50-69 years, APC from 2000-2002 was 28.05 ($p < 0.05$) and 0.3 from 2002-2021. For 70+ years, APC from 2000-2002 was 21.28 ($p < 0.05$) and 0.73 from 2002-2021. **Conclusions:** There was a statistically significant rise in incidence and incidence-based mortality across the board, underscoring the need for improved diagnostic and therapeutic modalities of management of pancreatic cancer. Research Sponsor: None.

TPS787 Trials in Progress Poster Session

Intra-tumoral CD40 antibody with irreversible electroporation (IRE) in locally advanced pancreas cancer. First Author: Rebekah Ruth White, University of California San Diego Moores Cancer Center, La Jolla, CA

Background: Irreversible electroporation (IRE) is a form of non-thermal tumor ablation that is currently in clinical use for selected patients with locally advanced pancreatic cancer (LAPC). Preclinical and clinical studies have suggested that IRE generates an in situ vaccination effect by releasing tumor neoantigens in the setting of inflammation. Our published data from syngeneic, orthotopic mouse models demonstrate that combination of IRE with local CD40 agonism induces systemic anti-tumor immune effects, including neoantigen-specific T-cell responses and inhibition of liver metastases (1). Mitazalimab, a second-generation CD40 agonistic IgG1 mAb, has demonstrated promising clinical antitumor activity when delivered systemically in combination with modified FOLFIRINOX in metastatic pancreatic cancer (2). **Methods:** This clinical trial (NCT06205849) is a phase I dose-escalation study of an agonistic CD40 antibody (mitazalimab) injected intratumorally at the time of surgical IRE in patients with LAPC who have not demonstrated distant disease progression after a minimum of 4 months of optimal standard of care (SOC) chemotherapy (modified FOLFIRINOX). The doses investigated are 25, 75 (starting dose), and 200 μg/kg, in an interval 3+3 design, with a maximum of 18 subjects. The primary endpoints are the rates of dose limiting toxicities (DLTs) and treatment-emergent adverse events (AEs). The secondary endpoints are progression-free survival (PFS) and overall survival (OS). At a liberal alpha of 25%, appropriate for this proof-of-concept early phase study, we have 80% power to detect an improvement in PFS from 12 months (based on prior studies of IRE alone) to 18 months, with an expected 16 evaluable subjects and an anticipated 11 observed events, using a one-sided log-rank test. Candidate neoantigens will be identified by profiling nucleic acids derived from tumor biopsies obtained intraoperatively prior to IRE using our Identification-Prioritization-Validation (IPV) bioinformatic pipeline (3). Blood samples will be collected pre-operatively and 12-weeks post-operatively to evaluate for evidence of neoantigen-specific T-cell reactivity and other neoantigen-independent analyses of immune response. We have enrolled two patients since the study opened in September 2024. 1. J. Shankara Narayanan et al., JITC 2023. 2. J. Van Laethem et al., Lancet Oncol 2024. 3. A. Miller et al., Sci Transl Med. 2024. Clinical trial information: NCT06205849. Research Sponsor: U.S. National Institutes of Health; R01 CA282439.

TPS788 Trials in Progress Poster Session

EVEREST-2: A seamless phase 1/2 study of A2B694, a logic-gated Tmod chimeric antigen receptor T-cell (CAR T) therapy, in patients with pancreatic cancer (PANC) or other mesothelin (MSLN)-expressing solid tumors and human leukocyte antigen (HLA)–A*02 loss of heterozygosity (LOH). First Author: Kristen Renee Spencer, Perlmutter Cancer Center, NYU Langone Health, New York, NY

Background: MSLN is a desirable candidate for targeted therapy given its limited expression in normal tissues and its overexpression in several malignancies, including PANC, mesothelioma, non-small cell lung, and ovarian cancers (TCGA 2022). MSLN-targeted CAR T and T-cell receptor therapies have shown promising clinical activity; however, on-target, off-tumor toxicity, including fatal events, have occurred (1-3). Autologous logic-gated Tmod CAR T therapies address the challenges of on-target, off-tumor toxicity by combining 2 CARs: activating and blocking receptors. The activator recognizes a tumor-associated antigen present on the surface of both tumor and normal cells; the blocker prevents CAR T activity when it binds HLA-A*02, which is present in normal cells and often lost in tumor cells. Thus, in patients with tumor-associated HLA-A*02 LOH, the blocker prevents on-target, off-tumor toxicity on normal cells due to retained HLA-A*02 expression (4). HLA-A*02 LOH is observed in various solid tumors, including ~20% of PANC (5). Two autologous CAR T therapies are currently under investigation: A2B530, targeting carcinoembryonic antigen, in the EVEREST-1 trial, and A2B694, targeting MSLN, in the EVEREST-2 trial, described herein. A2B694 may provide a therapeutic window to treat patients with PANC and other MSLN-expressing solid tumors exhibiting HLA-A*02 LOH due to its unique discriminatory mechanism. **Methods:** EVEREST-2 (NCT06051695) is a first-in-human, phase 1/2, open-label, non-randomized study to evaluate the safety and efficacy of A2B694 for recurrent unresectable, locally advanced, or metastatic cancers with MSLN expression, including PANC. Eligible patients must have exhausted options of potentially curative therapy and have recurrent disease. Enrollment to EVEREST-2 occurs through the prescreening study BASECAMP-1 (NCT04981119), in which patients with tumor-associated HLA-A*02 LOH are identified via next-generation sequencing (Tempus AI, Inc) and leukapheresis product is collected. A2B694 is manufactured from cryopreserved T cells when clinically appropriate for patients. The primary objective of phase 1 is to evaluate the safety and tolerability of A2B694 and identify the recommended phase 2 dose (RP2D). The dose-expansion phase will confirm RP2D and collect biomarker data to further characterize A2B694. Phase 2 will assess the primary endpoint overall response rate per RECIST v1.1 and secondary endpoints progression-free survival and overall survival. The first patient was dosed on April 24, 2024 and dose escalation is ongoing. 1. Beatty, et al. *Gastroenterology*. 2018. 2. Haas, et al. *Mol Ther*. 2023. 3. Hong, et al. ESMO 2021. Abstr 9590. 4. Hamburger, et al. *Mol Immunol*. 2020. 5. Hecht, et al. *J Clin Oncol*. 2022. Clinical trial information: NCT06051695. Research Sponsor: A2 Bio.

TPS789 Trials in Progress Poster Session

Phase II study of olaparib in patients with advanced pancreatic acinar cell carcinoma. First Author: Nebojsa Skorupan, National Cancer Institute, Bethesda, MD

Background: Pancreatic Acinar Cell Carcinoma (PACC) is a rare and biologically distinct subtype of pancreatic cancer, representing 0.5-1% of all pancreatic malignancies, with a median overall survival of 15-20 months. Significant serum CA 19-9 elevation is uncommon, while elevation of serum lipase is observed in at least 50% of patients. Most patients present at advanced stages and treatment with combination chemotherapies extrapolated from other GI malignancies leads to low response rates. No clinical treatment trials specific for PACC have ever been reported. PACC tumors from patients show evidence of high chromosomal instability, a hallmark of DNA repair deficiency. ID3 is ubiquitously downregulated and was recently shown to play a significant role in homologous recombination repair (HRR). Recent genomic profiling of larger PACC patient cohorts has revealed a high prevalence of mutations in HRR genes, including BRCA1, BRCA2, and PALB2. Olaparib is a Poly-ADP ribose polymerase (PARP)-1 inhibitor that has been FDA approved for treatment of BRCA-mutant HRR deficient cancers. We hypothesize that PACC will be sensitive to PARP inhibition with olaparib. **Methods:** This is an open label, single arm, single center, phase II study of olaparib in patients with advanced previously treated PACC. The primary objective of the trial is to evaluate anti-tumor activity of olaparib (objective response rate). Secondary objectives assess the safety (CTCAE v. 5.0) and alternative measures of efficacy such as clinical outcomes and blood biomarkers (disease control rate, median progression-free survival, median overall survival, best response in serum lipase). Key eligibility criteria include histologic diagnosis of PACC, receipt of at least one line of combination chemotherapy, presence of RECIST measurable disease and adequate organ and bone marrow function. A fixed dose of olaparib at 300 mg will be given orally continuously twice daily during a 28-day cycle, for up to 2 years or until disease progression or intolerable toxicities. In person visits to assess toxicity are required every 4 weeks, and response is determined through serial imaging every 8 weeks (CT or MRI). Exploratory objectives include acquisition of tumor tissue for establishment of new PACC preclinical models, assessment of genetic alterations through genetic mapping technique, assessment of HRR-related gene products as a predictive biomarker, assessment of ID3 protein levels, and assessment of treatment response to tumor profile. Up to 13 evaluable participants will be enrolled. Three participants have been enrolled as of Sep 19, 2024. Clinical trial information: NCT05286827. Research Sponsor: None.

TPS790 Trials in Progress Poster Session

A phase 2 study of amplitude-modulated radiofrequency electromagnetic fields (AM RF EMF) in metastatic pancreatic cancer. First Author: Ravi Kumar Paluri, Department of Hematology and Oncology, Wake Forest School of Medicine, Winston-Salem, NC

Background: Patients with advanced pancreatic adenocarcinoma (PDC) that progresses after first-line therapy face limited treatment options. Despite advances in systemic therapy, the prognosis for PDC is dismal, with a five-year overall survival (OS) of about 5% . Due to the high morbidity of this disease, there is a critical need for innovative treatments to improve patient outcomes. TheraBionic P1, a hand-held device, delivers tumor specific radio frequencies via a spoon-shaped antenna placed on the patient's tongue. The FDA and EMA have confirmed the safety of the device for amplitude-modulated electromagnetic field (AM-EMF) therapy. Improved efficacy outcomes were reported with AM RF EMF in other solid organ tumors including pancreatic cancer in preliminary studies. Preclinical and clinical evidence suggests that AM-EMF can inhibit tumor growth and was recently approved for use in hepatocellular cancer. The safety and clinical activity observed with AM RF EMF, paired with the unmet medical need for additional effective therapies in pancreatic cancer, provided a strong rationale for further clinical investigation. This protocol evaluates the efficacy and tolerability of AM RF EMF in combination with standard of care frontline therapy, gemcitabine and nab-paclitaxel for metastatic PDC. **Methods:** This is an ongoing, single-center, Phase II, non-randomized study using a Simon two-stage minimax design to assess the efficacy of AM-EMF in combination with gemcitabine/nab-paclitaxel as a first-line treatment in metastatic pancreatic cancer. Gemcitabine and nab-paclitaxel are administered per standard care. AM RF EMF are delivered to patients with a spoon-shaped applicator placed on the tongue three daily 60-minute treatments. Patients will be assessed for response every 8 weeks. Treatment is continued until progression or toxicity. The primary objective of this study is to evaluate the efficacy of the combination of nab-paclitaxel, gemcitabine, and AM RF EMF in improving 6-month overall survival rates in patients with metastatic PDC. Secondary objectives evaluate the safety, tolerability, profession free survival, objective response rate and disease control rate. Exploratory objectives evaluate change in CA19-9 and quality of life using FACT-General (FACT-G) and determine any possible correlation with clinical outcomes. Key eligibility includes treatment-naïve, histologically confirmed metastatic PC, age eighteen years old or greater, ECOG 0-1, and optimal organ function. Clinical trial information: NCT05776524. Research Sponsor: Wake Forest Baptist Health Comprehensive Cancer Center Boot Camp.

TPS791 Trials in Progress Poster Session

UNITEPANC: Using organoids to predict efficacy of adjuvant treatment to improve outcome in resectable pancreatic cancer—Prospective, multicenter, proof-of-concept trial of the AIO Pancreatic Cancer Group. First Author: Thomas Jens Ettrich, Department of Internal Medicine I, Ulm University Hospital, Ulm, Germany

Background: Pancreatic cancer (PDAC) is a highly heterogeneous disease and there is a lack of predictive biomarker to guide treatment. Patient tumor derived organoids (PDO) have been demonstrated to faithfully recapitulate morphological and genetic features of the parental tumor and also retain patient-specific heterogeneity. Retrospective studies show that PDO can be used to predict treatment sensitivity and can support the selection of an optimal treatment in terms of efficacy and also lower toxicity, e.g. by selecting a double rather than a triple combination. **Methods:** UNITEPANC is a prospective, proof of concept, multicenter IIT of the German AIO PDAC Group and funded by the German Cancer Aid. UNITEPANC examines the feasibility of an organoid-educated adjuvant treatment approach with respect to organoid establishment, expansion and characterization and its potential role for selecting an optimal adjuvant treatment in PDAC in a multicenter setting. Preparatory activities like harmonizing SOPs and round-robin-tests for generation of organoids and organoid-based pharmacotyping in the trial centers and the different Organoid Facilities has been completed and evaluated. UNITEPANC examines selection of adjuvant chemotherapy by pharmacotyping of tumor organoids. Options for adjuvant treatment are gemcitabine, gemcitabine/capecitabine, gemcitabine/nab-paclitaxel or mFOLFIRINOX. Major inclusion criteria are R0 or R1 resected, histologically confirmed PDAC and patients in principle, suitability for mFOLFIRINOX treatment in the adjuvant setting, postoperative Ca19-9$<$180 U/ml, and a timely PDO-based chemotherapy recommendation by the UNITEPANC Organoid Board within 11 weeks to guarantee a start of the adjuvant treatment within 12 weeks after resection. Design: UNITEPANC is an interventional, prospective, multicenter, single-arm trial. 95 patients shall be allocated to the trial for the generation of organoids, 38 patients shall be enrolled for PDO-based adjuvant treatment and 34 patients need to be analysed as ITT population in in the follow-up period. Efficacy: The efficacy endpoint of the trial is purely descriptive and intended to obtain a signal for further development of the strategy. This requires the proof of an efficient 1) generation and 2) expansion of PDO and 3) a clear sign that an organoid-educated treatment selection is superior to standard treatment with mFOLFIRINOX. For 1) and 2) we have chosen a rate of 75% and 60%, respectively, according to the literature. For 3): The approach would be considered promising, if the true 18-months-DFS rate is 77% (corresponding to a hazard ratio of 0.5 compared to Prodige-24, 80% power, one-sided type I error 0.1. HR-QoL data will be analyzed (QLQ-C30, Pan26). Additionally, an extensive translational program is enclosed. Research Sponsor: German Cancer Aid.

TPS792 Trials in Progress Poster Session

Phase I study of autologous CD8+ and CD4+ transgenic T cells expressing high-affinity KRAS G12V mutation-specific T cell receptors (FH-A11KRASG12V-TCR) in patients with metastatic pancreatic, colorectal, and non-small cell lung cancers with KRAS G12V mutations. First Author: Rachael A. Safyan, University of Washington, Fred Hutchinson Cancer Center, Seattle, WA

Background: KRAS, the most common driver oncogene in human malignancies, has predictable mutations. 85% of RAS mutations occur at the Glycine (G) 12 position with G12V being amongst the most common. Cancers driven by mutant KRAS include pancreatic (PDAC), colorectal (CRC), and non-small cell lung (NSCLC) malignancies. The prognosis of patients with activating KRAS mutations remains poor. Preliminary anti-tumor activity has been observed with T cell receptor (TCR) T cell therapies targeting KRAS mutations, but durable clinical benefit is lacking. FH-A11KRASG12V-TCR is an autologous CD8+ and CD4+ transgenic T cell product expressing (1) an HLA-A*11-restricted, high affinity KRASG12V mutation-specific T cell receptor (TCR) and (2) the wildtype CD8α/β coreceptor enabling CD4+ T cell recognition of KRAS G12V presented on an MHC class I allele mediating helper T cell activity and cytotoxicity. The TCR sequence is derived from the peripheral blood of a healthy HLA-A*11+ adult based on a strong pre-clinical dataset. **Methods:** This is an ongoing, investigator-initiated, phase 1, first-in-human, single-center, dose escalation study of FH-A11KRASG12V-TCR. The aims are to evaluate the safety, tolerability, maximum tolerated dose (MTD), and preliminary anti-tumor activity of FH-A11KRASG12V-TCR in patients with advanced PDAC, CRC, and NSCLC who express the HLA-A*11-restricted KRAS G12V mutation. Dose modification is guided by the Bayesian Optimal Interval design (BOIN). Patients are initially treated in cohorts of 3 at each dose level (DL). BOIN will be used to identify the recommended dose for subsequent patients. The planned target doses are 5 x 10^9 transgenic TCR T cells in DL 1 with dose escalation up to 15 x 10^9 transgenic TCR T cells in DL 3. This study enrolls patients who are 18 years or older, HLA-A*11:01-positive, with KRAS G12V mutated metastatic PDAC, CRC, or NSCLC. Patients must have progressed on or be intolerant of at least one line of prior therapies including any targeted therapies indicated for their current malignancy. Patients undergo leukapheresis prior to treatment and receive lymphodepleting chemotherapy with either cyclophosphamide intravenously (IV) and fludarabine IV on days -6, -5, and -4 or bendamustine IV on days -4 and -3. FHA11KRASG12V-TCR is administered IV on day 0. The dose-limiting toxicity observation period is 28 days. Patients may receive an additional FHA11KRASG12V-TCR IV infusion at the same or lower dose as soon as 28 days or up to 1 year after the first infusion. The study is actively recruiting. Clinical trial information: NCT06043713. Research Sponsor: Affini-T Therapeutics.

TPS793 Trials in Progress Poster Session

Randomized phase II trial of olaparib + pembrolizumab vs olaparib alone as maintenance therapy in patients with metastatic pancreatic cancer with germline BRCA1 or BRCA2 mutations: SWOG S2001 NCT04548752. First Author: Vincent Chung, City of Hope, Duarte, CA

Background: Olaparib was approved in 2019 as maintenance therapy for gBRCA1/2+ metastatic pancreatic cancer (mPDA) patients (pts). The POLO trial showed an improvement in median progression-free survival (mPFS) with olaparib compared to placebo (7.4 vs 3.8 months) for platinum sensitive gBRCA1/2+ mPDA pts. Preclinical studies have demonstrated that PARP inhibitors modulate the immune microenvironment by increasing genomic instability, PD-L1 expression and activating the immune inflammatory stimulator of interferon genes (STING) pathway. Results of the phase 2 pembrolizumab and olaparib (POLAR) maintenance trial in patients with germline or somatic BRCA1/2 or PALB2 mutations responding to platinum-based chemotherapy for at least 4 months was presented at ESMO 2024. In 33 patients, a 35% overall response rate with 90% disease control rate at 6 months was observed. S2001 is an important randomized study to better define the impact of the combination. **Methods:** S2001 was developed in collaboration with the Alliance and was activated in SWOG in October 2020. Key eligibility criteria include: mPDA pts with gBRCA1/2 mutations identified with standard of care germline genetic testing and progression-free after receiving at least 4 months of platinum-based chemotherapy (FOLFIRINOX, FOLFOX or gemcitabine/cisplatin +/- nab-paclitaxel). One cycle of gemcitabine and nab-paclitaxel is allowed while waiting for germline testing. Zubrod performance status (PS) 0 or 1 pts are eligible. Pts are stratified according to first line chemotherapy, PS 0 vs 1, and disease status after 1st line treatment. The primary objective of this study is to evaluate the PFS of mPDA pts treated with olaparib + pembrolizumab compared to olaparib alone as maintenance therapy. Based upon the POLO trial, we expect a mPFS of 7 months in the control arm. Targeting a mPFS of 11.7 months in the experimental arm (hazard ratio 0.6) and assuming 15 months follow-up, 80% power and a 1-sided alpha = 0.10, this design requires 78 evaluable pts with a total sample size of 88 pts. Prospective serial blood samples will be collected to bank DNA and RNA for future correlative studies. Support: NIH/NCI grants U10CA180888 and U10CA180819, U10CA180821 and U10CA180868. Clinical trial information: NCT04548752. Research Sponsor: NIH/NCI grants.

TPS794 Trials in Progress Poster Session

A phase I, single-center, open label, dose de-escalation and expansion study of Ivosidenib + mFOLFIRINOX in patients with resectable pancreatic adenocarcinoma. First Author: Ho Jun Lee, University Hospitals Seidman Cancer Center, Case Western Reserve University, Cleveland, OH

Background: Pancreatic ductal adenocarcinoma (PDA) carries a dismal prognosis, with a 34% 5-year survival rate for patients with localized disease. Guidelines recommend consideration of neoadjuvant approaches in localized PDA with the goals of controlling microscopic metastatic disease and increasing rates of microscopically margin-negative resections. One of the hallmark characteristics of PDA is a micronutrient poor, highly stromal microenvironment. Surviving in this environment requires enhanced mitochondrial function, which utilizes isocitrate dehydrogenase 1 (IDH1). In pre-clinical studies, ivosidenib, an FDA approved medication for IDH1 mutant AML, induced high levels of reactive oxygen species in PDA cells with wild-type IDH1, and caused tumor regressions and extended survival in implanted KPC mouse models. Here, we investigate the addition of ivosidenib to standard of care neoadjuvant mFOLFIRINOX in patients with resectable PDA to determine safety, pharmacodynamics, and early efficacy signals. **Methods:** This is a phase I, single-center, open label, dose de-escalation and expansion study in patients with resectable PDA. Eligible patients must have histologically confirmed and resectable right-sided PDA based on CT or MRI imaging. Patients receive approximately 10 weeks of neoadjuvant treatment composed of 2 weeks of ivosidenib monotherapy, followed by 6 weeks (3 cycles) of mFOLFIRINOX with ivosidenib, followed by up to 4 weeks of ivosidenib monotherapy until the day of surgery. Ivosidenib is administered once daily at 500mg; it is to be de-escalated to 250mg based on Bayesian Optimal Interval Design with Informative Prior and a target dose limiting toxicity (DLT) rate of 30%. The primary endpoint is to determine the safety and tolerability of ivosidenib in combination with mFOLFIRINOX. Secondary endpoints include RECIST version 1.1 response rates, major pathologic response rates, and biochemical (CA19-9, CEA) response rates. Correlative studies will be performed on surgical samples to evaluate metabolomic profiles. At the time of submission, 10 out of 16 planned patients have been enrolled. There have been no DLT at 500mg of ivosidenib. Clinical trial information: NCT05209074. Research Sponsor: Gateway for Cancer Research Foundation.

TPS795 **Trials in Progress Poster Session**

A phase II study of peri-operative NovoTTF-200T(P) in combination with gemcitabine and nab-paclitaxel for resectable pancreatic adenocarcinoma: Big Ten Cancer Research Consortium (BTCRC-GI21-500). First Author: Adam Khorasanchi, The Ohio State University Comprehensive Cancer Center – James Cancer Hospital and Solove Research Institute, Columbus, OH

Background: The recurrence rates and outcomes in resectable pancreatic ductal adenocarcinoma (R-PDA) is concerningly high. The benefit of neoadjuvant chemotherapy over traditional upfront surgery followed by adjuvant therapy is not clear. Electromagnetic fields generate bi-directional forces on highly polar intracellular components, causing abnormal microtubule polymerization during spindle formation and irregular cleavage furrow formation. We will leverage these anti-proliferative effects in managing R-PDA by tumor treating fields (TTF) with chemotherapy backed by strong in vitro and in vivo evidence. The PANOVA phase II trial (n=40) gave us safety and efficacy data of this combination in advanced PDA. No systemic effects were associated with TTF. The only notable safety issue was a small incidence of grade 3 device-related dermatitis. **Methods:** Our phase II single arm study will investigate perioperative use of TTF with gemcitabine and nab-paclitaxel (G-NP) combination. Eligible patients R-PDA (visible pancreatic mass, measurable disease, absence of arterial interface, venous interface $\leq$ 180°, patent portal splenic confluence, and no metastatic disease, including lymphadenopathy outside the surgical area) will wear TTF and receive three cycles of G-NP before restaging. Patients are expected to wear TTF for $>$ 80% of the time during this period. If they proceed with resection, additional 3 cycles of Gem-NP with TTF will be given to the patient. This study employs a Bayesian Optimal Phase 2 (BOP2) design with primary endpoints of overall survival (OS) at 2 years and resection rate. The null hypothesis posits a 40% OS rate compared to a target of 60%, and a resection rate null hypothesis of 60% versus a target of 75%. The trial aims to enroll 30 patients, with an interim futility analysis planned after 15 participants. Enrollment will be terminated early if 8 or fewer patients undergo resection in the initial stage, providing 83% power and a type I error rate of 10%. Secondary endpoints include adverse events, overall response rate, TTFields compliance rate, relative dose intensity, and overall survival. The trial started actively recruiting patients in April 2024. The accrual goal is 38 patients. At the time of submission, one patient was enrolled. Clinical trial information: NCT05624918. Research Sponsor: Big Ten Cancer Research Consortium.

796 **Poster Session**

Clinical characteristics and treatment outcomes for metastatic early-onset small bowel adenocarcinoma treated with first-line chemotherapy. First Author: Shotaro Yamaguchi, Multidisciplinary Treatment Cancer Center, Kurume University Hospital, Kurume, Japan

Background: Small bowel adenocarcinoma (SBA) is an aggressive rare cancer with its incidence on the rise. Although early-onset SBA (EO-SBA: diagnosed at < 50 years old) accounts for approximately 10% of all SBA, characteristics and survival outcomes of advanced EO-SBA remains under-evaluated. This study aims to evaluate clinical features and treatment outcomes of patients with advanced EO-SBA receiving first-line chemotherapy. **Methods:** We retrospectively analyzed the clinical data of patients who received fluoropyrimidine plus oxaliplatin as first-line treatment for recurrent or metastatic SBA at our hospital between January 2012 and July 2024 Patients were divided into two groups: those who were diagnosed at age of < 50 years old (EO-SBA) and those who were diagnosed at age of ≥50 years old (late-onset SBA, LO-SBA). Clinicopathological characteristics and treatment outcomes (objective response rate [ORR], progression-free survival [PFS], overall survival [OS], and adverse events [AEs]) were compared between two groups. Survival outcomes were estimated using the Kaplan–Meier method and compared by the log-rank test. Hazard ratios (HRs) and 95% confidence intervals (95%CIs) were calculated by univariable and multivariable Cox regression analyses. **Results:** A total of 69 patients (EO-SBA 18, LO-SBA 51) were included in this study. Compared to the LO-SBA group, the EO-SBA group were more likely to have non-duodenum tumor (89% vs. 49%, p = 0.002), lesser number of metastatic sites (≤2, 100% vs.78%, p = 0.007), and prior resection of primary tumor (83% vs 49%, p = 0.008). The EO-SBA group had longer PFS compared to the LO-SBA group (median PFS: HR = 0.32 [95% CI 0.12 - 0.85] p = 0.022). The EO-SBA group had a trend of longer OS compared to the LO-SBA group (median OS: HR = 0.39 [95%CI 0.15 - 1.01] p = 0.053). Age group was not identified as an independent prognostic factor for OS (HR = 0.63 [95% CI: 0.21 - 1.85], p = 0.40) and PFS (HR = 0.47 [95% CI: 0.16 - 1.37], p = 0.17) by multivariable analyses adjusted for previously reported prognostic factors (performance status, liver metastasis, primary site, histological classification, prior resection of primary tumor, and carcinoembryonic antigen [CEA] elevation). Among patients with measurable lesions (EO-SBA 9, LO-SBA 41), the EO-SBA group showed a numerically higher ORR compared with the LO-SBA group (63% vs 29%, p = 0.07). There were no significant differences of AE frequencies between two groups except for anemia (grade 3/4. EO-SBA 0% vs LO-SBA 22%, p < 0.001). **Conclusions:** Our study indicates the difference of clinical characteristics between two age groups in patients with advanced SBA. EO-SBA was not identified as independent prognostic factor for OS and PFS in patients treated with first-line chemotherapy. Research Sponsor: None.

797 **Poster Session**

Distribution of lymph node metastasis and prognosis in duodenal bulb tumors: A multicenter retrospective study. First Author: Kazuhiko Hisaoka, Department of Surgery, Keio University School of Medicine, Tokyo, Japan

Background: Duodenal tumors, especially duodenal bulb tumors, is a relatively rare malignancy of the gastrointestinal tract. The optimal surgical procedure for primary duodenal tumors has not been established because of its low incidence. The purpose of this study was to examine the distribution of lymph node (LN) metastasis in duodenal bulb tumors. Specifically, in T1a and T1b tumors, we aimed to evaluate the feasibility of distal gastrectomy with duodenal bulb resection combined with lymphadenectomy of regional gastric LNs. **Methods:** Data from patients who underwent surgery for either adenocarcinoma or neuroendocrine tumor located in the duodenal bulb between 2000 and 2020 were retrospectively analyzed from five high-volume centers in Japan. Patient background, clinicopathological factors, type of surgery, distribution of LN metastasis, and long-term outcomes were evaluated. **Results:** Of the 128 patients evaluated, LN metastasis was observed in 36%. The frequency of LN metastasis in T1b was 18%. Metastatic LNs were identified in #3, #5, #6, #8a, #8p, #12a, and #13 in T1b. The distribution of LN metastasis in T1b was similar between adenocarcinoma and neuroendocrine tumors. The 3 years overall survival rate in T1a, T1b, and T2–4 was 100%, 93%, and 64%, respectively. In T1b, while there were three patients with a recurrence, two cases were observed in distant organs without regional LNs, and one patient who underwent pancreaticoduodenectomy had metastasis in the gastric regional LNs. **Conclusions:** Based on the distribution of LN metastasis with pT1, distal gastrectomy with duodenal bulb resection and regional LN dissection is considered a curative treatment. Conversely, pancreaticoduodenectomy is recommended in tumors with T2 or deeper. Research Sponsor: None.

798 **Poster Session**

Observational study of the extent of lymph node metastasis among patients who underwent pancreatoduodenectomy for non-ampullary duodenal cancer using the National Clinical Database in Japan. First Author: Yasuhiro Kodera, Japan Hospital Organization Nagoya Medical Center, Nagoya, Japan

Background: Non-ampullary duodenal cancer (NPDC) is a rare cancer that occurs in 0.3% of all cancers of the gastrointestinal tract. The first Japanese guidelines for this entity were published in 2021, and resection by pancreatoduodenectomy was weakly recommended as a standard treatment. Although presence of regional lymph node metastasis was identified as a prognostic factor among patients who underwent resection with curative intent, the recommendation concerning the optimal extent of lymphadenectomy based on big data was lacking. **Methods:** We conducted an observational study of patients who underwent pancreatoduodenectomy for NPDC between 2018 and 2021 in Japan, using the National Clinical Data (NCD) database. The NCD database is web-based, is linked with the board certification systems of several domestic surgical societies and covers >95% of the operations carried out by surgeons at 5,728 institutions in Japan. All cases of pancreatoduodenectomy for NPSC were retrieved from the NCD database after rigorously checking appropriateness of the patient selection, and surgeons in charge of these patients were invited by e-mails to fill in some additional data for this study during a time slot of Oct 2022 ~March 2023. Data thus obtained were the analyzed with particular focus on the incidence of metastases to each lymph node station, stratified by the tumor location in relation to the papilla. In addition, safety of surgery was compared between high-volume (institutions conducting ≥30 major hepatobiliary-pancreatic surgery) and low-volume (<30) hospitals. The study was funded in part by the Japanese Ministry of Health, Labor and Welfare as a project to create guidelines for rare cancers. **Results:** Data of 881 patients from 600 institutions were obtained for the analysis. Male constituted 68% and the median age was 70 years. The depth of invasion and presence of nodal metastasis were significant prognostic factors. The median number of lymph node retrieval was similar between the high-volume and low-volume hospitals (19: 95%CI 12~27 versus 18: 95% CI 11~25). Distribution of metastatic nodes depended on the location of tumor. Incidence of nodal metastasis exceeded 10% in lymph node station Nos. 5, 6, 7, 8 and 9 in cancers located on the oral side of the papilla. Thus, stomach-conserving pancreatoduodenectomy was considered appropriate only for cancers located distal to the papilla. The incidences of Grade IIIb~IVb complications (4% versus 7.1%), Grade V complications (0.4% versus 1.9%), and mortality within 30 days (0.4% versus 2.6%) were lower in high-volume hospitals. **Conclusions:** Stomach-preserving pancreatoduodenectomy is recommended only for NDPC located distal to the papilla. Centralization of surgery may be desirable. Research Sponsor: Ministry of Health, Labor and Welfare, Japan; 20EA1021.

799 **Poster Session**

TROP2 expression in small bowel adenocarcinoma: A potential target for novel therapeutic strategies. First Author: Hiroyuki Fujii, Department of Gastrointestinal Medical Oncology, National Cancer Center Hospital, Tokyo, Japan

Background: Small bowel adenocarcinoma (SBA) is a rare cancer with limited chemotherapy options. To explore TROP2 as a potential therapeutic target for SBA and develop new treatment strategies, we examined TROP2 protein expression in SBA and analyzed associated clinicopathological features and prognosis, including PD-L1 expression levels in the same tumor samples from our previous studies. **Methods:** We retrospectively reviewed patients diagnosed with SBA who underwent adjuvant or palliative chemotherapy between July 2010 and July 2023 at our hospital. Immunohistochemistry staining of TROP2 (clone: SP295) was performed for pathological samples. The intensity of TROP2 membranous staining in tumor cells was classified as 0 (absent), 1 (weak to moderate), or 2 (strong). TROP2 positivity was defined as intensity 1 with ≥ 50% expression or intensity 2 with ≥ 10% expression. We also assessed overall survival (OS) in patients receiving palliative chemotherapy to examine the association between prognosis and TROP2, excluding microsatellite instability-high (MSI-H) patients treated with immunotherapy. **Results:** Pathological samples and clinical data were available for 51 patients. The median age was 63 years (range: 23–82), and 76% were male. Most patients (94%) had a performance status (PS) of 0–1. The primary lesion was in the duodenum in 49% of patients and in the jejunum or ileum in 51%. Surgery and adjuvant chemotherapy resulted in no recurrence in 12% of patients, while 88% received oxaliplatin-based palliative chemotherapy for unresectable SBA. Among those with unresectable SBA, 8% (4/51) had MSI-H and were treated with immunotherapy. TROP2 positivity was 84.3% (43/51). There were no differences in age, sex, or primary lesion between the TROP2-positive and TROP2-negative groups. In this cohort, the median CPS was 3 (range, 0–60). CPS was lower in TROP2-positive samples compared to negative samples (median CPS 7.9% vs. 23.4%, p = 0.030). However, all MSI-H cases were found to be TROP2-positive. Among patients receiving palliative chemotherapy, excluding those with MSI-H treated with immunotherapy, OS did not differ significantly between TROP2-positive and TROP2-negative patients (17.0 months vs. 9.2 months, HR 0.70, p = 0.480). Even after adjusting for age and PS, TROP2 expression was not identified as a prognostic factor. **Conclusions:** Our study highlighted that TROP2 is frequently expressed in SBA, suggesting it could be an effective therapeutic target for new treatment strategies. Further investigation is needed to confirm its potential. Research Sponsor: None.

800 **Poster Session**

Impact of homologous recombination deficiency on survival and chemotherapy response in small bowel adenocarcinoma. First Author: Yoshiyasu Kono, Department of Gastroenterology and Hepatology, Okayama University Graduate School of Medicine, Dentistry and Pharmaceutical Sciences, Okayama, Japan

Background: Small bowel adenocarcinoma (SBA) is a rare malignancy, and its genomic profile is distinct from that of gastric and colorectal cancers. However, the role of homologous recombination repair (HRR) gene mutations in SBA remains unexplored. This study aims to investigate the clinical significance of HRR gene mutations in SBA. **Methods:** We retrospectively analyzed data from the Center for Cancer Genomics and Advanced Therapeutics (C-CAT) utilization portal, in accordance with its policy. Between June 2019 and February 2024, patients with SBA who underwent comprehensive genomic profiling (Foundation One CDx, Foundation One Liquid, OncoGuide NCC Oncopanel, Guardant 360 CDx, and GenMineTOP) were included. The frequency of HRR variants (*ATM, BARD1, BRCA1, BRCA2, BRIP1, CHEK2, PALB2, RAD51C, RAD51D, RAD51, ATRX, ARID1A, BAP1, CDK12, MRE11, NBN, PTEN, FANCA, FANCC*) was analyzed. Homologous recombination deficiency (HRD) was defined as the presence of one or more alterations in HRR variants. Clinicopathological features, treatment outcomes, and survival were compared between HRD-positive and HRD-negative groups. **Results:** A total of 628 cases were analyzed, with a median age of 65 years, including 238 males and 390 females. The primary tumor sites were the duodenum in 332 cases and the jejunum in 296 cases. HRD was observed in 165 cases (26.5%) overall. The incidence of microsatellite instability-high (MSI-H) was significantly higher in the HRD-positive group compared to the HRD-negative group (15% vs. 1.1%, $P < 0.001$). Similarly, the median tumor mutational burden (TMB) values were significantly higher in the HRD-positive group compared to the HRD-negative group (5 Mut/Mb vs. 3.8 Mut/Mb, $P < 0.001$). The median overall survival (OS) was similar between HRD-positive and HRD-negative groups (1,243 days vs. 1,055 days, P = 0.22). However, in duodenal primary cases, the median OS was significantly longer in the HRD-positive group compared to the HRD-negative group (1,243 days vs. 702 days, P = 0.025). No significant difference in median OS was observed between HRD-positive and HRD-negative groups in jejunal primary cases (1,145 days vs. not reached, P = 0.51). The overall response rate (ORR) to platinum-based chemotherapy in first-line treatment was significantly higher in the HRD-positive group compared to the HRD-negative group in duodenal primary cases (34% vs. 18%, $P < 0.05$). Conversely, there was no significant difference in ORR between HRD-positive and HRD-negative groups in jejunal primary cases (23% vs. 20%, P = 0.70). **Conclusions:** Duodenal primary SBA with HRD showed better response rates to platinum-based chemotherapy and had a more favorable prognosis compared to non-HRD cases. In duodenal cancer, which typically has a worse prognosis than jejunal cancer, HRD may represent a novel prognostic and predictive marker for treatment efficacy. Research Sponsor: None.

801 **Poster Session**

Incidence and survival trends in small bowel adenocarcinoma (SBA) from 2000-2019 in the United States. First Author: James Jerod Sears, Washington University School of Medicine, St. Louis, MO

Background: SBA is a rare, poorly understood gastrointestinal malignancy. Adenocarcinomas are often grouped with other small bowel cancer histologies in epidemiologic and outcomes reports, which limits the ability to detect SBA-specific trends. Recent studies in European populations show increasing SBA incidence, but there are limited published data in American cohorts. Identifying trends in SBA can improve understanding of the population most at risk for developing SBA. This study describes U.S. trends in SBA incidence and outcomes from 2000 to 2019. **Methods:** Using the 2022 SEER research-limited dataset, we identified SBA cases and measured the incidence change from 2000-2019. We built linear regression models to evaluate changes in annual incidence by groups, including sex, race, and age of diagnosis, separately. We then compared the overall incidence and trends between subgroups. We investigated overall survival (OS) through the SEER research case listings for SBA patients diagnosed during the study period. **Results:** An average of 0.67 cases/100,000 SBA cases were reported annually from 2000-2019, with a median diagnosis age of 69 years. SBA incidence has increased annually by 0.3% per year from 2000-2019 (p=0.05). From 2000-2019, men were at a higher risk for SBA compared to women (p=0.02). In men, SBA cases increased at a rate of 0.004 cases/100,000/year from 2000-2019 (p=0.01) and a nonsignificant decrease in women of 0.00012 cases/100,000/year. There were no statistically significant differences in incidence or the change in incidence of SBA by race. The age-adjusted incidence was significantly higher each year between 2000 and 2019 in the $>$ 70 cohort compared to the $<$70 cohort (p=0.02). Cases of SBA increased at a rate of 0.02 cases/100,000/year in adults $>$70 years (p=0.02). There were no differences in incidence rates in patients $<$70 years. There was no significant difference in OS for patients diagnosed with SBA in 2000, 2005, 2010, 2015, and 2019 (p=0.21). **Conclusions:** The overall annual incidence of SBA is increasing in the U.S. population. The most significant rate increases are in men and adults older than 70. In the SEER population, trends in SBA contrast with the recent rise of colorectal cancer in the young adult population, highlighting the unique nature of SBA compared to other gastrointestinal malignancies. Additionally, survival outcomes in patients diagnosed with SBA have remained stable over the past 20 years, while other malignancies have shown improvement in the same period. The small cohort size and the lack of a universal U.S. cancer registry limited this evaluation. Improved reporting of SBA cases and further research to identify risk factors are urgently needed to reduce the growing impact of SBA. Research Sponsor: None.

802 Poster Session

Evaluation of social support, discrimination, and perceived health in patients with gastrointestinal tract cancers. First Author: Samuel Butensky, Yale University School of Medicine, Department of Surgery, New Haven, CT

Background: Health-related social conditions affect cancer outcomes. Gastrointestinal cancers (GIC) comprise a large proportion of cancers diagnosed in the United States. We hypothesized that a diagnosis of GIC negatively impacts Social Determinants of Health (SDOH) and perceived health (PH). **Methods:** Prospectively collected data from The All of Us database was queried for GIC patients who completed SDOH and PH surveys. Healthy, non-cancer (NC) participants were matched to GIC based on age, race, sex, deprivation index, and Charlson Deyo comorbidities using Propensity Score Matching. Regression was used to analyze the relationship between cancer status and survey scores. **Results:** A total of 231,033 patients met the criteria: 14,032 GIC and 217,001 NC. The majority of patients were white (69%%), female (59.2%), and non-Hispanic (97.2%). Colorectal cancer was most represented (81%), followed by anus (12%), stomach (3%), and pancreatic (1%) cancers. The median time between cancer diagnosis and survey response was 3.4 years (IQR 1.4-6.5 years). With respect to SDOH, GIC patients perceived more social support (β 0.07 [95% CI: 0.04, 0.10]), more neighborhood positivity (β 0.07 [95% CI: 0.05, 0.10]) and less healthcare discrimination (β -0.03 [95% CI: -0.06, -0.01]) (Table, NC is reference). GIC surveyed < 1 year after their diagnosis scored higher on social support than those surveyed > 1 year (β 0.13 [95% CI: 0.05, 0.21]). For PH, GIC were 15% more likely to report better mental health and 33% more likely to report better quality of life (QOL). Those diagnosed < 1 year before their survey had no difference in perceived physical and general health compared to NC, although their QOL (OR 1.3 [95% CI: 1.2-1.4]) and mental health (OR 1.1[95% CI: 1.0-1.2]) were similar to those surveyed > 1 year after the diagnosis. Chemotherapy-treated GIC patients reported better PH in all four domains than those not treated with chemotherapy, including QOL (OR 1.3, [95% CI: 1.2-1.5]) and mental health (OR 1.2, [95% CI: 1.1-1.4]). **Conclusions:** Findings of this survey study reveal that a GI cancer diagnosis has a favorable impact on SDOH and PH - GIC feel more supported, less discriminated, and perceive better health compared to NC, and time of cancer diagnosis and chemotherapy impact outcomes. A concerted societal effort is required to improve health related social conditions for individuals with and without cancer. Research Sponsor: None.

Impact of cancer status on perceived health and social determinants of health.

Social Determinants of Health*	Beta (95% CI)	P Value
Social Support	0.074 (0.04, 0.10)	< 0.001
Neighborhood Positive	0.074 (0.05, 0.10)	< 0.001
Healthcare Discrimination	-0.034 (-0.06, -0.01)	< 0.001
UCLA Loneliness	-0.013 (-0.03, 0.01)	0.196
Perceived Health Domain*	**Odds Ratio (95% CI)**	**P Value**
Quality of Life	1.32 (1.27, 1.38)	< 0.001
Physical Health	1.27 (1.21, 1.32)	< 0.001
Mental Health	1.15 (1.10, 1.20)	< 0.001
General Health	1.26 (1.21, 1.32)	< 0.001

*Comparison is gastrointestinal cancer vs. non-cancer, with non-cancer as the reference group.

803 Poster Session

Adequate racial and ethnic representation in cancer clinical trial enrollment: Real-world evidence from an NCI-designated cancer center in a minority-majority county. First Author: Farshid Dayyani, University of California, Irvine Chao Family Comprehensive Cancer Center, Orange, CA

Background: Despite strides to increase diversity in clinical trial enrollment, significant racial and ethnic disparities persist. This limits the generalizability of clinical trial data and consequently the application of novel therapeutics for diverse populations. We hypothesized that enrollment of underrepresented minorities (URM) is feasible, with improved access to clinical care and a tailored clinical trial portfolio addressing the unmet needs in URM. **Methods:** A retrospective cohort study was conducted at the University of California Irvine Chao Comprehensive Cancer Center (CFCCC), an NCI-Designated cancer center located in Orange County (OC), CA (a minority-majority county). Clinical trial enrollment data from 2015-2023 through the CFCCC clinical research database was included. Collected data on patient demographics, tumor types, and trial enrollment was compared to data provided by the NIH SEER reports in both OC and United States. **Results:** Between 2015-2023, 2317 subjects were enrolled in clinical trials at CFCC. Demographics were as follows: White, Asian, Black, American Indian/Alaska Native, mixed/unknown race (66.4%/20.1%/2.4%/0.7%/0.7%/9.3%). Non-Hispanic vs Hispanic ethnicity (77.8%/20.5%). Female vs Male sex (47.6%/52.3%). Age <70 vs >70 years (73.3%/26.7%). Study Phase: Phase I/II (35.6%), Phase II (25.4%), Phase II/III (2.8%), Phase III (26.6%), and Phase IV (0.3%). Study sponsor: Industry (61.1%), Institutional (23.6%), National (14.8%), and Externally Peer Reviewed (0.3%). Most notably, the Asian enrollment (20.1%) and Hispanic enrollment (20.6%) exceeded demographic representation of Asians (17%, RR 1.18) and Hispanics (19%, RR 1.07) at CFCCC. Lastly, Asian patients enrolled at significantly higher rates in lung and liver cancer trials, while Hispanic patients and Black patients enrolled at higher rates in breast cancer and prostate cancer trials respectively. **Conclusions:** Our results demonstrate that robust URM clinical trial enrollment, particularly among Asian and Hispanic populations, is feasible at an NCI-designated cancer center despite minority-specific barriers. These findings suggest that minority populations can be optimally enrolled in clinical trials when these trials are accessible and tailored to the population at hand. Further research is necessary to investigate factors that influence clinical trial participation in the pursuit of equitable cancer care. Research Sponsor: None.

804 Poster Session

Comparative analysis of racial disparities in social determinants of health among patients with GI cancer in the All of Us dataset. First Author: Manar Z Al Rubaye, University of Wisconsin School of Medicine and Public Health, Madison, WI

Background: Social Determinants of Health (SDoH) are key factors in understanding disparities in cancer treatment and outcomes. Examining SDoH across racial and ethnic groups helps uncover the mechanisms behind disparities and identify solutions. This study compared SDoH factors between White (non-Hispanic) and Non-White (Asian, African American, Hispanic) patients diagnosed with Gastrointestinal (GI) cancers using a national dataset. **Methods:** We analyzed data from the All of Us Research Program. Adult patients with GI cancers, identified via ICD-10 and SNOMED codes, who had completed the SoDH surveys were included. Comparative analyses were performed between White and Non-White participants across SDoH constructs, including social/community construct, neighborhood environment, economic stability, and healthcare access using Chi-squared, T-test, and Mann-Whitney U tests as appropriate. **Results:** Of the 6,620 GI cancer patients identified, 1,831 completed the SDoH survey, of which 13.9% were Non-White. The most common cancers were colon and liver, with pancreatic, stomach, rectal, esophageal, and biliary tract cancers also included. White participants were older than Non-White participants (71.0 vs. 64.2 years, SD = 10.7 and 11.4) and reported higher levels of social cohesion (Mean (SD) = 3.92 (0.70) vs. 3.64 (0.76), $p < 0.001$) and social support (Mean (SD) = 3.92 (1.00) vs. 3.68 (1.11), $p < 0.001$). Non-White participants experienced greater perceived discrimination (Mean (SD) = 1.73 (0.80) vs. 1.54 (0.57), p = 0.016) and reported more neighborhood physical disorder ($p < 0.001$). Food insecurity (23.6% vs. 5.8%, $p < 0.001$) and housing quality issues (41.8% vs. 21.8%, $p < 0.001$) were more common among Non-White participants, who also faced more delayed medical care (45.9% vs. 32.5%, $p < 0.001$). **Conclusions:** This study highlights racial disparities in SDoH among GI cancer patients. Non-White participants experienced greater economic instability and neighborhood disorder compared to White participants. The All of Us dataset offers valuable data for examining SDoH in this population, though it may not fully represent the broader US population. As the dataset expands, it will provide more opportunities for comparative analyses among cancer patients across racial and demographic groups. Research Sponsor: None.

Construct	White Non-Hispanic (n = 1489)	Non-White (n = 255)	P-Value
Gender: Female, n (%)	827 (55.5)	167 (65.5)	p = 0.010
Social Disorder; mean (SD)	2.10 (0.22)	2.17 (0.28)	p = 0.002
Perceived Stress; mean (SD)	11.36 (7.12)	12.85 (7.99)	p = 0.003
Global Physical Health T-score (SD)	37.16 (1.89)	36.26 (2.08)	p < 0.001
Global Mental Health T-score (SD)	38.30 (1.76)	38.22 (1.86)	p = 0.520
Food Insecurity; n (%)	170 (5.8)	117 (23.6)	p < 0.001
Poor Housing Quality; n (%)	326 (21.8)	127 (41.8)	p < 0.001
Delayed Medical Care; n (%)	484 (32.5)	117 (45.9)	p < 0.001
Cannot afford specialist; n (%)	47 (4.2)	20 (11.9)	p < 0.001

805 Poster Session

Disparities in clinical trial enrollment among LGBTQ+ individuals and its impact on cancer treatment and management: A systematic review. First Author: Akshit Chitkara, Thomas Jefferson University Hospital, Philadelphia, PA

Background: There is a significant gap in oncologic research, which arises from the exclusion of LGBTQ+ individuals from cancer clinical trials, compromising the development of personalized cancer therapies and limiting the generalizability of findings. Despite known health disparities in this population, including higher cancer risk factors and unique psychosocial stressors, data on clinical trial participation remains sparse. This study aims to elucidate the extent of underrepresentation of LGBTQ+ individuals in cancer clinical trials and examine its impact on treatment efficacy, safety, and overall management. **Methods:** We conducted a systematic review of literature and clinical trial databases, including PubMed, ClinicalTrials.gov, and other registries, focusing on cancer trials from 2010 to 2024. We extracted data on LGBTQ+ enrollment, analyzed demographic inclusivity criteria, and identified barriers to participation. Subgroup analyses evaluated the correlation between LGBTQ+ representation and treatment outcomes, considering variables such as trial design, recruitment strategies, and data collection on sexual orientation and gender identity. The review also examined policy changes and inclusivity initiatives aimed at improving trial access for this community. **Results:** The review identified a significant underrepresentation of LGBTQ+ individuals in cancer trials, with sexual orientation and gender identity data reported in only 6.2% of trials. Structural and sociocultural barriers, including heteronormative trial designs, lack of inclusive recruitment strategies, and pervasive discrimination, were frequently cited as impediments. Trials that included LGBTQ+ participants rarely conducted stratified analyses to assess differential treatment responses, leading to a paucity of data on drug safety, efficacy, and toxicity profiles specific to this population. The absence of these data may contribute to suboptimal therapeutic decision-making and potential adverse outcomes in clinical practice. **Conclusions:** The substantial underrepresentation of LGBTQ+ individuals in cancer clinical trials impairs the ability to develop inclusive, evidence-based oncology care. Addressing this gap requires systematic changes, including the integration of sexual and gender minority data collection, implementation of LGBTQ+-specific recruitment frameworks, and stratified analyses of treatment outcomes. Enhancing trial inclusivity will not only improve the external validity of oncologic research but also support the development of tailored therapeutic approaches that address the unique needs of LGBTQ+ cancer patients, ultimately advancing health equity in cancer care. Research Sponsor: None.

806 Poster Session

Racial disparities in cancer stage at diagnosis among older adults with gastrointestinal cancers in the southeastern United States. First Author: Apoorva Doshi, University of Alabama at Birmingham, Birmingham, AL

Background: Prior studies have documented racial disparities in survival among older adults with various gastrointestinal (GI) cancers, but mechanisms remain poorly understood. We hypothesized that delayed diagnosis leading to advanced cancer stage at presentation could be a contributing factor to these disparities. **Methods:** We included adults ≥ 60 years with newly diagnosed GI cancers presenting for initial consultation at University of Alabama at Birmingham from October 2017 to April 2024. Using self-reported race and ethnicity, we stratified patients into Non-Hispanic Whites, Non-Hispanic Black, Hispanics and other category. For this analysis, we limited our cohort to the first two categories. Cancer stage was abstracted from the Electronic Health Record and followed American Joint Commission on Staging (AJCC) 7th manual. We performed two-sided Pearson's chi square test to test the association between stage at diagnosis and race. We performed this analysis for the overall cohort as well as for specific GI cancer types. The level of significance was set at $p < 0.05$. **Results:** A total of 1714 patients were included; Out of which 74% were non non-Hispanic white and 26% were African American. The median age at presentation was 68 years (interquartile range: 64 - 74), and 42% were females. The most common cancer type was colorectal (28%) followed by pancreatic (26%), hepatobiliary (22%), other GI cancers (14%) and gastroesophageal (10%). Overall, around 45% patients had Stage IV at diagnosis, followed by Stage III (26%), Stage II (17%) and Stage I (11%). We found that as compared to non-Hispanic Whites, non-Hispanic Blacks were more likely to have advanced stage disease, stage III (29% vs 25%) and stage IV (47% vs 45%), this association was statistically significant (Table). Furthermore, this association was most prominent for pancreatic cancers, where non-Hispanic Blacks were more likely to have advanced stage disease, stage III (21% vs 16%), and stage IV (50% vs 47%; $p < 0.05$). **Conclusions:** We report that among older adults with GI cancers, African Americans were more likely to have advanced stage at diagnosis compared to non-Hispanic whites, specifically in patients with pancreatic cancer. This finding may contribute to some of the racial disparities observed in survival outcomes in this population. Research Sponsor: None.

Contingency table showing frequencies (%) of cancer stages for non-Hispanic Whites and African Americans.

GI Cancer Stage	Non-Hispanic White	African American	p value
Stage I	138 (11%)	51 (11%)	<0.05
Stage II	236 (19%)	57 (13%)	
Stage III	321 (25%)	129 (29%)	
Stage IV	572 (45%)	210 (47%)	

807 Poster Session

Provider perspectives on non-medical barriers to GI cancer care at a regional cancer center. First Author: Trisha Lal, University Hospitals Cleveland Medical Center, Department of Surgery, Division of Surgical Oncology, Cleveland, OH

Background: Disparities in care for patients with gastrointestinal (GI) cancers are well-documented, but solutions remain elusive. Understanding the barriers faced by healthcare providers is essential to improving care equity. This study explored the perceptions of healthcare providers at a regional academic health system regarding barriers to care for GI cancer patients. **Methods:** A cross-sectional survey was conducted among healthcare providers, including physicians, advanced care providers (APPs), nurses, dietitians, social workers, care coordinators, and physical/occupational therapists who directly care for patients with GI malignancies. Survey items assessed specific barriers encountered in clinical practice and suggestions for improving care. Descriptive statistical analysis evaluated the quantitative data from the closed-ended questions across professional roles. A thematic analysis was performed on open-response items and assessed to further identify key barriers to care based on respondents' clinical roles and experiences. **Results:** Seventy-nine healthcare providers responded to the survey. Initial treatment planning was consistently identified as the phase with the most significant barriers across all disciplines. Health literacy was cited as the most critical barrier to diagnosis, while insurance coverage, financial constraints, geographic location, lack of transportation, and social support were the primary obstacles to treatment. The importance of these barriers varied by profession: physicians noted geographic and transportation limitations, while nurses and social workers emphasized gaps in health literacy, insurance, and social support. Systemic issues such as limited access to specialized care and healthcare personnel shortages were also noted. Notably, 55% of physicians spent 10-30% of their time addressing non-medical barriers to treatment, while 38% of nurses spent 30-50%. Care coordination across disciplines was commonly cited as a challenge, with many calling for improved communication and establishing multidisciplinary clinics. Food insecurity, housing stability, and language/cultural barriers were less prominent concerns at this institution. **Conclusions:** Oncology providers at a large regional cancer center spend significant time addressing non-medical barriers to treatment for GI cancer patients, particularly during initial treatment planning. The most common challenges reported were insurance, transportation, health literacy, and social support. We will use these findings to guide in-depth provider and patient interviews and develop a team-based, equity-focused intervention. These efforts aim to improve care delivery at our institution and serve as a model enhancing GI cancer care in broader healthcare settings. Research Sponsor: None.

808 Poster Session

Association between skeletal muscle morphometrics and overall survival among older adults with gastrointestinal cancers. First Author: Apoorva Doshi, University of Alabama at Birmingham, Birmingham, AL

Background: Older adults with gastrointestinal (GI) cancers face a higher risk of treatment toxicities and adverse survival outcomes. However, the variability in these outcomes remains poorly understood. Age-related changes, including the progressive loss of skeletal muscle and gain in adipose tissue, contribute to this heterogeneity. Recent studies highlight an association between skeletal muscle gauge (SMG) and overall survival (OS) in older cancer patients, but its relevance among those with GI cancers is unclear. **Methods:** We included adults aged ≥ 60 years with newly diagnosed GI cancers, enrolled in the University of Alabama at Birmingham (UAB) Cancer and Aging Resilience Evaluation (CARE) registry between October 2017 and April 2024. Skeletal muscle index (SMI; skeletal muscle area in m^2/height in cm) and skeletal muscle radiodensity (SMD; Hounsfield units) were derived from single-slice computed tomography (CT) staging scans at the third lumbar vertebra (L3) using validated methods. SMG was calculated as the product of SMI and SMD, reported in arbitrary units (AU). Participants were followed until death or the study cutoff date of July 10, 2024. Kaplan-Meier survival analysis was used to estimate OS by SMG status (low: < median; high: ≥ median). Cox proportional hazards models were employed to assess the relationship between SMG and OS, adjusting for age, sex, race/ethnicity, and cancer stage. **Results:** A total of 568 patients were included in the study, with a median age at diagnosis of 68 years (interquartile range, IQR: 64-74), 55% male, 73% non-Hispanic White, and 43% were diagnosed with Stage IV disease. Colorectal (33%) and pancreatic (27%) cancers were the most common diagnoses. The median SMG was 1,529 AU (IQR: 1,151–1,967). Over a median follow-up of 60 months, 367 patients (64%) died. Median OS was 19 months for those with low SMG and 26 months for those with high SMG (log-rank test, $p < .05$). After adjustment for confounders, SMG remained independently associated with OS (adjusted Hazard Ratio: 0.678 per IQR increase; 95% CI: 0.57– 0.807; $p < .001$), indicating that each IQR increase in SMG is associated with a decrease in all-cause mortality risk by 32.2%. **Conclusions:** Skeletal muscle gauge, an integrated measure of SMI and SMD, is significantly associated with improved survival among older adults with GI cancers. Further research is needed to determine whether SMG has a causal impact on survival in this population. Research Sponsor: None.

809 Poster Session

Rate of gene fusion/rearrangement identification in GI cancers as sequencing evolved from limited panels to whole exome sequencing. First Author: Hilary Chan, Kaiser Permanente Vallejo Medical Center, Vallejo, CA

Background: Gene fusion/rearrangements occur at a frequency of 0-15% in GI system cancers. Detection and treatment of fusion genes such as NTRK, FGFR2, NRG1 and RET fusions are included in NCCN guidelines and are commonly detected using next generation sequencing (NGS). NGS panels have continued to expand over time, which should result in increases in the identification of gene fusions. The aims of this study were to describe the percent of patients with advanced GI tumors undergoing NGS with identified gene rearrangements across time as panels expanded. **Methods:** The study included 7237 patients in the Kaiser Permanente Northern California Integrated Health System receiving care for advanced GI cancers from 2017 to Aug 2024, whose tumors samples were evaluated using NGS (Stomach=926, Colorectal=2792, Gallbladder=569, Esophagus=761, liver=218, Pancreas=1971). **Results:** A total of 231 patients (3.2%) were identified with a gene rearrangement, including 4.6% in stomach(N=43), 2.8% in colorectal(N=78), 8.9% in gallbladder(N=24), 5.3% in esophagus(N=40), and 1.9% in pancreatic (N=38*). Rearrangements included NTRK1/2/3, FGFR2/3, NRG, ALK, ROS, CLDN18-ARHGAP, RSPO2/3, DNAJB1-PRKACA, BRAF, MET and RET*. The numbers per year were (see table). **Conclusions:** The use of more comprehensive NGS panels such as WES did not appear to increase the detection rate for fusion/rearrangements. In resource limited settings, limited panels may suffice. The number of GI cancer patients getting NGS testing has increased over time. Research Sponsor: None.

Gene rearrangements/fusions identified per year.

YEAR	2018	2019	2020	2021	2022	2023	2024 (Jan – Aug)
Gene Rearrangement identified	29	37	43	55	54	62	38
TOTAL GI Cancer Samples Sequenced	610	858	900	1047	1249	1181	785
Percentage	4.75%	4.31%	4.78%	5.25%	4.32%	5.25%	4.84%

810 **Poster Session**

Management trends and outcomes of appendiceal cancers: A National Cancer Database study. First Author: Ali Esparham, Mashhad University of Medical Science, Mashhad, Iran

Background: Appendiceal tumors encompass a variety of histological types that exhibit different behaviors. The current study aimed to investigate and compare overall survival, predictive factors for overall survival, and management of histological variations of appendiceal tumors. **Methods:** The National Cancer Data Bank (NCDB) database (2004–2020) was evaluated to include patients with appendiceal tumors histologies including goblet cell adenocarcinoma (GCA), neuroendocrine neoplasm (NEN), non-mucinous adenocarcinoma (NMA), and mucinous adenocarcinoma (MA). Univariate and multivariable analyses were conducted. **Results:** The GCA, MA, NEN, and NMA groups consist of 6,111, 16,471, 19,199, and 11,065 patients, respectively. The NMA and NEN groups had significantly lowest and higher overall survival compared to other types ($p<0.001$ for both) (GCC: 143.27 (140.23-146.30), MA: 111.91 (109.996-113.832), NEN: 170.88 (168.56-173.20), and NMA: 101.399 (99.132-103.666) months). The independent predicting factors of worse overall survival were age (HR: 1.03 (1.03-1.04)), black race (HR: 1.24 (1.08-1.43), ref: white race), fourth median income (HR: 0.84 (0.74-0.96), ref: first median), Charleson index comorbidity score (HR: 1.3 (1.22-1.38)), lymphovascular invasion (HR: 1.28 (1.15-1.42)), pathologic stage 2 (HR: 1.32 (1.12-1.57)), stage 3 (HR: 1.99 (1.66-2.40)), stage 4 (HR: 4.20 (3.47-5.07)), ref: stage 1, intraoperative systemic therapy (HR: 0.52 (0.39-0.70)), positive surgical margin (1.55 (1.39-1.73)), number of positive lymph nodes (HR:1.06 (1.05-1.07)), moderately differentiated tumors (HR:1.43 (1.25-1.63)), poorly differentiated (HR: 1.94 (1.69-2.24)), undifferentiated (HR: 1.48 (1.14-1.91)), ref: well differentiated, and NMA type (HR:1.30 (1.10-1.55)). Furthermore, in terms of type of surgery, significantly higher patients of GCA (54.6%) and NMA (55.2%) groups underwent subtotal colectomy compared to other groups. Regarding systemic therapy, significantly higher patients of MA group received intraoperative, and both neoadjuvant and adjuvant therapy compared to other groups. Also, NEN group had significantly higher patients who underwent surgery with laparoscopic approach compared to others. **Conclusions:** This study reveals that different types of appendiceal tumors have distinct characteristics in regard to overall survival outcomes and treatment approaches. NENs showed the highest survival rates, while NMAs had the lowest. These results underscore the need for personalized treatment strategies for each tumor type to improve patient outcomes. Research Sponsor: None.

811 **Poster Session**

Treatment patterns in patients with advanced gastrointestinal stromal tumor in Japan: An administrative claims database study. First Author: Yoshito Komatsu, Division of Cancer Center, Hokkaido University Hospital, Sapporo, Japan

Background: Gastrointestinal stromal tumors (GIST) are the most common mesenchymal tumors of the gastrointestinal tract. Tyrosine kinase inhibitors, including imatinib, sunitinib, and regorafenib, and the HSP90 inhibitor, pimitespib have been approved for advanced GIST in Japan, but few studies have examined the nationwide trend in treatment patterns. This study aimed to evaluate the real-world treatment patterns for advanced GIST in Japan. **Methods:** This retrospective cohort study used the Japanese hospital-based administrative claims database provided by Medical Data Vision Co., Ltd. (Tokyo, Japan). Patients diagnosed with advanced GIST according to ICD-10 codes who received any approved standard drug in Japan, such as imatinib, sunitinib, regorafenib, or pimitespib (Approved in June 2022) between July 2013 and March 2023 were included. The primary endpoint was the proportion of patients with GIST who were treated with each approved standard drug. Secondary endpoints included treatment patterns, time to treatment failure (TTF), overall survival, adverse events, and the proportion of patients who underwent surgical intervention for GIST while taking standard drugs. **Results:** This study included 2702 patients with advanced GIST. The median age was 70.0 years, 17% of the patients were ≥80 years, and 59% were male. The most common primary tumor sites were the stomach (37%), small intestine (25%) and large intestine (12%). The proportions of patients with GIST treated with imatinib, sunitinib, regorafenib, and pimitespib were 94% (2536 patients), 26% (702 patients), 14% (376 patients) and 1% (37 patients), respectively. The most common first-line regimen was imatinib (2485/2702 patients; 92%), second-line regimen was sunitinib (560/672 patients; 83%), third-line regimen was regorafenib (200/317 patients; 63%), and fourth-line regimen was imatinib (55/119 patients; 46%). The median TTF was 30.4 months (95% CI 27.4–32.9) with imatinib, 6.4 months (95% CI 5.6–7.7) with sunitinib, 5.2 months (95% CI 4.4–6.2) with regorafenib and 2.5 months (95% CI 1.7–3.1) with pimitespib. The proportions of patients who underwent surgical intervention for GIST was 15% (403/2702 patients), and curative surgery was 10% (273/2702 patients). **Conclusions:** This is the first real-world study to provide insights into the patient characteristics and current treatment patterns for advanced GIST in Japan. The treatment patterns are consistent with the Japanese Clinical Practice Guidelines. Research Sponsor: Taiho Pharmaceutical.

812 **Poster Session**

Current status and treatment outcomes of intraductal papillary mucinous carcinoma in Korea: A multi-center cohort study. First Author: Hee-Eon Lim, Seoul National University Bundang Hospital, Seongnam, South Korea

Background: The spectrum of intraductal papillary mucinous neoplasm (IPMN) varies from benign cysts to intraductal papillary mucinous carcinoma (IPMC). There is a lack of large-scaled data on IPMC and high-grade dysplasia. We aimed to study the real-world status and treatment outcomes of IPMC in South Korea. **Methods:** This is a retrospective, multicenter study conducted in South Korea. We reviewed the electronic medical records of patients who were histologically diagnosed with IPMC or high-grade (HG) dysplasia between 2011 and 2020, and analyzed the baseline characteristics, treatment patterns, and survival. **Results:** A total of 388 patients were eligible from six institutes (145 female [37%]; median age 67.8; 259 IPMC [67%] vs 129 HG dysplasia [33%]). The median overall survival for IPMC patients was 75 months, compared to 133 months for HG dysplasia. In IPMC group, 217 (84%) were underwent surgical resection and 42 (16%) did not underwent surgery due to general condition or advanced stage whose median overall survival was 83 months and 49 months respectively. In resected IPMC, 104 (48%) received adjuvant chemotherapy (gemcitabine single 48%, 5FU+leucovorin 33%, others 19%) and 66 (30%) experienced recurrence and median recurrence-free survival (RFS) was 20 months. Among IPMC group, 11 (4.2%) was in advanced stage and eight of those received palliative chemotherapy (FOLFIRINOX 62.5%; gemcitabine single 25%; gemcitabine with erlotinib 12.5%). In HG dysplasia, 108 (84%) patients underwent surgery, and 21 (16%) patients did not undergo surgery whose OS were 171 and 133 months, retrospectively. In patients with resected HG dysplasia, 29 (27%) experienced recurrence and median RFS was 21 months. **Conclusions:** IPMC generally showed longer survival and a lower recurrence rate than those of historical data in PDAC, with HG dysplasia demonstrating even better outcomes. In real world, similar chemotherapy regimens used for PDAC were applied to IPMN patients. Research Sponsor: None.

Treatment status of intraductal papillary mucinous carcinoma and high-grade dysplasia.

	IPMC(259)	IPMN-HG(129)	Total(388)
Median overall survival (month)	75	133	85
Treatment OP	217(84%)	108(84%)	325(84%)
Adjuvant chemotherapy	104(48%)	19(18%)	123(38%)
Recurrence rate	66(30%)	29(27%)	95(29%)
Median recurrence-free survival (month)	20	19	21
Palliative chemotherapy	8(3%)	NA	
Progression rate	5(63%)	NA	
Progression-free survival (month)	13	NA	

813 **Poster Session**

Patient-reported outcomes in GI cancer: Leading change to meet the needs of the gastrointestinal cancer community. First Author: Martha Raymond, GI Cancers Alliance, Dennisport, MA

Background: According to the World Health Organization International Agency for Research on Cancer, globally gastrointestinal (GI) cancers account for 1 in 4 cancer cases and 1 in 3 cancer deaths, with an estimated 4.8 million new cases and 3.4 million GI cancer deaths reported annually. Understanding the global burden of GI cancers and the importance of understanding the real-world impact of a GI cancer diagnosis, the GI Cancers Alliance (GICA) conducted a 12-month patient-reported outcomes (PRO) research initiative that included a workshop at the 2023 American Society of Clinical Oncology (ASCO) annual meeting. In addition, our PRO research was published in the Journal of Clinical Oncology (J Clin Oncol 42, 2024 suppl 3; abstr 731) with a poster presented at the 2024 ASCO Gastrointestinal Cancers Symposium. **Methods:** Based on our 2022-2023 PRO research of 1,122 patients highlighting the unmet needs and critical gaps in services of the GI cancers community, we developed a series of follow-up initiatives. Nutrition education 17%, biomarker/precision medicine education 19%, patient-centered care education 11% and caregiver support 16% were four of the highest levels of unmet need based on our global research. Subsequently, in 2023 we created the 'GI Cancers Patient & Caregiver Advisory Board' representing all GI cancers. The board hosts educational forums supporting patients and caregivers throughout the care continuum. Our pilot nutrition program launched at the 2024 ASCO annual meeting, consisting of in-person and online video components, and an extensive nutrition guide with recipes tailored for GI cancer patients. In 2024, we partnered with our gastric cancer advocates to develop 'Test Your Biomarkers' enhancing education and awareness of biomarker testing and precision medicine. **Results:** Implementing direct patient feedback from our published research, we are creating actionable solutions to unmet needs and gaps in services. Our 'Patient & Caregiver Advisory Board' continues to amplify the patient and caregiver voice participating in global engagement opportunities. GICA provided 10,000 nutrition guides to community oncology centers, and our online messaging reached 250,000 individuals across digital platforms and partner websites in the first three months from launch. In the first six months of the 'Test Your Biomarkers' campaign our messaging reached 1.7 million individuals across all digital platforms, campaign, and partner websites. **Conclusions:** Our PRO research provides the roadmap for new initiatives and programming to meet the needs of our GI cancer community. The GI Cancers Alliance will proceed to collaborate with our 100+ partner organizations to activate change by targeting the needs of our patient community while eliminating critical gaps in services. By doing so, we will continue to provide greater patient impact through the amplification of the patient voice. Research Sponsor: None.

814 **Poster Session**

Impact of next-generation sequencing in advanced gastrointestinal cancer. First Author: Aarushi Grover, Comprehensive Hematology-Oncology, St Petersburg, FL

Background: Next-generation sequencing (NGS) is a cutting-edge technology that allows for rapid DNA and RNA sequencing, identifying key mutations in cancer patients. It provides both diagnostic and predictive data for treatment decisions. Approximately 75.6% of US oncologists use NGS testing for patients with advanced solid tumors.This study retrospectively evaluates the impact of NGS testing on treatment outcomes for advanced GI cancer patients at a community-based cancer center in Florida. **Methods:** Between March 2019 and September 2023, 267 patients with stage 3 or 4 solid malignancies underwent NGS testing. Of these, 77 patients with advanced GI cancer were included in the study. Nearly half (48%) of the patients were diagnosed with advanced-stage colon cancer.NGS testing was performed using tissue samples (79.22%), blood samples (16.88%), or both (3.90%). Demographic and clinical outcomes, including time to diagnosis, treatment initiation, and survival, were collected and analyzed. **Results:** The median age of the study population was 70, with 88.31% of patients insured through Medicare. The cohort was composed of 61.04% males. Targetable mutations were identified in 59.74% of patients, leading to FDA-approved targeted therapies for 44.16%, and clinical trial eligibility for 79.22%. 52 patients were alive at the time of this analysis; those with biomarkers identified through NGS testing had a median treatment duration of 11 months regardless of whether testing impacted choice of therapy; whereas, patients without identified biomarkers had a median duration of 7.5 months. NGS testing impacted first-line therapy in only 6.49% of all patients; however, 80% of those patients were alive at the time of this analysis with a median duration of response of 5 months. **Conclusions:** NGS testing has become a cornerstone of oncologic care for advanced solid tumors; however, in this cohort of advanced GI malignancies, NGS results influenced first-line treatment decisions in only 6.49% of cases. Although the study included a small sample size, it reflects a real-world community setting where access to therapies and insurance coverage varies. For example, though patients may have targetable novel therapies available for their disease, the indications for such therapies may not be in the first line setting or covered by insurance. Advancements in NGS targeted therapies has greatly impacted the care of other solid malignancies such as non-small cell lung cancer; however, this study highlights that in the field of GI oncology, there is a large clinically unmet need of identifying newer biomarkers that are clinically relevant and impactful in earlier lines of oncologic care. Research Sponsor: None.

815 **Poster Session**

The efficacy of anamorelin in gastrointestinal cancer with cancer cachexia: Single center experience. First Author: Yuka Suzuki, Department of Gastrointestinal Medical Oncology, National Cancer Center Hospital, Tokyo, Japan

Background: Cancer cachexia is a catabolic syndrome that causes weight loss and loss of appetite in cancer patients. Anamorelin (ANAM) is an oral drug approved in Japan to treat cancer cachexia by selectively activating ghrelin receptors. There is limited real-world data available on the efficacy of ANAM for patients with gastrointestinal cancer. We assessed its efficacy and identified associated clinical features. **Methods:** We retrospectively collected data from medical records and questionnaires on gastric and colorectal cancer patients who received ANAM between April 2021 and March 2023. We evaluated the change in body weight, appetite, and oral intake from the time of initiation of ANAM. We judged the responder who gained more than 0.5 kg in body weight, improved appetite, and increased oral intake, respectively. The association between clinical features and the responders was also assessed using multivariate logistic regression analysis. **Results:** 72 patients were included in this study. Characteristics of the patients: median age: 68 years old (range 38-85), male/female: 35/37, gastric/colorectal cancer: 39/33, PS 0-1/2 or more: 63/9, median body mass index (BMI): 18.0 kg/m^2 (range 12.3-27.9), number of prior chemotherapy regimens 1/2/3 or more: 17/22/33, ascites none/mild/moderate/severe: 42/19/7/4, median serum albumin levels (Alb): 3.2 g/dl (range 2.0-4.2). The median treatment duration of ANAM was 47.5 days (3-450 days). 57 patients (79.2%) continued ANAM for more than 3 weeks, and 9 patients for more than 12 weeks. The main reasons for discontinuation were progressive disease (39.7%), ineffective (12.7%), effective (11.1%) and adverse events of ANAM (11.1%). 33 patients (45.8%) gained weight, 41 patients (56.9%) improved appetite and 34 patients (47.2%) increased oral intake. There was a correlation found between BMI $<$ 18.0 kg/m^2 and the responders (OR 3.43, p = 0.041). **Conclusions:** In this study, 68.0% of patients who received ANAM for showed response, that is, gained body weight, or improved appetite, or increased oral intake. And these observations suggested that BMI was associated with effects of ANAM. Further studies involving more patients needed. Research Sponsor: None.

816 **Poster Session**

Utilizing an automated, high-throughput platform to generate comprehensive clinical & molecular datasets to enable a real-world evidence approach for improving care of patients with gastrointestinal malignancies. First Author: Mahmoud M.G. Yousef, The University of Texas MD Anderson Cancer Center, Houston, TX

Background: More than 44,000 patients are seen at MD Anderson Cancer Center annually. However, using the diverse, mostly unstructured, data from these patient encounters has been challenging and required manual chart reviews. In 2018, MD Anderson and Palantir Technologies (Denver, CO) began developing a unified, cloud-based, graphical user interface clinical informatics platform to extract, structure, and integrate data from the different data sources that comprise the electronic health record (EHR). Here, we describe our experience using this novel platform to apply a real-world evidence (RWE) approach to study patients with gastrointestinal (GI) malignancies. **Methods:** Institutional Review Board approval for retrospective chart review of patients with GI malignancies was previously obtained. The Foundry platform was used to incorporate more than 150 datasets, including structured data elements like lab values, unstructured data as the full text of clinical notes, and natural language processing (NLP) derived datasets. The datasets include unique patient identifiers to allow the merging of demographic, clinical, molecular, and outcomes information. The platform allows processing of the note text through NLP to extract non-discrete data elements into a discrete form. In addition, it continuously updates new data on daily bases, allowing the inclusion of new patients' information in an automated fashion. **Results:** From 2,013,048 patients with date of diagnosis ranging from 1944 to 2024, we have created datasets for colorectal adenocarcinoma (CRC, $>$50,000 patients, $>$8,000 with molecular data), pancreatic adenocarcinoma (PDAC, $>$13,000 patients), and appendiceal adenocarcinoma (AA, $>$3,000 patients). More than 50 variables have been integrated, including demographic information, stage, grade, overall survival, and molecular information. Focused manual validation of the automated extraction across the cohorts consistently demonstrated an accuracy of over 94%. Work is underway to extract additional features including DFS and PFS, sites of metastasis, and to build out additional cohorts for biliary tract, upper GI, and neuroendocrine tumors. Initial discovery efforts have already led to multiple publications including discovery of molecular causes of racial and ethnic disparities in CRC, survival impact of *KRAS* and co-mutations in PDAC, and prognostic utility of serum tumor markers in AA. **Conclusions:** Utilizing an automated, highly dynamic platform allowed integration of comprehensive datasets for multiparameter oncology data in patients with GI malignancies. This resulted in dramatic acceleration of cohort identification, outcomes analysis, and enabled utilizing a data-driven approach to guide decision making in an effort to enhance and optimize outcomes. Research Sponsor: national cancer institute; K22 CA234406; the Cancer Prevention & Research Institute of Texas; RR180035; the Cancer Prevention & Research Institute of Texas; RP240392; Conquer Cancer, the ASCO Foundation; CDA-7604125121.

817 **Poster Session**

The impact of prediabetes on cerebrovascular risk among patients with gastrointestinal malignancies. First Author: Kanishka Uttam Chandani, New York Medical College/Landmark Medical Center, Woonsocket, RI

Background: Cancer screening ingastrointestinal (GI) malignancies has translated into an improvement in overall survival due to detection at an earlier stage. The majority of survivors (up to 67%) are over the age of 65. The impact of metabolic profiles on cerebrovascular events amongst this population of cancer survivors is unclear. **Methods:** Data from the NIS database from 2016 to 2020 were reviewed for patients with seven GI malignancies. Analysis was performed based on the presence or absence of prediabetes using ICD-10 codes. Baseline characteristics, comorbidities, and cerebrovascular outcomes were studied in these 2 cohorts. **Results:** 1,532,250 hospitalizations of patients with seven GI malignancies, namely colorectal cancer (CRC), pancreatic cancer, hepatocellular carcinoma (HCC), gastric cancer, small intestinal cancer, cholangiocarcinoma, and esophageal cancer, were evaluated and divided into two subgroups: prediabetics (15,610) and non-prediabetics (1,516,640). Prediabetics with GI malignancies as compared to non-prediabetics with GI malignancies were noted to be older (67.7 years vs. 65.64 years), with higher total hospitalization charges (USD 87,267 vs. USD 80,173), greater rates of elective admissions (34.92% vs. 25.89%) but a lower length of stay (6.018 days vs. 6.53 days). Prediabetic patients with GI malignancies were noted to have a higher prevalence of obesity (18.64% vs. 7.69%), dyslipidemia (52.99% vs. 26.16%), and hypertension (49.59% vs. 39.37%). Prediabetics with CRC, as compared to non-prediabetics, had higher adjusted odds ratios (aORs) of occlusion/stenosis of precerebral or cerebral arteries not leading to cerebral infarction, and cerebral infarction, with aORs of 2.03 (95% CI 1.24-3.30, p 0.004), and 1.63 (95% CI 1.01-2.63, p 0.045) respectively. Additionally, prediabetics with pancreatic cancer had a higher aOR of cerebral infarction as compared to non-prediabetics, with aOR of 2.42 (95% CI 1.70-3.44, p$<$0.001). **Conclusions:** Prediabetic patients with CRC had higher aORs of cerebral infarction and occlusion/stenosis of precerebral arteries not leading up to cerebral infarction, and prediabetics with pancreatic cancer had twice the risk of cerebral infarction as compared to non-prediabetics. This study highlights the impact of prediabetes, a modifiable risk factor, on cerebrovascular outcomes in patients with GI malignancies. Research Sponsor: None.

818 **Poster Session**

Surgical treatment in patients with advanced gastrointestinal stromal tumours (GISTs) in second-line and later stages: A single-centre cohort study. First Author: Xinhua Zhang, Department of Gastrointestinal Surgery, The First Affiliated Hospital, Sun Yat-sen University, Guangzhou, China

Background: Gastrointestinal stromal tumour (GIST) after imatinib treatment failure usually progresses rapidly with limited overall survival. Previous studies have suggested that surgery is beneficial for advanced GISTs in disease control to imatinib, but the role of surgery in > = 2 line patients remains unclear. **Methods:** Imatinib-resistant patients with advanced GIST between May 2009 and Nov. 2023 were included. Surgical treatments included tumour reduction, radiofrequency ablation and TACE. The primary endpoint was overall survival (OS) and secondary endpoints were time to second-line treatment failure (TTF) and postoperative morbidity. To balance clinicopathological characteristics, 1:1 propensity score matching (PSM) was used. **Results:** A total of 242 patients with advanced GIST treated with 2nd-line or later drugs were included, including 125 in the surgical group and 117 in the non-surgical group. The median follow-up was 30.3 months. OS was significantly better in the surgical group than in the non-surgical group (pre-PSM: 51.7m vs 35.9m , p = 0.001; post-PSM: 50.8m vs 38.8m, p = 0.043). Univariate analysis showed that surgical treatment (HR 0.487, 95% CI 0.320-0.740, p = 0.001), age ≤60 years (HR 0.640, 95% CI 0.411-0.997, p = 0.049) and low tumour burden (HR 0.491, 95% CI 0.293-0.822, p = 0.007) were associated with prolonged OS. Multivariate analysis indicated that surgical treatment (HR 0.541, 95% CI 0.348-0.843, p = 0.007) and low tumour burden (HR 0.510, 95% CI 0.300-0.865, p = 0.012) were significantly associated with OS benefit. Patients (n = 84) who had surgical treatment during second-line treatment had longer TTF compared to patients (n = 158) who had TKI only (14.2m vs 8.1m , p = 0.003). There were 29 (14.8%) postoperative complications with Clavien-Dindo classification ≥3, including 5 deaths. **Conclusions:** Surgical treatment can provide a survival benefit for patients with second-line and later advanced GIST, but further studies are needed for a more detailed indication of the surgical population. Research Sponsor: None.

819 **Poster Session**

Risk of second primary cancer among patients with a first primary appendiceal adenocarcinoma. First Author: Andreana Natalie Holowatyj, Vanderbilt University Medical Center, Nashville, TN

Background: Incidence rates of rare appendiceal cancers are increasing across the United States. Although this is suggestive of a growing population of appendiceal cancer survivors over time, the risk of second primary cancers specific to patients with a first primary appendiceal adenocarcinoma remains unknown. **Methods:** We conducted a population-based study of adults (age ≥18 years) diagnosed with a first primary appendiceal adenocarcinoma from 1992 to 2021, using data from the Surveillance, Epidemiology, and End Results (SEER) Program. We estimated cumulative incidence of second primary cancer, defined as any primary cancer diagnosed at least 3 months after appendiceal adenocarcinoma, with Fine and Gray methods to account for the competing risk of death. For comparison to the general population, we also estimated standardized incidence ratios [SIRs], or the observed number of second cancers relative to the expected number based on age-, sex-, and race-specific incidence rates. **Results:** A total of 5,827 adults diagnosed with a first primary appendiceal adenocarcinoma (mean age 58.0 years; 51.8% female; 68.5% non-Hispanic White) were identified during the study period, of whom 418 developed a second primary cancer. The most common types of second primary cancer were prostate (17.7%), colorectal (14.4%), female breast (10.5%), lung (8.4%), and melanoma (6.0%). At 10 and 20 years after appendiceal adenocarcinoma diagnosis, the cumulative incidence of second primary cancer was 7.5% (95%CI 6.8%-8.4%) and 11.7% (95%CI 10.5%-12.9%), respectively. Cumulative incidence significantly differed by sex ($p<0.01$) and age at appendiceal adenocarcinoma diagnosis (early-onset [age<50] *vs* late-onset; $p<0.01$). For example, at 10 years post-diagnosis, the cumulative incidence of second primary cancer was 6.4% (95%CI 5.4%-7.4%) for females and 8.8% (95%CI 7.6%-10.1%) for males. The overall SIR for any second primary cancer was 1.12 (95%CI, 1.02-1.23)—corresponding to 14.8 excess cancers per 10,000 person-years. SIRs for common types of second primary cancers were: 1.07 (95%CI 0.84-1.35) for prostate, 2.12 (95%CI 1.63-2.70) for colorectal, 0.91 (95%CI 0.68-1.19) for female breast, 0.81 (95%CI 0.58-1.09) for lung, and 1.20 (95%CI 0.79-1.73) for melanoma. **Conclusions:** Approximately one in every 13 adults diagnosed with a first primary appendiceal adenocarcinoma in the United States will be diagnosed with a second primary cancer within 10 years. After diagnosis of an appendiceal adenocarcinoma, patients have a 12% excess incidence in second primary cancers and double the excess incidence in second primary colorectal cancers compared to the general population. Implementation of cancer prevention and early detection strategies (e.g., colonoscopy screening, genetic testing) for patients with an appendiceal adenocarcinoma is a clinical priority. Research Sponsor: Appendix Cancer Pseudomyxoma Peritonei (ACPMP) Research Foundation; National Cancer Institute/U.S. National Institutes of Health; P50 CA236733; National Institutes of Health/Eunice Kennedy Shriver National Institute of Child Health and Human Development; K12 HD043483.

820 **Poster Session**

Evaluating the clinical benefits of anti-cancer drugs approved by Food and Drug Administration for gastrointestinal neoplasms for the past two decades. First Author: Fares Jamal, Mayo Clinic Comprehensive Cancer Center, Phoenix, AZ

Background: The number of FDA-approved anticancer drugs have increased in recent years with a more use of surrogate endpoints. However, it is unclear whether the same trend applies to GI cancers. Therefore, we aimed to evaluate the trend in FDA-approved GI cancer drugs over the past two decades and the associated clinical benefit. **Methods:** The FDA's online database was searched for drugs approved for GI cancers from 2006 until 2024. Primary and secondary endpoints along with trial characteristics were extracted. ESMO Magnitude of Clinical Benefit Scale (ESMO-MCBS) was assessed by two blinded investigators with a third resolving disagreements. We divided the study into period A (01/2006-12/2014) and period B (01/2015-08/2024). **Results:** Since 2006, FDA approved 62 regimens for GI cancer based on 70 trials, mostly (n= 45;72.6%) in period B. Most trials (n=44) were phase III with only 26 phase II: 22 single arm, and 4 randomized. All the 22 single-arm phase II trials were approved in period B with two randomized phase II trials approved in each period. Approvals based on phase III trials dropped by 33.5% (from n=15 to n=29) between the two periods. Half (n=34) of the approvals were based on overall survival (OS) benefit, while the use of objective response rate (ORR) as an endpoint has increased from 11.8% (n=2) in period A to 45.3% (n=24) in period B. Quality of life (QoL) was assessed only in 27% of the studies and found benefit only in 22.7%. The mean gain in OS was similar across both periods; 2.3 months [1.2-4.7]. Out of the 19 accelerated approvals, 17 were in period B, of which 23% (4/17) were based on secondary endpoint, 18% (3/17) were withdrawn, and 59% (10/17) received full approval. Substantial clinical benefit (ESMO score 4-5) was seen in 11.8% of the trials in period A and 19.6% in period B, with 17.4% overall from 2006-2024. **Conclusions:** Despite rising FDA approvals for GI cancer drugs since 2006, only 17.4% demonstrated substantial clinical benefit based on ESMO-MCBS. Recent approvals rely more on single-arm phase II studies and ORR, often without OS or QoL benefit. These trends underscore the need for more rigorous clinical trials that prioritize meaningful endpoint to ensure that new therapies provide real clinical value to the patient. Research Sponsor: None.

821 **Poster Session**

Frailty assessed by a 10-item index from comprehensive geriatric assessment (FI-CGA-10) versus oncologist-assessed performance status (PS) in older adults with gastrointestinal (GI) cancer. First Author: Kohei Arimizu, Geriatric Oncology Service/Department of Gastrointestinal and Medical Oncology, NHO Kyushu Cancer Center, Fukuoka, Japan

Background: A FI-CGA-10 is a recently developed measure of frailty in the geriatric oncology setting [Oncologist, 26, e1751 (2021)]. Our objective was to compare the multidimensional frailty assessment by FI-CGA-10 with PS assessed by a primary oncologist in older adults with cancer. **Methods:** This study included 790 older adults with GI cancer who underwent a CGA before cancer treatment decisions at a geriatric oncology service between September 2018 and May 2024, and whose PS was documented in the electronic medical record by a primary oncologist. Fitness and frailty level were evaluated using the FI-CGA-10, which assesses 10 domains: cognition, mood, communication, mobility, balance, nutrition, basic and instrumental activities of daily living, social support, and comorbidity. Deficits in each domain were scored as 0 (no problem), 0.5 (minor problem), and 1.0 (major problem). FI-CGA-10 scores (range 0-1) were calculated by dividing the sum of the scores for each domain by 10 and then categorized as fit (<0.2), pre-frail (0.2–0.35), and frail (>0.35). The strength of the ordinal association between the frailty category and each CGA domain score was assessed using Spearman's non-parametric correlation coefficient (rho). **Results:** The median age was 79 years; 60% were male, 62% had GI tract cancer (esophageal, gastric, colorectal), 20% had hepatobiliary cancer, 15% had pancreatic cancer, and 46% had stage 4 disease. Overall (n=790), 25% of patients were fit, 40% were pre-frail, and 35% were frail (Table). Among patients with PS 0-1 (n=633), 30% were classified as fit, 48% as pre-frail, and 22% as frail. Among PS 0-1 patients, Spearman's rho was greater than 0.5 for the association between the frailty category (fit, pre-frail and frail) and cognition, IADL, and mobility domains. In this subcohort (PS 0-1), the proportion of patients with a cognition domain score of 0.5 (mild cognitive impairment) or 1.0 (dementia) was 7%, 30%, and 78% in the fit, pre-frail, and frail groups, respectively; the proportion of those with an IADL domain score of 0.5 (OARS IADL = 12-13) or 1.0 (OARS IADL ≤ 11) was 5%, 46%, and 92%, respectively; the proportion of those with a mobility domain score of 0.5 (0.8 m/s ≤ gait speed < 1.0 m/s) or 1.0 (gait speed < 0.8 m/s) was 45%, 85%, and 98%, respectively. **Conclusions:** In this study, 70% of older adults with GI cancer whose PS was scored as 0-1 by a primary oncologist were classified as pre-frail or frail when assessed using FI-CGA-10. Among patients with PS 0-1, there were strong correlations between the frailty levels and impairment levels in the cognition, IADL, and mobility domains, indicating that these domains are major contributors to the frailty classification (fit, pre-frail, and frail). Research Sponsor: None.

All patients (n=790)	Fit	Pre-frail	Frail
PS 0 (n=351), n (%)	145 (41)	163 (46)	43 (12)
PS 1 (n=282), n (%)	45 (16)	141 (50)	96 (34)
PS 2-3 (n=157), n (%)	4 (3)	18 (11)	135 (86)

822 Poster Session

A qualitative study of the home-based time burdens faced by persons affected by GI cancer. First Author: Preethiya Sekar, University of Minnesota, Minneapolis, MN

Background: As cancer care is increasingly delivered at home, more tasks and responsibilities fall on patients and informal care partners. This is especially relevant for people with GI cancer who often have specific home-based care needs such as feeding, ostomy care, 5-fluorouracil chemotherapy, and complex medication management. Home-based time costs are undercounted in current measures of time toxicity that only include care received in formal healthcare settings. **Methods:** We conducted semi-structured interviews with patients with GI cancer and their care partners at a single tertiary cancer center in MN, USA from March-Oct 2023. Interviews explored cancer care tasks at home, associated time burdens, and how they compare to in-facility care. We analyzed interview transcripts using a grounded theory approach to identify themes. **Results:** We included 33 individuals (11 aged >60y, 21 female, 25 white) including 15 patients (8 colorectal cancer, 8 ECOG 0) and 18 care partners (15 lived with the patient, 7 working at least part-time). We identified 5 themes (Table). **Conclusions:** This foundational work characterizes time burdens of home-based cancer care for persons affected by GI cancer. In addition to providing the first input on incorporating home-based time burdens into objective measures of time toxicity, we highlight the complexity of assessing time burden which is context and role-dependent. Research Sponsor: Minnesota Cancer Clinical Trials Network; American Cancer Society; Institutional Research Grant #129819-IRG-21-049-61-IRG138; Pancreatic Cancer Action Network; Minnesota Colorectal Cancer Research Foundation.

Theme	Subtheme	Quote
Unexpected home-based care is time burdensome	Challenge with home as a site of healthcare, difficult transition from hospital to home	"What they would do at the clinic behind the scenes she was doing on my kitchen table." (Patient)
Other burdens compound time burdens	Logistic/administrative, emotional, symptom	"The time spent on administrative things and coordinating [...] are more stressful than going to the clinic." (Patient)
Time burdens evolve over the disease course and differentially impact patients and care partners	Cancer care constantly changes, discomfort with some tasks at home, less support provided if more competent, home-based care allows for connection	"They're in a season of remission and [...] we want to live as normal a life as we can. And then there are other seasons where it's the absolute opposite." (Care partner)
Factors influencing the choice of home-based care	Need to manage family duties, supportive care partners matter, value connection with nurses, decreased travel	"There is a different cost when you start doing things at home, especially when you have small children because if they're witnessing it, that's emotional." (Patient)
Home-based care is generally perceived as less time-burdensome than similar cares provided in-facility	Different opinions on whether home-based care should be included in time toxicity measure	"If you're just talking about video visits I wouldn't consider that time toxic. But if I'm going [to the clinic], I've pretty much lost the whole day." (Patient)

823 Poster Session

A systematic review and meta-analysis on evaluating the efficacy and safety of netupitant/palonosetron compared to aprepitant for the prevention of chemotherapy-induced nausea and vomiting in patients undergoing emetogenic chemotherapy. First Author: Bibi Maryam, University of Oklahoma Health Sciences Center, Oklahoma City, OK

Background: NEPA stands as the exclusive oral fixed combination antiemetic agent, comprising the highly selective NK1 receptor antagonist (RA) netupitant (300 mg) and the clinically distinct 5-HT3 RA, palonosetron (0.5 mg). Administered in a single dose, NEPA concurrently modulates two critical antiemetic pathways, offering a more convenient and straightforward alternative for sustained protection against chemotherapy-induced nausea andvomiting (CINV). Numerous studies have assessed NEPA's efficacy relative to other Neurokinin (NK) 1 receptor antagonists (RAs), notably aprepitant (APR), widely utilized for CINV. Our study critically examines the efficacy and safety of NEPA versus APR for preventing CINV in patients receiving emetogenic chemotherapy, drawing insights from recent studies. **Methods:** Studies comparing NEPA versus APR for the prevention of CINV in patients receiving emetogenic chemotherapy until August 2024 were systematically searched on Medline, Embase, and Cochrane Library. Notably, no systematic review or meta-analysis of pooled data from existing studies is currently present in the literature. **Results:** Following screening, four studies were deemed eligible for inclusion in the meta-analysis, encompassing a total of 2079 patients. The NEPA group comprised 1047 individuals, while the APR group comprised 1032 subjects. The primary outcome included the comparison of subjects experiencing no nausea in overall phases. Secondary outcomes involved subjects with no emesis and no nausea in the acute and delayed phases. A significant reduction in the risk of overall nausea in the NEPA group was observed compared to the APR group after Moderately Emetogenic Therapy (MEC) or Highly Emetogenic Therapy (HEC) [Risk Ratio (RR) of 1.07, 95% Confidence Interval (CI) = 1.01-1.13, P = 0.01). A trend of reduced emesis during both acute [RR = 1.08, 95% CI = 0.95-1.22, P = 0.24] and delayed periods [RR = 1.03, 95% CI = 0.99-1.07, P = 0.15] was observed in the NEPA group, but the results were not statistically significant. Significant nausea in the acute and delayed post-chemotherapy periods exhibited a similar trend (RR = 1.02, 95% CI = 0.98-1.06, P = 0.43) and (RR = 1.04, 95% CI = 0.98-1.10, P = 0.23) respectively, with non-significant statistical outcomes. **Conclusions:** Our analysis suggests that the use of NEPA results in better control of nausea in overall phases compared to Aprepitant for the prevention of CINV in patients receiving emetogenic chemotherapy. However, NEPA demonstrates comparable efficacy to aprepitant in achieving control of nausea and emesis when the symptoms are stratified in acute and delayed phases. Nevertheless, further studies are warranted for a comprehensive comparison between NEPA and Aprepitant for the prevention of CINV. Research Sponsor: None.

824 Poster Session

The presenting symptoms of patients with appendiceal malignant tumors. First Author: Andreana Natalie Holowatyj, Vanderbilt University Medical Center, Nashville, TN

Background: Timely detection of appendiceal malignant tumors is a clinical priority because of the propensity for this malignancy to metastasize to the peritoneal cavity. Up to one in every 2 patients with this rare tumor type will present with distant metastatic disease, owing to there being no standardized screening tests for early detection of primary appendiceal cancers. Case reports have pointed to broad/non-specific symptoms of patients diagnosed with appendiceal malignant tumors, yet information is needed from large cohorts to support prompt/accurate clinical diagnoses. **Methods:** We analyzed data from patients with a pathologically-confirmed first primary appendiceal malignant tumor (AC) who prospectively enrolled in the nation-wide Genetics of Appendix Cancer [GAP] clinical cohort study (Clinicaltrials.gov, NCT05734430) between November 2022 and May 2024. The primary outcome was presenting symptoms that led to AC diagnosis. Symptom effects and interactions with diagnosis age/year, sex, body mass index (BMI), and tumor histology, were quantified with negative binomial regression models and presented as incidence rate ratios (IRRs) and 95% confidence intervals. **Results:** Of the 352 patients included in our analyses (median [IQR] age, 51.0 [42-59] yr), 77.6% were female and 96.6% had an adenocarcinoma histology. Seventy-seven percent of patients (n=270) reported one or more presenting symptoms, of whom 55.2% (n=149) experienced these symptoms for 3+ months prior to diagnosis. On average, patients reported 3 symptoms (mean: 2.9, SD 3.0). The most prevalent symptoms among males and females were abdominal pain (57.1%), bloating/distension (32.1%), pelvic pain (18.5%), and abdominal/pelvic mass (18.2%). Overall, patients with early-onset AC [age<50] more commonly presented with symptoms versus cases with late-onset AC (82.9% *vs* 71.7%, p=0.01). This finding persisted in adjusted models—the number of symptoms was highest among younger patients, rapidly decreased and plateaued around age 50 (IRR 0.97, 95%CI 0.95-0.99, p=0.002), and slowly increased again with older age (IRR 1.03, 95%CI 1.00-1.05, p=0.037). Patient sex was also significantly associated with symptom number, as males had a lower rate of symptoms versus females (IRR 0.63, 95%CI 0.48-0.83, p=0.001). In contrast, histology (IRR 1.03, 95%CI 0.53-2.02), diagnosis year (IRR 1.01, 95%CI 0.99-1.04, p=0.35) and BMI (IRR 1.01, 95%CI 0.99-1.02) were not associated with symptom number in adjusted models. **Conclusions:** In a large nation-wide cohort, three of every 4 patients were symptomatic prior to primary AC diagnosis. Compared to the rapid time course of acute appendicitis, over 40% of this population presented with symptoms for 3+ months prior to AC diagnosis. The higher symptom burden among young patients and females supports the need for clinical providers to keep occult appendiceal tumors in the differential diagnosis of patients presenting in this manner. Research Sponsor: Dalton Family Foundation; National Institutes of Health/Eunice Kennedy Shriver National Institute of Child Health and Human Development; K12 HD043483.

825 Poster Session

Phase II trial of sintilimab in cancer of unknown primary. First Author: Ryan W Huey, The University of Texas MD Anderson Cancer Center, Houston, TX

Background: CUP is an aggressive rare malignancy accounting for 2-4% of all cancers. Beyond frontline therapy, predominantly systemic cytotoxic chemotherapy, no clear standard of care exists. Preliminary data suggests that immune-checkpoint inhibitors have encouraging activity in patients with CUP. Prior anti-PD1 agents have demonstrated response rates of 21-23%. We studied the role of sintilimab, an anti-PD1 antibody, in CUP. **Methods:** In this phase II study (NCT05024968)10 patients (of planned 40) were enrolled prior to closure of the study due to sponsor funding. Patients were eligible if they were refractory or intolerant to one line of systemic therapy or had a contraindication for cytotoxic chemotherapy. Patients received sintilimab 200 mg IV every 3 weeks. Paired tumor biopsies were collected for biomarker analysis. The primary endpoint was confirmed objective response rate (cORR) by RECISTv1.1 assessed by independent radiology review. Key secondary endpoints were safety, disease control rate (DCR), duration of response (DOR), progression-free survival (PFS), overall survival (OS), and Quality of Life (QOL). Median PFS and OS were calculated according to the Kaplan-Meier method. Adverse events (AEs) were assessed according to CTCAE v5.0. **Results:** Between 10/2021 and 2/2023, 10 patients were enrolled and treated. Median age was 65 years and 30% of patients were female. Seven (70%) had poorly differentiated carcinoma. Patients received a median of 1 line of therapy prior to enrollment. Median follow-up was 19.8 months. All patients were evaluable for toxicity and response. The cORR was 40% (95% CI, 12.2 to 73.8); one (10%) patient with a complete response and 3 (30%) with partial responses. The median DOR was 15.7 months. Three patients (30%) had Grade ≥ 3 (no grade 4/5) treatment-related AEs including 2 patients with sepsis and 1 patient with leukocytosis. Median PFS and OS were 4.4 months (95% CI, 2.0 to 25.4) and 25.4 months (95% CI, 4.0 to 27.4), respectively. **Conclusions:** Anti-PD1 therapy with Sintilimab demonstrated significant anti-tumor activity for select patients with refractory CUP with safety profile consistent with other anti-PD1 agents. Ongoing correlative studies may inform the role of biomarker selection for immunotherapy in CUP. Clinical trial information: NCT05024968. Research Sponsor: None.

826 **Poster Session**

Peak part 1 summary: A phase 3, randomized, open-label, multicenter clinical study of bezuclastinib (CGT9486) and sunitinib combination versus sunitinib in patients with gastrointestinal stromal tumors (GIST). First Author: Jonathan C. Trent, University of Miami – Sylvester Comprehensive Cancer Center, Miami, FL

Background: After initial response to first line therapy with imatinib, GISTs commonly progress due to secondary resistance mutations in KIT. As the KIT mutation targeting profiles of bezuclastinib (type I TKI) and sunitinib are distinct, when combined they inhibit a broad spectrum of secondary KIT mutations. Herein we report extended experience of patients treated with 2nd line combination bezuclastinib and sunitinib, including response assessment and safety. **Methods:** Peak (NCT05208047), a global randomized Phase 3, open-label study, aims to evaluate efficacy and safety of bezuclastinib + sunitinib vs sunitinib in GIST pts with imatinib intolerance or resistance. In Part 1a of the 3-part study, the dose of an optimized formulation of bezuclastinib was escalated in serial cohorts to achieve a target exposure comparable to that achieved at the RP2D established in the prior Phase 1b/2a study. Part 1b evaluated the interaction between bezuclastinib + sunitinib. Key inclusion: adult with locally advanced, metastatic and/or unresectable GIST, ≥1 measurable lesion according to modified RECIST v1.1, and ECOG PS 0 to 2. Part 1 completed enrollment in Apr 2023. Based upon PK and safety, a dose of bezuclastinib 600 mg QD + sunitinib 37.5 mg QD was selected for Part 2. All Part 1 data reported herein are as of Aug 2023; updated safety and efficacy will be presented. **Results:** Part 1 has completed enrollment with 19 pts in Part 1a and 23 in 1b. In Part 1a, Cohort 1 included 5 pts starting at bezuclastinib 300 mg QD + sunitinib 37.5 mg QD; Cohort 2 included 14 pts at bezuclastinib 600 mg QD + sunitinib 37.5 mg QD. Pt characteristics in Part 1a+1b: median age - 60 yrs (range: 33-77); 81% men; 98% ECOG PS 0-1; 98% metastatic and 2% locally advanced. The majority of TEAEs were low grade and reversible with low rate (38%) of Grade 3+ events. There were limited (24%) dose reductions and infrequent (n=2) discontinuations due to TEAEs. Three pts experienced serious AEs possibly associated with study medications. In pts evaluable for response (received ≥1 cycle of study treatment and had assessments at baseline and post-baseline [n=40]), the objective response rate (ORR) was 20% (6 confirmed PRs, 2 unconfirmed PRs). Data remain immature to estimate median PFS for Part 1 pts. In a subset of pts with prior imatinib only (n=6), the closest approximation for pts enrolling in Part 2, the ORR was 33% (2 confirmed PRs). The majority of 2nd-line pts remain on treatment past 12 months. **Conclusions:** Data from Peak Part 1 show an encouraging safety and tolerability profile generally consistent with published sunitinib monotherapy experience. ORR in evaluable pts from Part 1 was 20%; ORR in 2nd line pts was 33%. Part 2 of the Peak study is actively enrolling pts globally at the selected dose of bezuclastinib 600 mg QD + sunitinib 37.5 mg QD versus sunitinib 37.5 mg QD. Clinical trial information: NCT05208047. Research Sponsor: Cogent Biosciences.

828 **Poster Session**

A phase 1/2 study evaluating the safety and efficacy of autologous TAC T cells in subjects with claudin 18.2+ advanced solid tumors. First Author: Ecaterina Elena Dumbrava, The University of Texas MD Anderson Cancer Center, Houston, TX

Background: Claudin18.2 (CLDN18.2) is a tight junction protein found in gastric epithelial cells. It can become abnormally expressed in gastric cancer and other solid tumors, rendering it a candidate for targeted therapy. There are no approved therapeutic agents against CLDN18.2. The T cell antigen coupler (TAC) technology modifies T cells *ex vivo*, allowing cytotoxicity of tumor cells by co-opting the natural T cell receptor. TAC T cells demonstrate a safer profile than chimeric antigen receptor T cells. TAC01-CLDN18.2 is an autologous T-cell product comprising T cells expressing CLDN18.2 TAC. **Methods:** In this first-in-human study (NCT05862324), subjects will undergo leukapheresis and may receive bridging anticancer therapy during cell manufacturing. Prior to TAC01-CLDN18.2 infusion, subjects will undergo lymphodepletion chemotherapy. In Phase I, TAC01-CLDN18.2 will be administered at increasing doses (3 cohorts) in adult subjects with ≥2 lines of prior therapy (1 for subjects with pancreatic ductal adenocarcinoma (PDAC)) using the classic 3+3 dose escalation study design. CLDN18.2 expression levels will be determined centrally using a validated clinical trial assay. Dose-limiting toxicities (DLTs) will be assessed up to 28 days after the infusion. A second dose may be administered according to preidentified clinical and safety criteria. In Phase II, dose expansion groups will evaluate the efficacy, safety, and pharmacokinetics of the optimal TAC01-CLDN18.2 dose, with the option of redosing. Indications will include gastric and esophageal adenocarcinoma (group A), PDAC (group B) and ovarian and non-small cell lung cancer (group C) in subjects with <4 lines of prior therapy. **Results:** The first two dose cohorts of Phase I have been completed, with no reported dose-limiting toxicities (DLT). All of the TAC-related adverse events were low-grade. No cytokine release syndrome was reported. One subject in cohort 2 experienced a grade 1 neurotoxicity, which resolved the same day without intervention. Six subjects reported a total of 6 serious adverse events, none related to TAC01-CLDN18.2. An 83.3% disease control rate was observed at first tumor assessment (Day 29). One subject in cohort 1 with heavily pre-treated PDAC (3 prior lines) with high CLDN18.2 expression has an ongoing, confirmed partial response. The subject received a second dose and is still on treatment (6 months). **Conclusions:** Treatment with the lower doses of TAC01-CLDN18.2 is safe and shows promising clinical activity in a heavily pre-treated cancer population. Treatment of cohort 3 has begun. Clinical trial information: NCT05862324. Research Sponsor: Triumvira Immunologics.

829 **Poster Session**

AGX101: A TM4SF1-directed tubulin inhibitor conjugate in ongoing first-in-human trial including GI cancers. First Author: Paul Jaminet, Angiex, Inc., Cambridge, MA

Background: TM4SF1 (Transmembrane-4 L-Six-Family-Member-1) is an endothelial marker with critical roles in angiogenesis, as well as a tumor cell antigen that contributes significantly to invasion and metastasis. TM4SF1 is upregulated 20-fold in angiogenic tumor vascular endothelium compared to normal vasculature endothelium and exhibits a unique nuclear internalization pathway. AGX101 is a novel tubulin inhibitor conjugate specifically directed against TM4SF1, delivering a potent maytansinoid payload directly to the nucleus of cells within the tumor microenvironment. Delivery to both the vascular and tumor cell compartments of the tumor results in three mechanisms of action (MoAs): (1) activation of tumor immune surveillance, (2) tumor blood supply deprivation, and (3) direct tumor cell killing. **Methods:** TM4SF1 expression was assessed using immunohistochemistry. Safety and pharmacokinetics of AGX101 were evaluated in non-human primates (NHP), with escalating doses to determine the highest non-severely toxic dose (HNSTD). Efficacy studies were also conducted in mouse models, enabling assessment of the minimum effective dose (MED) needed to engage each of the three MoAs. The combination of HNSTD and MED enables calculation of a therapeutic index (TI). The potential for synergy with immune checkpoint inhibitors (ICIs) was also investigated. A first-in-human study of AGX101 in patients with advanced solid tumors with the primary objective of safety and dose-limiting toxicities (DLTs) has initiated. **Results:** Notable cancers with high TM4SF1 scoring intensity include pancreatic adenocarcinoma, hepatocellular carcinoma, cholangiocarcinoma, and gastric cancer. In esophageal cancer, 20% of patients have a TM4SF1 copy number amplification, and these have worse prognosis. In NHP, AGX101 exhibited a favorable safety profile. In preclinical efficacy studies, monotherapy demonstrated robust efficacy through each of the three MoAs in syngeneic models in mice including CT26 colon carcinoma, and human tumor xenograft models including MIA PaCa-2 pancreatic cancer. Measured by exposure, TI was large. In syngeneic mouse models including CT26, the effects of ICIs were potentiated, suggesting a potential synergistic approach in cancer therapy. The first two dose levels of AGX101 monotherapy in humans has cleared without DLTs. Three patients with pancreatic adenocarcinoma have been treated thus far. **Conclusions:** AGX101, targeting the TM4SF1 antigen, represents a promising new approach in cancer therapy. The preclinical data suggest that AGX101 could provide a significant therapeutic benefit by novel and differentiated mechanisms of action, namely selectively targeting the tumor vasculature and potentiating immune-based therapies. Further clinical development of AGX101 is ongoing and initial outcome data will become available by the conference. Clinical trial information: NCT06440005. Research Sponsor: Angiex, Inc.

830 **Poster Session**

A 20-year population-wide cohort study on association of aspirin and risk of gastrointestinal and non-gastrointestinal cancer. First Author: Kelvin KF Tsoi, The Chinese University of Hong Kong, Hong Kong, Hong Kong

Background: Aspirin has been shown to reduce the risk of various gastrointestinal (GI) and non-GI cancers[1,2]. However, little was known about at what age aspirin should be started for maximal chemoprotective better on GI cancer. Furthermore, the randomised controlled trial on aspirin for primary prevention use in healthy elderly (ASPREE) showed no association between aspirin and cancer incidence despite the limitation of short follow-up duration[3]. The current study aims to investigate the 20-year risk of cancer using aspirin in a territory-wide Hong Kong population cohort. **Methods:** The study included all aspirin users from 2000 to 2019, and non-aspirin users matched by age and sex at a ratio of 1:2. Enrolled subjects with a history of cancer at enrolment, cancer incidence or death within 6 months were excluded. The incidence of individual GI and non-GI cancer was presented as the primary outcome. Baseline characteristics between aspirin and non-aspirin users were adjusted in the survival analysis by inverse probability of treatment weighting (IPTW). The fine-grey model has been used to address bias from competing risk of death. The sub-distribution hazard ratio was presented for the association of aspirin use and risk of GI and non-GI cancers. **Results:** The current study included 538,147 aspirin users and 968,378 non-users with a mean age of 64.8 years. A total number of 36,683 cases of GI cancer (2.4%) and 47,196 cases of non-GI cancers (3.1%) were observed. Aspirin was associated with a lower risk of several common individual GI cancers, including colorectal cancer (SHR 0.78, 95% CI 0.76-0.81), liver cancer (SHR 0.67, 95% CI 0.64-0.70), stomach cancer (SHR 0.79, 95% CI 0.75-0.84) and pancreatic cancer (SHR 0.85, 95% CI 0.79-0.91), but not oesophageal cancer. On the other hand, aspirin was associated with a lower risk of prostate cancer (SHR 0.95, 95% CI 0.91-1.00) and breast cancer (SHR 0.76, 95% CI 0.73-0.79), but not lung cancer and kidney cancer. In overall, aspirin was associated with a 24% lower risk of GI cancers (SHR 0.76, 95% CI 0.74-0.78) and a 3% lower risk of non-GI cancers (SHR 0.97, 95% CI 0.95-0.99). **Conclusions:** Aspirin was associated with a lower risk of most GI cancers, including colorectal cancer, liver cancer, stomach cancer and pancreatic cancer, but not most non-GI cancers. In general, results on the effect of aspirin on GI cancer prevention were consistent with the previously presented 10-year cohort. 1. Bosetti C, Santucci C, Gallus S, Martinetti M, La Vecchia C. Aspirin and the Risk of Colorectal and Other Digestive Tract Cancers: An Updated Meta-analysis through 2019. Ann Oncol. 2020;31(5):558-568. 2. Santucci C, Gallus S, Martinetti M, La Vecchia C, Bosetti C. Aspirin and the risk of nondigestive tract cancers: An updated meta-analysis to 2019. Int J Cancer. 2021;148(6):1372-1382. 3. McNeil JJ, Gibbs P, Orchard SG, et al. Effect of Aspirin on Cancer Incidence and Mortality in Older Adults. J Natl Cancer Inst. 2021;113(3):258-265. Research Sponsor: Health and Medical Research Fund, by Health Bureau Hong Kong.

831 Poster Session

Uses of VEGF inhibitor bevacizumab in appendiceal adenocarcinoma. First Author: Jacquelyn McCullough, University of Texas Health Science Center at Houston McGovern Medical School, Houston, TX

Background: Systemic treatment of metastatic appendiceal adenocarcinoma (AA) has been challenging due to a lack of well-established AA chemotherapy regimens and historical reliance on colorectal cancer (CRC) protocols. In the United States, but less commonly in Europe, bevacizumab is frequently incorporated in treatment of AA given its approved use for CRC. However, previously there have been no studies evaluating if there is a survival benefit attributable to bevacizumab in patients with AA. **Methods:** The Palantir Foundry system was used to extract data from the MD Anderson Cancer Center (MDA) electronic medical record for patients with biopsy-proven AA who received chemotherapy at MDA from 2015 to 2024. Only patients with complete staging and histopathology data were included in the analysis. **Results:** We identified 203 patients with AA who received chemotherapy at MDA and had complete tumor data. Of these, 43% (n=87) received bevacizumab at some point in their treatment. Patients who received bevacizumab were demographically similar to patients who received non-bevacizumab chemotherapy in terms of age at diagnosis, race, and sex; mucinous histology was more frequent in the bevacizumab cohort (52%, n=45 vs. 39%, n=45 in non-bevacizumab cohort). The 87 patients who received bevacizumab were treated with 1-6 lines of therapy per patient (mean = 1.57). The most frequently used bevacizumab-containing regimens were FOLFOX + bevacizumab (34%, n=46), FOLRIRI + bevacizumab (29%, n=40), 5-FU + bevacizumab (15%, n=21), and atezolizumab + bevacizumab (11%, n=15), with less common irinotecan + bevacizumab, TAS-102 + bevacizumab, and FOLFOXIRI + bevacizumab. Three regimens containing bevacizumab were discontinued due to complications: one instance of hand-foot syndrome attributed to xeloda, one instance of cardiomyopathy attributed to immunotherapy, and one bowel perforation with abdominal wall abscess attributed to bevacizumab. To determine if bevacizumab was associated with overall survival (OS) benefit we performed Kaplan-Meier analysis, with OS calculated from start of first line chemotherapy to death, in patients with metastatic AA who received either 5-FU doublet therapy alone or 5-FU doublet therapy with bevacizumab. Patients who received bevacizumab trended toward better OS (median 52 vs 25 months for doublet therapy alone, HR: 0.53, p= 0.059). The effect of bevacizumab was particularly notable in patients with signet ring cell histology; where bevacizumab use was associated with significantly longer OS (median OS 50 vs 14 months for doublet therapy alone, HR: 0.24, p= 0.034). **Conclusions:** Bevacizumab is frequently added to cytotoxic chemotherapy in the treatment of AA. These retrospective data suggest an OS benefit to adding bevacizumab to doublet chemotherapy, particularly in the case of signet ring cell histology. Research Sponsor: None.

832 Poster Session

A comparative study of dosage strategies in HIPEC: BSA-based vs fixed dose methods. First Author: Harsimran Kaur Multani, University of Vermont Larner College of Medicine, Burlington, VT

Background: Peritoneal metastases commonly occur in ovarian, gastric, and colorectal cancers, significantly contributing to patient morbidity and mortality. Hyperthermic Intraperitoneal Chemotherapy (HIPEC) is a promising approach for managing peritoneal surface cancers. This procedure involves cytoreductive surgery to remove visible tumors, followed by the perfusion of heated chemotherapy (41-42°C) directly into the peritoneal cavity. HIPEC addresses the limitation of systemic chemotherapy, which often fails to penetrate the peritoneal cavity effectively. HIPEC dosing may either be a flat-fixed dose or be calculated based on Body Surface Area (BSA). BSA can be determined using three methods: actual BSA (based on weight and height), ideal BSA (based on sex and height), and adjusted BSA (considering lean and fat mass). **Methods:** A retrospective chart review was conducted on 151 metastatic cancer patients at UVMMC. Patient BMI, weight, and height at diagnosis were collected to calculate BSA values (actual, adjusted, and ideal) for estimating a hypothetical chemotherapy dosage. Statistical analysis using SPSS involved linear mixed models and paired t-tests to compare Mitomycin-C dosages between a fixed dose (40 mg) and a BSA-derived dose (30 mg/m²). **Results:** The average dosage of Mitomycin-C differed significantly depending on whether the dose was calculated using actual, adjusted, or ideal BSA. The largest discrepancy in dosage was observed between the actual BSA method and the flat-fixed dose, with an average difference of 16 mg. The smallest difference in dose was between the adjusted and ideal BSA methods. When stratifying by BMI, the greatest variation in Mitomycin-C dosage was found among overweight and obese patients (BMI $\geq$ 25). For these patients, the average dosage difference between the flat fixed dose and the actual BSA dose was 20 mg. In comparison, in patients with a BMI under 18.5 (underweight) the actual BSA dose was comparable to the fixed dose. **Conclusions:** Different institutions utilize a variety of dosing strategies for the HIPEC procedure. Differences in BSA calculation methods and perfusate volumes (literature values of 2, 3 and 4 L is most common) contribute to the complexity of establishing standardized dosing. The significant dosing variations observed, especially in overweight and obese patients, are of concern, as 70% of the U.S. population has a BMI $\geq$ 25. Our study demonstrates significant variability in HIPEC chemotherapy dosing, especially among patients with higher BMI. The discrepancy between BSA-based methods and flat-fixed doses may impact treatment efficacy and toxicity. Standardizing dosing approaches is important to ensure consistent therapeutic outcomes and patient safety. Research Sponsor: None.

833 Poster Session

Personalised use of repurposed non-anticancer drugs in colorectal cancer based on up-regulated signalling pathways: A real world cohort feasibility study. First Author: Andrew M. Gaya, Cromwell Hospital, London, United Kingdom

Background: Targeting signaling pathways in colorectal cancers is a common therapeutic strategy. In addition to targeted drugs, non- anticancer drugs approved for use in other health conditions can inhibit key pathways. De novo tumor transcriptome analysis can identify patient-specific up- and down-regulated transcripts, based on which the corresponding repurposed drugs can be used in synergistic combinations with standard of care (SoC) regimens. **Methods:** This retrospective study was conducted on a real-world cohort of 459 colorectal cancer cases (218 females, 241 males; median age 56 years) who had taken Exacta multi-omics tumor profiling for selection of personalized treatments. Transcriptome analysis was performed as part of the evaluations to determine up and down-regulated gene transcripts for selecting targeted, endocrine, and cytotoxic anticancer agents as well as non-anticancer (repurposed) drugs and phytochemicals that could be used for tandem targeting of the dysregulated signaling pathways, as indicated in the table below. **Results:** Transcriptome analysis of this cohort revealed dysregulation of $\geq$1 pathway in 391 patients (85.2%). MAPKs (n = 244, 53.2%) and MMPs (n = 210, 45.8%) were most frequently upregulated. Celecoxib (n = 292, 63.6%), Atorvastatin (n = 255, 55.6%), and Doxycycline (n = 208, 45.3%) were the most frequently indicated repurposed drugs for targeting the identified dysregulated pathways. Potential indications for $\geq$1 repurposed drugs were seen in 353 patients (76.9%). **Conclusions:** Several cellular signaling pathways are known or potential therapeutic targets in cancers. The present study indicates that it may be possible to use multi-omics analyses to create personalized treatment strategies using a synergistic combination of repurposed drugs to potentiate the action of SoC systemic anticancer agents in colorectal cancer. Research Sponsor: None.

Biomarkers for non anti-cancer drugs selection.

Drug	Biomarker/Gene Expression
Aspirin	PTGS2 (COX2)
Atorvastatin	HMGCR, MAPK
Bromelain	PTGS2 (COX2)
Celecoxib	PTGS2 (COX2), FZD, WNT, MAPK
Chloroquine	HMGB1
Curcumin	MMP, BIRC5, BCL2
Doxycycline	MMP
Metformin	MMP, BCL2, MAPK
Quercetin	FZD, WNT
Resveratrol	MMP, BCL2, PCNA, PTGS2 (COX2)
Vitamin C	SLC2A1 (GLUT1)
Artesunate	FZD, WNT, MMP, BCL2
Cannabidiol	MMP
Epigallocatechin gallate	MMP, MAPK
Mebendazole	XIAP, MMP, MAPK

834 Poster Session

CK2 expression gastrointestinal cancer: Exploring clinical characteristics and cancer genetics. First Author: Nicholas Wilson, Chobanian and Avedisian School of Medicine, Boston University, Boston, MA

Background: The expression of protein kinase CK2 subunits (CSNK2A1, CSNK2A2, CSNK2A3, and CSNK2B) in gastrointestinal cancers has not been fully characterized. In this study, we seek to evaluate how changes in CK2 subunit expression are associated with GI cancer clinical characteristics and genetics. **Methods:** Data analysis was performed with data from The Cancer Genome Atlas (TCGA). R version 4.2.2 was used for analysis. We provide an analysis of several cancer cohorts looking at CK2 subunit expression, as well as cancer mutations, copy number alterations, and clinical characteristics. **Results:** From the TCGA datasets containing mRNA expression data, we assessed hepatocellular carcinoma (n=361), colorectal adenocarcinoma (n=339), gastric adenocarcinoma (n=306) and diffuse type gastric adenocarcinoma (n=69), pancreatic adenocarcinoma (n=169), esophageal adenocarcinoma (n=89) and squamous cell carcinoma (n=95), and cholangiocarcinoma (n=36). We explored associations between CK2 subunit expression and GI cancer-specific known driver mutations, of which we report the following significant findings ($p < 0.05$ for all). Mutations in TP53 were associated with increased expression of CSNK2A1 in hepatocellular carcinoma and stomach adenocarcinoma, increased CSNK2A2 in colorectal adenocarcinoma, and CSNK2B in colorectal adenocarcinoma, pancreatic adenocarcinoma, and stomach adenocarcinoma. Mutations in KRAS were associated with increased expression of CSNK2A1 and CSNK2B in pancreatic adenocarcinoma. Mutations in BRAF were associated with decreased expression of CSNK2A2 and CSNK2B in colorectal adenocarcinoma. We also noted significant changes in CSNK2 subunit expression when looking at mutations and copy number alterations in signaling pathways such as Wnt, NOTCH, PI3K, TGFβ, and more. We also explored associations between CK2 subunit expression and GI cancer clinical characteristics, of which we report the following significant findings ($p < 0.05$ for all). In colorectal adenocarcinoma, increased CSNK2A2 expression was associated with lymph node invasion and advanced clinical stage. In hepatocellular carcinoma, increased CSNK2A1 and CSNK2A3 expression was associated with greater tumor grade, tumor size, and more advanced clinical staging. Increased CSNK2A2 expression was associated with greater tumor size, more advanced clinical staging, and vascular invasion. Increased CS2NKB expression was associated with increased tumor grade, size, and vascular invasion. In pancreatic adenocarcinoma, increased CSNK2A3 expression was associated with a worse response to the first treatment course. **Conclusions:** Our results suggest that changes in protein kinase CK2 expression are associated with different cancer clinical characteristics as well as cancer-specific mutations and signaling pathways. Research Sponsor: None.

835 Poster Session

Characterization of the tumor immune microenvironment (TIME) and somatic landscape in gastrointestinal (GI) malignancies with *MTAP* deletions (del). First Author: Jun Gong, Cedars-Sinai Medical Center, Los Angeles, CA

Background: PRMT5 is a synthetic lethality target in patients (pts) with *MTAP* del and early phase trials are underway with PRMT5 inhibitors. Additionally, *MTAP* del are associated with a less immunogenic TIME and reduced efficacy of immunotherapy, but research has primarily not been focused on GI malignancies. Thus, we investigated the TIME and somatic landscape in GI malignancies with *MTAP* del. **Methods:** From the Tempus Database, we retrospectively analyzed de-identified next-generation sequencing data from pts across GI malignancies, including pancreatic (n=11,217), gastroesophageal (GEJ, n=5,803), cholangiocarcinoma (CCA, n=3,244), and colorectal (CRC, n=17,537) cancers. Tumors were sequenced with the Tempus xT DNA (648-gene panel) and xR RNA assays. *MTAP* del were defined as two-copy losses. Somatic alterations (alt), immune cell infiltration predicted from gene expression patterns, PD-L1 from IHC, TMB, and MSI were evaluated. Chi-squared/Fisher's Exact tests or Kruskal-Wallis tests were used to assess statistical significance ($p<0.05$, $q<0.05$ for false discovery rate correction for multiple testing). **Results:** *MTAP* del were identified in 14.8%, 11.3%, 0.9%, and 7.3% of pancreatic, CCA, CRC, and GEJ cancers, respectively. Of these, 98/93%, 99/91%, 92/89%, and 98/89% had co-occurring *CDKN2A/B* loss. In pancreatic, CCA, and CRC pts, *MTAP* del was associated with a reduced proportion of B cells and CD4 T cells, and there were higher percentages of macrophages vs pts with *MTAP* WT status ($p<0.001$ for all). Reductions in proportion of CD8 T cells were also associated with *MTAP* del in pancreatic and CCA pts ($p<0.001$ for both). Lower TMB (p=0.036) and MSI-H status (p=0.009) were found in CRC pts with *MTAP* del. GEJ with *MTAP* del also exhibited significantly lower MSI-H status than *MTAP* WT GEJ (p=0.014). *SMAD4 alterations*, a marker of reduced immune infiltrates, were more prevalent in pts with *MTAP* del across GI malignancies ($q<0.005$). In the CCA cohort, there was a higher percentage of *BRAF* alt and FGFR2 fusions in pts with *MTAP* loss ($q<0.001$, q=0.028), while *KRAS* alt were higher in pancreatic cases with *MTAP* loss ($q<0.001$). **Conclusions:** This is the largest analysis of the TIME and somatic landscape of *MTAP* loss across GI malignancies. In pts with *MTAP* del and pancreatic cancer, CCA, and CRC, we observed a less immunogenic TIME pattern, indicating the evaluation of immunotherapy implications in these GI malignancies with *MTAP* del is warranted. Our findings are hypothesis-generating, providing further rationale to study synthetic lethality and novel combinatorial therapeutic strategies in GI malignancies with *MTAP* del. Research Sponsor: Tempus AI, Inc.

836 Poster Session

Short interval circulating tumor DNA (ctDNA) kinetics as a predictor of tumor response in patients with gastrointestinal (GI) cancer receiving immune checkpoint inhibitor (ICI)-based treatment. First Author: Sakti Chakrabarti, University Hospitals Seidman Cancer Center, Case Comprehensive Cancer Center, Cleveland, OH

Background: Circulating tumor DNA (ctDNA) has a short half-life (<2 hours) which may permit real-time monitoring of tumor status. This single-institution study aimed to assess the feasibility of rapid treatment response evaluation through serial short-interval ctDNA testing. **Methods:** Patients with gastrointestinal (GI) cancer undergoing immune checkpoint inhibitor (ICI)-based therapy were included. A personalized, tumor-informed ctDNA assay (Signatera, Natera, Inc.) was used in this study. We collected samples to measure the baseline ctDNA levels on cycle 1, day 1 (C1D1) of treatment or within 14 days prior to C1D1. A second ctDNA sample was obtained after 14-21 days depending on the cycle length of the treatment regimen employed (on cycle 2, day 1 [C2D1]). Changes in ctDNA levels between baseline and C2D1 were recorded, as well as the tumor response at the first radiographic assessment. Patients achieving partial response (PR) or better, or stable disease (SD) with tumor shrinkage supporting continuation of same treatment were considered responders. **Results:** The study cohort consisted of 14 patients aged 36-89 years (median age 66); 5 patients were female. Tumor types included advanced colorectal (n=5) and gastroesophageal (n=9) cancers. Treatments consisted of pembrolizumab alone (n=5), ICI + chemotherapy combinations (n=8), and regorafenib plus nivolumab (n=1). Baseline ctDNA levels were obtained on C1D1 in 9 patients, with the remaining 5 patients having baseline measurements within 14 days before C1D1. All patients had the second ctDNA level drawn on C2D1. The median interval between C1D1 and the first radiographic response assessment was 61 days (range, 45-84). A reduction in ctDNA levels by 50% or more (range: 66-100%) was observed in 9 patients (64%) after one cycle of treatment. All 9 patients were classified as responders at the time of first radiographic assessment: 7 with PR and 2 with SD but significant tumor shrinkage leading to continuation of the same treatment. In 3 patients (21%), ctDNA levels increased (range: 20-500%), correlating with progressive disease (PD) on first assessment scan. Two patients showed increased ctDNA levels of 82% and 22% after one cycle of treatment but achieved PR and SD, respectively. In both patients, however, the 3rd ctDNA level drawn before the first assessment scan dropped by >90%. Overall, short-interval ctDNA kinetics correlated with treatment response in 12 of 14 patients (86%), with a binomial test p = 0.0065. **Conclusions:** Short-interval ctDNA kinetics demonstrates a strong correlation with radiographic tumor response and clinical benefit in GI cancer patients receiving ICI-based treatments, providing early signal of tumor response. This approach has the potential to guide treatment decisions if validated in larger prospective studies. Research Sponsor: None.

838 Poster Session

Use of gene expression profiling to examine association between MUC1 and claudin 18 in gastrointestinal cancers: Implications for targeted therapies. First Author: Aditya V. Shreenivas, Division of Hematology and Medical Oncology, City of Hope Cancer Center, Duarte, CA

Background: Gastrointestinal tract cancers exhibit considerable molecular heterogeneity, which requires a detailed understanding of their distinct genetic profiles. Understanding the gene expression of other genes along with Claudin 18 (CLDN18) could provide insights into their interactions. **Methods:** We conducted a retrospective analysis of NGS profiling on 1,043 GI cancer tissue samples from three cohorts: Upper and Lower Gastrointestinal Cancer Cohort (GICC; n=651), Pancreaticobiliary Cancer Cohort (PBCC; n=316), and Hepatocellular Carcinoma Cohort (HCC; n=76). Immunohistochemistry data for PD-L1, Her2 and MMR proteins, NGS based Tumor mutation burden (TMB) and gene expression profiling (GEP) of 20,802 genes including CLDN18 was examined in a subset by targeted transcriptome profiling. We explored associations of these biomarkers with expression of CLDN18. **Results:** In GICC, frequent mutations were seen in TP53 (68%; 442), APC (33.5%; 218) and KRAS (33.3%; 217). PBCC, showed mutations in TP53 (62%; 197), KRAS (47%; 149) and CDKN2A (16%; 51). HCC showed mutations in TP53 (42%; 32), ARID1A (18%; 14) and KRAS (9%; 7). Amplifications were predominantly observed in MYC (13%; 120), ERBB2 (5%; 49) and CCND1 (5%; 44). Fusions were infrequent. All MSI-H cases (11/462; 2.4%) were observed in GICC. High TMB (≥10 muts/mb) was seen in 23% (111/481), stratified as 28% (90/318) in GICC, 13% (18/139) in PBCC and 10% (3/30) in HCC. Out of 321 samples analyzed for gene expression profiling, CLDN18 was significantly upregulated in 65 (20%) samples. The expression of CLDN18 did not show statistically significant correlation with Her2 (n=184), PD-L1 (n=231), TMB (n=228), or MSI (n=229); ($p>0.05$). In subset where both CLDN18 and MUC1 expression were analyzed (n=118), all the CLDN18 positive tumors showed overexpression of MUC1. This finding was statistically significant ($p<0.05$). Increased expression of MUC1 promotes cancer cell mobility promoting metastases. Expression of MUC1 can play an important role in the development of resistance to chemotherapy. MUC1 expression in cancer cells is associated with avoidance of immune defenses. Thus, this finding of CLDN18 and MUC1 association is especially relevant as the CLDN18 targeted therapies may be the potential avenue for the aggressive and chemo-resistant MUC1 expressing tumors. **Conclusions:** This study identifies potential association between MUC1 and CLDN18. GEP could enhance biomarker detection and support personalized treatment approaches. Future research should integrate these findings with clinical data to improve precision in GI cancer care. Research Sponsor: None.

839 Poster Session

Clinical validation of the Northstar Response: A novel quantitative methylation ctDNA monitoring assay for advanced GI cancer treatment response. First Author: Ilyas Sahin, University of Florida Health Cancer Center, Gainesville, FL

Background: Circulating tumor DNA (ctDNA) is a promising tool for monitoring treatment response but is challenged by pan-cancer quantification and consistent outcome correlations. Northstar Response (NSR) quantifies >500 cancer specific methylated loci, offering a reflection of disease burden and clinical response. **Methods:** We present a prospective observational study to evaluate the clinical utility of NSR. Patients with advanced GI cancers had serial ctDNA assessed at baseline and during systemic treatment with correlations to radiographic or clinical response. Tumor methylation scores (TMS) were expressed as Log10 values and a mixed-effects model evaluated clinical assessment correlations. **Results:** 73 patients with locally advanced or metastatic GI cancers were analyzed. Among these, 35 (47.9%) had at least four ctDNA assays and all had radiographic or clinical response assessments. The median age was 66 years (± 8.5), with most being male (67.1%) and White (74.0%). Table 1 shows the case distribution. Baseline TMS had significant variation (p=0.023), but was measurable across all tumor types. CRC exhibited the highest mean TMS-log10 while biliary and pancreatic cancers had the lowest. Cases with both primary and metastatic lesions had a higher mean TMS-log10 than cases with either primary tumors alone or metastatic lesions alone (3.4 vs. 3.0). Compared to stable and responsive disease, TMS changes correlated with progressive disease, with a coefficient of 0.48 at the third collection time point (p=0.05). However, 23/73 (32%) cases were without an on-treatment TMS timepoint (OTT) due to rapid progression (average time to death of 24 vs. 103 days for patients with OTT). In exploratory modeling of TMS relative to RECIST-like treatment response for those with OTT, NSR evaluations were associated with response assessments ($p<0.04$, Fisher's Exact Test). **Conclusions:** TMS was consistently measurable but varied across tumor types and correlated with disease burden. TMS changes appeared associated with response assessments when accounting for rapid progressors. Additional analyses are ongoing and will be presented. Research Sponsor: BillionToOne, Inc.; University of Florida Health Cancer Center; P30CA247796.

Baseline study cohort characteristics.

Characteristic	HCC N = 25 (34%)	BTC N = 12 (16%)	EGA N = 12 (16%)	CRC N = 11 (15%)	PDAC N = 8 (11%)	Other GI N = 5 (7%)	p-value
Age Median (25% to 75%)	67 (63 - 76)	68.5 (62.3 – 75.5)	73.5 (62.5 – 75.3)	54 (39.5 – 60)	67.5 (55.5 – 76.3)	60 (57 – 63)	0.008
Baseline TMS Log10 Median (25% - 75%)	3.2 (2.3 – 3.9)	2.5 (2.1 – 3.4)	3.2 (2.2 – 3.8)	4.4 (3.5 – 5)	2.5 (1.9 – 3.3)	4 (3.4 – 5.2)	0.023
Primary tumor present	12 (48%)	8 (66.7%)	12 (100%)	6 (54.5%)	7 (87.5%)	4 (80%)	0.015
First line therapy vs. not	23 (92%)	5 (41.7%)	5 (41.7%)	6 (54.5%)	3 (37.5%)	5 (100%)	0.001
Progressive disease	16 (64%)	4 (33.3%)	9 (75%)	5 (45.5%)	3 (37.5%)	4 (80%)	0.2
Death	13 (52%)	3 (25%)	9 (75%)	7 (63.6%)	3 (37.5%)	2 (40%)	0.2

Kruskal-Wallis rank sum test; Fisher's exact test.

840 Poster Session

Universal screening for mismatch repair deficiency in gastrointestinal cancers in a reference center in Mexico. First Author: Jorge Humberto Hernandez-Felix, Instituto Nacional de Ciencias Medicas y Nutrición Salvador Zubirán, Mexico City, DF, Mexico

Background: Gastrointestinal (GI) cancers have a poor prognosis. With the availability of immunotherapy for tumors with deficient mismatch repair (dMMR), identifying these patients has become increasingly important. Universal dMMR screening across all GI cancers is uncommon, and its prevalence has not been well studied prospectively. In Mexico, the prevalence of dMMR is unknown, even though GI cancers account for a large portion of new cancer cases and cancer-related deaths. Our objective was to prospectively determine the prevalence of dMMR in GI cancers at a referral center in Mexico. **Methods:** Prospective universal screening for dMMR began in March 2024. Primary tumor sites included were esophagus, stomach, pancreas, biliary tract, small intestine, colon, and rectum. Histologies included were adenocarcinoma, squamous cell, or mixed. All consecutive patients were included regardless of age, clinical stage or medical history. MMR status was determined by immunohistochemistry, analyzing expression of MLH1, PMS2, MSH2, and MSH6. This abstract presents the results from the first six months of this universal screening. **Results:** A total of 161 GI cancers were included (see Table). Prevalence of dMMR was 9.3% (n=15). The highest prevalence was in gastric cancer (13.3%, 2/15), followed by colon (12.1%, 7/58), pancreatic (11.1%, 4/36), and bile duct (8.3%, 2/24) cancer. MMR status was not performed in 9.3% (n=15) of cases. Combined deficiency was found in 60% (9/15) for MLH1/PMS2 and 20% (3/15) for MSH2/MSH6. Isolated deficiency occurred in 20% (3/15) for PMS2 and 13.3% (2/15) for MSH2. No difference in age or clinical stage was observed between MMR status groups. **Conclusions:** The high prevalence of dMMR, along with the poor prognosis and availability of immunotherapy based treatments, supports the need for universal screening. The high prevalence of dMMR in pancreatobiliary cancers was notable. Implementation efforts in our institution are essential to obtain MMR status in all cases. Research Sponsor: None.

Characteristics of population.

		pMMR n=131 (%)	dMMR n=15 (%)	Not performed n=15 (%)	Total n=161	p value
Gender	Male	63 (48.1)	10 (66.7)	7 (46.7)	80 (49.7)	0.38
	Female	68 (51.9)	5 (33.3)	8 (53.3)	81 (50.3)	
Age	Median (IQR)	65 (54 -72)	52 (43 -80)	65 (57 -71)	65 (53 -72)	0.52
Histology	Adenocarcinoma	130 (99.2)	15 (100.0)	14 (93.3)	159 (98.8)	N/A
	Mixed	1 (0.8)	0 (0.0)	0 (0.0)	1 (0.6)	
	Squamous Cell	0 (0.0)	0 (0.0)	1 (6.7)	1 (0.6)	
Clinical Stage	Unknown	12 (9.2)	1 (6.7)	1 (6.7)	14 (8.7)	0.49
	Localized	24 (18.3)	2 (13.3)	2 (13.3)	28 (17.4)	
	Locally advanced	40 (30.5)	8 (53.3)	3 (20.0)	51 (31.7)	
	Metastatic	55 (42.0)	4 (26.7)	9 (60.0)	68 (42.2)	
Tumor site	Esophageal	9 (6.9)	0 (0.0)	1 (6.7)	10 (6.2)	N/A
	Gastric	13 (9.9)	2 (13.3)	0 (0.0)	15 (9.3)	
	Small B.	1 (0.8)	0 (0.0)	0 (0.0)	1 (0.6)	
	Bile duct	14 (10.7)	2 (13.3)	8 (53.3)	24 (14.9)	
	Pancreas	29 (22.1)	4 (26.7)	3 (20.0)	36 (22.4)	
	Colon	50 (38.2)	7 (46.7)	1 (6.7)	58 (36.0)	
	Rectal	15 (11.5)	0 (0.0)	2 (13.3)	17 (10.6)	

841 Poster Session

Can an alteration of the gut microbiome increase risk of gastrointestinal malignancies? Insights from the National Inpatient Sample. First Author: Sweta Lohani, Brooklyn Hospital Center, Brooklyn, NY

Background: The gut microbiome is composed of trillions of microorganisms, inhabiting the lining of the gastrointestinal (GI) tract. The gut microbiome plays multiple roles in upholding human health including maintaining metabolism and regulating immune homeostasis. Dysregulation of the gut flora is known as gut dysbiosis. Studies have shown that gut dysbiosis is linked to the development of malignancies such as colorectal cancer. The purpose of this study is to assess the correlation between an altered gut flora and development of GI malignancies. **Methods:** The National Inpatient Sample was queried between 2016-2020 for all adult patients with altered gut flora, defined by history of gastrointestinal/bariatric surgeries, history of clostridium difficile infection, long-term antibiotic use, and inflammatory bowel disease. Rates of various cancers were compared between this cohort and all other patients utilizing chi-squared tests. Binary logistic regressions were utilized to calculate odds ratios of developing the various cancers, adjusted for age, sex, Charlson Comorbidity Index (CCI), and smoking status. **Results:** A total of 7,758,168 hospitalizations with altered gut flora were identified. Esophageal cancer (aOR 3.16, 95% CI 3.08 - 3.24) and colorectal cancer (aOR 3.15, 95% CI 3.11 - 3.18) were observed to have the highest odds of occurring in patients with altered gut flora. Gastric, hepatobiliary, pancreatic, and neuroendocrine cancers also all had significantly higher odds of occurring in patients with altered gut flora. **Conclusions:** Our study shows that conditions altering the gut flora are associated with increased odds of malignancies of the GI tract. This suggests that the gut microbiome may play an intricate role in regulation of tumor cell proliferation. Optimization of gut health and restoration of gut flora may prove beneficial to patients at high risk for malignancies of the GI tract. Research Sponsor: None.

	Altered Gut Flora (%)	Intact Gut Flora (%)	Adjusted Odds Ratio (CI)	p-value
Esophageal	0.4	0.1	3.16 (3.08 - 3.24)	**<.001**
Gastric	0.3	0.1	1.67 (1.64-1.69)	**<.001**
Colorectal	2.5	0.6	3.15 (3.11 - 3.18)	**<.001**
Hepatobiliary	0.4	0.3	1.04 (1.02 - 1.07)	**<.001**
Pancreatic	0.6	0.3	1.34 (1.31 - 1.37)	**<.001**
Neuroendocrine	0.2	0.1	1.38 (1.33 - 1.44)	**<.001**

842 Poster Session

Comparing outcomes for early and late drainage of malignant ascites. First Author: Suriya Baskar, Brooklyn Hospital Center, Brooklyn, NY

Background: Malignant ascites (MA) is a common sequela of peritoneal carcinomatosis resulting in significant morbidity and poses substantial management challenges for clinicians. Large volume paracentesis remains the gold standard treatment. The impact of time to drainage of MA is poorly studied. The purpose of this study was to compare clinical outcomes of hospitalized patients with MA that had drainage performed within 48 hours of admission and after 48 hours of admission. **Methods:** The NIS was queried between the years 2016 and 2020 to identify all adult patients hospitalized with malignant ascites (ICD-10 code R18.0) and had a drainage procedure performed (ICD-10 Procedure code 0W9Gx). These patients were then stratified into an early cohort (<48 hours after admission) and late cohort (>48 hours after admission). Baseline characteristics and clinical outcomes were collected and analyzed using chi squared tests, independent sample t-tests, and binary logistic regression (adjusted for age, gender, and Charlson comorbidity index or CCI). Adjusted Odds ratios (aOR) are presented with 95% Confidence intervals (CI). **Results:** A total of 129,040 patients were analyzed, with 89,905 (69.7%) hospitalizations in the early cohort and 39,135 (30.3%) in the late cohort. Baseline characteristics were similar between the groups, with no significant differences in age (63.8 vs 63.6, p = .06) or gender distribution (63.6% vs 63.1%, p = .09). However, patients in the early paracentesis group exhibited a lower CCI (9.8 vs 10.2, p<.001). Notably, early paracentesis was associated with significantly improved outcomes, including lower mortality (aOR 0.67, 95% CI 0.64–0.69), reduced incidence of sepsis (aOR 0.67, 95% CI 0.65–0.69), decreased need for transfusions (aOR 0.69, 95% CI 0.66–0.71), lower rates of mechanical ventilation (aOR 0.44, 95% CI 0.42–0.47) and vasopressor use (aOR 0.79, 95% CI 0.73–0.87). **Conclusions:** Early paracentesis in patients with malignant ascites was found to be associated with significantly improved clinical outcomes, independent of comorbidity burden. These findings underscore the importance of timely intervention to mitigate complications and enhance quality of life. Research Sponsor: None.

	Late Paracentesis N=39,135	Early N=89,905	p-value
Age (years)	63.6 ± 13.1	63.8 ± 13.0	.06
Gender (% female)	24665 (63.1%)	57120 (63.6%)	.09
CCI	10.2 ± 3.1	9.8 ± 3.2	**<.001**
Mortality	5720 (14.6%)	9030 (10.0%) aOR .67 (.64 - .69)	**<.001**
Sepsis	7675 (19.6%)	12525 (13.9%) aOR .67 (.65 - .69)	**<.001**
Transfusion	5275 (13.5%)	8605 (9.6%) aOR .69 (.66 - .71)	**<.001**
Ventilation	2130 (5.4%)	2190 (2.4%) aOR .44 (.42 - .47)	**<.001**
Vasopressor	770 (2.0%)	1385 (1.5%) aOR .79 (.73 - .87)	**<.001**

TPS843 Trials in Progress Poster Session

EQUITY GI: A prospective study to enhance quality, inclusivity, and trial participation in Black patients with gastrointestinal cancer. First Author: Sakti Chakrabarti, Department of Oncology, University Hospitals Seidman Cancer Center, Case Western Reserve University, Cleveland, OH

Background: Black individuals, including African Americans, experience a disproportionate cancer burden and exhibit the lowest survival rates among all racial groups for gastrointestinal (GI) cancers. The EQUITY GI study seeks to address the significant health disparities encountered by Black patients with GI cancers in the areas of biomarker testing, evidence-based care, clinical trial participation, and health literacy. Our central hypothesis postulates that a comprehensive approach—encompassing appropriate biomarker testing, ensuring evidence-based treatments, support for clinical trial participation, and health literacy initiatives—is essential to effectively mitigate health disparities among Black patients with GI cancers. **Methods:** The primary objective of this study is to promote equitable care for Black patients with GI cancers by ensuring appropriate biomarker testing and the delivery of evidence-based care. The secondary objectives include promoting clinical trial participation and enhancing health literacy. The specific goals are to increase the rate of evidence-based care, including biomarker testing, to ≥80%, increase the clinical trial enrollment rate from 5% to 15%, and improve the Cancer Health Literacy Test-30 (CHLT-30) score by 30%. Following a comprehensive assessment of the care gap, several initiatives will be implemented: 1. Tracking biomarker testing through a clinical information tracking tool (EQUITY-GI Oncotracker), 2. Providing guidance to treating providers through a molecular tumor board, 3. Facilitating referrals for clinical trials through navigators, and 4. Implementing health literacy interventions measured by CHLT-30 score. We aim to enroll 200 patients over 18 months. With this expected enrollment, the power to detect a ≥60% absolute improvement in the biomarker testing rate (from the current 20% to the expected ≥80%) is over 98%, using a two-sided chi-square test with a significance level of 0.05. To demonstrate an improvement in the clinical trial enrollment rate from 5% to 15%, 18 black patient will be required to get enrolled in a clinical trial through the referrals generated from the current project. To detect a 30% improvement in the mean CHLT-30 score (17 to 22) following health literacy interventions, at least 120 patients will be required, using a two-sided paired t-test with a significance level of 0.05. Research Sponsor: Merck.

TPS844 Trials in Progress Poster Session

A phase 1/2 study to evaluate CHM-2101, an autologous cadherin 17 (CDH17) chimeric antigen receptor (CAR) T cell therapy for the treatment of relapsed or refractory gastrointestinal cancers. First Author: Jennifer Rachel Eads, University of Pennsylvania, Abramson Cancer Center, Philadelphia, PA

Background: Patients with advanced gastrointestinal (GI) malignancies have poor prognoses and limited treatment options. Cadherin 17 (CDH17) is a cell membrane-associated protein important for GI cell-cell interactions that is expressed in proximity to epithelial tight junctions. In some GI malignancies, CDH17 expression is also expressed diffusely across the cancer cell membrane, thus providing access to CDH17 directed CAR T-cells. CHM CDH17 is a third generation autologous CAR T-cell product candidate that was designed to target and eradicate CDH17+ solid tumors. Notably, expression of CDH17 varies across indications in the cancer setting CDH17– TABLE 1. **Methods:** Clinical Trial NCT06055439 is a seamless Phase 1/2 clinical trial of autologous CHM CDH17 CAR T-cells for patients with advanced gastric cancer (GC), colorectal (CRC) cancer or neuroendocrine tumors (NETs) of the midgut or hindgut. As CDH17 expression in GC is heterogenous, GC subjects will be screened for CDH17 expression. In preparation for receiving CDH17 CAR T-cells, subjects receive 3-days of lymphodepleting intravenous (IV) chemotherapy (fludarabine and cyclophosphamide). The Phase 1 portion of the clinical trial uses a 3+3 dose escalation and a starting dose of 50 million CAR-T cells in a single IV infusion. Decisions regarding dose escalation/de-escalation are based on real-time dose-limiting toxicity evaluation during the first 28 days of treatment during dose escalation. After establishing a recommended Phase 2 dose, three indication-specific Simon 2-Stage cohorts will be initiated in (1) GC (2) CRC and (3) NETs of the midgut or hindgut to further characterize the safety and assess the efficacy of CHM CDH17 CAR T-cells. This is an ongoing first-in-human multi-center clinical trial of CHM CDH17, a CDH17 directed autologous CAR T-cell product candidate, enrolling subjects with advanced GI cancers that express CDH17. Clinical trial information: NCT06055439. Research Sponsor: Chimeric Therapeutics Ltd.

Expression of CDH17 in selected gastrointestinal cancer indications.

Indication	Sample size	CDH17 Expression %*
Colorectal cancer	821	96%
Neuroendocrine tumors of the midgut and hindgut	119	97%
Gastric cancer	1,409	64%

*Expression of total tumors tested for CDH17 by immunohistochemistry (CHM data on file).

TPS845 Trials in Progress Poster Session

MOONRAY-01, a phase 1 study of LY3962673, a potent, orally bioavailable, and selective KRAS G12D inhibitor in *KRAS G12D*-mutant solid tumors. First Author: Nehal J. Lakhani, The START Center for Cancer Research, Grand Rapids, MI

Background: *KRAS G12D* mutations occur in ~35%, 13%, and 4% of pancreatic, colorectal, and non-small cell lung cancers, respectively and less commonly in other cancers. Currently, there are no approved therapies for patients with *KRAS G12D*-mutant solid tumors. LY3962673 is a potent and orally bioavailable inhibitor of GDP-bound *KRAS G12D* that is highly selective for *KRAS G12D* against wild-type KRAS and other RAS mutations or amplifications. Preclinically, LY3962673 demonstrated robust anti-tumor activity *in vivo* as a single-agent and in combination with other agents in multiple KRAS G12D PDX models (Gong X, et al. Poster presented at AACR 2024). **Methods:** MOONRAY-01 is a global, open-label, multicenter, first-in-human phase 1 study of oral LY3962673 in participants (pts) with *KRAS G12D*-mutant solid tumors conducted in 2 phases: phase 1a dose escalation (Part A) to evaluate LY3962673 monotherapy and phase 1b dose expansion/randomized dose optimization (Parts B-E) to evaluate LY3962673 monotherapy and in combination with other anticancer therapies (Table). In phase 1a, additional pts will be allowed to backfill to previously cleared dose levels that demonstrate therapeutically relevant exposures or direct evidence of clinical activity. In phase 1b, each combination cohort will include a safety lead-in of 3-6 pts. Phase 1a/b escalation will utilize the mTPI-2 method. The DLT evaluation period will be 28 days. Eligible pts (≥18 years) must have locally advanced, unresectable, and/or metastatic cancer, measurable disease per RECIST v1.1, evidence of *KRAS G12D* mutation in tumors or ctDNA, and an ECOG PS of 0-1; asymptomatic CNS disease is allowed. In phase 1a, pts must have received ≥1 prior line of systemic chemotherapy for advanced or metastatic disease. In phase 1b, prior KRAS therapy is not allowed; other eligibility requirements are outlined in the Table. Key exclusion criteria include known active CNS metastases and/or carcinomatous meningitis, or significant cardiovascular disease. Key objectives are to determine the RP2D/optimal dose and assess the safety, tolerability, PK properties, and antitumor activity of LY3962673 per RECIST v1.1. The study is currently enrolling pts. Clinical trial information: NCT06586515. Research Sponsor: Eli Lilly and Company.

Phase 1b Parts	Key Eligibility	Cohorts
B: PDAC	· B1: ≥1 prior systemic therapy · B2, B3, B4: treatment naïve allowed · B5: treatment naïve only	· *B1: LY3962673 · B2: LY3962673 + gemcitabine (G) + nab-paclitaxel (P) · *B3: LY3962673 + G + P · B4: LY3962673 + mFOLFIRINOX · *B5: LY3962673 + mFOLFIRINOX
C: CRC	· C1, C2: ≥1 prior systemic therapy · C3, C4, C5: treatment naïve allowed	· C1: LY3962673 + cetuximab (C) · *C2: LY3962673 + C · C3: LY3962673 + C + FOLFOX · C4: LY3962673 + C + FOLFIRI · *C5: LY3962673 + C + FOLFIRI / FOLFOX
D: NSCLC	· ≥1 prior systemic therapy	· *D1: LY3962673
E: Other solid tumors	· ≥1 prior systemic therapy	· E1: LY3962673

*Cohorts randomized to various LY3962673 doses.

TPS846 Trials in Progress Poster Session

Phase 1/2 study of the trophoblast cell surface antigen 2 (TROP2) antibody-drug conjugate (ADC) sacituzumab tirumotecan (sac-TMT) as monotherapy or combination therapy in gastrointestinal cancers: LIGHTBEAM-02A. First Author: Michael J. Pishvaian, Sidney Kimmel Comprehensive Cancer Center, Johns Hopkins University, Washington, DC

Background: There is a significant need for new tolerable and effective therapeutic options for colorectal cancer (CRC), pancreatic ductal adenocarcinoma (PDAC), and biliary tract cancer (BTC) that can provide survival benefit. TROP2 is expressed broadly in several cancers, including CRC, PDAC, and BTC. Sac-TMT (formerly MK-2870/SKB264) is an ADC consisting of a humanized antihuman TROP2 monoclonal antibody, a linker, and a cytotoxic belotecan–derivative topoisomerase I inhibitor. The nonrandomized, open-label, phase 1/2 LIGHTBEAM-02A study (NCT06428409) is being conducted to examine the safety and efficacy of sac-TMT in gastrointestinal cancers. **Methods:** Approximately 130 patients will be assigned to 1 of 3 cohorts based on their disease. Patients in cohort 1 (CRC; n ≤ 50) must have previously treated locally advanced or metastatic colorectal adenocarcinoma; patients in cohort 2 (PDAC; n ≅ 40) must have previously treated locally advanced or metastatic PDAC; patients in cohort 3 (BTC; n ≅ 40) must have previously treated locally advanced or metastatic BTC. Cohort 1 is included in a dose-escalation phase to determine the recommended phase 2 dose of sac-TMT when used in combination with fluorouracil and leucovorin and an efficacy phase. In the dose-escalation phase, patients will receive sac-TMT 3 mg/kg or 4 mg/kg intravenously (IV) every 2 weeks (Q2W) on days 1 and 15 of each 4-week cycle plus 2400 mg/m^2 fluorouracil IV infused over 46-48 hours Q2W and 400 mg/m^2 leucovorin IV Q2W; in the efficacy phase, patients will receive the recommended phase 2 dose of sac-TMT with the same chemotherapy regimen as the dose-escalation phase. Patients in cohorts 2 and 3 will receive sac-TMT 4 mg/kg IV monotherapy Q2W on days 1 and 15 of each 4-week cycle. The primary objectives are to evaluate safety and tolerability and objective response rate per RECIST v1.1 by blinded independent central review (BICR). Secondary objectives are to evaluate duration of response and progression-free survival per RECIST v1.1 by BICR and overall survival. Enrollment is ongoing. Clinical trial information: NCT06428409. Research Sponsor: Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA.

TPS847 Trials in Progress Poster Session

A phase 1 dose escalation/expansion study of GSK5764227 (GSK'227), a B7-homolog 3 (B7-H3) protein targeted antibody-drug conjugate (ADC), in patients with advanced solid tumors, including gastrointestinal (GI) cancers. First Author: Wasif M. Saif, Barbara Ann Karmanos Cancer Institute, Wayne State University, Detroit, MI

Background: B7-H3 is an immune checkpoint protein overexpressed in multiple solid tumors with limited expression in normal tissues. GSK'227 (HS-20093), a novel B7-H3-targeted ADC, is composed of a human anti–B7-H3 monoclonal antibody linked to a topoisomerase I inhibitor via a protease-cleavable linker, and has shown acceptable safety and promising antitumor activity in patients of Asian origin with advanced solid tumors (NCT05276609; NCT05830123). The current study will evaluate the safety, tolerability, efficacy, and pharmacokinetics (PK) of GSK'227 in patients with solid tumors, including GI cancers, in a global population. **Methods:** This two-part (dose-escalation [1a] and expansion [1b]) global, open-label, Phase I study (NCT06551142) will enroll ~260 patients, with enrollment currently ongoing. Key eligibility includes: aged ≥18 years, histologically confirmed advanced solid tumors, including those with colorectal cancer, esophageal squamous cell carcinoma, and pancreatic cancer, with measurable disease per Response Evaluation Criteria in Solid Tumors version 1.1 (RECIST v1.1), Eastern Cooperative Oncology Group (ECOG) performance status of 0–1, and no prior B7-H3 treatment. Eligible patients will receive intravenous GSK'227 every 3 weeks (Q3W) until progression, toxicity, loss to follow-up, or death. For Phase 1a, a Bayesian optimal interval design will be used to determine the maximum tolerated dose. Phase 1a primary endpoints are safety and tolerability, including incidences of adverse events (AEs) and serious AEs. Secondary endpoints include objective response rate (ORR), disease control rate, duration of response, immunogenicity, and PK. The Phase 1b primary endpoint is progression-free survival (extensive-stage small cell lung cancer cohort) or ORR (GI and other solid tumors cohort). Secondary endpoints include additional efficacy assessments, PK, safety and tolerability. For Phase 1a and 1b, efficacy will be assessed per RECIST v1.1, with imaging conducted Q6W from first dose then Q12W after 24 weeks. Safety follow-up will be conducted at 30, 60, and 90 (±7) days after the last dose. Safety, tolerability, and efficacy analyses will be conducted using descriptive statistics and, for efficacy analyses, point estimates with 2-sided 95% confidence intervals. This abstract was previously submitted to the European Society for Medical Oncology (ESMO) Immuno-Oncology Congress 2024 and is submitted on behalf of the original authors with their permission. Funding: GSK (Study 223054). Medical writing support was provided by Avalere Health, funded by GSK. Clinical trial information: NCT05830123. Research Sponsor: GSK.

TPS848 Trials in Progress Poster Session

INSIGHT: A phase 3, randomized, open-label study of ripretinib vs sunitinib in patients with advanced gastrointestinal stromal tumor previously treated with imatinib with *KIT* exon 11 + 17/18 mutations. First Author: Michael C. Heinrich, Portland VA Health Care System and OHSU Knight Cancer Institute, Portland, OR

Background: Gastrointestinal stromal tumor (GIST) is the most common gastrointestinal sarcoma, with approximately 80% of cases driven by *KIT* mutations. Most patients (pts) with advanced GIST experience disease progression following first-line treatment with imatinib due to *KIT* secondary resistance mutations occurring most commonly in the ATP-binding pocket (exons 13/14) and/or activation loop (exons 17/18). Sunitinib is approved as second-line therapy for advanced GIST. Ripretinib is a switch-control tyrosine kinase inhibitor approved for pts with GIST who received prior treatment with 3 or more kinase inhibitors, including imatinib. In the INTRIGUE phase 3 study (NCT03673501) in second-line therapy for advanced GIST, ripretinib was not superior to sunitinib in terms of progression-free survival (PFS); however, a more favorable safety profile was observed with ripretinib vs sunitinib (Bauer S et al. *J Clin Oncol*. 2022). Exploratory mutational analysis from INTRIGUE using baseline circulating tumor DNA (ctDNA) demonstrated that pts harboring primary *KIT* exon 11 mutations with secondary resistance mutations exclusively in *KIT* exons 17/18 derived PFS benefit with ripretinib vs sunitinib (median, 14.2 vs 1.5 months; HR, 0.22; 95% CI, 0.11 to 0.44; nominal $P < 0.0001$; Heinrich MC et al. *Nat Med*. 2024). Here, we describe an ongoing phase 3 study for pts with advanced GIST previously treated with imatinib harboring *KIT* exon 11 + 17/18 mutations. **Methods:** INSIGHT (NCT05734105) is a phase 3, randomized, open-label study designed to evaluate the efficacy of ripretinib vs sunitinib in pts with advanced GIST previously treated with imatinib and who harbor *KIT* exon 11 + 17/18 mutations. Eligible pts must be ≥18 years old with histologically confirmed GIST and co-occurring *KIT* exon 11 + 17/18 mutations confirmed by ctDNA analysis. Pts must also have advanced disease with ≥1 measurable lesion per modified Response Evaluation Criteria in Solid Tumors (mRECIST) v1.1, radiologic progression on imatinib, and an Eastern Cooperative Oncology Group performance status ≤2. Key exclusion criteria include a *KIT* exon 9, 13, or 14 mutation confirmed via ctDNA analysis at screening and prior treatment with another line of therapy in addition to imatinib for advanced GIST. A total of 54 pts will be randomized (2:1) to receive ripretinib 150 mg once daily (QD; continuous) or sunitinib 50 mg QD (4 weeks on/2 weeks off) in 6-week cycles. The primary endpoint is PFS by independent radiologic review (IRR) per mRECIST v1.1; key secondary endpoints are objective response rate by IRR using mRECIST v1.1 and overall survival. Safety and patient-reported outcome measures will also be assessed. Pts randomized to the sunitinib arm may cross over to the ripretinib arm upon disease progression. This abstract was originally presented at ASCO 2023. Clinical trial information: NCT05734105. Research Sponsor: Deciphera Pharmaceuticals, LLC.

TPS849 Trials in Progress Poster Session

A prospective multi-site translational study investigating the association of gut microbiome (GM) diversity with pathological complete response (pCR) after neoadjuvant treatment in early stage rectal and esophageal cancer. First Author: Rachel Clarke, Beaumont RCSI Cancer Centre, Dublin, Ireland

Background: The diversity of the GM is defined as the number and relative abundance distribution of distinct types of microorganisms colonizing within the gut. Studies have suggested that dysbiosis of the GM confers a predisposition to certain malignancies and impacts response to immunomodulating therapies. The influence of the GM diversity on the pathological response after neoadjuvant chemotherapy and radiotherapy is unclear. Some studies have suggested that the GM may offer predictive biomarkers for response to chemoradiation in rectal cancer. Other studies in early-stage rectal cancer patients indicated an association between GM diversity and pathological outcomes following neoadjuvant therapy (NAT). We hypothesize that a more diverse GM constitution at baseline leads to an improved pathological response at the time of definitive surgery. **Methods:** We designed a cross-institutional translational study investigating the impact of the GM diversity on the efficacy of NAT in GI cancers by assessing its association with pathological response. The study population includes patients with early-stage rectal or esophageal cancer due to commence NAT (including chemotherapy and chemoradiation) who are planned for definitive surgery. Exclusion criteria includes prior allogenic tissue/solid organ transplantation and prior receipt of anti-cancer therapy. The study assessments include fecal sampling of the GM prior to NAT, upon completion and again six months post completion of therapy. Fecal samples are analysed by 16S RNA sequencing. Pathological response will be examined at time of surgery and patients will be classified as responders (complete pathological response) or non-responders. The primary endpoint of the study is to examine the association between the GM diversity and pathological response. Exploratory analysis will include the assessment of the association between cf-DNA and the GM diversity as well as an assessment of the association between cf-DNA at baseline and pCR. Species richness (Alpha Diversity) will be analysed using the Shannon diversity index and Jaccard similarity index will be used to calculate beta diversity. Following planned study recruitment, classification and clustering analysis will be performed with Principal Component Analysis (PCA) and Random Forest analysis. Logistic regression analysis adjusting for potential confounding factors will be employed to assess the primary endpoint of the association between GM and complete pCR in the final statistical analysis. Adjusted odds ratios (OR) and 95% confidence intervals will be presented. This trial accrued 11 patients between May 2023 and Sept 2024. Out of the 11 patients enrolled, 9 patients have undergone their planned surgery. We are expecting to have 30 patients accrued prior to Jan 2025. Research Sponsor: Science Foundation Ireland and Breakthrough Cancer Research.

TPS850 Trials in Progress Poster Session

Evaluating novel therapies and circulating tumor DNA (ctDNA) as a marker of response in curatively treated gastrointestinal cancers with microscopic residual disease. First Author: Reetu Mukherji, Ruesch Center for the Cure of Gastrointestinal Cancers, Lombardi Comprehensive Cancer Center, Georgetown University Medical Center, Washington, DC

Background: ctDNA can identify the earliest sign of relapse (MRD) after curative therapies with specificities of 93-100%, positive predictive values over 95%, and median lead times of 8-9 months before radiographic relapse. ctDNA has identified a "new stage" of high-risk patients (pts) with no evidence of disease radiographically (rNED) and MRD who lack treatment (tx) strategies as, traditionally, they wait for radiographic progression of disease (POD) before starting therapies. Treating MRD is a promising strategy supported by observational studies where ctDNA clearance is linked with improved survival and on-tx kinetics predict drug efficacy**.** We designed a pilot investigator-initiated trial to 1) determine the feasibility of using ctDNA as a rapid signal for positive therapeutic activity and 2) obtain pilot data on the efficacy of a novel combination therapy (atezolizumab [A] + bevacizumab [B]) in eradicating MRD in previously curatively treated pts with GI cancers. **Methods:** We will enroll 20 pts with any-stage cancers divided over 4 cohorts (n=5 colorectal [CRC], n=5 hepatocellular/biliary tract carcinoma, n=5 upper GI, n=5 pancreatic adenocarcinoma) who have a positive Signatera ctDNA test and rNED any time after completing standard curative therapies. Pts are treated with A 1200 mg IV + B 15 mg/kg IV on day 1 of a 21-day cycle for up to 1 year with imaging every 12 weeks and serial ctDNA testing every 3 weeks. Co-primary endpoints are rates of pt enrollment in 12 months and rates of ctDNA responses at a 12-week landmark after tx initiation. Pts with ctDNA POD (ctDNA doubling on each of 2 sequential tests, any rate of rise in 3 sequential tests, or radiographic relapse) will stop while those with ctDNA complete response (CR; clearance on 2 sequential tests + ongoing rNED) or ctDNA partial/stable response (PR; not ctDNA CR or POD + ongoing rNED) will continue tx. Each cohort's Bayesian predictive probability of positive therapeutic activity will be calculated. If at least 60% of a cohort experiences ctDNA CR, it will be expanded by 5 pts. This scenario is equivalent to Simon's 2-stage Minimax design testing a null hypothesis of 20% CR vs. an alternative hypothesis of at least 60% CR at 5% alpha with 80% power. If at least 5/10 pts in an expanded cohort experience ctDNA CR, this will generate interest in developing a larger confirmatory trial to evaluate pt survival outcomes with A+B and ctDNA as a surrogate for survival. Secondary endpoints are toxicity and enrollment barriers. Exploratory endpoints are associations between ctDNA conversion and disease-free survival, tumor whole exome sequencing data, and peripheral blood immune profiles collected at baseline and on tx. Enrollment began in Q1 2023, paused in 7/2023 due to funding changes, and re-opened 8/2024 with 5 patients enrolled currently (n=4 CRC, n=1 biliary). Clinical trial information: NCT05482516. Research Sponsor: Genentech, Inc; Natera, Inc.